Informatica University

Training Guide

Multidomain MDM SaaS: Manage Multidomain Data

Version: MDMSaaS-JUL24-MMD-202407

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Multidomain MDM SaaS: Manage Multidomain Data

Version: MDMSaaS-JUL24-MMD-202407

July 2024

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This guide uses the following formatting conventions:

If you see…	It means…	Example
>	Indicates a sub menu to navigate to.	Click Repository > Connect. In this example, you should click the Repository menu or button and choose Connect.
boldfaced text	Indicates text you need to type or enter.	Click the Rename button and name the new source definition **S_EMPLOYEE**.
UPPERCASE	Database tables and column names are shown in all UPPERCASE.	T_ITEM_SUMMARY
italicized text	Indicates a variable you must replace with specific information.	Connect to the Repository using the assigned *login_id*.
Note:	The following paragraph provides additional facts.	**Note:** You can select multiple objects to import by using the Ctrl key.
Tip:	The following paragraph provides suggested uses or a Velocity best practice.	**Tip:** The m_ prefix for a mapping name is…

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Introduction

Multidomain MDM SaaS is an industry-first, cloud-native, enterprise-scale solution for managing all domains of master data.

It is a reimagined, efficient, and modern Master Data Management (MDM) system that is a simple yet intelligent service. It benefits from the Informatica Intelligent Data Management Cloud, which provides capabilities to move data from source systems, process it, and publish a version from a trusted master to downstream consuming systems. This service helps you share and reuse clean and up-to-date data and provides a trusted 360-degree view of the master data, that organizations can use to make informed business decisions.

The first module of this guide gives an overview of Informatica Intelligent Data Management Cloud (IDMC). The second module covers the Multidomain MDM and Business 360. The third module assists you in defining business entities that suit the business requirement. The fourth module discusses migrating assets and how you can export and import the assets. The fifth module walks you through the process of defining hierarchies and relationships. The sixth module demonstrates the steps to design the user interface for the Multidomain MDM application. The seventh module covers the topic of configuring data quality rules. The eighth module helps in creating the custom user roles and assigning privileges to them. The ninth module guides you in onboarding the data into the Multidomain MDM application. The tenth module talks about managing hierarchies and analyzing the data model through the relationship graph. The eleventh module discusses the process of configuring the match model and adding declarative rules to it. The twelfth module explains how to define a job and create business events. The thirteenth module demonstrates the process of creating custom reports and adding these reports to the dashboard.

About the Guide

This is a comprehensive guide to the Multidomain MDM SaaS service and its functionalities. This guide includes concepts along with the hands-on exercises to be performed by the learners.

Lab Environment

You will be provided with a virtual lab environment where you will execute the lab exercises. Use the credentials provided to access the lab environment.

Module 0: Getting Started

Lab 0-1: Starting the Environment

Overview:

Multidomain MDM SaaS: Manage Multidomain Data has one Virtual Machine (VM). For the lab environment to work, we need to have the VM and the Cloud environment up and running. In this lab, you will start the VM and the IICS environment.

Objectives:

- Start the VM in the lab environment
- Start IICS from the browser
- Create a user with multiple roles

Duration:

20 minutes

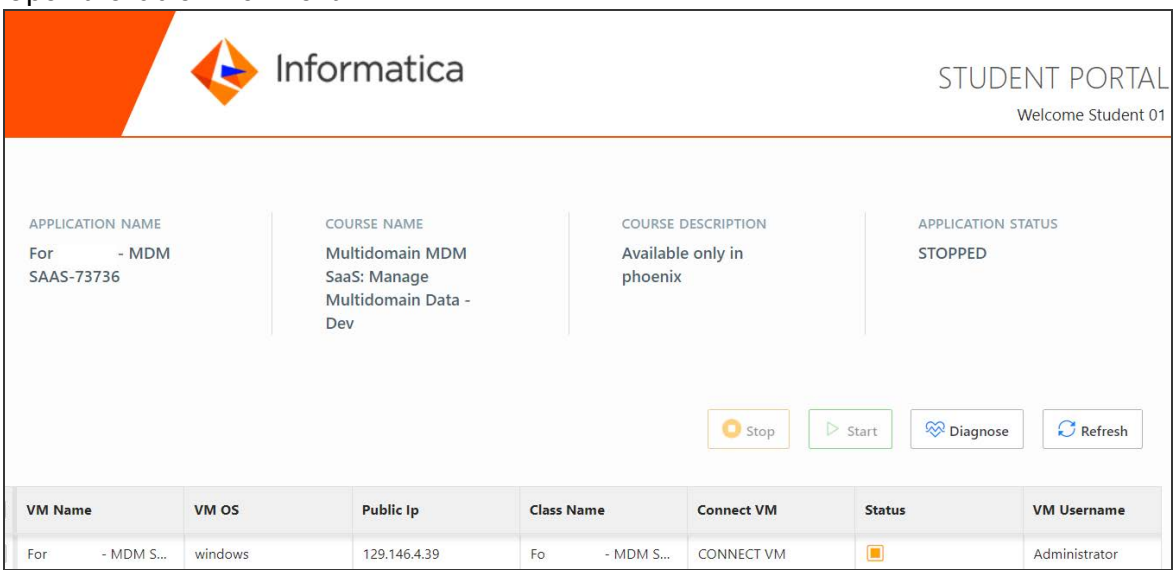
Tasks:

- Starting the environment
- Create Multidomain MDM admin user role with multiple privileges

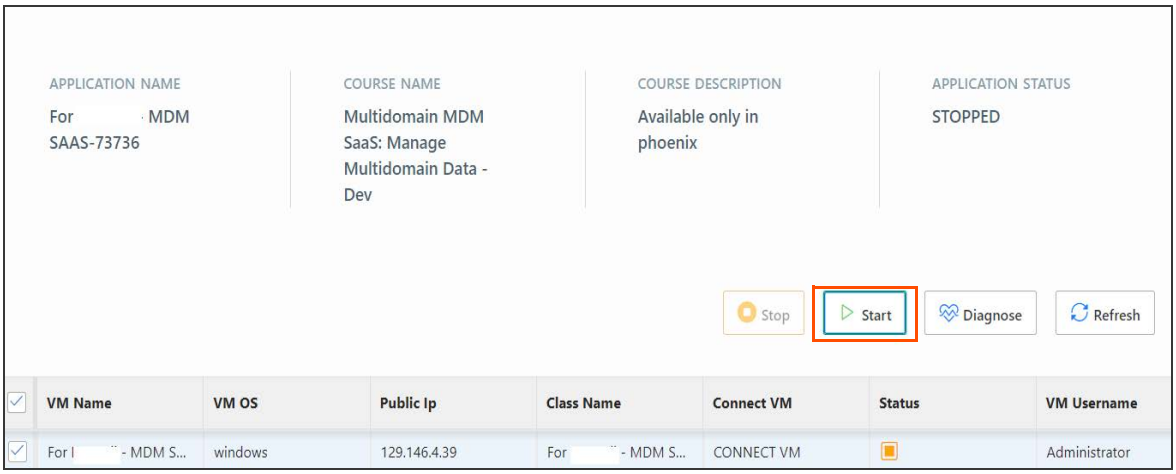
Starting the Environment

You will initially launch the virtual environment where you will conduct your labs.

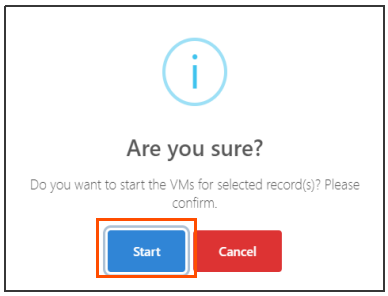
1. Open the lab environment link.



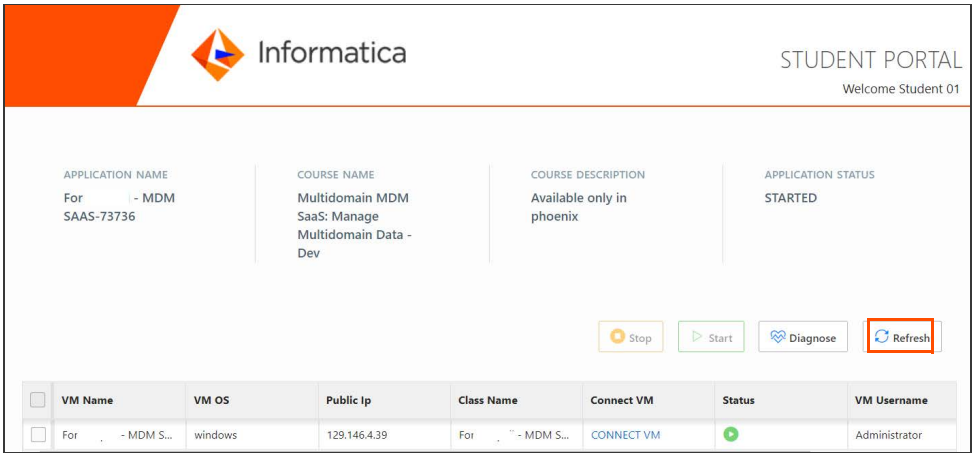
2. Select the VM and click **Start**.



3. A confirmation window appears. Click **Start**.

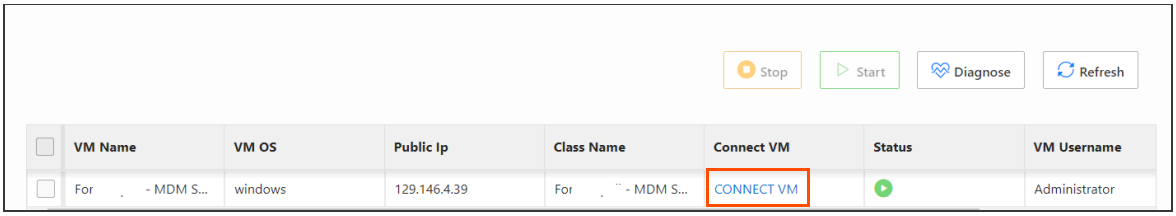


It will take a few minutes for the VM to start. Refresh the browser or click the Refresh button on the page.

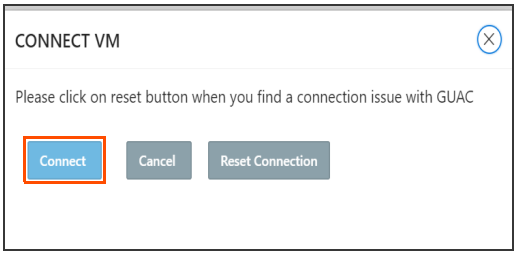


4. You can see that the VM is started.

5. Click the **CONNECT VM** link of the **Multidomain MDM** VM.

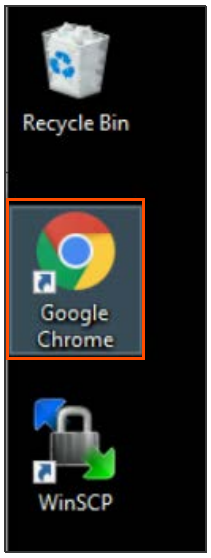


6. Click **Connect**.

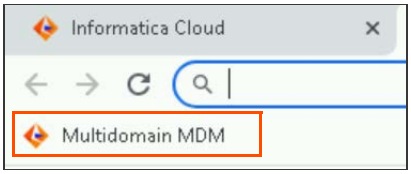


VM opens in a new tab.

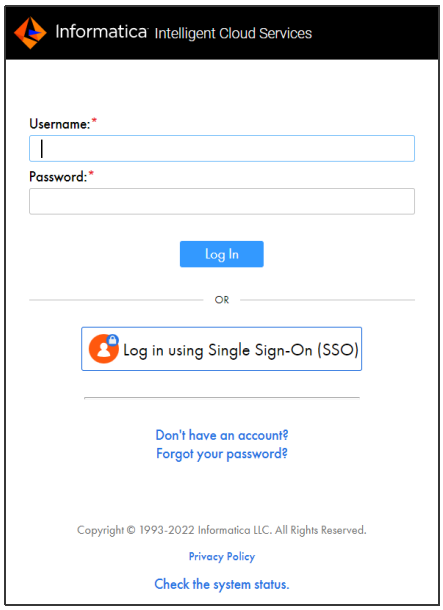
7. Open the **Chrome** browser from the desktop.



8. Click the bookmark **Multidomain MDM**.



IICS login page opens.

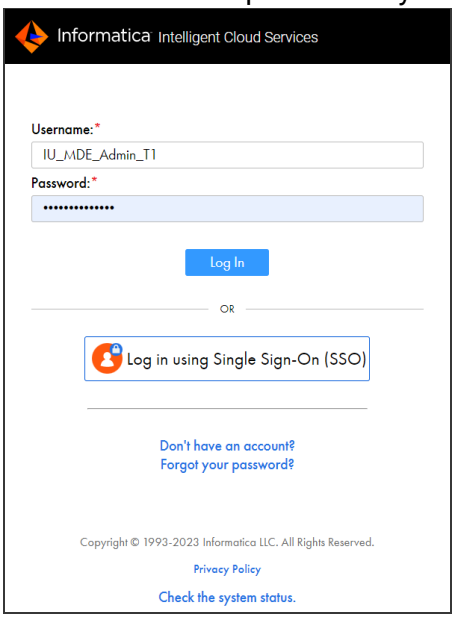


You are now all set to execute the labs of **Multidomain MDM SaaS: Manage Multidomain Data** course. Prior to that, you will be creating Multidomain MDM SaaS user role with multiple privileges to create and execute Multidomain MDM related assets.

Create Multidomain MDM Admin User Role with Multiple Privileges

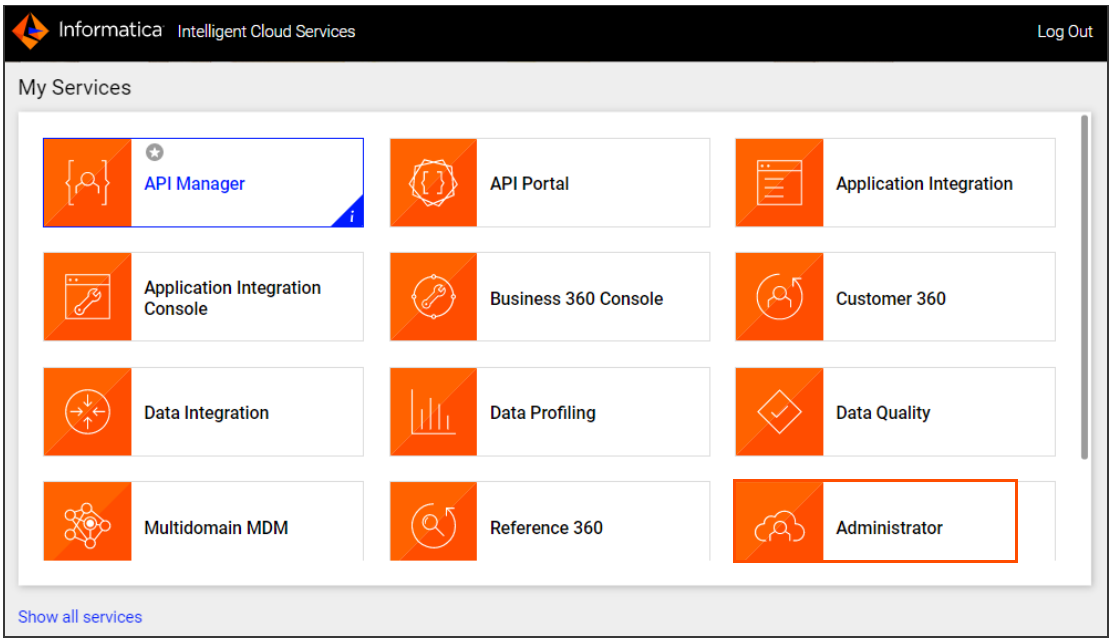
You will now create a user with admin role and assign other privileges to it.

9. Enter the username and password as **IU_MDE_Admin_T1 / Multidomain360**. (Use the IICS admin credentials provided to you by the instructor during class).

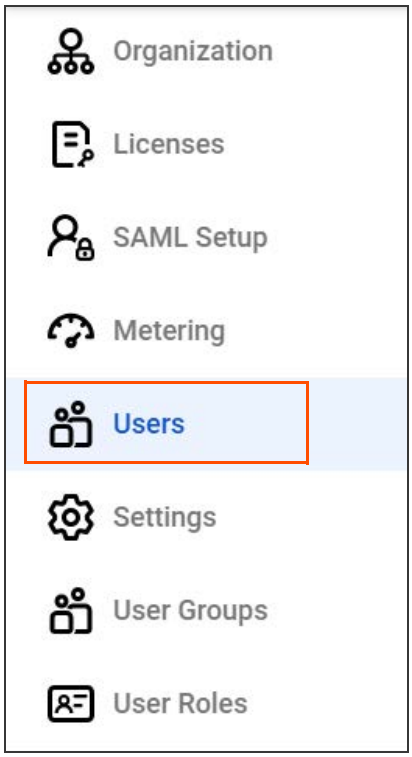


IU_MDE_Admin_T1 is the IICS admin user created by the instructor. You will login as this user to create the Multidomain MDM users.

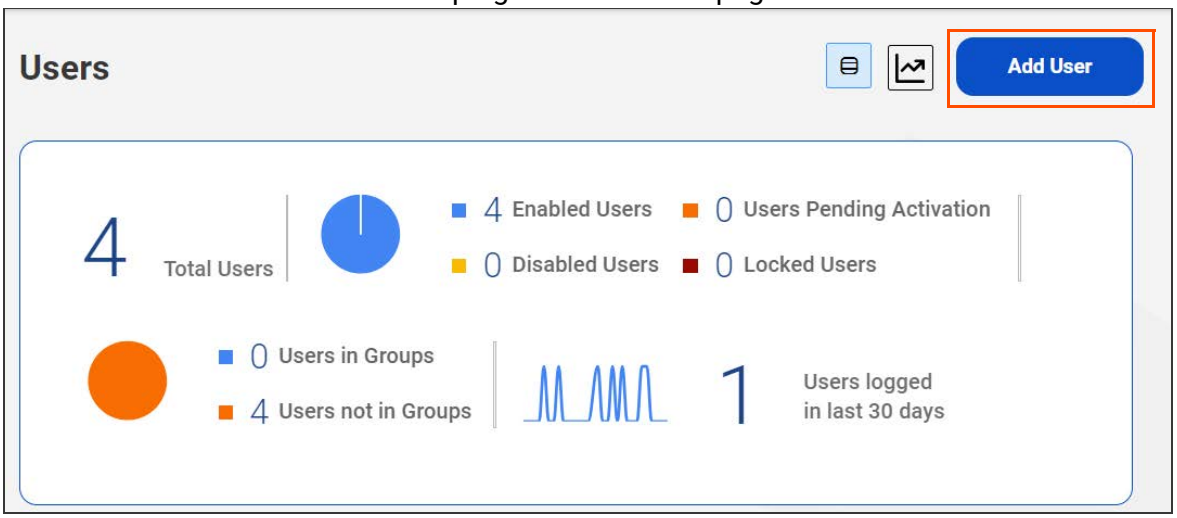
10. Click **Administrator**.



11. Click **Users**.



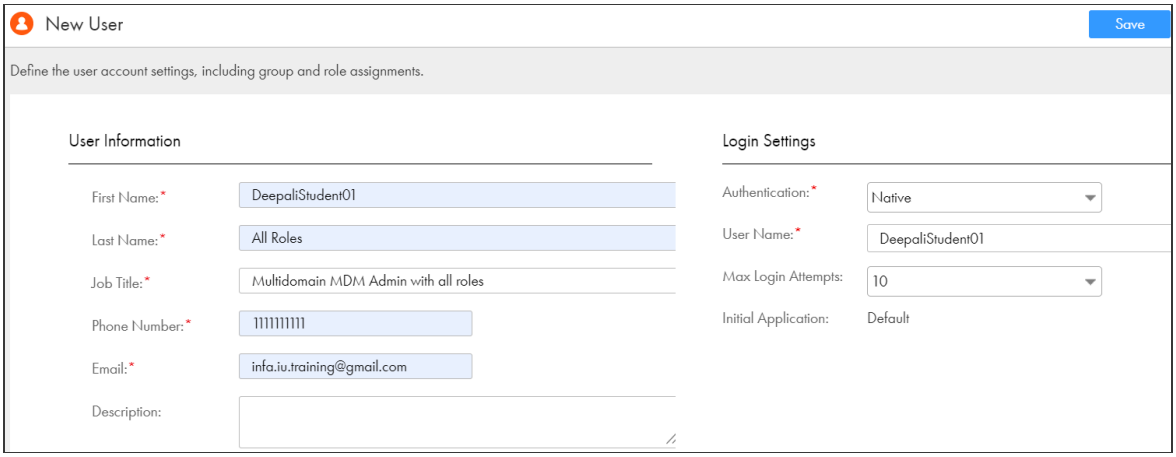
12. Click the **Add User** button at the top right corner of the page.



As a student, you need to create all the Multidomain MDM roles required for the class. The format can be **StudentnameStudent<no>**, such as, JohnStudent01, LisaStudent02, and so on. In the screenshots you may see a different name, but make sure you use your name and number while creating these users. You will first create a Multidomain MDM Admin user.

13. Enter the following details:

Field	Value
First Name	StudentnameStudent<no.>
Last Name	All Roles
Job Title	Multidomain Admin with all roles
Phone Number	<Your Phone Number>
Email	<Your email ID>
Description	Multidomain MDM Administrator Role
Authentication	Native
Username	Studentnamestudent<no.>
Max Login Attempts	10



14. Scroll down to the **Assigned User Groups and Roles** section.

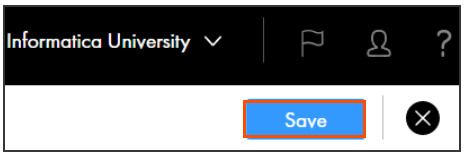
15. Select the **Admin, Business360ProcessExecutor, Designer, Job Executer, MDM Designer, Monitor, Operator, Service Consumer** and **MDM Business User** role. Also select all the Reference 360 roles that are available.

Enabled	Role Name	Description
☑	Admin	Role for performing administrative tasks for ...
☐	Application Integration Business ...	Role used for business managers
☐	Application Integration Data Vie...	Role used for granting access for data
☐	Application_Administrator	null
☑	Business360ProcessExecutor	null
☐	Business_Steward	null
☐	Customer 360 Analyst	Customer 360 role for Analysts.
☐	Customer 360 Data Steward	Customer 360 role for Data Stewards.
☐	Customer 360 Manager	Customer 360 role for Managers.
☐	Data Integration Data Previewer	Role to preview data
☐	Data Integration Task Executor	Role to run Data Integration tasks
☐	Dataloader Admin	Role for the Dataloader Admin User.
☐	Deployer	Role used by deployer
☑	Designer	Role for creating assets, tasks, and process...
☐	Governance Administrator	Governance Administrator role for Metada...
☑	Job Executor	null

☑	Operator	Role used for monitoring exe...
☑	Reference 360 Administrator	Role for configuring the Refer...
☑	Reference 360 Business Analyst	Role for viewing and exporti...
☑	Reference 360 Business Steward	Role for creating and managi...
☑	Reference 360 Planner	Role for creating and managi...
☑	Reference 360 Primary Owner	Role for defining and configu...
☑	Reference 360 Stake Holder	Role for contributing changes...
☑	Reference 360 User	Role for logging into Referen...
☐	Sales_Operation_Analyst	null
☐	Sales_Operation_Manager	null
☑	Service Consumer	Role for running tasks, taskflo...

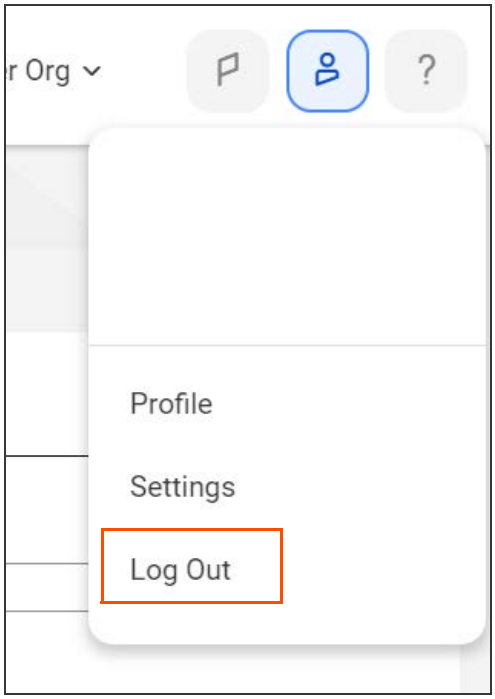
☑	Designer	Role for creating assets, tasks...
☐	Governance Administrator	Governance Administrator ro...
☐	Job Executor	null
☐	Key Admin	Role to manage key rotation ...
☑	MDM Business User	Role that provides access to ...
☑	MDM Designer	MDM Designer
☐	Model Serve System Role	Model Serve role for perfor...
☑	Monitor	Role used for application mo...

16. Click **Save**.

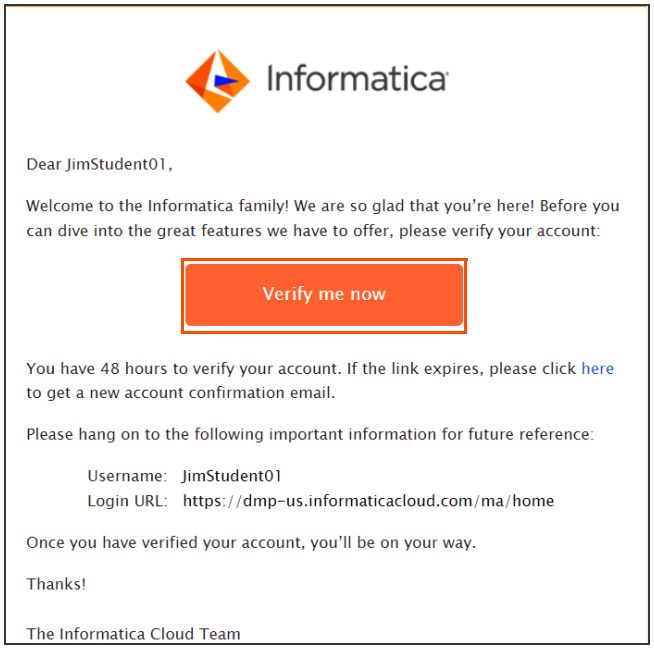


17. Close the New User window.

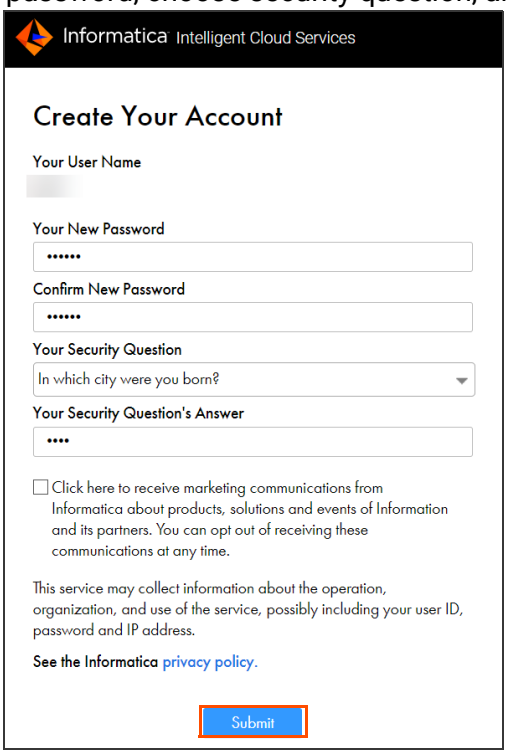
18. Log out of **IU_MDE_Admin_T1** profile.



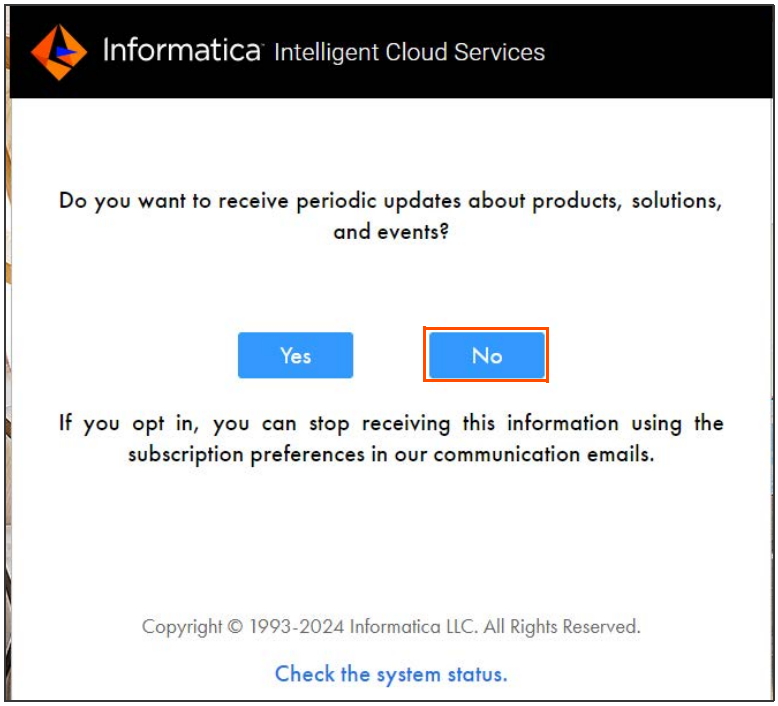
19. You will get an email notification to set the user password. Click **Verify me now**.



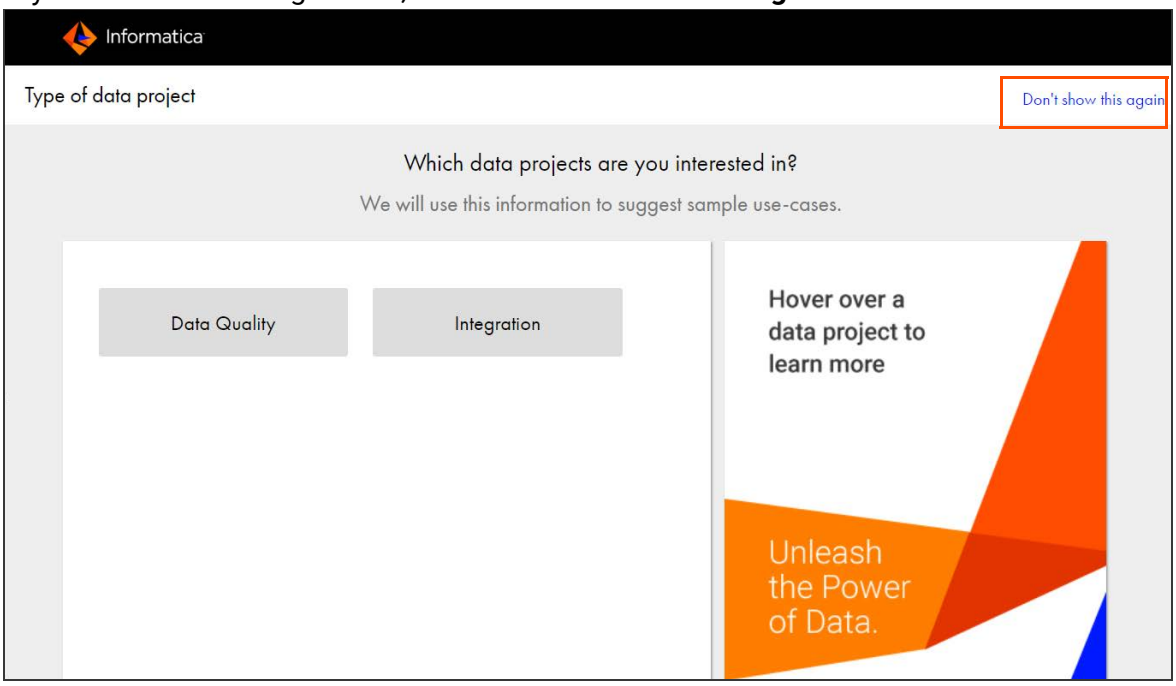
20. In the Set Up Your Password page, set the password as **Multidomain360**, Verify the password, choose security question, and enter answer for that and click **Submit**.



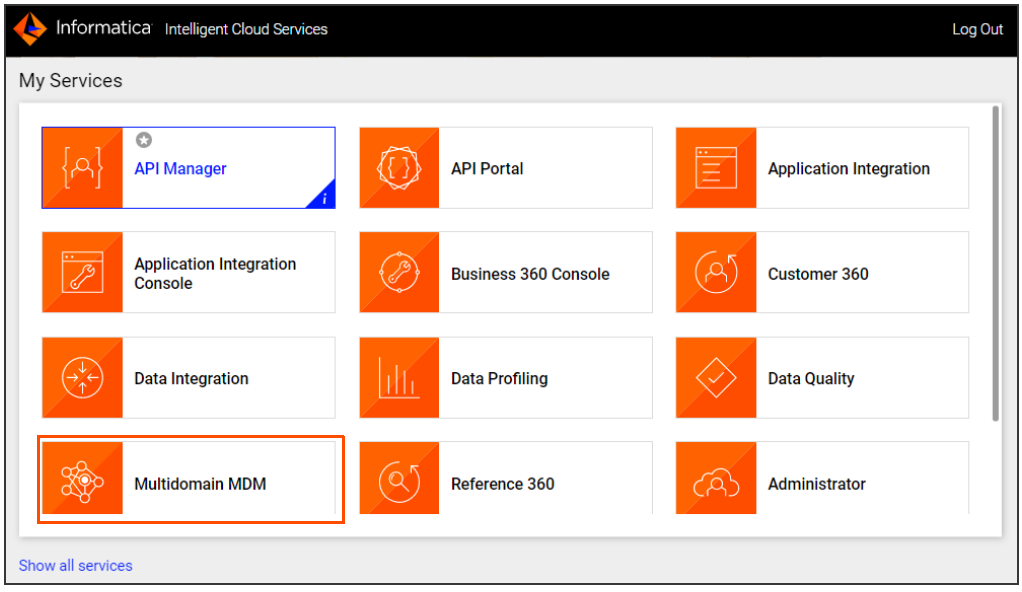
21. Click **No**.



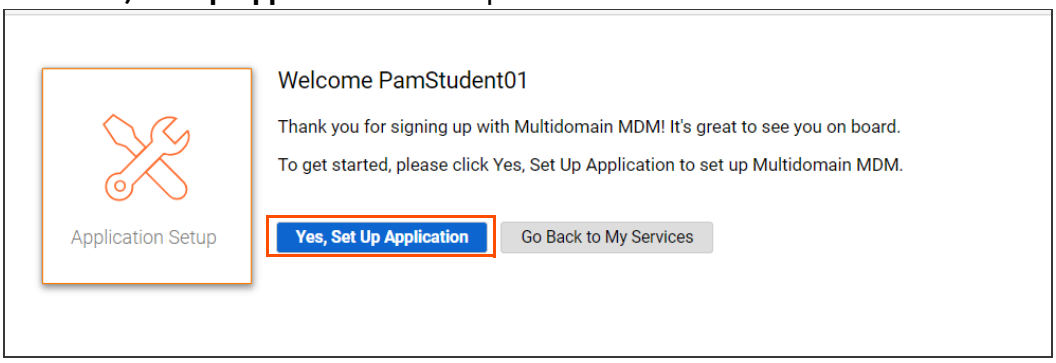
22. If you see the following screen, then click **Don't show this again**.



23. You are now logged in as StudentnameStudent<no>, which is an admin user. Click **Multidomain MDM** to launch the service.

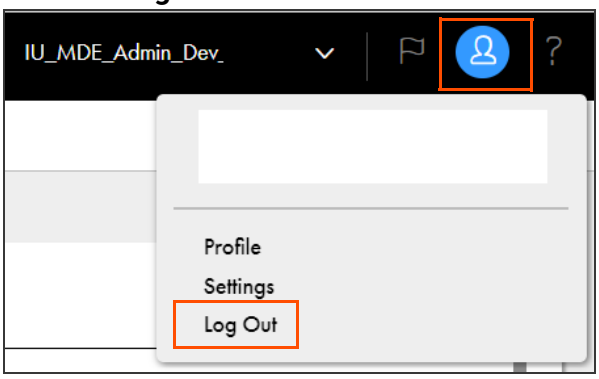


24. Click **Yes, Set Up Application** to set up the Multidomain MDM service.



Setting up the application may take 5 to 7 minutes.

25. You will land into the **Multidomain MDM** home page. Click user icon on the top right corner and click **Log Out**.



This concludes the lab.

Module 1: Informatica Intelligent Data Management Cloud (IDMC) Overview

Scenario

Homekart: Gearing up for the Holiday Season

It's the holiday season and shoppers are eager to spend money! Homekart appliance is gearing up for the busy shopping season but understands that the careful inventory management is critical to profitability. This year especially due to pandemic related shortages, appliances are in short supply and back orders can take months to fulfill. Lack of Inventory means fewer sales, lower profits, and possibly poor customer satisfaction.

Homekart is using a Multidomain MDM system to get a 360-degree understanding of :

- Storefronts
- Products and inventory per store
- Suppliers for each product
- They use a relationship graph to monitor their operational activities, inventory and suppliers

Hierarchical Structure for StoreFronts

Homekart has stores all over the GTA(Greater Toronto Area). These are modelled as DNB style hierarchical relationships in the Multidomain MDM system. Each site offers a selection of their appliances to end users. The Data Stewards job is to ensure there are no imbalances in the system which may impact sales.
These include checking:

- Low inventory in some stores
- Excess inventory in other stores
- Delivery dates and units delivered from suppliers

Introduction to IDMC

Informatica Intelligent Data Management Cloud (IDMC) is a comprehensive cloud-based data management platform provided by Informatica. It is designed to help organizations manage data across multiple cloud and on-premises environments. The platform offers a suite of tools and capabilities for a wide range of data management operations. Informatica IDMC is targeted at businesses looking to streamline their data management processes, ensuring data reliability and availability across varied systems and environments. This platform is particularly beneficial in complex data landscapes and for enterprises undergoing digital transformation.

IICS stands for Informatica Intelligent Cloud Services. It is a next-generation iPaaS solution that can be used to exchange data between applications or business partners. You can use IICS to integrate, synchronize, and relate data, applications, and processes residing on-premise or in your cloud environment. IICS is the term used for the services in Informatica Data Management Cloud (IDMC).

You can use IDMC to perform the following business processes:

- **Importing or Migrating Data** – Businesses often need to import data from an external system or migrate data from one contact management system to another.
- **Object Synchronization** – Many organizations use IDMC to synchronize copies of data in multiple systems. For example, if you have “Account” and “Contact” data in Salesforce, and the same information in an Accounting system, then you can use IICS to synchronize such information in each of these systems.
- **Process Integrations** – This helps you connect one process with another and update both the systems with appropriate information. For example, you have a Salesforce system where you track all the sales-related activities, such as opportunities. When an opportunity closes, you can update that information in your order management system.
- **Replication and Archiving** – This involves taking a backup of your data regularly or at scheduled intervals. You can also replicate the data to an on-premise database to run analytics.

Figure 1-1 shows the Informatica Intelligent Cloud Services (IICS) User Interface(UI) that provides a consistent look and feel across all intelligent cloud services and experiences tailored to user roles such as Designer, Operator, and Administrator, using a common user interface shell and service switcher. The new design also introduces the concept of workspaces which allows users to keep multiple tabs open within a cloud service.

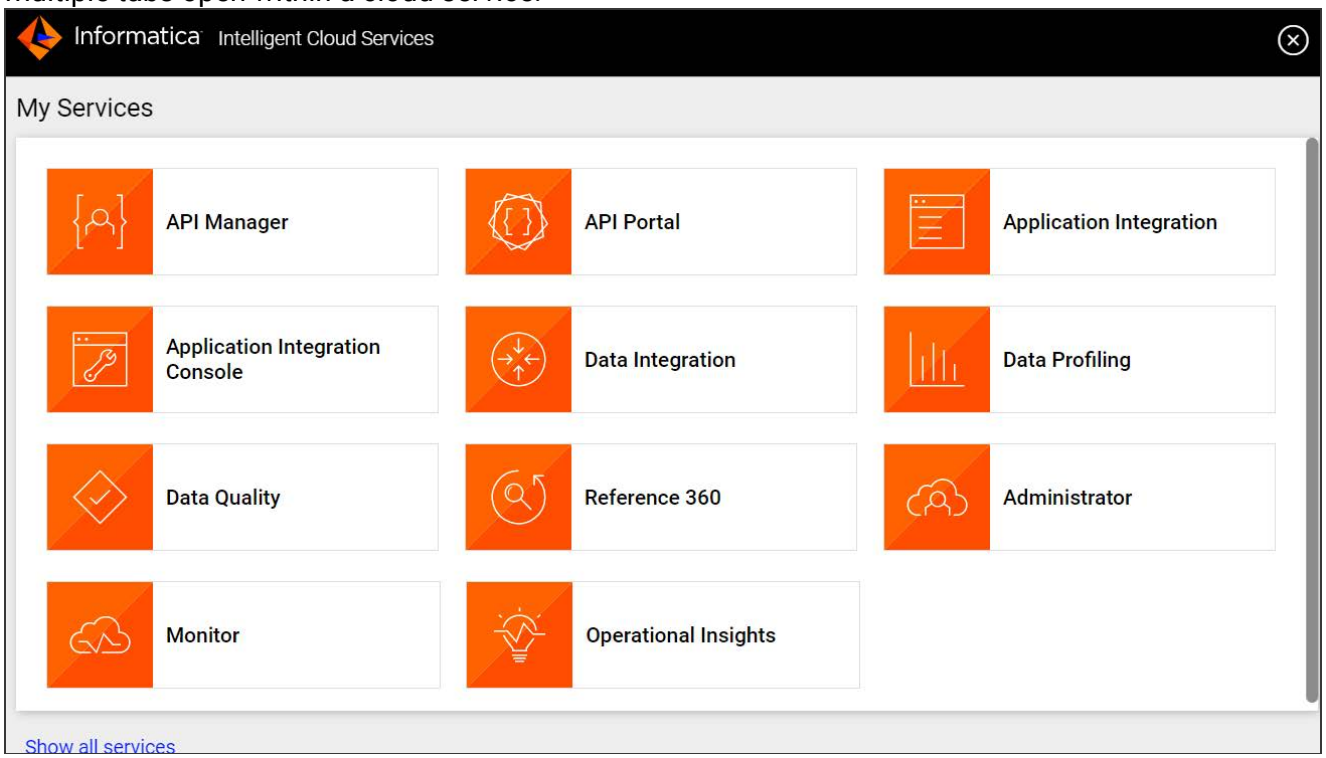


Figure 1-1: IICS Interface

Terminologies Used in IDMC

- **Connection**: A Connection provides the information that IICS needs to connect to your data sources and targets. For this training, you would need to create connections to connect to your data source and bring the metadata from them into the catalog.

- **Org**: An Org is an IICS organization. Generally, you have one Org per company where all users log in and create objects. However, in some cases, a company may have a Production and a Sandbox Org. Large companies may have separate business units which require separate Production Orgs.
- **Secure Agent**: The Secure Agent runs behind the company firewall and ensures that scanners can access your data sources and perform tasks such as ingestion of metadata into the catalog, profiling of data, and so on. You will learn more on the Secure Agent in the Runtime Environment module.
- **Asset**: An Asset can be described as a specific piece of IICS functionality, such as mappings, connections, and so on. In terms of Data Governance and Catalog, assets are classified into two – Business Assets and Technical Assets. A Business Asset includes business terms, policies, rules, and so on, while a technical asset includes catalog sources, tables, columns, and so on.

IDMC Architecture

IICS is an iPaaS-based application. This means that you can access it from any machine with internet access and a web browser application.

When a user accesses the application, the web browser connects to the Informatica Cloud Services through a secure HTTP. The Informatica Cloud Services include the IICS repository that stores various information about the task.The Data Quality functionality can be found in the Data Quality and Profiling Services. Mappings can be built and run in the Data Integration Service. The application has multiple wizards that guide users to build and test Data Quality tasks. IICS also allows you to schedule and monitor the jobs.

IDMC is the modular solutions that meet your needs—all on a single platform. IDMC combines Informatica's 260+ intelligent cloud offerings, including but not limited to Data Catalog, Data Quality, MDM, Data Integration and Data Governance.

Figure 1-2 shows the IDMC architecture.

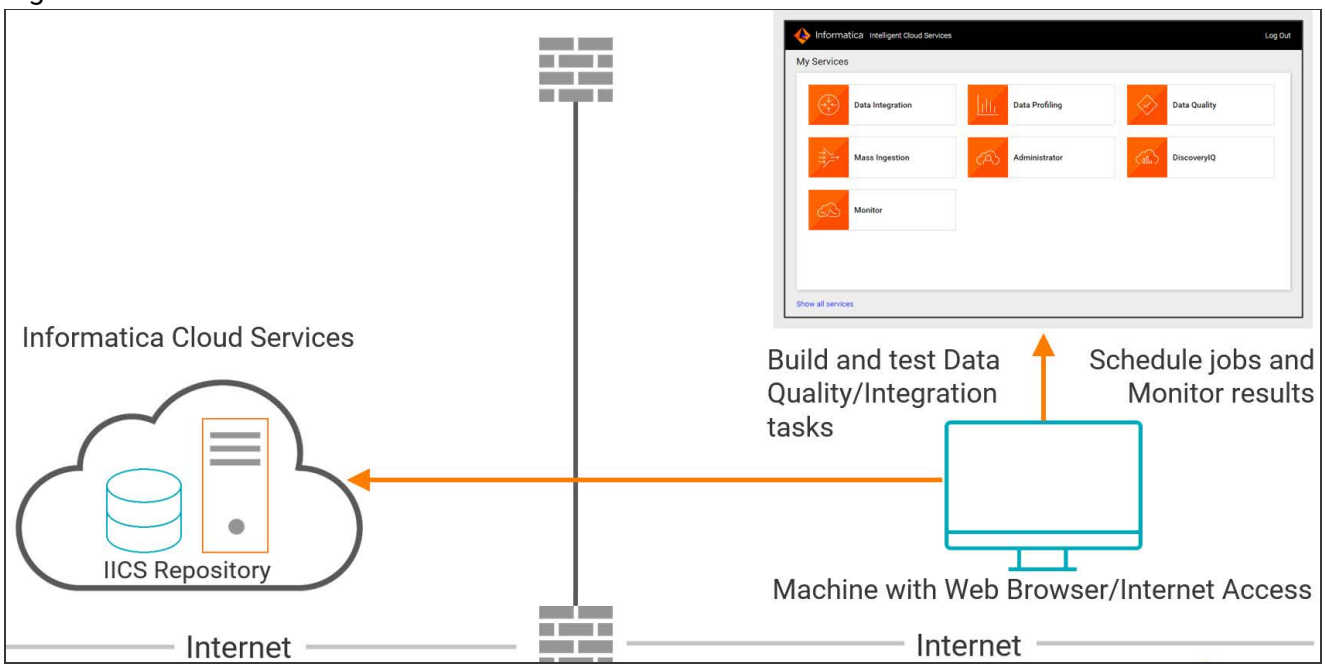


Figure 1-2: IDMC Architecture

Capabilities of the IDMC Platform

Informatica IDMC provides complete, comprehensive cloud-native and AI-powered data management capabilities:

- Data catalog intelligently scans metadata to discover and understand your enterprise data. Y
- Data integration accesses and integrates data at scale using elastic and serverless computing.
- API and application integration connect your applications and automate end-to-end business processes and workflows.
- Data quality automates insights and delivers consistent, trusted and governed data across the enterprise.
- Master data management (MDM) and 360 applications enable you to innovate with 360-degree views of business data and trusted intelligent insights.
- Governance and privacy automates data intelligence to govern, protect and democratize your data.
- A data marketplace offers the industry's only cloud-native, AI-powered solution for data sharing

The platform holds all these services together with the help of visual tools for the technical users and self-service wizards for the business users. The platform also encompasses Data Governance and ensures that administration and security are seamless across the network. You can expose the dataset using REST API and utilize reusable codes from Informatica Data Marketplace. The Informatica Data Marketplace is an open platform to host solutions that support all phases of the Data Integration lifecycle. It is important to note that the IDMC platform allows you to use OEM Embedded Solutions.

Figure 1-3 shows various capabilities the IDMC platform provides.

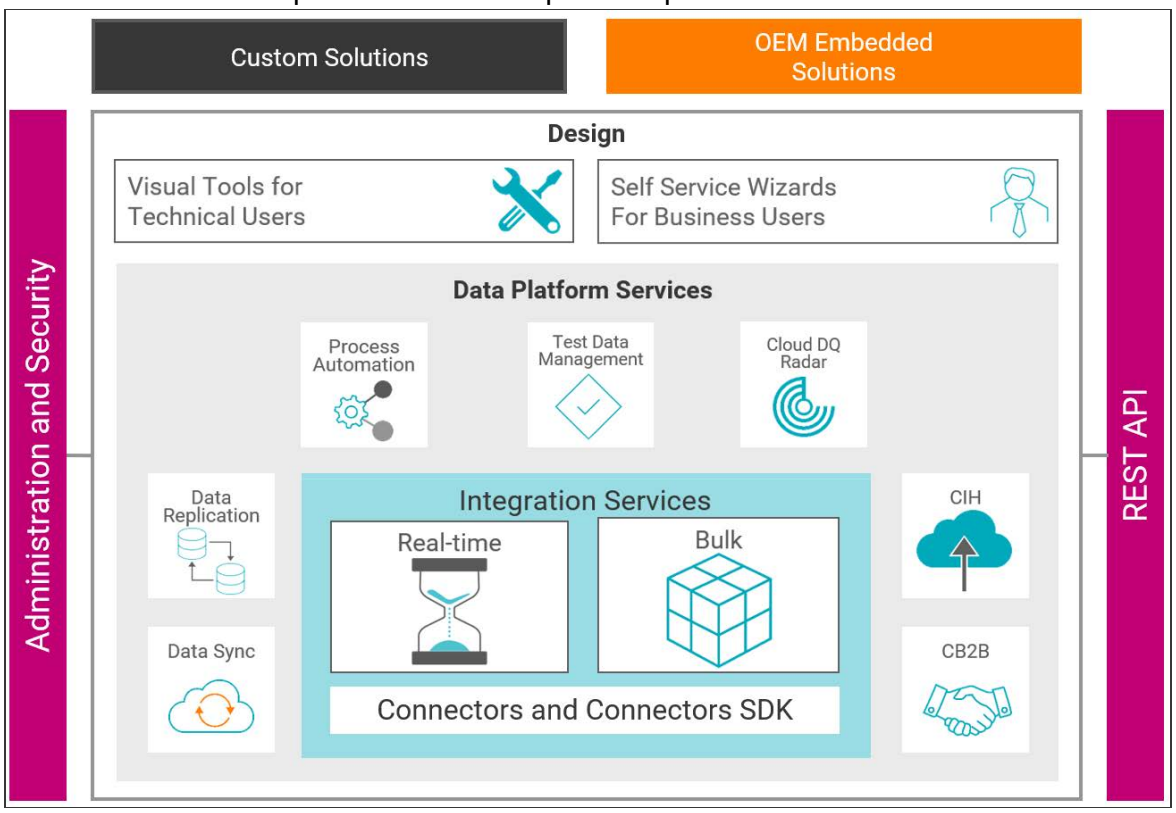


Figure 1-3: IICS Platform Capabilities

Cloud Administrator

The Cloud Administrator is used to configure Org settings. Administrator provides organization management capabilities across Informatica Intelligent Cloud Services. Depending on the users permissions, they may or may not have access/limited access to the Administrator. It can be used to configure a range of settings including Users and Groups, Runtime Environment, Connections, and Schedules.

You can use Administrator to manage the following aspects of your organization:

- Organization and sub-organizations
- Licenses
- Metering
- Ecosystem and SAML single sign-on
- General and security settings
- Users, user groups, and user roles
- Permissions
- Runtime environments
- Serverless runtime environments
- Secure Agent services
- Advanced clusters

- Schedules
- Add-on bundles
- Data services repository
- Event monitoring
- File transfer

For this Multidomain MDM course you will be using the Administrator services like: Runtime Environments, Users, user groups, and user roles and Organization and sub-organizations.

Informatica Cloud Runtime Environments

A Runtime Environment is the execution platform that runs a Data Integration or Application Integration task. To run tasks in your organization, you must have at least one Runtime Environment set-up in your organization. A runtime environment consists of one or more Secure Agents.

Informatica Cloud supports two runtime environments:

- Informatica Cloud Hosted Agent
- Informatica Cloud Secure Agent

You can set up runtime environments in the following ways:

- **Use the Informatica Cloud Hosted Agent**: When you use the Hosted Agent, you run tasks within the Informatica Cloud hosting facility. Informatica maintains the Hosted Agent runtime environment and agents. For more information about the Informatica Cloud Hosted Agent, see Hosted Agent.
- **Create one or more Secure Agent groups**: You can install one or more Secure Agents to run within your network or in a cloud computing services environment such as AWS, Google Cloud, Microsoft Azure, or Oracle Cloud Infrastructure. You can install one Secure Agent on each physical or virtual machine. When you install a Secure Agent, it is added to its own group by default. You can add multiple agents to one Secure Agent group. For more information about Secure Agent groups, see Secure Agent groups.
- **Configure a serverless runtime environment**: If your cloud environment is AWS, you can configure a serverless runtime environment. This environment is hosted by Informatica, so you don't need to configure a Secure Agent or Secure Agent group. For more information about serverless runtime environments, see Serverless runtime environments.

Secure Agents

The Informatica Cloud Secure Agent is a lightweight program that runs all tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services. When the Secure Agent runs a task, it connects to the Informatica Cloud hosting facility to access task information. It connects directly and securely to sources and targets, transfers data between them, orchestrates the flow of tasks, runs processes, and performs any additional task requirement. If the Secure Agent loses connectivity to Informatica Intelligent Cloud Services, it tries to reestablish connectivity to continue the task. If it cannot reestablish connectivity, the task fails.

You can install and run one Secure Agent on a physical or virtual machine. After you install a Secure Agent, all users in the organization share the Secure Agent. You can configure the Secure Agent

properties and move it to a different Secure Agent group. To improve scalability, you can also add multiple agents to a Secure Agent group.

Secure Agent Architecture

Figure 1-4 shows Secure Agent Architecture.

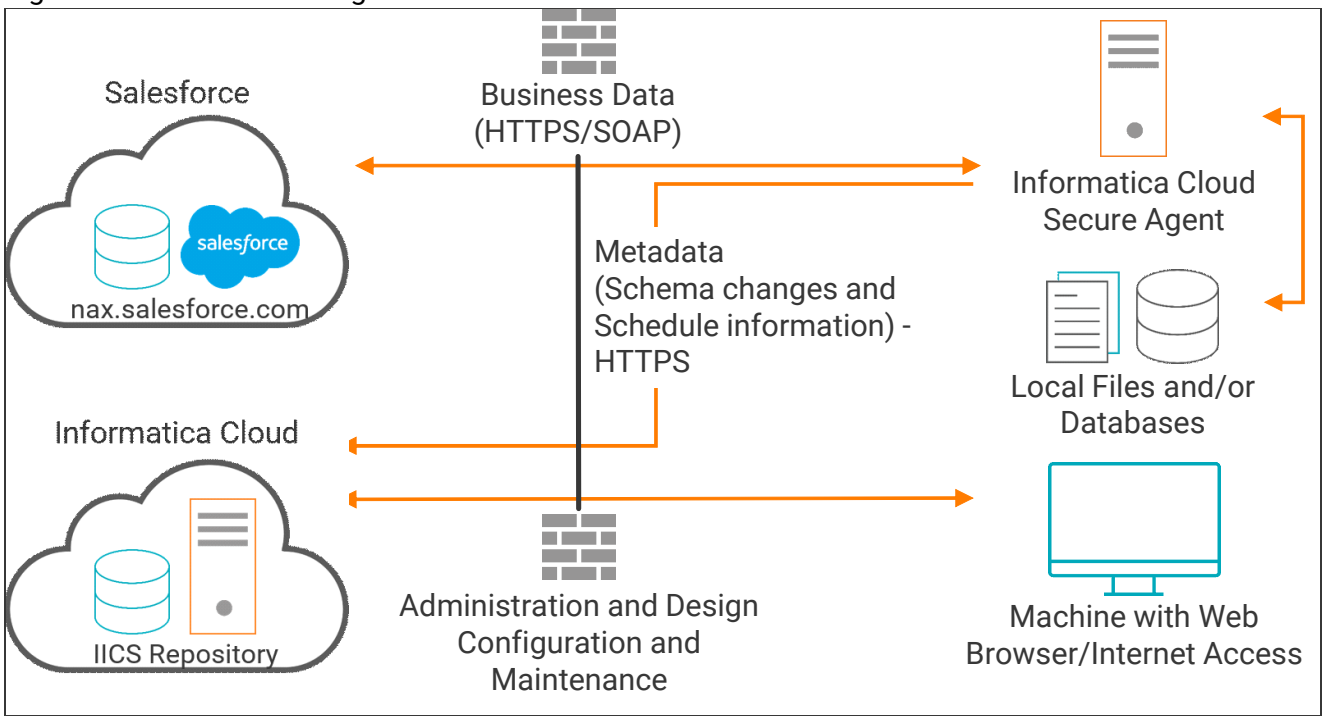


Figure 1-4: Secure Agent Architecture

Connections

Connections can be configured in the Administrator. They provide access to data in cloud and on-premise applications, platforms, databases, and flat files.They specify the location of sources, lookup objects, and targets. A connection becomes available to all users in the Org. A secure agent must be installed in the Org before creating any connection.

You can create connections in IDMC by selecting “Connections” from the Administrator Services. You can also create connections on-the-fly using the wizard when you configure a task. When you create a connection, the connection becomes available to all the users in the Org. Except for Software-as-a-Service (SaaS) applications, all connections depend on the Secure Agent. Therefore, you must first install the Secure Agent in your IDMC Org before creating any connection.

Lab 1-1: Installing Secure Agent

Overview:

The Secure Agent is a lightweight, self-upgrading program that runs inside your network. It is responsible for moving data from the source to a target. IICS Secure Agent runs all tasks and enables secure communication between your organization and Informatica Cloud. Secure Agent is needed for this training, as you would be ingressing/onboarding data from SQL Server and other source systems.

In this lab, you will set up the Secure Agent.

Objectives:

- Download the Secure Agent
- Install the Secure Agent

Duration:

20 minutes

Scenario:

Mark is employed by Homekart Appliances as a Data Architect. He is experiencing some challenges at Homekart due to the lack of an MDM system. He wants to build a data model, but before proceeding with the same, Mark should complete preliminary tasks in administrative services. This includes downloading and installing a secure agent.

Tasks:

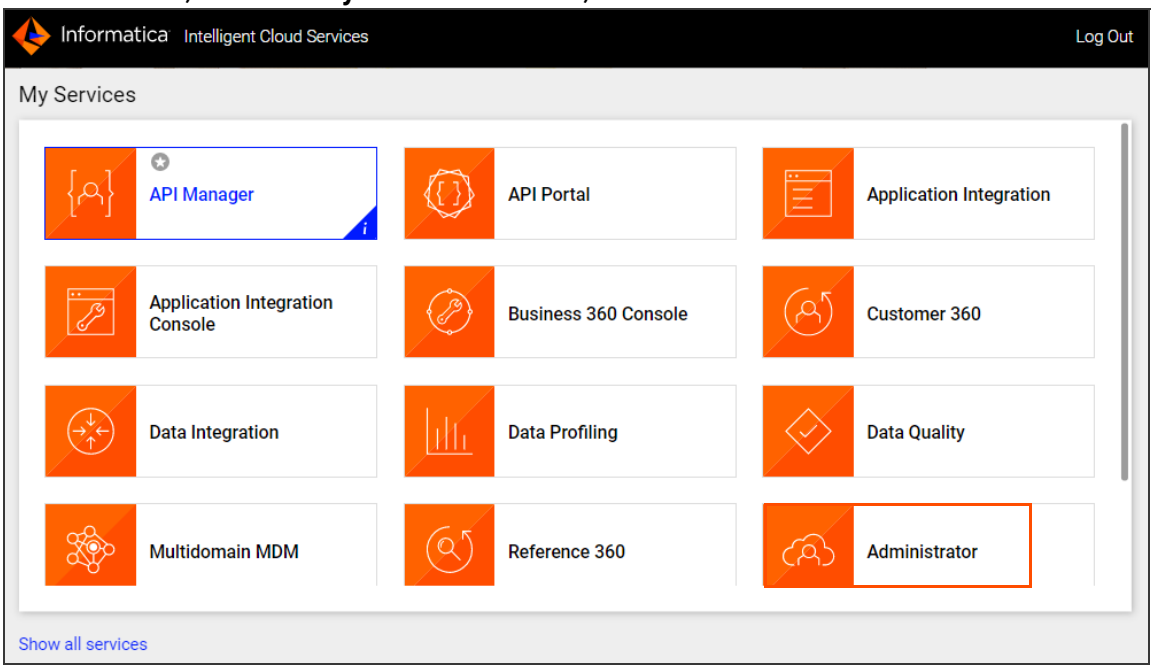
- Download the Secure Agent
- Install the Secure Agent
- Enable Services and Connectors on Secure Agent

Download the Secure Agent

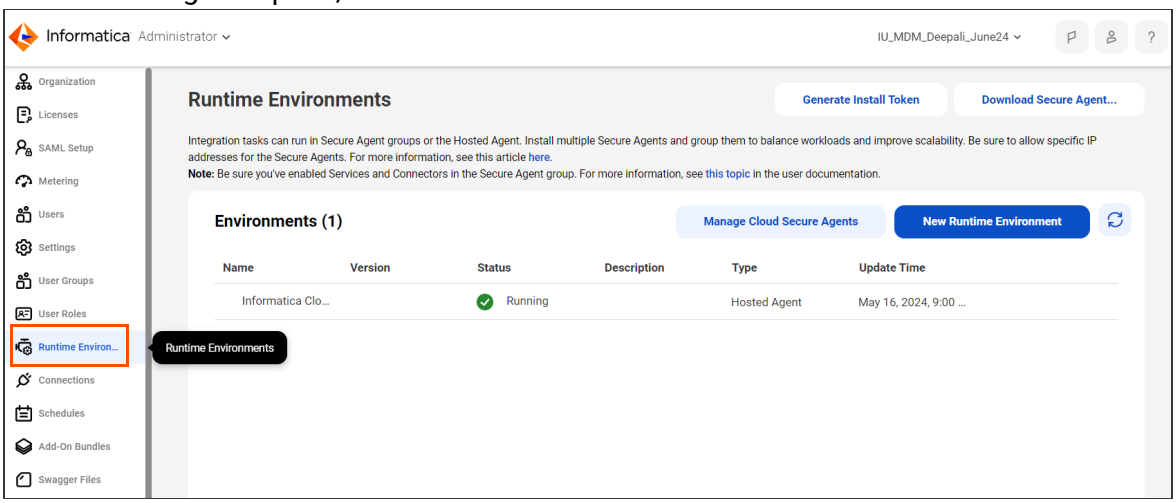
As the first task, you will download the executable file from the Runtime Environments.

1. Log in to IICS using Multidomain MDM admin credentials that you created in the previous lab. **StudentnameStudent<no> / Multidomain360**. (Ex: JimStudent01)

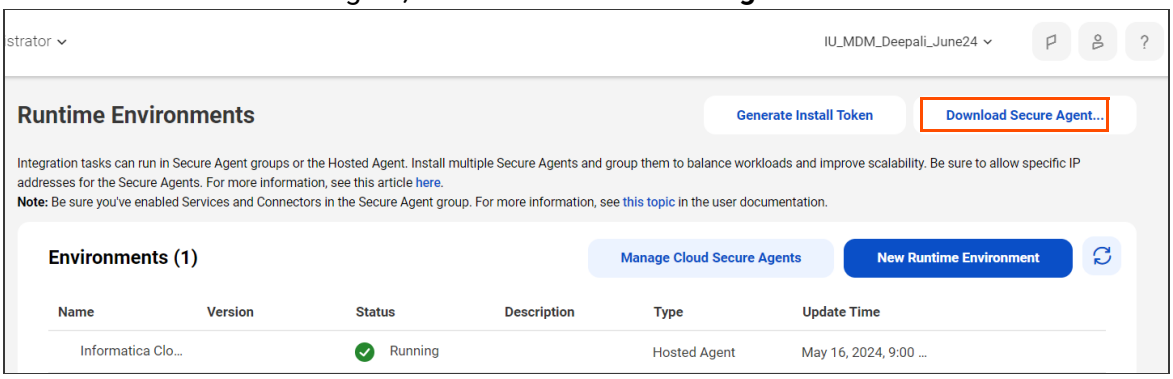
2. In the IICS tab, from the **My Services** window, click **Administrator**.



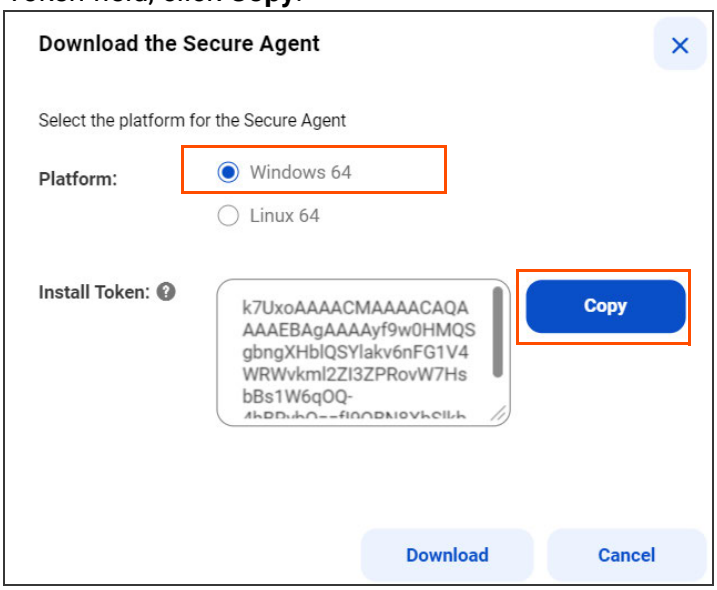
3. From the navigation pane, click **Runtime Environments**.



4. To download the Secure Agent, click **Download Secure Agent**.



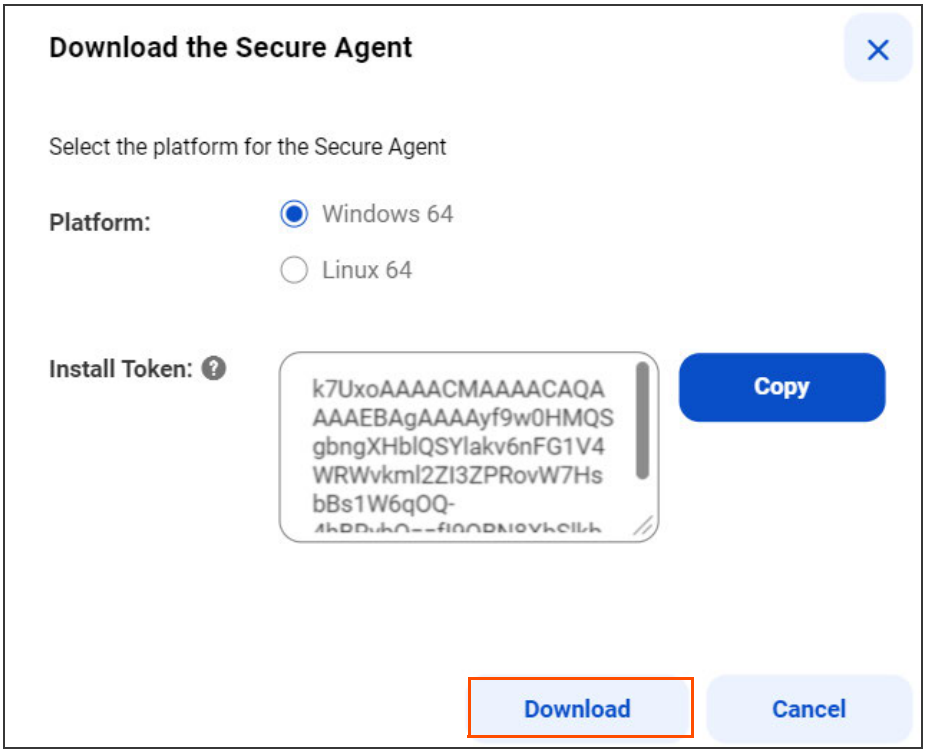
5. From the **Download the Secure Agent** window, select **Windows 64**, and from the Install Token field, click **Copy**.



This will copy the token to your clipboard.

6. Paste the copied token in a text file. You will need this token later.

7. Click **Download**.



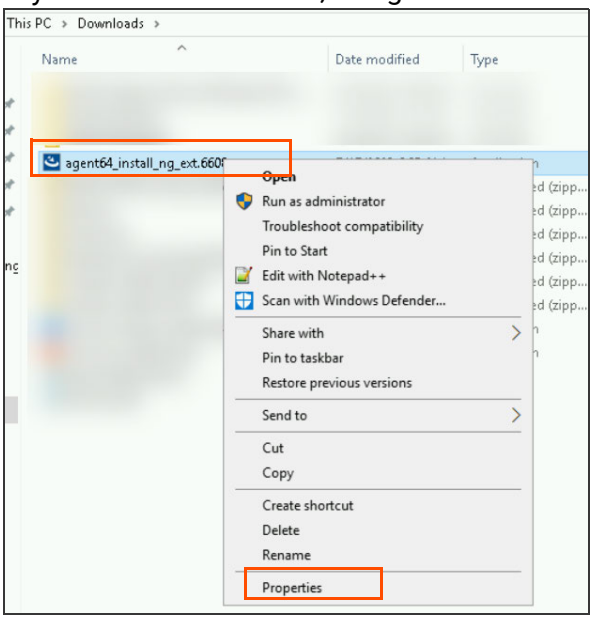
Install the Secure Agent

Locate the downloaded executable file and install the secure agent.

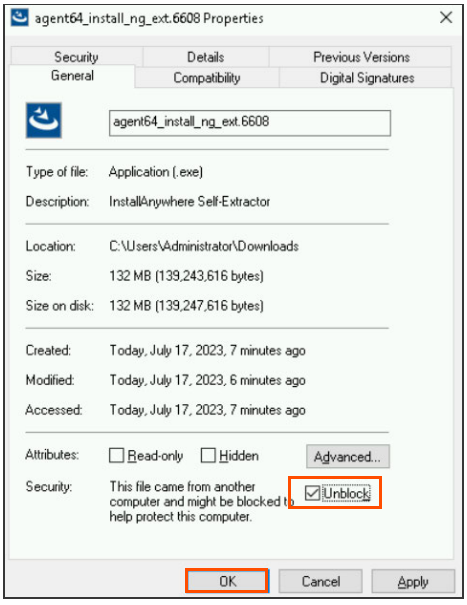
8. Go to the download directory on your machine and locate the agent installation file.

9. To install the Secure Agent, run the executable file **agent64_install_ng_ext.exe**. Proceed with the step 13 if you can run the executable file. If not, follow the troubleshooting steps below.

10. If you cannot run the file, navigate to the executable file, right click and select **Properties**.

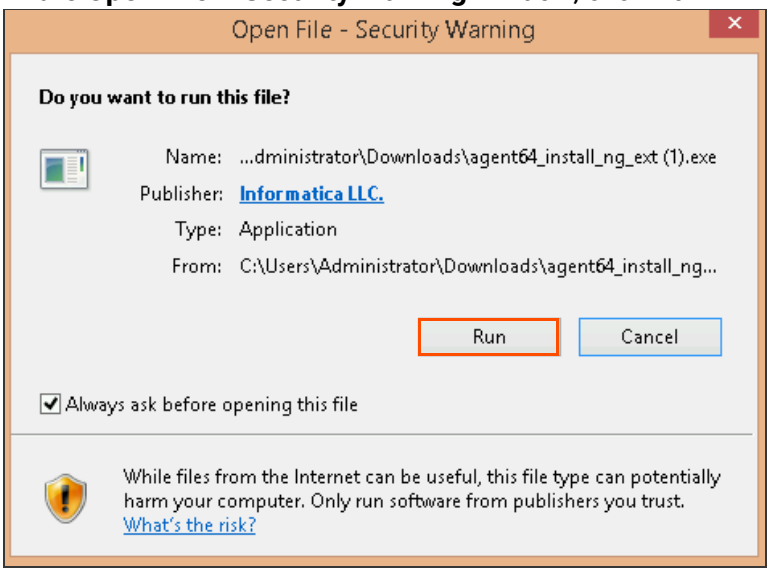


11. In the General tab, select **Unblock** check box and click **OK**.

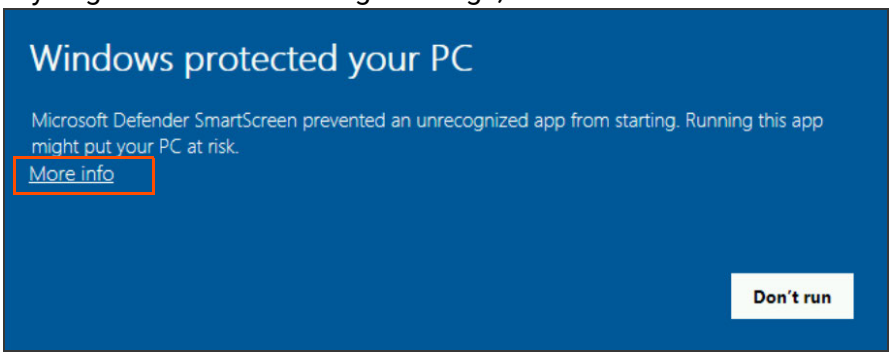


12. Double click on the executable file **agent64_install_ng_ext.exe** to run it.

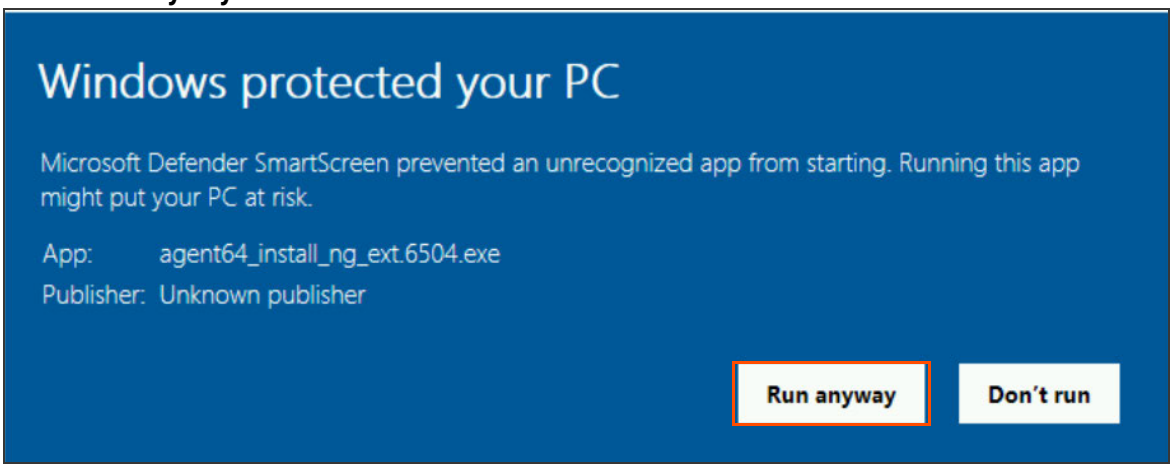
13. In the **Open File – Security Warning** window, click **Run**.



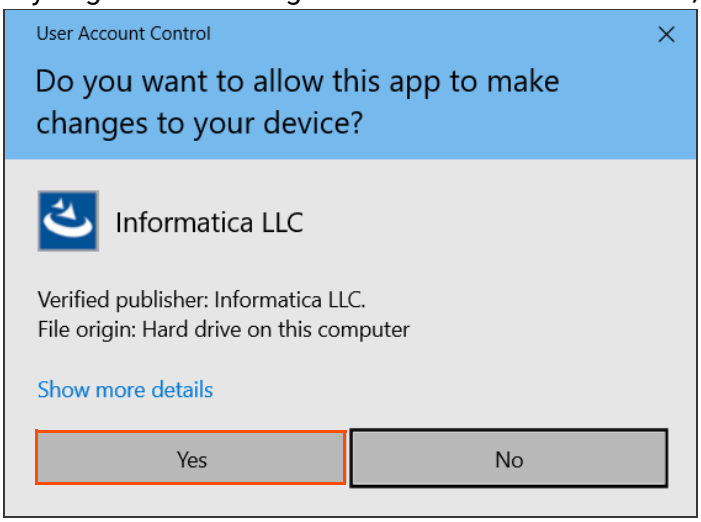
14. If you get the below warning message, click **More info**.



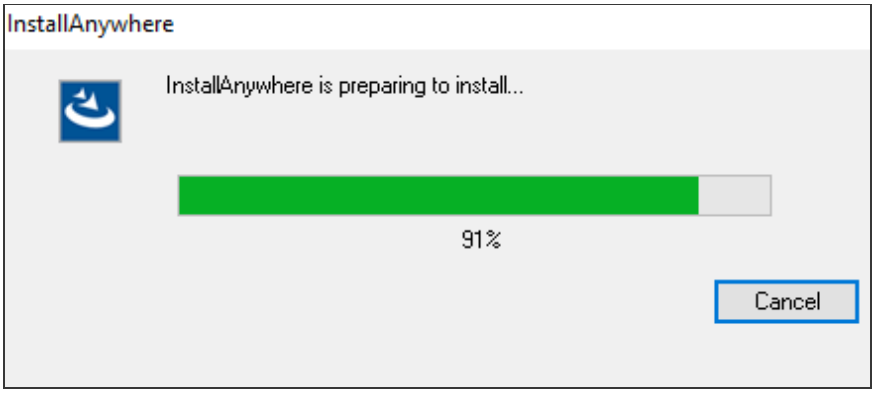
15. Click **Run anyway**.



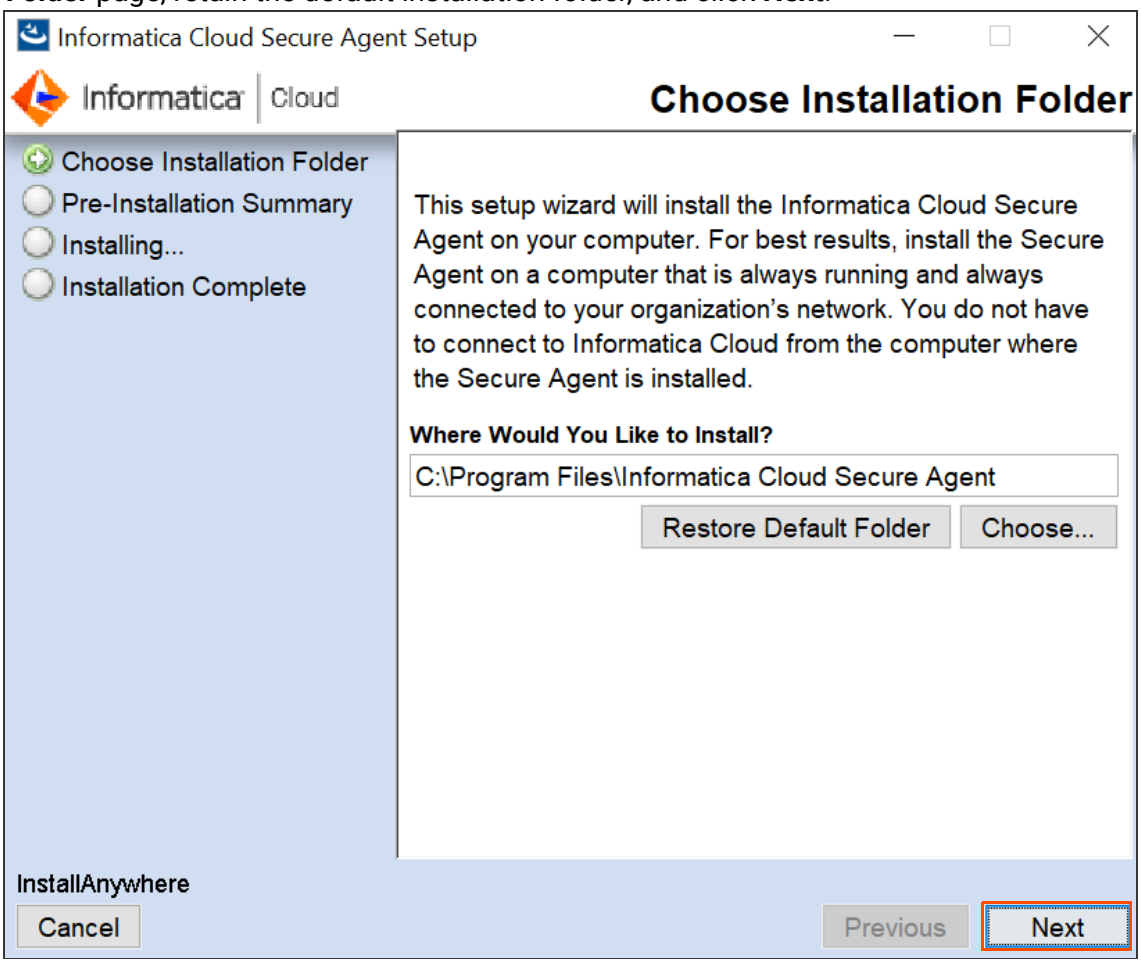
16. If you get the following User Account Control window, click **Yes**.



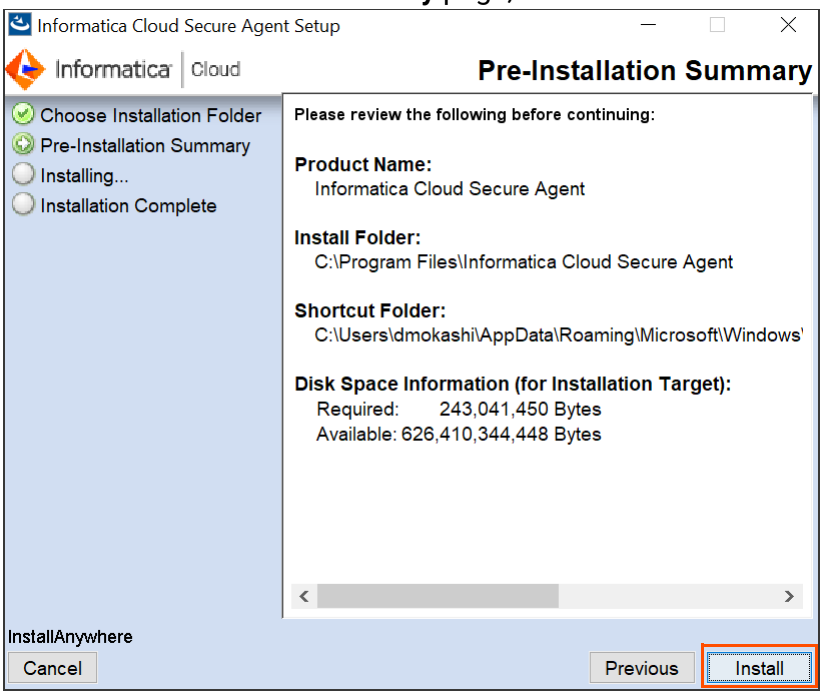
When the Secure Agent initiates the installation, it displays the **InstallAnywhere** window with the installation progress.



17. The **Informatica Cloud Secure Agent Setup** wizard opens. In the **Choose Installation Folder** page, retain the default installation folder, and click **Next**.

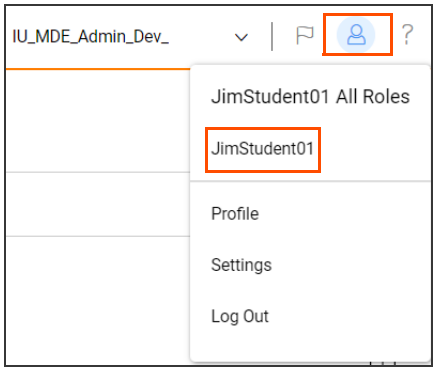


18. On the **Pre-Installation Summary** page, click **Install**.

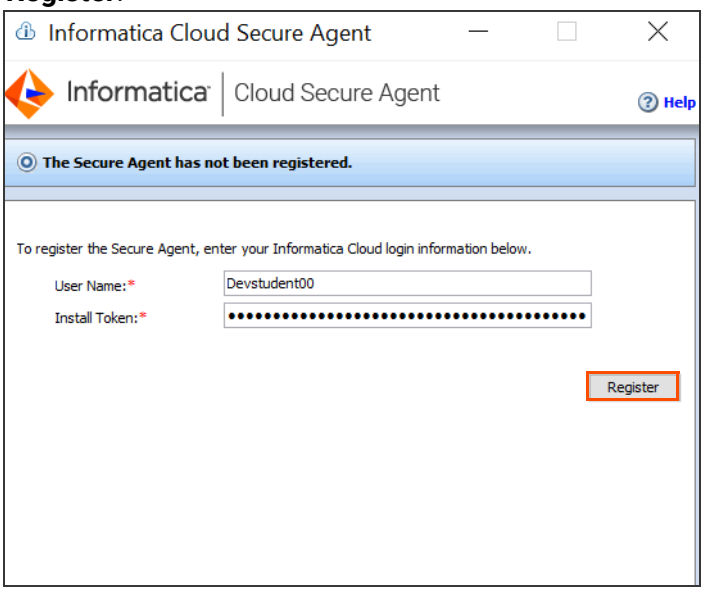


19. To register your secure agent, in the User Name text box, enter your IICS user name.

If you do not remember your username, click the User Settings icon to find it.

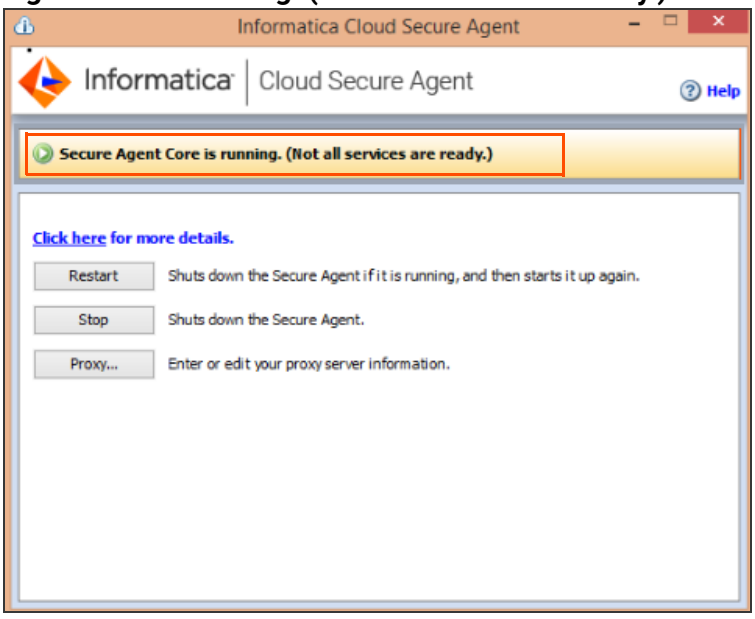


20. In the **Install Token** text box, paste the Install Token copied earlier in the notepad and click **Register**.



After successful registration, the Secure Agent will download the necessary files for the connectors.

21. When the configuration of the Secure Agent services is in progress, it displays the "**Secure Agent Core is running. (Not all services are ready.)**" message.



It takes about 10 to 15 minutes for the secure agent to come up and for all the services to be in a running state.

Enable Services and Connectors on Secure Agent

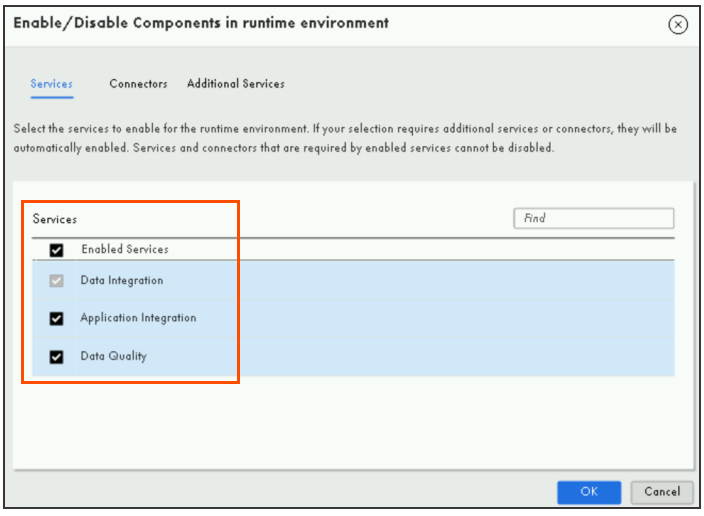
After registering the secure agent with your organization, you should enable the necessary services that are essential to carry out specific tasks such as ingress or egress.

22. Back in the browser, from the Administrator service navigation pane on the left, click **Runtime Environments**.

23. Click on the three dots on the right hand side of the Secure Agent group and select **Enable or disable Services, Connectors**. Click refresh if you can't see the secure agent.

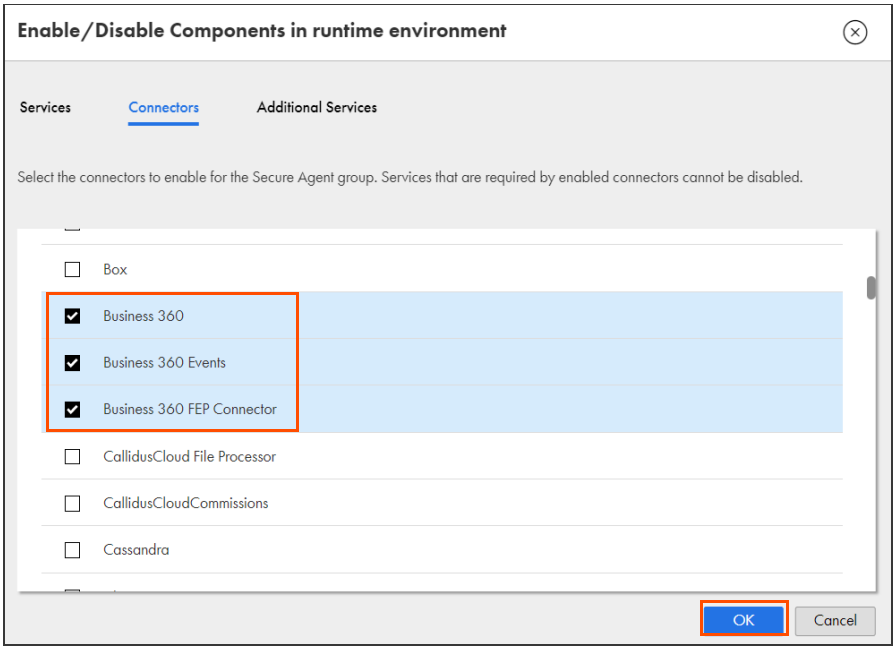
24. To perform lab exercises for this course, you must select the following services to run on the agent.

- Data Integration
- Application Integration
- Data Quality

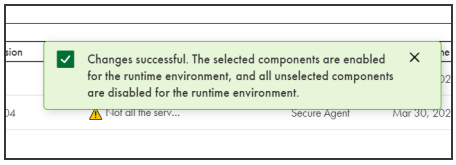


25. In the same window, select the **Connectors** tab, and make sure that the following connectors are selected to perform lab exercises for this course. Click **OK**.

- Business 360
- Business 360 Events
- Business 360 FEP Connector

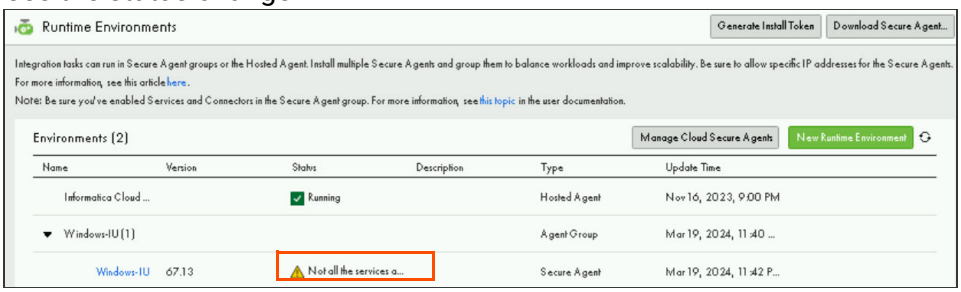


You will see the following message:



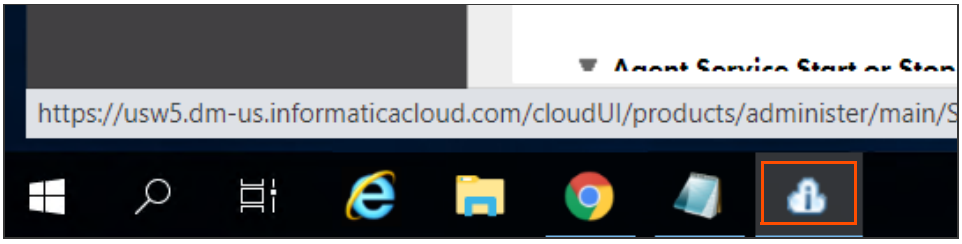
You can also enable all the connectors available in the list to be available for the selected Secure Agent.

26. Refresh the page. Notice that status is showing as “Not all services are up and running”. Wait some more time for the services to be up and running. Keep refreshing the page to see the status change.

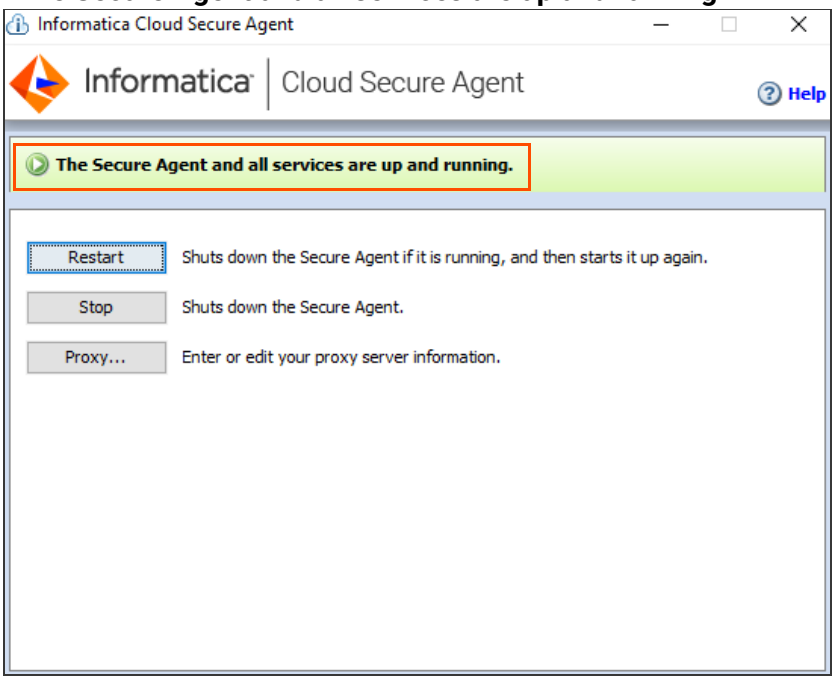


It takes about 10 to 15 minutes for the secure agent to come up and for all the services to be in a running state. By default, the Secure Agent takes the name of the computer it is installed on. If the secure agent does not appear on the **Runtime Environments** page, refresh the web page.

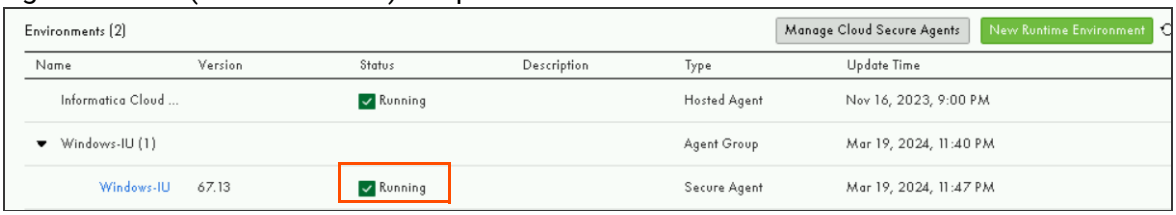
27. Navigate back to the secure agent window.



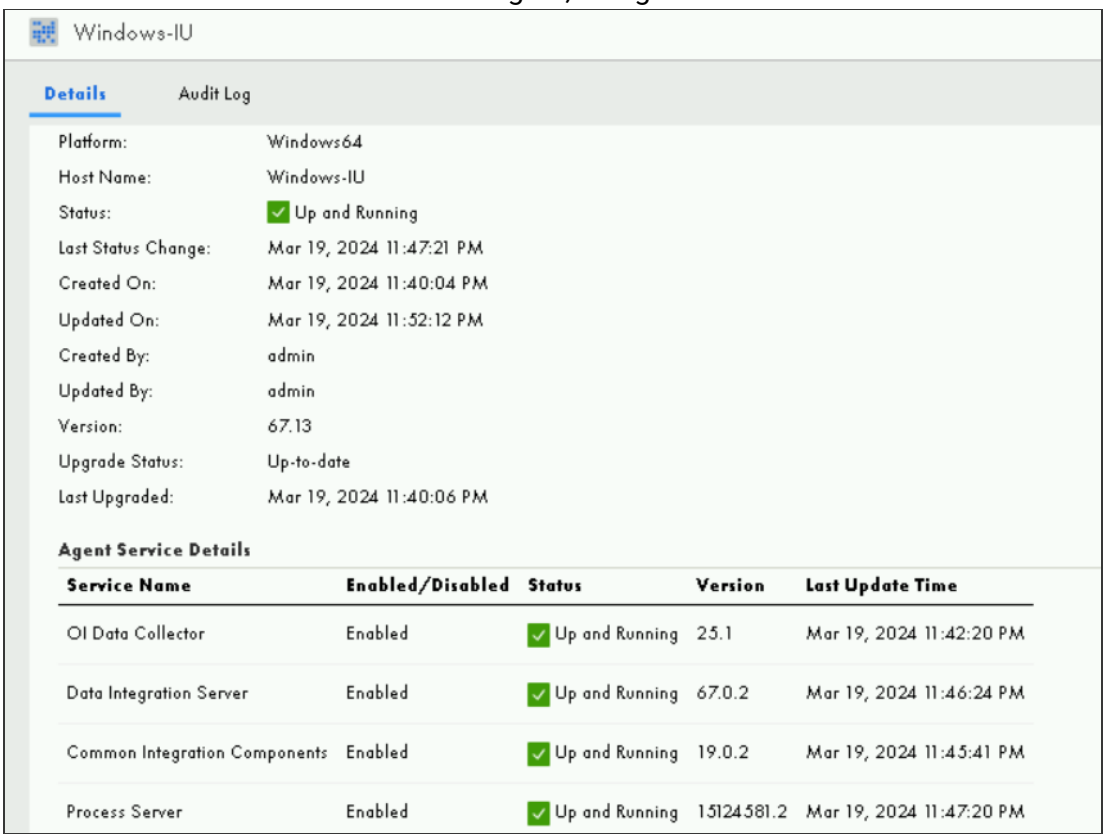
28. After the Secure Agent configures all the services, the agent status message changes to “**The Secure Agent and all services are up and running**”.



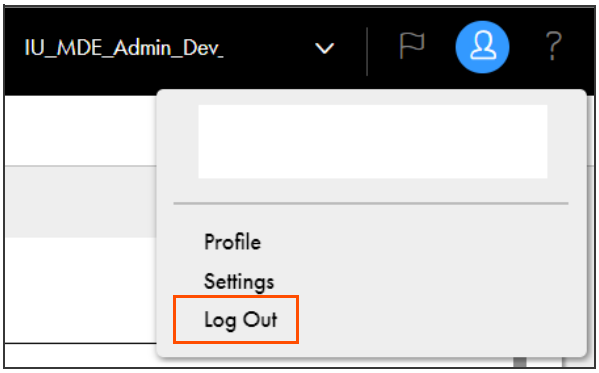
29. Once the status changes to 'Up and Running', back in the browser window, click the secure agent's name (the one in blue) to open it.



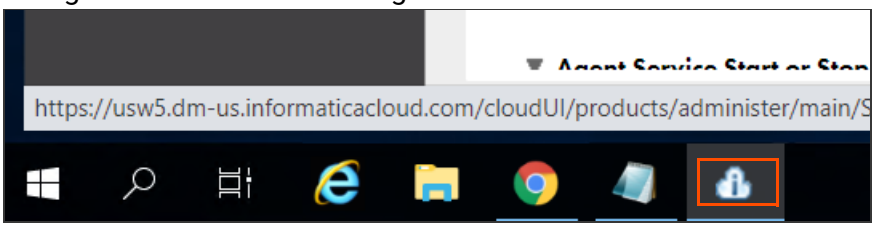
You can see the details of the secure agent, along with the microservices and their status.



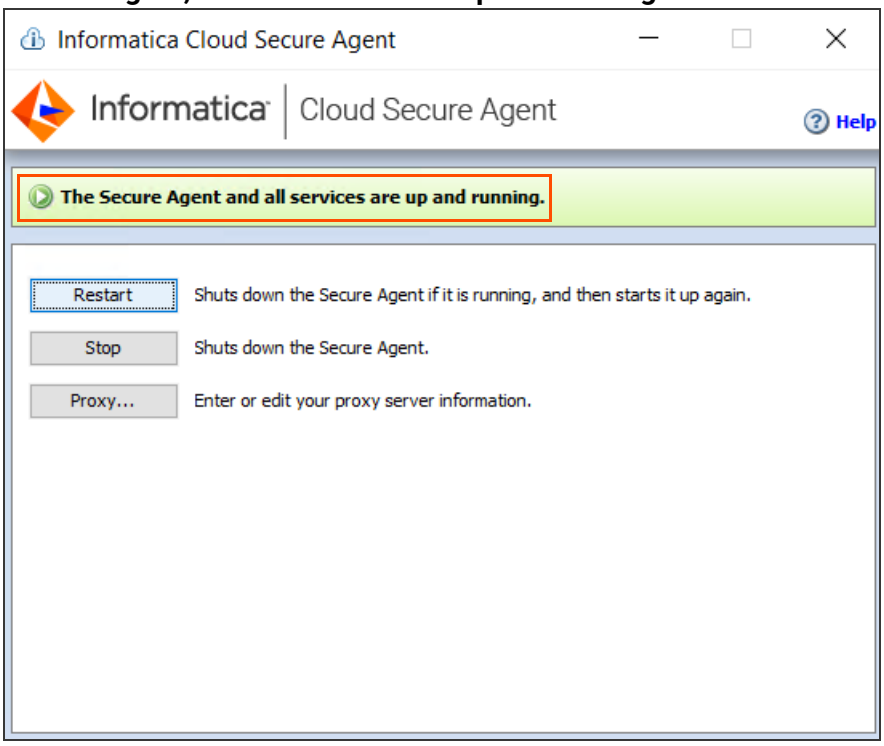
30. To log out from the Org, click the **User** icon, and select **Log Out**.



31. Navigate back to the secure agent window.



After the Secure Agent configures all the services, the agent status message changes to **The Secure Agent, and all services are up and running**.



This concludes the lab.

Lab 1-2: Creating a SQL Server Connection

Overview:

In IICS, you can connect to the SQL Server Database cloud service through a SQL Server connection. In this lab, you will create an SQL Server Connection in IICS.

Objectives:

- Create a SQL Server connection

Duration:

5 minutes

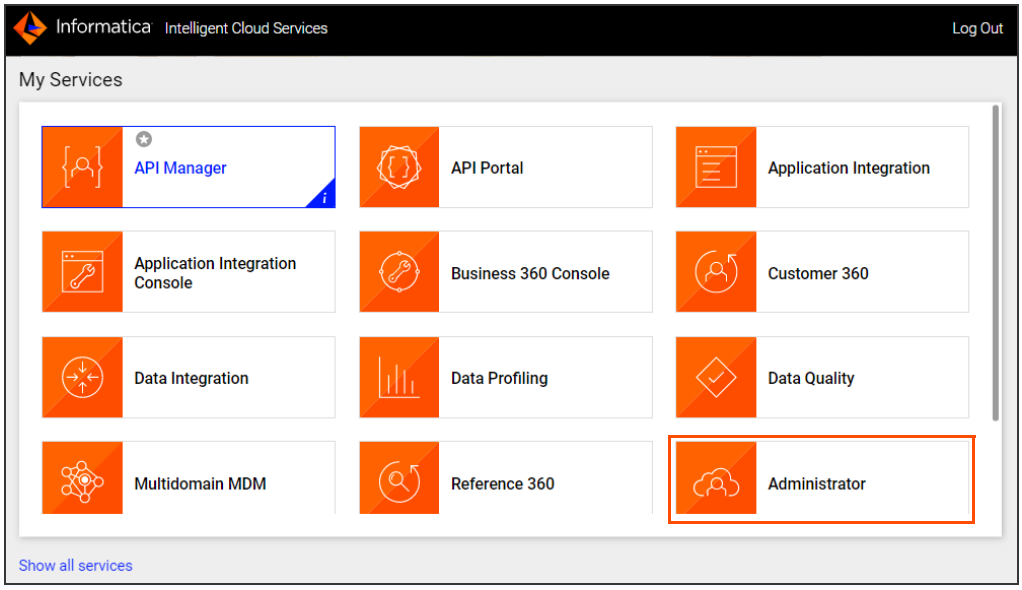
Tasks:

Create SQL Server Connection

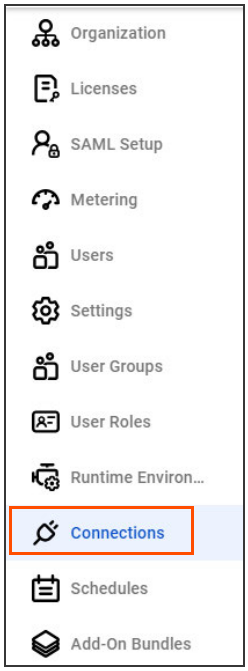
You will now create a SQL Server connection that will be useful while ingressing data.

1. Log in to IICS using MDM MDE admin credentials(StudentnameStudent<no> / Multidomain360).

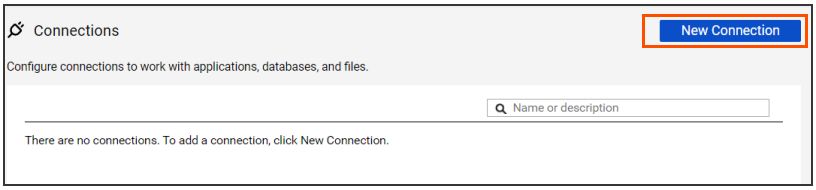
2. If you have logged out of IICS, log in to it, and navigate to the **Administrator** service.



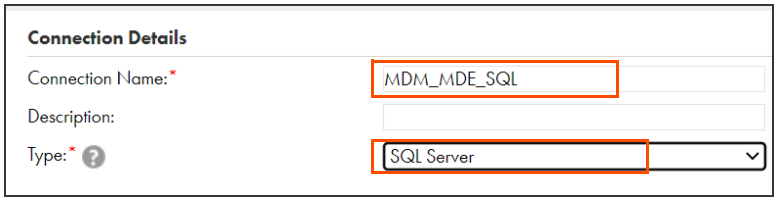
3. From the navigation panel, go to **Connections**.



4. Click **New Connection**.



5. In the Connection Name field, enter **MDM_MDE_SQL**.
6. From the Type drop-down, select **SQL Server**.

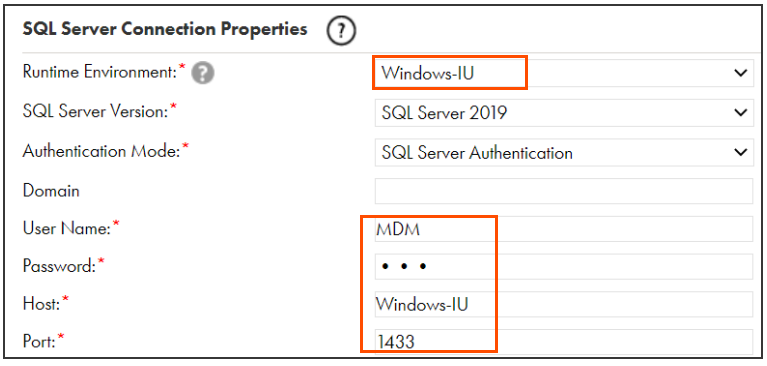


7. From the **Runtime Environment** drop-down, select your runtime environment.
8. From the **SQL Server Version** drop-down select **SQL Server 2019**.
9. From the **Authentication Mode** drop-down select **SQL Server Authentication**.
10. In the Username and Password fields, enter **MDM / MDM**.

The Password is case-sensitive.

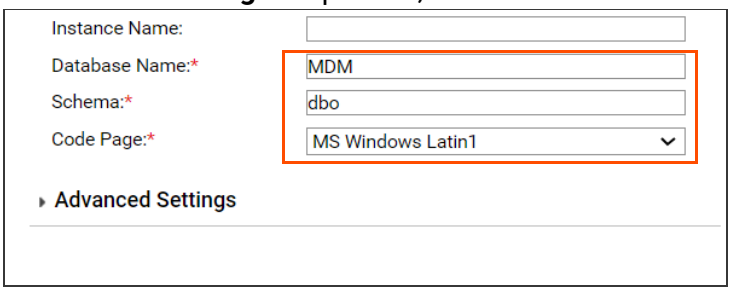
11. In the **Host** field, enter **Windows-IU**, and enter the **Port** value as **1433**.

Important: Make sure to enter the exact same name as the secure agent in the Host field. Notice the name of the Runtime Environment and the Host in the following image.

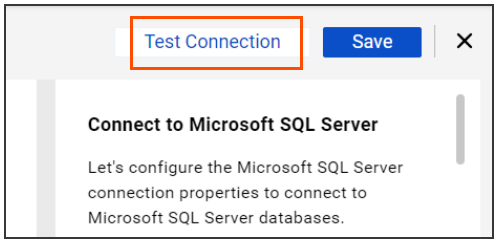


12. In the **Database Name** field, enter **MDM**. Enter the **Schema** as **dbo**.

13. From the **Code Page** drop-down, select **MS Windows Latin 1**. Retain the other fields.



14. Click **Test Connection**.



You will see the test connection successful message.

15. **Save** your changes.
A message - “Connection saved successfully” appears on the screen. Close the connection.

This concludes the lab.

Lab 1-3: Creating a Business 360 Connection

Objectives:

A Business 360 connection reads data from or writes data to the Business 360 data store. You can use the Business 360 connection in a Source transformation to read data or use the same connection in a Target transformation to write data. You can create a Business 360 connection on the Connections page and use the connection in Cloud Data Integration (CDI) to create a mapping.

In this lab, you will create a Business 360 Connection in IICS.

Overview:

- Create a Business 360 connection

Duration:

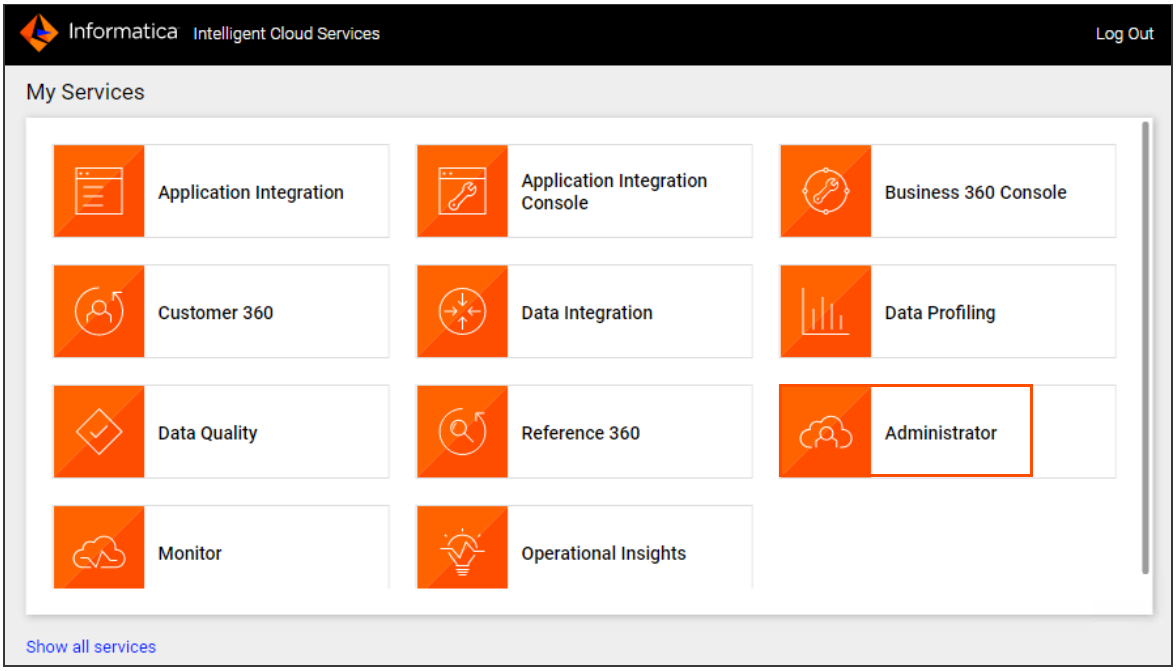
5 minutes

Tasks:

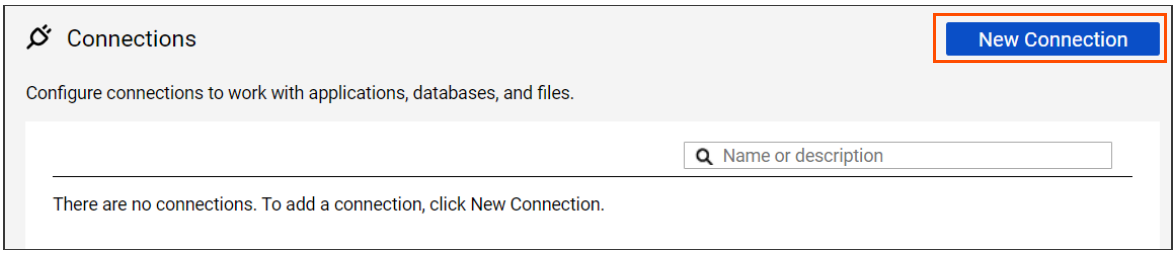
Create Business 360 Connection

You will now create a Business 360 connection that you will using while ingressing and egreesing data.

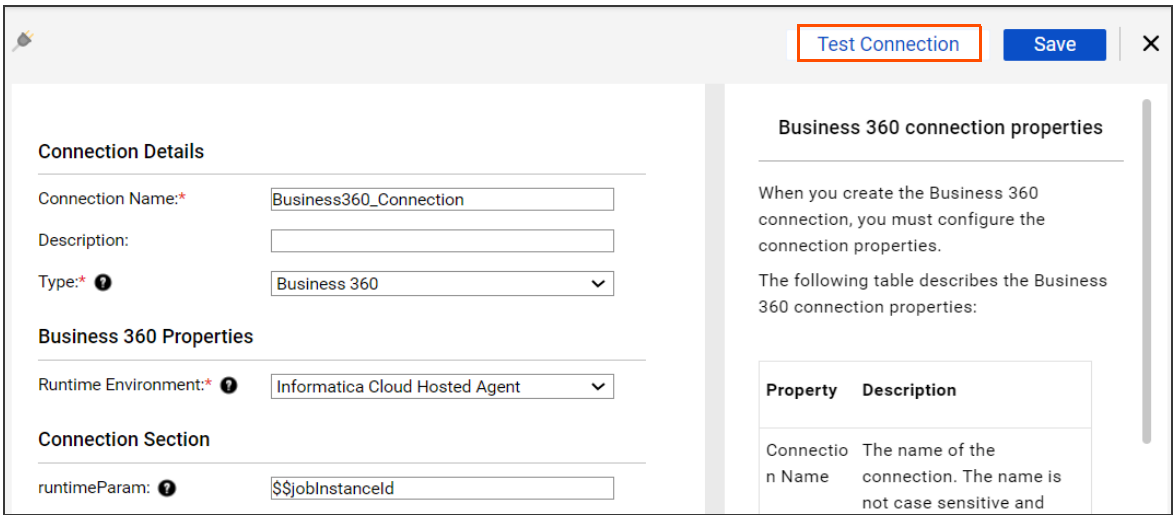
1. Log in to IICS, and from the My Services window, select **Administrator**, if you have logged out.



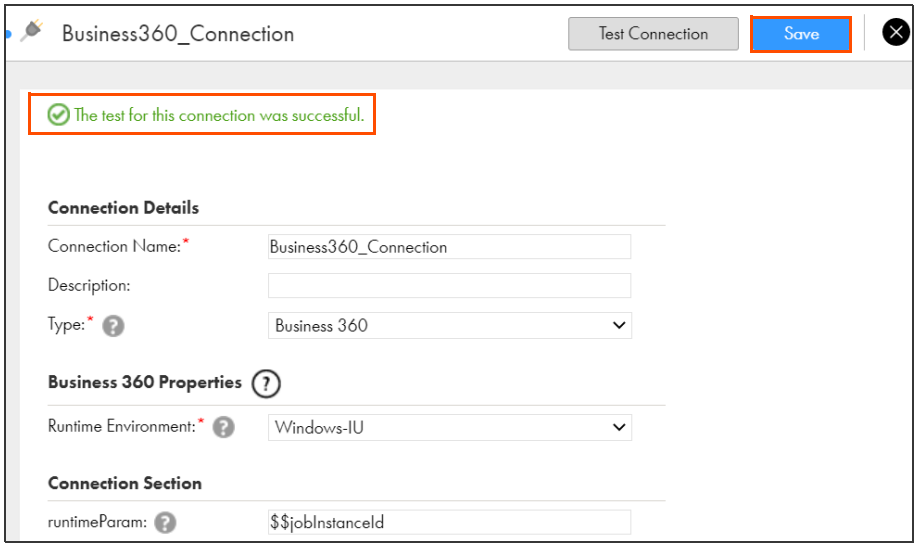
2. From the Connections tab, click **New Connection**.



3. In the Connection Name field, enter **Business360_Connection**.
4. From the **Type** drop-down, select **Business 360**.
5. From the **Runtime Environment** drop-down, select your secure agent group.
6. Retain the default value for **runtimeParam**.
7. **Test** the connection.



8. Upon successful connection, Click **Save** to save the changes, and click **Close**.



This concludes the lab.

Lab 1-4: Creating a Flat File Connection

Overview:

Connections define the location and the credentials of the data source or the data target to which multidomain MDM connects when a source application publishes data, or when a target application consumes published data.

In this lab, you will create a Flat File connection.

Objectives:

- Create a Flat File connection

Duration:

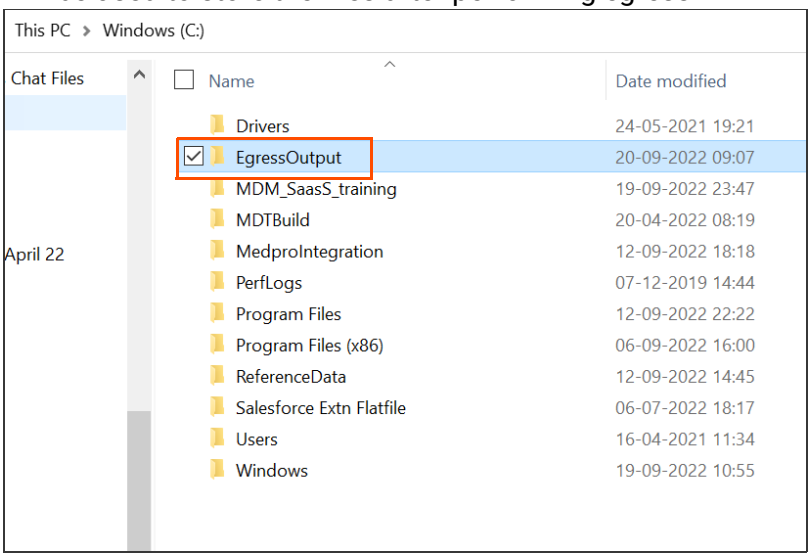
5 minutes

Tasks:

Create a Flat File Connection in IICS

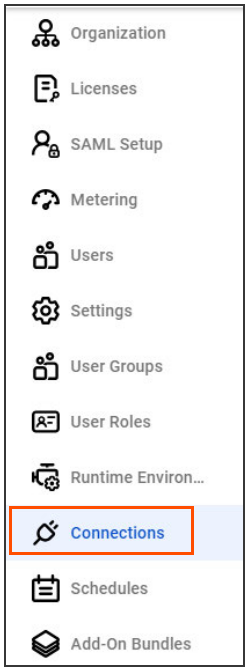
You will now create a SQL Server connection that will be useful while egreesing data.

1. On your local machine, go to C drive and create a folder named **EgressOutput**. This folder will be used to store the files after performing egress.

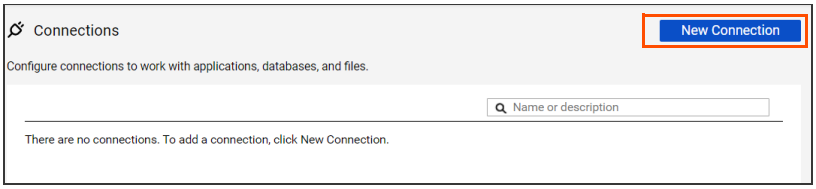


2. Navigate back to IICS and go to the **Administrator** service tab.

3. To create connections, click **Connections**.

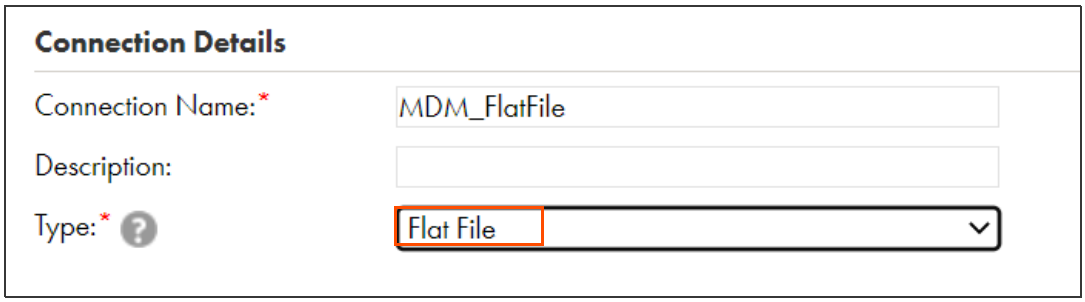


4. Click **New Connection**.



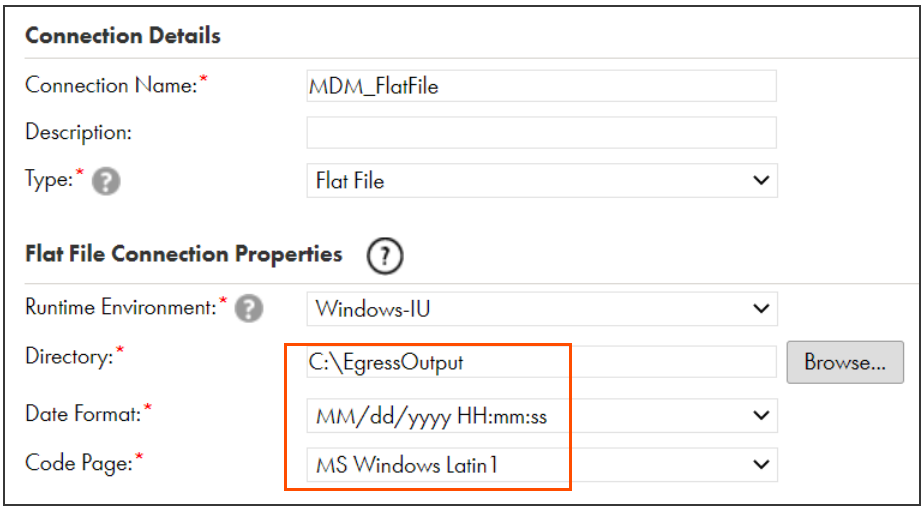
5. In the **Connection Name** field, enter **MDM_FlatFile**. You can optionally enter a description in the Description field.

6. Set the connection Type to **Flat File**.



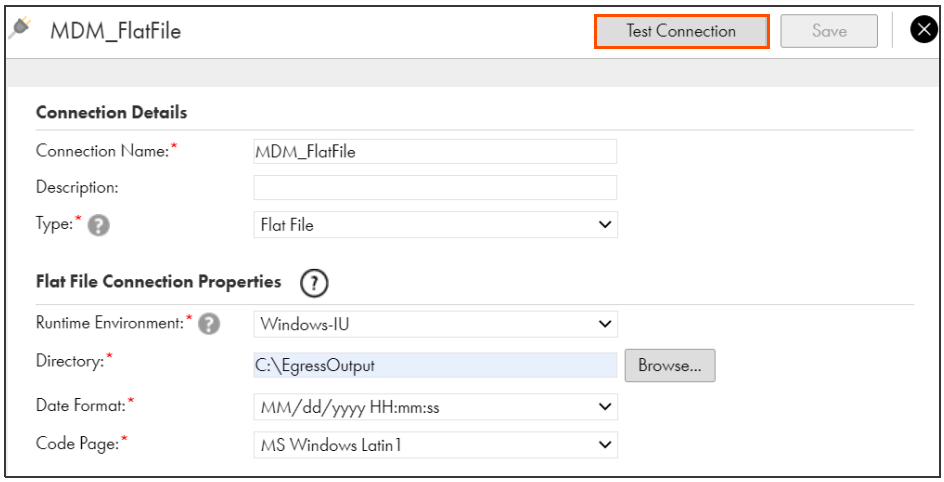
7. For the **Runtime Environment**, select your Secure Agent.

8. In the Directory field, enter the path **C:\EgressOutput** you created earlier. Retain the **Date Format**. From the **Code Page** drop-down menu, select **MS Windows Latin1**.

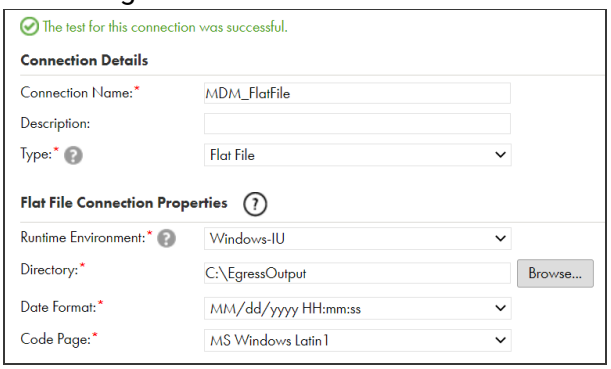


You will receive an error if you try to select the directory using the browse button. This is due to not setting the administrative rights for the secure agent. Since you are using separate suborgs for the class, it is not required to setup the administrative rights for the secure agent. The directory path should always be copied and pasted into the Directory field.

9. Click **Test Connection** to test the connection.



A message '**The test for this connection was successful**' appears.



10. If the test connection fails, check the connection details, and test it again.
11. Click **Save** to save the connection.

    A confirmation message '**Connection saved successfully**' appears.
12. Click **Close** to close the connection.

This concludes the lab.

Key Takeaways

- Informatica Intelligent Data Management Cloud (IDMC), designed to help businesses innovate with their data on any platform, any cloud, multi-cloud, and multi-hybrid.
- A Connection provides the information that IICS needs to connect to your data sources and targets.
- IDMC combines Informatica's 260+ intelligent cloud offerings, including but not limited to Data Catalog, Data Quality, MDM, Data Integration and Data Governance.
- A Runtime Environment is the execution platform that runs a Data Integration or Application Integration task
- The Cloud Administrator is used to configure Org settings.

Module 2: Multidomain MDM SaaS and Business 360 Overview

What is SaaS?

Software-as-a-Service is a cloud computing offering that provides users with access to a vendor's cloud-based software.

SaaS provides an intriguing alternative to standard software installation in the business environment (traditional model), where you have to build the server, install the application, and configure it. Instead, the applications reside on a remote cloud network accessed through the web or an API, and it works like a rental. You and your organization have the authorization to use it for a period of time and pay for the software that you are using.

The advantages of SaaS include:

- Reduced time to benefit
- Lower costs
- Scalability and integration
- New releases (upgrades)
- Easy to use and perform proof-of-concepts

Informatica MDM SaaS Applications

The Business 360 Console functions as a configuration tool and is placed above four MDM SaaS essential services. To use any of these four services, your organization must have the appropriate license along with a Business 360 Console license.

Informatica provides the following MDM SaaS services that assist in designing the MDM system for your business requirements:

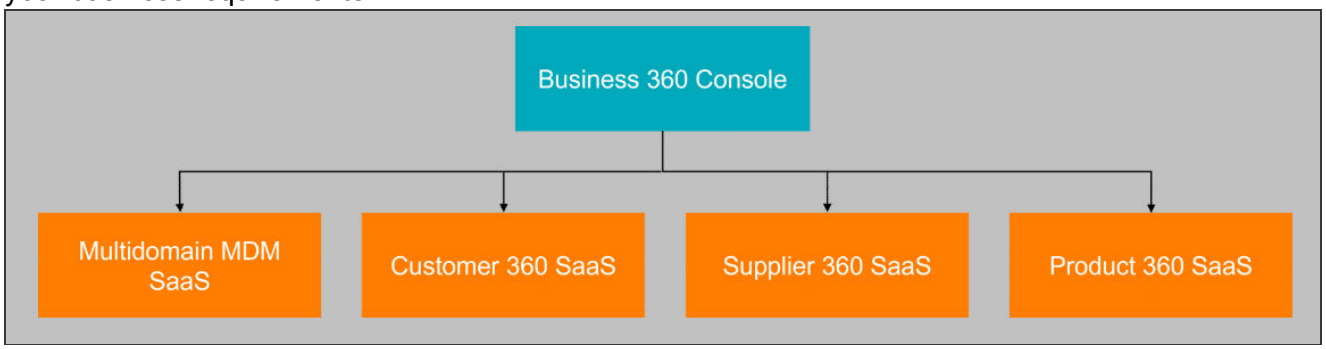


Figure 2-1: MDM SaaS Applications

- **Business 360:** The Business 360 Console is used as a configuration tool. This console allows for the creation and management of data management projects, where you can establish assets like business entities, define their relationships, and set up events and workflows. Additionally, it enables the management of service integrations and the execution and monitoring of batch jobs. Once you design your MDM system using the Business 360 console, you will then manage your projects using the following MDM SaaS client applications.

- **Customer 360 SaaS** : When managing customer data, you have the option to utilize the Customer 360 service and benefit from its features. Customer 360 serves as a front-end application that aids in creating business entity records, managing reports, and overseeing dashboards.
- **Product 360 SaaS**: When you need to manage your product data, opting for the Product 360 service can be beneficial. Product 360 functions as a front-end application that facilitates the creation of business entity records and the management of reports and dashboards
- **Supplier 360 SaaS**: When it comes to managing your supplier data, the Supplier 360 service offers a useful option with significant benefits. Supplier 360 operates as a front-end application, assisting in the creation of business entity records and the management of reports and dashboards.
- **Multidomain MDM SaaS**: When managing business data that does not fall into the categories of customer, product, or supplier data, Multidomain MDM is the ideal choice. Multidomain MDM serves as a flexible platform on which you can create and customize your own business entities, effectively accommodating a variety of data types and needs.

The Challenge

Digital leaders understand that data is fundamental to fueling insights for all aspects of digital transformation. This includes strategic decisions, identifying opportunities and risks, optimizing costs and efficiencies, and enhancing customer experiences.

Number of factors are hindering organizations' capability to transform for the future, including:

- Data chaos
- An inability to innovate due to legacy technologies and manual processes
- Security concerns
- Talent shortages
- Organizational silos

Solution: Multidomain MDM

Informatica Cloud Multidomain MDM is a cloud-based solution that helps organizations manage master data across systems and applications. The term master data refers to critical business information, such as customer details, product details, employee details, and more. Enterprises can ensure the consistency, accuracy, and reliability of their data by centralizing and managing this information.

Multidomain MDM from Informatica provides data profiling, quality management, governance, integration, and synchronization capabilities. Through aggregating and consolidating data from multiple sources, businesses can create a single, unified view of master data. As a result of this unified view, organizations are able to make better-informed decisions, improve operational efficiency, and improve the customer experience.

With multidomain MDM the organizations can overcome the above challenges. Following are the advantages of Cloud Multidomain MDM SaaS:

- **Quick Time to Value:** Informatica Cloud Multidomain MDM SaaS enables organizations to

start managing their master data quickly and efficiently. The cloud-based deployment eliminates the need for complex infrastructure setup and software installation. As a result, organizations can realize the benefits of master data management faster and reduce the implementation time.

- **Scalability and Flexibility**: As a SaaS solution, Informatica Cloud Multidomain MDM SaaS offers the flexibility to adapt to the changing data volumes and evolving business requirements. Depending on your organization's needs, you can scale up or down without having to invest in additional hardware or infrastructure.
- **Lower Total Cost of Ownership (TCO)**: By adopting Informatica Cloud Multidomain MDM as a SaaS model, organizations can eliminate upfront hardware costs as well as ongoing maintenance expenses. Organizations can avoid infrastructure procurement, software licensing, and maintenance efforts, resulting in a lower total cost of ownership over time.
- **Data Governance and Data Quality** : Informatica Cloud Multidomain MDM offers comprehensive data governance capabilities, including rules for defining and enforcing data quality, data validation, and data stewardship workflows. Data governance policies can be established, data quality can be monitored, and regulatory compliance can be ensured.
- **User-Friendly Interface:** Informatica Cloud Multidomain MDM provides a user-friendly interface that makes data management activities accessible to business users without technical expertise. It enables business users to take ownership of their master data and make data-driven decisions by defining data models, configuring data attributes, and establishing data hierarchies.
- **Seamless Integration**: Informatica Cloud Multidomain MDM facilitates the synchronization and exchange of master data across the enterprise. Integration capabilities ensure data consistency and accuracy across organizations, connecting with a variety of data sources and target systems.

Cloud Multidomain MDM SaaS Architecture

Informatica MDE, also known as Master Data Edition, is a component of Informatica Cloud MDM that provides a comprehensive set of tools for managing and maintaining master data. It offers a user-friendly interface that allows business users to define, enrich, and govern master data without any technical expertise.

As shown in the below architectural diagram, with Multidomain MDM SaaS, business users consolidate data from various sources, such as marketing, customer, and product data, and store it in a central master database. This master data is then utilized by data consumers for analytics and making informed business decisions.

Figure 2-2 shows the Architecture of Multidomain MDM.

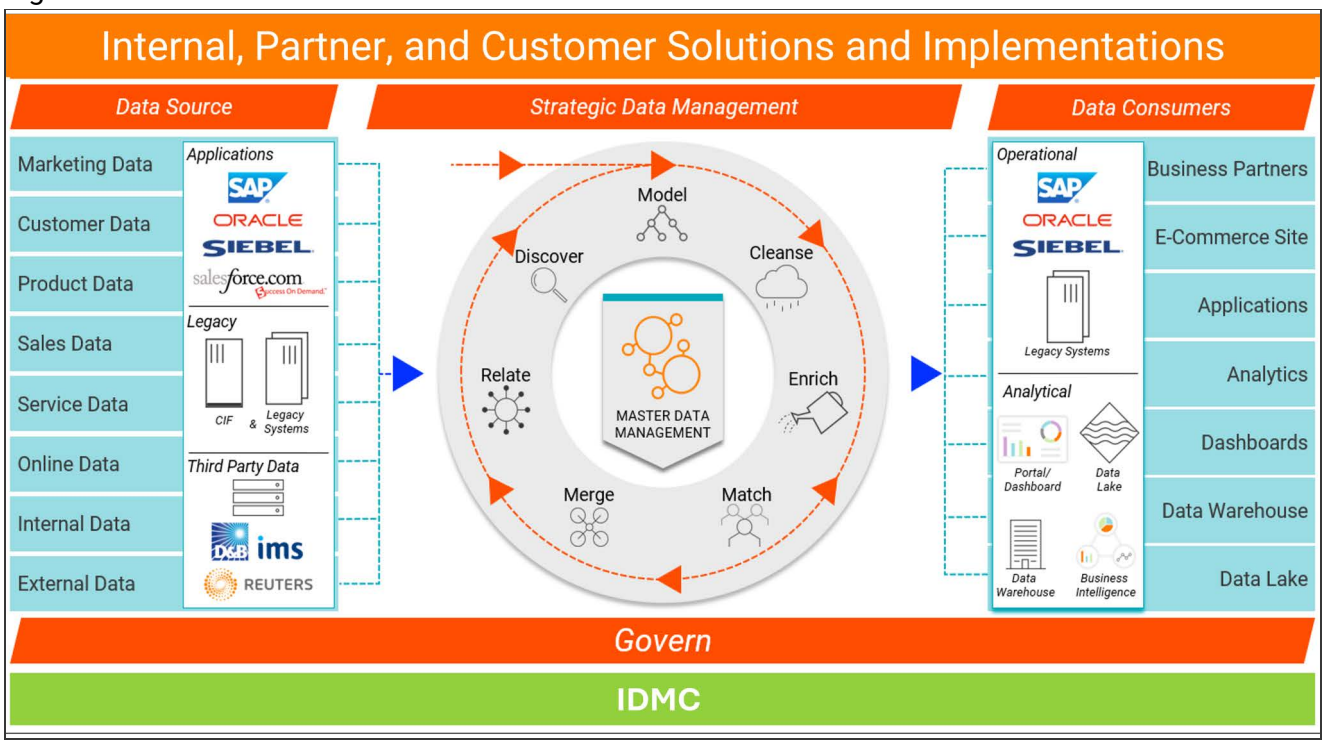


Figure 2-2: Multidomain MDM Architecture

Key Features and Benefits

- **Combine the Power of Informatica MDM with the Speed and Agility of Cloud:** Informatica MDM Cloud Edition accelerates the speed of MDM deployment and MDM 360 applications, including Customer 360, Product 360, and Supplier 360. MDM running on the cloud infrastructure allows you to be agile and innovative, and leverages a simple, pay-as-you-go pricing approach, which removes any upfront investment in infrastructure. This pricing model—based on paying only for the capacity you use—allows you to access all the benefits of Informatica MDM, regardless of whether you're a large, medium, or small business.
- **Auto-Upgrade to Leverage Latest Features:** Organizations receive all the capabilities of Informatica MDM, as well as automated updates and fixes, with the Informatica MDM Cloud Edition. As a result, you won't need to bother about updating Informatica MDM. This frees up the time so you can concentrate on achieving business results from your MDM installation because the security upgrades are included. This enables you to take advantage of the most recent improvements, features, and functionality that Informatica offers with each release.
- **Lower Maintenance Costs and IT Expenses** : With the introduction of Informatica MDM Cloud Edition, IT managements is now Informatica's responsibility. With an affordable monthly charge, you can eliminate the staffing costs for infrastructure. This is the most efficient way to utilize, upgrade, and maintain your MDM system, which will drastically reduce your business's IT costs. The award-winning customer care staff of Informatica will have your best interests in mind.
- **Enjoy End-to-End MDM Capabilities** : All the features of an Informatica MDM on-premises installation are available in the cloud edition. To support the supply of a holistic view of customers, products, suppliers, and other master data, it embeds Informatica's industry-leading Data

Quality, Data Integration, and Business Process Management solutions. You can guarantee the integrity and safety of your data with the help of secure cloud technology from Informatica and its numerous awards for data security.

User Interfaces

Multidomain MDM user interface is simple and Intuitive which makes it easy to learn and helps the user to pick it up quickly and easily.

Home Page

After you log into the Multidomain MDM SaaS, you will see the Home page. The Home page is a dashboard that can display charts.

The following are the components of the Home Page:

- **Dashboard**: You can view a visual summary of your data as charts on your dashboard. You can add the predefined and custom charts as widgets to the dashboard. You can create multiple dashboards on the Home page based on your requirements and business needs. You can create up to six different dashboards using the predefined dashboard layouts. **For example**, you can create a dashboard to visualize record details and another one for record counts. You can view a visual summary of your data on your dashboard as charts.
- **Charts**: Charts summarize and display data visually. Charts make it easier to analyze data and make crucial business decisions. You can add predefined and custom charts to the dashboard as widgets. **For example**, a user can view area-wise distributors for a supplier in a geographical region.
- **Widget**: A widget is a placeholder user interface element for reports. You can add, edit, or remove a widget to customize the dashboard layout. Each widget displays the last updated time and estimated next run time of the associated report.
- **Reports**: A report is a tabular representation of data. You can visualize the reports as charts. You can use predefined reports or create a custom report to display information about your records.

Multidomain MDM Home page also provides a search box that helps you to find a record. You can use this search box to perform a full-text search. When you use the search box, you can search within a specific business entity or across multiple business entities.

You can access the following pages from the navigation bar for Multidomain MDM:

- **New** : Create Multidomain MDM console assets, such as records, hierarchies, and, reports.
- **Home**: Return to the Home page.
- **Search**: Allows you to search for a record by providing a search box or to perform a filtered search.
- **File Import** : Allows you to import data, such as business entity records, relationship records, and hierarchies, from a CSV file into Multidomain MDM SaaS.
- **Workflow Inbox** : Shows the assigned tasks to the logged-in user.
- **Hierarchies** : Allows you to create hierarchy instances.
- **Reports** : Create reports to display information about your records.

- **My Jobs**: Allows you to monitor jobs that run as a result of the actions that you perform in Multidomain MDM SaaS.

New Visual Experience

Multidomain MDM SaaS has a new look and feel that allows you to visualize and interact with data with greater clarity.

When you log in to Multidomain MDM SaaS, the classic experience is enabled by default. To explore the new look and feel of your business application, turn on New visual experience. To revert to the classic experience, you can turn off the New visual experience.

Figure 2-3 shows the Multidomain MDM Home page.

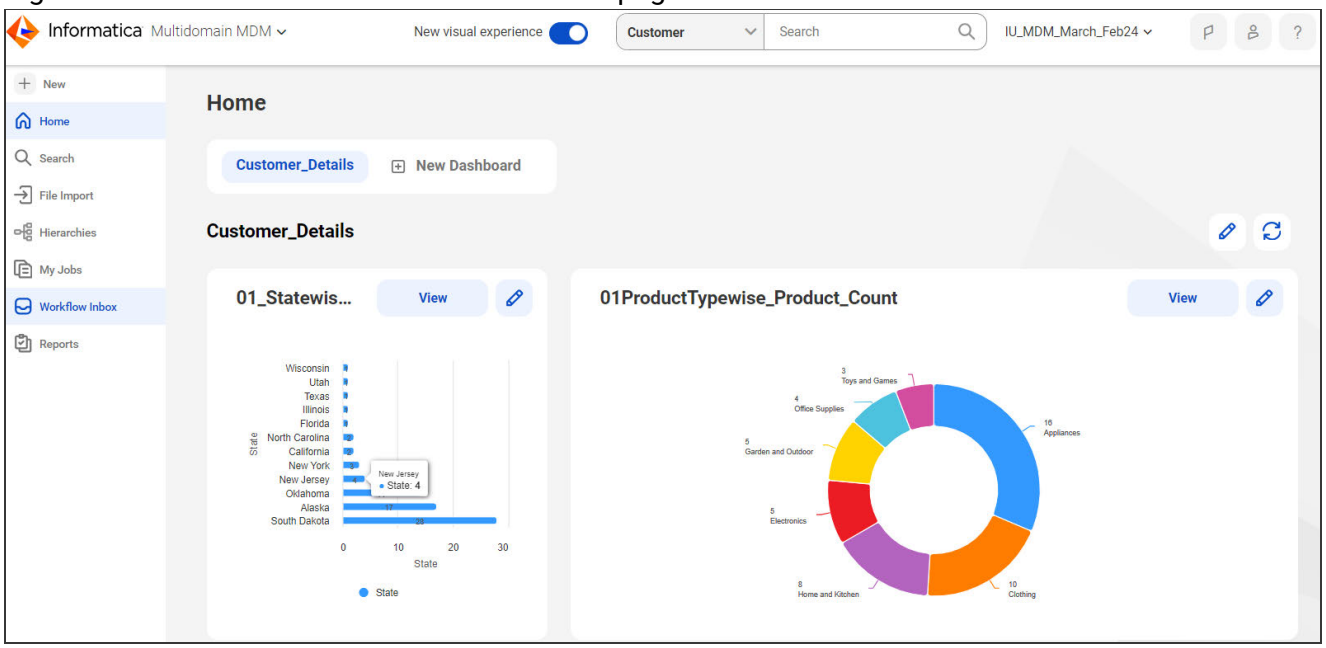


Figure 2-3: Multidomain MDM Home Page

Key Concepts

To work effectively in the Multidomain MDM SaaS, you need a basic understanding of the master data management concepts. These include:

Business Entities

A business entity is an entity of importance to your organization and consists of fields that support your data needs. To define a business entity, decide on the fields based on the data that your source systems will contribute to the master data.

For example, a company that manufactures cars might define business entities for some or all of the following entities: customers, employees, suppliers, factories, materials, and products. A charitable organization might define donors, households, and projects. A healthcare provider might define doctors, care sites, and insured services.
Multidomain MDM comes as a blank slate that can be configured as per the business requirement. You can create and configure the business entities and add required fields to them. This process is called

as Configuring the Data Model. You will learn more about the data model in the next module.

Source Record

A source record is a record from a source system. Source records from multiple source systems can contribute data to the master records.

Patch Record

MDM SaaS creates a patch record when you edit a master record through the user interface, API, or batch jobs. A patch record acts as a source record, but it contains only the master record fields that are edited. If the same master record is edited multiple times, the patch record contains the latest changes.

Related Records

These are the records that have relationships with the first-level nodes. Relationships describe the connection between two records. For example, if John Smith is an employee of Informatica, the person record of John Smith can be related to the organization record of Informatica.

Workflow and Tasks

A workflow consists of a sequence of tasks and ensures approvers review and approve the data before it becomes master data. When a user creates or updates a record, the action can trigger a workflow. A workflow consists of multiple tasks linked together. A task is an activity that must be completed, such as accepting or rejecting an edited record.

Hierarchy

A hierarchy contains hierarchy relationships between records. Create hierarchies to show how records are related to one another. A record can appear in multiple hierarchies.

Business 360 Console Overview

As mentioned earlier, Informatica Cloud Business 360 Console is one of the key services offered by Informatica Intelligent Cloud Services. You can use the Business 360 Console to create data management projects and run data integration tasks.

When you select Business 360 Console from the My Services page, the Business 360 Console Home page appears. Figure 2-3 shows the Business 360 Console Home page.

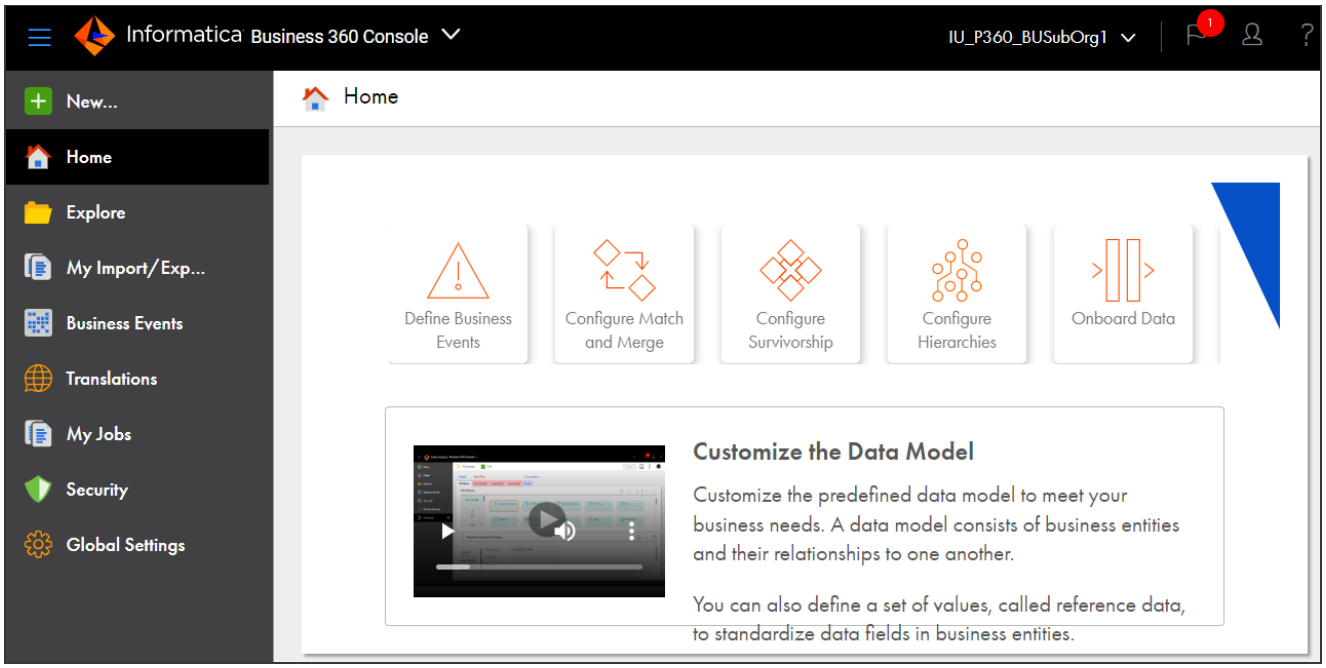


Figure 2-4: Business 360 Console

You can access the following pages from the navigation bar of the Business 360 Console:

- **New**: Create Business 360 Console assets, such as business entities, workflows, and applications.
- **Home**: Return to the Home page.
- **Explore**: Create and manage projects and assets.
- **My Import/Export Logs**: Displays the imported and exported files.
- **Business Events**: Create business events to trigger workflow.
- **My Jobs**: View, schedule, and monitor jobs.
- **Security**: List all custom user roles that you create in Administrator.
- **Global Settings**: Set license keys for DaaS services.

When you switch from Business 360 Console to another service, such as Administrator or Monitor, the options in the navigation bar change as per the selected service.

Business 360 Console Tools

Informatica Intelligent Cloud Services provides the following tools that you can use to meet your master data management business needs:

- Projects
- Assets
- Explore Page

Projects
Informatica Intelligent Cloud Services assets are organized into projects. You can create multiple folders within a project to contain the assets that you need for your project. You can copy, move, or delete any custom project or asset. You can't add, update, or delete any asset in the predefined projects. You can't copy or move predefined MDM SaaS assets, such as pages, components, or job definitions. The types of assets that you use in your projects depend on the services you use. **For example**, the Business 360 Console project includes assets such as business entities, source systems, rule specifications, and business applications.

Multidomain MDM contains predefined assets specific to MDM. You can see this project only if you have the Multidomain MDM license. You can create folders within this project and add custom assets to them.

Assets
In the Business 360 Console, the core assets are business entities, source systems, business applications, pages, and components. You can create rule specification assets in Data Quality for operations such as data standardization, data cleansing, and data validation. Also, you can create process assets in Application Integration to automate business processes such as file import, task triggers, and match and merge.

You can copy, move, or delete any custom asset. You can't add, update, or delete any predefined MDM SaaS asset in the predefined projects. You can't copy or move predefined MDM SaaS assets, such as pages, components, or job definitions. You can create the assets you need to satisfy your business requirements. You can use the assets in multiple projects. All assets must be the part of at least one project. If you create an asset without specifying a project, the asset is created in the Default project.

Explore Page
You can use the Explore page to work with your Informatica Intelligent Cloud Services projects and assets. You can perform actions on projects, folders, and assets on the Explore page.

You can use any of the following methods to find your projects and assets on the Explore page:

- **Explore by projects and folders:** View all projects or select a particular project.
- **Explore by asset types:** View all assets or view assets of a particular type.
- **Explore by tag:** View assets associated with a particular tag.
- **Search for projects or assets:** To search all projects, folders, and assets in the organization, view the Explore page by All Projects and then enter a name or description in the Find box. Or, to narrow your search, view the Explore page by Asset Types and select an asset type from the All Assets list. Then, in the Find box, enter a name or description in full or part.
- **Sort the search results:** Sort the Explore page by name, asset type, last update date, create date, description, or status. When you sort by type, the Explore page groups assets by asset type. It does not list the asset types in the alphabetical order.
- **Filter the objects on the page:** To filter objects, click the Filter icon. To apply a filter, click Add Field, select the property to filter by, and then enter the property value. The filters available depend on how you view the page. You can specify multiple filters.

Language Translation

Different languages are supported by your business applications. You can translate the labels of assets associated with your application into other languages. The translation files contain translations for all predefined asset labels, so you can get started quickly. Before you translate your business applications, ensure that you define your data model and configure your business applications.

To manage translations, you need one of the following user roles:

- Admin
- Designer
- MDM Designer

You can translate asset labels into the following languages:

- French
- German
- Japanese
- Portuguese
- Spanish

Lab 2-1: Multidomain MDM UI Walkthrough

Overview:

Multidomain MDM SaaS help to you to create, find, and view records in Multidomain MDM SaaS. With Multidomain MDM, you can also import data, manage hierarchies, review tasks, and monitor jobs that you trigger. In this lab you will know about the home page of Multidomain MDM and various elements that are present on the home page of Multidomain MDM.

Objectives:

- Log in to Multidomain MDM with the admin credentials
- Walkthrough the UI Multidomain MDM

Duration:

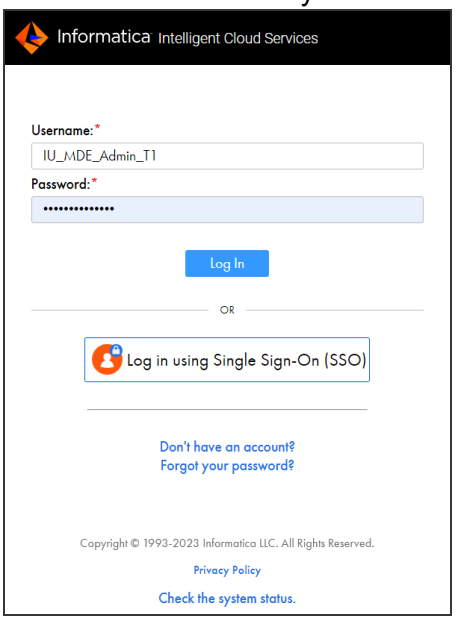
15 minutes

Tasks:

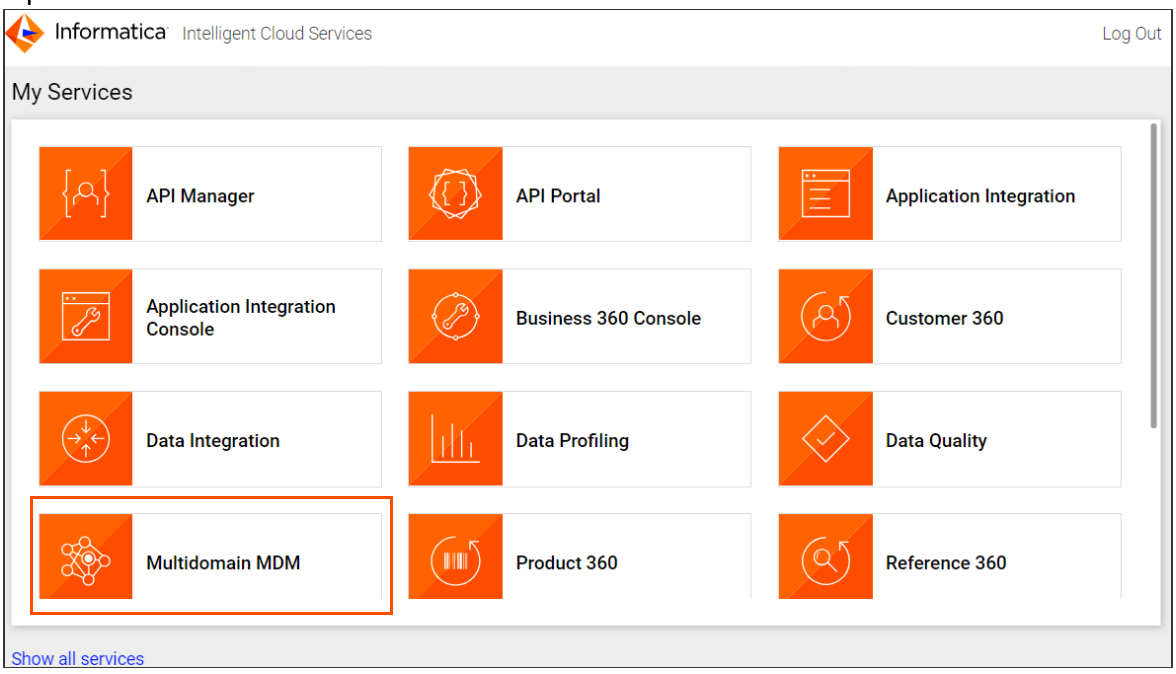
Walk through the UI of Multidomain MDM

You will now login to Multidomain MDM service and explore the its user interface.

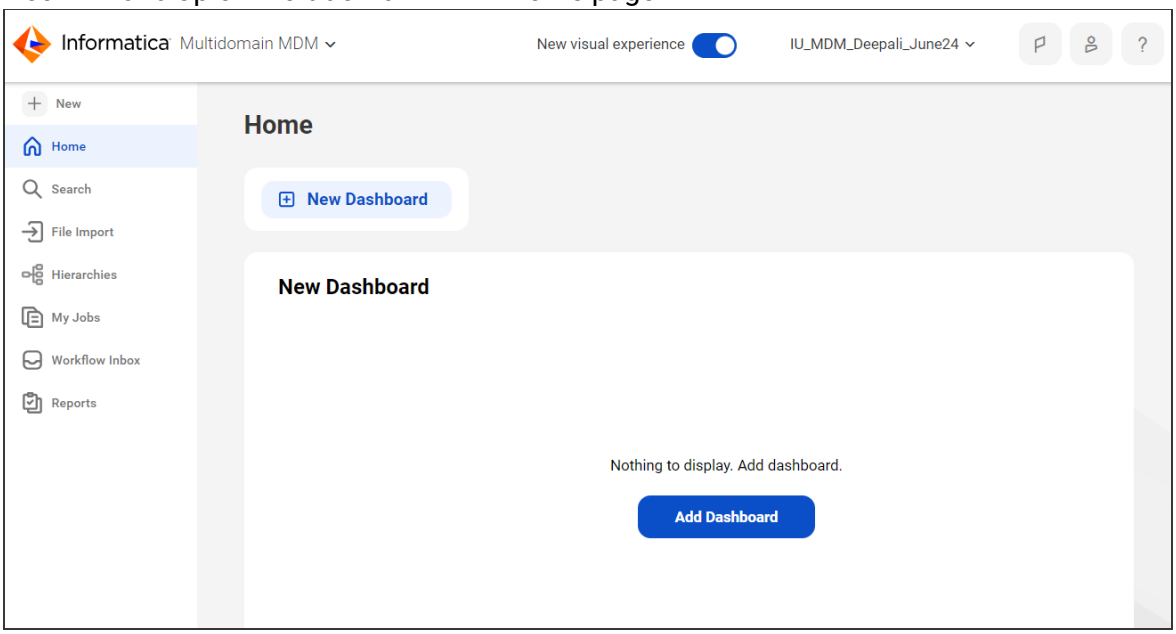
1. Enter the username and password as IU_MDE_Admin_T1 / Multidomain360. (Use the admin credentials that your instructor has shared with you).



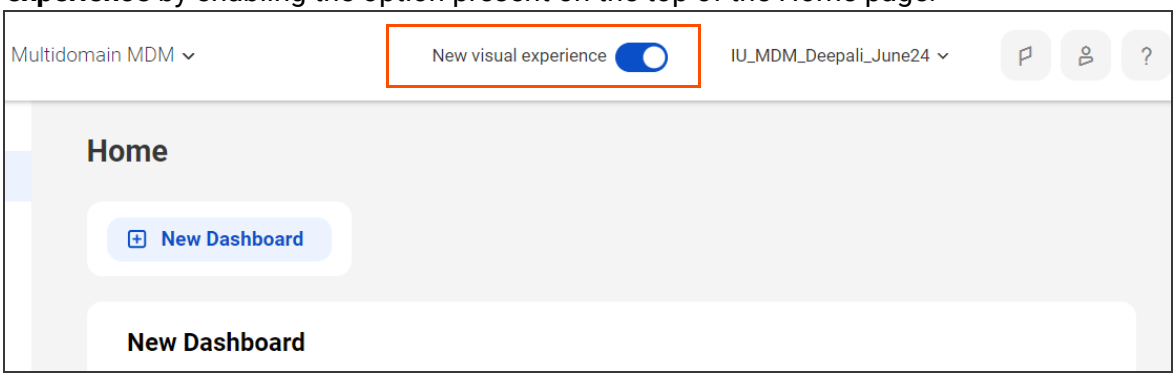
2. Open **Multidomain MDM** service from the list of service.



You will land up on Multidomain MDM Home page.



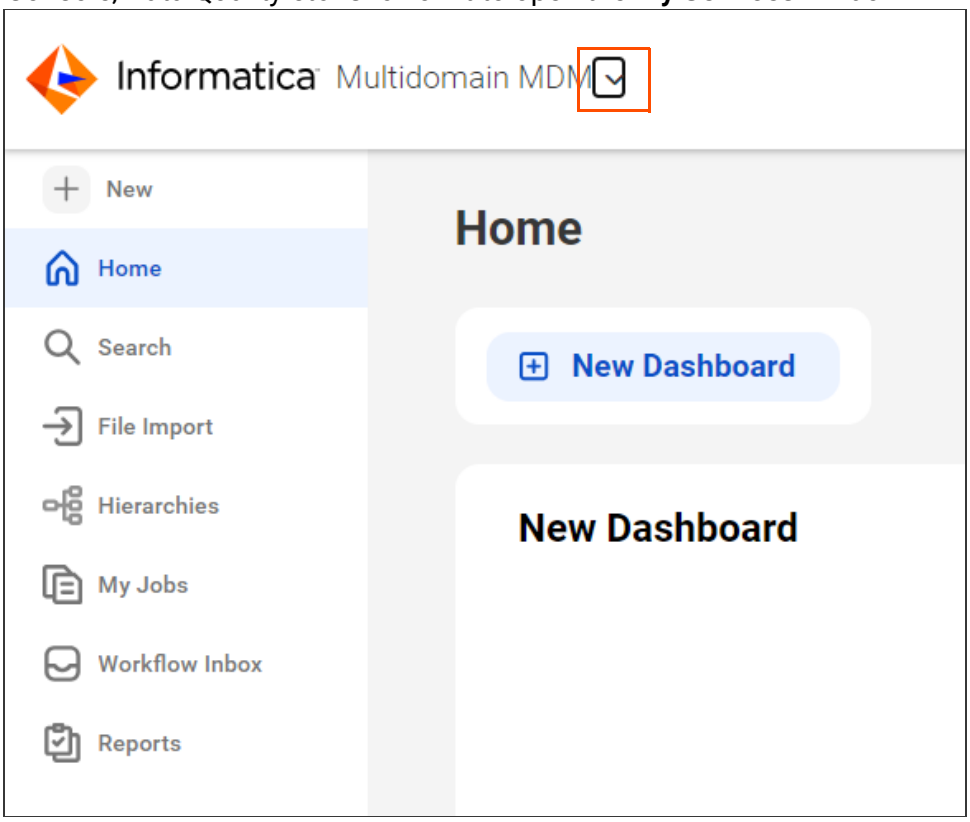
3. In the current release of Multidomain MDM, you will see an option to switch to **New visual experience** by enabling the option present on the top of the Home page.



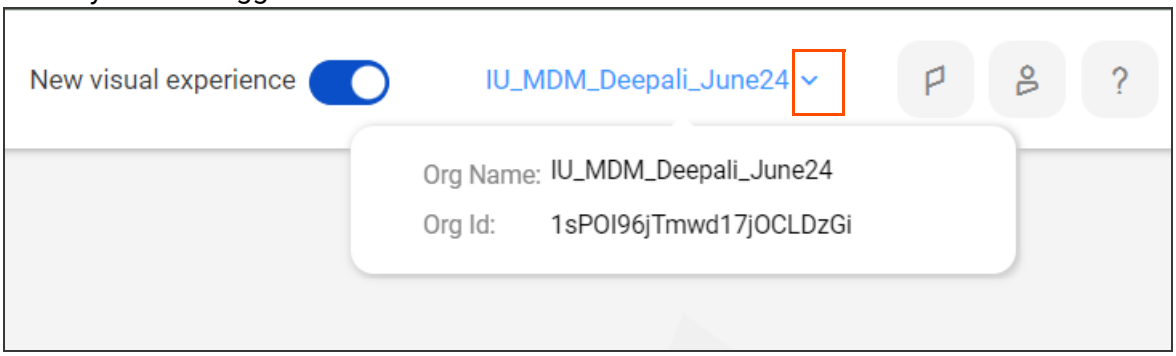
You can access the following pages from the navigation bar for Business 360 Console:

- **New:** Create Multidomain MDM console assets, such as records, hierarchies, and, reports.
- **Home**: Return to the Home page.
- **Search**: Allows you to search for a record by providing a search box or to perform a filtered search.
- **File Import:** Allows you to import data, such as business entity records, relationship records, and hierarchies, from a CSV file into Multidomain MDM SaaS.
- **Workflow Inbox:** Shows the assigned tasks to the logged-in user.
- **Hierarchies:** Allows you to create hierarchy instances.
- **Reports:** Create reports to display information about your records.
- **My Jobs**: Allows you to monitor jobs that run as a result of the actions that you perform in Multidomain MDM SaaS.

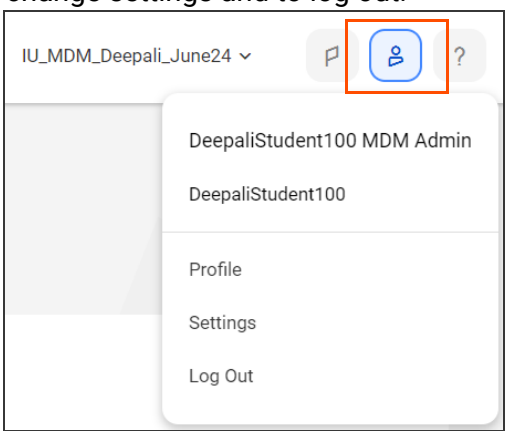
4. You can click on the downward facing arrow next to the Multidomain MDM which acts as a product switcher. This helps you to open other services like Administrator, Business 360 Console, Data Quality etc. Click on it to open the **My Services** window



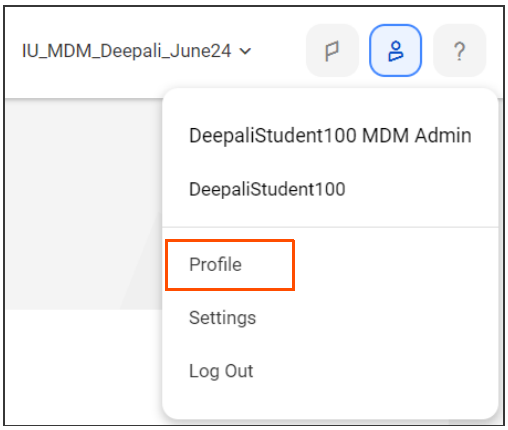
5. Click on the downward facing arrow next to the org to know the Org Name and the Org Id to which you have logged in.



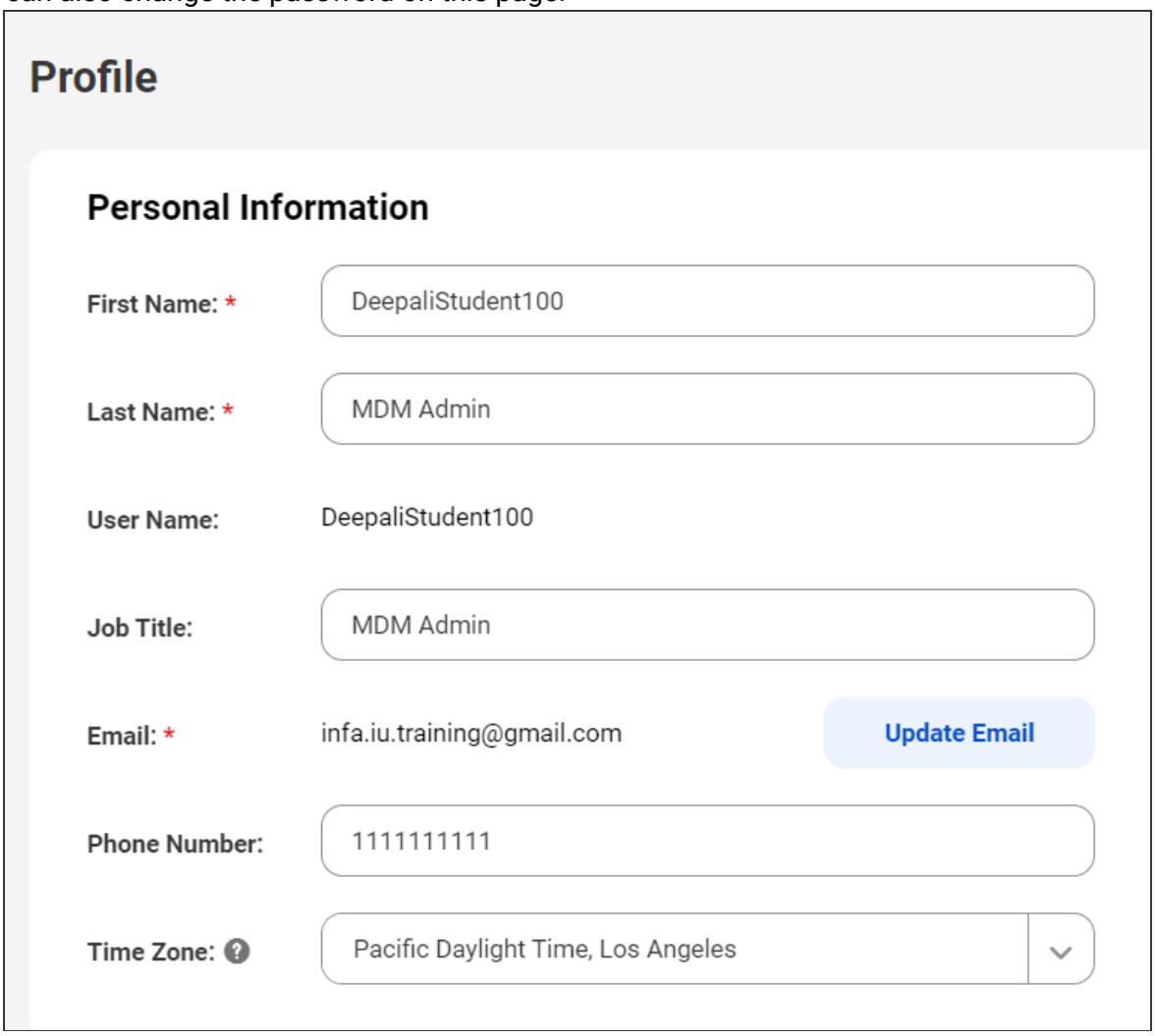
6. Click on the **User** icon to see the details such as the username, to view profile details, change settings and to log out.



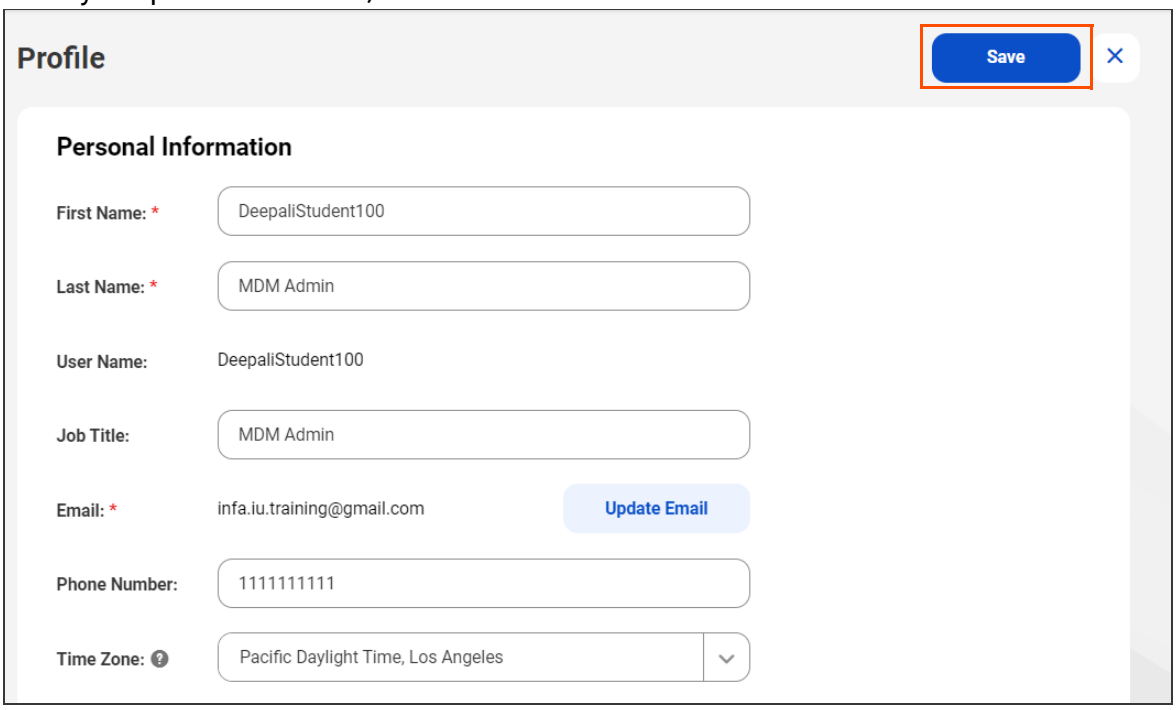
7. Select **Profile** from the menu.



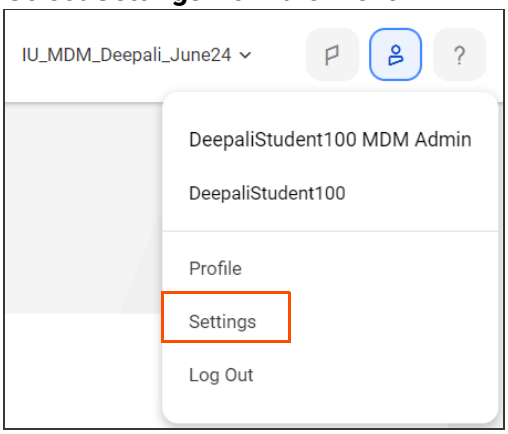
Here you can change the personal Information like First Name, Last Name, Job Title etc. You can also change the password on this page.



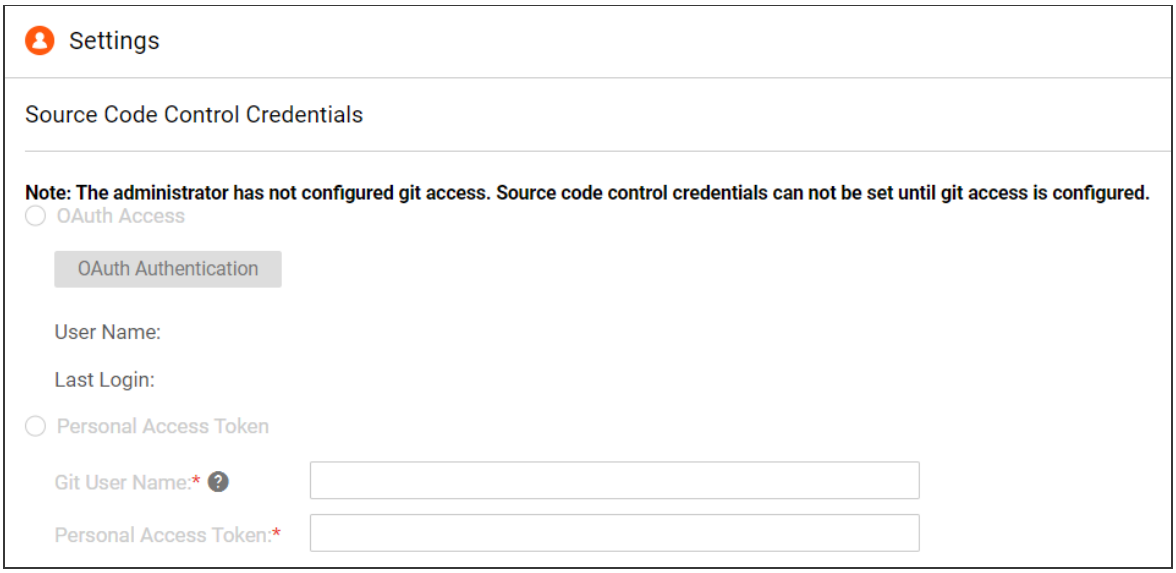
8. Once you update the details, click **Save** to save the details.



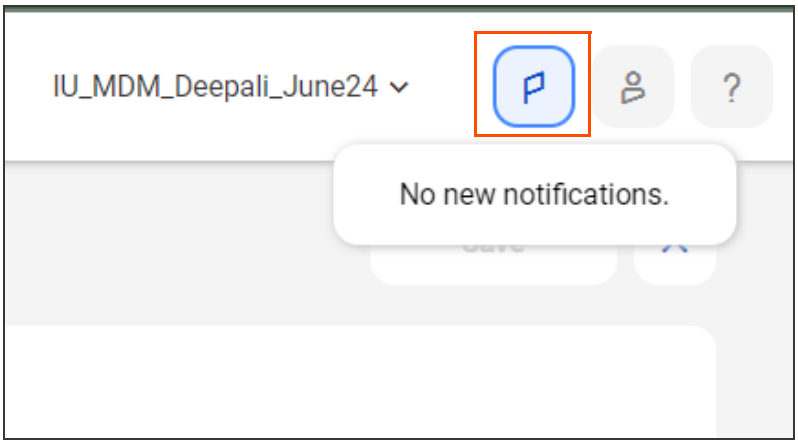
9. Select **Settings** from the menu.



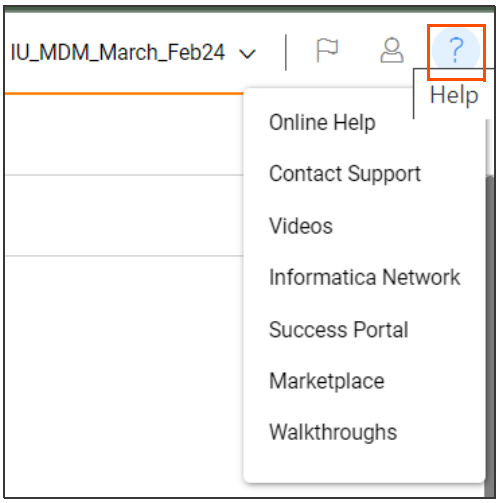
Here you will be able to change **Source Code Control Credentials** if appropriate privileges are given to the user.



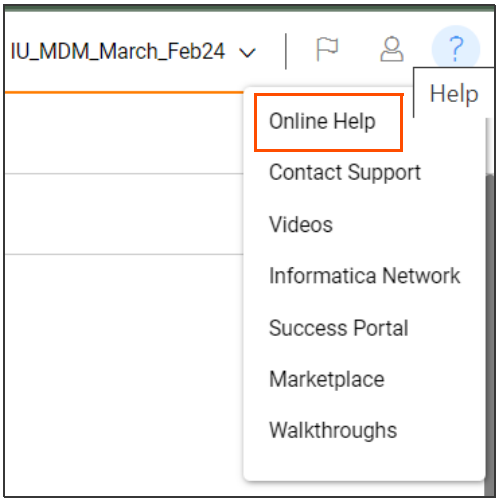
10. Click on the **Notification** icon. Here you can see the details of all the jobs that were run by the user.



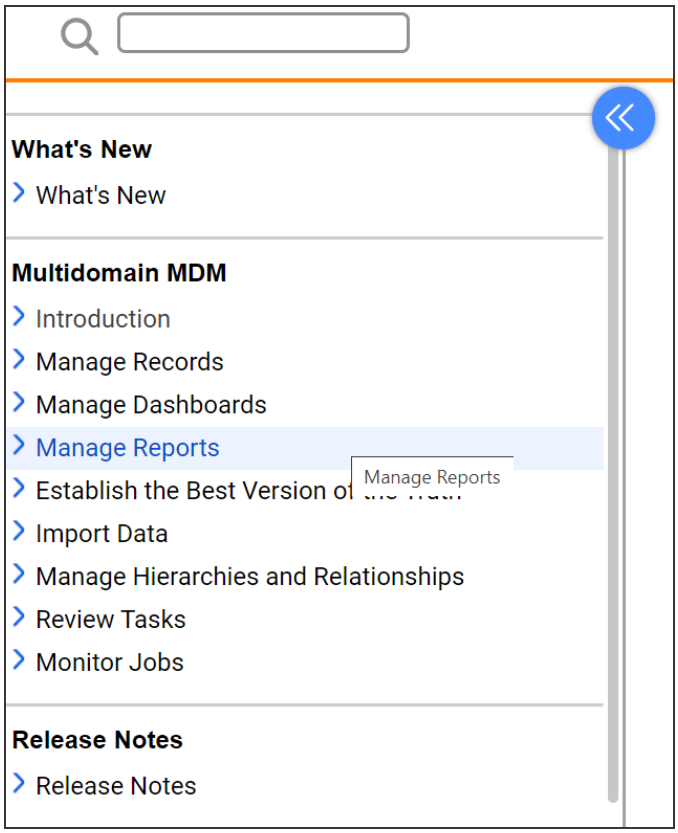
11. To get help on several topics, click the '**?**' to access the **Help**. You will see the list of menu items.



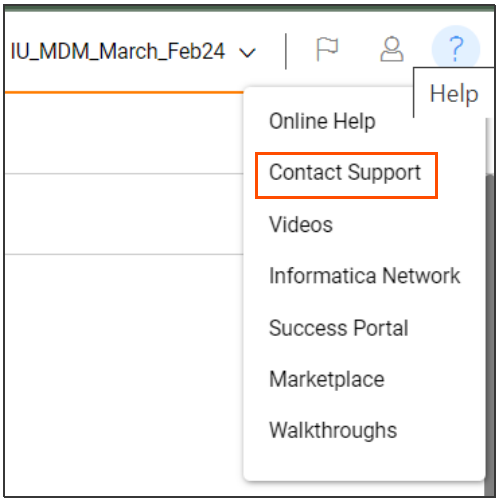
12. Select the **Online Help** option from the menu.



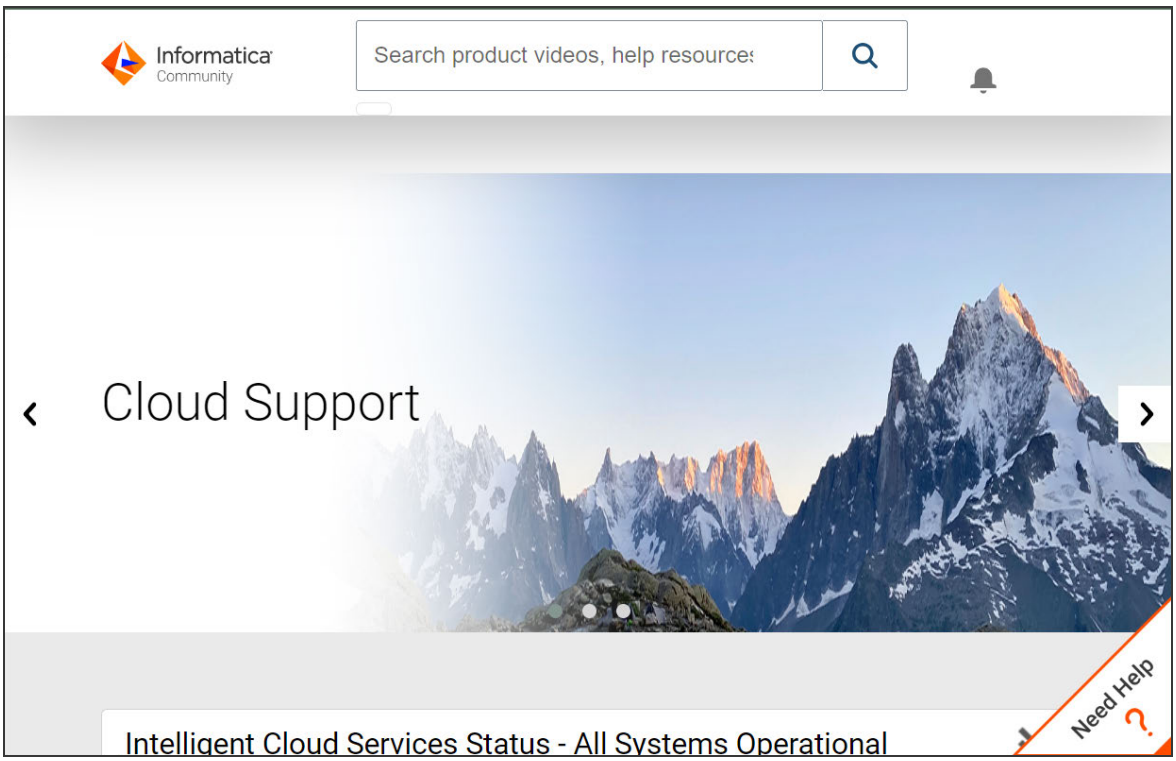
Here you will see all the topics/features related to Multidomain MDM. You can choose any topic and learn about it in detail.



13. Select the **Contact Support** option from the menu to get in touch with the support team of Informatica.



This includes **Cloud Support** and help with regards to various regions like West 1 Services, East 2 Services etc. and other contact details.



14. Similarly there are other options like Videos, Success Portal etc. You can click on the each option to access help and resources related to Multidomain MDM. Refer to the following table to know about each menu option:

Option	Description
Videos	Here you can watch demos, Webinars, and other videos
Success Portal	Here you can see Informatica Experience Lounge, explore the learning path of various Informatica products
Informatica Network	Here you can post your questions, issues with the Informatica community
Marketplace	Here you can browse through hundreds of Apps and Services to find what you need.
Walkthroughs	Here you can watch simulations on use case scenarios of Informatica product features.

This concludes the lab.

Lab 2-2: Business 360 Console UI Walkthrough

Overview:

Informatica Cloud Business 360 Console is one of several services available in Informatica Intelligent Cloud Services. Business 360 Console is an on-demand subscription service that provides a platform for cloud master data management. You can use Business 360 Console to create a variety of master data management projects to fit your business needs.

Objectives:

- Log in to Business 360 Console with the admin credentials
- Walkthrough the UI of Business 360 Console

Duration:

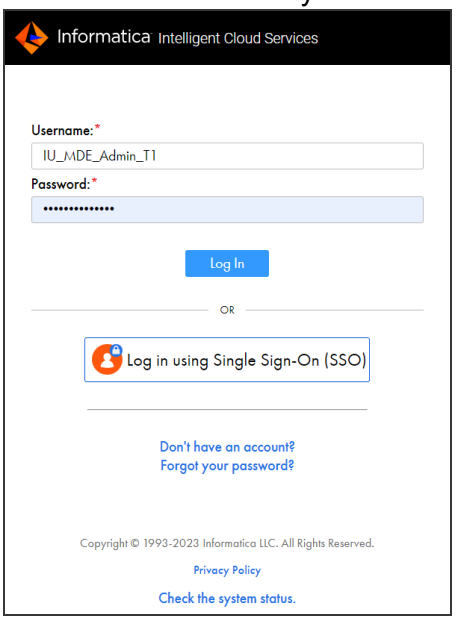
15 minutes

Tasks:

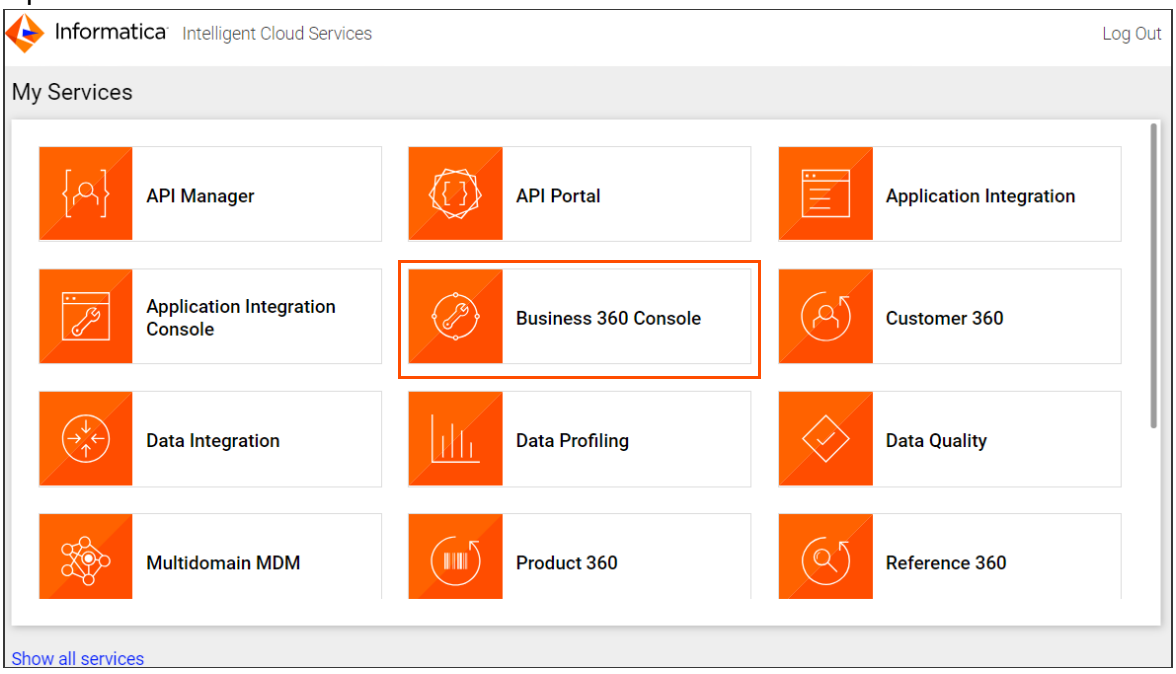
Walkthrough the UI of Business 360 Console

You will now login to Business 360 Console service and explore the its user interface.

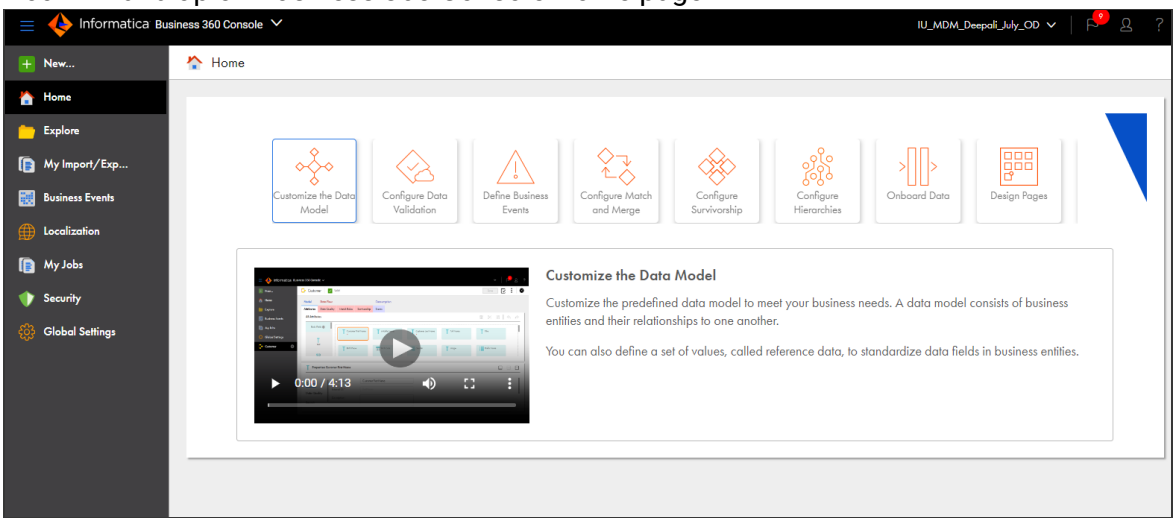
1. Enter the username and password as IU_MDE_Admin_T1 / Multidomain360. (Use the admin credentials that your instructor has shared with you).



2. Open **Business 360 Console** from the list of service.



You will land up on Business 360 Console Home page.

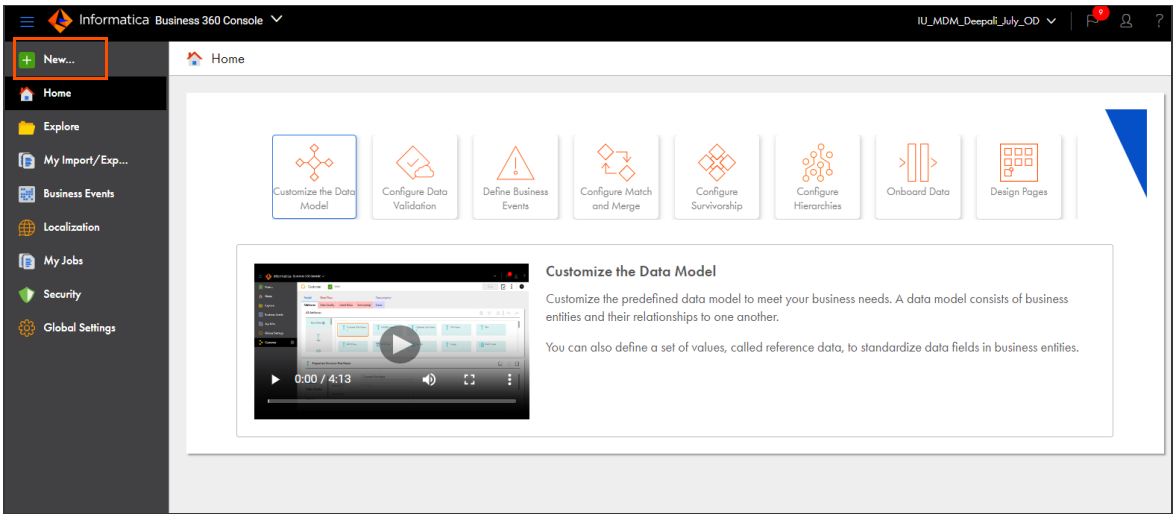


3. You can access the following pages from the navigation bar for Business 360 Console:

- **New**: Create Business 360 Console assets, such as business entities, workflows, and applications.
- **Home**: Return to the Home page.
- **Explore**: Create and manage projects and assets.
- **My Import/Export Logs**: View and download the logs of your import and export jobs.
- **Business Events**: Create business events to trigger workflows.
- **Translations**: Manage the translations of asset labels that appear in business applications.
- **My Jobs**: View, schedule, and monitor jobs.

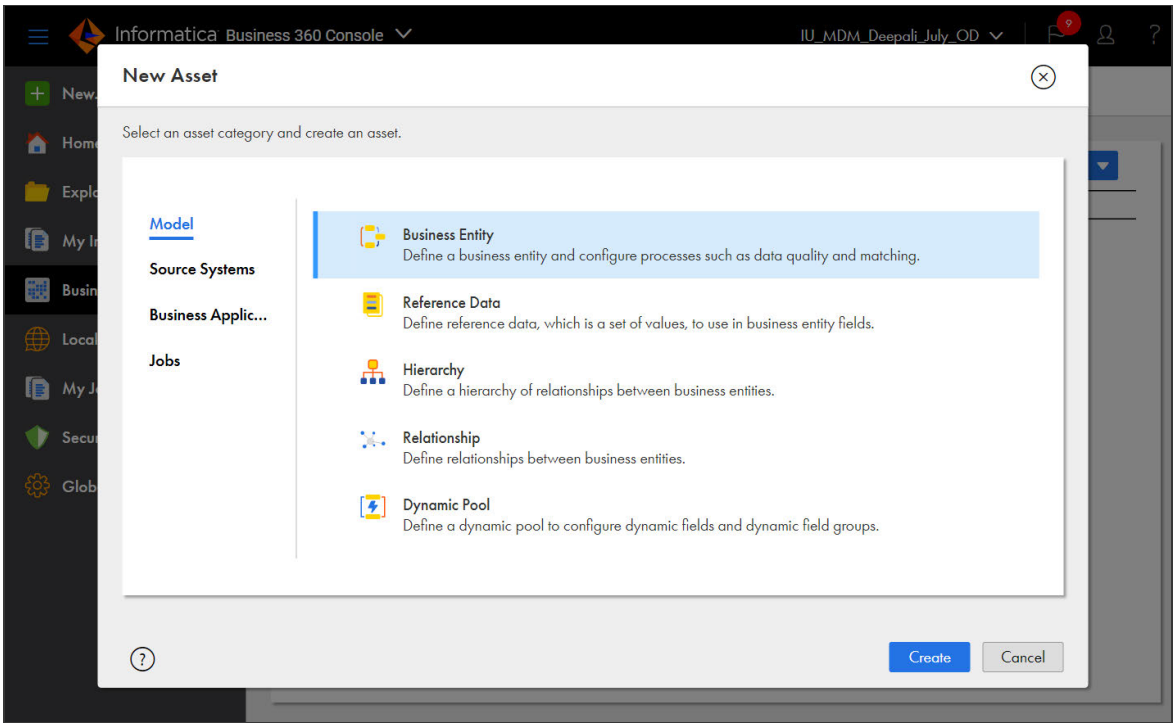
- **Security**: Configure privileges and data access rules for custom user roles.
- **Global Settings**: Configure the settings for license keys, Informatica Global Customer Support access, and data purge.

4. Click **New**.

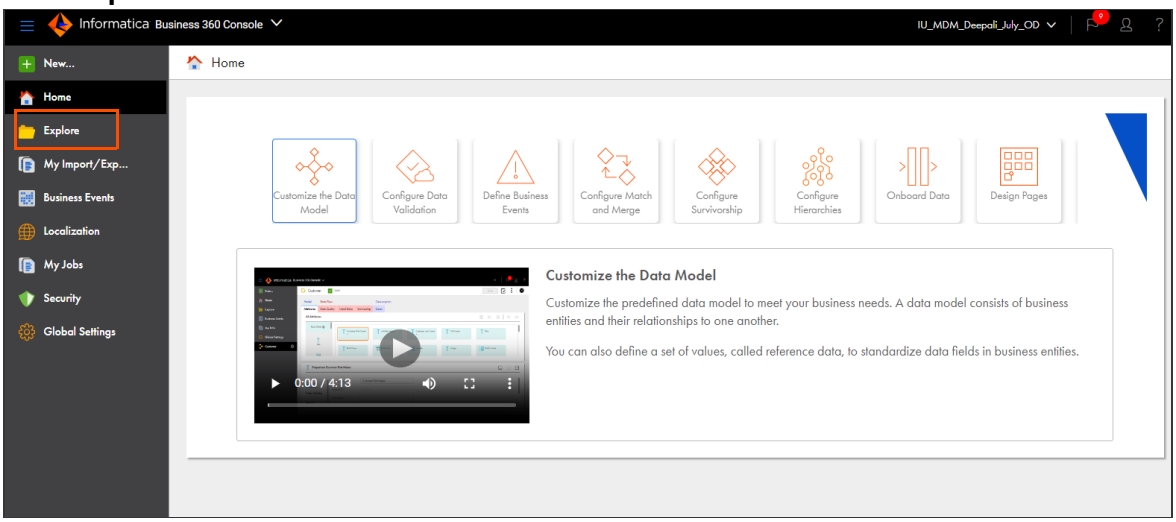


5. Here you can create new assets. These are multiple options that you can choose while creating the asset. These inlcude:

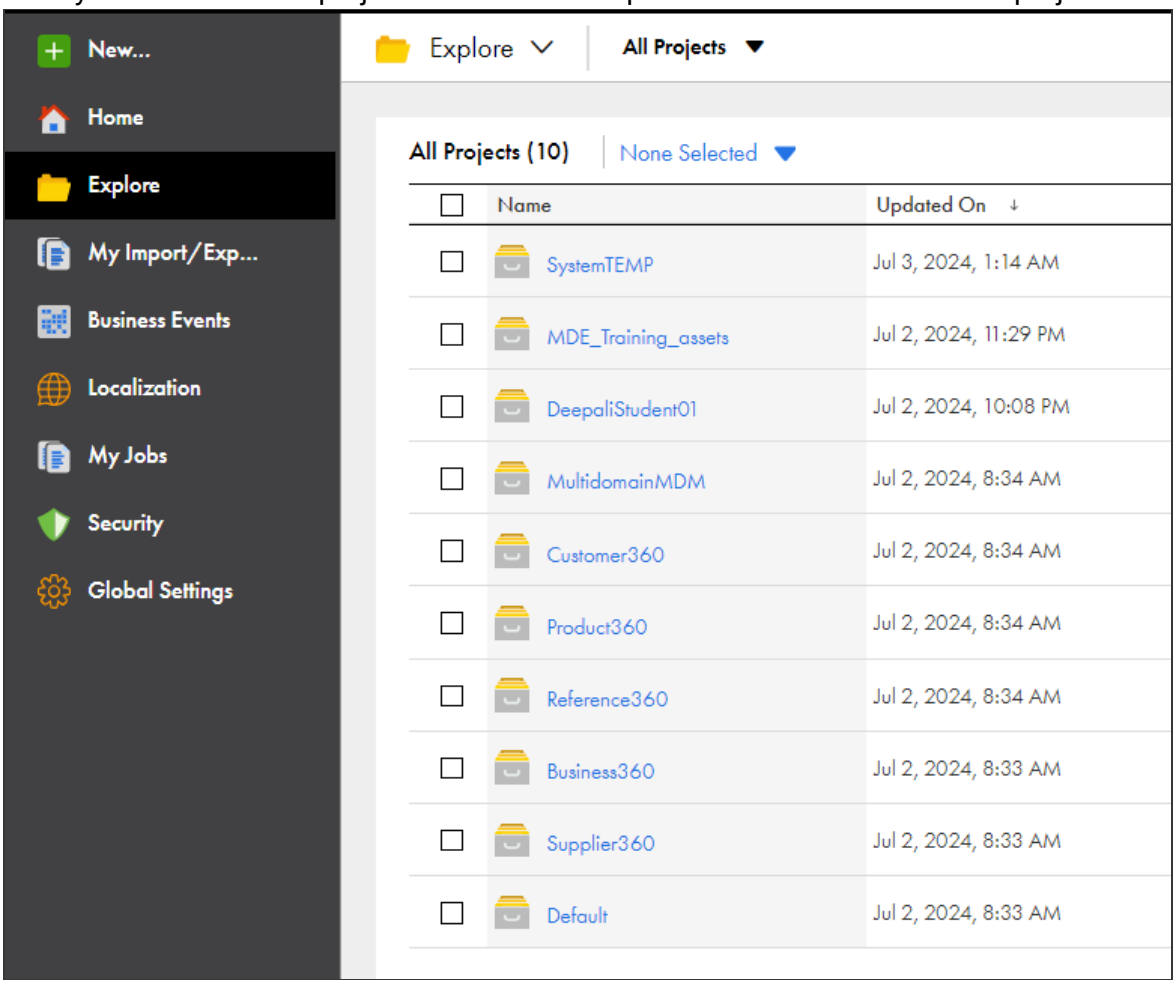
- **Model**: Here you can create Business entity, reference data, Hiearachy, relationshio and dynamic pool.
- **Source System**: Here you can create a source system.
- **Business Application**: Here you can create a user interface component or custom page.
- **Jobs**: Here you can create jobs like ingress,match and merge, egress, generate merge task etc.



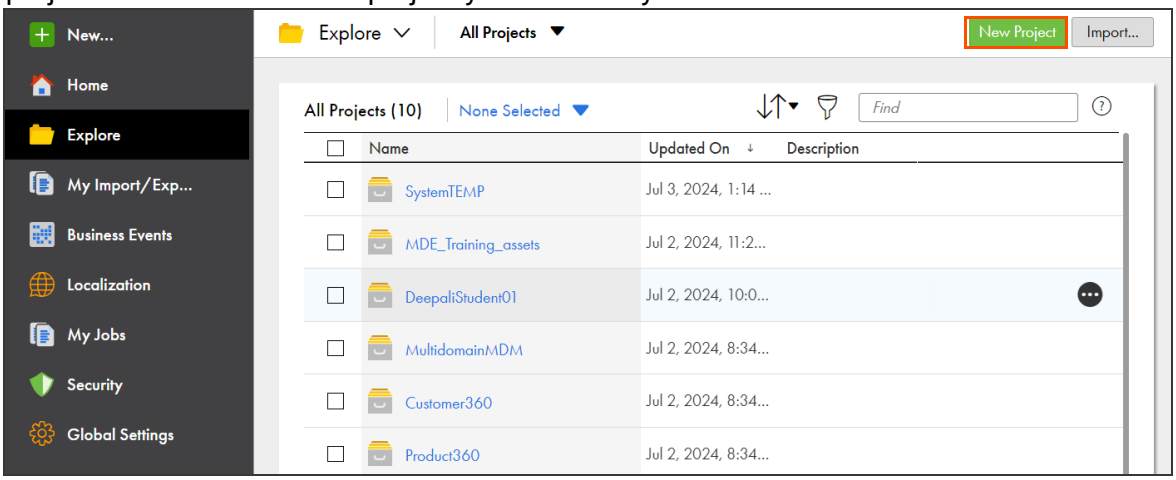
6. Click **Explore**.



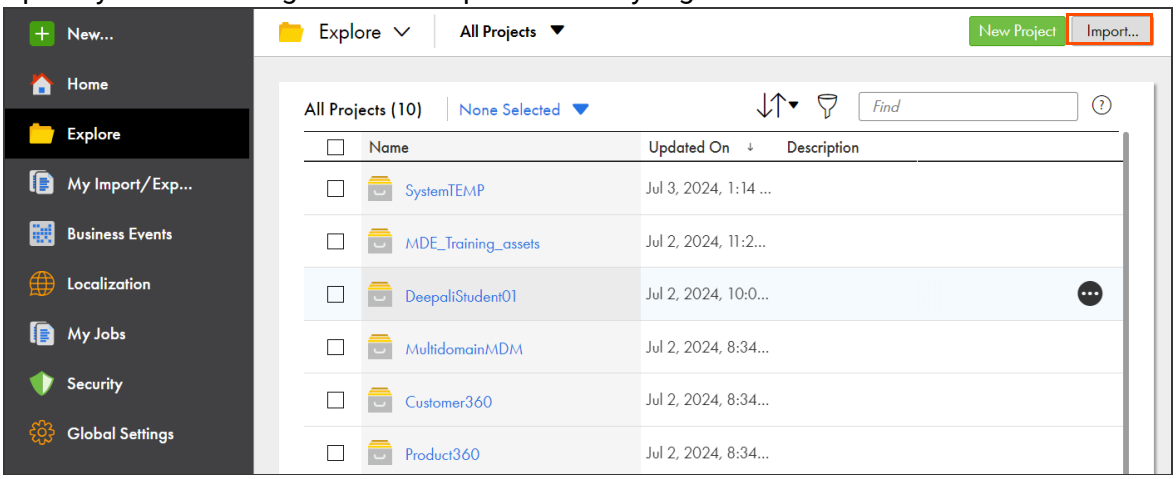
7. Here you will see all the projects. These include predefined as well as custom projects.



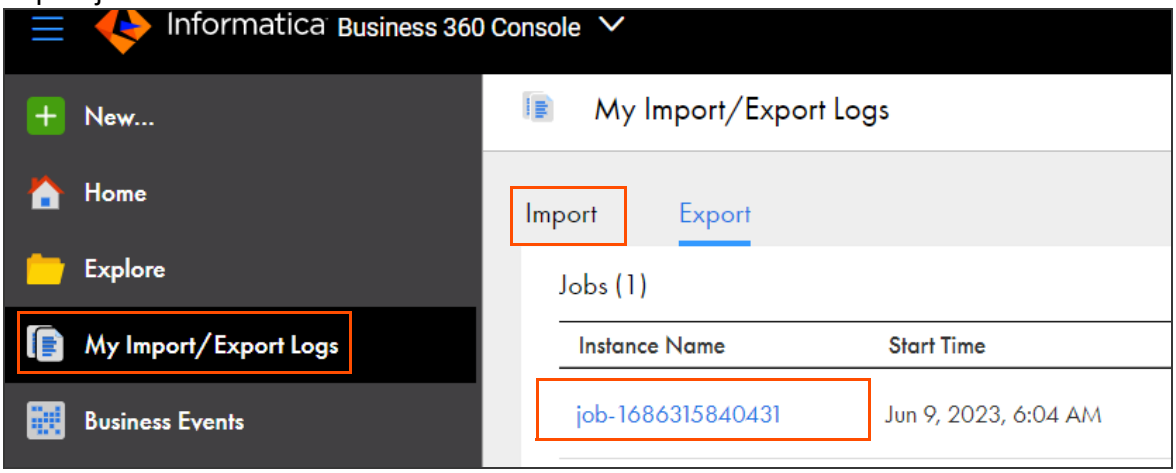
8. On your right hand side, you will see **New Project** option. You can create your student project here. In this student project you can save your custom assets.



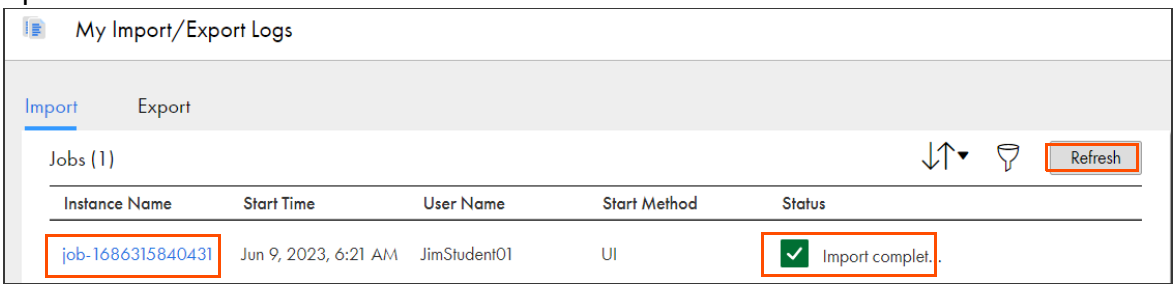
9. The **Import** option will allow you to import assets from any a zip file. With this import option you can leverage the assets present in any org.



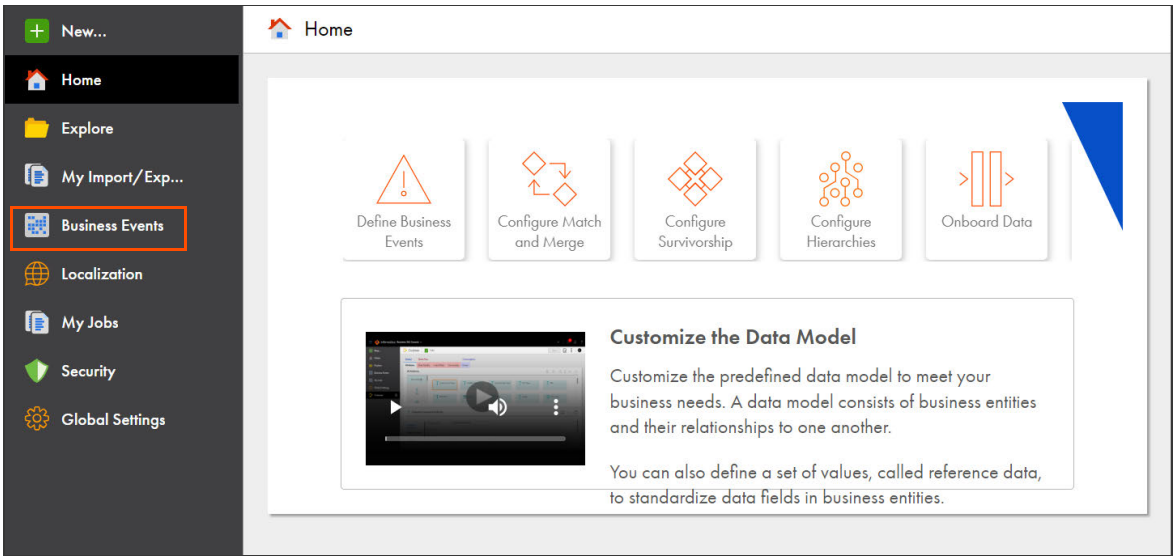
10. Click **My Import/Export Logs**. Here you can view and download the logs of your import and export jobs.



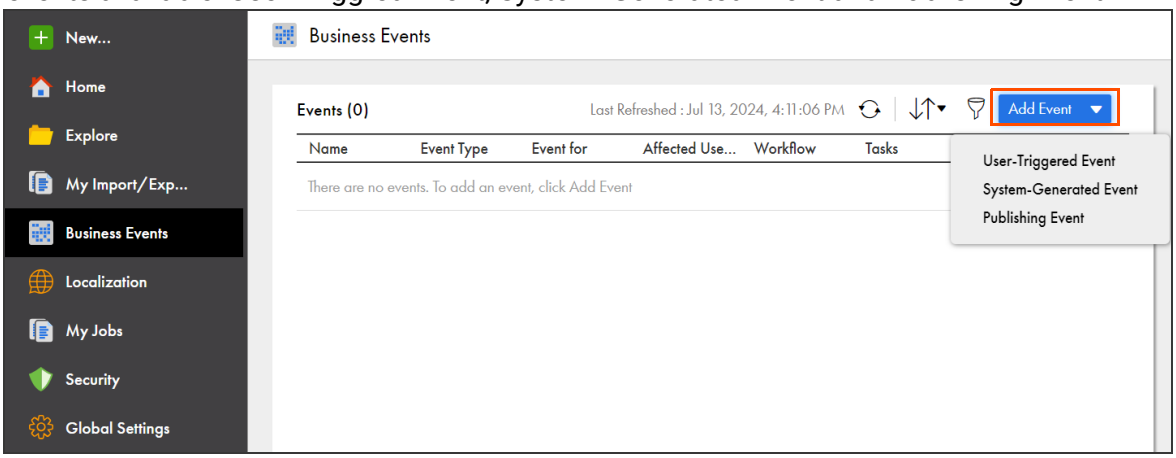
11. If you open the job instance, you will view the details. You can click **Refresh** to see the updated status



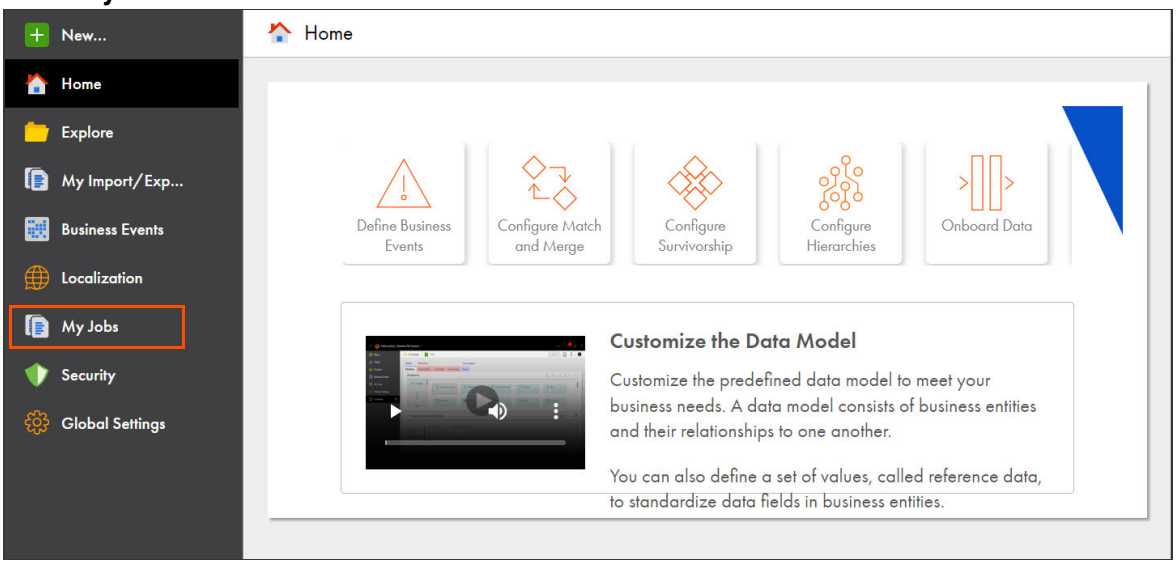
12. Click **Business Events**.



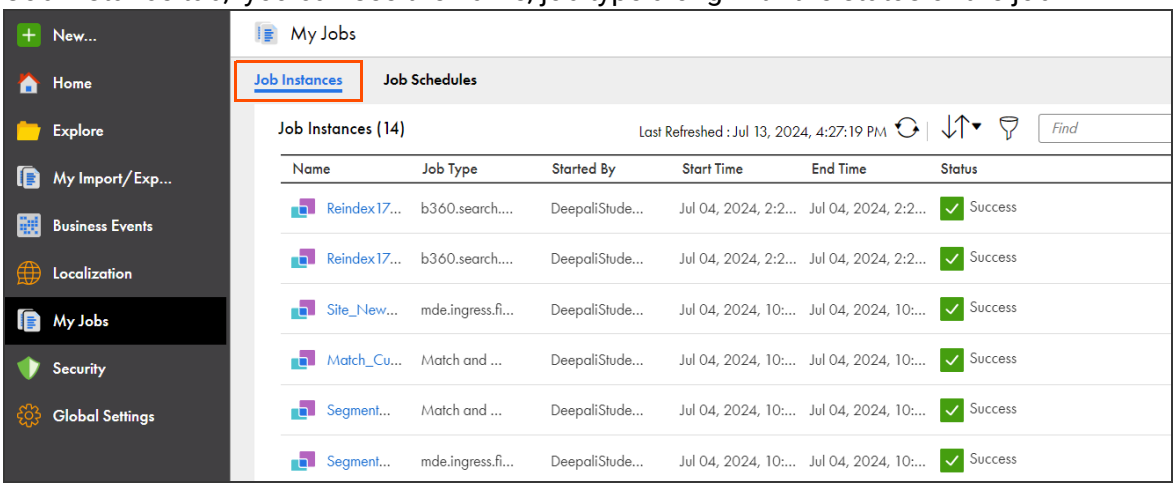
13. Click **Add Event**. Here you can create business events to trigger workflows. Three types of events available: User-Triggred Event, System- Generated Event and Publishing Event.



14. Click **My Jobs**.



15. Here you can see view, schedule, and monitor jobs that you have created and run. Under Job Instance tab, you can see the name, job type along with the status of the job.



This concludes the lab.

Key Takeaways

- Informatica MDM Cloud Edition empowers companies with reliable data about customers, products, suppliers, locations, and more by providing a 360-degree view of business-critical data.
- Multidomain MDM user interface is simple and Intuitive that is easy to learn and helps the user to pick it up quickly and easily.
- Informatica Cloud Business 360 Console is one of the key services offered by Informatica Intelligent Cloud Services.
- IICS provides various tools like: Projetcs, Assets, Explore Page.

- A business entity is an entity of importance to your organization and consists of fields that support your data needs.

Module 3: Defining Business Entities

Define the Data Model

To define a data model, you need to analyze, strategize, and define your business and data requirements by collaborating with various stakeholders.

The following are the prerequisites that you need to consider before you start defining the data model:

- Understand the business objectives and requirements for your data management project.
- Provide insights into what various stakeholders want and need from the data management project.
- Identify the source systems that contribute to the master data.
- Analyze the attributes and quality of the data that you want to consolidate into master data.
- Identify the business entities and the relationships between business entities that are critical to your data model.
- Identify reference data, such as lists of countries, currencies, products, genders, and salutations, that your organization uses.

To define your data model, perform the following tasks:

1. Customize predefined business entities, or define custom business entities and configure data quality, match model, and survivorship.
2. Define reference data, which is a set of values, to use in picklists that you include in business entities.
3. Define relationships between business entities.

Configuring a Data Model

To be able to define and use a custom business entity, ensure that you follow the steps in the configuration process.

Figure 3-1 shows the custom business entity configuration process:

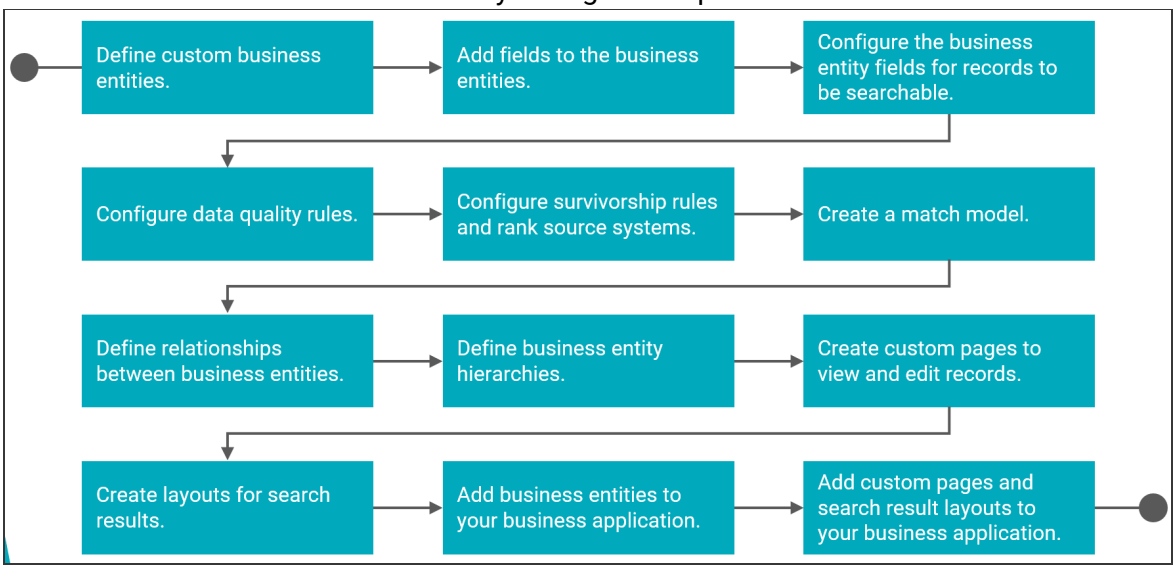


Figure 3-1: Configuration Process

Defining custom business entities: To define a custom business entity, perform the following tasks:

- Create a custom business entity
- Add fields to a business entity
- Configure the field properties

Add fields to a business entity
After creating a custom business entities, you will add fields to it. Then configure the field properties.

Configure business entity properties
You can configure business entity properties that suit your business requirements. The following table describes the business entity properties:

- Display Name
- Internal ID
- Description
- Location
- Icon

Configure data quality rules and survivorship
You can associate data quality rules with the business entity fields to cleanse, standardize, or validate the field values. Survivorship is the process of creating a master record or the most trusted version of a record from multiple matching source records. The privileges specify whether users can read and update the survivorship configuration of the business entity and fields.

Create a match model
The match model determines how your data is matched. You can use the predefined match model to identify and resolve duplicate records.

Define relationship between business entities
You can define relationships between business entities. The users of business applications, can then define and visualize the relationships between business entity records.

Defining hierarchy models
A hierarchy model consists of a collection of hierarchy relationships between business entities.

Creating a custom page
You can create and design custom pages for your application. You can add one or more tabs, split the page into sections, and add components. The components differ based on the business entity and page type you select.

Create a search results layout component
You can configure search results layout components for the business entities based on your business needs and assign them to the user roles defined in your application. Each search result in the application displays the field values based on the search results layout components assigned to a user role.

Add business entities to an application
You can add business entities and configure the display fields for each business entity. The values of the selected display fields appear in the left navigation bar when you open a record. You can view the list of components, such as search box, notifications, and tabs, that are enabled for the application. The enabled components appear on the application header and left navigation bar.

Configure search for an application
You can configure the search properties, such as search suggestions and search history, for an application. For example, you can configure the number of previous searches to display when you click the search box.

Defining Business Entities

A business entity is an entity of importance to your organization and consists of fields that support your data needs. To define a business entity, decide on the fields based on the data that your source systems will contribute to the master data.

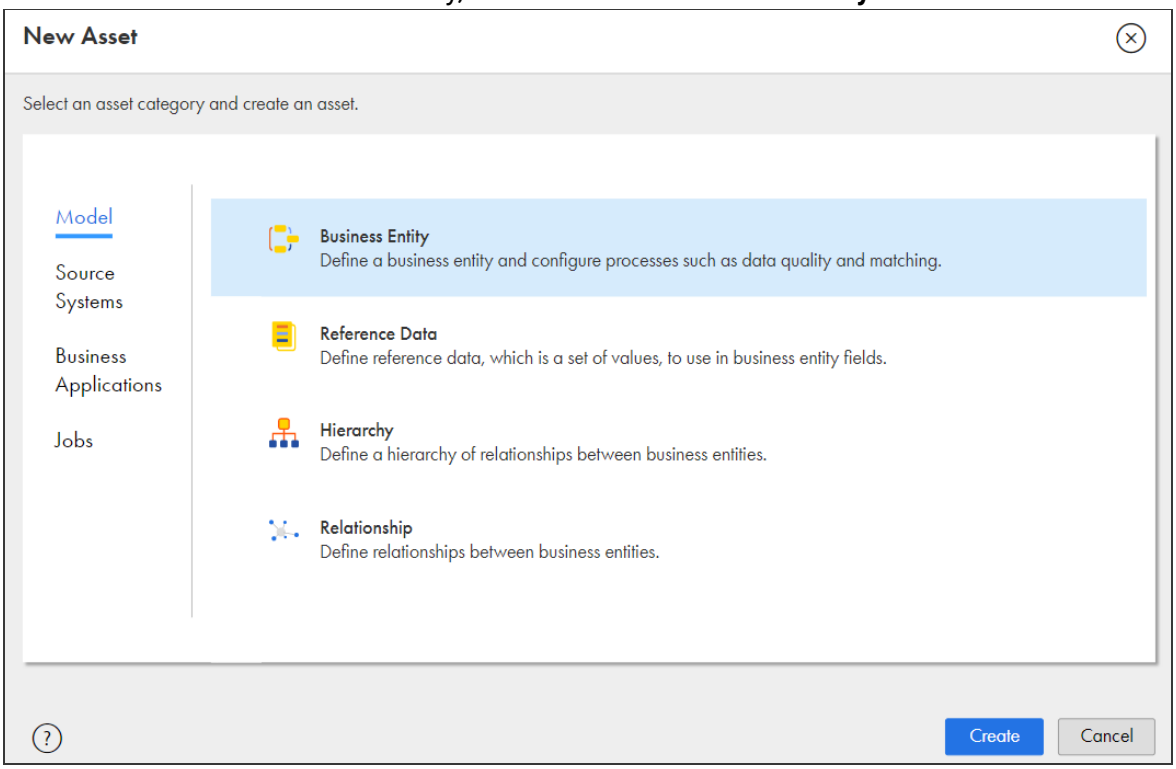
For example, a company that manufactures cars might define business entities for some or all of the following entities: customers, employees, suppliers, factories, materials, and products. A charitable organization might define donors, households, and projects. A health care provider might define doctors, care sites, and insured services. To define a business entity, you often use an iterative process. You can come back later to add more fields and to modify or add configurations, such as data quality, search, and survivorship. Before you define a business entity, determine the type of business entity that you want to use.

You can define and use the following types of business entities:

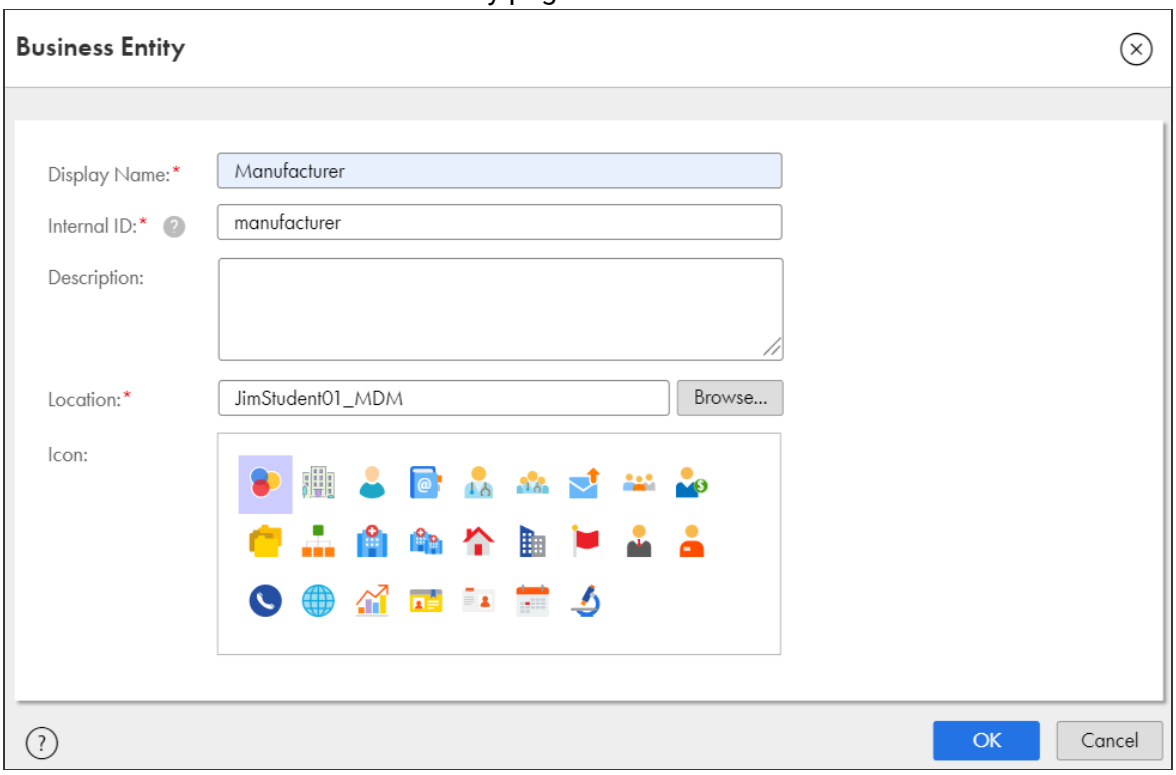
- Predefined business entities
- Custom business entities

Refer to the following steps to leanr the process of creating custom business entities:

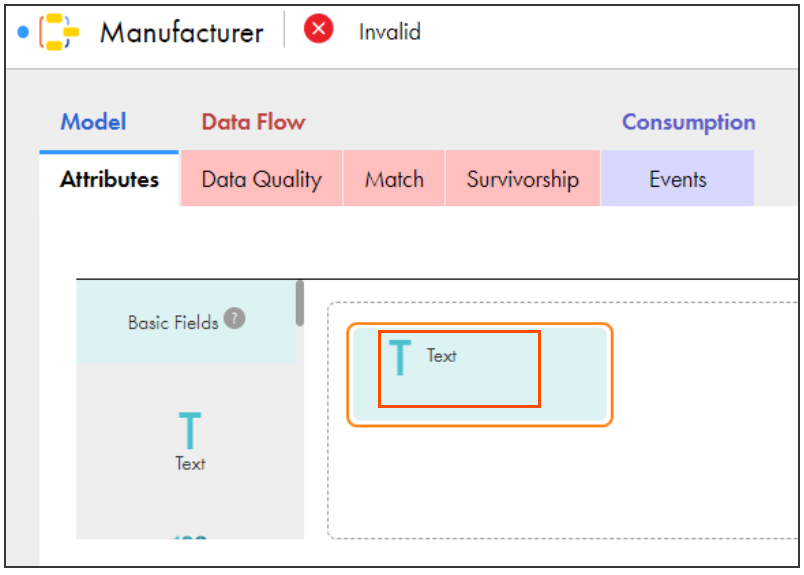
1. To create a custom business entity, select **Model** > **Business Entity** and click **Create**.



2. Enter the details in the Business Entity page.

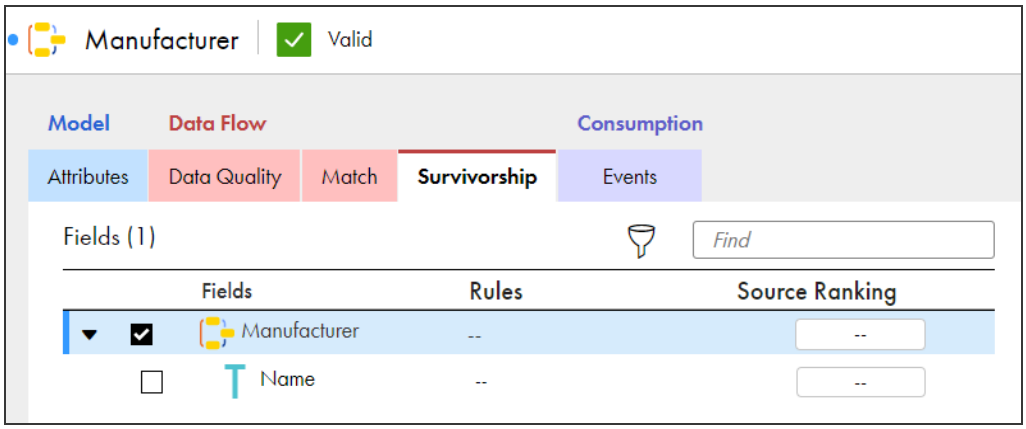


3. Adding Fields to the custom business entity.



4. Make the fields searchable.

5. Configure survivorship.



6. Click **Save**.

Relationships

You can establish associations between business entities that are connected in some manner. Users of Business 360 applications like Customer 360 SaaS can then map out and visualize the interconnections between business entity records. Multiple associations can be defined for each business entity, allowing for relationships not only with other business entities but also with themselves.

For example: A relationship can be delineated between the Organization and Person business entities to signify employment, or a relationship can be made from the Person to Person business entity to denote a spousal connection.

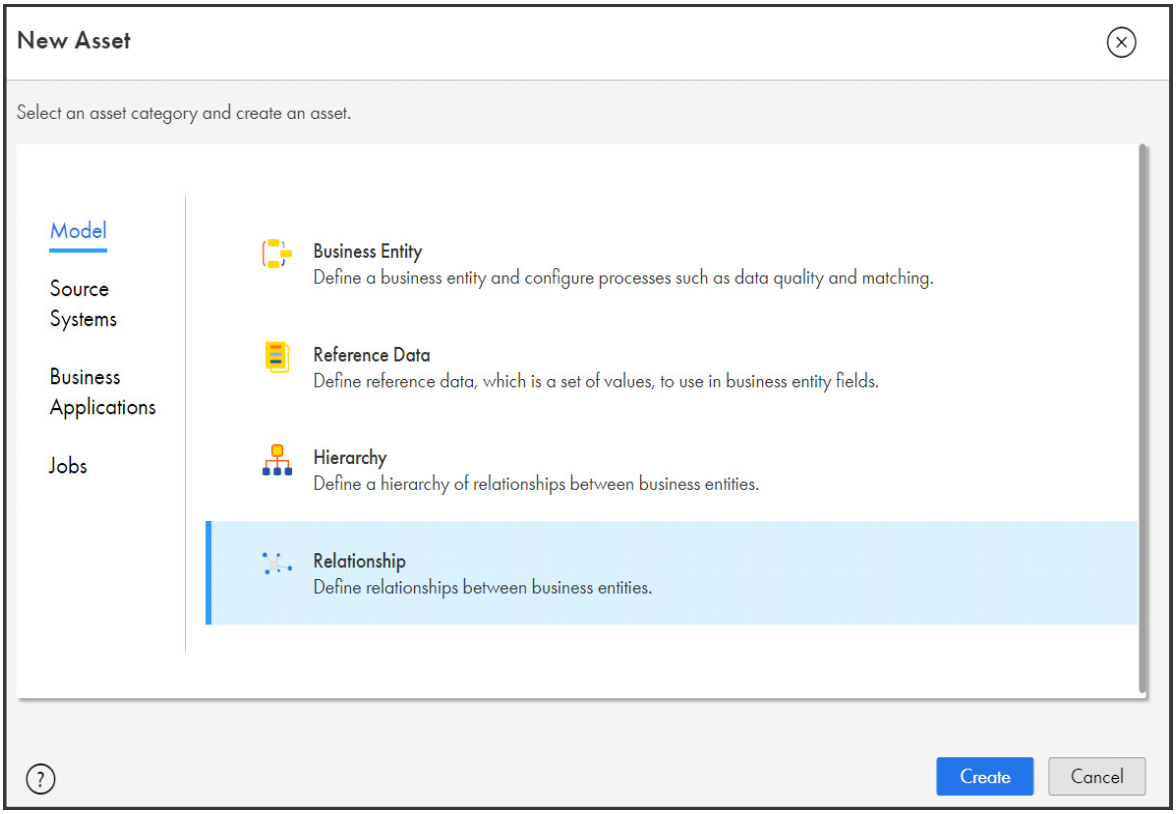
Relationship Types

Records can be interconnected through various types of relationships, including hierarchical relationships, associations between related records, and relationships based on fields within business entity records.

MDM SaaS supports one-to-one, one-to-many, and many-to-many relationships between business entities. Configure one-to-one relationships using hierarchies or relationships. Configure one-to-many and many-to-many relationships using business entity record fields. If a user has the necessary permissions, they can view, add, edit, and manage relationships.

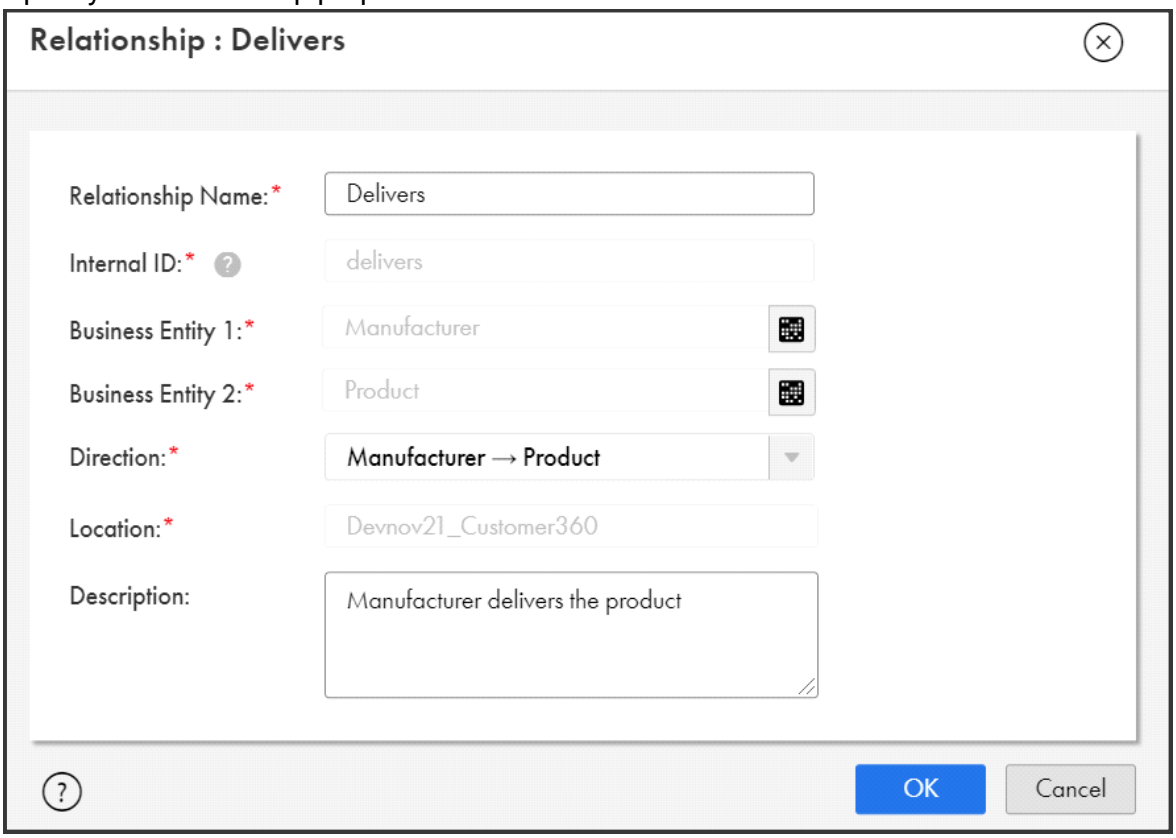
To create a relationship, perform the following steps:

1. Click **New**.
2. Click **Model** > **Relationship**.



3. Click **Create**.

4. Specify the relationship properties.



5. Click **Save**.

Lab 3-1: Creating and Configuring Custom Business Entities

Overview:

A business entity is an entity of importance to your organization and consists of fields that support your data needs. To define a business entity, decide on the fields based on the data that your source systems will contribute to the master data. To define a business entity, you often use an iterative process. You can come back later to add more fields and to modify or add configurations, such as data quality, search, and survivorship.

In this lab, you will design a data model from beginning.

Objectives:

- Create a new project
- Create a source system
- Create custom business entities
- Configure custom business entities
- Define relationships

Duration:

50 Minutes

Scenario:

After Mark has finished analyzing, strategizing, and defining the business and data requirements through collaboration with various stakeholders, he should move on to defining the data model. This involves creating project folders to organize the assets, identifying the source systems where his data is stored, creating custom business entities related to his business, and configuring relationships between these entities.

Tasks:

Perform the following tasks to define the data model for Homekart scenario:

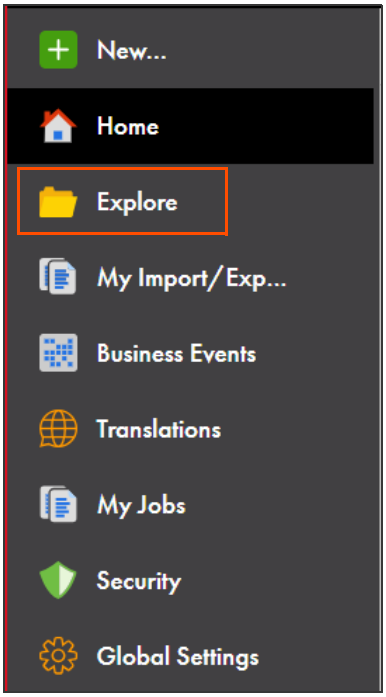
- Create a new project
- Create a new source system
- Creating Manufacturer custom business entity
- Adding fields to Manufacturer BE
- Creating Product business entity and add fields to it
- Add a picklist

Create a New Project

It is recommended that every student names their project after their username. (For example, **MarkStudent01**), and save the assets in the respective projects. This avoid naming conflicts and aids in the better organization of folders and assets that you create.

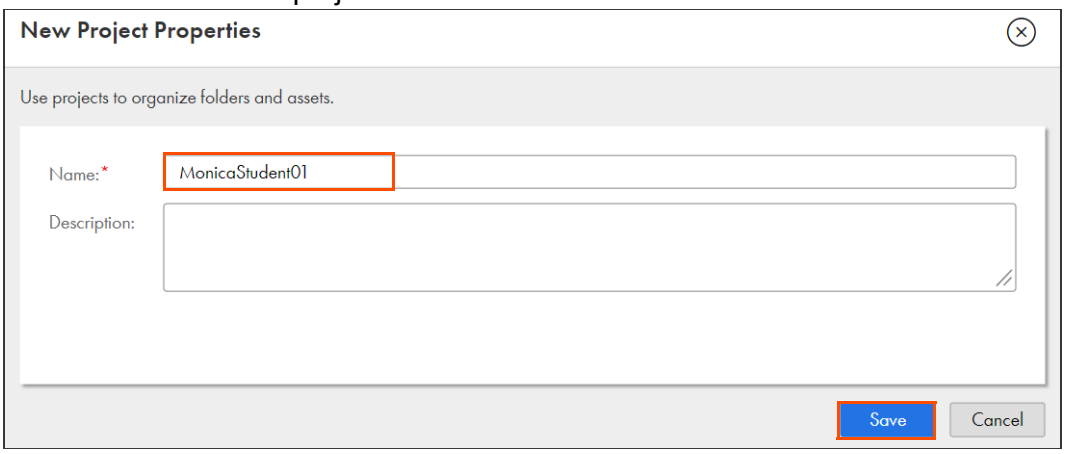
1. If you are navigated away from the previous lab, log in to **Business 360 Console** as the admin. **<studentName>student<studentNumber> / Multidomain360** (for example **MarkStudent01** / **Multidomain360**).

2. In the **Business 360 Console** window, from the left panel, click **Explore**.

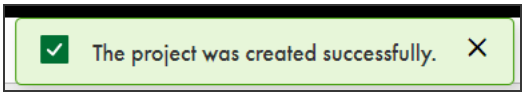


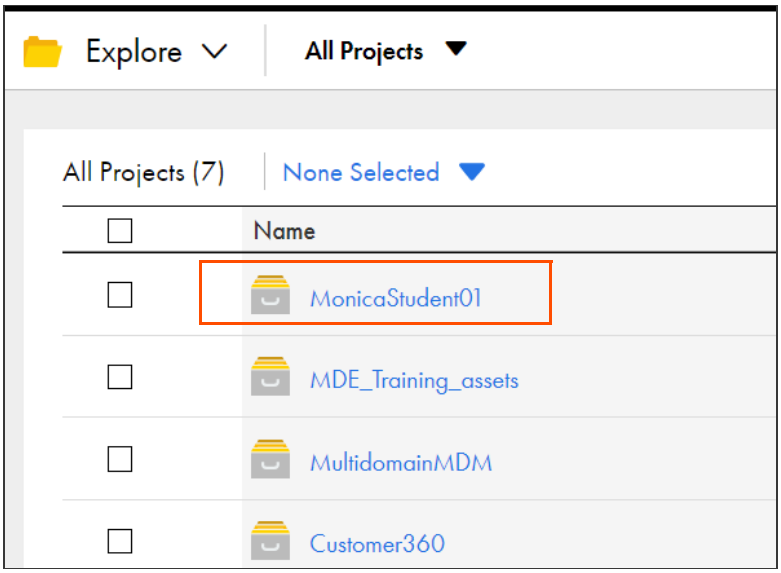
3. Click **New Project**.

4. Enter the name of the project as **StudentNameStudent<no.>** and click **Save**.



A notification that the project is created appears. The project now appears under **All Projects**. In the screenshots you may see MonicaStudent01 as the student project. It will be different in your case.

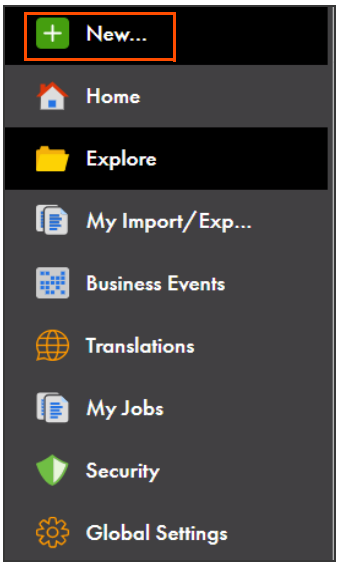




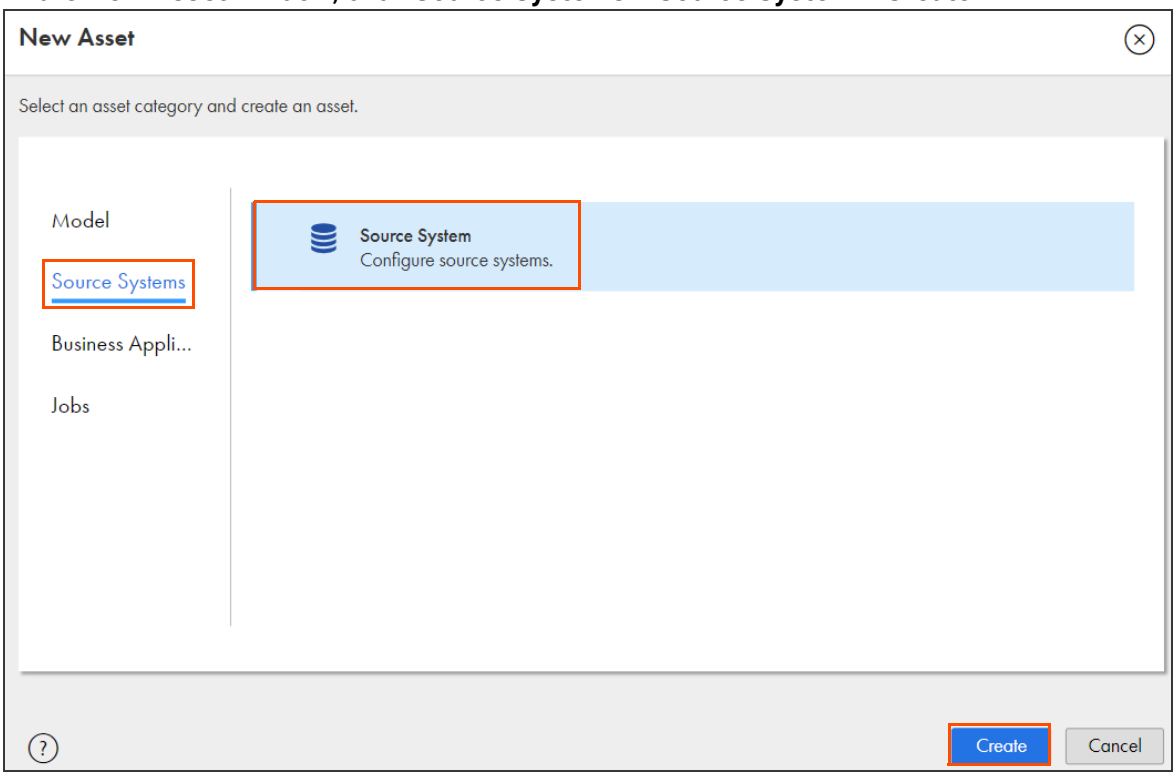
Create a New Source System

Before you start creating the custom business entities, you will first create a source system that act as data sources for the business applications to use.

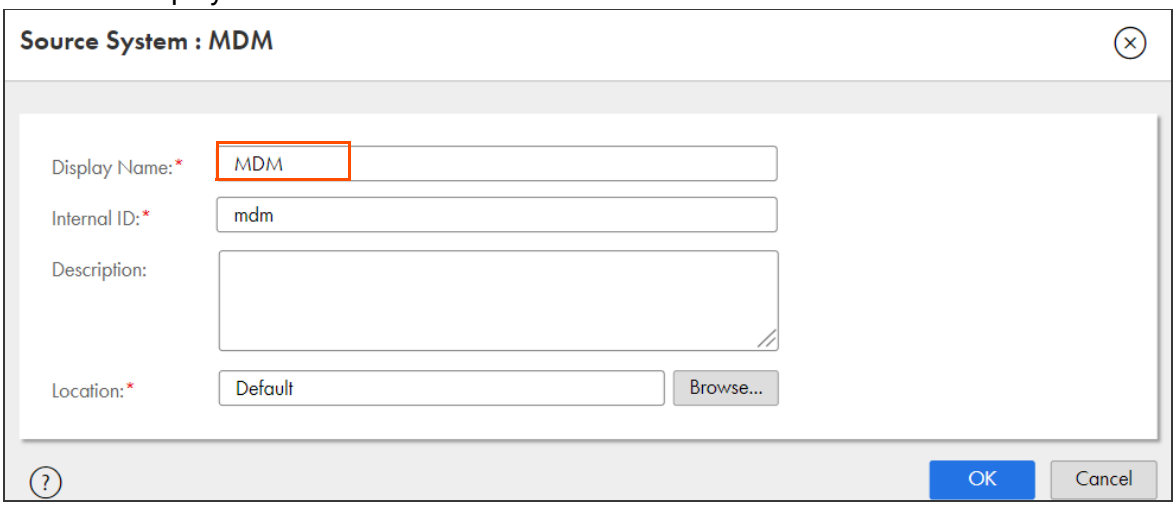
5. Click **New**.



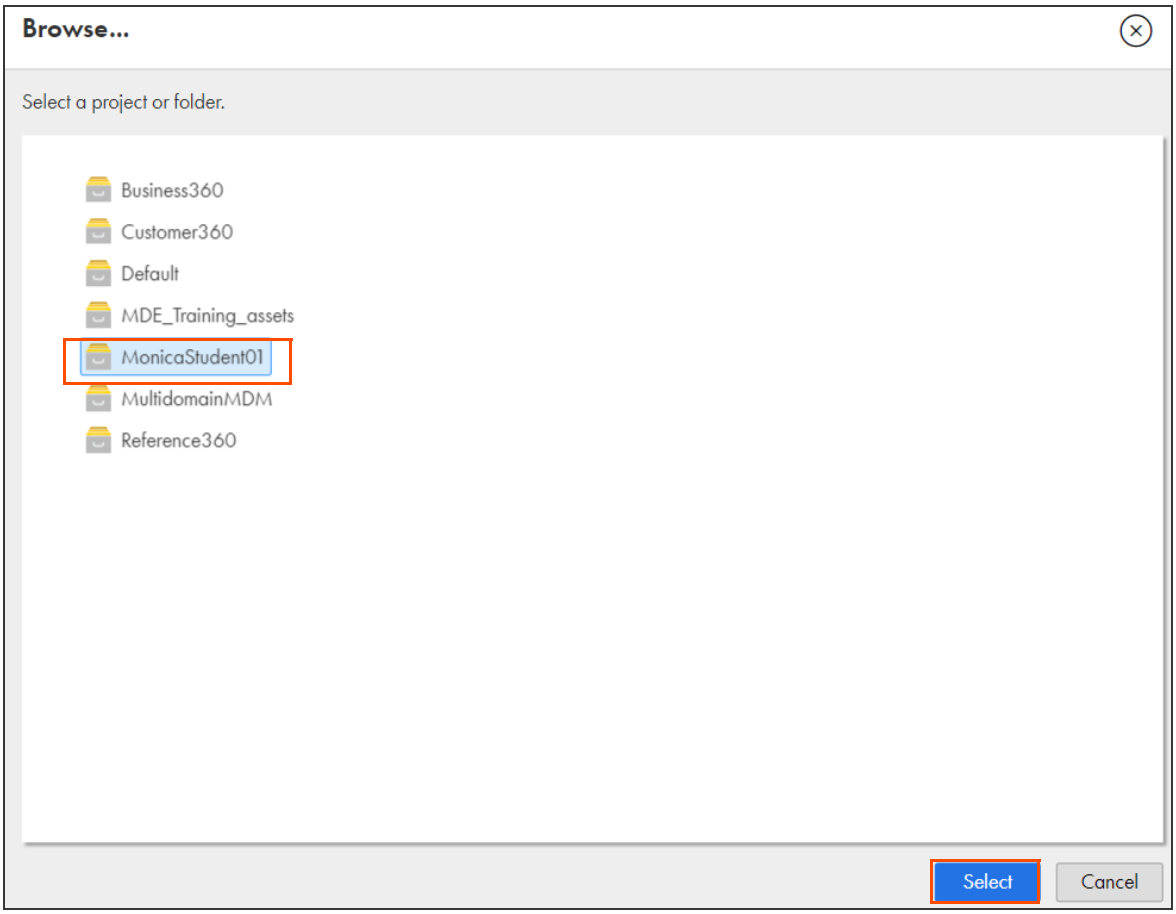
6. In the **New Asset** window, click **Source Systems** > **Source System** > **Create**.



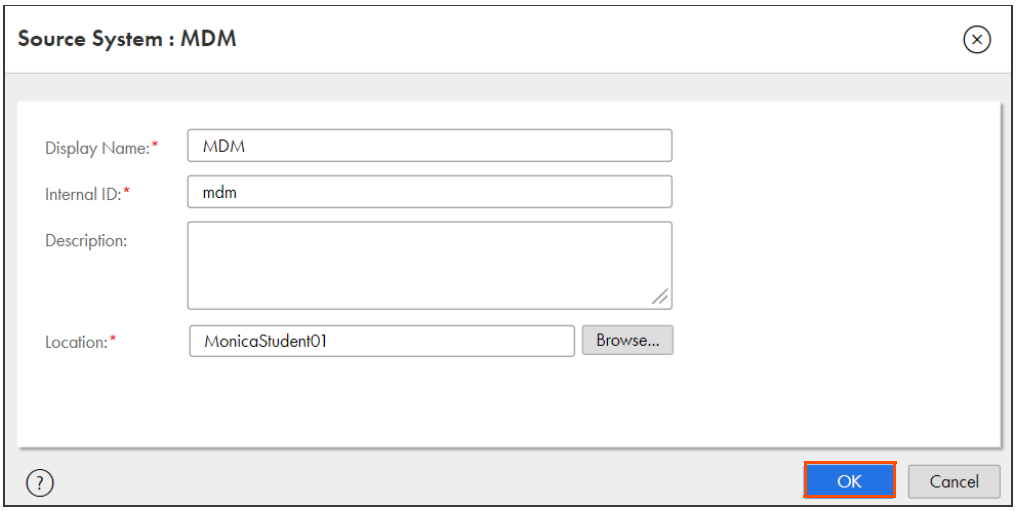
7. Enter the Display Name as **MDM**.



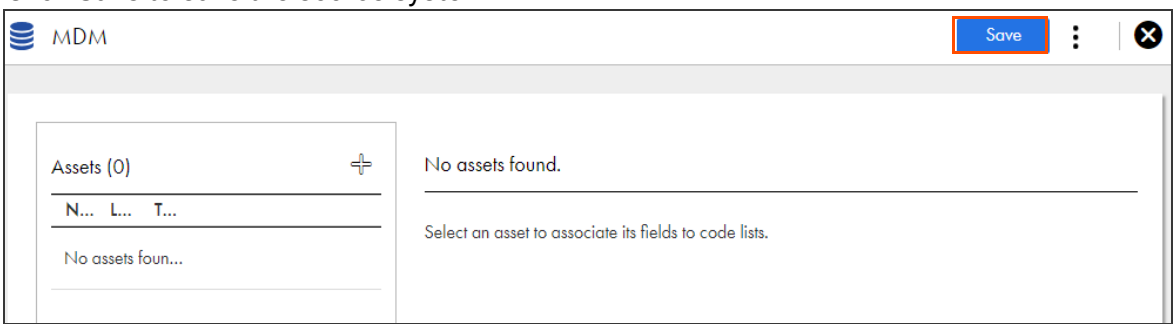
8. Click browse and select the student project that you created in previous steps. Click **Select**.



9. Click **OK**.



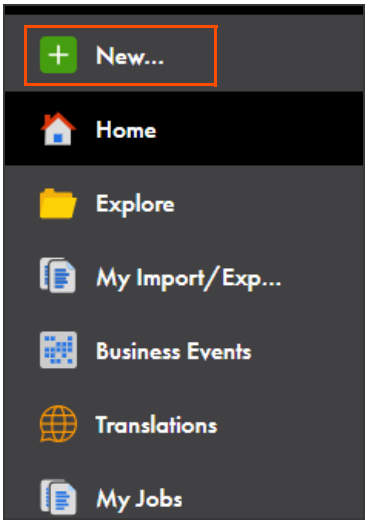
10. Click **Save** to save the source system.



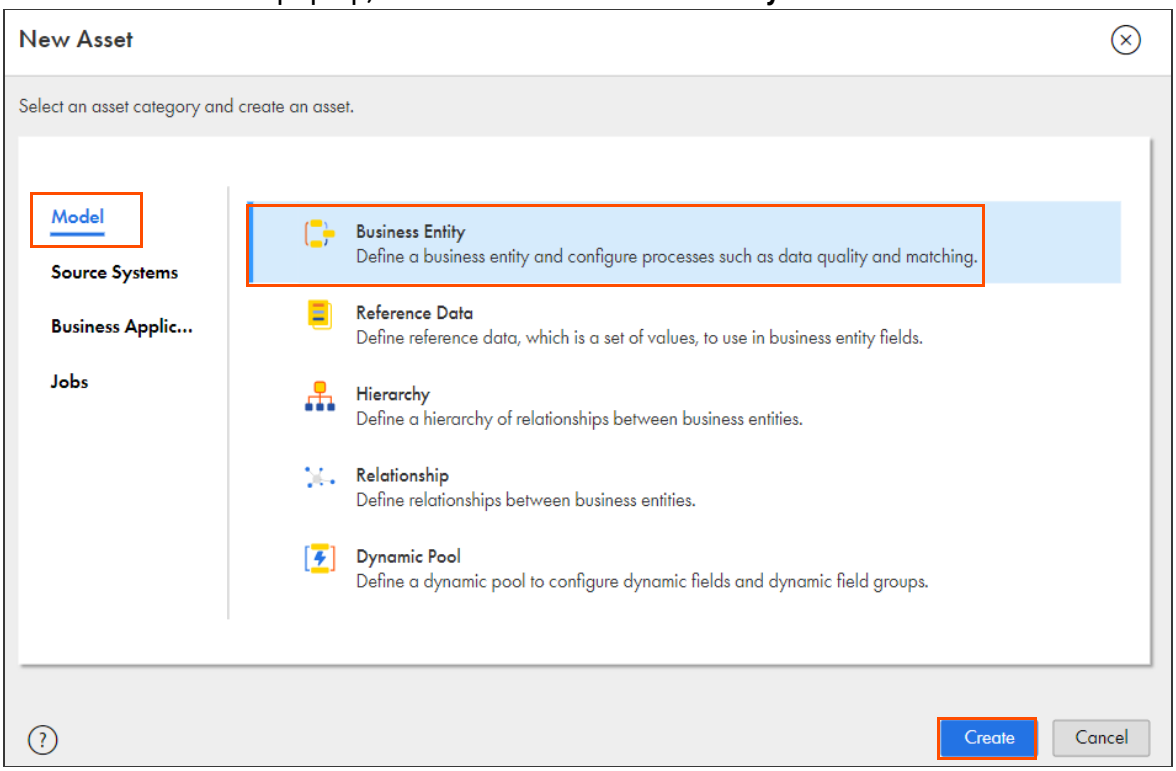
Creating Manufacturer Custom Business Entity

As a part of this task, you will create two custom business entities: **Manufacturer** and **Product**.

11. To create a new custom business entity named Manufacturer, from the left side menu, click **New**.

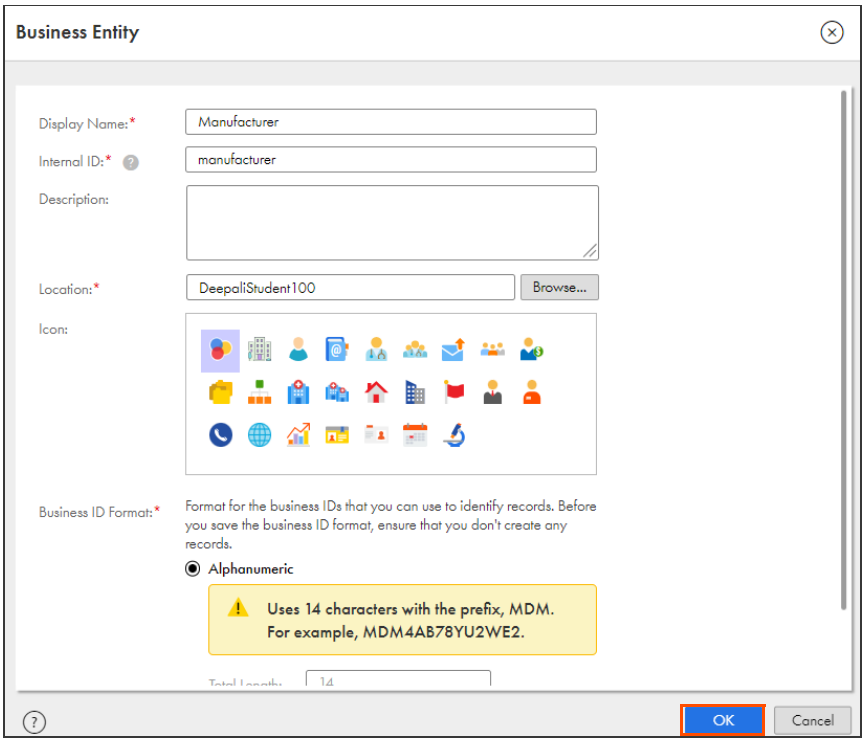


12. From the New Asset pop-up, select **Model** > **Business Entity** and click **Create**.



13. Enter the following details in the Business Entity page. Click **OK**.

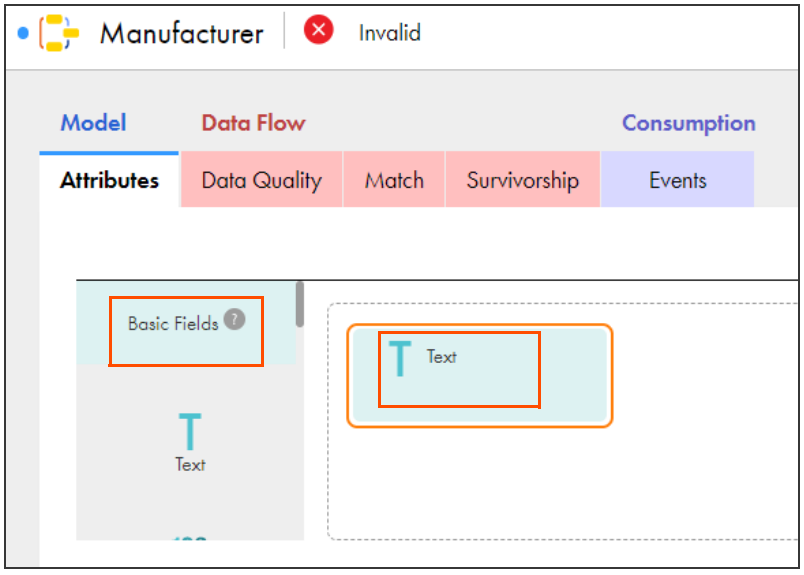
Field	Value
Display Name	Manufacturer
Location	<Select your student project>
Icon	



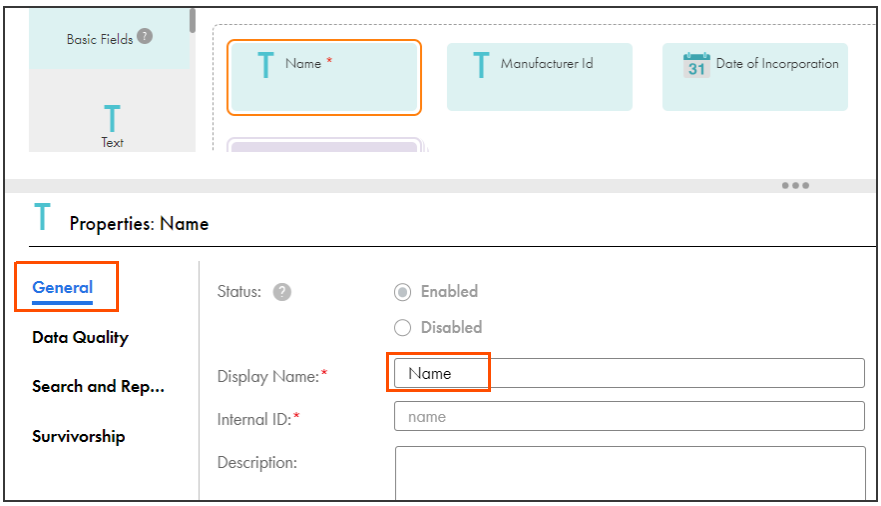
Adding Fields to Manufacturer BE

You will add basic fields to the business entities to configure the field properties such as General, Data Quality, Search, and Survivorship. Then you will add smart fields like Address and Email.

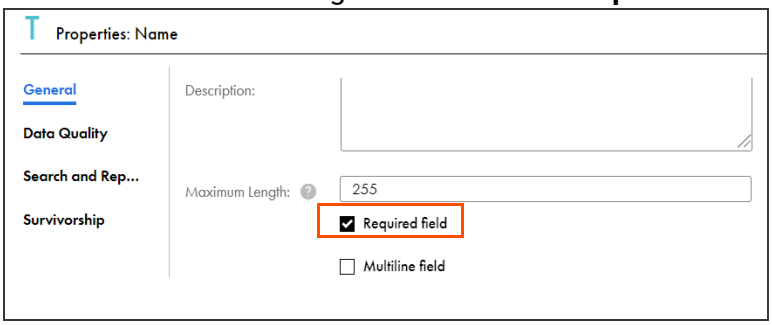
14. In the Manufacturer business entity page, expand the **Basic Field**, double click **Text**.



15. An empty field of Text type is added. From the **Properties** > **General** section, enter Display Name as **Name**.

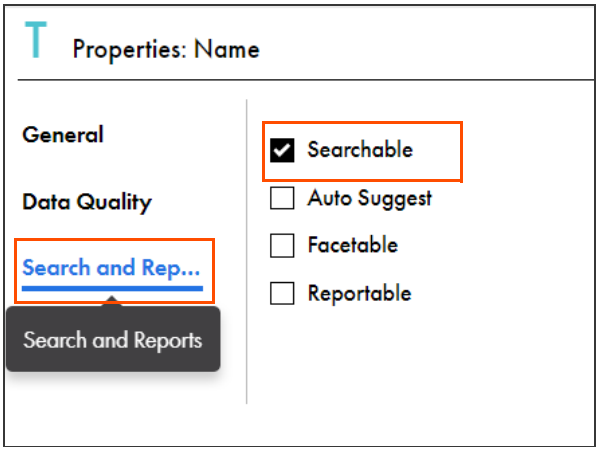


16. Retain the Maximum Length and select the **Required Field** check box.

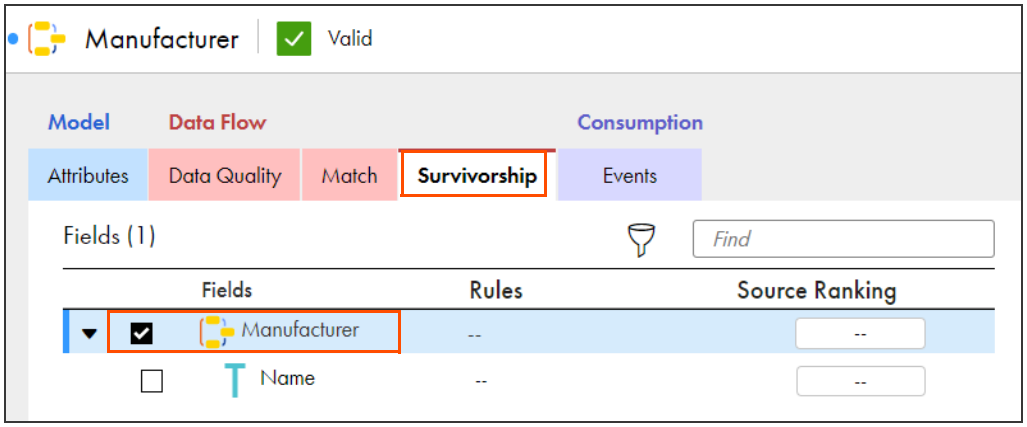


17. In the Data Quality tab, you can add basic, advanced, or DaaS rule associations based on the field you select. you will not configure any data quality rules for this business entity.

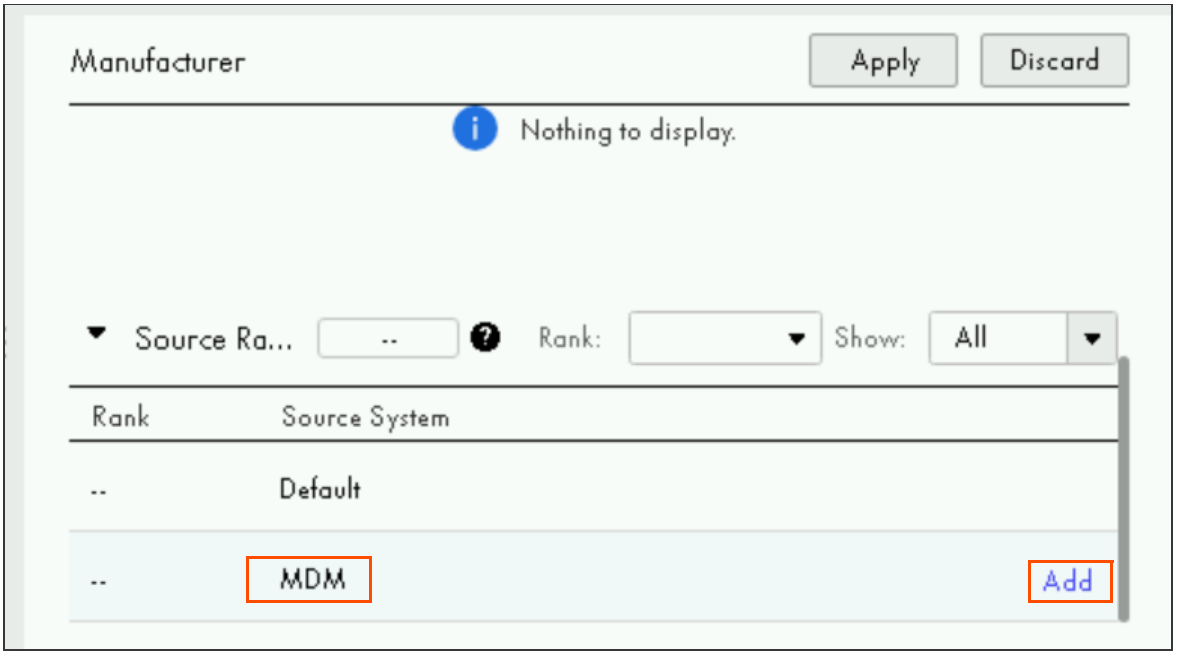
18. To make the field searchable, click the **Search and Reports** tab and select **Searchable**.



19. To configure survivorship, click **Survivorship** tab on top of the Business Entity page and select **Manufacturer**.

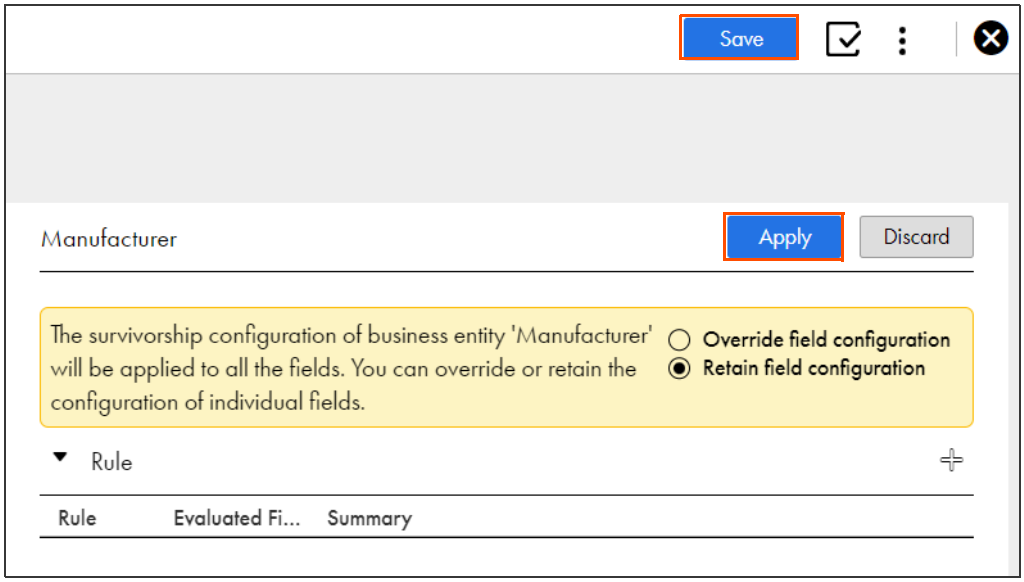


20. In the Source Ranking section of the survivorship page, hover over **MDM** data source and click **Add**.

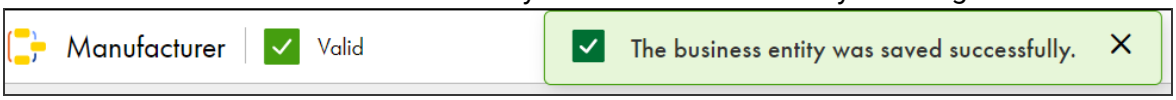


If you don't see the source system, refresh the page.

21. Click **Apply** and then click **Save**.



You can see that the 'The business entity was saved successfully' message.



22. Like the Name field you have added above, you will add few more basic fields to the Manufacturer business entity. Refer to the following table to add the fields.

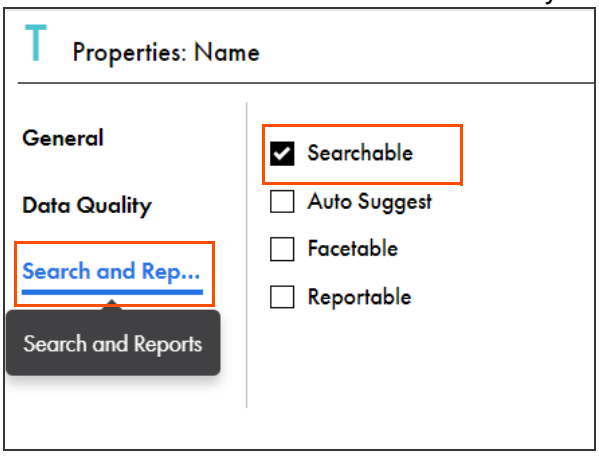
Navigate to the **Attributes** tab to add the fields. You are not required to add survivorship configuration to the below fields.

Field	Type
SIC Code	Integer
Manufacturer Id	Text
Date of Incorporation	Date

23. Similarly, add two more smart fields: Address and Phone Number. Refer to the following table to add the fields:

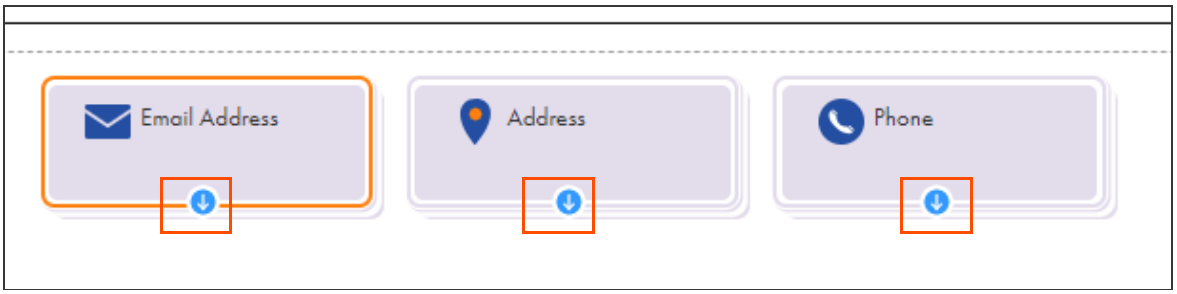
Field	Type
Address	Postal Address
Phone Number	Phone
Email Address	Email Address

24. Make all the fields searchable, click the **Search and Reports** tab and select **Searchable**. Make sure to do this for all the fields that you created.



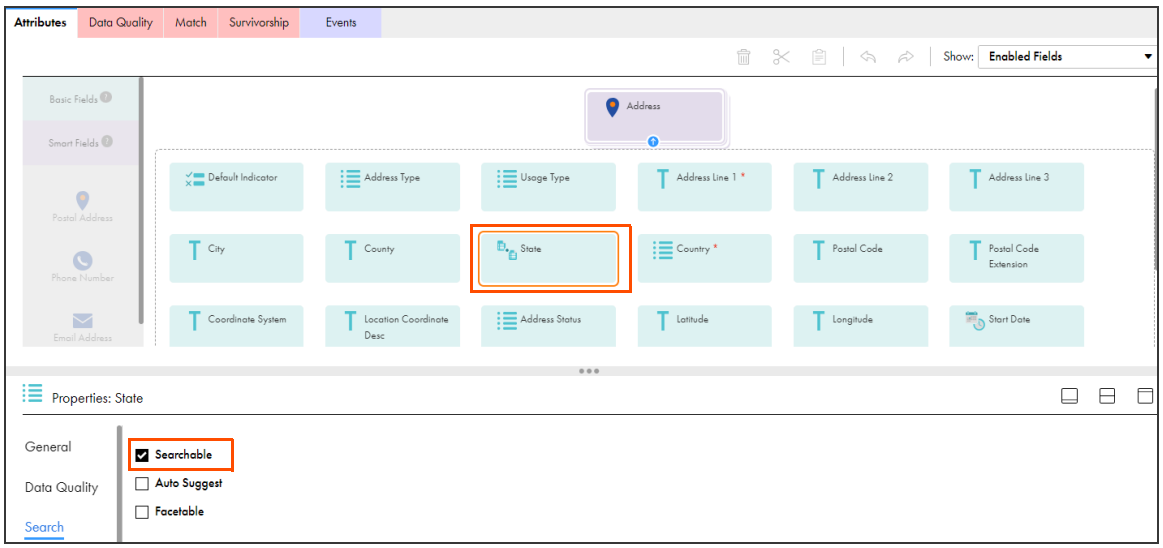
Smart Fields are group of required and non-required individual fields. You cannot see the Search tab property for the smart fields. To make them searchable, you will have to expand the each field and make the required fields searchable.

25. To see the fields that are listed under the Smart Fields, click the down arrow below to it.



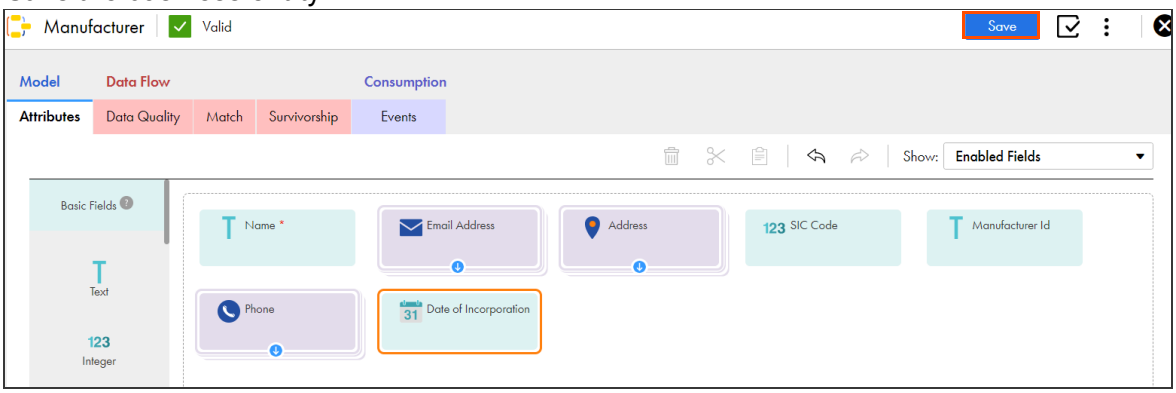
26. Refer to the following table to see which sub fields to make searchable:

Field	Sub field to be made searchable
Phone Number	Phone Number
Email Address	Electronic Address
Address	Address Line 1, City, State, Country



IMPORTANT: Make sure to make all the fields searchable. This will make the fields accessible while creating the custom pages for the business entity.

27. **Save** the business entity.

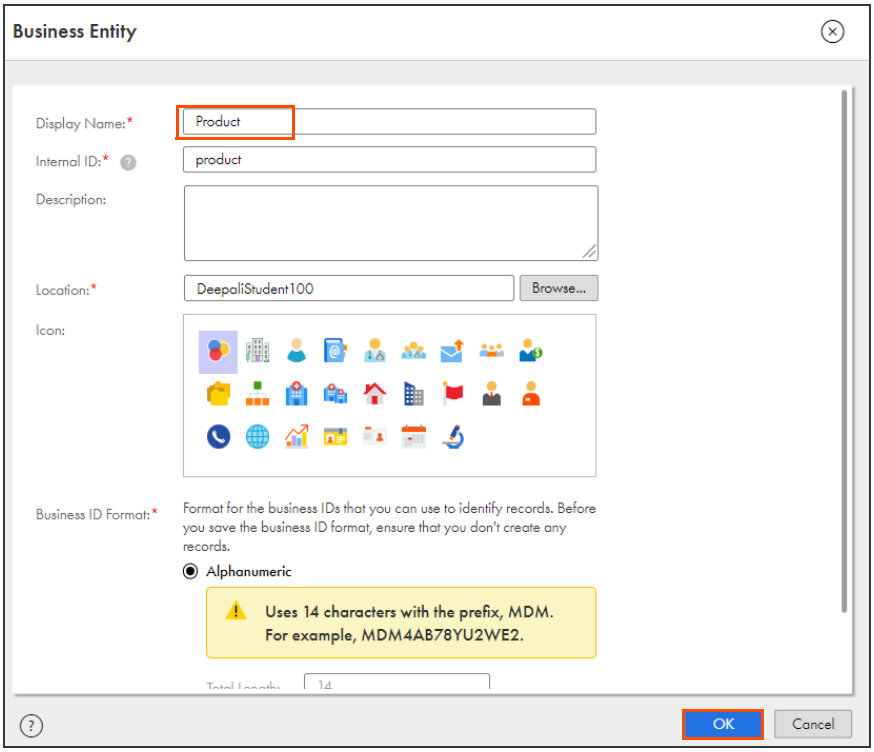


Creating Product Business Entity and Add Fields to it

You have seen how to create a custom business entity, adding fields, and configuring the field properties. Apply the above knowledge and create Product business entity, add fields, and configure the field properties to it.

28. Create **Product** business entity using the following properties:

Field	Value
Display Name	Product
Location	<Select your student project>
Icon	



29. You will add following basic fields to the **Product** business entity. Once you create, make these fields as **Required Fields** and **Searchable**. Refer to the following table to add the fields:

Field	Type
Name	Text
Availability Date	Date
Expiration Date	Date
Product Description	Text
Selling Price	Text

30. Make all the fields searchable. This is very important.

31. **Save** the business entity.

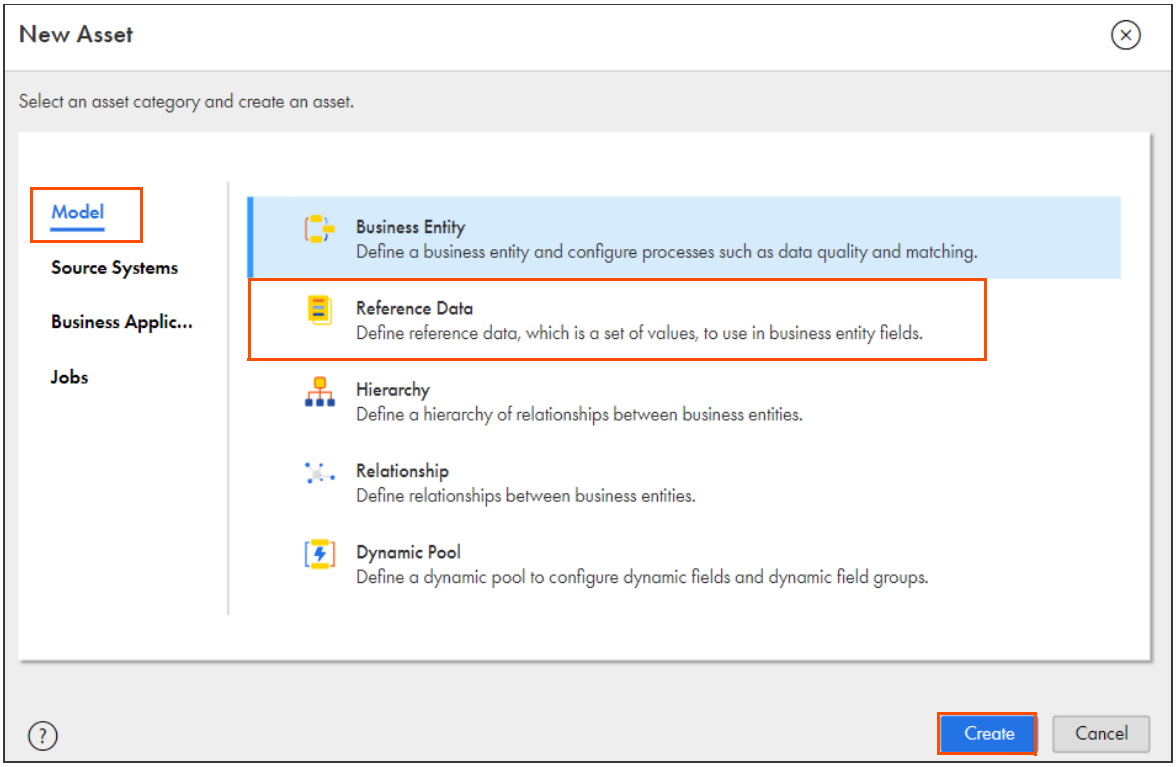
Add a Picklist

You will add another field called **Product Type** which will be of type Picklist. To provide a list of values for users to select from, you can add a picklist field to a business entity. But, before you add a picklist field to a business entity, you will have to configure the reference data asset to use in the picklist field.

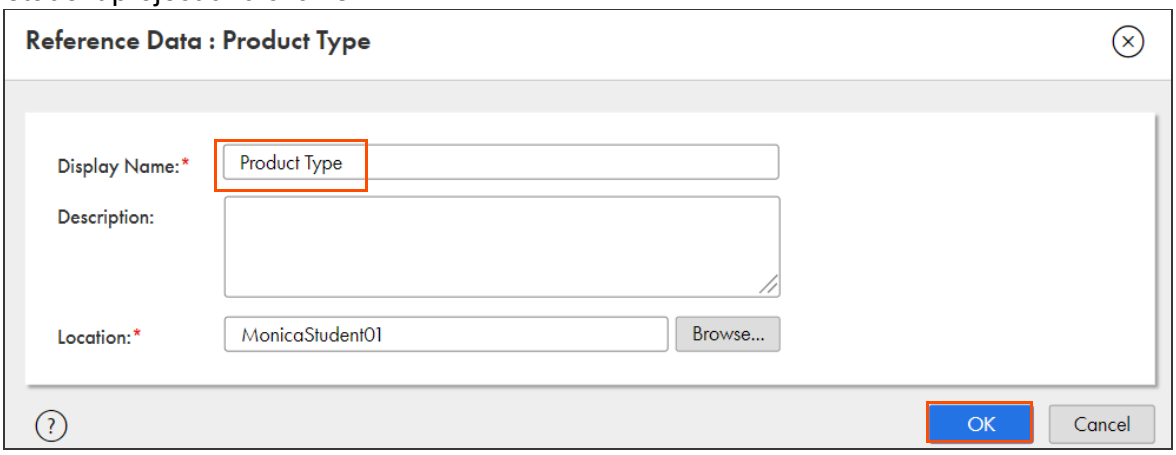
Configure a Reference Data Asset

A reference data asset contains a list of values that you can use in picklist fields. You will now create a reference data asset.

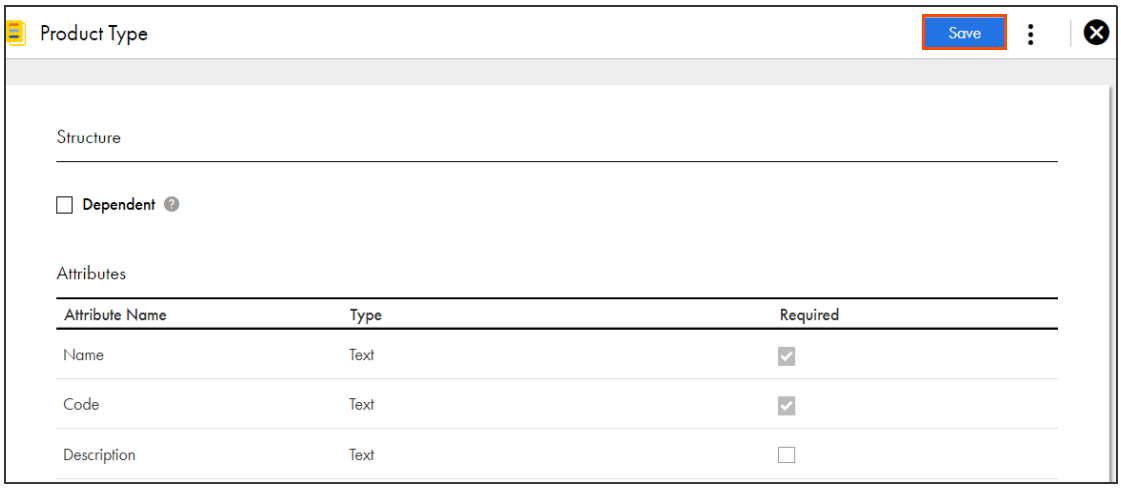
32. In the Business 360 Console, select **New** > **Model** > **Reference Data**. Click **Create**.



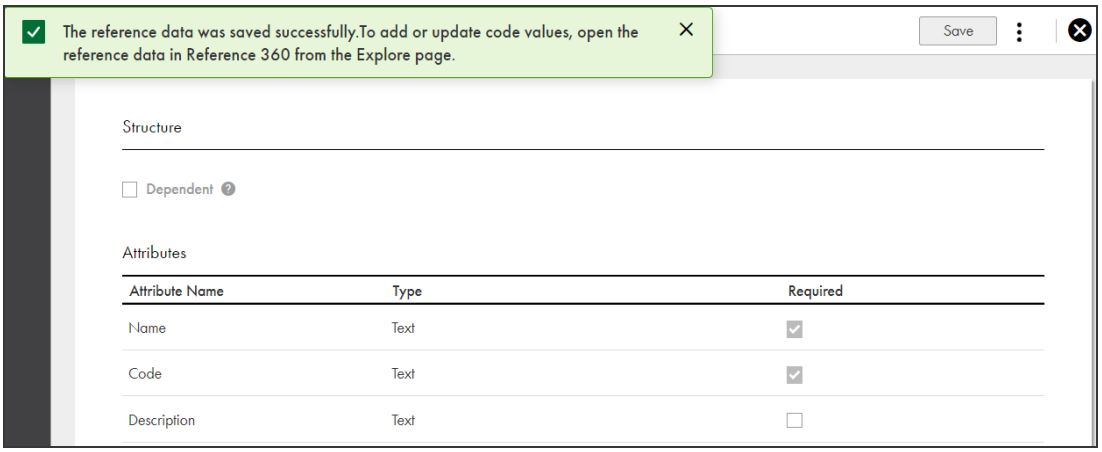
33. In the new Reference Data window, enter the Display Name as **Product Type**, select your student project and click **OK**.



34. Click **Save**.

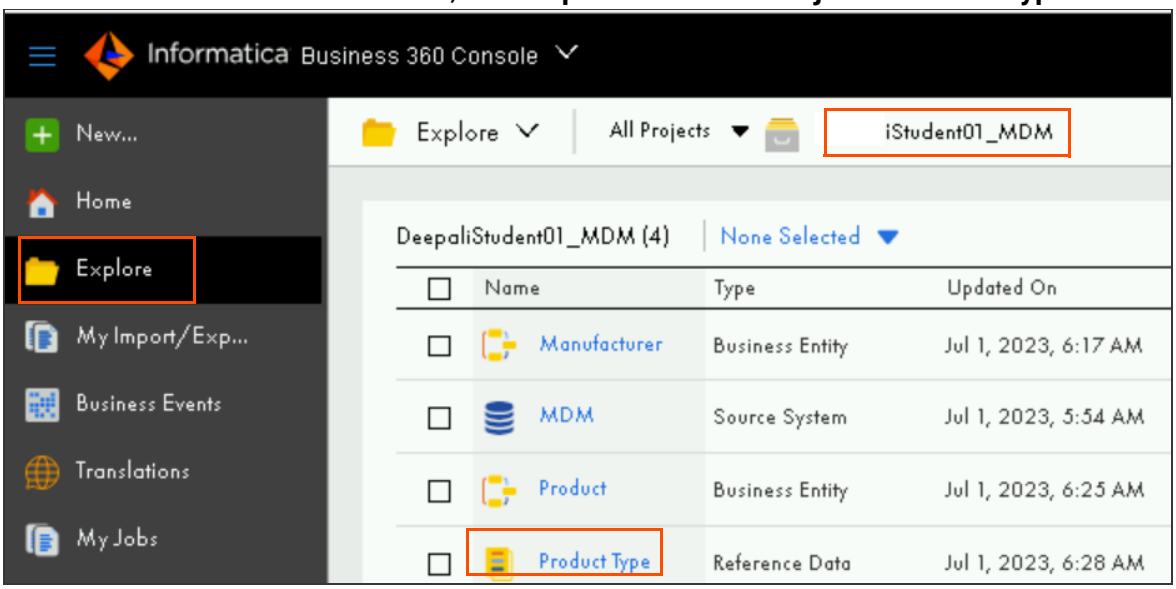


The reference data will be saved successfully.



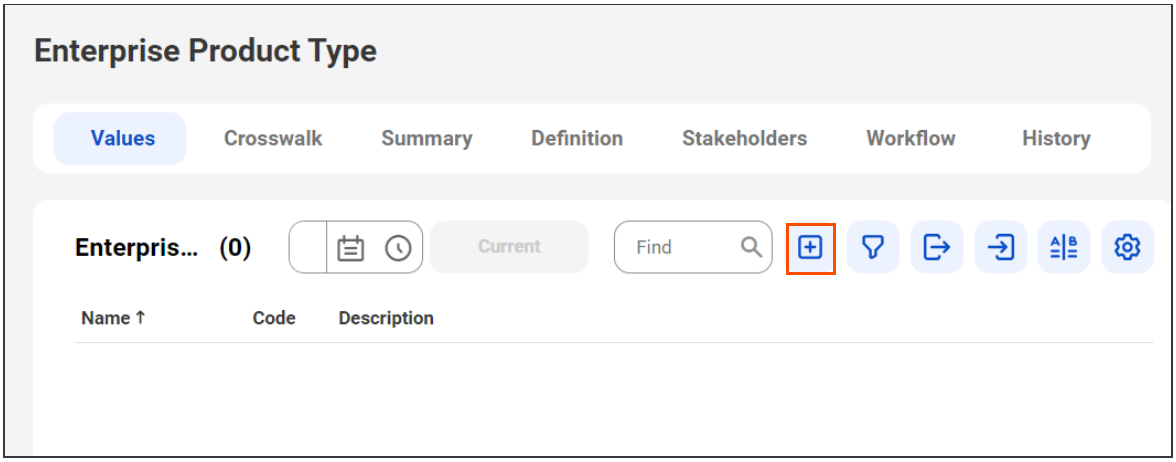
You have created a reference data set. Now you will add reference data to it. we also call this reference data as code values in the system and these r the values that may appear in the picklist

35. In the same Business 360 console, click **Explore** > **Student Project** > **Product Type**.

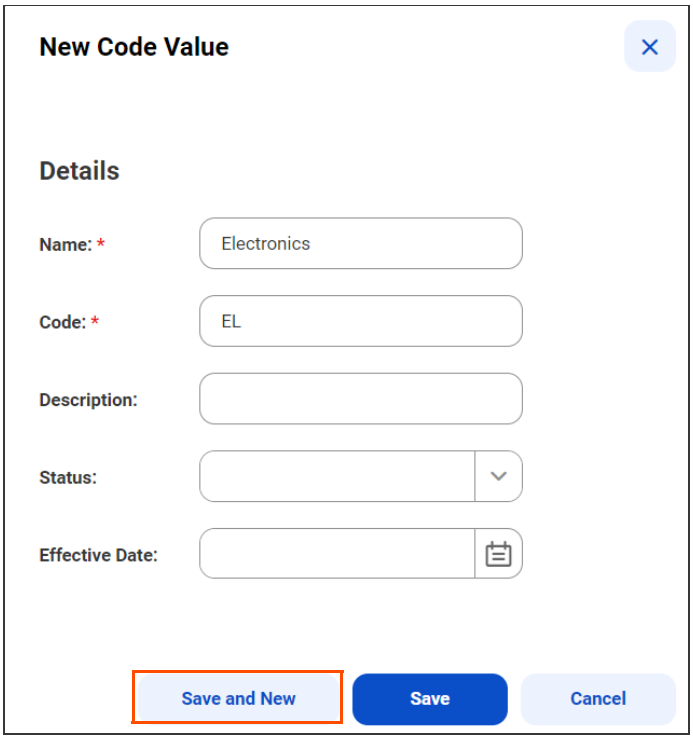


Important: This will take you to the Reference 360 service.

36. In the **Values** tab, click the **+** icon to add code values.



37. In the New Code Value, enter the Name as **Electronics**, Code as **EL** and click **Save and New**.

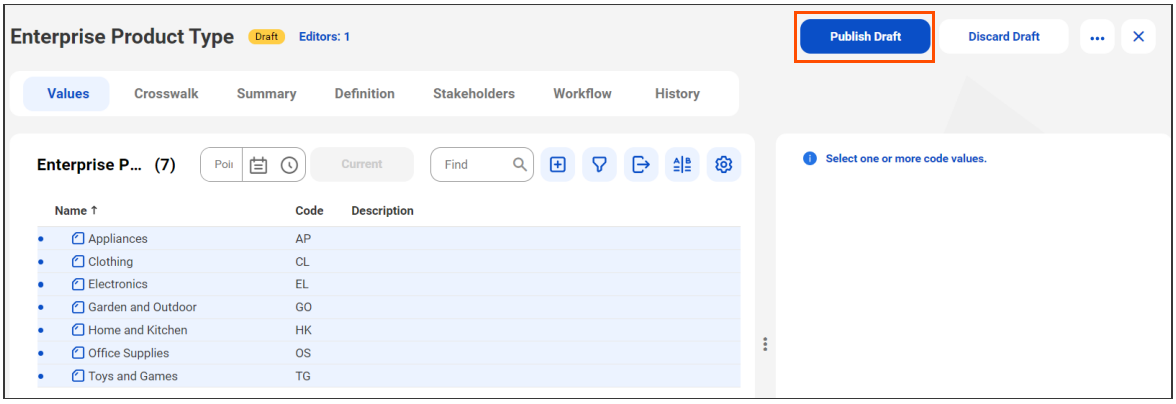


38. Similarly, add some more code values to the reference data set. Refer to the following table to do the same:

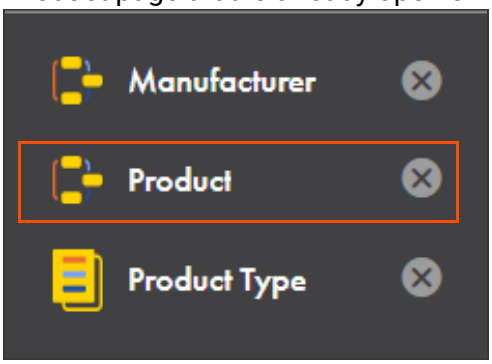
Name	Code
Appliances	AP
Clothing	CL
Toys and Games	TG
Office Supplies	OS
Home and Kitchen	HK
Garden and Outdoor	GO

When you add the last code value, click **Save.**

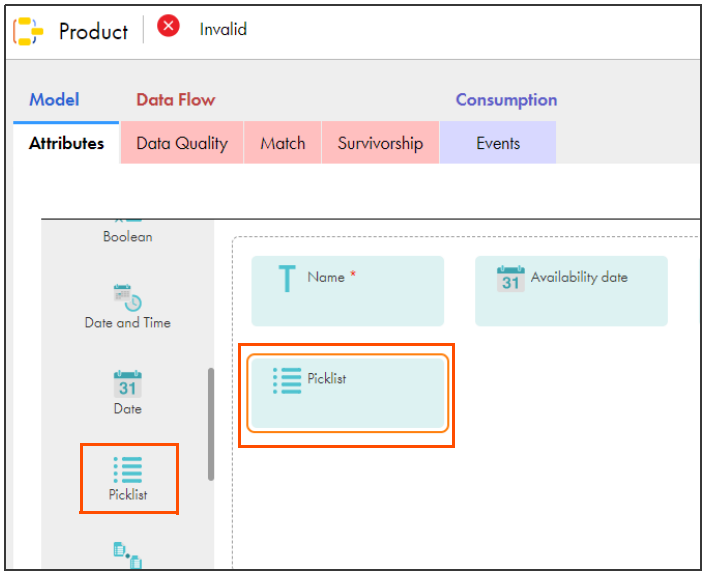
39. Click **Publish Draft**. Publishing the draft will make the reference data set available for the users of the org.



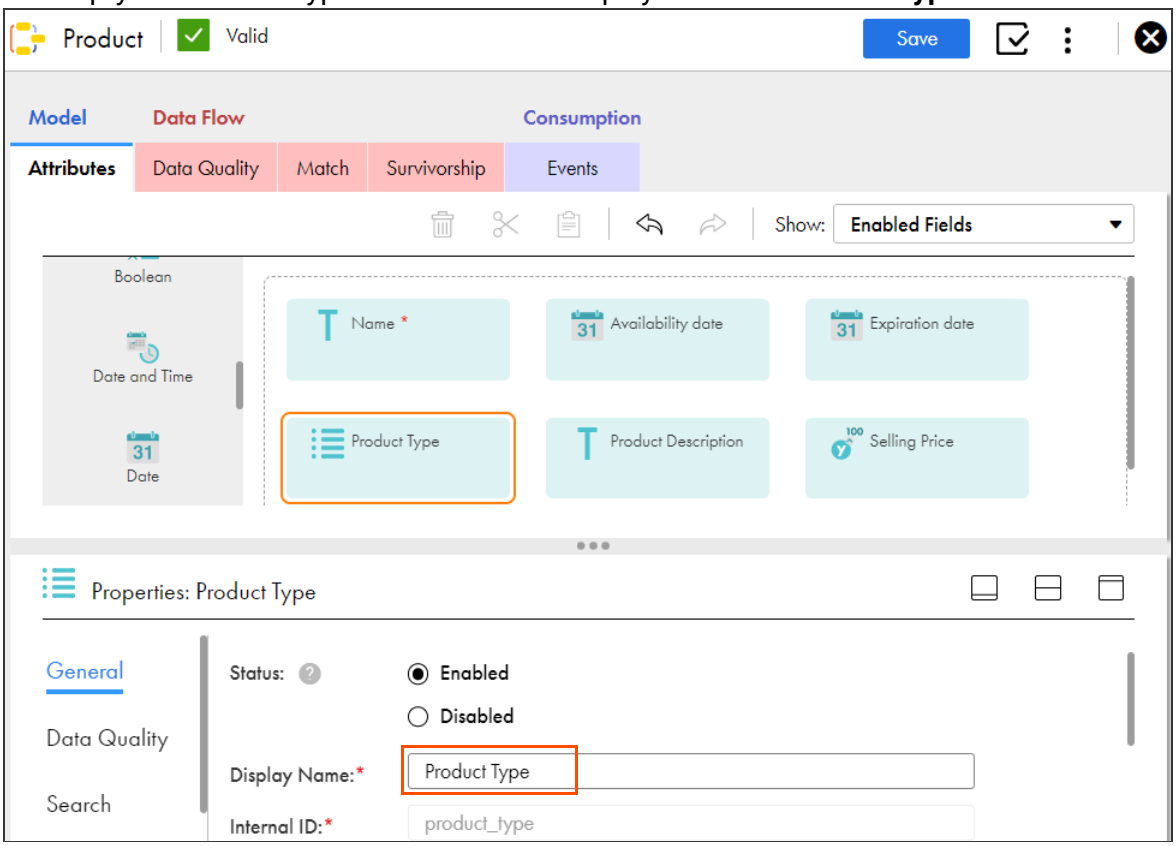
40. You will now add this reference data set to the picklist field. To do that, navigate back to the **Business 360** console and open the **Product** business entity page by clicking the Product page that is already open on the left side bar.



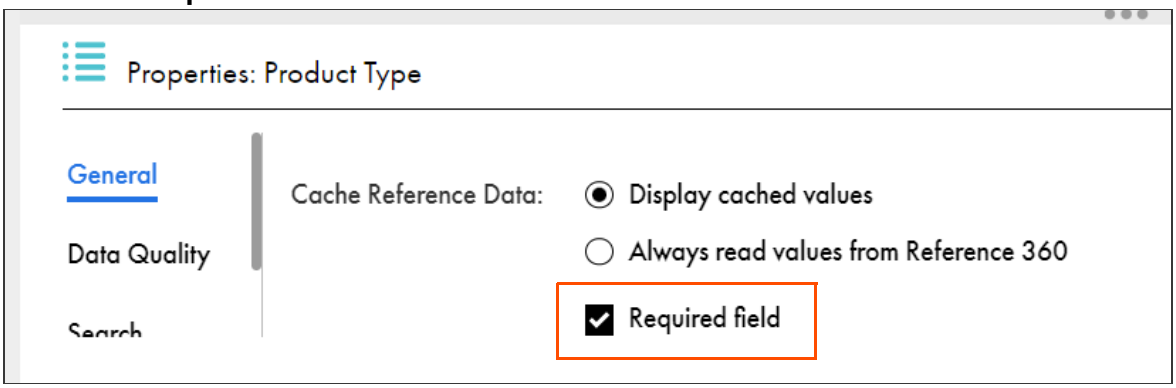
41. Double click **Picklist** field.



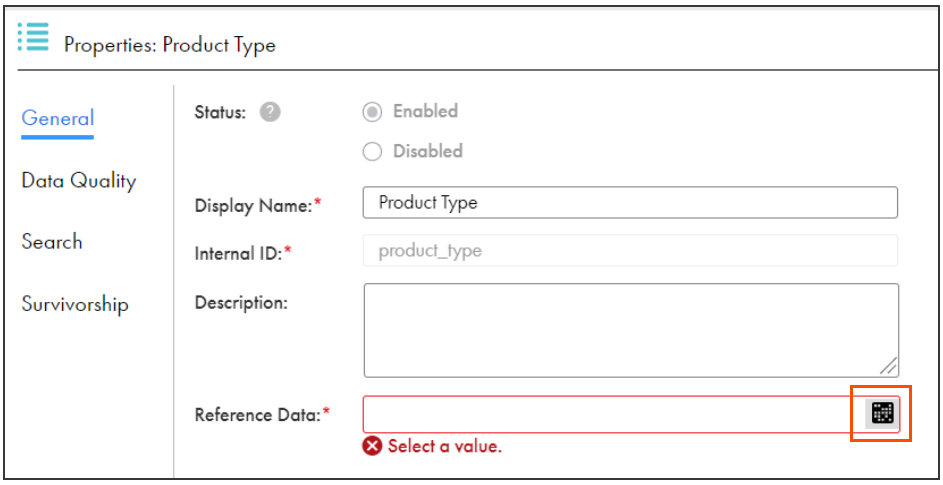
42. An empty field of Text type is added. Enter Display Name as **Product Type**.



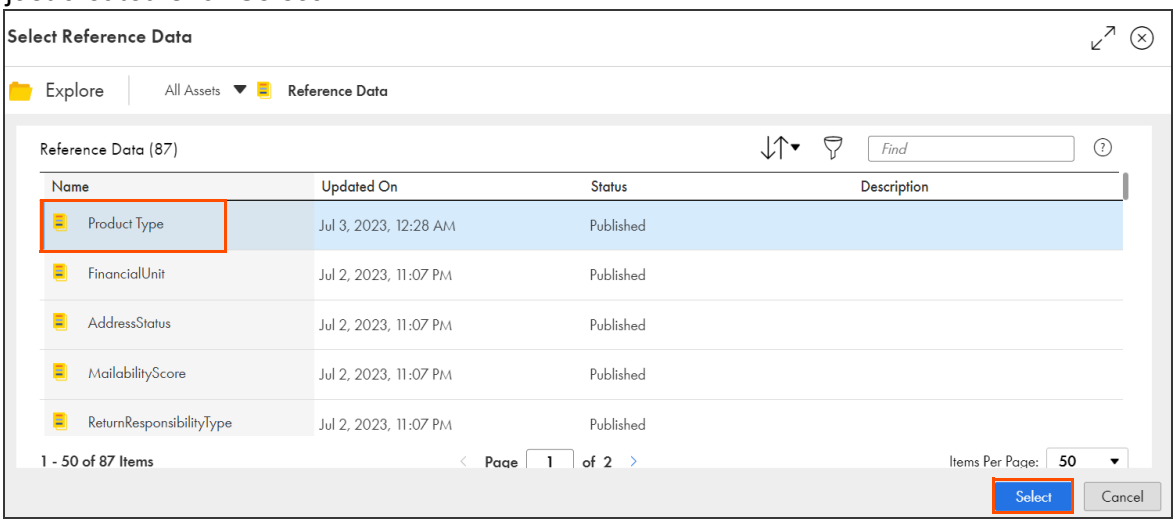
43. Select the **Required Field** check box.



44. In the Reference Data field, click the asset picket icon to select a reference data set.

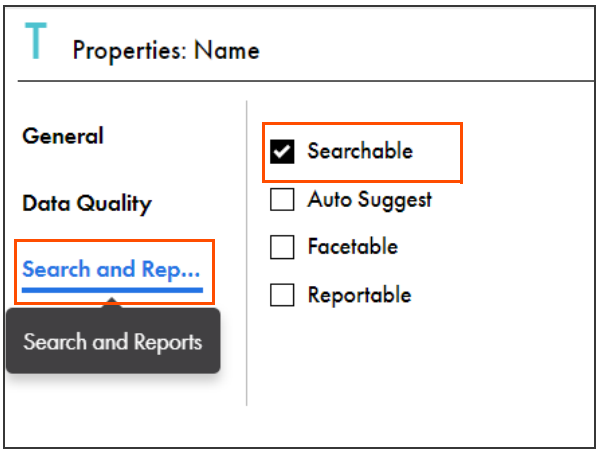


45. In the Select Reference Data window, select the **Product Type** reference data set that you just created.Click **Select**.

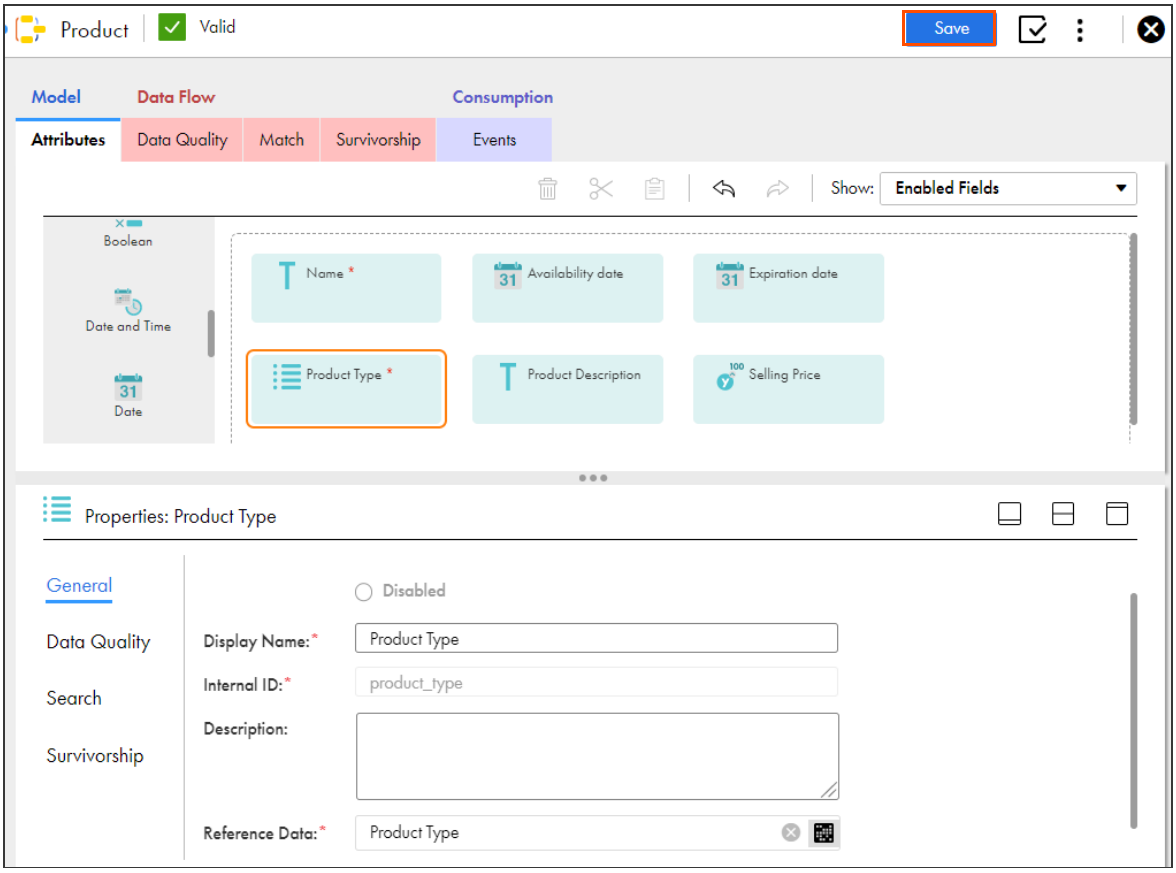


The **Product Type** picklist field is now configured.

46. Make sure that you configured all the fields as searchable. If not, select each field of the Product business entity, click the **Search and Reports** tab and select **Searchable** check box.

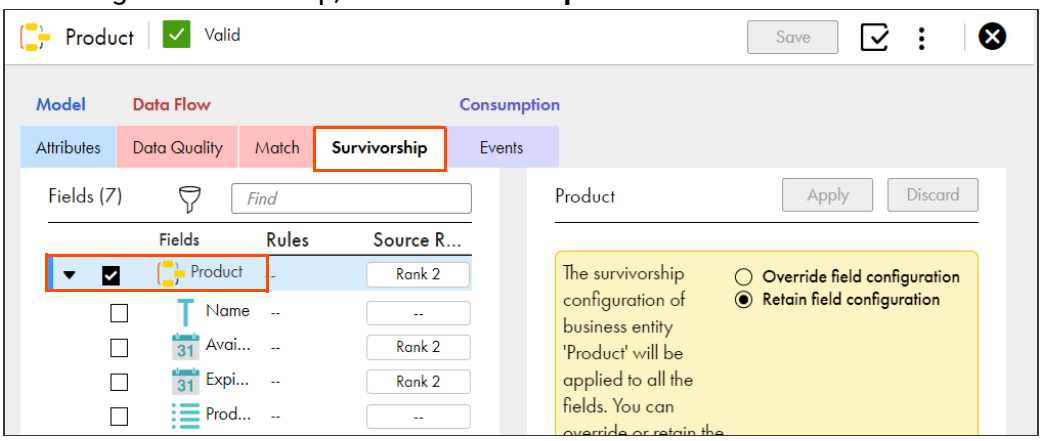


47. Click **Save**.

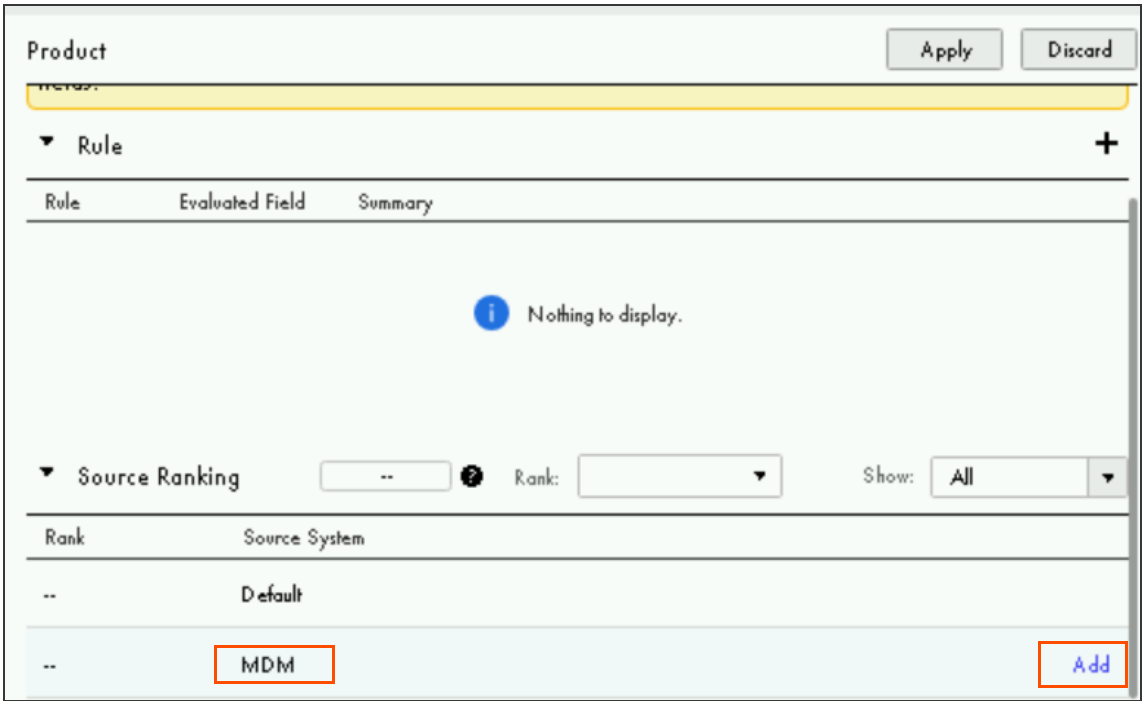


48. In the Data Quality tab, you can add basic, advanced, or DaaS rule associations based on the field you select. you will not configure any data quality rules for this business entity.

49. To configure survivorship, click **Survivorship** tab and select **Product**.

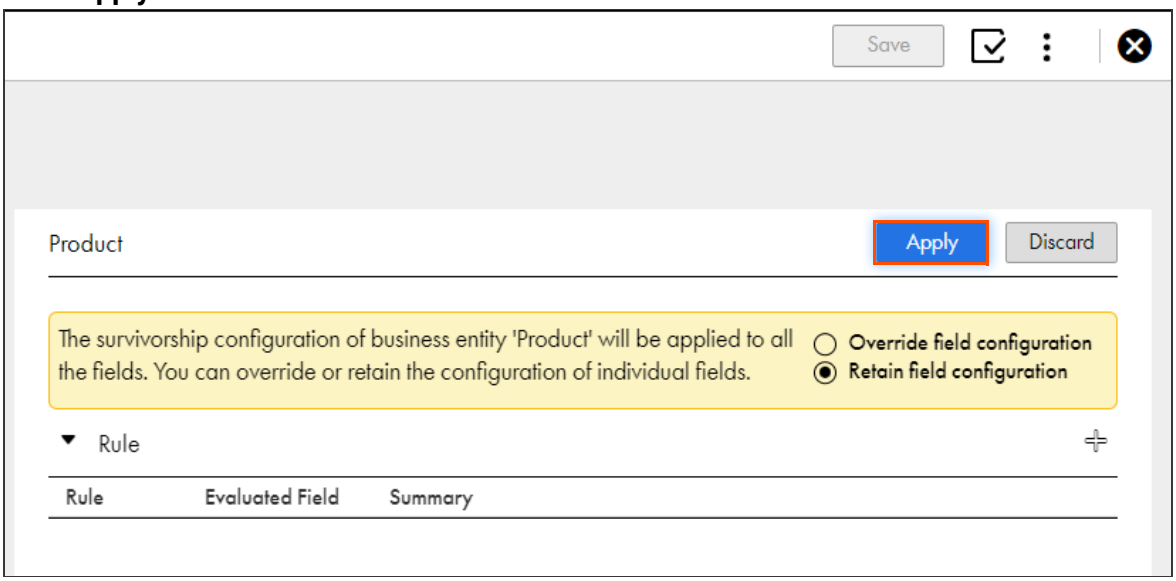


50. Hover over **MDM** data source and click **Add**.

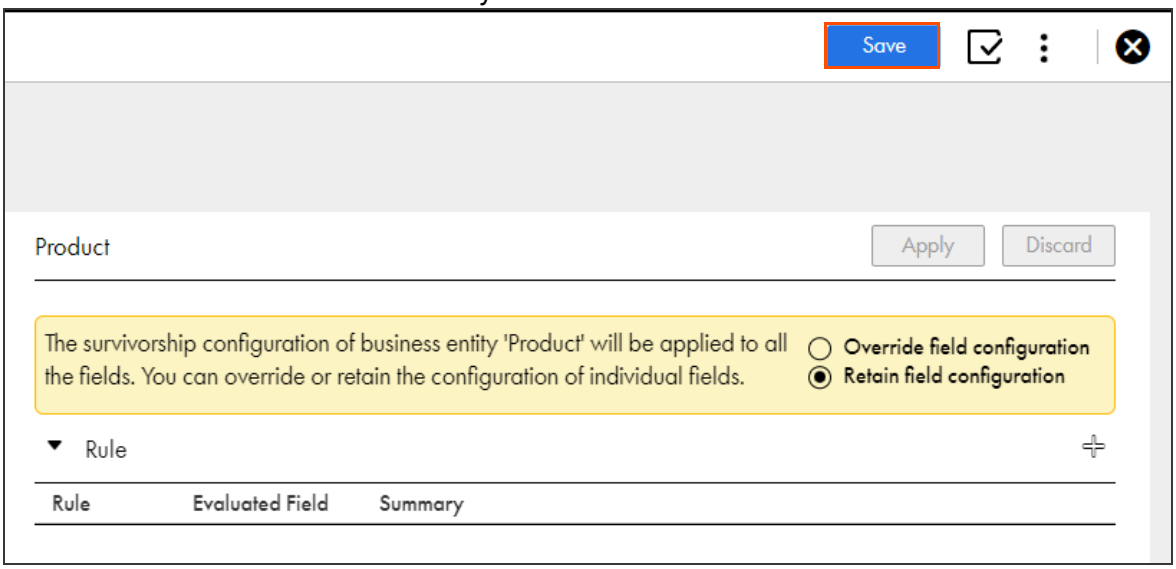


Refresh the page if you are unable to see the source systems.

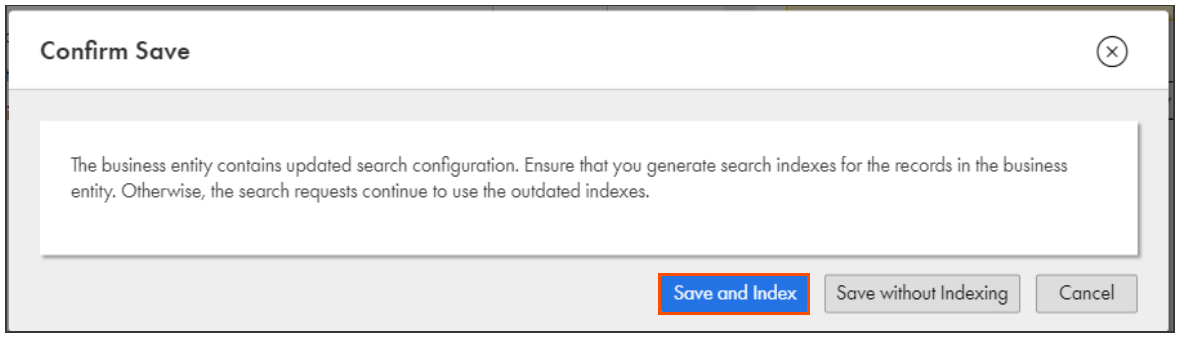
51. Click **Apply**.



52. Click **Save** to save the business entity.



If you get the following message while saving, then click **Save and Index**.



You have successfully created two business entities.

This concludes the lab.

Key Takeways

- A business entity is an entity of importance to your organization and consists of fields that support your data needs.
- To define a data model, you need to analyze, strategize, and define your business requirements by working with stakeholders.
- To be able to define and use a custom business entity, you need to follow the steps in the configuration process.

Module 4: Manage And Migrate Assets

Managing Projects

Projects and assets can be handled on the 'Explore' page of the Business 360 console. This page is a feature of IDMC and is accessible for the majority of services. If you employ various services, it's likely that projects, folders, and assets from all your services will be visible on the 'Explore' page.

You can manage your IDMC projects and assets in the following ways:

- Copy the contents of projects or folders to other locations.
- Move folders to other locations.
- Delete custom projects or folders.
- Rename custom projects or folders.
- Configure permissions for projects, folders, or assets.
- Migrate assets between organizations.

Copying Projects and Folders

On the Explore page, you can copy projects and folders. Predefined projects can only be copied to a new project by copying their contents. You might want to copy an object to use as a template, or you might want to create a backup copy.

To copy projects and folders, follow the steps mentioned below:

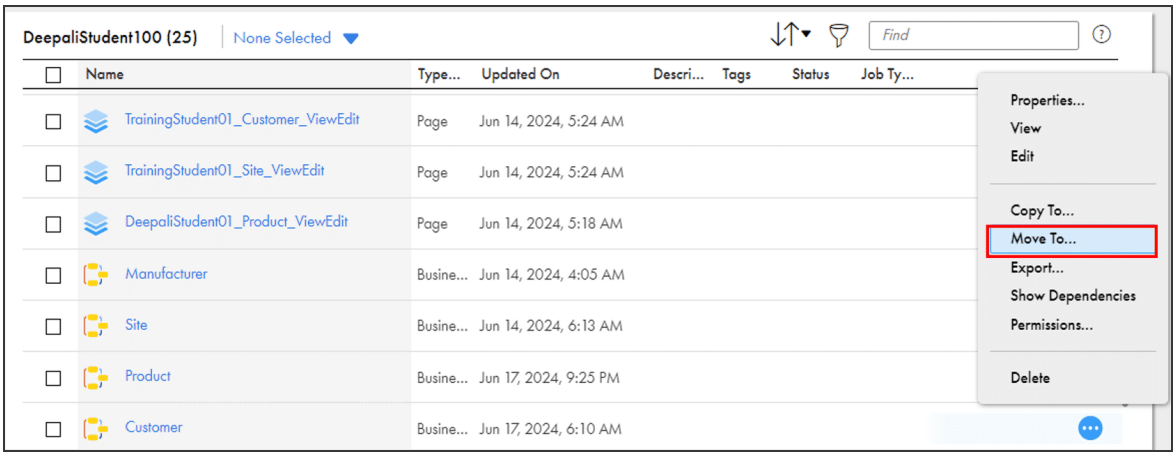
1. On the Explore page, navigate to the object that you want to copy.
2. In the row that contains the object that you want to copy, click **Actions** and select **Copy To**.
3. Specify the name of the new project or folder and click **Save**.

Moving Folders

You can move folders on the Explore page. The folders can be moved across different projects or within the folders inside the same project.

Refer to the following steps to move folders:

1. On the Explore page, navigate to the folder that you want to move.
2. To move a folder, in the row that contains the folder, click **Actions**, and select **Move To**.



3. Browse to the new location and click **Select**.

Renaming Projects and Folders

You can rename projects and folders. You can't use the following characters in projects, folders, or tag names: # ? ' | { } " ^ & [ ] / \. You can't use special characters in a name or use the same name as another object that is in the same folder.

To rename a project or folder, refer to the steps below:

1. On the Explore page, navigate to the project or folder that you want to rename.
2. To rename a project or folder, in the row that contains the project or folder, click **Actions** and select **Properties**.
3. Enter the new name and click **Save**.

Deleting Projects, Folders, and Custom Assets

You can delete a project or folder if you no longer need it. However, before you delete it, verify that no users in the organization plan to use it. To delete custom assets, contact Informatica Global Customer Support. You can't retrieve projects, folders, or custom assets after you delete them.

Migrating Projects and Assets

You can migrate Informatica Intelligent Cloud Services projects and assets from one organization to another organization. To migrate projects or assets, you export the projects or assets from the source organization and then import the projects or assets into the target organization. Before you define a business entity, determine the type of business entity that you want to use.

You can export a single project or asset, groups of projects or assets, or export all the assets in a project. When you export all the assets in a project, the export fails for the unsupported assets. If you export a project or folder, the file structure remains intact so that when you perform the import in the target organization, you can duplicate the original structure.

Prerequisites for Migrating Assets

Before you migrate assets from one organization to another, ensure that you meet the following prerequisites:

- The source and target organizations have the same version and use the same application license.
- You have user accounts in the source and target organizations with roles that have export and import privileges, such as the Administrator role or a combination of Designer, MDM Designer user roles.
- You have the required permissions to migrate the assets and folders from the source to the target organization.
- You have a backup of the existing assets of the target organization. If the import fails, you can import the backup into your target organization.
- Target connection is created for the target transformation in the Cloud Data Integration application to migrate assets, such as mappings, mapping tasks, and taskflows.

Required roles and permissions to migrate assets

To import an asset, you need the following roles and permissions:

- User with the Admin role or a combination of Designer and MDM Designer roles.
- To import new assets into the target organization, you need Create permission.
- To overwrite the existing assets in the target organization, you need the metadata permission for the assets.

Guidelines for migrating assets

To migrate assets between organizations, consider the following rules and guidelines:

- In order to import or export asset types, such as authorization configurations, you must have administrative privileges.
- Once you migrate an asset, you must export and import the modified asset to the target organization if the asset has been updated in the source organization. Do not manually modify the target organization's assets.
- Assets must be migrated by the user role that updates them.
- The MDM Designer must migrate assets to a target organization if they are updated in a source organization. The migration will fail if this is not done.
- You can't delete reference data sets created in the Business 360 Console from MDM - Reference 360.
- After you migrate the reference data assets between organizations, don't modify the name of reference data assets in the source organization.
- If you reimport any renamed reference data asset, a new asset is created instead of updating the existing asset.

Supported Assets

You can export and import the following types of assets:

- Predefined and custom source systems
- Predefined business entities
- Business events
- Customer 360 business application with custom pages
- Code lists of reference data
- Hierarchies
- Job definitions, such as match and merge, generate merge tasks, and find lost records
- Rule specifications such as advanced data quality

You can also export and import predefined business entities that you have customized. The customization includes adding custom fields, custom field groups, custom smart fields, custom match rules, custom basic and advanced data quality rule associations, or custom DaaS rule associations.

Unsupported Assets

You cannot export and import the following types of assets:

- Ingress and export job definitions
- Processes

Dependent Assets

Dependent objects are assets that other assets require. For example, a business entity uses a source system in its configuration. A source system is a dependent object for the business entity. When you set up an export, you can include or exclude dependent objects in the export file.
You can include dependent objects if:

- They do not exist in the target organization.
- You want to replace the dependent objects in the target organization with updated versions from the source organization. You can exclude dependent objects of an asset if the objects exist in the target organization, and you do not want to replace them.

Exporting Assets

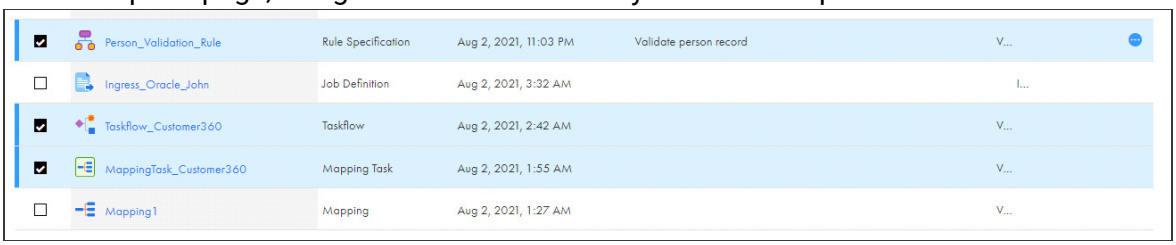
When you export assets, Informatica Intelligent Cloud Services creates a ZIP file containing the assets that you selected for export. You can export individual assets, or an entire project or folder. When you export a project or folder, the export file includes all the assets in the project or folder.

To export an asset, you need the following roles and permissions:

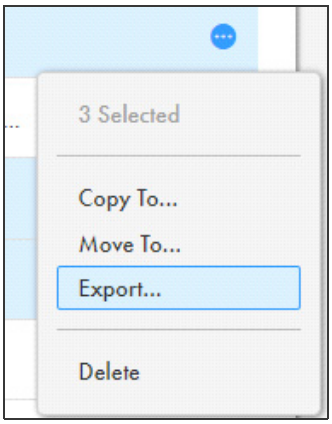
- User with the Admin role or a combination of Designer, MDM Designer, and MDM Business User roles.
- Read permission to access the asset.

Refer to the following steps to export an asset:

4. Log in to the source organization.
5. On the Explore page, navigate to the assets that you want to export.

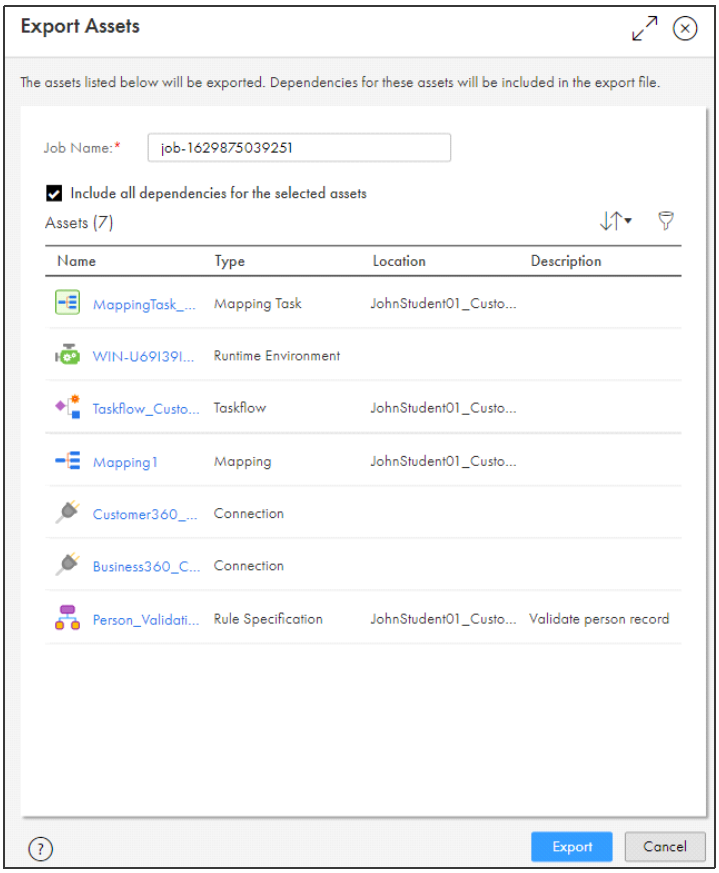


6. Click **Actions**, and then select **Export**.

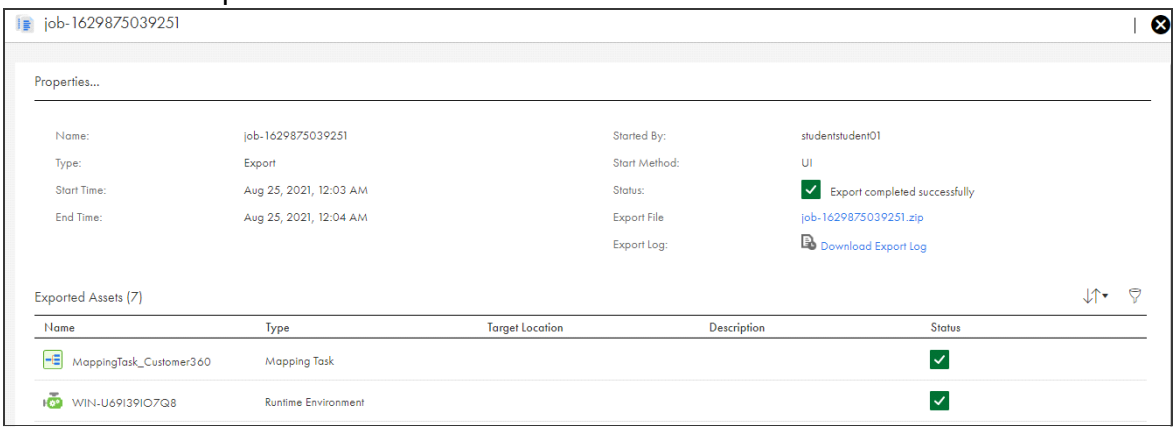


7. On the Export Assets page, specify the job name.
8. Select whether to include dependent objects for the assets.

9. Click **Export**.



10. View progress of the job from the **My Import/Export Logs** page.
11. Download the export file.



Import Assets

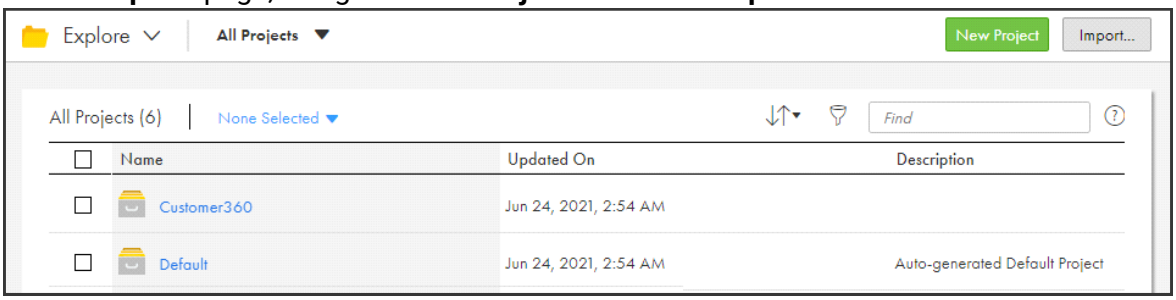
You can import all the assets from the export file or select the assets that you want to import. To import an asset, you need the following roles and permissions:

- User with the Admin role or a combination of Designer, MDM Designer, and MDM Business User roles.
- To import new assets to the target organization, you need Create permission.
- To overwrite the existing assets in the target organization, you need the Update and Read permissions for the assets.

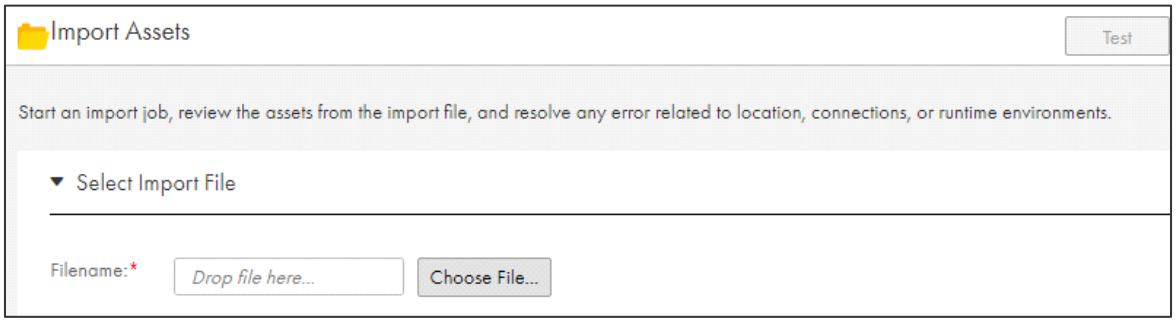
The **Import Assets** page lists the assets that are available in the export file. You can use the same project name as the source project or select a different project. If the project does not exist in the target organization, Informatica Intelligent Cloud Services creates it. If the project exists in the target organization, Informatica Intelligent Cloud Services overwrites it.

Refer to the following steps to import an asset:

1. Log in to the target organization.
2. On the **Explore** page, navigate to **All Projects** and click **Import**.

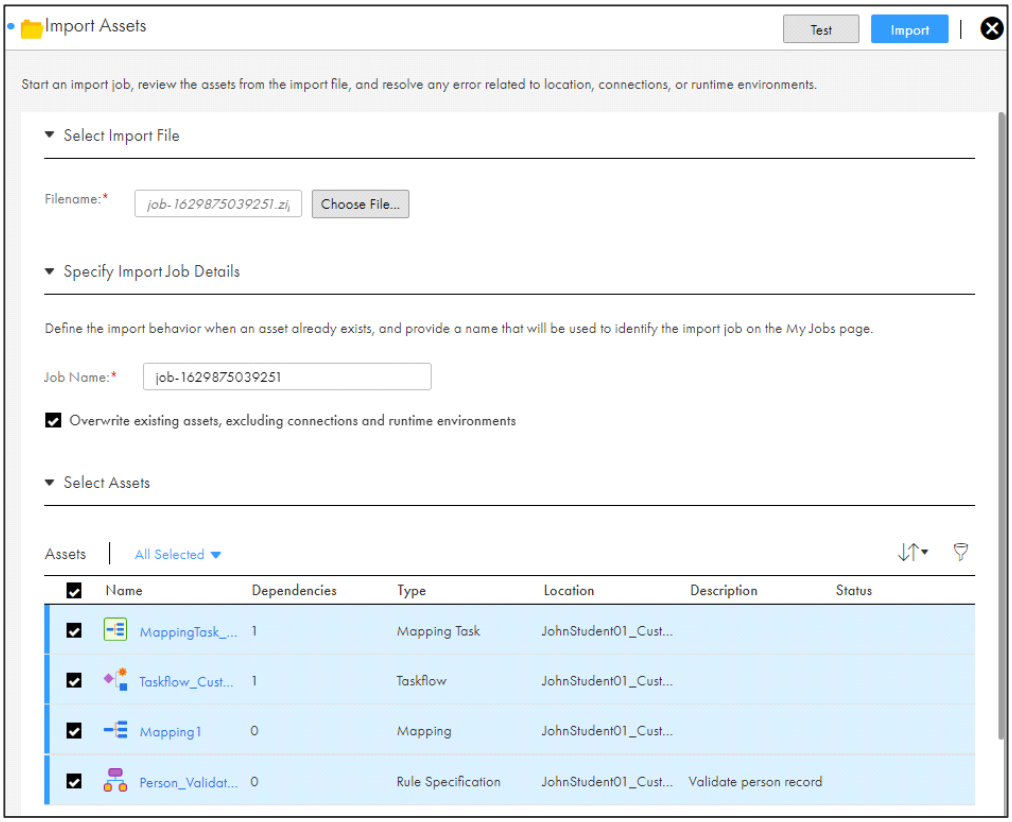


3. On the **Import Assets** page, attach the export file.

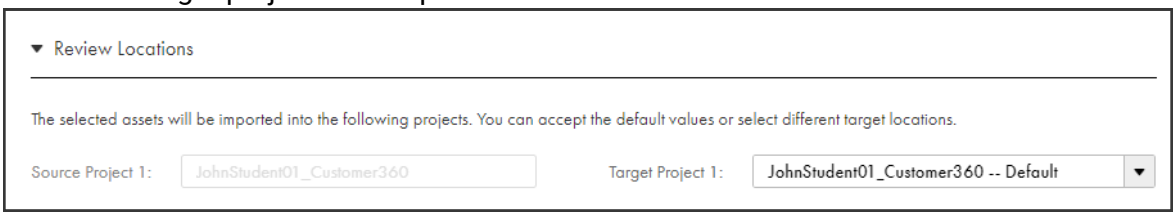


4. In the **Import Assets** window, you can choose to change the import job name or retain the default value.

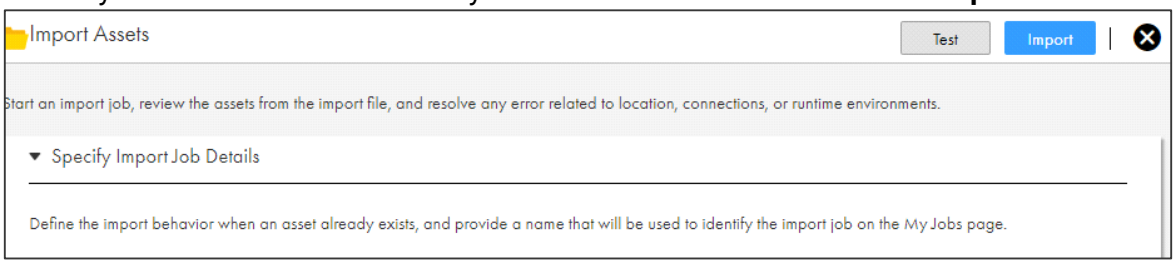
5. You can choose to overwrite the existing assets with the assets in the import file.



6. Select the target project or accept the default.



7. Click **Test** to see the potential results of the import.
8. Revise your selections to resolve any issues in the test results and click **Import**.



9. View the progress of the job from the **My Import/Export Logs** page.

Lab 4-1: Importing and Exporting Custom Business Entities

In the last module you created two custom business entities: Manufacturer and Product. In this lab, you will import other two custom business entities: Customer and Site that are required for Homekart data model. The import feature in Multidomain MDM allows you to import various types of assets, including custom pages, search layouts, and more.

Objectives:

- Import an asset containing custom business entities
- Export the custom business entities

Duration:

10 Minutes

Scenario:

In the previous lab, Mark created two business entities (BEs) and now needs to add additional business entities to support the HomeKart data model. Along with his team, who have been working on creating other business entities, Mark must gather these assets from his team and import them into his organization. Similarly, Mark needs to export the assets he has worked on from his organization and send them to his team. This will enable them to import these assets into their organization and proceed with development effectively.

Tasks:

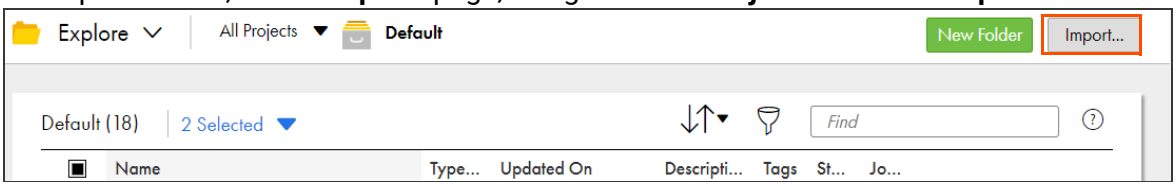
Perform the following tasks to import the business entities for Homekart scenario:

- Import the Custom Business Entities
- Verify the Imported Business Entities
- Export the custom business entities

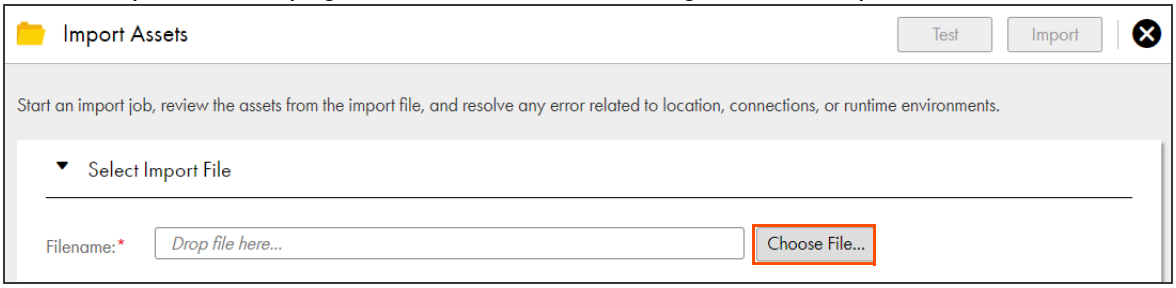
Import the Custom Business Entities

The custom business entities are already pre-created in your VM. Now, as the first step, you will import the custom business entities to the Multidomain MDM application. You can also import custom pages or search layouts.

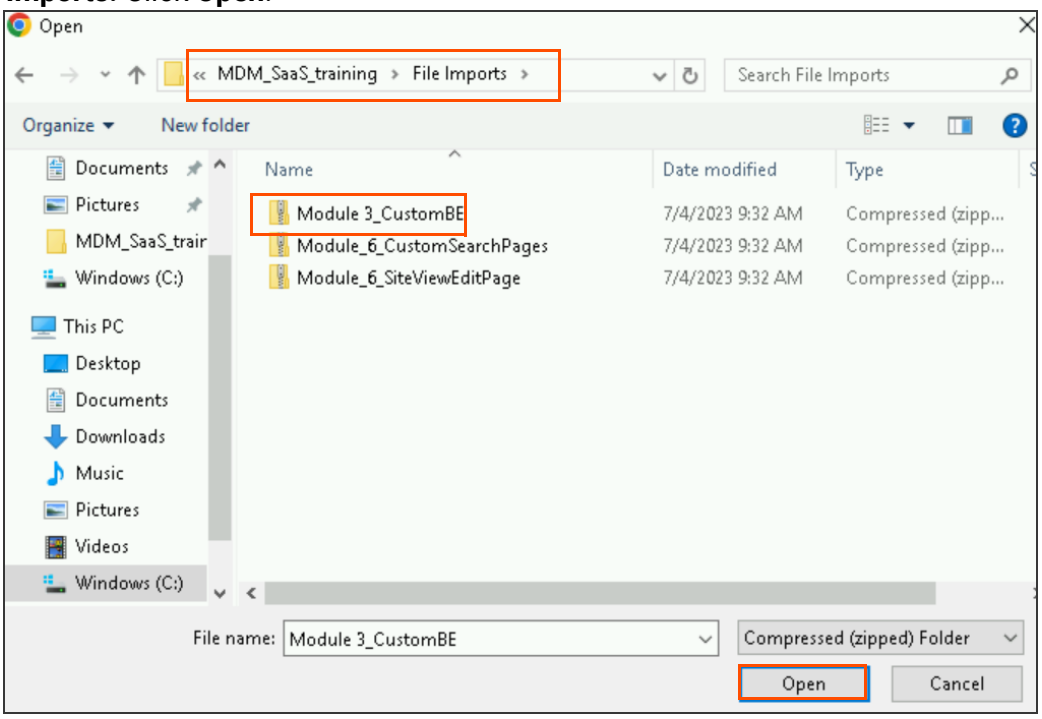
1. Log in to the IICS using your admin credentials. For example, **Jimstudent01**.
2. Open **Business 360 Console** service.
3. To import assets, on the **Explore** page, navigate to **All Projects** and click **Import**.



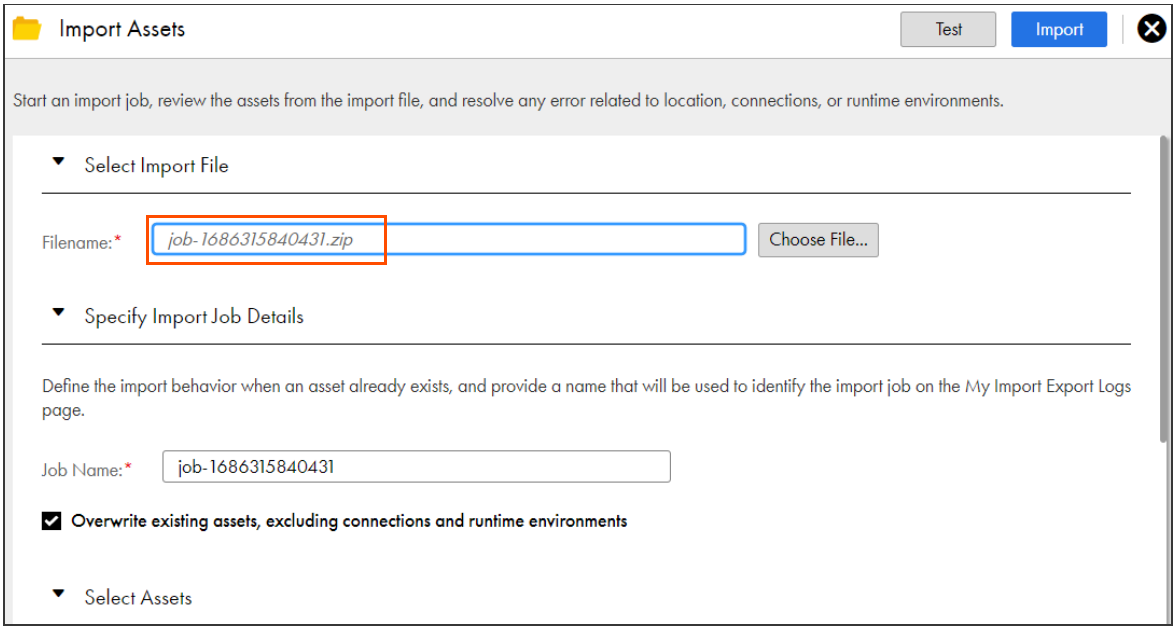
4. On the Import Assets page, click **Choose File** to navigate to the export file.



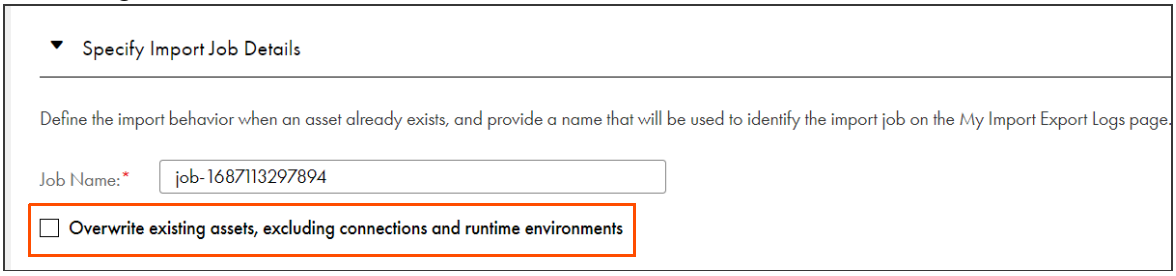
5. Choose the ZIP file **Module 3_CustomBE** from the location **C:\MDM_SaaS_training\File Imports.** Click **Open**.



A page will appear where you can optionally, change the import job name. You can choose to retain the same name.

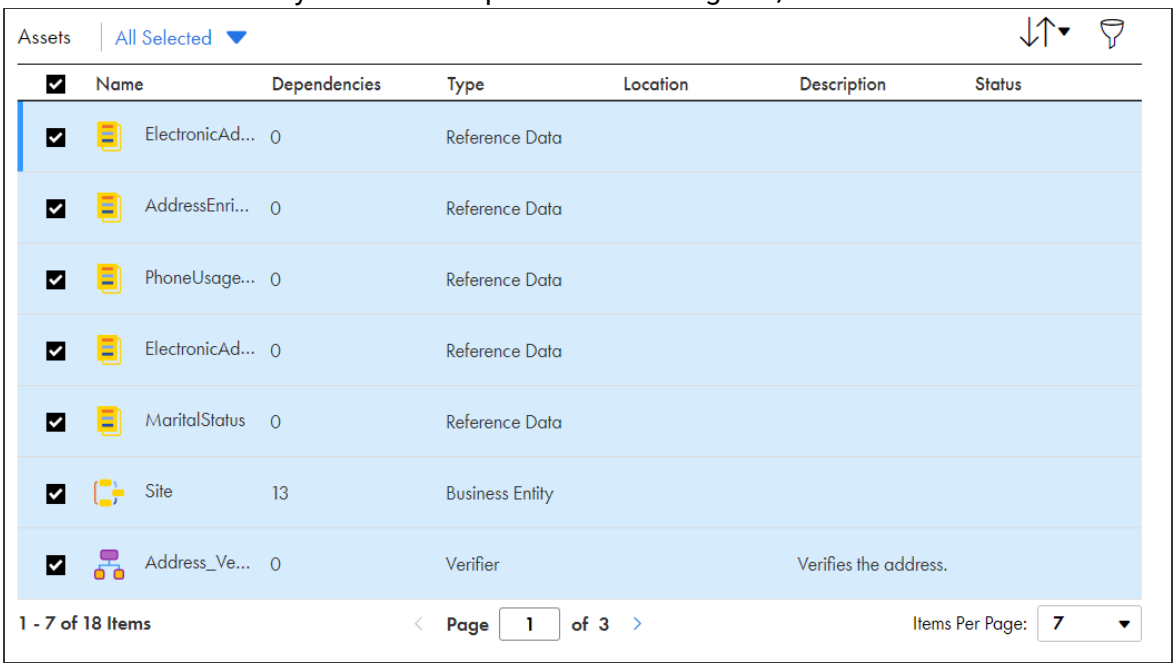


6. In the Specify Import Job Details, deselect the check box **Overwrite existing assets, excluding connections and runtime environments**.

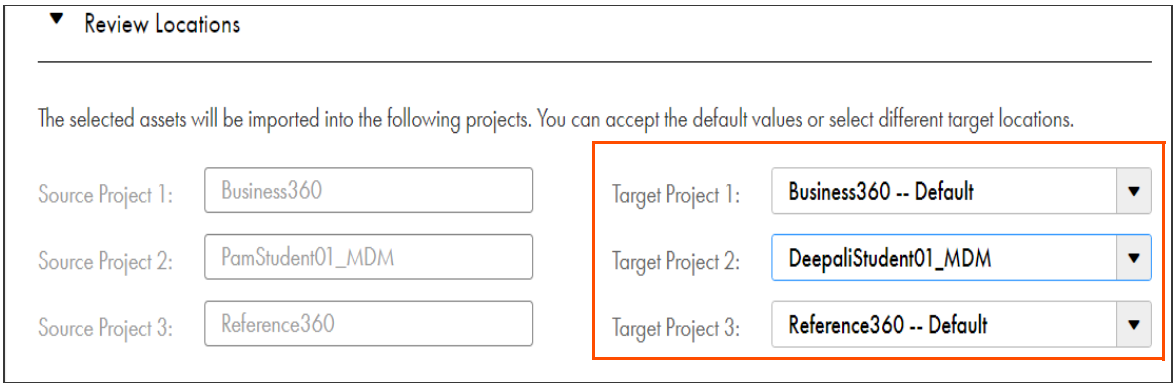


IMPORTANT: Make sure to deselect the check box. If you choose to overwrite existing assets, when an asset has the same name and internal ID as an asset in the target project, the asset replaces the existing asset in the target project. We don't want this to happen. Hence you need to uncheck the box.

7. Select the assets that you want to import. For this lab guide, retain the default selection.

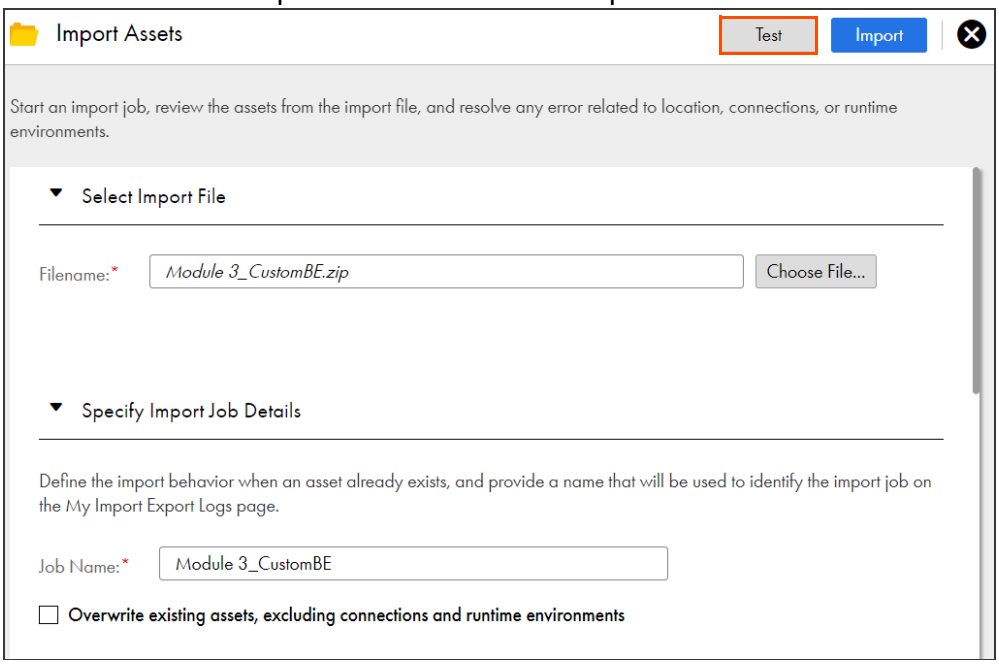


8. In the Review Locations sections, for the **Source Project 2,** you will see a project (Pamstudent01_MDM) from which you will be importing the assets. In the **Target Project 2**, select your student project. This ensures that the assets that were there in the source project will save into your student project. Also ensure that the source project and target project system match. **Ex**: If the Source Project 1 is Business 360, then the Target Project 1 has to be Business360 --Default.

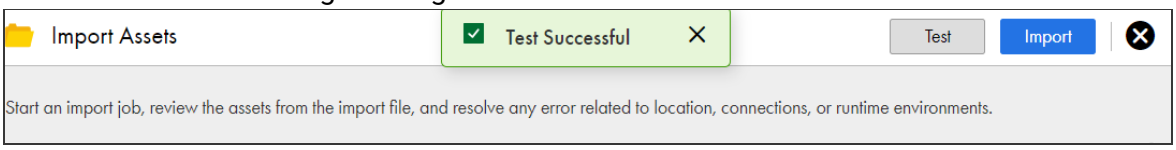


IMPORTANT: Make sure your student project is selected as Target Project 2. If this is not the case, a new project folder will be created as Source Project 2, which contains all assets. If a asset is created in one folder, you cannot move, copy or delete the asset. This may cause hinderance in the upcoming labs.

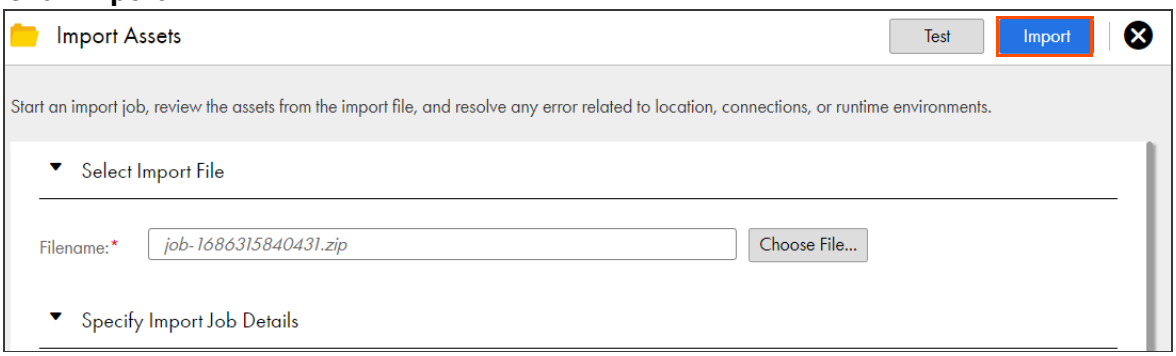
9. Click **Test** to see the potential results of the import.



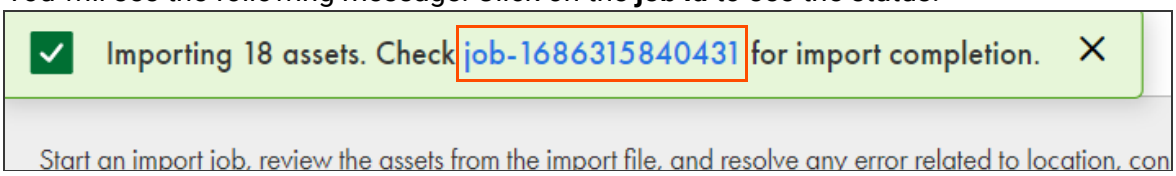
You will see the following message:



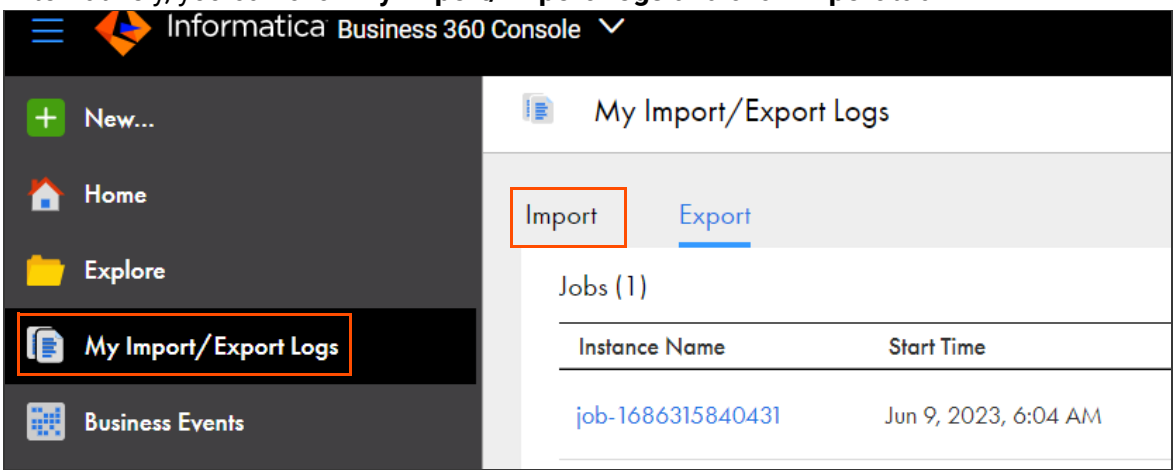
10. Click **Import**.



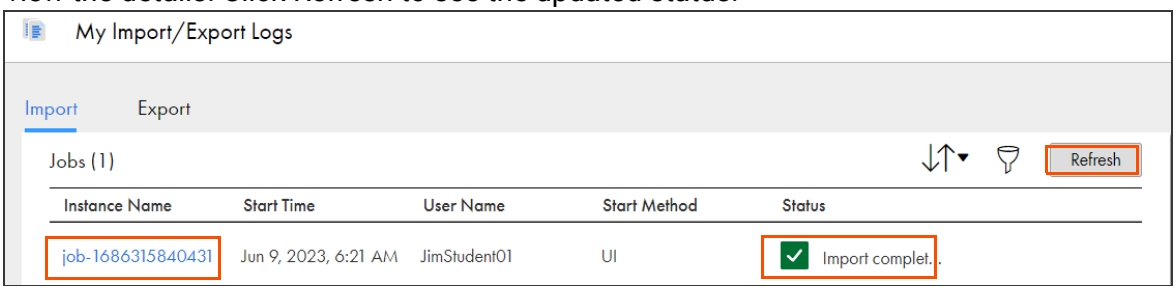
11. You will see the following message. Click on the **job id** to see the status.



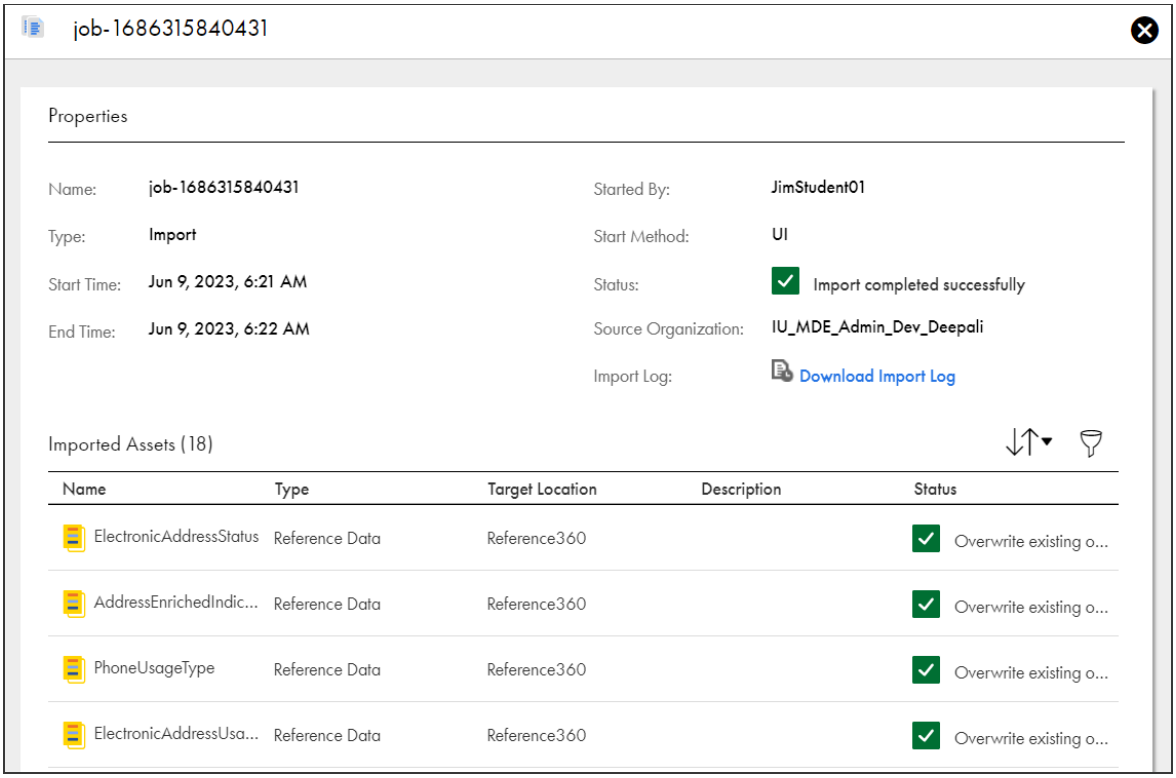
12. Alternatively, you can click **My Import/ Export Logs** and click **Import** tab.



13. You can see that the import has successfully completed. Click on the **job instance name** to view the details. Click Refresh to see the updated status.



You will see the details as follows:

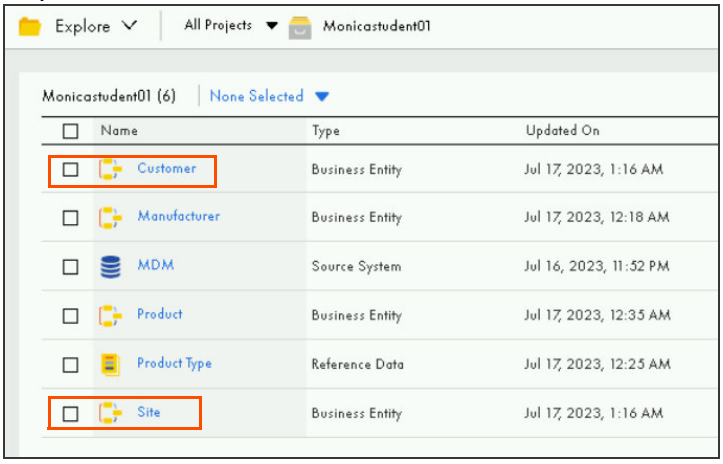


Verify the Imported Business Entities

Now, you will verfiry whether the business entities have been imported successfully.

14. Click **Explore** in the Business 360 Console.

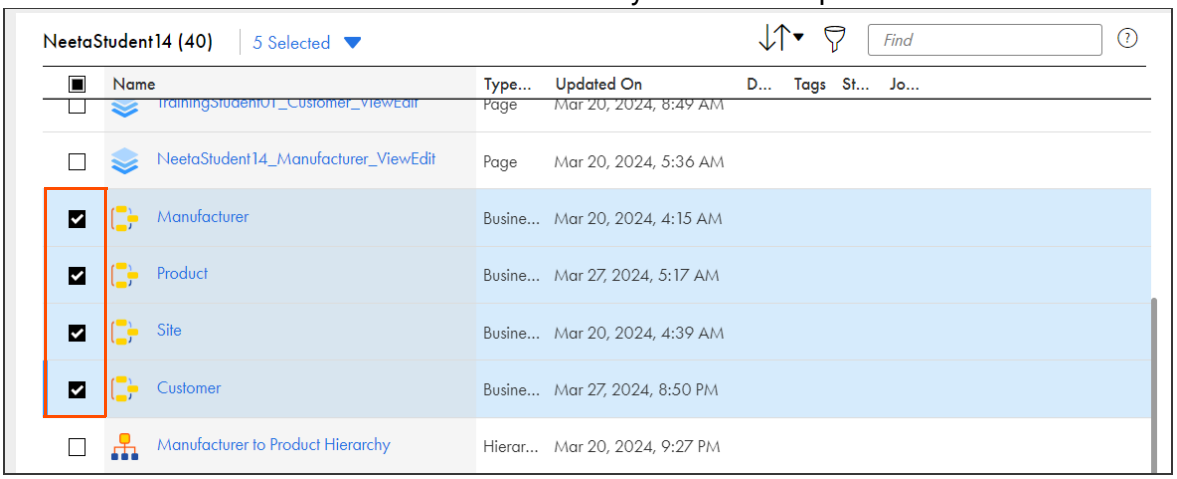
15. Go to your student project and confirm if the business entities **Site** and **Customer** are imported.



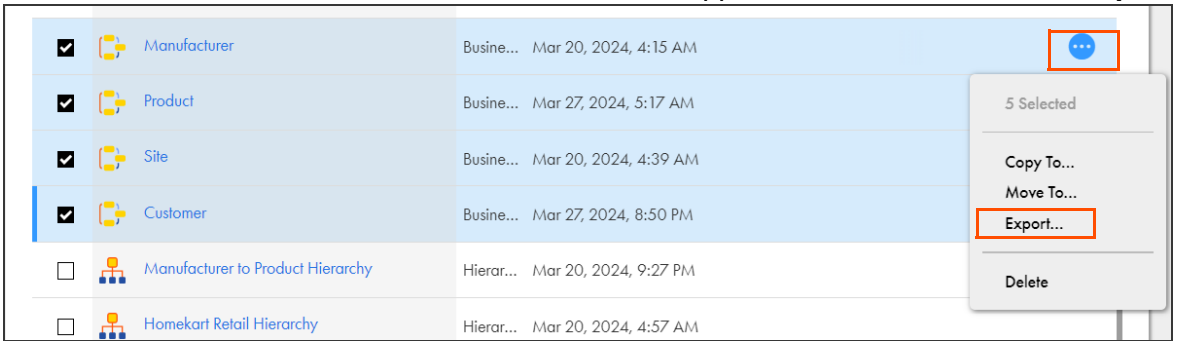
Export the Custom Business Entities

You will now export these assets into a zip file. This file can then be imported into any other org.

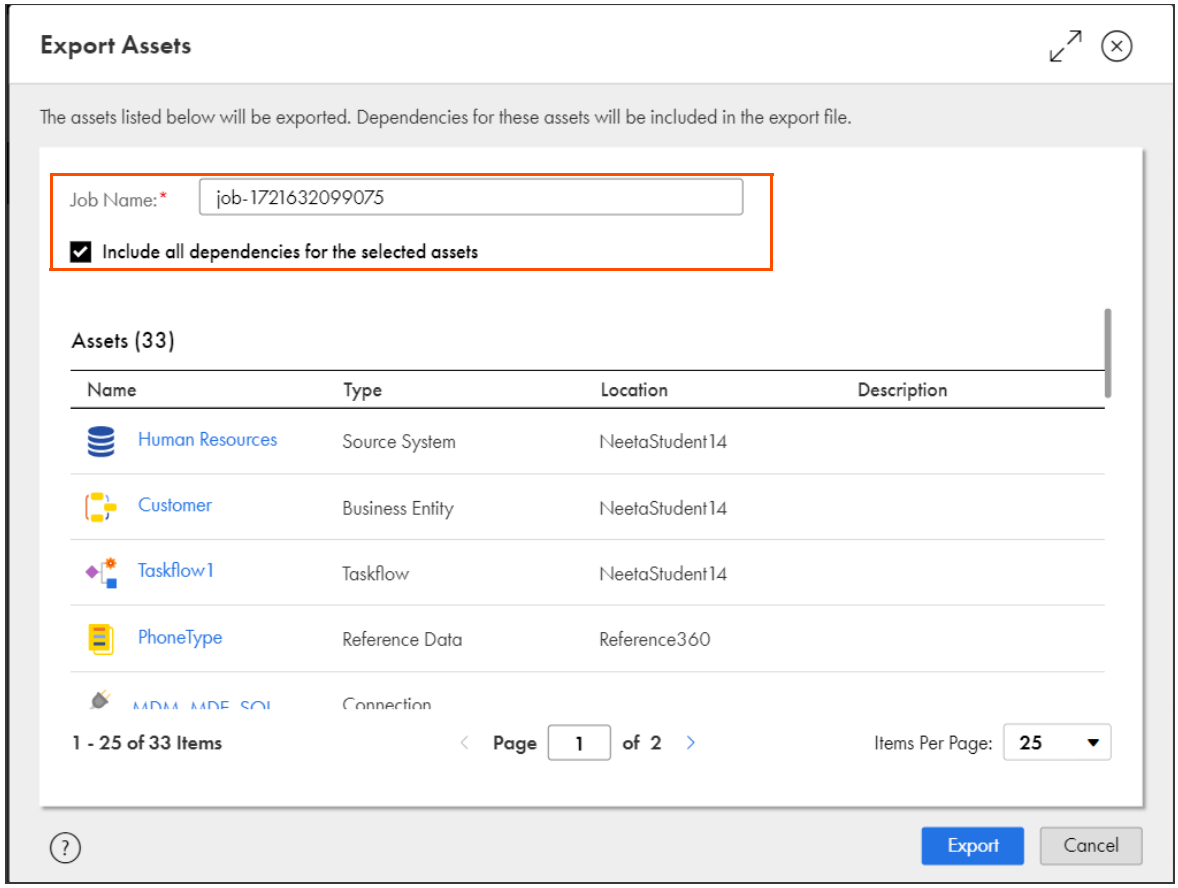
16. Select all the four custom business entities that you want to export.



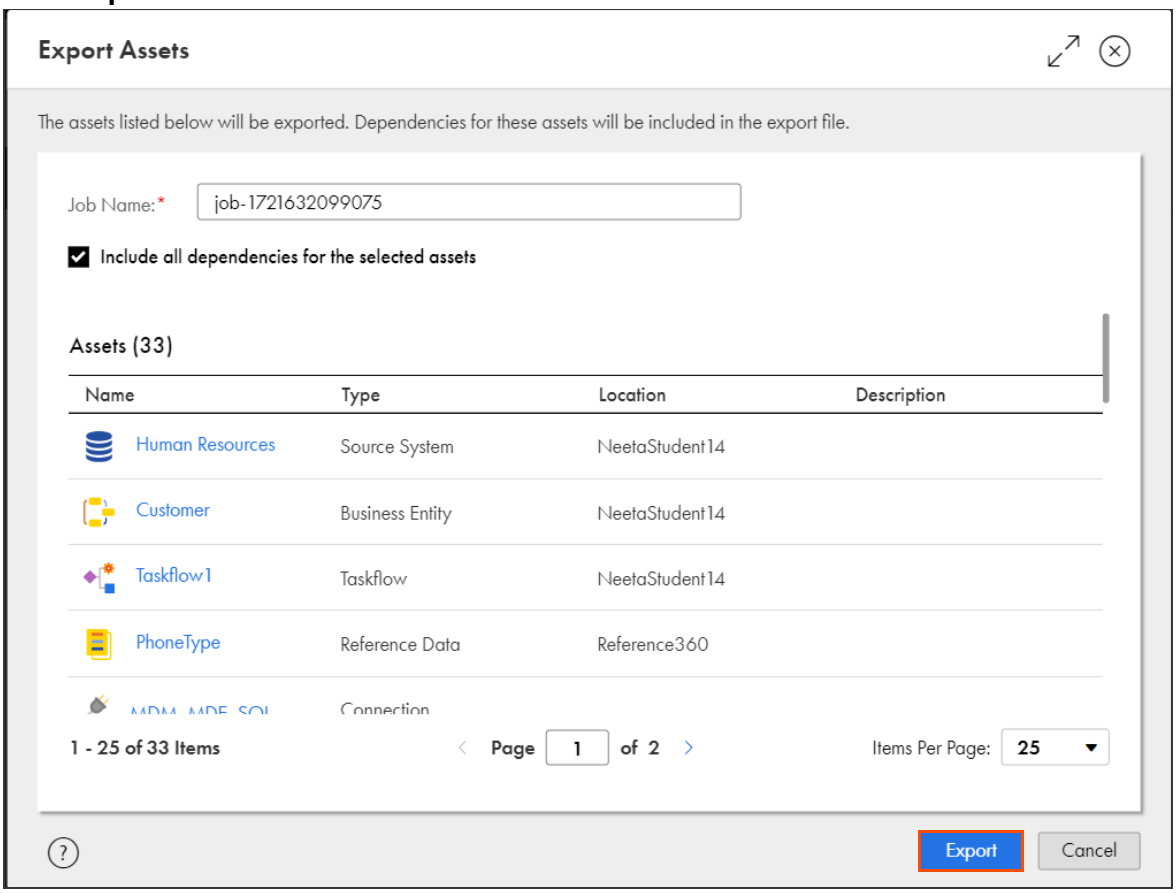
17. Hover over the asset and click on the three dots that appear next to the asset. Click **Export**.



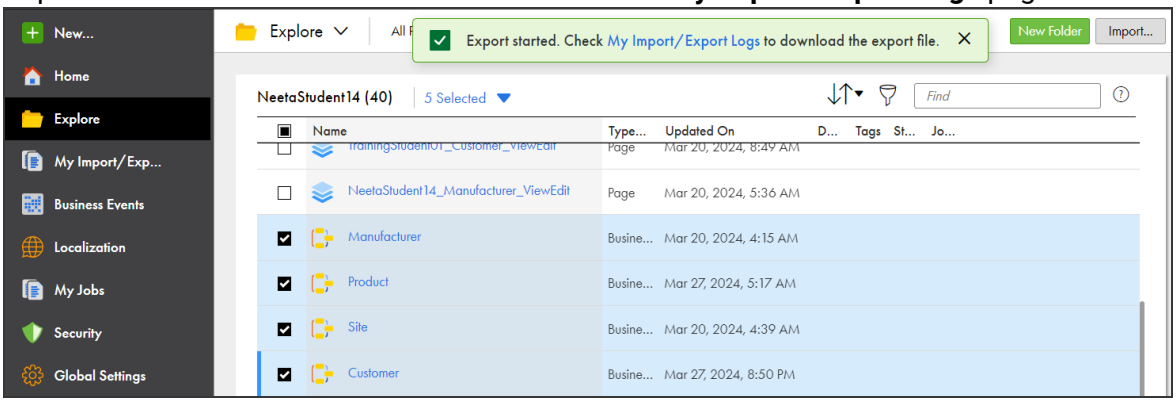
18. On the Export Assets page, either keep the existing job name or select a relevant job name that suits your needs. Select the checbox **Include all dependencies for the selected assets**. This will ensure that all the dependencies associated with the assets that you are exorting, are included.



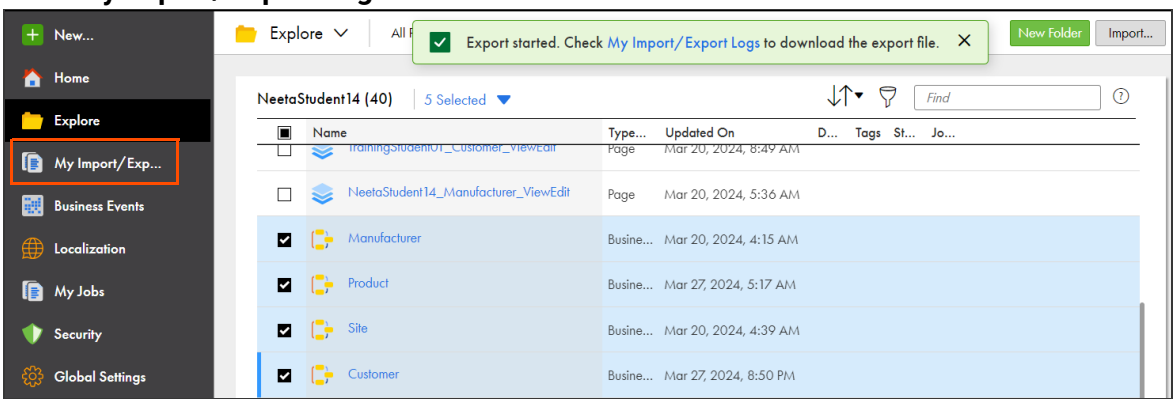
19. Click **Export**.



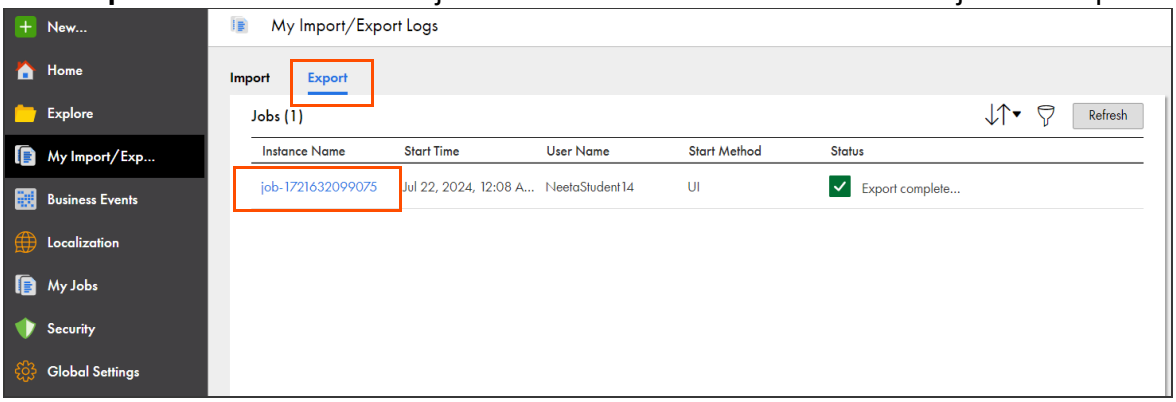
20. Export will start and the file will be available on the **My Import/ Export Logs** page.



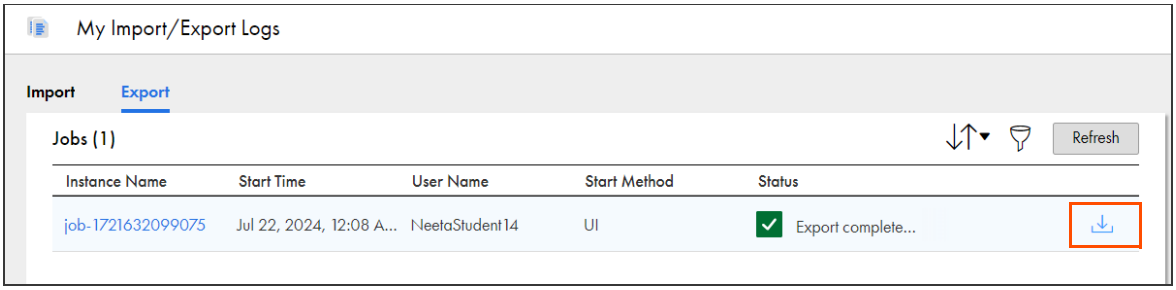
21. Click **My Import/ Export Logs**.



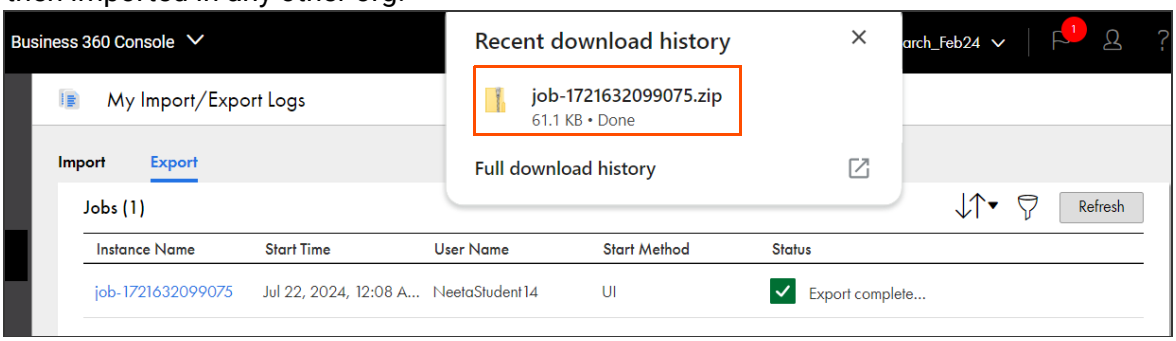
22. Click **Export** tab. You will see the job instance with its status. Wait for the job to complete.



23. Click the **download** icon.



24. The zip file will be downloaded in your VM that contains all the assets. The same can be then imported in any other org.



This concludes the lab.

Module 5: Defining Hierarchies and Relationships

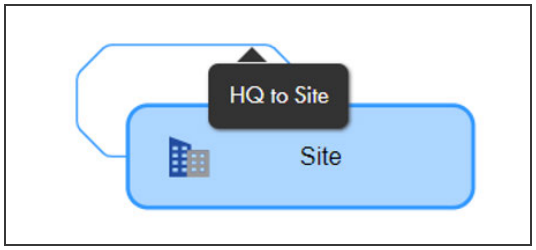
Defining Hierarchy Model

A hierarchy model consists of a collection of hierarchy relationships between business entities. For example, Manufacturer hierarchy model shows the relationships from the Manufacturer business entity to the Customer business entity.

A hierarchy model consists of a top-level business entity, additional business entities, and hierarchy relationships that define parent-child relationships between business entities. Each business entity in a hierarchy model must be connected to the top-level business entity directly or through other business entities. You can create as many hierarchy relationships as you need to represent the hierarchy model. You can also define attributes of relationships to allow users to provide additional information about relationships.

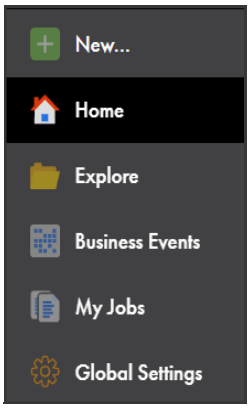
High-level Steps to Define a Hierarchy Model

1. Create a HQ hierarchy model.
2. Add the **Site** business entity as the top-level business entity.
3. Create a recursive relationship from the **Site** business entity to the **Site** business entity.

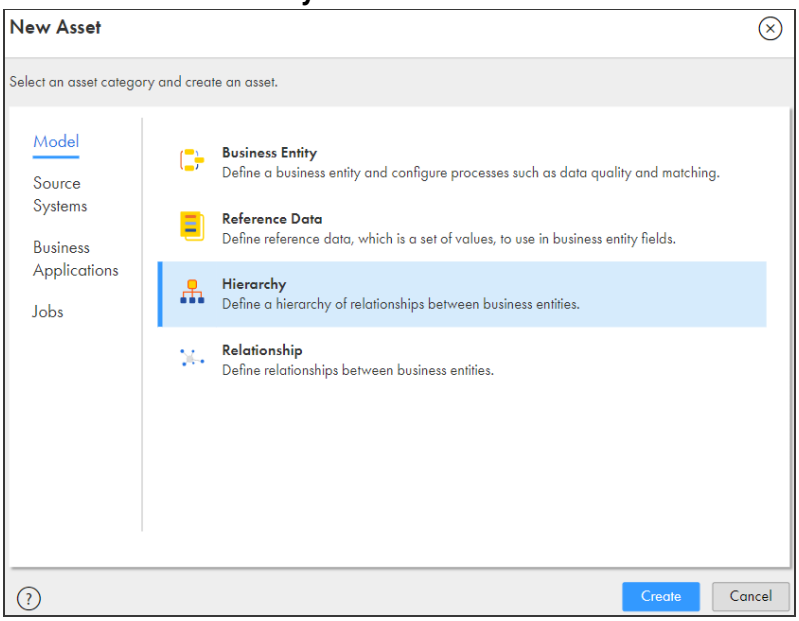


Step 1 – Create a Hierarchy Model

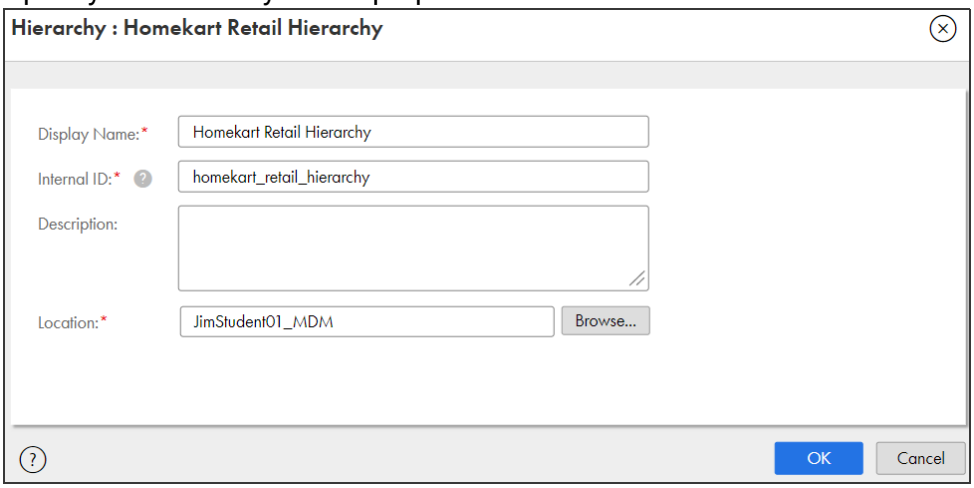
1. Click **New**.



2. Click **Model** > **Hierarchy** > **Create**.



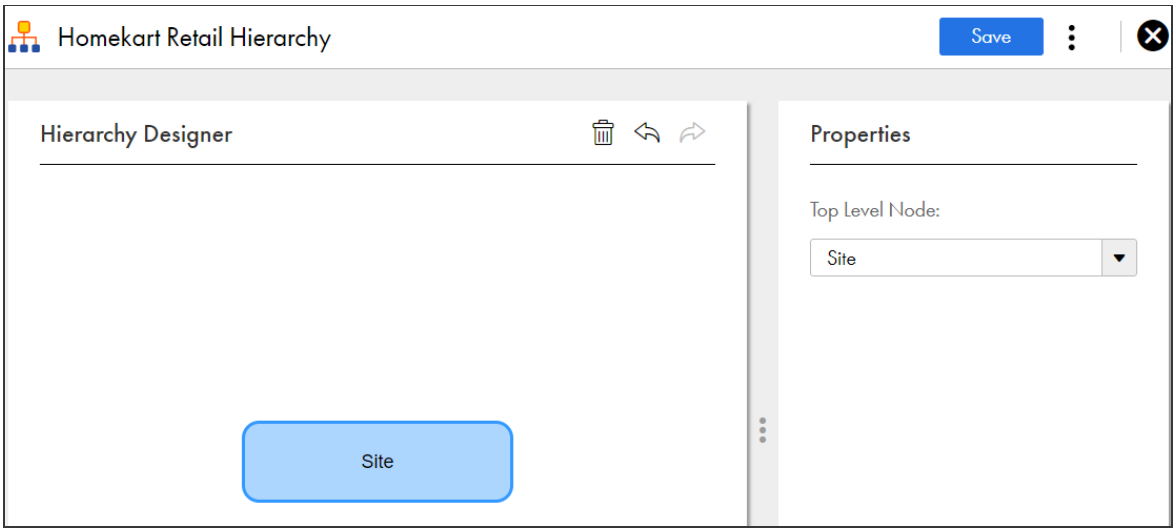
3. Specify the hierarchy model properties.



4. Click **OK**.

Step 2 – Configure the Top-level Node

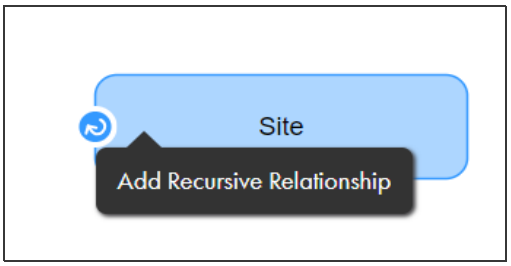
5. Specify a business entity as the top-level node in the hierarchy model.
   For example, in the HQ hierarchy model, you define the Site business entity as the top-level node.



6. In the **Properties** panel, select a business entity.
   The selected business entity appears as the top-level node on the canvas.

Step 3 – Add a Node

7. Hover over the node from which you want to create a relationship and click **Add Child**.
8. From the Properties panel, select a business entity for the undefined node.
9. The Parent Relationships field populates based on the selected business entity.
10. To add a recursive relationship, hover over the node for which you want to create a recursive relationship and click **Recursive**.



11. Click **Save**.

Define Relationships

You can define relationships between business entities that have some type of association. The users of Business 360 applications, such as Multidomain MDM, can then define and visualize the relationships between business entity records.

You can define multiple relationships for a business entity. A business entity can have a relationship with other business entities or with itself. For example, you can define a relationship between the Product and Customer business entities to indicate that a customer orders product.

Relationship Types

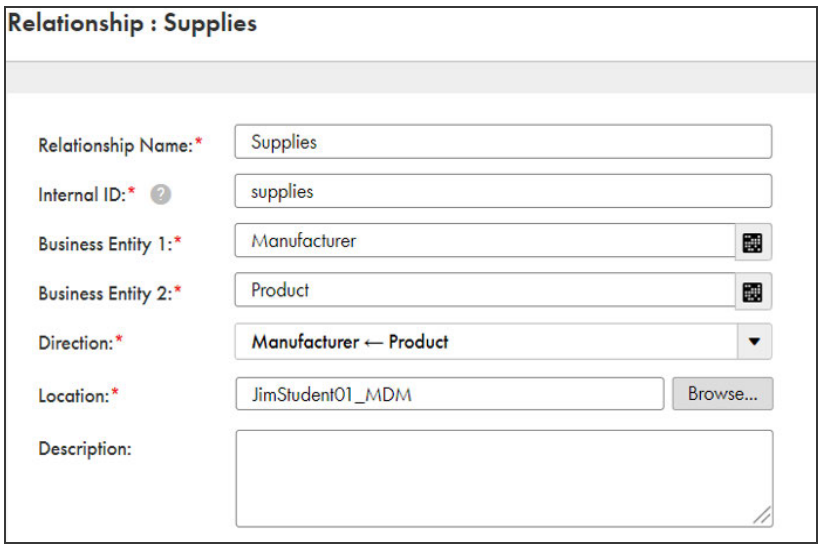
Records can be related based on hierarchical relationships, related record relationships, and business entity record field relationships.

MDM SaaS supports one-to-one, one-to-many, and many-to-many relationships between business entities. Configure one-to-one relationships using hierarchies or relationships. Configure one-to-many and many-to-many relationships using business entity record fields. If a user has the necessary permissions, they can view, add, edit, and manage relationships. Records can be related in the following ways:

- **Hierarchical relationships**: Hierarchical relationships connect two business entity records in a hierarchy. For example, the Informatica record is the parent of the John Smith record. You can view these relationships in a hierarchy or relationship graph for the record.
- **Related record relationships**: Related record relationships connect related business entity records by using relationships. For example, the John Smith record is related to the PX100 Keyboard record through the person to object relationship. You can view these relationships in a relationship graph or related records component for the record.
- **Business entity field relationships**: Business entity record field relationships use a field to relate to business entity records. The users can search for records and set them as field values. Business entity record field relationships are maintained when you ingress or egress data.

Create a Relationship

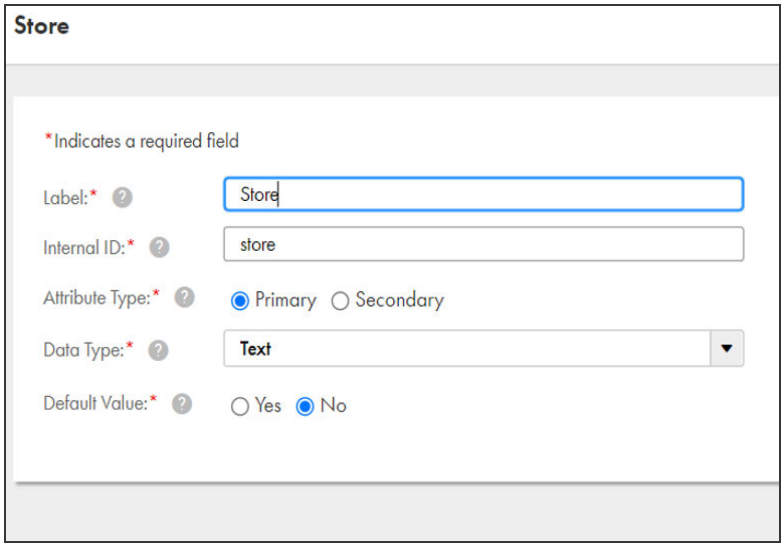
1. Click **New**.
   The New Asset dialog box appears.
2. Click **Model Relationship**.
3. Click **Create**.
   The Relationship dialog box appears.
4. Specify the relationship properties.
5. Click **OK**.
   The relationship appears on the Properties page.
6. Click **Save**.
   The relationship appears on the Explore page.



Configure Relationship Attributes

Define attributes for custom relationships so that users in Business 360 applications can provide additional information about relationships.

1. From the Explore page, find and open the relationship.
2. In the Attributes section, click **Add Attribute**.
   The Add Attribute dialog box appears.
3. Specify the properties of the attribute.
4. To reorder the attributes, click **Move up** or **Move down** beside the attribute that you want to move.
5. To define additional attributes for the relationship, repeat steps 2 and 3.
6. Click **Save**.



Lab 5-1: Creating a Hierarchy Model and Defining Relationships

Overview:

A hierarchy model consists of a collection of hierarchy relationships between business entities. For example, a manufacturer hierarchy model shows the relationships from the manufacturer business entity to the product business entity, and product to site, and so on.

Objectives:

- Create a hierarchy model in Business 360 console
- Define relationships

Duration:

20 minutes

Scenario:

In the previous session, Mark configured the Custom Business Entities and added fields to them. He also managed the import/export of assets among his team members to facilitate parallel work on the data model. Now, Mark needs to create a hierarchy model with Homekart Canada as the top-level (parent) element and all Homekart branches as child elements. This structure will simplify tracking and provide unified information about all stores in one place.

Tasks:

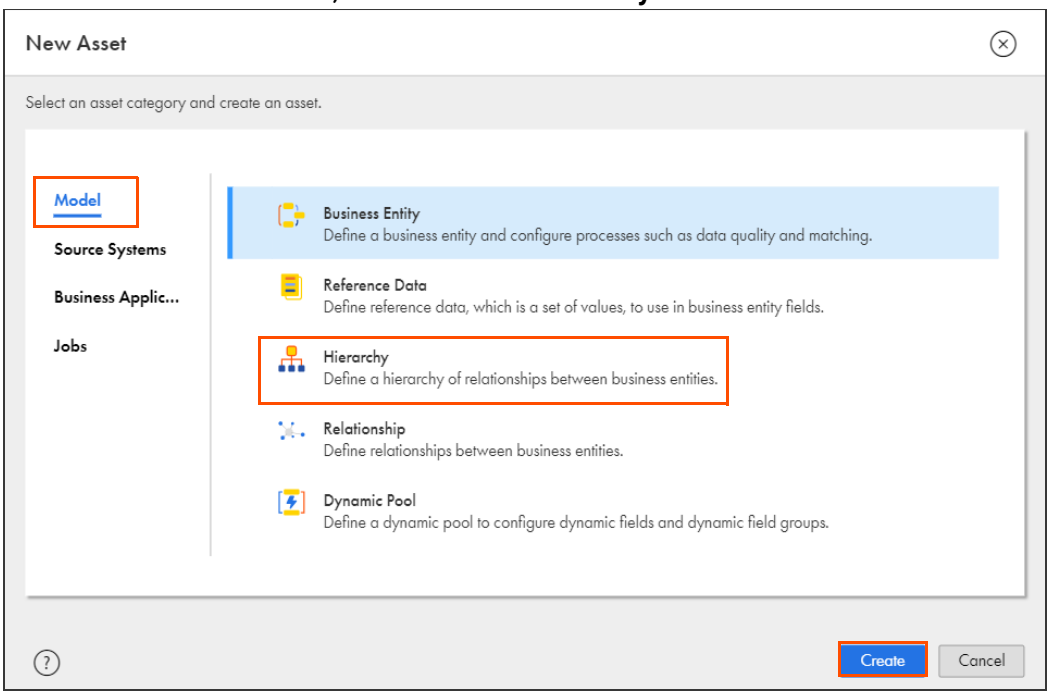
Perform the following tasks to define a hierarchy model and relationships for Homekart scenario:

- Define a hierarchy model
- Creating relationship between business entities
- Creating another hierarchy model in Business 360

Define a Hierarchy Model

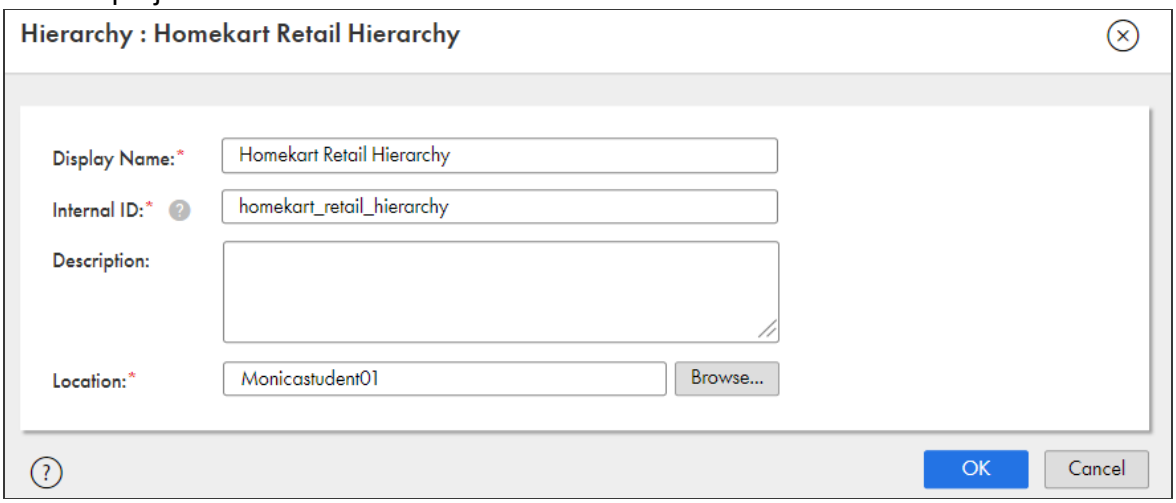
Homekart has stores all over the GTA (Greater Toronto Area). You will add these stores as hierarchical relationships in the Multidomain MDM system. We define GTA as the parent store in the hierarchy where all the other individual sites would act as the child under this parent store.

1. Log in to IICS using your admin credentials. For example, **Jimstudent01**.
2. Navigate to **Business 360 Console**.
3. Click **New**.
4. In the **New Asset** window, click **Model** > **Hierarchy** and click **Create**.



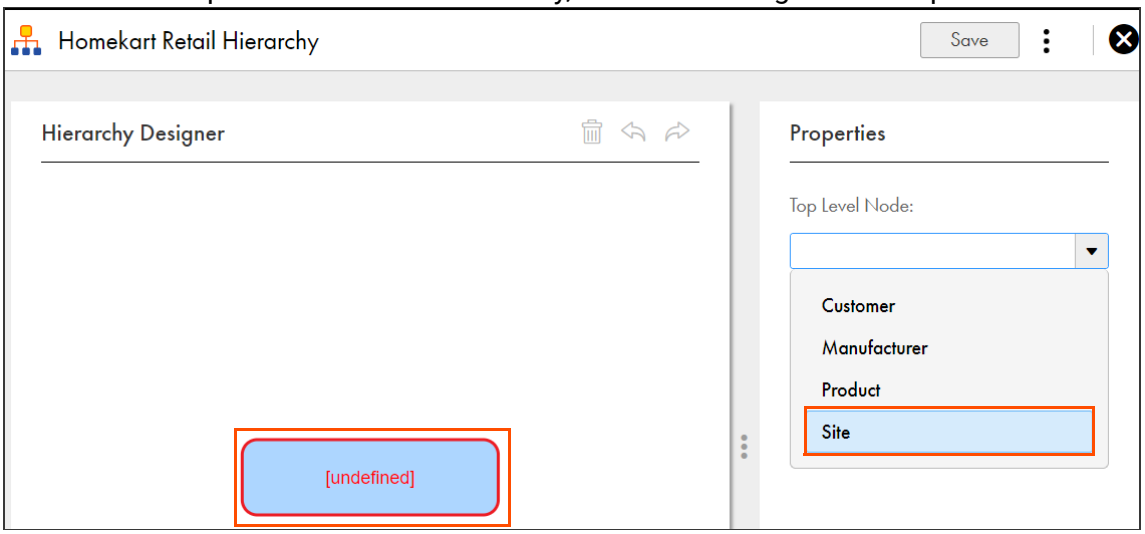
5. In the Hierarchy window, Enter the name as **Homekart Retail Hierarchy** and select your

student project. Click **OK**.

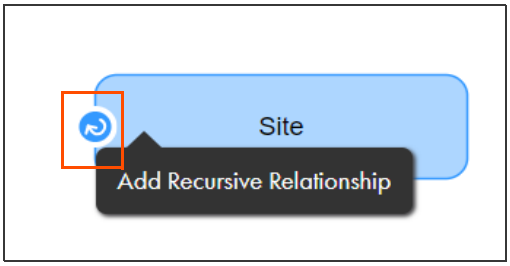


The student project in the screenshot may vary.

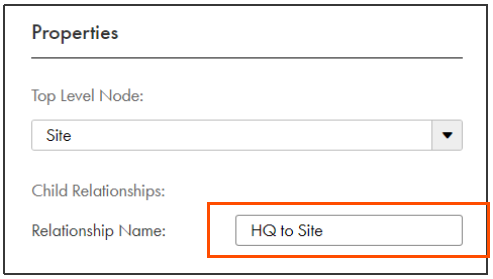
6. In the Hierarchy Designer page, click the **undefined** node, and from Properties panel on the right, from the **Top-Level Node** drop down, select **Site**. As we already discussed, Homekart GTA will be the parent node in the hierarchy, we are selecting **SITE** as top level node.



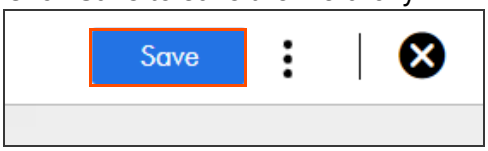
7. Hover the mouse over Site entity and click **Add Recursive Relationship** icon. To ensure that all Homekart branches are child entities under the top-level node and that these branches belong to the SITE Business Entity, a recursive relationship must be added.



8. In the Hierarchy Designer page, click the **Site** node, and from Properties panel on the right, Enter the Child Relationship Name as **HQ to Site**. This represents the name of the relationship between Homekart GTA and the child nodes within the Homekart hierarchy.



9. Click **Save** to save the hierarchy.



Creating Relationship Between Business Entities

You will create relationships between the business entities and define attributes for custom relationships. Users in Business 360 application can provide additional information about relationships.

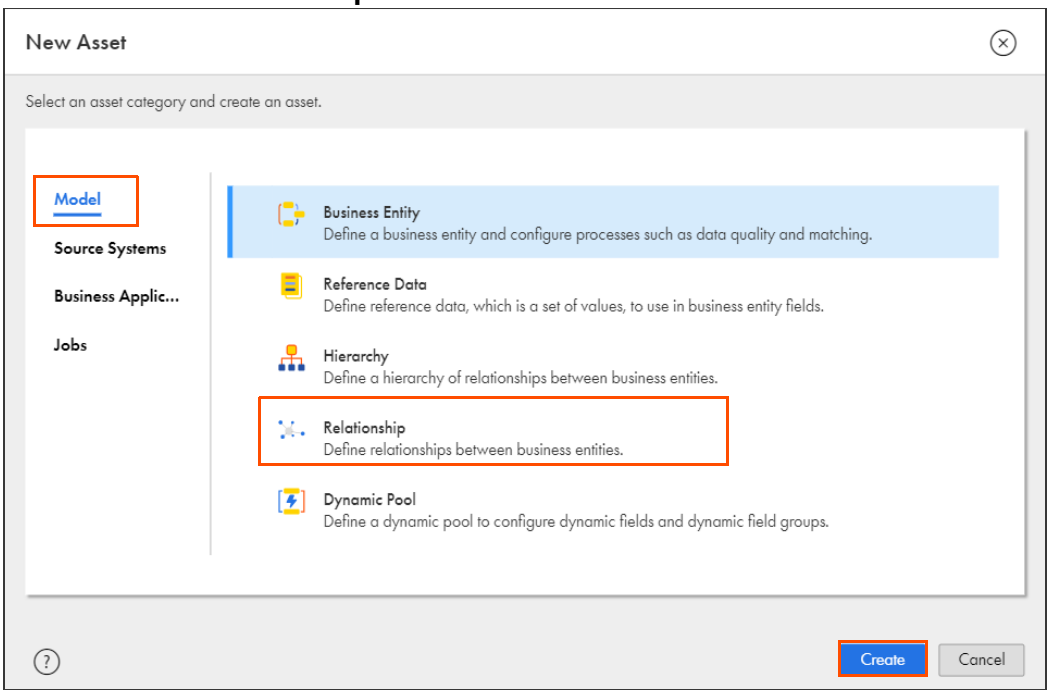
You will be creating following relationships:

Relationship Name	Business Entities
Supplies	Manufacturer -> Product
Orders	Customer -> Product
Delivers	Manufacturer -> Product
Sells	Site -> Product

First, you will create a relationship named **Supplies** between Manufacturer and Product.

10. In Business 360 console, click **New**.

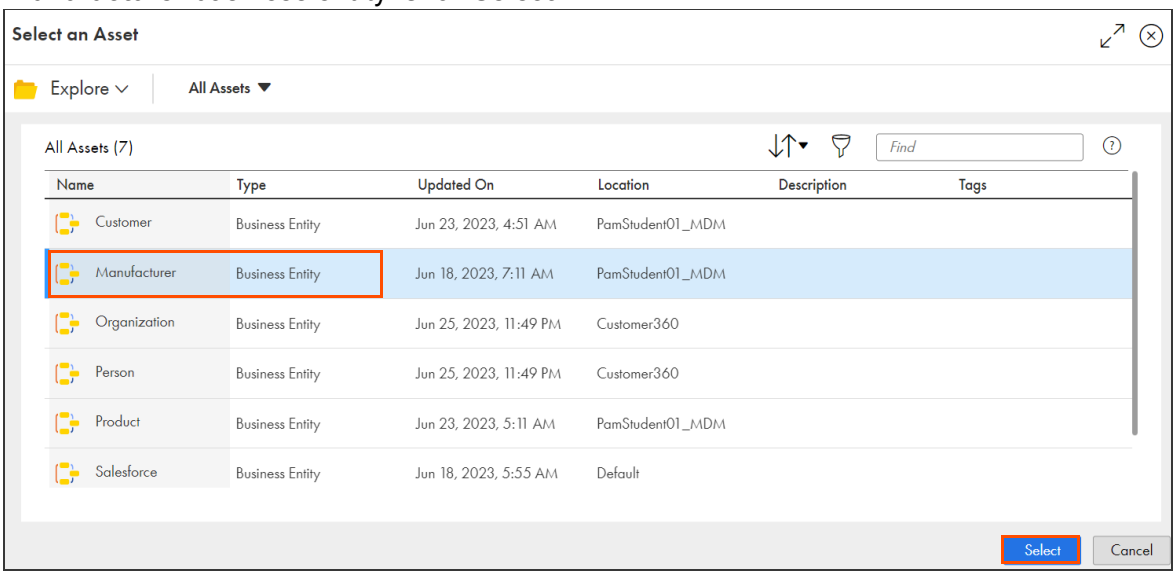
11. Select **Model** > **Relationship** and click **Create**.



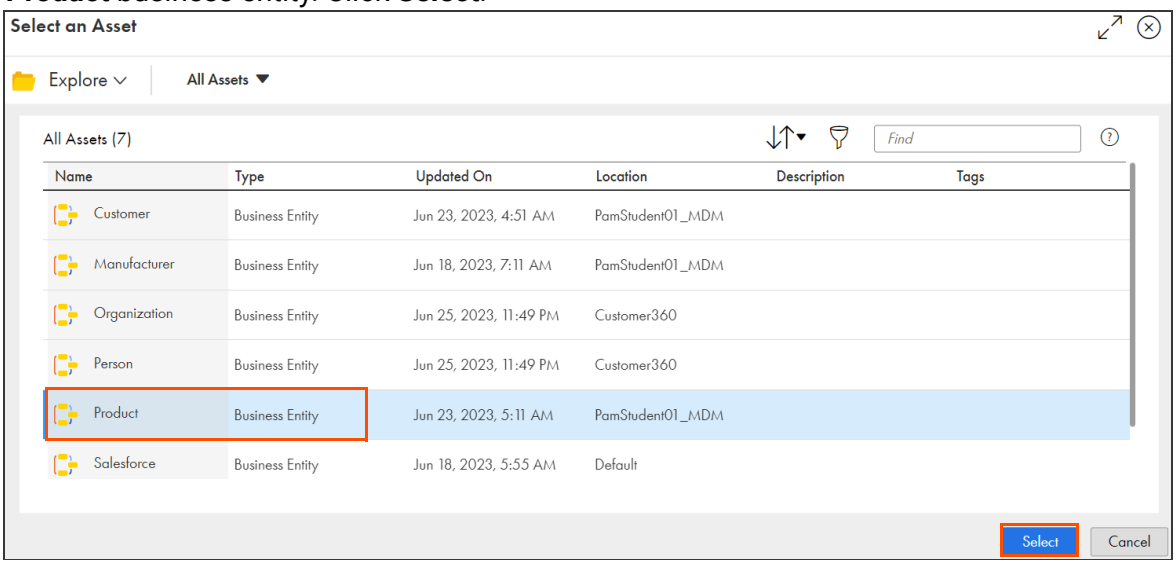
12. Enter the following details:

Field	Value
Relationship Name	Supplies
Business Entity1	Manufacturer
Business Entity2	Product
Direction	Manufacturer -> Product
Location	<Student project>

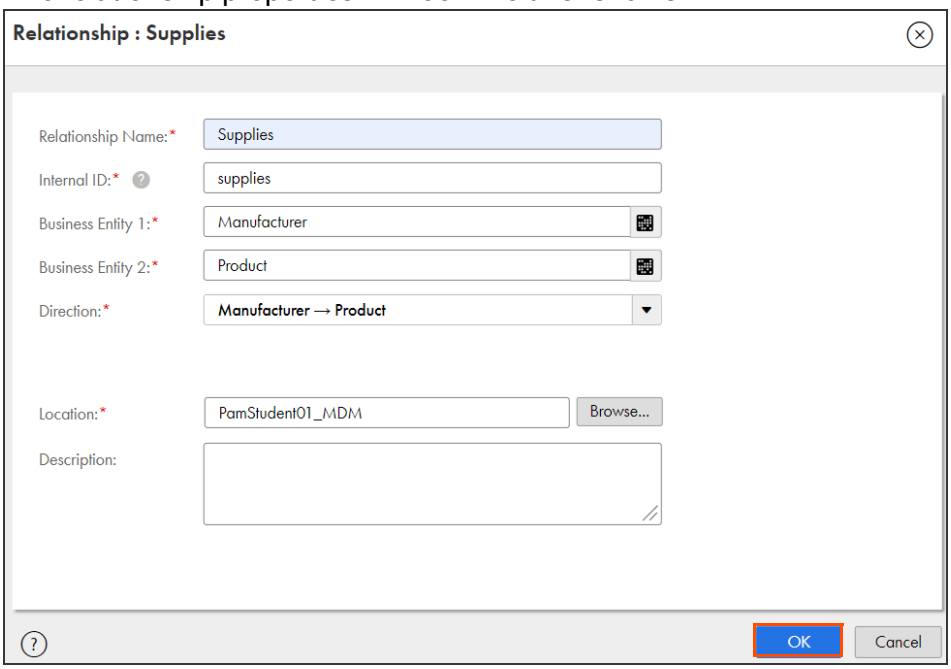
13. To select the Business Entity 1, select the asset picker icon next to it and select the **Manufacturer** business entity. Click **Select**.



14. To select the Business Entity 2, select the asset picker icon next to it and select the **Product** business entity. Click **Select**.

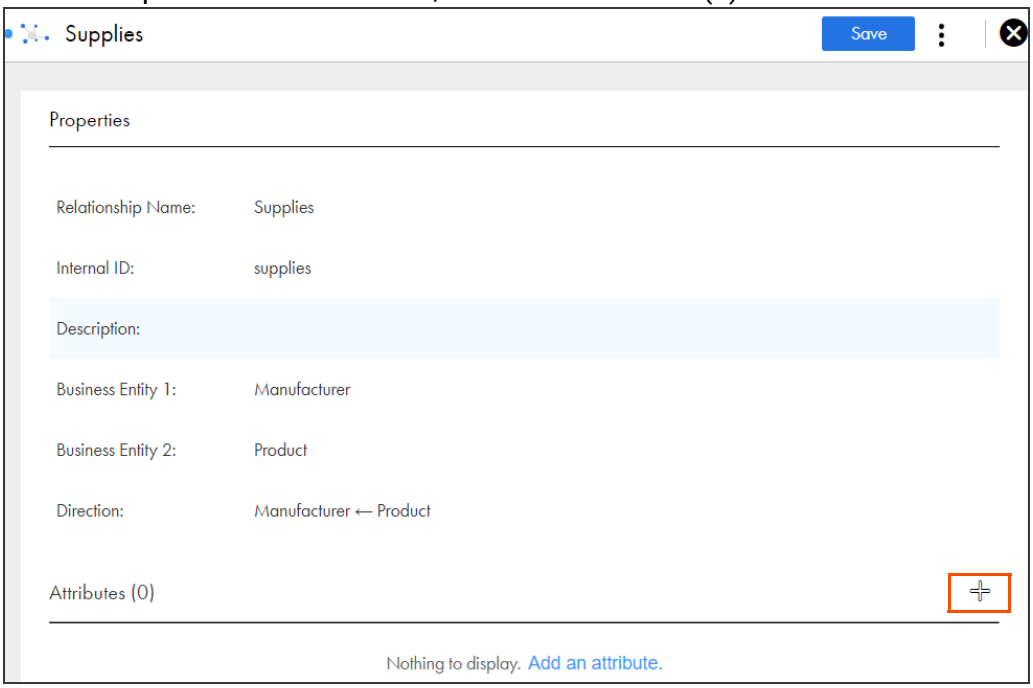


15. The relationship properties will look like this. Click **OK**.

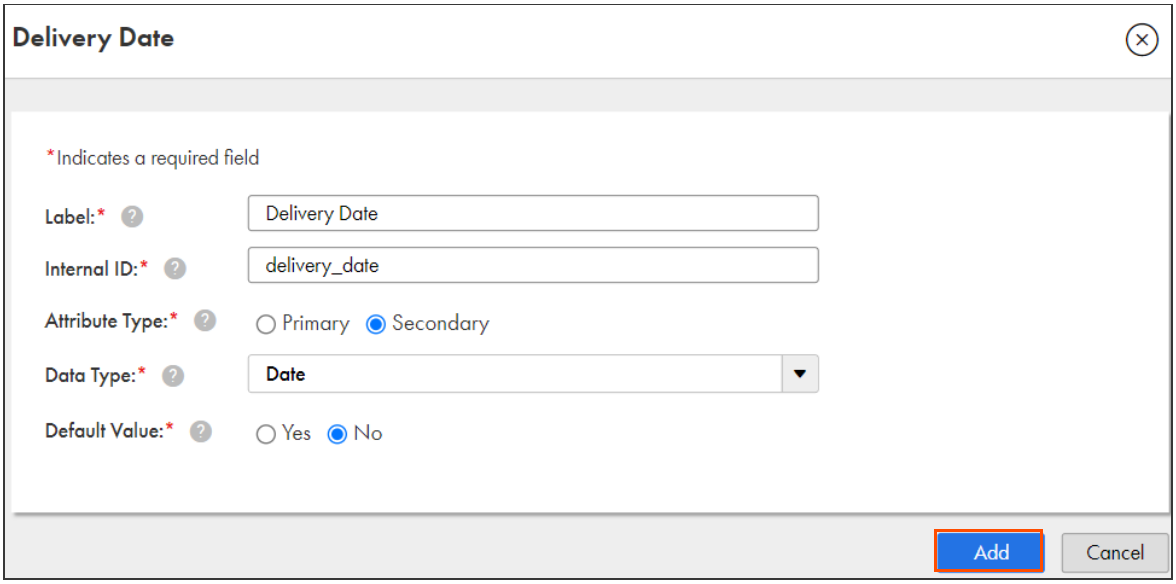


The student project in the screenshot may vary.

16. You can define attribute values of relationships to provide more information about relationships. To add an attribute, click **New Attribute** (+) icon under Attributes tab.

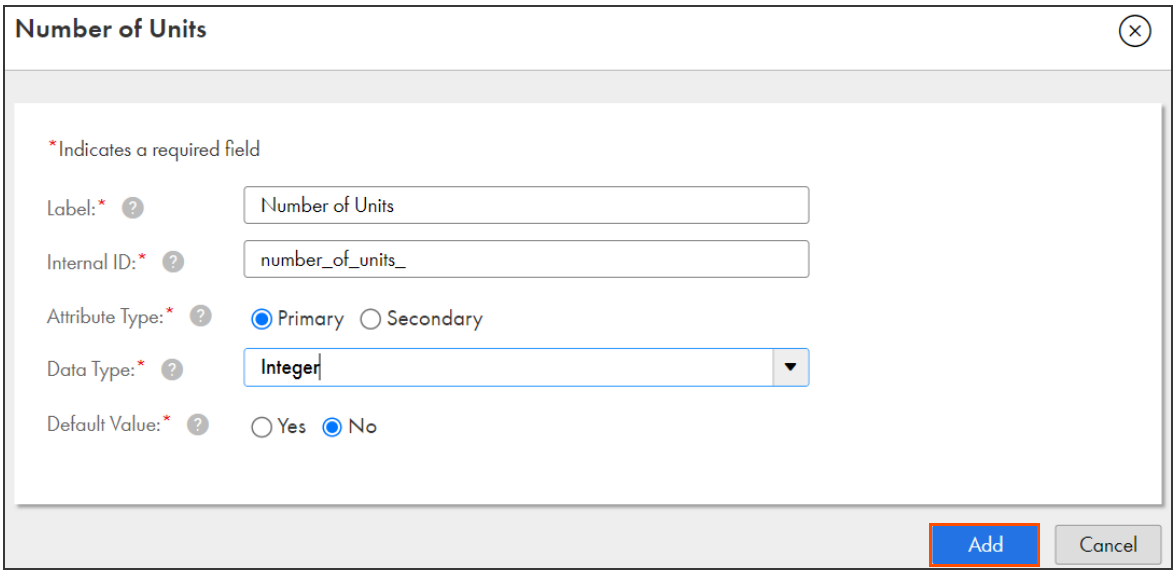


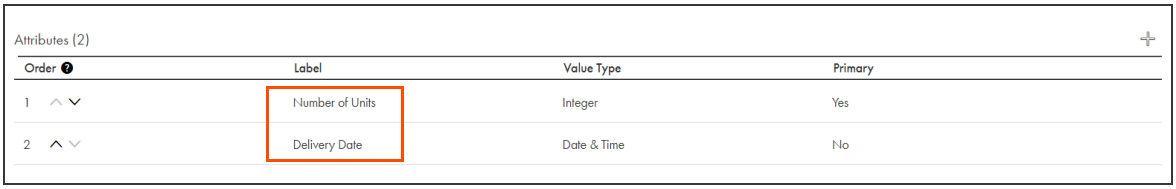
Field	Value
Label	Delivery Date
Data Type	Date
Attribute Type	Secondary
Default Value	No



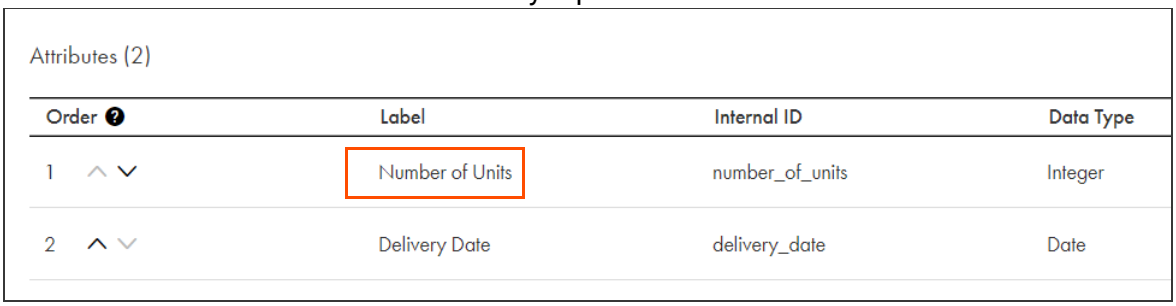
17. To add an another attribute, click **New Attribute** (+) icon under Attributes tab and add the following:

Field	Value
Label	Number of Units
Data Type	Integer
Attribute Type	Primary
Default Value	No

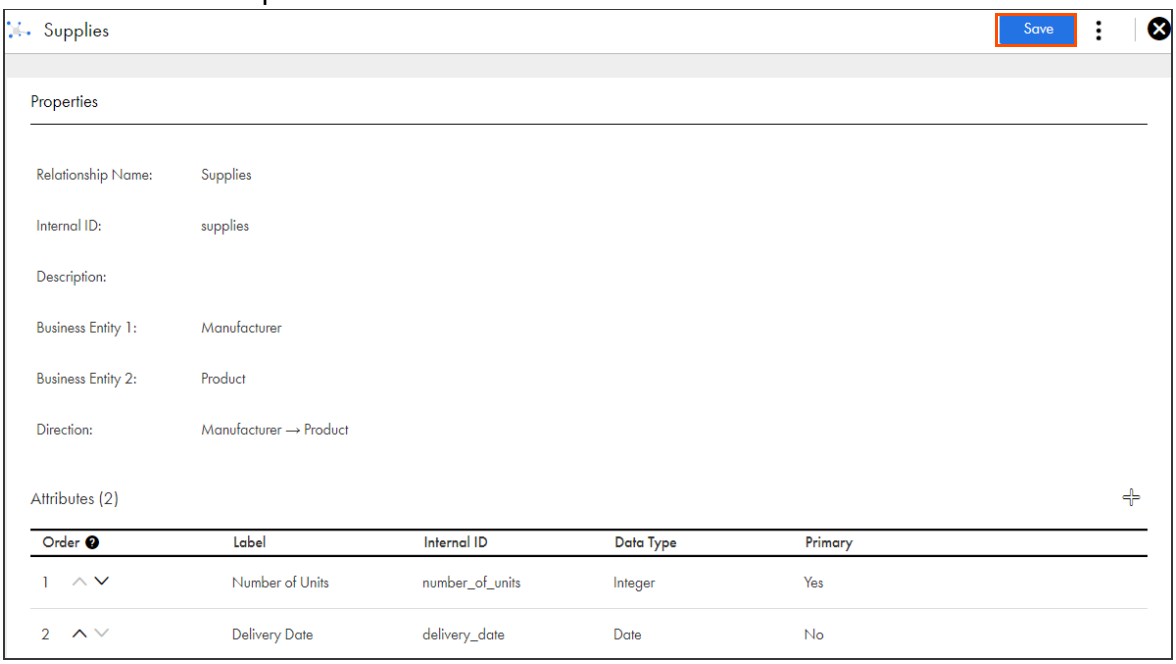




18. Make sure **Number of Units** attribute stay top in the row.



19. **Save** the relationship.



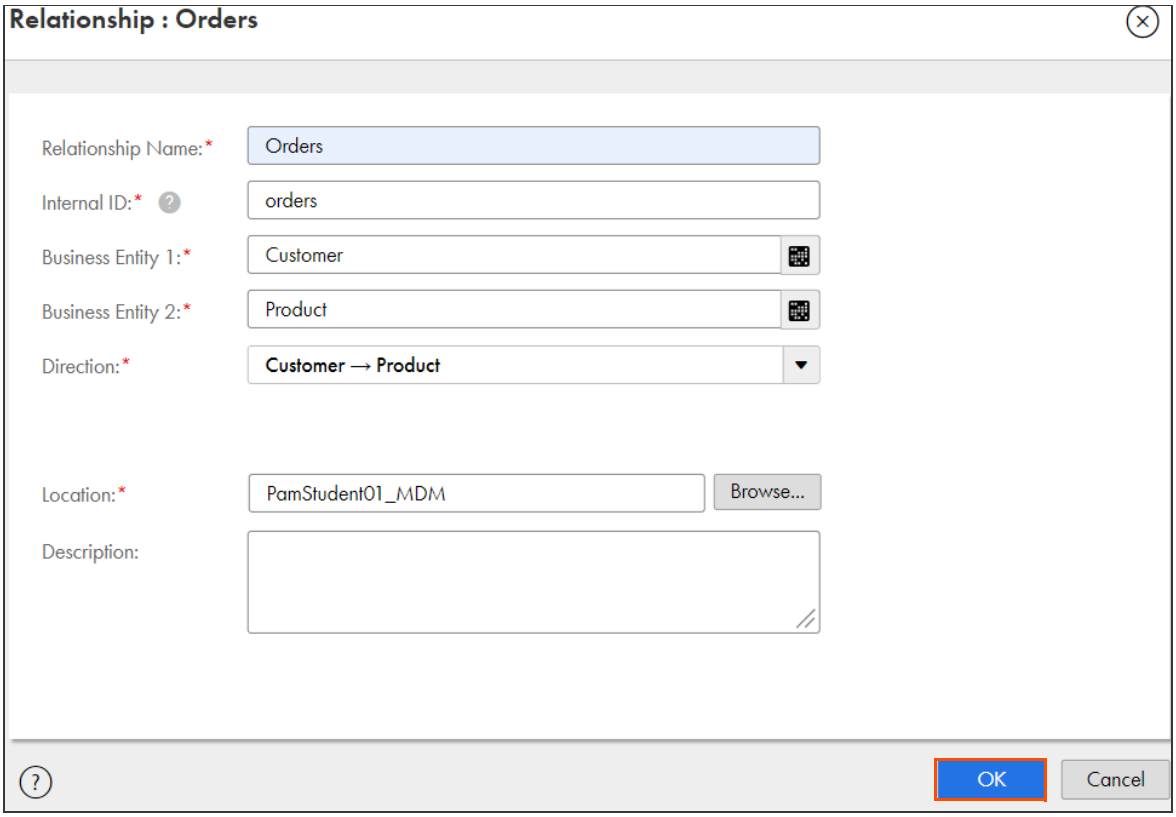
Creating the Relationship Named Orders Between Customer and Product.

20. In Business 360 console, click **New**.

21. Select **Model** > **Relationship** and click **Create.**

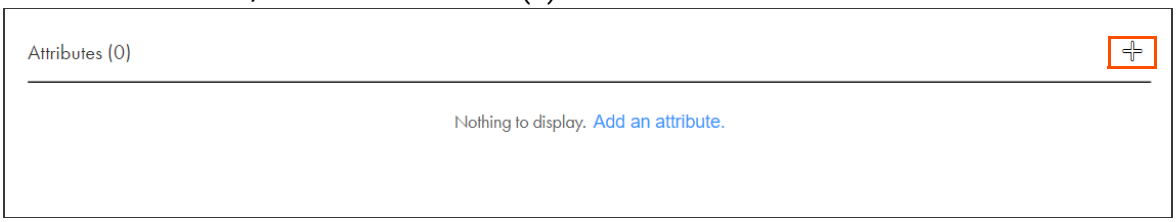
22. Enter the following details and click **OK**.

Field	Value
Relationship Name	Orders
Business Entity1	Customer
Business Entity2	Product
Direction	Customer → Product
Location	<Student project>



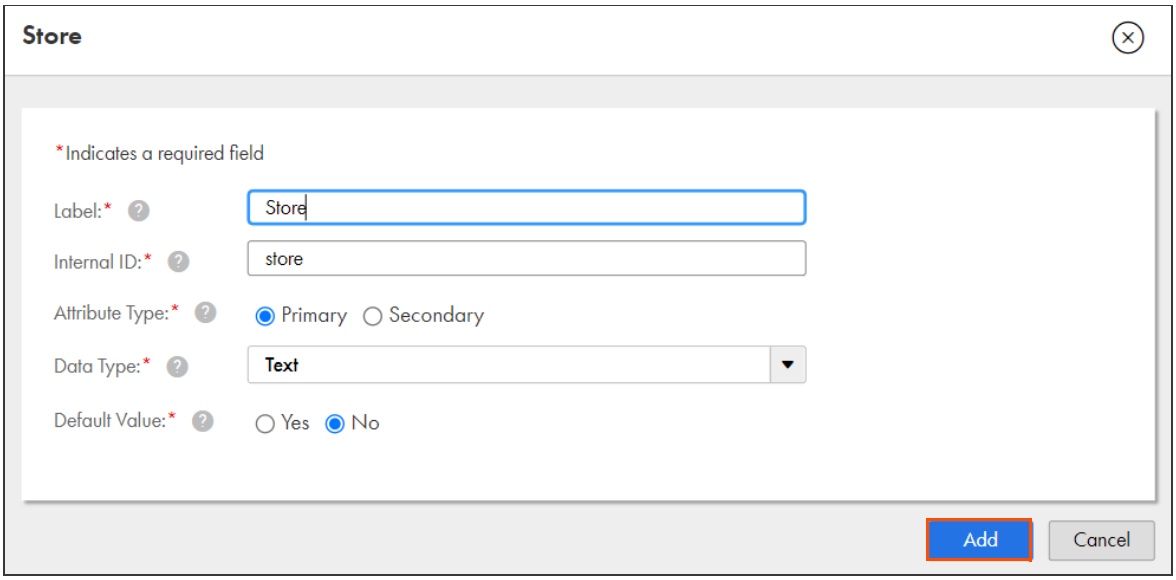
The student project in the screenshot may vary.

23. To add an attribute, click **New Attribute** (+) icon under Attributes tab.

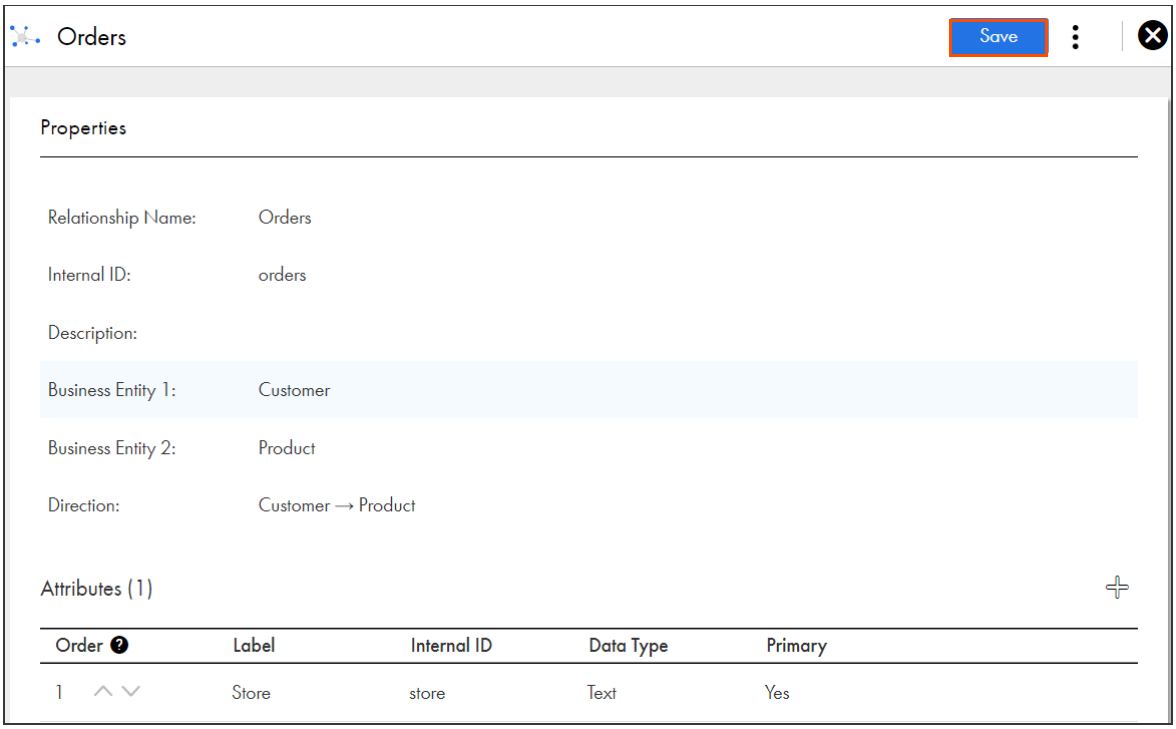


24. In the **Add Attribute** page, enter the following and click **Add**.

Field	Value
Label	Store
Data Type	Text
Attribute Type	Primary
Default Value	No



25. Click **Save**.



Self-Try Exercise (Mandatory)

Create Delivers relationship between Manufacturer and Product.

Add Number of Units and Delivery Date attributes to this.

26. In Business 360 console, click **New**.

27. Select **Model** > **Relationship** and click **Create**.

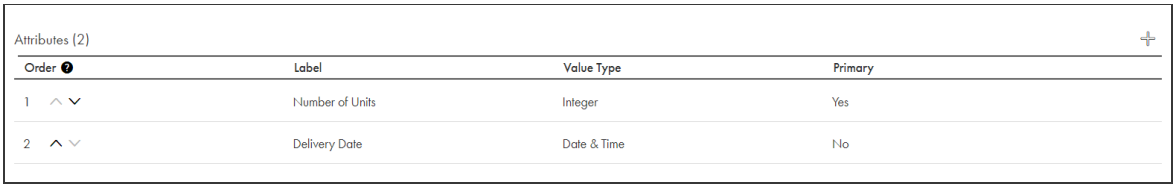
28. Specify the relationship properties.

Field	Value
Relationship Name	Delivers
Business Entity1	Manufacturer
Business Entity2	Product
Direction	Manufacturer → Product
Location	<Student project>
Description	Manufacturer Delivers the Product

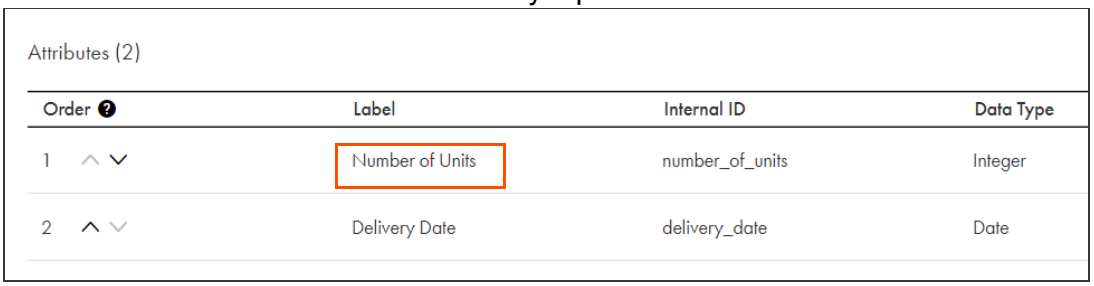
29. Add below attributes:

Field	Value
Label	Number of Units
Data Type	Integer
Attribute Type	Primary
Default Value	No

Field	Value
Label	Delivery Date
Data Type	Date
Attribute Type	Secondary
Default Value	No



30. Make sure **Number of Units** attribute stay top in the row.



31. **Save** the relationship.

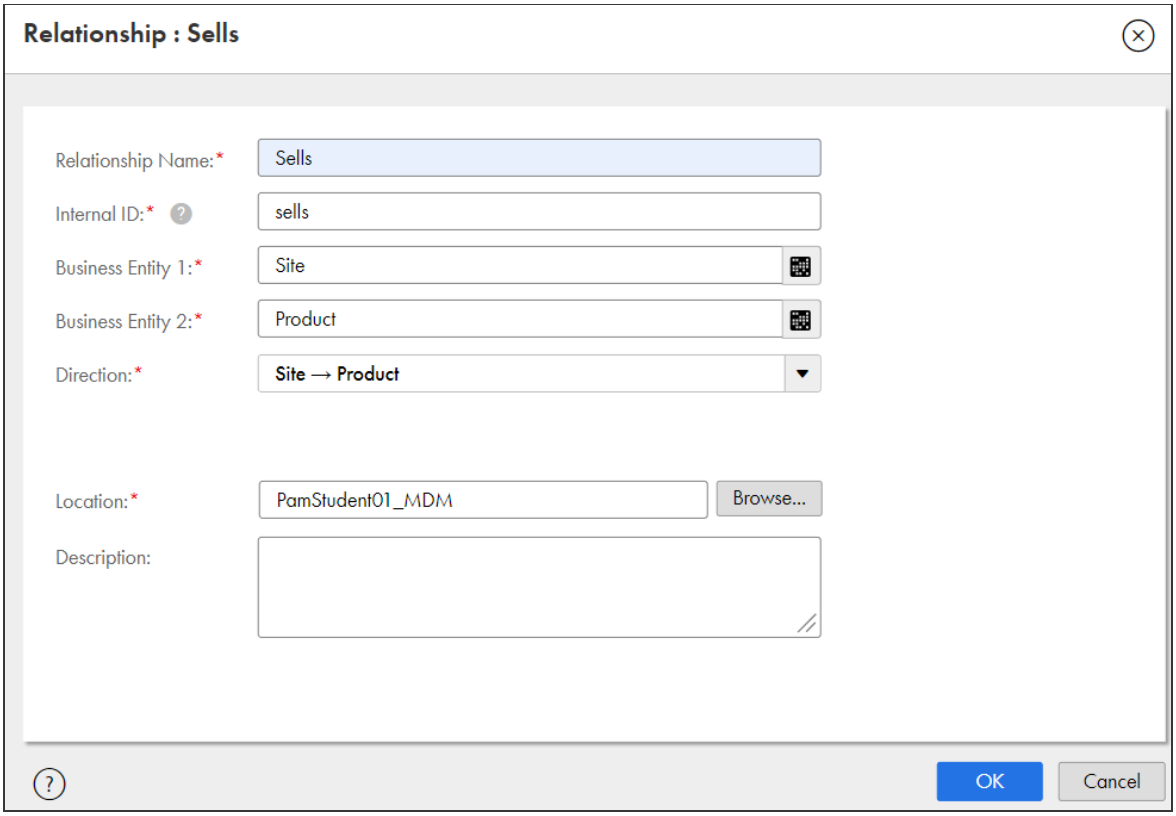
Self-Try Exercise (Mandatory)

Create Sells relationship between Site and Product.

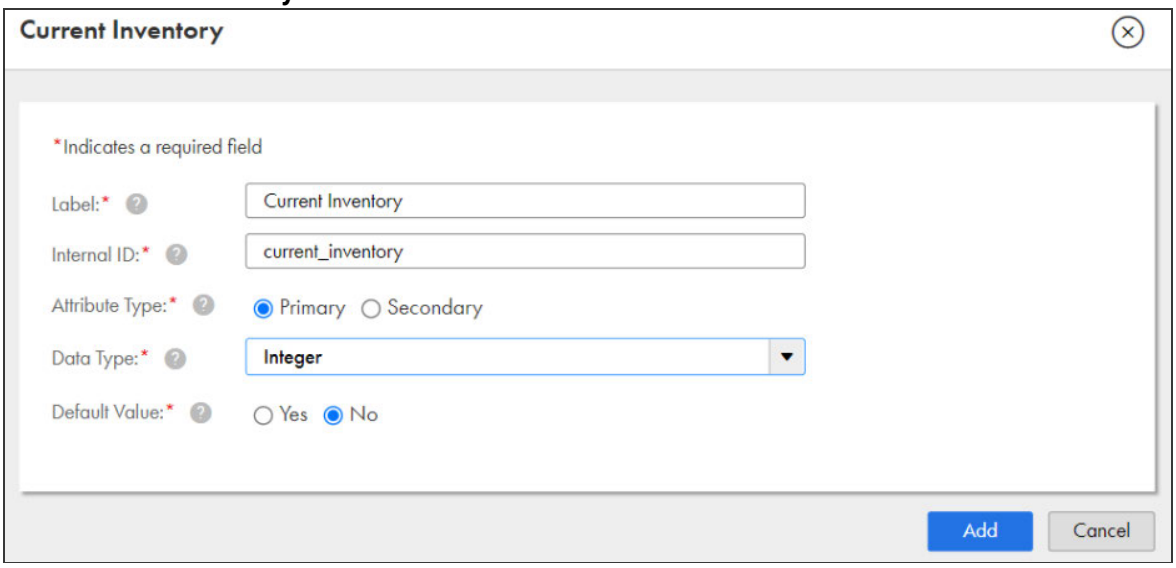
Add Current Inventory attributes to this.

High Level Steps

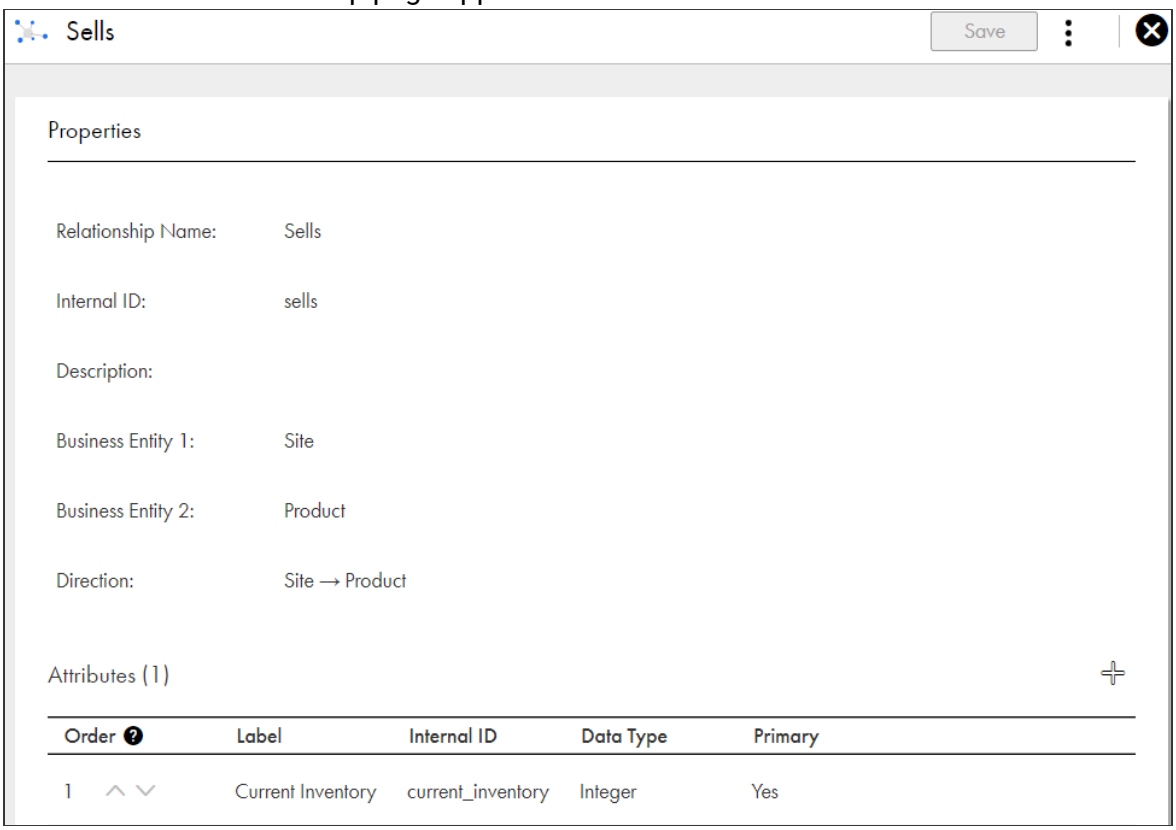
- Create **Sells** Relationship and using the below properties.



- Add **Current Inventory** attribute to it.



- Click **Save**. The relationship page appears as below.



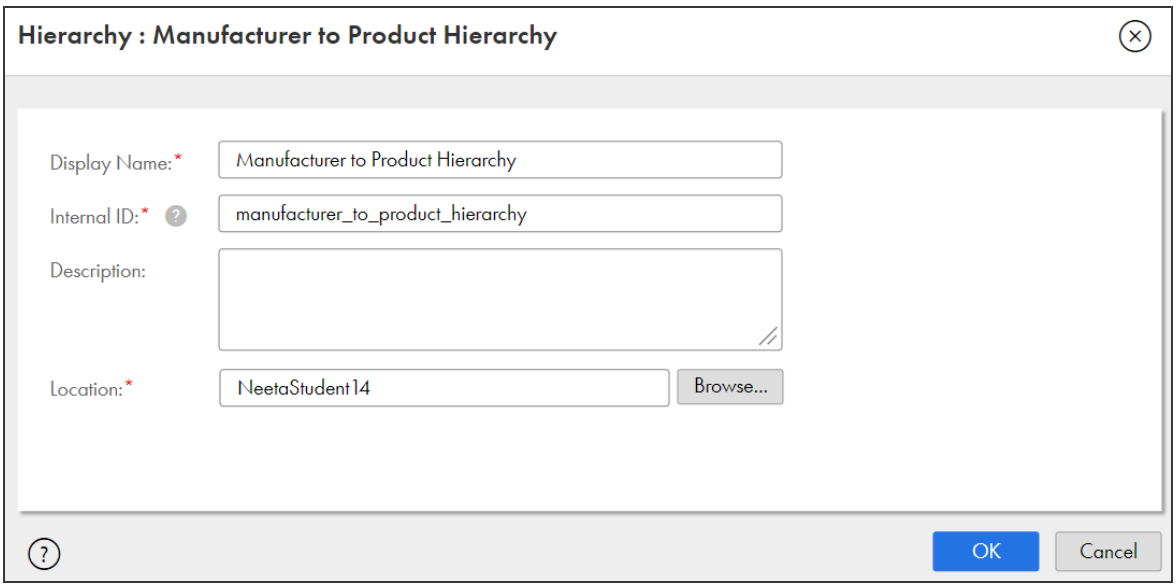
Self-Try Exercise (Mandatory)

Creating Another Hierarchy Model in Business 360

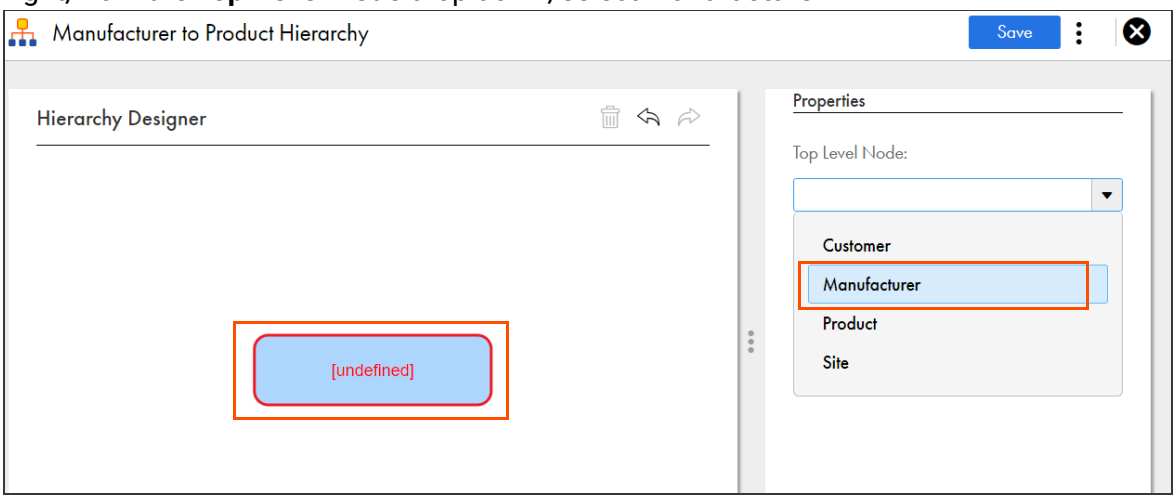
Now will create another hierarchy model in Business 360.

32. In Business 360 Console, click **New** > **Model** > **Hierarchy**. Click **Create**.

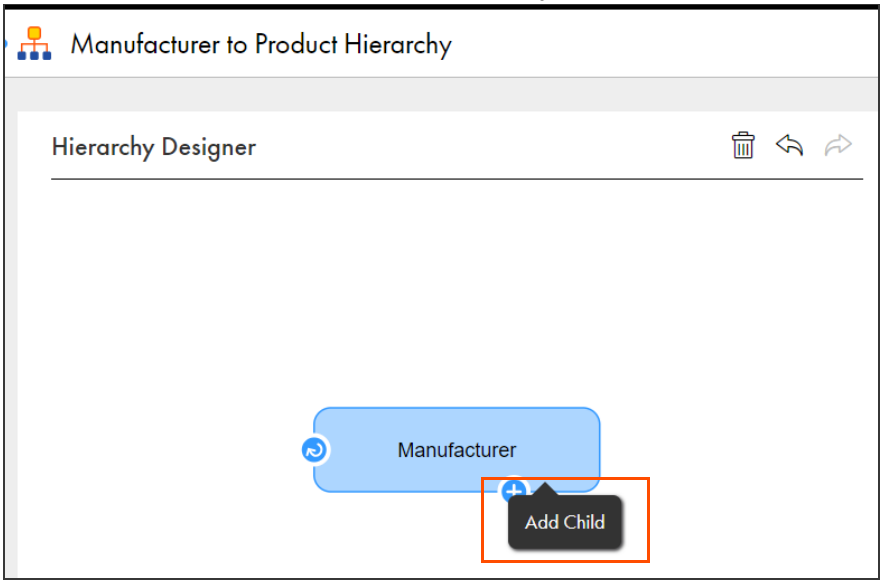
33. Enter the name as **Manufacturer to Product Hierarchy**. Select the Student folder and click **OK**.



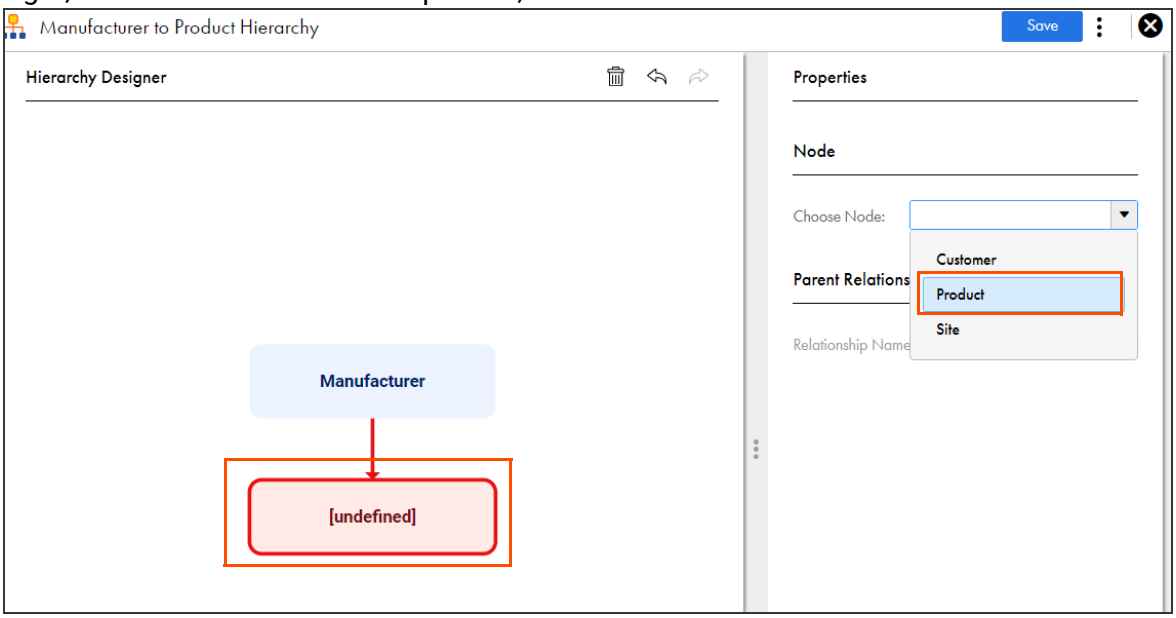
34. In the Hierarchy Designer page, click the **undefined** node, and from Properties panel on the right, from the **Top-Level Node** drop down, select **Manufacturer**.



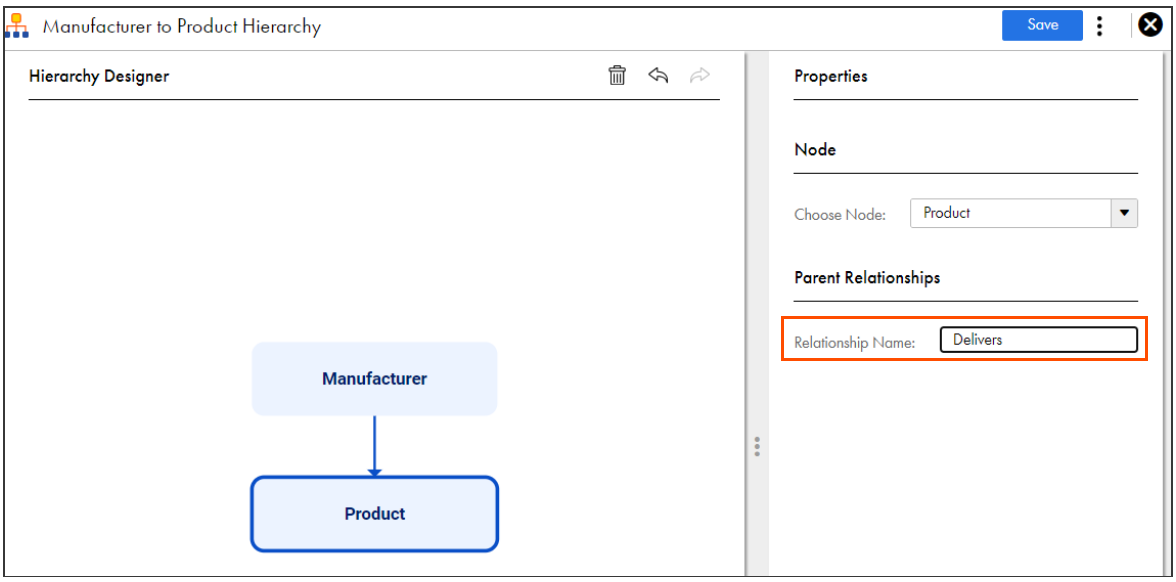
35. Hover the mouse over Manufacturer entity and click **Add Child** icon.



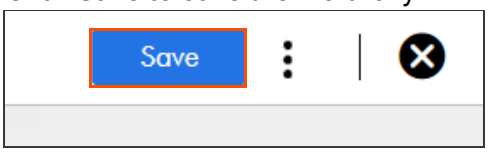
36. In the Hierarchy Designer page, click the **undefined** node, and from Properties panel on the right, from the **Choose Node** drop down, select **Product**.



37. In the Hierarchy Designer page, from Properties panel on the right, Enter the Relationship Name as **Delivers**.



38. Click **Save** to save the hierarchy.



This concludes the lab.

Key Takeaways

- A hierarchy model consists of a collection of hierarchy relationships between business entities.
- A hierarchy model consists of a top-level business entity, additional business entities, and hierarchy relationships that define parent-child relationships between business entities.
- MDM SaaS supports one-to-one, one-to-many, and many-to-many relationships between business entities.
- You can define multiple relationships for a business entity.

Module 6: Designing User Interface

Challenges

When working with the Multidomain MDM application, you may encounter some issues related to the predefined pages provided for your use. These challenges include:

- Too many undesired fields displayed on the Search page or view and edit page.
- Unable to view hierarchies on the Search page.
- Search page to only display your first name, last name, full name, address, email ID, birthplace, and education.

Solution: Designing a Custom User Interface

A user interface includes pages, icons, buttons, and other elements that help you interact with your business application. You can design the user interface based on your business requirements. You can create custom pages and add the predefined components. The users can view the customized pages when they create or open a record in the application. You can assign pages to user roles, associate business entities with your application, and configure search properties and search layouts for your application.

For example, in your business application, you want to view suggestions while searching for a record. When you open a record, you want to view the record name at the top and the history of updates. To meet your requirements, you configure the search properties to enable suggestions. You create a custom page and add the header component that can contain the record name. You add a new tab to the custom page and add the history component that displays the change history of the record.

User Interface

A user interface includes pages, icons, buttons, and other elements that help you interact with your business application. You can design the user interface based on your business requirements. You can create custom pages and add the predefined components. The users can view the customized pages when they create or open a record in the application. You can assign pages to user roles, associate business entities to your application, and configure search properties and search layouts for your application. **For example**, in your business application, you want to view suggestions while searching for a record. When you open a record, you want to view the record name at the top and view the history of updates. To meet your requirements, you can configure the search properties to enable suggestions. You create a custom page and add the header component that can contain the record name.

Designing Custom Pages

You can create and configure custom pages for your application. Assign the custom page to your application and view the user interface as per your requirement. You can configure the layout of the page by adding tabs, panels, and sections. You can split each panel into sections of different sizes. In each panel, you can add components, and select fields and field groups. When you complete creating your page, assign the page to your application.

Sample Custom Page

The Figure 6-1 shows a sample custom page:

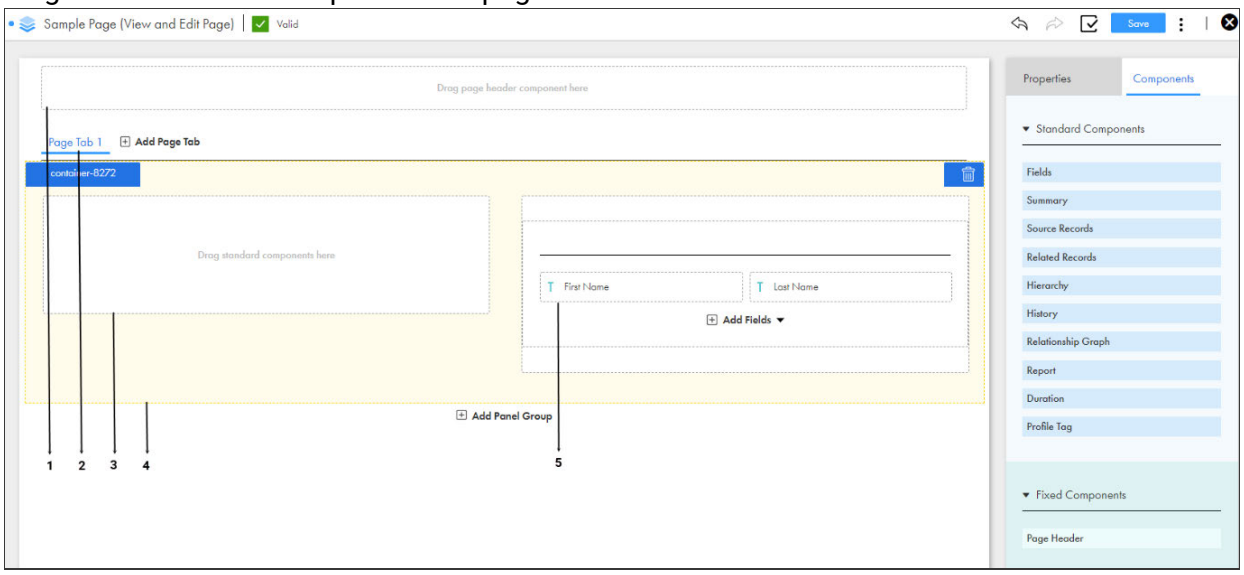
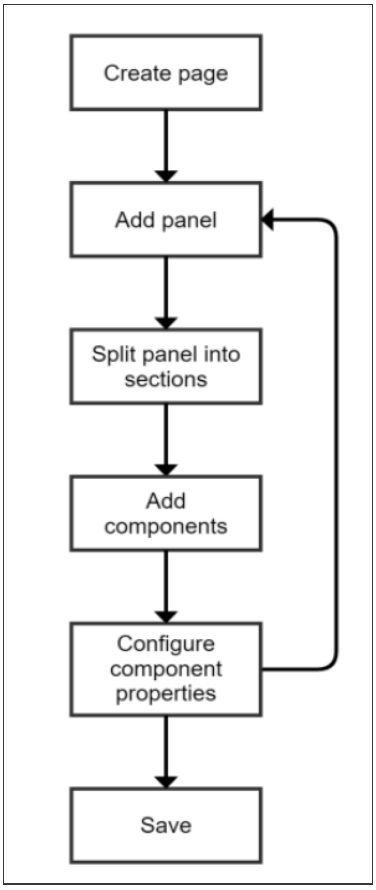


Figure 6-1: Sample Custom Page

1. Page Header
2. Page Tab
3. Section
4. Panel Group
5. Field within a section

Custom Page Design Process

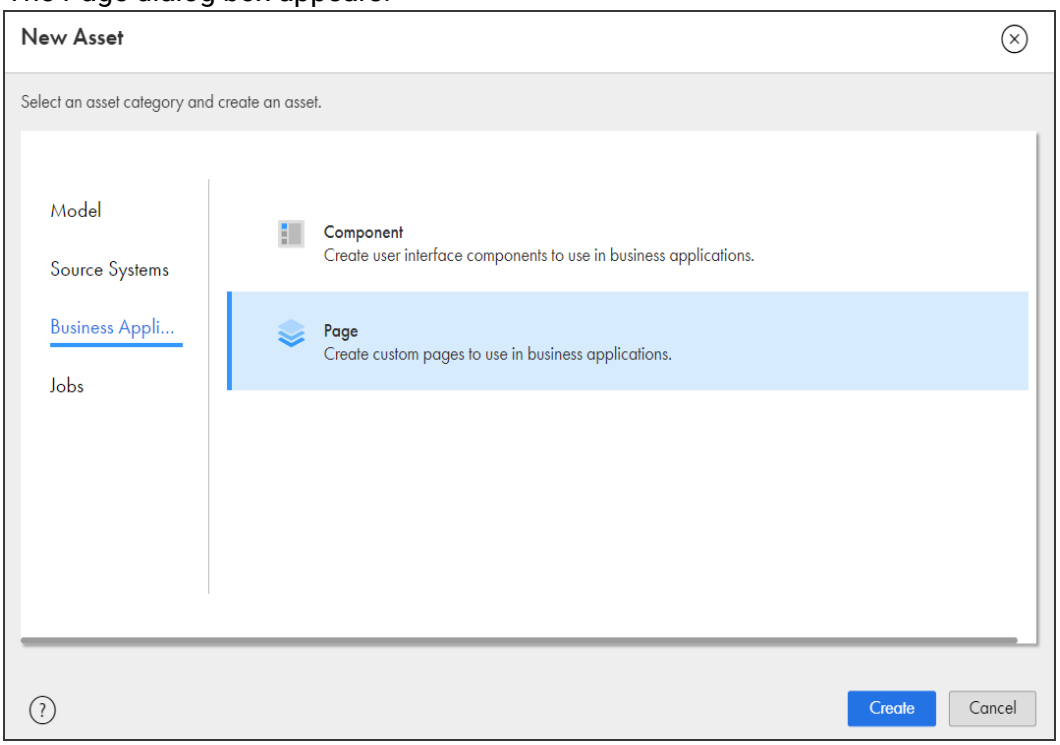
The following flow chart shows the process of designing a custom page:



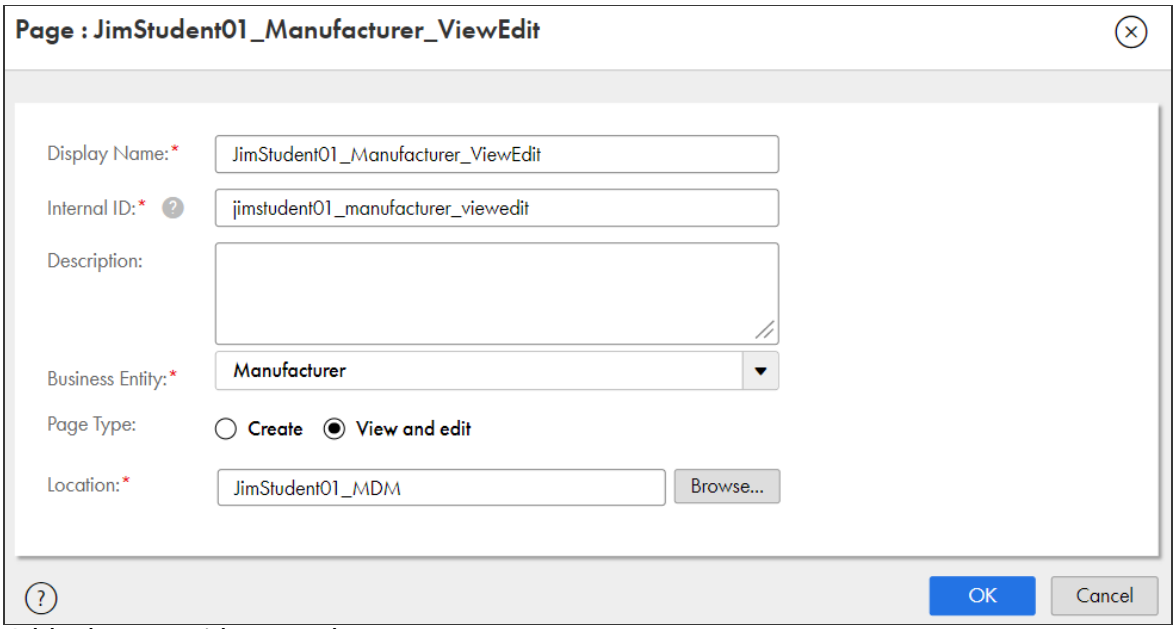
Creating a Custom Page

You can create and design custom pages for your application. You can add one or more tabs, split the page into sections, and add components. The components differ based on the business entity and page type you select.

1. Click New Business Applications Page, click **Create**.
   The Page dialog box appears.

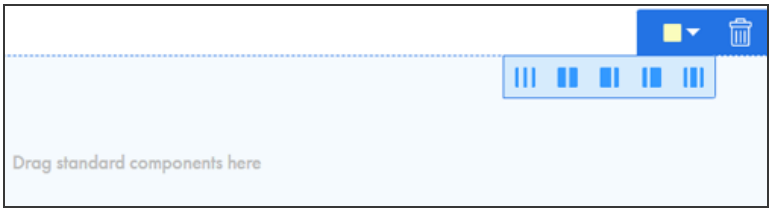


2. Specify the page properties.
3. Click **OK**.



   A blank page with a panel appears.
4. To split the panel into sections, click the panel, click the down arrow, and select the required option.
   The following options are available:

- 1:1:1. Splits into three equal sections.
- 1:1. Splits into two equal sections.
- 2:1. Splits into two sections. The first section is double the size of the second.
- 1:2. Splits into two sections. The first section is half the size of the second.
- 1:2:1. Splits into three sections. The first and the third sections are half the size of the second.



5. On the Page Components tab, drag the required component to a panel or section. Based on the components you choose, configure the component properties.
6. To add additional panel, click **Add Panel Group**.
7. To add a page tab, click **Add Page** Tab.

Configuring Page Components

The components are the building blocks of the custom page. You can configure components, such as fields, page header, and history when you create or edit a page.

Page Component Types

The standard components include the following components:

- **Fields**: Contain business entity fields and field groups.
- **Summary**: Displays fields that summarize the record data.
- **Source Records**: Contain data that is used to create the master records.
- **Related Records**: Contain data associated with a record.
- **Hierarchy**: Contains hierarchical data of a record.
- **History**: Displays historical data of a record.
- **Relationship Graph**: Displays network of relationships of a record.
- **Report**: Displays a widget through which you can visualize a chart.
- **Duration**: Displays the duration between the selected fields for a record.
- **Profile Tag**: Displays data that is used to tag a record profile.

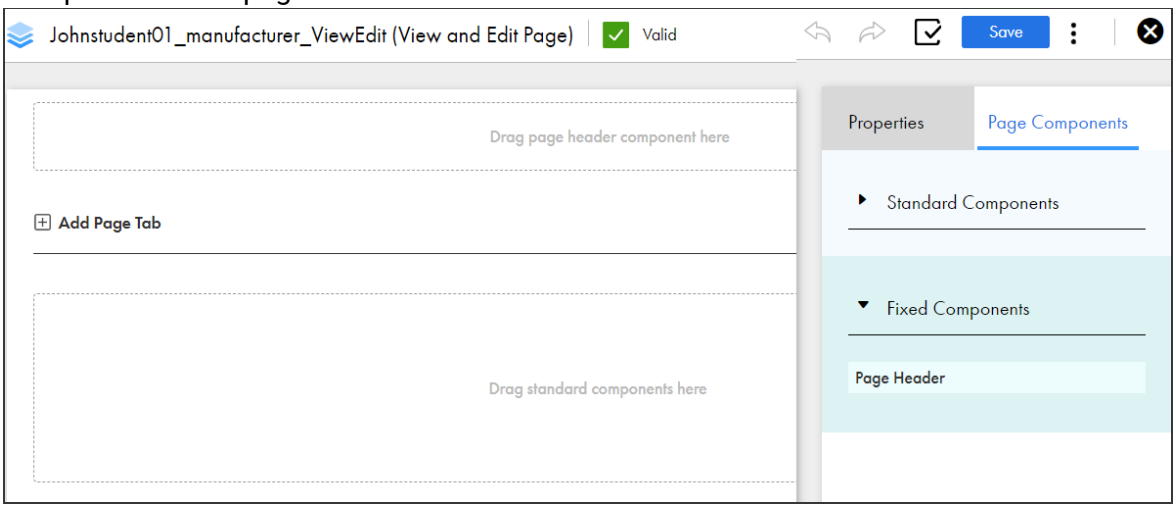
The fixed components include the following component:

- **Page Header**: Contains data that is displayed on the page header.

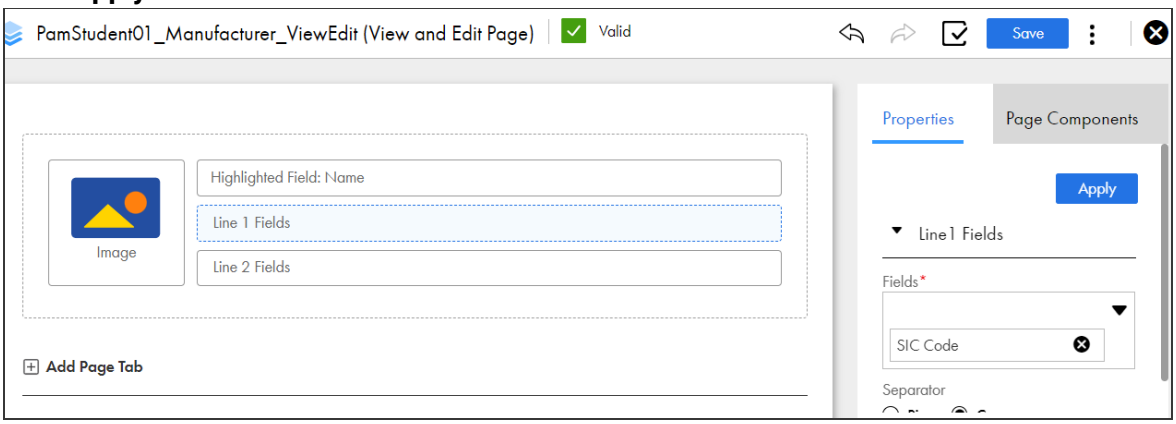
Configuring Page Header

You can add a header to a custom page to display the field data that you want to highlight for a record. For example, you are working with a large insurance company that employs numerous insurance agents. The agents often open customer records to send emails regarding insurance premium. You configure the page header to display the photo of the customer, customer name, insurance policy number, and email address. When the agents open a record, they find the required details on the header.

1. On the Components tab, in the Fixed Components section, drag the **Page Header** component to the page header section.



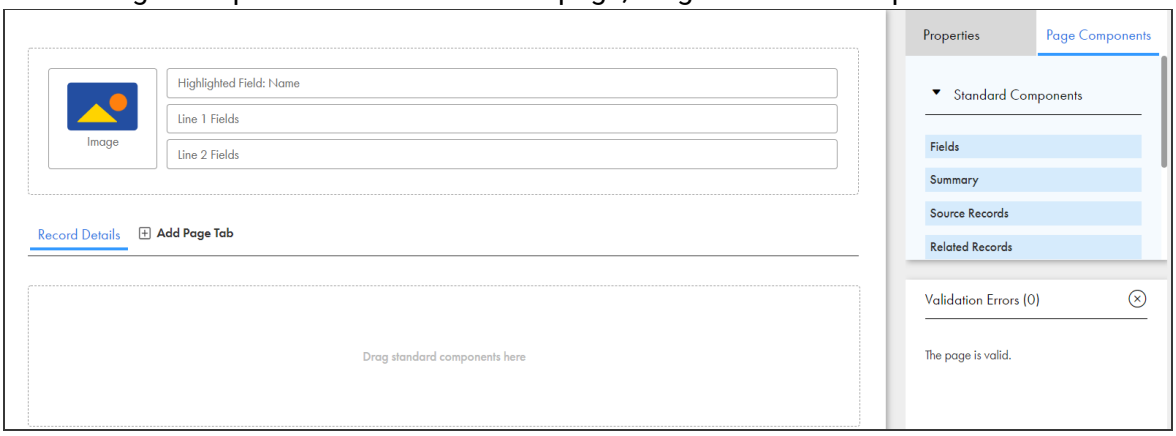
2. Select each item in the page header, configure the properties on the Properties tab, and click **Apply** and **Save**.



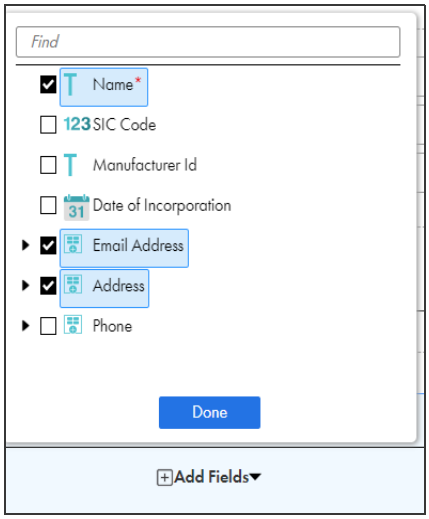
Configuring a Fields component

When you add the Fields component to a custom page, you can add fields and field groups. You can configure properties for the fields or field groups that you add.

1. On the Page Components tab of a custom page, drag the Fields component to a section.



2. On the Properties tab, configure the properties.
3. Click **Apply**.
4. On the canvas, click **Add Fields,** select the required fields or field groups, and click **Done**. The selected fields and field groups appear in the panel.
5. Select a field or field group, and configure the properties in the Properties tab.



6. Click **Apply**, and then click **Save**.

Configuring the Related Records Component

You can configure the Related Records component to display related records of a record. You can also use the component to relate a record with other records through relationships.

Configuring a Relationship Graph Tab

You can configure a Relationship Graphs tab to visualize and explore the relationships of a record. You create a page tab, add the Relationship Graph component, and then configure the component properties. Before you configure a relationship graph tab, create a custom page

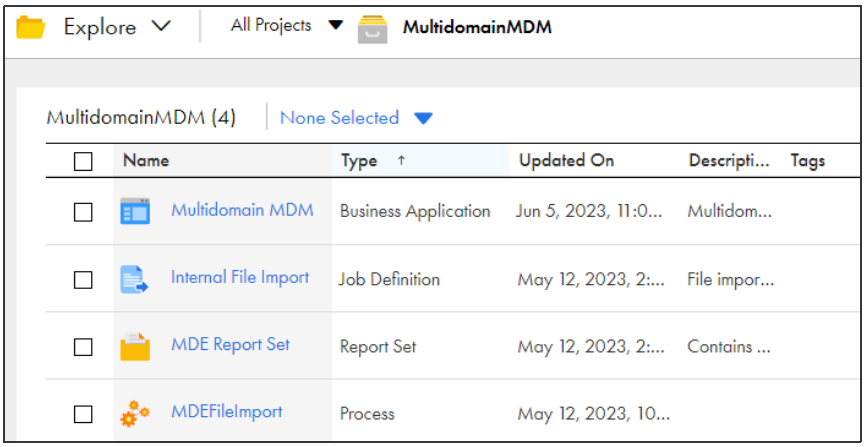
Configuring Applications

Once you done creating the business entities, pages and search layouts for the data model, you need to add these to the Business Application in order to see the configurations in the front end applications as such as Multidomain MDM, C360, P360 and S360. You can configure your business application based on your business requirements. For example, you can add business entities and custom pages, and configure search properties.

Add Business Entities to an Application

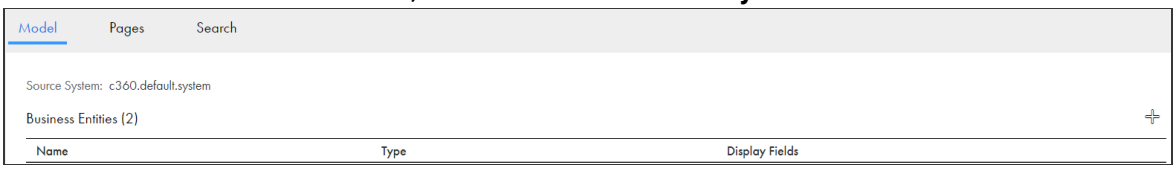
You can add business entities and configure the display fields for each business entity. The values of the selected display fields appear in the left navigation bar when you open a record. You can view the list of components, such as search box, notifications, and tabs that are enabled for the application. The enabled components appear on the application header and left navigation bar.

1. On the Explore page, find and open the business application.



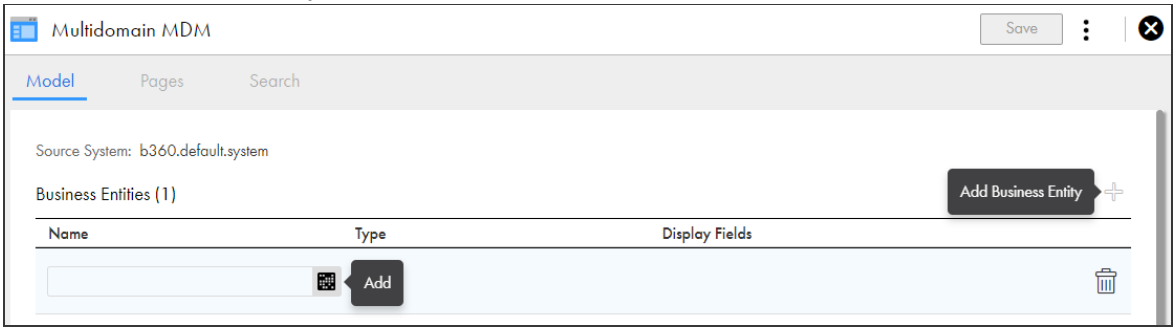
   The Model tab appears.

2. In the Business Entities section, click **Add Business Entity.**



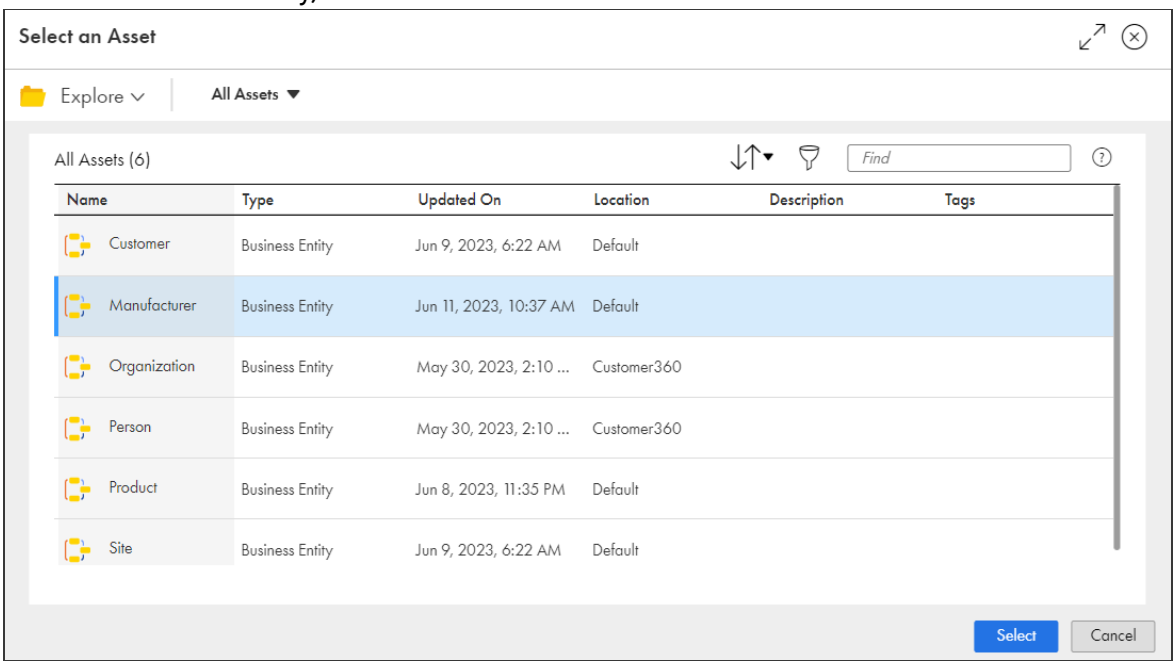
   A blank row appears.

3. To add a business entity to the business application, click the asset picker icon.



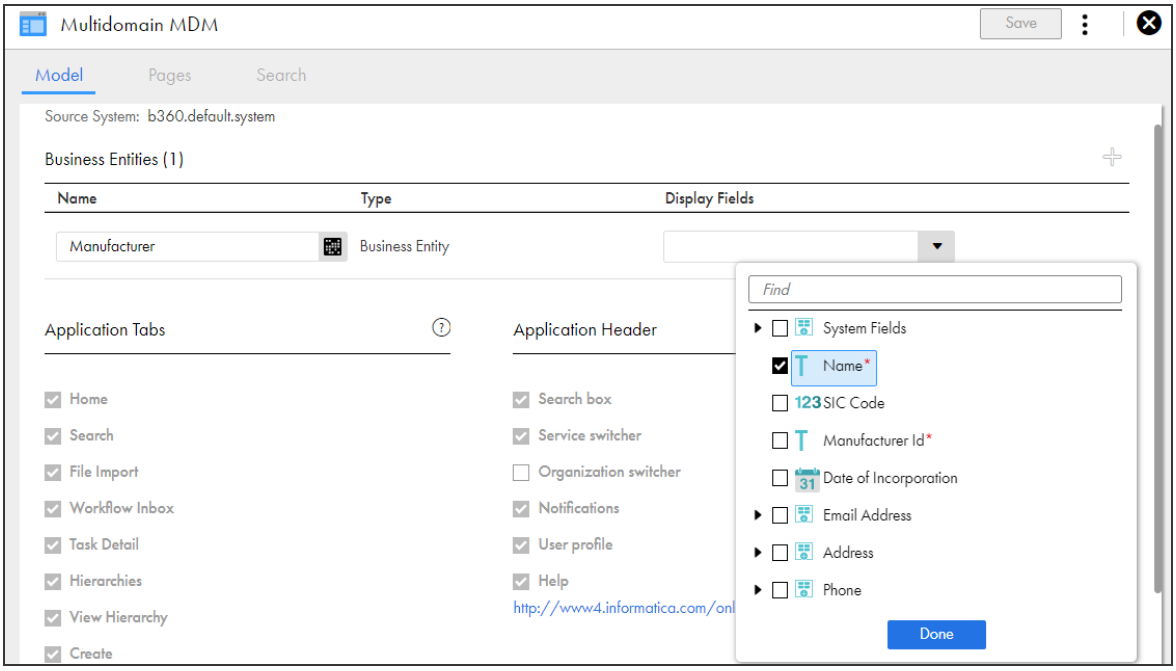
The Select an Asset dialog box appears.

4. Select a business entity, and click **Select**.



The selected business entity appears in the row.

5. Select the display fields that a record can display in the left navigation bar when you open the record, and click **Done**.

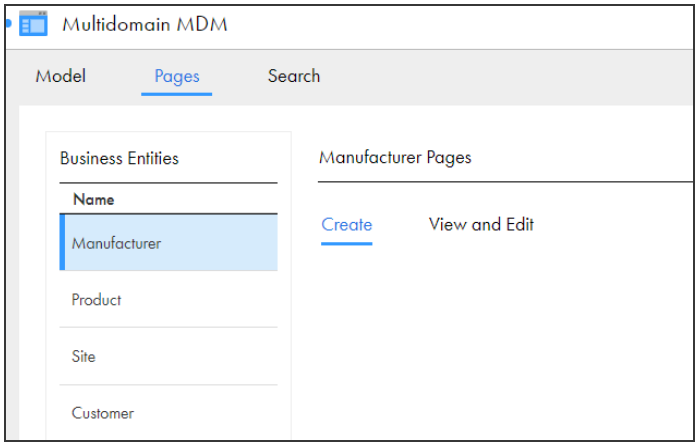


6. Click **Save**.

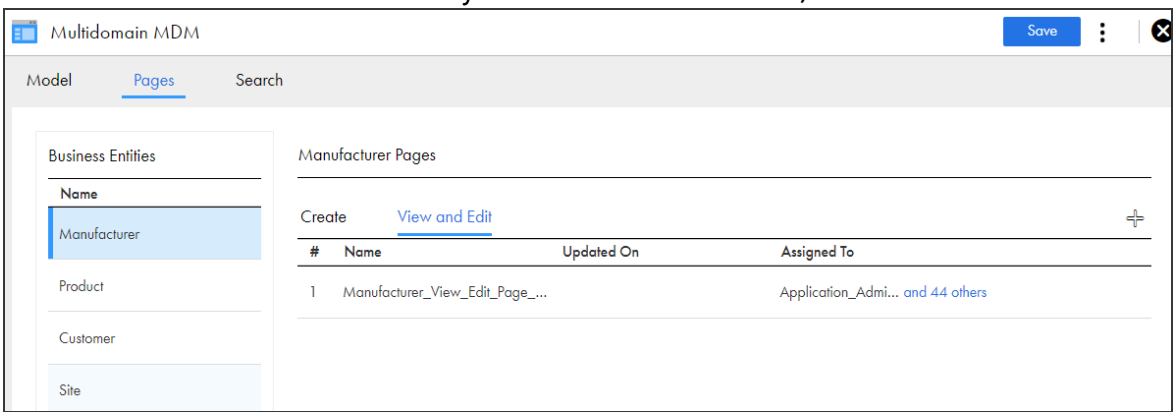
Assign Pages to the Multidomain MDM Application

You can assign pages to the user roles defined for your application. You can assign different pages for different roles. The users can view the business application interface based on the pages assigned to the user roles.

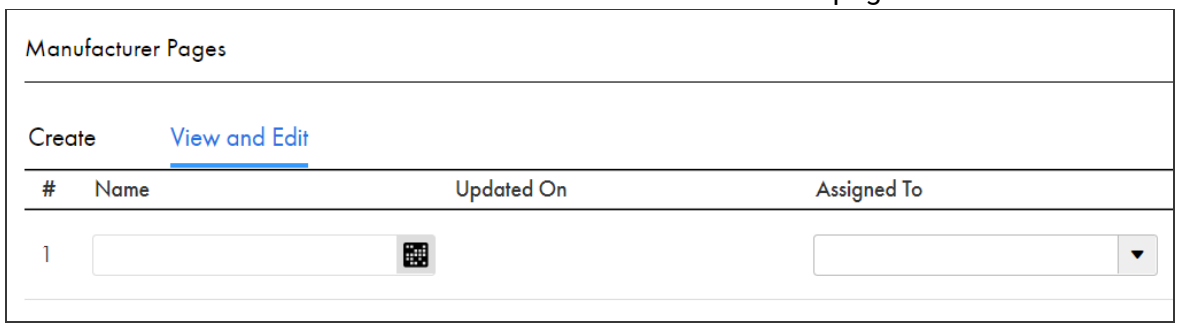
1. Click **Pages** tab. Here, you can add custom view and edit page.

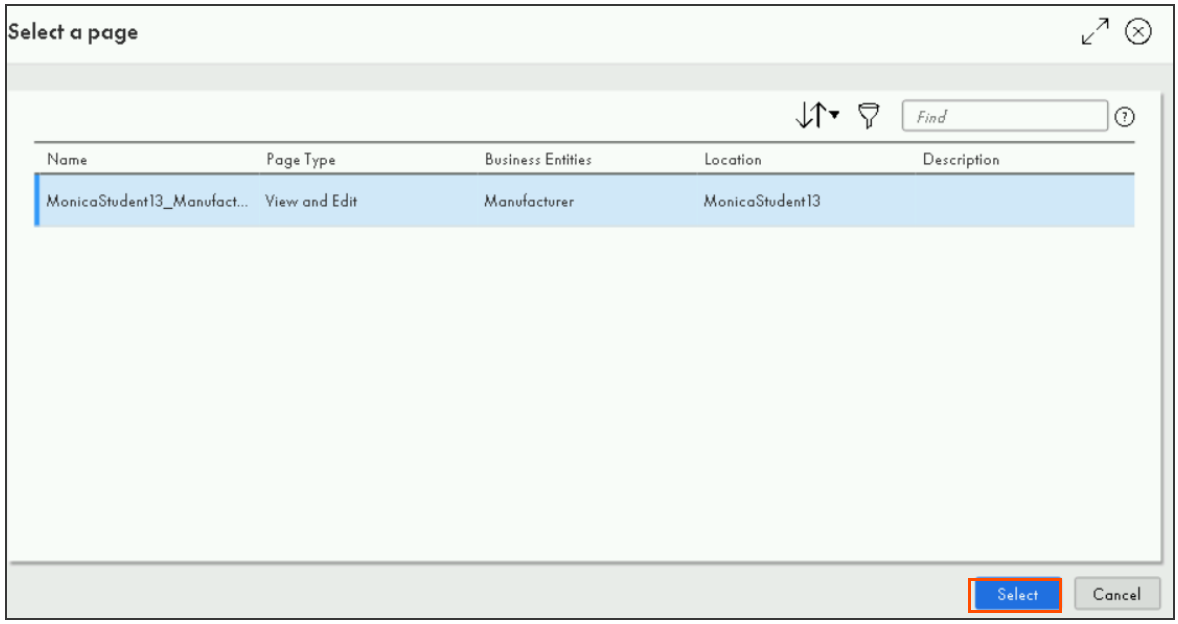


2. Select **Manufacturer** business entity. Under **View and Edit** tab, Click **+** icon.



3. Click the Asset Picker icon and select the custom view and edit page. Click **Select**.





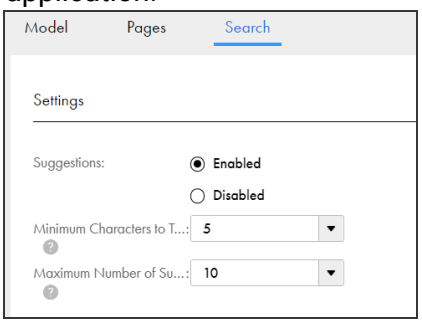
4. Click **Save**.
   This way, you can add the custom page to your business entity.

Add Search Layout to the Application

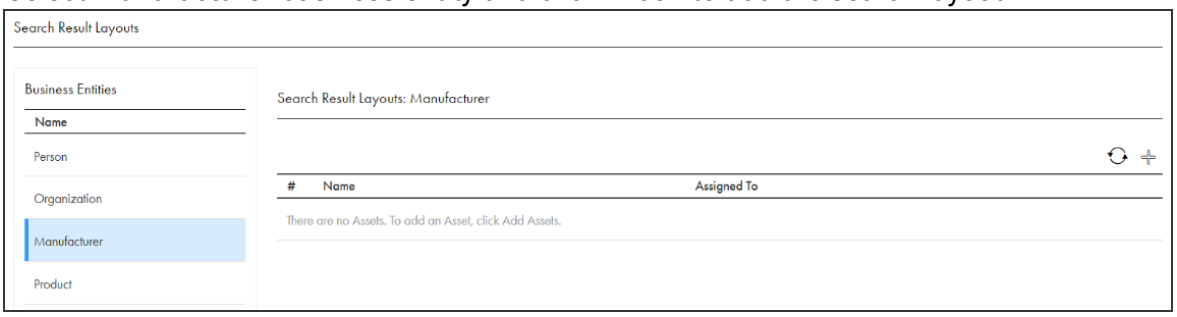
You can configure the search properties, such as search suggestions and search history, for an application. For example, you can configure the number of previous searches to display when you click the search box.

Based on your business needs, you can assign the search result layout components to the user roles defined for your application. You can view the search results of the business entities in your application based on the search layout components assigned to the user roles.

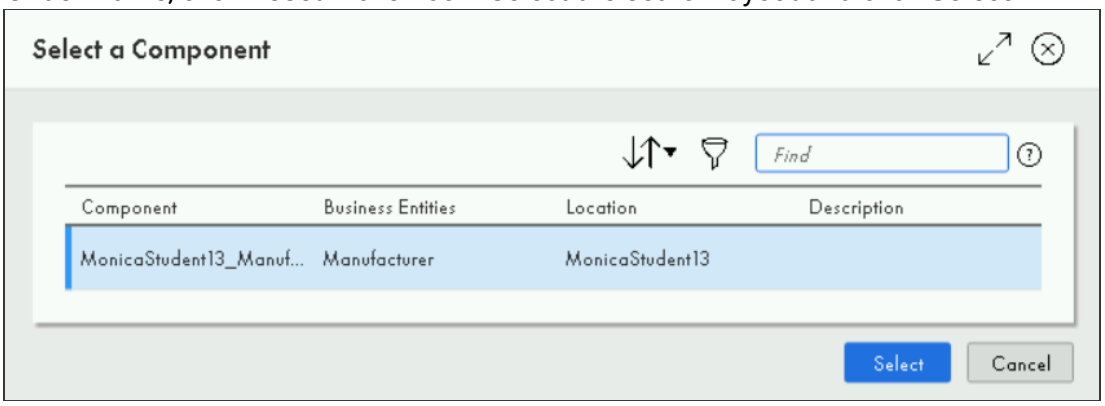
1. Click the **Search** tab. Here, you will add the custom search layout to the Multidomain MDM application.



2. Select **Manufacturer** business entity and click **+** icon to add the search layout.



3. Under Name, click Asset Picker icon. Select the search layout and click **Select**.



4. Click **Save**.

Customize the Brand

You can customize the logo and favicon that appear on your business application. Use Administrator to configure custom branding for your parent organization and apply them to your sub-organizations. You can also configure custom branding for each sub-organization. To change the branding, you must have the Custom Logo & Color Themes license. For more information about licenses, see Licenses in the Administrator help. You can use PNG, JPG, and GIF formats for the custom logo, and PNG format for the favicon.

The branding settings do not apply to notifications, tooltips, custom components, and the following user interface screens:

- File Import
- My Jobs
- Reports
- Source Records

Lab 6-1: Creating Page Layouts

Overview:

You can create and configure custom pages for your application. Assign the custom page to your application and view the user interface as per your requirement. You can configure the layout of the page by adding tabs, panels, and sections. You can split each panel into sections of different sizes. In each panel, you can add components, and select fields and field groups. When you complete creating your page, assign the page to your application. You can edit a custom page based on your business needs. Your user role must have the required privileges to edit the custom page. Multidomain MDM provides create and view and edit pages. Search layouts needs to be created.

In this lab, you will create page layouts for view and edit, and search. Though Multidomain MDM provides view and edit pages, to understand the process of page creation you will create view and edit page for one of the business entities.

Objectives:

- Create custom pages to view and edit records

Scenario:

Mark has established Business Entities, Hierarchies, and Relationships as components of the data model. He should now create page layouts to display the assets he previously created. Once he completes this, he must integrate these pages into the Multidomain MDM application to view these configurations in the front-end application. With these steps, Mark will be nearing the completion of the data model creation phase.

Duration:

20 Minutes

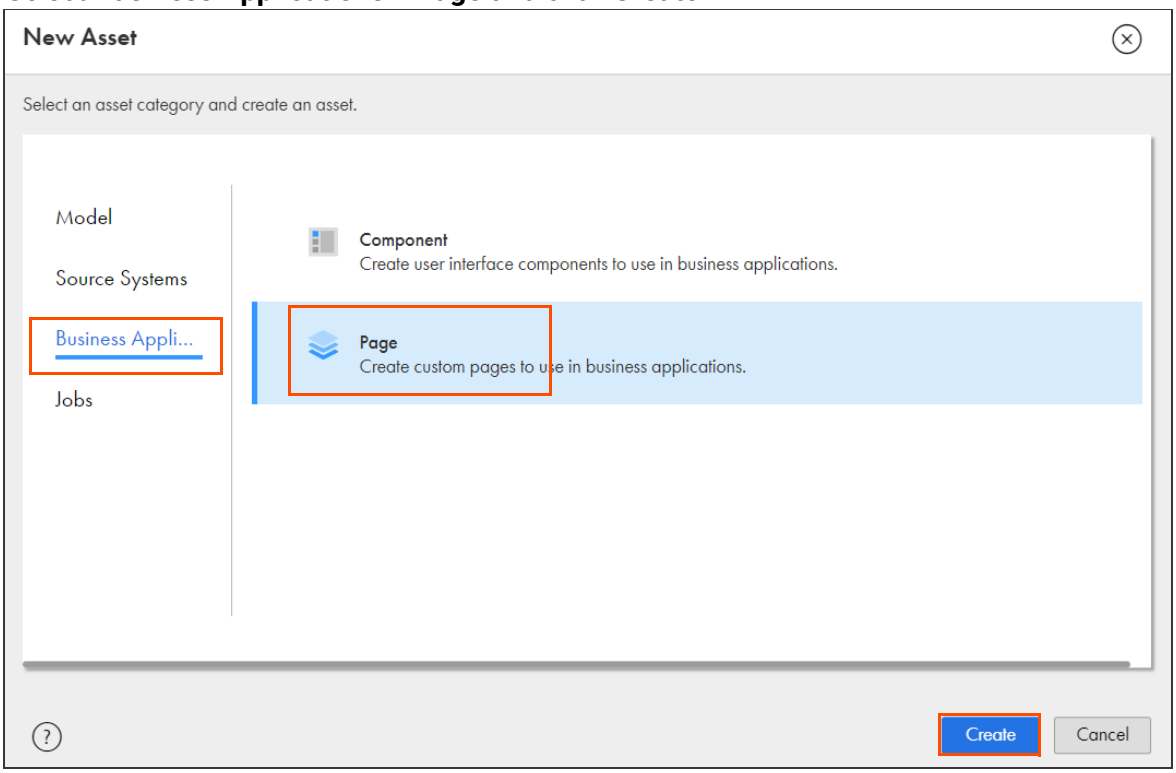
Tasks:

- Creating View and Edit page for Manufacturer
- Creating View and Edit page for Product
- Create layouts for search results

Creating View and Edit Page for Manufacturer

You will first create a View and Edit page for the Manufacturer business entity.

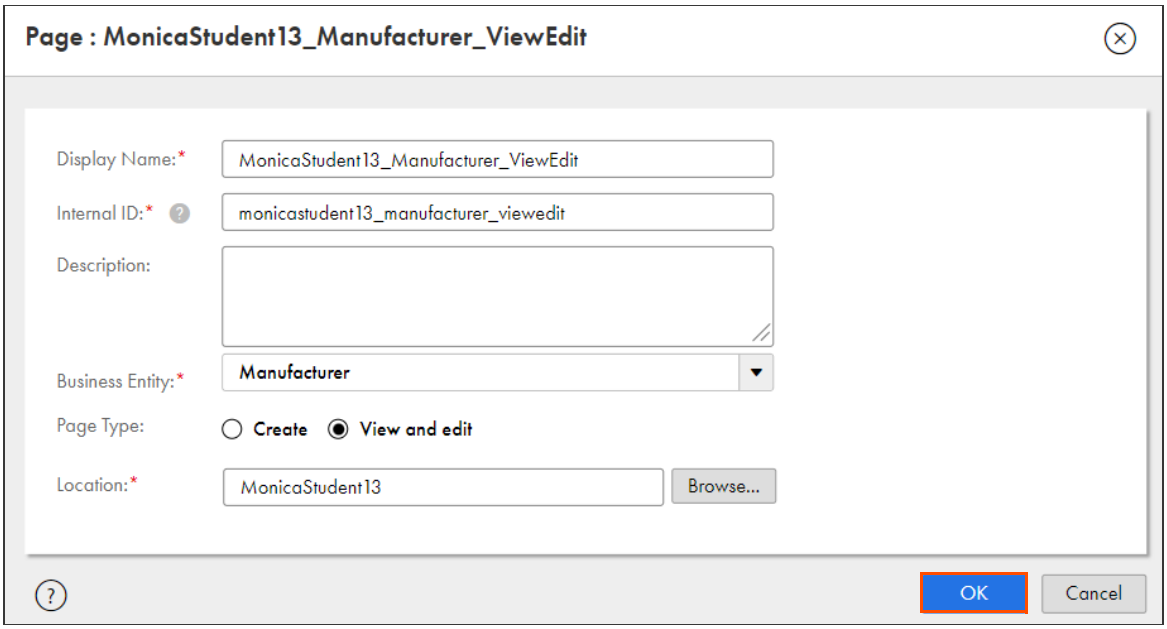
1. Log in to IICS using your admin credentials if you have logged out.
2. From the Business 360 console menu, click **New**.
3. Select **Business Applications** > **Page** and click **Create**.



4. In the Page properties, enter the following values and click **OK**:

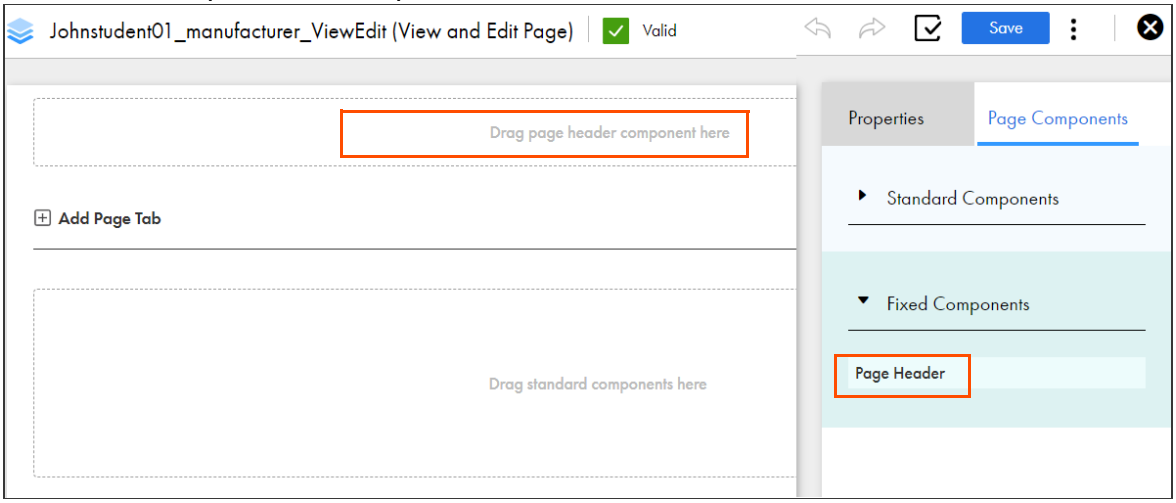
Field	Value
Display Name	<studentNamestudentNumber>_Manufacturer_ViewEdit
Business Entity	Manufacturer
Page Type	View and Edit
Location	Your student project

The student project name in the screenshot may vary.

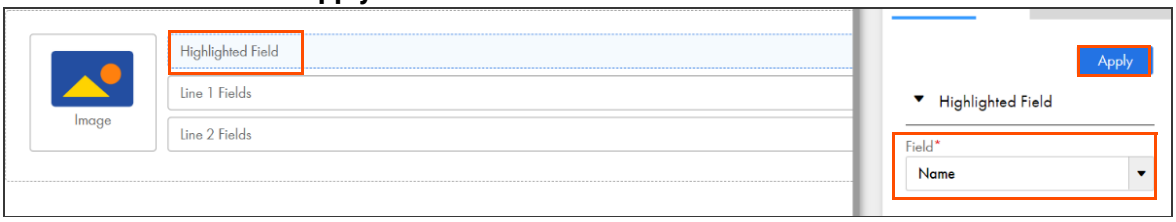


Refresh the browser if you do not find the business entity from the drop-down.

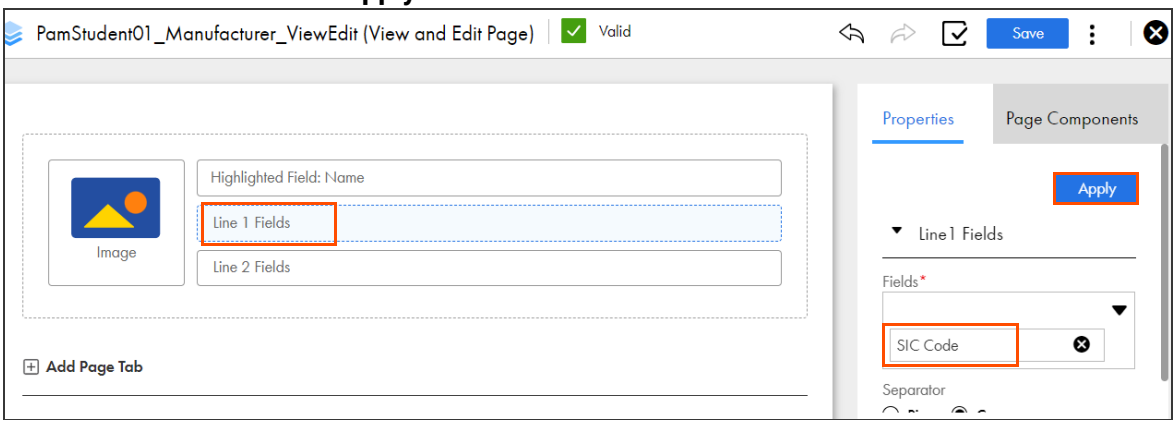
5. From the **Page Components** tab, under the **Fixed Components** section, drag the **Page Header** and drop it onto the topmost dotted box.



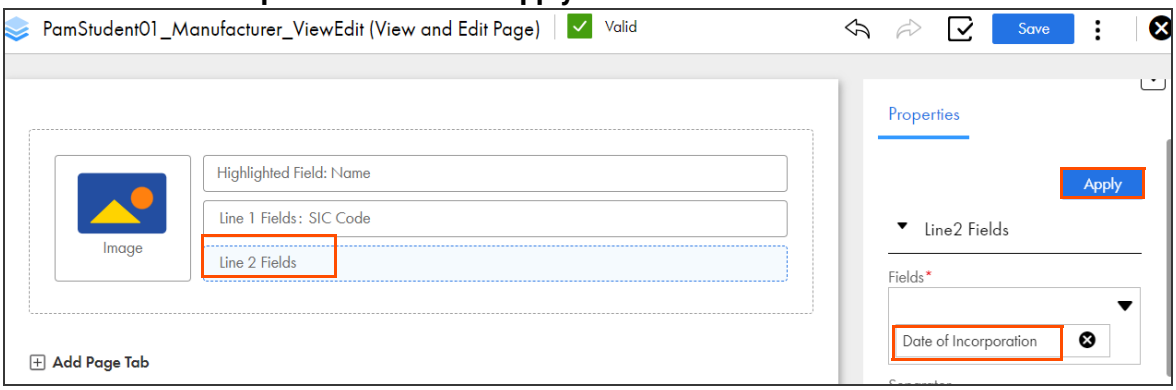
6. Select **Highlighted Field**, and from the **Properties** tab in the right, from the **Field** drop-down, select **Name** and click **Apply**.



7. Select the **Line 1 Fields**, from the **Properties** tab in the right, from the **Field** drop-down, select **SIC Code** and click **Apply**.

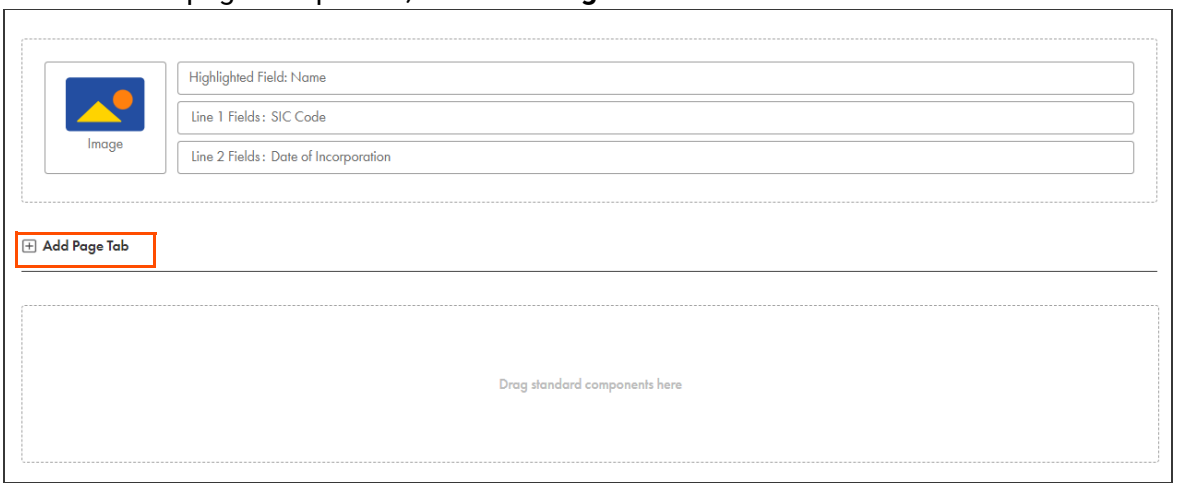


8. Select the **Line 2 Fields**, from the **Properties** tab in the right, from the **Field** drop-down, select **Date of Incorporation** and click **Apply**.

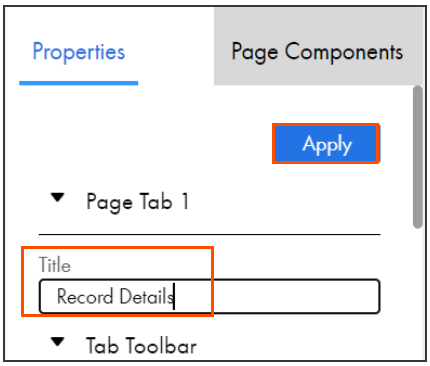


You have finished setting up the headers on the page. Next, you will configure different tabs on the pages. These tabs will include the Business Entity fields that you wish to display.

9. To add a new page component, click **Add Page Tab**.

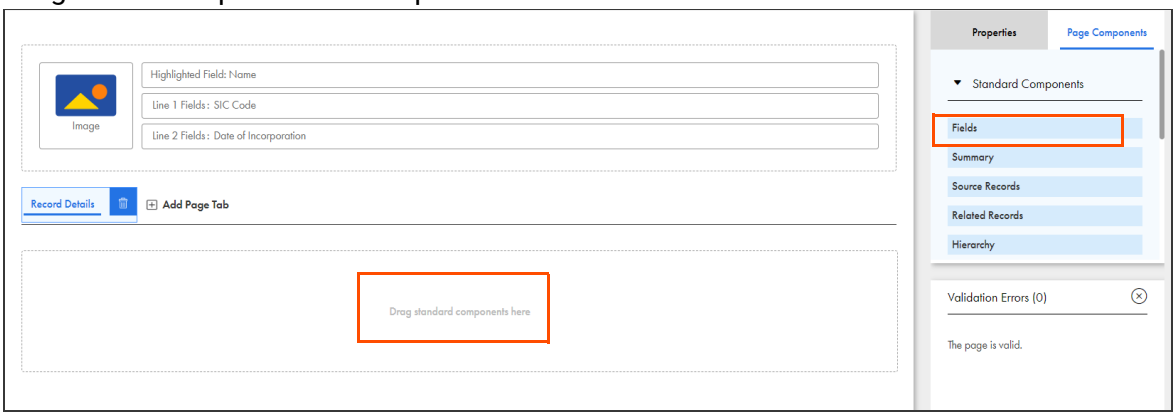


10. From the properties tab on the right, enter **Title** as **Record Details**. Click **Apply**. Your page tab is renamed as **Record Details**.

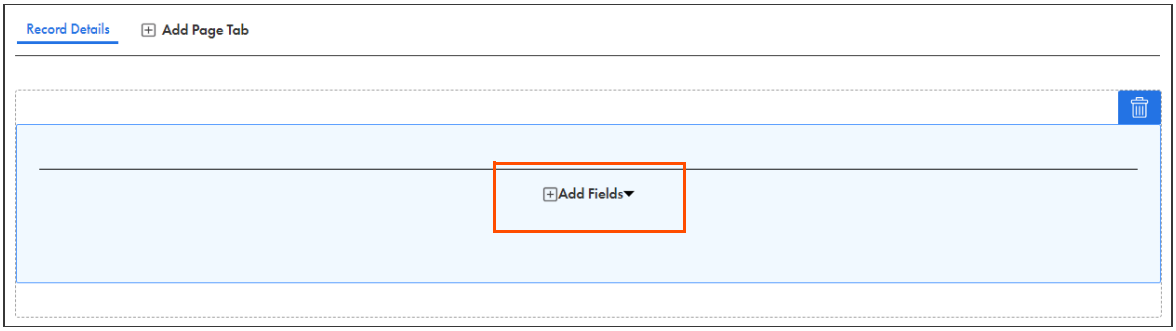


11. Select **Page Components** Tab.

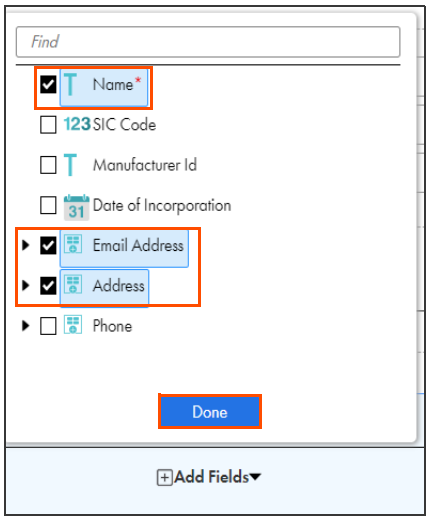
12. Drag **Fields** component and drop it onto the **Record Details** tab.



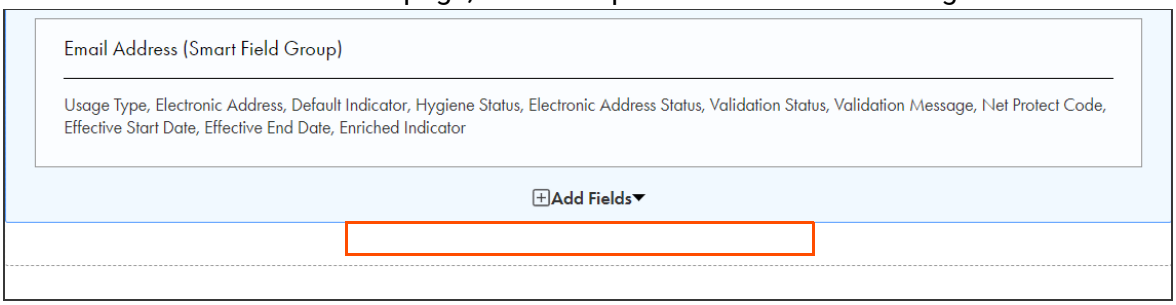
13. Click **Add Fields**.



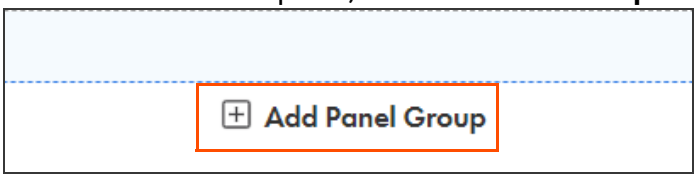
14. From the Add Fields pop-up, select **Name**, **Email Address**, **and Address** fields and click **Done**.



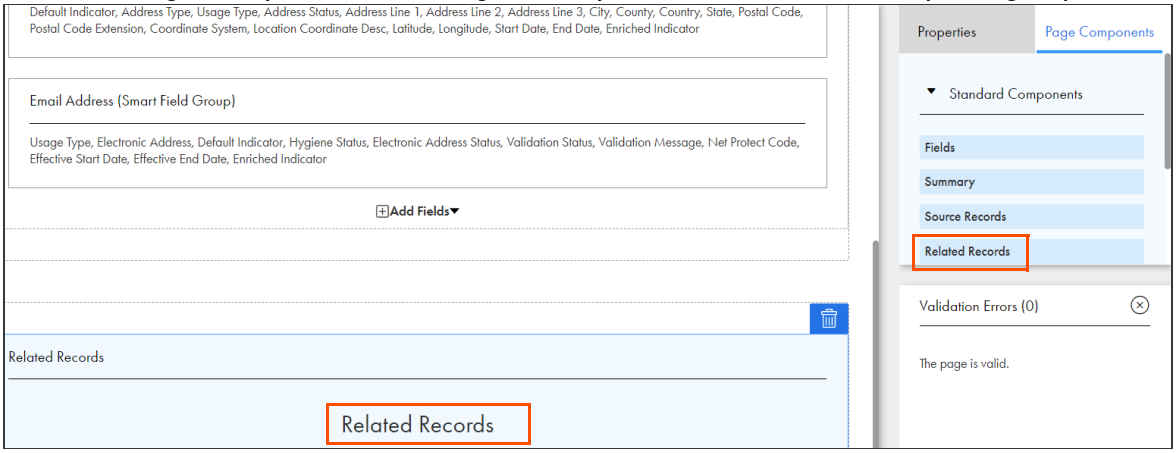
15. To add Related Records to the page, click the space as shown in the image.



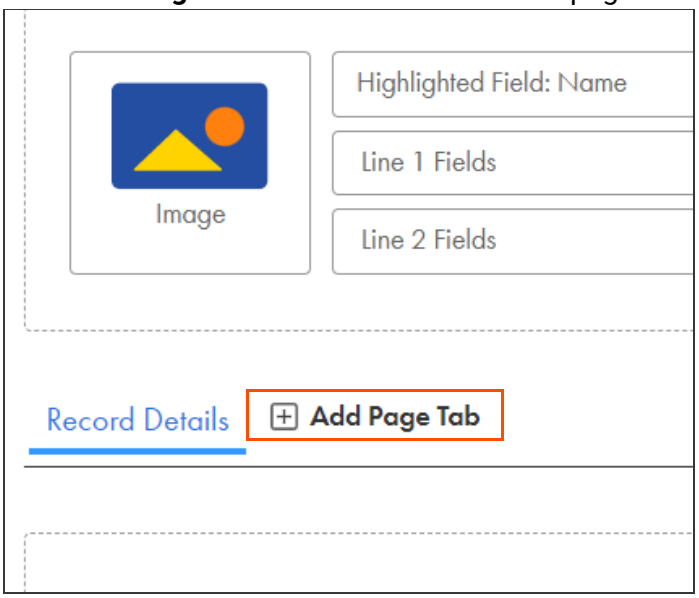
16. To add an additional panel, click **Add Panel Group**.



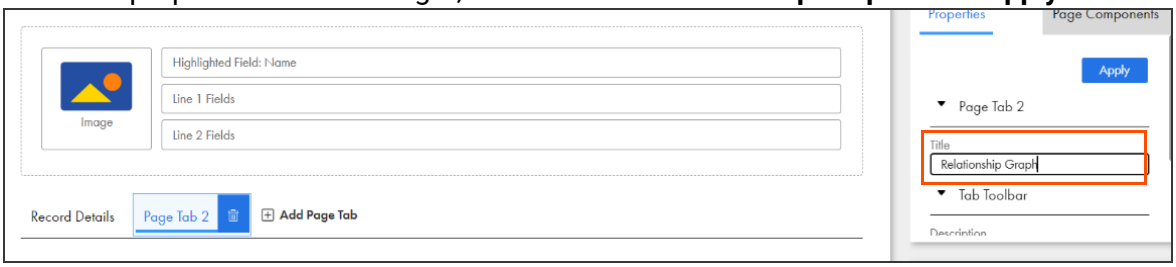
17. From the Page Components tab, drag and drop **Related Records** to the panel group.



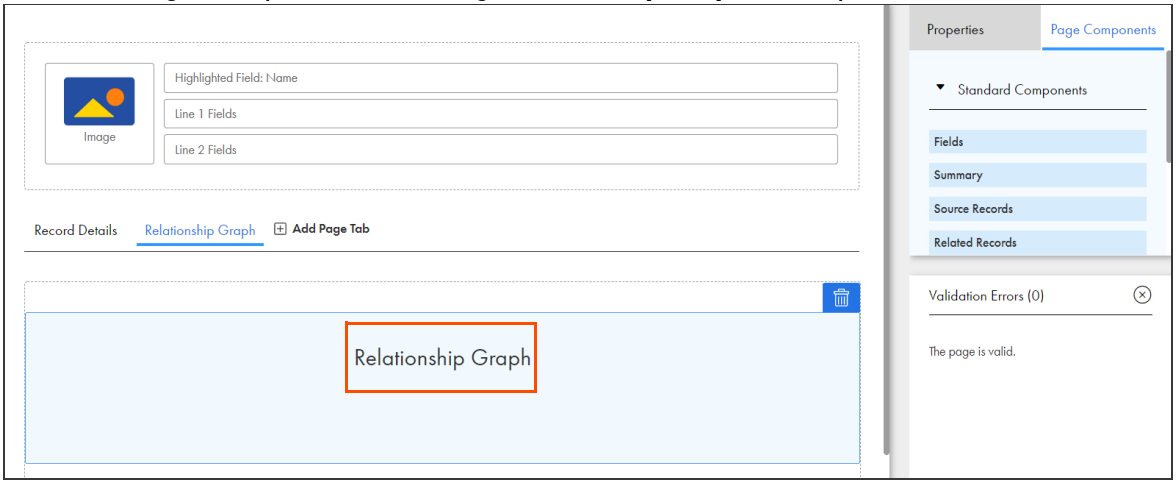
18. Click **Add Page Tab** to add a new tab to the page.



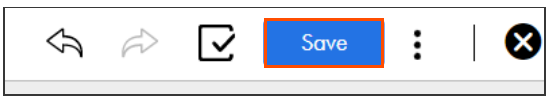
19. From the properties tab in the right, enter **Title** as **Relationship Graph**. Click **Apply**.



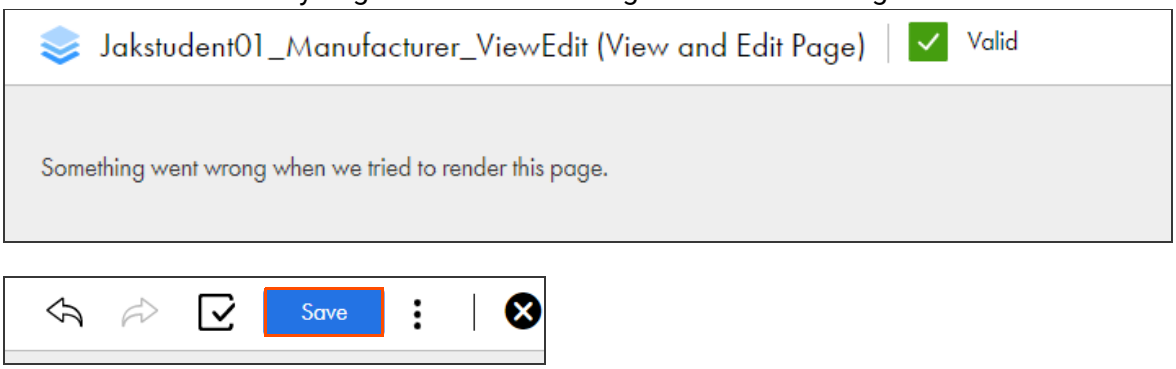
20. From the Page Components tab, drag **Relationship Graph** to the panel.



21. Click **Save**.



22. Refresh the browser if you get the below message and click **Save** again.



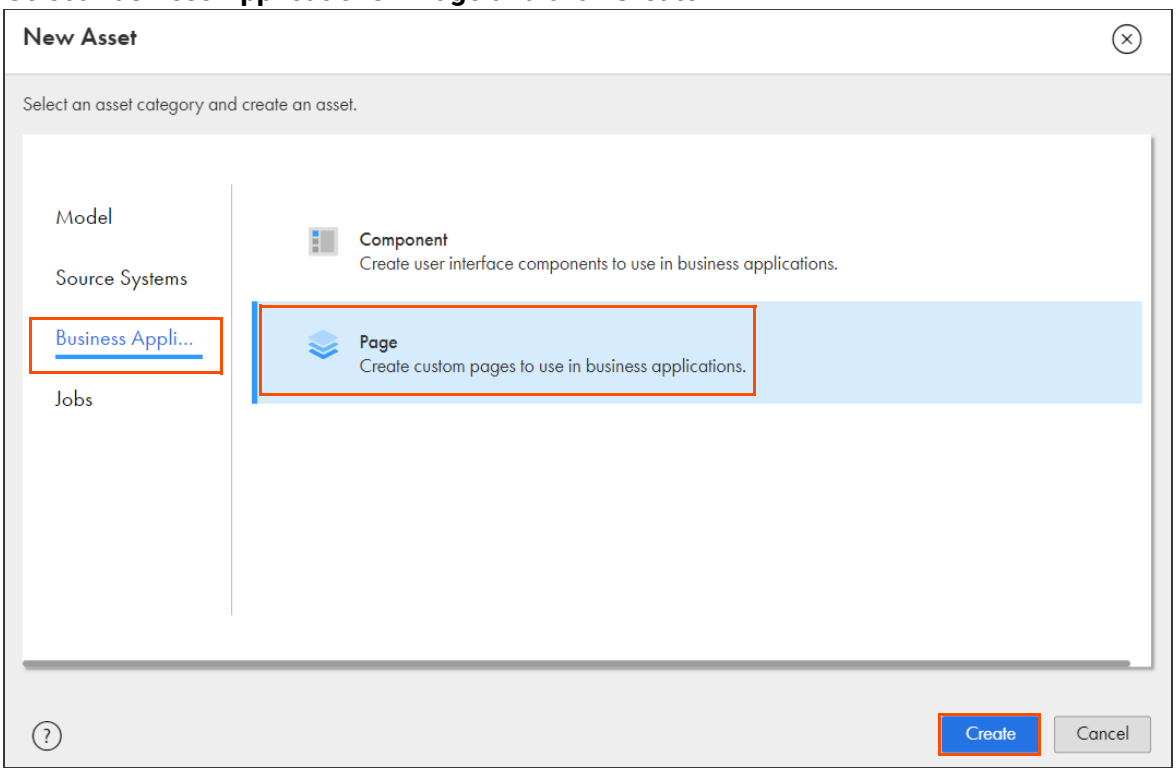
Self-Try Exercise (Mandatory)

Creating View and Edit Page for Product

You will create view and edit page for the Product business entities.

23. From the Business 360 console menu, click **New**.

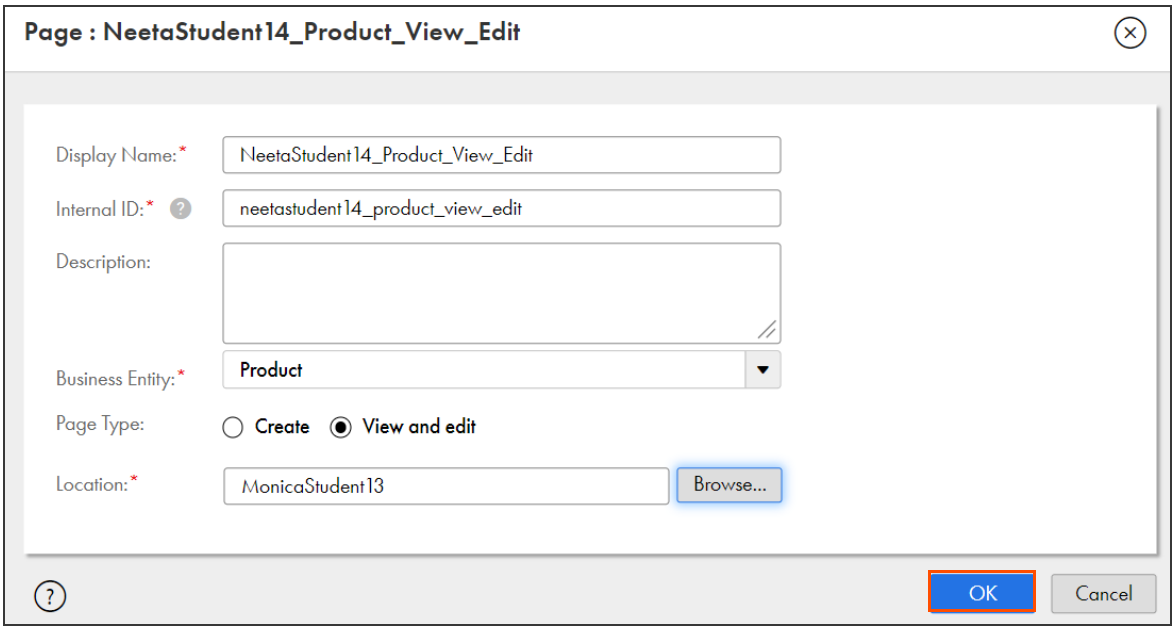
24. Select **Business Applications** > **Page** and click **Create**.



25. In the Page properties, enter the following values and click **OK**:

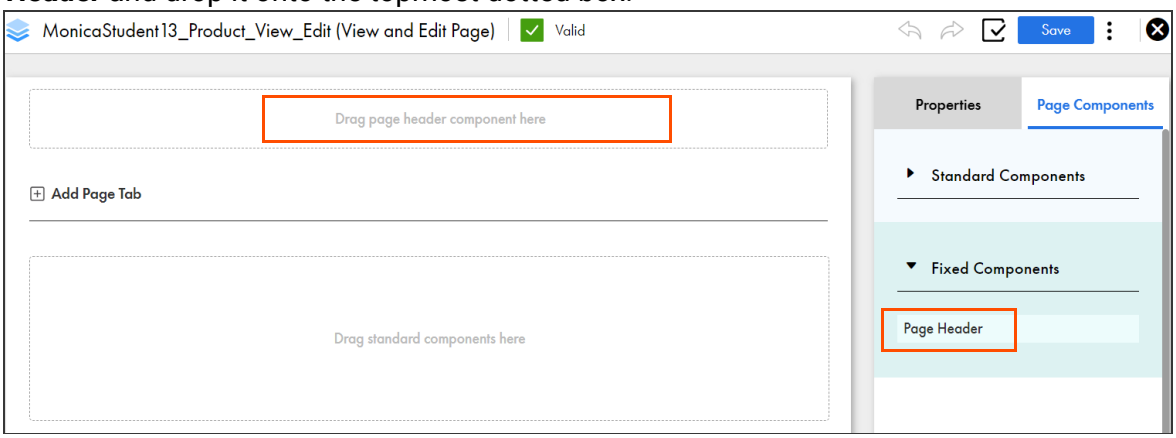
Field	Value
Display Name	<studentNamestudentNumber>_Product_ViewEdit
Business Entity	Product
Page Type	View and Edit
Location	Your student project

The student project name in the screenshot may vary.

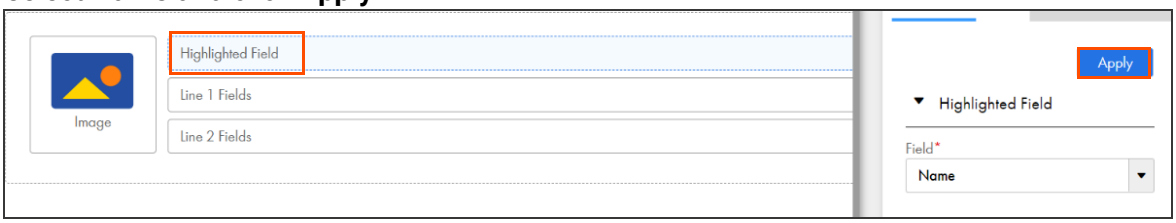


Refresh the browser if you do not find the business entity from the drop-down.

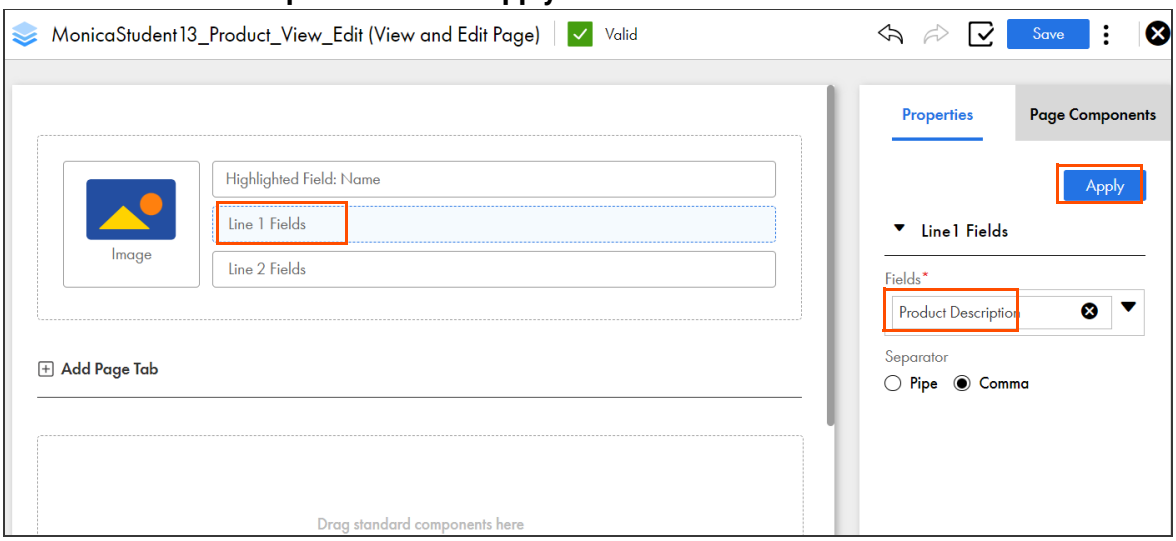
26. From the **Page Components** tab, under the **Fixed Components** section, drag the **Page Header** and drop it onto the topmost dotted box.



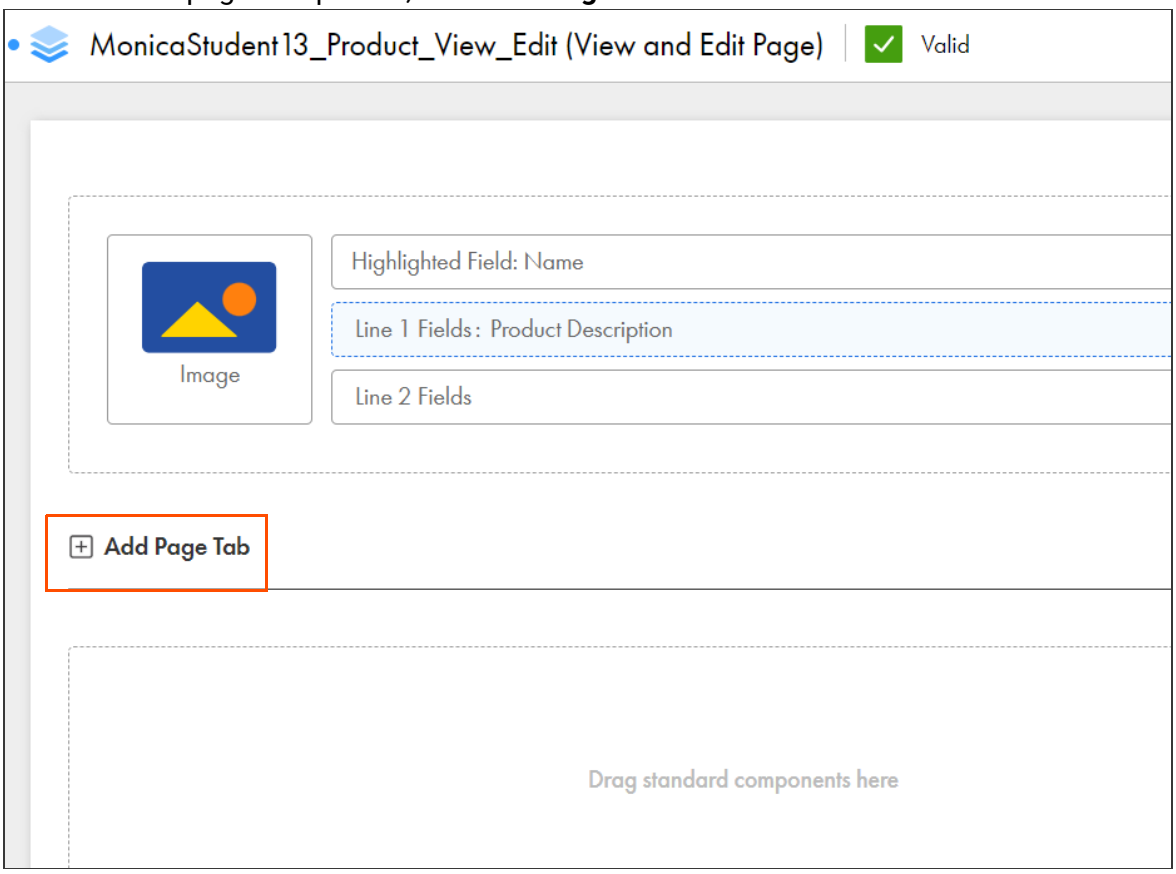
27. Select **Highlighted Field**, and from the **Properties** tab in the right, from the **Field** drop-down, select **Name** and click **Apply**.



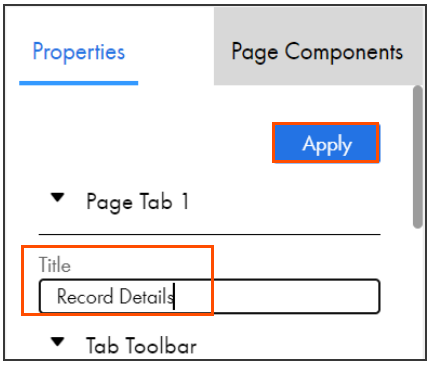
28. Select the **Line 1 Fields**, from the **Properties** tab in the right, from the **Field** drop-down, select **Product Description** and click **Apply**.



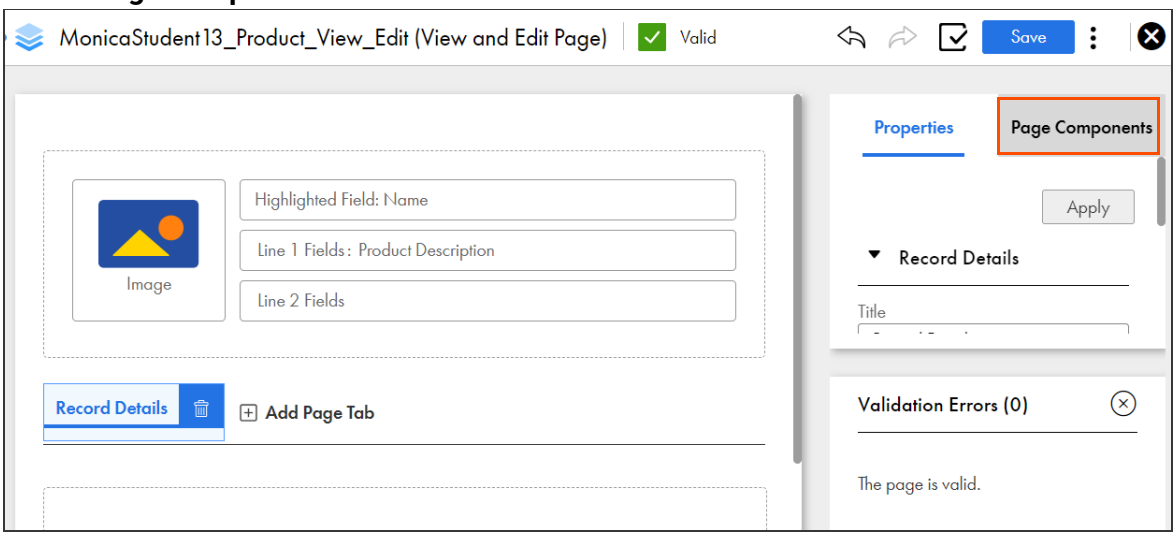
29. To add a new page component, click **Add Page Tab**.



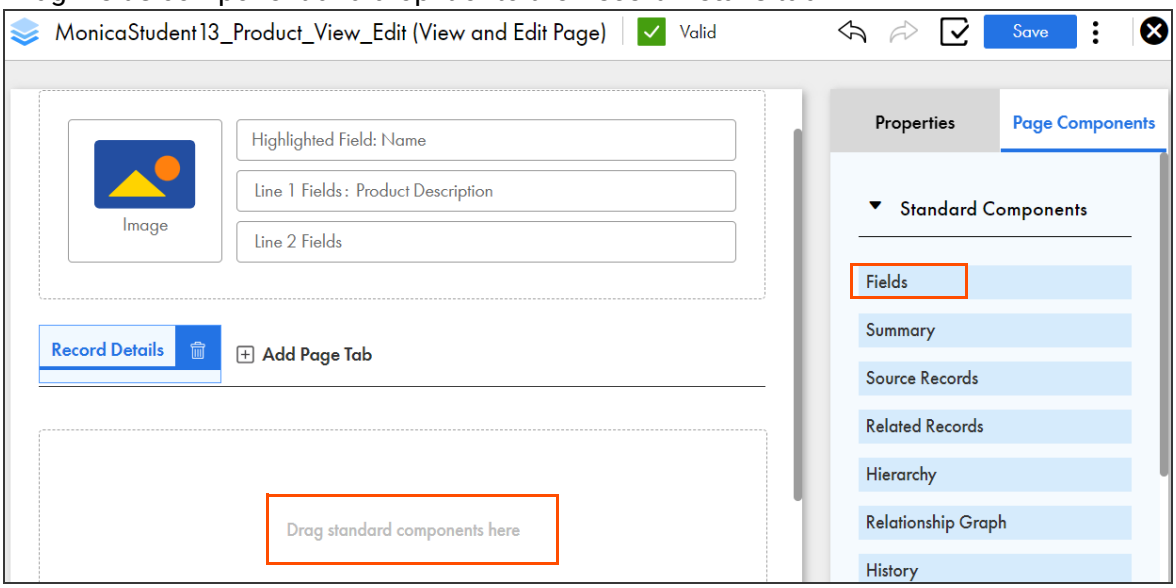
30. From the properties tab on the right, enter **Title** as **Record Details**. Click **Apply**.



31. Select **Page Components** Tab.

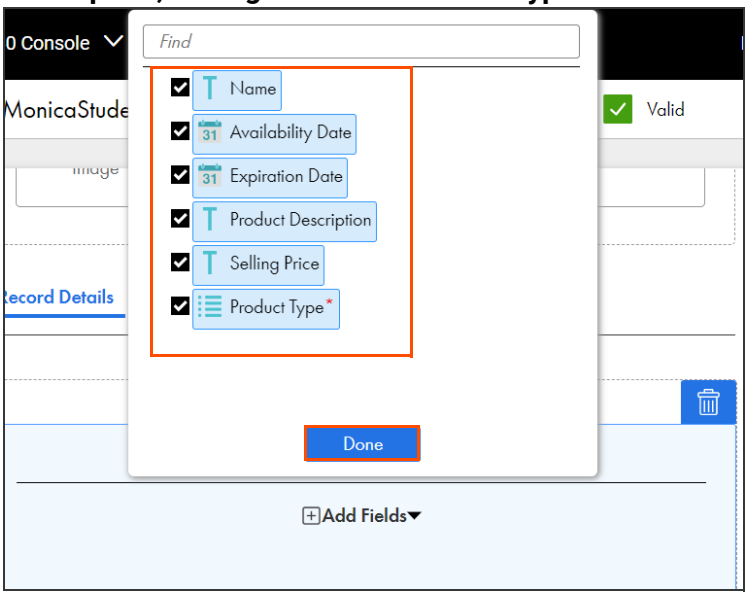


32. Drag **Fields** component and drop it onto the **Record Details** tab.

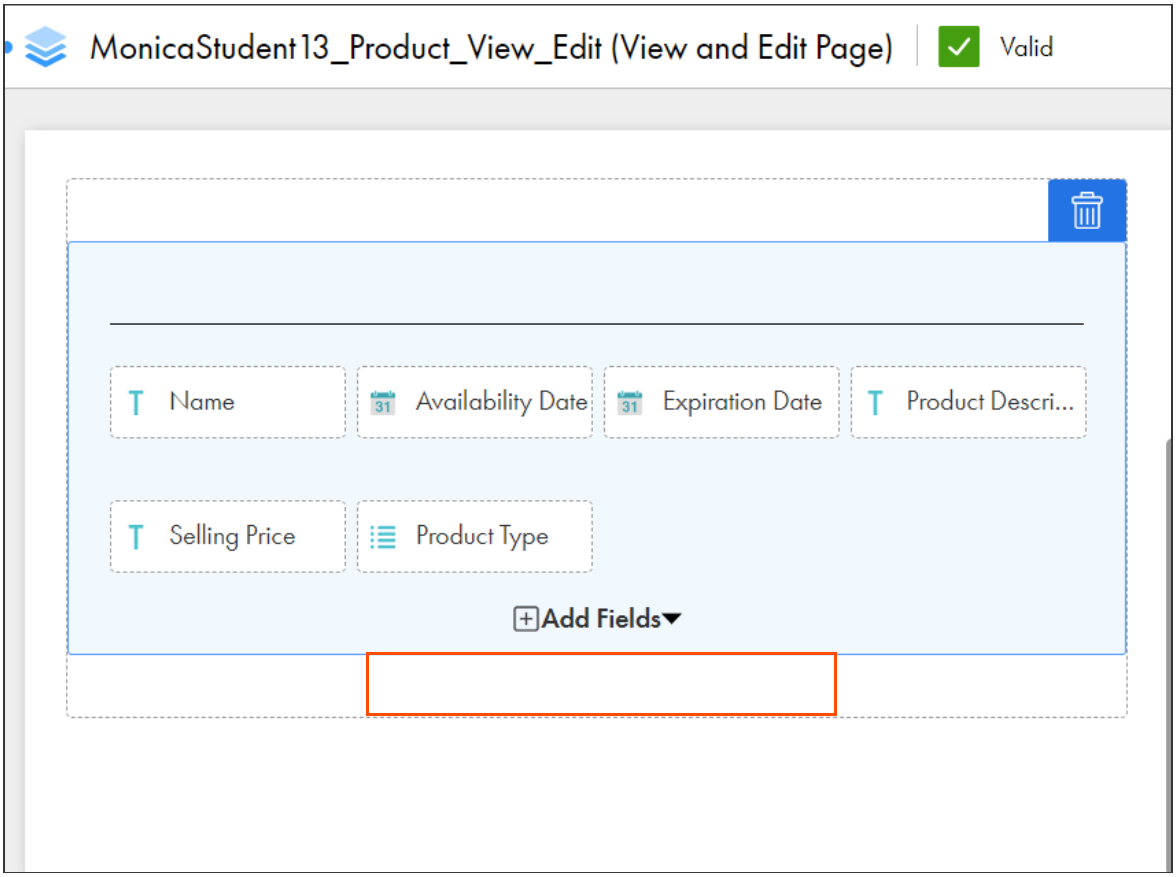


33. Click **Add Fields**.

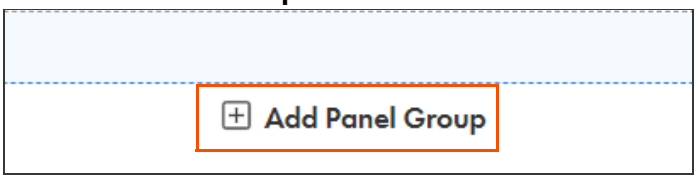
34. From the Add Fields pop-up, select **Name**, **Availability Date, Expiration Date, Product Description, Selling Price and Product Type** and click **Done**.



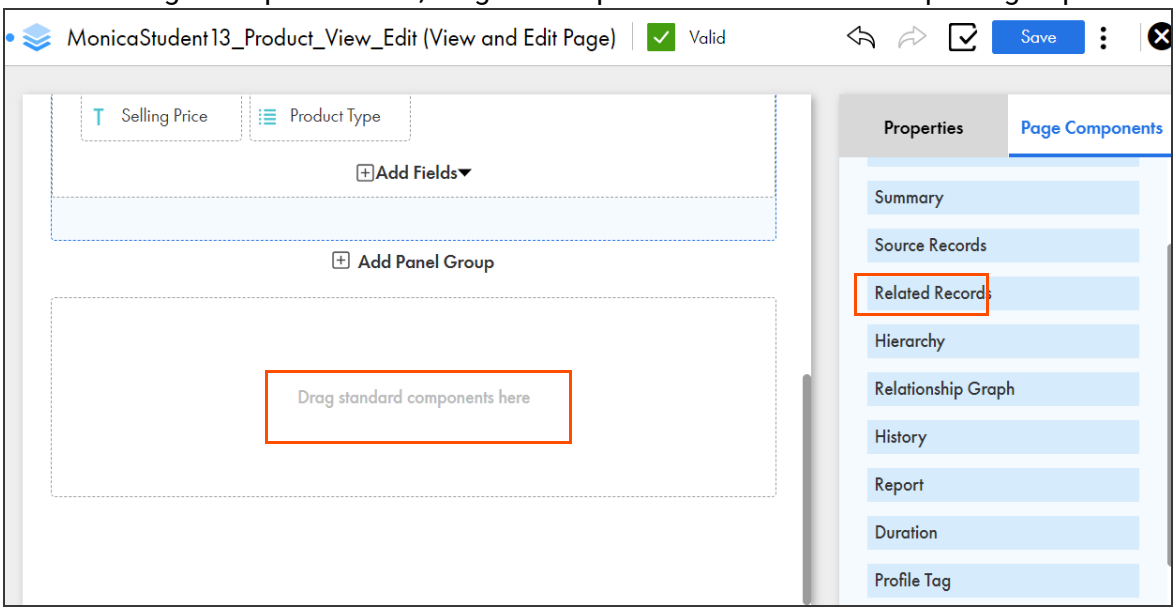
35. To add Related Records to the page, click the space as shown in the image.



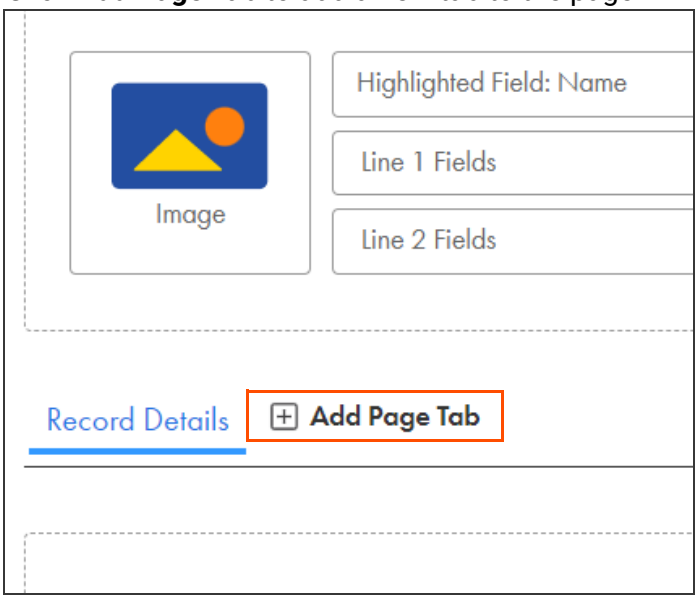
36. Click **Add Panel Group**.



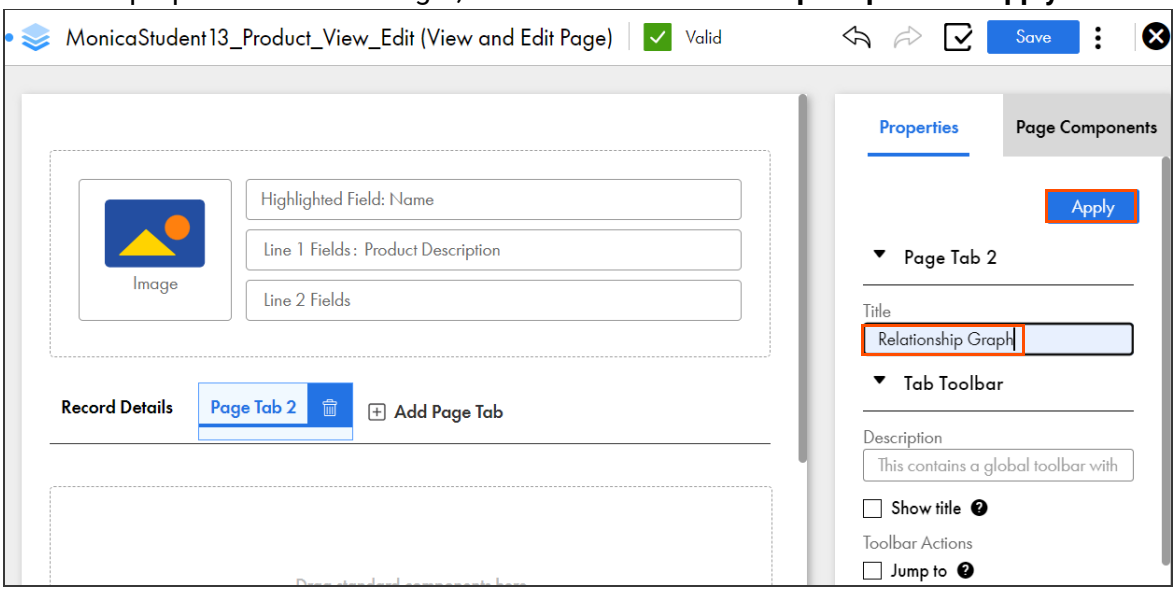
37. From the Page Components tab, drag and drop **Related Records** to the panel group.



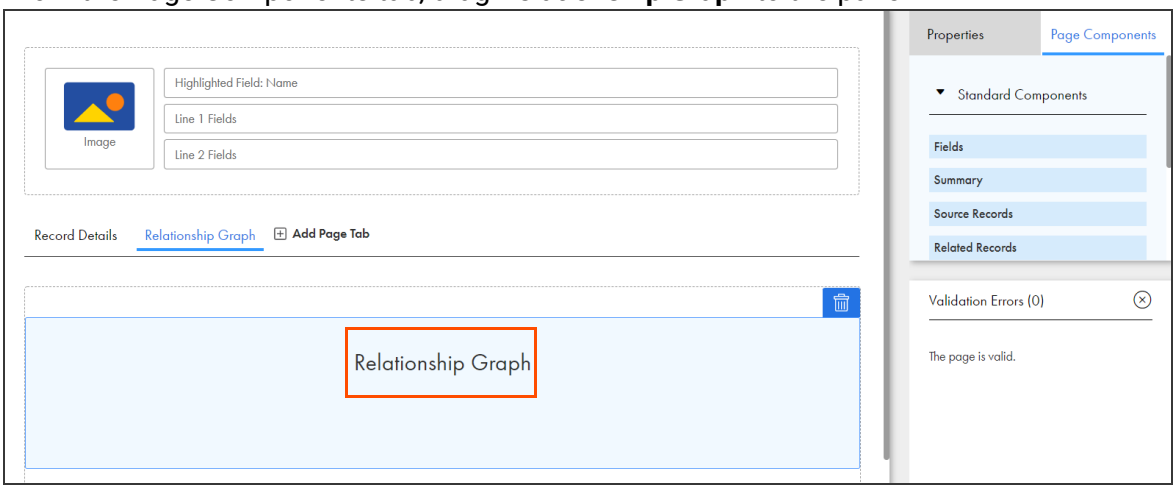
38. Click **Add Page Tab** to add a new tab to the page.



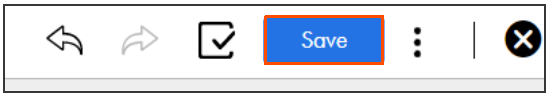
39. From the properties tab in the right, enter **Title** as **Relationship Graph**. Click **Apply**.



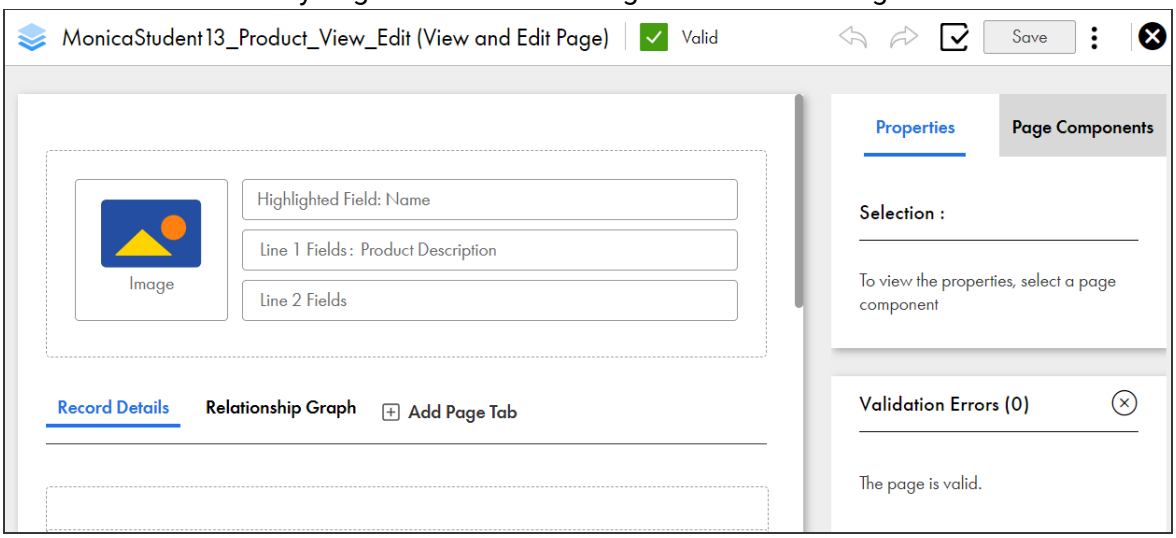
40. From the Page Components tab, drag **Relationship Graph** to the panel.

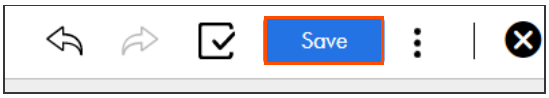


41. Click **Save**.



42. Refresh the browser if you get the below message and click **Save** again.



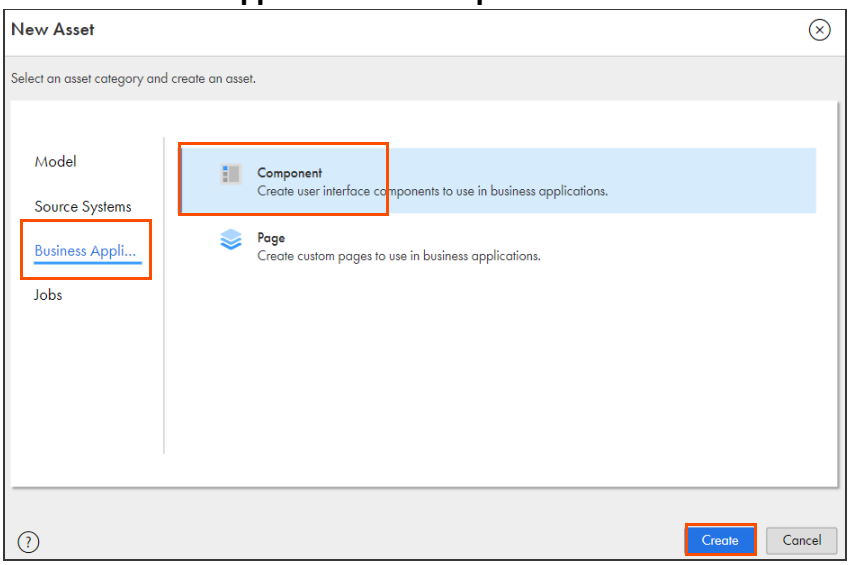


Create Layouts for Search Results

Based on your business needs, you can assign the search result layout components to the user roles defined for your application. You can configure the search properties, such as search suggestions and search history, for an application. You will now create a search layout for the Manufacturer business entities.

43. Click **New**.

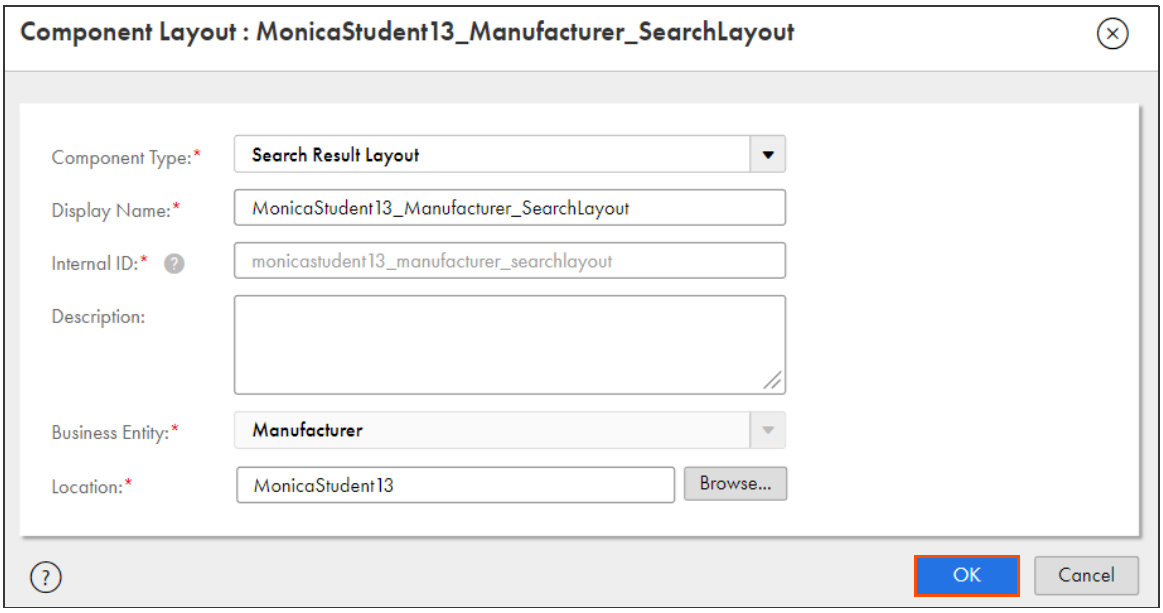
44. Select **Business Applications > Component** and click **Create**.



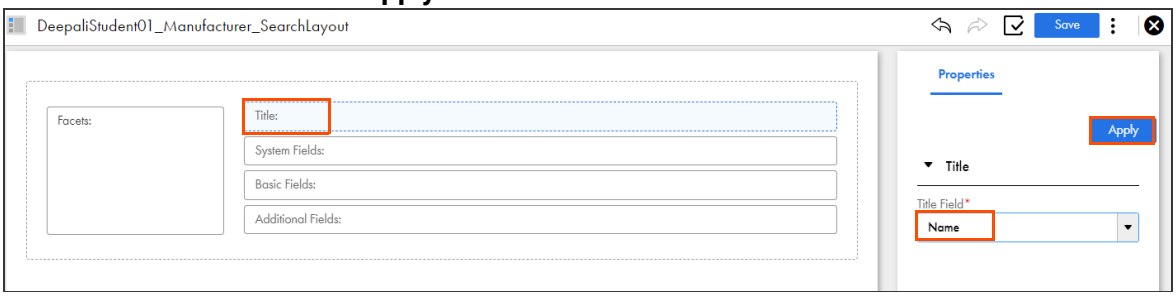
45. In the Component Layout page, enter the following values and click **OK**:

Field	Value
Display Name	<studentName>student<studentNumber>_Manufacturer_SearchLayout
Business Entity	Manufacturer
Location	Your student project

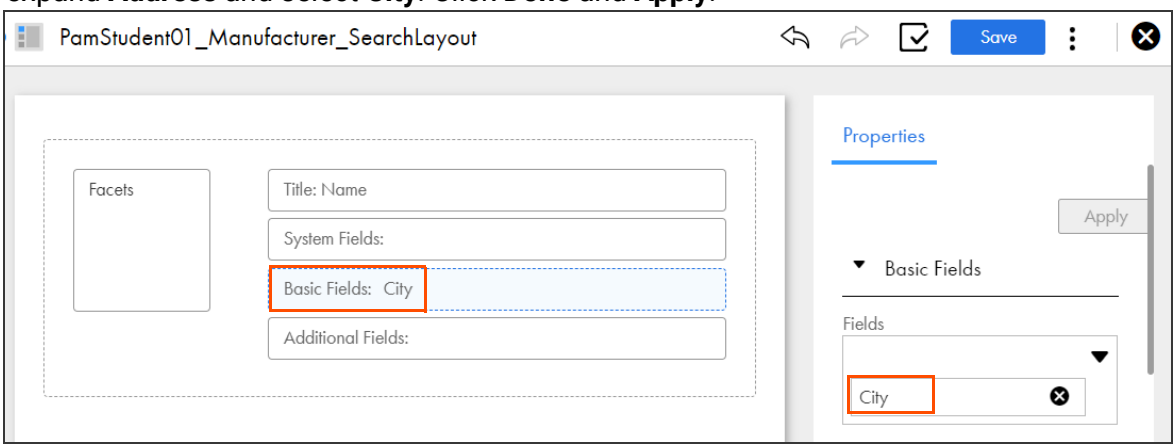
The student project name in the screenshot may vary.



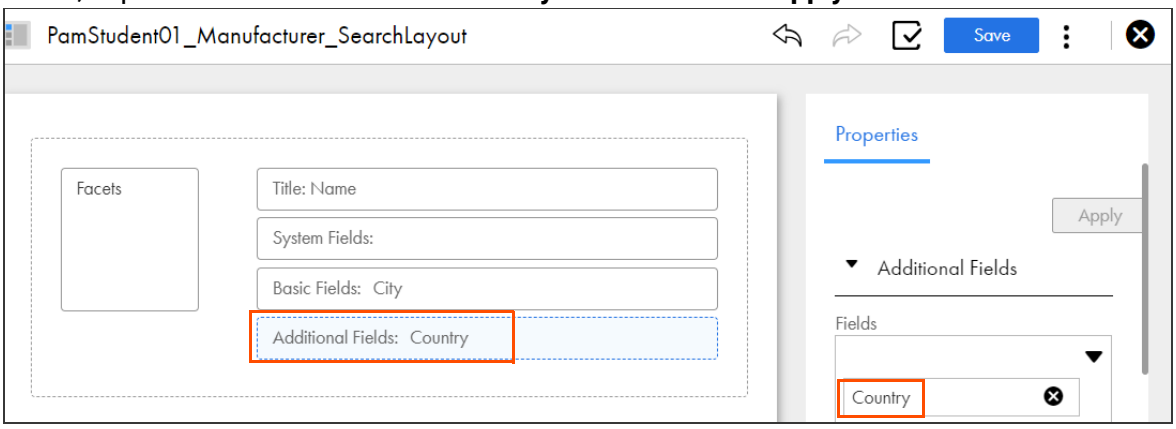
46. Select **Title,** and from the **Properties** tab in the right, from the **Field** drop-down, select **Name.** click **Done** and then **Apply**.



47. Select **Basic Fields,** and from the **Properties** tab in the right, from the **Fields** drop-down, expand **Address** and select **City**. Click **Done** and **Apply**.



48. Select **Additional Fields,** and from the **Properties** tab in the right, from the **Fields** drop-down, expand **Address** and select **Country**. Click **Done** and **Apply**.



49. Click **Save**.

This concludes the lab.

Lab 6-2: Importing Page Layouts

Overview:

In the previous lab, you have you have created a two custom pages, one for the search layout and other for the view and edit. Since Multidomain MDM provides the import asset option, you will utilize this feature and import the search layout for the remaining three business entities.

Objectives:

- Import search layouts for the business entities
- Import view and edit page for site business entity

Duration:

10 Minutes

Scenario:

Previously, while Mark was developing several page layouts, his team was simultaneously creating additional pages and search layouts. They will now share all the page layouts they have created by exporting and then importing them into the organization. As you may already know, there are two types of pages for each Business Entity: Create and ViewEdit. For Create-type pages, Mark uses the default ones generated during the creation of the Business Entity. For ViewEdit-type pages, he employs those he either created or imported in the labs.

Tasks:

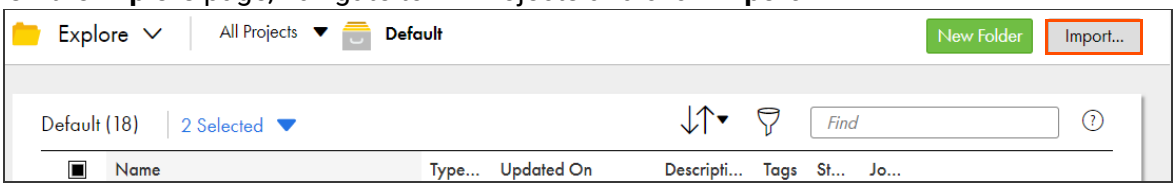
Perform the following tasks to import the search layouts and View/Edit pages for Homekart scenario:

- Import the search layouts
- Verify the imported search layouts
- Import View and Edit page for the Site business entity

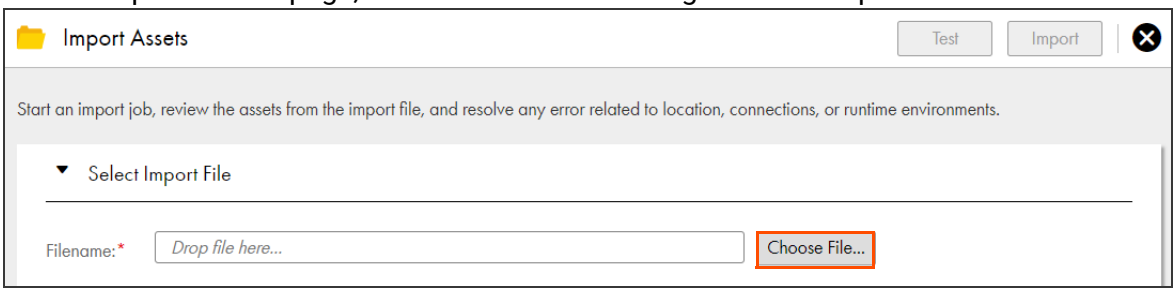
Import the Search Layouts

First, you will import the search layout to the Multidomain MDM.

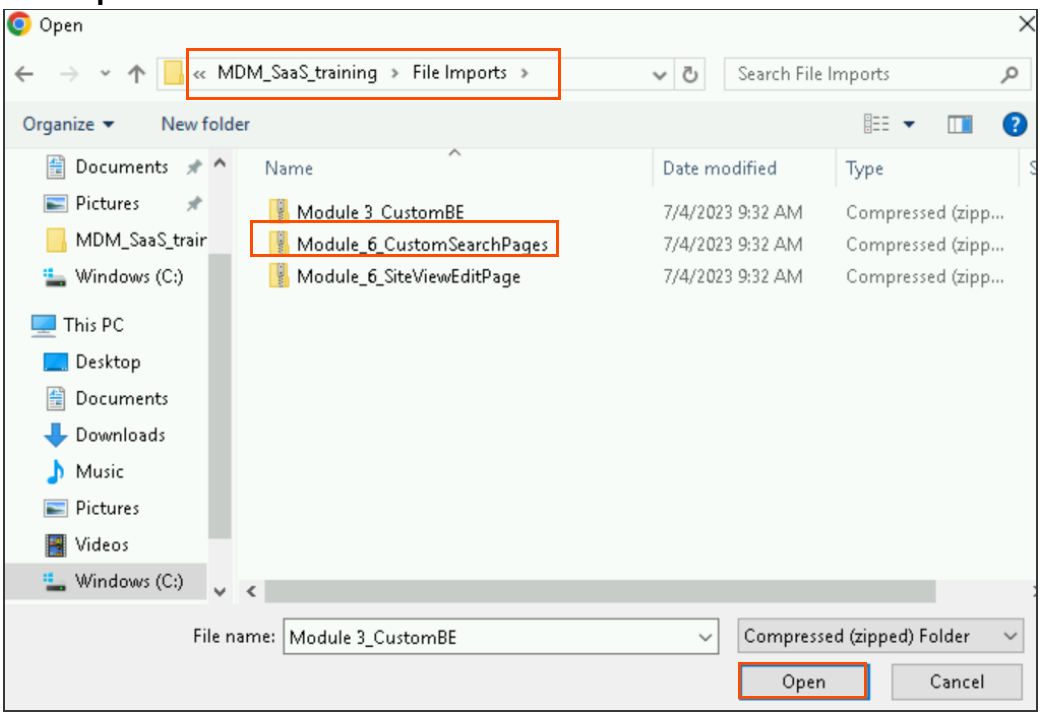
1. Log in to the IICS using your admin credentials.
2. Open Business 360 Console service.
3. On the **Explore** page, navigate to All Projects and click **Import**.



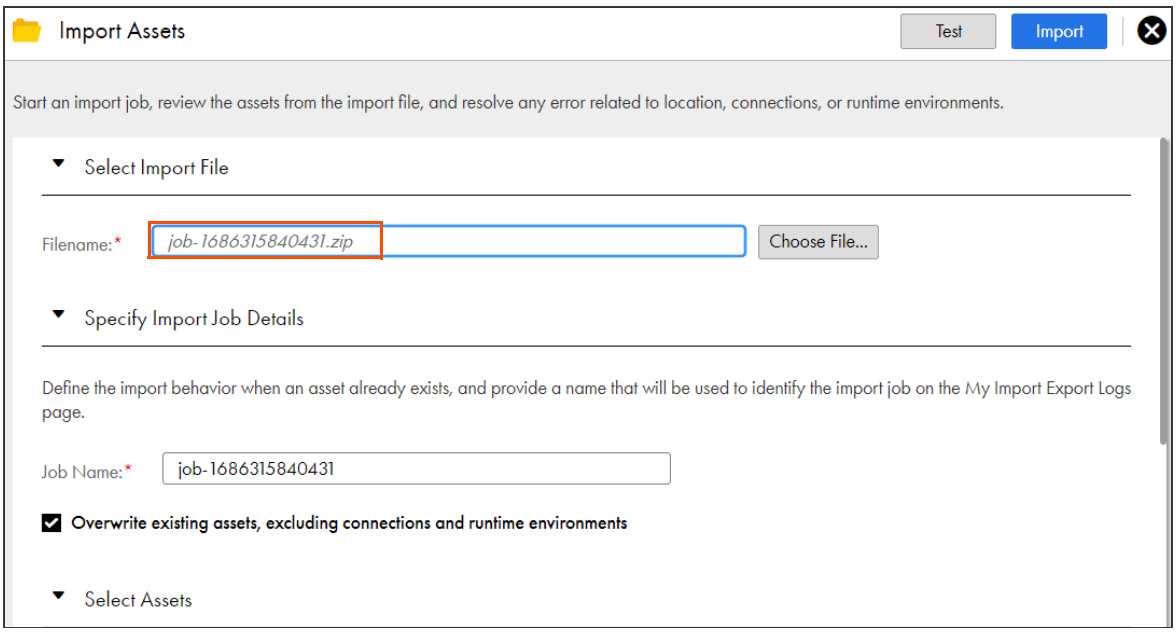
4. On the Import Assets page, click **Choose File** to navigate to the export file.



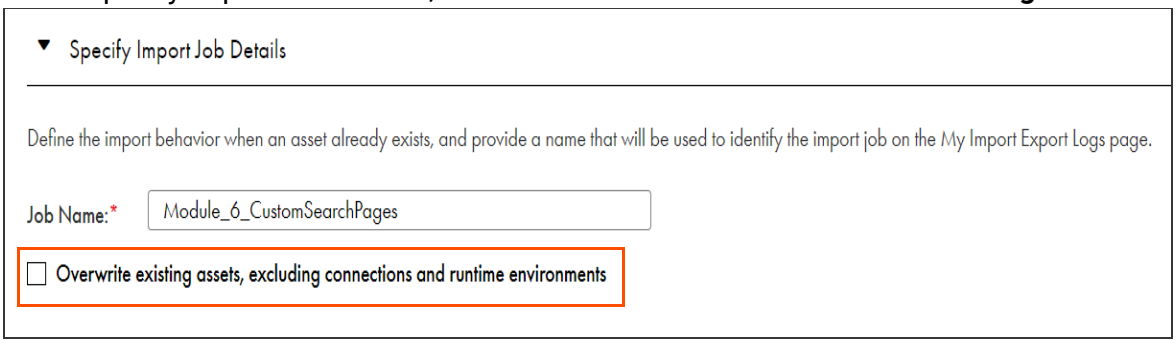
5. Select the **MDM_SaaS_training\File Imports\Module_6_CustomSearchPages** ZIP file. Click **Open**.



6. A page will appear where you can optionally, change the import job name. You can choose to retain the same name.

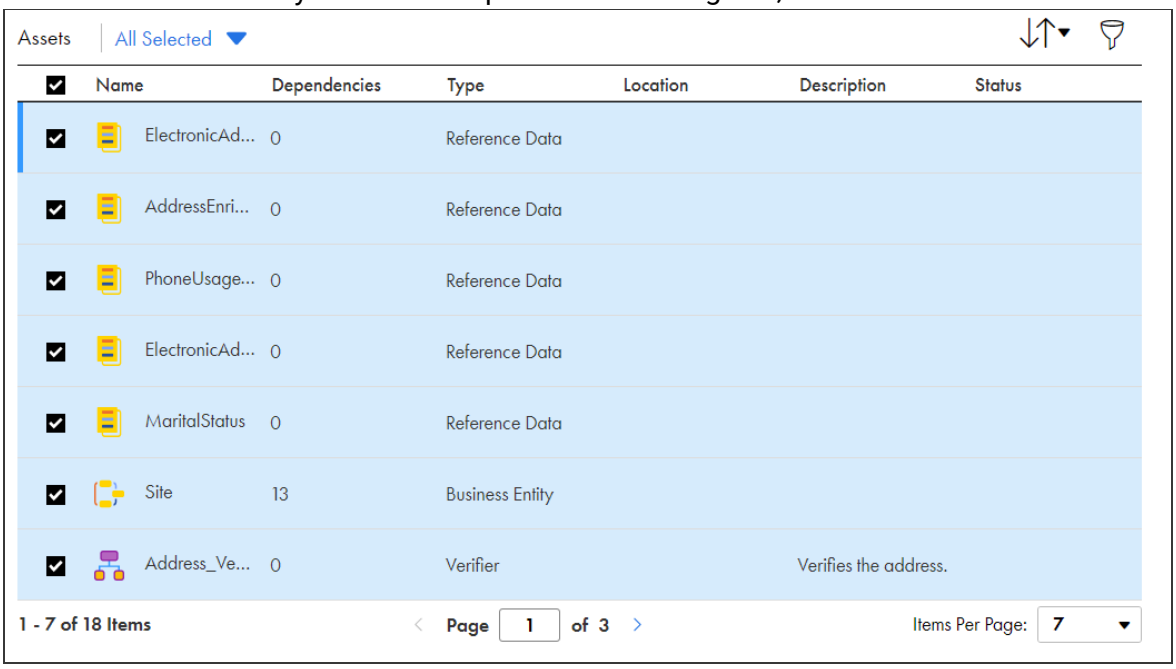


7. In the Specify Import Job Details, deselect the check box **Overwrite the existing assets**.



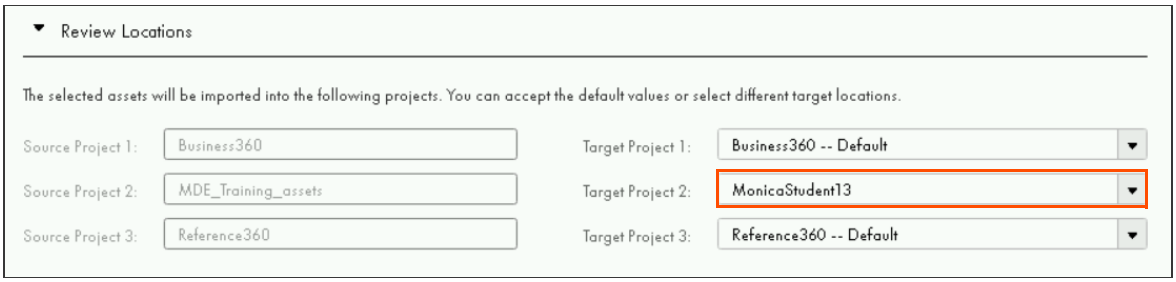
If you choose to overwrite existing assets, when an asset has the same name and internal ID as an asset in the target project, the asset replaces the existing asset in the target project.

8. Select the assets that you want to import. For this lab guide, retain the default selection.

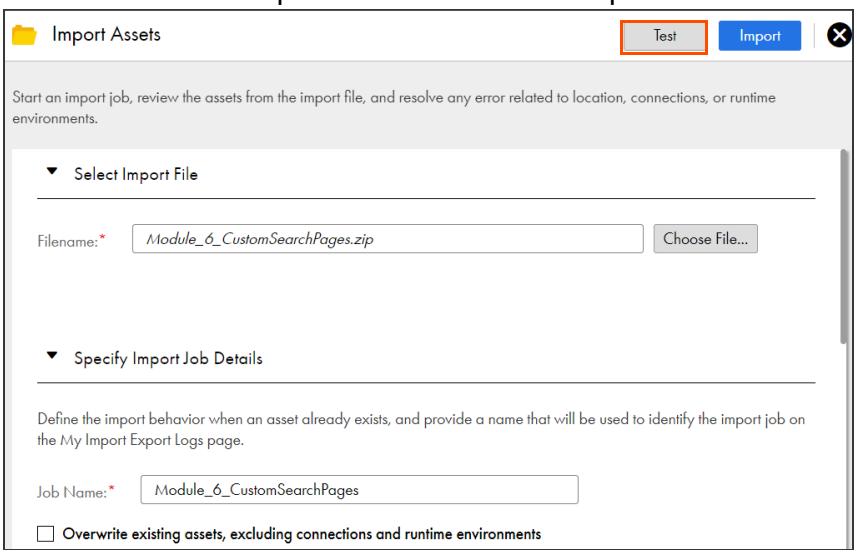


9. Select the **Target Project 2** as you student project. This is to make sure that the assets are imported to your student folder.

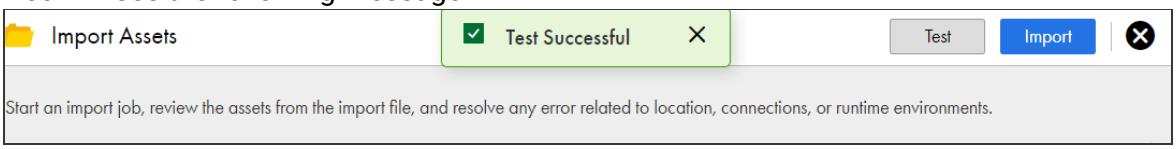
Make sure you select the your student project. Otherwise a new folder will be created and pages will be imported in that folder.



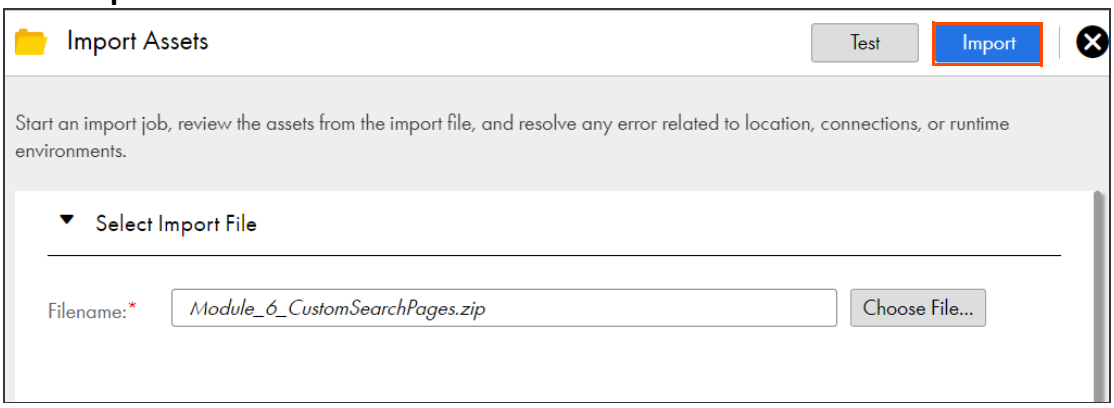
10. Click **Test** to see the potential results of the import.



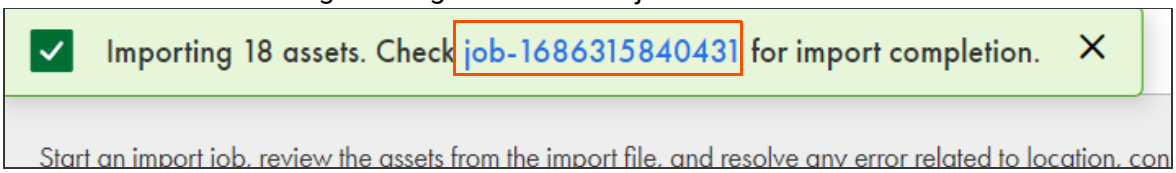
You will see the following message:



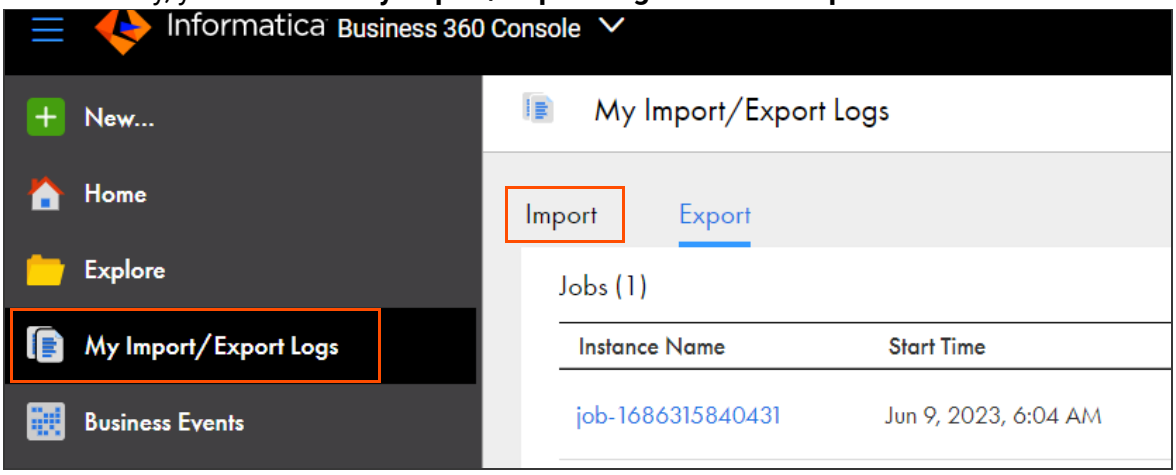
11. Click **Import**.



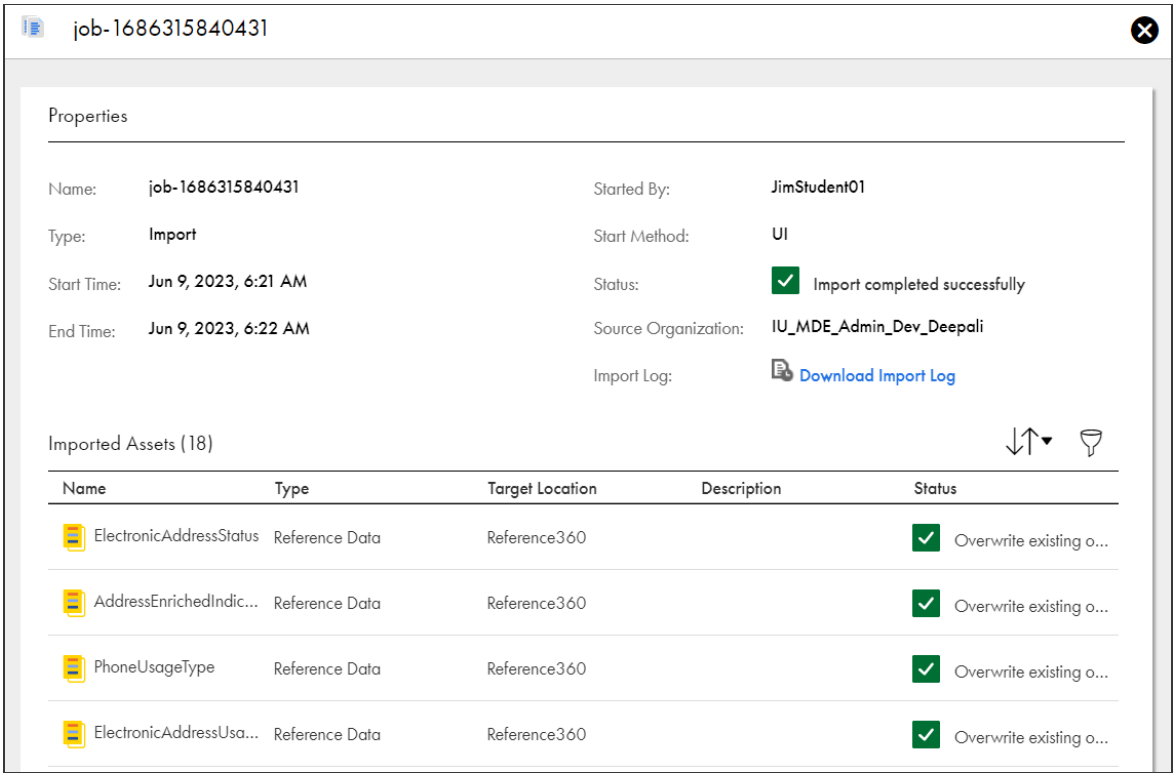
12. You will see the following message. Click on the job id to see the status.



13. Alternatively, you can click **My Import/ Export Logs** and click **Import** tab.



14. You can see that the import has successfully completed. Click on the **job instance name** to view the details.

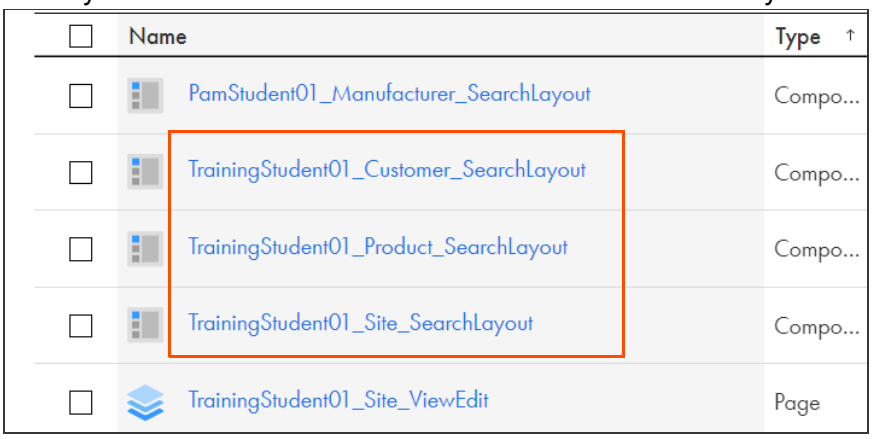


Verify the Imported Search Layouts

Now, you should check to see if the search layouts were imported successfully.

15. Click **Explore** in the Business 360 Console.

16. Go to your student folder and confirm if the three search layouts are imported.



The names of the imported page have the prefix TrainingStudent01.

Self-Try Exercise (Mandatory)

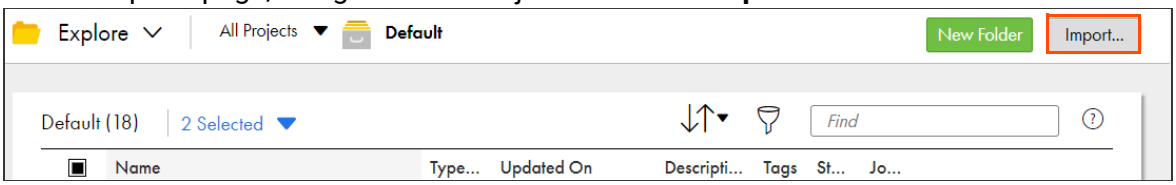
Import View and Edit Page for the Site Business Entity

Now you will import a view and edit page for the Site business entity.

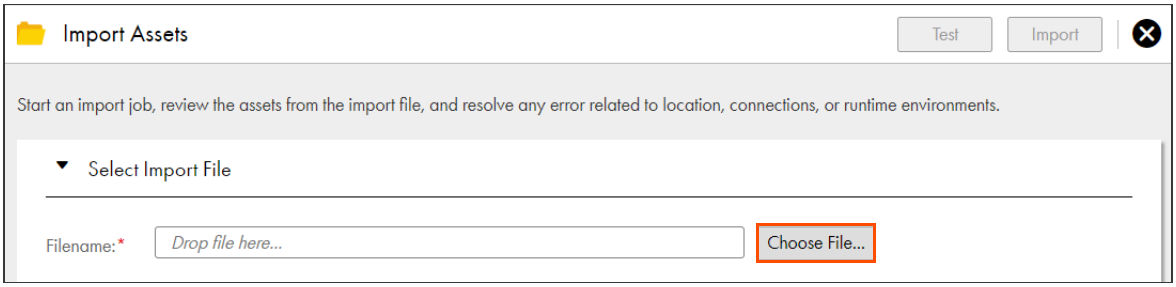
17. Log in to the IICS using your admin credentials.

18. Open Business 360 Console service.

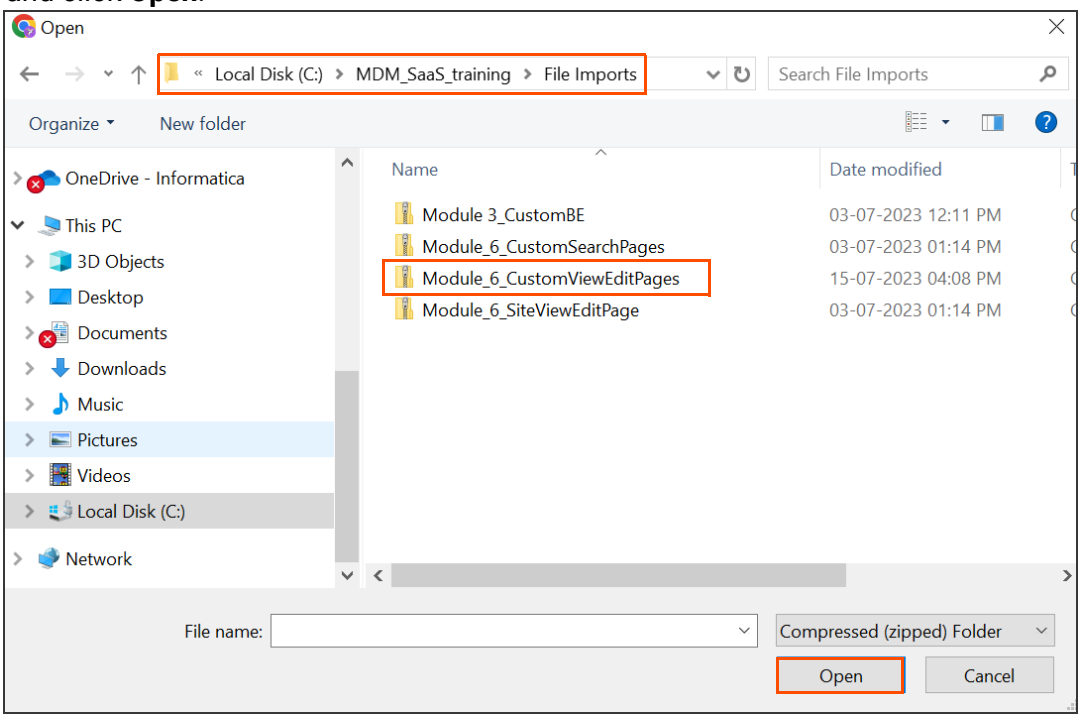
19. On the Explore page, navigate to All Projects and click **Import**.



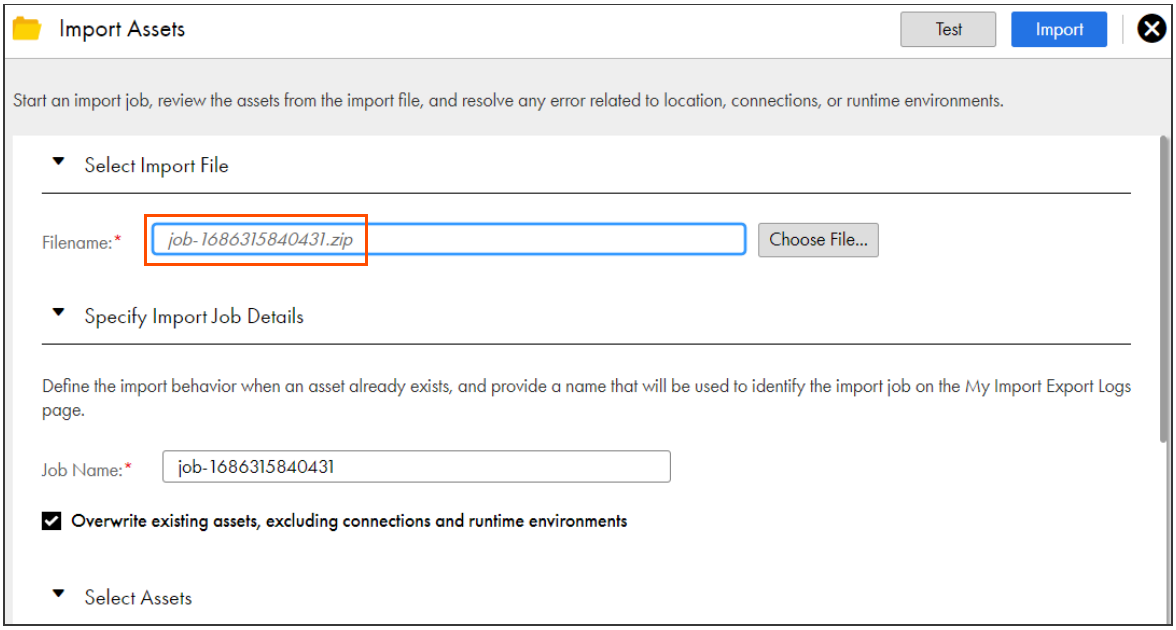
20. On the Import Assets page, click **Choose File** to navigate to the export file.



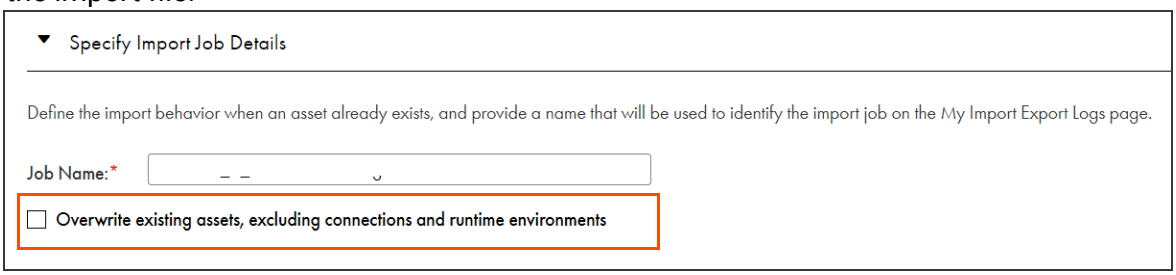
21. Select the **MDM_SaaS_training\File Imports\Module_6_CustomViewEditPages** ZIP file and click **Open**.



22. A page will appear where you can optionally, change the import job name. You can choose to retain the same name.

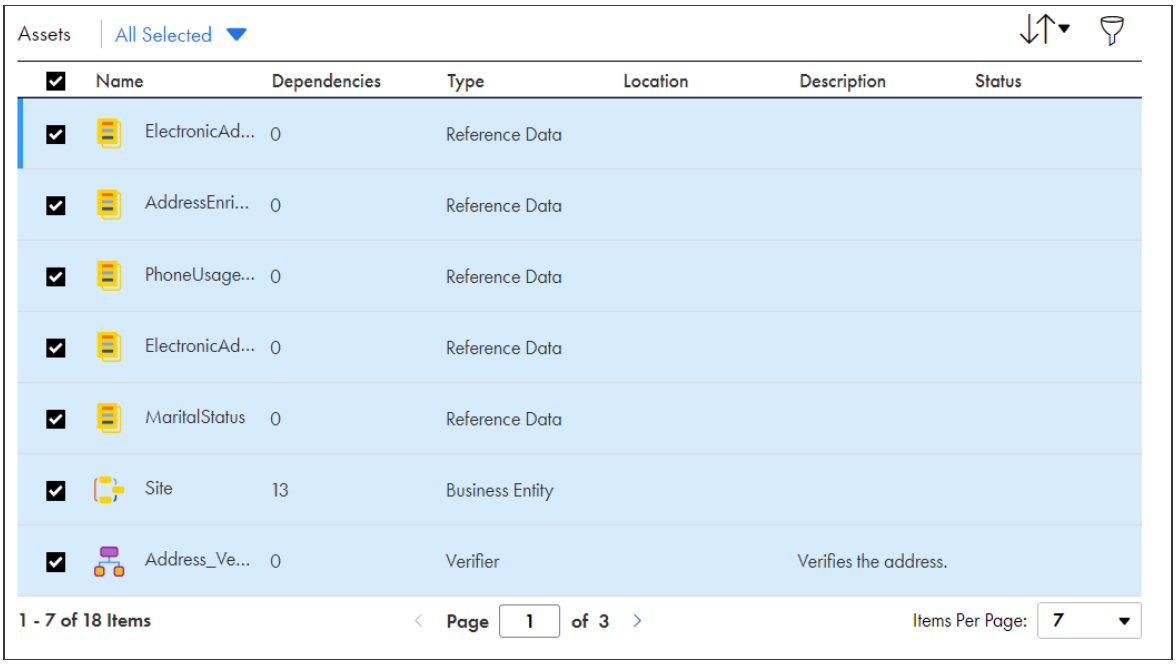


23. In the Specify Import Job Details, choose to overwrite the existing assets with the assets in the import file.

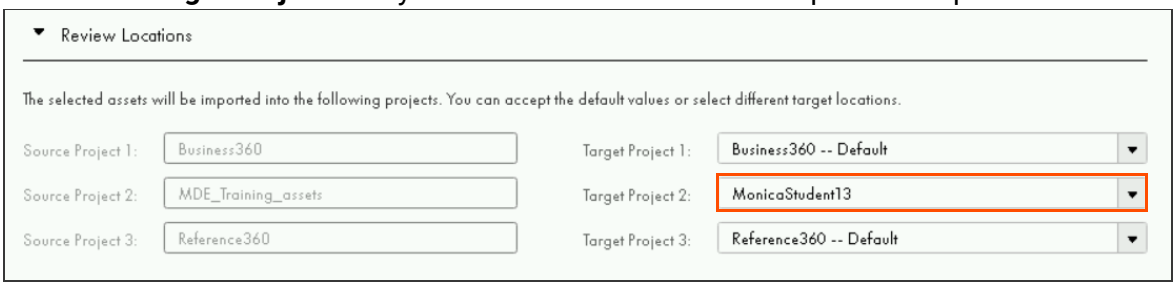


If you choose to overwrite existing assets, when an asset has the same name and internal ID as an asset in the target project, the asset replaces the existing asset in the target project.

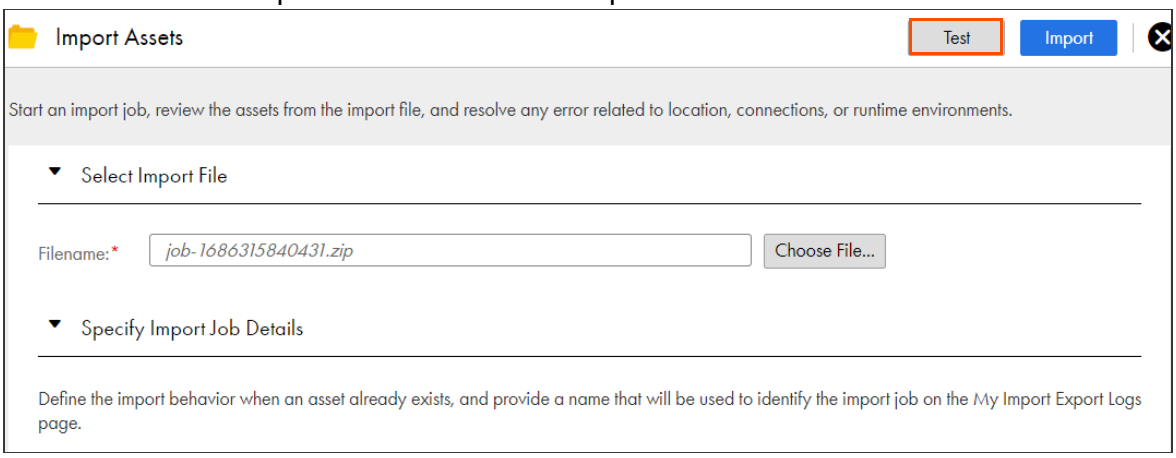
24. Retain the default selection for assets.



25. Select the **Target Project 2** as you student folder. This is an important step.

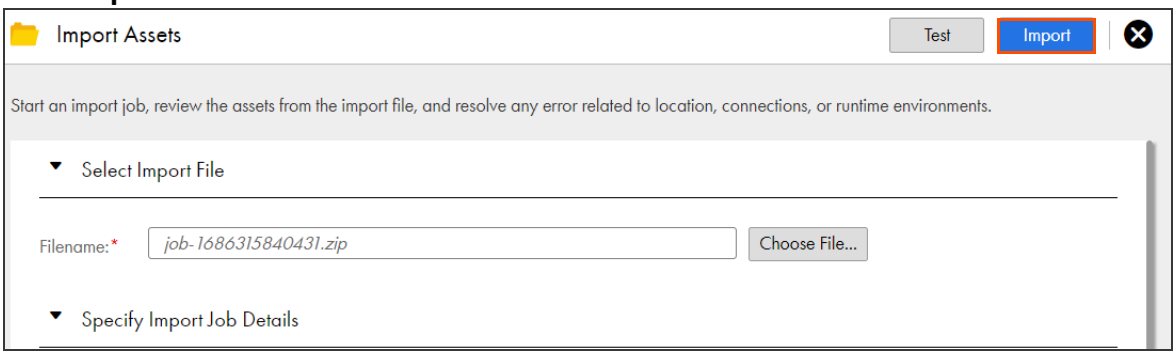


26. Click **Test** to see the potential results of the import.

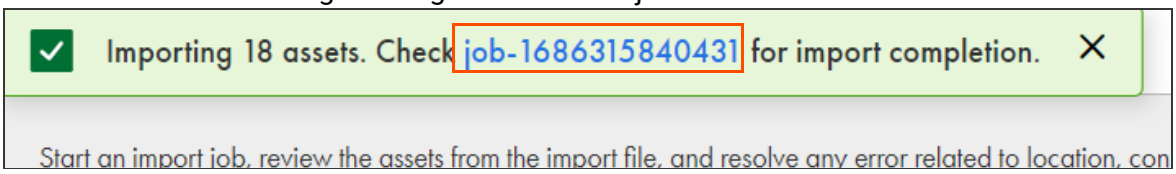


You will see that the test is successful.

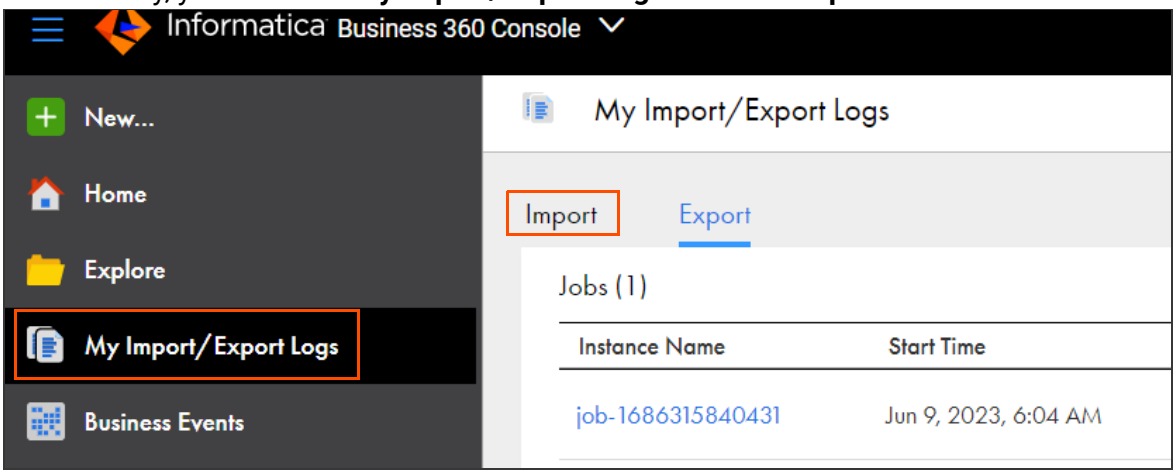
27. Click **Import**.



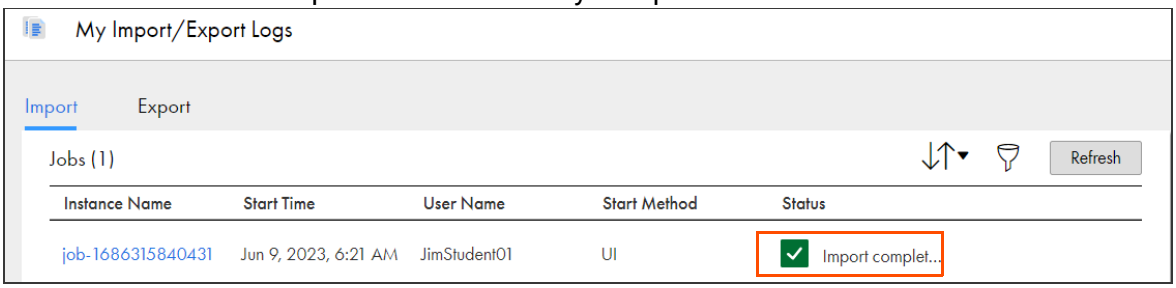
28. You will see the following message. Click on the job id to see the status.



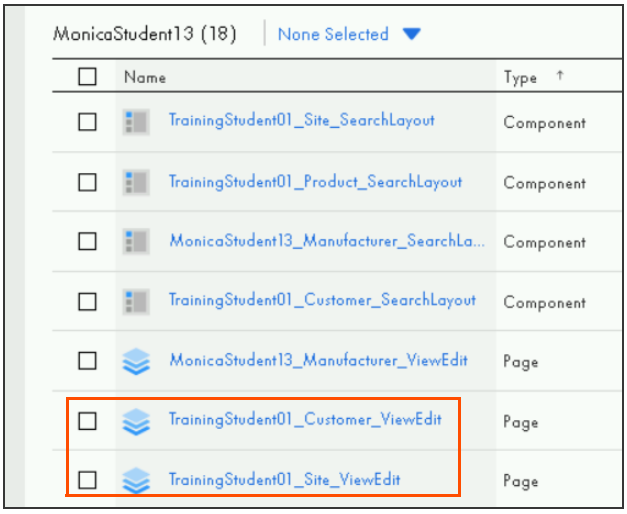
29. Alternatively, you can click **My Import/ Export Logs** and click **Import** tab.



You can see that the import has successfully completed.



30. You can confirm if the assets are imported successfully by checking your student project.



This concludes the lab.

Lab 6-3: Adding Objects to Multidomain MDM Application

You can add business entities and configure the display fields for each business entity. The values of the selected display fields appear in the left navigation bar when you open a record. You can view the list of components, such as search box, notifications, and tabs, that are enabled for the application. The enabled components appear on the application header and left navigation bar.

In the previous lab, you have created business entities, relationships, custom pages, hierarchies, and search layouts. Here, you will assign those objects to the Multidomain MDM application.

Objectives:

- Add business entities to Multidomain MDM application
- Add pages to Multidomain MDM application
- Add search layouts to Multidomain MDM application

Scenario:

Mark and his team have developed numerous assets for the Homekart data model, including Business Entities, custom layouts, hierarchies, search layouts, and relationships. To display these objects in the client application, such as Multidomain MDM, they need to be added to the Multidomain MDM application. Additionally, Mark can configure which users have access to these pages in the front end.

Duration:

30 Minutes

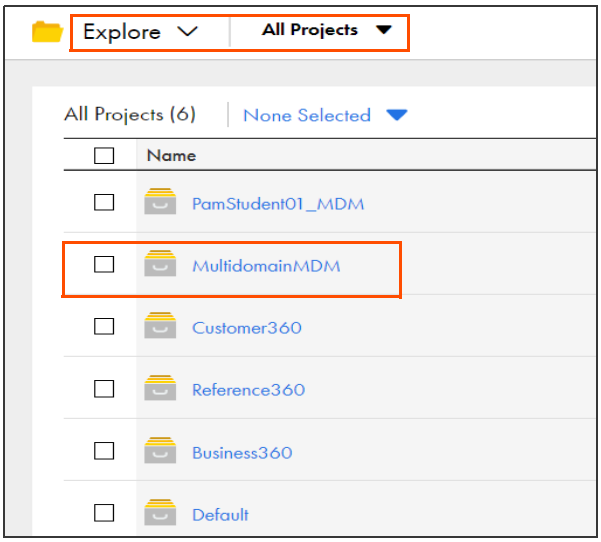
Tasks:

- Add Business entities to Multidomain MDM application
- Add pages to the Multidomain MDM application
- Add custom search layout to the Multidomain MDM application

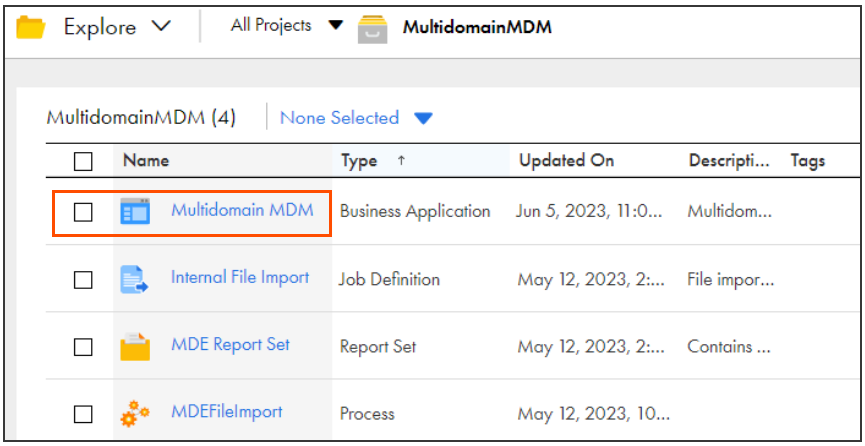
Add Business Entities to Multidomain MDM Application

You will add the custom business entities that you have created to the Multidomain MDM application.

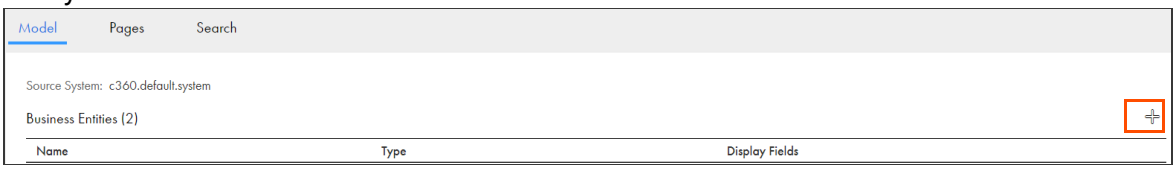
1. Navigate to **Explore > All Projects > MultidomainMDM** folder.



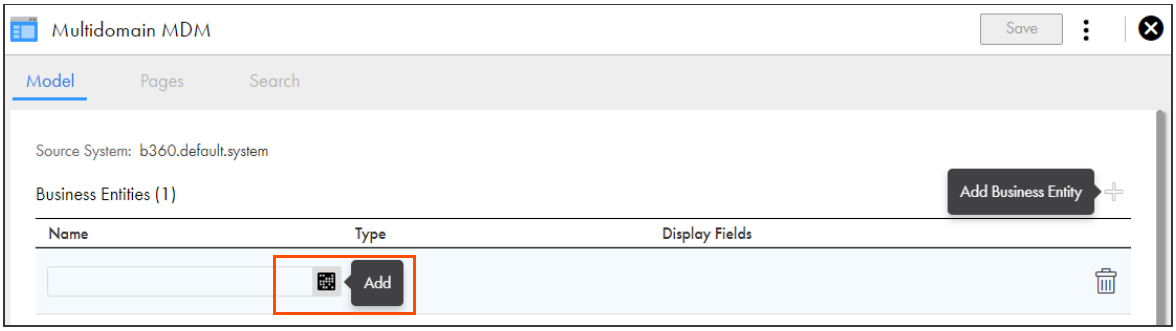
2. Click to open **Multidomain MDM** business application.



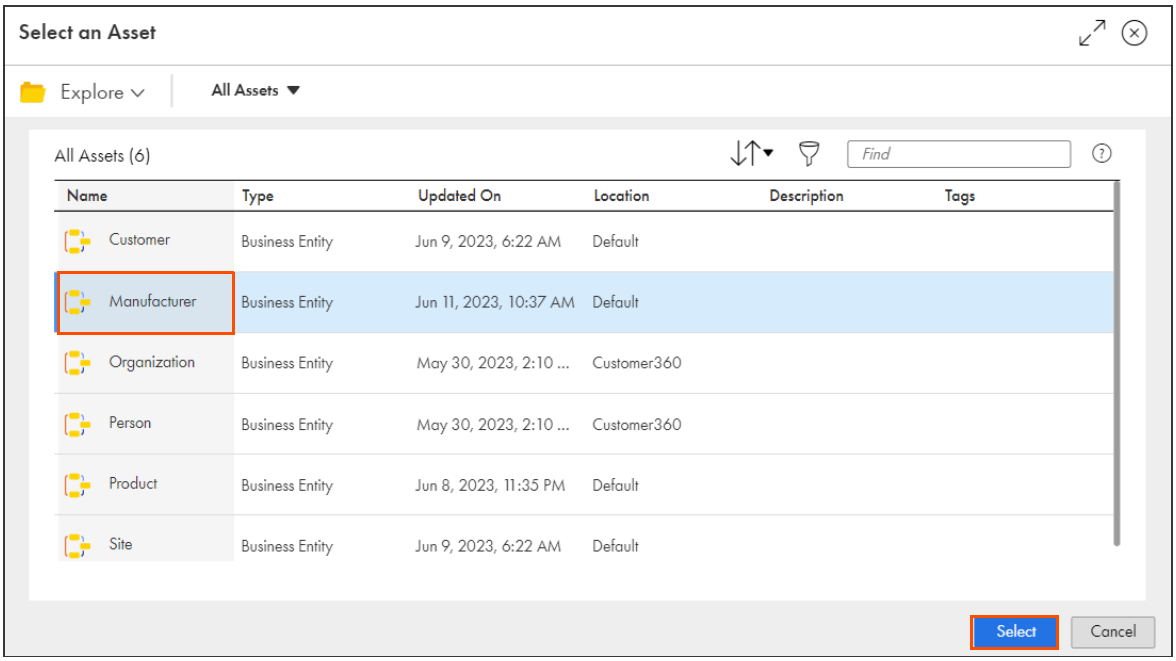
3. You will add custom business entities that you have created in the previous lab to the application. On the Model tab, click **+** (Add Business Entity) icon to add a custom business entity.



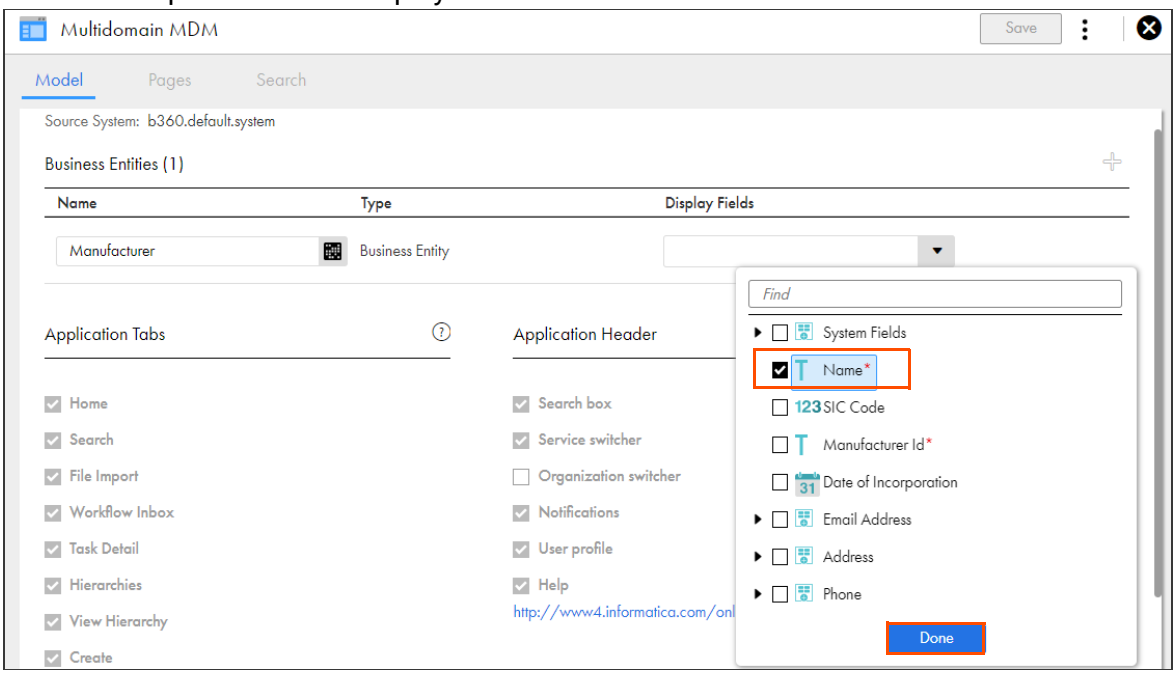
4. To select the asset, click the **Asset Picker** icon.



5. Select **Manufacturer** from the list of business entities and click **Select**.

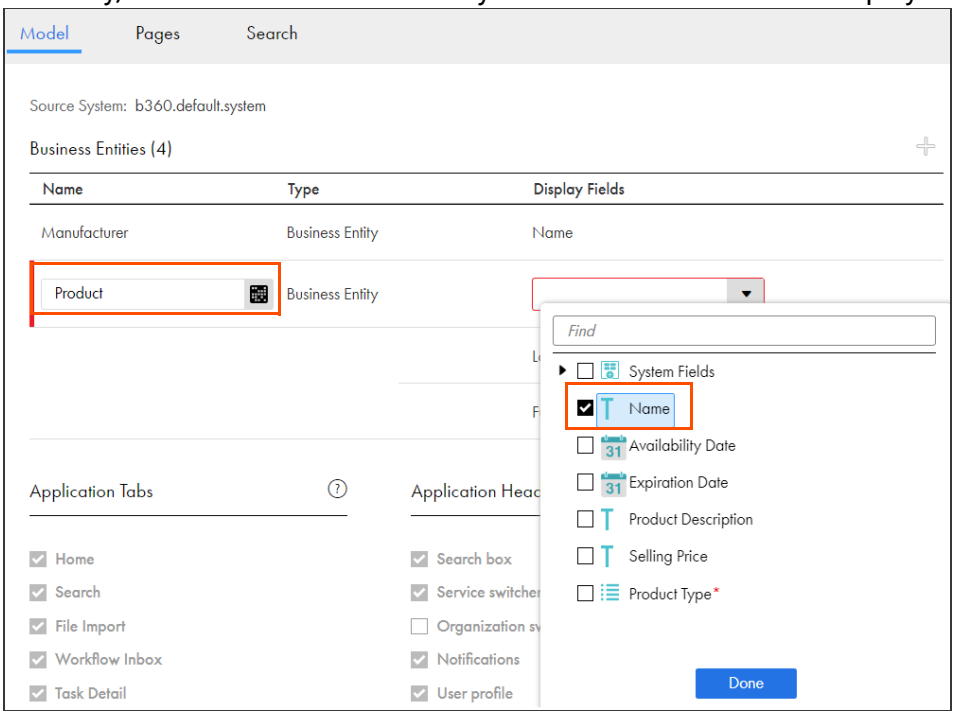


6. Click the drop-down under Display Fields and select **Name** field. Click **Done**.

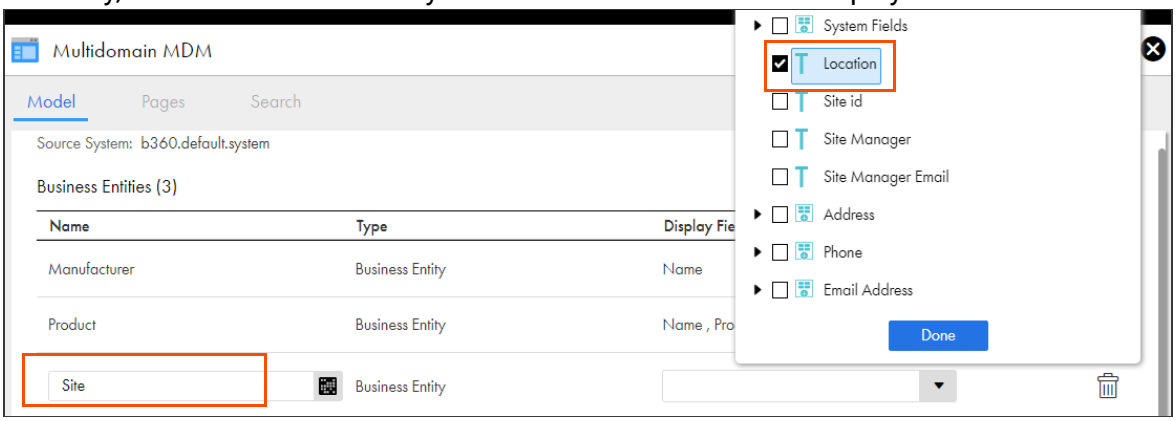


7. Click anywhere on the page to save the changes.

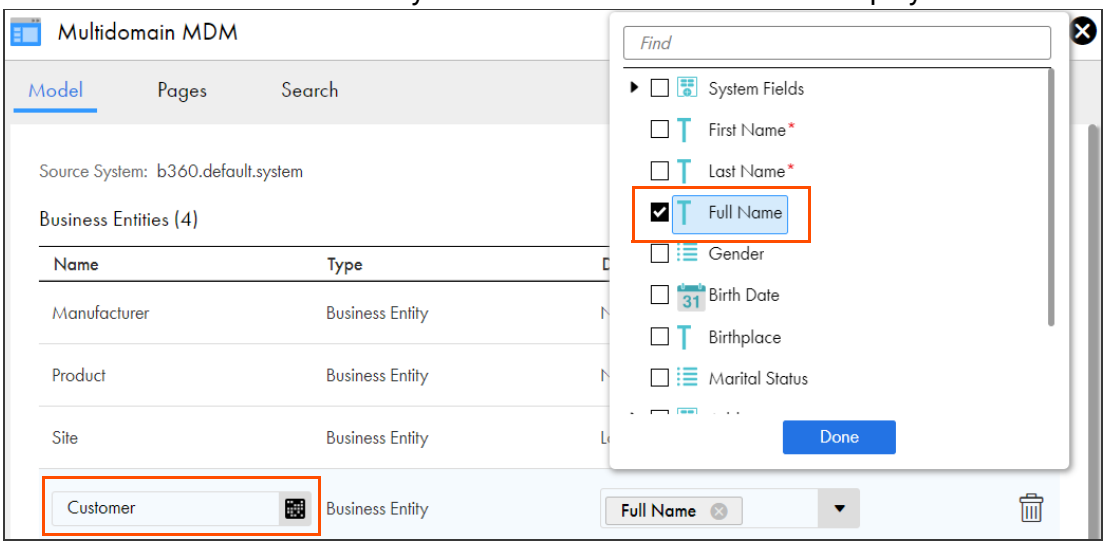
8. Similarly, add **Product** business entity and select **Name** field as display fields.



9. Similarly, add **Site** business entity and select **Location** field as display field.



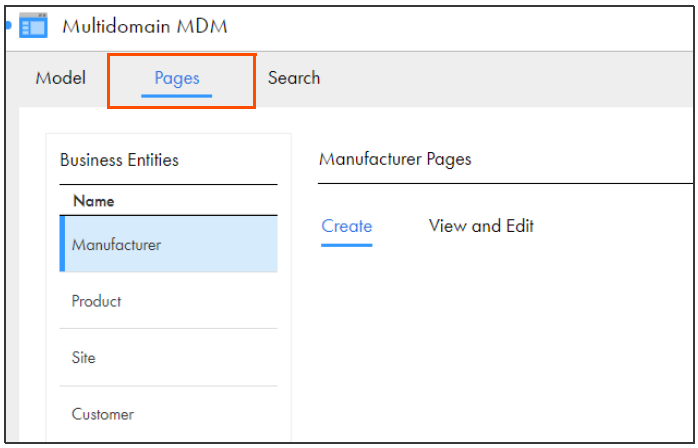
10. Select **Customer** business entity and select **Full Name** fields as display fields.



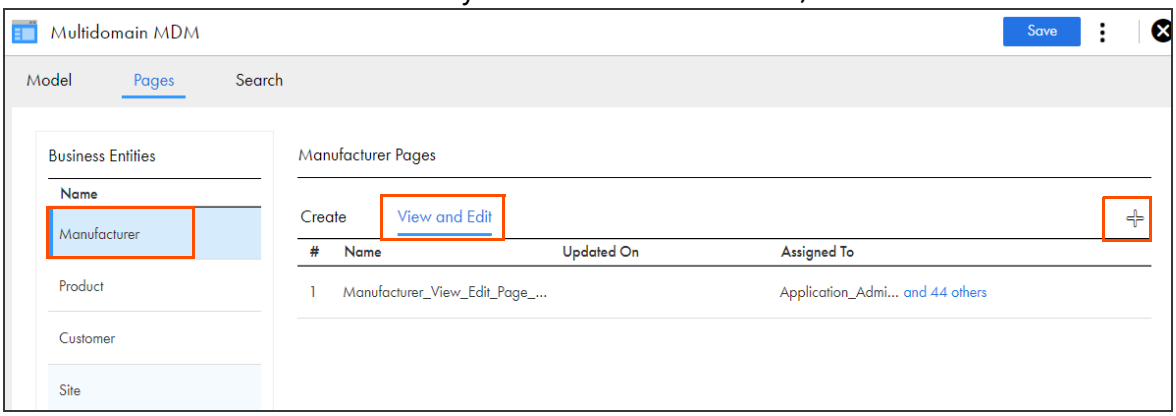
Add Pages to the Manufacturer Business Entity

As the next step, you will add custom view and edit page to the Manufacturer business entity.

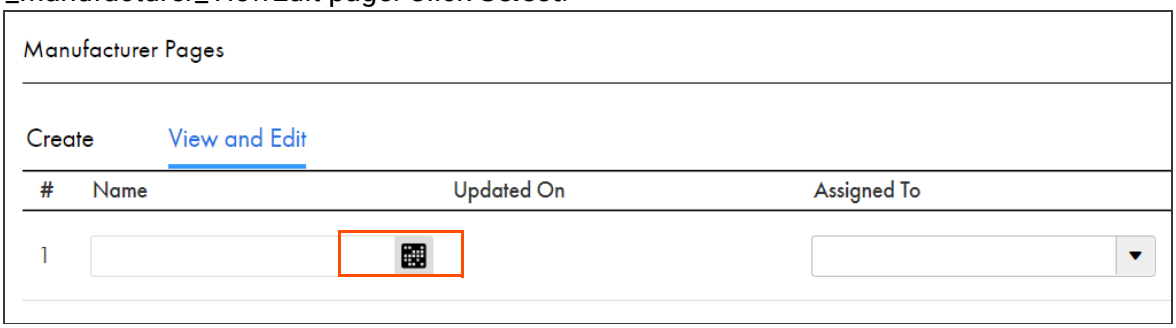
11. Click **Pages** tab. Here, you will add custom view and edit pages that you have created in the previous lab.

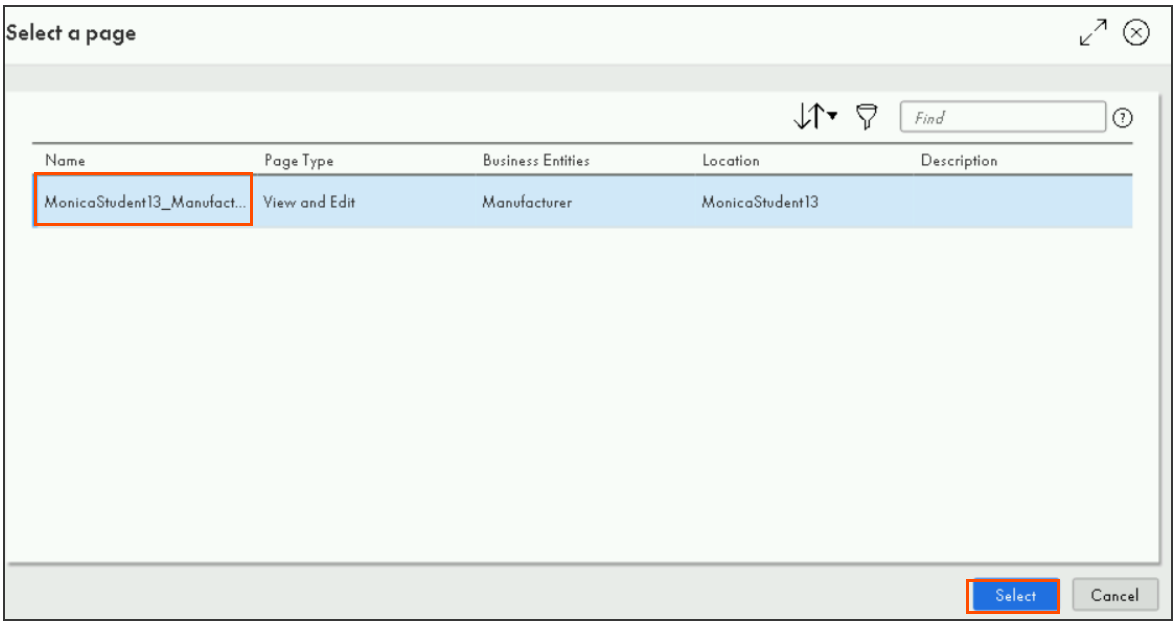


12. Select **Manufacturer** business entity. Under **View and Edit** tab, Click **+** icon.



13. Click the Asset Picker icon and select <StudentNumberStudentName> _Manufacturer_ViewEdit page. Click **Select**.





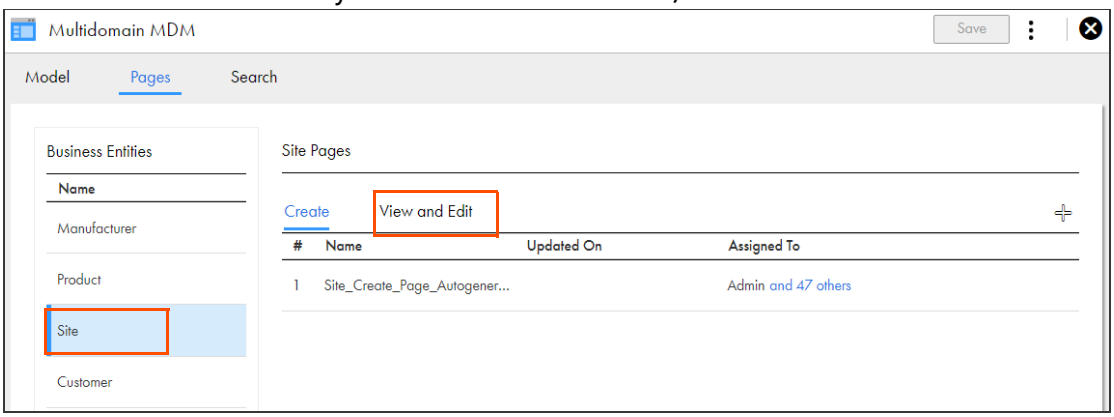
The prefix added to the view and edit page may vary in the screenshot. Select the one created by you.

Self-Try Exercise (Mandatory)

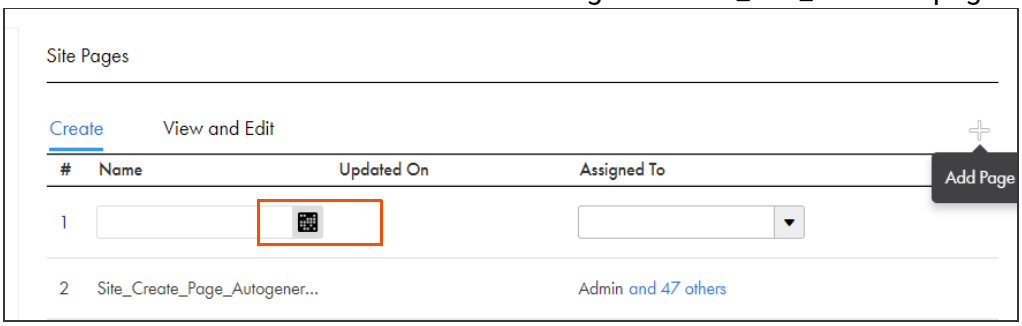
Add View and Edit Page to the Site Business Entity

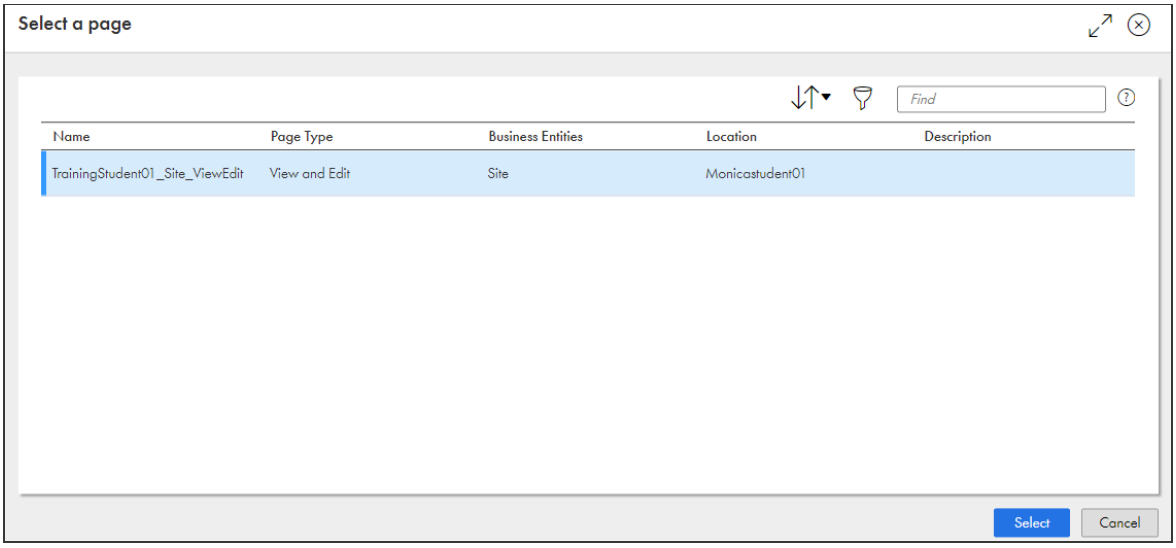
Now, you will add custom view and edit page to the Site business entity.

14. Select **Site** business entity. Under **View and Edit** tab, Click **+** icon.



15. Click the Asset Picker icon and select TrainingStudent01_Site_ViewEdit page. Click **Select**.

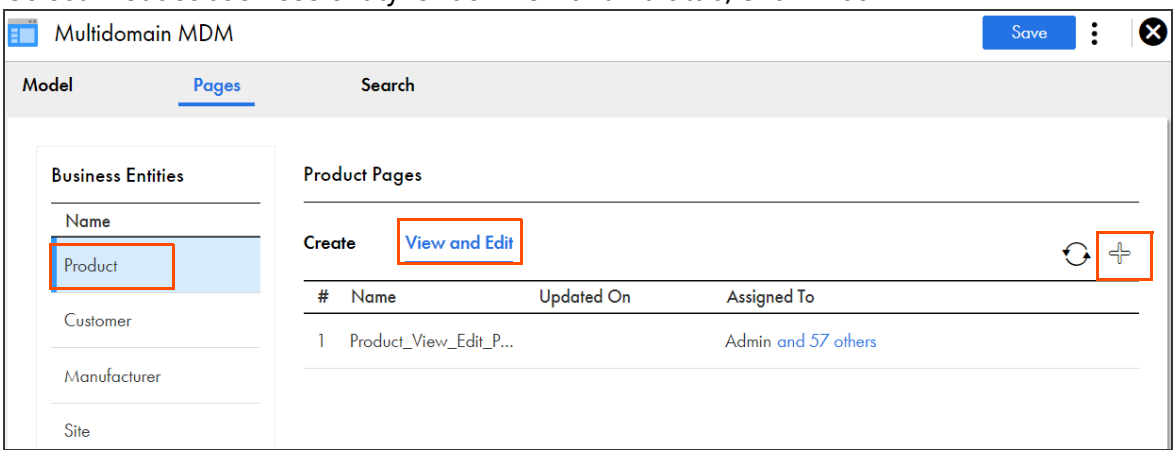




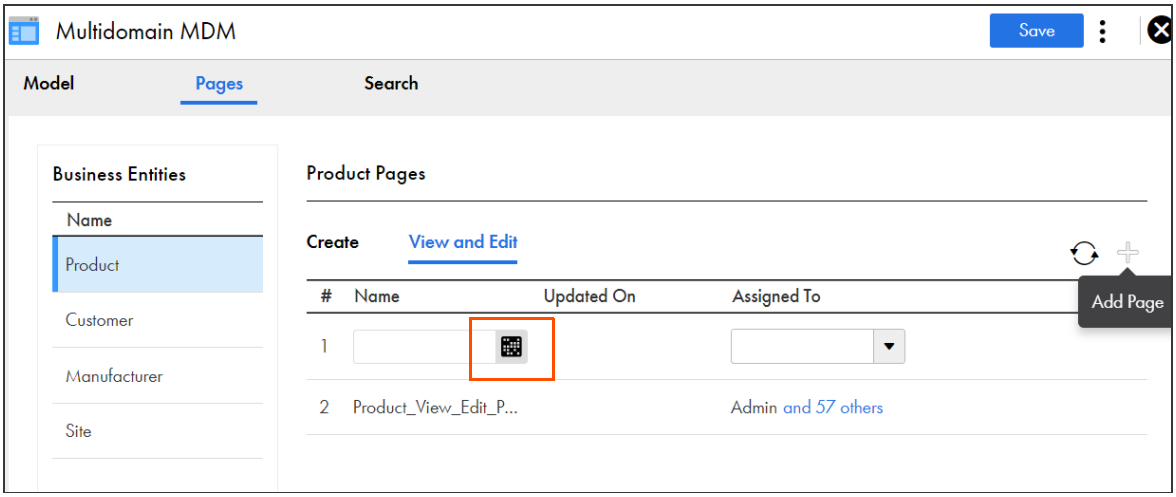
Add View and Edit Page to the Product Business Entity

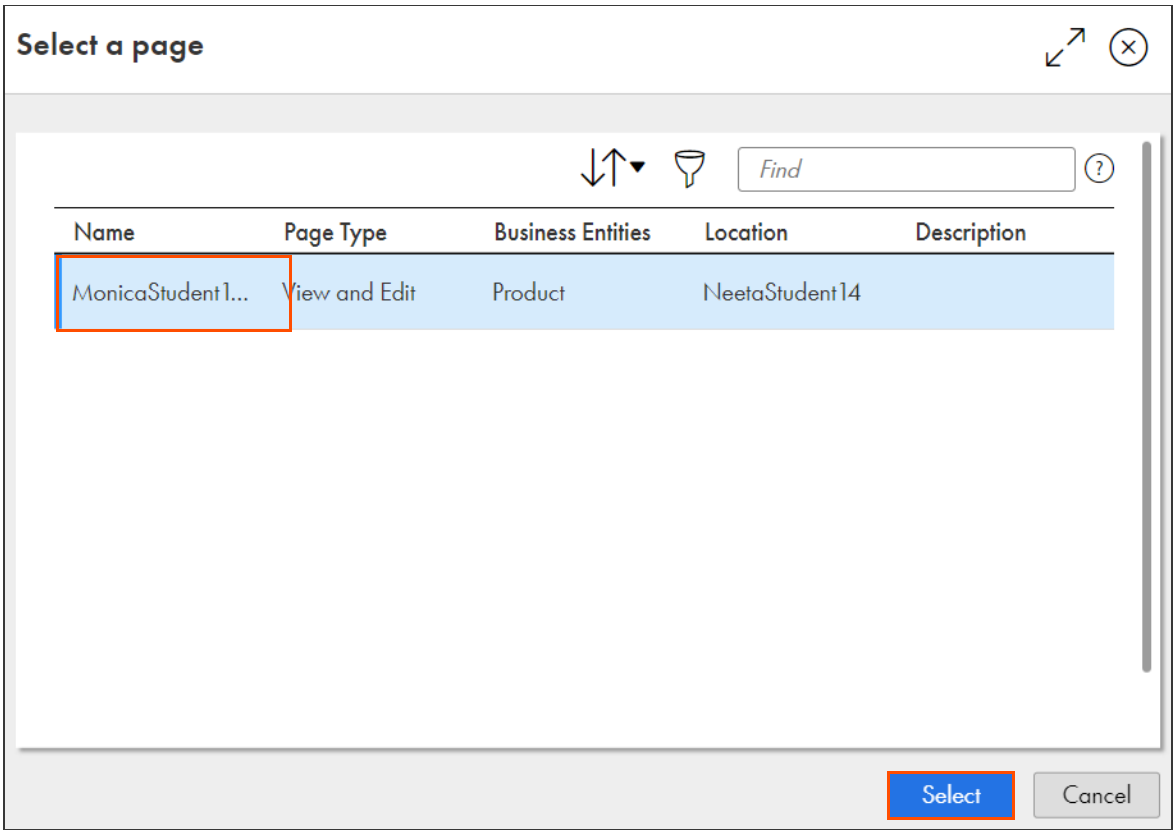
Next, you will add custom view and edit page to the Product business entity.

16. Select **Product** business entity. Under **View and Edit** tab, Click **+** icon.



17. Click the Asset Picker icon and select TrainingStudent01_Product_ViewEdit page. Click **Select**.

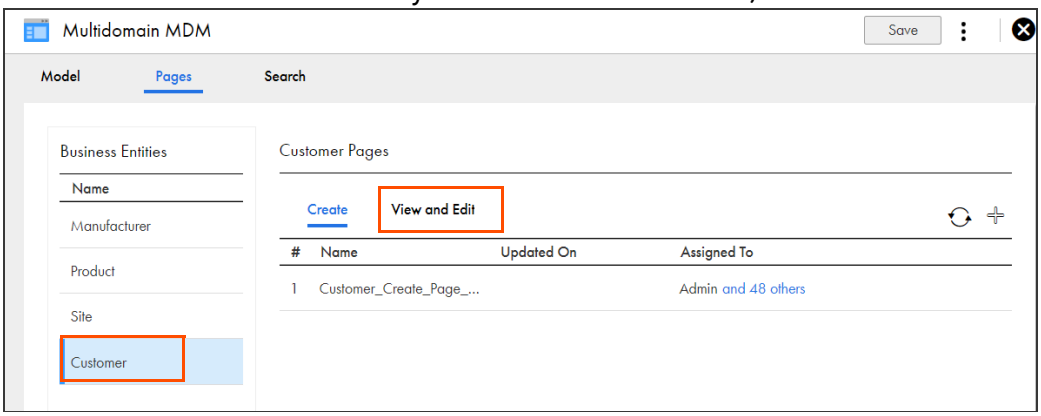




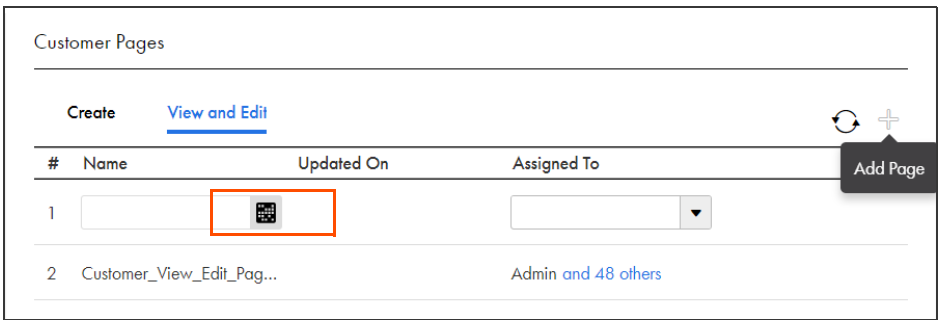
Add View and Edit Page to the Customer Business Entity

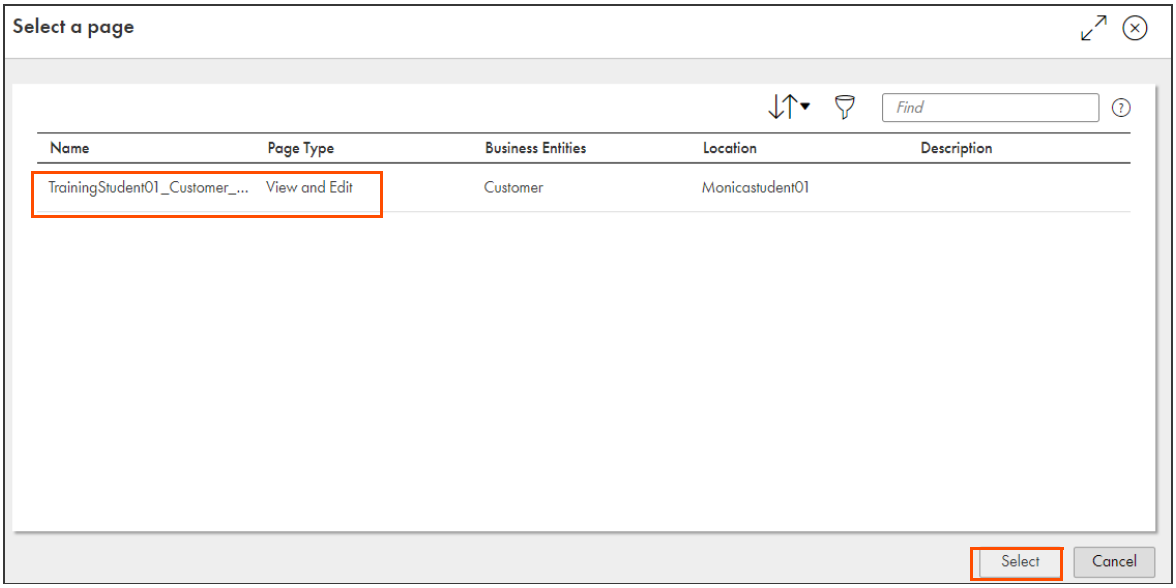
Next, you will add custom view and edit page to the Customer business entity.

18. Select **Customer** business entity. Under **View and Edit** tab, Click **+** icon.



19. Click the Asset Picker icon and select TrainingStudent01_Customer_ViewEdit page. Click **Select**.

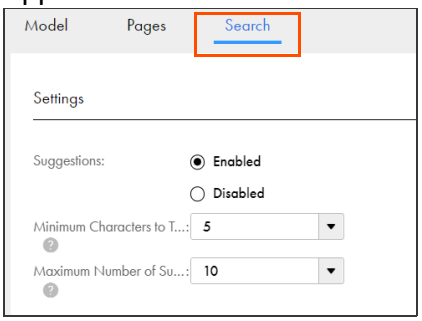




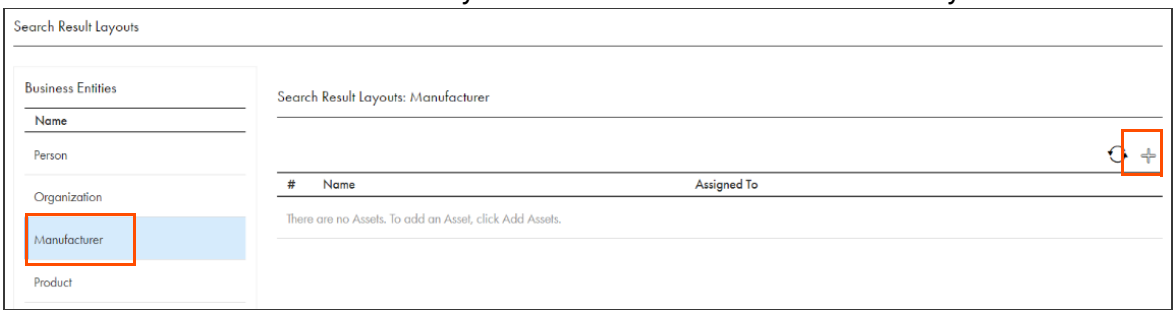
Add Custom Search Layout to the Multidomain MDM Application

Next, you will add custom search layout to the Multidomain MDM Application.

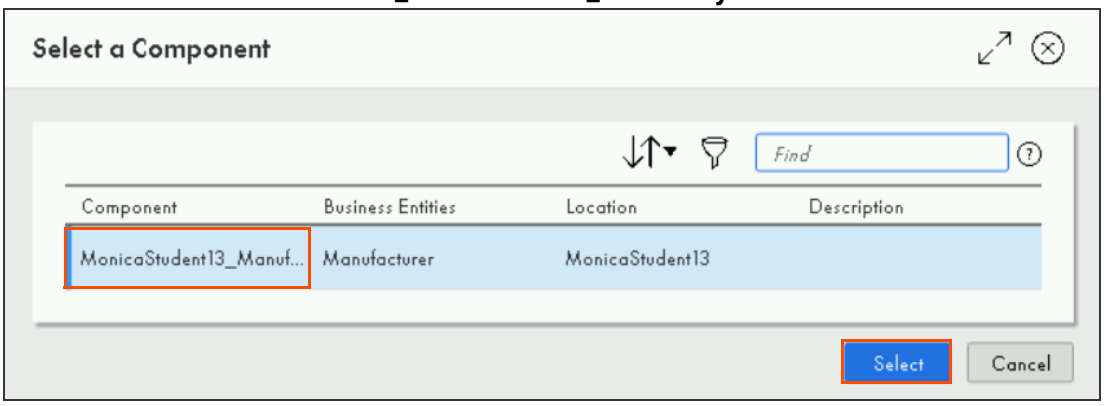
20. Click the **Search** tab. Here, you will add the custom search layout to the Multidomain MDM application.



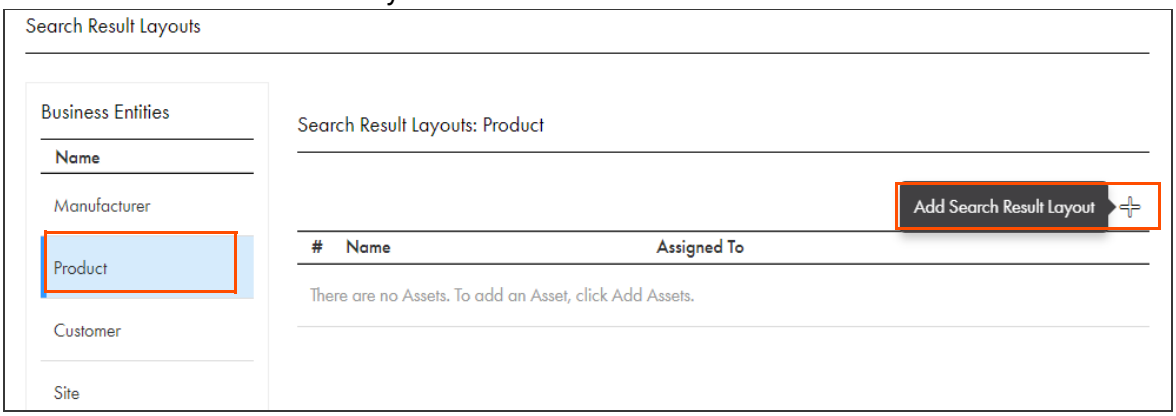
21. Select **Manufacturer** business entity and click **+** icon to add the search layout.



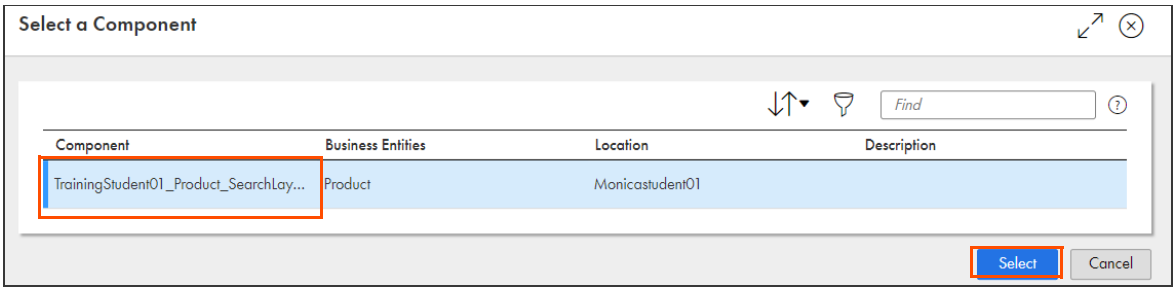
22. Under Name, click Asset Picker icon. Select **StudentNameStudentNumber_Manufacturer_Searchlayout** and click **Select**.



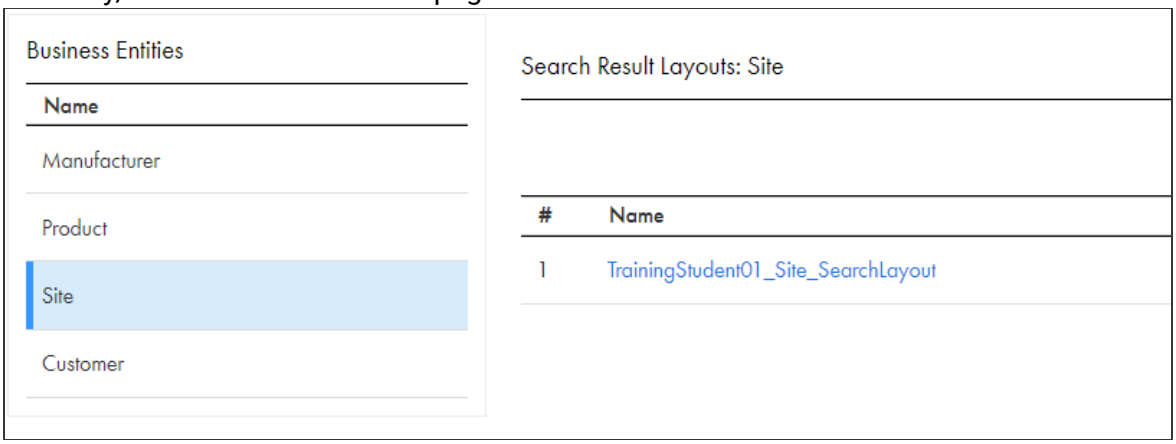
23. Select **Product** business entity and click the **+** icon.

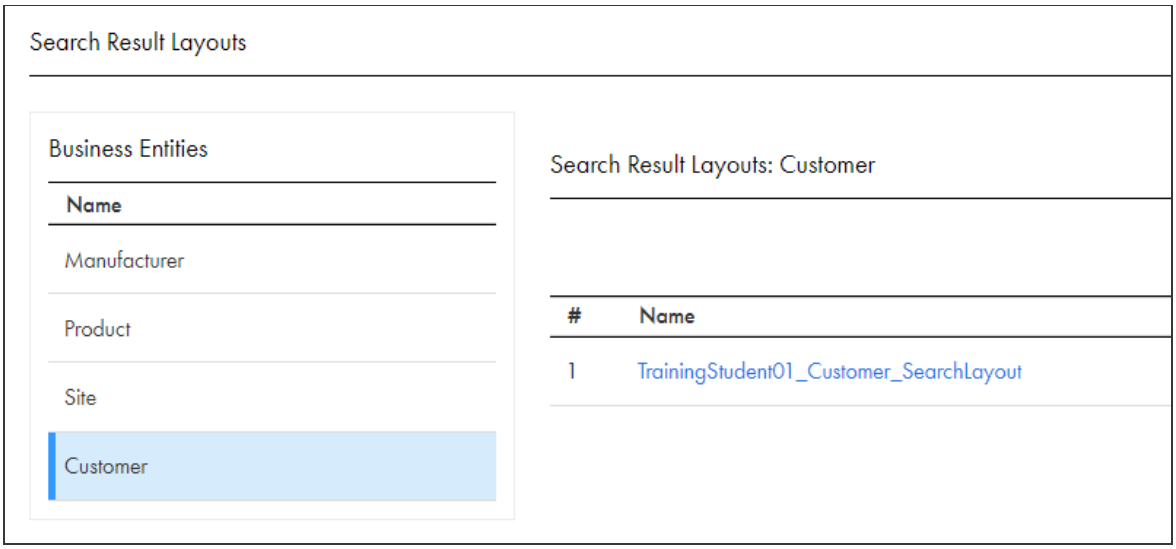


24. Click the Asset Picker icon, select **Trainingstudent01_Product_search_layout** page and click **Select**.

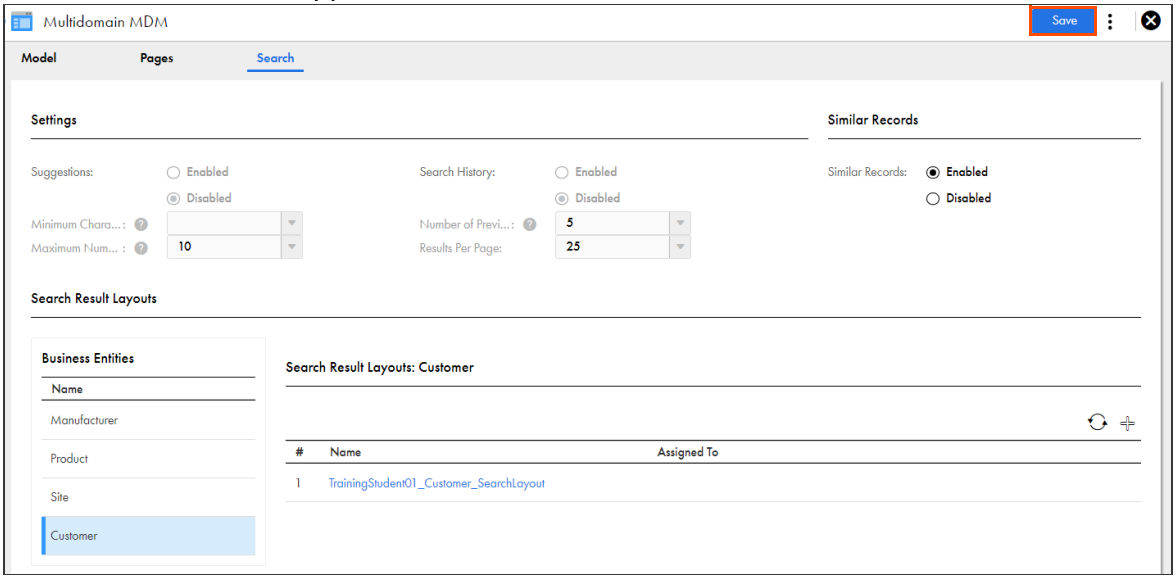


25. Similarly, add the custom search pages to the **Site** and **Customer** business entities.

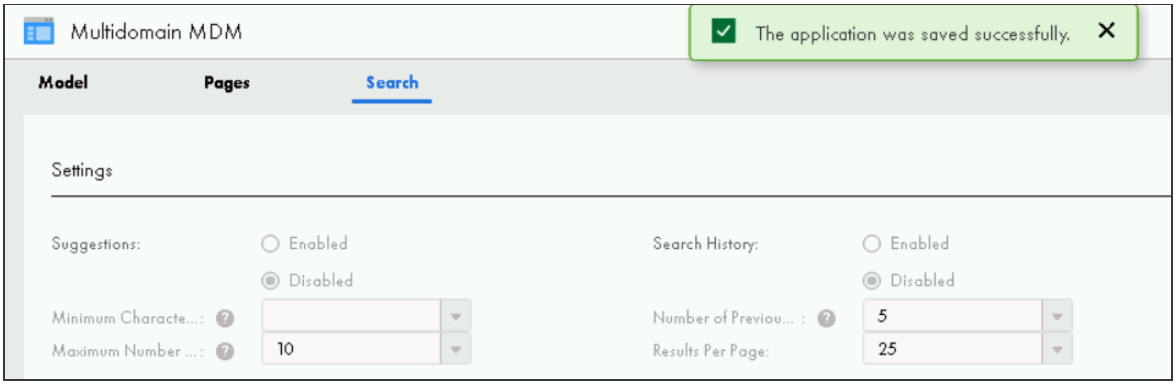




26. Click **Save** to save the application.



You will see the following message:



This is how you add the objects to the Multidomain MDM application.

This concludes the lab.

Key Takeaways

- A user interface includes pages, icons, buttons, and other elements that help you interact with your business application.
- You can create and configure custom pages for your application.
- The components are the building blocks of the custom page.
- You can add a header to a custom page to display the field data that you want to highlight for a record.
- When you add the Fields component to a custom page, you can add fields and field groups.

Module 7: Configuring Data Quality and Survivorship

Data Quality Overview

Master data should always be a clean, trustworthy, and a good quality data. To achieve this, you can associate data quality rules with the business entity fields to cleanse, standardize, or validate the field values. For example, you can remove unwanted punctuation marks or spaces, standardize the date format, or verify the phone numbers. Data quality rules help improve the quality and consistency of data.

You can associate data quality rules based on the simple conditions, Cloud Data Quality, and DaaS with business entity fields. You can also add integrity checks to business entity fields only. When you add or import records, the rule associations validate and transform the records, and store them in the data store.

When validation fails, you can downgrade the trust score of the field, and that helps determine the confidence level of the source data.

Types of rule associations

You can create different types of rule associations for business entity fields. You can use the following rule association types:

- Basic rule associations
- Advanced rule associations
- DaaS rule associations

Basic rule associations

A basic rule association is an association between a business entity field and a simple condition-based rule. You can create basic rule associations using the predefined functions such as Regular Match and Concatenate.

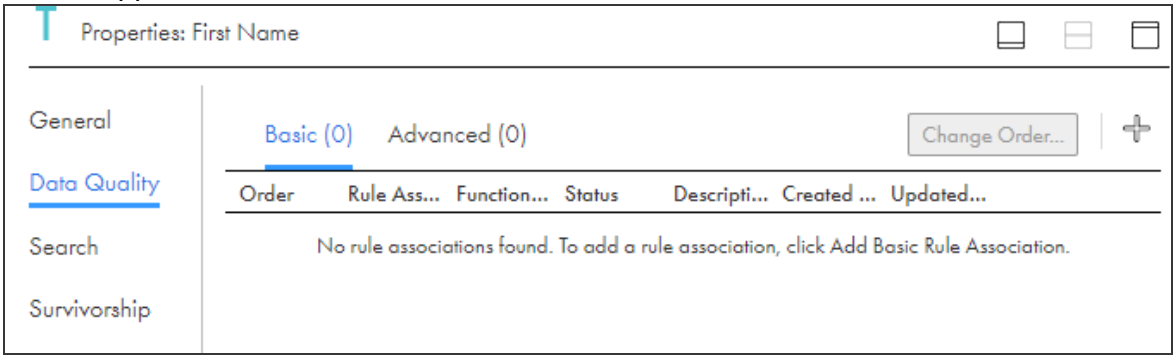
For example, Smith Store wants to master data from a newly acquired business. Smith Store wants to combine the first name and last name of each customer to generate full name to print on the receipt. Smith Store can associate a basic rule to the Full Name field and concatenate the first name and last name values.

You can group the predefined functions into the following categories:

- Validation: Validates whether data matches the specified condition. When you use a validation function, you must configure a validation message. **Examples**: Maximum Length, Minimum Length, Starts With, Ends With, No Spaces, Contains, Does Not Contain, Allowed Characters, Regular Match, Date Range, Greater Than Date, Less Than Date, No Future Date, Maximum Decimal Scale, Minimum Value, Maximum Value, and Maximum Precision
- Transformation: Transforms source data. **Examples**: Concatenate, Concatenate With Spaces, Uppercase, Lowercase, Title Case, Left Trim, Right Trim, Replace

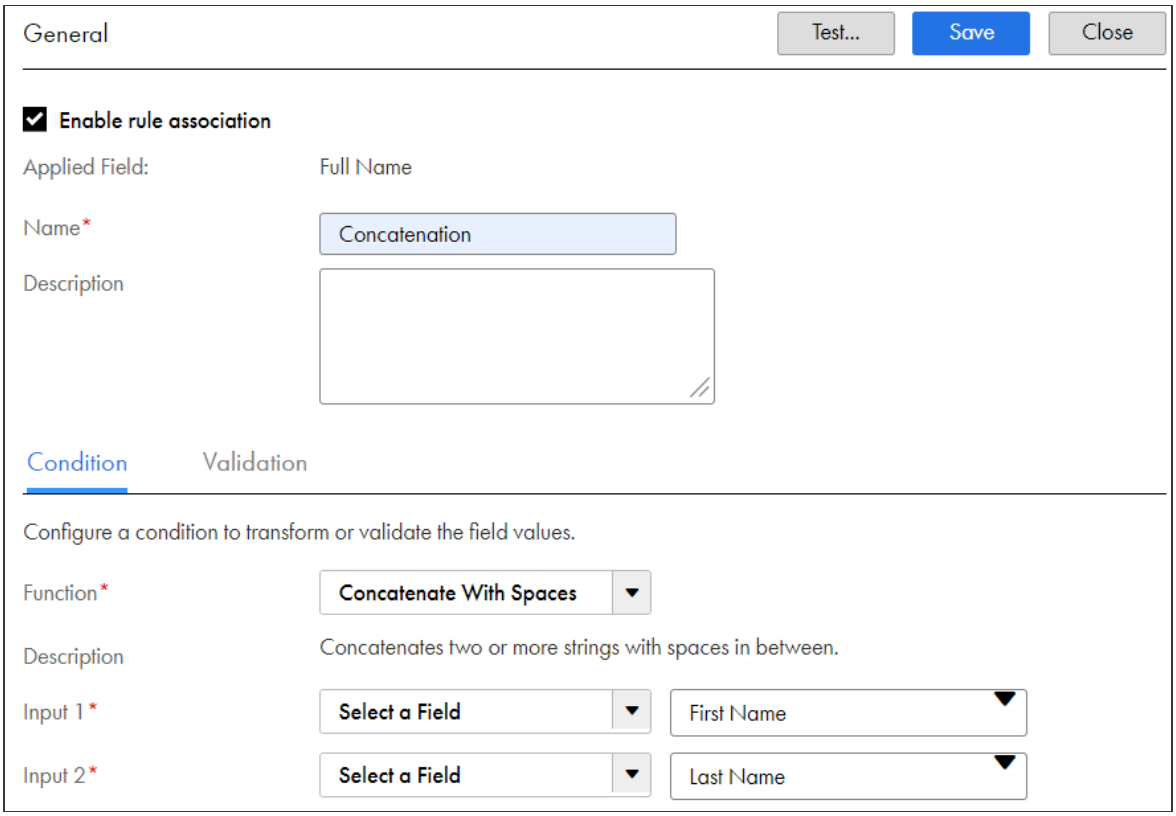
Steps to Add Basic Rule Association

1. On the business entity page, select a field to associate rules.
2. In the Properties section, click **Data Quality > Basic.**
3. Click Add **Basic Rule Association**.
   A form appears.



4. In the General section, enter a unique name for the rule association.
5. Optionally, enter a description.

6. On the Condition tab, perform the following tasks:
   - Select a function.
   - For the input field, select an option from the list.
   - If you select Select a Field, select a business entity field from the list.
   - If you select Enter a Value, type a value for validation.



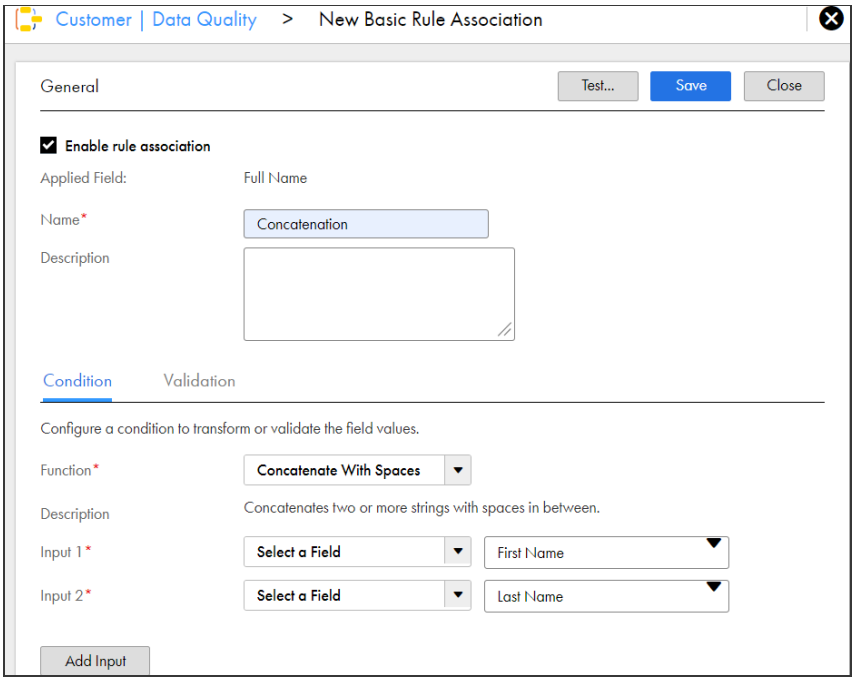
7. If you select a validation function, on the Validation tab, perform the following tasks:

- Select one of the following severity levels for the validation failure:
  - Information: For minor data quality issues. The users can save the data.
  - Warning: For less severe data quality issues. The users can save the data.
  - Error: For severe data quality issues. The users cannot save the data.
- Enter an error message to display.
- Enter the percentage of trust score to downgrade when the validation fails.

8. To verify the created rule, perform the following tasks:

- Click **Test**.
  The Test dialog box appears.
- Enter values in the Input column, and click **Run**.
  The validation result appears in the Output column.
- Click **Close**.

9. Click **Save**.

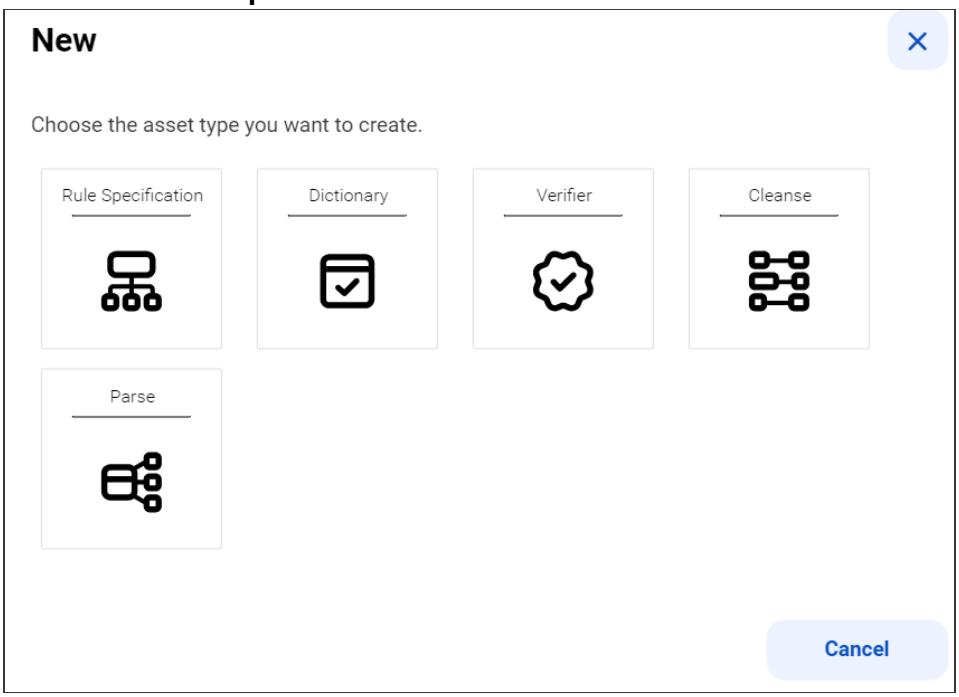


Adding Advanced Rule Associations

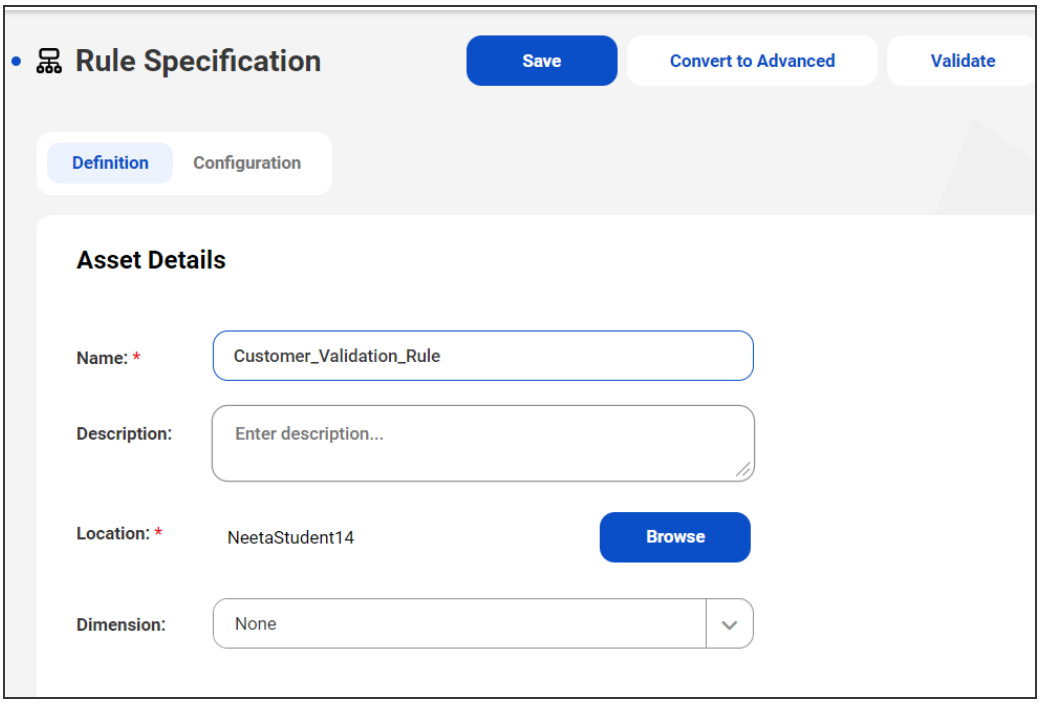
An advanced rule association is an association between business entity fields and a Cloud Data Quality rule specification. You can use the predefined rule specifications or create custom rule specifications in Cloud Data Quality and use them. The supported predefined rule specification is DUNS SSN Validation that validates the DUNS and Social Security numbers.

1. Navigate to Data Quality from the IICS homepage.

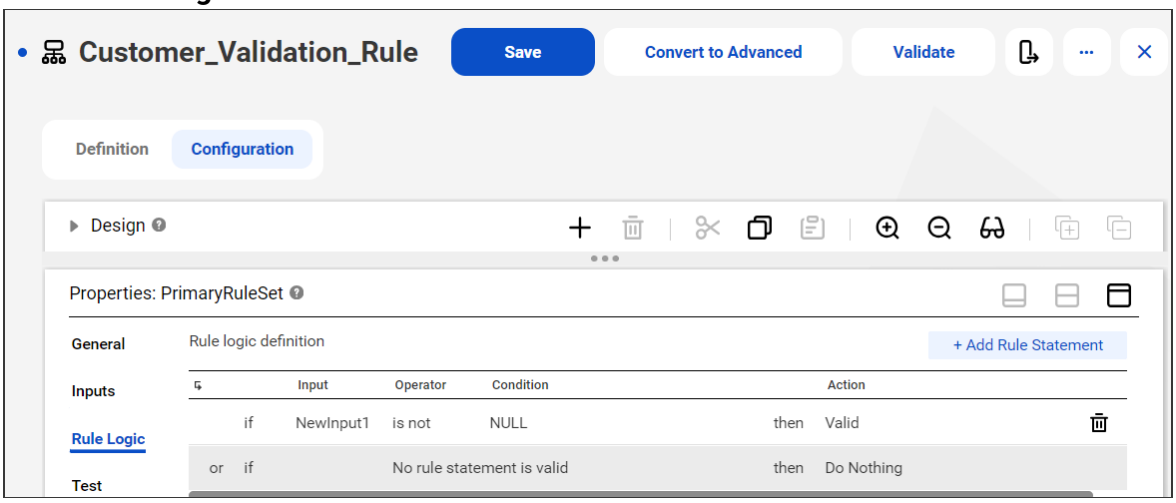
2. Click **New** > **Rule Specification**.



3. In the Definitions tab, enter the details.

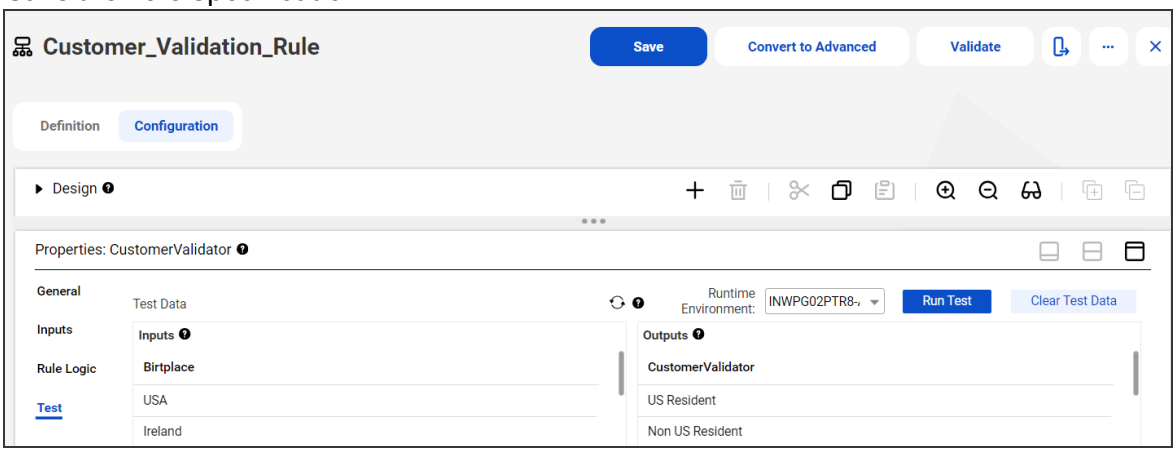


4. Click the **Configurations** tab.



5. Create a Rule Logic.

6. **Save** the Rule Specification.



Uniqueness Criteria

You can configure the de-duplication of multivalued field groups for a custom field group. Customers can add uniqueness criteria based on which de-duplication will happen. Currently, users can set up uniqueness criteria based on the mandatory fields of a field group. This feature supports exact matching and not fuzzy matching.

In the uniqueness criteria, multiple field group values are qualified as unique based on the required

fields in a field group. Based on the fields that you configure with the uniqueness criteria, duplicate values are merged.

Survivorship for a Business Entity

Survivorship is the process of creating a master record or the most trusted version of a record from multiple matching source records. The survivorship process begins after the matching process identifies the matching source records. You must run a match and merge job to create master records based on the survivorship configuration. To configure survivorship, you define survivorship rules and rank source systems from which records originateee.

Key Survivorship Concepts

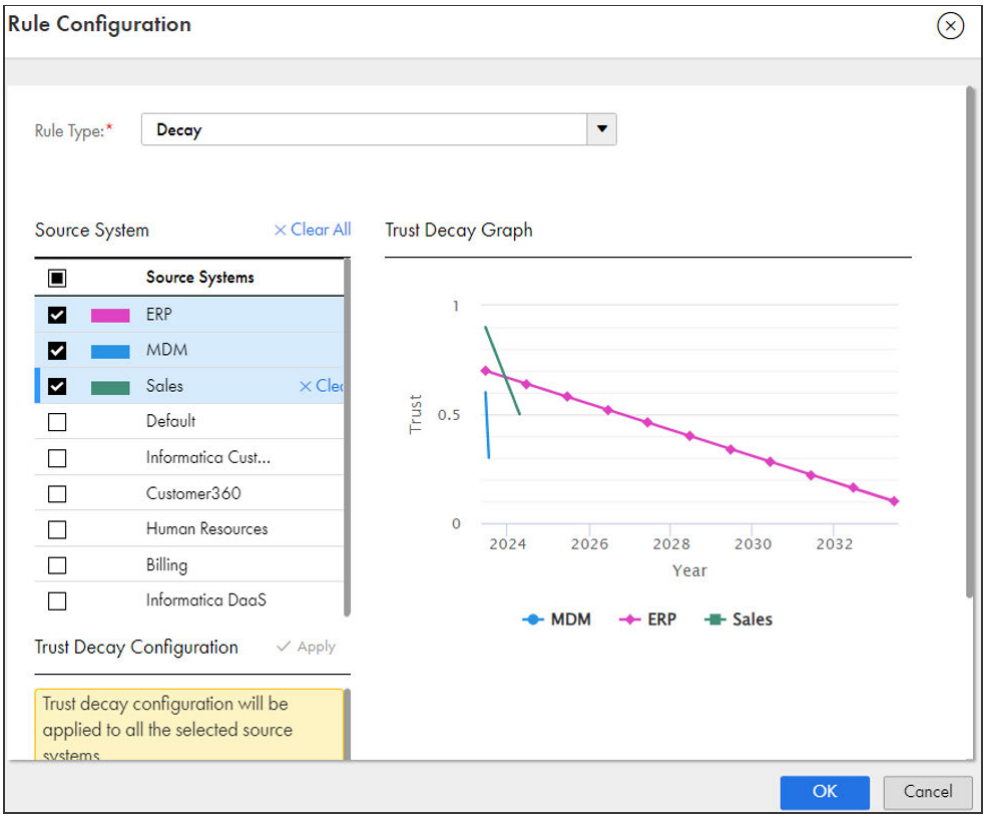
Survivorship configurations contain rules, source system ranks, or both. You must configure survivorship for business entities before users create records.
Before you configure survivorship, understand key concepts, such as source ranking, rules, and the source last updated date.

- Source Ranking
- Rule
- Decay

Source ranking: Determines the reliability of source systems. Based on the accuracy and credibility of data that source systems provide, you can assign ranks. A higher rank indicates that the data from that source system is more reliable than the data from a source system with a lower rank. Consider the Informatica MDM source system as the most reliable source system and rank it as 1. Data from the MDM source system becomes the reliable data for the master record.

Rule: Defines the conditions for one or more fields and field groups or all the fields in the business entity to survive. Based on the survivorship configuration, a rule identifies the trusted value for a field. You can add, edit, or delete survivorship rules.

Decay : Identifies the trusted value based on the trust levels and the decay rate of the field values. The trust levels indicate the confidence you have in the source system. The decay rate indicates the rate at which the trust level must decrease over a period of time. You can configure different trust levels and decay rates for each source system. When the trust level of a source system decreases, the rule identifies another source system with a higher trust level during that time period.

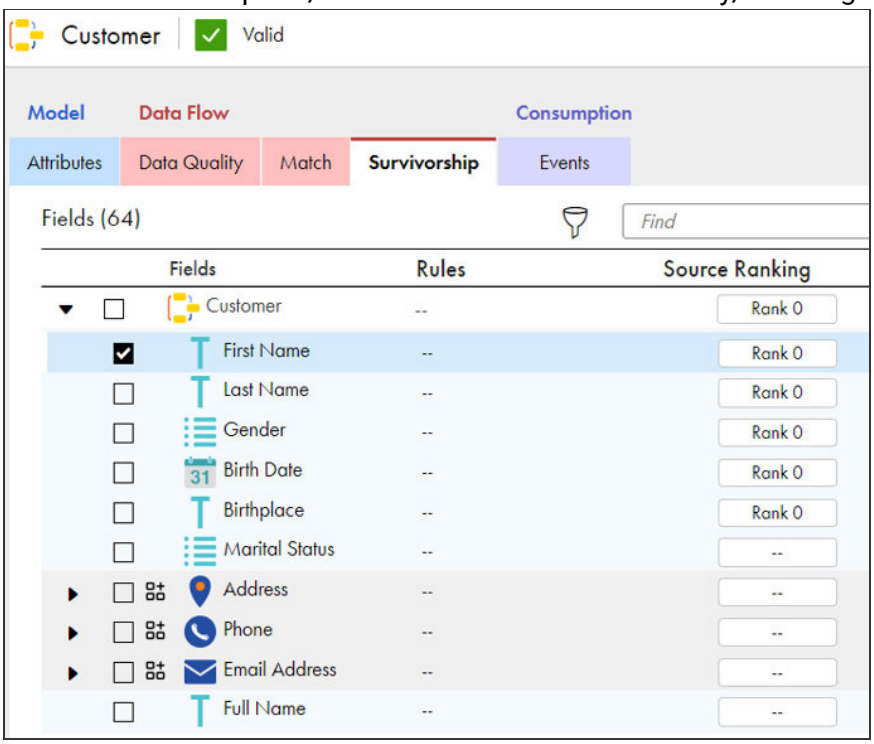


Configuring Survivorship for a Business Entity

You can configure survivorship for one or more fields in a business entity.

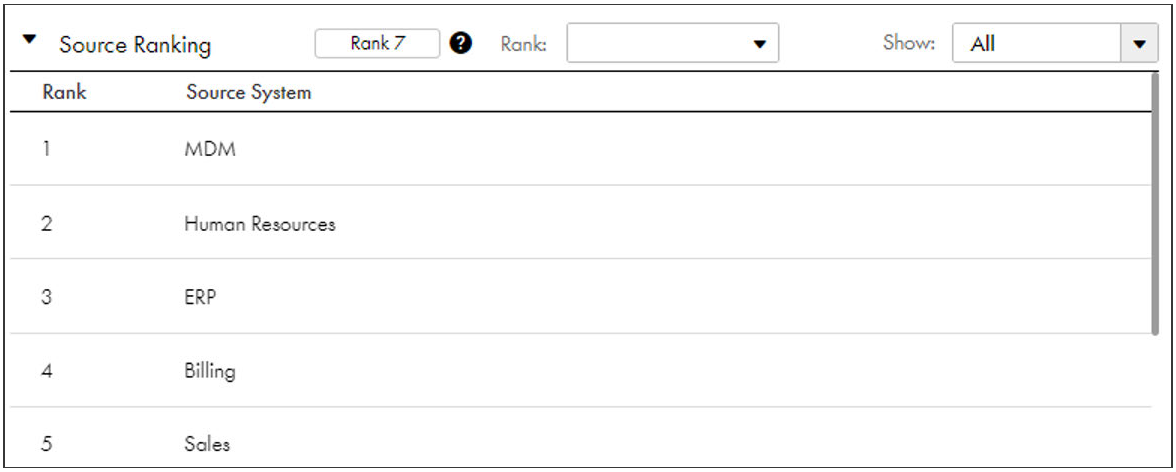
1. On the **Explore** page, choose the required business entity.

2. On the Survivorship tab, select either the Business entity, or a single field or multiple fields.



3. To rank the source systems, in the Source Ranking section, perform the following steps:

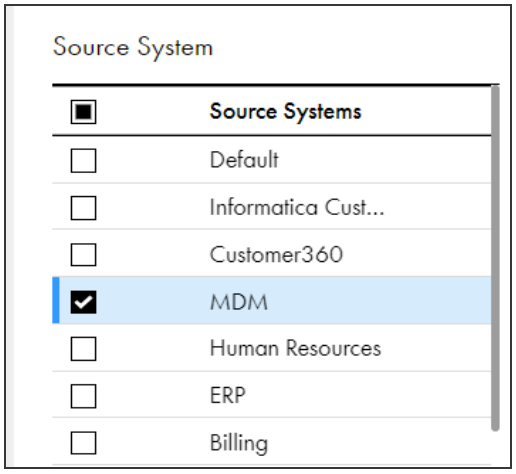
- To add or remove a source system from source ranking, select the source system, and click **Include in Ranking** or **Exclude from Ranking**.
- To change the ranking order, select the source system, and use the up or down arrows as required.
- To use an existing source ranking, click **Change**, and choose an available source ranking.



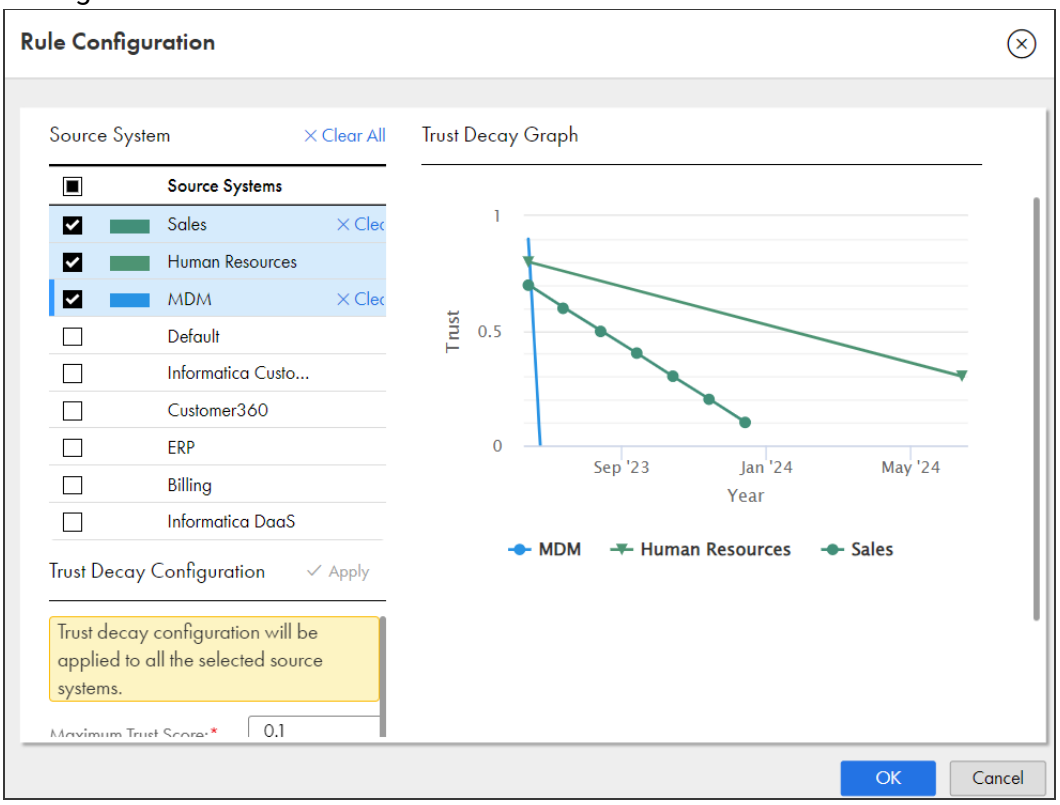
4. Click **Save**

Decay Rule for Source Systems

1. To set a survivorship rule for the selected objects, in the Rule section, select **Decay**.
2. To configure or edit the decay settings, click **Configure**.
3. Select one or more source systems from the table.



4. To set the trust decay configuration, enter the trust scores and click **Apply**. Save the Rule Configuration.



Lab 7-1: Adding Basic and Advanced Data Quality Rules

Overview:

You can associate data quality rules to the business entity fields to cleanse, standardize, or validate the field values according to your business requirements. In this lab, you add basic and advanced rule associations and import the advance rule into Business 360.

Objectives:

- Add Basic rule association
- Add Advanced rule association

Scenario:

Mark from Homekart wants to standardize the data that he will be onboarding later. He decides to cleanse, standardize, and enrich the data by using some data quality rules. Data quality rules help improve the quality and consistency of data. As there are some IICS microservices coupled with Multidomain MDM, Mark uses Cloud Data Quality to enhances the quality of data.

Duration:

40 Minutes

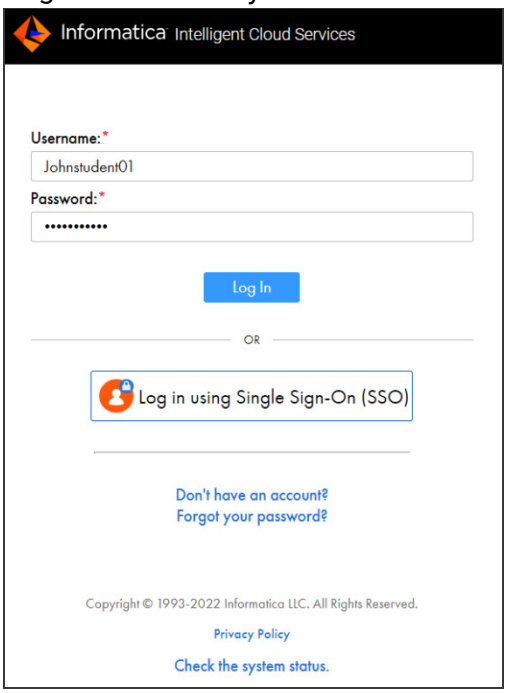
Tasks:

- Create a Basic rule association
- Test the rule in Multidomain MDM
- Add Advanced rule association
- Import rule specification to Business 360
- Test the Advanced rule association in Multidomain MDM

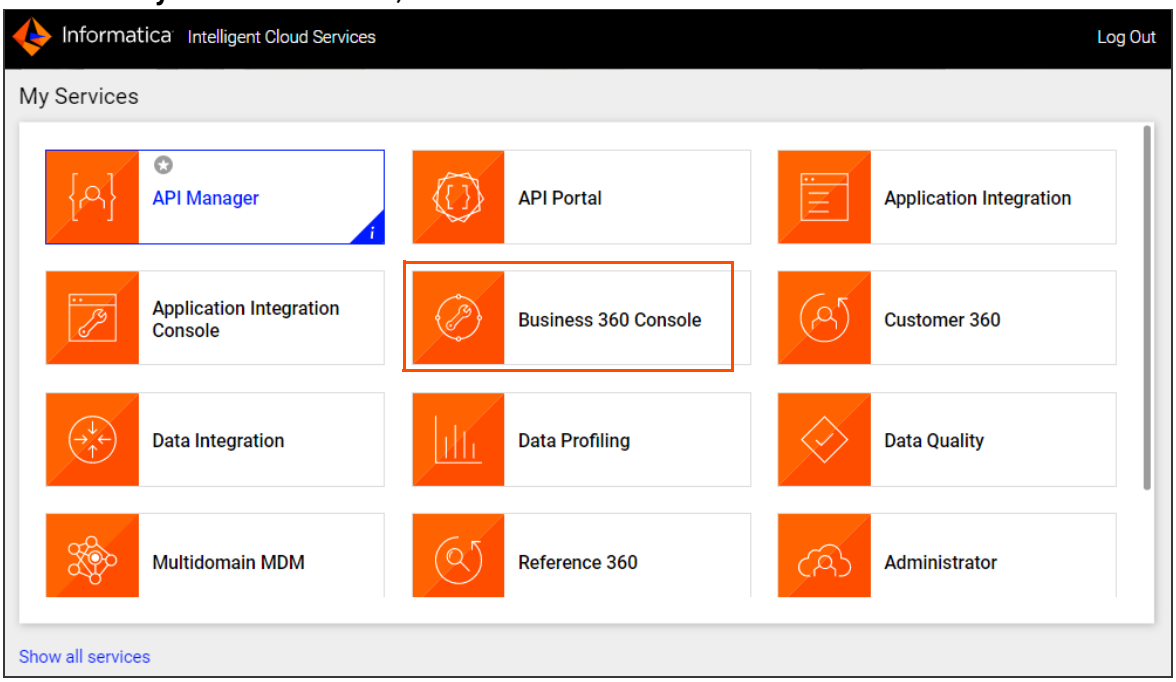
Create a Basic Rule Association

A basic rule association is an association between a business entity field and a simple condition-based rule. You can create basic rule associations using some predefined functions. In You will select a field on the business entity page and add basic rule association to it.

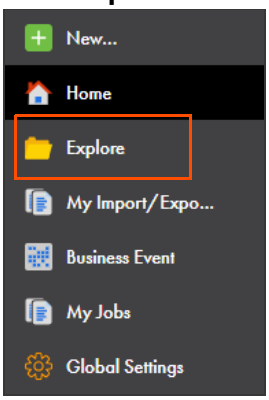
1. Log in to IICS with your admin credentials (**studentNamestudent<#> / Multidomain360**).



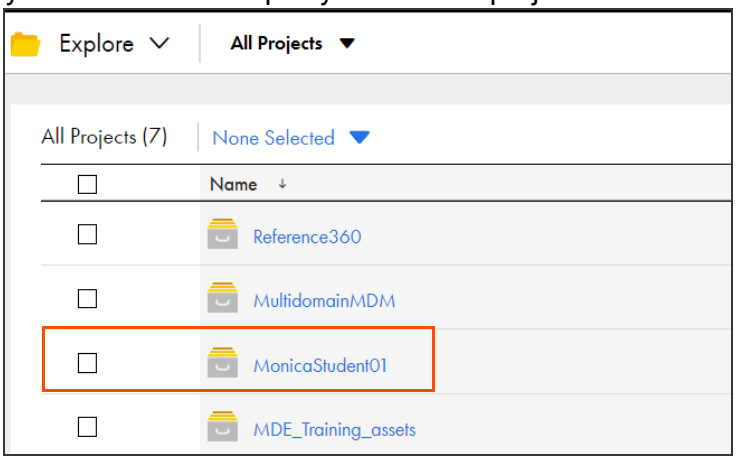
2. From the **My Services** window, click **Business 360 Console**.



3. Click **Explore**.

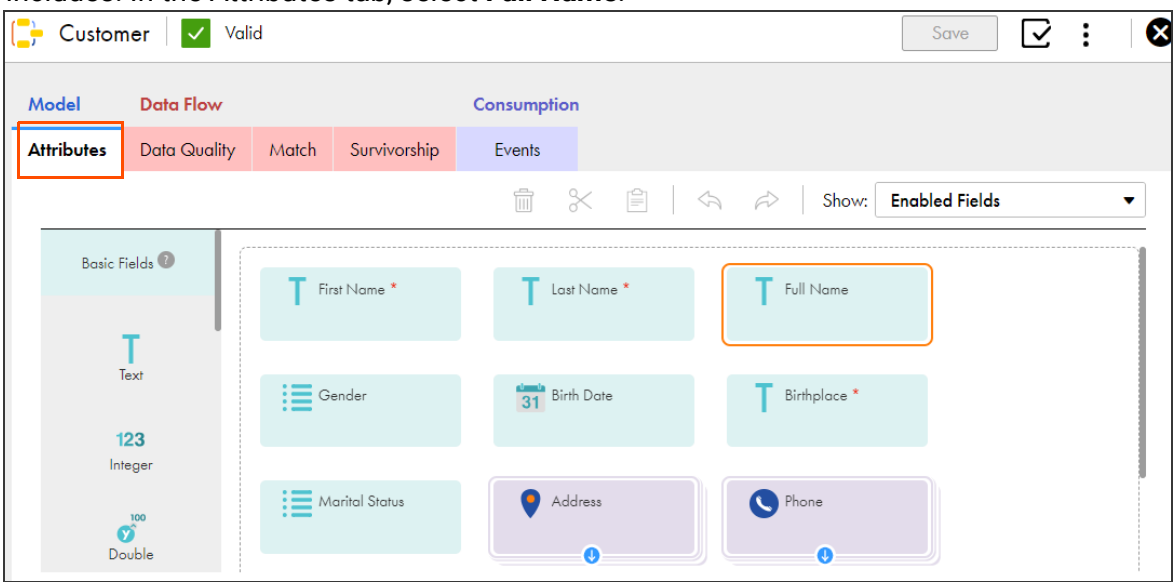


4. In the Explore page, you can see the projects, folders, and assets for all the services that you use. Click and open your student project.



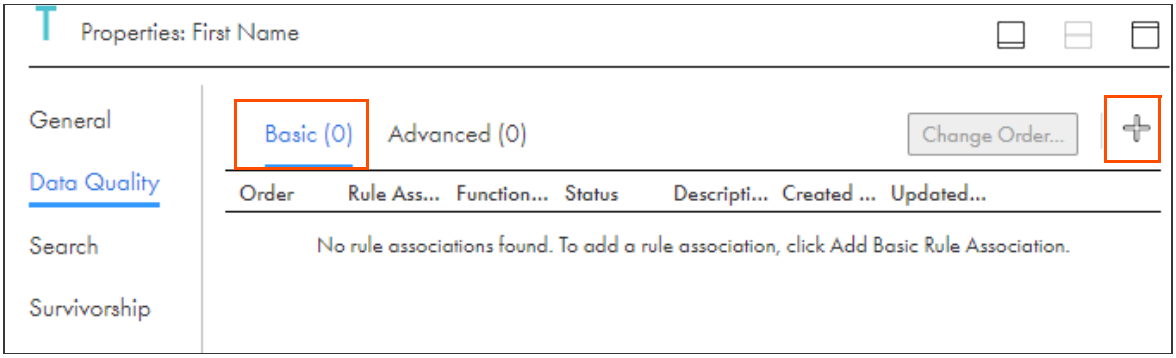
5. Open the **Customer** business entity.

6. The business entity page opens. The canvas displays all the fields that the business entity includes. In the Attributes tab, select **Full Name**.

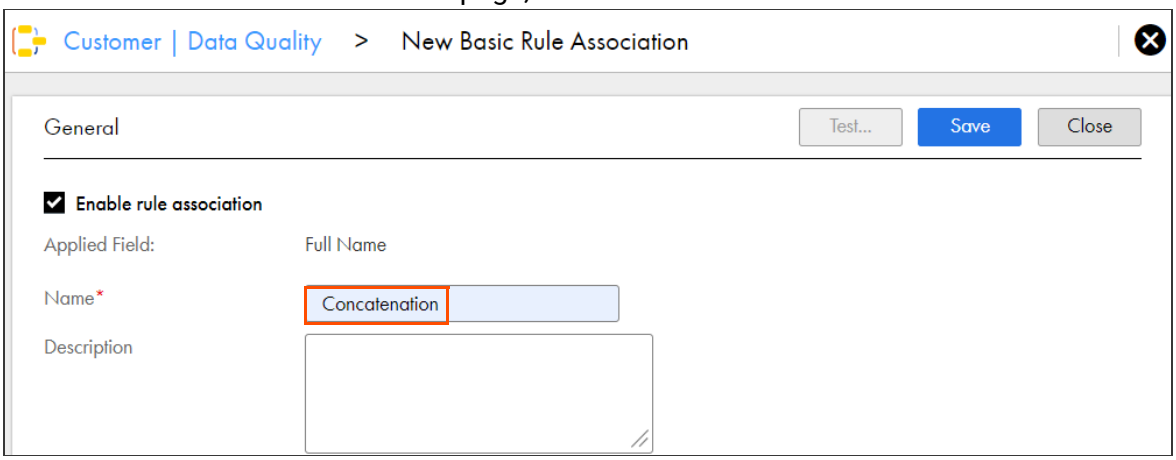


7. In the **Properties** section, click **Data Quality**.

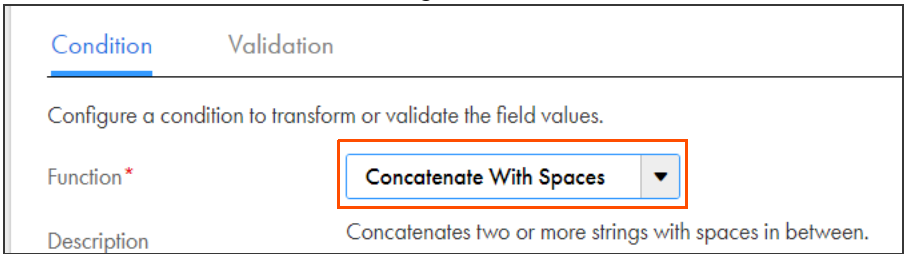
8. From the **Basic** tab, click **+** to add a new basic rule.



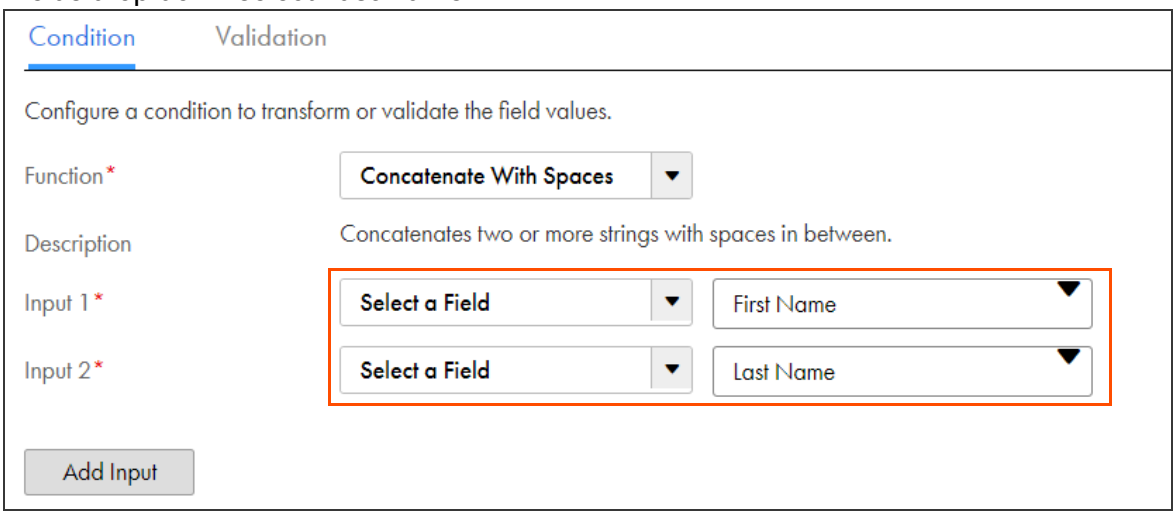
9. In the **New Basic Rule Association** page, enter **Concatenation** for Name.



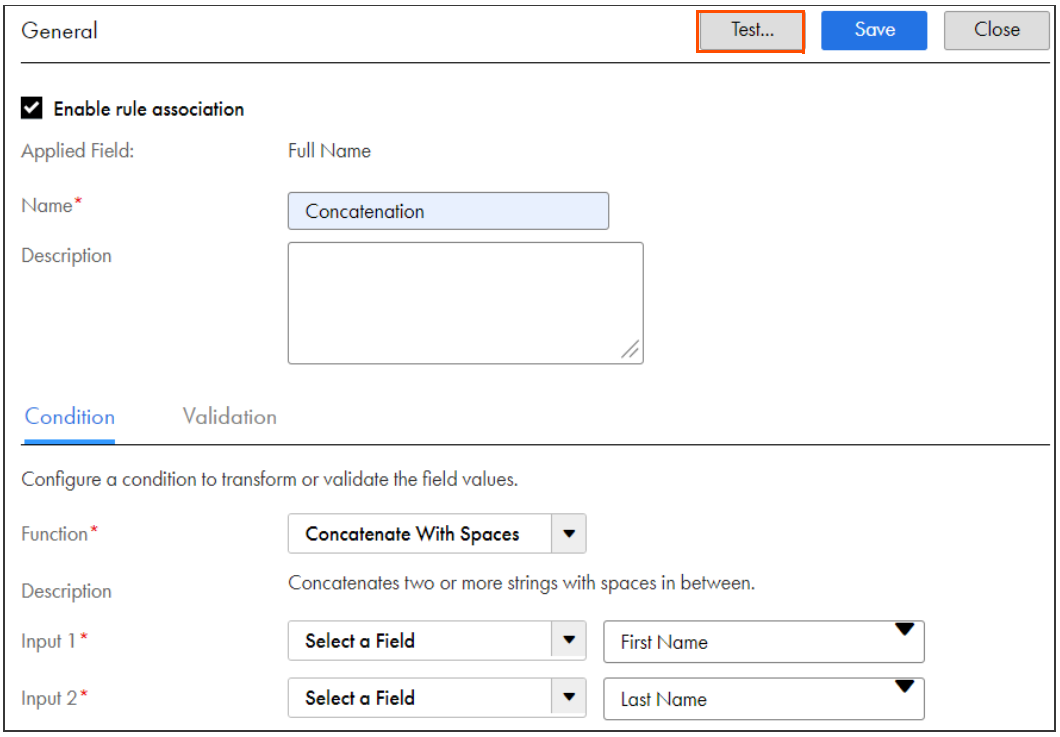
10. In the **Condition** tab, from the Function drop-down, select **Concatenate With Spaces.** This function concatenates two strings.



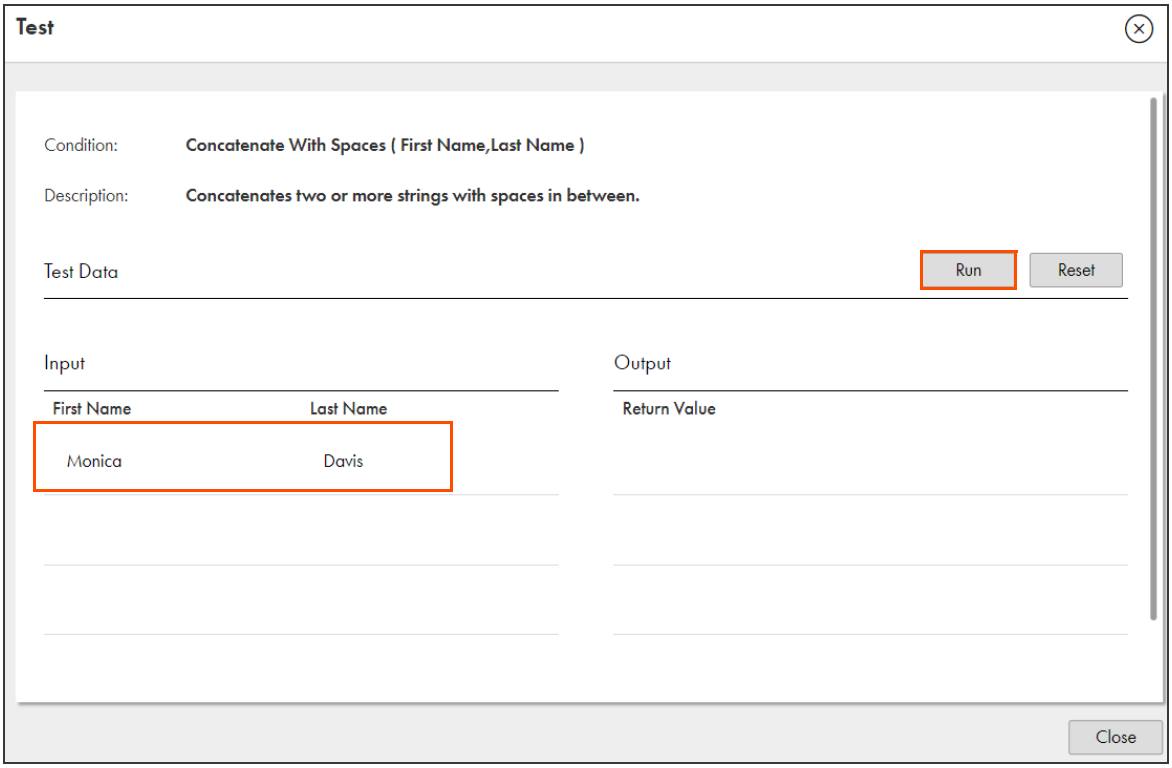
11. In the **Input1**, from the drop-down, select **Select a Field** and in the Fields drop-down select **First Name**. Similarly, in the **Input2**, from the drop-down select **Select a Field** and in the Fields drop-down select **Last Name**.



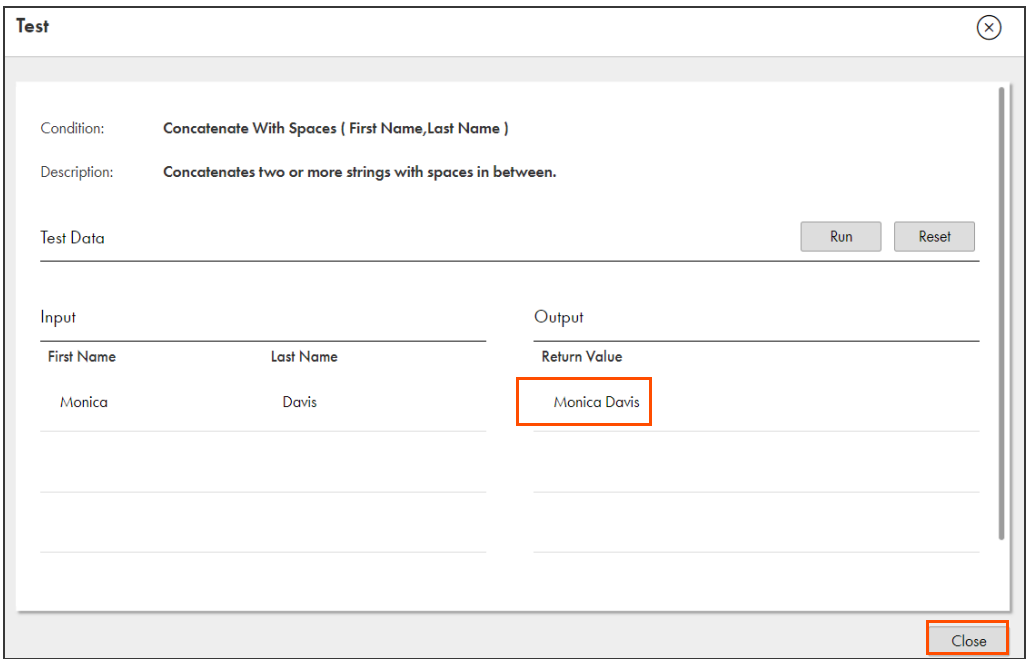
12. Click **Test**.



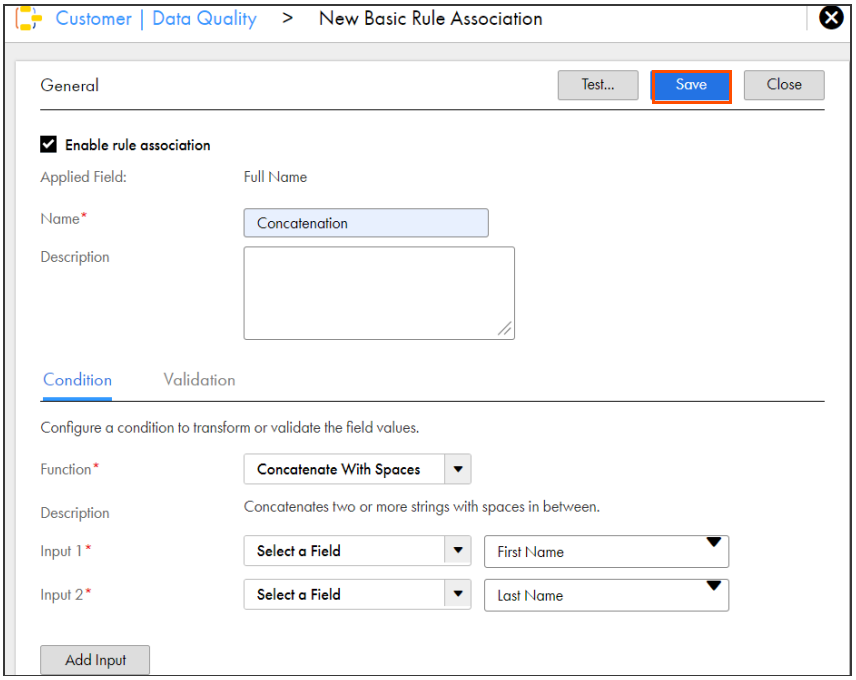
13. In the **Test** page, under the Input tab, in the First Name field, enter **Monica** and in the Last Name field, enter **Davis** and Click **Run**.



14. Notice that the first name and last name are concatenated. Click **Close** to close the window.

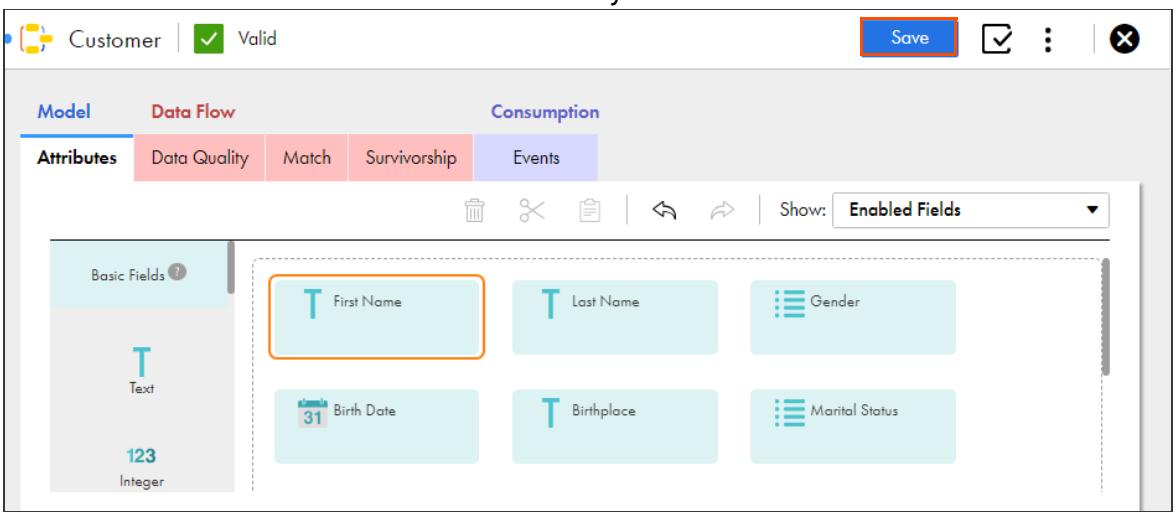


15. Click **Save**.



A notification with the message “Rule association is added successfully” is displayed.

16. Click **Save** to save the customer business entity.

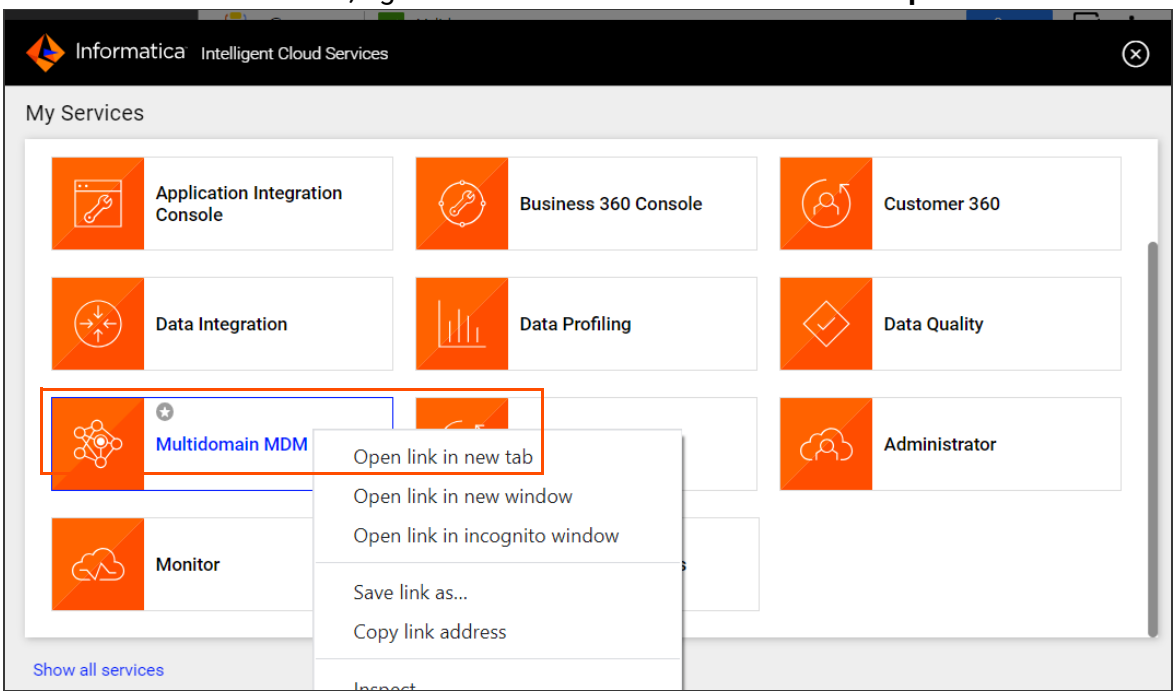


Test the Rule in Multidomain MDM

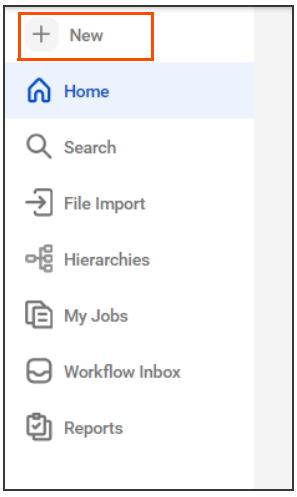
You will now add a new record in Multidomain MDM to test if the basic rule created works as expected.

17. To switch to Multidomain MDM, click **Business 360 Console** at the top of the page.

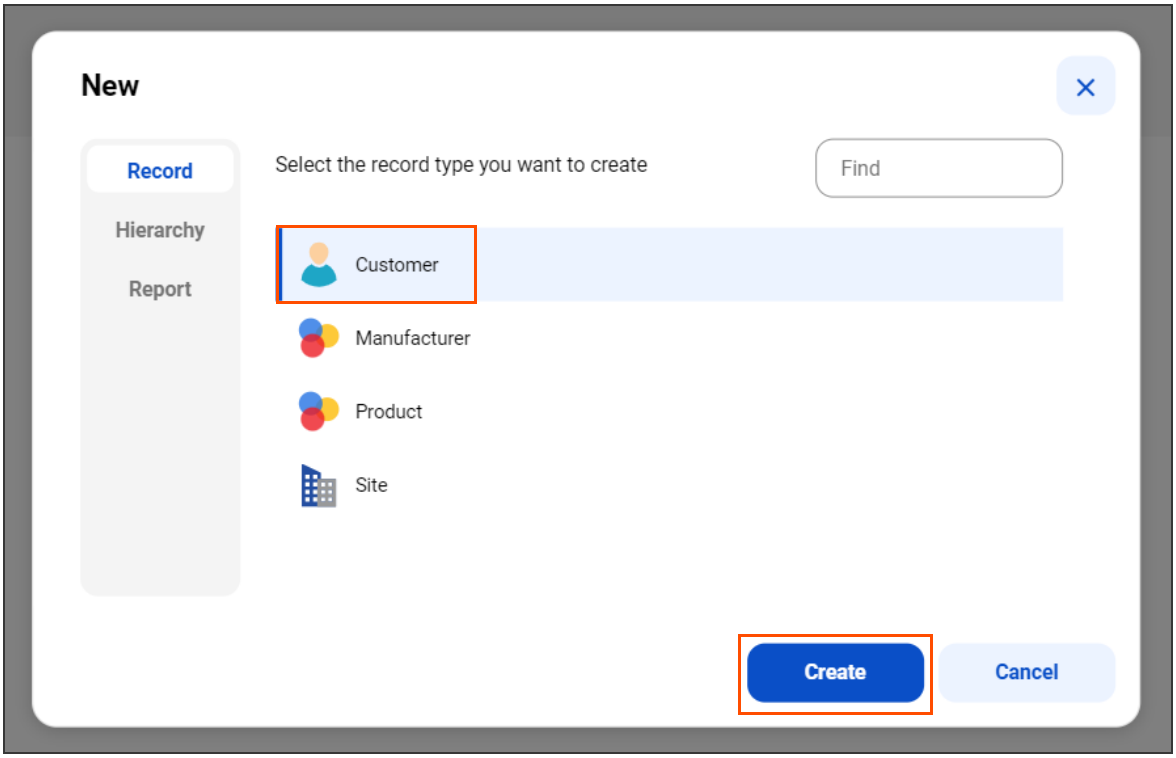
18. From the **Services** window, right-click **Multidomain MDM** and select **Open link in new tab**.



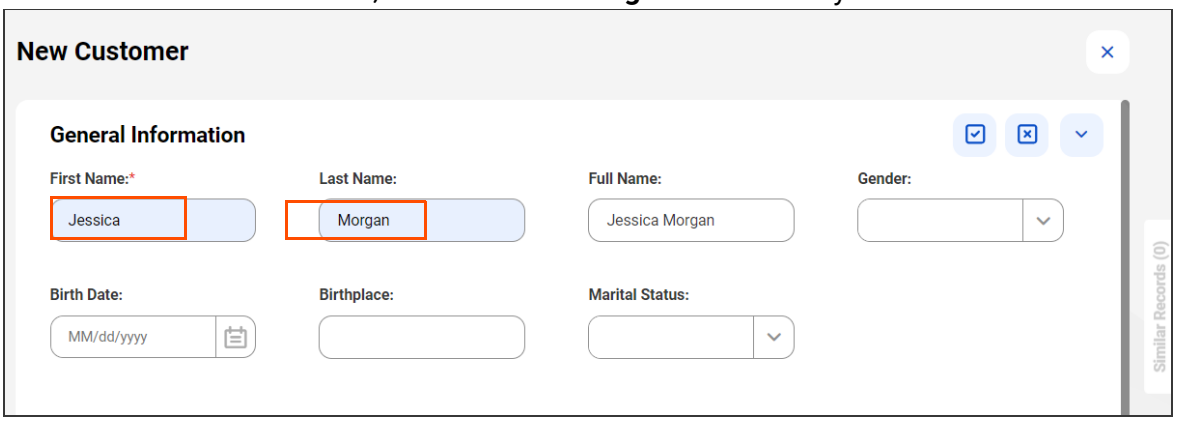
19. In the Multidomain MDM user interface, click **New**.



20. Under Record, select **Customer** and click **Create**.

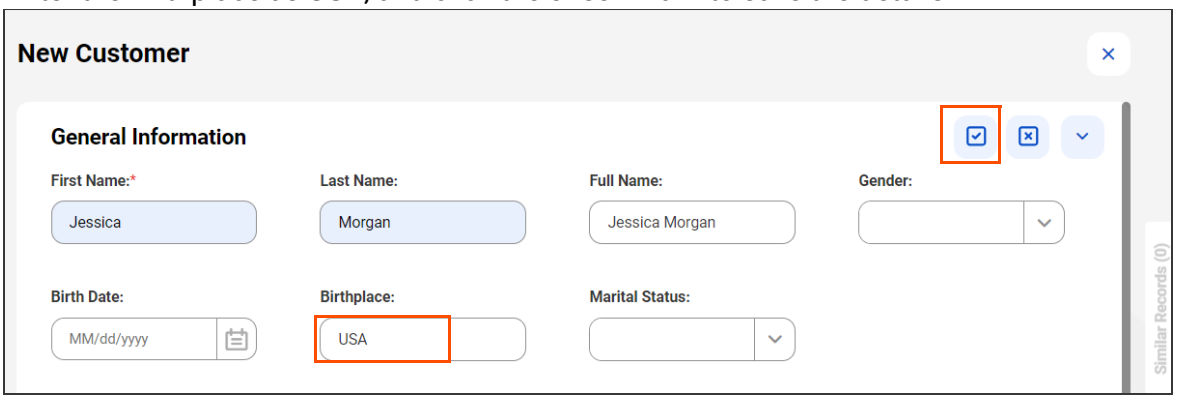


21. In the First Name as **Jessica**, Last Name as **Morgan** and click anywhere else in the window.

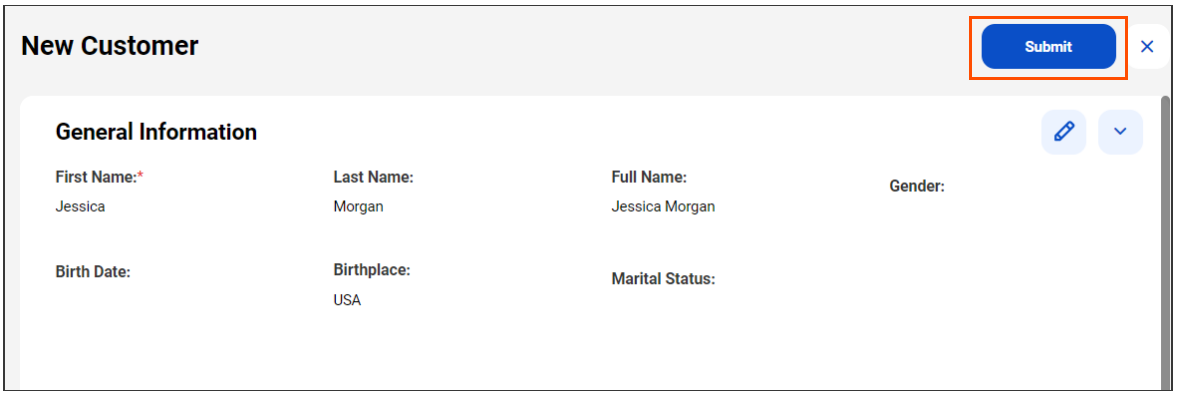


You will see that the First Name and Last Name are concatenated and displayed in the Full Name field.

22. Enter the Birthplace as **USA**, and click the **check mark** to save the details.



23. Click **Submit** to submit the record.



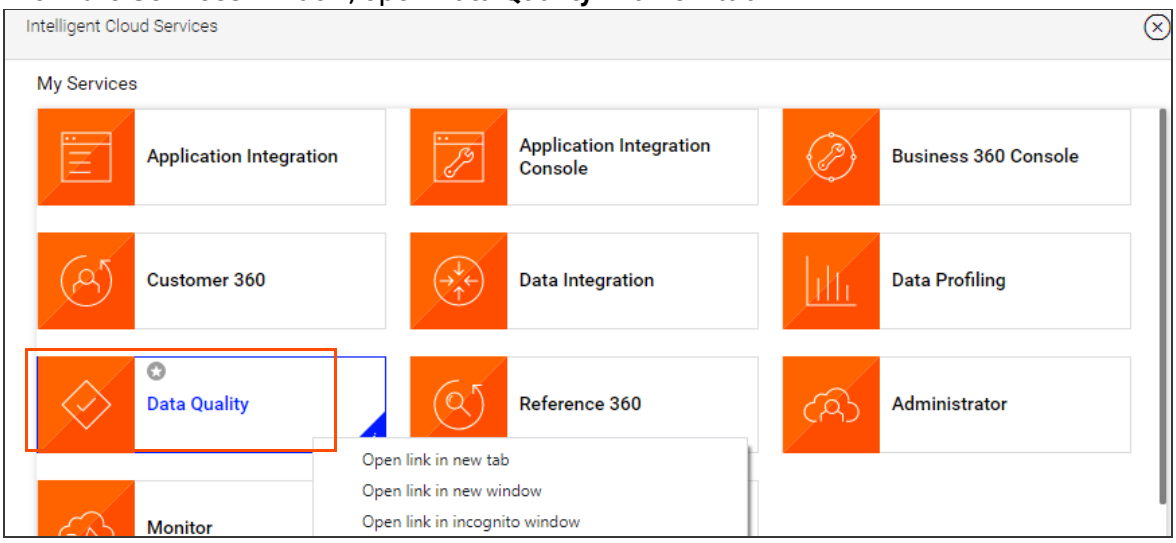
The record will be created successfully.

Add Advanced Rule Association

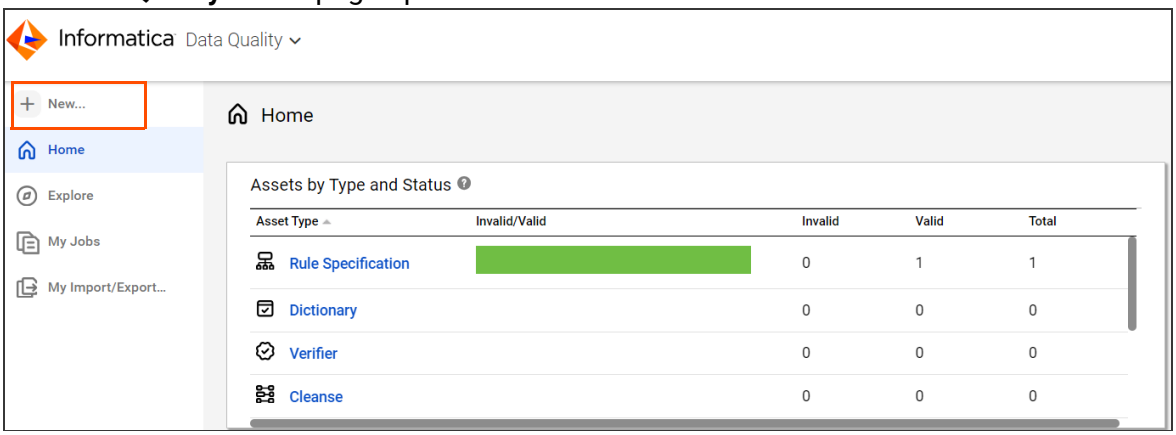
An advanced rule association is an association between business entity fields and a Cloud Data Quality rule specification. You can use the predefined rule specifications or create custom rule specifications in Cloud Data Quality and use them.

24. Switch to Data Quality. To do this, click **Multidomain MDM Console** at the top of the page.

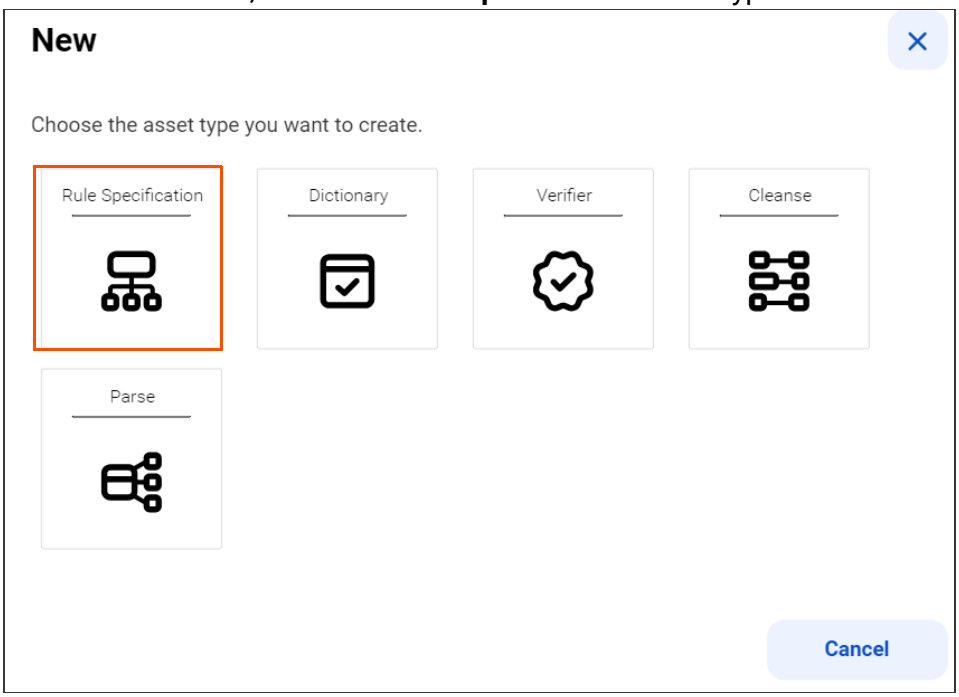
25. From the **Services** window, open **Data Quality** in a new tab.



26. The **Data Quality Home** page opens. Click **New**.

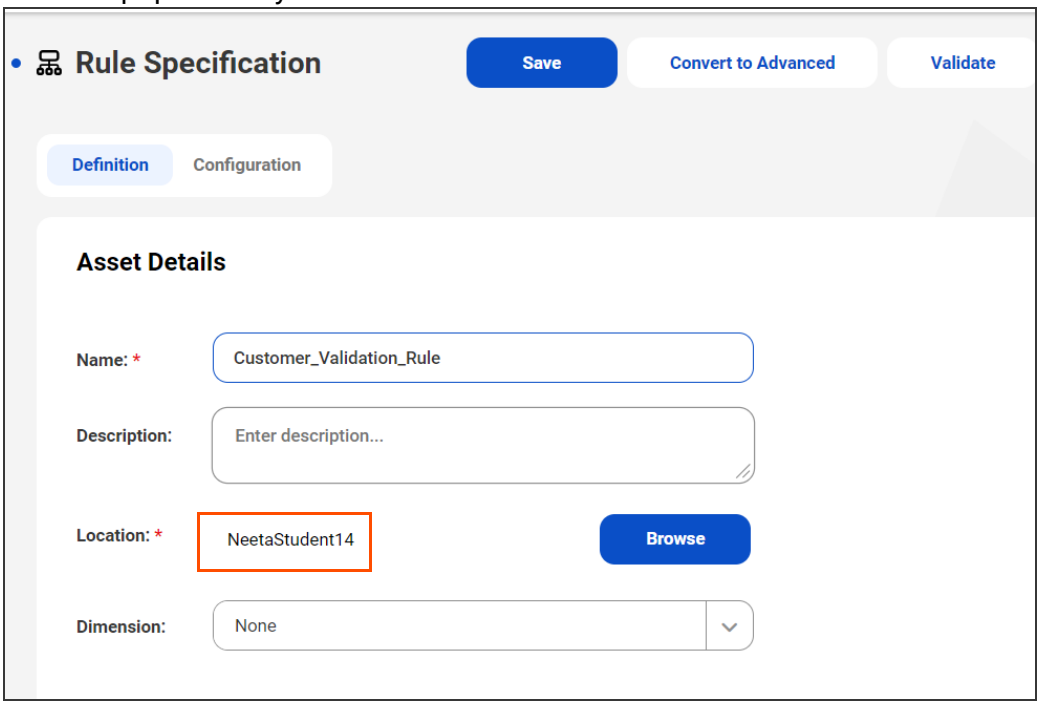


27. In the **New** window, select the **Rule Specification** asset type.

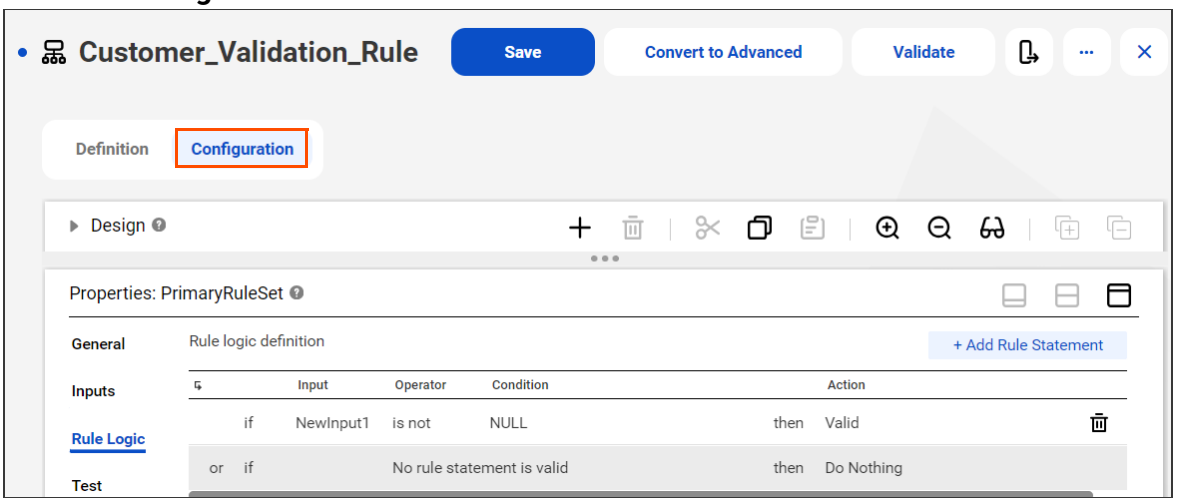


28. In the Definition tab, enter **Customer_Validation_Rule** for Name and write a valid description.

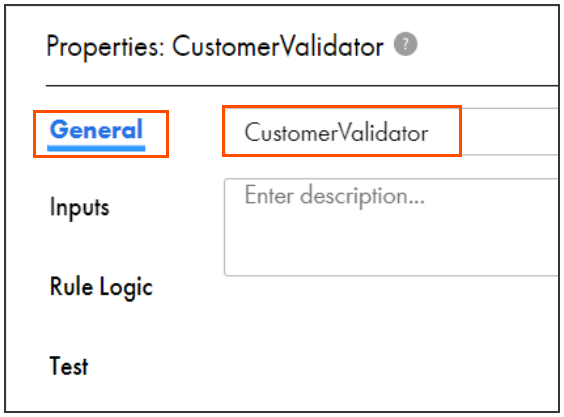
29. Select the location as your student project. Browse and select your student project if it is not auto populated by default.



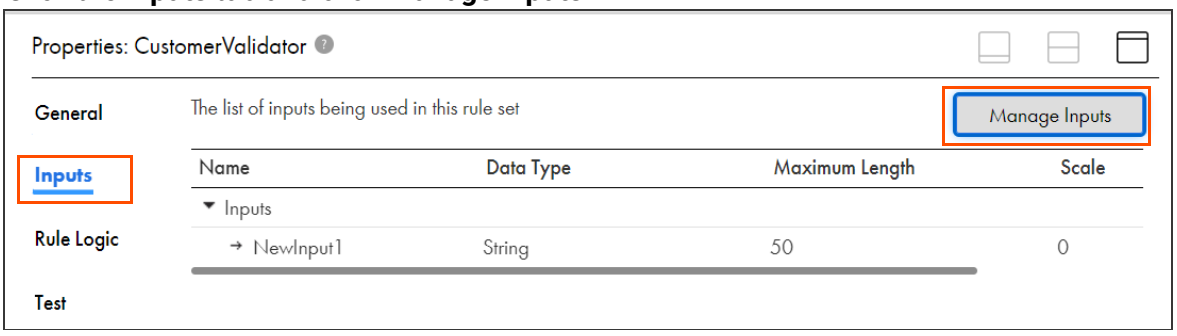
30. Click the **Configuration** tab.



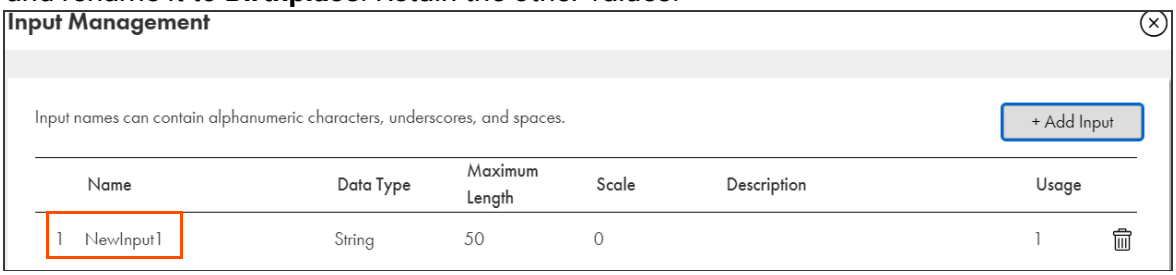
31. In the Properties section, in the **General** tab, enter the Name as **CustomerValidator** and enter a valid description.



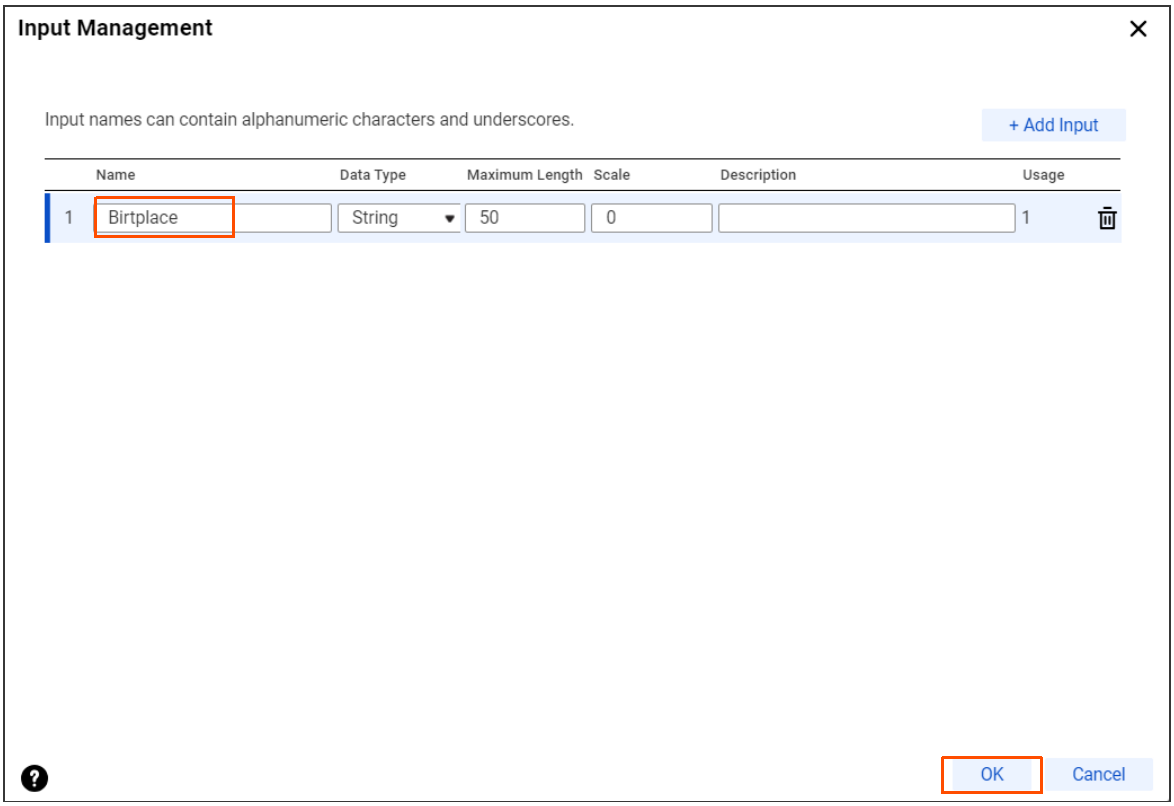
32. Click the **Inputs** tab and click **Manage Inputs**.



33. In the **Input Management** window, double click on the NewInput1 that is already present and rename it to **Birthplace**. Retain the other values.



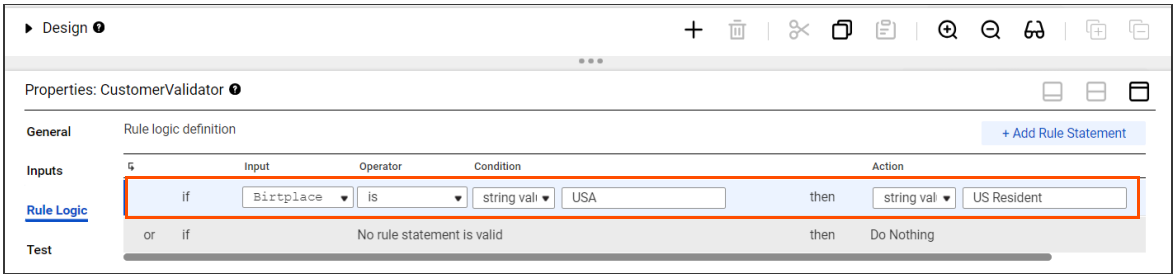
34. Click **OK** to save and close.



35. Click the **Rule Logic** tab.

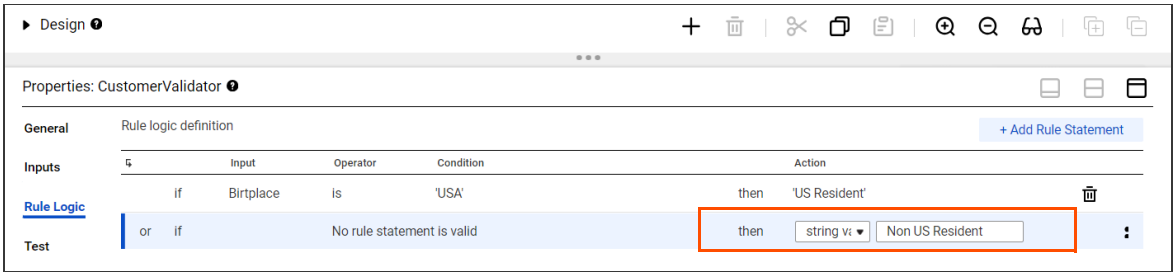
36. Select the following values:

Field	Value
Input	Birthplace
Operator	Is
Condition	string value
String value	USA
Action	string value
String value	US Resident



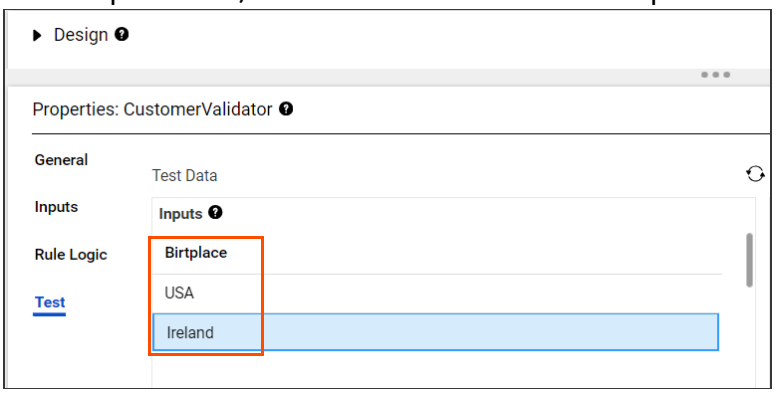
37. For the OR statement, enter the following:

Field	Value
Action	string value
Action value	Non-US Resident

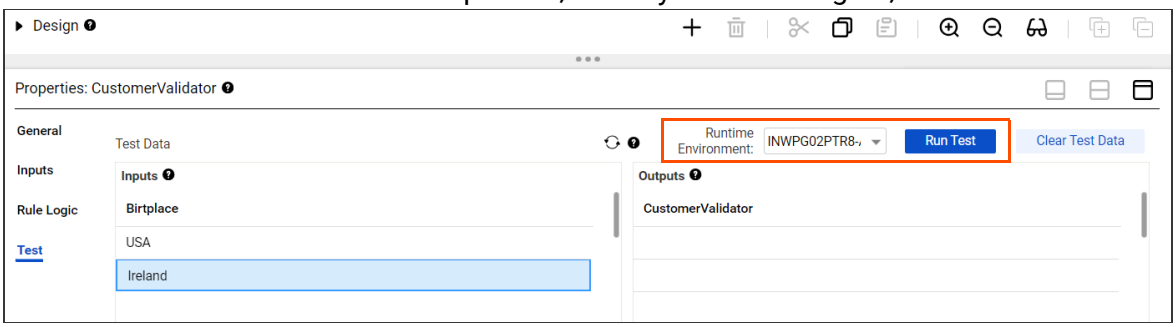


38. Click the **Test** tab.

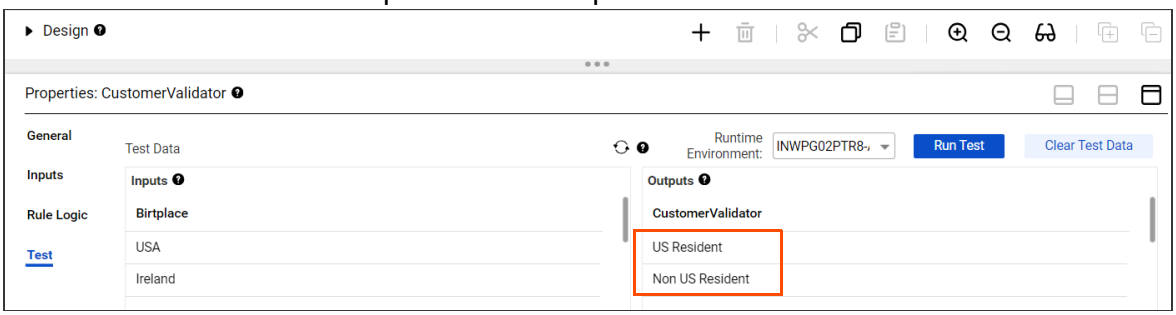
39. In the Inputs table, enter **USA** and **Ireland** for Birthplace.



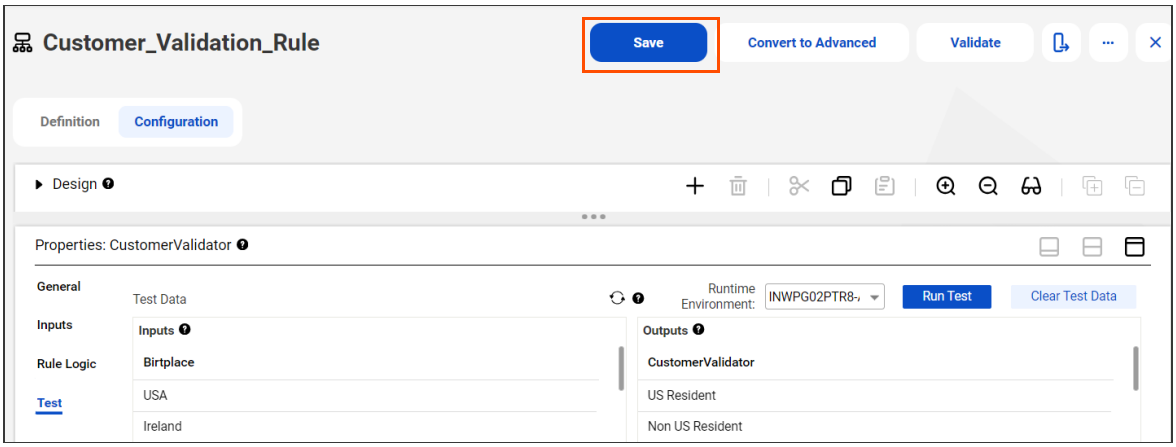
40. From the **Runtime Environment** drop-down, select your secure agent, and click **Run Test**.



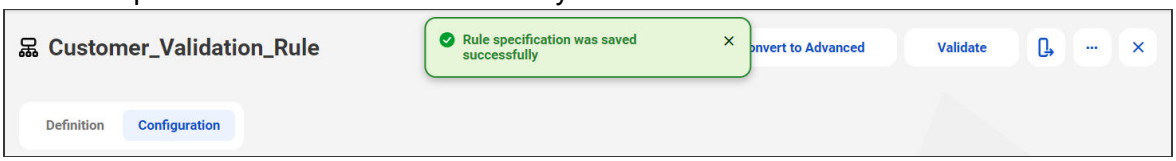
The test result shows the output values as expected.



41. Click **Save**.



The rule specification is saved successfully. Close the window.



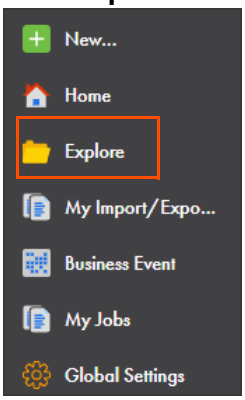
Import Rule Specification to Business 360

Now, you will import the advanced rule that you created to Business 360.

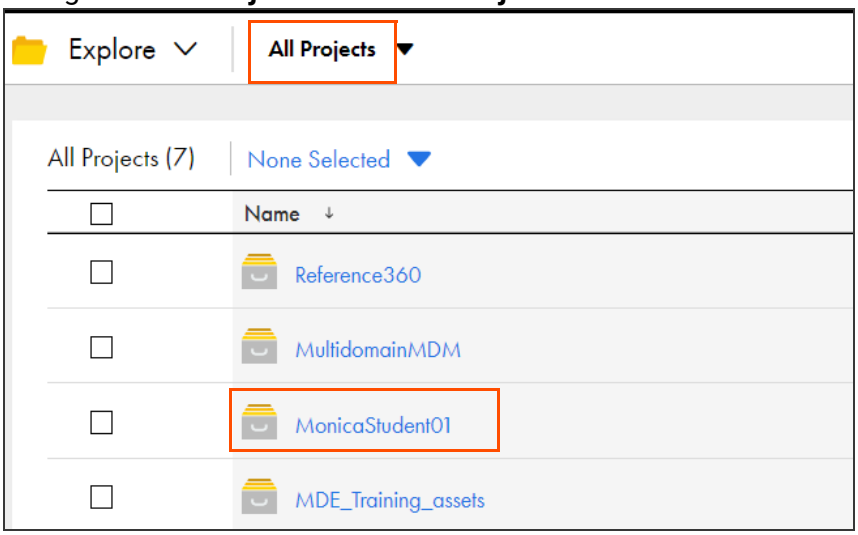
42. Open Business 360 in a different tab.

To open, click Data Quality from the top of the page, and from the Services window, click Business 360 Console.

43. Click **Explore**.

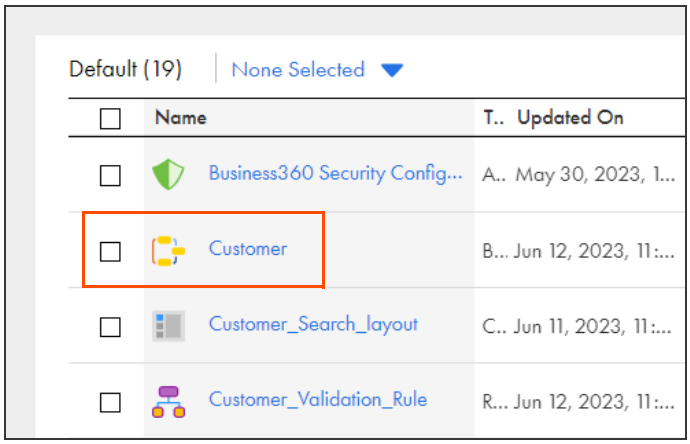


44. Navigate to **All Projects** > **Student Project**.

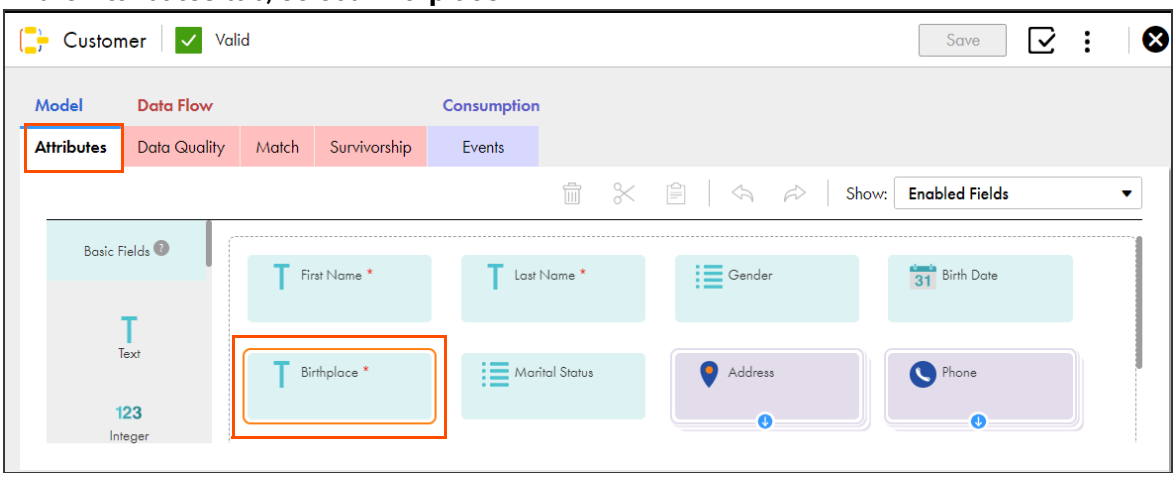


You should refresh the browser if you are trying to access a page that has already opened.

45. Scroll down and click the **Customer** business entity.

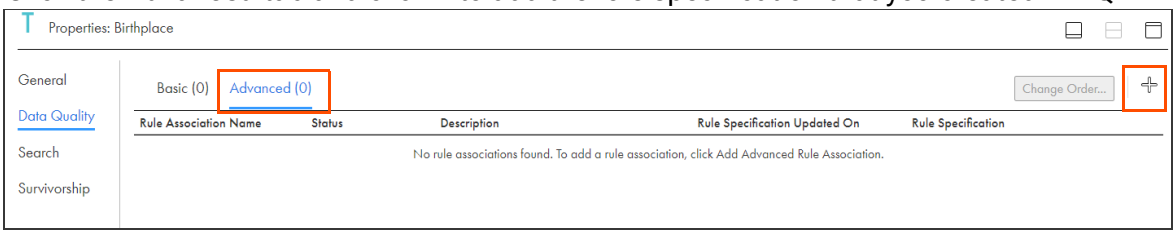


46. In the **Attributes** tab, select **Birthplace**.

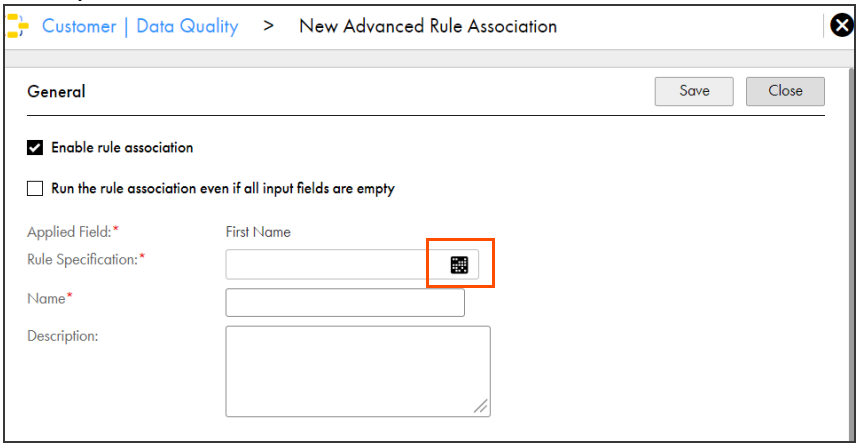


47. In the Properties section, click **Data Quality**.

48. Click the **Advanced** tab and click **+** to add the rule specification that you created in DQ.

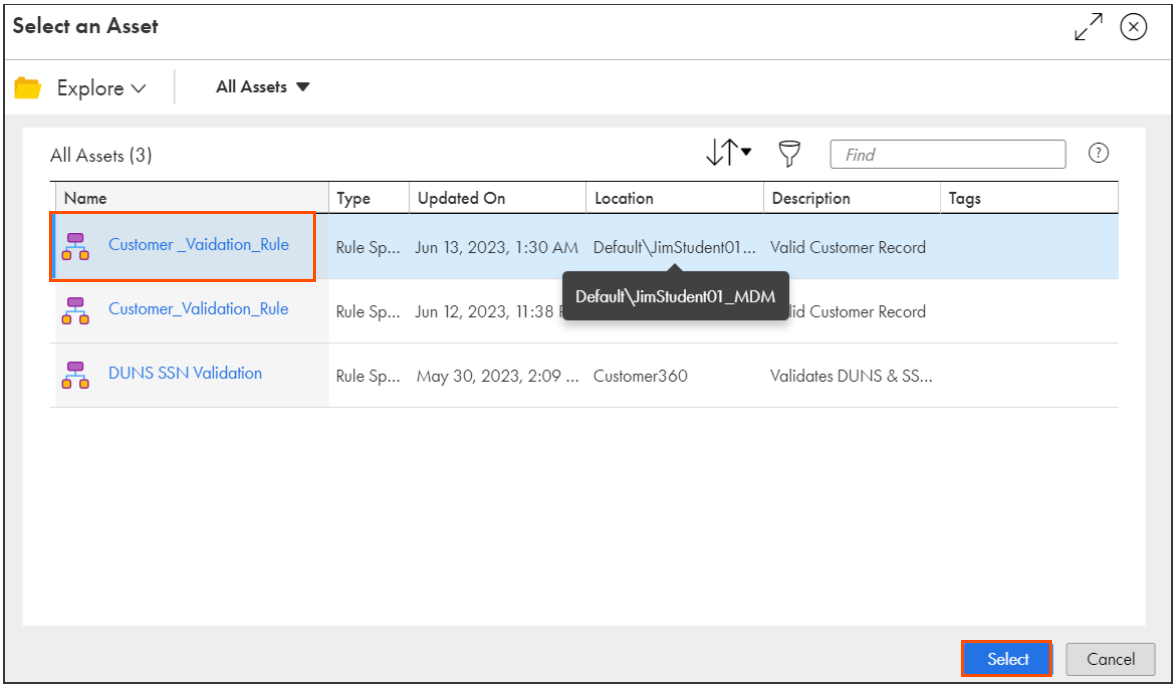


49. In the **New Advanced Rule Association** window, click the **Select Asset** button to select the rule specification.

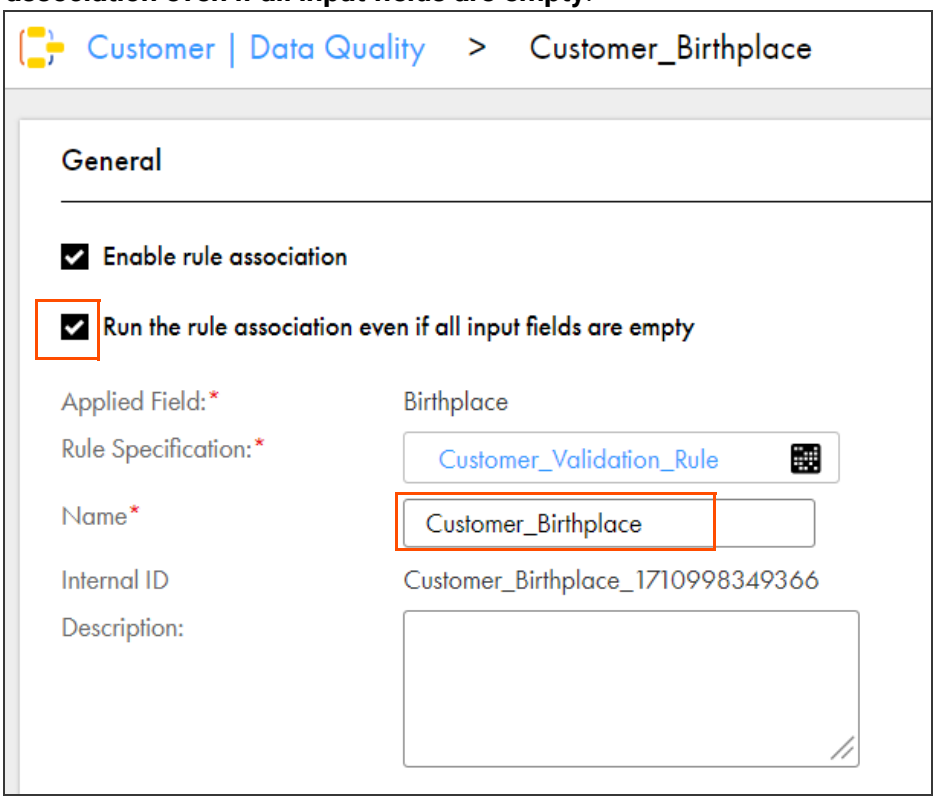


50. Select the **Customer_Validation_Rule** from the list and click **Select**.

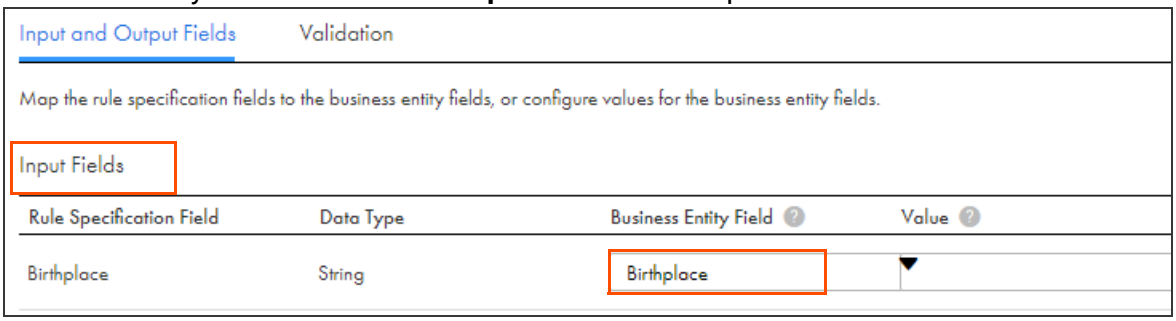
Ensure that you select the asset from your student project, for example, JimStudent01_MDM.



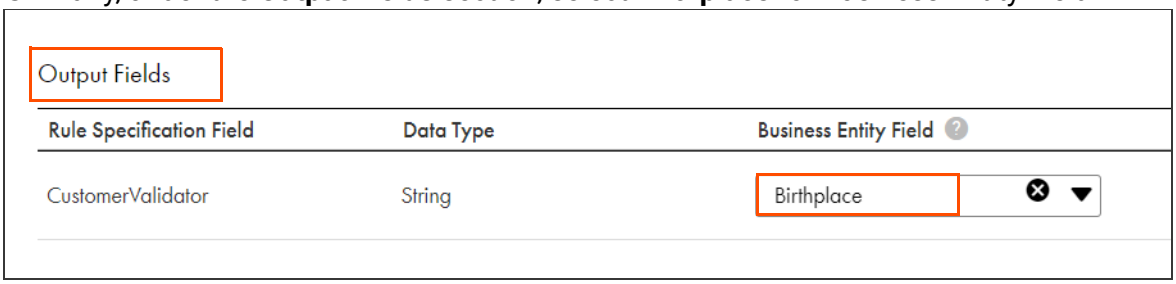
51. Enter the name as **Customer_Birthplace**. Also enable the check box **Run the rule association even if all input fields are empty**.



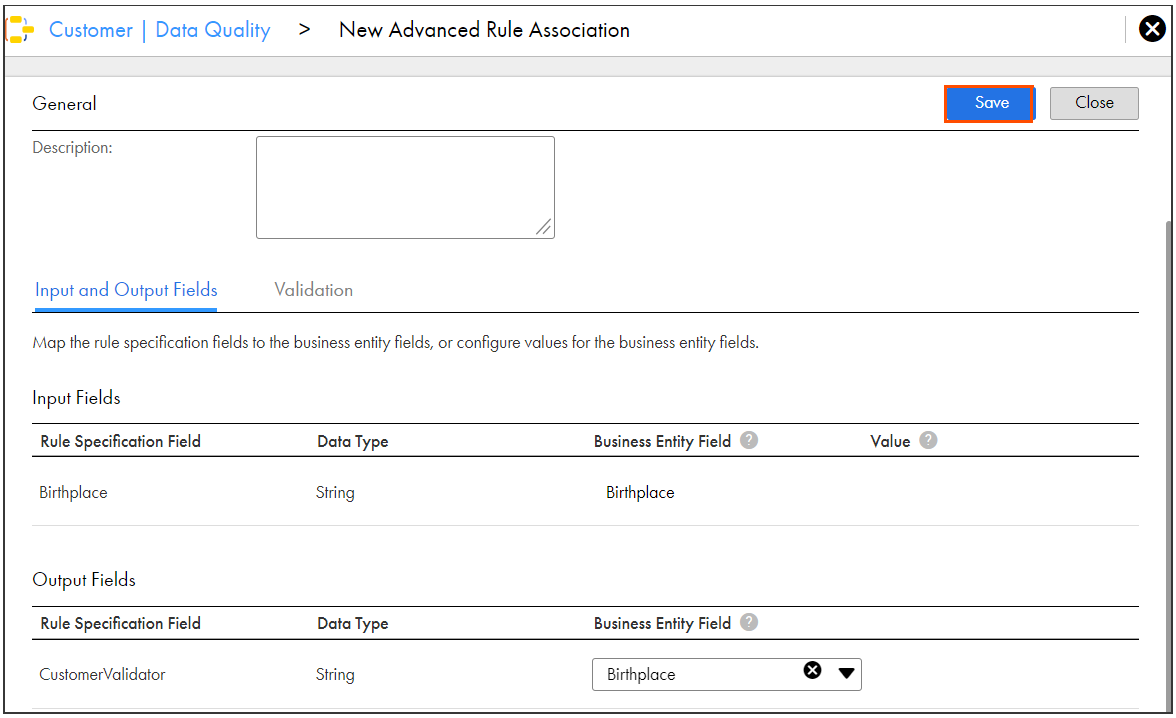
52. In the **Input and Output Fields** tab, under the **Input Fields** section, click the empty box for Business Entity Field and select **Birthplace** from the drop-down.



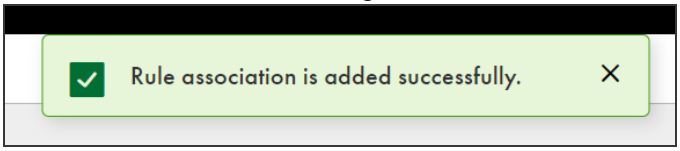
53. Similarly, under the **Output Fields** section, select **Birthplace** for Business Entity Field.



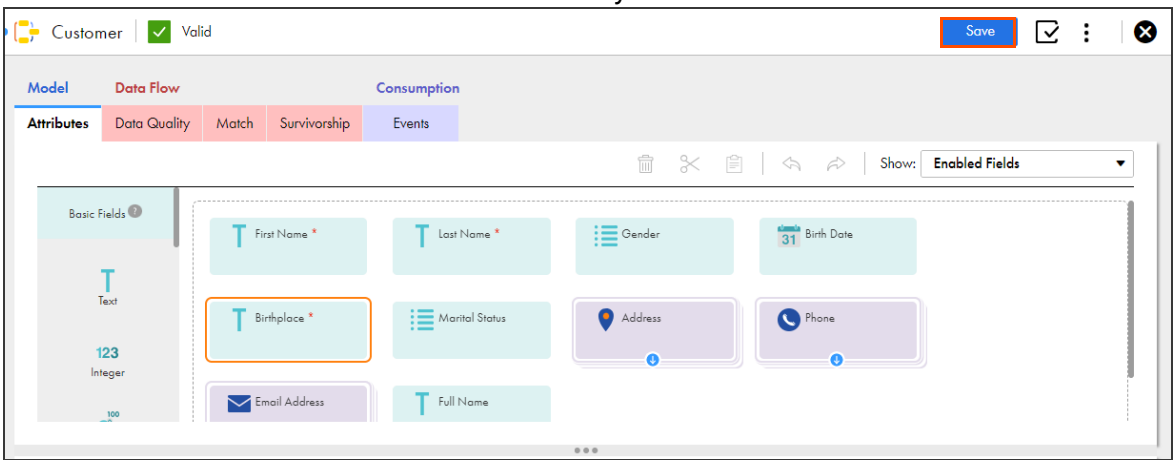
54. Click **Save**.



A notification with a message “Rule association is added successfully” is displayed.



55. Click **Save** to save the Customer business entity.



A notification with a message “The business entity was saved successfully” is displayed.

Test the Advanced Rule Association in Multidomain MDM

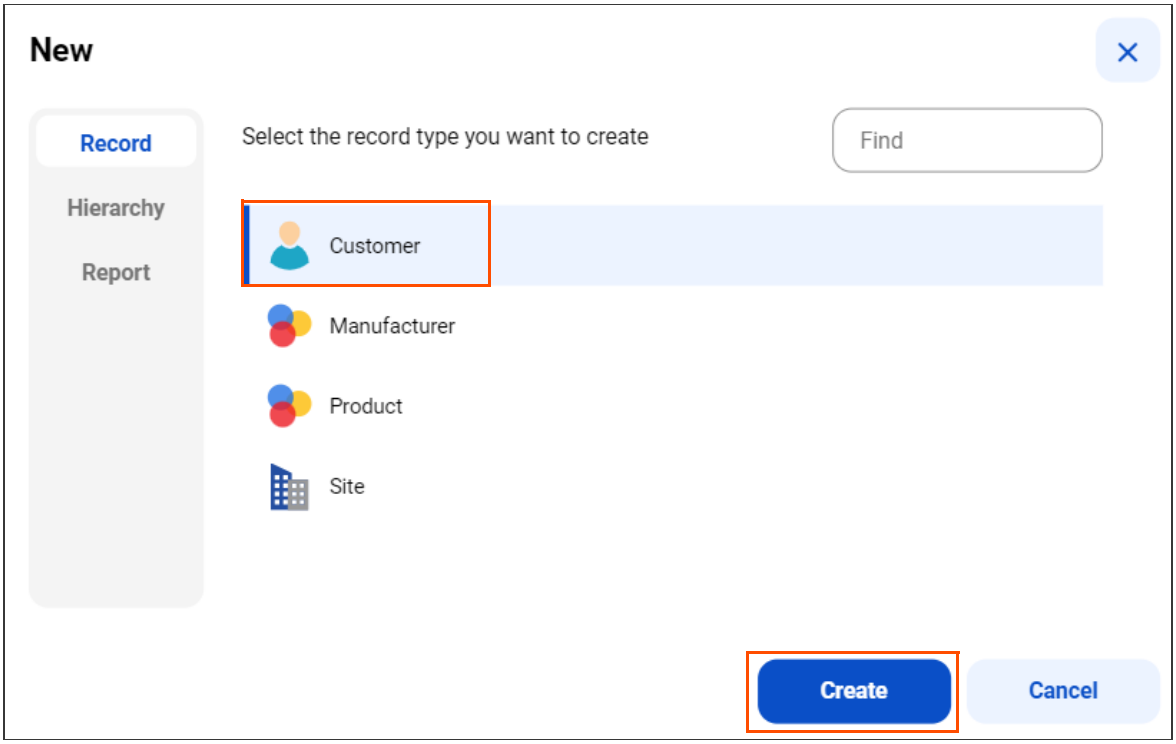
You will now test the advance rule in Multidomain MDM.

56. Switch to **Multidomain MDM** user interface in a separate tab.

You should refresh the browser if you are trying to access a page that has already opened.

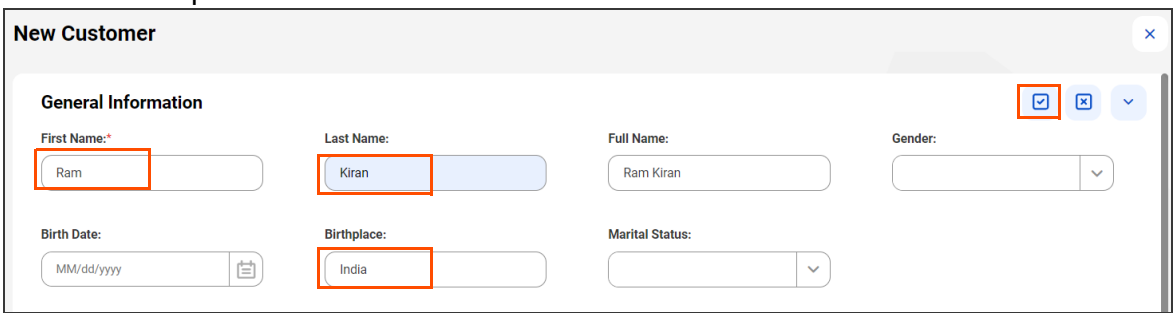
57. Click **New**.

58. Select **Customer** and click **Create**.

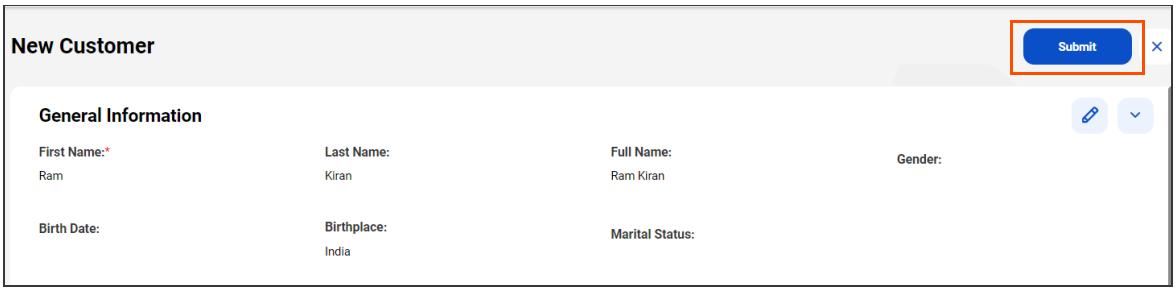


59. Enter the First Name as **Ram**, and Last Name as **Kiran**.

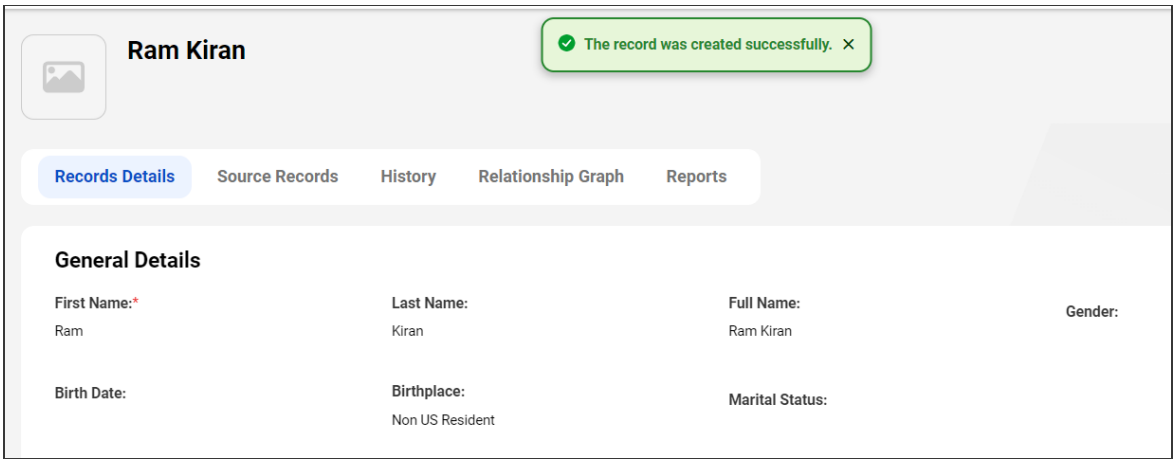
60. Enter the Birthplace as **India** and click the check mark.



61. Click **Submit**.



You will be notified by message that the record was created successfully. Notice that the **Birthplace** field shows **Non-US Resident**.



You have successfully created a rule specification in DQ and imported it to Business 360.

62. Log out of all the services.

This concludes the lab.

Lab 7-2: De-duplication of Multivalued Field Groups

Overview:

You can configure the de-duplication of multivalued field groups for a custom field group. Customers can add uniqueness criteria based on which de-duplication will happen. Currently, users can set up uniqueness criteria based on the mandatory fields of a field group. This feature supports exact matching and not fuzzy matching.

In the uniqueness criteria, multiple field group values are qualified as unique based on the required fields in a field group. Based on the fields that you configure with the uniqueness criteria, duplicate values are merged.

Objectives:

- Configure uniqueness criteria
- Create customer records to perform de-duplication of the data

Scenario:

Mark from Homekart speculates that the customer data store of Homekart may contain duplicate address information. Mark decides to use the Uniqueness criteria of the Multidomain MDM application to de-duplicate the address data.

Duration:

15 minutes

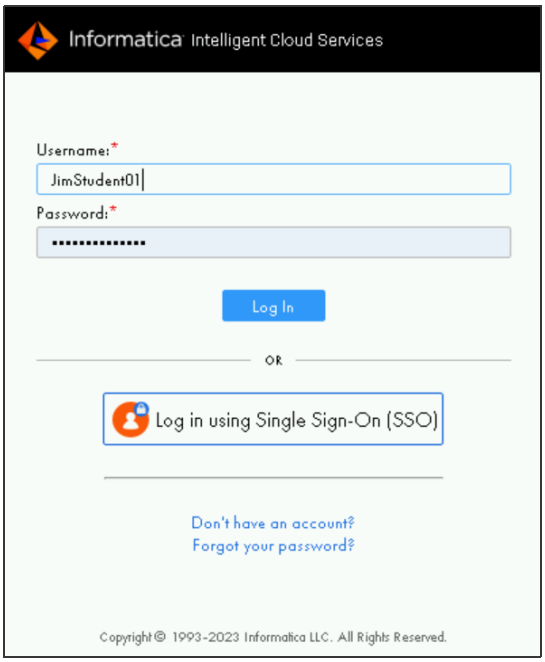
Tasks:

- Create a new customer eecord with duplicate address data
- Configuring uniqueness criteria to the Customer business entity
- Checking uniqueness criteria by adding duplicate address values

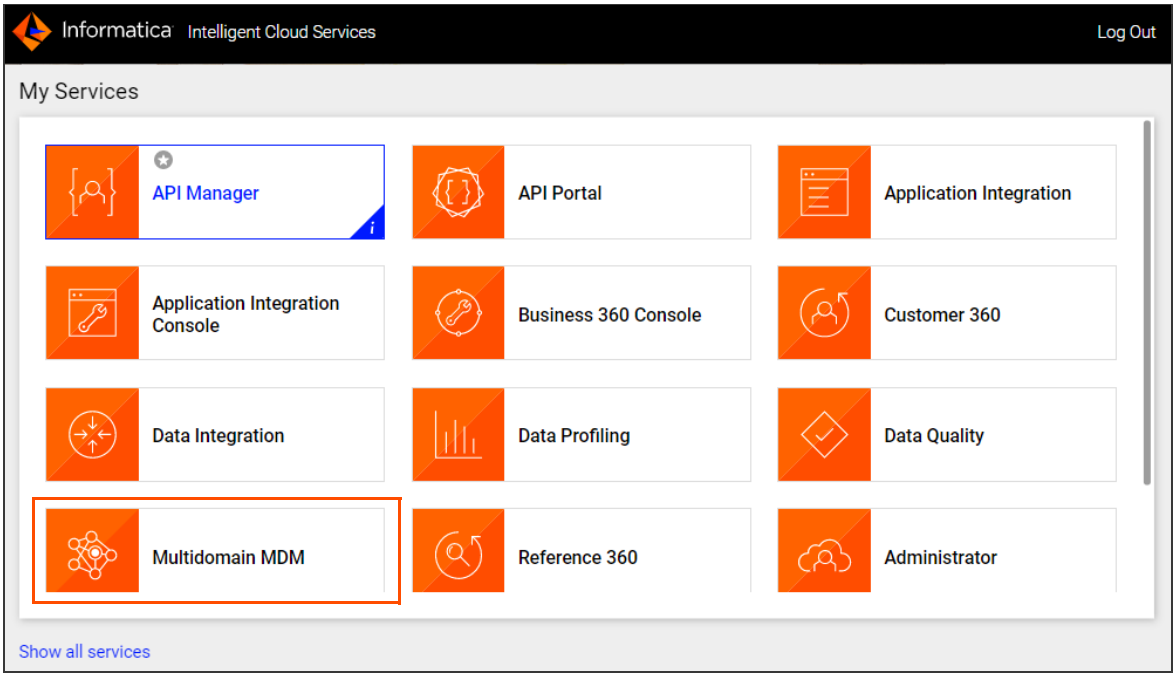
Create a New Customer Record with Duplicate Address Data

You will now create a customer record with the duplicate address information.

1. Login to IICS with your credentials (**studentNamestudent<#> / Multidomain360**) if you have logged out.

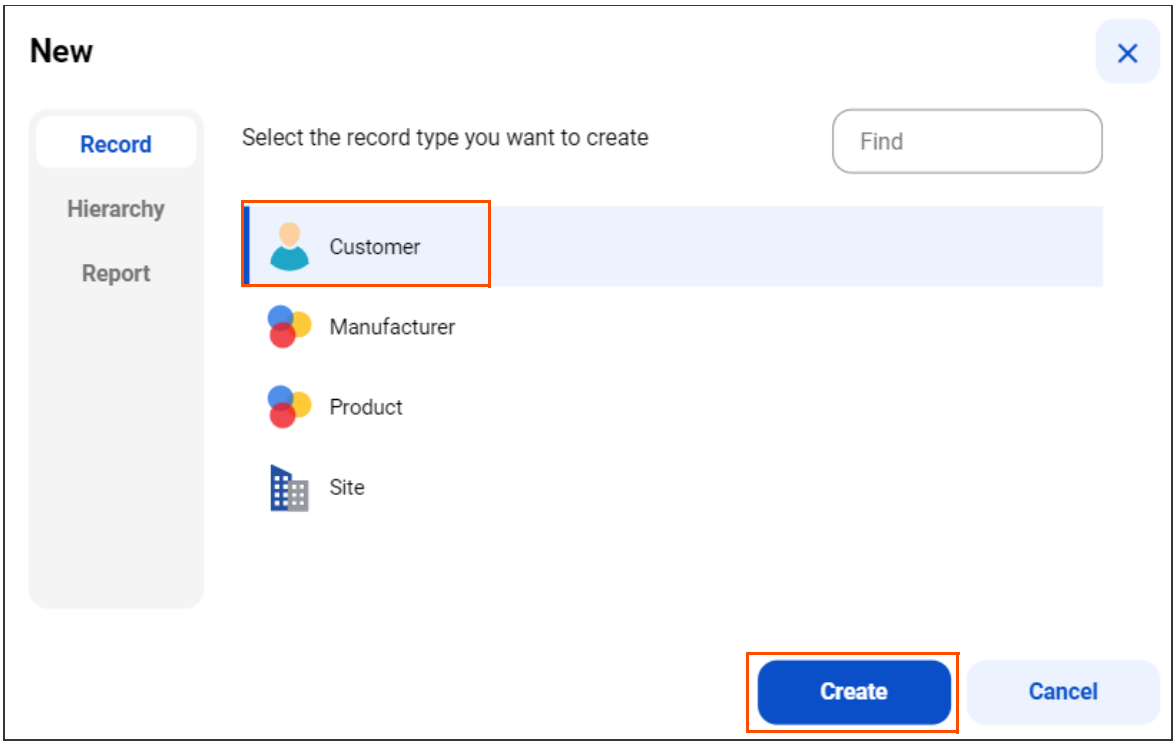


2. To launch the Multidomain MDM service, click **Multidomain MDM** IICS Service.



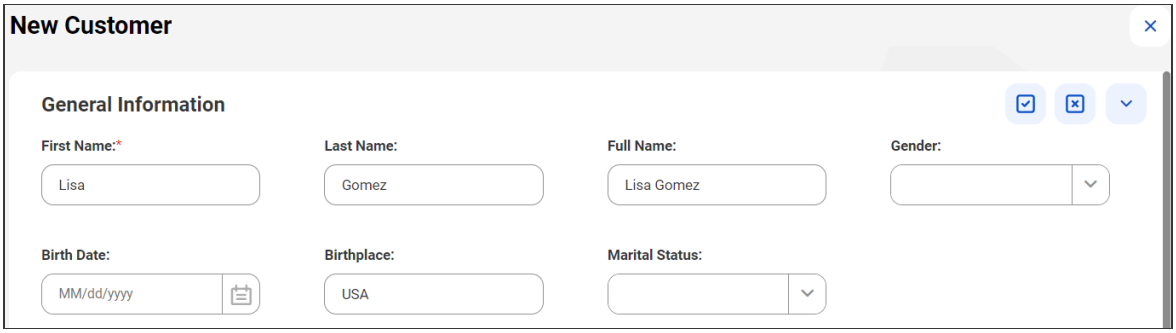
3. In the **Multidomain MDM** page, click **New**.

4. To create a new customer record, select **Record** > **Customer** and click **Create**.



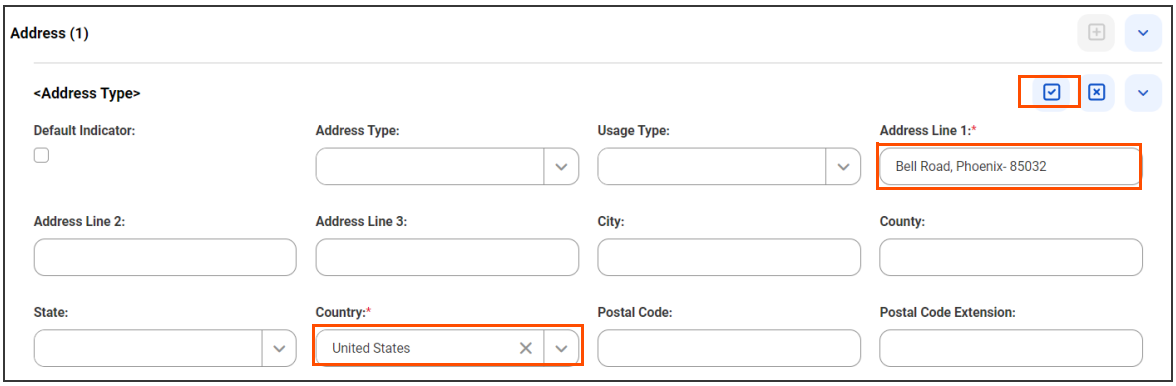
5. On the **New Customer** page, enter the following details and click tick icon to save the details.

Field	Value
First Name	Lisa
Last Name	Gomez
Birthplace	USA



6. Click the **+** icon for **Address** and enter the following details and save your changes by clicking the tick icon.

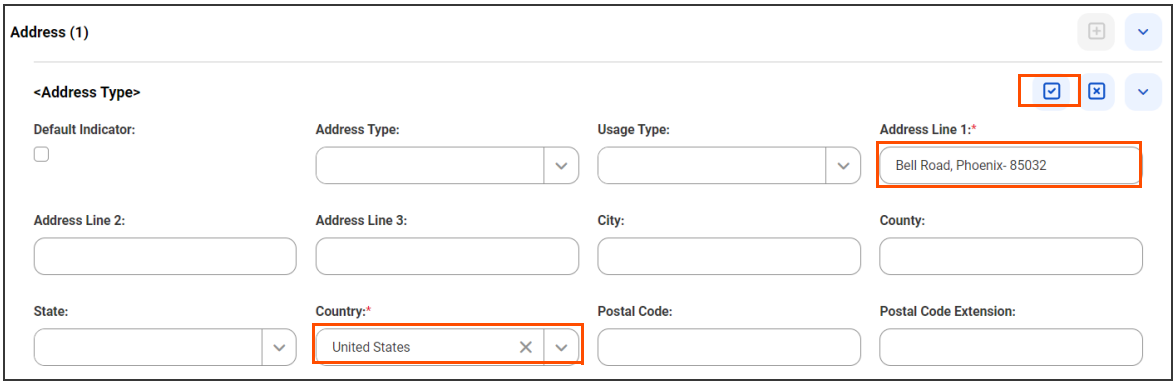
Field	Value
Address Line1	Bell Road, Phoenix- 85032
Country	United States



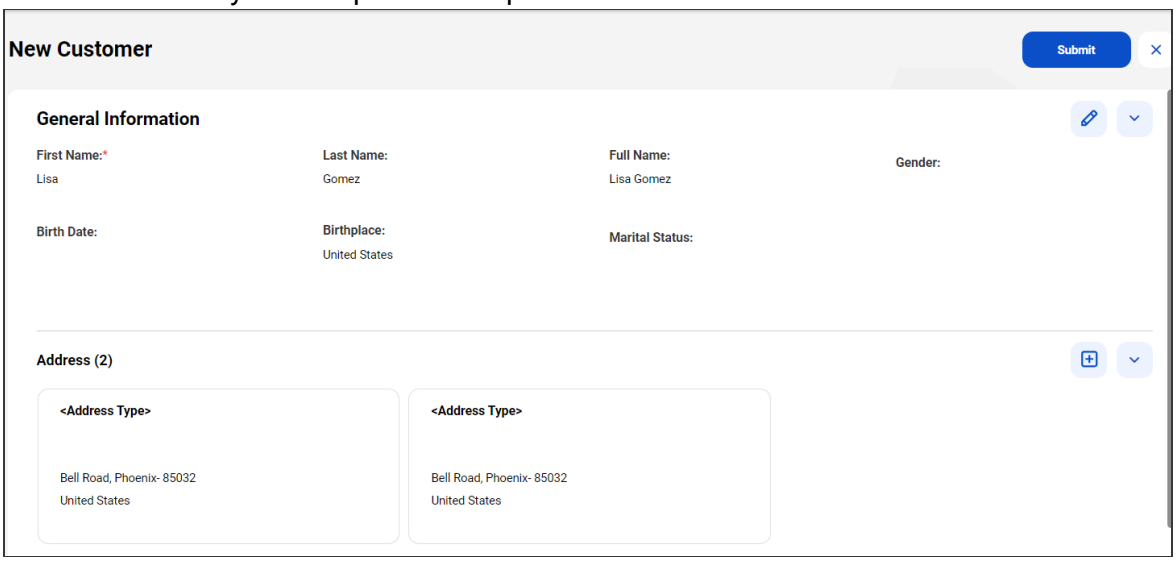
You should ignore the Internal error notification when selecting a country. Similarly, you can scroll down the page and enter other details as per your requirement.

7. To add duplicate address, click the **+** icon again for **Address** and enter the following details and save your changes.

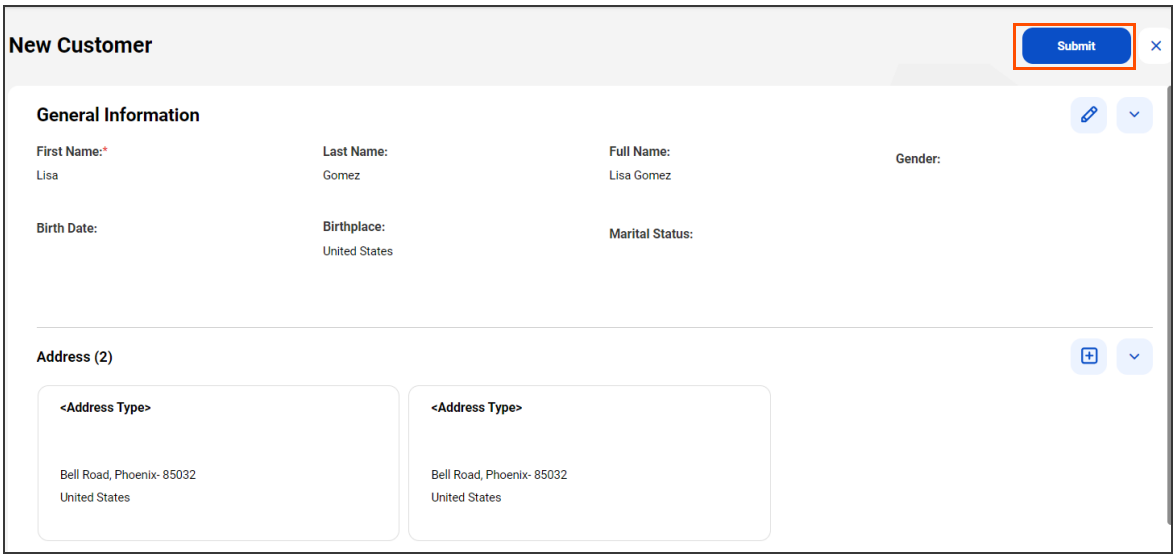
Field	Value
Address Line1	Bell Road, Phoenix- 85032
Country	United States



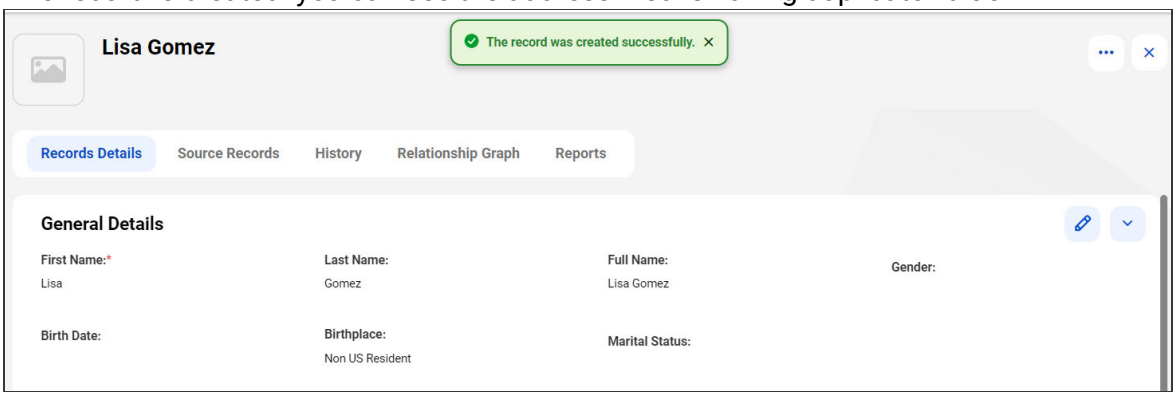
You can see that you have provided duplicate values to the address field.



8. Click **Submit**.



The record is created. you can see the address filed is having duplicate value.

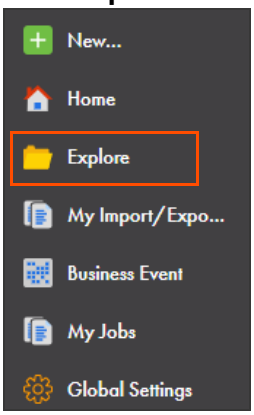


Configuring Uniqueness Criteria to the Customer Business Entity

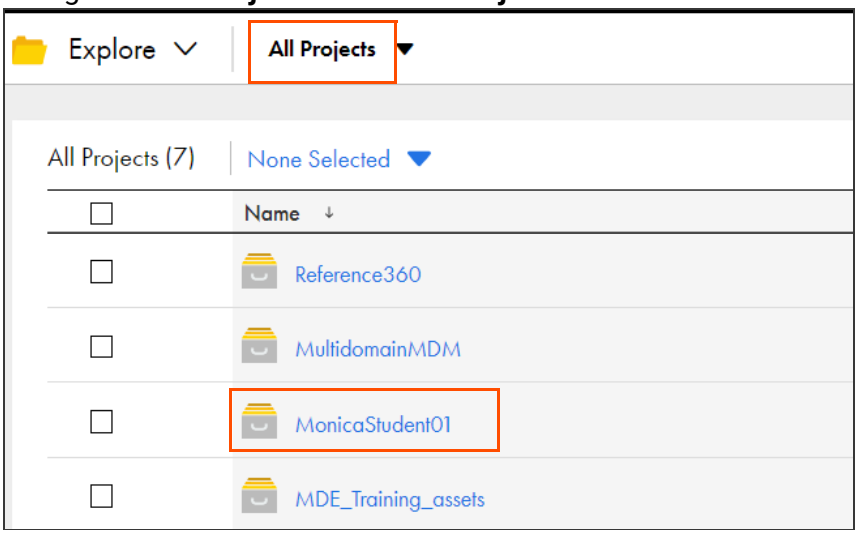
You will now configure a uniqueness criteria to the address field of Customer business entity.

9. Open Business 360 console in a different tab by using the product switcher

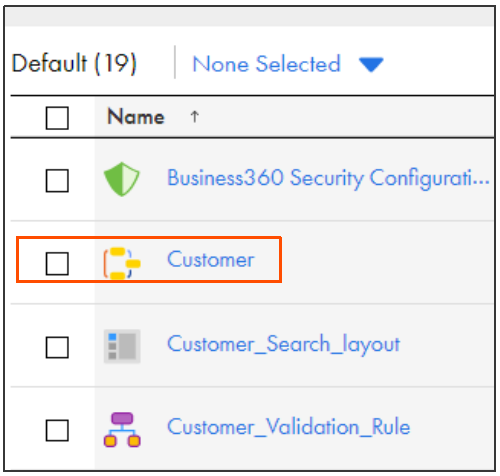
10. Click **Explore**.



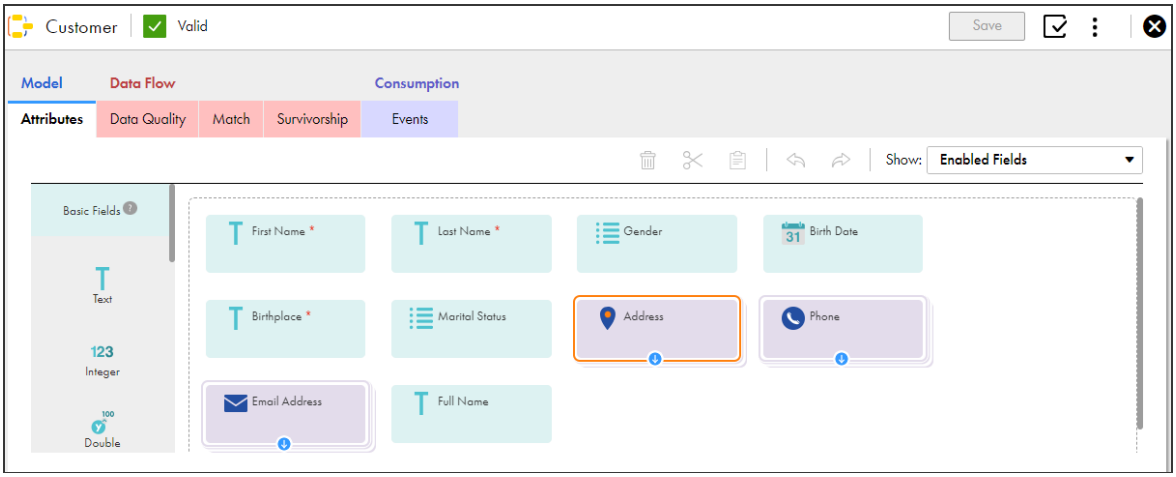
11. Navigate to **All Projects** > **Student Project**.



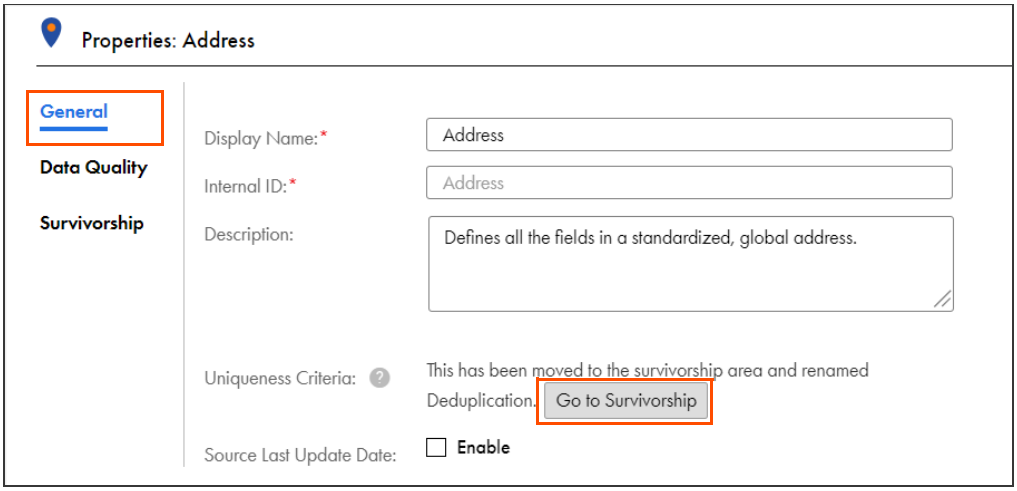
12. Scroll down and click the **Customer** business entity.



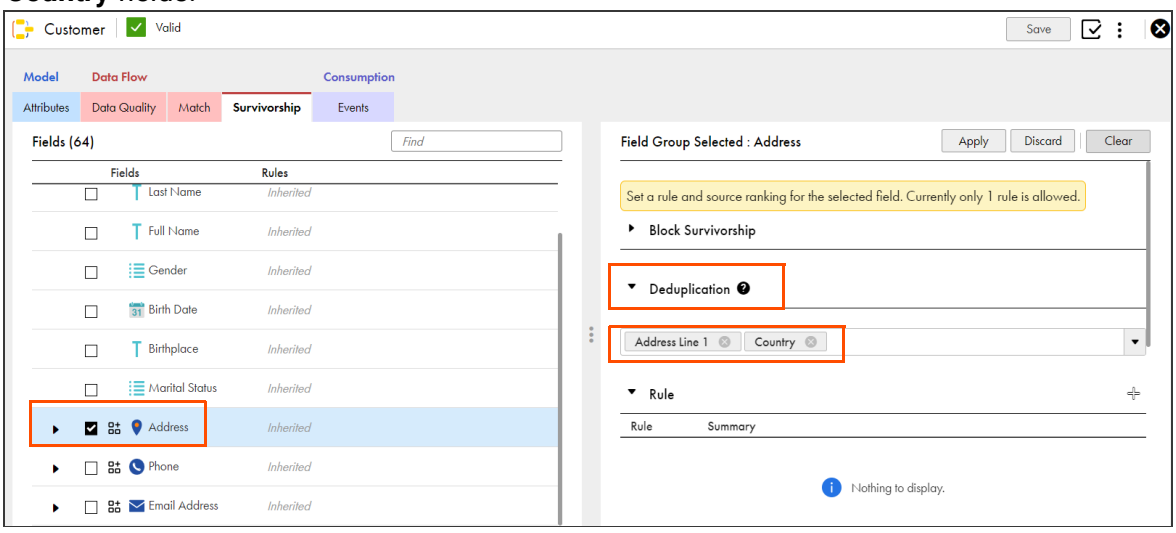
13. The business entity page opens. The canvas displays all the fields that the business entity includes. In the Attributes tab, select **Address**.



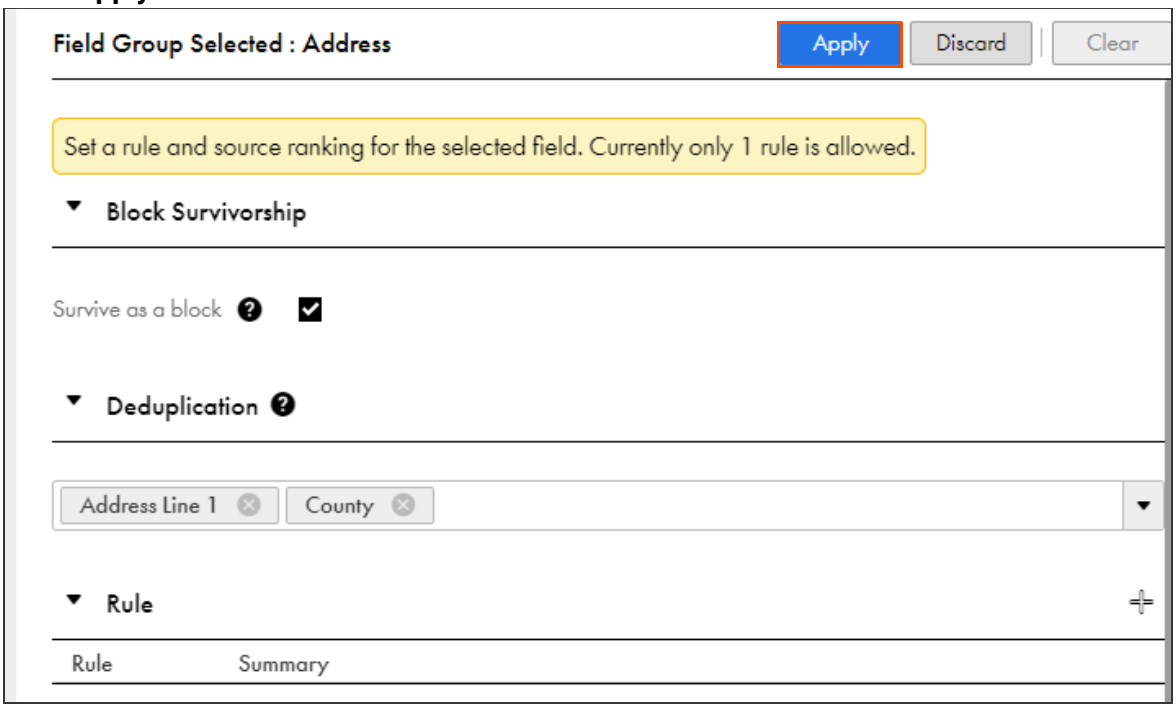
14. In the **Properties** > **General** section, scroll down and click **Go to Survivorship**.



15. Select the **Address** field, under the **Deduplication** drop-down, select **Address Line1** and **Country** fields.



16. Click **Apply**.



17. Click **Save** to save the changes.

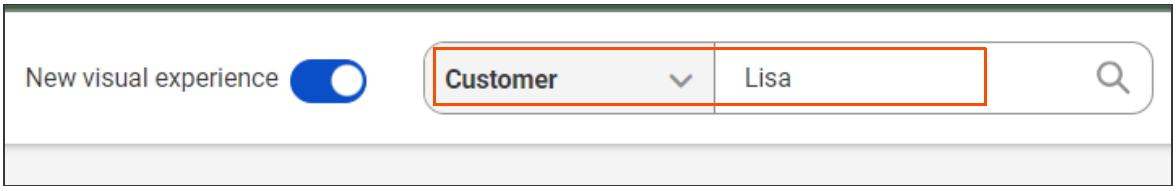
Checking Uniqueness Criteria by Adding Duplicate Address Values

Now you will test the uniqueness criteria by adding duplicate address values to a customer record.

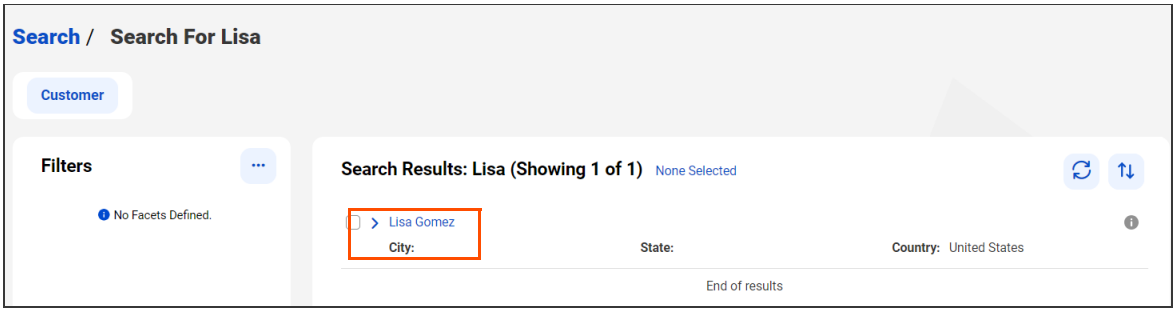
18. Navigate to **Multidomain MDM** service opened in a different tab.

19. **Refresh** the browser.

20. In the Multidomain MDM page, in the search tab, select Customer business entity and enter the search string as "**Lisa**".

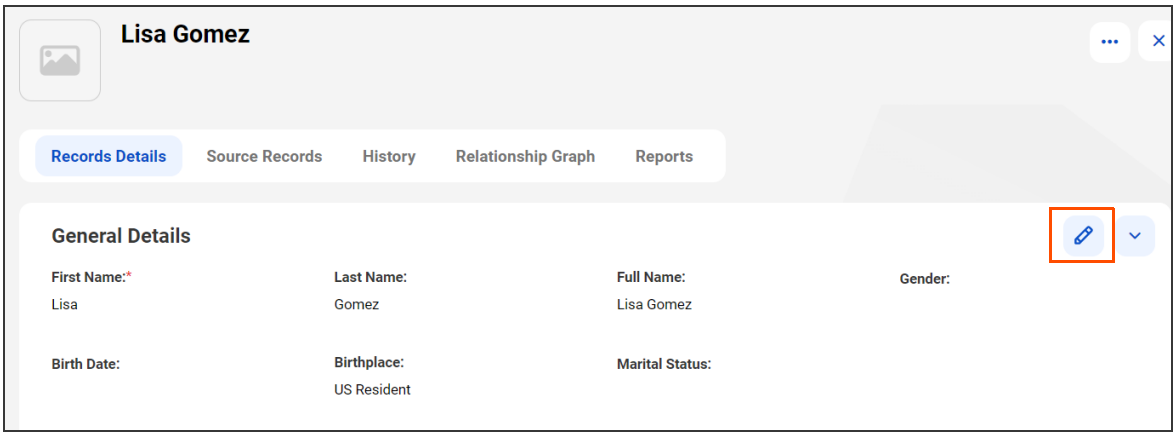


21. Open the **Lisa** record.

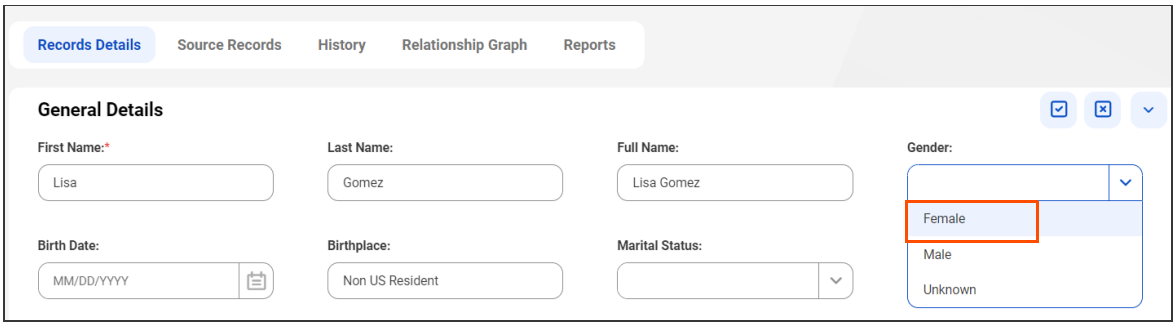


The Lisa record already has duplicate address. Let us edit the record and submit it again. The duplicate address should be merged.

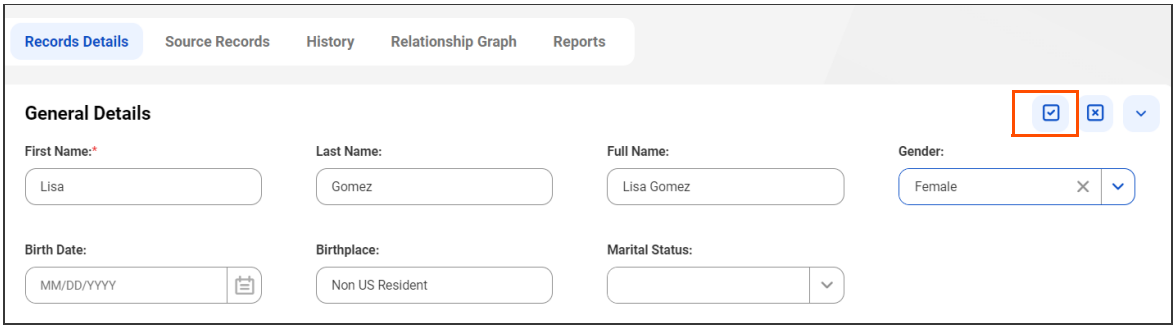
22. Click on the **edit** icon.



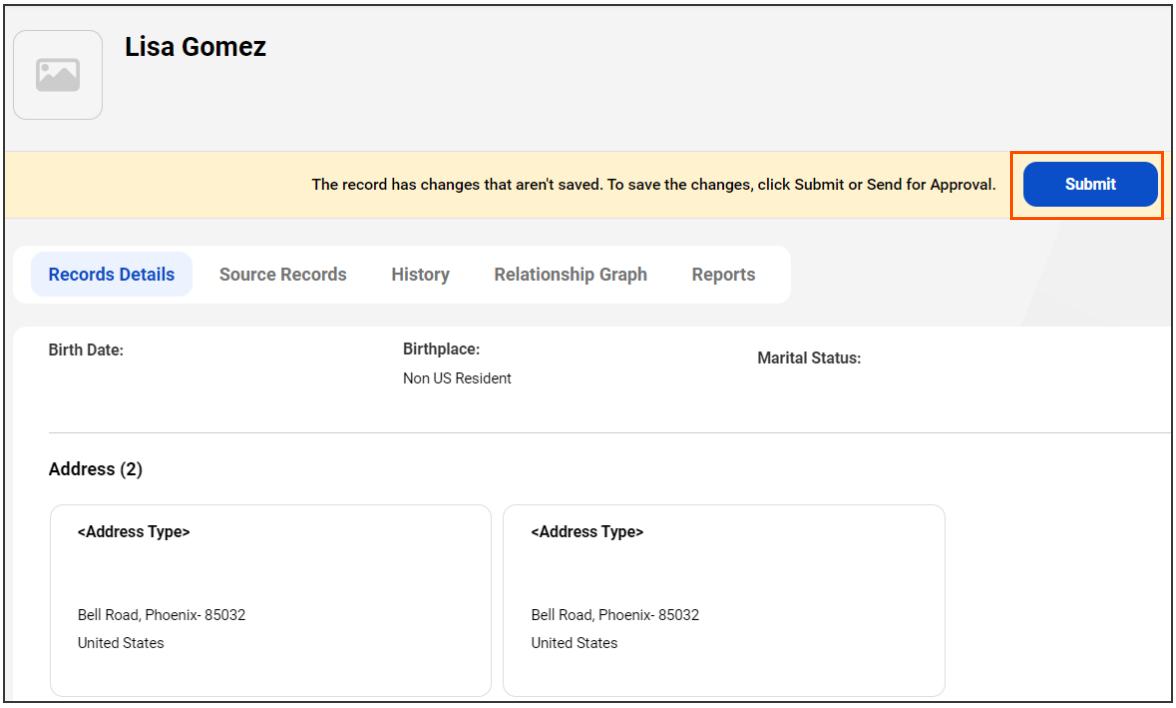
23. Select Gender as **Female**.



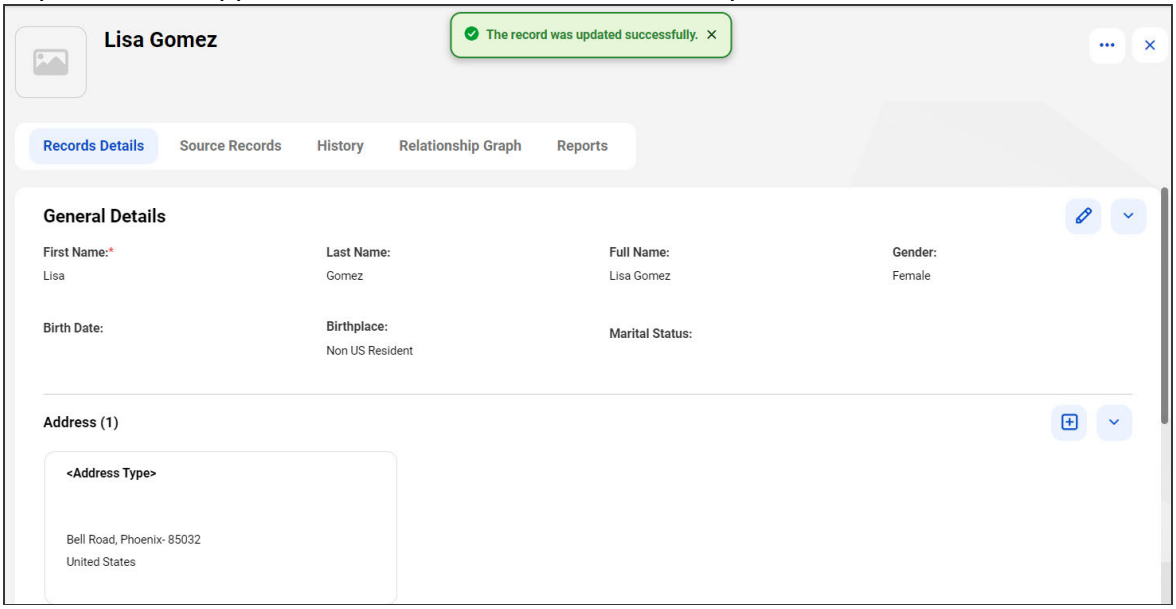
24. Select the check box to save the details.



25. **Submit** the record.



26. The record will be saved successfully. Notice the address section. You can see that the de-duplication is happened to the address value and the duplicate values are eliminated.



This concludes the lab.

Lab 7-3: Configuring Survivorship

Overview:

A source system is an external application or a system that acts as data source for Business 360 data store. For example, Salesforce or Workday can provide data to Business 360 data store. You can use survivorship to select the most accurate data from multiple match pair records to create master records. The survivorship process begins after the matching process is complete. After the matching process identifies the match pairs records, the survivorship configuration is applied to create the master records.

Objectives:

- Define source systems
- Configure Survivorship
- Configure validation rules to downgrade trust score

Scenario:

Mark wants to onboard data from different source systems. These source systems include data from CRM, Sales, HR, Billing, and other systems. Now, the main task is to build the confidence in the source systems and to rank them. Only the records that are accurate will survive. Multidomain MDM provides a feature called “survivorship”, that enables Mark to configure ranking and decay rules. This configuration will be used later when the Match and Merge jobs are run.

Duration:

50 minutes

Tasks:

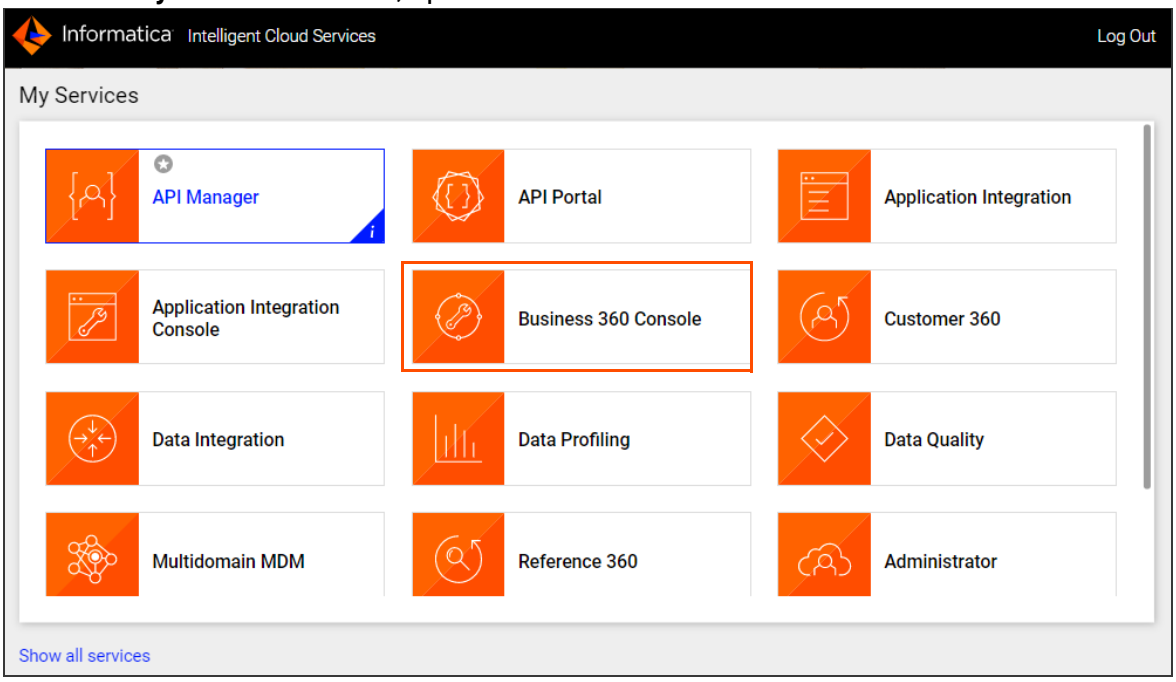
- Create source systems
- Configure Survivorship
- Configure decay rule
- Create validation rules to downgrade trust score
- Testing the validation rule in Multidomain MDM

Create Source Systems

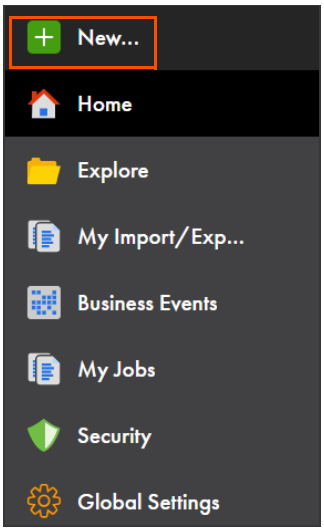
Before configuring the survivorship you will create source systems.

1. Log in to IICS with your credentials (studentNamestudent<#> / Multidomain360).

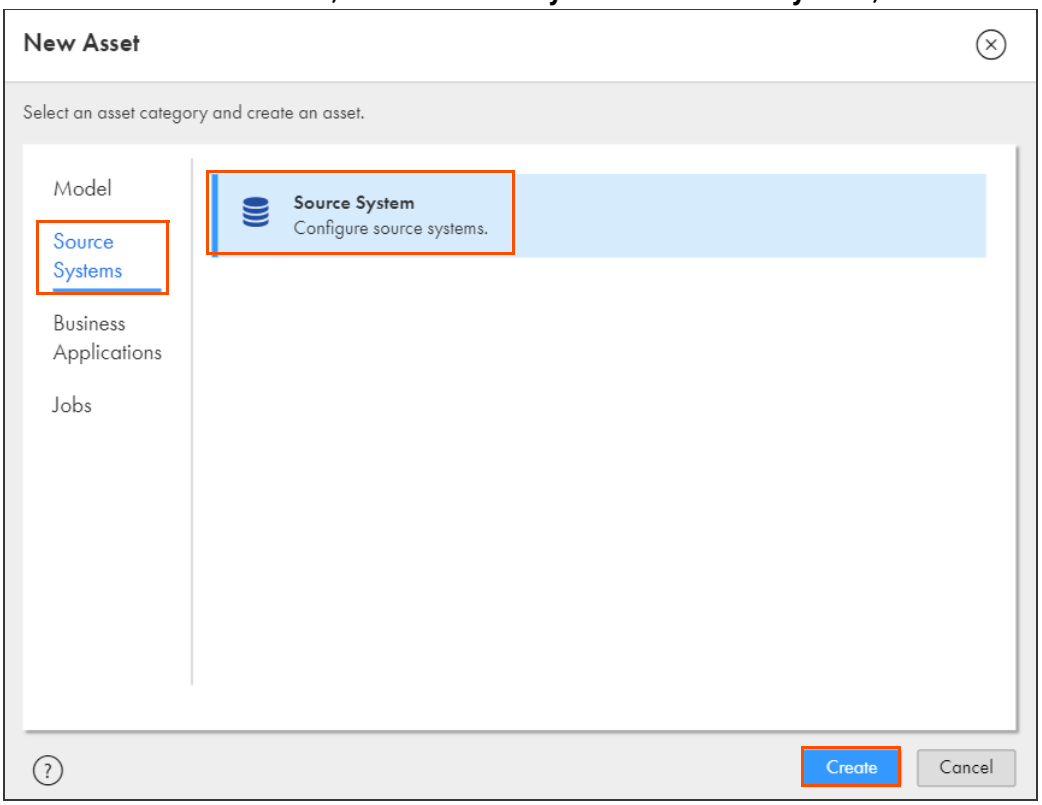
2. From the **My Services** window, open **Business 360 Console** in a new tab.



3. Click **New**.

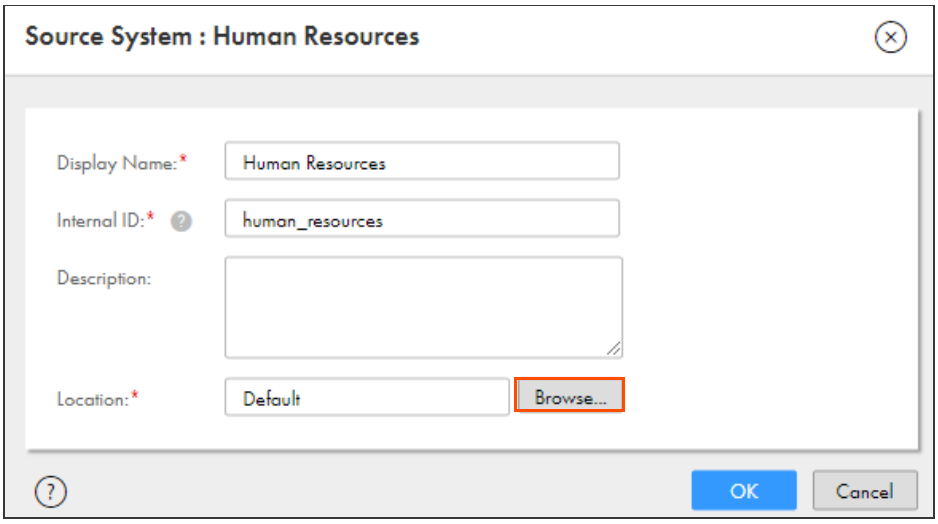


4. In the New Asset window, select **Source Systems** > **Source System**, and click **Create**.

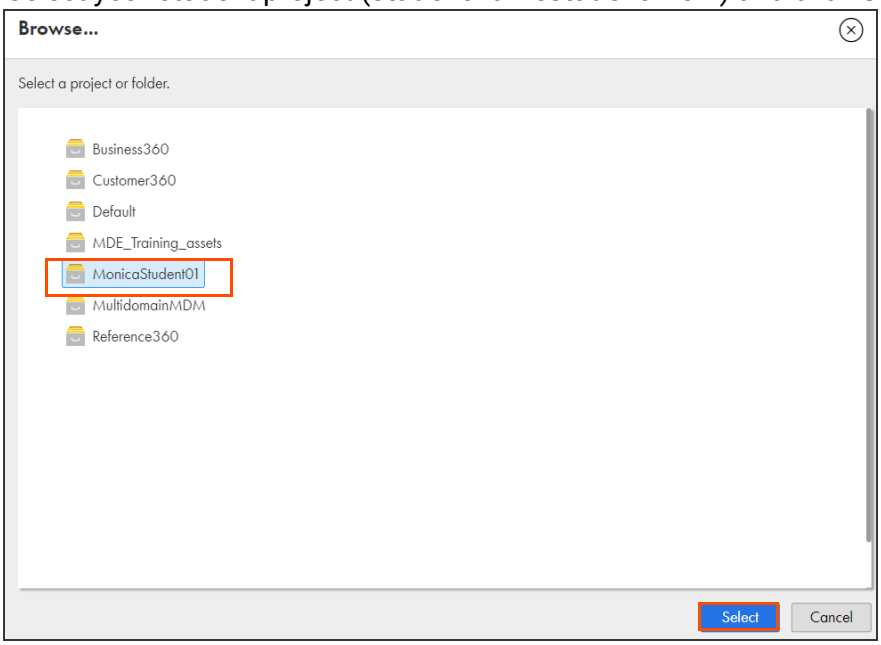


5. Enter the Display Name as **Human Resources**. The Internal ID is populated, by default.

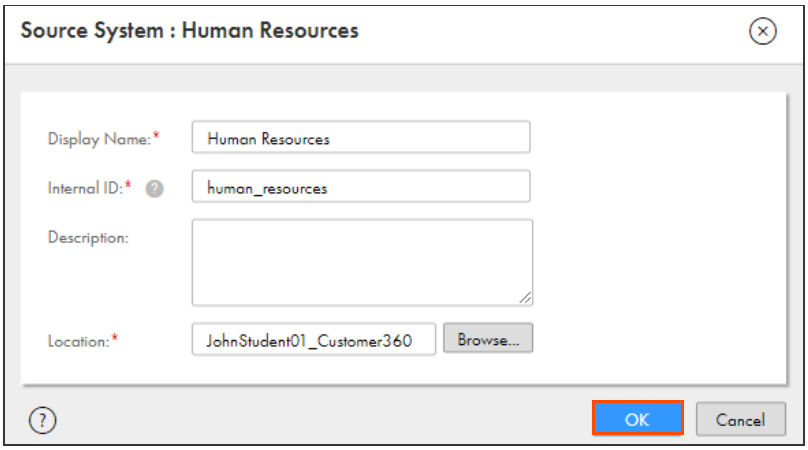
6. For Location, click **Browse**.



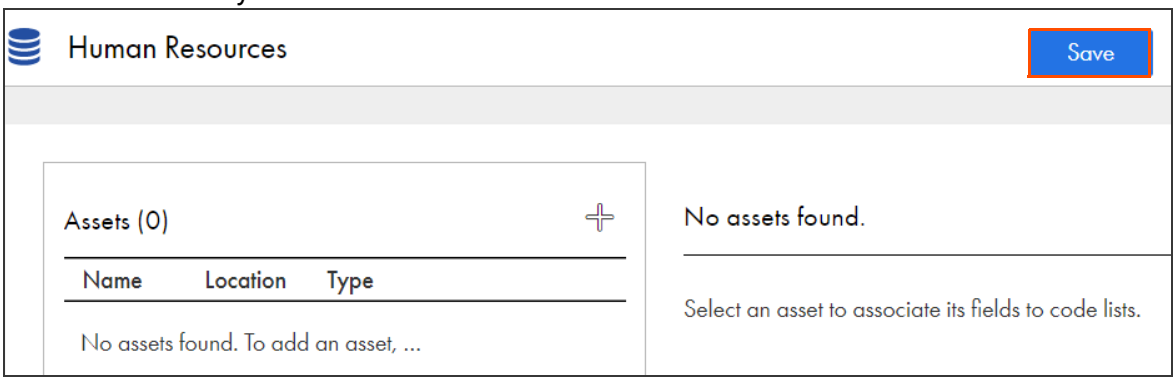
7. Select your student project (**studentNamestudent<no.>**) and click **Select**.



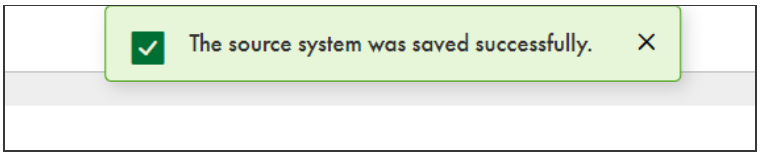
8. Click **OK**.



9. **Save** the source system.



The source system will be saved successfully. Close the window.



Self-Try Exercise (mandatory)

Create the following source systems from which data will be onboarded to Multidomain MDM:

- ERP (Entity Resource Planning)
- Billing
- Sales
- Informatica DaaS

High-Level Steps

10. In Business 360 Console, click **New** > **Source Systems** > **Source System** > **Create**.
11. Enter the display name for the source system. Ex: **ERP**.

12. For the location, choose the student project.

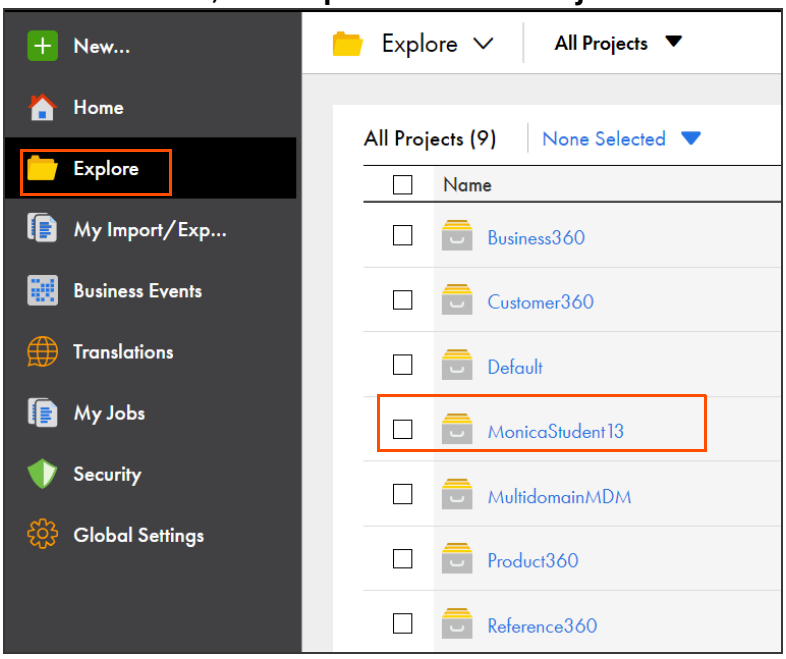
13. Click **OK** and **Save**.

Repeat the above steps to create all the other source systems mentioned in the self-try exercise.

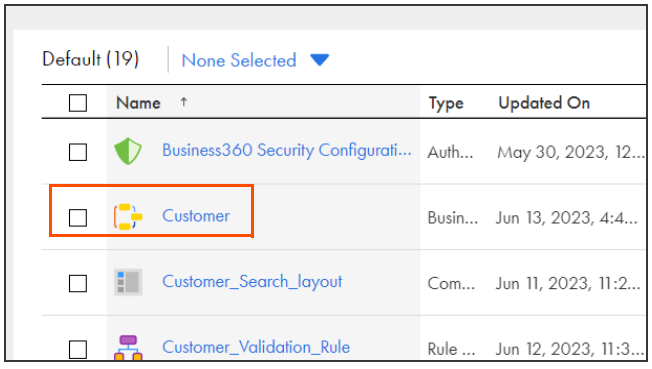
Configure Survivorship

You will now configure the survivorship to the Customer business entity.

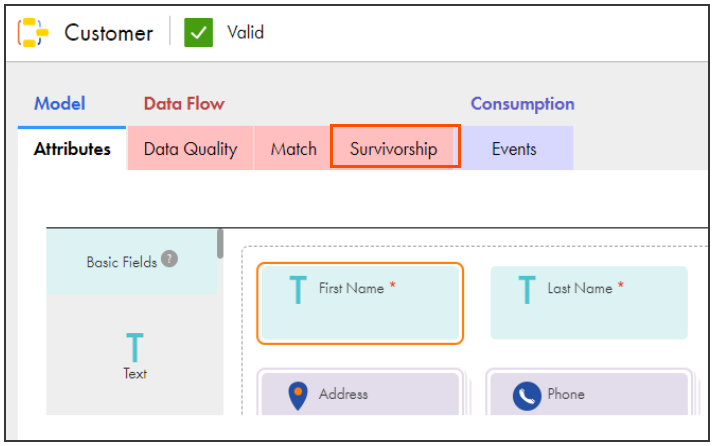
14. In Business 360, click **Explore** > **Student Project**.



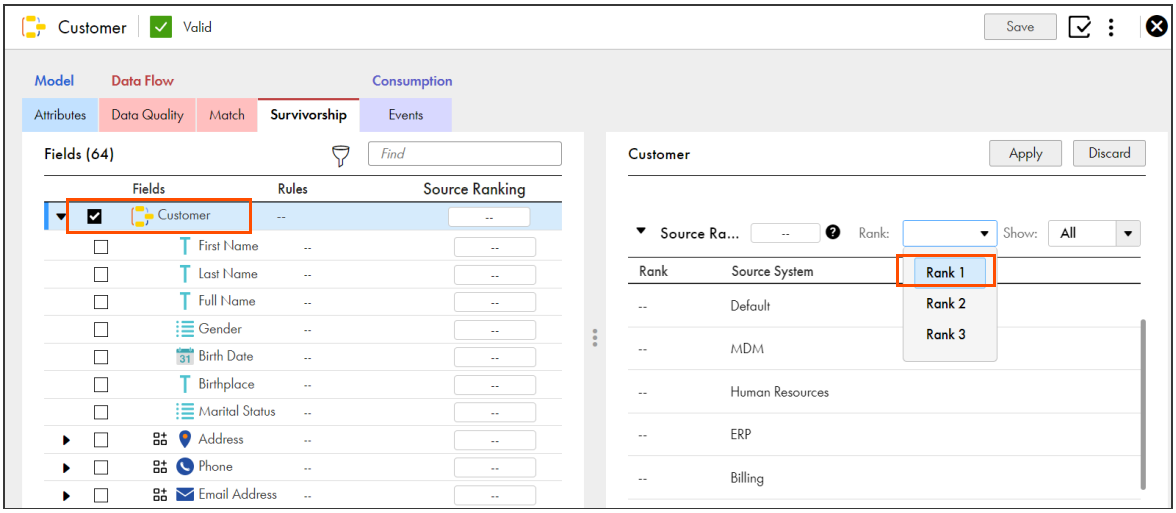
15. Open the **Customer** Business entity.



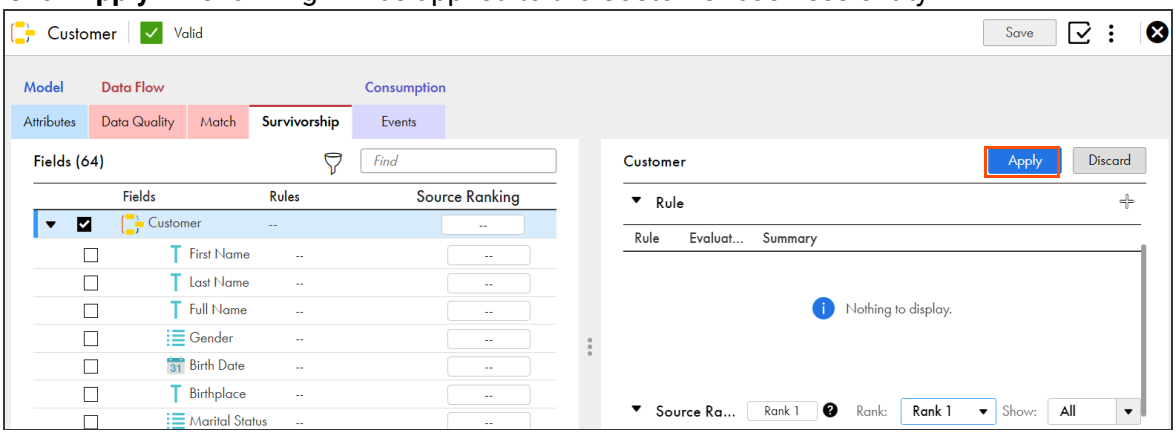
16. Click the **Survivorship** tab.



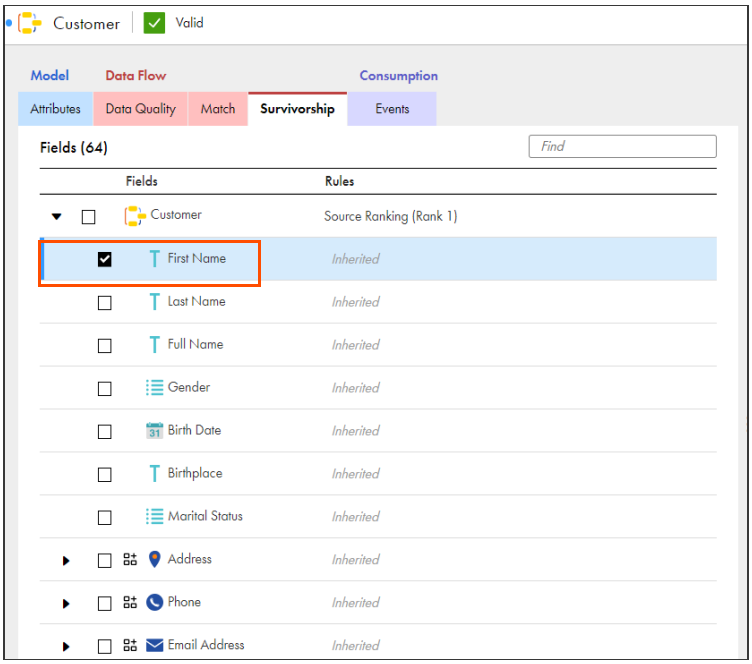
17. Click Customer check box and on the right hand side, from the Rank drop-down, select **Rank 1**.



18. Click **Apply**. The ranking will be applied to the Customer business entity.



19. Now de-select Customer and click the **First Name** field check box



Trust Levels

The trust levels indicate the confidence you have on the source system. The decay rate indicates the rate at which the trust level must decrease over a period. You can configure different trust levels and decay rate for each source system. First, you will provide ranks to the source systems. In this case, you can trust Informatica Multidomain MDM source system the most, and then the HR, ERP, Billing, and Sales.

20. In the **Source Ranking** section, hover the mouse on each Source System listed, and click **Add** and provide the ranks either by using the upward and downward arrows or by entering the ranks in the **Rank** field.

Refresh the page if you are unable to see the source systems.

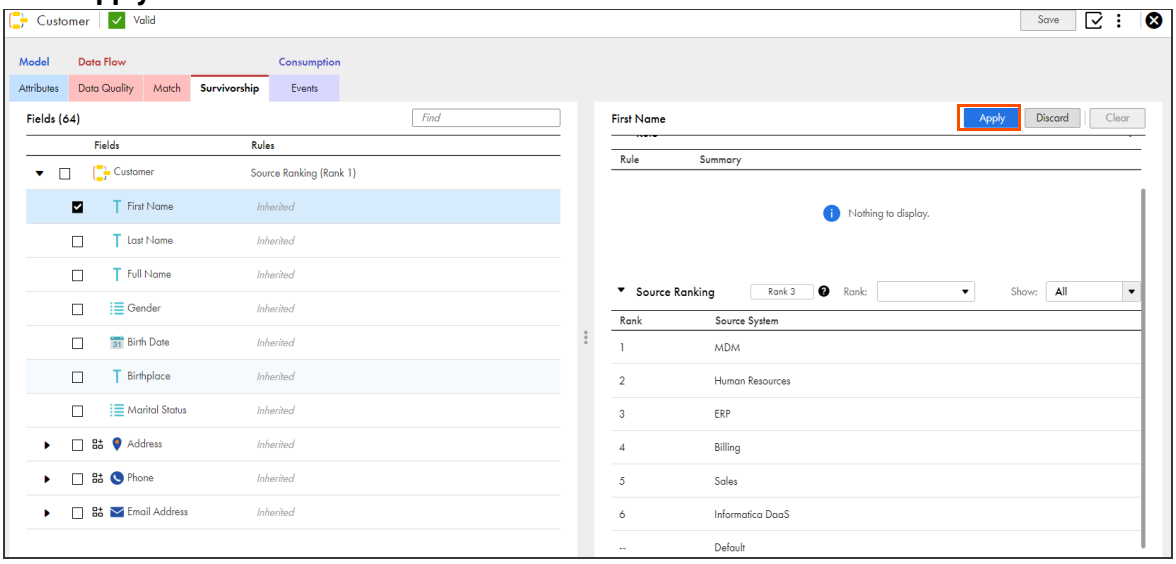
21. The ranking should be in the following order. This is the order of survivorship.

- MDM
- Human Resource
- ERP (Entity Resource Planning)
- Billing
- Sales
- Informatica DaaS

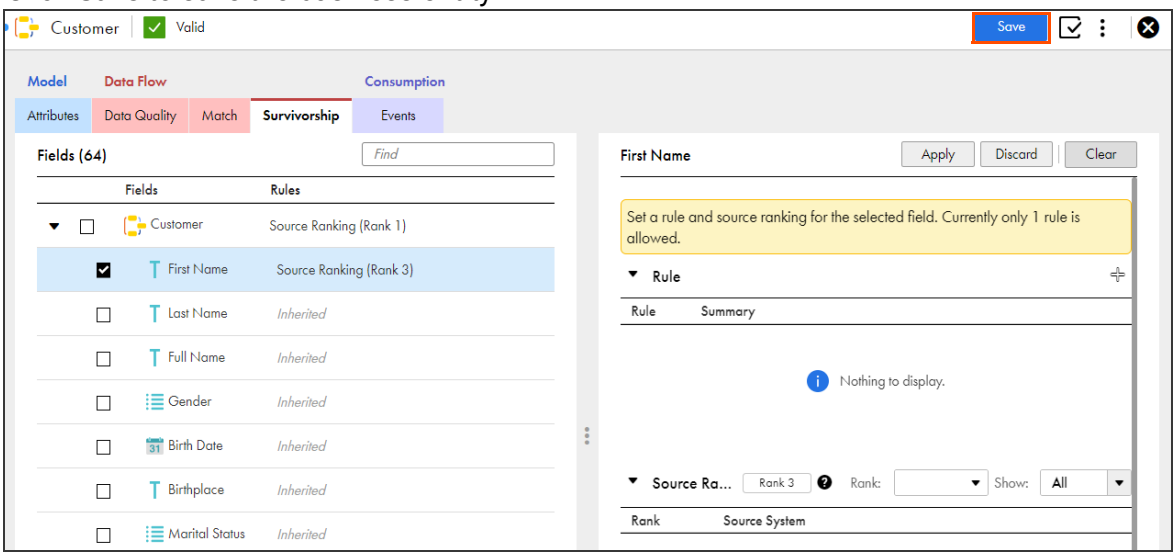
Source System	Rank
MDM	1
Human Resources	2
ERP (Entity Resource Planning)	3
Billing	4
Sales	5
Informatica DaaS	6

This means that if the First name **Smith** is coming from MDM, it wins over the same name **Smyth** or **Smith** coming from Human Resources or ERP. Informatica MDM becomes the preferred source (highly trusted source), and the others follow.

22. Click **Apply**.



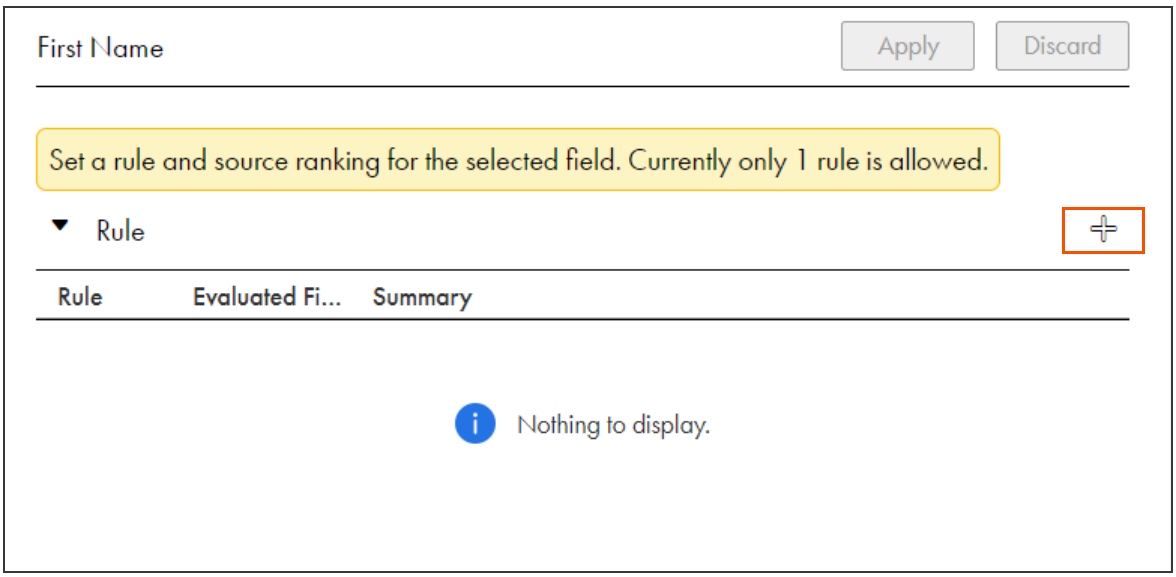
23. Click **Save** to save the business entity.



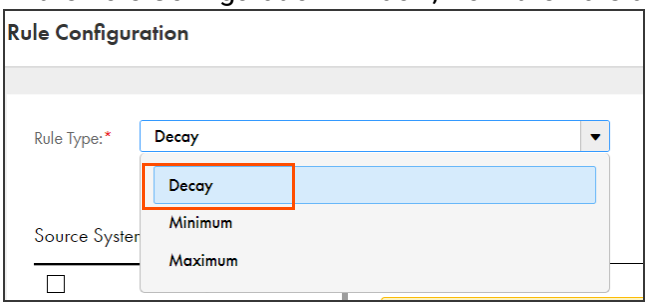
Configure Decay Rule

After defining the trust levels or rankings, you should now define the decay rates for different fields that indicate the rate at which the trust level must decrease over a period.

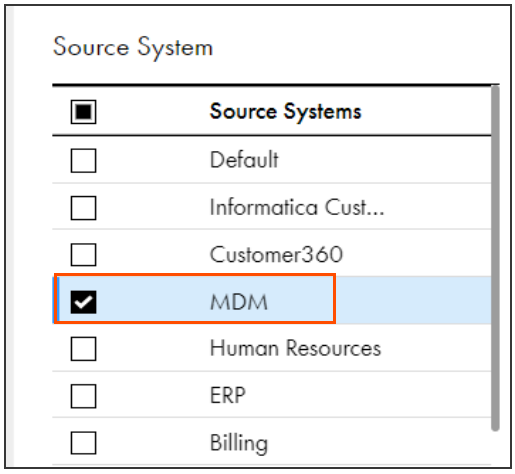
24. Under **Survivorship** tab, de-select any field if selected and select the First Name. Click the **+** icon to add a new rule.



25. In the Rule Configuration window, from the **Rule** drop-down, select **Decay**.



26. Select **MDM**, the highly trusted source.

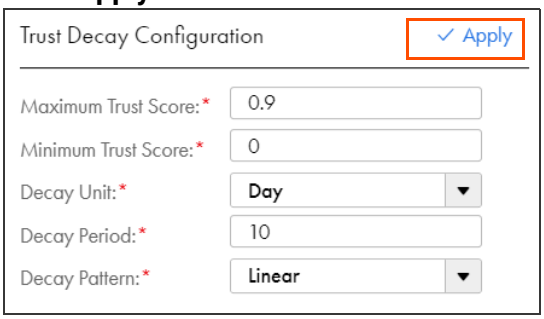


27. In the **Trust Decay Configuration** section, enter the following values:

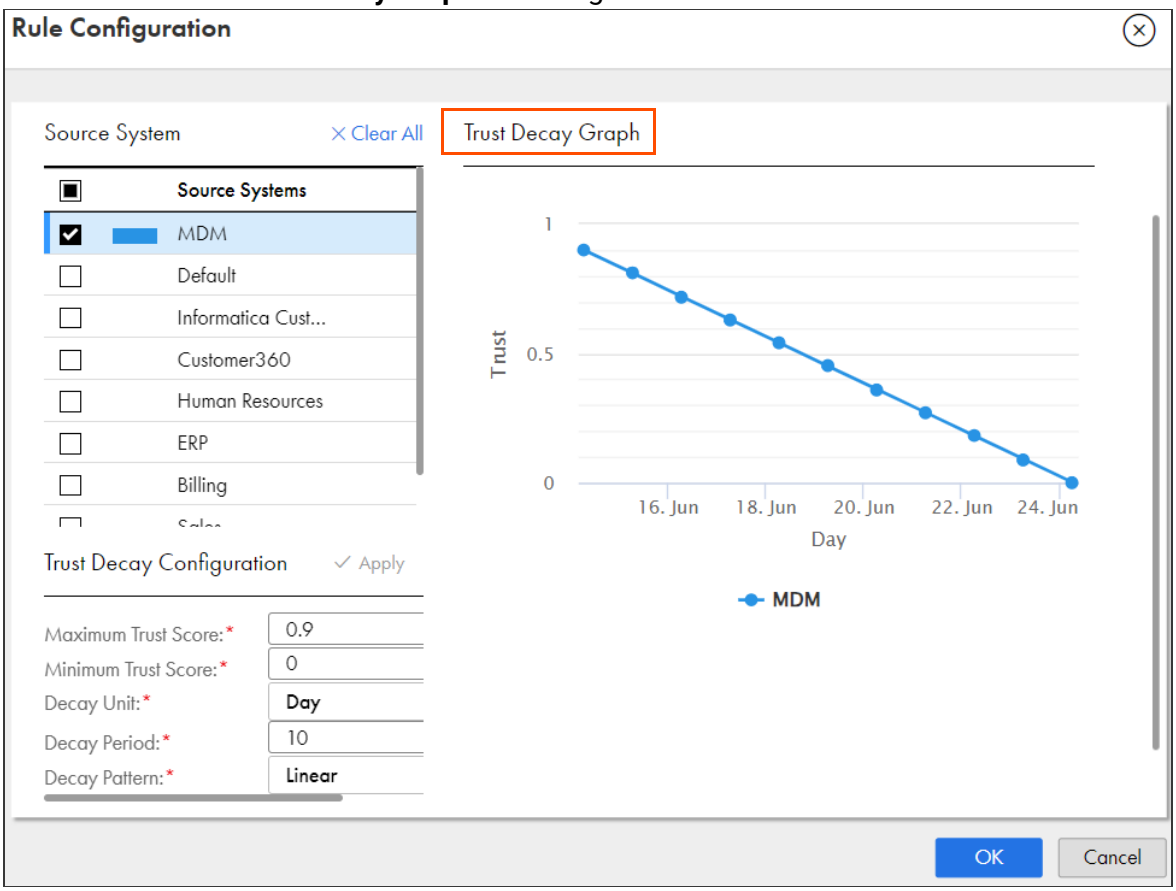
Field	Value
Maximum Trust Score	0.9
Minimum Trust Score	0
Decay Unit	Day
Decay Period	10
Decay Pattern	Linear

This means that if the system is not updated in 10 days, the trust score decreases from 0.9 to 0.

28. Click **Apply**.

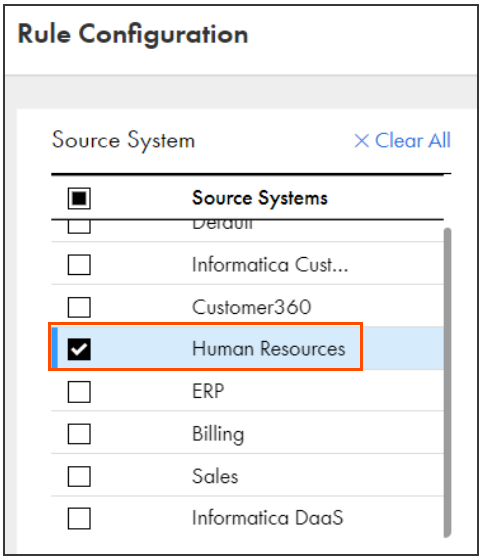


29. You can view the **Trust Decay Graph** on the right side.



30. In the Source System tab, clear any existing selections.

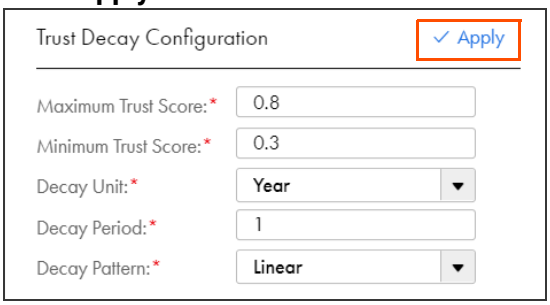
31. Select **Human Resource** from the Source System.



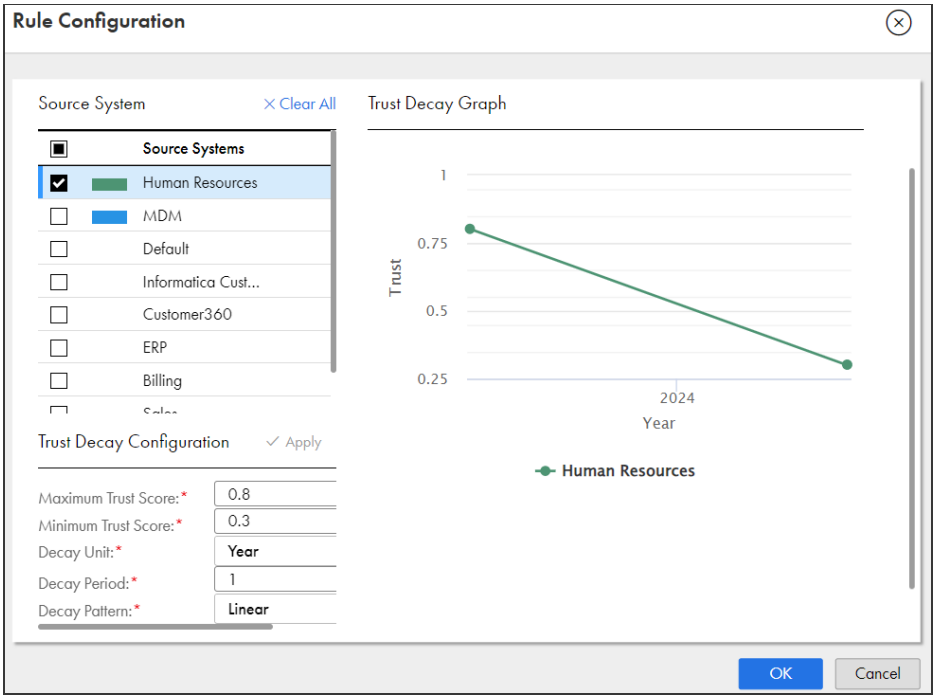
32. In the **Trust Decay Configuration** section, enter the following values:

Field	Value
Maximum Trust Score	0.8
Minimum Trust Score	0.3
Decay Unit	Year
Decay Period	1
Decay Pattern	Linear

33. Click **Apply**.



Notice the **Trust Decay Graph** on the right side.



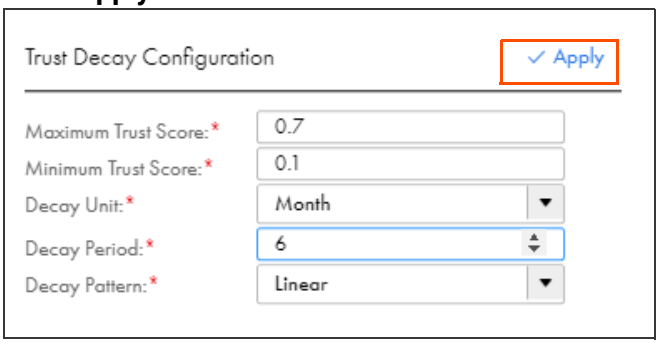
34. Similarly, clear previous selections (if any).

35. Select **Sales** Source System and enter the following values in the **Trust Decay**

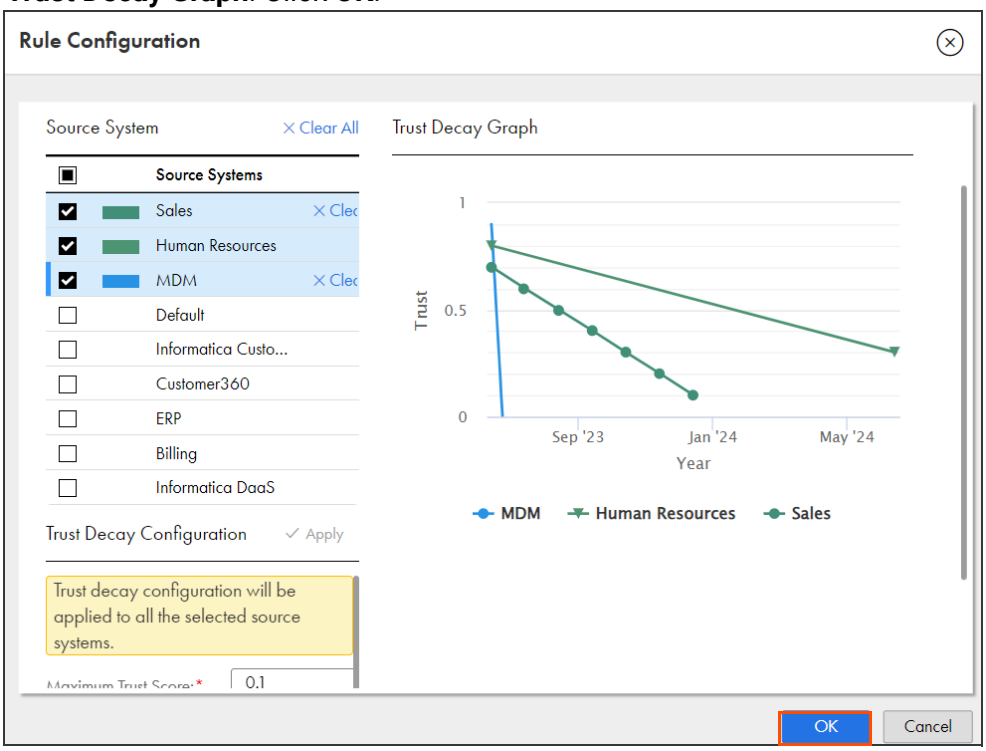
Configuration section:

Field	Value
Maximum Trust Score	0.7
Minimum Trust Score	0.1
Decay Unit	Month
Decay Period	6
Decay Pattern	Linear

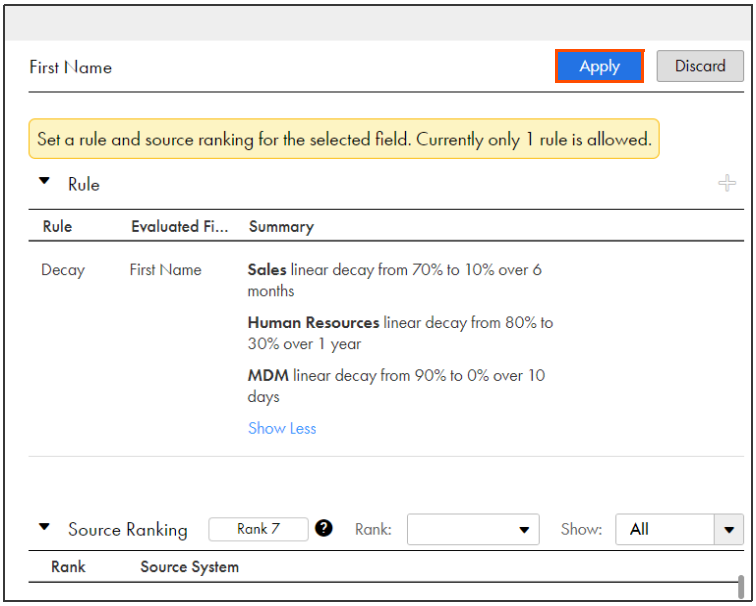
36. Click **Apply**.



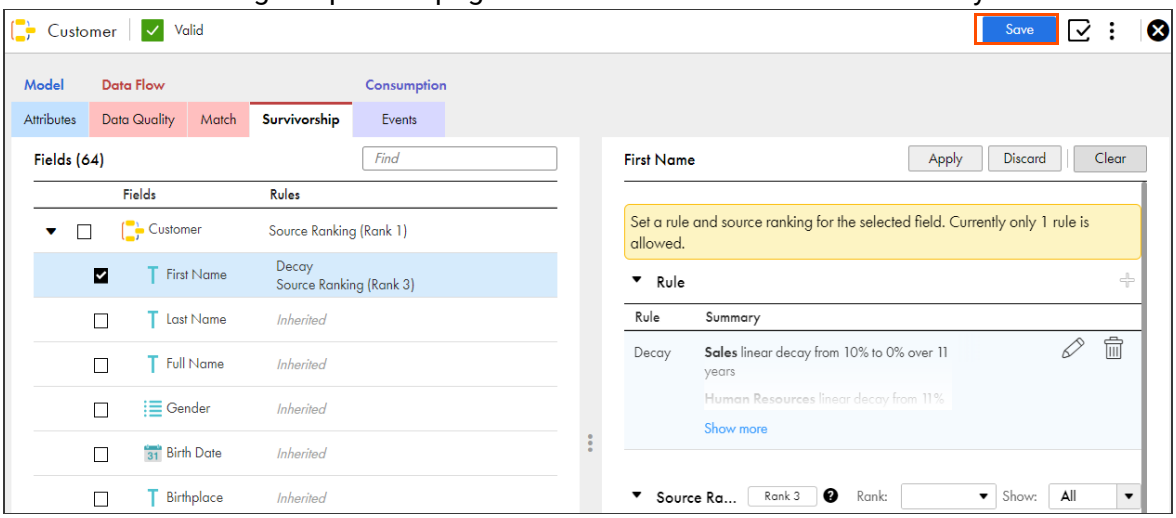
37. Select all the three source systems for which you entered the trust score and view the **Trust Decay Graph**. Click **OK**.



38. On the entity page, click **Apply** to save the First Name field.



39. Click **Save** on the right top of the page to save the Customer business entity.

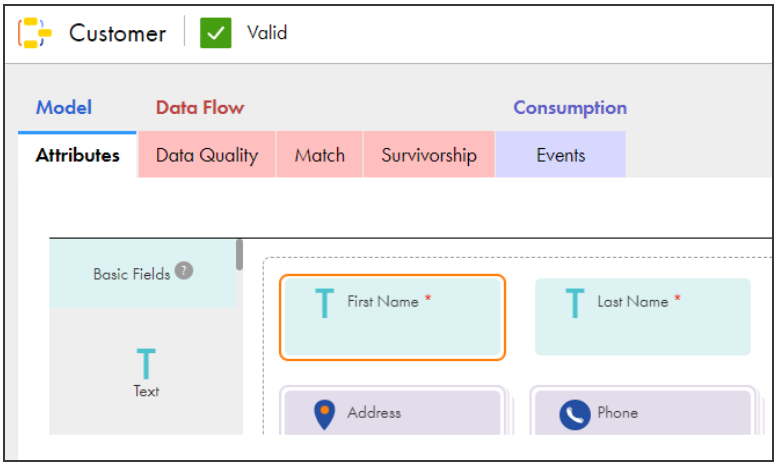


Create Validation Rules to Downgrade Trust Score

Validation follows survivorship. For example, if the highly trusted MDM has a NULL value, the less trusted ERP will win over the highest trusted source. Hence, you will need validation rules to downgrade the scores.

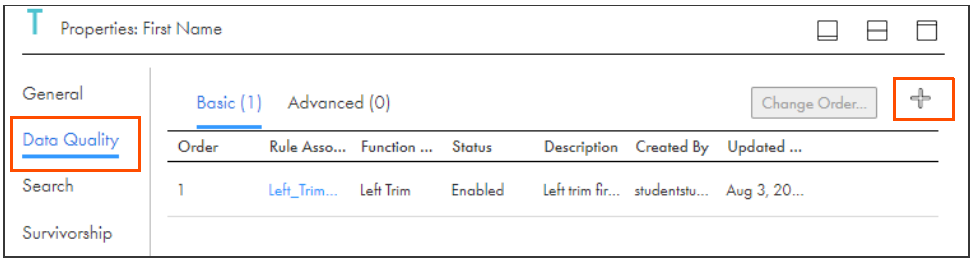
40. As soon as you click Save in the Decay Configuration page, you will be navigated back to the Customer entity page. If you have navigated elsewhere, click **Explore** > **Student Project** > **Customer** entity page.

41. Click the **Attributes** tab and select **First Name**.



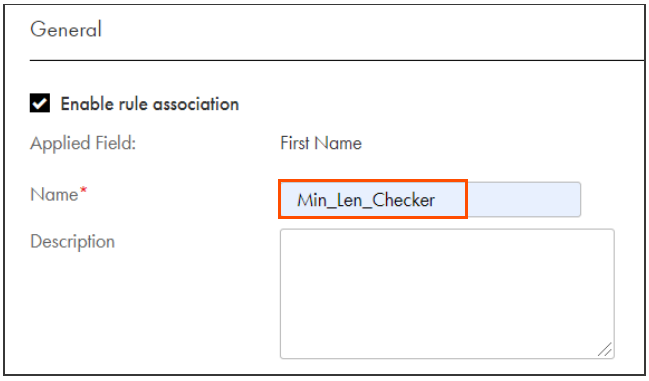
42. In the Properties section, click **Data Quality**.

43. From the Basic tab, click **+** icon to add a Basic rule association.



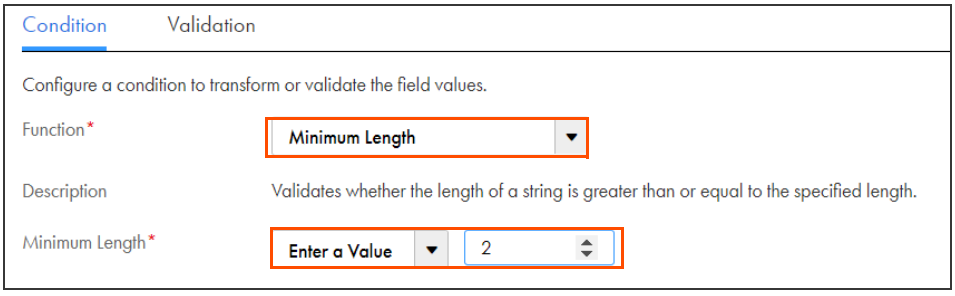
Rule Logic: Minimum length of the First Name cannot be less than or equal to two characters. Minimum length function validates whether the length of a string is greater than or equal to a specified length. If the minimum length is less than 2 characters, the information is not valid and hence you will need to downgrade the score by 90%.

44. Enter the Name of the rule as **Min_Len_Checker**.



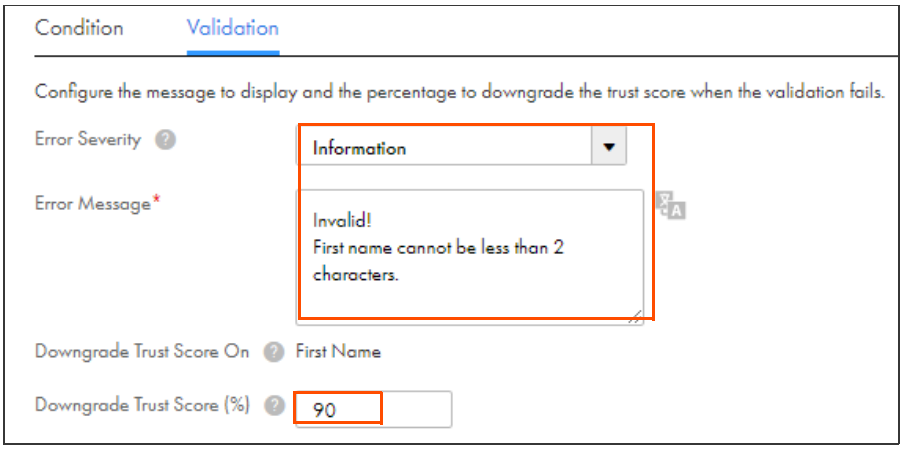
45. In the Condition section, enter the following values:

Field	Value
Function	Minimum Length
Minimum Length	Enter a Value
Value	2

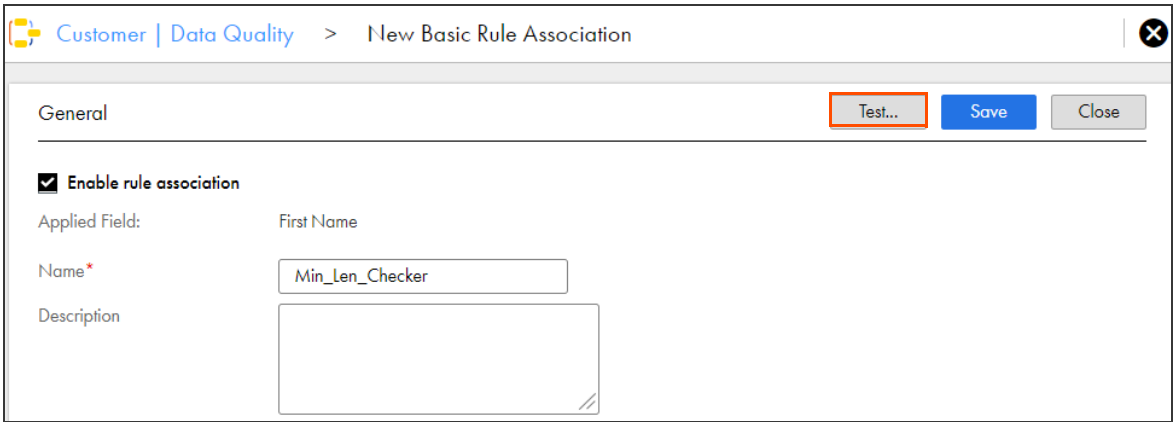


46. Click the **Validation** tab and enter the following values:

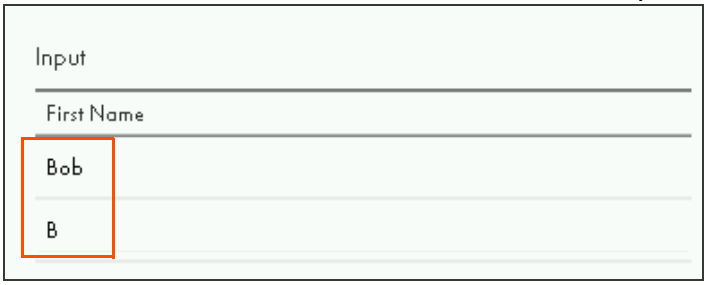
Field	Value
Error Severity	Information
Error Message	Invalid! First name cannot be less than 2 characters.
Downgrade Trust Score (%)	90



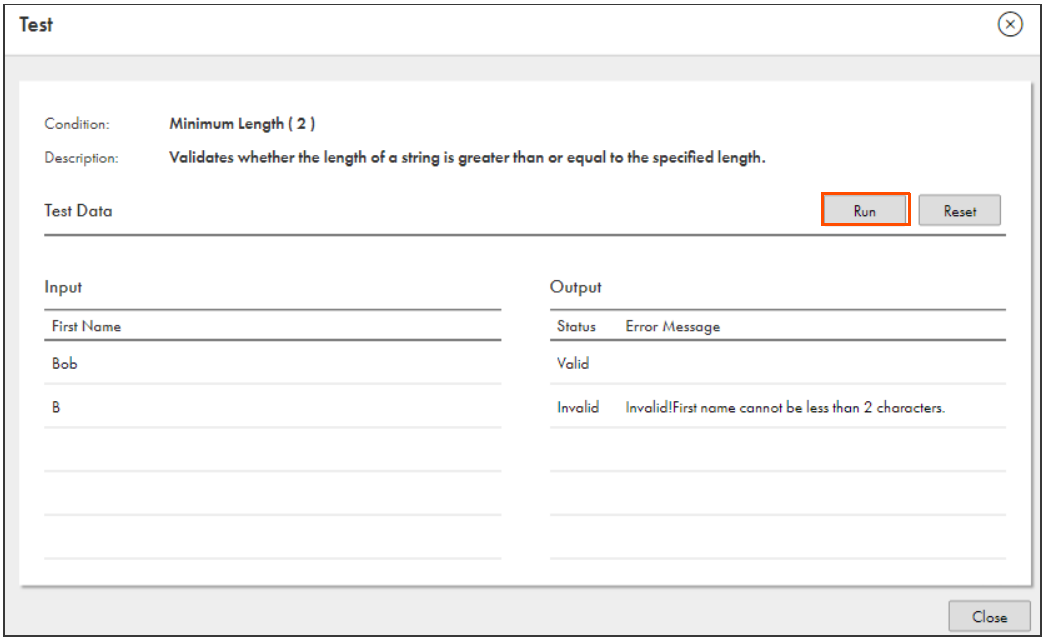
47. Click **Test**.



48. In the **Test** window, enter any two values in the **Input** table – one with two characters and another with more than two characters, for example, **Bob** and **B**.

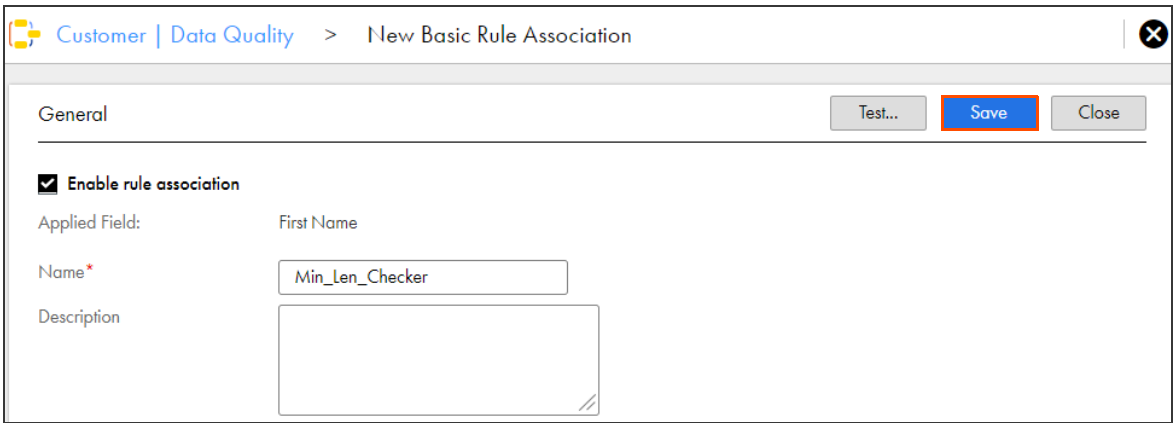


49. Click **Run** and notice the Status and the Error Message columns under Output section.

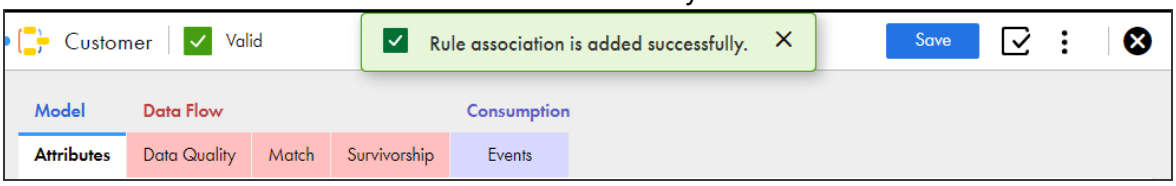


50. **Close** the Test window.

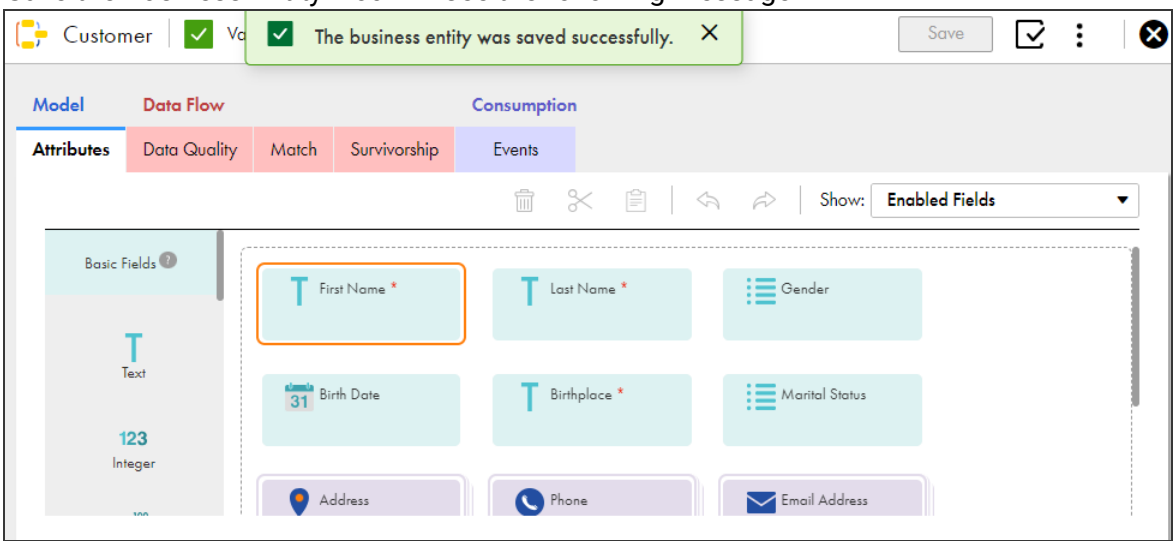
51. Click **Save**.



Notice that the rule association is added successfully.



52. Save the Business Entity. You will see the following message:



Self-Try Exercise (Mandatory)

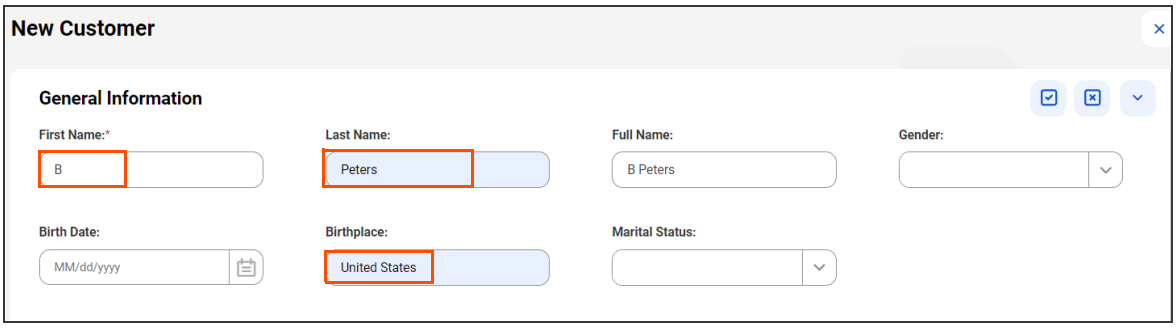
Testing the Validation Rule in Multidomain MDM

You will now test the validation rule in Multidomain MDM.

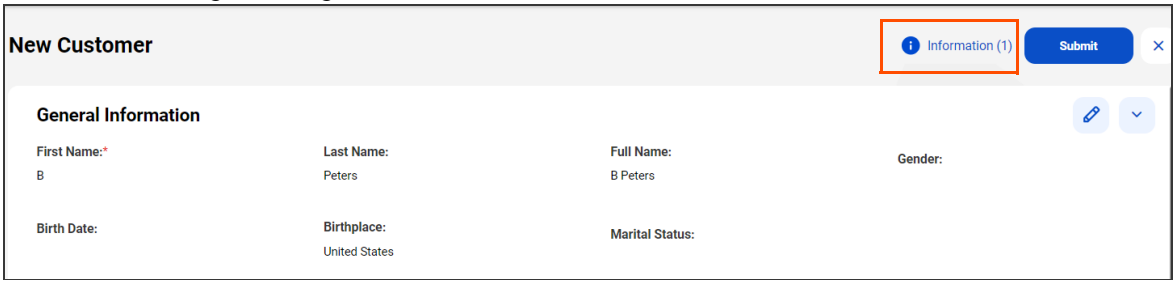
53. Open the Multidomain MDM Service in another browser if you have closed it. Refresh the browser.

54. Click **New** > **Customer**.

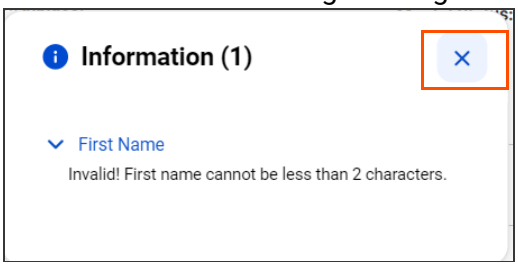
55. Enter the First Name as "**B**", Last Name as **Peters** and Birthplace as **United States**. Click the check mark.



56. A warning message is displayed on the top right side of the page. Click the **Information** to view the warning message.

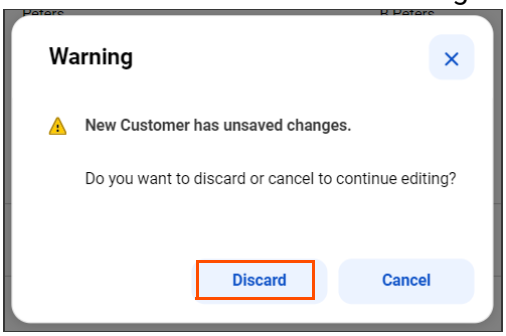


57. You will see a message that you have configured while creating the rule. This is neither a warning nor an error. It is just an intimation to the user that the value entered may be invalid. Close the warning message.



58. **Close** the record.

59. You will see the Invalid Asset dialog box. Click **Discard** to discard the changes.



This concludes the lab.

Instructor Demo 7-1: Enabling DaaS Rule for Phone and Address Fields

Overview:

A DaaS rule association is an association between a field group and Informatica Data as a Service (DaaS). You can use the DaaS rule association only on the fields that store postal addresses, email addresses and phone numbers. To use the required DaaS services, add valid license keys in the Global Settings page.

Objectives:

- Enabling DaaS rule association for Phone and Address fields
- Test the DaaS Rule Association in Multidomain MDM

IMPORTANT: This lab is NOT for execution, it is only for INSTRUCTORS DEMO.

Duration:

10 minutes

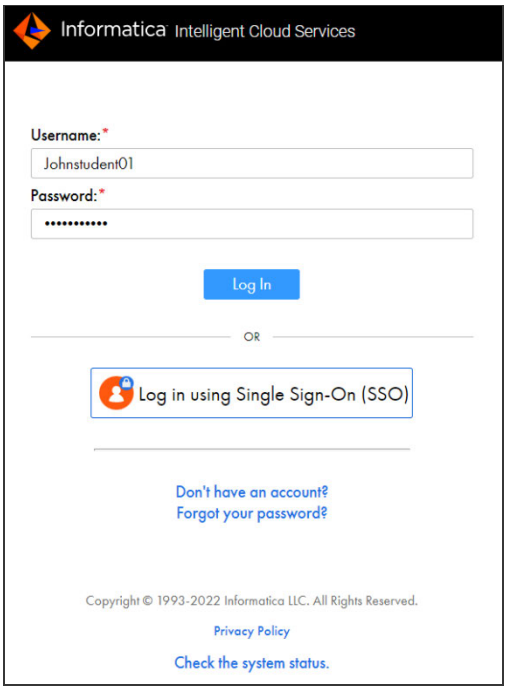
Tasks:

Creating DaaS Rule Association

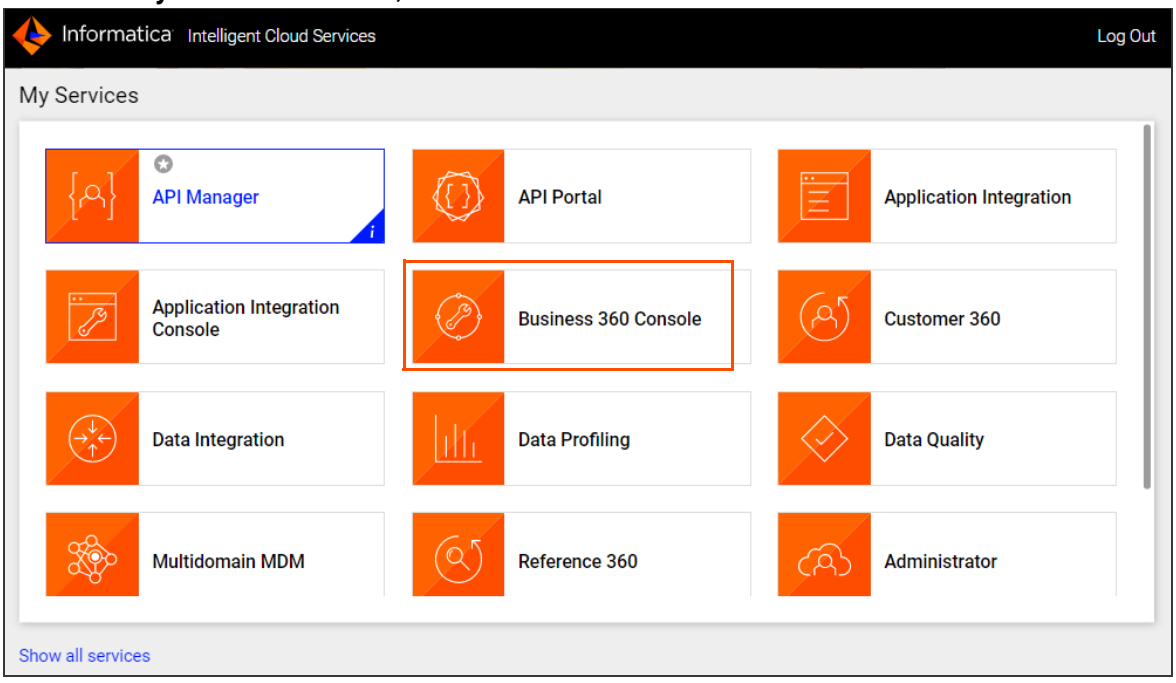
IMPORTANT: The CD team has already configured the following steps. Please walk the students through these steps accordingly.

1. Log in to IICS with the credentials that are given in the course set up guide. Do not share these credentials with the students.

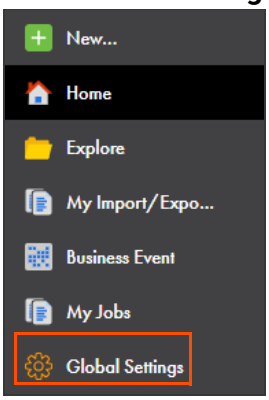
IMPORTANT: The Instrustors will perfom these steps in the org using the credentials provided in course set up guide.



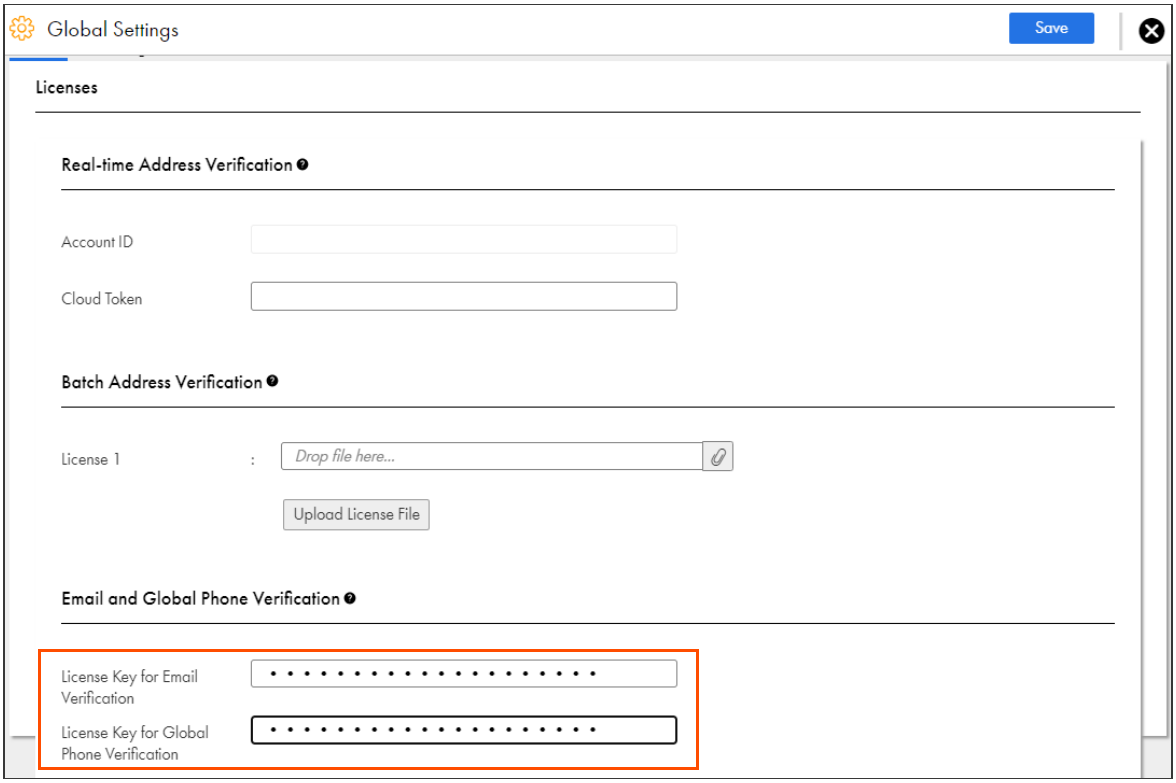
2. From the **My Services** window, click **Business 360 Console**.



3. Click **Global Settings**.



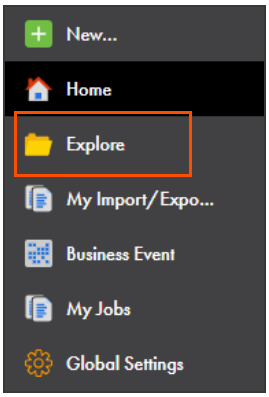
4. You can see the License Key for Email Verification and License Key for Global Phone Verification in the respective fields. Students can procure these keys from their organization.



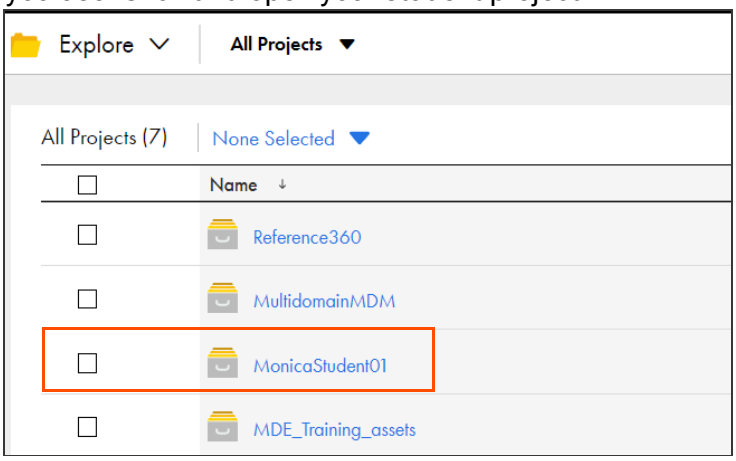
Enabling DaaS Rule

Enable DaaS rule association in Business 360 Console.

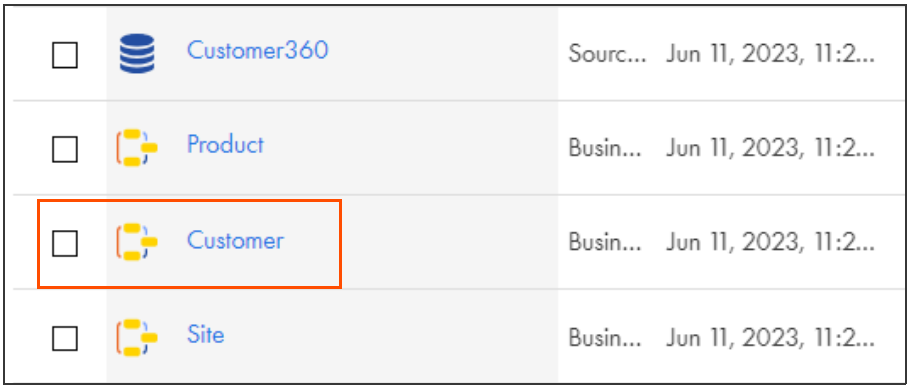
5. To create DaaS rule association, click **Explore**.



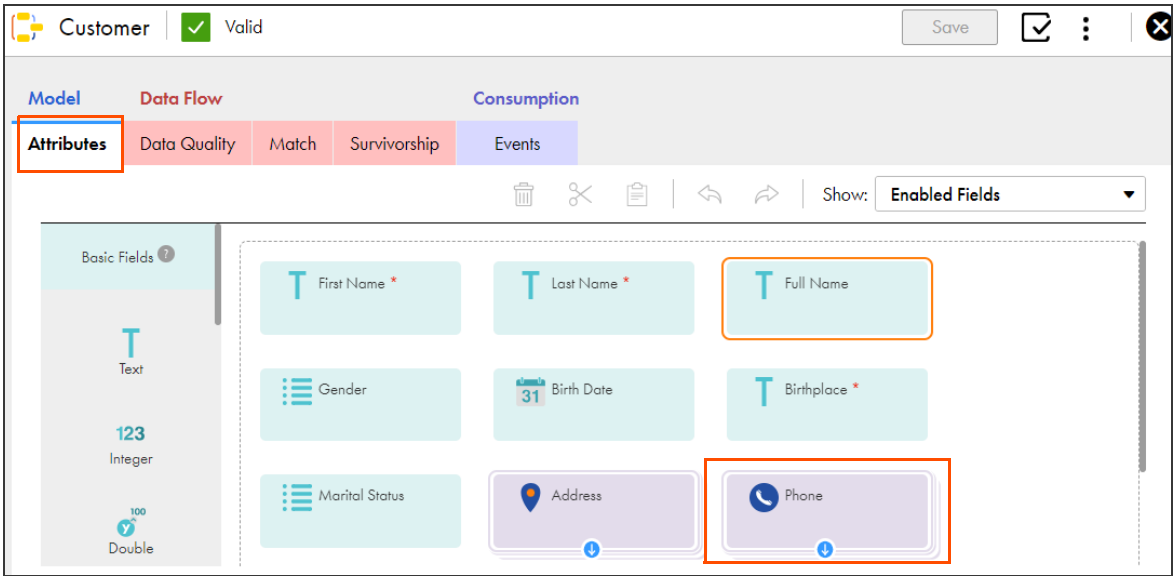
6. In the Explore page, you can see the projects, folders, and assets for all the services that you use. Click and open your student project.



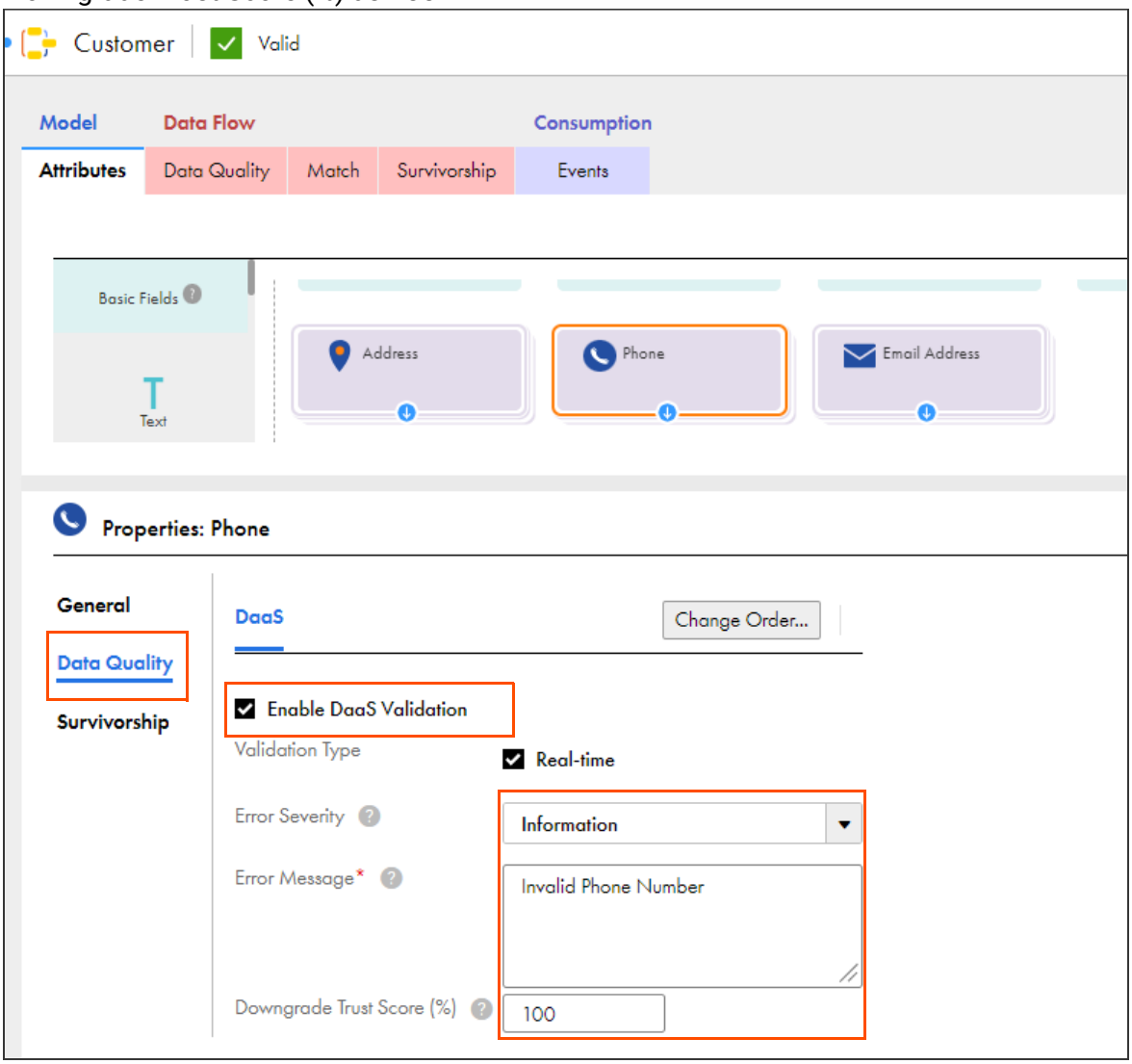
7. Open the **Customer** business entity.



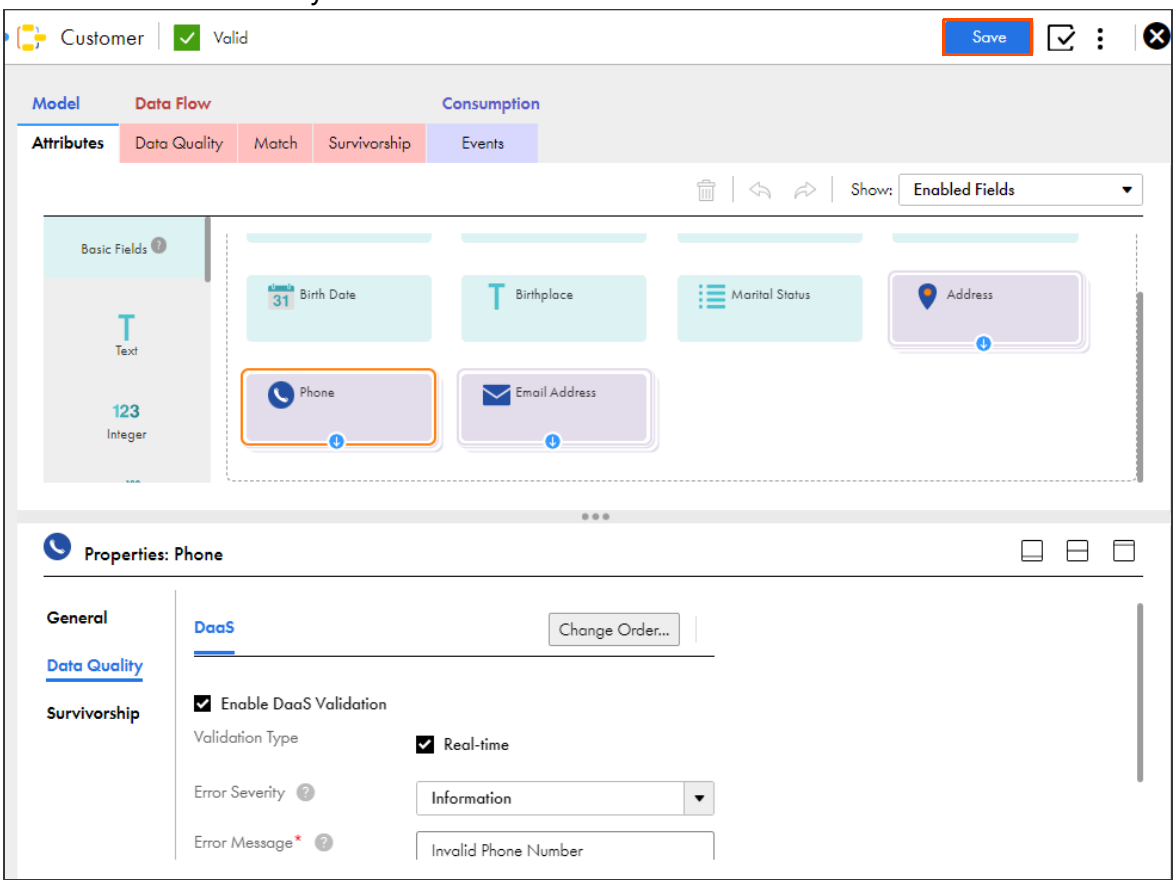
8. The business entity page opens. The canvas displays all the fields that the business entity includes. In the Attributes tab, select **Phone**.



9. In the **Properties** section, click **Data Quality** and check the **Enable DaaS Validation** field. Enter the Error Severity as **Information**, Error Message as **Invalid Phone Number** and Downgrade Trust Score (%) as **100**.



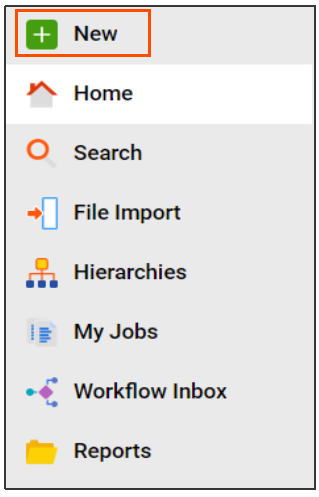
10. **Save** the Business Entity.



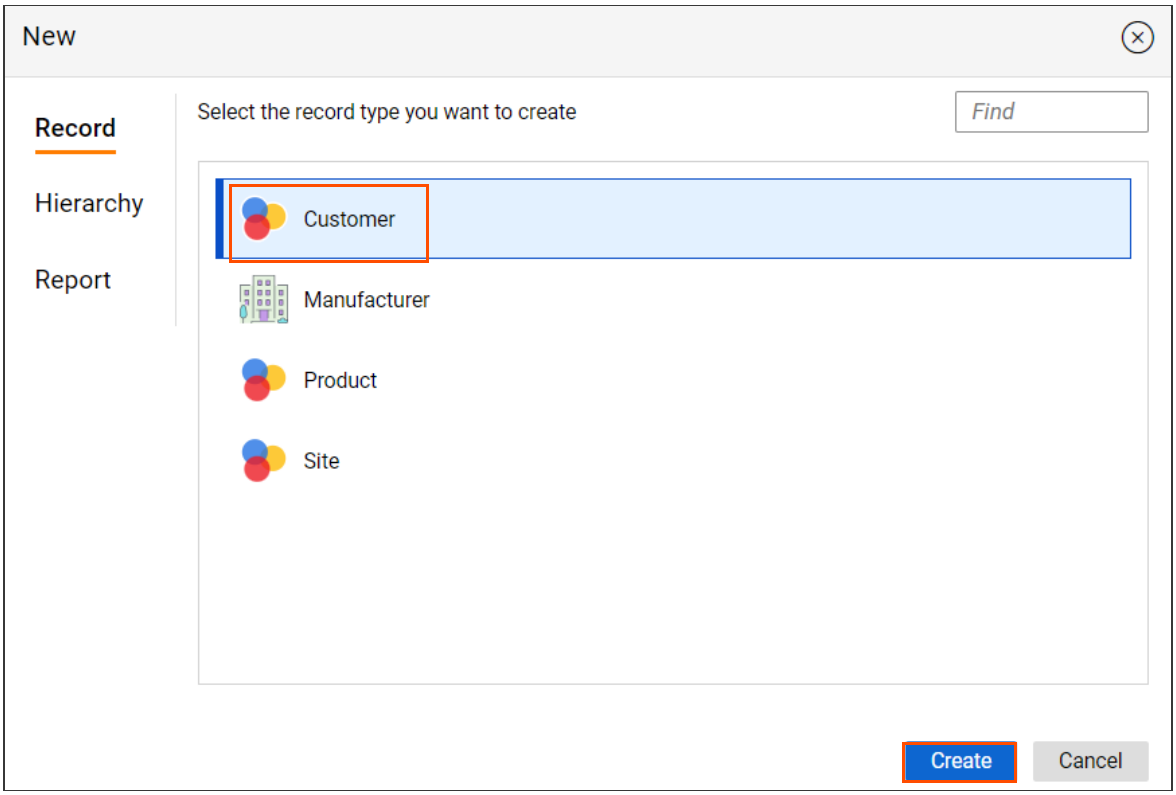
Test the DaaS Rule Association in Multidomain MDM

11. Using product switcher, open **Multidomain MDM** application, or refresh the Multidomain MDM page if it is already open.

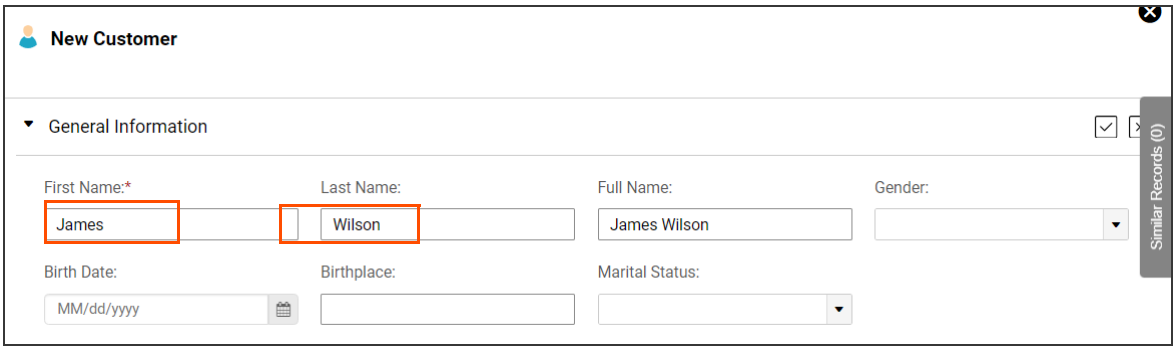
12. In the Multidomain MDM user interface, click **New**.



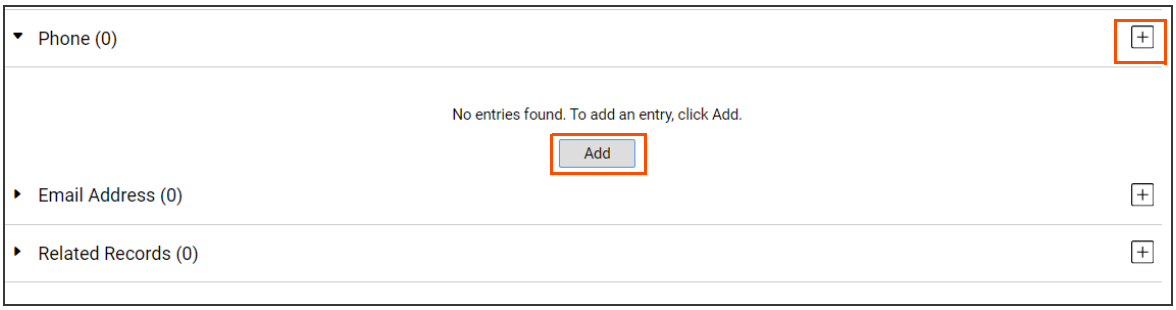
13. Select **Customer** and click **Create**.



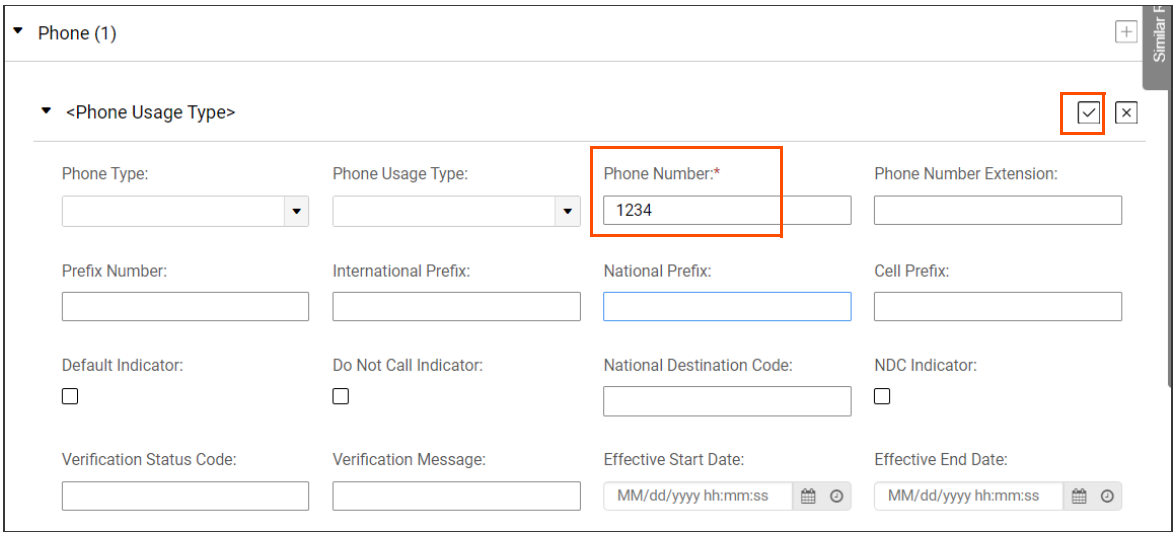
14. In the First Name as **James**, Last Name as **Wilson** and click on the check mark to save the details



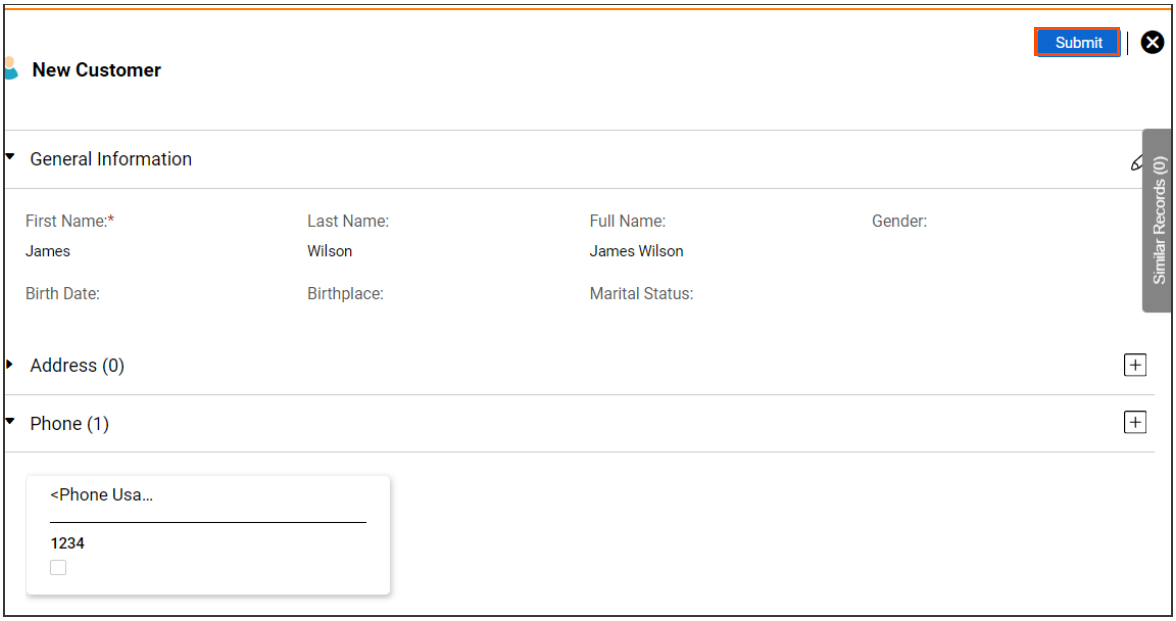
15. Click the **+** icon to add the Phone number and click **Add**.



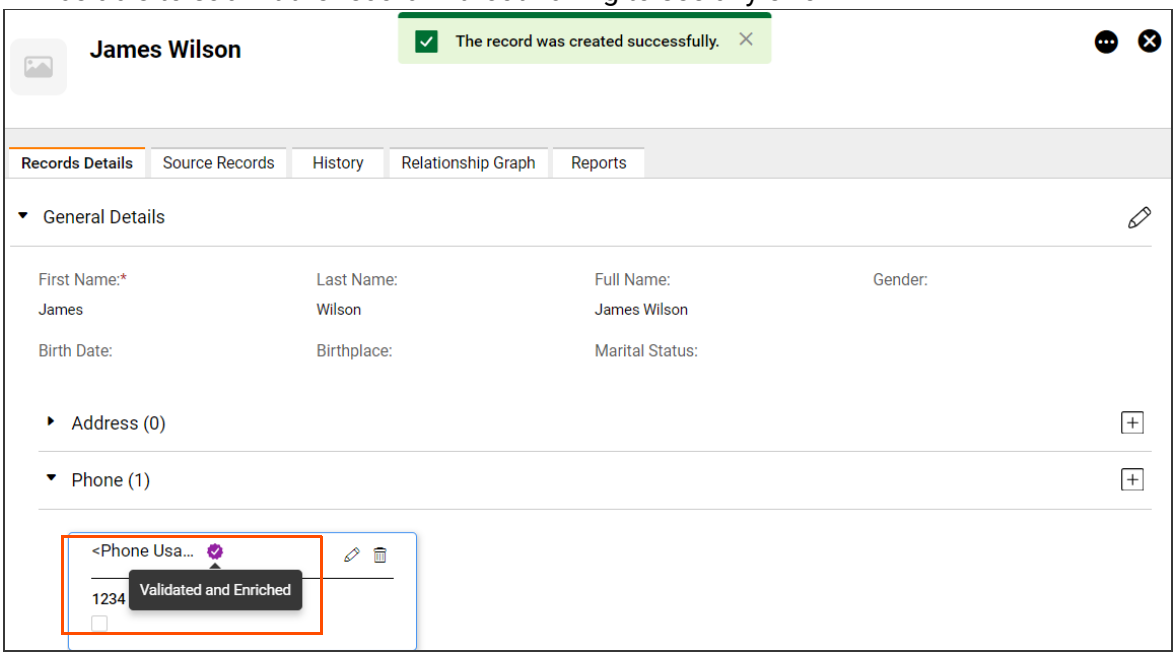
16. Enter the Phone Number as **1234** and click the check mark to save the details.



17. **Submit** the record.



The record will be saved. You can see that the Phone Number field has been Validated and Enriched. Since while creating the DaaS, we have selected Information as Error Severity, you will be able to submit the record without having to see any error.



This concludes the lab.

Key Takeaways

- Data quality rules are configured on the fields of the business entity fields to cleanse, standardize, or validate the field values.
- You can improve the exact match results by automatically including the uniqueness criteria fields as exact match fields in the declarative rules.
- Survivorship is the process of creating a master record or the most trusted version of a record from multiple matching source records.
- Decay rule Identifies the trusted value based on the maximum and minimum trust scores of a source system and the rate at which the scores decay.

Module 8: Managing Custom User Roles and Privileges

User and User Roles

A **user** is an individual account in Informatica Intelligent Cloud Services. Multidomain MDM and Business 360 Console users must have Informatica Intelligent Cloud Services user accounts. You can create and manage users in Administrator. A **user role** is a collection of privileges that you assign to users to allow access to Multidomain MDM and Business 360 Console.

Pre-defined User Roles

Predefined user roles are a collection of predefined privileges that you can assign to users. The predefined user roles define the privileges for different types of assets and service features. You cannot edit, rename, delete, or customize predefined user roles.

Assign the following types of roles to provide privileges for MDM SaaS users:

- Business 360 Roles
- Customer 360 Roles
- Product 360 Roles
- Supplier 360 Roles

Create a Custom User Role

You can create and assign user roles to users and user groups in your organization. A user is an individual account that allows secure access to an organization. A user group is a group of user accounts in which all members of the group can perform the same tasks and have the same access rights for different types of assets. Users and groups can perform tasks and access assets based on the roles that you assign to them.

You can create custom user roles and configure privileges for them in your organization. You can control access to assets, attributes, and features of business applications for each custom user role in your organization. **For example**, you can control a custom user role to view records but not update or edit the records. The system-defined roles include predefined access privileges in the Administrator, and you cannot change them.

The custom user roles are not pre-defined and you can create them based on the specific requirements of your organization. You cannot delete system-defined roles but you can delete custom roles. You can also control the specific tasks that a custom user role can perform in your organization. **For example**, you can authorize a custom user role to import data into Multidomain MDM.

You can use the Administrator to create custom user role and assign permissions at the asset level. You can then fine-tune the permissions in the Security page of Business 360 Console. The authorizations that you configure in the Administrator works in conjunction with the authorizations that you configure on the Security page. Unlike predefined roles, custom user roles can be created based on the specific requirements of your organization. You can control access to assets, attributes, and features of business applications for each custom user role in your organization.

Cloning Pre-defined Roles

You can clone a predefined user role in Administrator to create a custom user role. However, these custom user roles can't perform all the tasks that a predefined user role can perform in your organization.

Example: Designer or MDM Designer user roles can configure business applications but custom user roles can't configure business applications. If you clone the Designer or MDM Designer user role to create a custom user role, the cloned custom user role can't configure business applications.

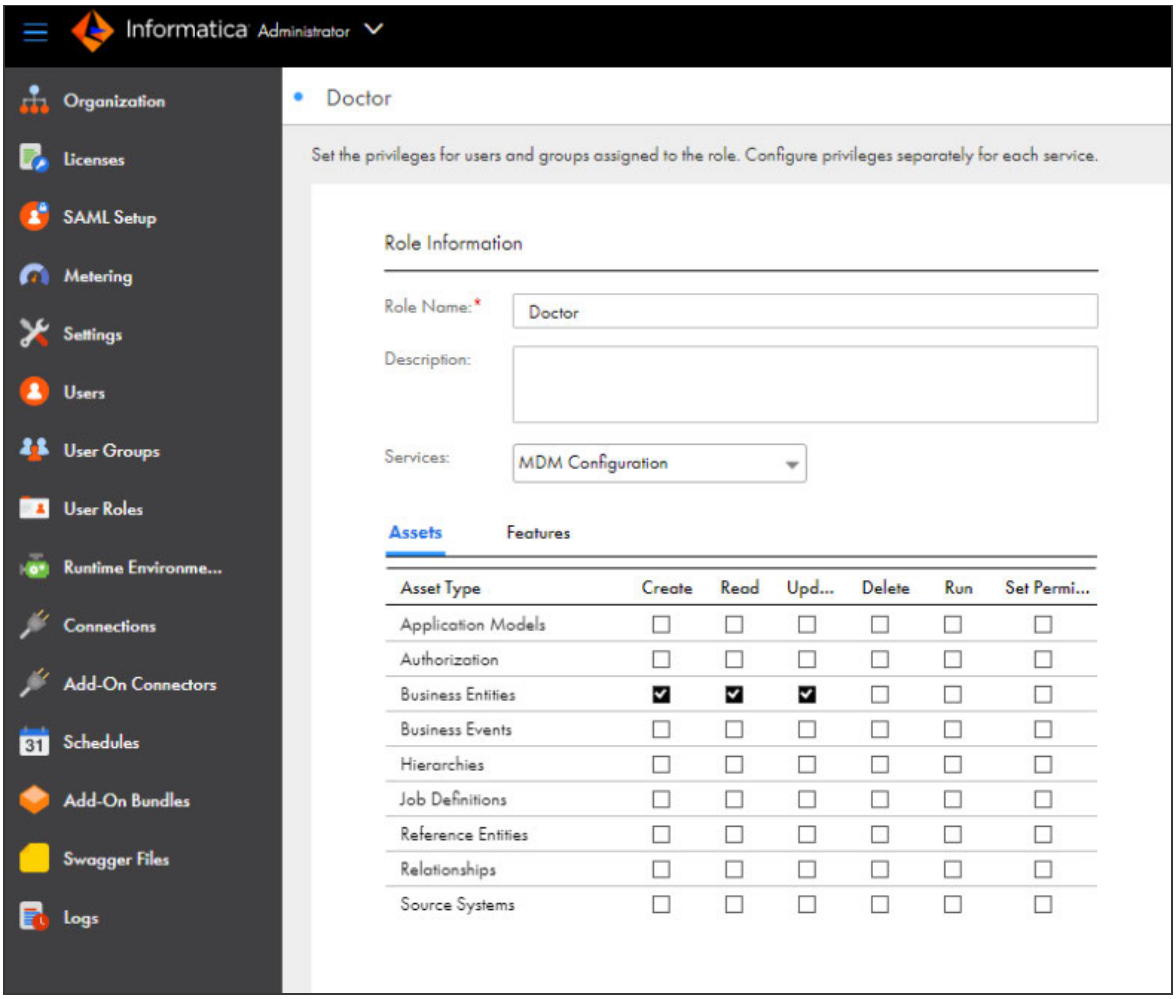
Similarly, if you manually assign all the privileges of predefined user roles to custom user roles, the custom user roles might not be able to perform all the tasks that these predefined user roles can perform. To enable a user to perform a specific task, ensure that you assign the custom user role and other required user roles to the user along with the required privileges to perform that task.

Example: To enable a user to perform file import, assign the custom user role and the Business360ProcessExecutor user role along with the required privileges on the Security page and Administrator service.

Refer to the following steps to create a custom role:

1. In Administrator, click **User Roles**.
2. Select the custom user role for which you want to configure the privileges.
   The role details page appears.
3. Select required service and assign relevant privileges for the custom user role on the Assets and Features tabs.

4. Click **Save**.



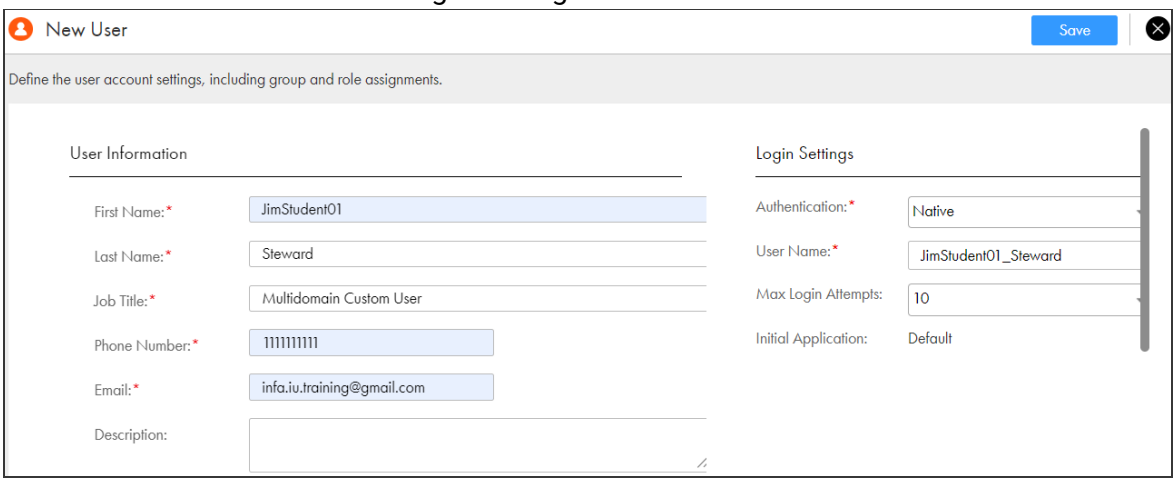
Creating a User

Refer to the following steps to create a custom role:

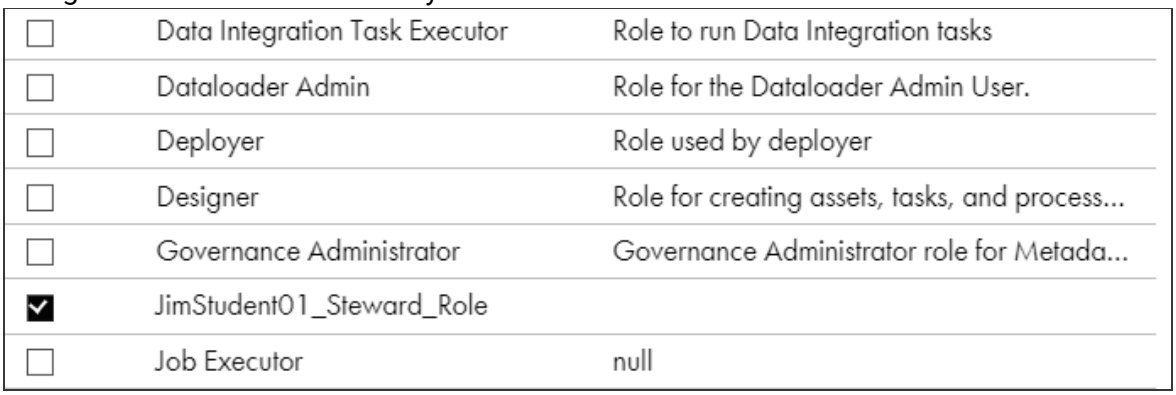
1. Open Administrator service from My Services page.

2. Click **Users** from the navigation menu and then click **Add User**.

3. Enter the user information and login settings.



4. Assign the custom user role hat you created from the list of roles.

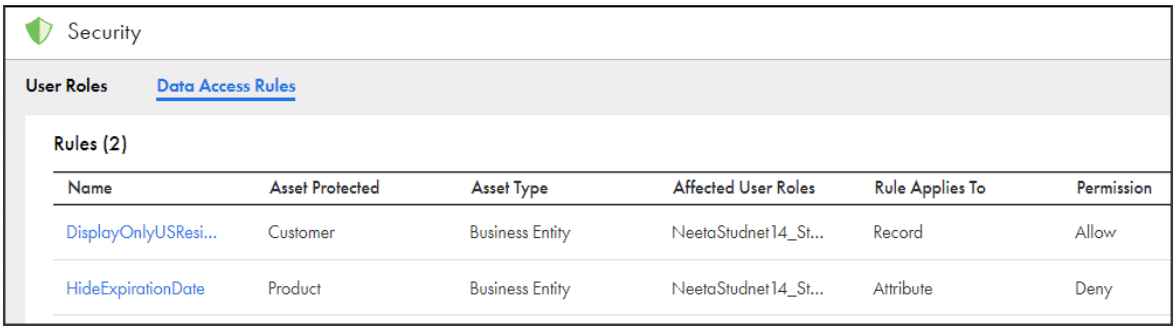


5. Click **Save**.

Role Based Access Control

Once you are done configuring privileges for custom user roles in Administrator service, you need to configure role-based privileges and create data access rules on the Security page of Business 360 Console.

The User Roles tab on the Security page lists all the custom user roles that you create in Administrator. You can select the custom user roles to configure privileges. The Data Access Rules tab on the Security page lists the existing data access rules for custom user roles.



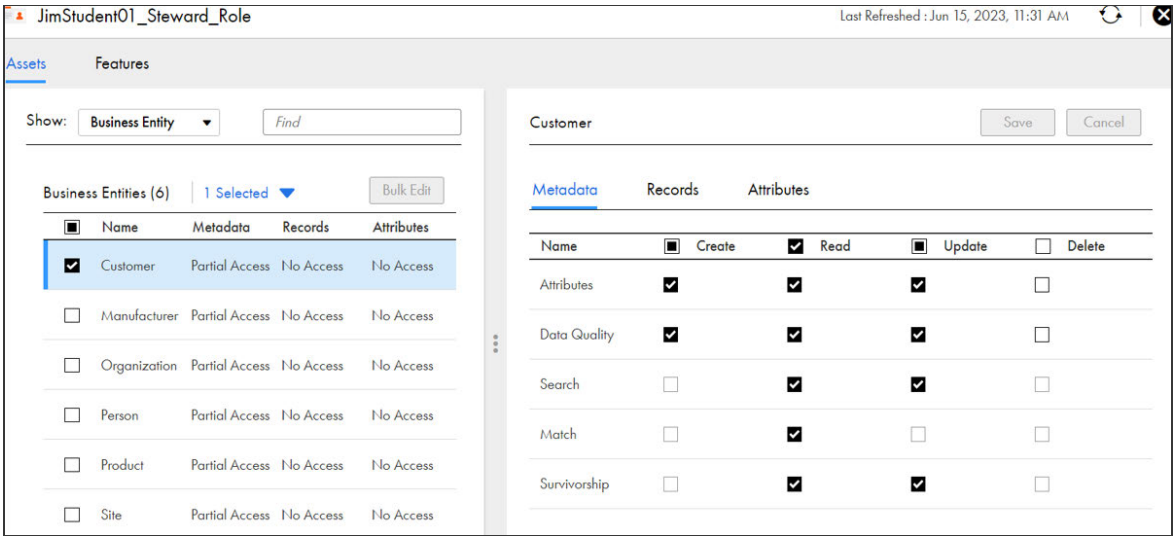
After you assign privileges on the Security page, don't rename the custom user role in Administrator. If you rename the custom user role, the following errors occur:

- The existing privileges on the Records tab of business entities, relationships, and hierarchies are cleared.
- The workflow privileges assigned on the Features tab are cleared.
- The asset-level privileges for business entities assigned for the custom user role in Administrator override the existing privileges on the Metadata tab of the business entity.

Configure Privileges for a Business Entity

You can configure privileges related to a single business entity or multiple business entities for a custom user role. The configured privileges are your basis to configure data access rules for selected business entities. Ensure you configure appropriate base privileges for the business entities to all the roles that you intend to configure data access rules.

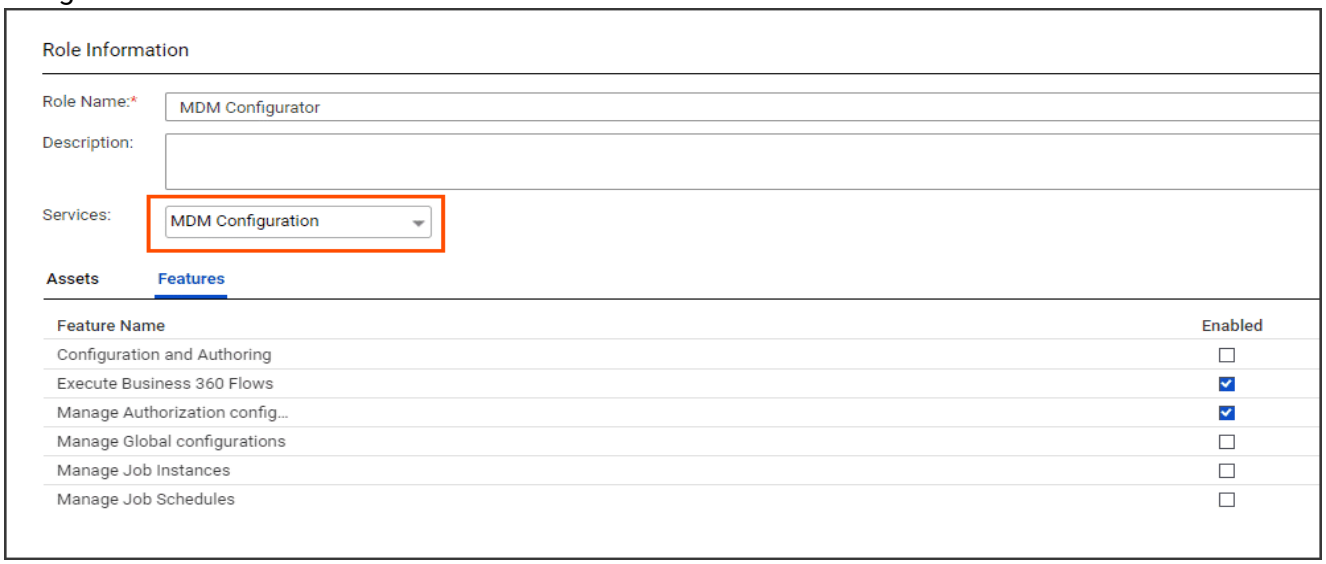
1. In Business 360 Console, on the Security page, click the custom user role for which you want to configure privileges.
   The user role details page appears and lists the business entities for which the user role has access permissions.
2. Select the business entity for which you want to configure privileges.
3. On the following tabs, select required permissions:
   - **Metadata**: Lists privileges related to field properties of the business entity. This tab is available only for the Business 360 Console custom user roles. For more information, see Configure metadata privileges for Business 360 Console user roles.
   - **Records**: Lists privileges that allow users to create, read, update, delete, and unmerge records in the business applications. You can specify whether users can perform tasks that triggers jobs. You can also specify whether users can access some components in the business applications. For more information, see Configure privileges for business application user roles .
   - **Attributes**: Lists privileges that allow users to create, read, update, and delete values for every field and field group in the selected business entity. For more information, see Configure field privileges for business application user roles .



Configure Privileges for a Relationship

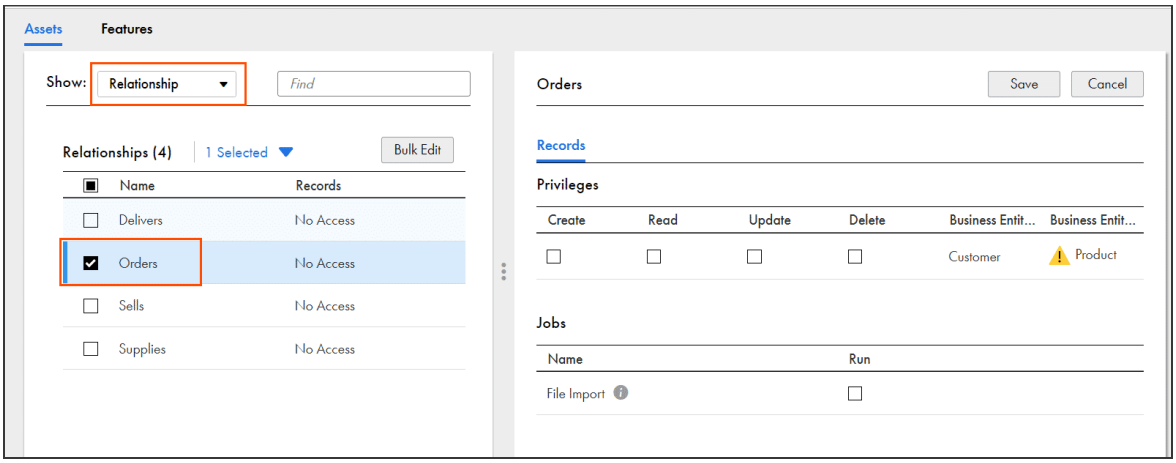
You can configure privileges related to a single relationship or multiple relationships for a custom user role. You can assign required permission for relationships on the Assets tab of the MDM Configuration service in Administrator for a custom user role.

The following image shows the user role details page in Administrator with relationship privileges assigned for a custom user role:



Refer to the following steps to configure relationship for a custom user role:

1. In Business 360 Console, on the Security page, click the custom user role for which you want to configure privileges.
2. On the Relationship tab, select the relationship for which you want to configure privileges. The following image shows the Relationship tab with the list of predefined and custom relationships:

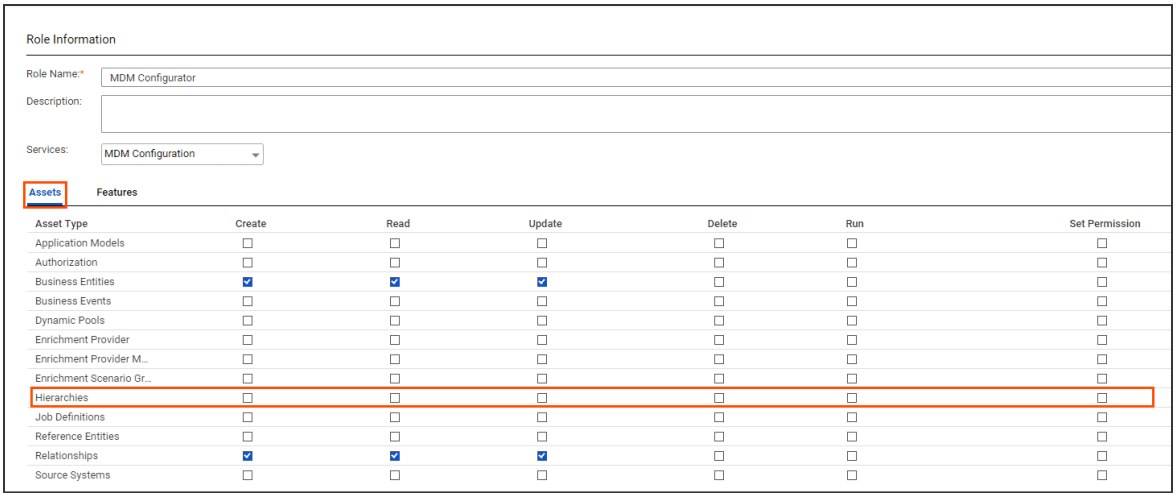


3. On the Records tab, select **Create**, **Read**, **Update**, and **Delete** privileges pertaining to related records in the business applications. You can also specify whether users can import relationship data.

Configure Privileges for a Hierarchy

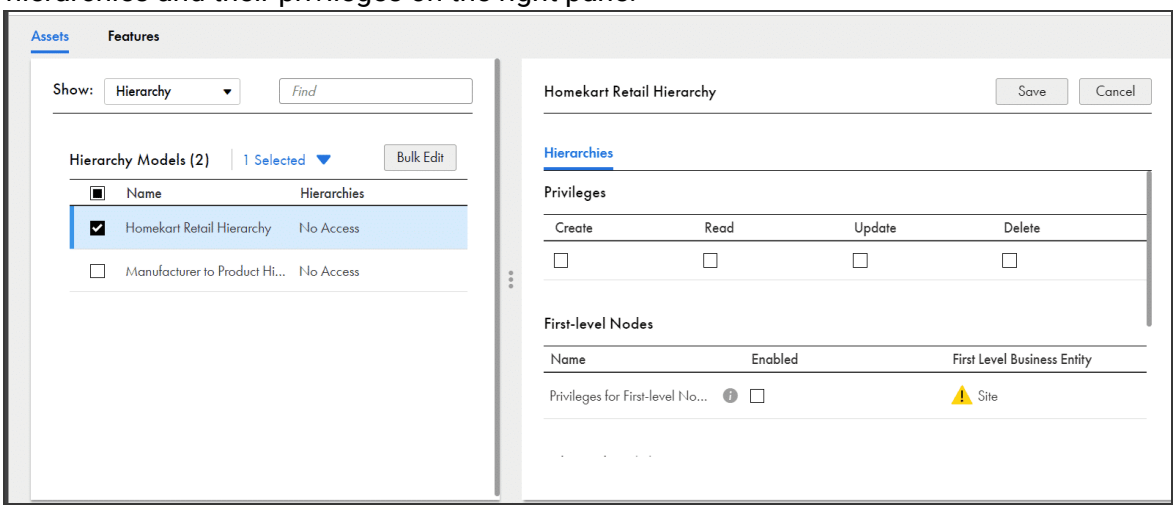
You can configure privileges related to hierarchies for a custom user role. Before you assign privileges for hierarchies on the Security page, you can assign privileges for hierarchy models on the Assets tab of the MDM Configuration service in Administrator for a custom user role if required.

The following image shows the user role details page in Administrator with the hierarchy privileges assigned for a custom user role:



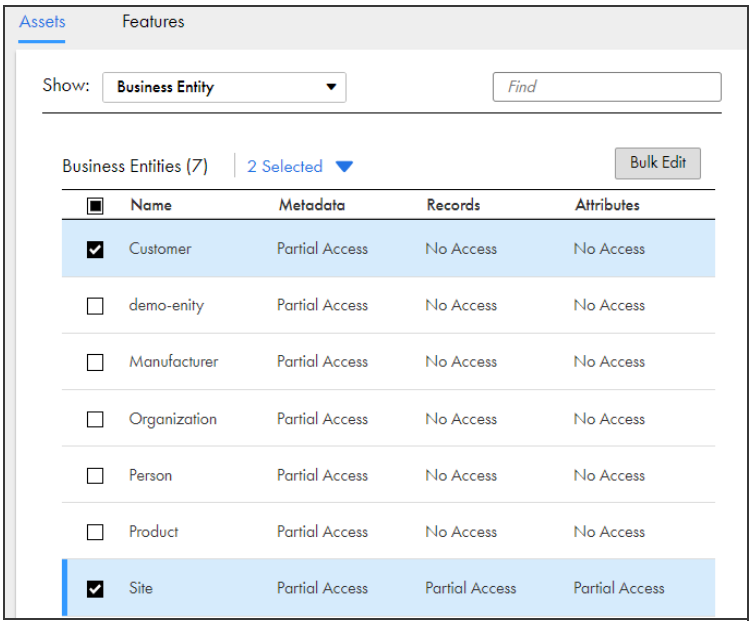
1. In Business 360 Console, on the Security page, click the custom user role for which you want to configure privileges.
2. On the Relationship tab, select the relationship for which you want to configure privileges.
3. Select the required hierarchy model. The privileges for the hierarchies based on the selected hierarchy model appear on the right pane.

The following image shows the list of hierarchy models on the left pane and the corresponding hierarchies and their privileges on the right pane.

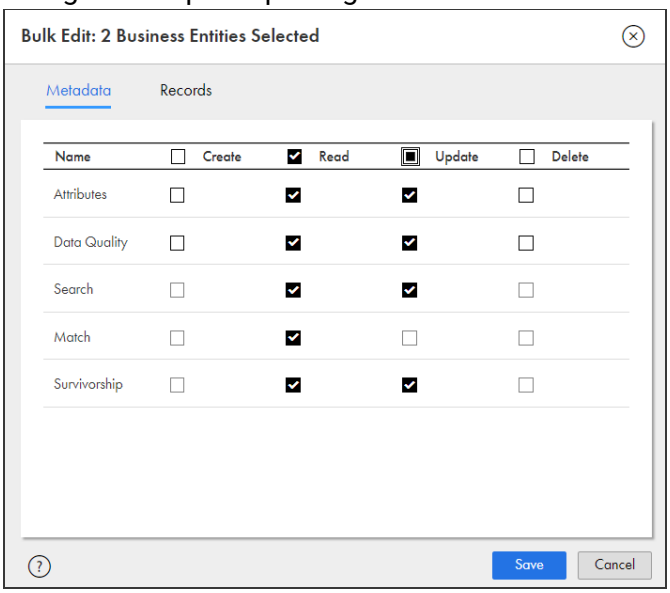


Configure Privileges for Multiple Assets

1. Select the custom user role for which you want to configure privileges.
2. On the Business Entity tab, select multiple business entities.
3. Click **Bulk Edit**.



4. Assign the required privileges on the Metadata and Records tabs.



5. Click **Save**.

Lab 8-1: Creating Custom Users and Configuring Privileges to Them

Overview:

You can create custom user roles and configure privileges for them in your organization. You can control access to assets, attributes, and features of business applications for each custom user role in your organization. For example, you can control a custom user role to view records but not update or edit the records.

The system-defined roles include predefined access privileges in the Administrator, and you cannot change them. The custom user roles are not predefined, and you can create them based on the specific requirements of your organization. You cannot delete system-defined roles, but you can delete custom roles. You can also control the specific tasks that a custom user role can perform in your organization. For example, you can authorize a custom user role to import data into Multidomain MDM.

Objectives:

- Create a new custom user role
- Configure privileges to the custom user role

Scenario:

Mark is planning to hire a new employee on contract to work on the Multidomain MDM. The new employee requires few privileges to work on the application. Hence, Mark wants to create a new custom user called steward and provide necessary privileges required for the new employee to work on the application.

Duration:

40 minutes

Tasks:

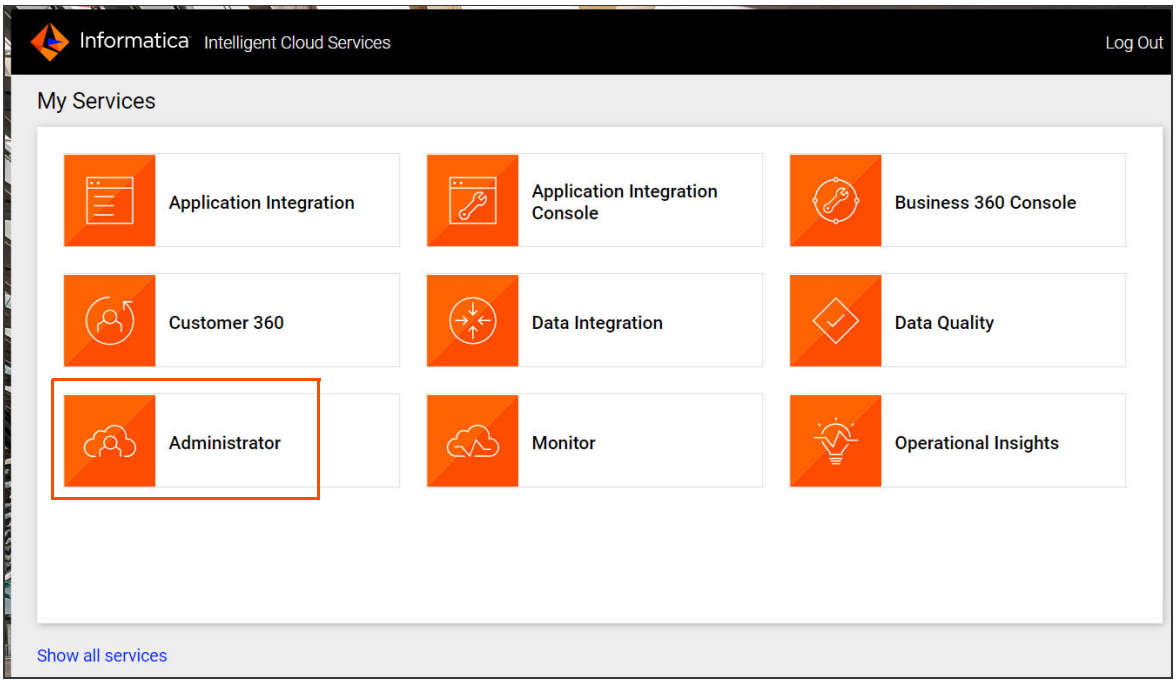
- Complete the prerequisites necessary for creating custom user roles for steward and manager users
- Create two custom users for steward and manager
- Configure privileges for a business entity for the custom roles
- Configure privileges for multiple business entities

Create a Custom User Role

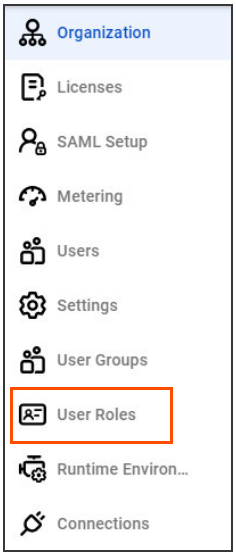
You will now create a custom user role named Steward and configure privileges for them in the Administrator service.

1. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**), if you have logged out.

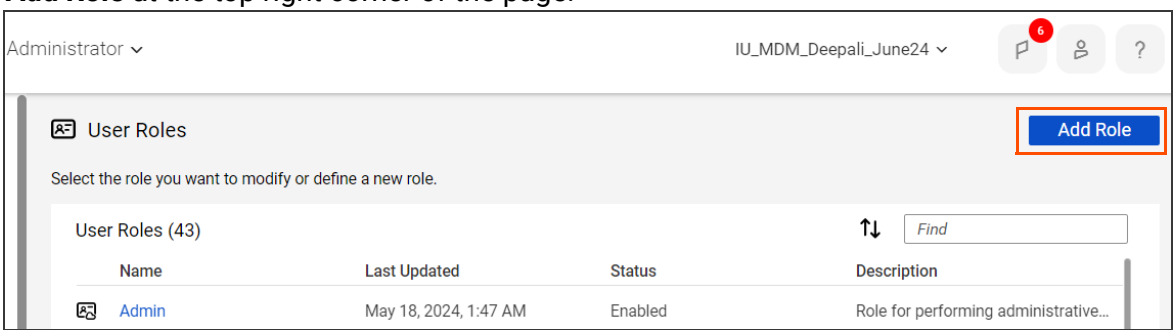
2. From the Services window, click **Administrator** Console.



3. Click **User Roles**.



4. You can see the multiple pre-defined user roles here. To create a new custom role, click **Add Role** at the top right corner of the page.

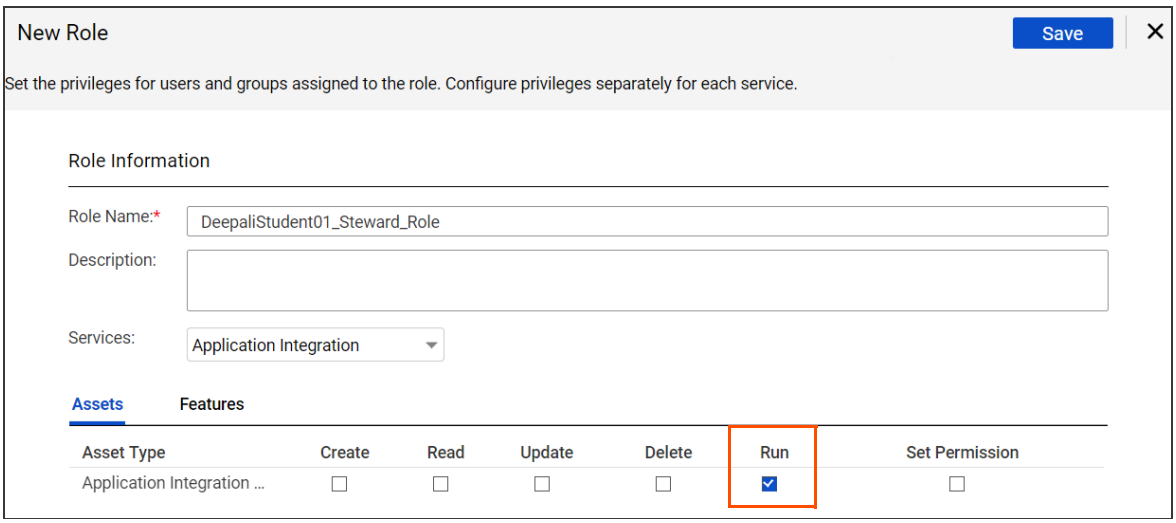


5. Enter the Role Name as **studentNamestudent<#>_Steward_Role**.

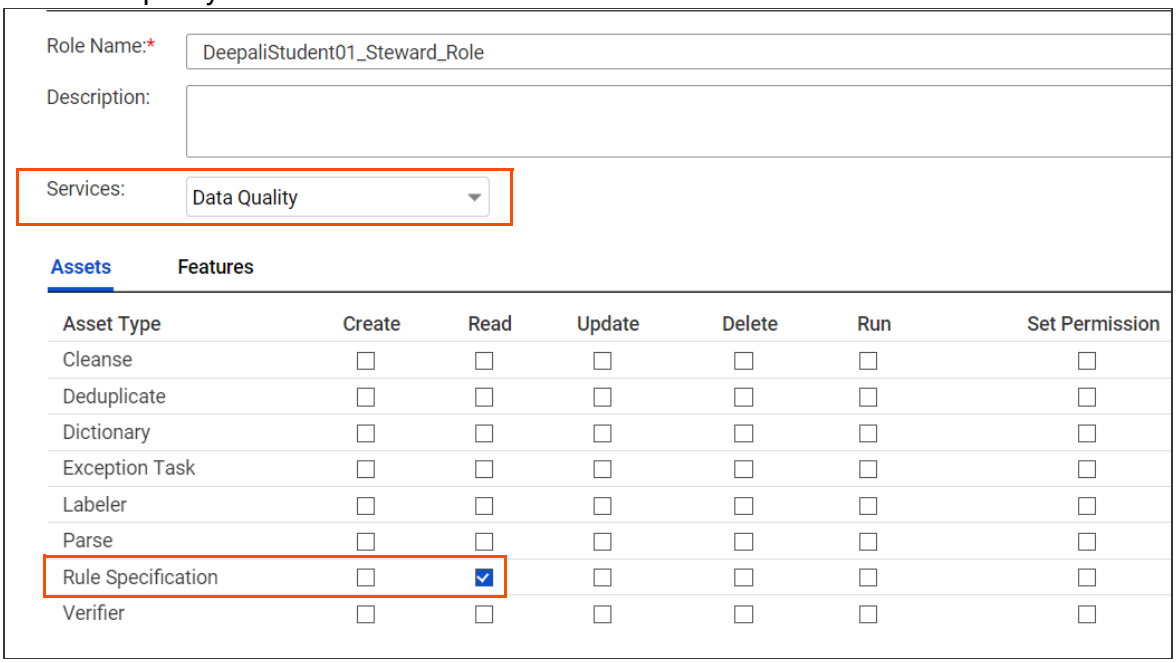
Now, you must assign relevant privileges for the custom user role.

6. From the Services drop down, select **Application Integration**.

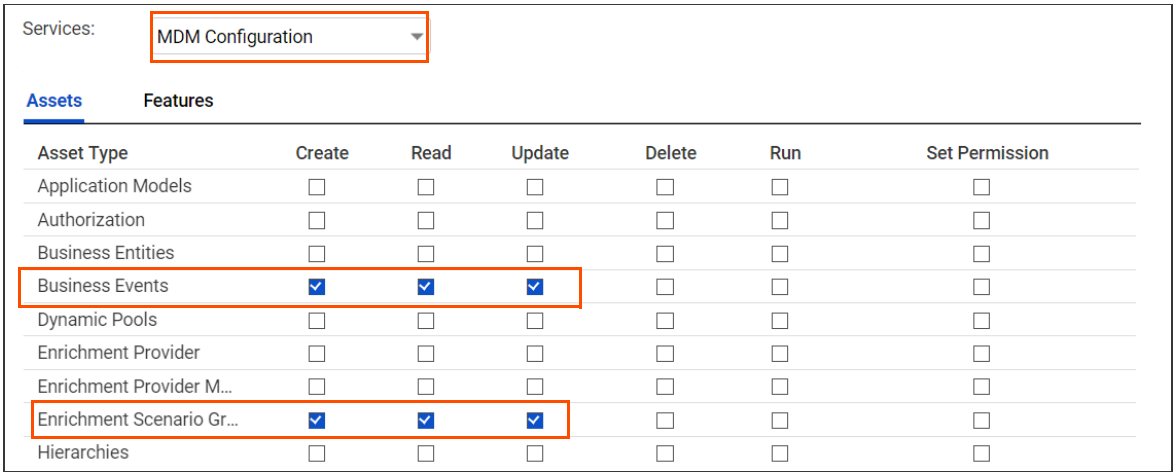
7. Under the Assets tab, select **Run**. This will ensure that the role can create, update, and delete metadata of the business entities in the Business 360 Console.



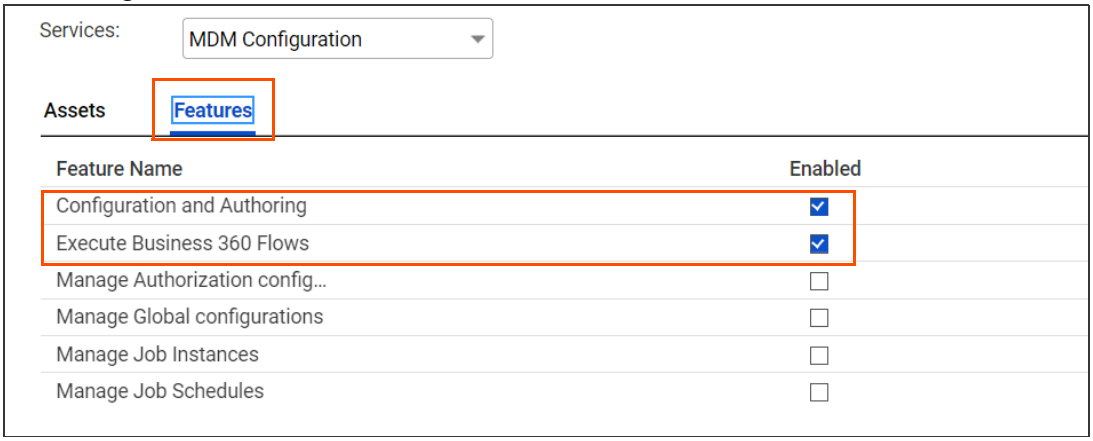
8. From the Services drop-down, select **Data Quality**. Under the Assets tab, select **Read** for the **Rule Specification** asset type. It will allow users to import data for use in data models with data quality rule associations.



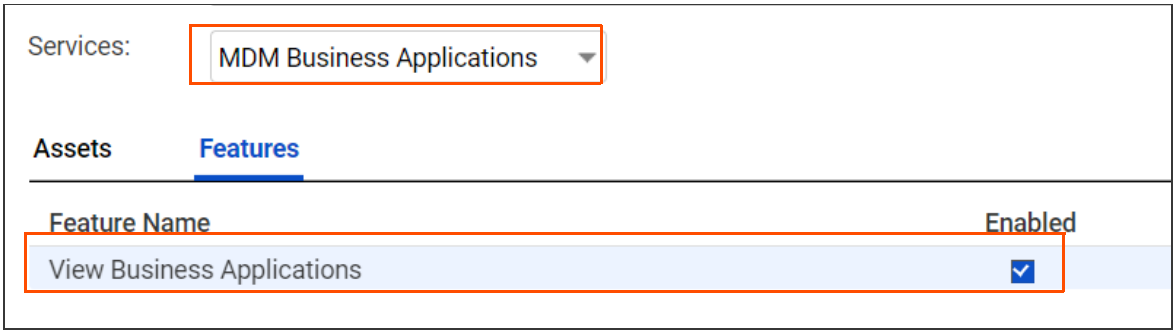
9. From the Services drop-down, select **MDM Configuration**. Under the Assets tab, select **Create**, **Read and Update** for the **Business Events and Enrichment Scenario Group** asset type. This indicates whether the user can create, update, and delete business entities and relationships in Business 360 Console.



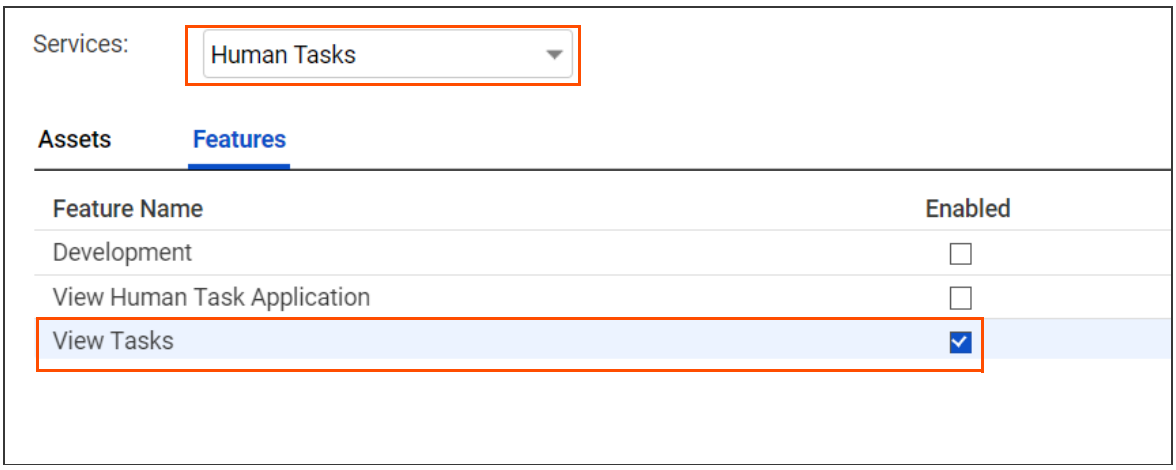
10. Under the Features tab, enable **Execute Business 360 Flows** and **Configuration and Authoring**.



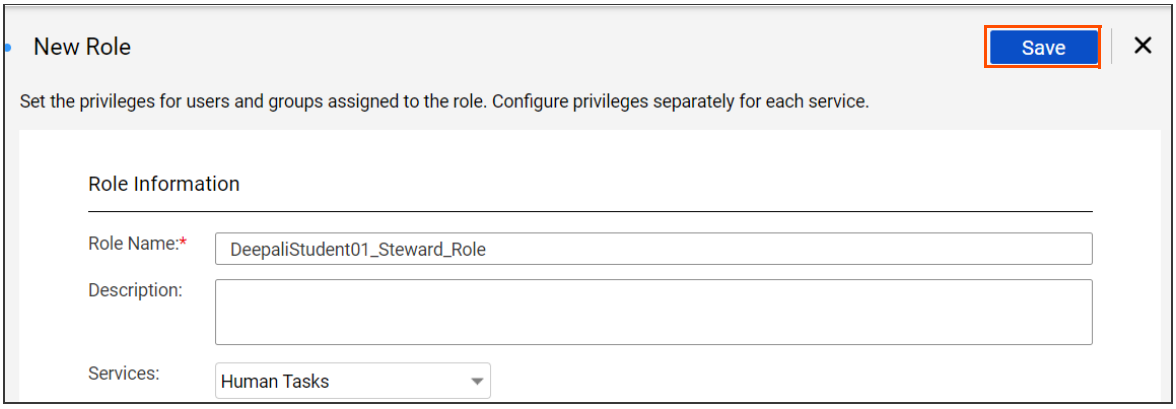
11. From the Services drop-down, select **MDM Business Applications**. Under the Assets tab, enable **View Business Applications**. This indicates whether user can access MDM applications, such as Multidomain MDM, on the informatica IICS platform.



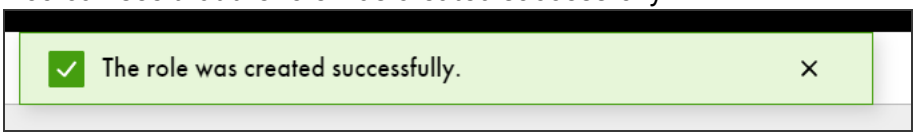
12. From the Services drop-down, select **Human Tasks**. Under the Features tab, enable **View Tasks**.



13. Click **Save**.



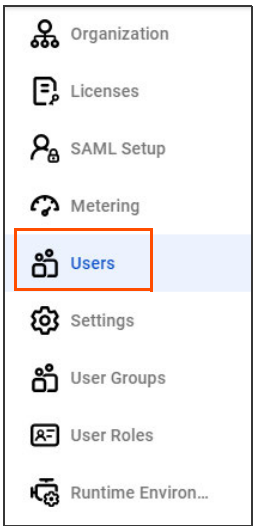
You can see that the role was created successfully.



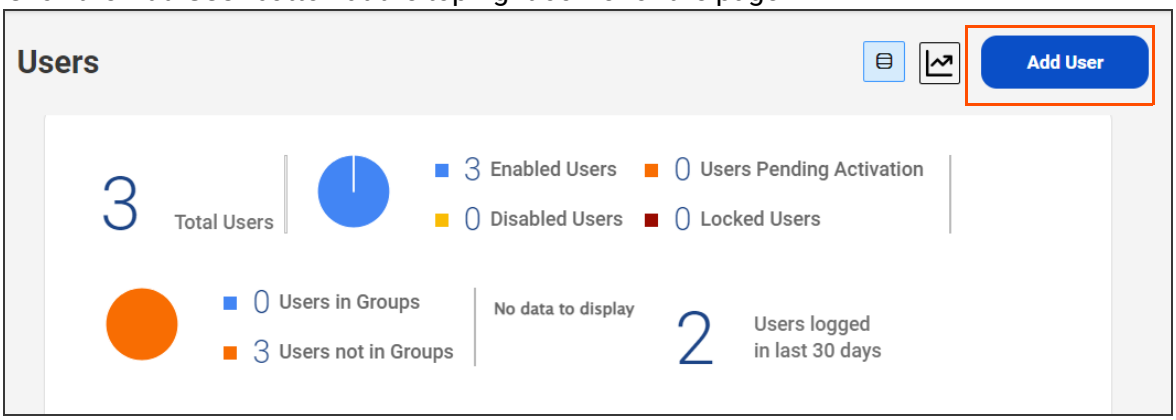
Create a User and Assign the Custom Role to it

You will now create a user and assign the custom role created above.

14. Click **Users**.

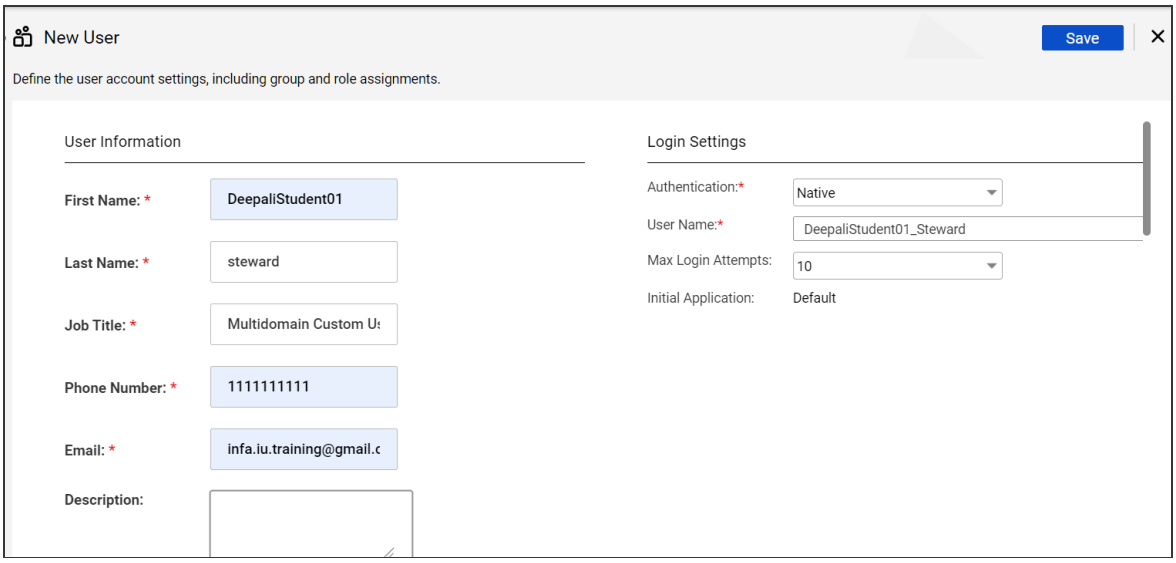


15. Click the **Add User** button at the top right corner of the page.

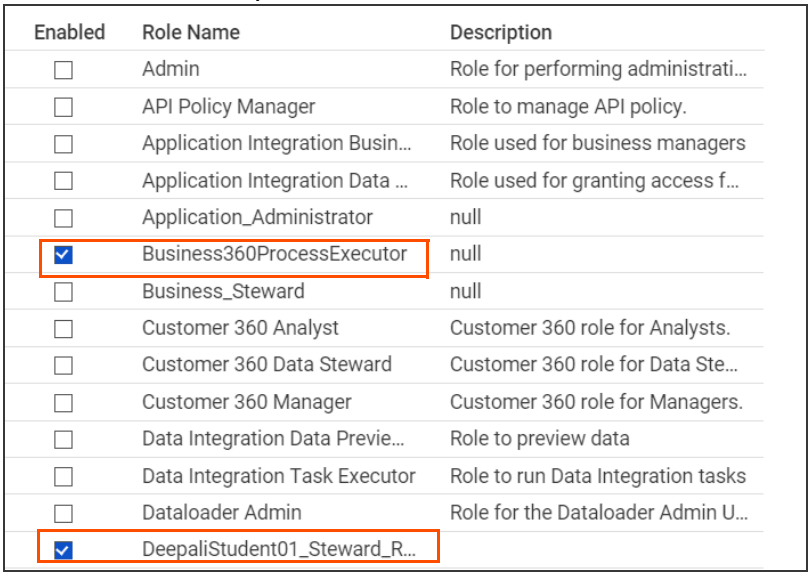


16. Enter the following details:

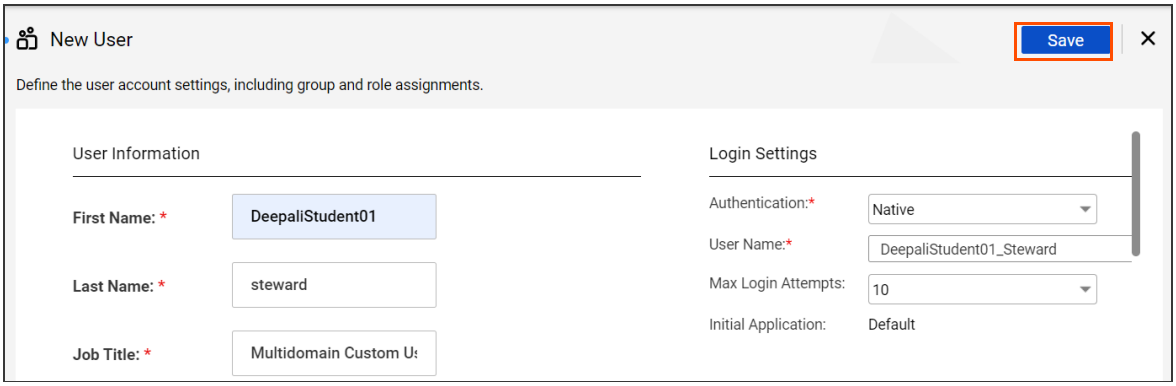
Field	Value
First Name	studentNamestudent<#>
Last Name	steward
Job Title	Multidomain Custom User
Phone Number	<Your Phone Number>
Email	<Your email ID>
Description	Multidomain MDM Custom Role
Authentication	Native
User Name	studentNamestudent<#>_Steward
Max Login Attempts	10



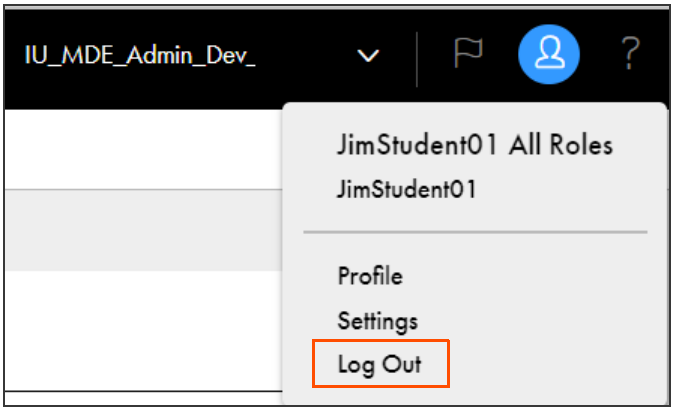
17. Select **Business360ProcessExecutor** and **studentNamestudent<#>_Steward_Role** from the list of the roles provided.



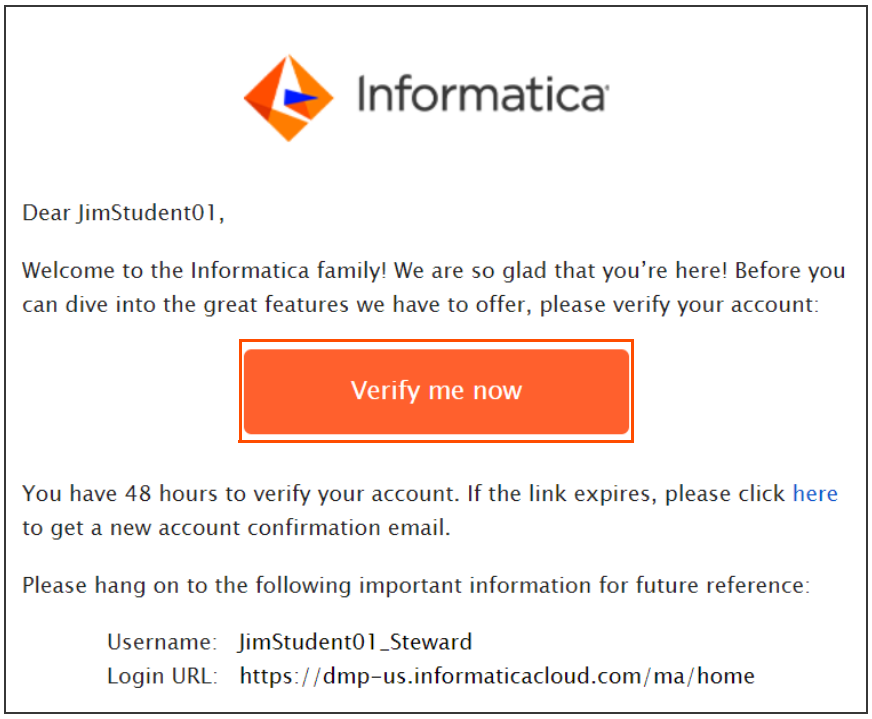
18. Click **Save**.



19. Log out of your profile.



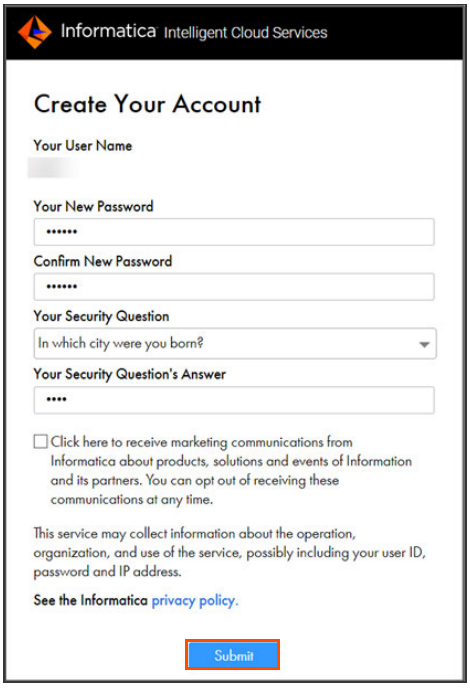
You will get a notification email to set the user password.



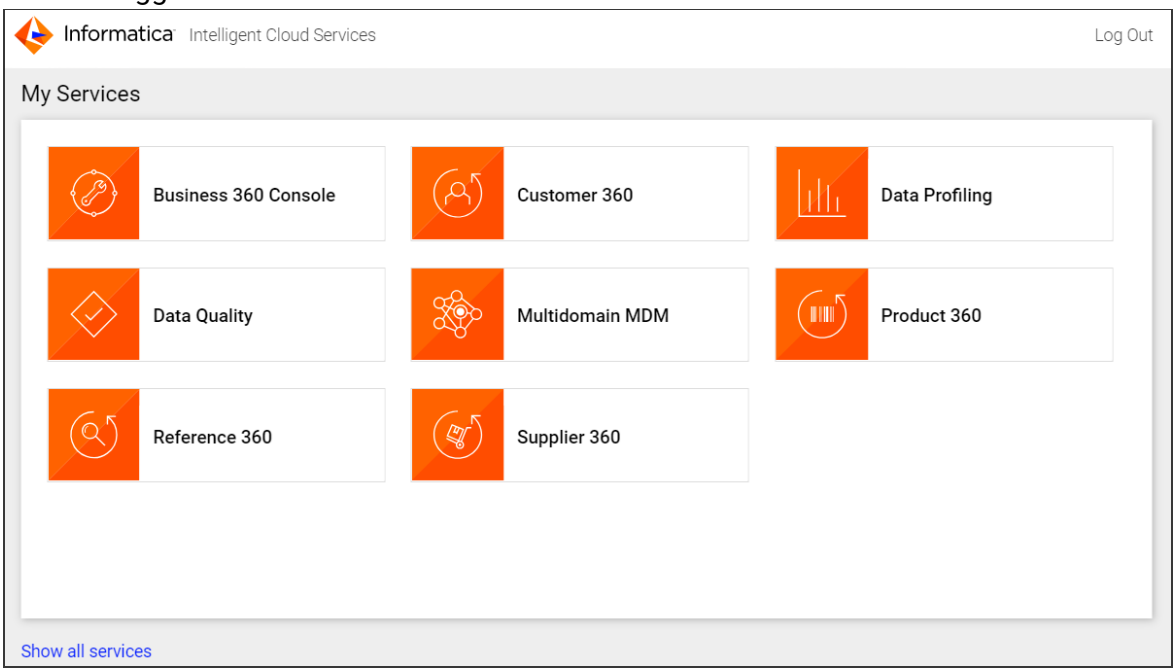
20. Click **Verify me now** link.

21. Enter the Password as **Multidomain360**, Confirm Password, and answer a Security question too, in the set password page.

22. Click **Submit**.



23. You are logged in as Steward now.



24. **Log out** as Steward.

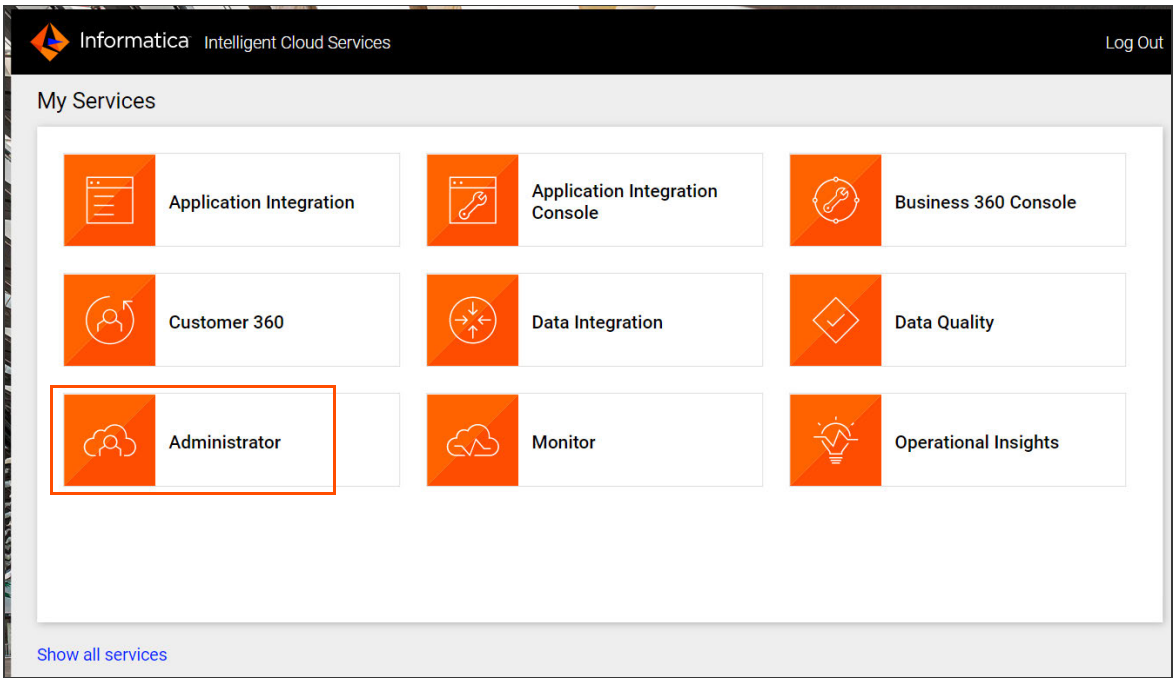
Self Try Excercise (Mandatory)

Create a Custom User Role as Manager

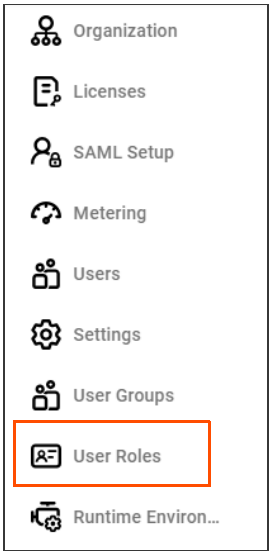
You will now create a custom user role named Manager.

25. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**), if you have logged out.

26. From the Services window, click **Administrator** Console.

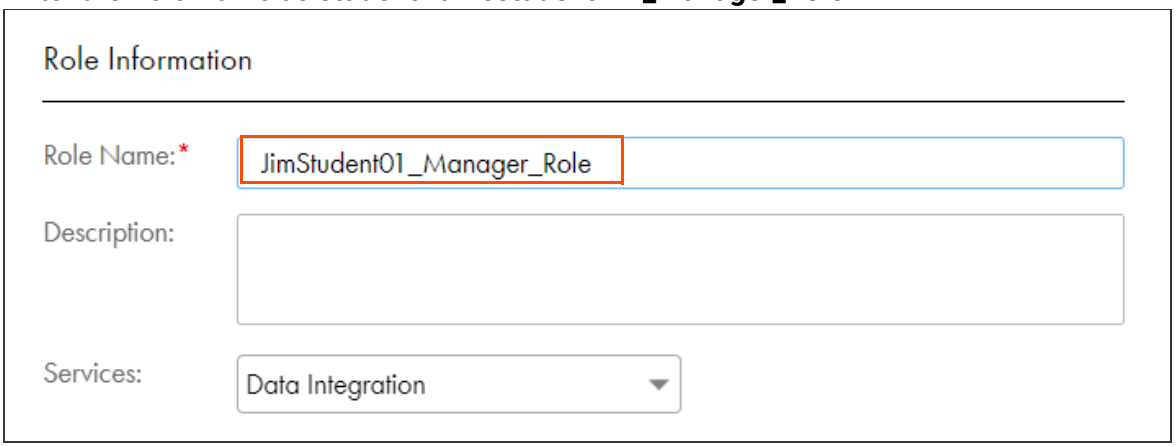


27. Click **User Roles**.



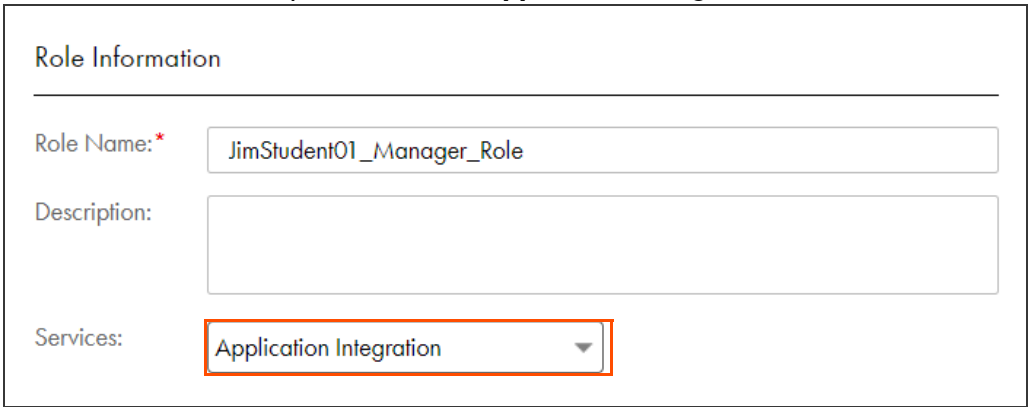
28. You can see the multiple pre-defined user roles here. To create a new custom role, click **Add Role** at the top right corner of the page.

29. Enter the Role Name as **studentNamestudent<#>_Manager_Role**.

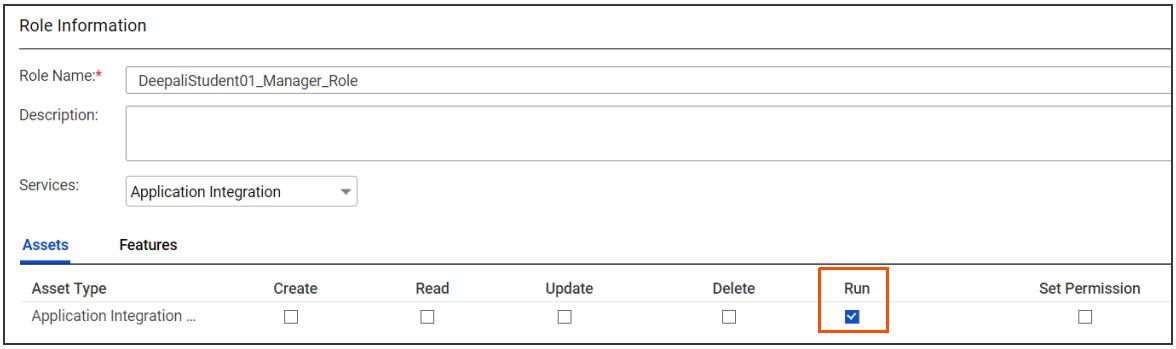


Now, you must assign relevant privileges for the custom user role.

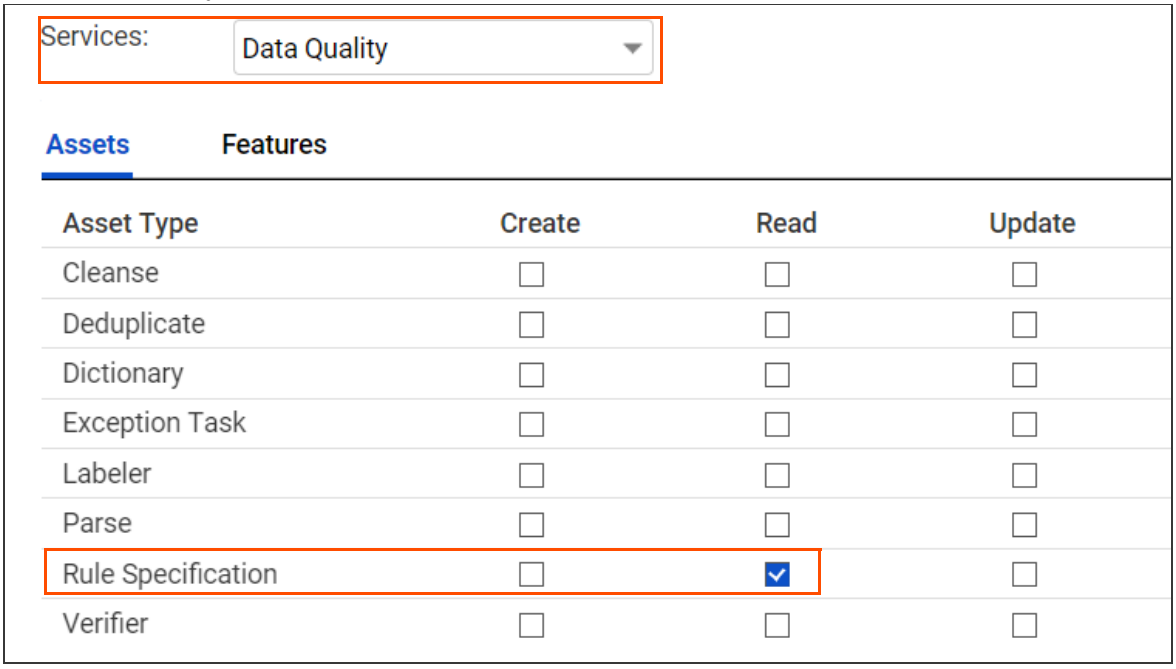
30. From the Services drop down, select **Application Integration**.



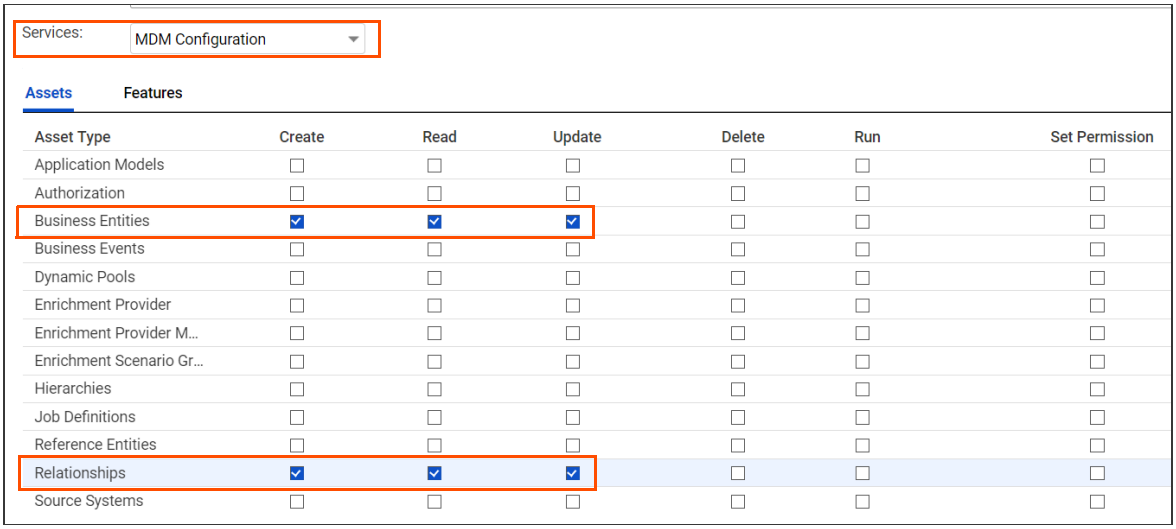
31. Under the Assets tab, select **Run**. This will ensure that the role can create, update, and delete metadata of the business entities in the Business 360 Console.



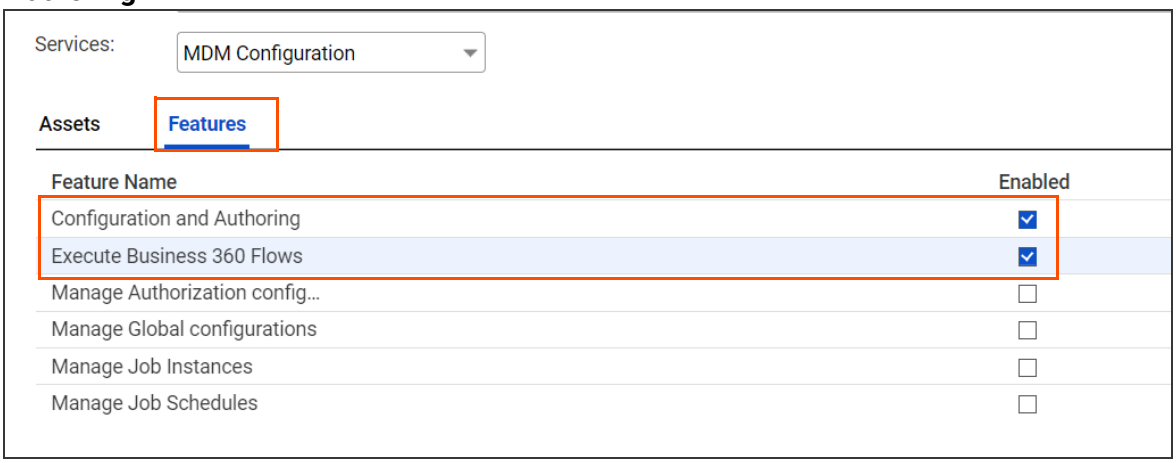
32. From the Services drop-down, select **Data Quality**. Under the Assets tab, select **Read** for the **Rule Specification** asset type. It will allow users to import data for use in data models with data quality rule associations.



33. From the Services drop-down, select **MDM Configuration**. Under the Assets tab, select **Create**, **Read and Update** for the **Business Entities and Relationship** asset type. This indicates whether the user can create, update, and delete business entities and relationships in Business 360 Console.



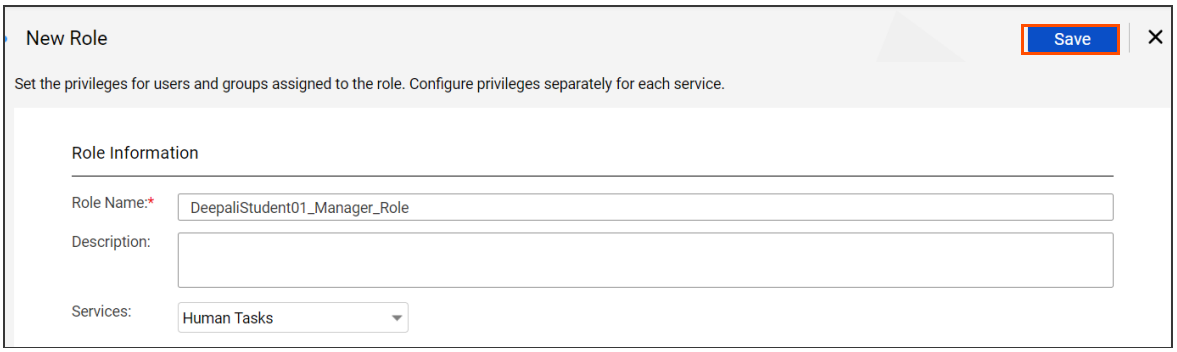
34. Under the Features tab, enable **Execute Business 360 Flows** and **Configuration and Authoring**.



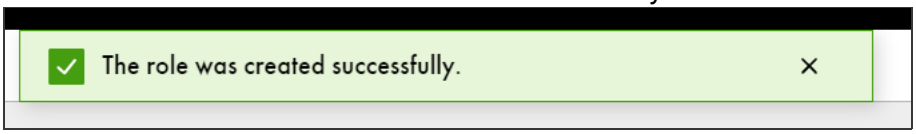
35. From the Services drop-down, select **MDM Business Applications**. Under the Assets tab, enable **View Business Applications**. This indicates whether user can access MDM applications, such as Multidomain MDM, on the informatica IICS platform.

36. From the Services drop-down, select **Human Tasks**. Under the Features tab, enable **View Tasks**.

37. Click **Save**.



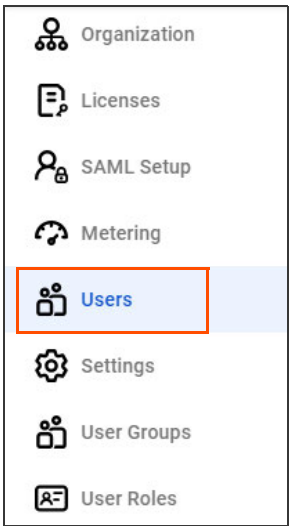
You can see that the role was created successfully.



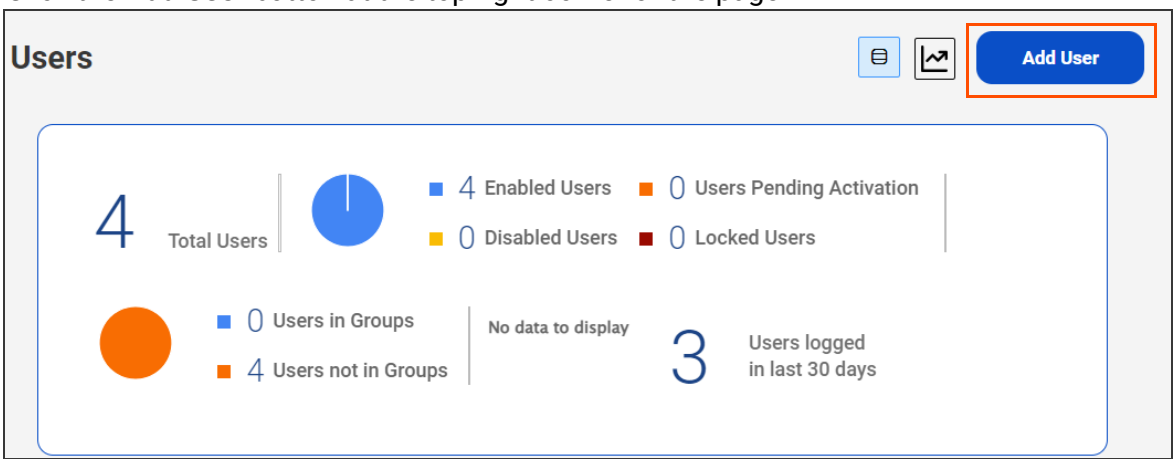
Create a User and Assign the Custom Role to It

You will now create a user and assign the custom role created above.

38. Click **Users**.

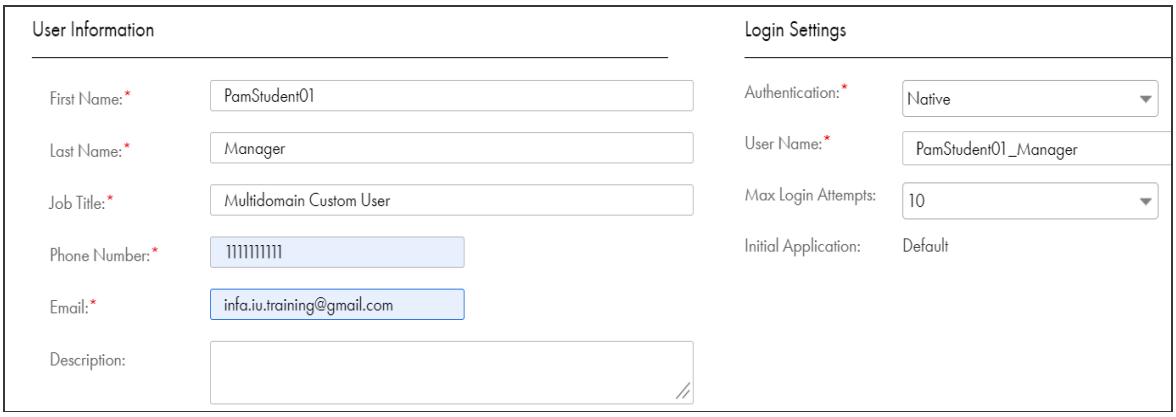


39. Click the **Add User** button at the top right corner of the page.

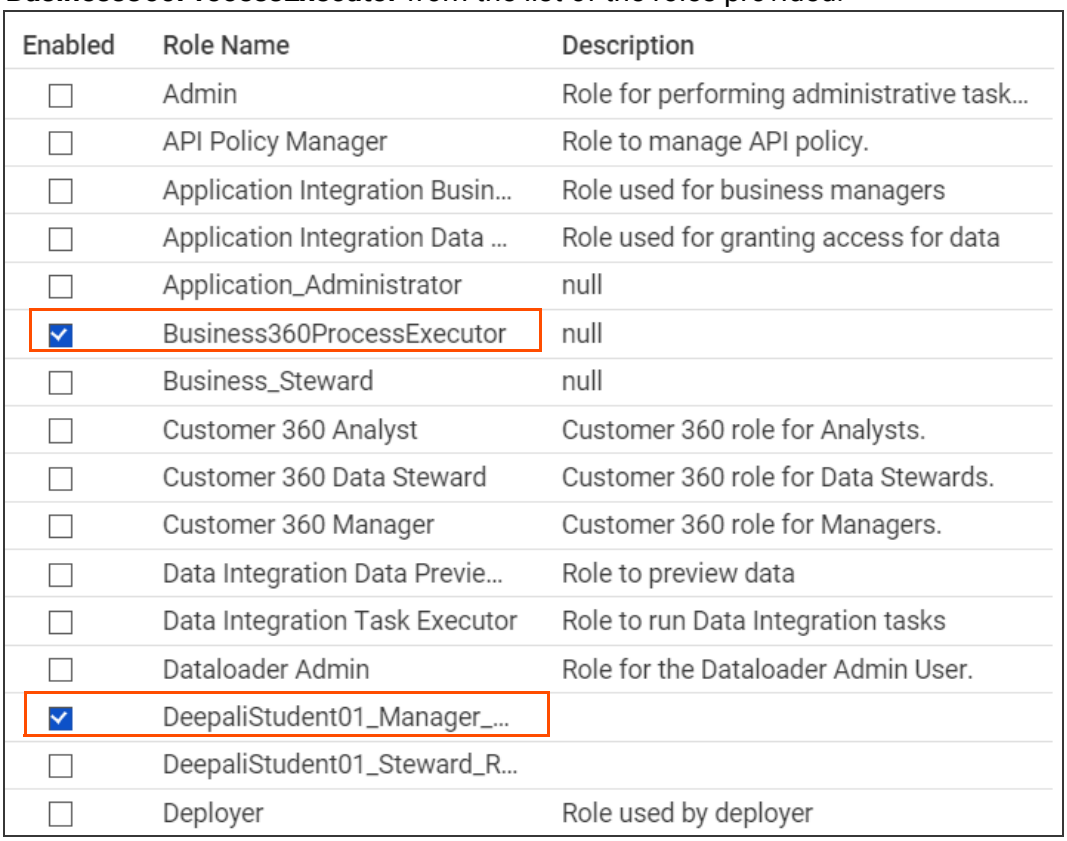


40. Enter the following details:

Field	Value
First Name	studentNamestudent<#>
Last Name	Manager
Job Title	Multidomain Custom User
Phone Number	<Your Phone Number>
Email	<Your email ID>
Description	Multidomain MDM Custom Role
Authentication	Native
User Name	studentNamestudent<#>_Manager
Max Login Attempts	10

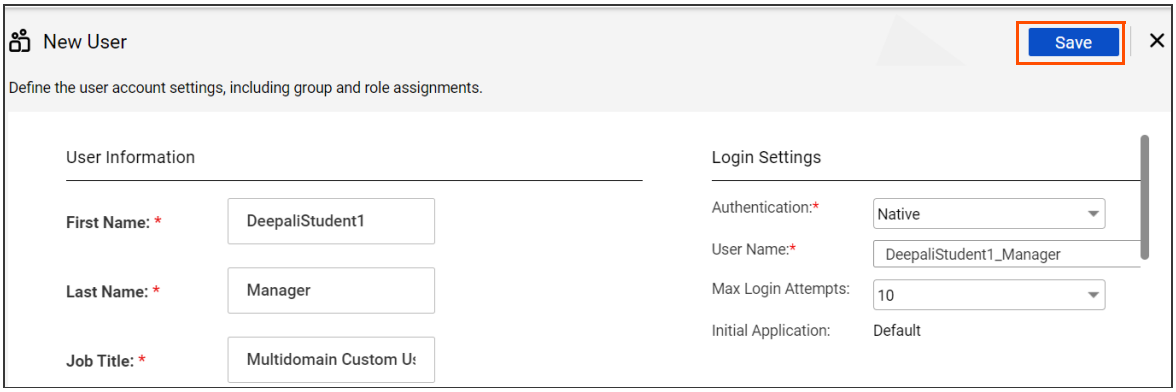


41. Scroll down and select **studentNamestudent<#>_Manager_Role** and **Business360ProcessExecutor** from the list of the roles provided.

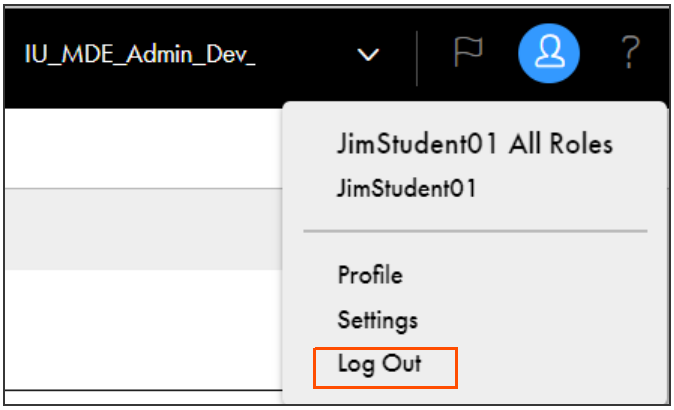


Enabled	Role Name	Description
☐	Admin	Role for performing administrative task...
☐	API Policy Manager	Role to manage API policy.
☐	Application Integration Busin...	Role used for business managers
☐	Application Integration Data ...	Role used for granting access for data
☐	Application_Administrator	null
☑	Business360ProcessExecutor	null
☐	Business_Steward	null
☐	Customer 360 Analyst	Customer 360 role for Analysts.
☐	Customer 360 Data Steward	Customer 360 role for Data Stewards.
☐	Customer 360 Manager	Customer 360 role for Managers.
☐	Data Integration Data Previe...	Role to preview data
☐	Data Integration Task Executor	Role to run Data Integration tasks
☐	Dataloader Admin	Role for the Dataloader Admin User.
☑	DeepaliStudent01_Manager_...	
☐	DeepaliStudent01_Steward_R...	
☐	Deployer	Role used by deployer

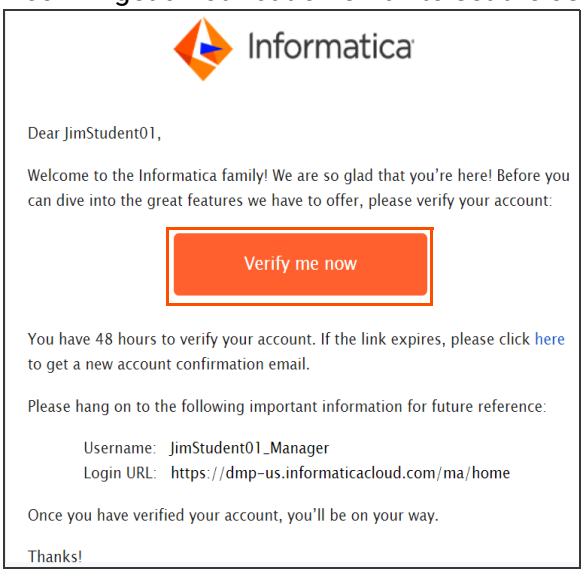
42. Click **Save**.



43. Log out of your profile.



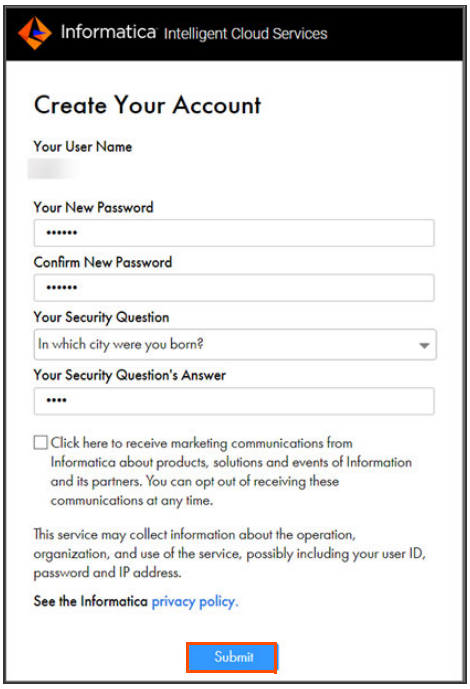
You will get a notification email to set the user password.



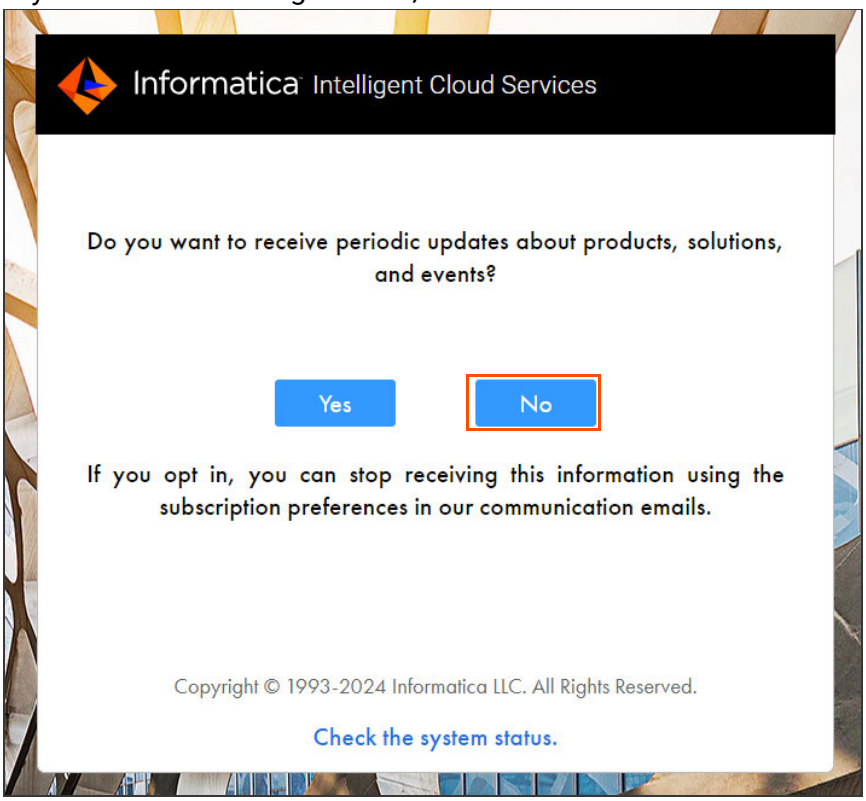
44. Click **Verify me now** link.

45. Enter the Password as **Multidomain360**, Confirm Password, and answer a Security question too, in the set password page.

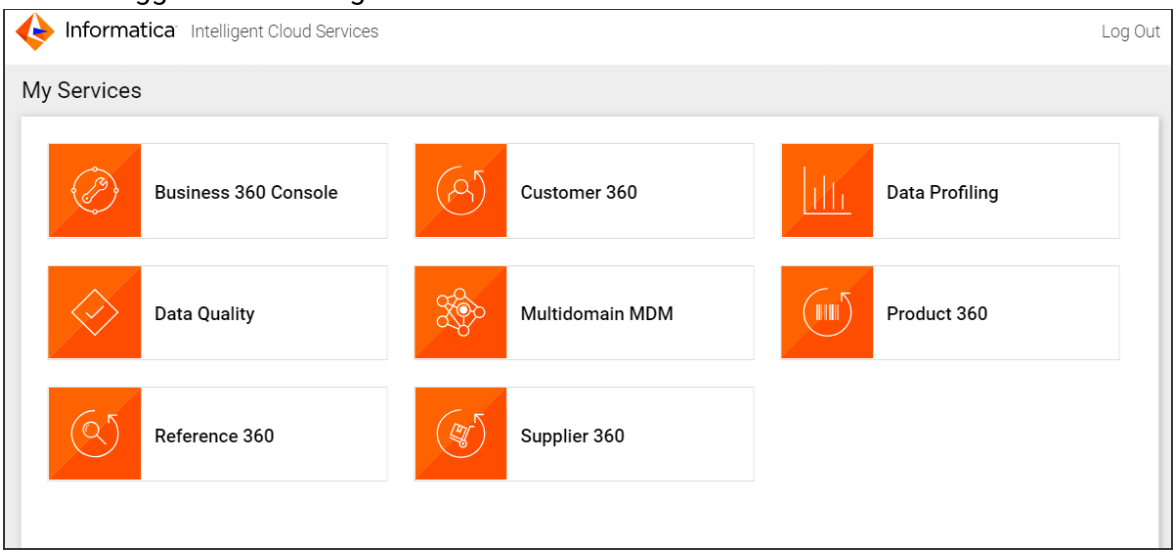
46. Click **Submit**.



47. If you see the following window, click No.



48. You are logged in as Manager now.



49. **Log out** as Manager.

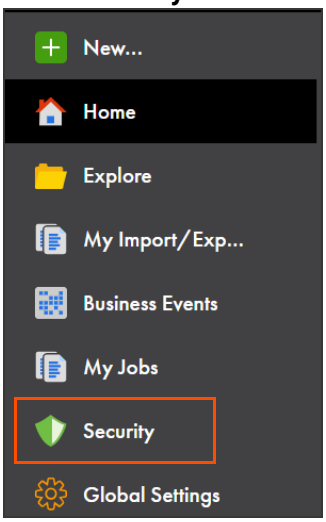
Configure Privileges for Steward Role in Business 360 Console

You can configure privileges for custom user roles on the Security page of Business 360 Console. The Security page lists all the custom user roles that you create in Administrator.

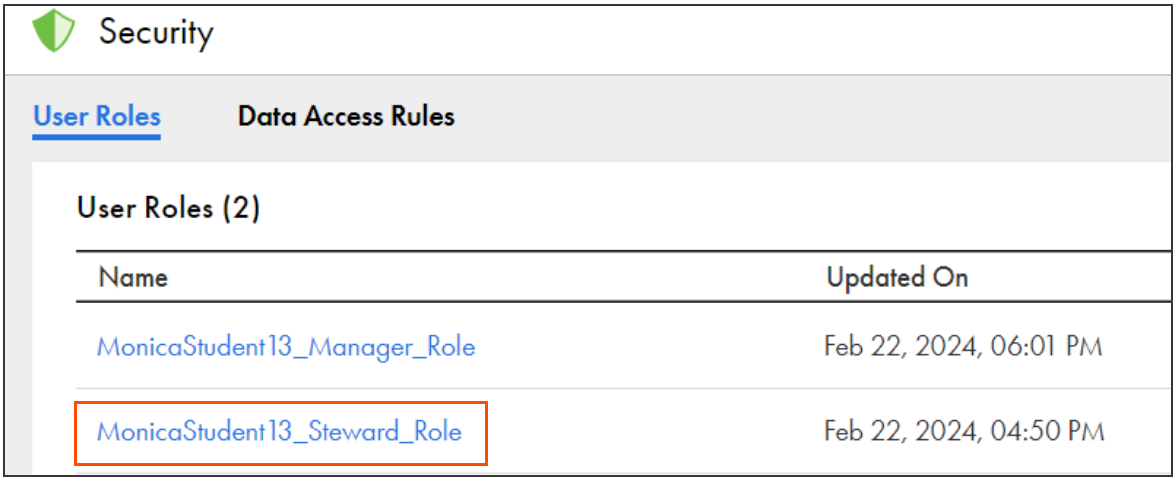
50. Log in to IICS with your credentials (**studentNamestudent<#>/Multidomain360**).

51. Open the **Services** window and open **Business 360 Console**.

52. Click **Security**.



53. You can see the list of custom roles created. Click the **Steward** custom role you have created.

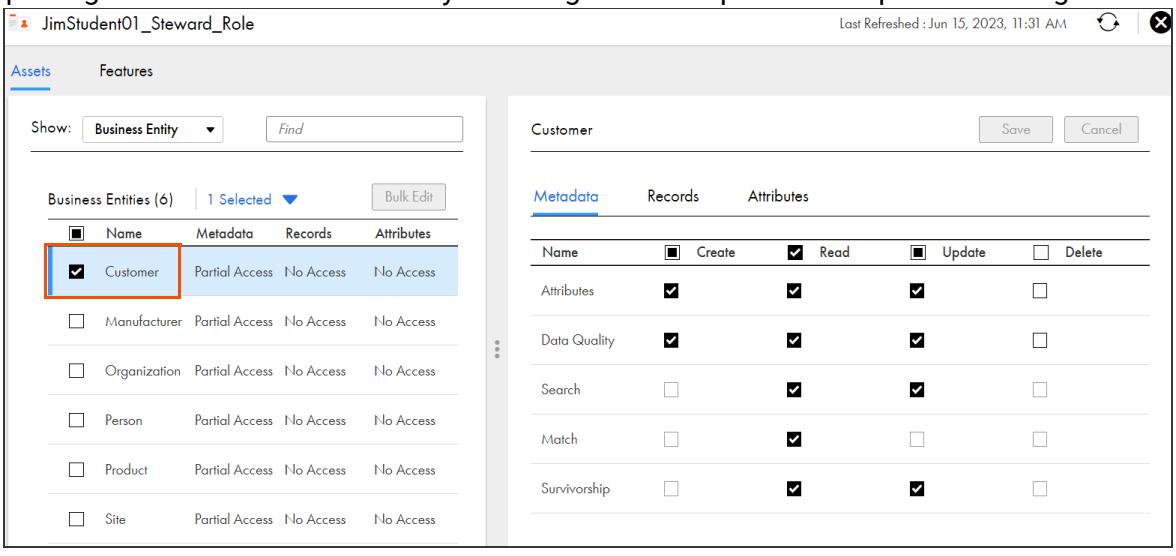


If you cannot see the custom user, refresh the page using the Refresh button. Also, the names of the custom users may vary.

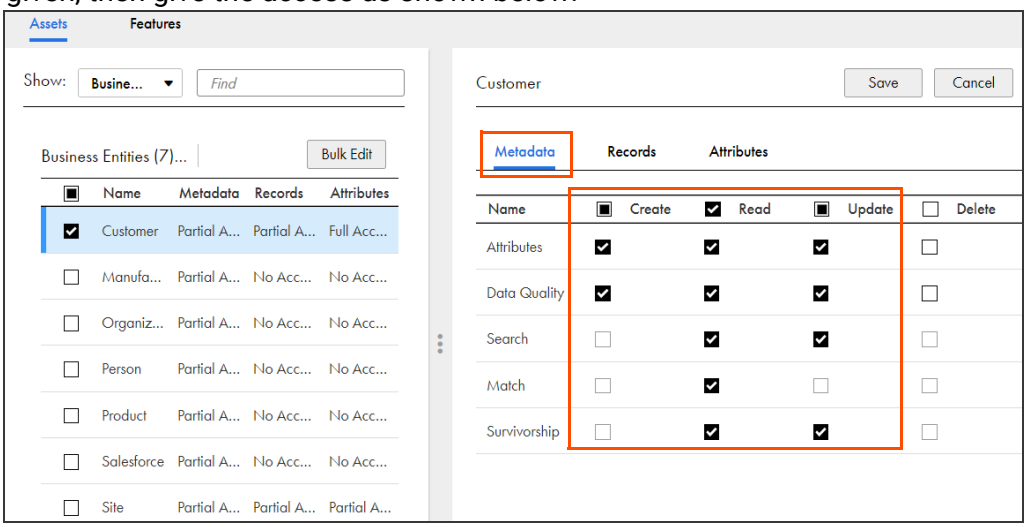
Configure Privileges for a Business Entity

Now, you will configure privileges related to a single business entity for a custom user role.

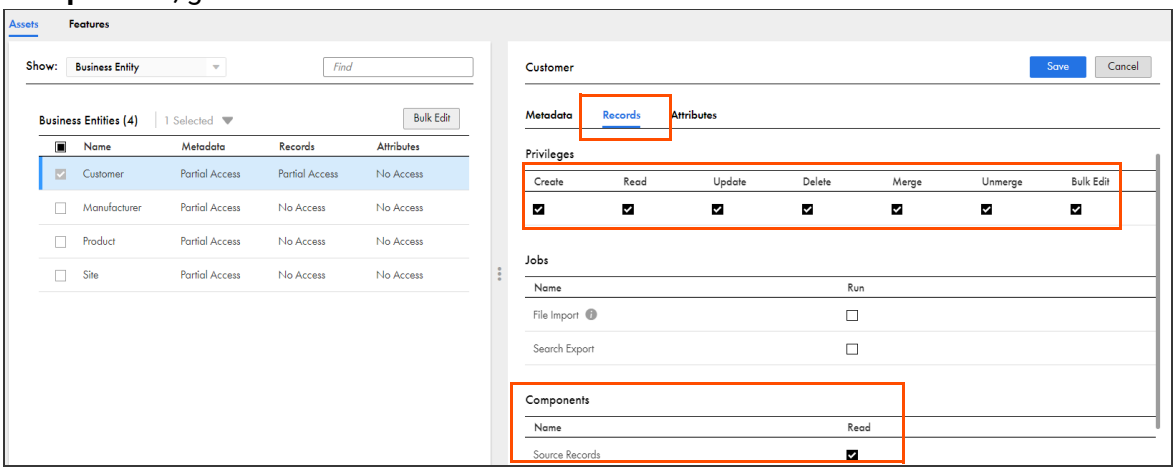
54. Under Assets, select **Customer** business entity. You can see that under Metadata, some of the privileges are assigned already for the business entity. This happens due to the IICS privilege for the business entities you have given in the previous steps of this lab guide.



55. Under **Metadata**, retain the access given to the Customer business entity. If no access given, then give the access as shown below:

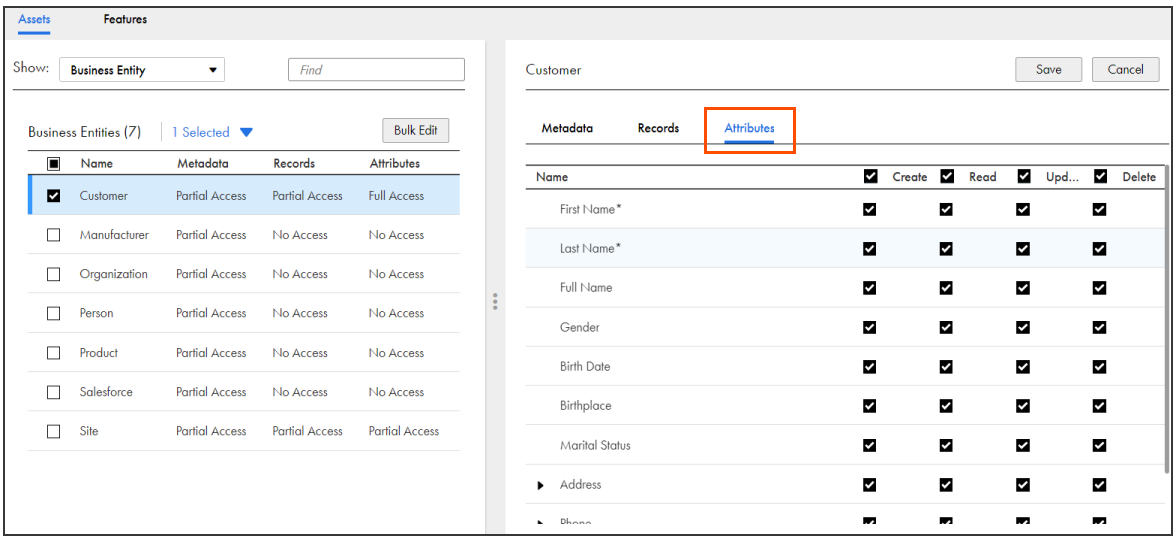


56. Click **Records** and under **Privileges**, give all the access as shown below. Also, under **Components**, give **Read** access to the Source Record.

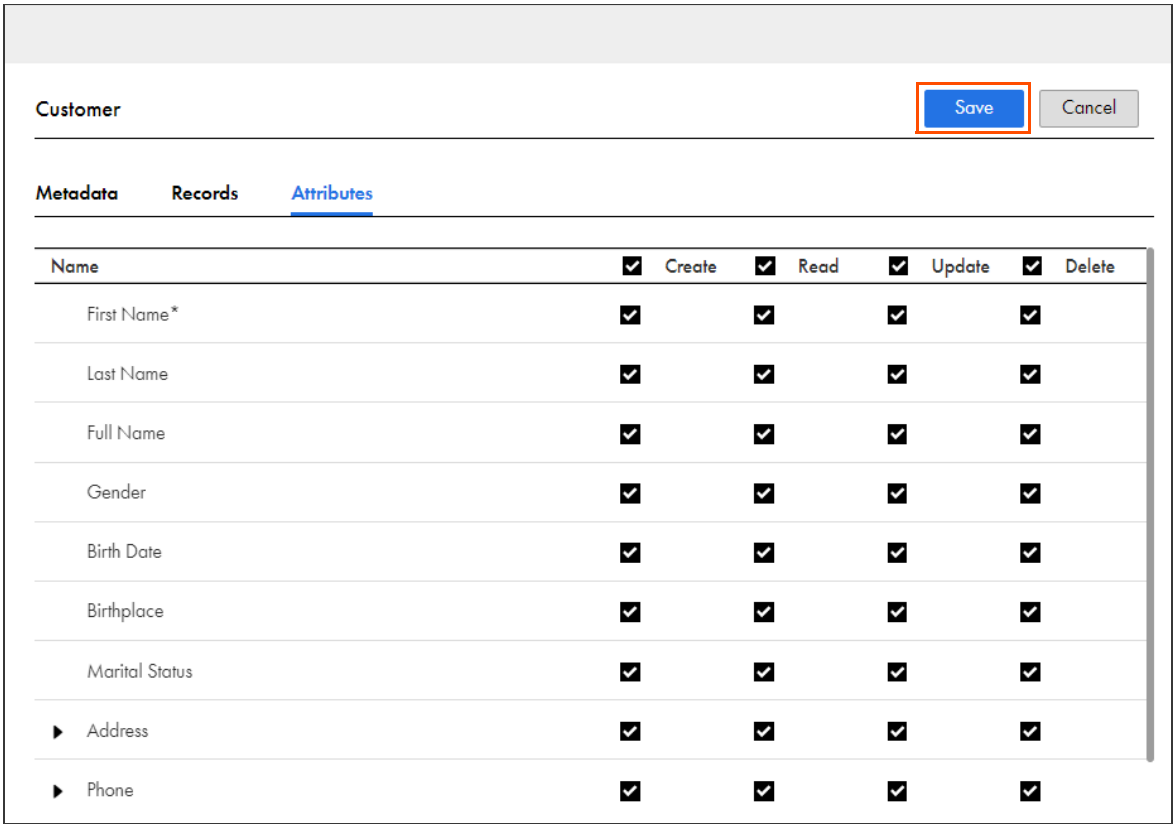


The **Source Records** check box must be enabled in order to execute merge and unmerge labs. Please ensure that you perform this step without fail.

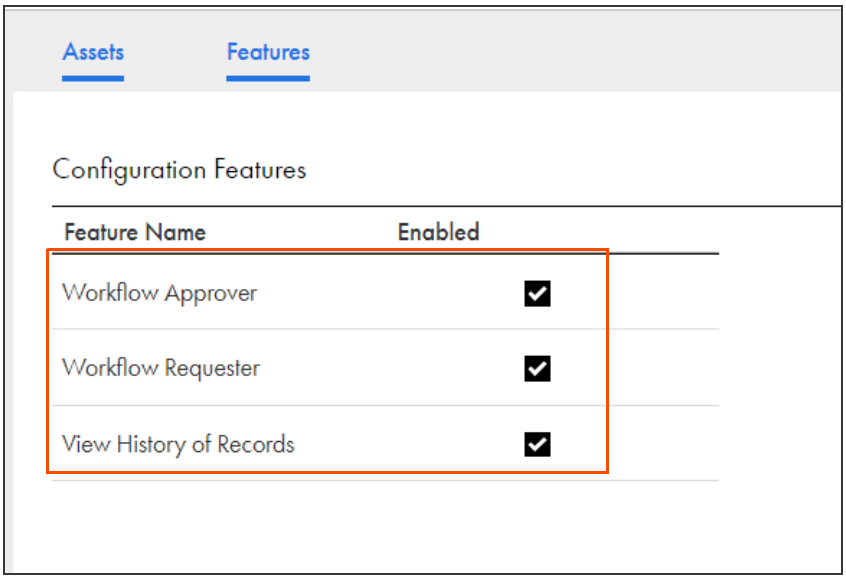
57. Click **Attributes** and make sure to give all the access under **Create**, **Read**, **Update** and **Delete**.



58. Click **Save**.

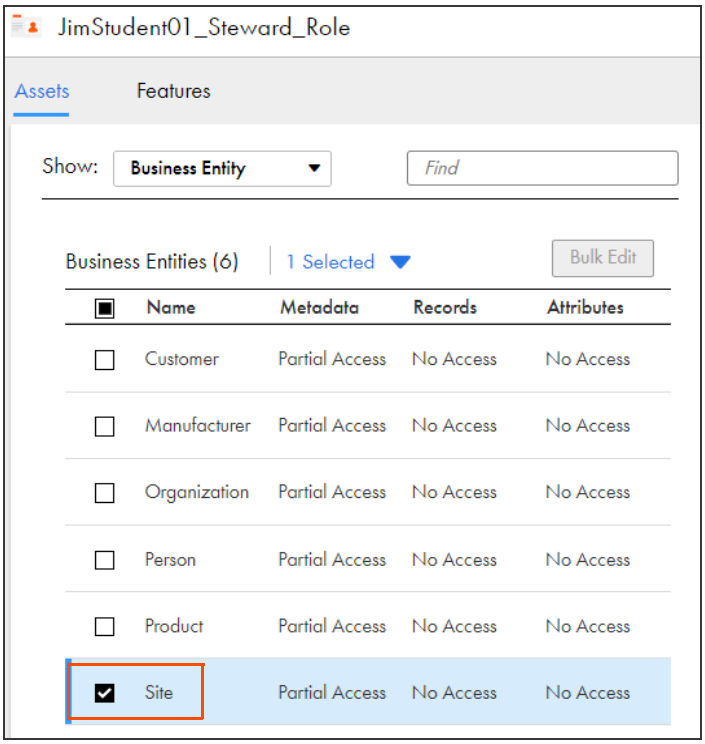


59. Click **Features** and enable all the three features as shown below:

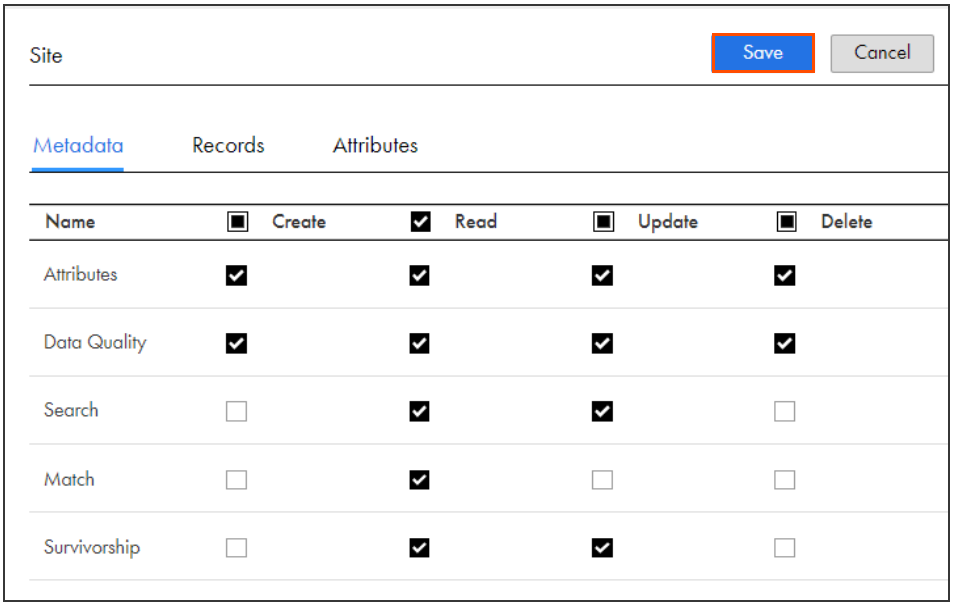


60. Click **Save** again.

61. Click the **Assets** tab again and clear the Customer business entity and select **Site** business entity.

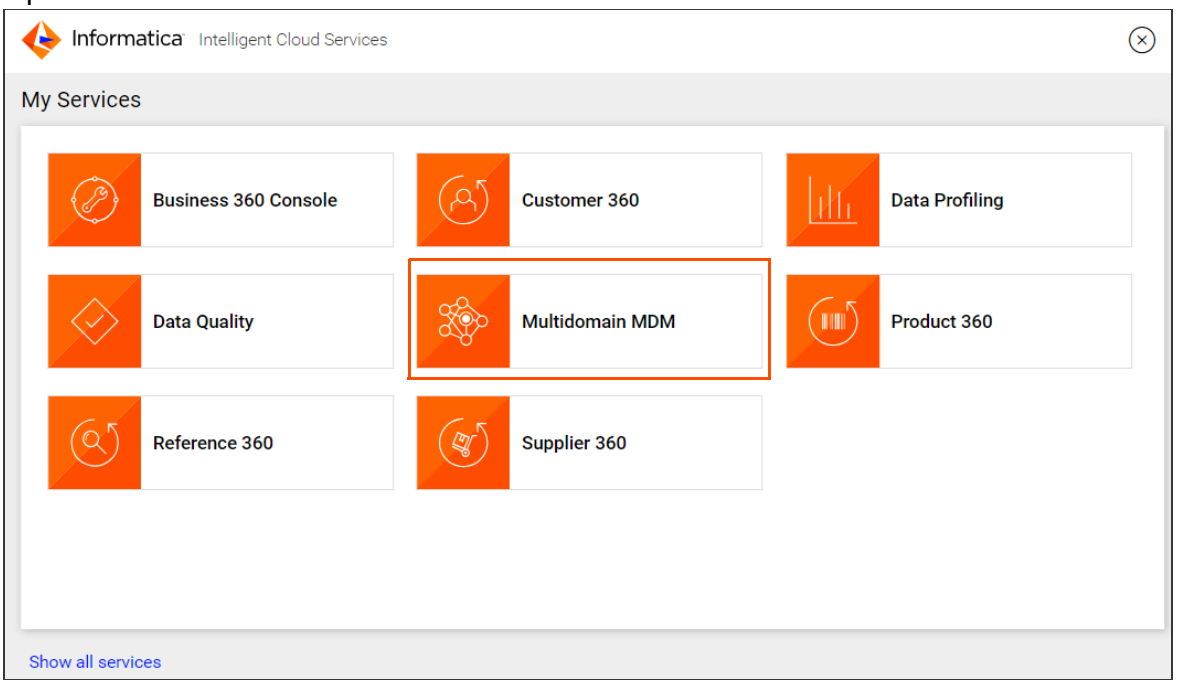


62. Under Metadata, ratain the existing access and provide **Delete** access to Attributes, Data Quality and click **Save**.

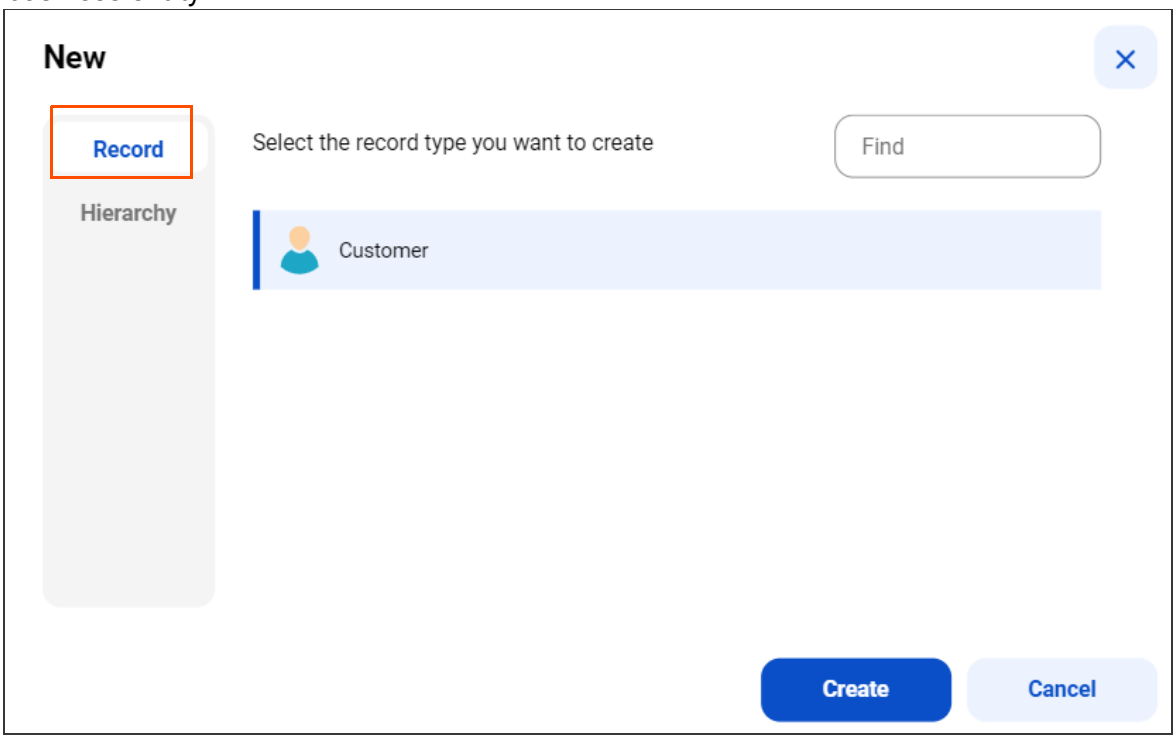


63. Log out as the admin and log in as Steward and check if the privileges are working as intended or not.

64. Open the Multidomain MDM serivce.



65. Click **New** > **Record**. You can see that only **Customer** business entity is visible as the access is given in such a way that steward user can only create records of Customer business entity.



66. Similarly you can check if the privileges are working as intended or not.

Self-Try Excercise (Mandatory)

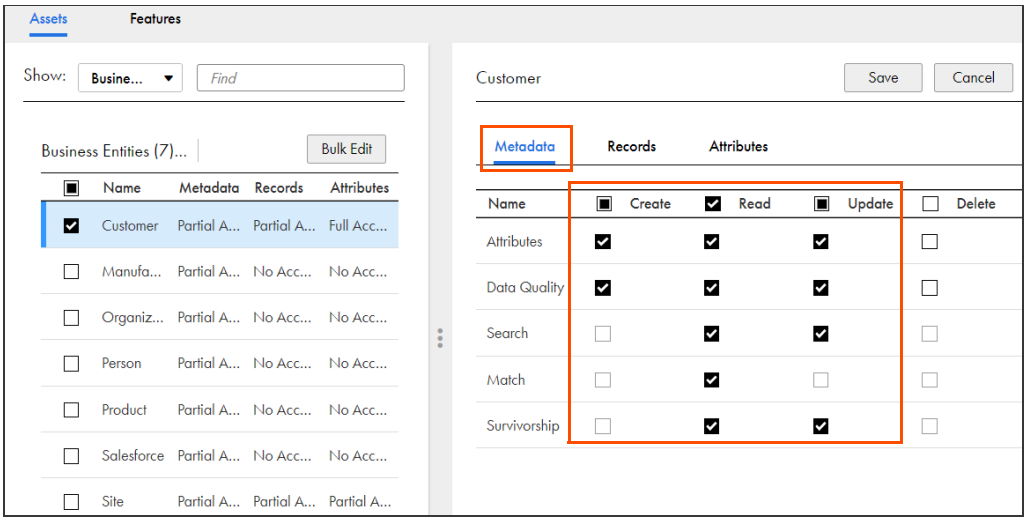
Configure Privileges for Manager Role in Business 360 Console

Refer to the above steps that you performed to configure privileges for Steward role and configure the same privileges to the Manager role.

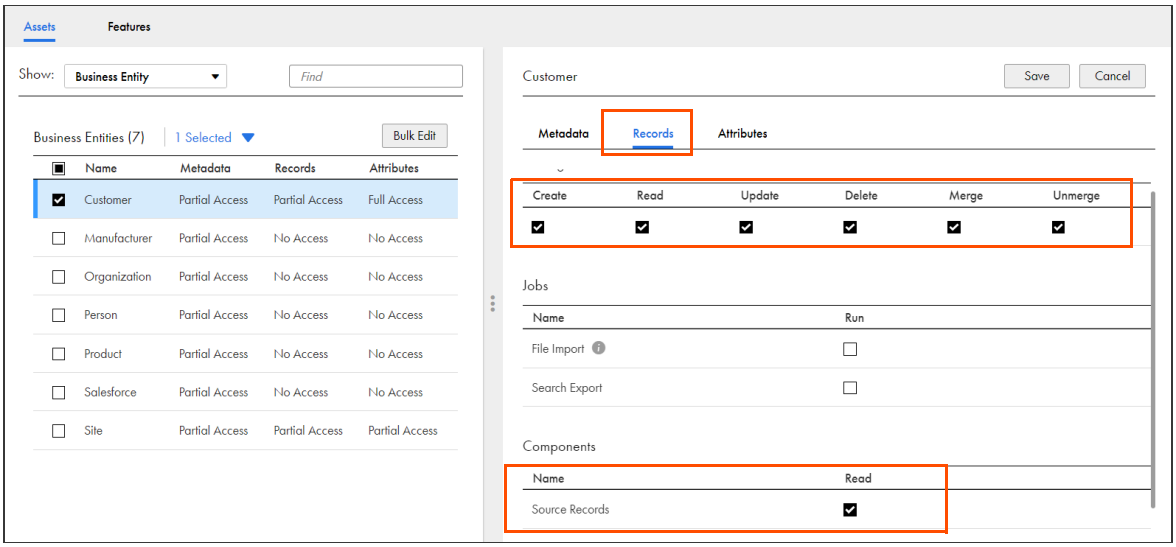
High-level steps

- In Business 360, cick Security and open Manager custom role.

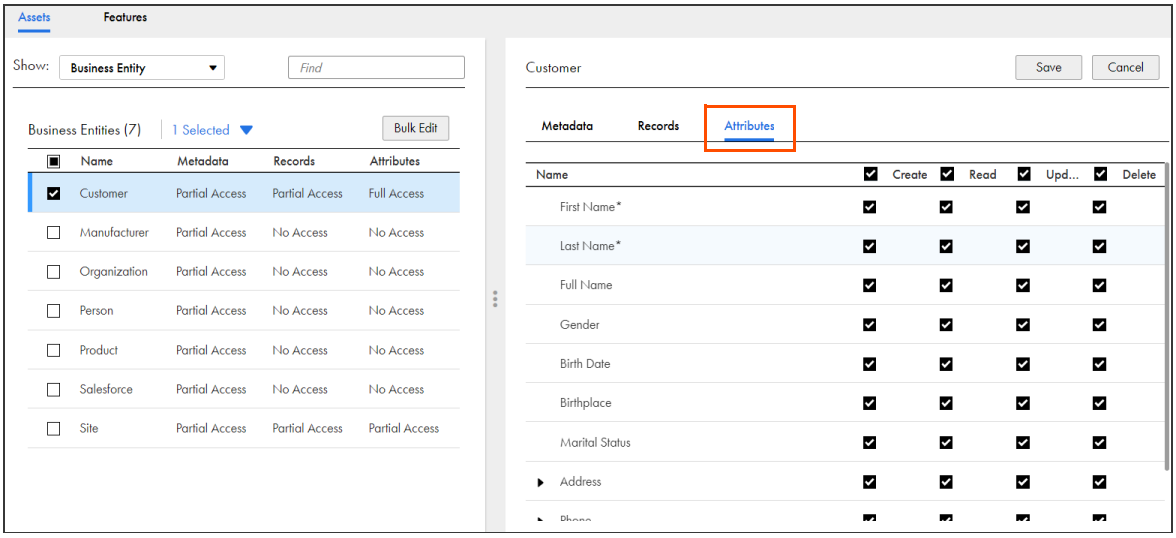
- Configure privilesges to the Customer business entity by providing access under Metadata, Records and Attributes.
- **Metadata:**



- **Records**:

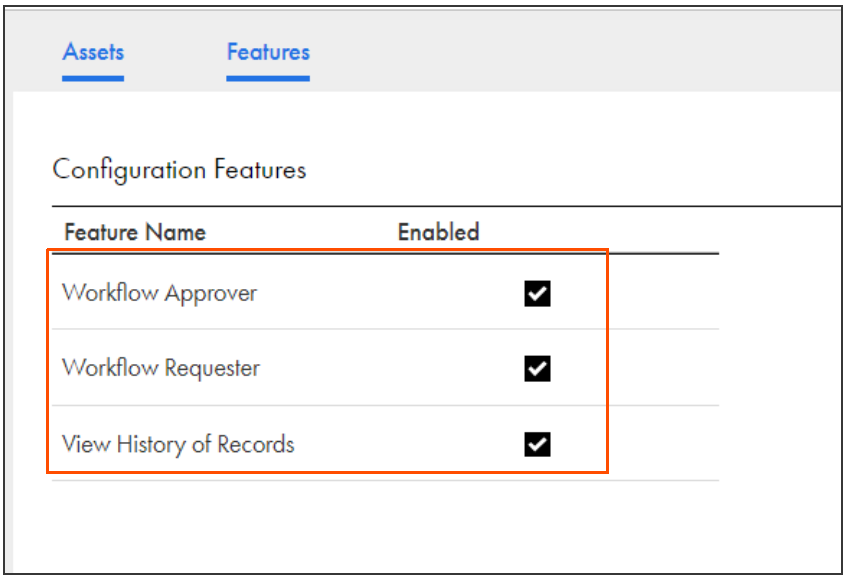


- **Attributes**:



- **Save** the configuration.

- Enable all the three features under the Features tab.



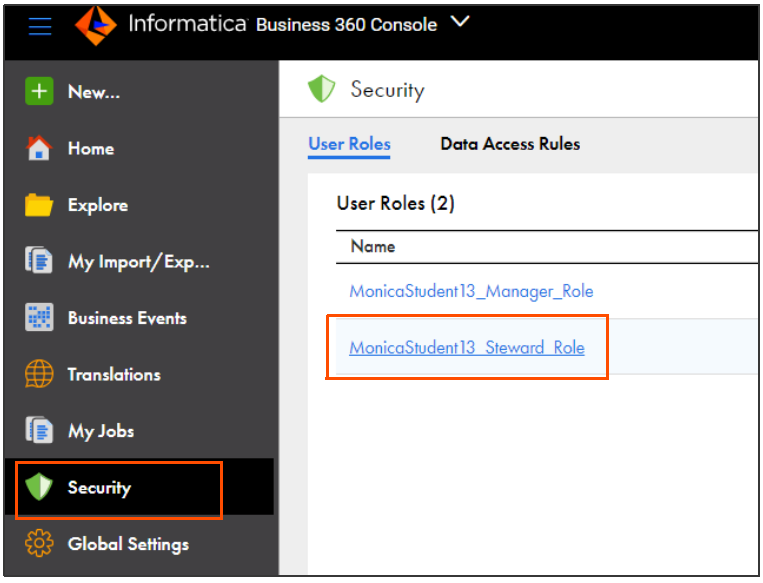
- **Save** the configuration.

Configure Privileges for Multiple Business Entities

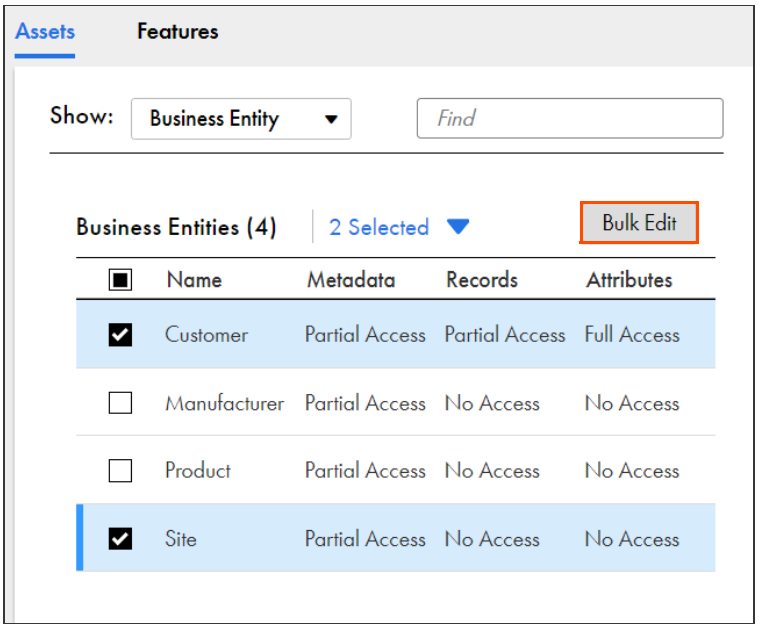
You can configure privileges for multiple business entities instead of configuring privileges for individual asset. You can configure metadata privileges and privileges related to the user roles. You cannot configure privileges to specific fields.

67. Switch back to main browser window where you have logged in as the admin user.

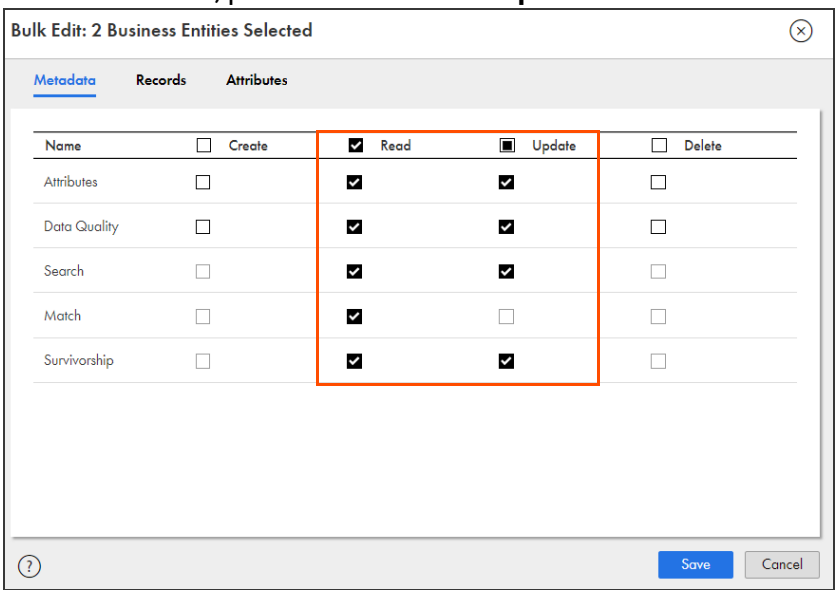
68. Click Security and select the Steward role that you have created.



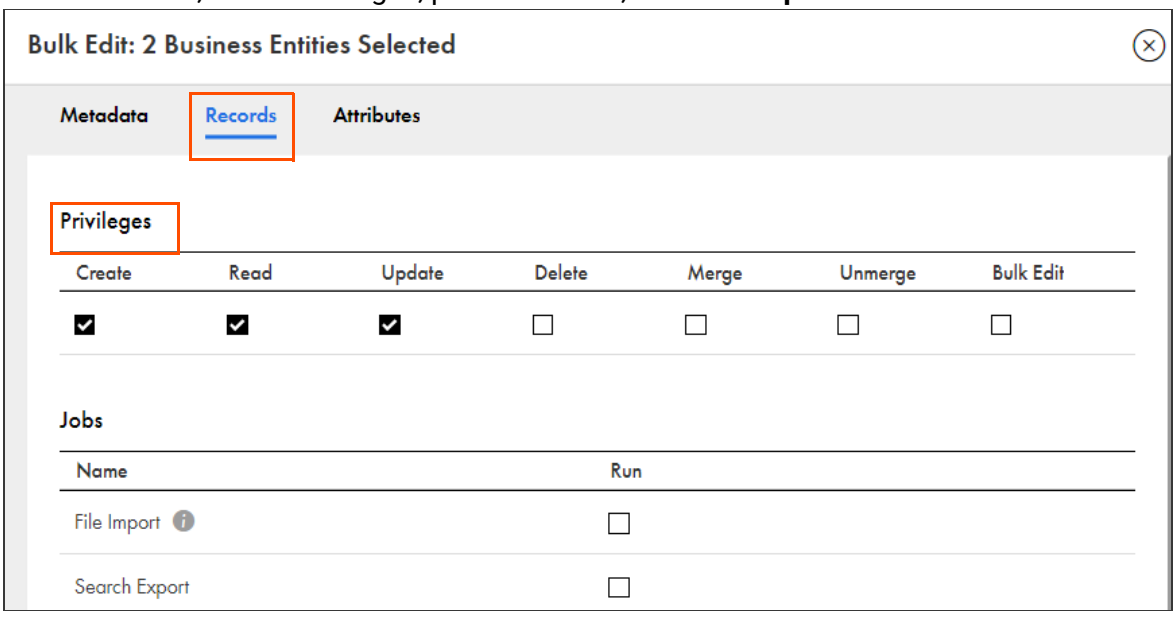
69. Select **Site** and **Customer** business entities and click **Bulk Edit**.



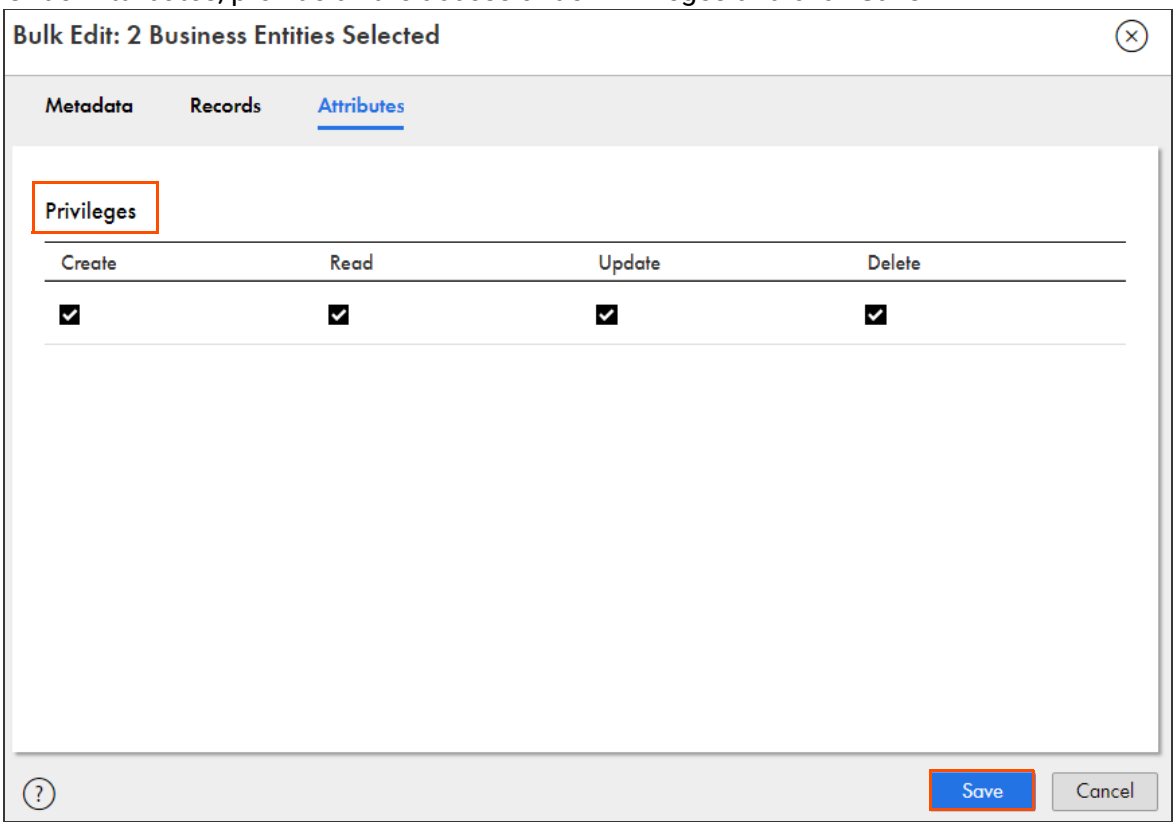
70. Under Metadata, provide all **Read** and **Update** access.



71. Under Records, under Privileges, provide **Create**, **Read and Update** access.



72. Under Attributes, provide all the access under Privileges and click **Save**.



73. You can log in as custom user and check the privileges are working as intended or not.

This concludes the lab.

Lab 8-2: Defining Attribute Based Access Control Rule (ABAC)

Overview:

You can allow or deny user access to specific attributes or field groups of records. An attribute-level data access rule controls access to the selected attributes and field groups of all the records that meet the conditions in the data access rule.

You can define data access rules in terms of who is denied access, who is allowed access, or both.

Objectives:

- Create an Attribute Level Data Access (ABAC) rule

Scenario:

The new employee that Mark has hired to work as a data steward will be working only with US customers. So, Mark wants to define ABAC rule to this user to allow to see only US customers within the Multidomain MDM application. Once he defines this rule, he will also test the same in Multidomain MDM application.

Duration:

10 minutes

Tasks:

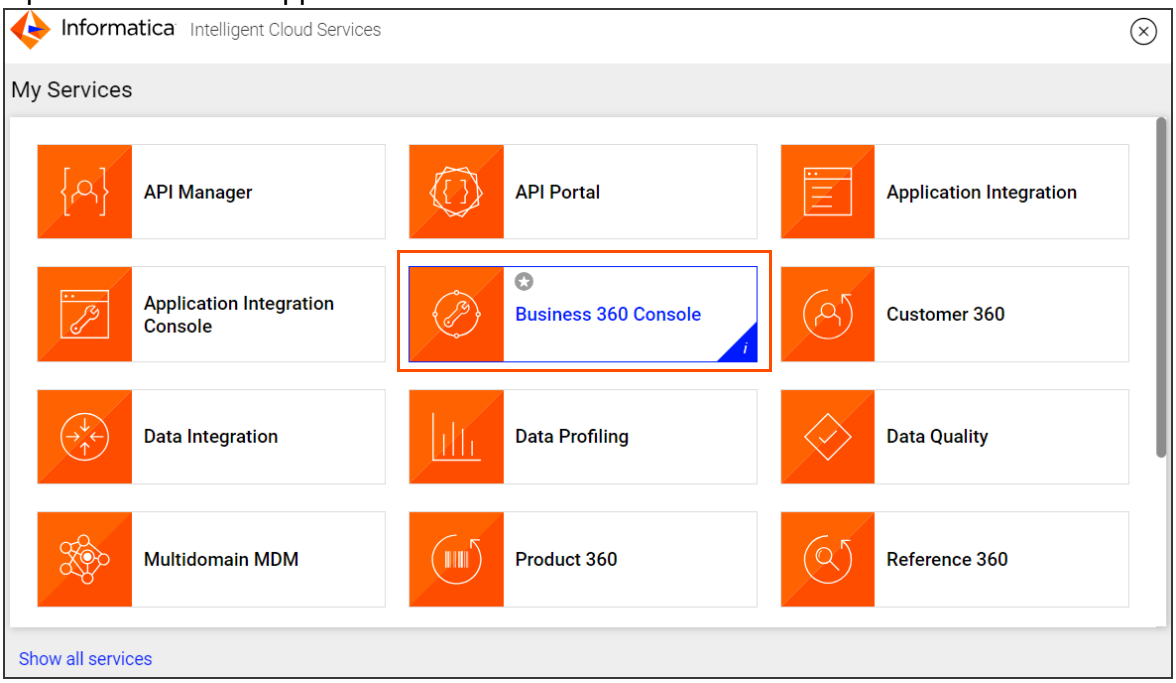
- Create a record level data access rule
- Create an attribute level data access rule

Create a Record Level Data Access Rule

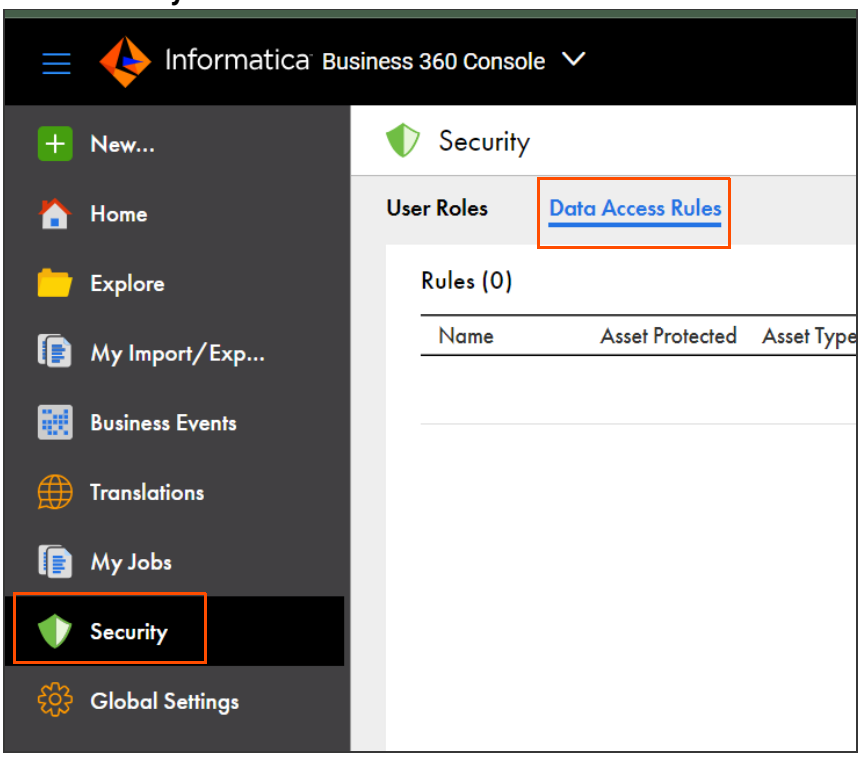
You will create a record level data access rule to display only US citizens.

1. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**), if you have logged out.

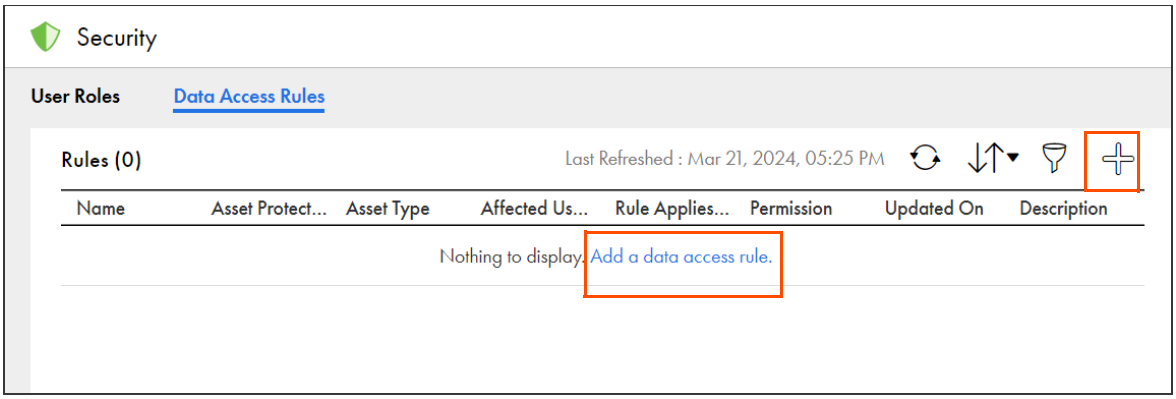
2. Open **Business 360** application from the list of services.



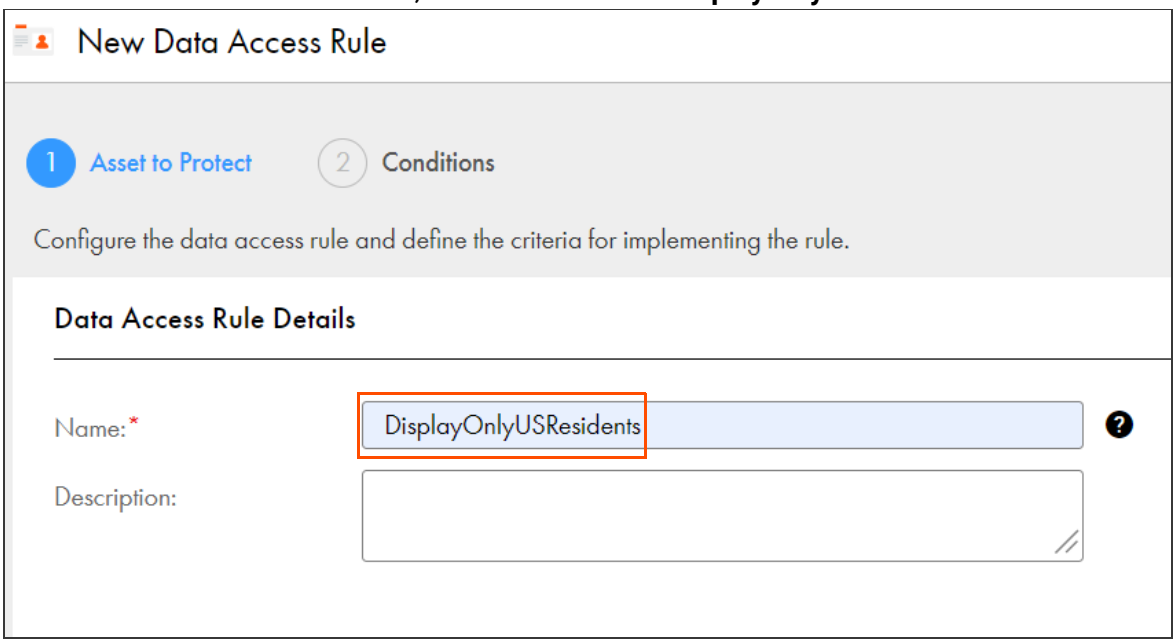
3. Click **Security** from the menu and then select **Data Access Rules** tab.



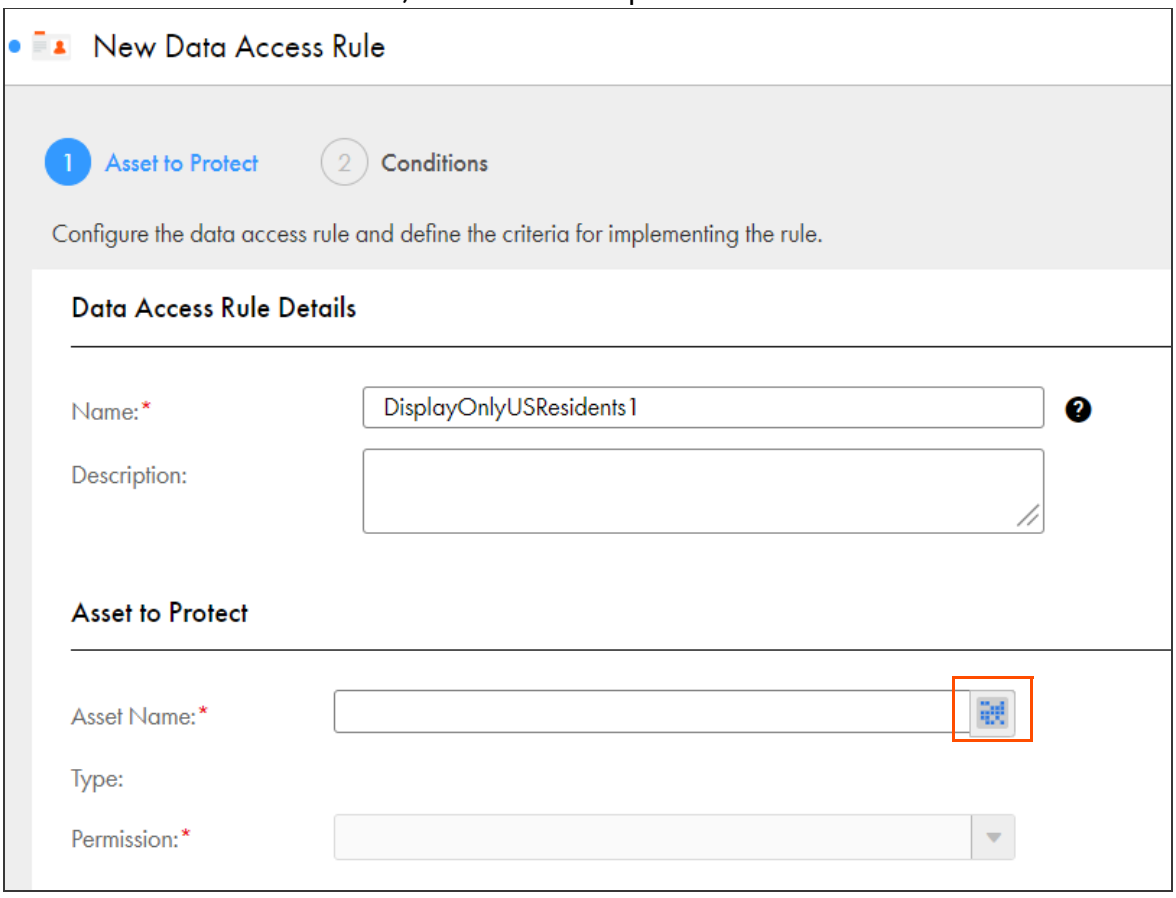
4. Click the **+** icon or **Add a data access rule**.



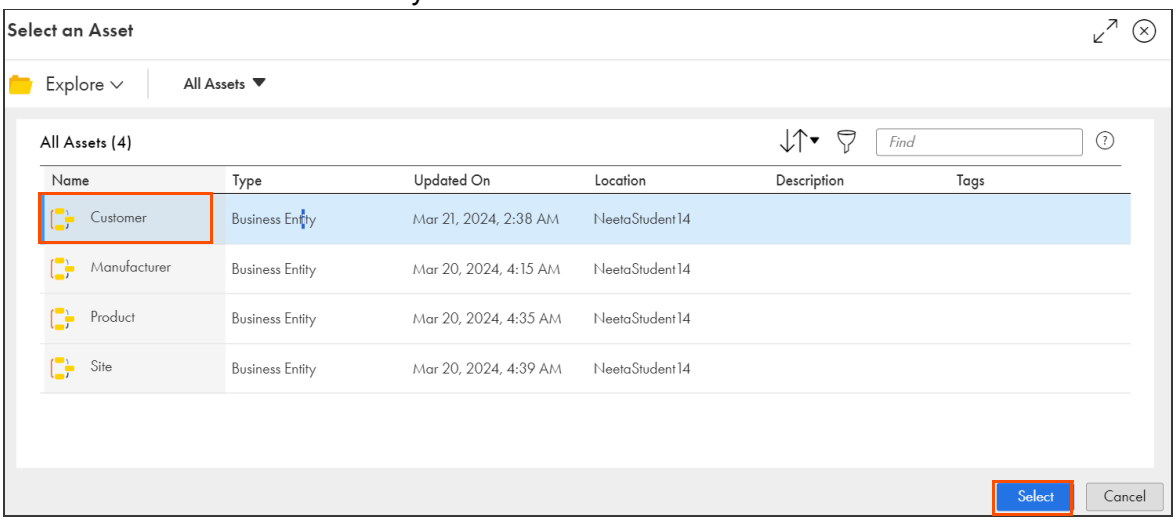
5. Under Data Access Rule Details, enter the name as **DisplayOnlyUSResidents**.



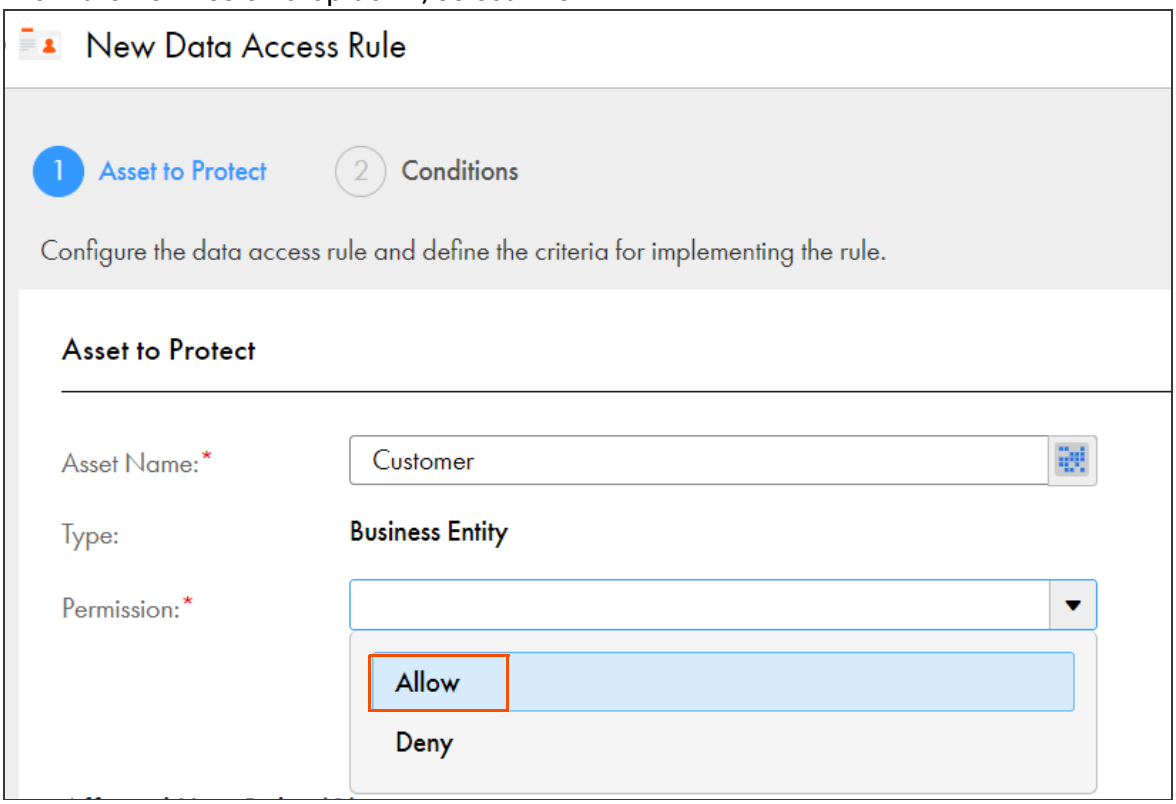
6. Under Asset to Protect section, select the asset picker icon under Asset Name.



7. Select **Customer** business entity and click **Select**.

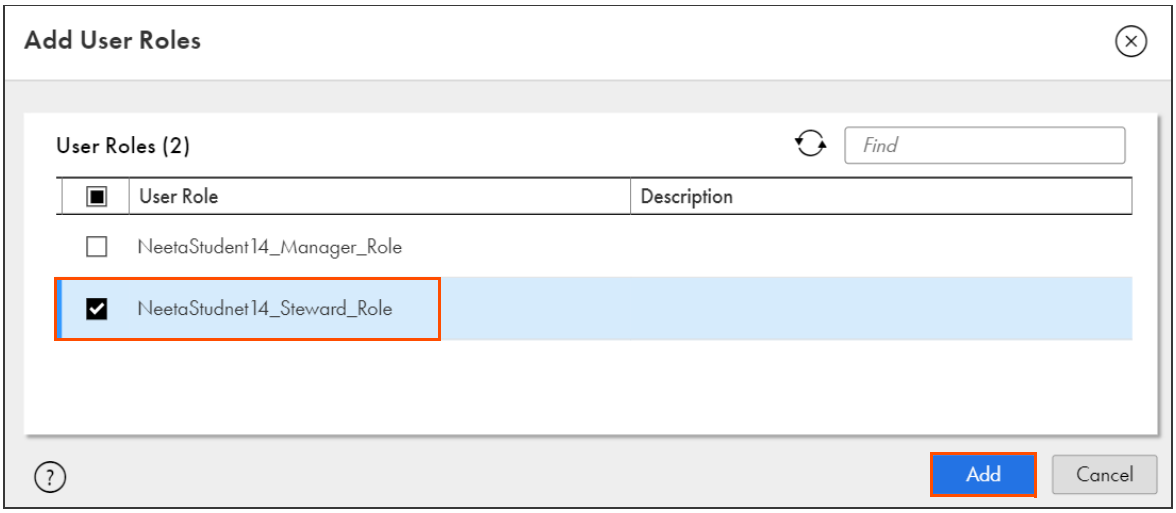


8. From the Permission drop-down, select **Allow**.

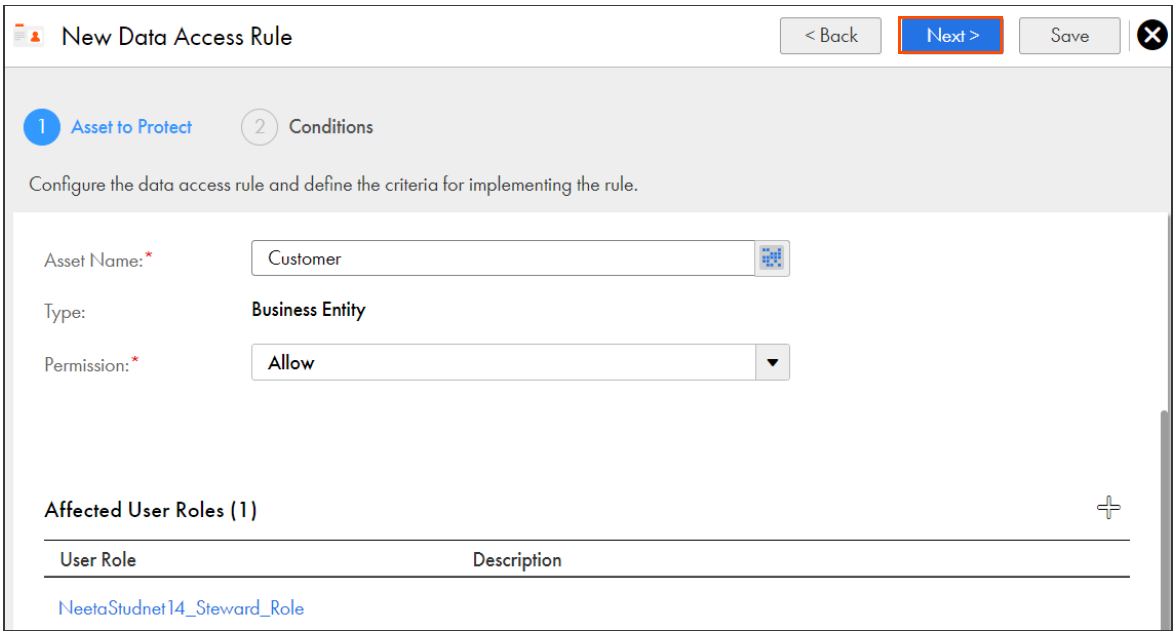


9. Under Affected User Roles, click **Add a user role** and select the custom Steward role and

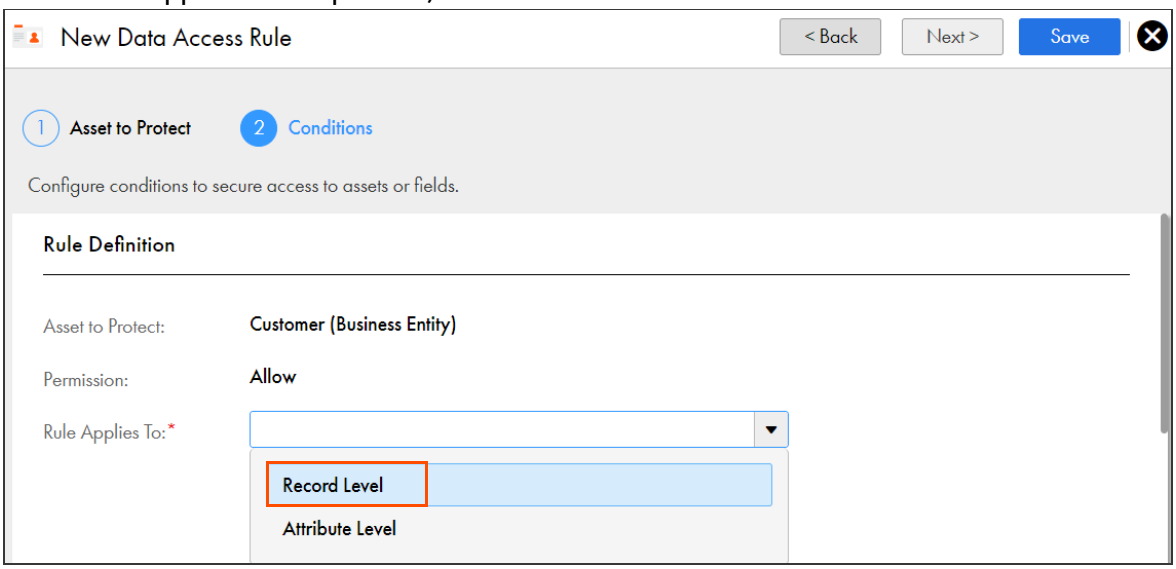
click **Add**.



10. Click **Next**.

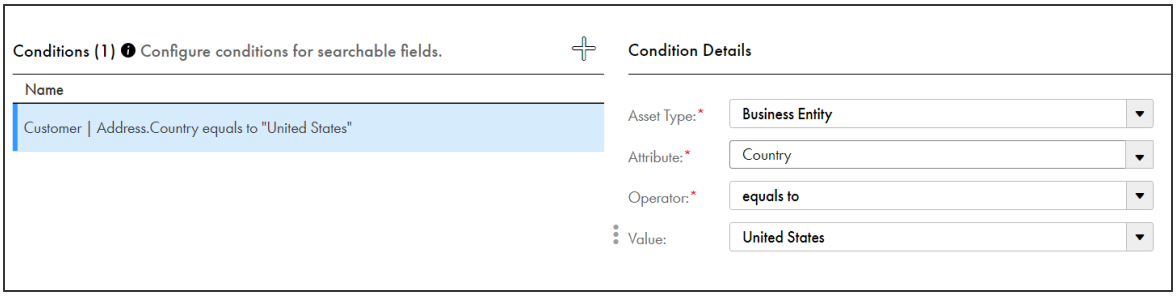


11. From Rule Applies To drop-down, select **Record Level**.

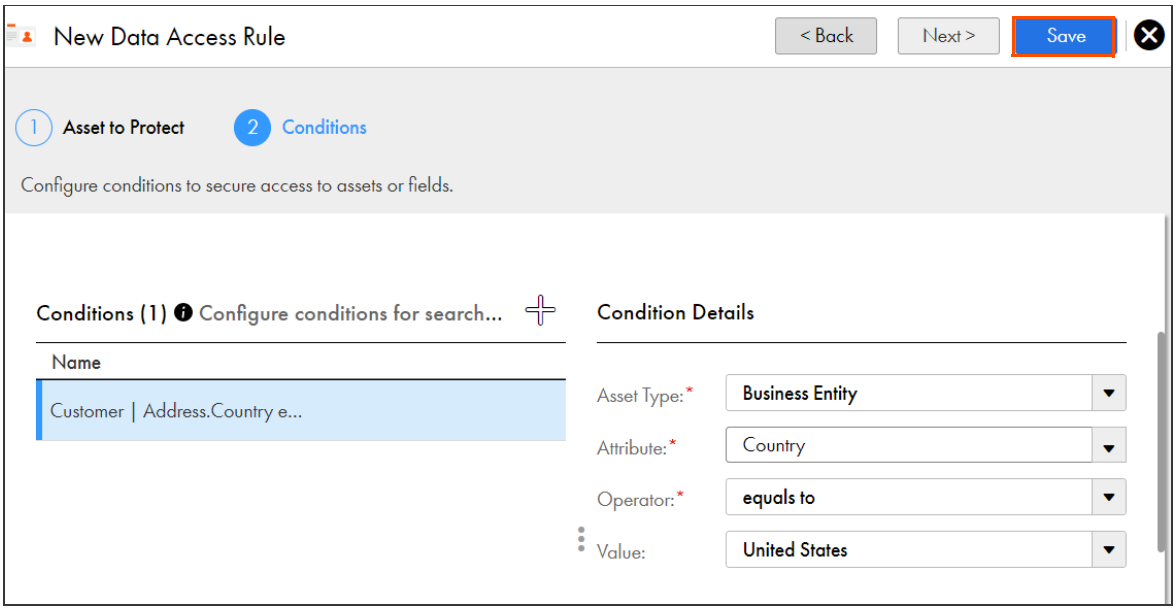


12. Refer to the following table and enter the fields under **Condition Details** section.

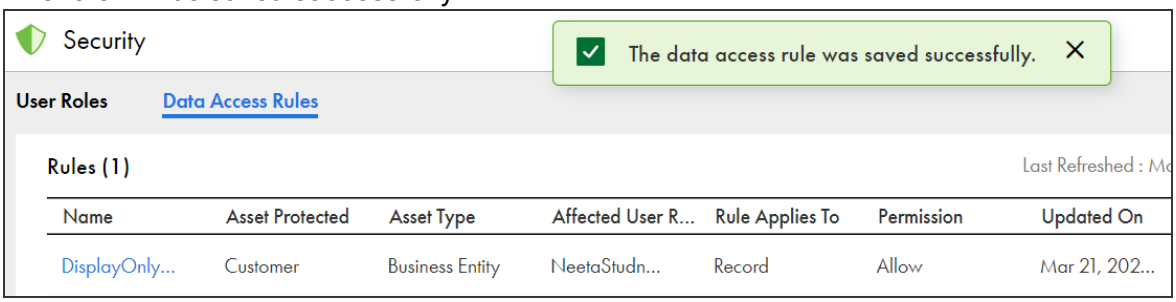
Field	Value
Asset Type	Business Entity
Attribute	Country
Operator	equals to
Value	United States



13. Click **Save**.

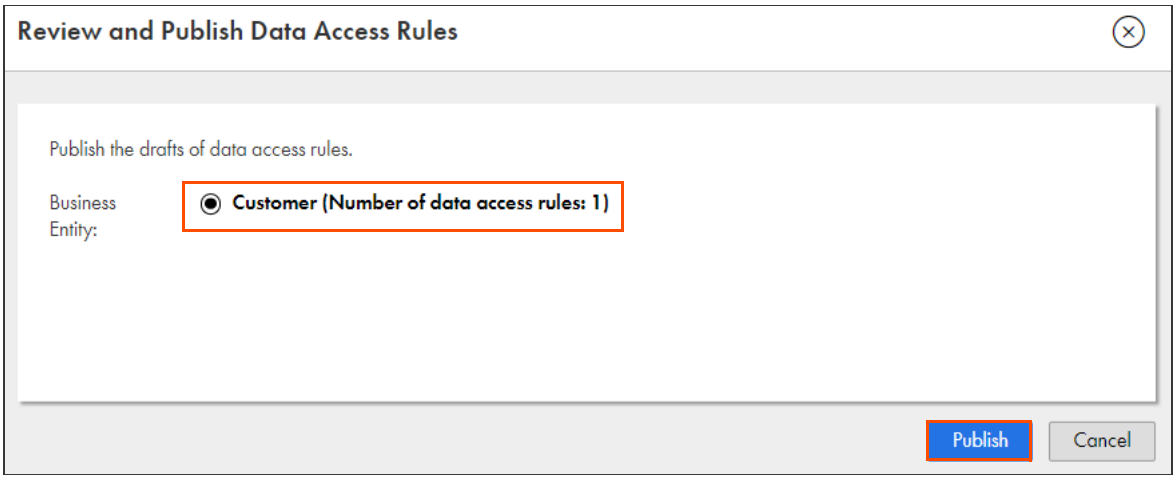


The rule will be saved successfully.

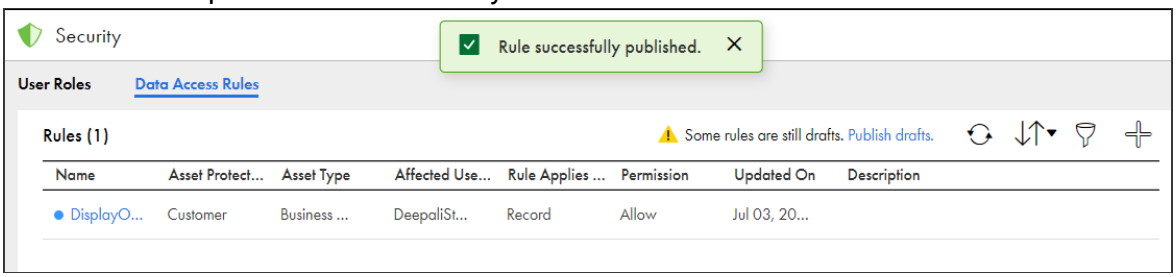


14. Click **Publish drafts** to publish the rule.

15. Select the rule and click **Publish**.



The rule will be published successfully.



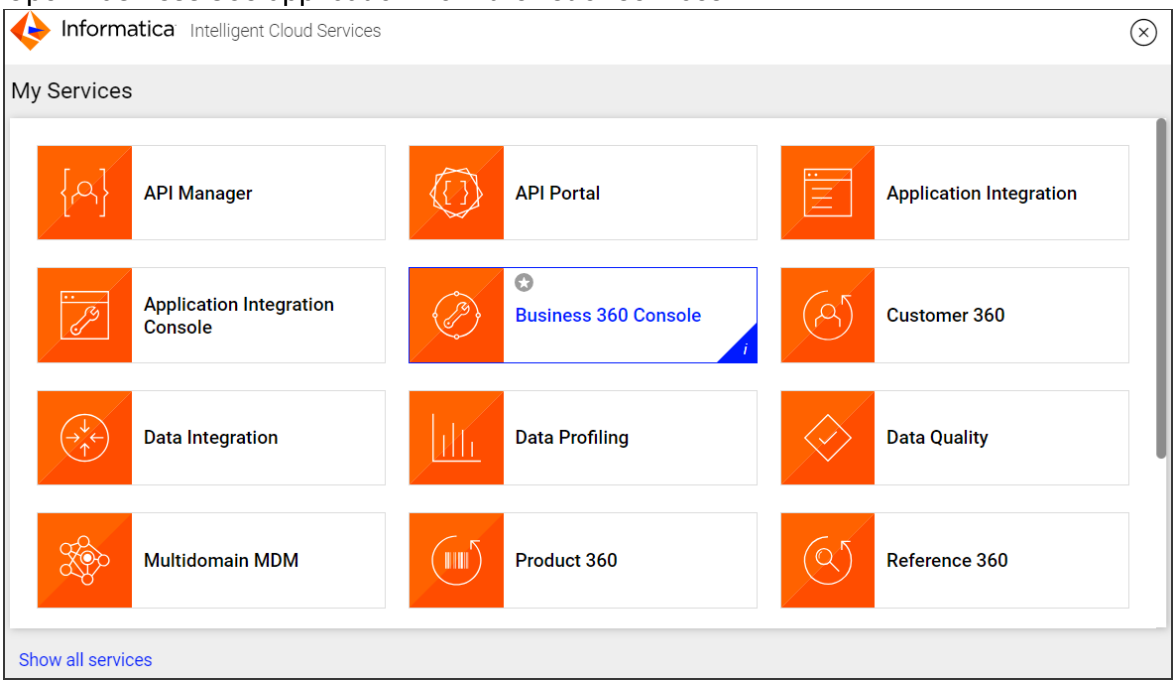
Self Try Excercise (Mandatory)

Create an Attribute Level Data Access Rule

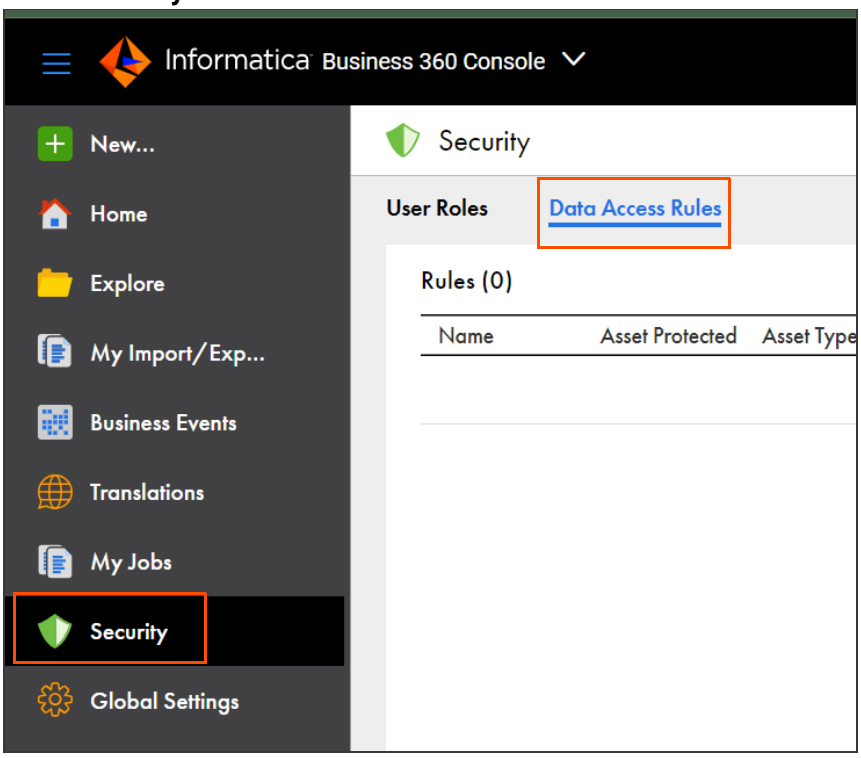
A new requirement has come up where Mark's customers who are from the state of Alasks would be excluded from the search result for the Steward user. Hence, Mark wants to create a new rule for the data steward user, so that the steward will not be able to see the address if the customer belongs to Alaska.

16. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**).

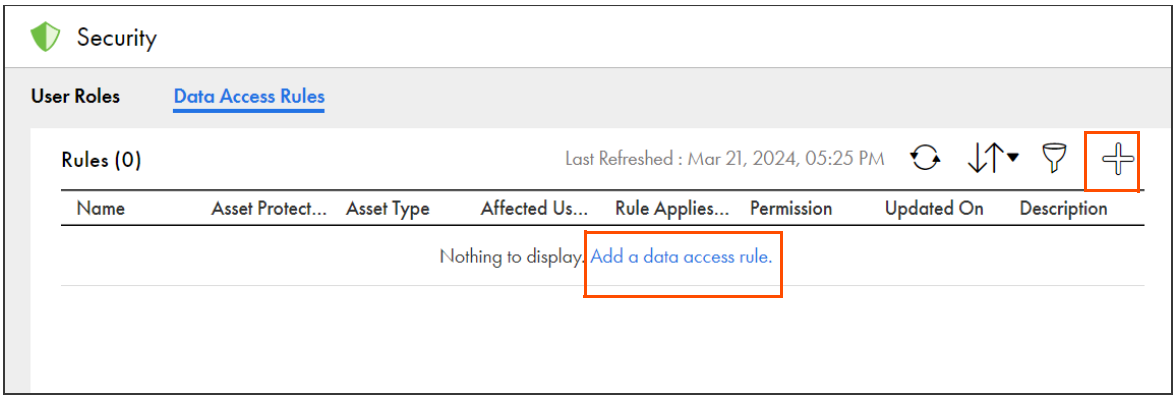
17. Open **Business 360** application from the list of services.



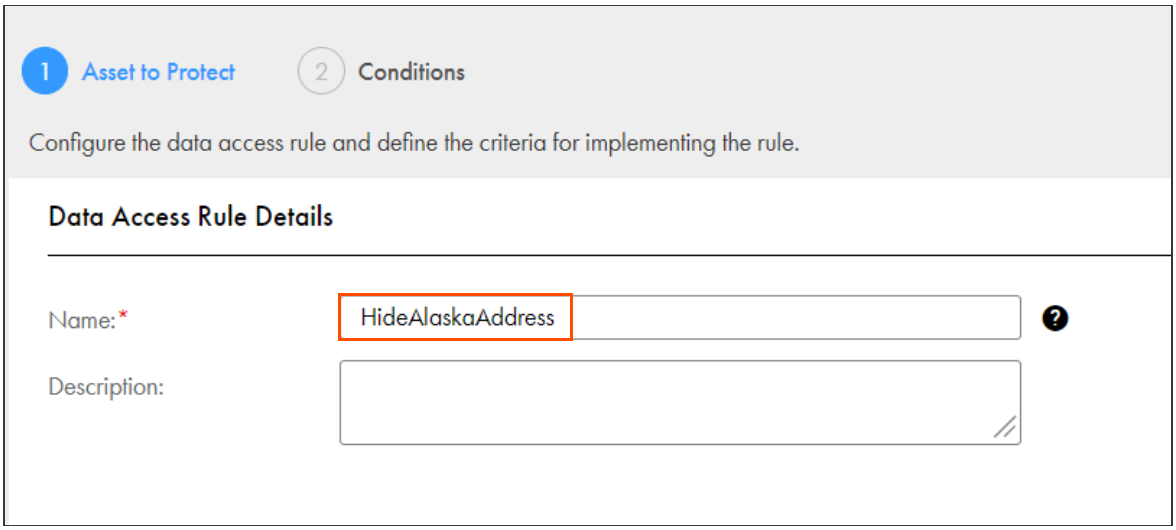
18. Click **Security** from the menu and then select **Data Access Rules** tab.



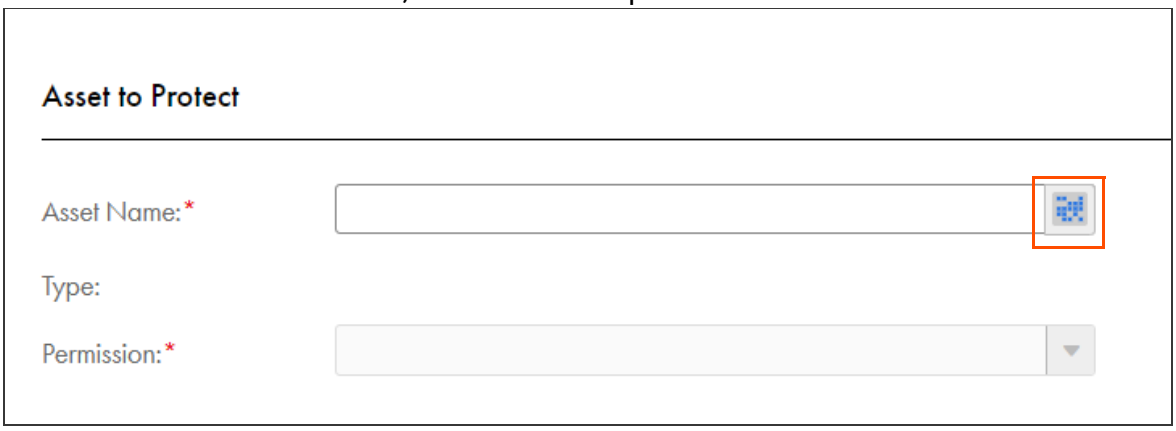
19. Click the **+** icon or **Add a data access rule**.



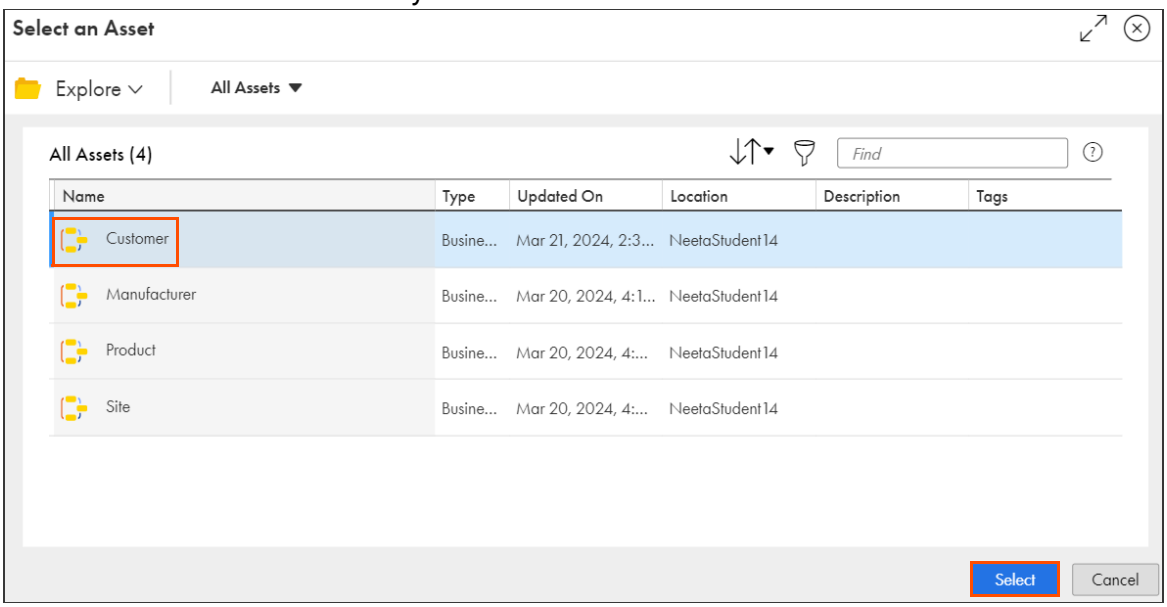
20. Under Data Access Rule Details, enter the name as **HideAlaskaAddress**.



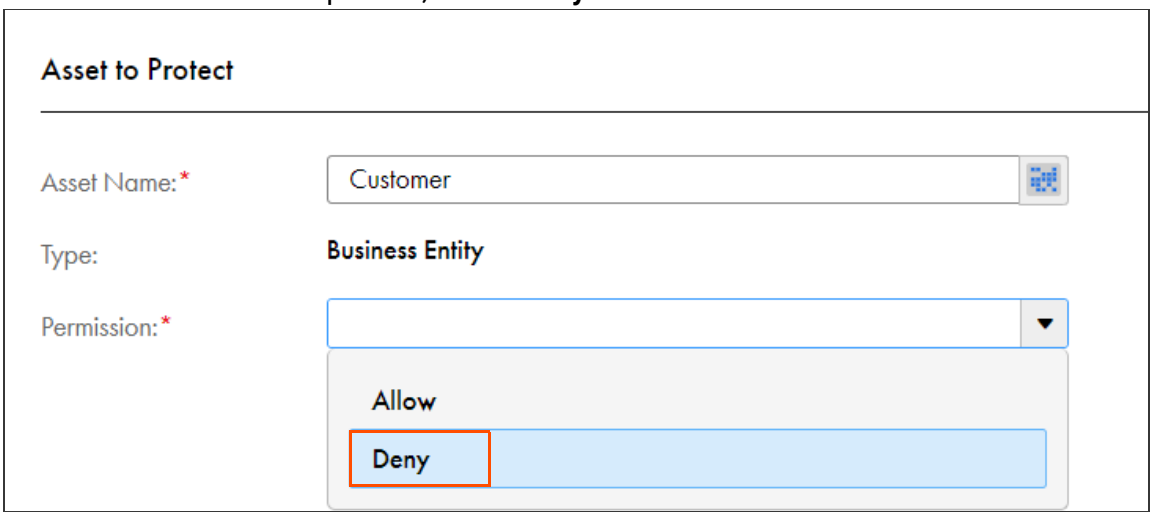
21. Under Asset to Protect section, select the asset picker icon under Asset Name.



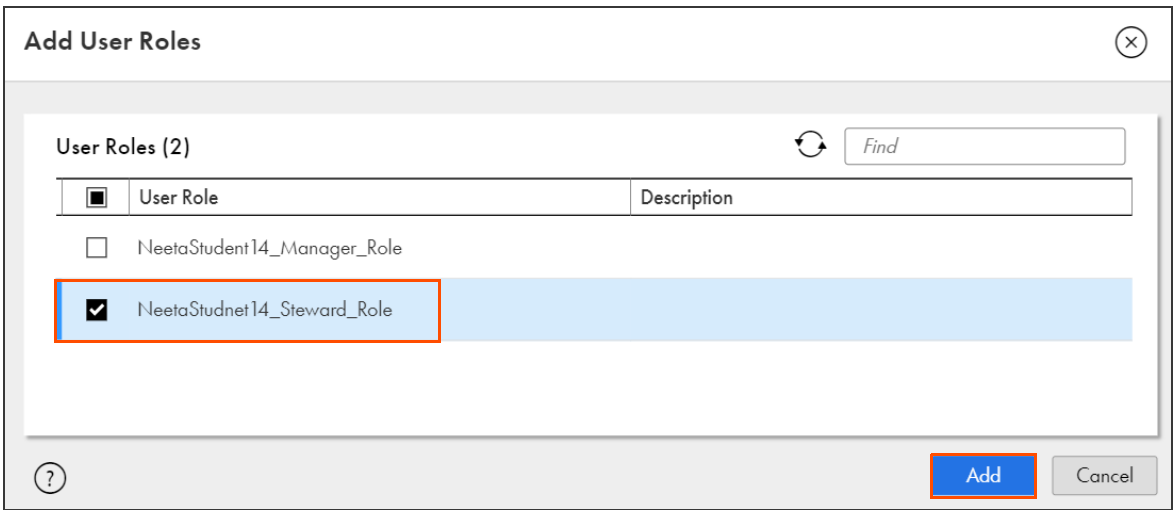
22. Select **Customer** business entity and click **Select**.



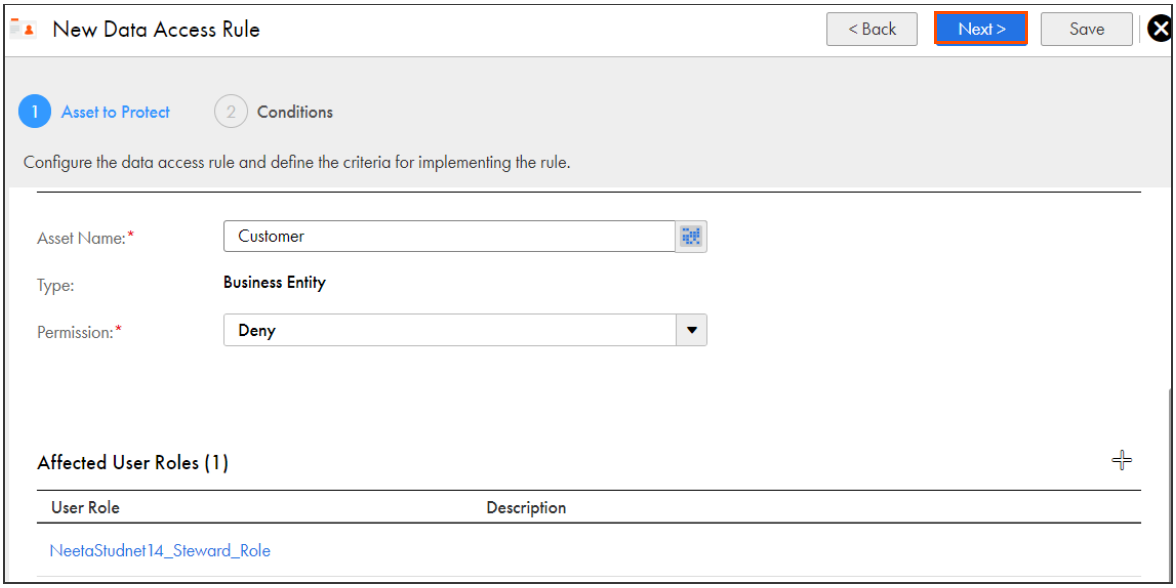
23. From the Permission drop-down, select **Deny**.



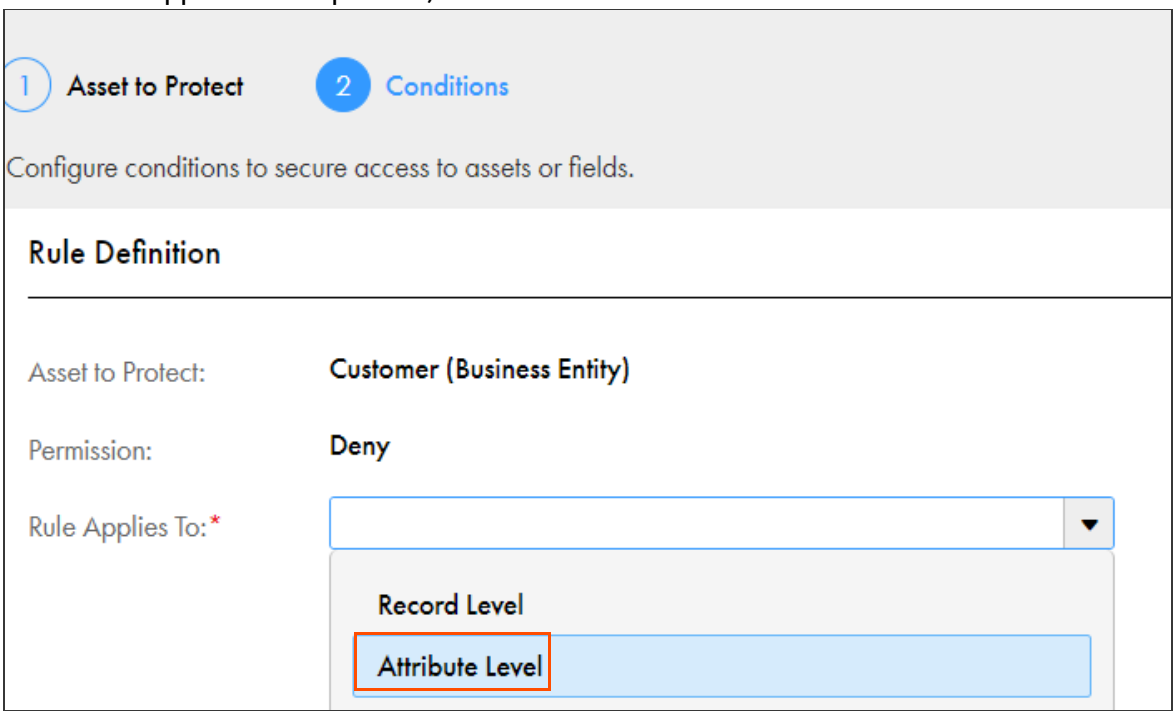
24. Under Affected User Roles, click **Add a user role** and select the custom steward role and click **Add**.



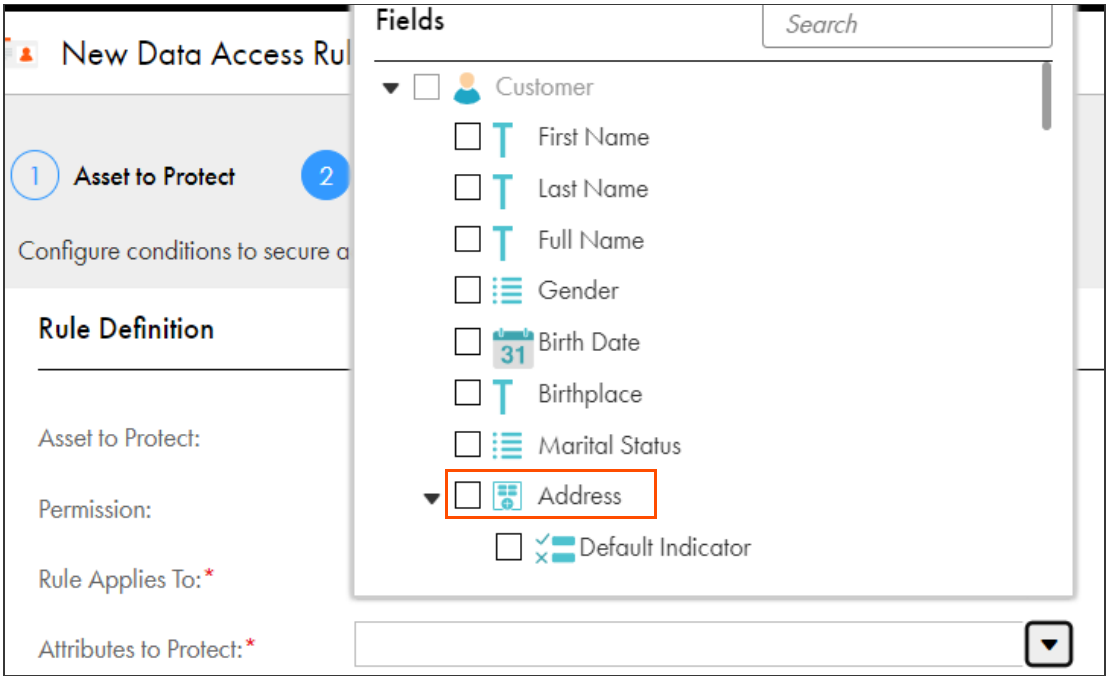
25. Click **Next**.



26. From Rule Applies To drop-down, select **Attribute Level**.

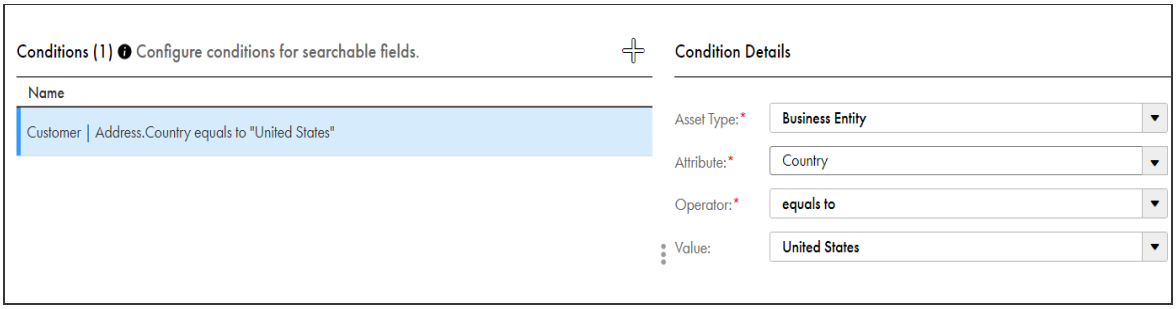


27. From the **Attributes to Protect** drop-down, select **Address**.

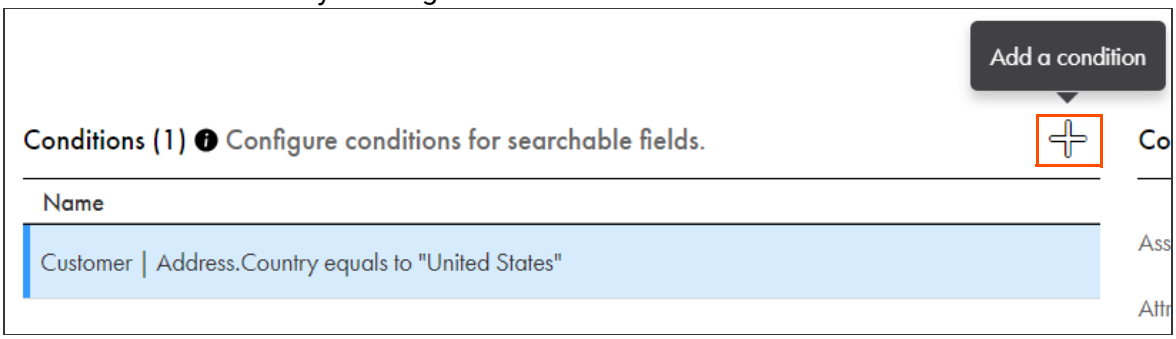


28. Refer to the following table and enter the fields under **Condition Details** section.

Field	Value
Asset Type	Business Entity
Attribute	Country
Operator	equals to
Value	United States

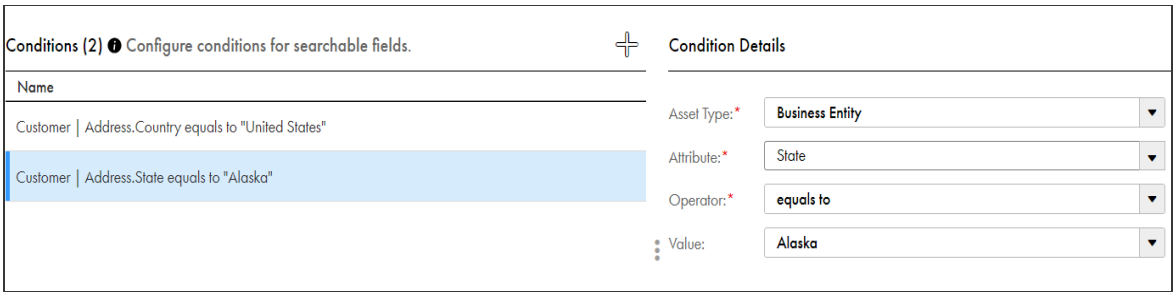


29. Add another condition by clicking the **+** icon.

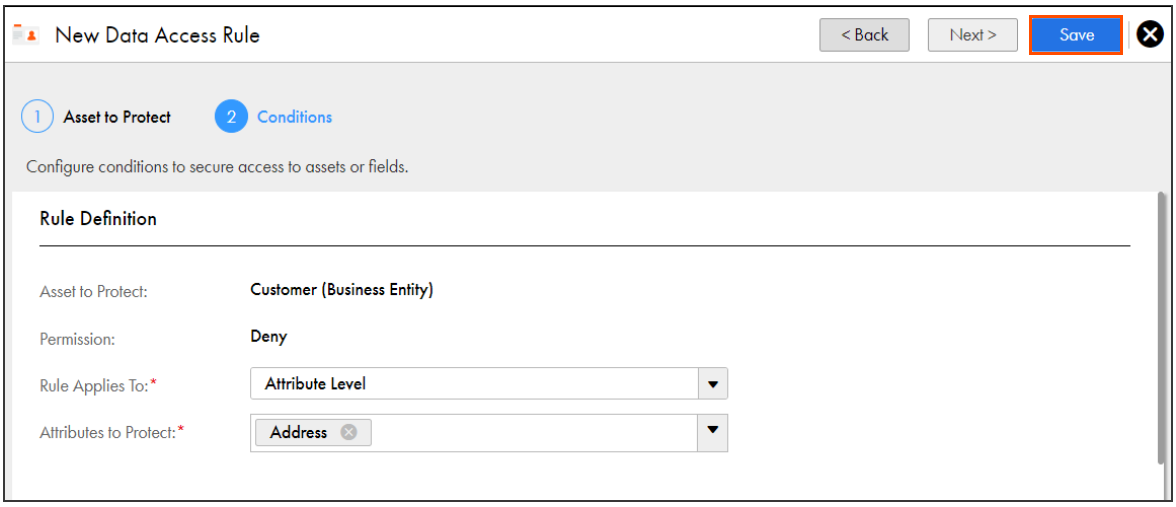


30. Refer to the following table and enter the fields under **Condition Details** section.

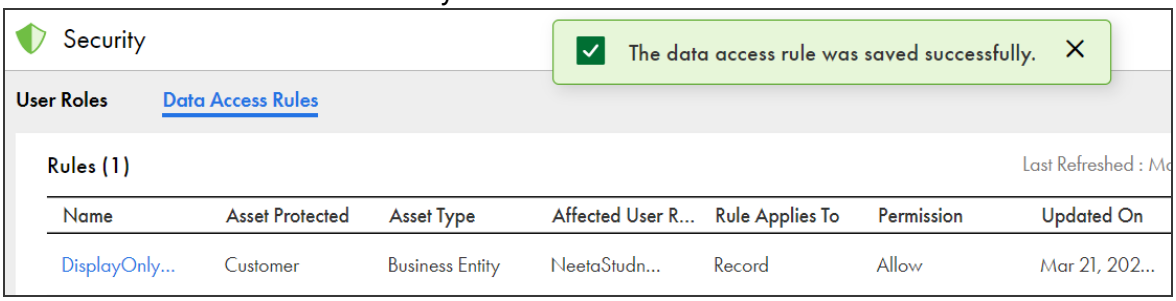
Field	Value
Asset Type	Business Entity
Attribute	State
Operator	equals to
Value	Alaska



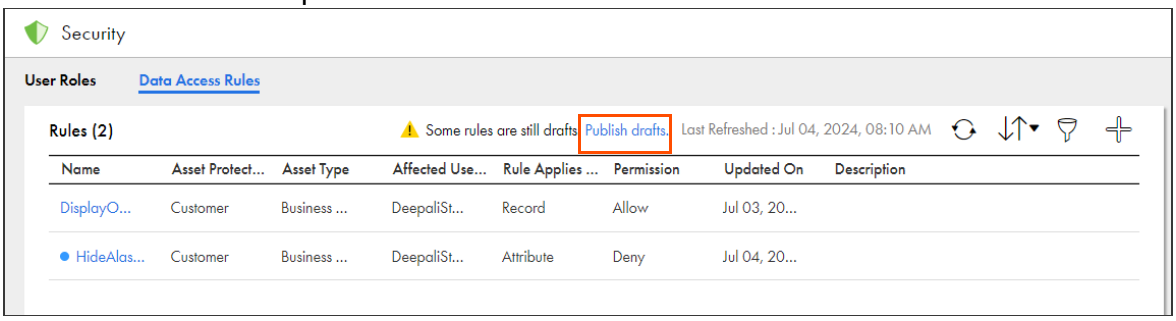
31. Click **Save**.



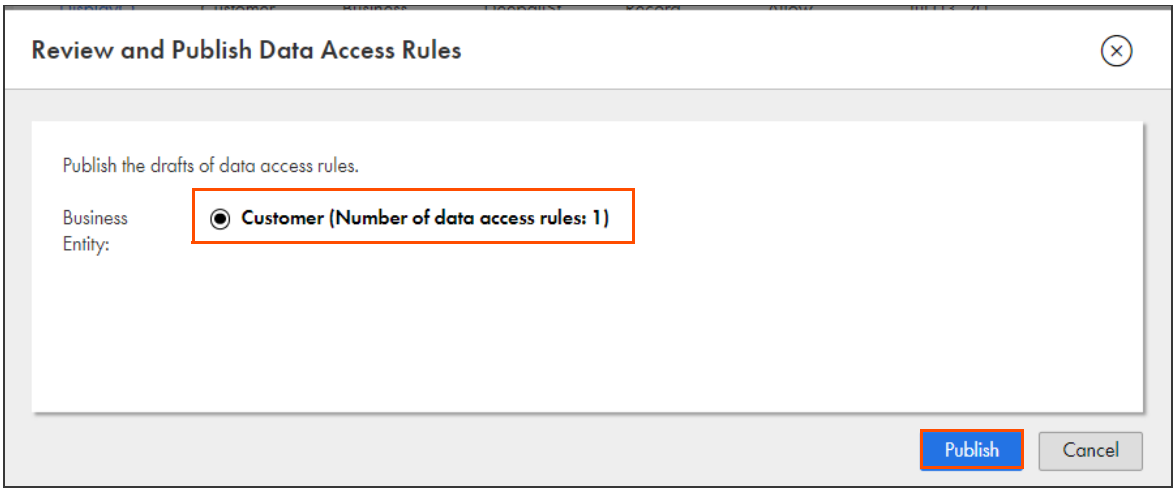
The rule will be saved successfully.



32. Click **Publish drafts** to publish the new rule.



33. Select the rule and click **Publish**.



This concludes the lab.

Key Takeaways

- User is an individual account in Informatica Intelligent Cloud Services.
- You can create and assign user roles to users and user groups in your organization
- You can configure privileges related to a single business entity or multiple business entities for a custom user role.

Module 9: Onboarding Data

Manage Records

Multidomain MDM supports, creation of records or managing existing records. It also allows you search for records that you want to manage. To add or change the information in records, you can edit the records. You can also delete the records that are not needed anymore. Once you have the Business Entities/Relationships/Hierarchies created, you can create records to the Business Entities/ Relationships/Hierarchies or manage existing records. You can search for records that you want to manage. To add or change the information in records, you can edit the records.

For example, you can create a person record and add person data, such as the personal information and contact details of a person. You can then edit information in the record.

To find a record, you can enter a search text in a search box or perform a filtered search.

Add New Records

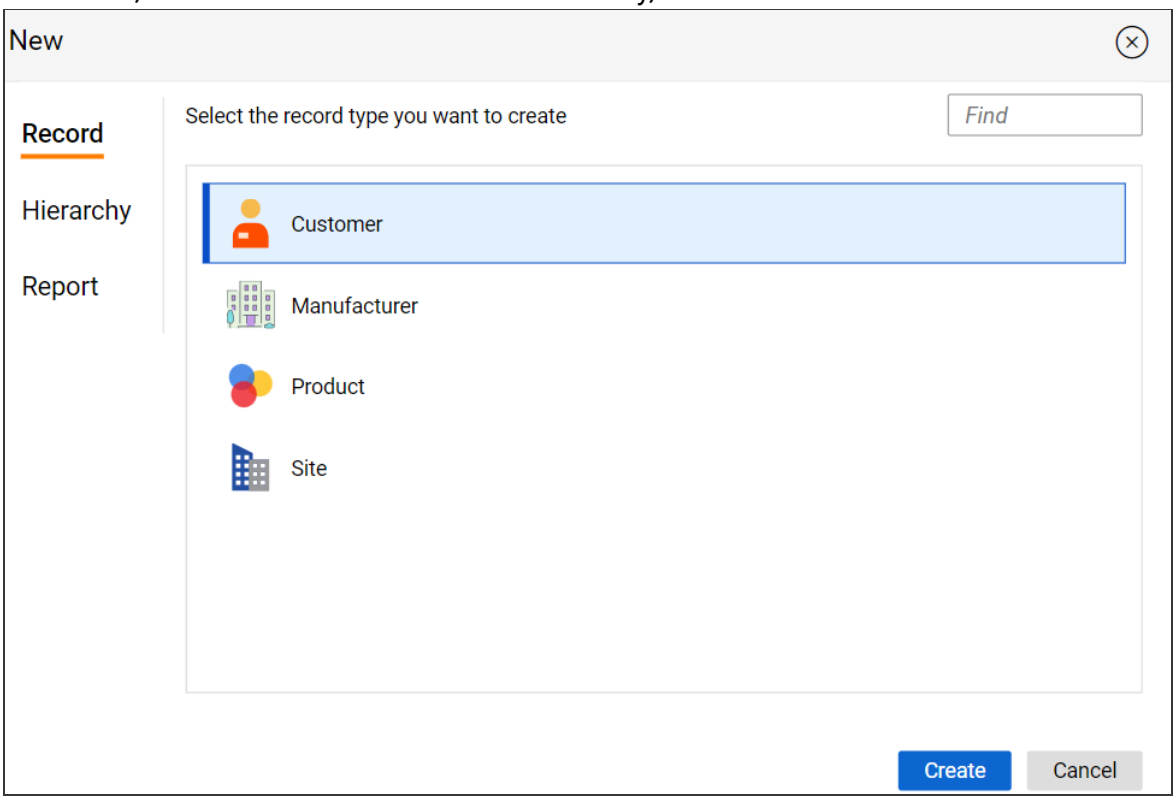
You can add business entity records to the Multidomain MDM database. Multidomain MDM allows you to create business entity as per your requirement. **For example**: Monica from Homekart Appliances has created business entities like Customer, Manufacturer, Site and Product. She can use these business entitites to save the details. Customer business entity can be used to add customer data, such as personal information and contact details of a customer. Manufacturer business entity can be used to add manufacturer data, such as name, its contact details. Product business entity can be used to add product data, such as name, product type, price and Site business entity can be used to add site data, such as location, warehouse and location id. She can also add related records to these business entity records. Multidomain MDM validates some of the fields before saving a record to the database.

Creating a Customer Scenario

To create a new customer record, Anthony Davis refer to the following steps:

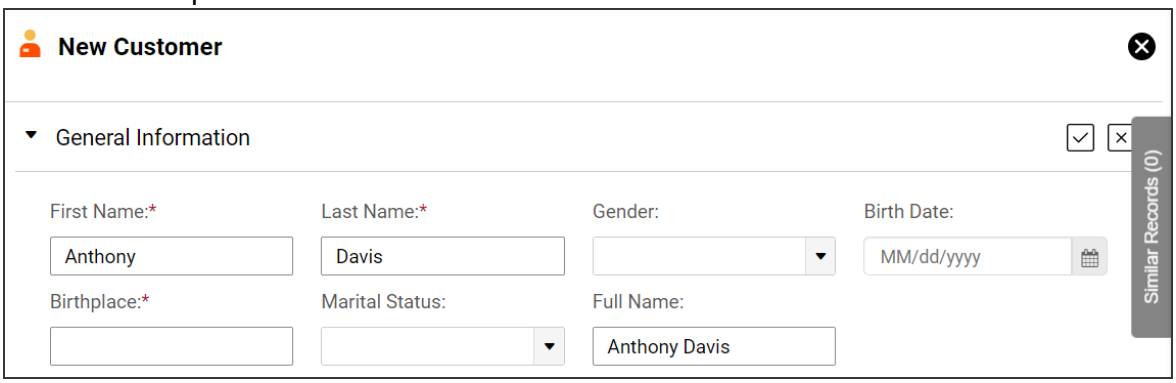
1. Log in to Multidomain MDM SaaS.

2. Click **New**, select Customer as the business entity, and then click **Create**.



3. Type Anthony as the first name and Davis as the last name.

4. Fill in other required fields and save the record.

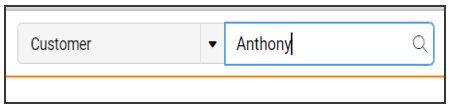


Similarly, you can create other records by selecting respective business entitites and save them into Multidomain MDM application.

Related Records

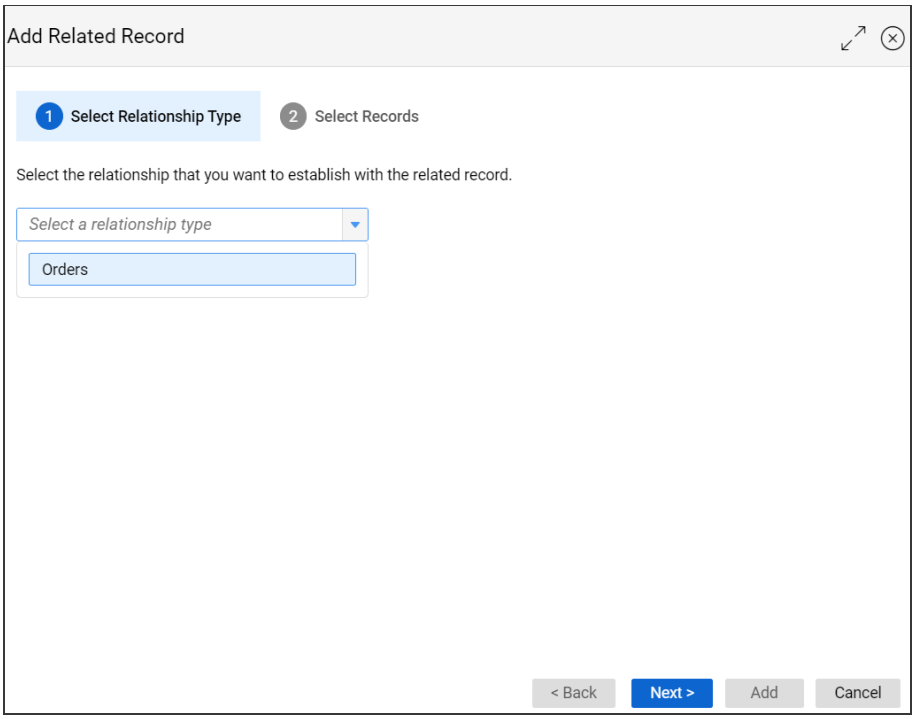
Related records are the records that contain data associated with a record. You can establish relationship between two records within a business entity or across business entities.
Refer to the following steps to add a related record:

1. Click **Multidomain MDM** Service.
2. Click **Search**, select the **Customer** business entity and enter the search string as **Anthony**.

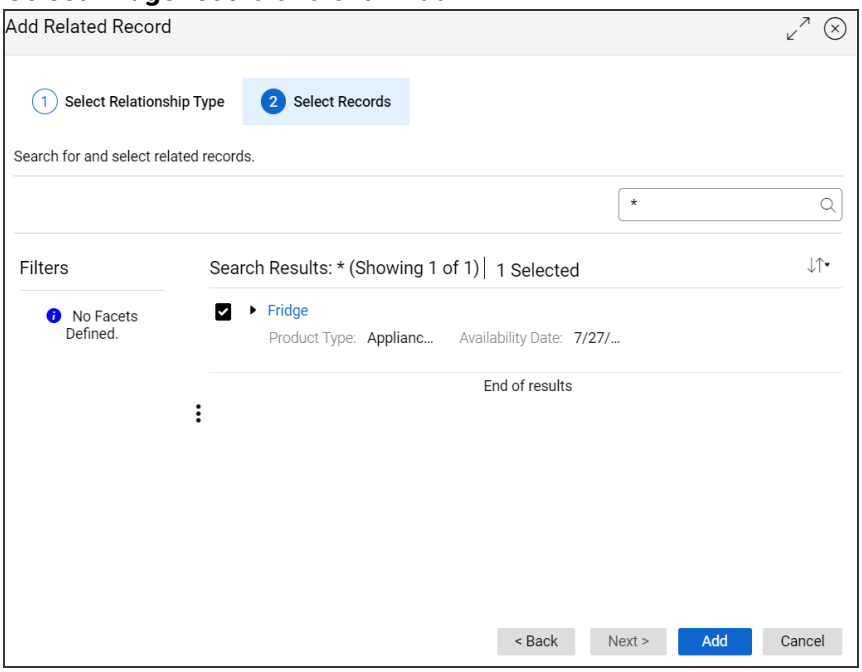


3. Click **Anthony Davis** and scroll down to find the Related Records section at the bottom of the page.
4. Click the **+** icon to the right of the Related Records to add product record.

5. The **Add Related Record** page opens. Here, you will select a relationship type that you want to establish between the records. From the **Select a relationship type** drop down, select **Orders.** Click **Next**.



6. In the Select Records tab, enter * in the search box and click the Search icon to search. Select **Fridge** record and click **Add**.



7. Enter the Store as Burlington. Click **Submit**.

Search Records

To find a record, you can either enter a search text in a search box or perform a filtered search. A search can return multiple matching records. You can sort and filter the search results to refine the search results. Use the search box to perform a full-text search. When you use the search box, you can search within a specific business entity or across multiple business entities. Use filtered search to specify a criteria for your search. To repeatedly perform the same search with the configured filter values, you can save a search and run it at any time. To repeatedly perform a search with the predefined fields, you can save a filtered search as a template. You can use different values in the search template and perform the search multiple times. You can open any record and view the complete details of it.

If a record has multiple entries for a field group, the search result displays the first entry even when the search term might match another entry. For example, when you search for California, the search result might display New York as the city. In this scenario, the record might have multiple addresses, and the search result displayed the first address that has New York as the city.

If you use a source business ID to search for a record, the search returns the master record to which the source record contributes. You can view the fields based on which the search results match the search term.

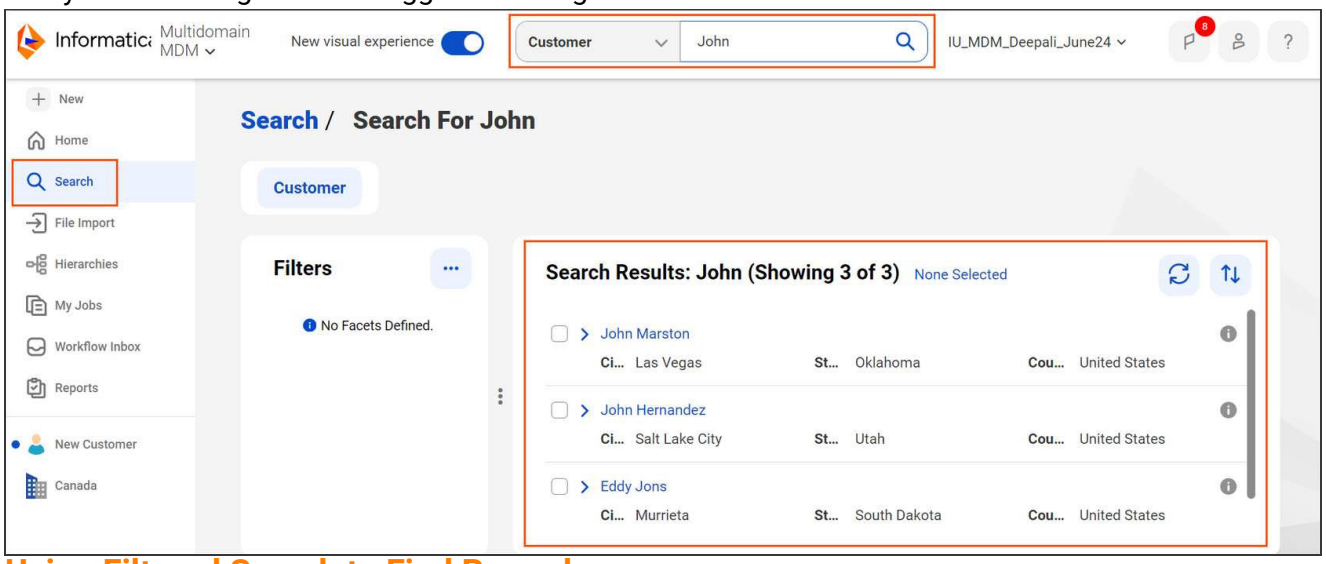
Select one or more business entities and use the search box to search for records. The search results show the most important details of the records. You can see the complete details by opening the records.

Using The Search Box to Find Records

Select one or more business entities and use the search box to search for records. The search results show the most important details of the records. You can see the complete details by opening the records. You can use a complete or partial text as a search string. When you start typing a search string, you can select any matching string that appears as a suggestion. You can also select a recently used search string.

When you type a string to search for records across multiple business entities, Customer 360 SaaS lists the strings based on the previous search suggestion. Customer 360 SaaS doesn't include business

entity names along with the suggested strings.

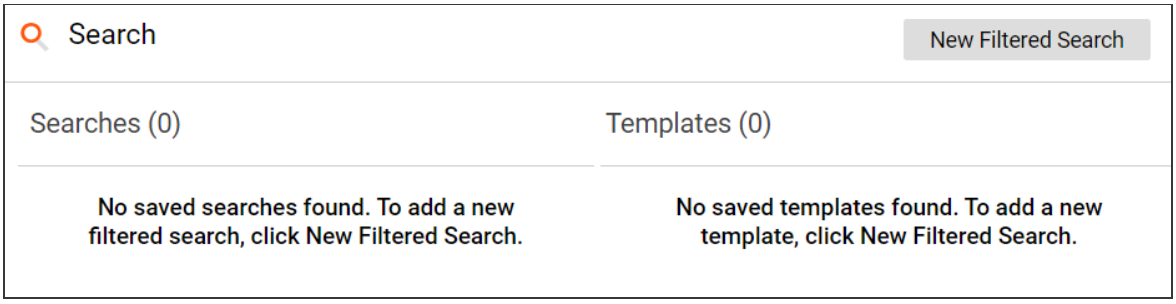


Using Filtered Search to Find Records

You can use filtered search to search for records based on one or more field values. You can also use business IDs to search for records. To repeatedly search for records with the configured filter values, you can save the search and run it at any time. You can save a search as a template, and enter different field values to perform a search. You can't use search terms that contain special characters for filtered search.

Steps to perform a filtered search:

1. Click **Search** > **New Filtered Search**.
   The New Filtered Search panel appears.

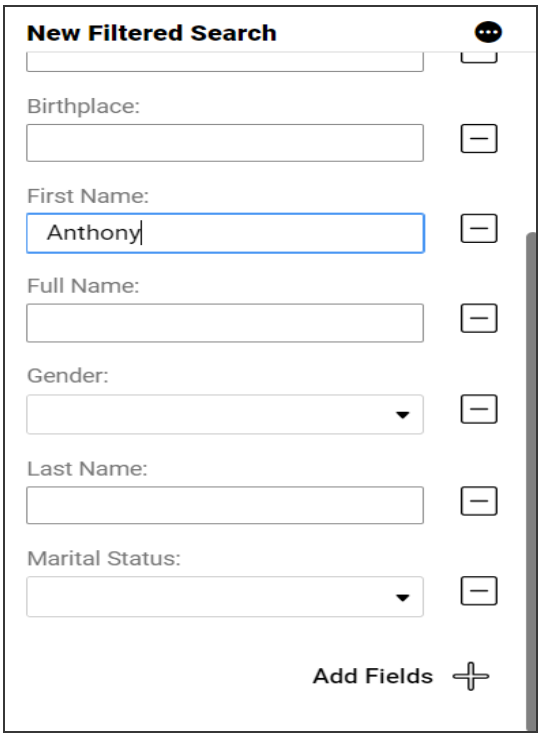


2. Select a business entity.
   The fields related to the business entity appears.

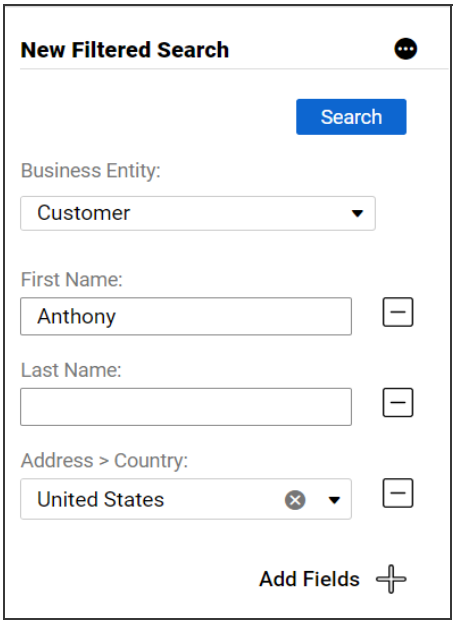
3. Enter search terms for the fields that you want to include in the search.

4. To add fields, perform the following tasks:
   - Click **Add Field**.
     The Add Field dialog box appears.

5. Select the fields, enter values to include in your search, and click **OK**.



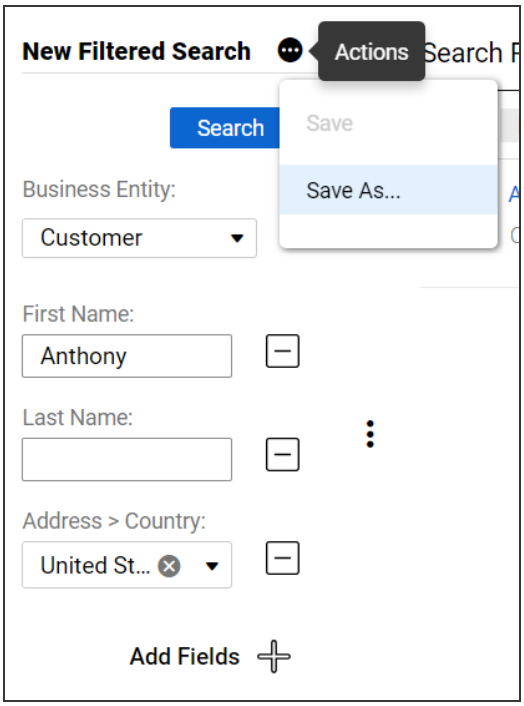
6. Click **Search**.



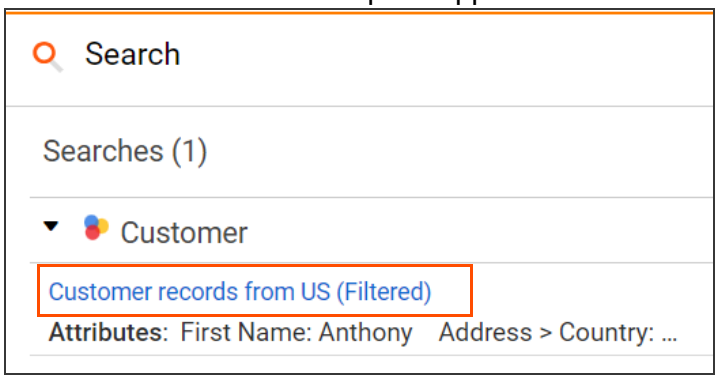
The matching records appear in the Search Results section.

7. To export the search results to a CSV file, select the records you want to export, and click **Selected Export**.

8. To save a search, perform the following tasks:
   - Click Actions **Save As**.



   - In the Save As dialog box, enter a unique name for the search.
   - To save the search as a template, select **Save search as a template**.
   - Click **Save**.
     The saved search or template appears on the Search page.

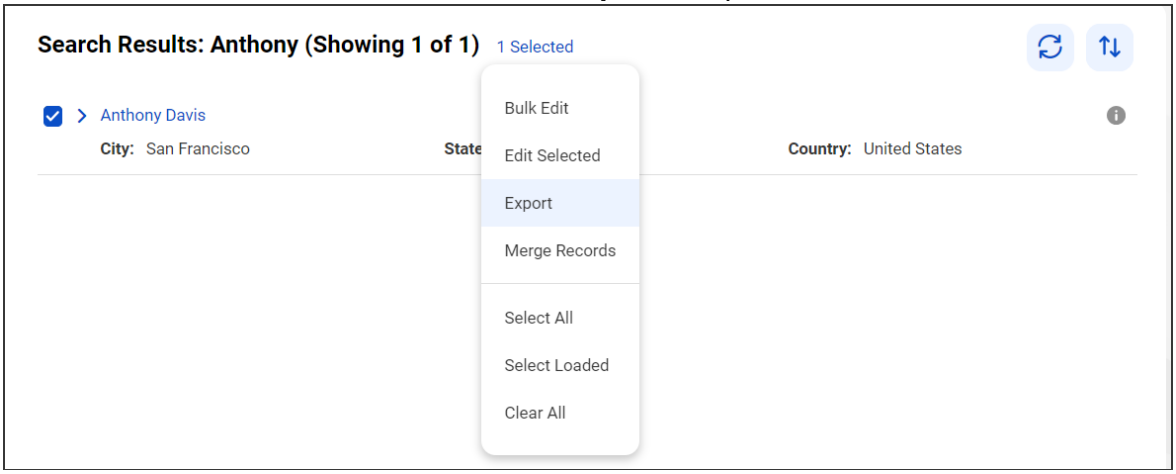


Exporting Search Results

You can export the search results to a CSV file. The exported file contains the values of fields that are set to be searchable and the values of fields that appear as filters.

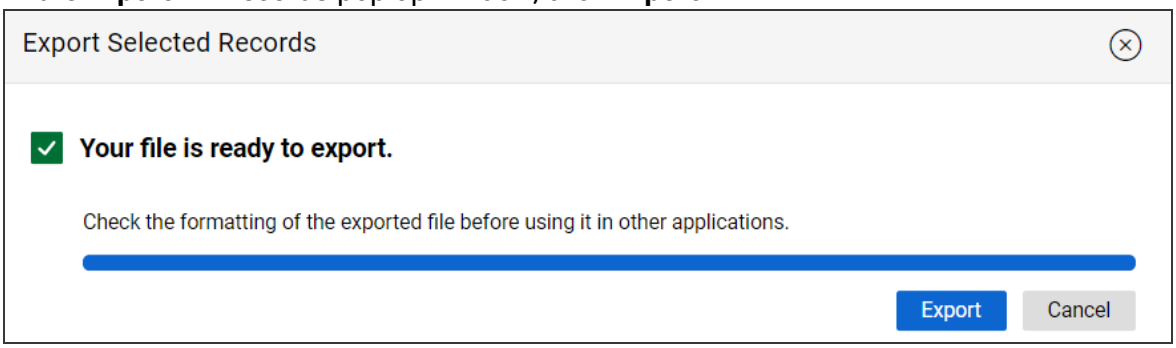
1. Select a business entity and use the search box to search for the records. The Search Results page appears.

2. On the Search Results page, search for the record that you want to export. Select the record and click the **Selected** button**.** Click **Export** to export the record.

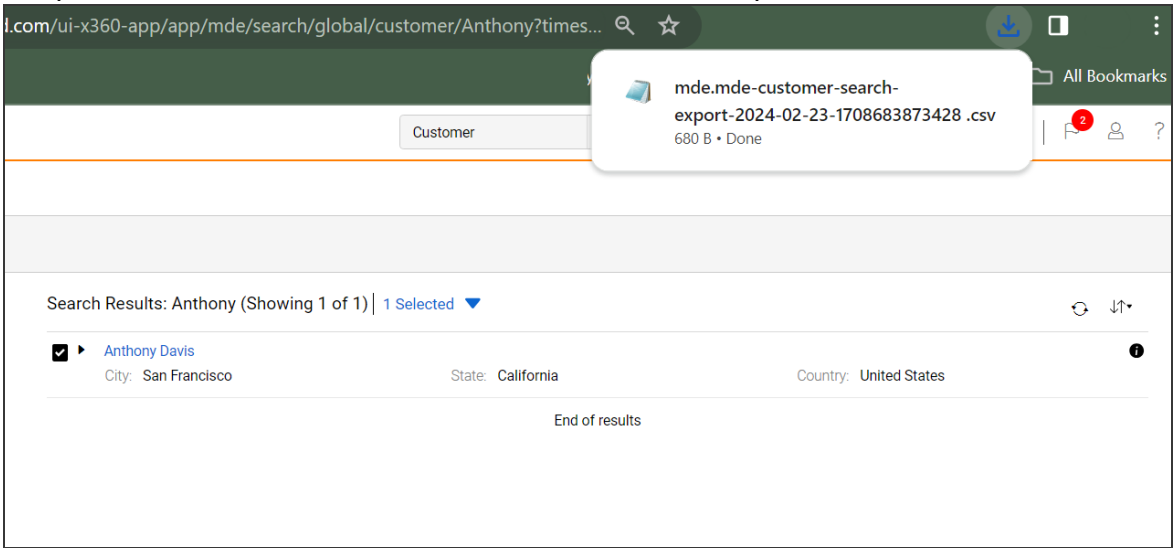


In this case, we have only one record. After onboarding data, we will have many records populated after performing a search.

3. In the **Export All Records** pop-up window, click **Export**.



4. A .csv file will be downloaded. You can see a notification on your screen. Click the .csv file to open and view the contact details. You can choose to open the file in Microsoft Excel.



5. The CSV file opens. View the customer details and close the file.

A	B	C	D	E	F	G	H
_meta.id	First Name	Last Name	Full Name	Gender	Birth Date	Birthplace	Marital St
6491a7b0(	Anthony	Davis	Anthony D	Male		US Resident	

Import Data

You can import data, such as business entity records, relationship records, and hierarchies, from a CSV file into Multidomain MDM SaaS. The CSV file can have a maximum size of 20 MB. The import process parses the data in the CSV file and maps the columns of the CSV file to the fields of the selected business entity or relationship.

You can import parent and child hierarchy records in bulk from a CSV file. **For example**, if you have a hierarchy with a relationship defined from the Customer to the Product business entity, you can import the related records in bulk from a CSV file into the hierarchy.

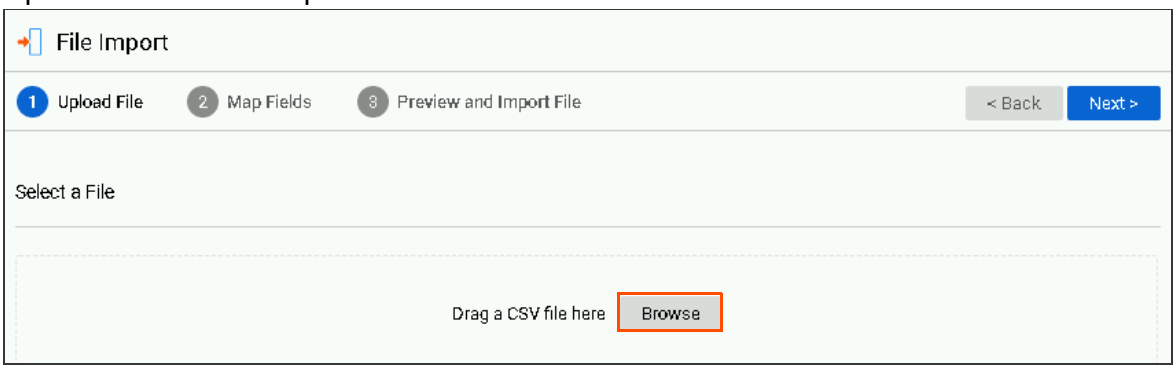
When you import hierarchies, map the source columns with the required target fields. The automatically mapped fields display confidence indicators to indicate the accuracy of the mappings. If fields have low confidence scores or incorrect mappings, you can manually map the source columns with the required target fields.

Before you import hierarchy data, ensure that you have a CSV file with the parent and child records that you want to import.

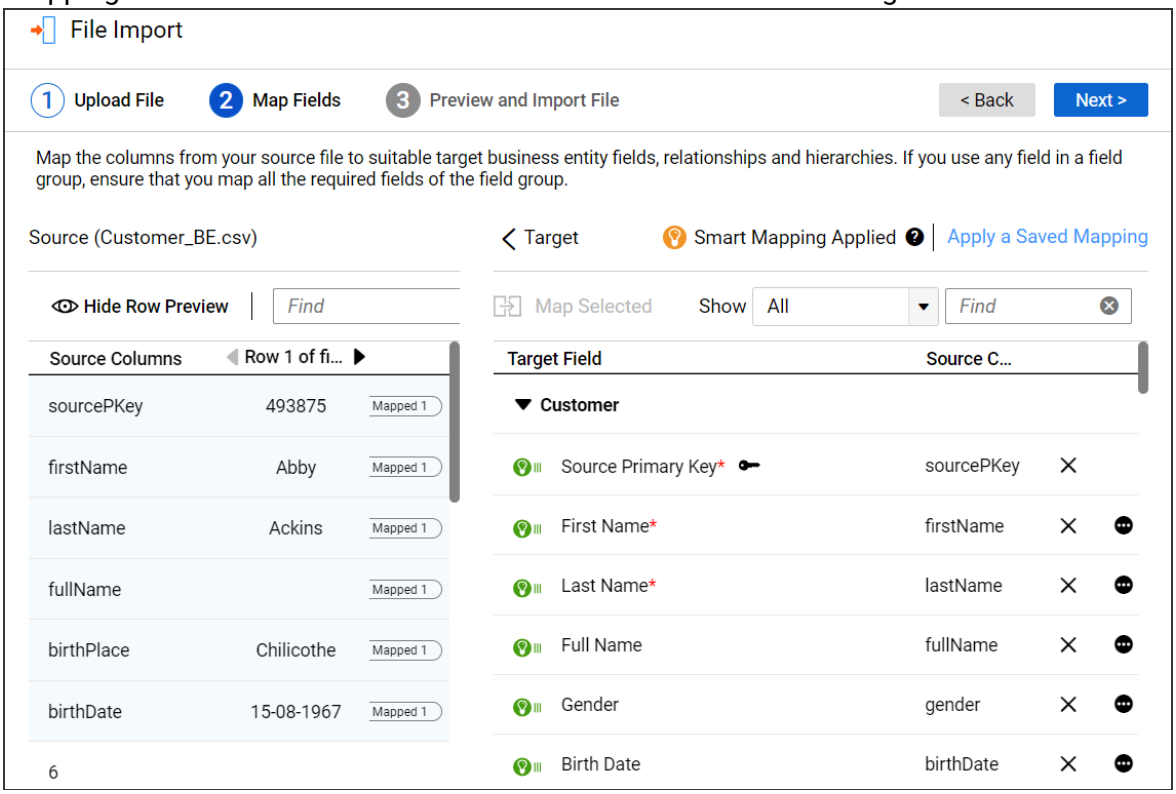
Import Process

Your user role must have permission to import data.
To import data, perform the following tasks:

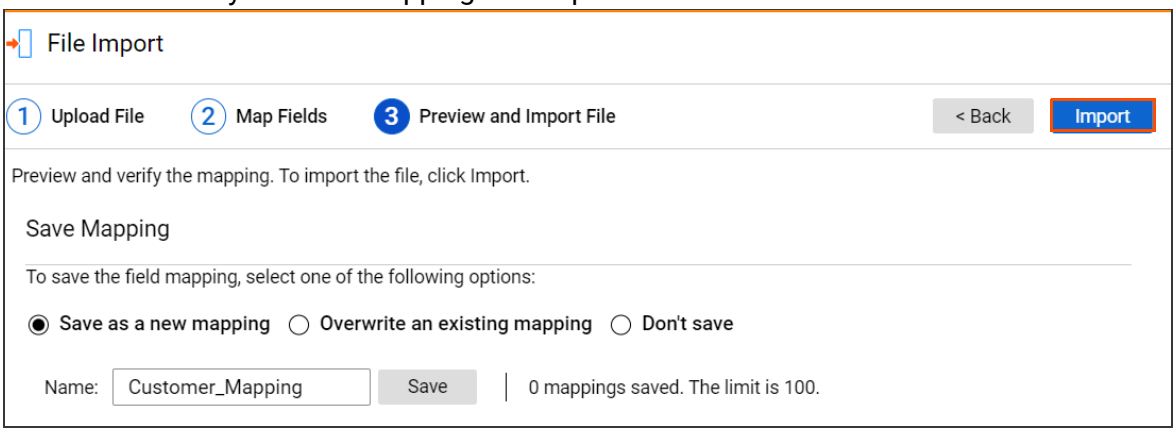
1. Upload a CSV file to import.



2. Verify the automatically mapped fields. Manually map the fields that have incorrect mapping to ensure that the columns of the source file match the target fields.



3. Preview and verify the field mapping and import the file.



Ingress Overview

Ingress is a process that imports data from external source systems, masters the data, and stores it in MDM SaaS assets. When you get started with MDM SaaS for the first time, to onboard your initial data, you must ingress it. You can continue to ingress data incrementally to be in sync with the source systems.

For example, you work for an enterprise that manages customer data in Salesforce. You subscribe to MDM SaaS to master the customer data in Salesforce. To get started with MDM SaaS, ingress the initial data from Salesforce. You can then continue to ingress data incrementally into MDM SaaS so that the data in MDM SaaS is in sync with the Salesforce data.

You can use business IDs, a unique identifier that MDM SaaS uses to identify records, to ingress data. When you use business IDs to ingress data, the records merge directly with the existing records. Otherwise, the records go through the match and merge process.

Pre-requisites for Data Ingress

Before you ingress data, ensure that you have defined the following assets:

- **Source system**: Define the data sources from which you want to ingress data. Example are Flatfiles, Oracle, Salesforce, Finance 360, SQL Server, and so on.
- **Secure Agent:** Specify a Secure Agent that is up and running.
- **Mapping**: Create a mapping with the source connection as the source transformation and the Business 360 connection as the target transformation.
- **Mapping Task**: Create a mapping task by selecting the runtime environment and adding the mapping you created.
- **Taskflow**: Create the required taskflows. Publish the taskflows to use in your reference data import job. In Reference 360, you will import the Taskflow and then start the Ingress job.

Before you import reference data, ensure that you complete the following prerequisites:

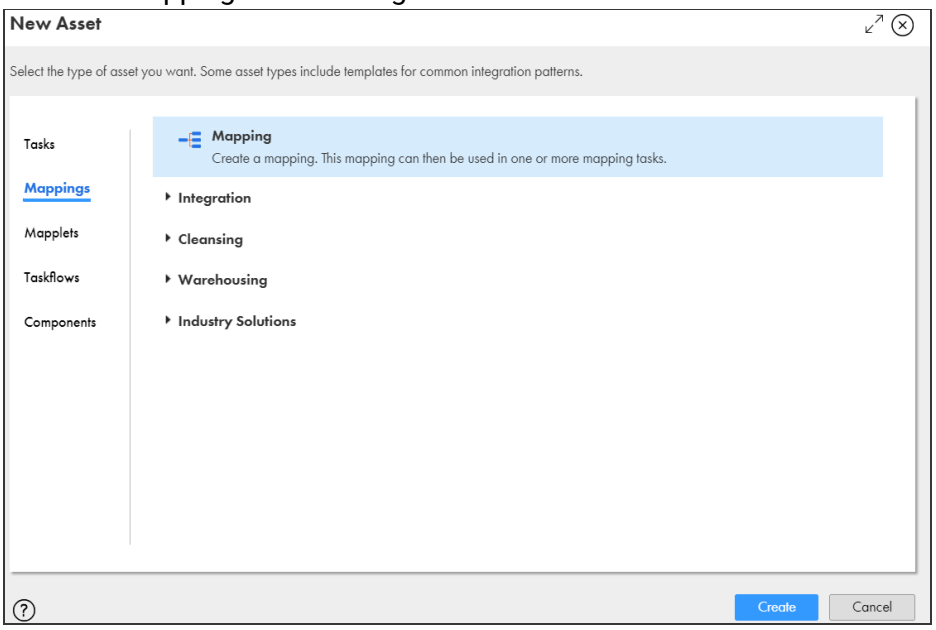
1. In Administrator, create a source connection to read data from source and a Business 360 connection to write to the Reference 360 assets.
2. In Administrator, specify a Secure Agent that is up and running as the runtime environment

to run a mapping task.

3. In Cloud Data Integration, create a mapping with the source connection as the source transformation and the Business 360 connection as the target transformation. Configure the target by selecting the code list from Reference 360 and define field mappings for the source and target fields. Ensure that you create an in-out parameter jobInstanceId in the Parameters panel. The name of the in-out parameter is case sensitive.
4. In Cloud Data Integration, create a mapping task by selecting the runtime environment and adding the mapping that you create.
5. In Cloud Data Integration, create the required taskflows. Publish the taskflows to use in your reference data import job. For more information about creating taskflows, see Taskflows in the Data Integration help.

Following steps summarizes the Ingress Process:

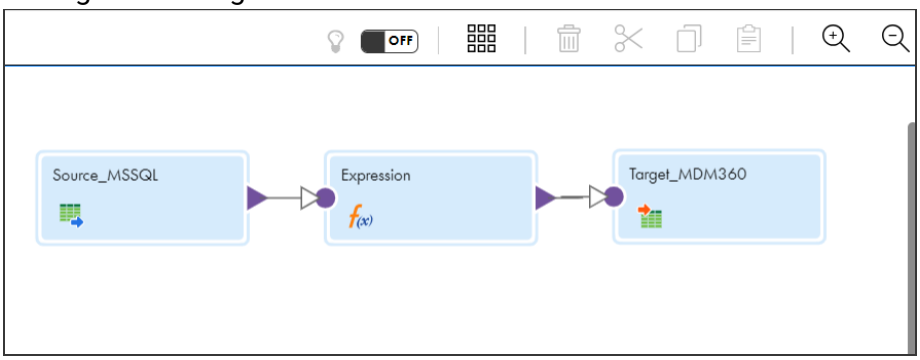
1. Create a Mapping in Data Integration.



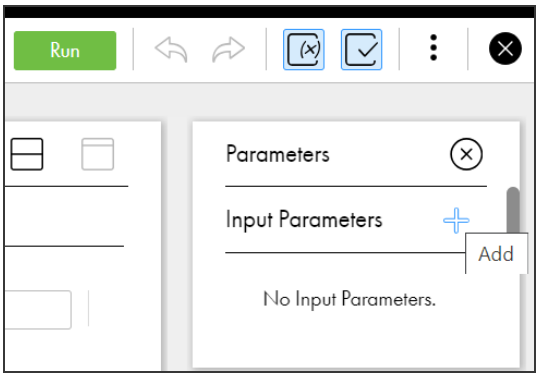
2. Add an Expression Transformation.

3. Configure the Source Connection.

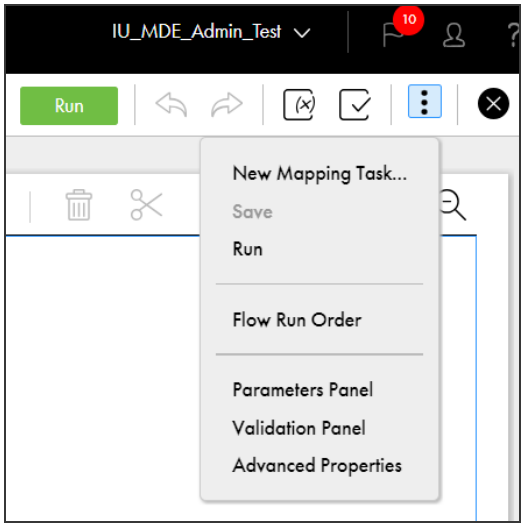
4. Configure the Target Connection.



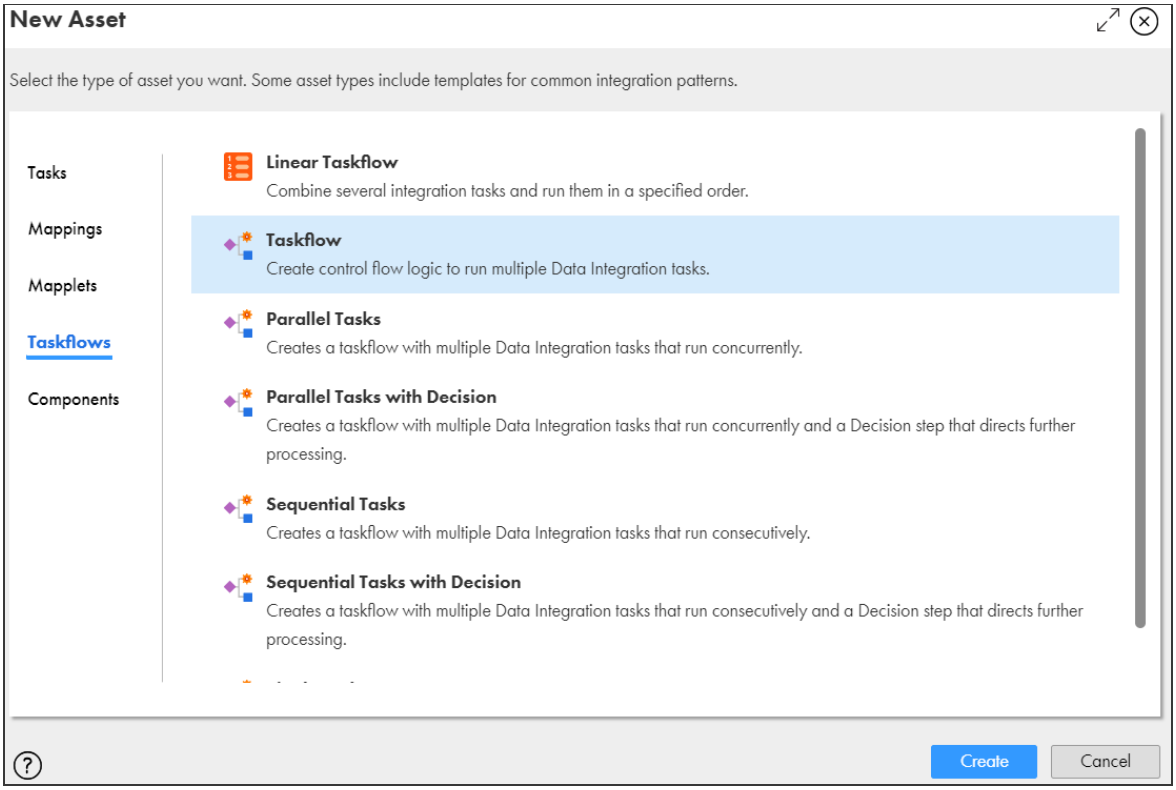
5. Create In-Out Parameter.



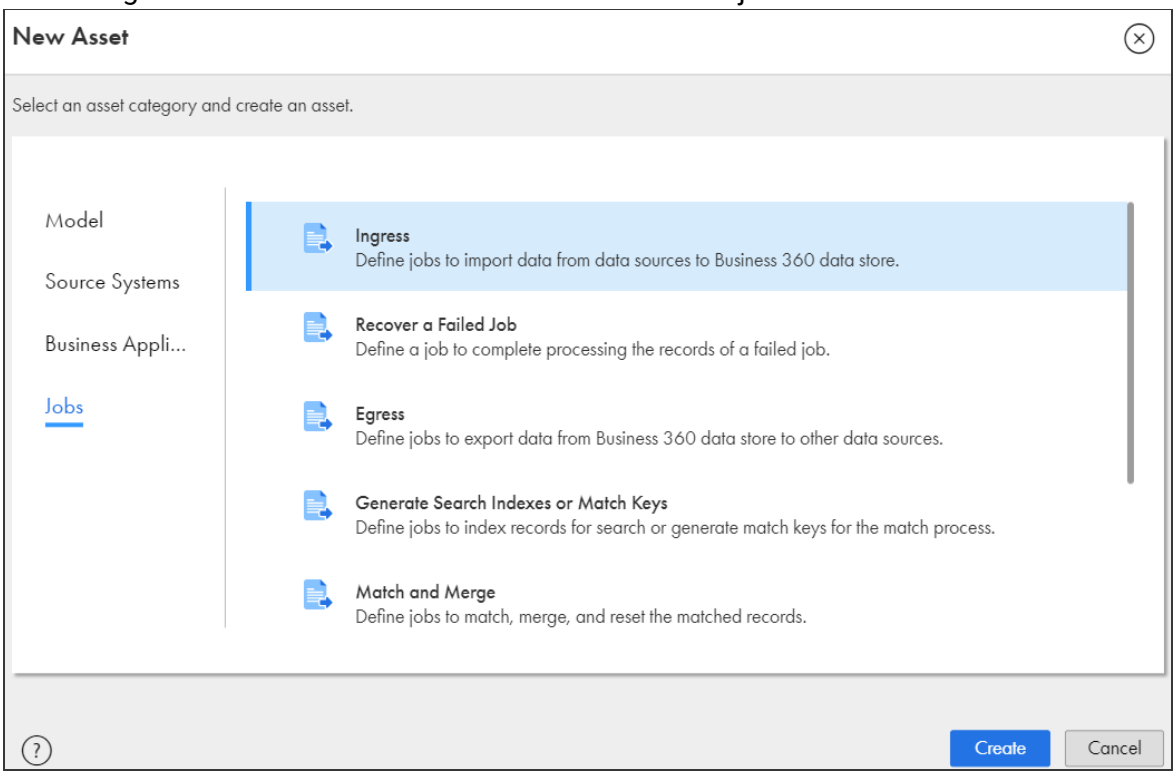
6. Create a Mapping Task.



7. Create a Taskflow.



8. Create Ingress Job in Business 360 Console and run the job.



Egress Overview

You can export data from Business 360 data store to an external data source. For example, your organization uses Multidomain MDM to master customer data. To analyze the purchase behavior of the customers, you want to load the mastered customer data to Tableau. You can export data from Multidomain MDM to Tableau.

Before you Egress data, ensure that you have defined the following assets:

- **Source system**: Create a Business 360 connection to read data from Reference 360 and a target connection to write the data to an external data source
- **Secure Agent** : Specify a Secure Agent that is up and running as the runtime environment to run a mapping task.
- **Mapping** : Create a mapping with the source connection as the Business 360 connection and flat file as the target transformation.
- **Mapping Task**: Create a mapping task by selecting the runtime environment and adding the mapping you created.
- **Taskflow:** Create the required taskflows. Publish the taskflows to use in your reference data export job.

Export Job Types

- **Standard export**: Exports all the records in the first run of the job and then exports the records added or updated after the first run incrementally based on the job schedule.
- **Custom export**: Exports all the records added or modified after the specified date in the first run of the job, and then exports the records added or updated after the first run incrementally.
- **Export all**: Exports all the records in each run of the job.

Supported Assets

You can egress data to the following types of assets:

- Predefined and custom business entities
- Predefined and custom relationships
- Hierarchies
- Code lists of reference data
- The process of creating an egress job is similar to that of an ingress job.

Lab 9-1: Adding Records

Overview:

You can add business entity records to the Multidomain MDM application. Use the “Customer” business entity to add a customer’s data such as personal information and contact details. Multidomain MDM validates some of the fields before saving a record to the database.

In this lab, you will create business entity records, and perform a search of those records.

Objectives:

- Create a new Customer record
- Create a new Manufacturer record
- Create a new Site record
- Create a new Product record
- Perform search

Scenario:

The data model for HomeKart is now complete, and it's time to begin onboarding data into the MDM system. Mark discovered various methods to onboard data, including manual record entry, CSV import, and ingress. He has assigned different data onboarding tasks to each team member, and together, they will collaborate to populate the system with data. In this lab, you will manually add the data into the Multidomain MDM application.

Duration:

40 minutes

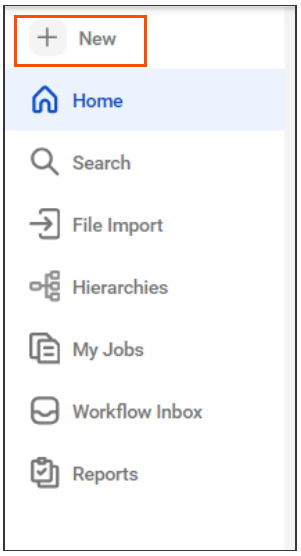
Tasks:

- Create a new Customer record
- Create a new Site record
- Create Manufacturer record
- Create Product record
- Search for records
- Export search result

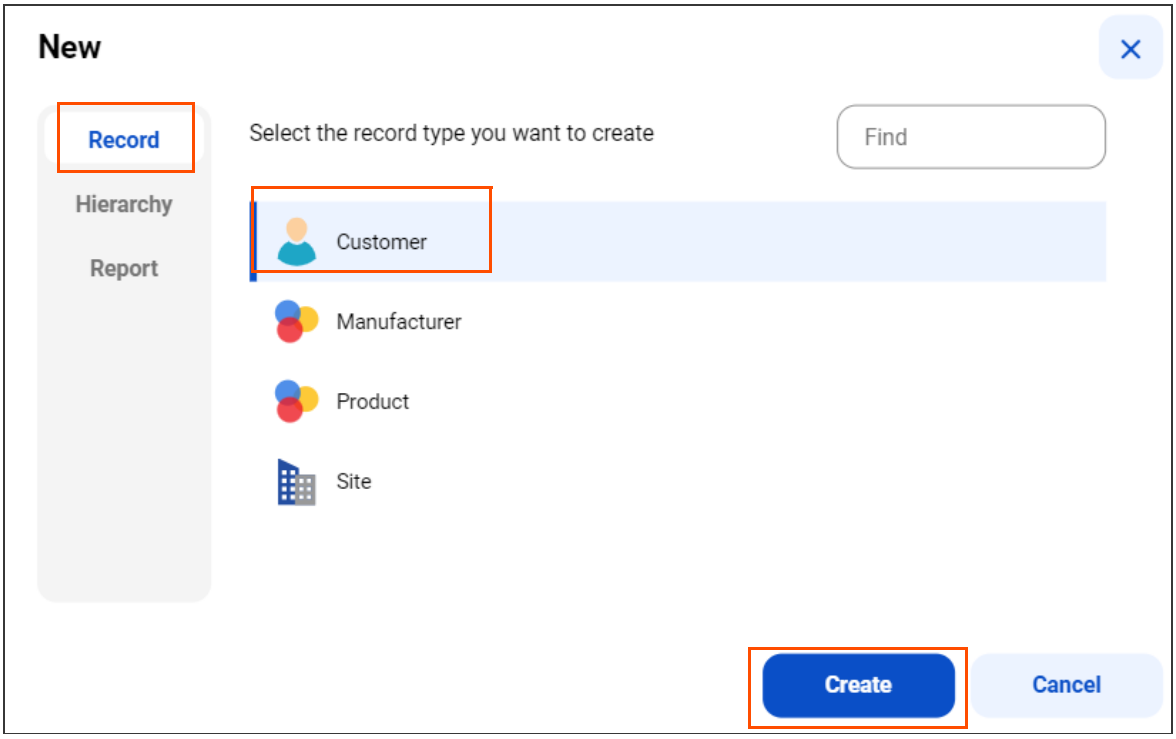
Create a New Customer Record

You will now create a Customer record and enter the required information for the record.

1. Log in to the IICS using your admin credentials. For example, Jimstudent01.
2. Open **Multidomain MDM** application.
3. In the Multidomain MDM user interface, click **New**.

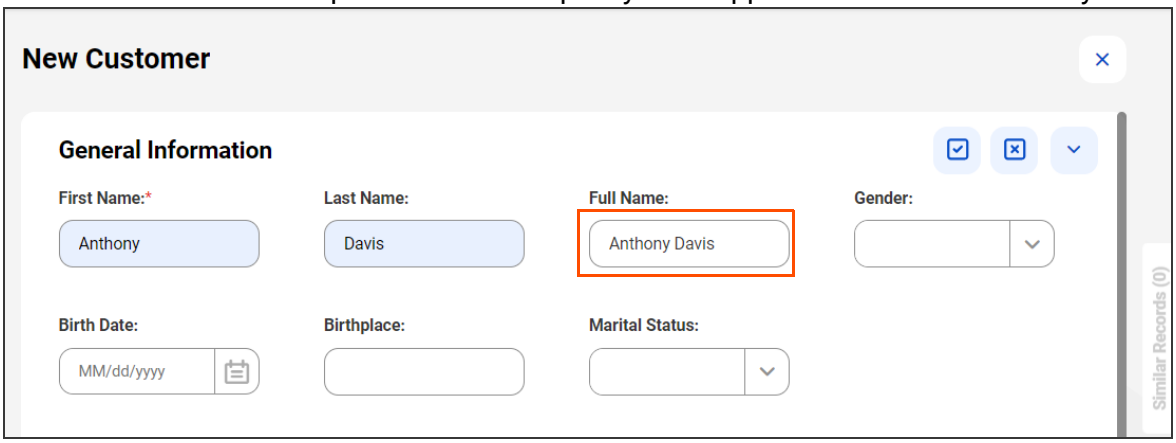


4. To create a new customer record, select **Record** > **Customer** and click **Create**.



5. Enter the First Name as **Anthony**, and the Last Name as **Davis**.

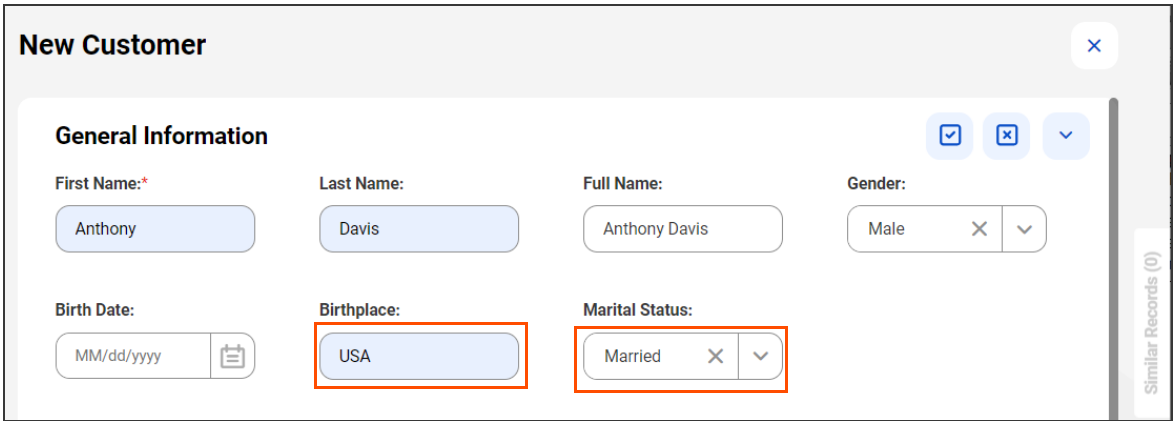
6. Observe that the **Full Name** field is auto populated. It concatenates the First Name and the Last Name based on the pre-defined data quality rules applied to the business entity.



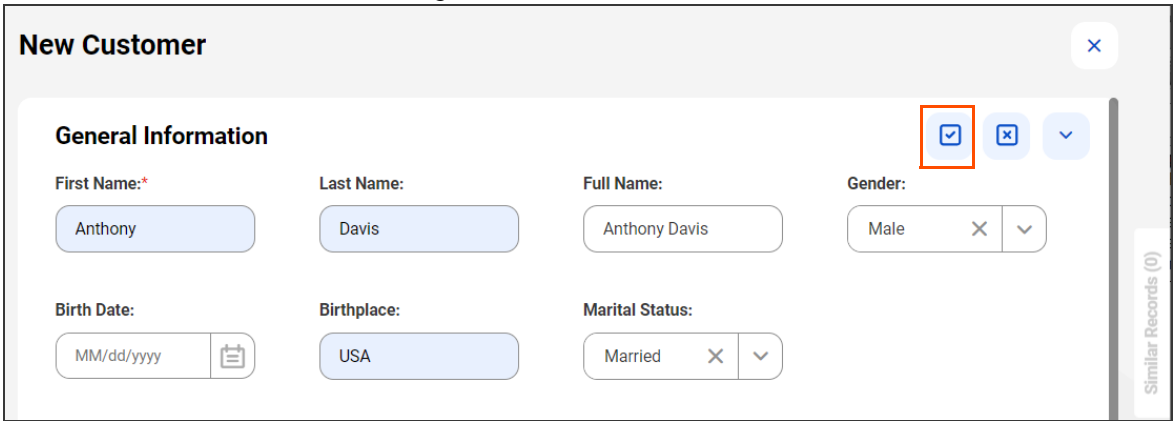
The concatenation happens if you click anywhere on the page.

7. Enter the Gender as **Male** and Birthplace as **USA**.

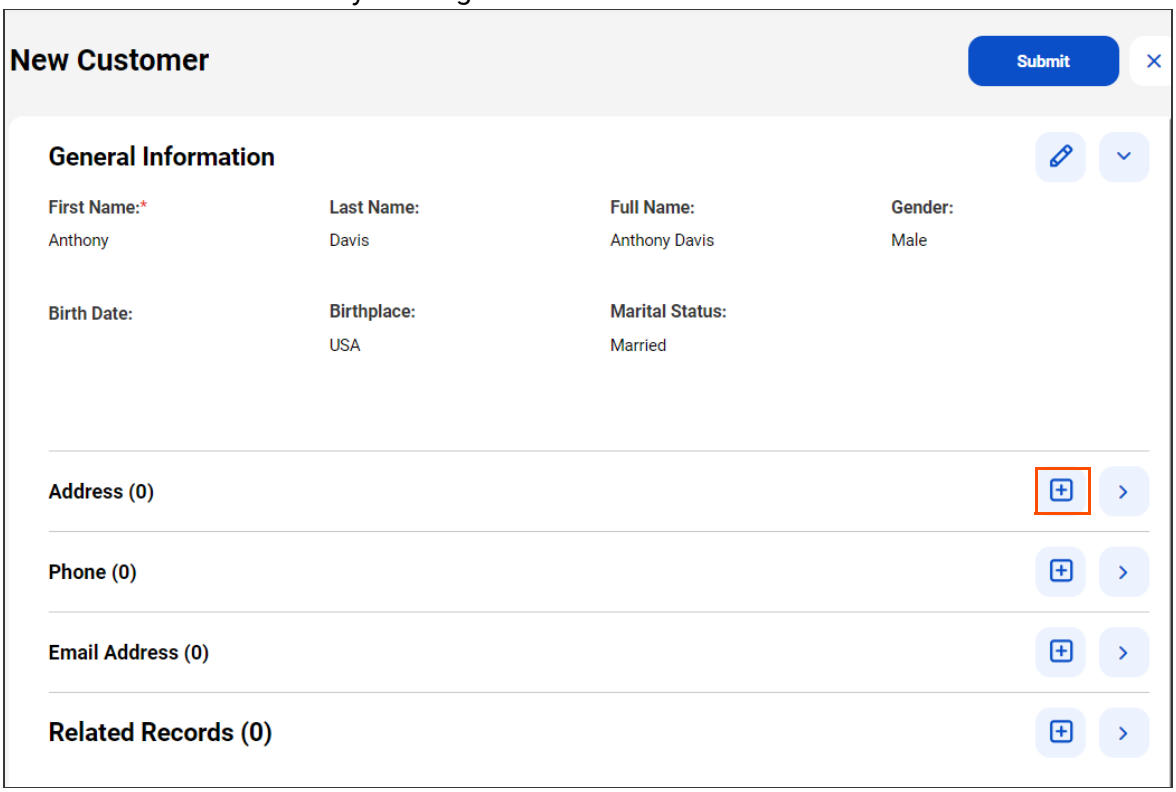
8. You can choose to enter marital status and birth date field.



9. Click the check mark to save the general information of the record.

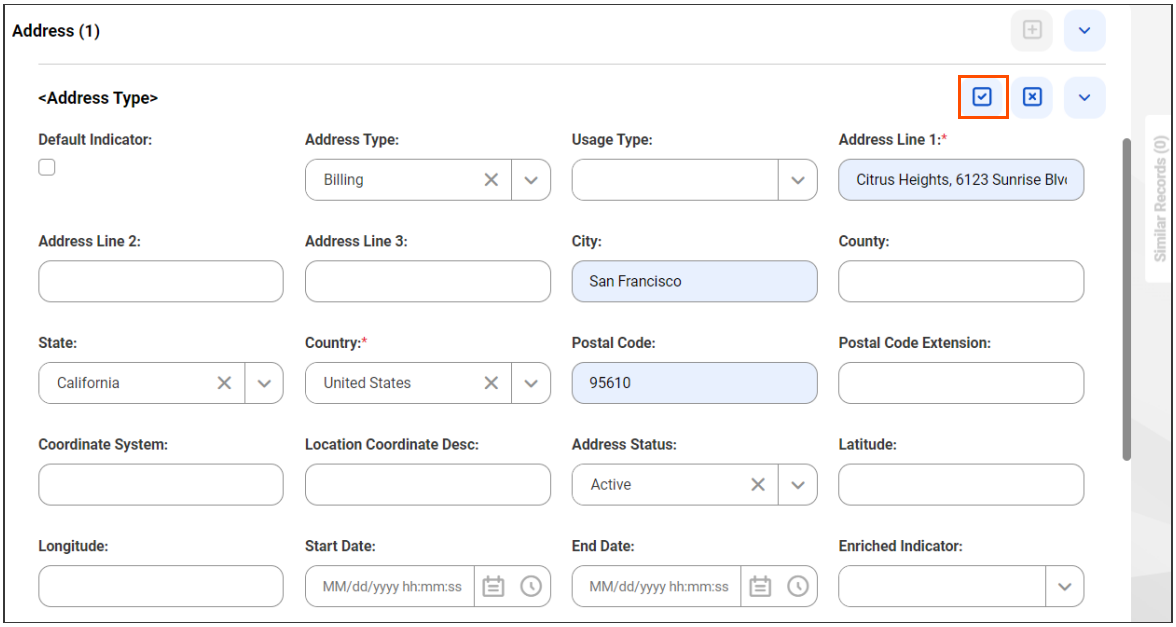


10. Add address information by clicking the **+** icon next to **Address**.

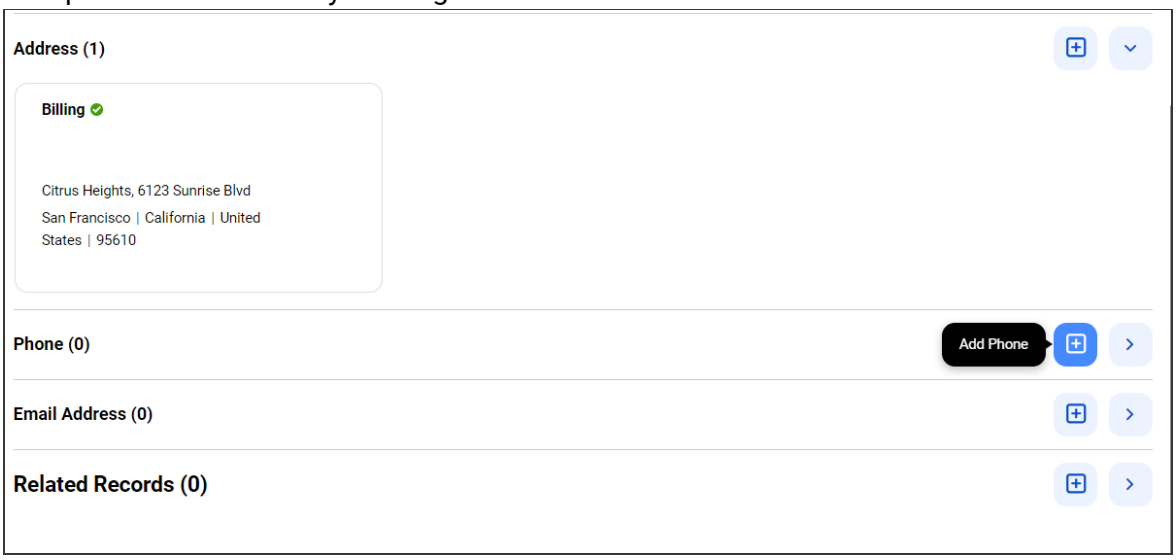


11. Enter the following details and click the tick icon to save the address details:

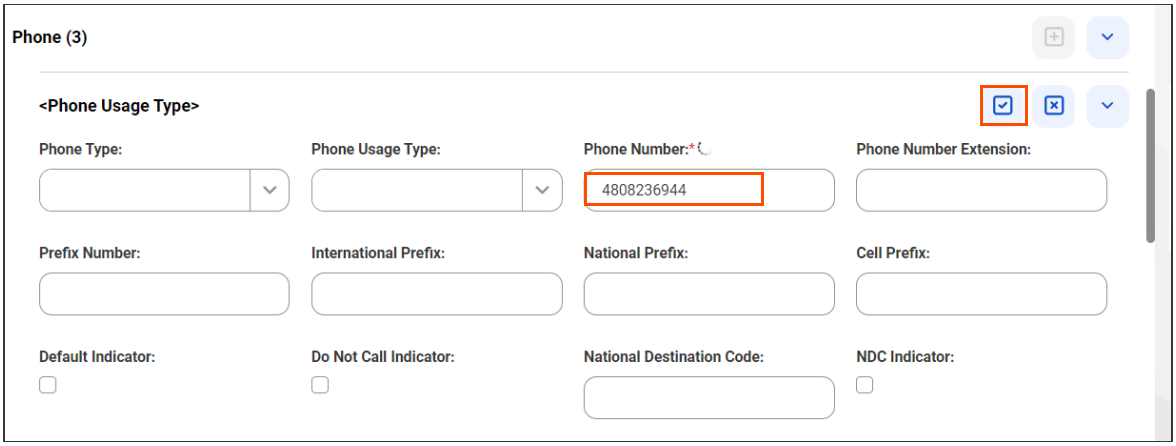
Field	Value
Default Indicator	<select check box>
Address Type	Billing
Address Line 1	Citrus Heights, 6123 Sunrise Blvd
City	San Francisco
Country	United States
State	California
Postal Code	95610
Address Status	Active



12. Add phone information by clicking the **+** icon next to **Phone**.



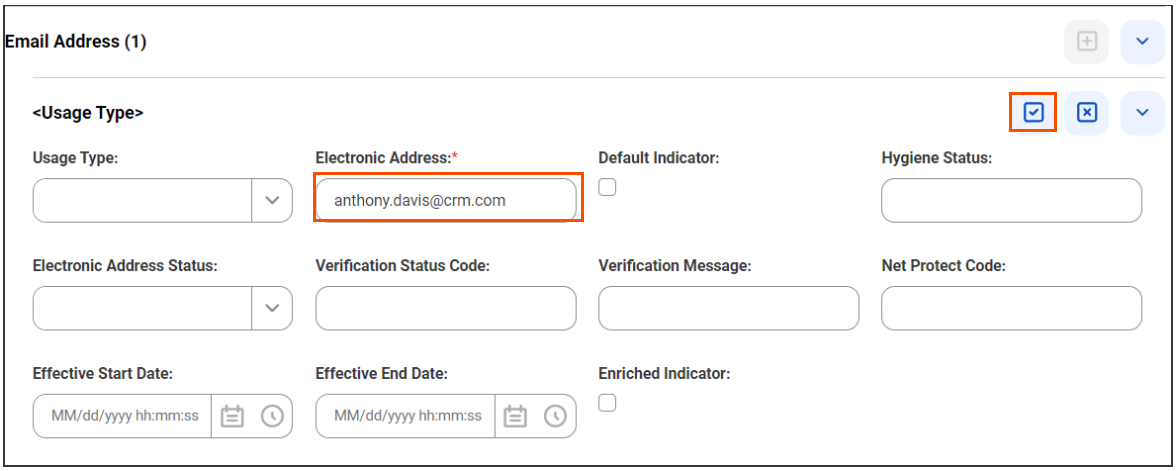
13. Select the Phone Type as **Cell** and enter the Phone Number as **4808236944**. Click the check mark to save the phone number details.



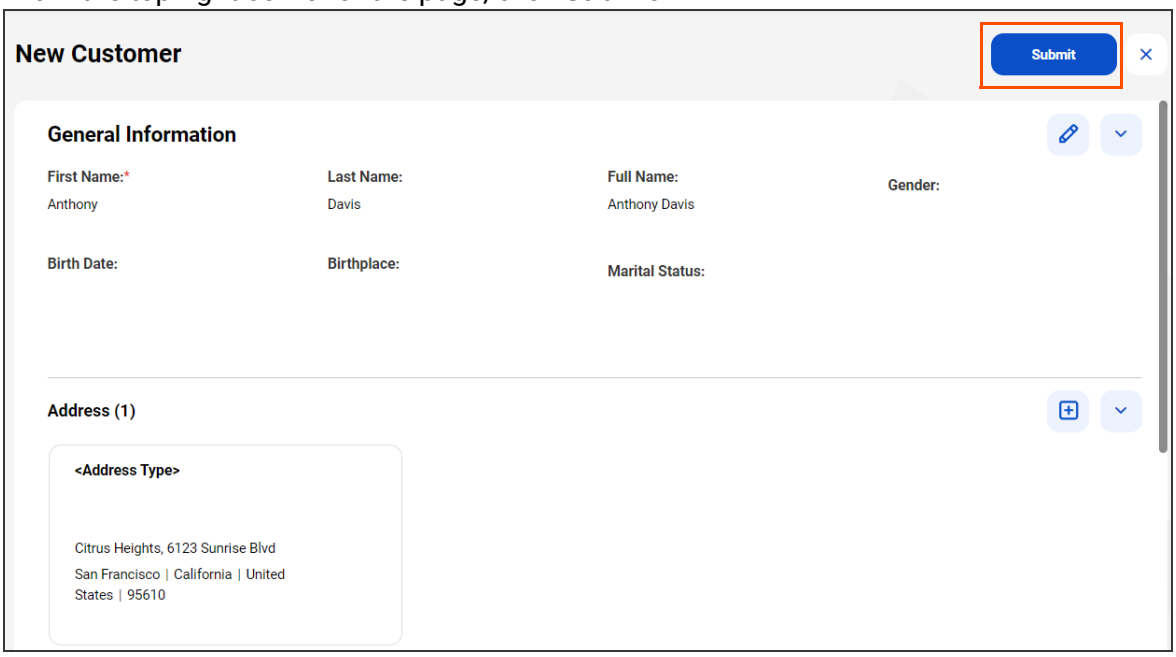
14. Add email information by clicking the **+** icon next to the Email.

15. Enter **anthony.davis@crm.com** for **Electronic Address**. Click the check mark to save the

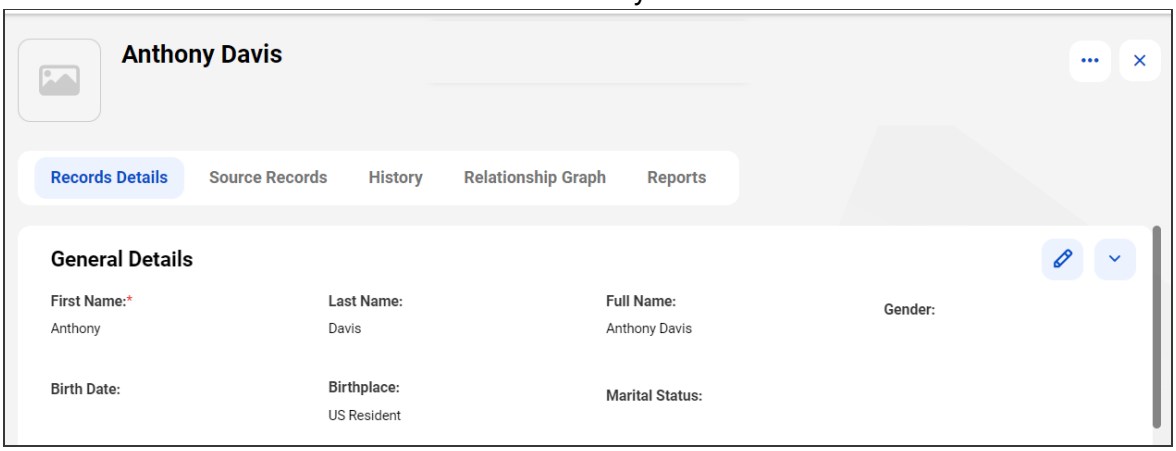
email address.



16. From the top right corner of the page, click **Submit**.



The Customer record is now created successfully.



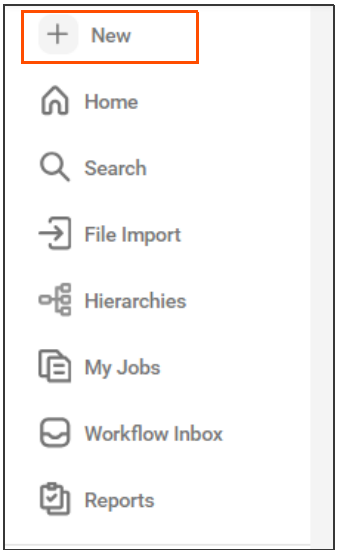
Notice the Birthplace has been replaced with US Resident, as you have defined a data quality rule on the Birthplace field.

17. Click **X** to close.

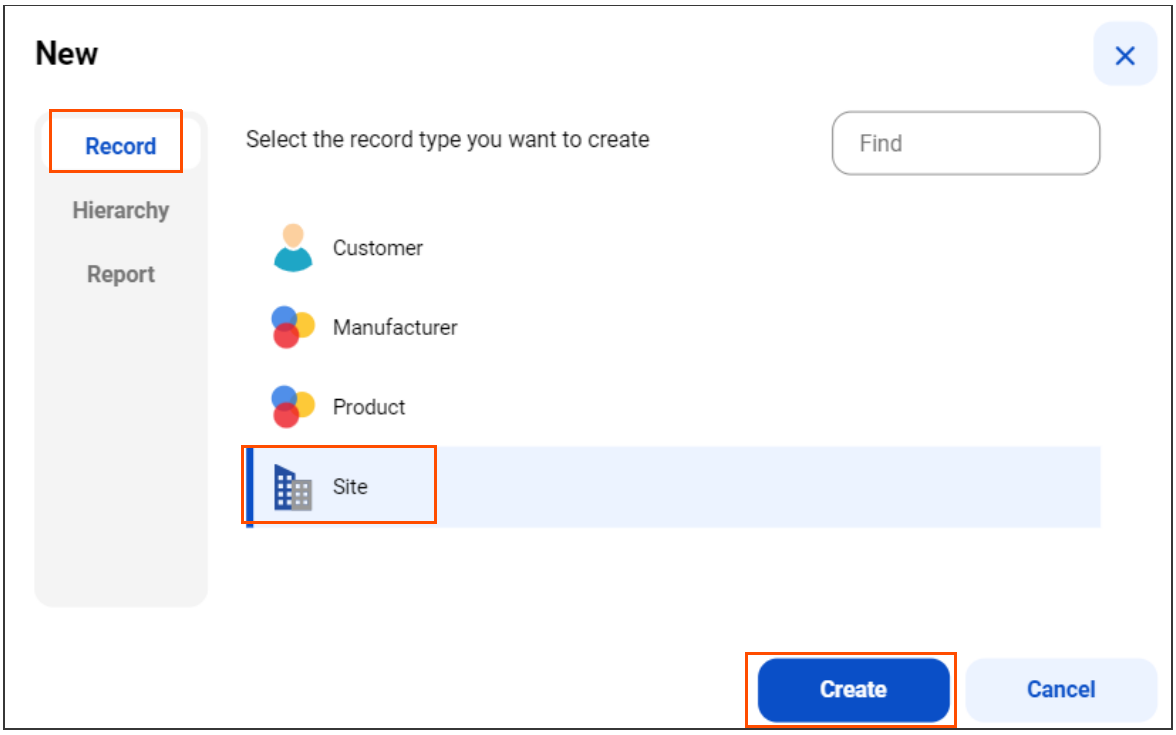
Create a New Site Record

The next step is to use the **Site** business entity to create a site record.

18. From the left side menu, click **New**.

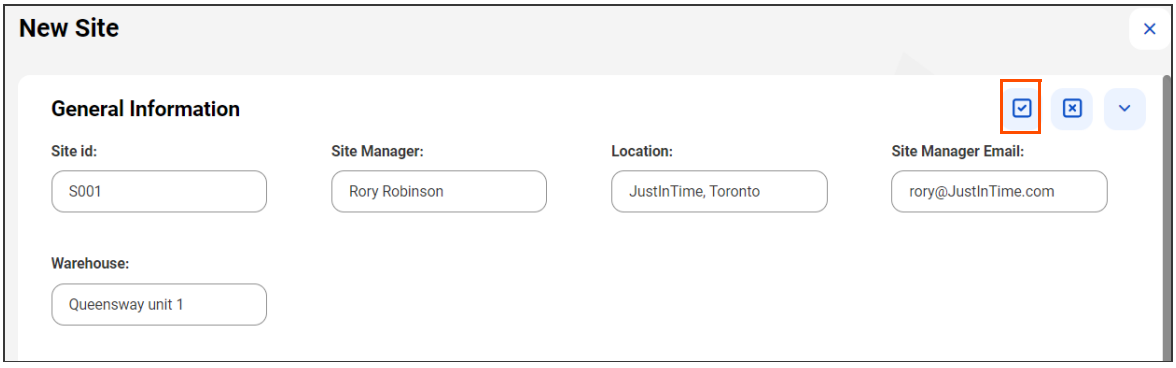


19. Select **Record** > **Site** and click **Create**.



20. On the **New Site** page, enter the following details and click tick icon to save the details.

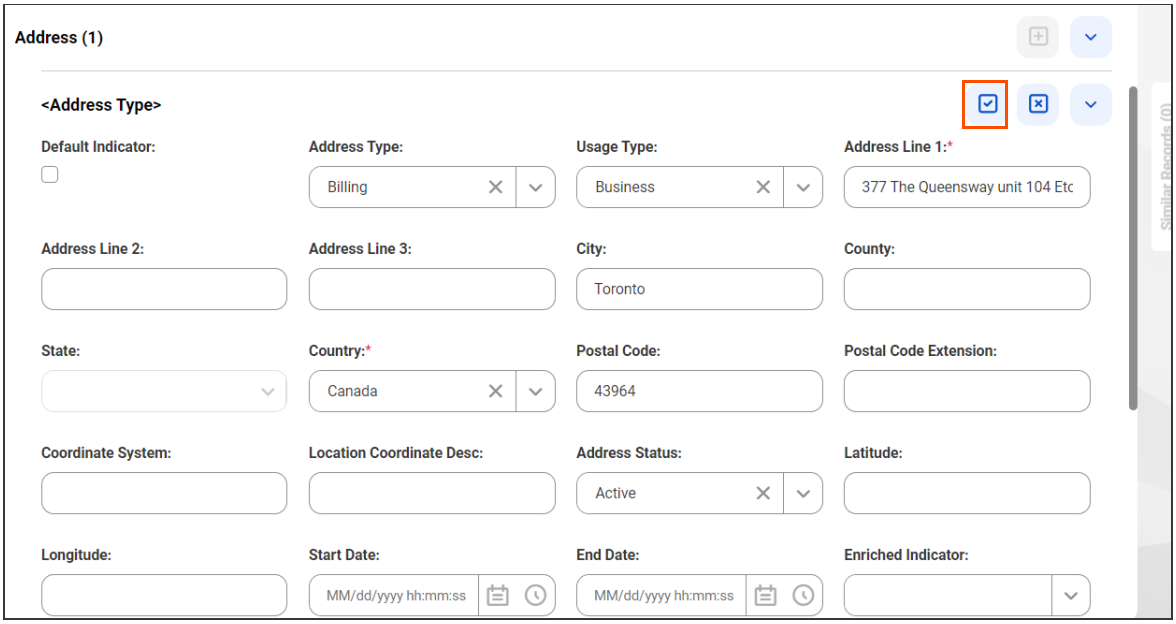
Field	Value
Site id	S001
Site Manager	Rory Robinson
Location	JustInTime, Toronto
Site Manager Email	rory@JustInTime.com
Warehouse	Queensway unit 1



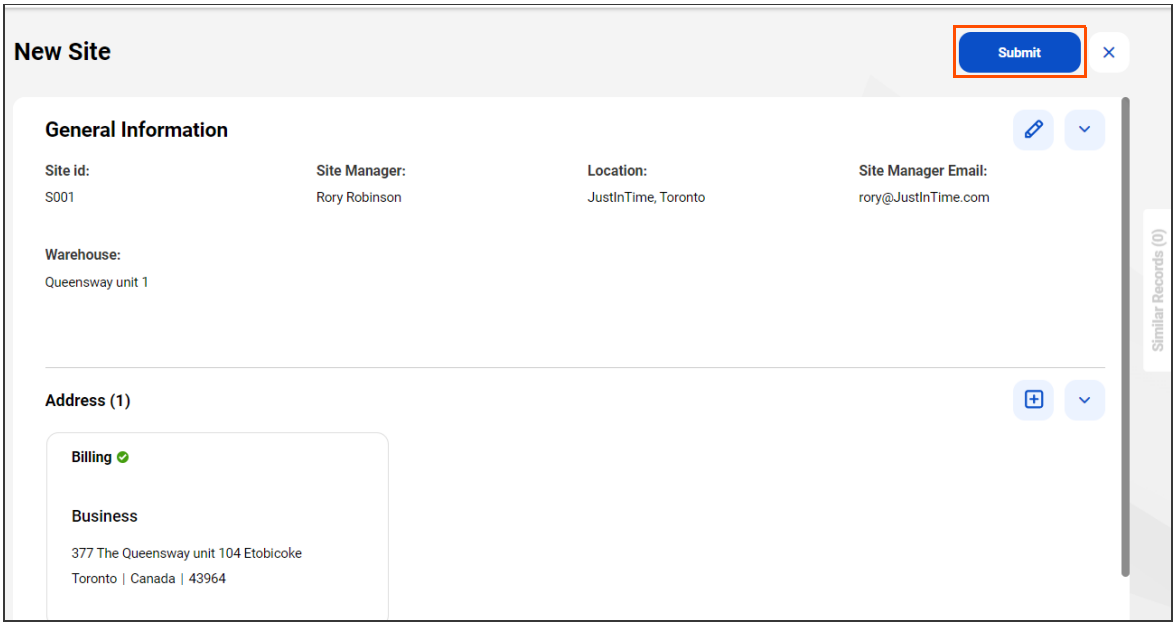
The fields in the screenshot may have different order than you see on your page.

21. Click the **+** icon for **Address** and enter the following details and click tick icon to save your changes.

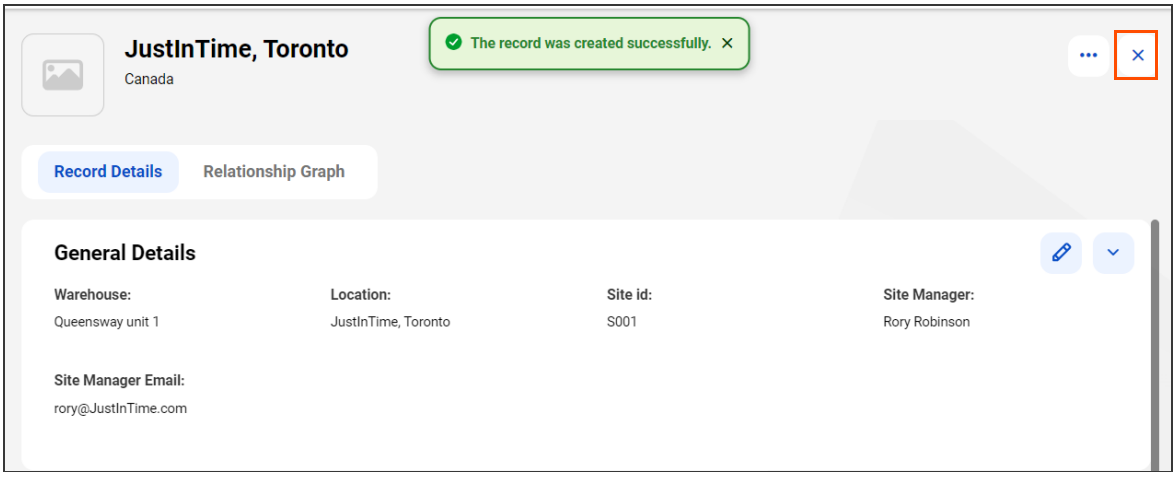
Field	Value
Default Indicator	<select check box>
Address Type	Billing
Usage Type	Business
Address Status	Active
Address Line1	377 The Queensway unit 104 Etobicoke
City	Toronto
Country	Canada
Postal Code	43964



22. Click **Submit**.



23. The record is created. **Close** the record.



Self-Try Exercise (Mandatory)

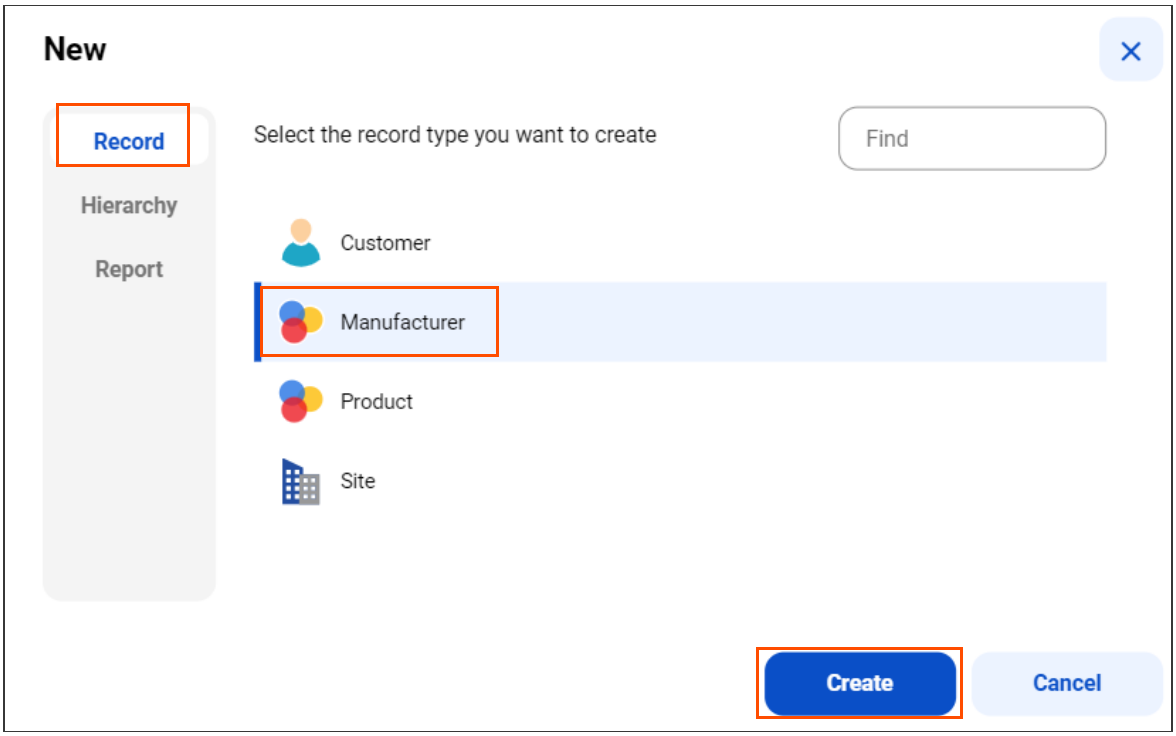
Create Manufacturer Record

The next step is to use the Manufacturer business entity to create a manufacturer record.

High Level Steps:

24. In the Multidomain MDM application page, click **New**.

25. Select **Record** > **Manufacturer** and click **Create**.



26. On the **New Manufacturer** page, enter the following details and click tick icon to save the details.

Field	Value
Name	Whirlpool
SIC Code	1090
Manufacturer Id	M001
Date of Incorporation	02/07/1990

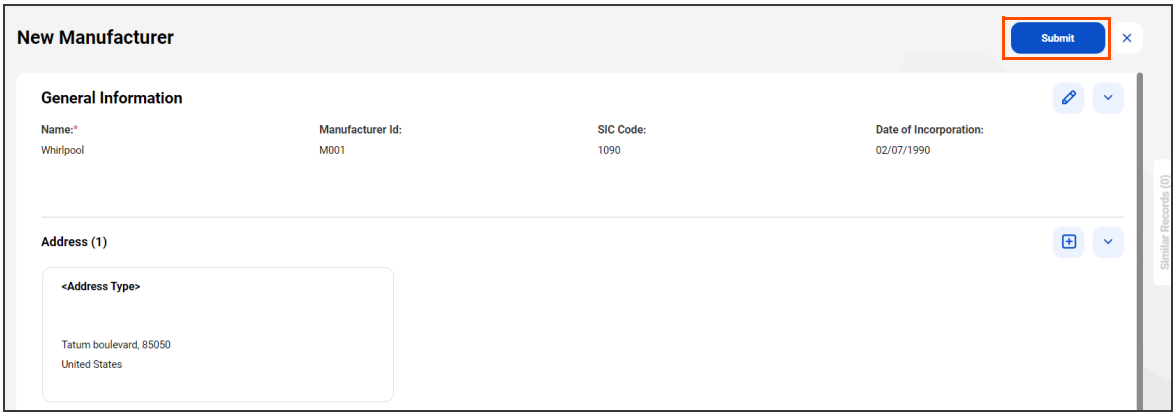
27. Click the **+** icon for **Email Address** and in the Electronic Address field, enter **w1@whirlpool.com**. Click the tick icon to save your changes.

28. Click the **+** icon for **Address** and enter the following details and click tick icon to save your changes.

Field	Value
Address Line 1	Tatum boulevard, 85050
Country	United States

You can choose to enter other fields like Phone Number and Related Records.

29. Click **Submit**.



30. The record is created. Close the record.

Self-Try Exercise (Mandatory)

Create Product Record

The next step is to use the Product business entity to create a product record.

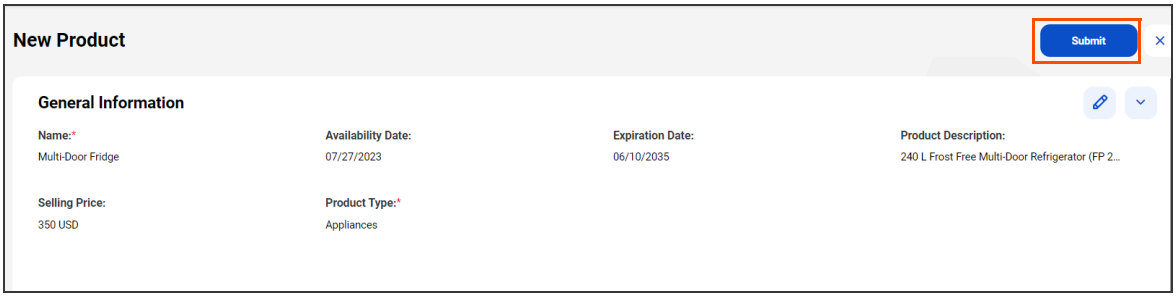
High Level Steps:

- In the Multidomain MDM application page, click **New**.
- Select **Record** > **Product** and click **Create**.

- On the **New Product** page, enter the following details and click tick icon to save the details.

Field	Value
Name	Multi-Door Fridge
Availability date	07/27/2023
Expiration date	06/10/2035
Product Description	240 L Frost Free Multi-Door Refrigerator (FP 263D Proton Roy, Alpha Steel)
Selling Price	350 USD
Product Type	Appliances

- Click **Submit**.

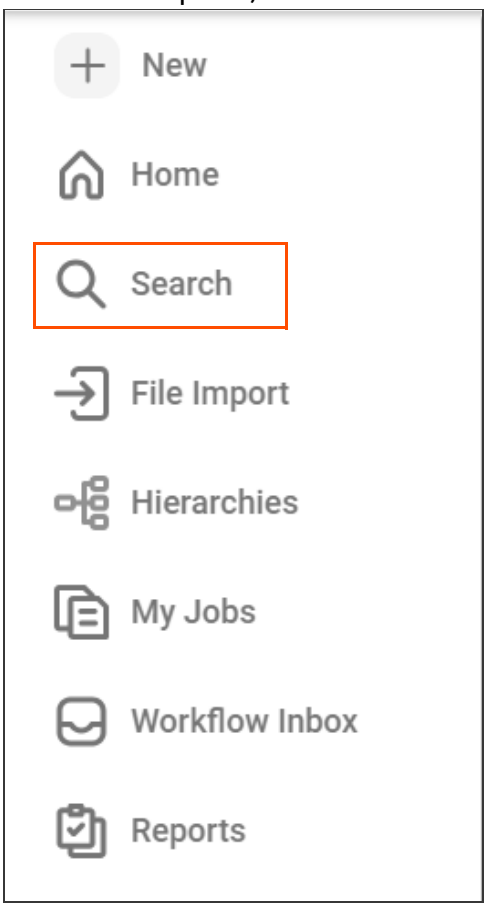


- The record is created. Close the record.

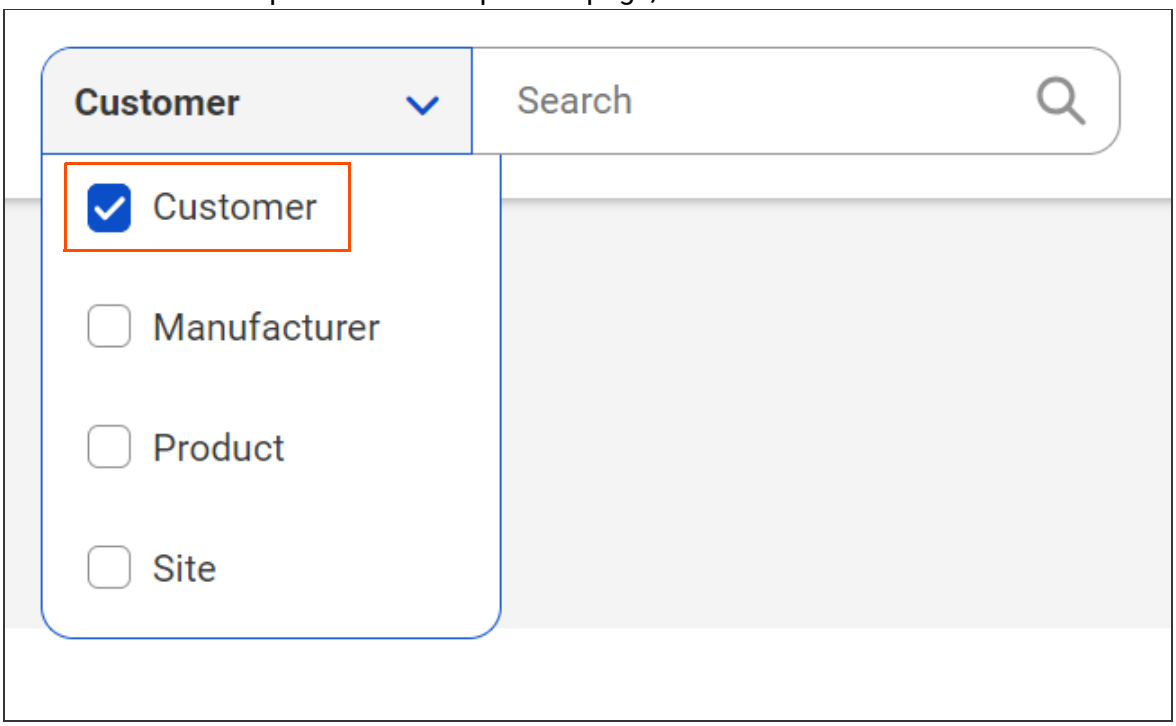
Search for Records

You will now search for the records that you have created.

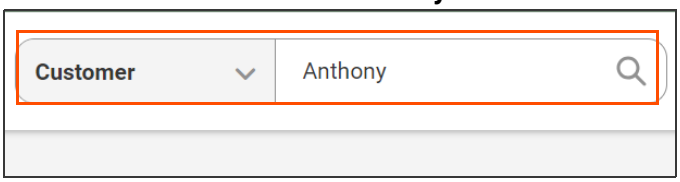
31. From the left panel, click **Search**.



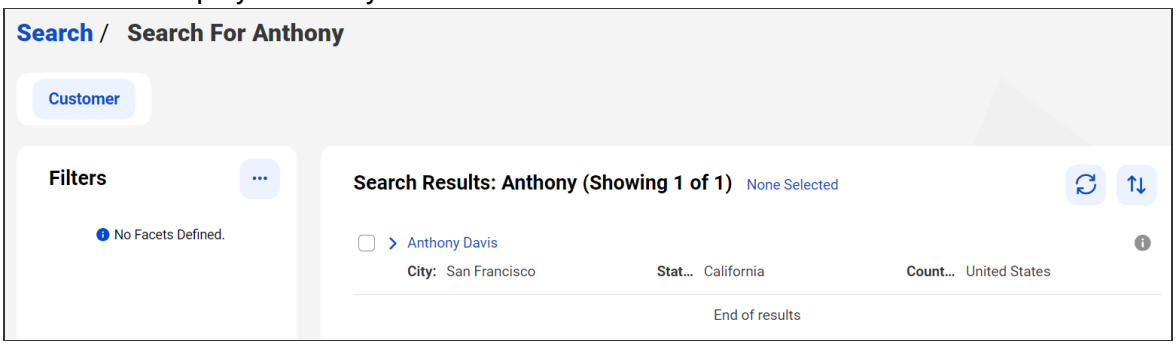
32. From the Search drop-down at the top of the page, and select **Customer**.



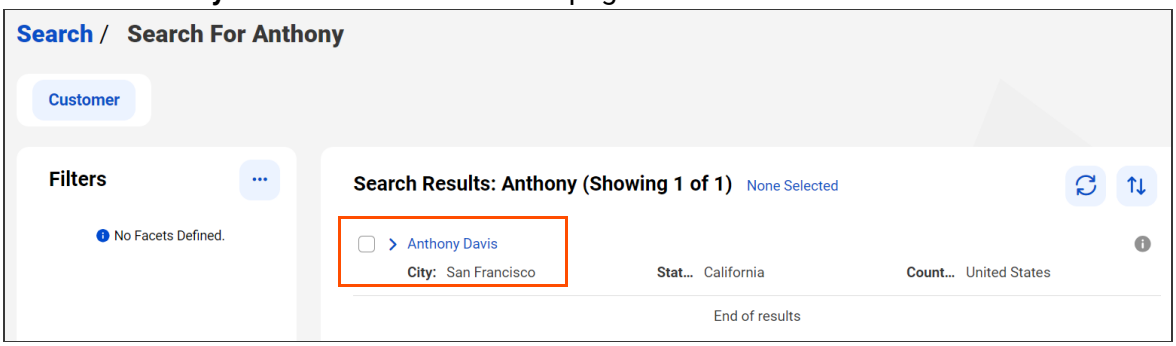
33. Enter the search name as **Anthony** and click the **Search** icon.



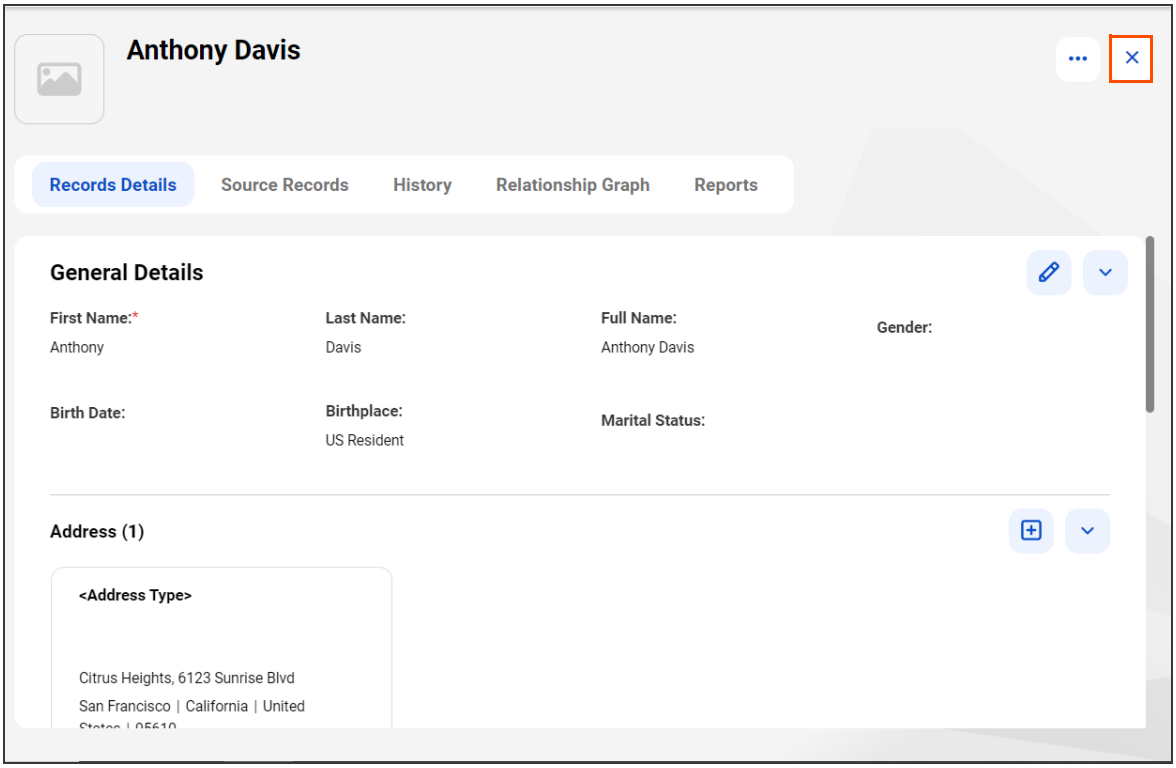
The search displays Anthony's information.



34. Click on **Anthony Davis** on the main search page.



35. You can view all the details of the customer. **Close** the record.



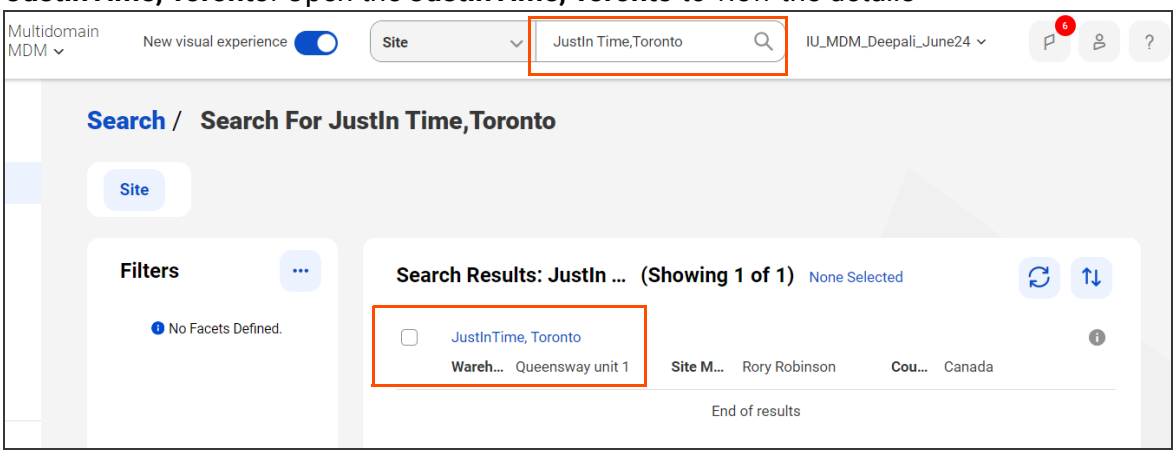
Self Try Exercise (Optional)

Search for the Site Record

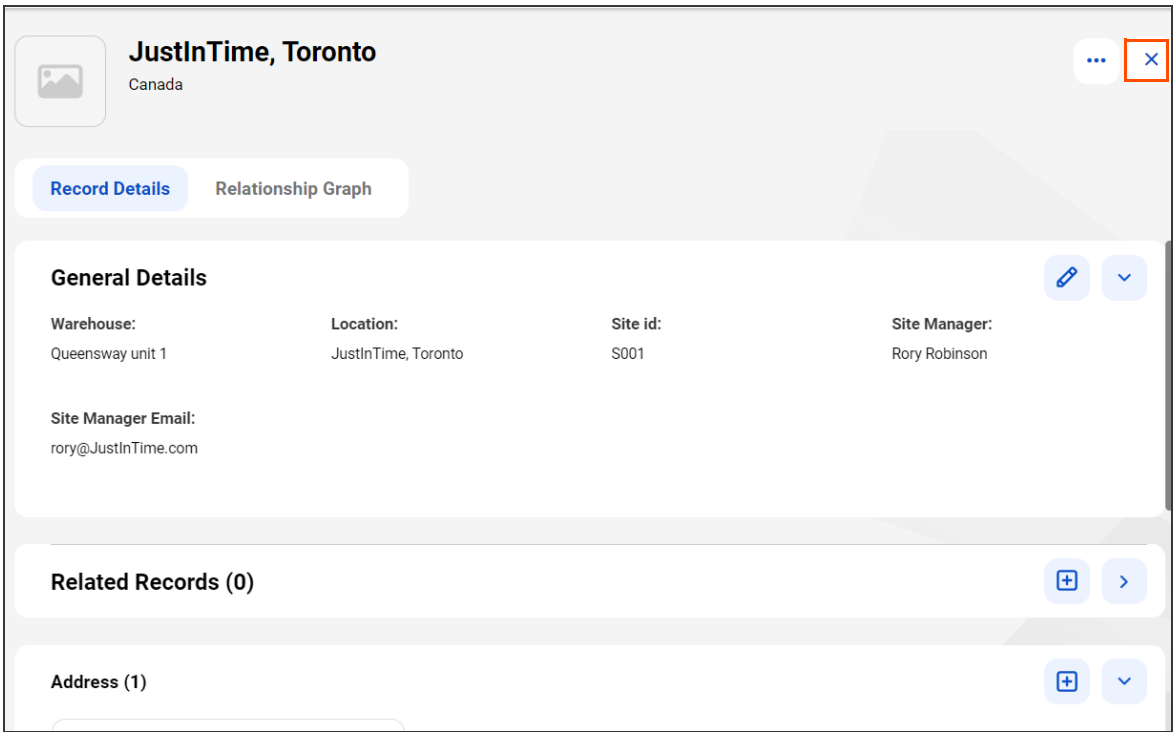
Search for the Site record.

36. From the left panel, click **Search**.

37. From the search drop-down at the top of the page, select **Site** and enter the string as **JustInTime, Toronto**. Open the **JustInTime, Toronto** to view the details



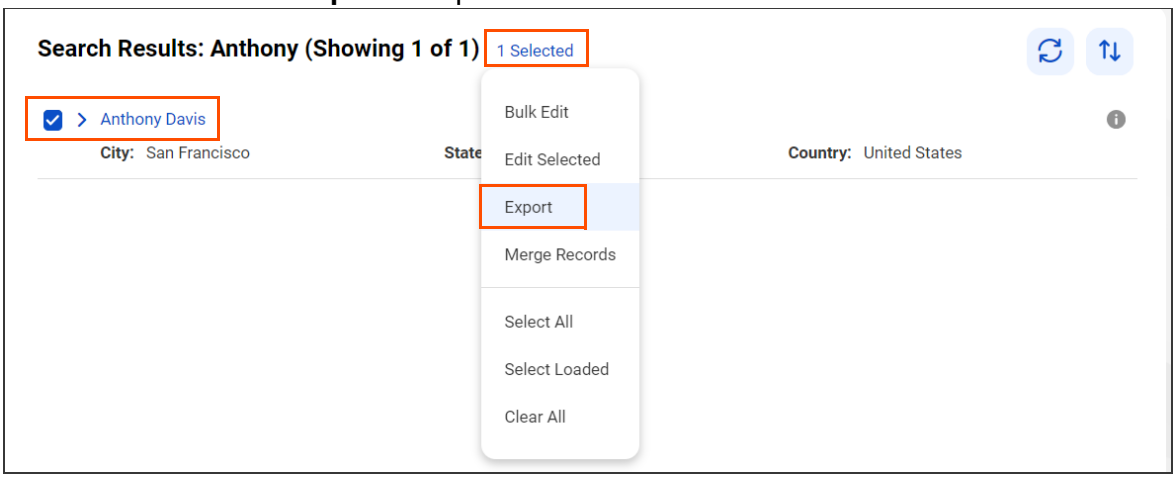
38. View the details and **close** the record.



Export Search Result

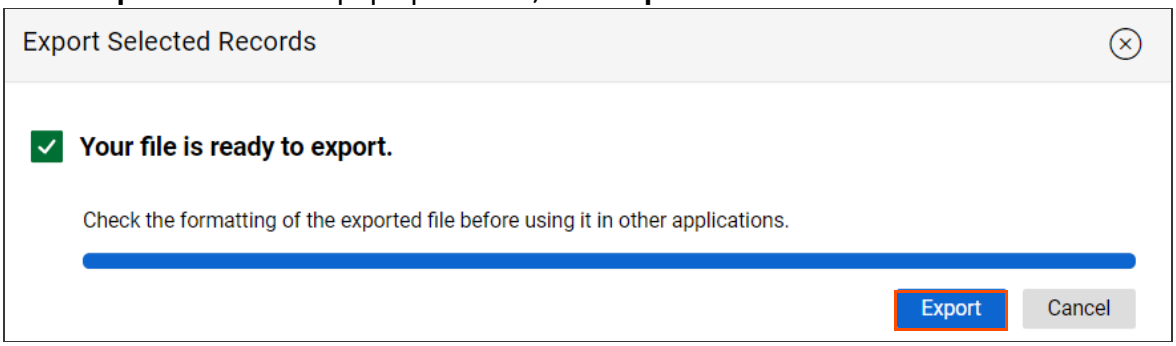
In this section, you will export the search results and save them to a CSV file.

39. On the Search Results page, search for the **Anthony** record. Select the record and click the **Selected** button**.** Click **Export** to export the record.

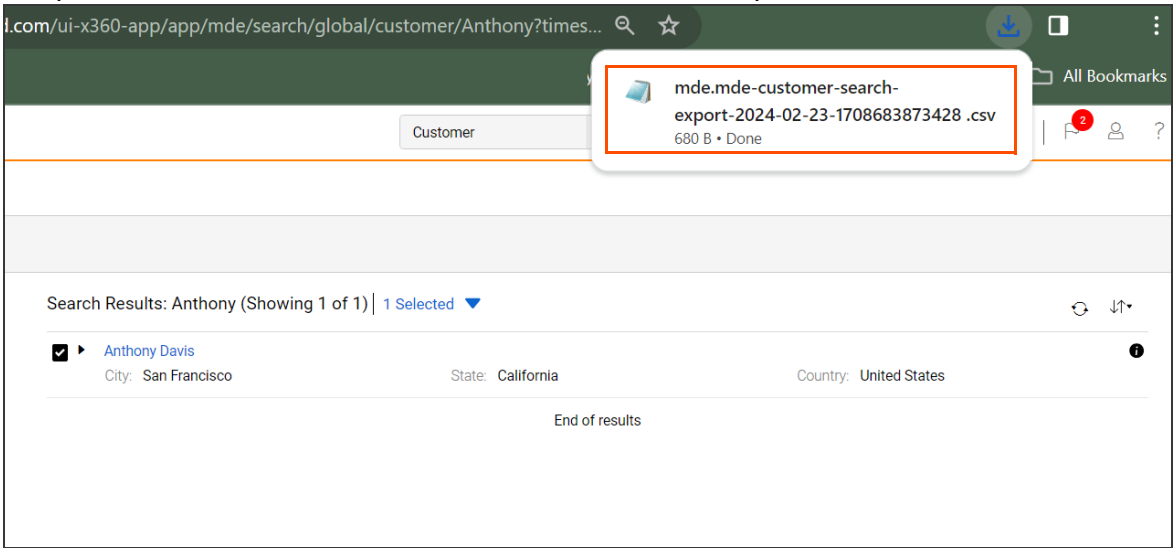


In this case, we have only one record. After onboarding data, we will have many records populated after performing a search.

40. In the **Export All Records** pop-up window, click **Export**.

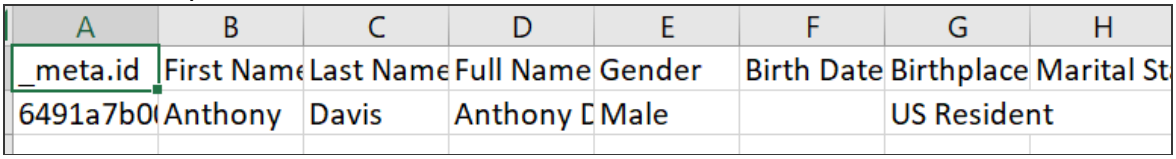


41. A .csv file will be downloaded. You can see a notification on your screen. Click the .csv file to open and view the contact details. You can choose to open the file in Microsoft Excel.



42. If the Microsoft Office Activation Wizard pop-up window appears, click **Cancel**.

43. The CSV file opens. View the customer details and close the file.



A	B	C	D	E	F	G	H
_meta.id	First Name	Last Name	Full Name	Gender	Birth Date	Birthplace	Marital St
6491a7b0(	Anthony	Davis	Anthony D	Male		US Resident	

This concludes the lab.

Lab 9-2: Adding Related Records

Overview:

Related records are the records that contain data associated with a record. You can establish a relationship between two records within a business entity or across business entities.

In this lab, you will add related records to the existing records that you created in the previous lab.

Objectives:

- Add a related record for Customer record
- Add a related record for Manufacturer
- Add a related record for Site

Scenario:

Mark has created some relationships like orders, sells and delivers. He wants to use these to represent relationship between the business entities. This will help him in visualizing the relationship among various business entities.

Duration:

20 minutes

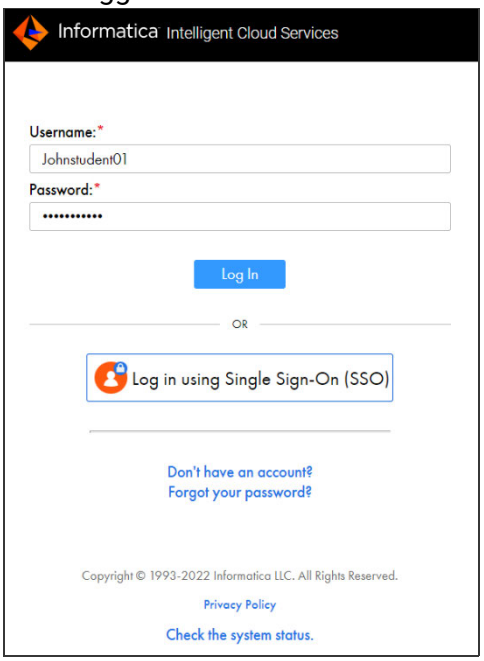
Tasks:

- Add Product as the related record to the Customer record
- Add Product as the related record to the Manufacturer record
- Add Product as the related record to the Site record

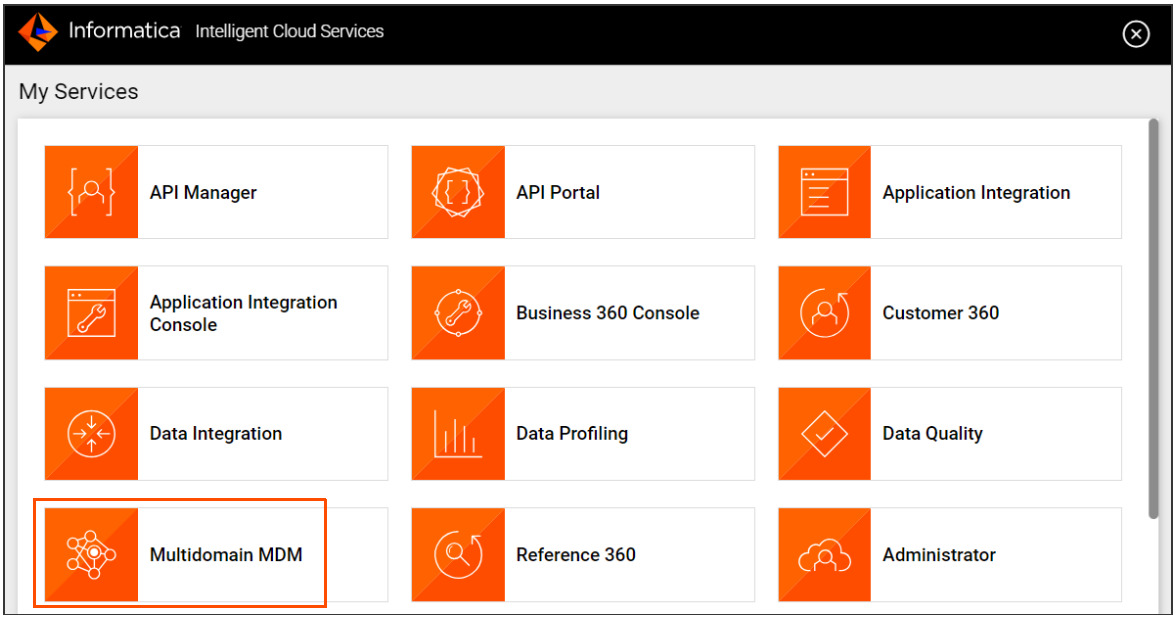
Add Product as the Related Record to the Customer Record

You will add Product record as the related record to the Customer business entity.

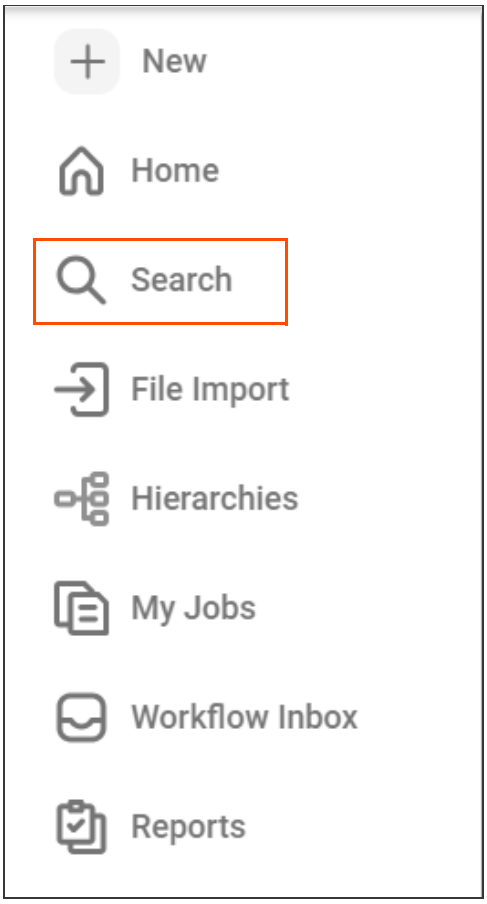
1. Login to IICS with your credentials (**studentNamestudent<#> / Multidomain360**) if you have logged out.



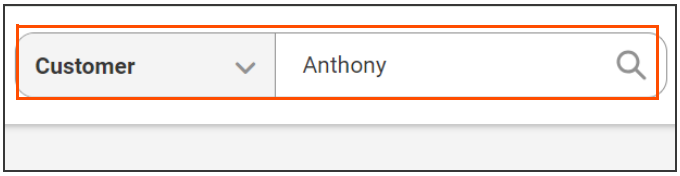
2. Click **Multidomain MDM** Service.



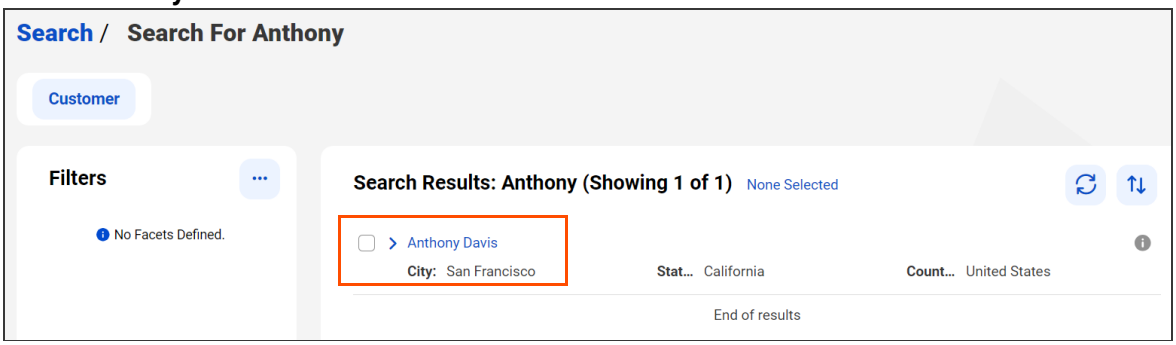
3. Click **Search**.



4. Select the **Customer** business entity and enter the search string as **Anthony**.

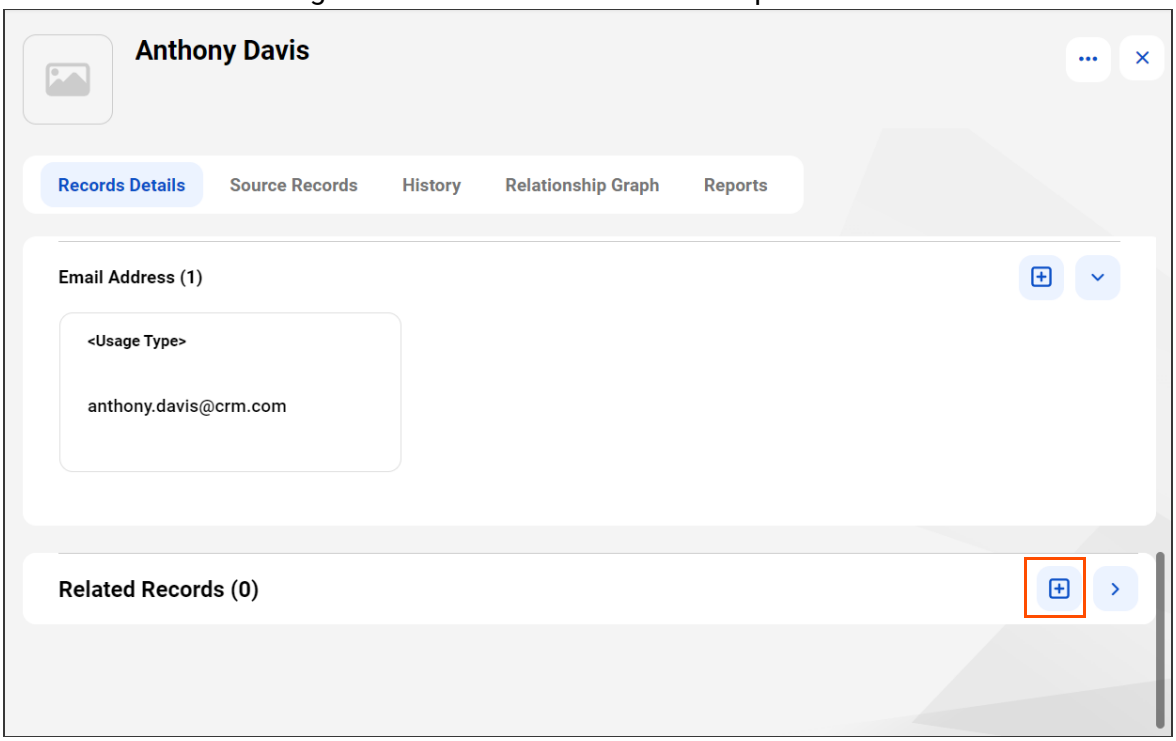


5. Click **Anthony Davis** from the list of records.

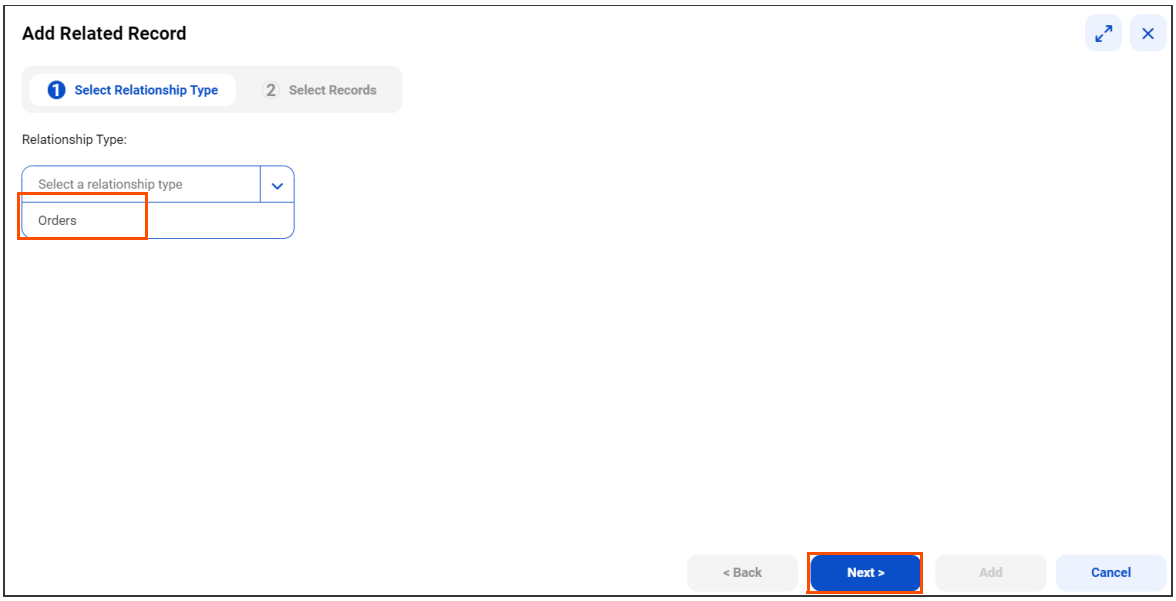


6. Scroll down to find the **Related Records** section at the bottom of the page.

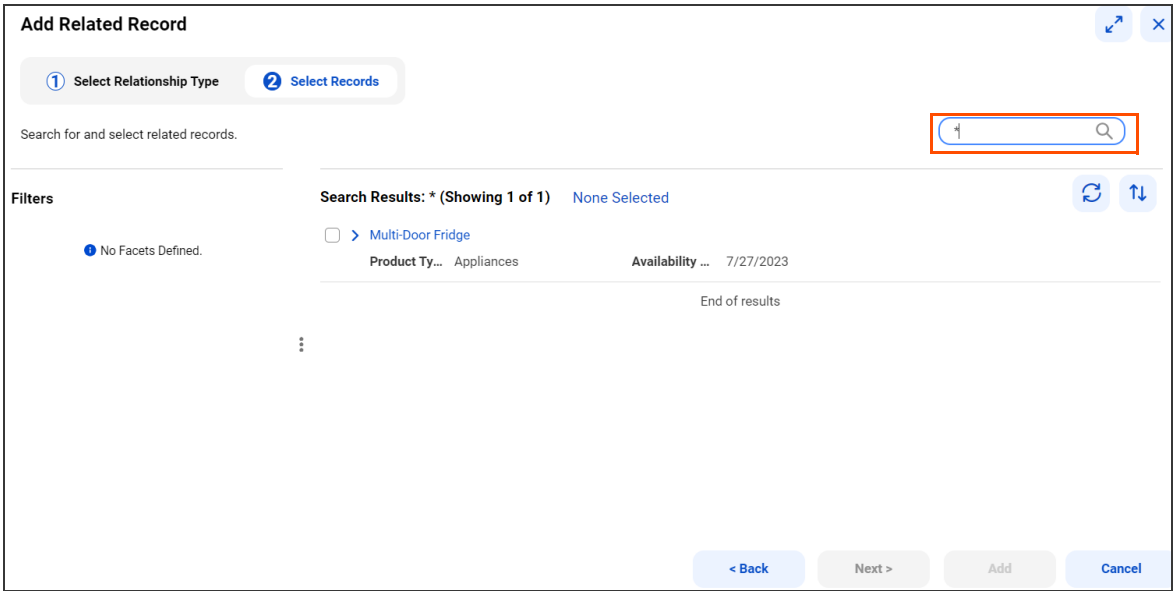
7. Click the **+** icon to the right of the **Related Records** to add product record.



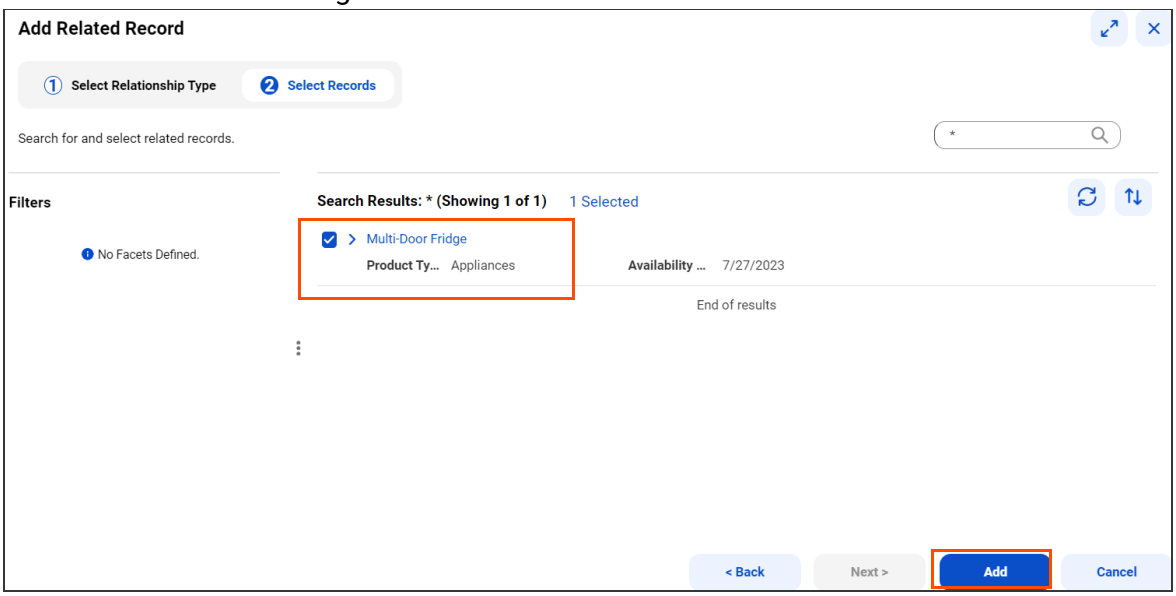
8. The **Add Related Record** page opens. Here, you will select a relationship type that you want to establish between the records. From the **Select a relationship type** drop down, select **Orders**. Click **Next**.



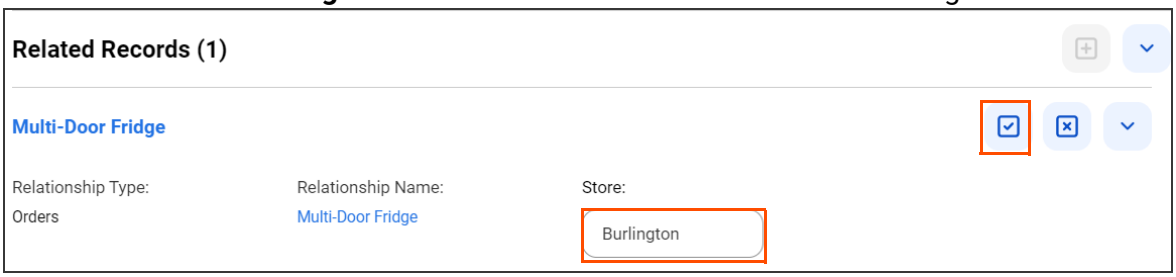
9. In the **Select Records** tab, enter * in the search box and click the **Search** icon to search.



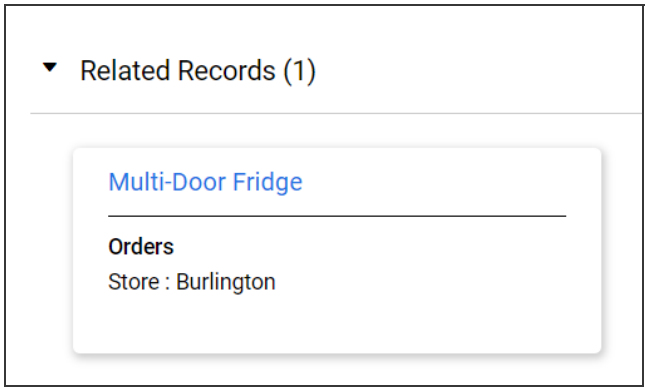
10. Select the Multi-Door Fridge record and click **Add**.



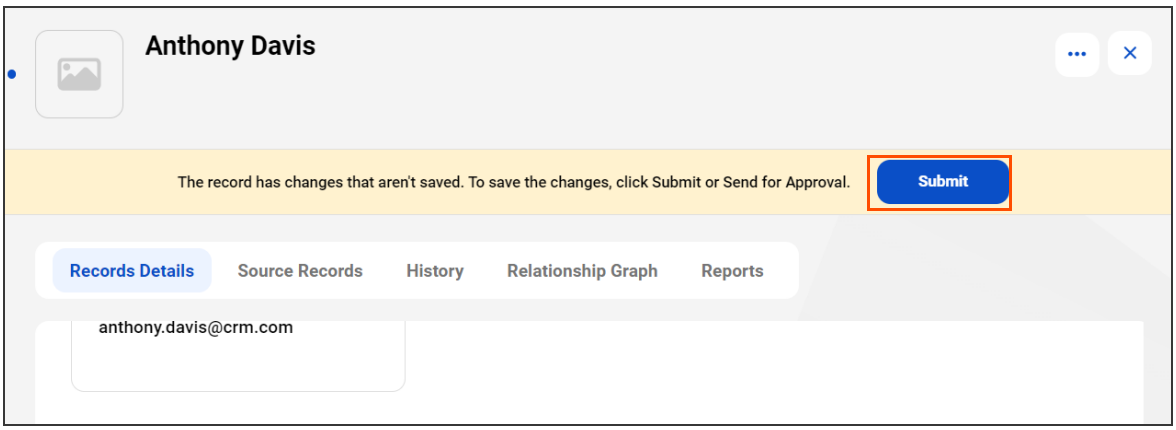
11. Enter the Store as **Burlington** and click the check mark icon to save changes.



You can see that related record has been added.



12. You can see that the record has been updated, at the top of the page. To save the changes, click **Submit**.

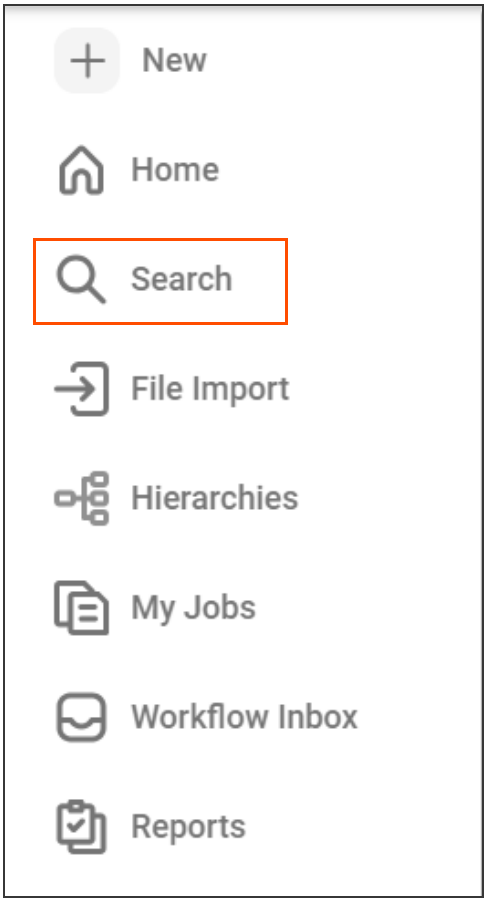


13. The record will be updated successfully. Close the record.

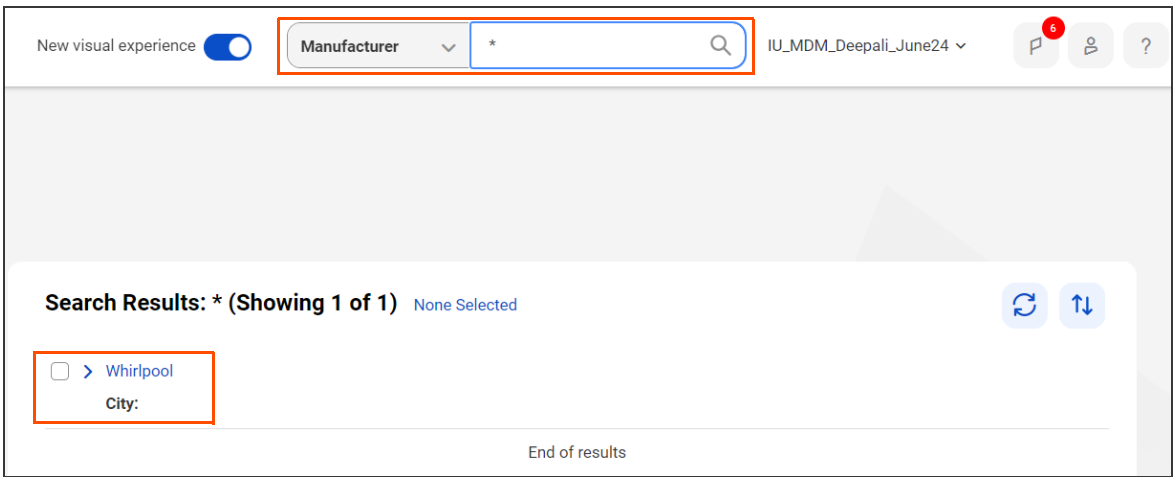
Add Product As the Related Record to the Manufacturer Record

You will add Product record as the related record to the Manufacturer business entity.

14. Click **Search**.

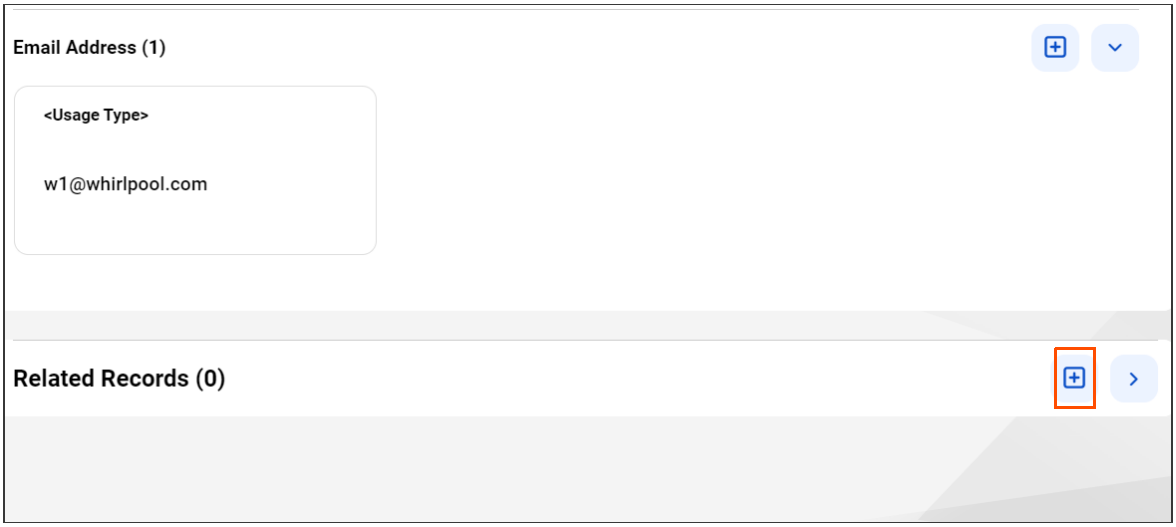


15. Select the **Manufacturer** business entity and enter the search string as *. Click **Whirlpool** from the list of records.

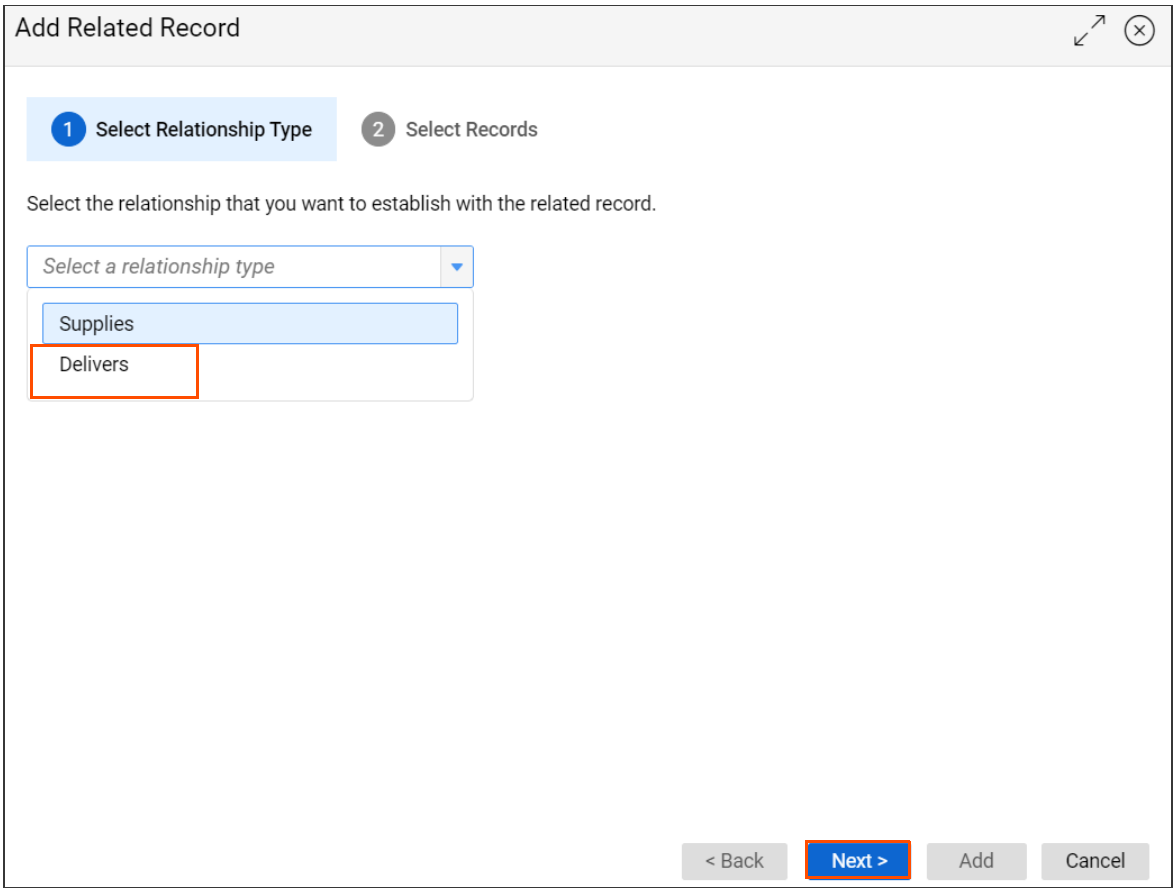


16. Scroll down to find the **Related Records** section at the bottom of the page.

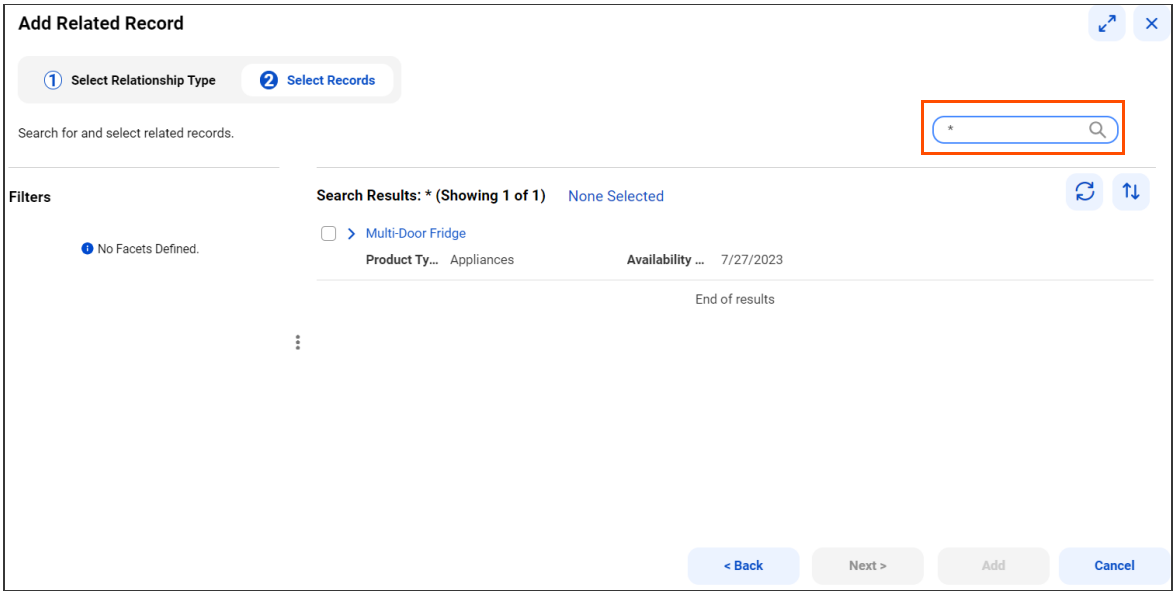
17. Click the **+** icon to the right of the **Related Records** to add product record.



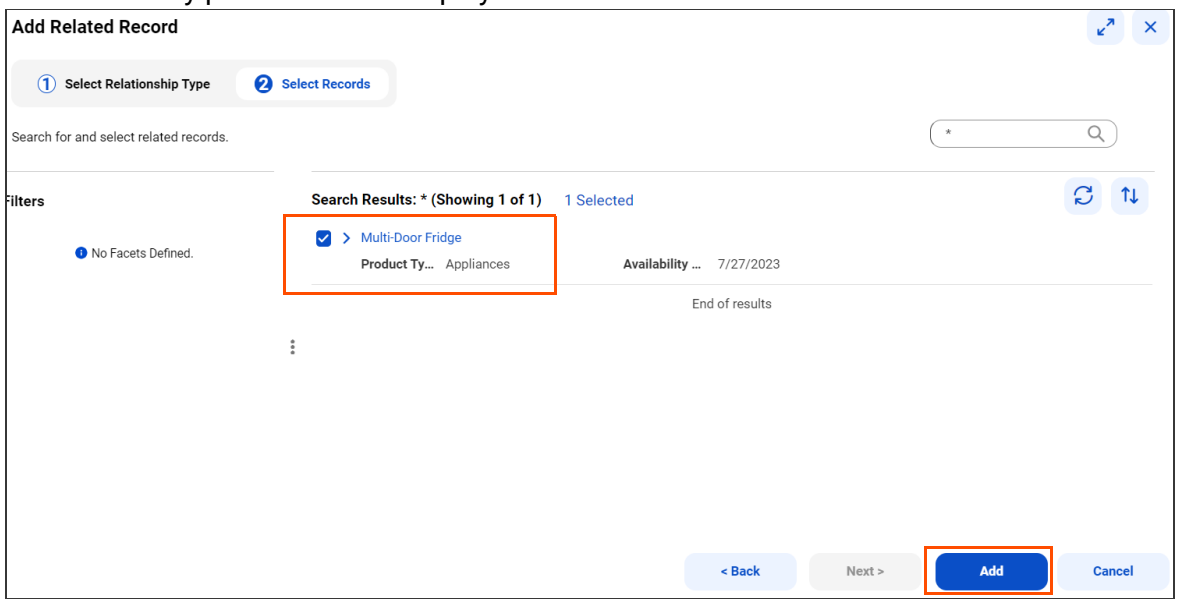
18. The **Add Related Record** page opens. Here, you will select a relationship type that you want to establish between the records. From the **Select a relationship type** drop-down, select **Delivers**. Click **Next**.



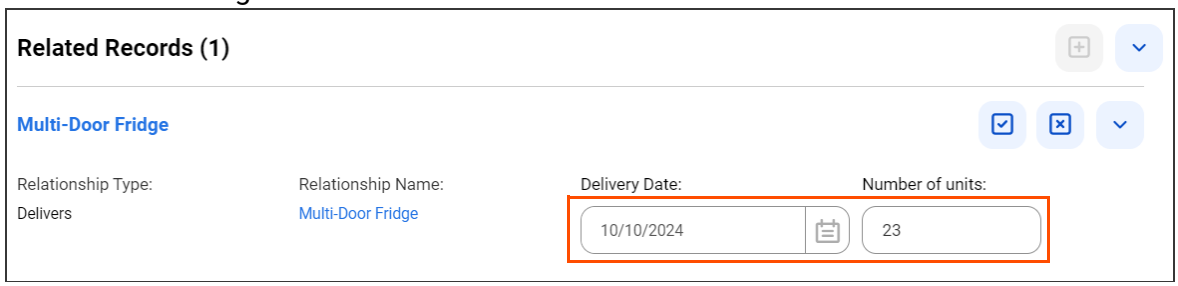
19. In the **Select Records** tab, enter * in the search box and click the **Search** icon to search.



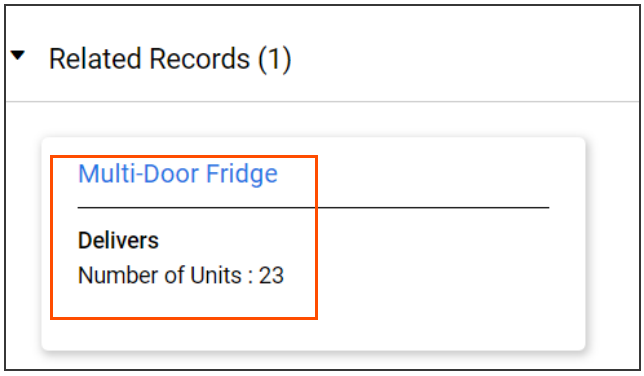
20. Select the only product that is displayed and click **Add**.



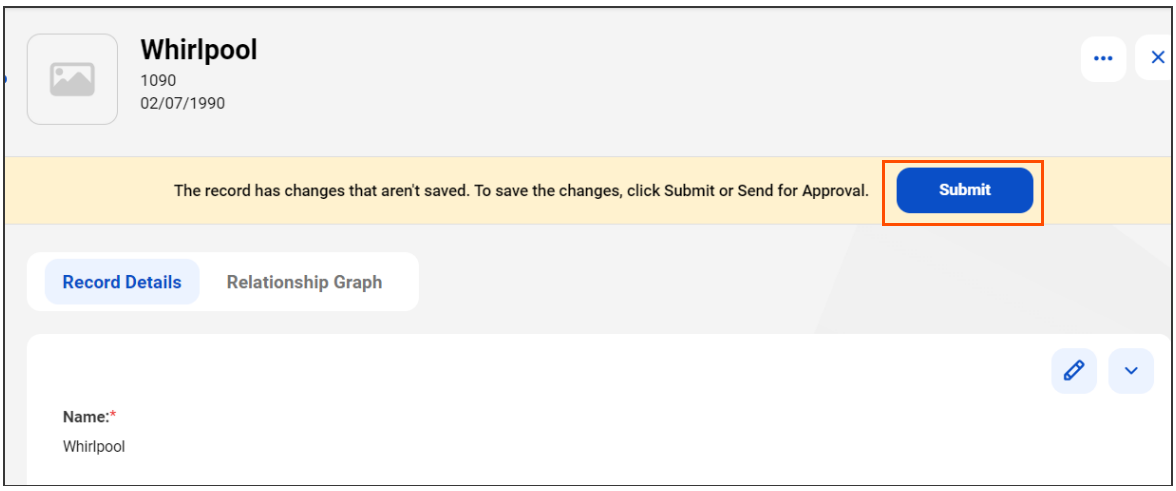
21. Enter the Number of Units as **23** and Delivery Date as **10/10/2024.** Click the check mark icon to save changes.



You can see that related record has been added.



22. You can see that the record has been updated, at the top of the page. To save the changes, click **Submit**.



23. The record will be saved successfully. Close the record.

Self-Try Exercise (Mandatory)

Add Product as the Related Record to the Site Record

You will add Product record as the related record to the Site business entity.

High Level Steps:

- Search for the Site record **JustInTime** and open it.
- Go to **Related Record** section and click **Add**.
- Select the relationship type as **Sells**.
- In the Select Records step, search for any of the product record and click **Add**.
- Enter the Current Inventory as **15** and click the check mark to save the details.
- **Submit** the record.

This concludes the lab.

Lab 9-3: Performing Filtered Search

Overview:

Filtered search is used to perform searches based on one or more field values. You can configure fields and field values to include in your search. In this lab, you will use filtered search to search for records.

Objectives:

- Search records using Filtered Search
- Save searches and templates

Scenario:

Mark wants to perform search for some records and wants to refine the search results. He can achieve this by applying filters to them.

Duration:

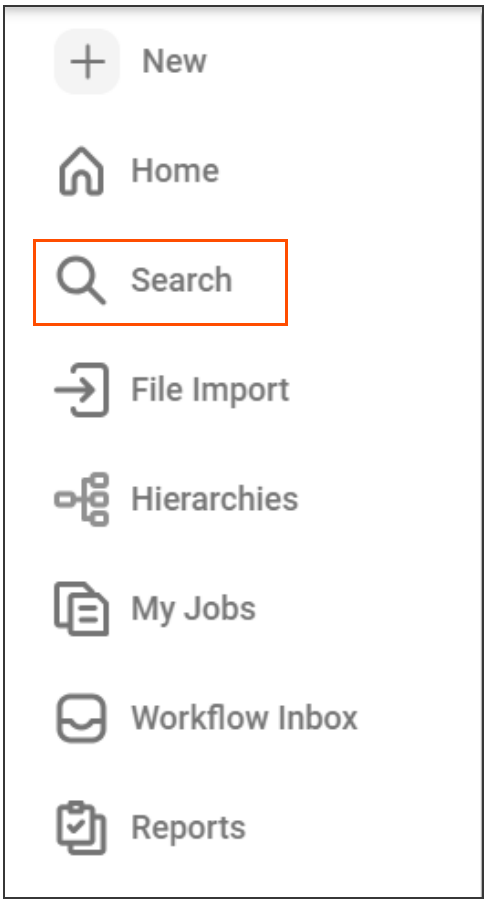
20 minutes

Tasks:

Perform Filtered Search

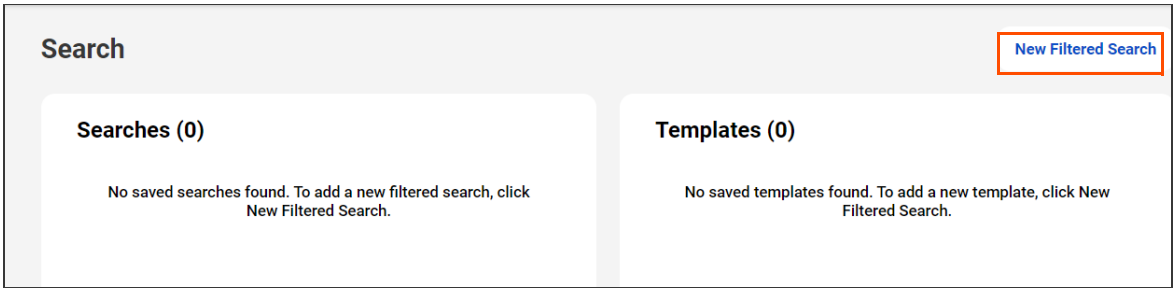
You will now use the filtered search function to apply filters and search for records. You will also save the search query so that you can reuse it in future.

1. From the left side menu, click **Search**.

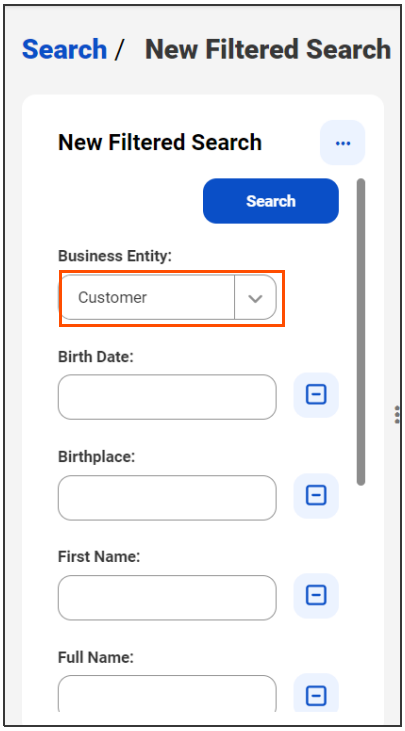


2. If you still see the previous search result, click **Search** from the Multidomain MDM search page to reset the page to normal.

3. In the search page, click **New Filtered Search** at the top right corner.

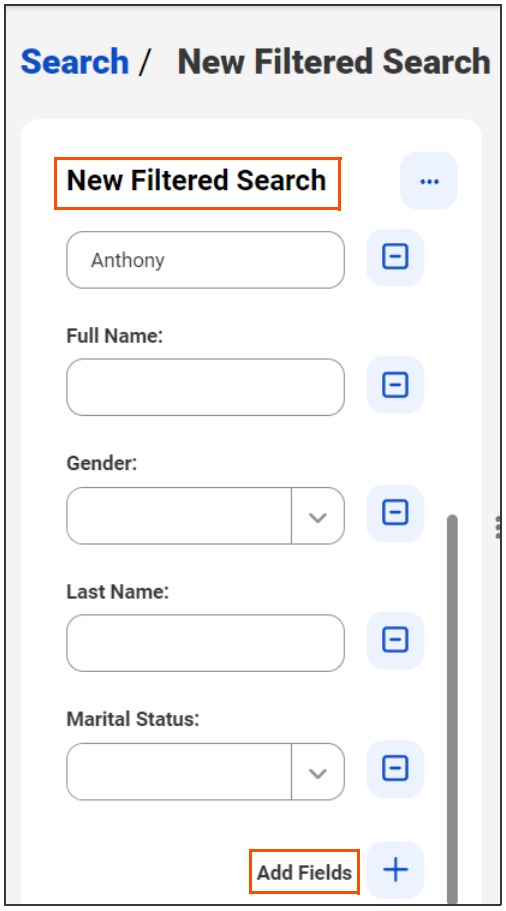


4. The **New Filtered Search** page opens. From the **Business Entity** drop-down, select **Customer**.

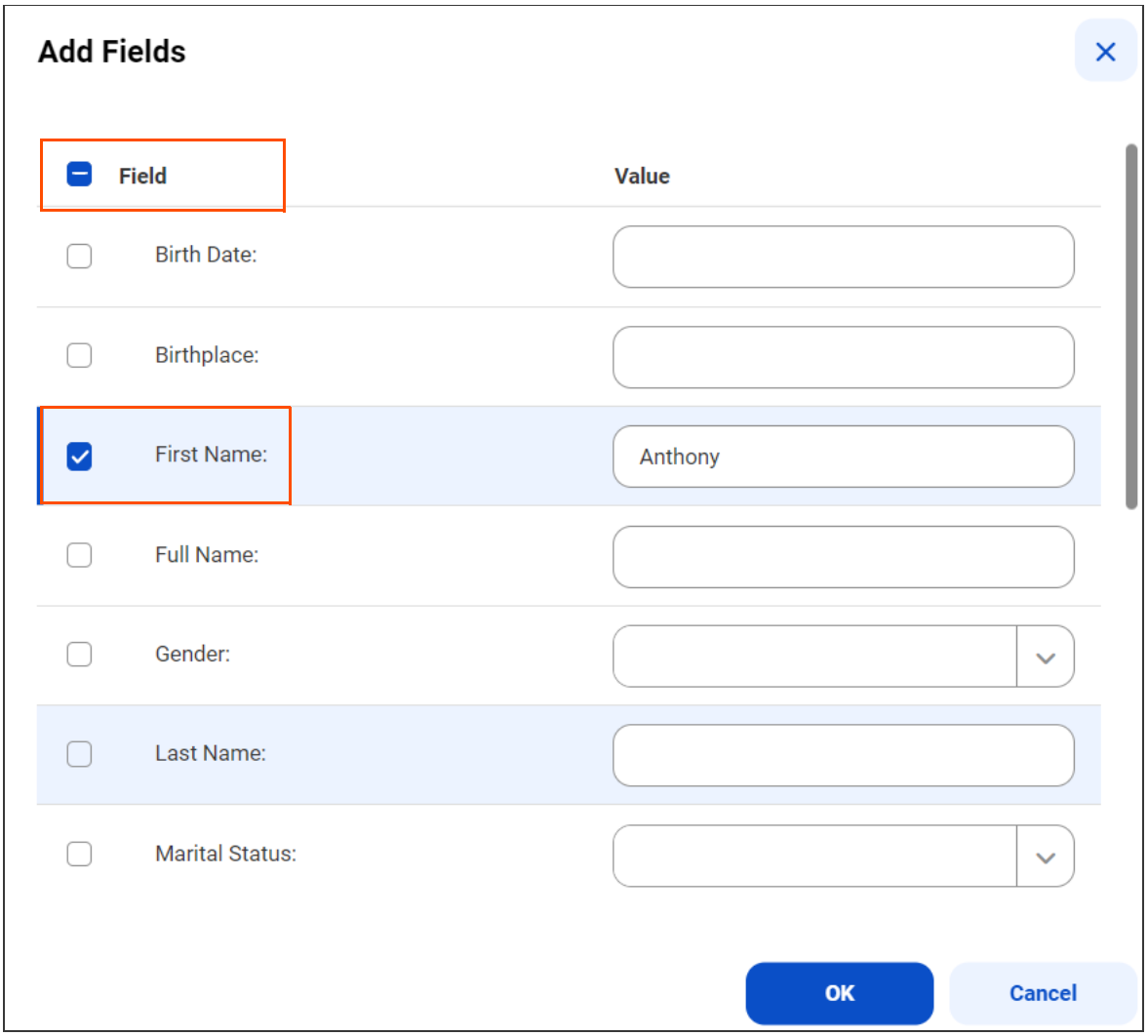


5. Enter **Anthony** as the First Name.

6. Click the **+** icon beside to **Add Fields** to have more filter conditions.

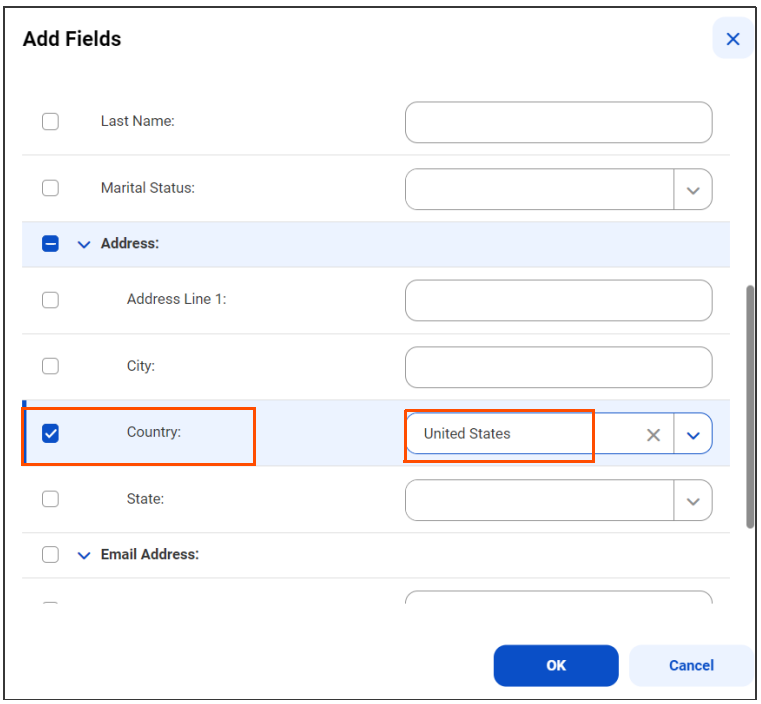


7. De-select all the field and select the **First Name** field.



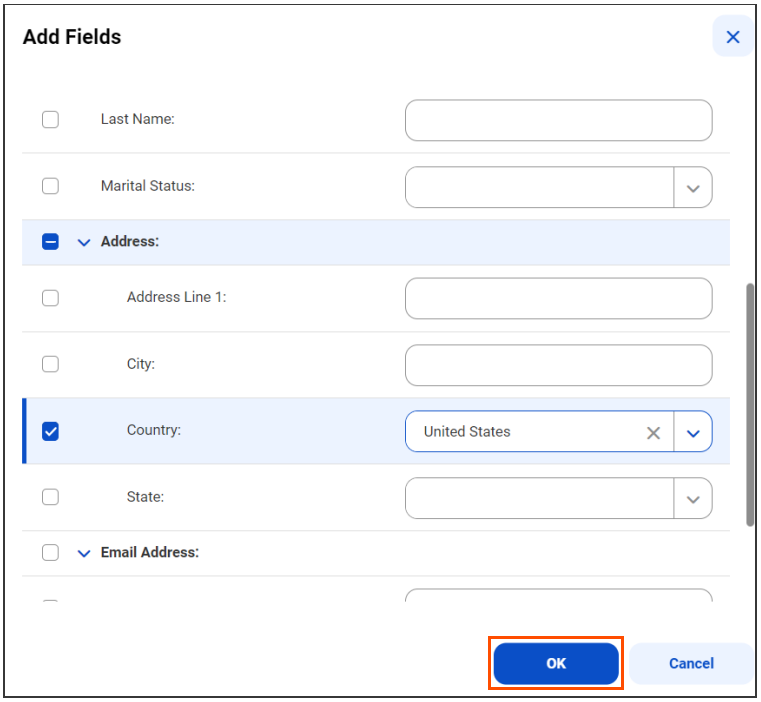
8. In the **Add Fields** page, expand the **Address** field.

9. Select **Country** field. From the drop-down menu next to the Country field and select **United States**.

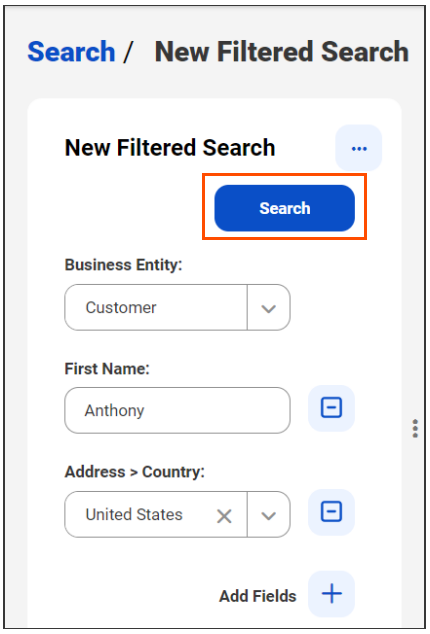


ideally, only First Name, Last Name and Country fields to be selected.

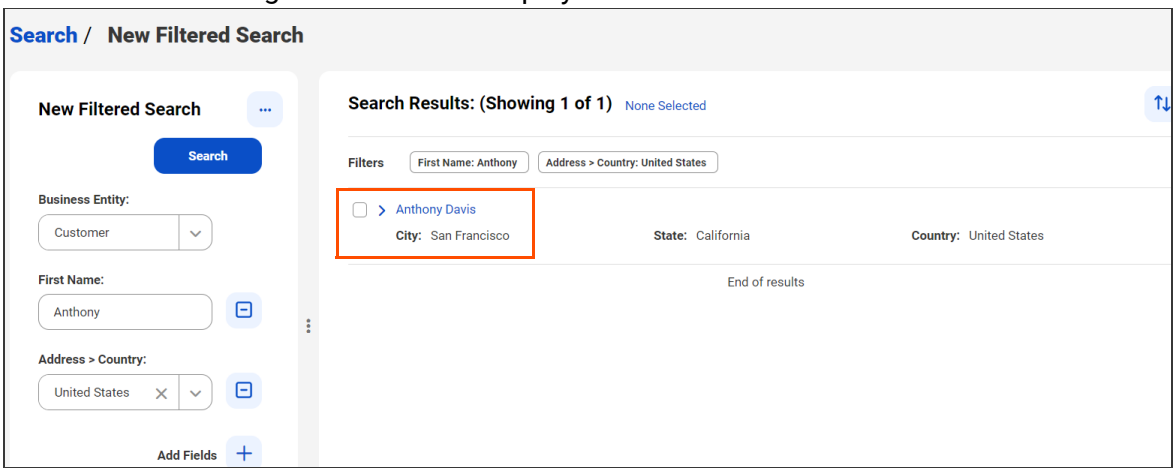
10. Click **OK**.



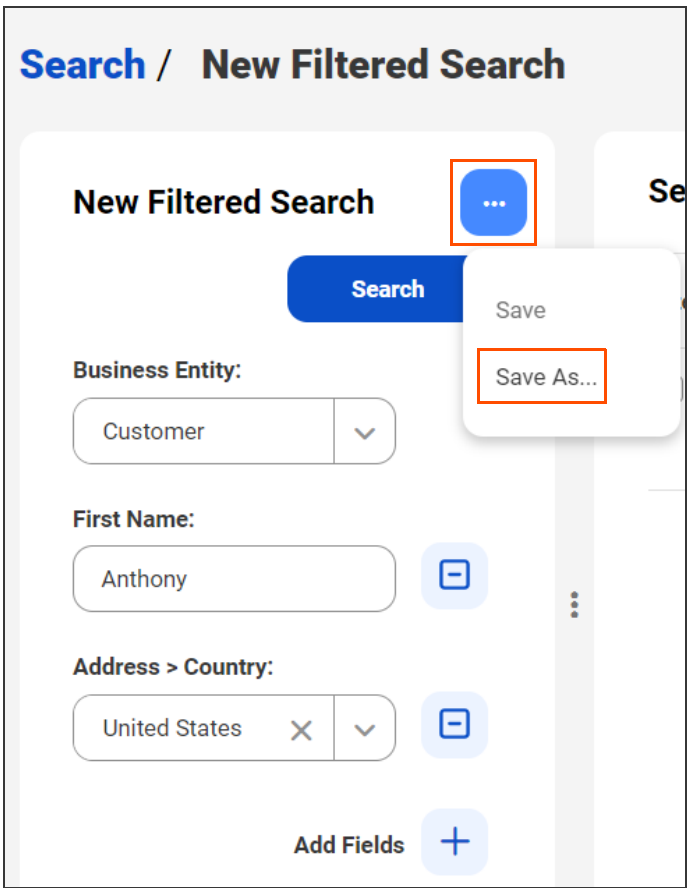
11. The fields are added. Click **Search**.



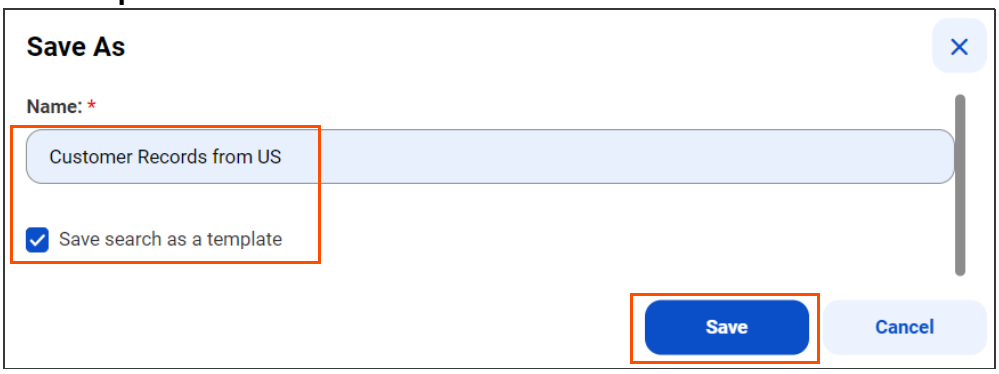
The results matching the criteria are displayed.



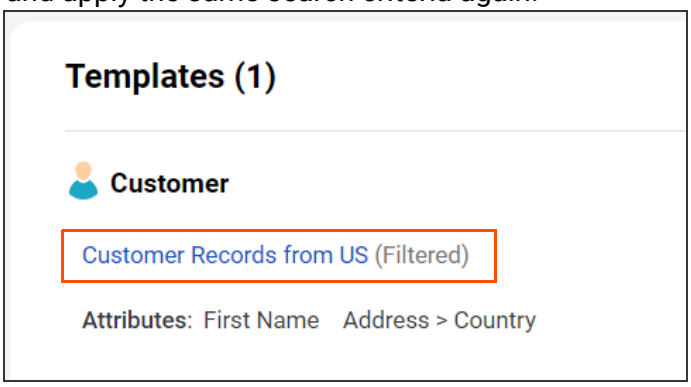
12. You can now save this search query. In the **New Filtered Search**, click the **Actions** icon and select **Save As**.



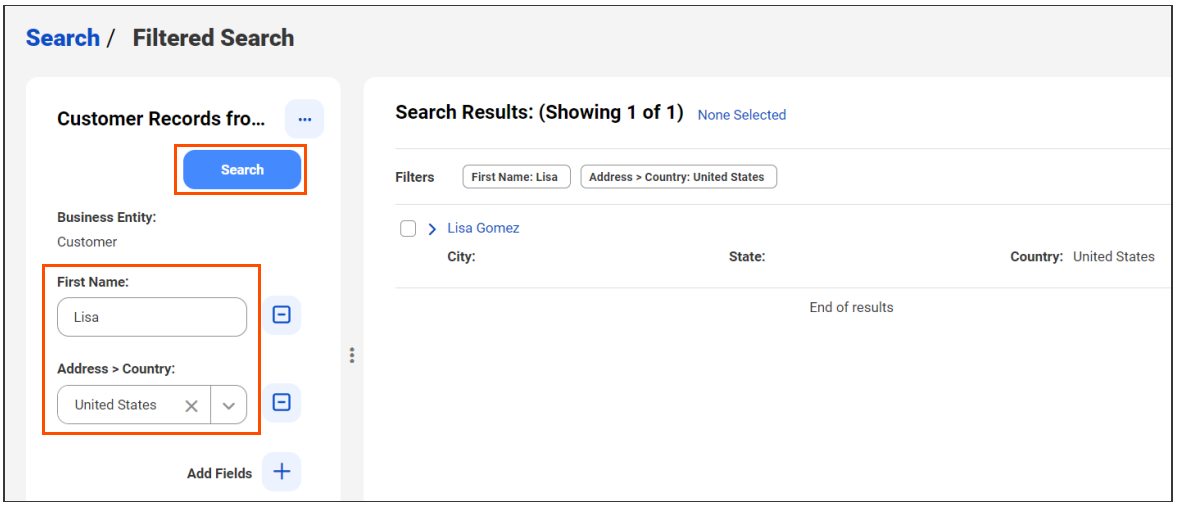
13. Enter a unique name for the search. To save the search as a template, select **Save search as a template.** Click **Save**.



14. The saved template appears under **Templates** section. Click the saved template to view and apply the same search criteria again.



15. Provide the input values and click **Search**. You can see the matching records related to the search criteria.



This concludes the lab.

Lab 9-4: Import Data From a Flat File

Overview:

You can import customer data from a CSV file into Multidomain MDM. The data can be business entity records or relationship records. You can import a CSV file of up to 20 MB at a time.

The import process parses the import data and automatically maps the columns of the source file with the fields of the selected business entity or relationship. The automatically mapped fields display confidence indicators to indicate the accuracy of mapping. For the fields with low confidence score or for the fields with incorrect field mapping, you can manually map the source columns with the required target fields. You can preview and verify the field mapping before you import the file.

In this lab, you will import customer, product, site and manufacturer records from a flat file.

Objectives:

- Import Customer.csv file
- Import Product.csv file
- Import Site.csv file
- Import Manufacturer.csv file

Scenario:

HomeKart intends to incorporate all its stakeholders' data into Multidomain MDM, with some of this data stored in a flat file. They have tasked Mark with this job. To facilitate this, Mark employs the File Import option in Multidomain MDM to import the data from a .csv file.

Duration:

20 minutes

Tasks:

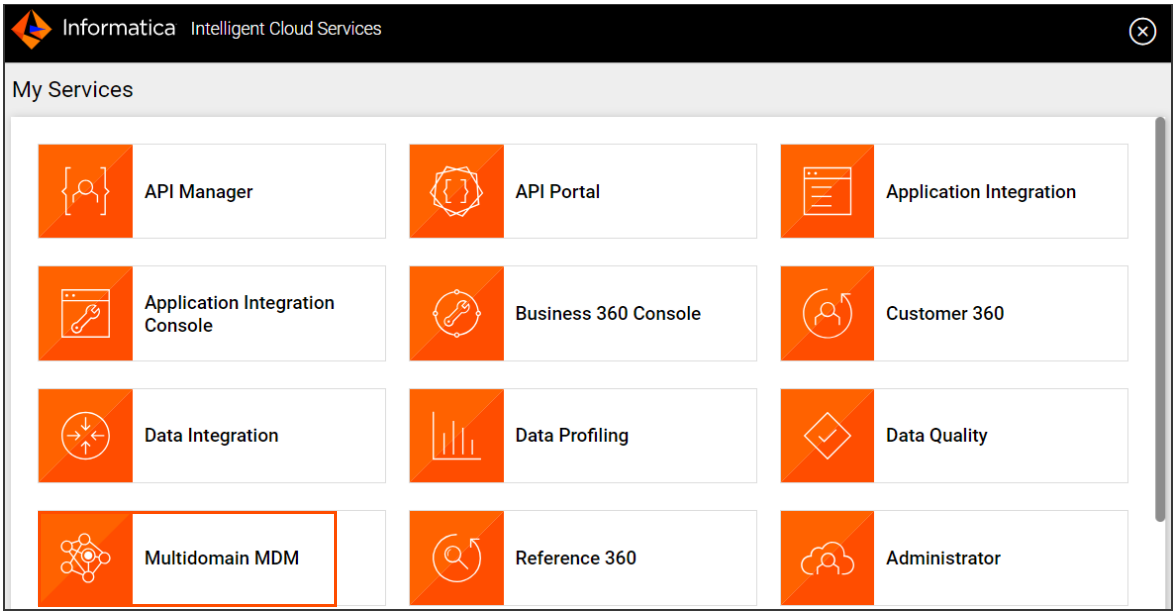
- Import data from a flat file to Customer, Product, Site and Manufacturer business entity
- Import data into the hierarchy
- Import data into the relationships

Import Data from a Flat File to Customer

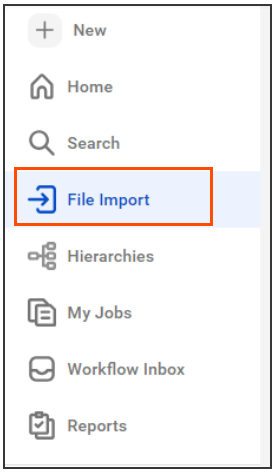
You will import a CSV file into Customer business entity.

1. Login to IICS with your credentials (**studentNamestudent<#> / Multidomain360**) if you have logged out.

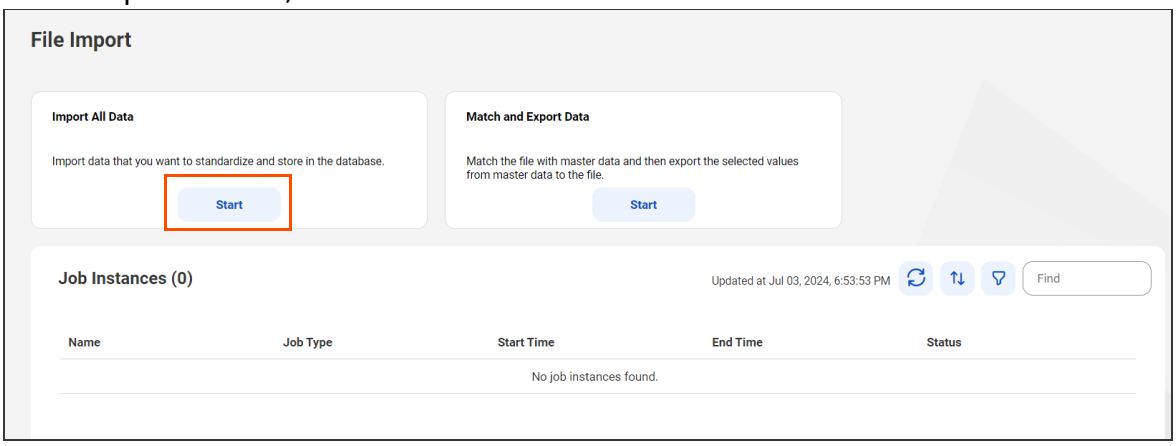
2. Click **Multidomain MDM**.



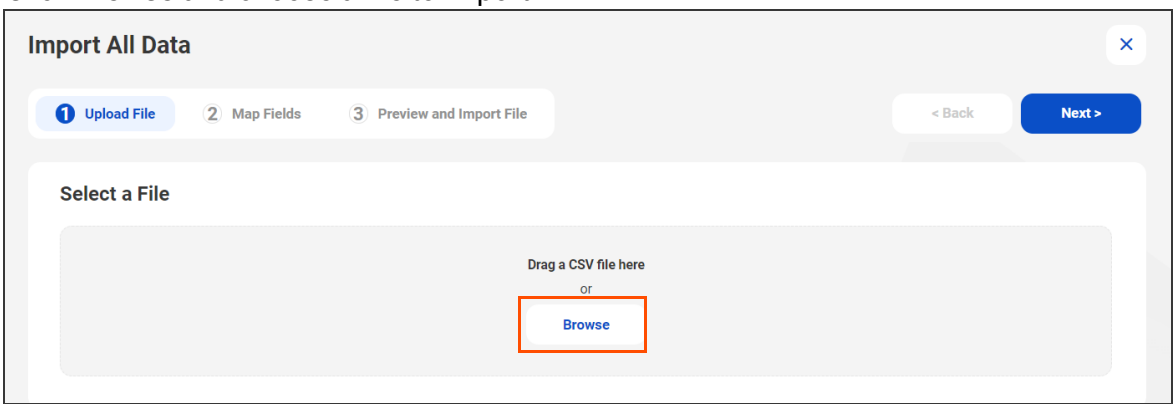
3. In the Multidomain MDM page, click **File Import**.



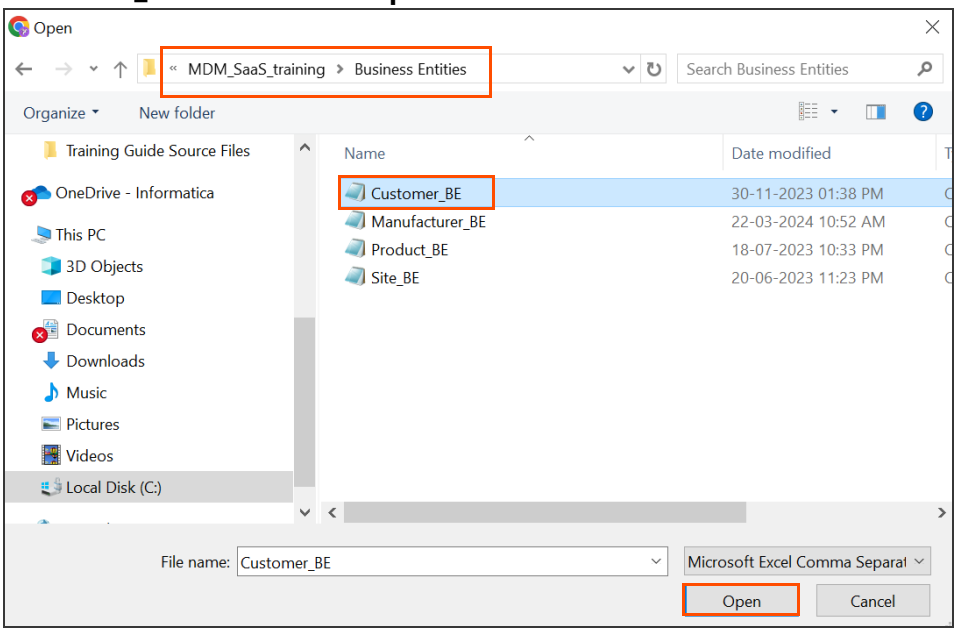
4. Under Import All Data, click **Start**.



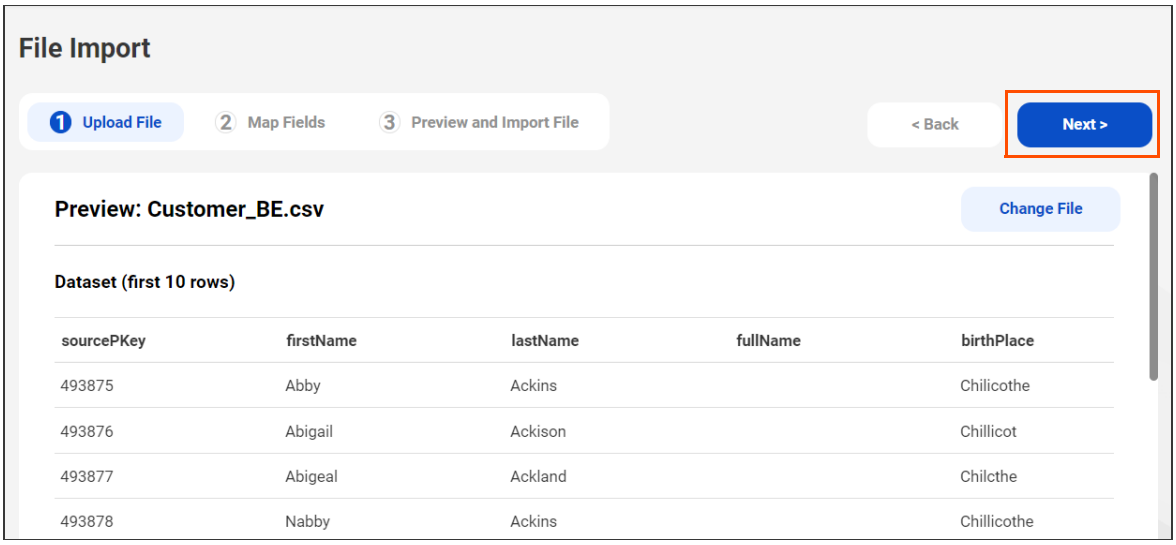
5. Click **Browse** and choose a file to import.



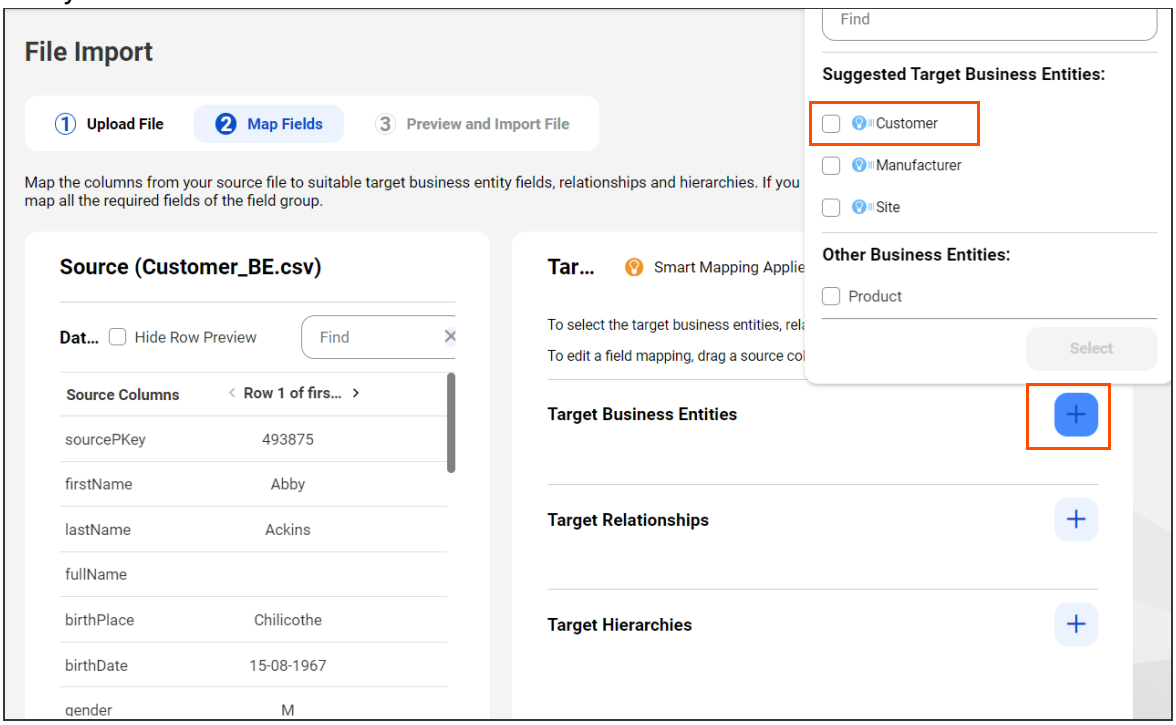
6. Navigate to **C:\MDM_SaaS_Training\BusinessEntities** on the VM and select **Customer_BE.csv** file. Click **Open**.



7. Preview the file and click **Next**.

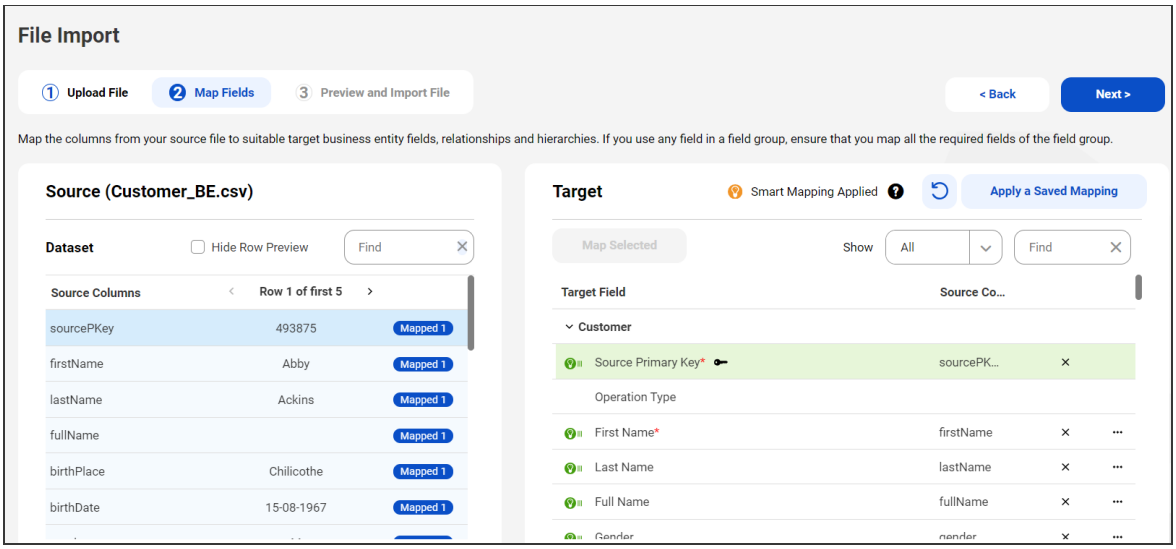


8. The Target section in the **Map Fields** step allows you to select an entity, relationship, or hierarchy as the target. For this lab exercise, you will load data into target business entity. Click the **+** icon to the right of the **Target Business Entities** Select the **Customer** business entity .



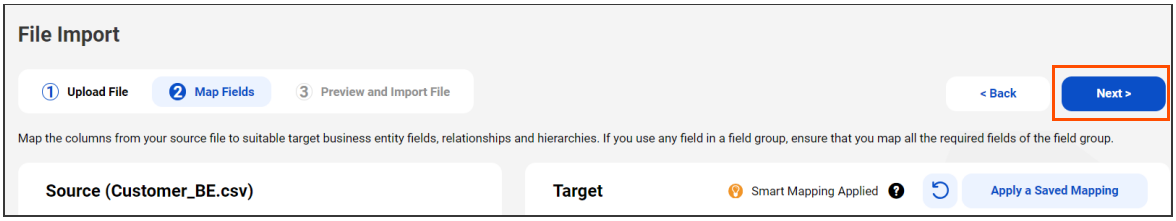
Based on the target business entity you have selected in the previous step, the import process automatically maps the column headers of your source file to suitable target fields. Verify the mapping of all fields and ensure that all the required fields are mapped.

If you are not in agreement with the system-suggested field mapping, you can un-map the field by clicking the **X** icon and manually map the fields by dragging a source column to a target field.

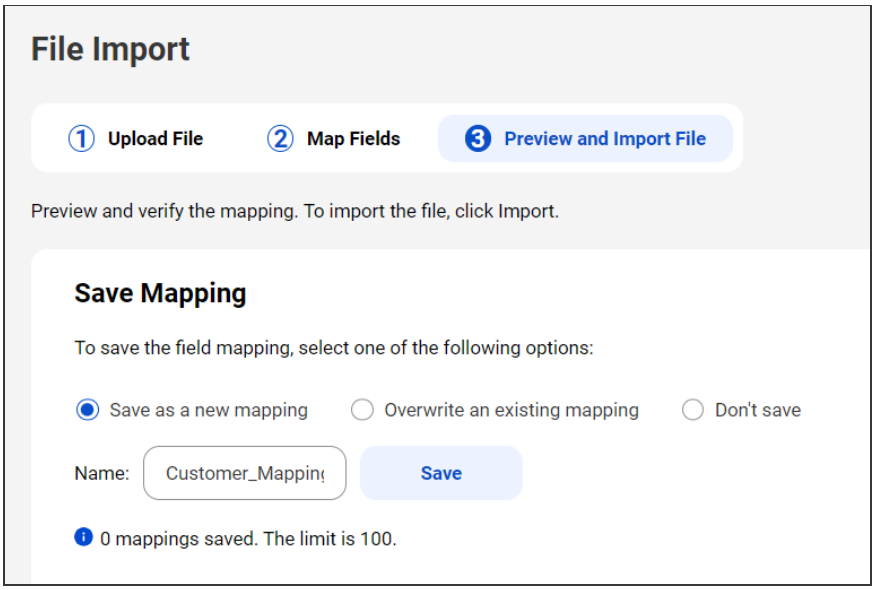


If you use any field in a field group, ensure that you map all the required fields of the field group. For File import to work, even if one field is mapped in a Field Group then all mandatory ones in that group should be mapped. As we are not mapping Employment Source PKey or any other fields, we need to delete designation

9. Click **Next**.

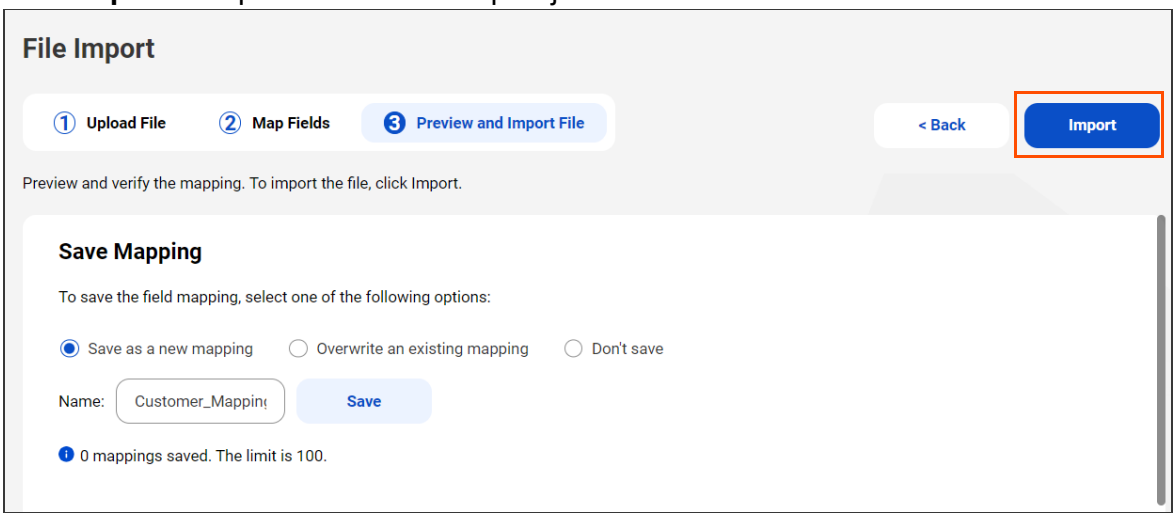


After you map the fields, you can save the mapping for future use. Use the **Preview** section to preview and verify the mapping, and then import the data. In the **Preview and Import File step**, with the **Save as new mapping** option selected, enter the **Name** of the mapping as **Customer_File_Import** and click **Save**.

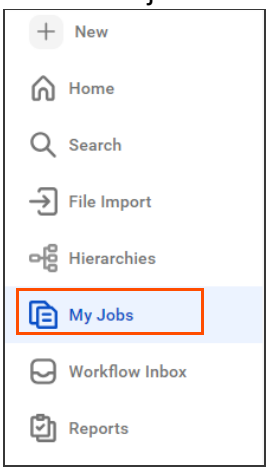


A notification is displayed with the message “The mapping is saved successfully.”

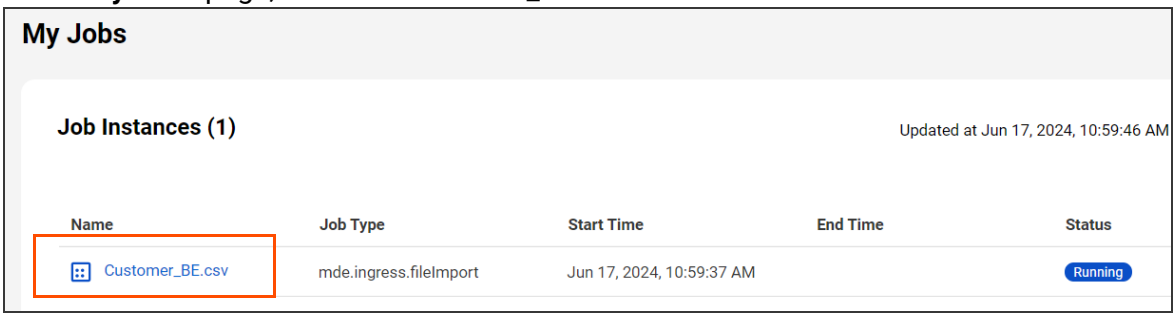
10. Click **Import** to import the file. The import job starts.



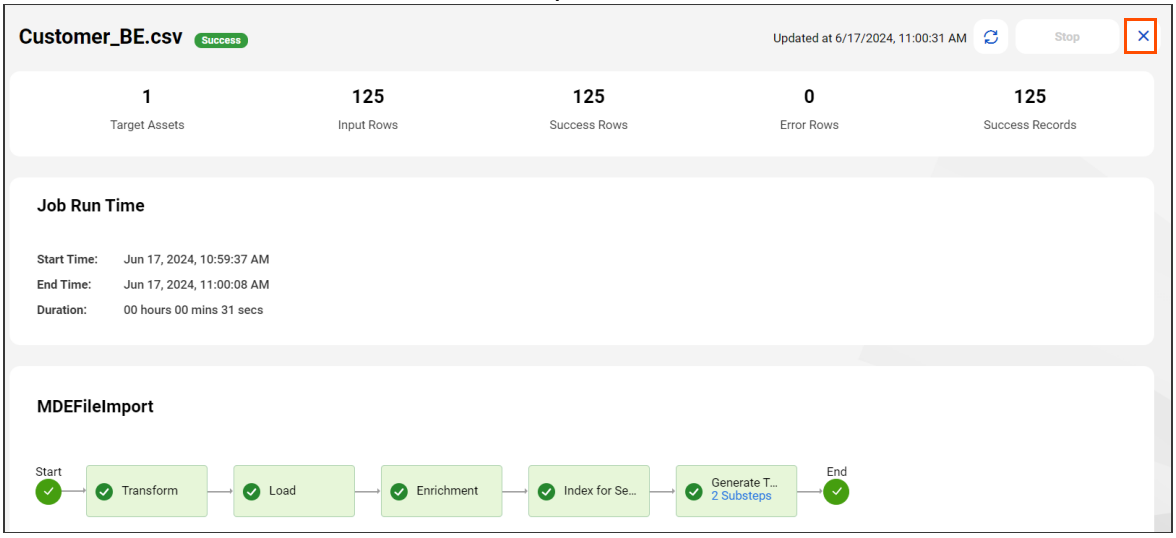
11. To see the job running status, click **My Jobs** from the left side menu.



12. In the **My Jobs** page, click the **Customer_BE.csv**.

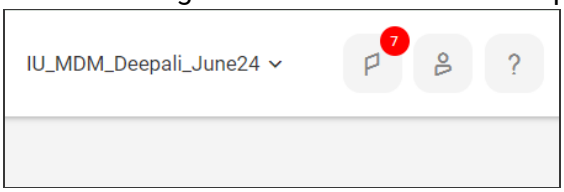


13. If the job is still running, wait till it completes all the steps – Transform, Load, Index for Search, and Generate Tokens. Once it completes, click the **X** button to close.

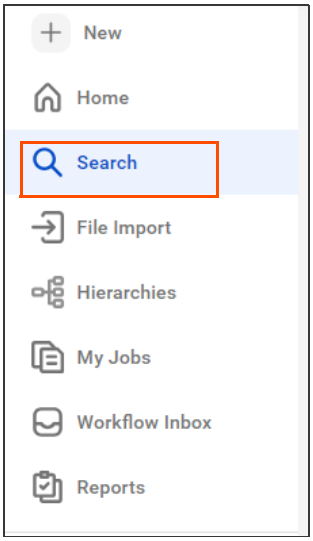


It takes 1-2 mins to import all the records in the Customer-Data.csv file. Click **Refresh** icon to see the updated status.

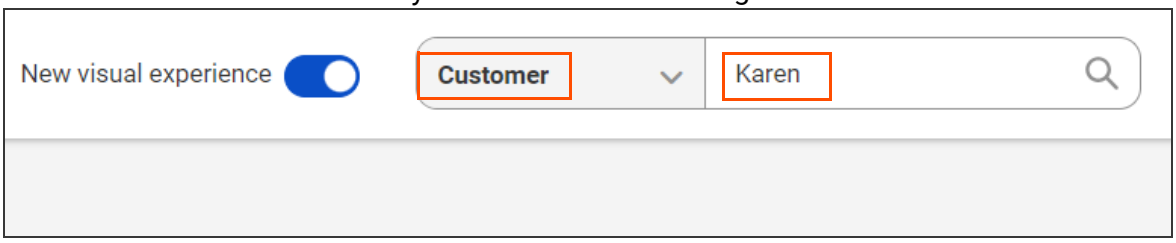
You will also get a notification after the import process is complete.



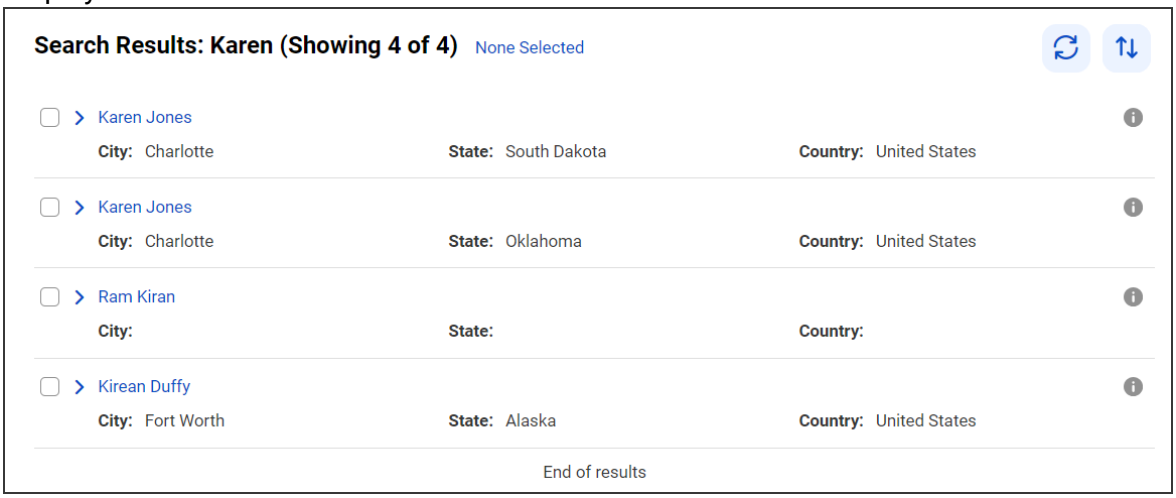
14. Click **Search** from the left menu to check if the records have been imported.



15. Select **Customer** business entity and search for the string **Karen**.



All records for the name Karen, that were imported from the **Customer_BE.csv** file are displayed.



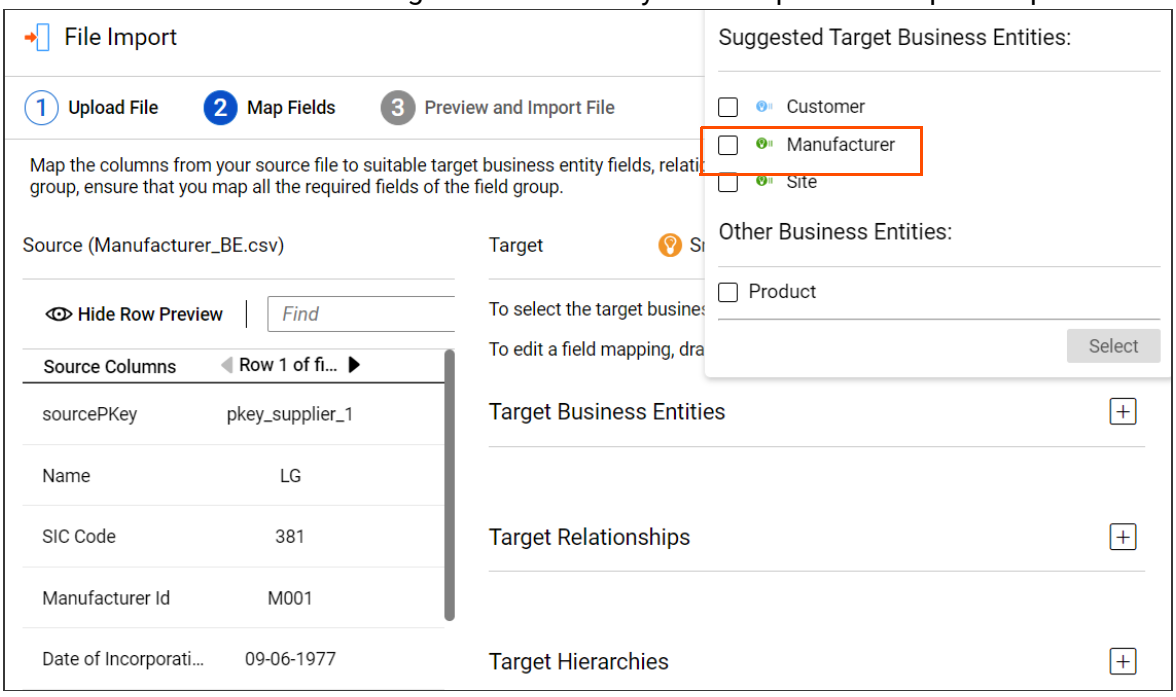
Self-Try Exercise (Mandatory)

Import Data from a Flat File to Manufacturer

Follow the above steps and import Manufacturer_BE.csv file into Multidomain MDM.

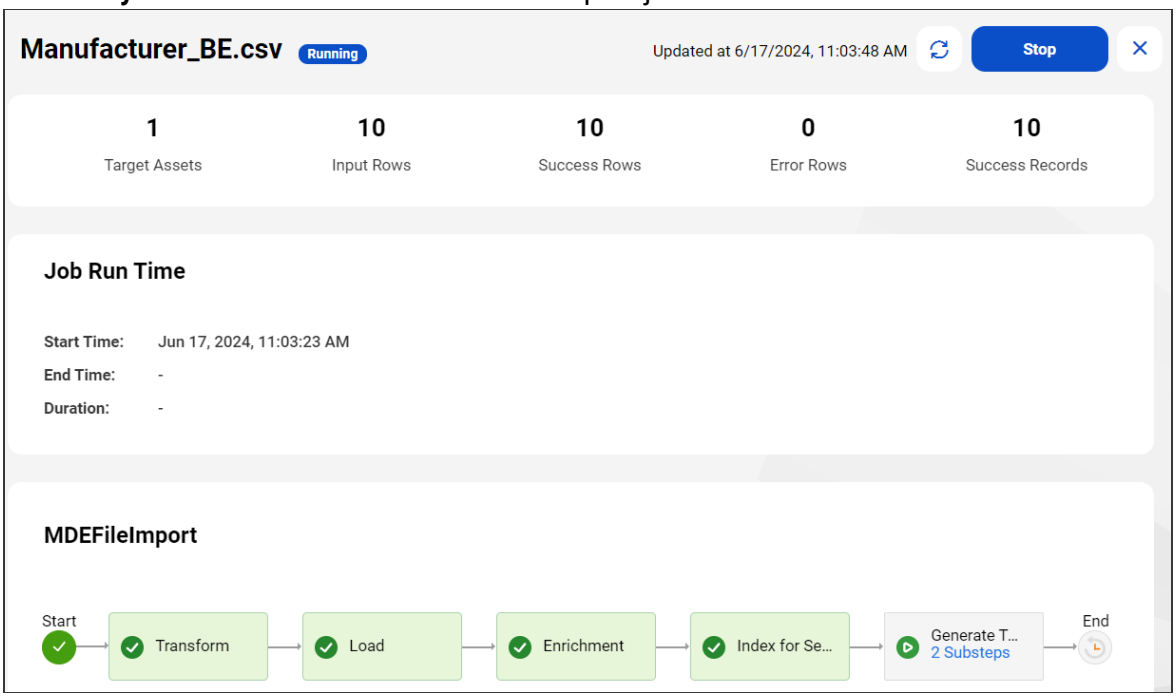
16. Choose the file from the file location **C:\MDM_SaaS_Training\BusinessEntities\Manufacturer_BE.csv**.

17. Select **Manufacturer** as the Target Business Entity and complete the Import steps.

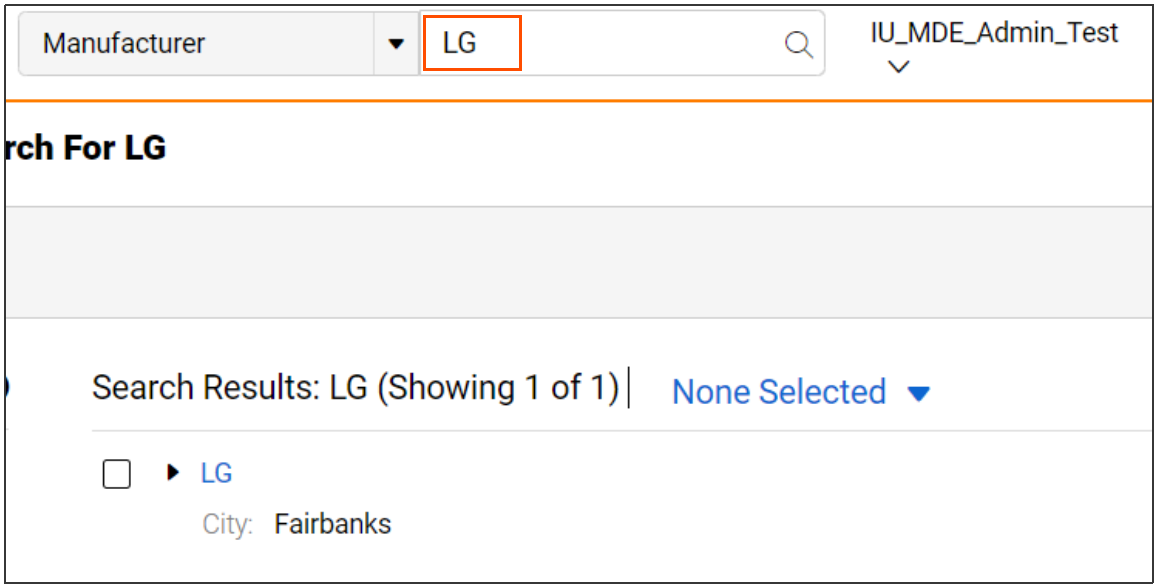


It is possible that you will receive an error when mapping. To manually map fields from the source column to the target column, drag and drop the fields from the source column and drop them into the target field.

18. Check **My Jobs** to check the status of the import job.



19. Perform searches for **LG** or **Whirlpool** to validate successful import.



Self-Try Exercise (Mandatory)

Import Data from a Flat File to Product and Site business entity

High Level Steps

Follow the above steps and import **Product_BE.csv** and **Site_BE.csv** files into Multidomain MDM.

- File location for **Product**: C:\MDM_SaaS_Training\BusinessEntities\ Product_BE.csv
-

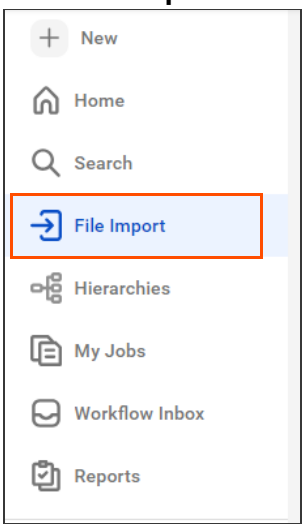
- File location for **Site**: C:\MDM_SaaS_Training\BusinessEntities\ Site_BE.csv
- Choose the respective files from the file location C:\MDM_SaaS_Training\BusinessEntities.
- Select the Target Business Entity and complete the Import steps.
- Check **My Jobs** to check the status of the import job.
- Perform searches for Fridge and Washer for the Product BE and to validate successful import.
- Similarly, Perform searches for Homekart, HQ for the Site BE and to validate successful import.

Self-Try Exercise (Mandatory)

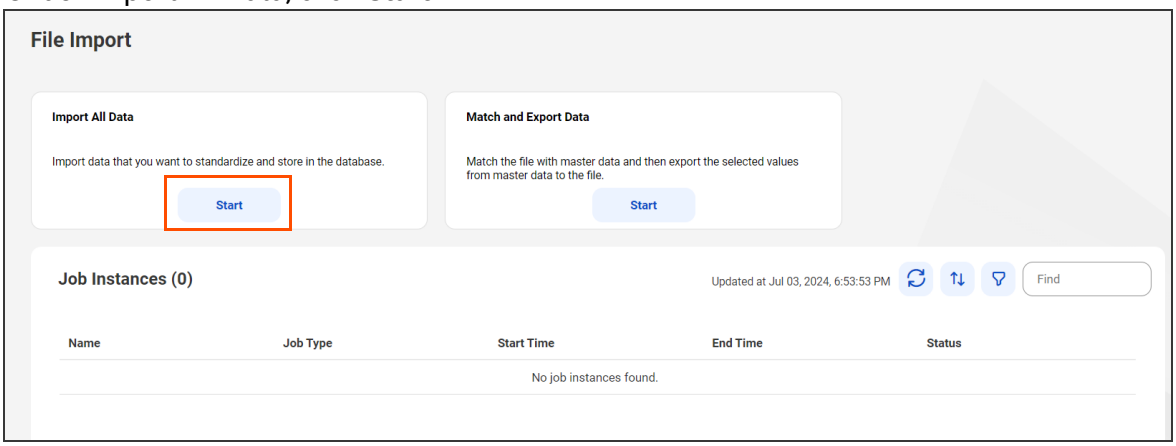
Import Data into the Hierarchy

Now that you know how to how to import data into the business entity. Apply the knowledge to import the data into the hierarchy and relationships.

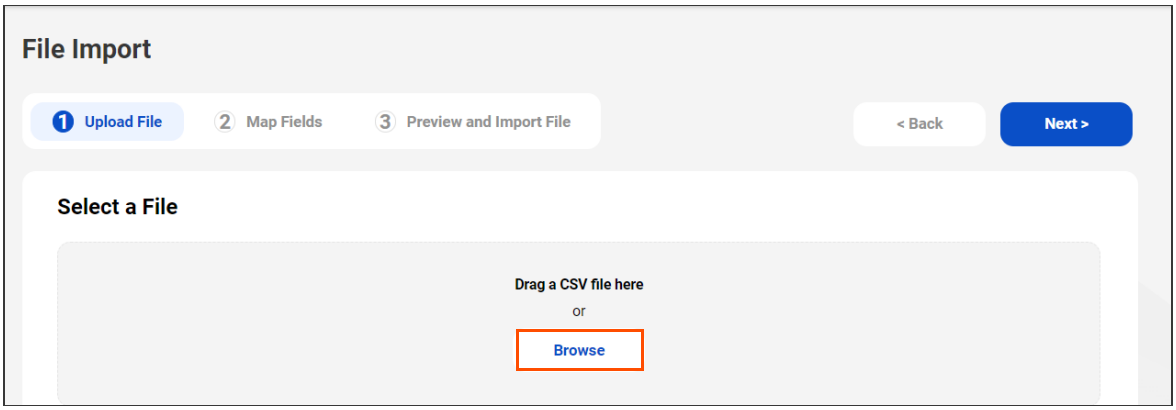
20. Click **File Import**.



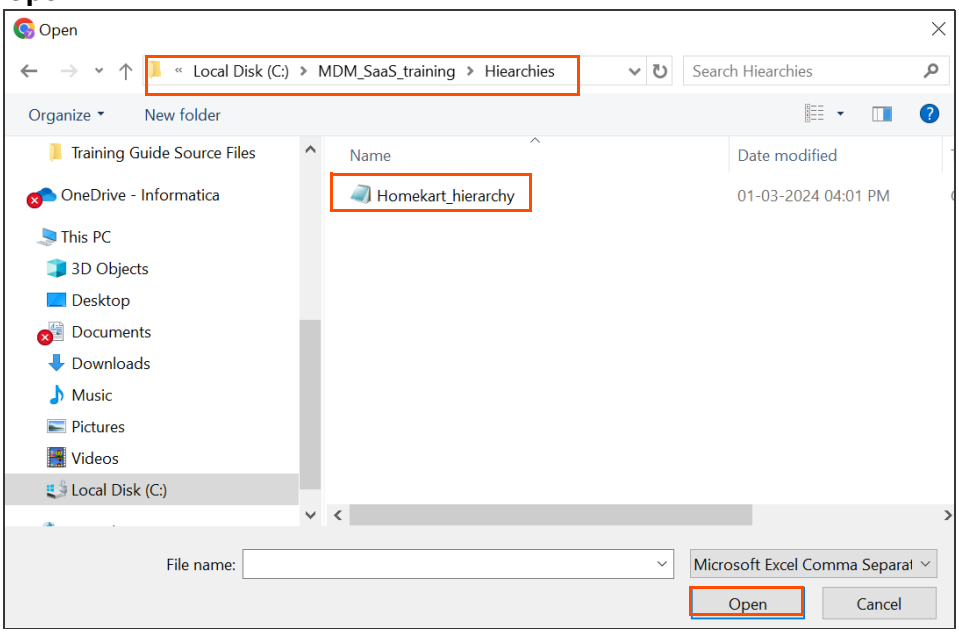
21. Under Import All Data, click **Start**.



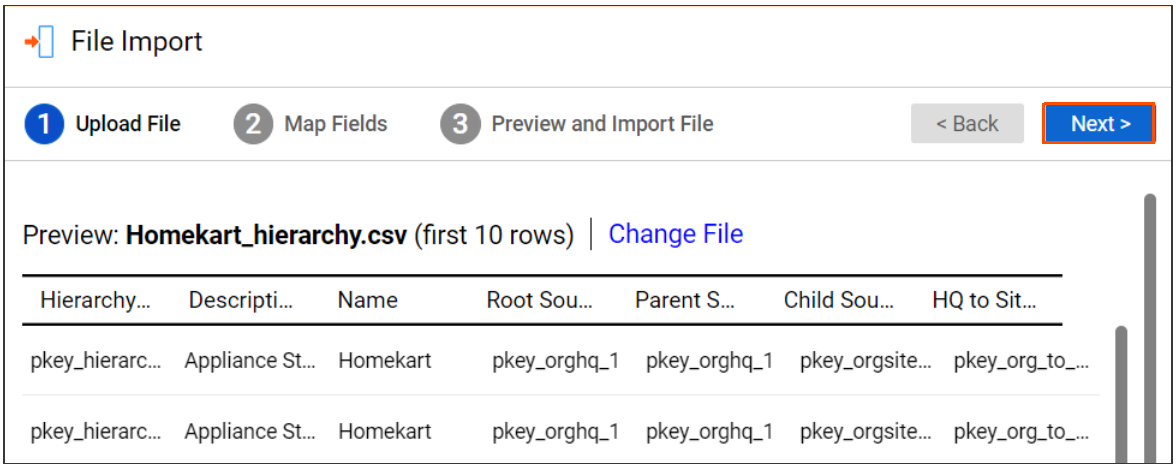
22. Click **Browse**.



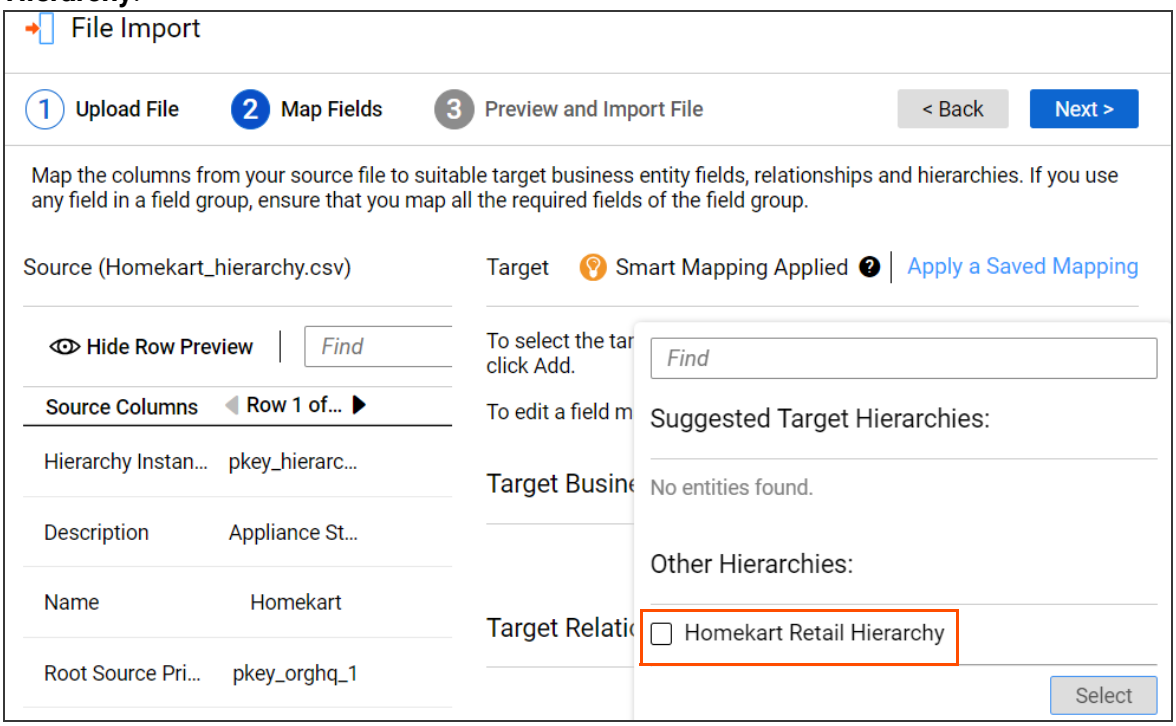
23. Navigate to C:\MDM_SaaS_Training\Hierarchies\ Homekart_hierarchies file and click **Open**.



24. Click **Next**.

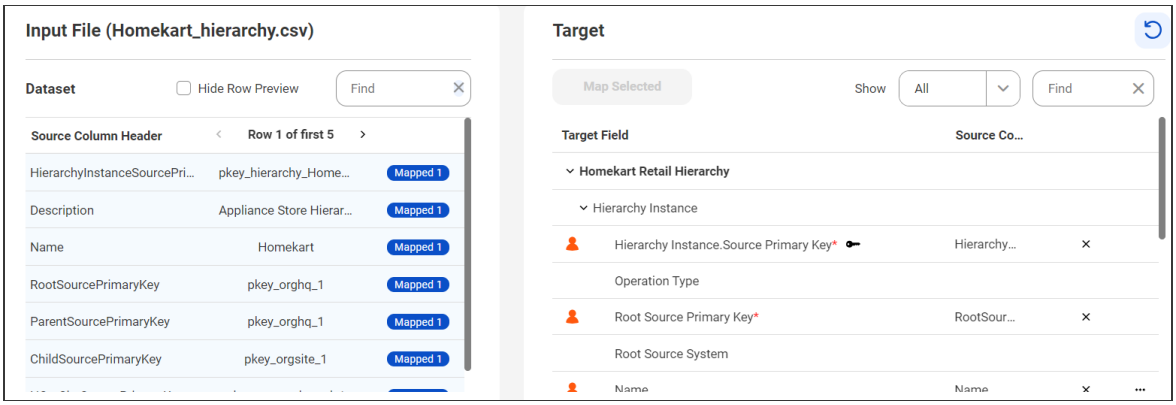


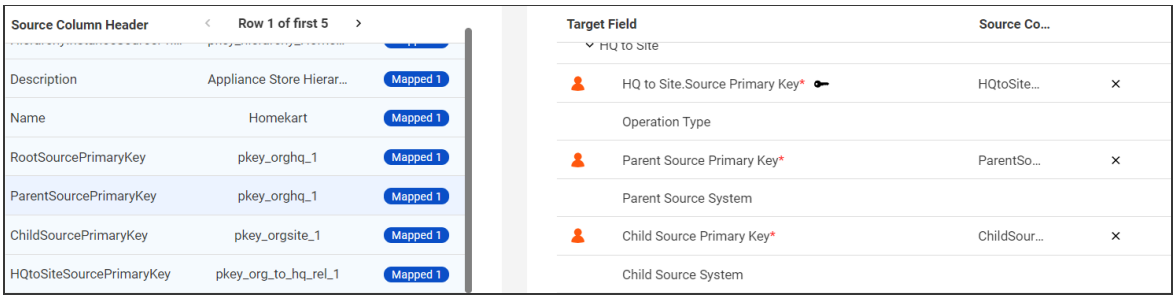
25. In the Map Fields section, Click the + icon for **Target Hierarchies.** Select **Homekart Retail Hierarchy**.



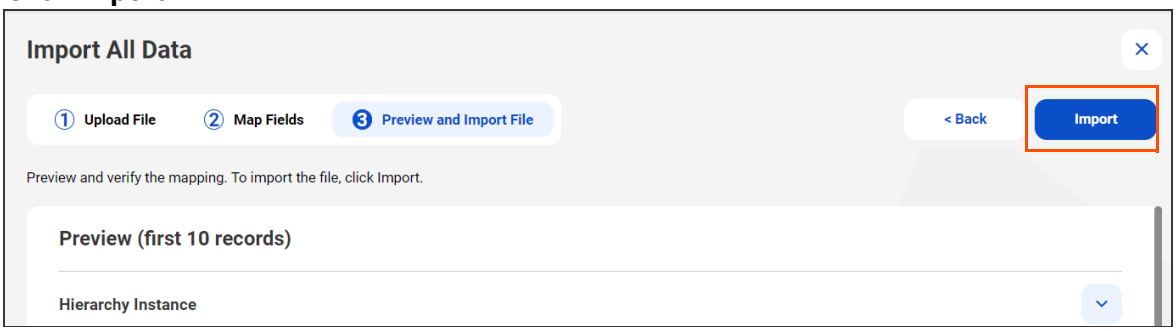
26. Map source and Target fields as below. To do the mapping, **drag** the source fields and **drop** them on to the respective target fields. Refer to the following table to perform the mapping. and click **Next**.

Source Columns	Target Field
Root Source Primary Key*	Root Source Primary Key*
Parent Source Primary Key*	Parent Source Primary Key*
Child Source Primary Key*	Child Source Primary Key*
HQ to Site.Source Primary Key*	HQ to Site.Source Primary Key*
Hierarchy Instance.Source Primary Key*	Hierarchy Instance.Source Primary Key*
Description	Description
Name	Name

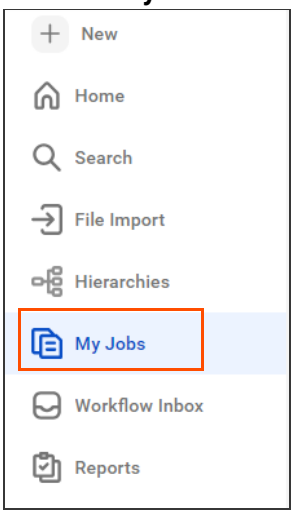




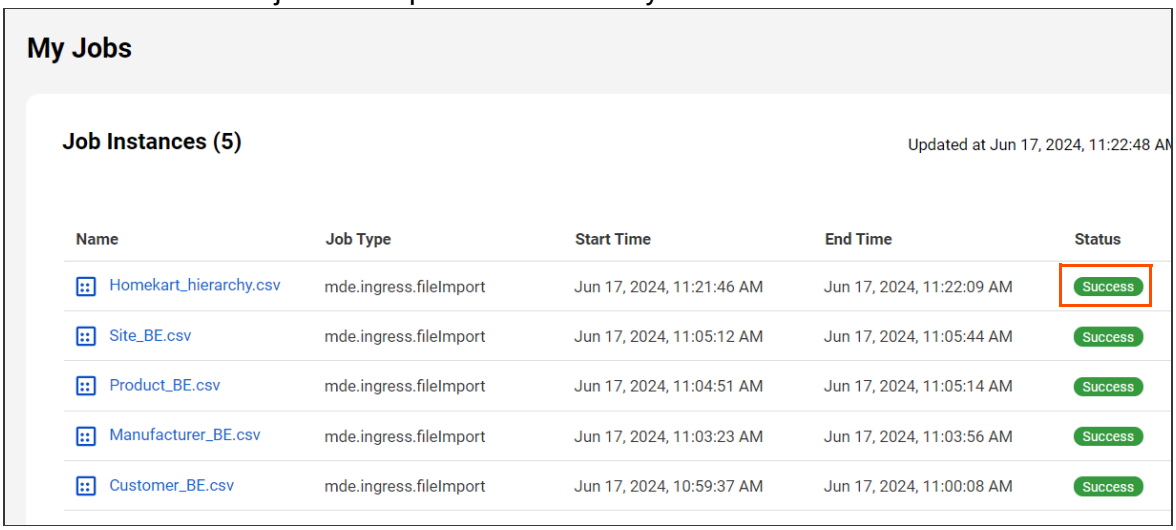
27. Click **Import**.



28. Click on **My Jobs** to see the job status.



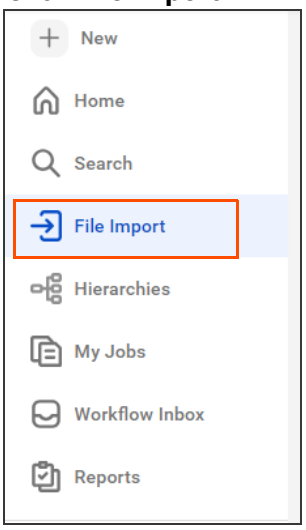
You will see that the job is completed successfully.



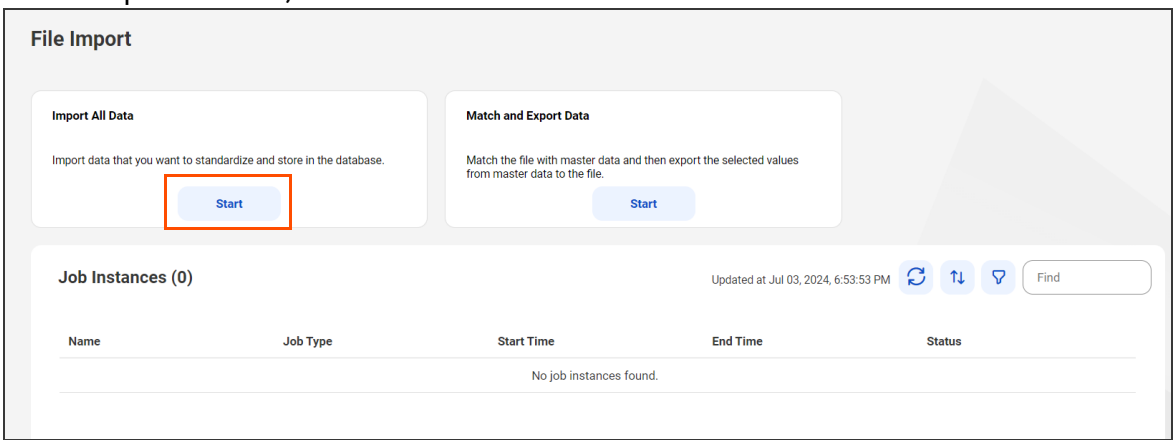
Import Data into the Relationships

Now, you will import the data into the relationships that you created between the business entities.

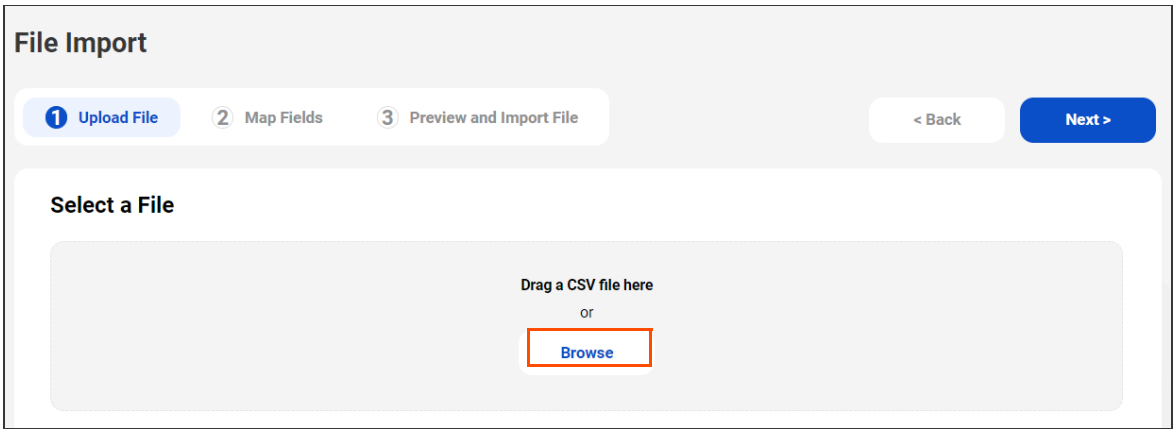
29. Click **File Import**.



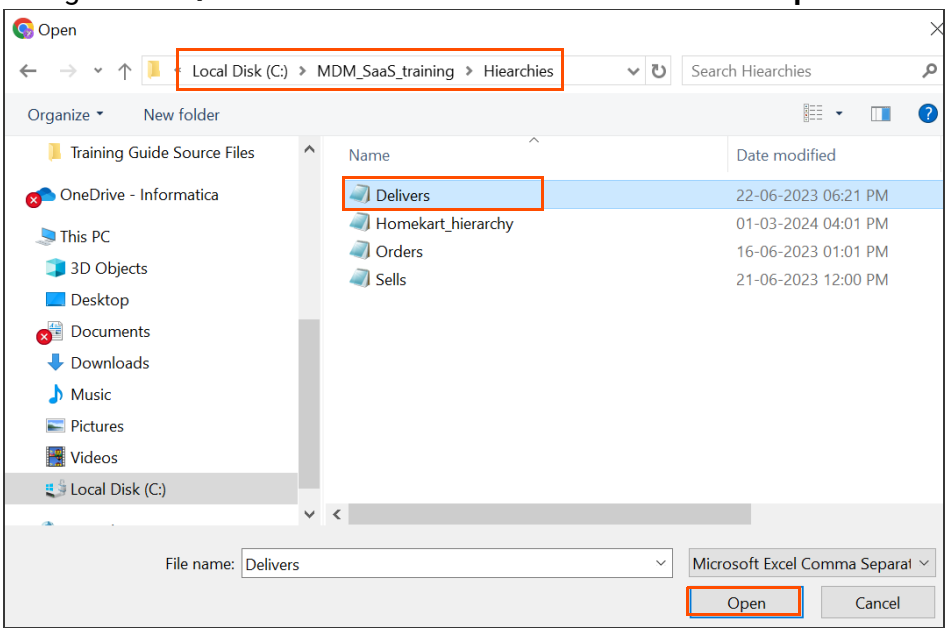
30. Under Import All Data, click **Start**.



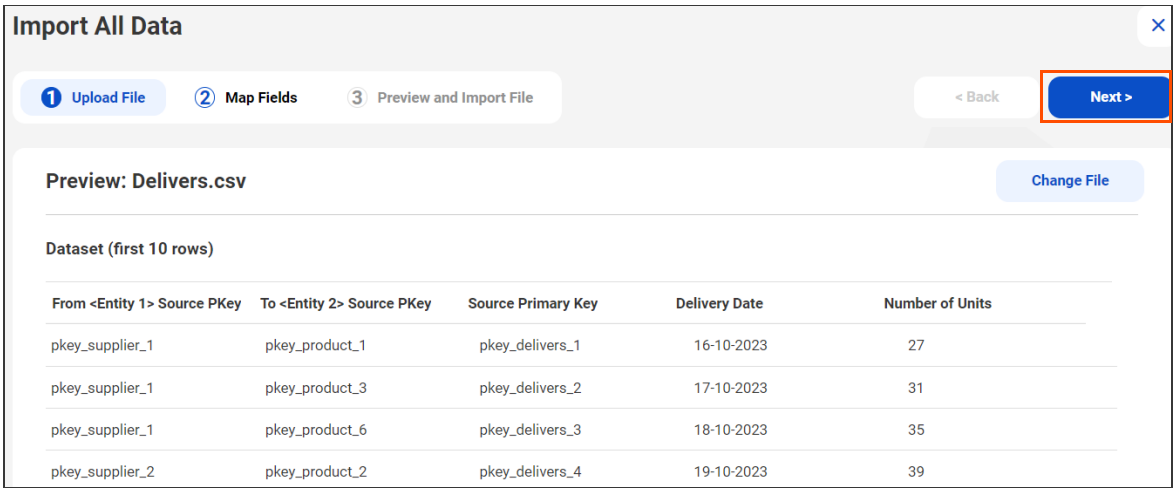
31. Click **Browse**.



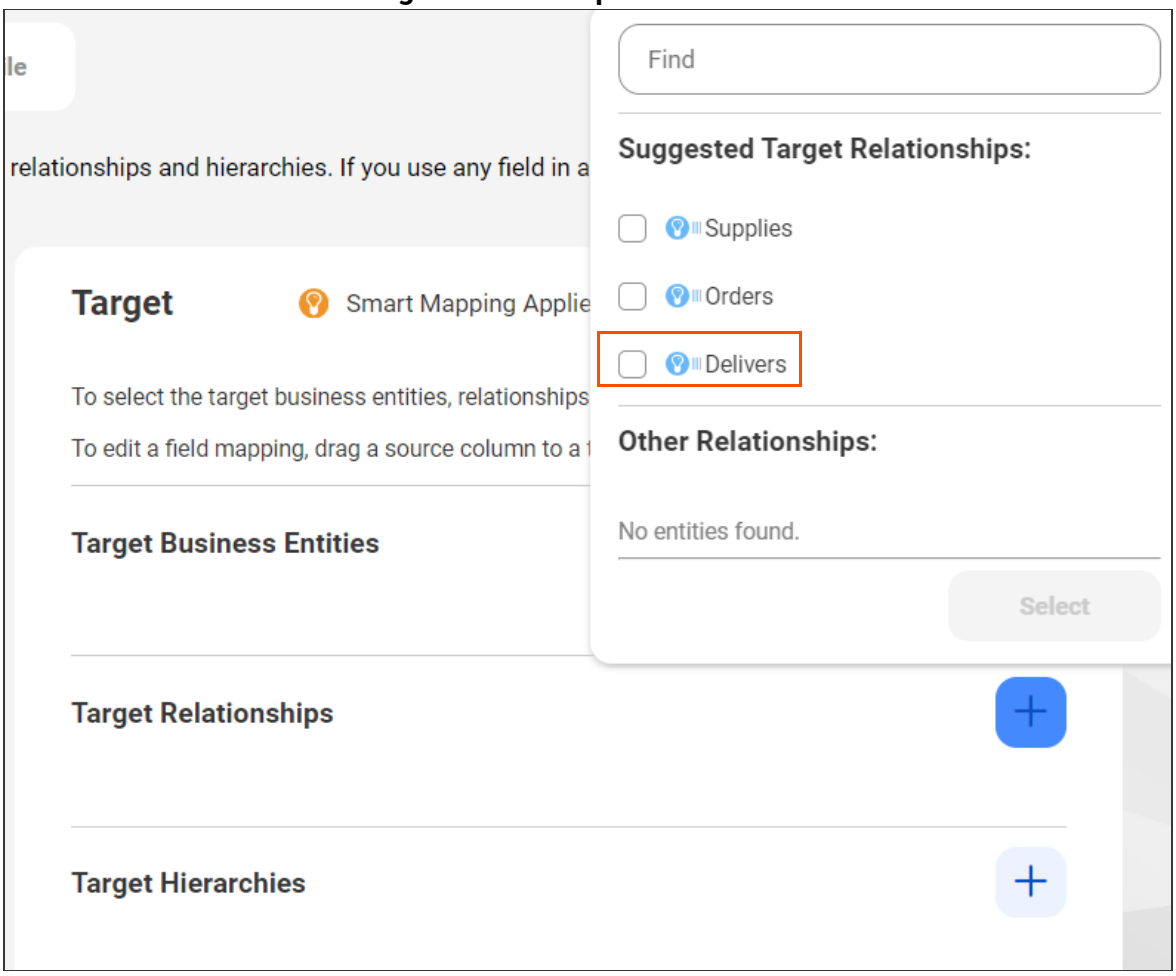
32. Navigate to **C:\Data Files** and select **Delivers.csv** file. Click **Open**.



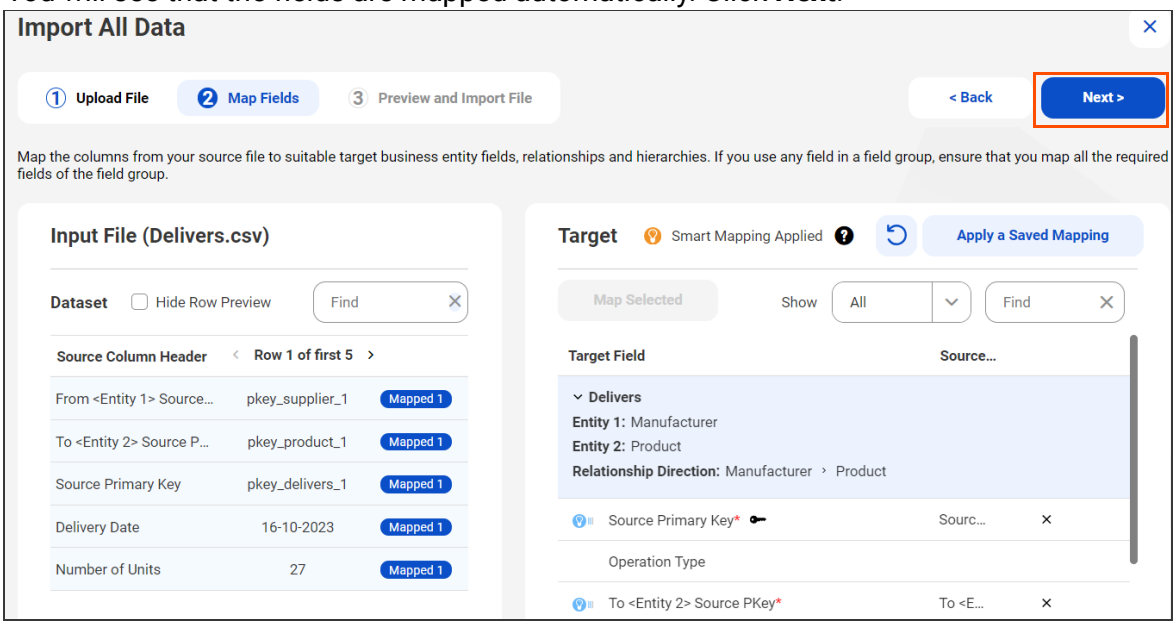
33. Click **Next**.



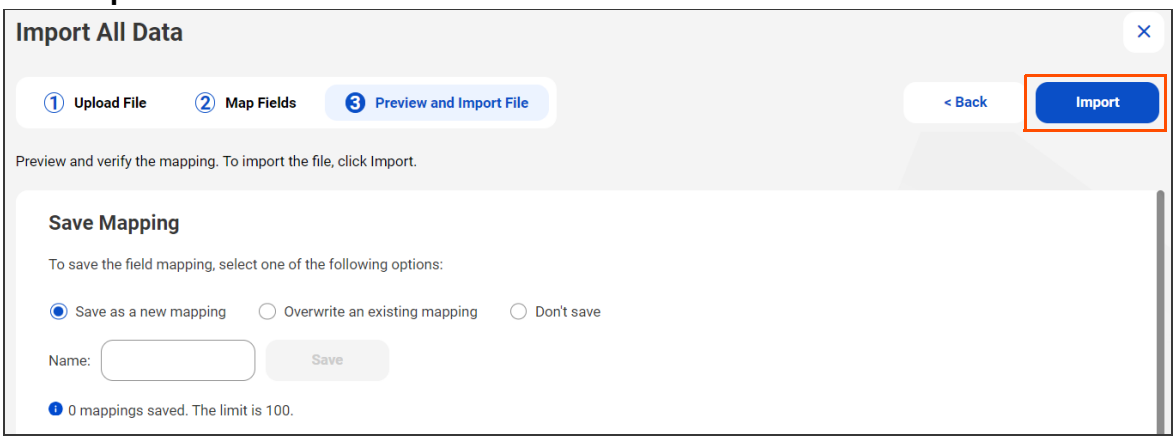
34. Click the **+** icon next to the **Target Relationships** and select **Delivers**.



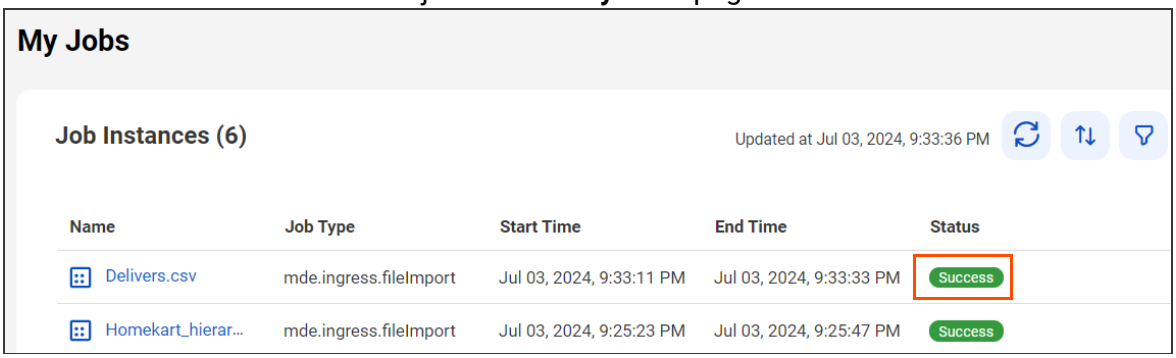
35. You will see that the fields are mapped automatically. Click **Next**.



36. Click **Import**.



37. You can check the status of the job onn the **My Jobs** page.



Self-Try Exercise (mandatory)

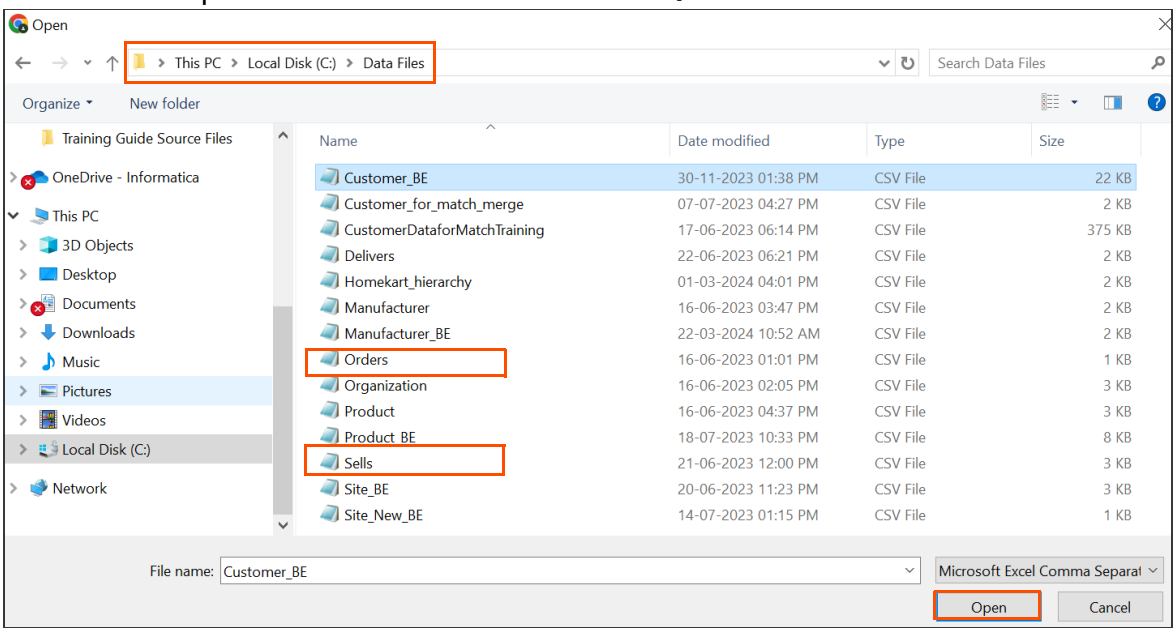
Import Data into **Orders** and **Sells** relationships.

High-level steps:

Follow the import steps and import **Orders.csv and Sells.csv** files into Multidomain MDM.

- File location for **Orders**: C:\MDM_SaaS_Training\Hierarchies\Orders.csv.
- File location for **Sells:** C:\MDM_SaaS_Training\Hierarchies\Sells.csv.

38. Choose the respective files from the file location **C:\Data Files**.



39. Select the Target Relationships as **Orders** and **Sells** receptively and complete the import steps.

40. Check **My Jobs** to check the status of the import job.

This concludes the lab.

Lab 9-5: Ingressing Data From SQL Server

Overview:

Ingress is a process that imports data from external source systems, masters the data, and stores it in Business 360 assets. When you get started with Business 360 for the first time, to onboard your initial data, you must ingress it. You can continue to ingress the incremental data to be in sync with the source systems.

For example, you work for an enterprise that manages customer data in Salesforce. You subscribe to Business 360 to master the customer data in Salesforce. To get started with Business 360, ingress the initial data from Salesforce. You can continue to ingress the incremental data to Business 360 so that the data in Business 360 is in sync with the Salesforce data.

In this lab, you will import customer records from SQL Server.

Objectives:

- Create a new project
- Create a Data Integration mapping
- Create a mapping task
- Create a taskflow
- Create and monitor an Ingress job in Business 360

Scenario:

Mark, representing HomeKart, has identified customer records within an SQL Server table that he wants to onboard into the Informatica Multidomain MDM data store. To achieve this, Mark sets up an Ingress job to transfer data from SQL Server. Moving forward, whenever new customers are added to SQL Server, all Mark needs to do is re-run the Ingress job to keep the data updated.

Duration:

40 minutes

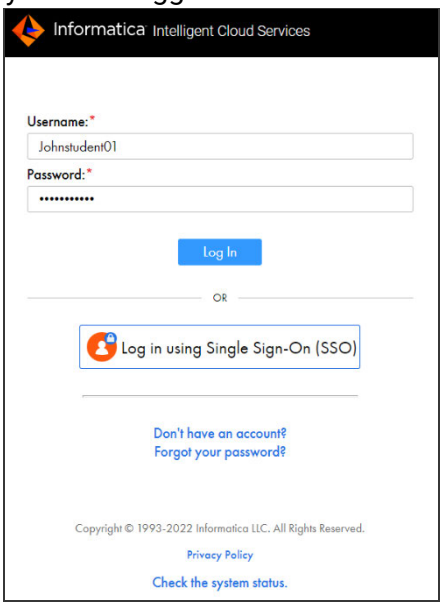
Tasks:

- Create a DI Mapping
- Add an expression transformation
- Configure the target connection
- Create In-Out parameter
- Create a mapping task
- Create a taskflow
- Create ingress job in Business 360 Console
- View the imported records in Multidomain MDM

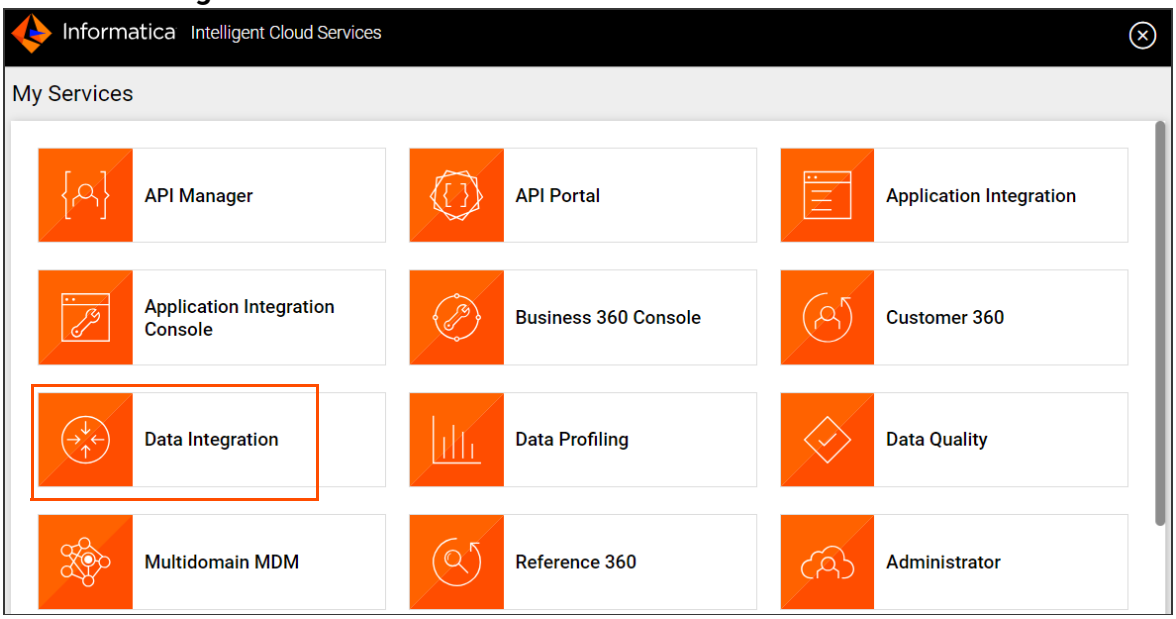
Create a DI Mapping

As the first step, you will create a mapping in Data Integration service.

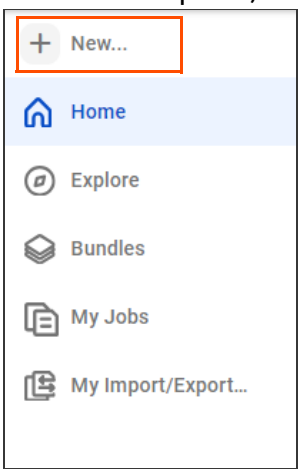
1. Log in to IICS with your credentials as **<studentNamestudent<no.> / Multidomain360**), if you have logged out.



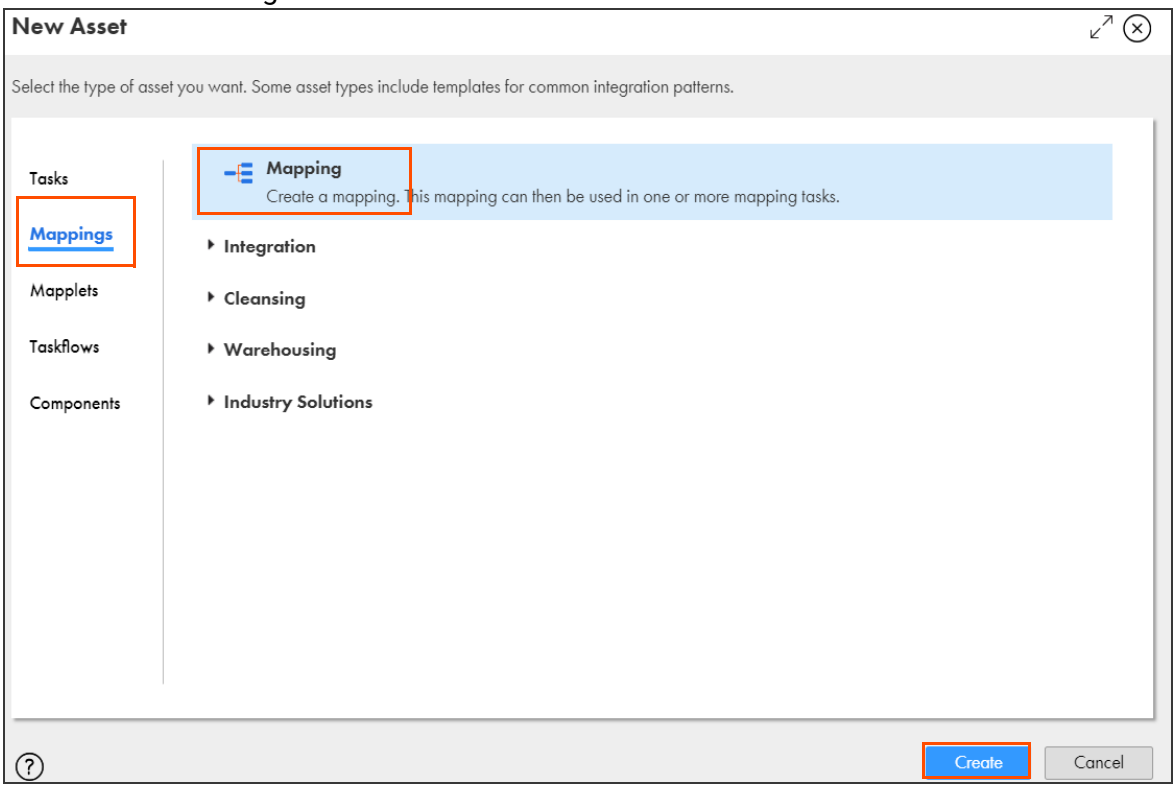
2. Click **Data Integration**.



3. From the left panel, click **New**.

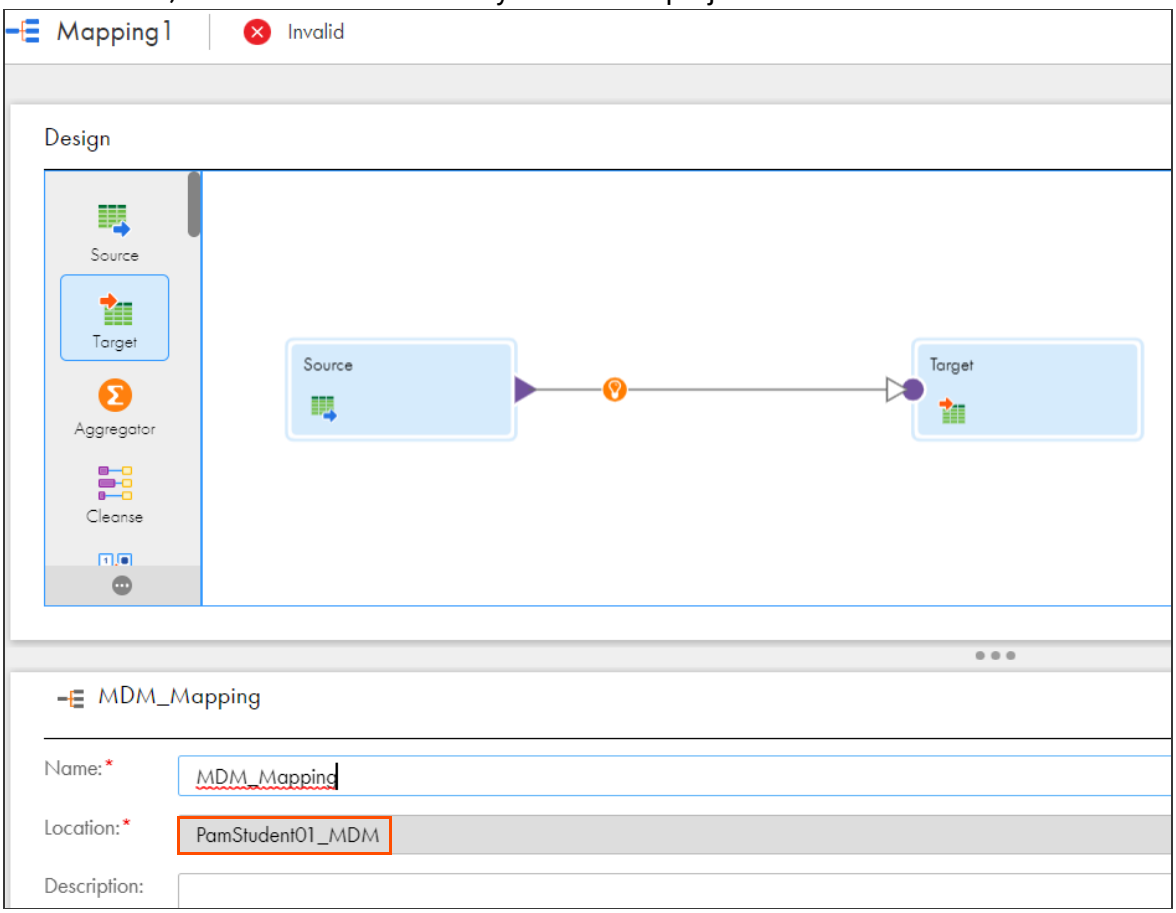


4. In the New Asset window, select **Mappings** > **Mapping**, and click **Create**. This will open a new mapping designer page. When you configure a mapping, you describe the flow of data from source and target.

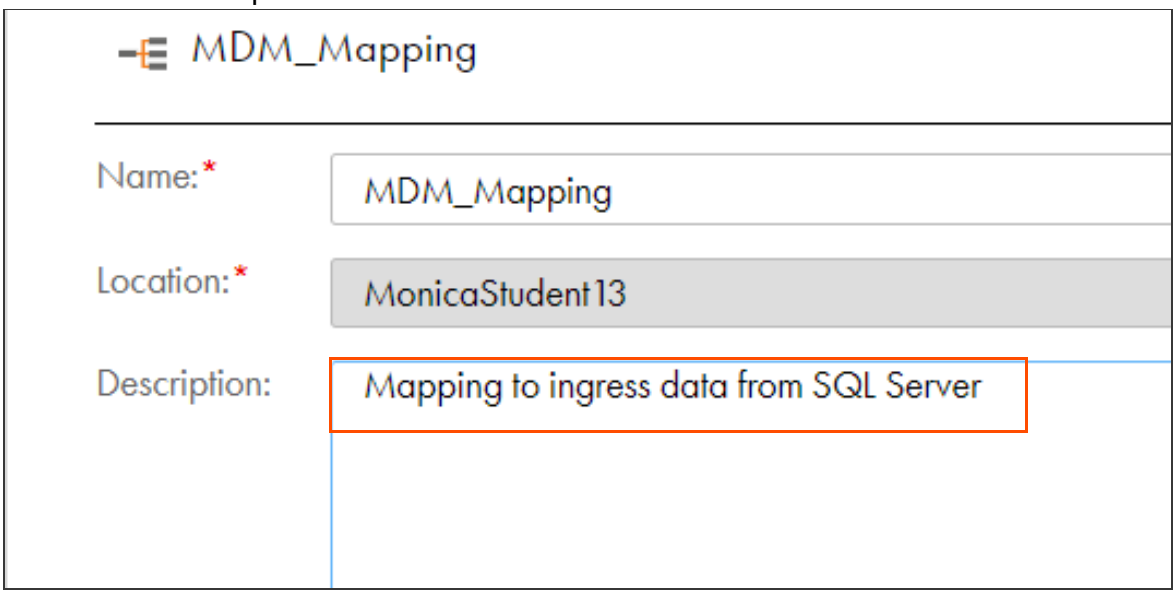


5. The mapping canvas opens with default Source and Target assets in the canvas. In the Properties section, enter the mapping name as **MDM_Mapping**.

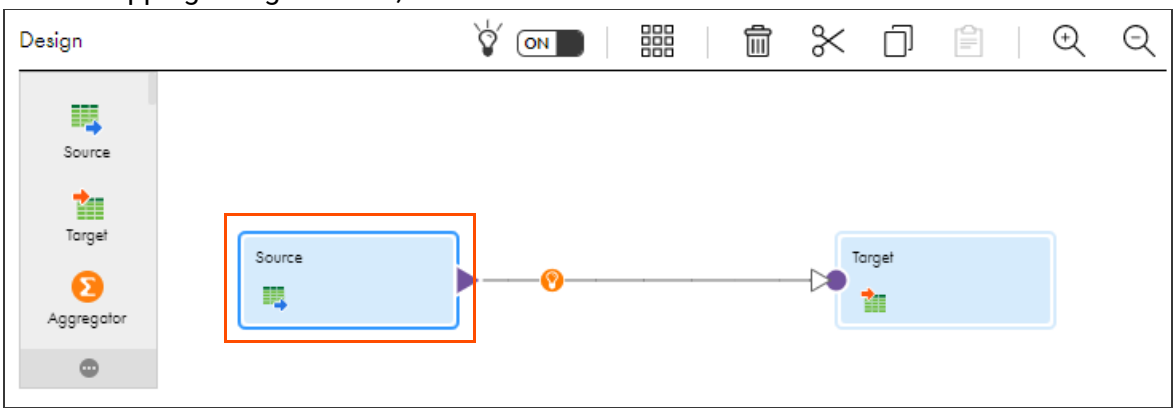
6. For **Location**, click **Browse** and select your student project.



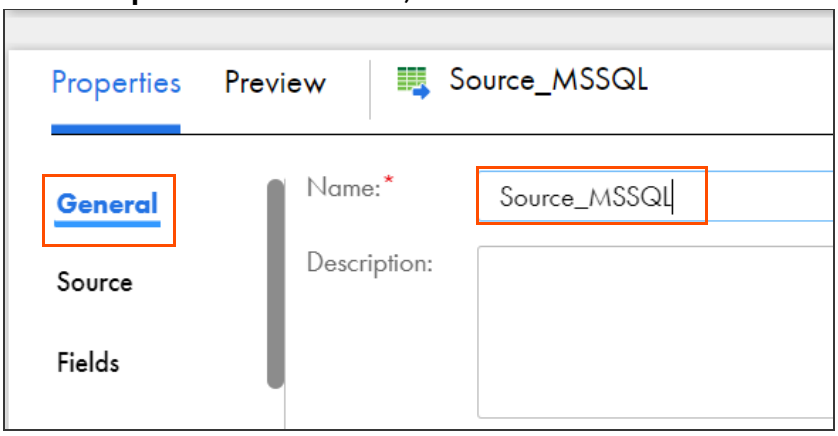
7. Enter a valid description.



8. In the Mapping Design canvas, select the **Source**.

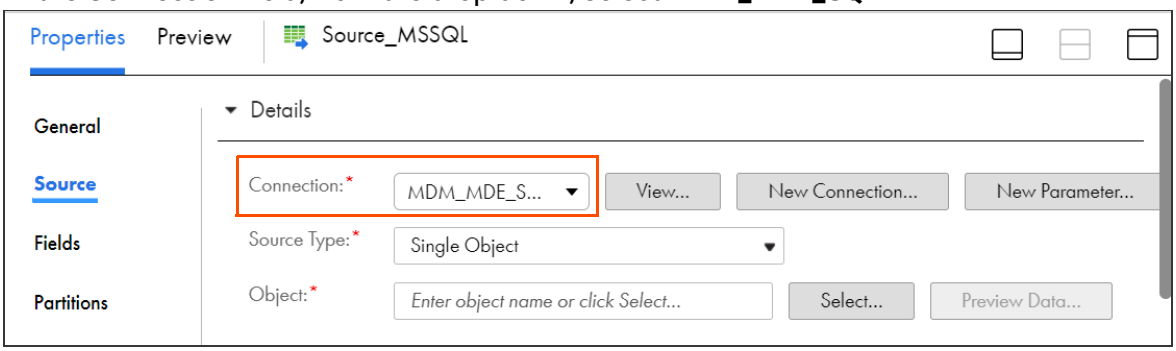


9. In the **Properties** > **General** tab, name the source transformation as **Source_MSSQL**.

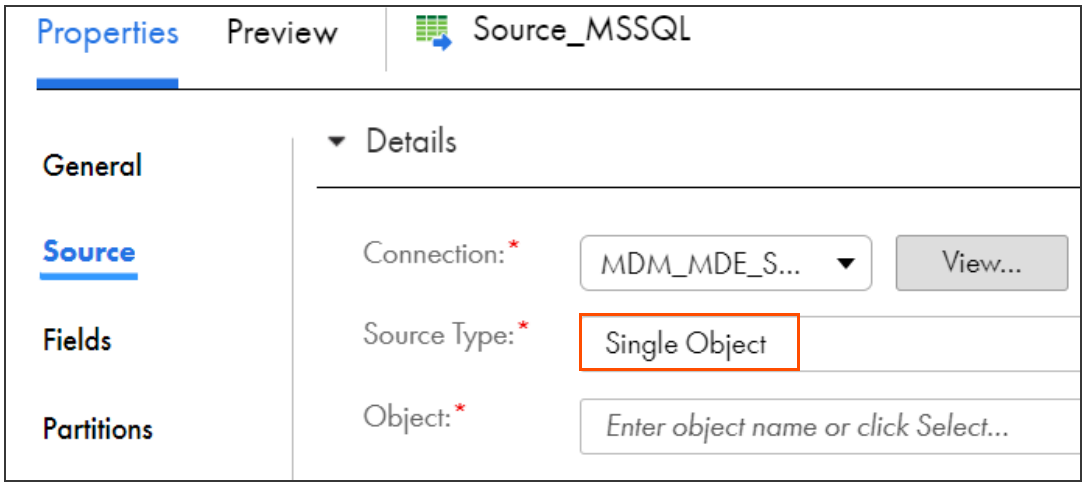


10. Click the **Source** tab.

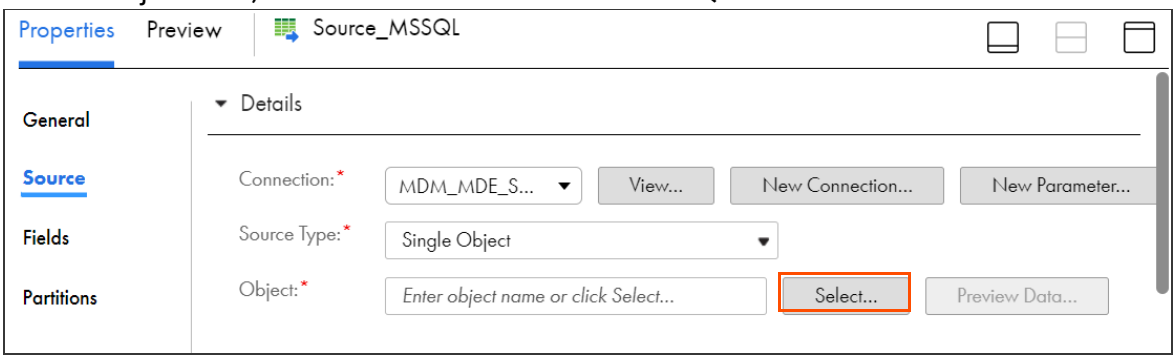
11. In the **Connection** field, from the drop-down, select **MDM_MDE_SQL**.



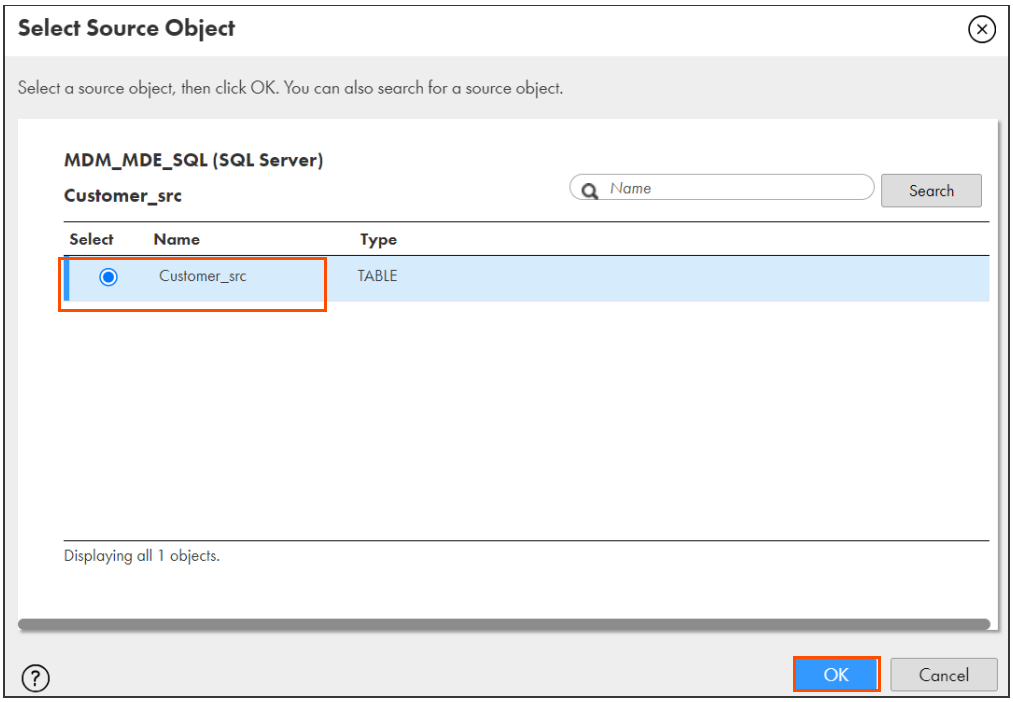
12. In this lab, you will use only one sql table. Hence, retain the Source Type as **Single Object**.



13. For the Object field, click **Select** button to select the SQL Server table.

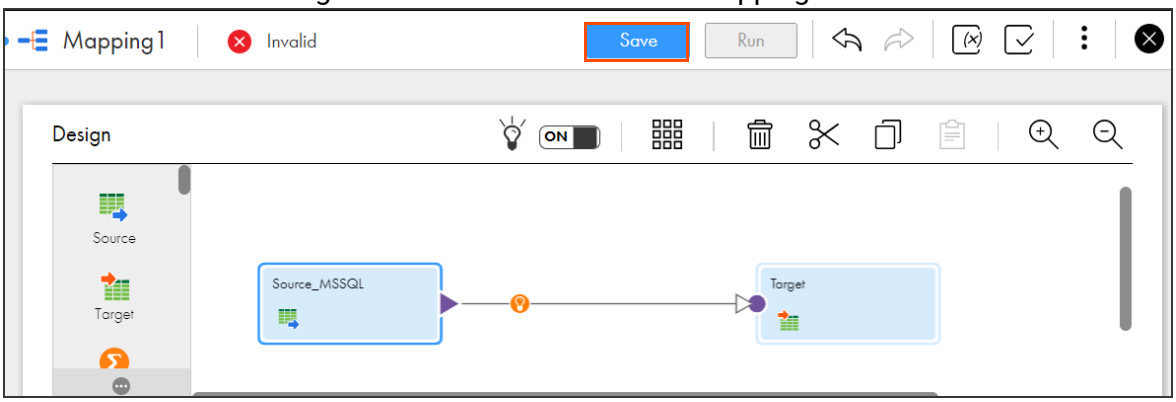


14. Select the **Customer_src** table and click **OK**.

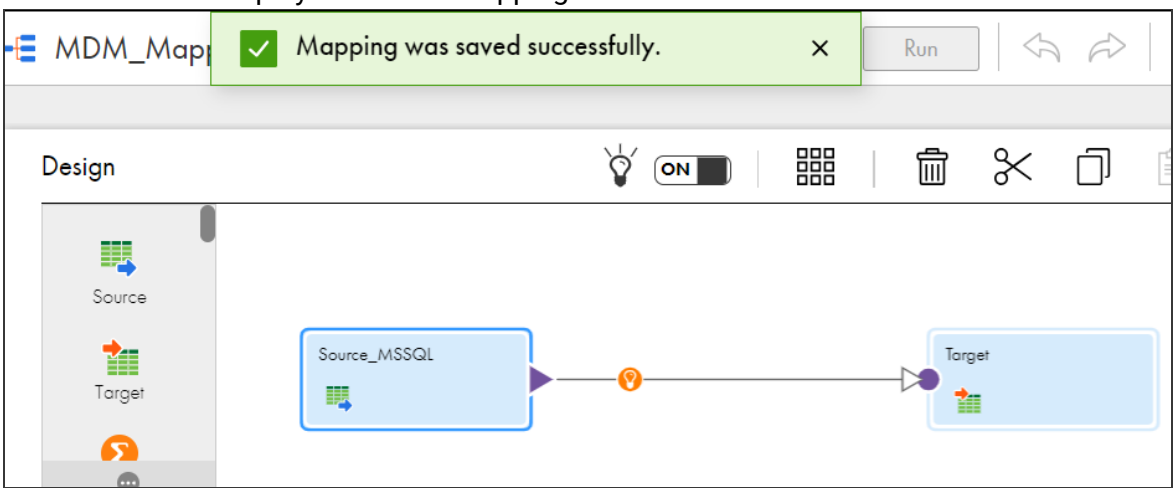


Number of tables may vary from screenshot.

15. The source is now configured. Click **Save** to save the mapping.



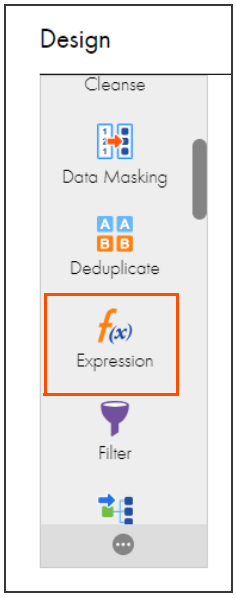
A notification is displayed that the mapping is saved.



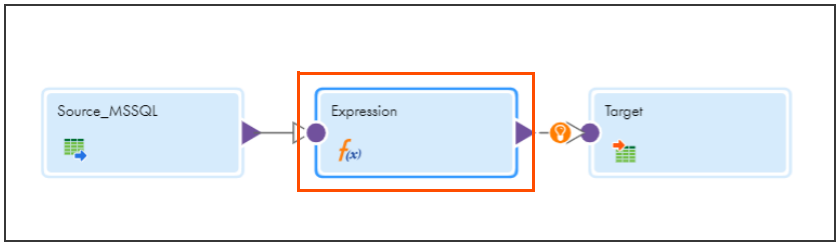
Add an Expression Transformation

An Expression transformation should be included here to notify that the source is SQL Server.

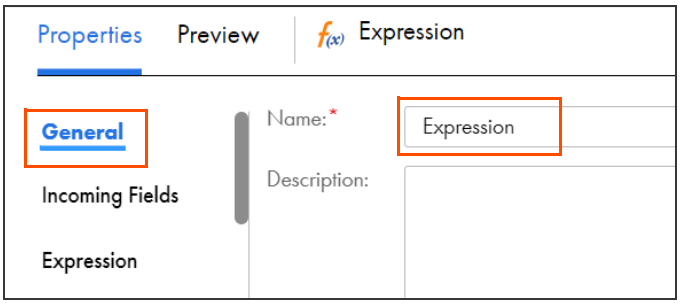
16. In the Design window, select the **Expression** transformation from the list.



17. Drag and drop it between the Source and the Target.

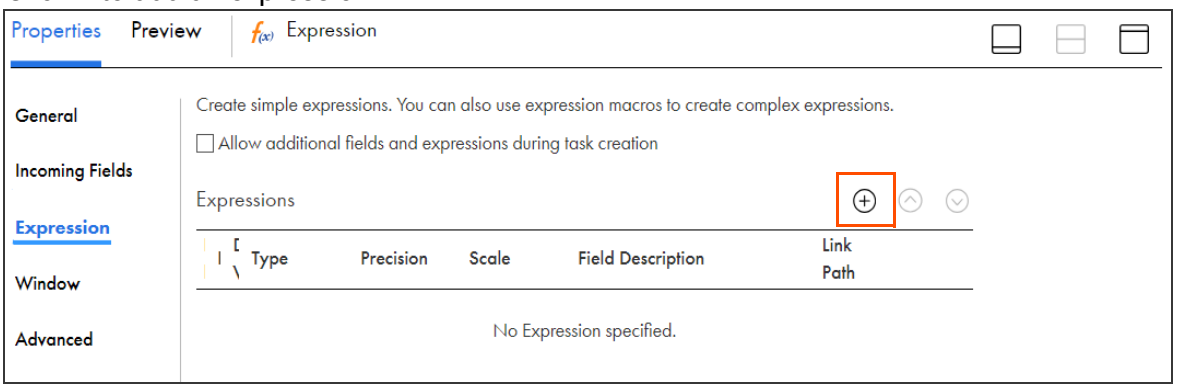


18. Click the **Expression** transformation in the mapping canvas. In the **Properties** > **General** tab, retain the Name as **Expression**.



19. Click the **Expression** tab.

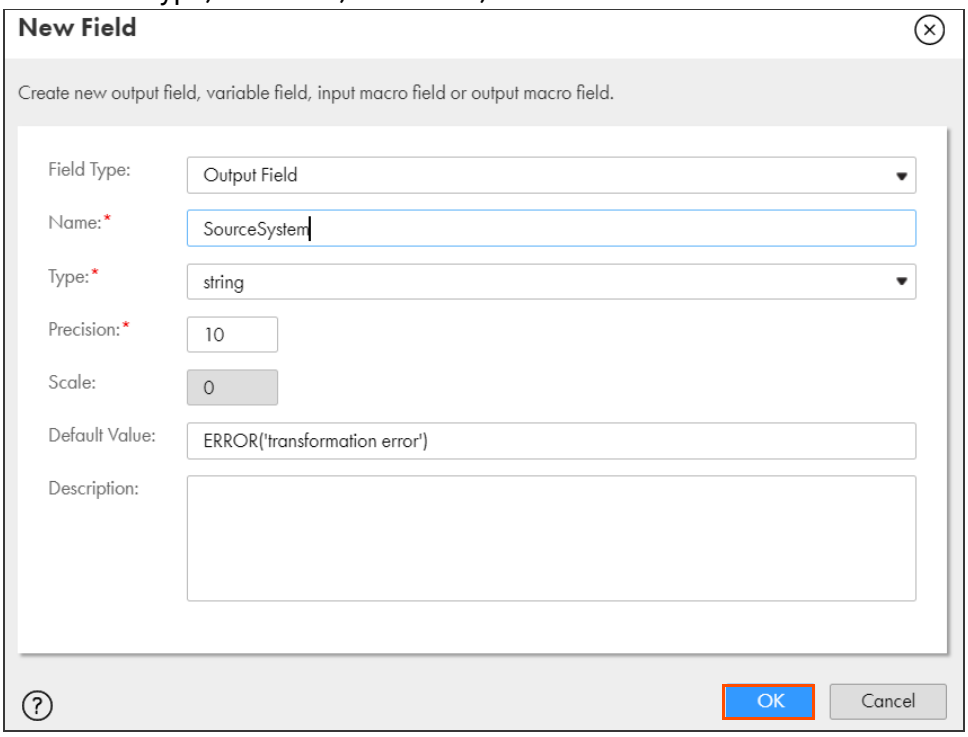
20. Click **+** to add an expression.



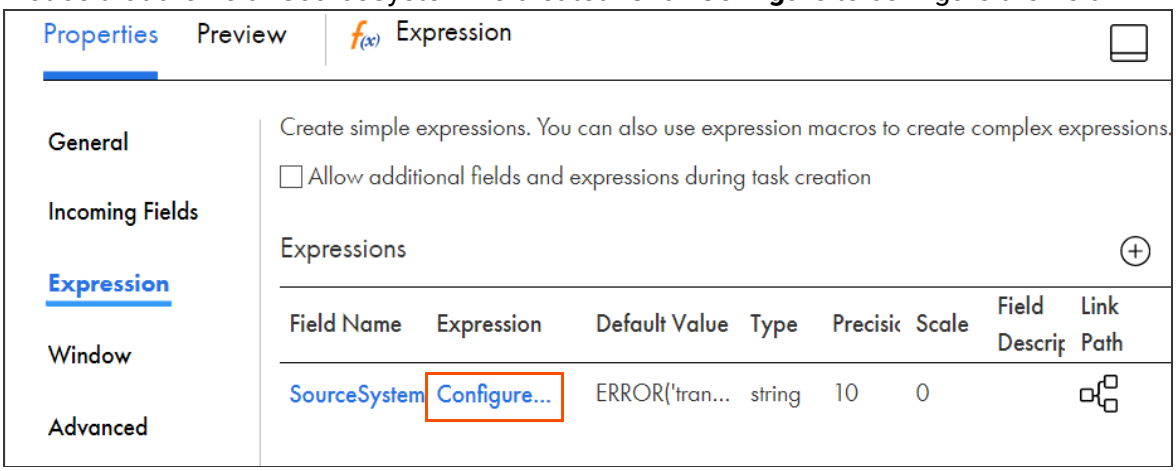
21. Here, you will create a field which notify that the source is SQL Server, also, it acts as the output field for the results. Retain the Field Type as **Output Field**.

22. Replace the Name with **SourceSystem**.

23. Retain the Type, Precision, and Scale, and click **OK**



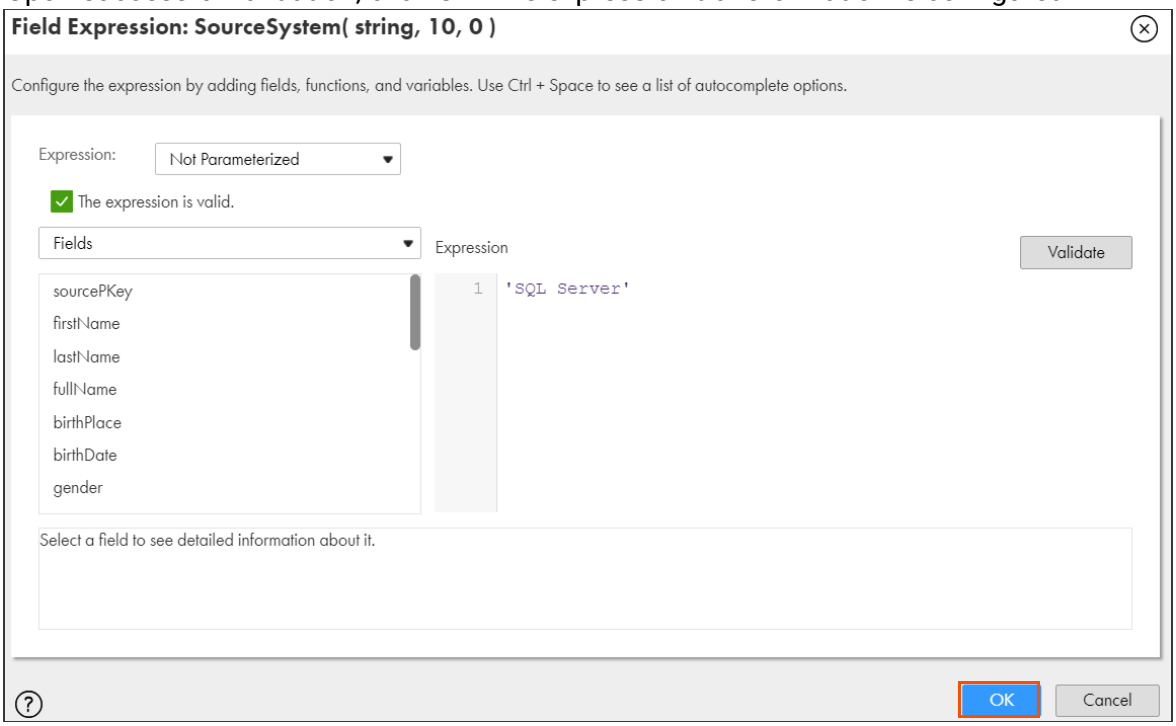
24. Notice that the field "SourceSystem" is created. Click **Configure** to configure the field.



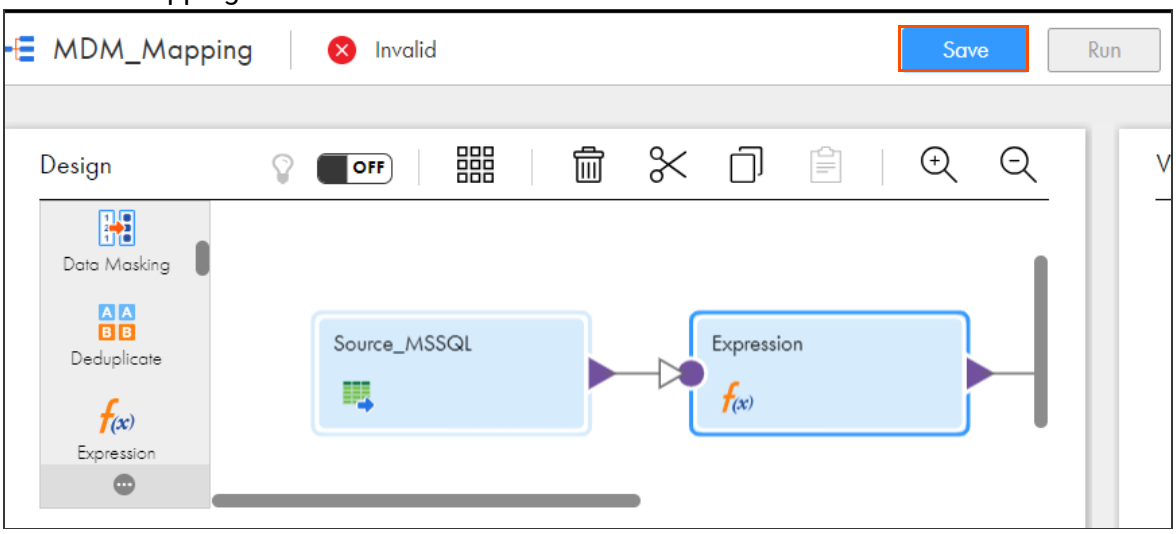
25. Enter the expression string as **'SQL Server'**. Type the string to avoid any validation errors.

26. Click the **Validate** button to validate the expression.

27. Upon successful validation, click **OK**. The expression transformation is configured.



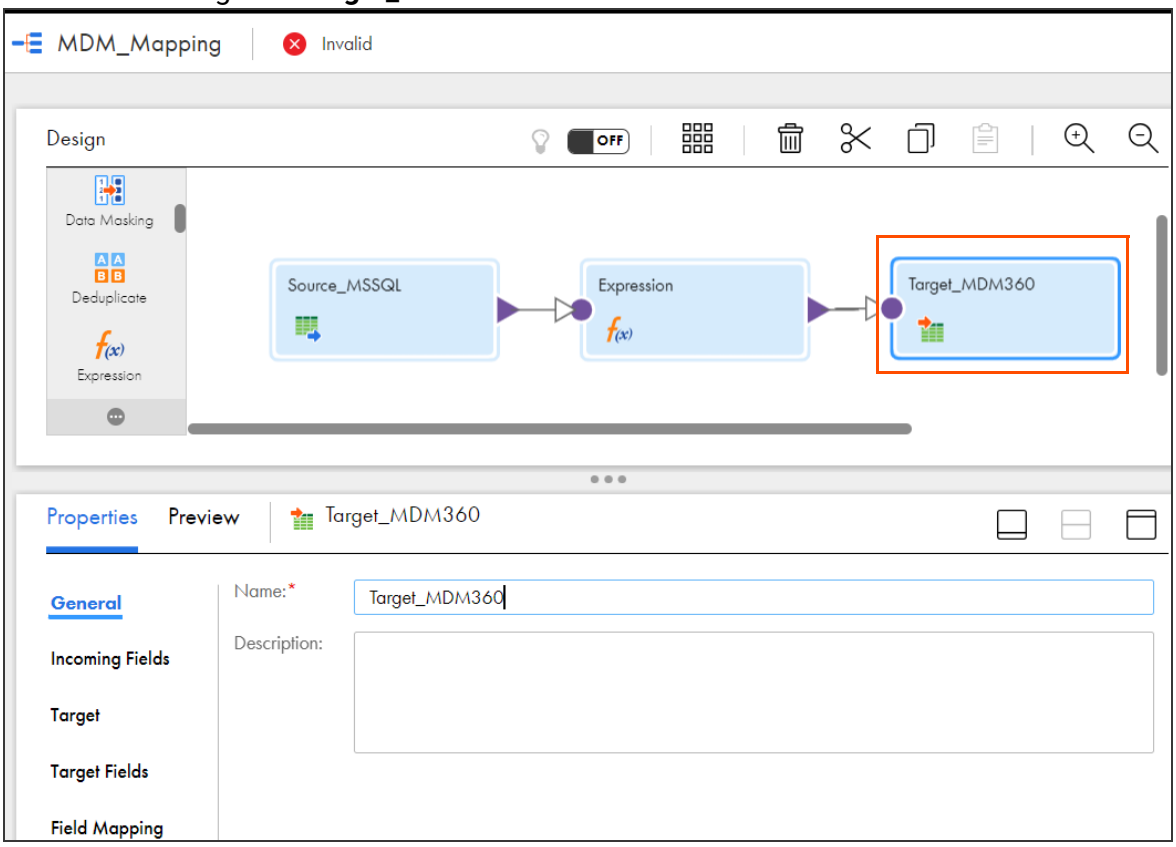
28. **Save** the mapping.



Configure the Target Connection

Now, you will configure the target connection from which you will be ingressing the data.

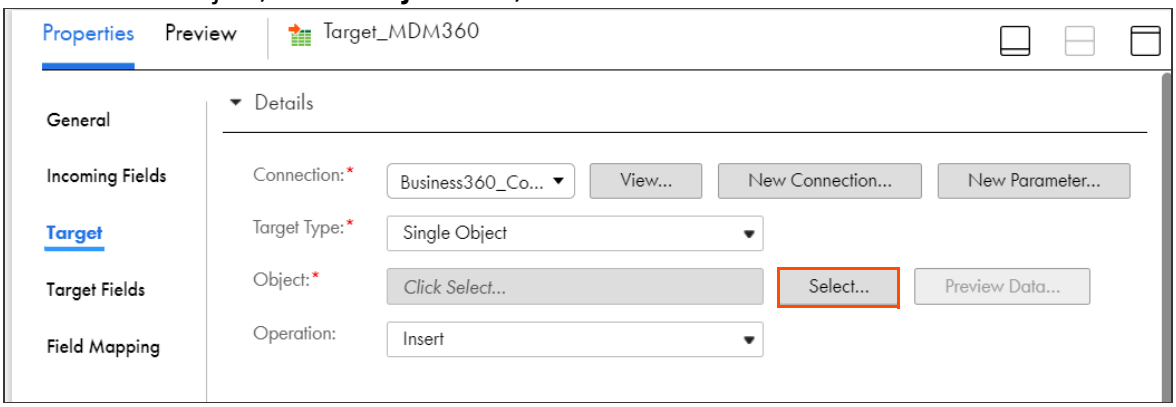
29. Select the **Target** in the mapping canvas. From the **Properties** > **General** tab, enter the Name of the Target as **Target_MDM360**.



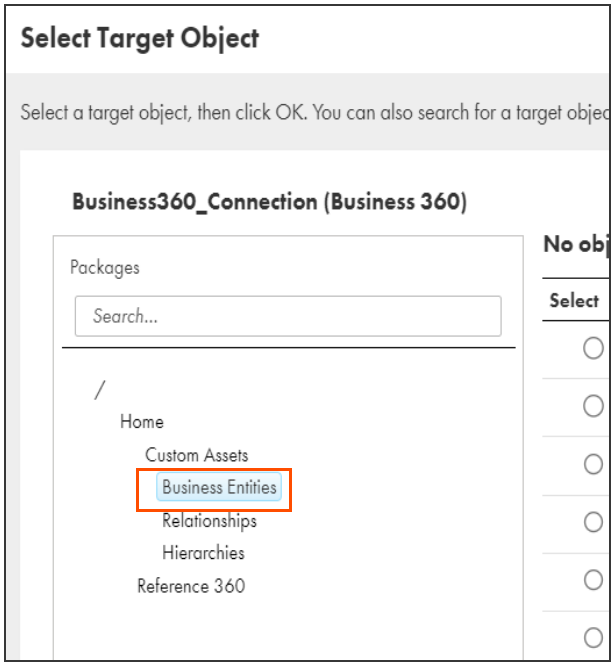
30. Click the **Target** tab.

31. In the Connection field, from the drop-down, select your Business 360 connection **Business360_Connection**.

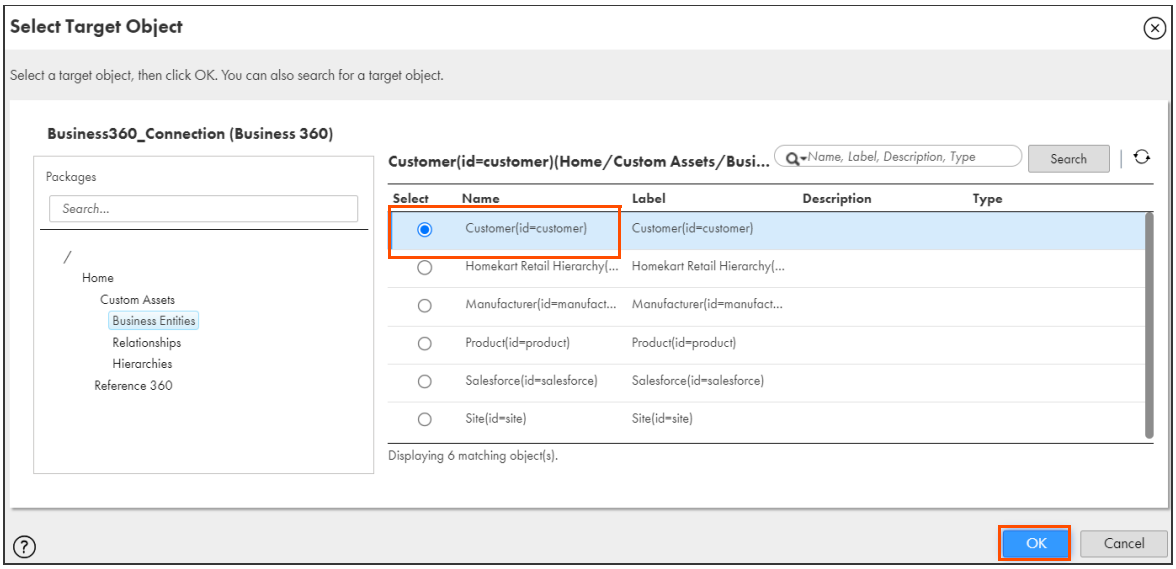
32. To select the object, for the **Object** field, click the **Select** button.



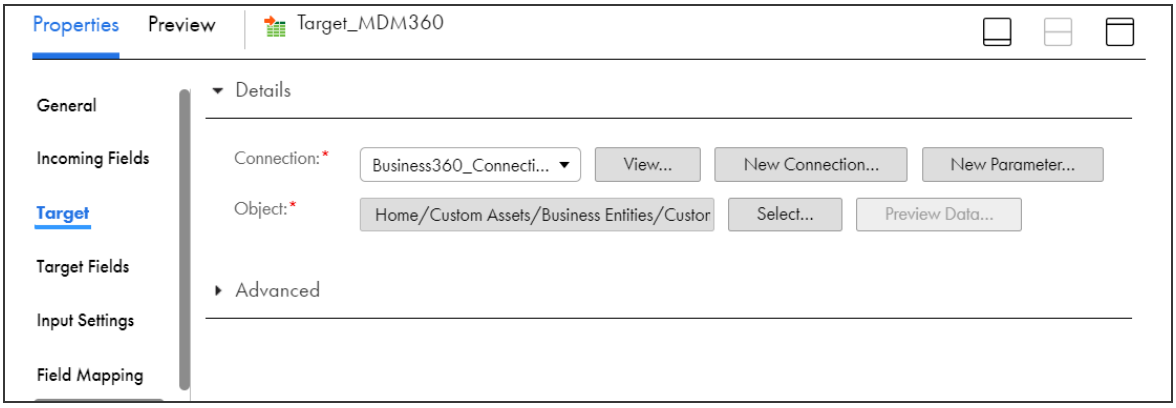
33. Click **Home** > **Custom Assets** and select **Business Entities**.



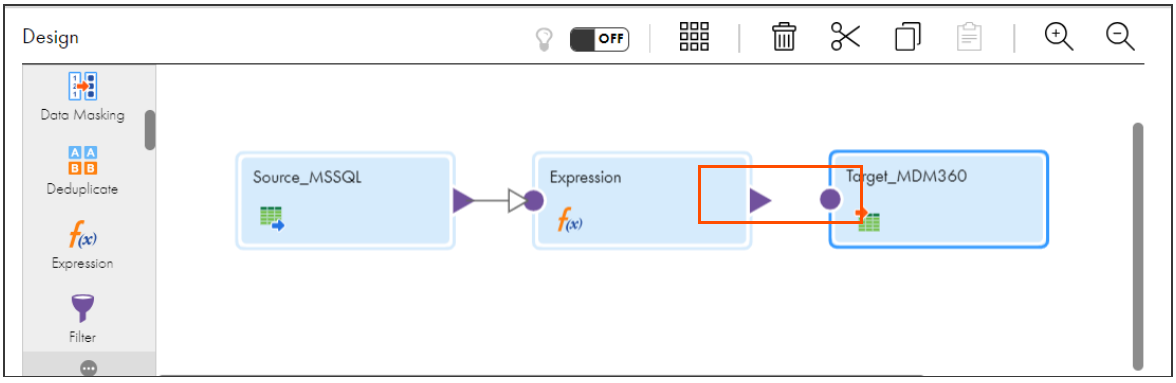
34. Select **Customer** and click **OK**.

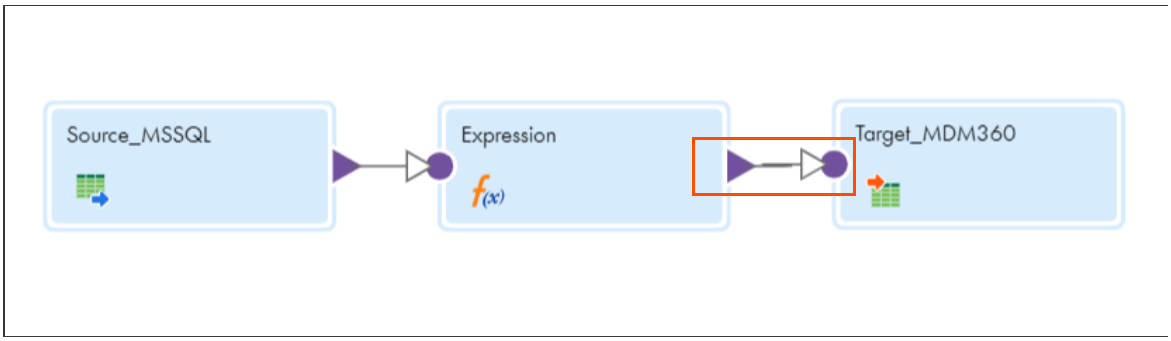


Data will be written to the Customer object.

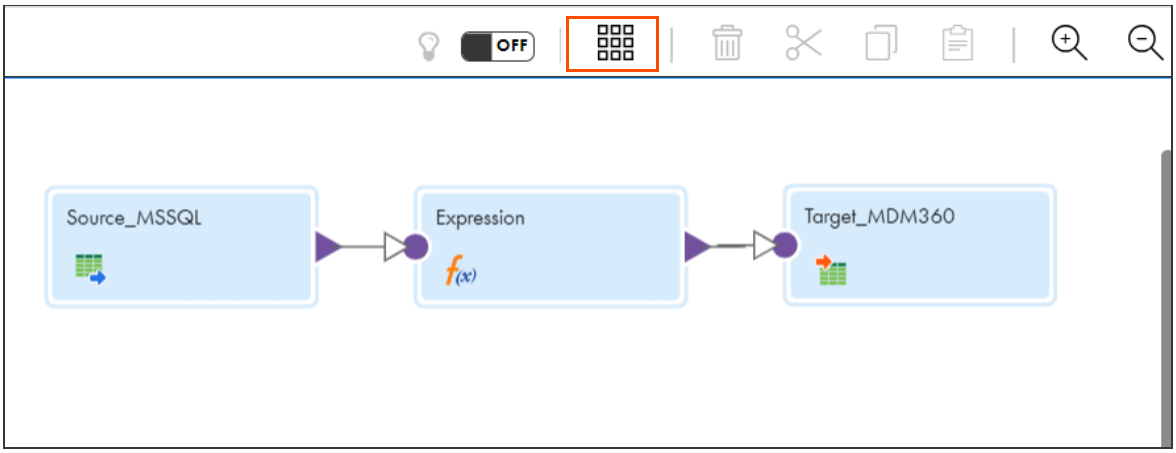


35. In the mapping canvas, notice that the connections are disconnected either between Source to Expression Transformation or from Expression Transformation to Target. Connect them, as needed.



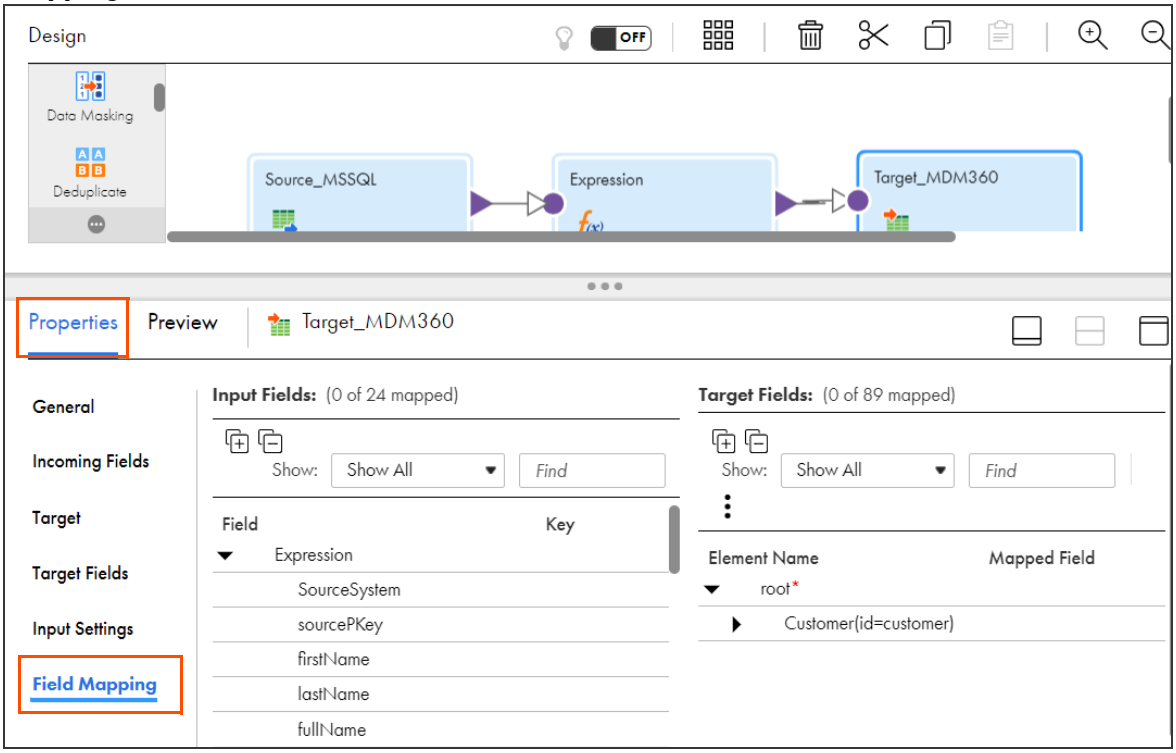


Click the **Arrange All** icon on the top right side of the canvas to arrange the objects in the mapping.

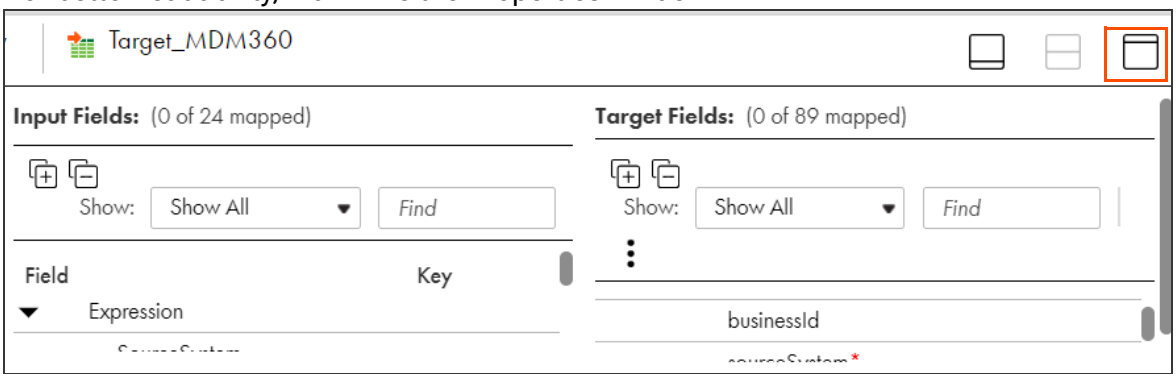


36. **Save** the mapping.

37. Click the target in the mapping canvas, and from the **Properties** section, click the **Field Mapping** tab.



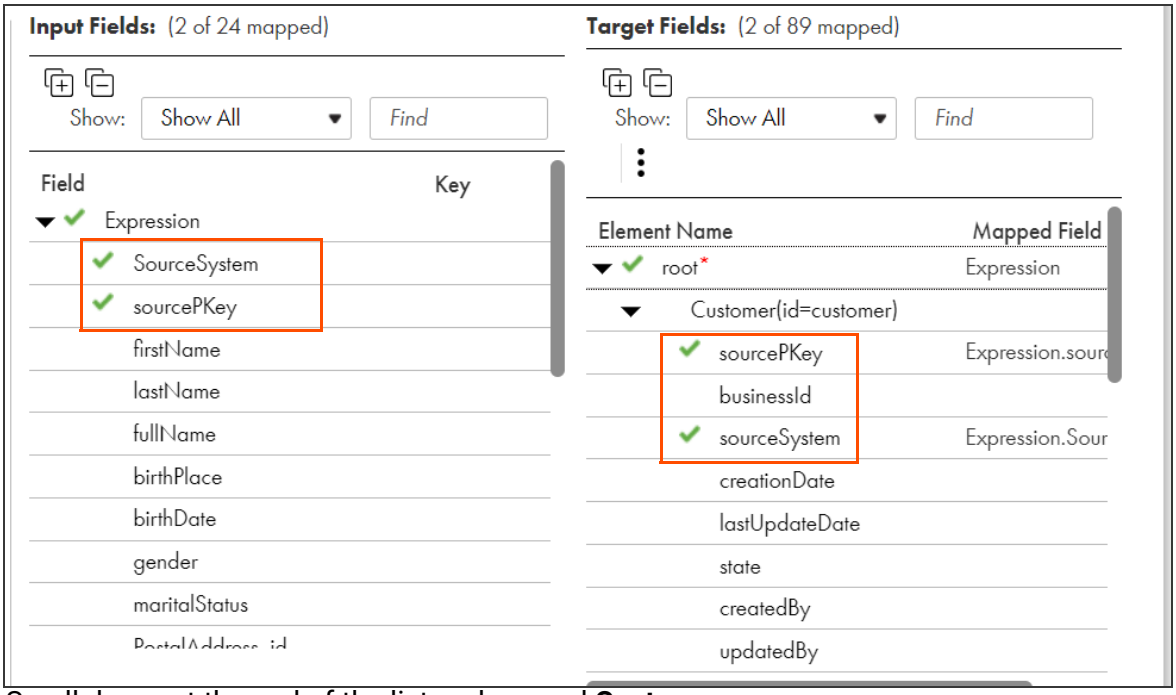
38. For better readability, maximize the Properties window.



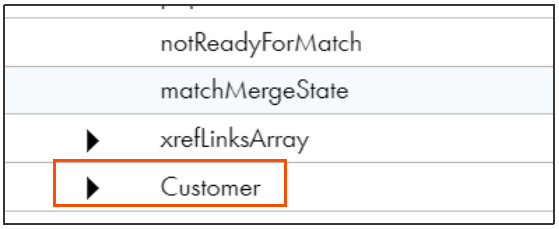
39. From the **Target Fields** section, expand **root** > **Customer(id=customer)**.

40. Drag and drop the following fields to map the fields from **Source fields** to the **Target fields**.

Source Field	Target Field
SourceSystem	sourceSystem
SOURCEPKEY	sourcePKey

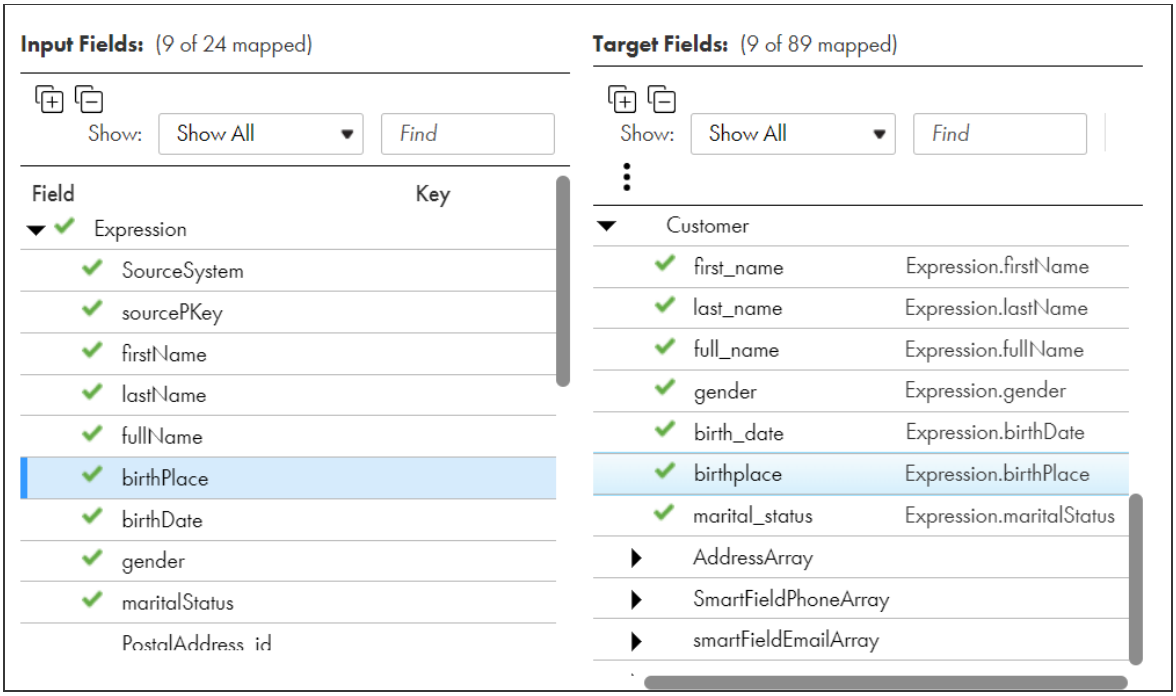


41. Scroll down, at the end of the list and expand **Customer**.

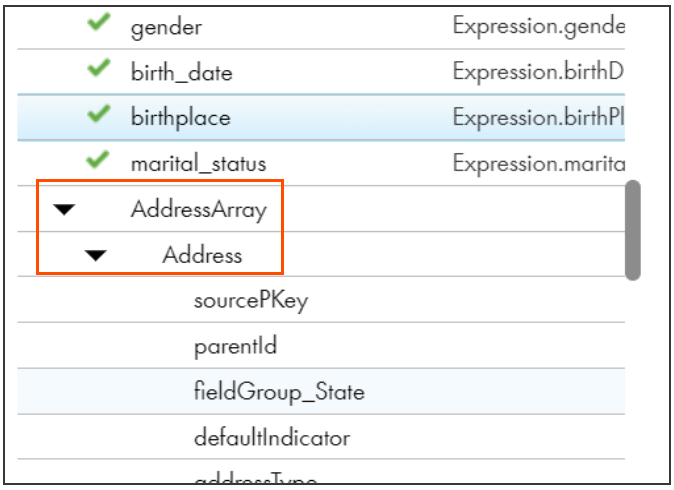


42. Perform the mapping as below:

Source Field	Target Field
firstName	first_name
lastName	last_name
fullName	full_name
birthPlace	birthplace
birthDate	birth_date
gender	gender
maritalStatus	marital_status

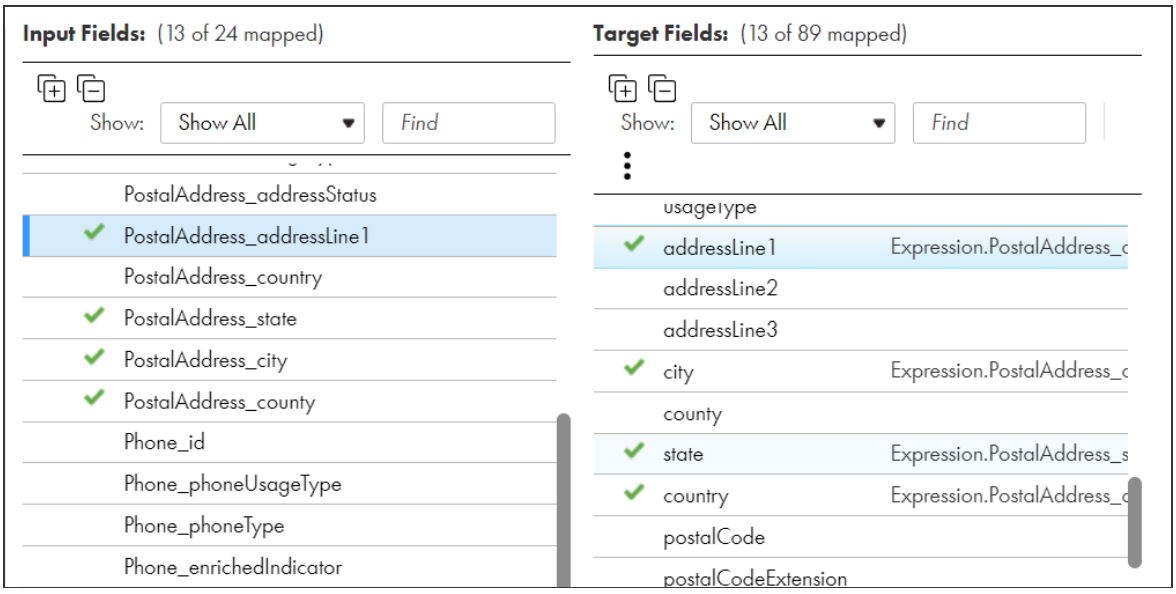


43. Expand **AddressArray** > **Address**.

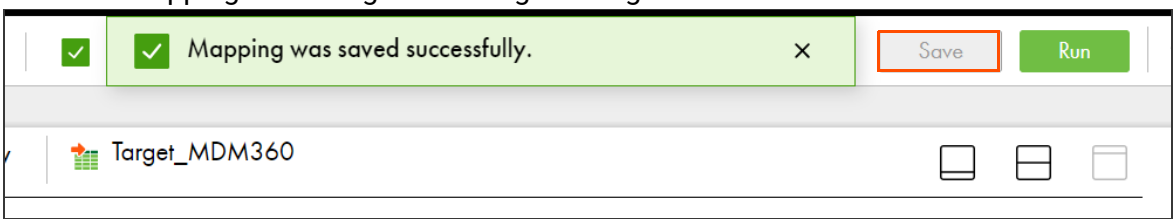


44. Perform the following mapping under the address field:

Source Field	Target Field
PostalAddress_addressLine1	addressLine1
PostalAddress_city	city
PostalAddress_state	state
PostalAddress_country	country



45. **Save** the mapping. You will get following message.

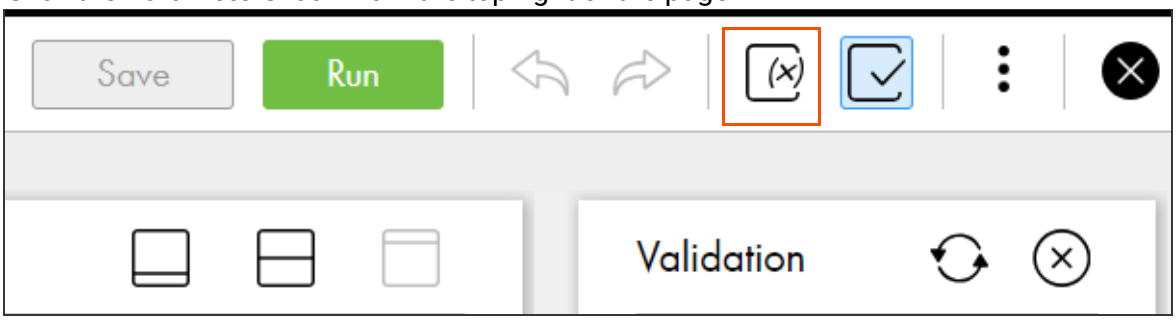


46. Click **Restore** icon to retain the properties window to its default size.

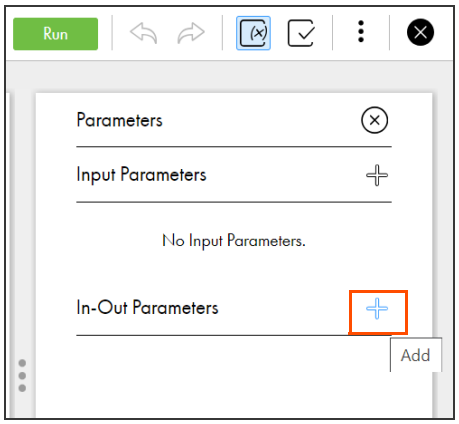
Create In-Out Parameter

In-out parameters are created when the value of a particular field keeps changing at runtime. You will now create an in-out parameter named **jobInstanceID** that reads the job instance ID at runtime.

47. Click the **Parameters** icon from the top right of the page.

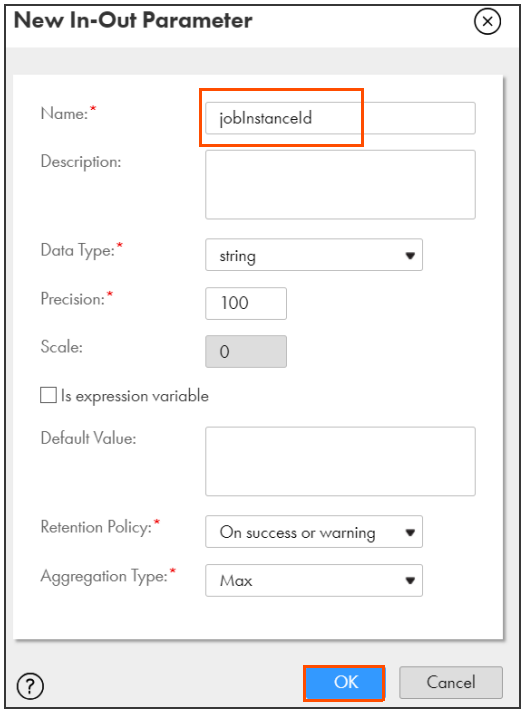


48. In the **Parameters** section on the right pane, click the **+** icon for the **In-Out Parameters**.



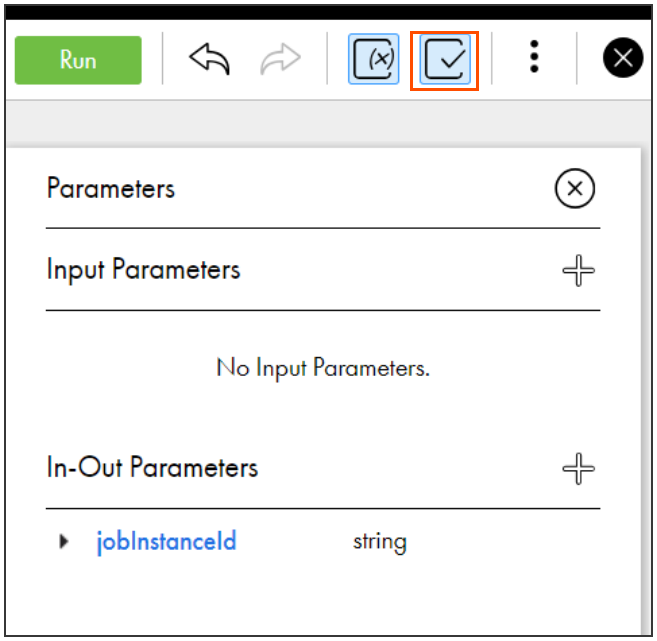
49. Enter **jobInstanceId** as the Name.

50. Set the Precision to **100** and click **OK**.

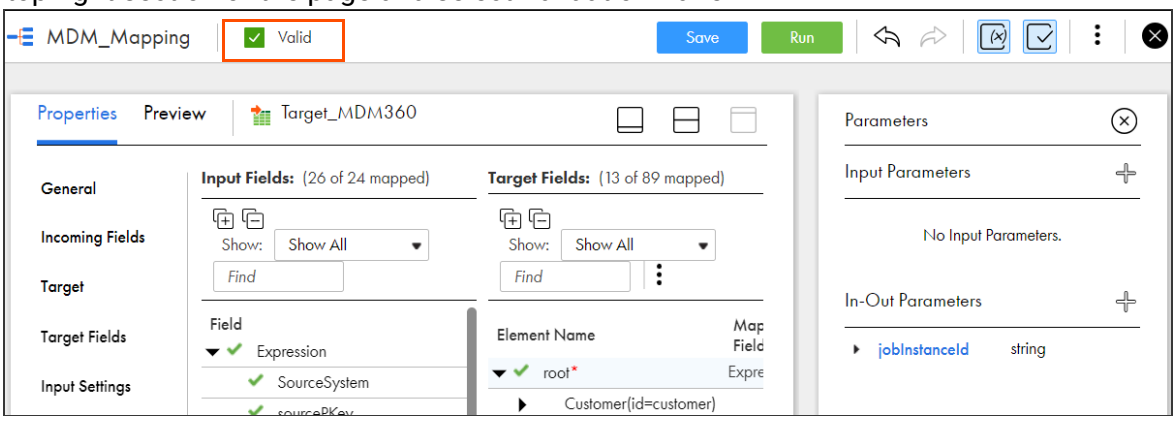


jobInstanceId is case sensitive. So please enter it as is while creating it.

51. The in-out parameter now appears in the Parameters panel. Click the **Validate** icon.



52. Notice that the mapping is valid in the Validation section. You can also view Valid check mark at the top of the page. If you cannot see the validation panel, click Action menu at the top right section of the page and select Validation Panel.

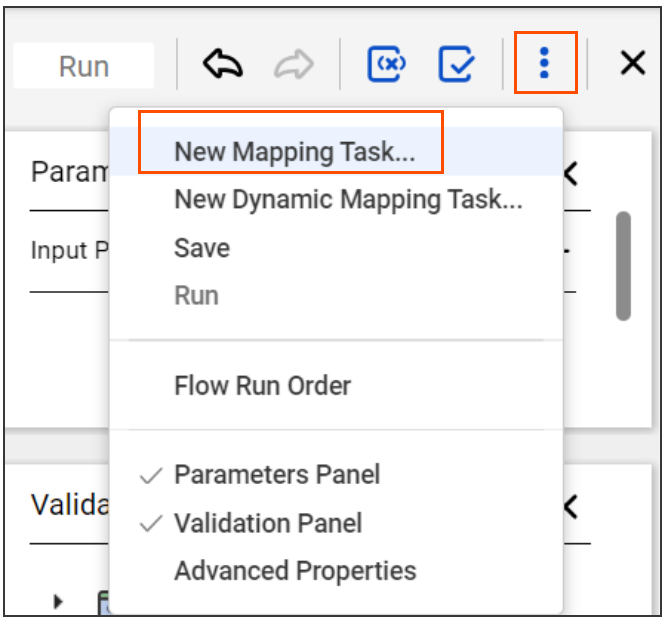


53. **Save** the mapping.

Create a Mapping Task

As the next step, you will create a mapping task and configure the properties.

54. To create a mapping task, click the 3 dots next to the Valid icon and click **New Mapping Task**.

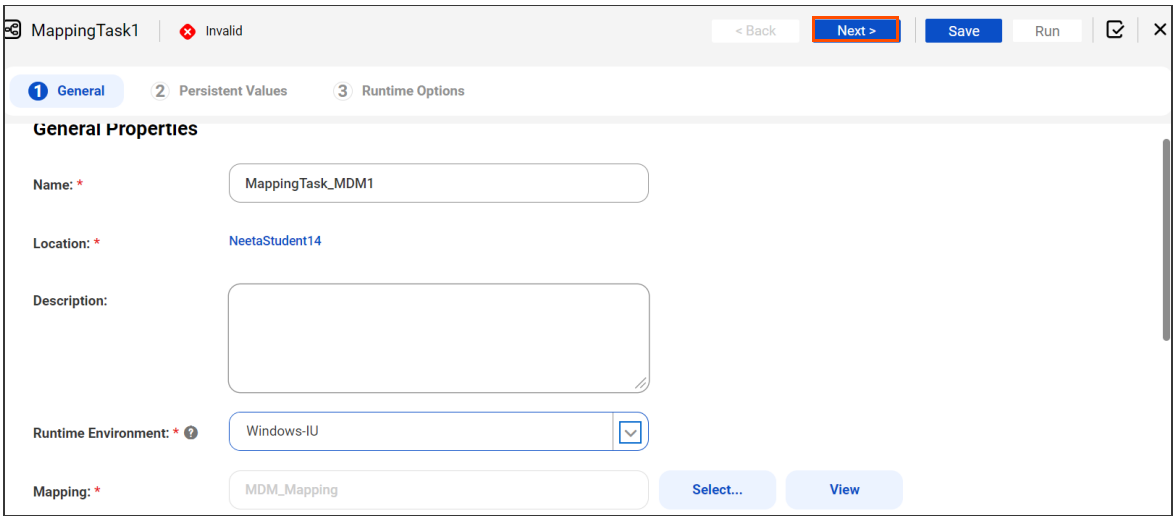


55. In the **New MappingTask** window, enter the Task Name as **MappingTask_MDM**.

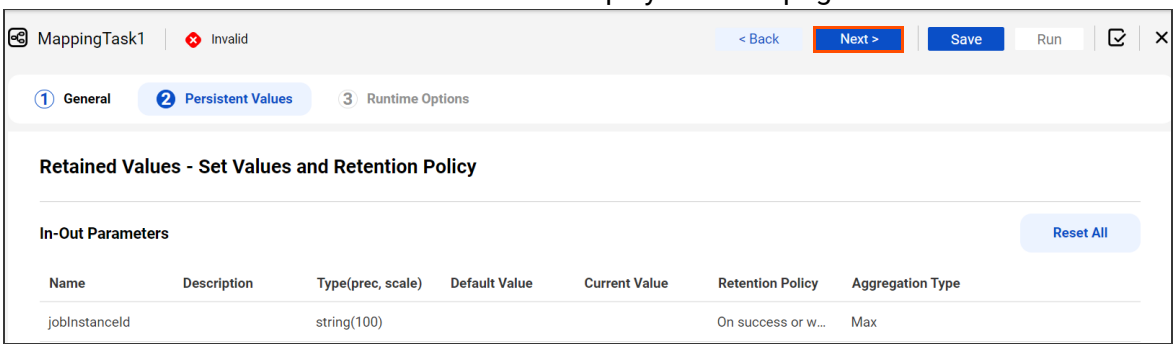
56. The Location and the Mapping are auto populated. Retain the same.

57. Select the **Runtime Environment** (SecureAgent that you have installed).

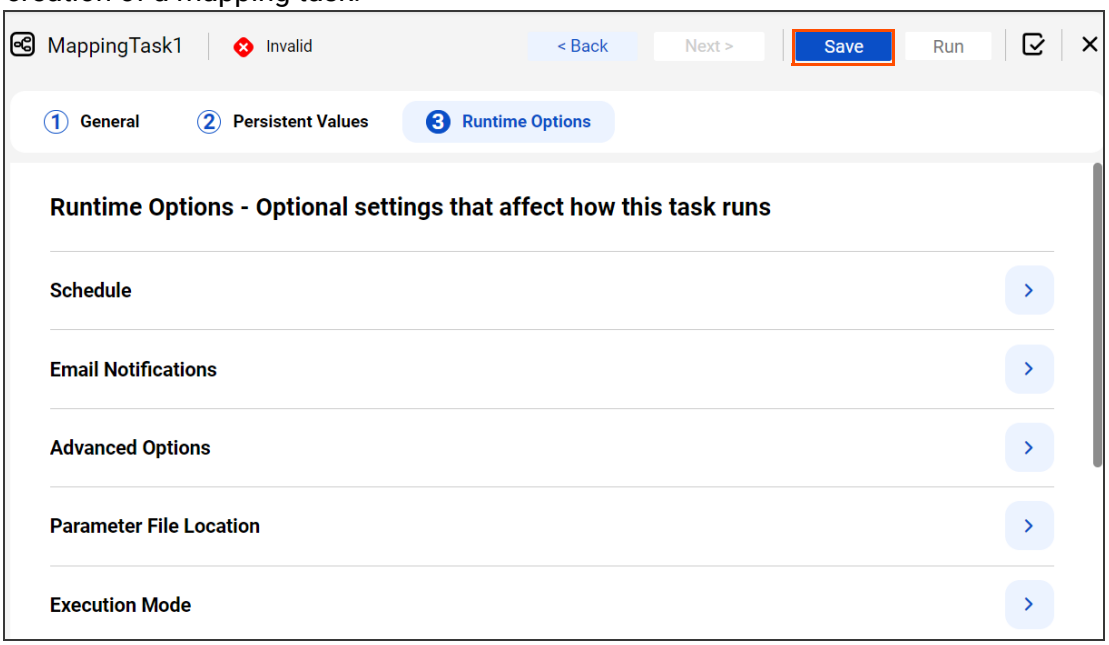
58. Click **Next**.



59. The **In-Out Parameters** that was created is displayed in this page. Click **Next**.



60. Do not make any changes in the **Schedule Details** section. Click **Save** to complete the creation of a mapping task.

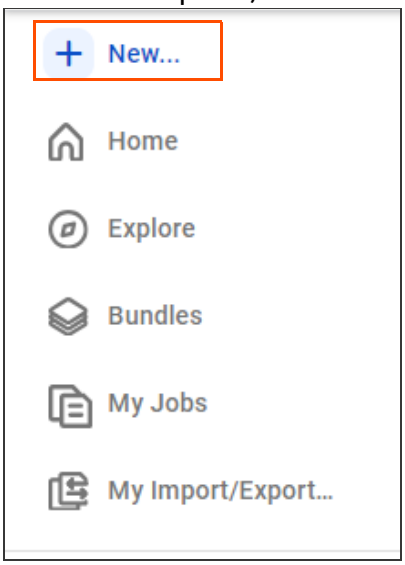


61. The mapping task is saved successfully. Scroll down to see all details.

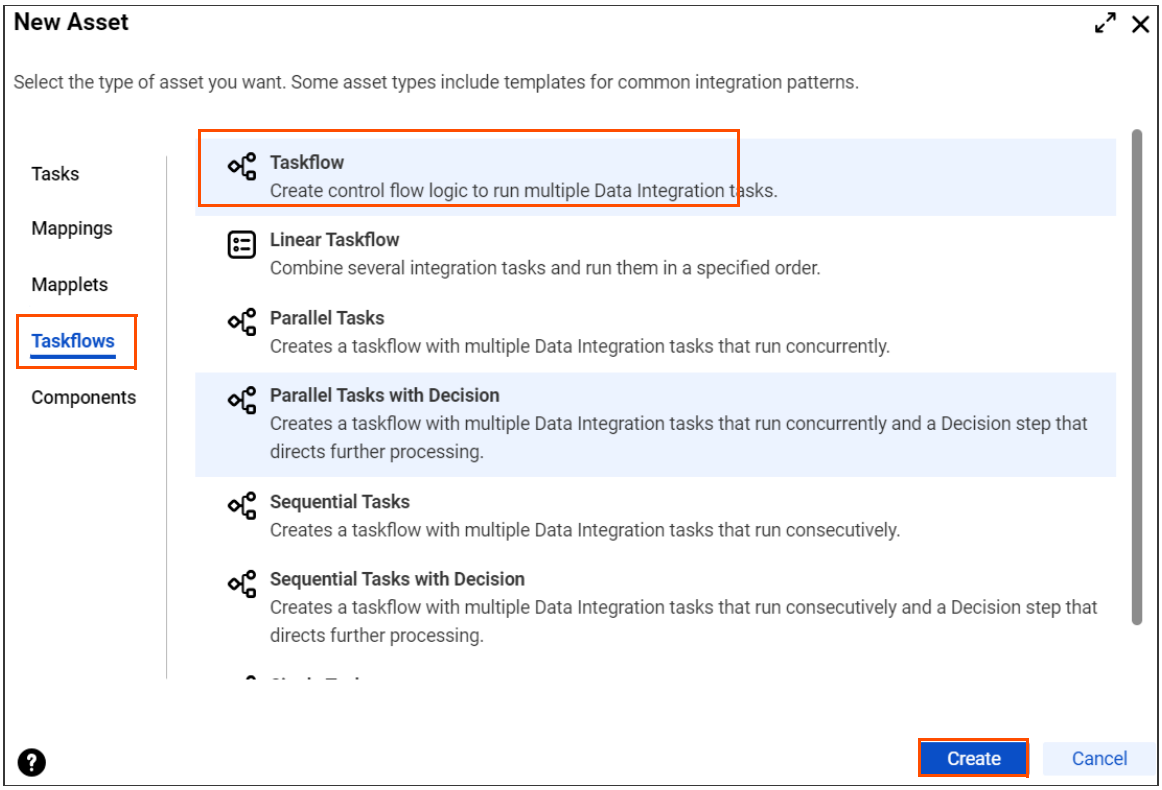
Create a Taskflow

You will now create a taskflow, that you will be publishing later.

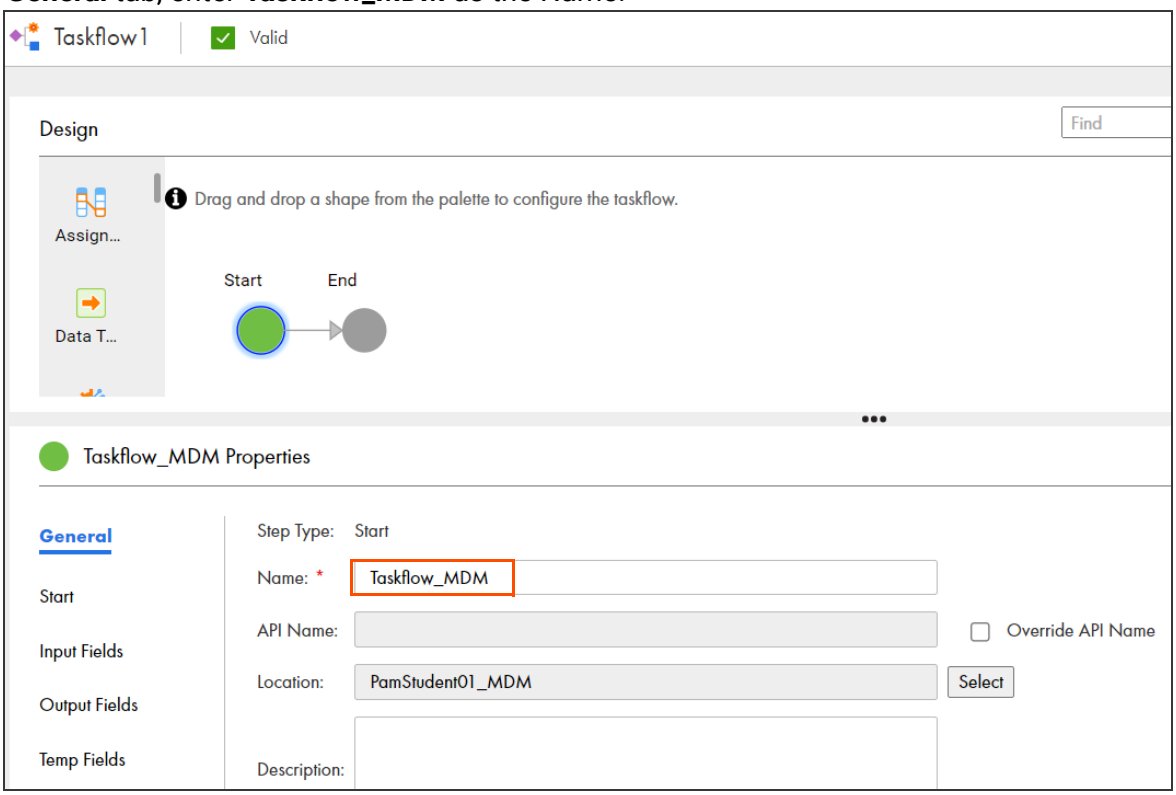
62. From the left panel, click **New**.



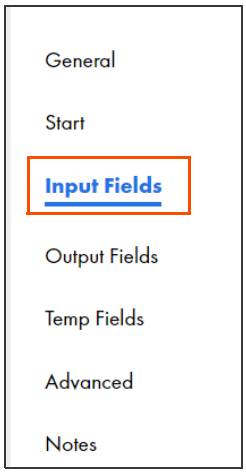
63. Select **Taskflows** > **Taskflow** and click **Create**.



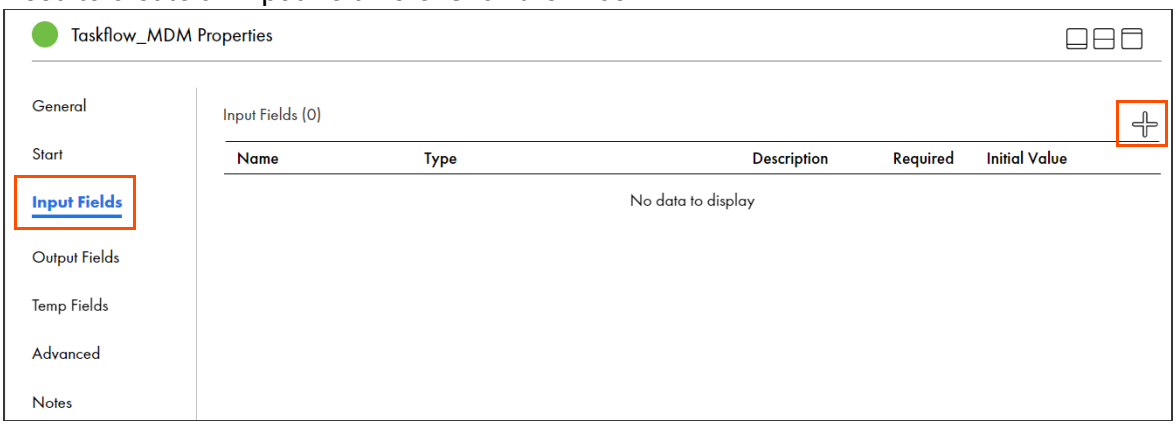
64. The taskflow appears in the canvas. The Start node is selected by default. From the **General** tab, enter **Taskflow_MDM** as the Name.



65. Click the **Input Fields** tab.

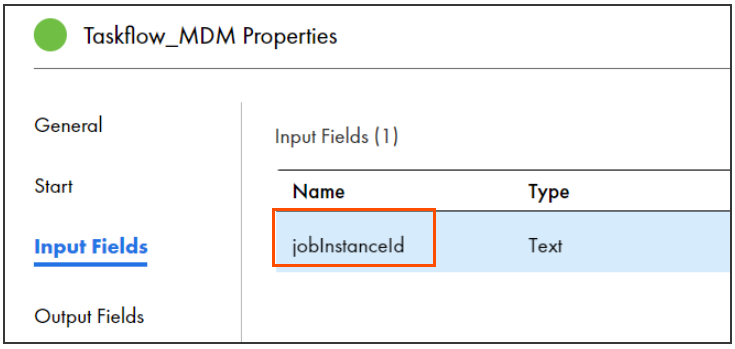


66. The parameter that was created while creating the mapping will be used as an input field here. The mapping will send that instance id as an input to the taskflow. Hence, you will need to create an input field here. Click the **+** icon.

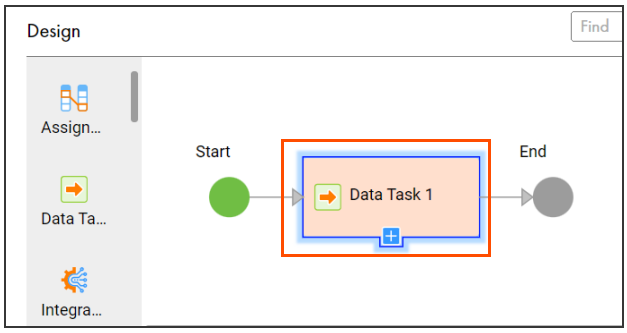


67. Double-click the field and enter the Name as **jobInstanceId**.

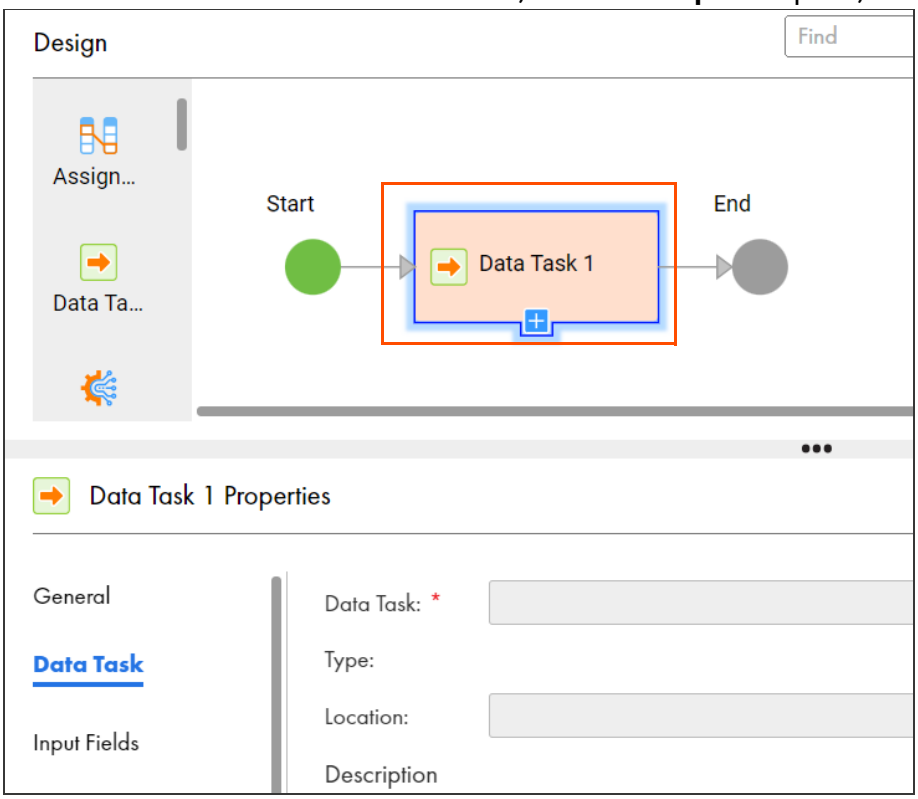
68. Retain the Type as **Text**.



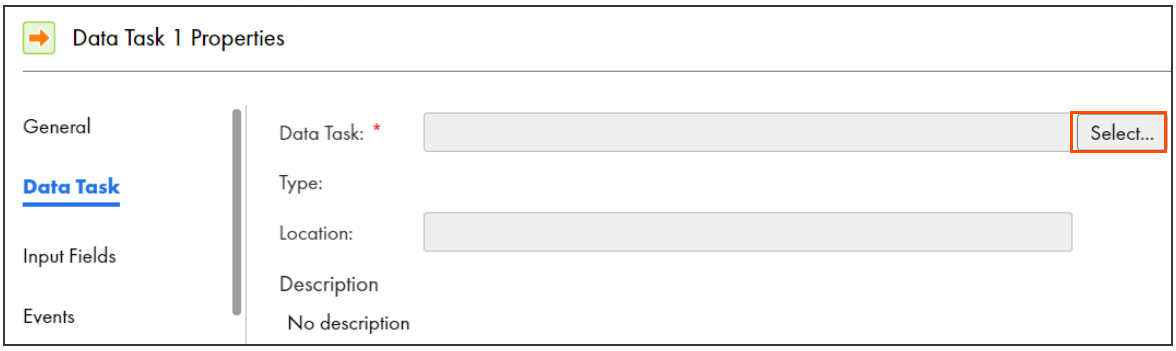
69. To call a mapping task, drag and drop a **Data Task** between the Start and End nodes.



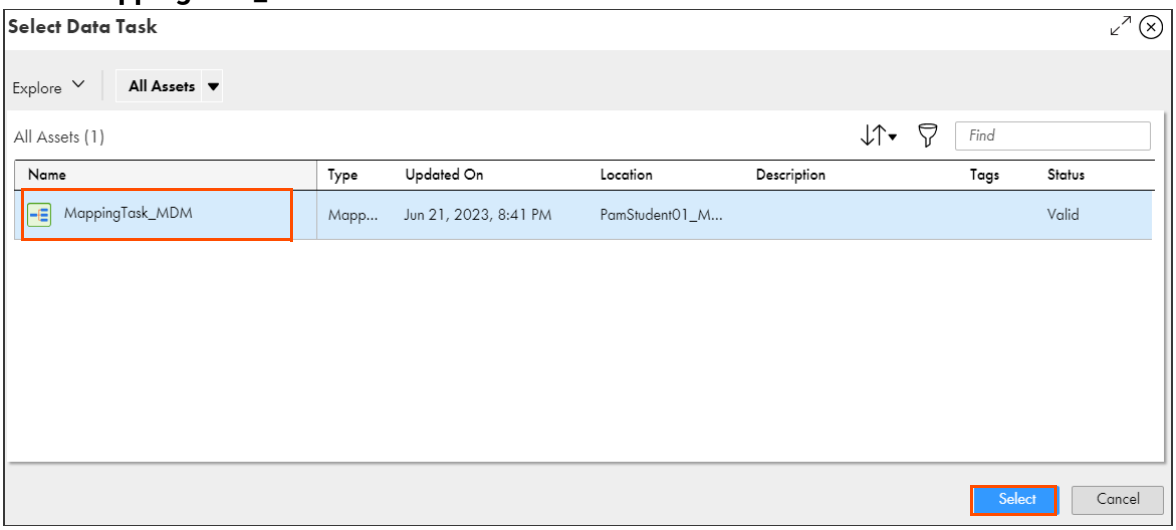
70. With Data Task 1 selected in the canvas, from the **Properties** pane, click the **Data Task** tab.



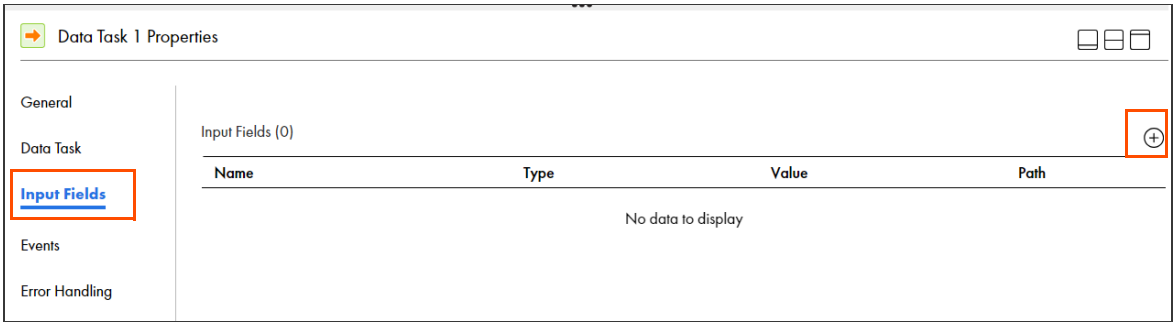
71. For the Data Task field, click **Select**.



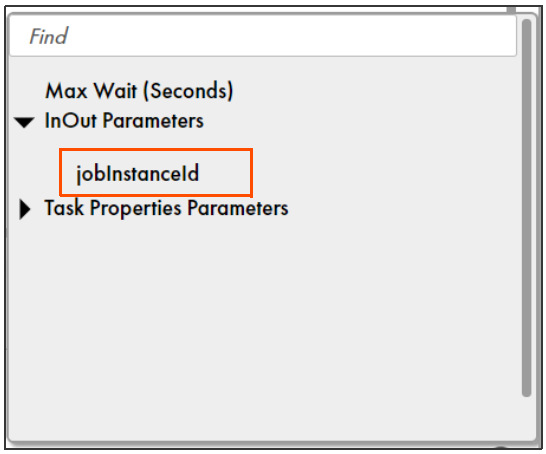
72. Click **MappingTask_MDM** and click **Select**.



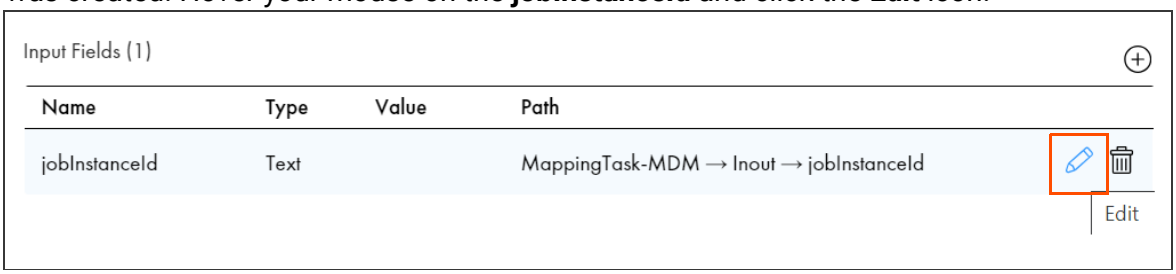
73. Click the **Input Fields** tab and click the **+** icon to add an input parameter.



74. Expand **InOutParameters** and select **jobInstanceId**.

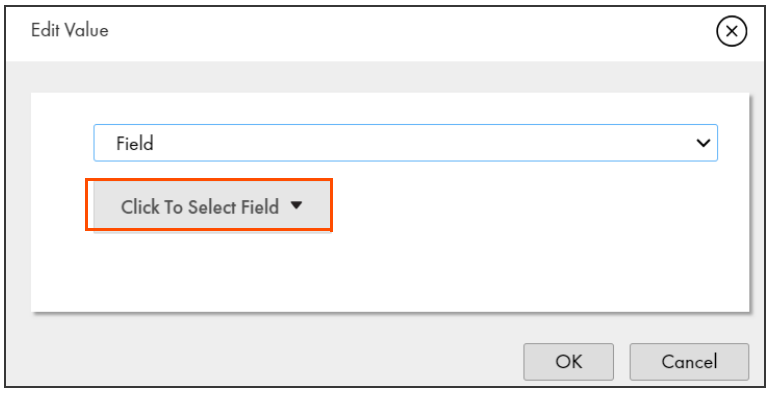


75. By default, the type is **Text**, but "jobInstanceId" is not of type text. It is the input field that was created. Hover your mouse on the **jobInstanceId** and click the **Edit** icon.

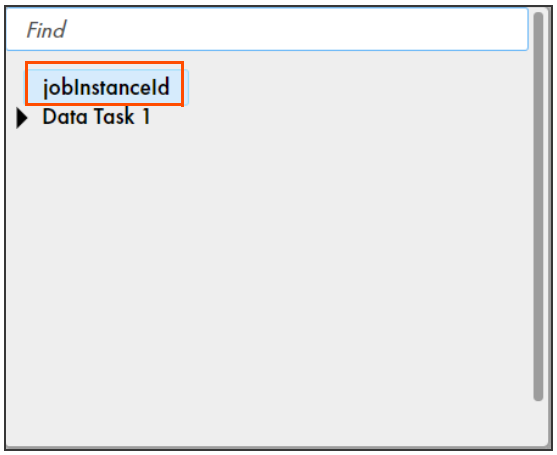


76. In the **Edit Value** window, select the type as **Field** from the drop down.

77. Click the **Click To Select Field** button.

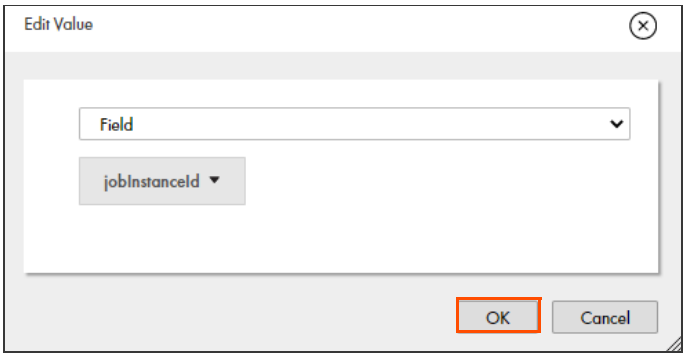


78. Click **jobInstanceId**.

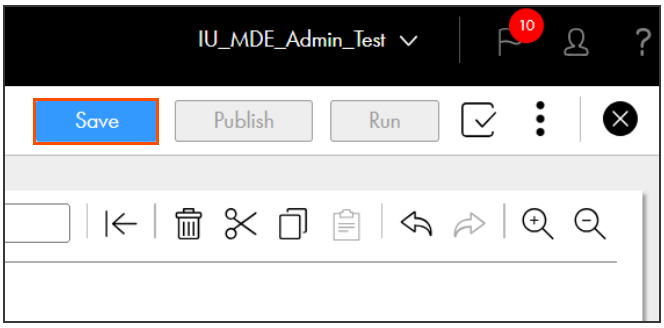


Make sure that you select the jobInstanceId as shown in the above image.

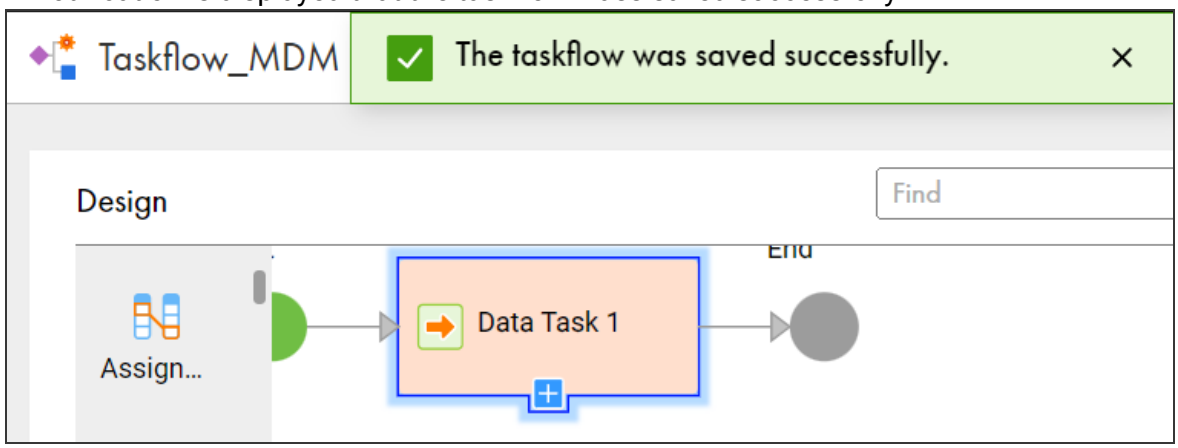
79. Click **OK**.



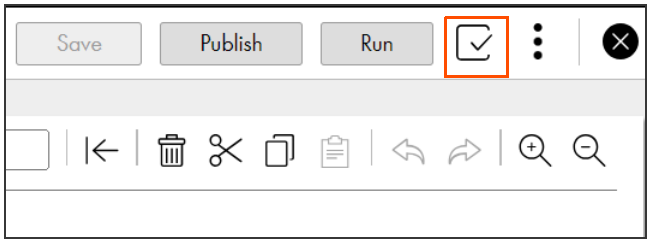
80. **Save** the Taskflow.

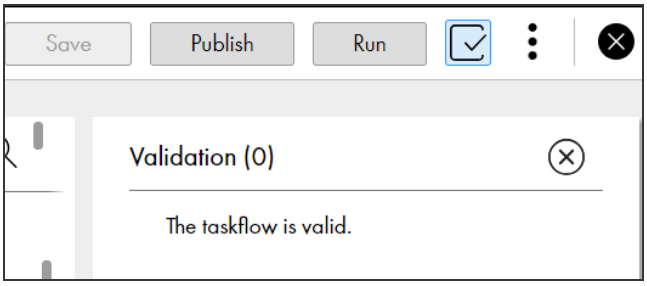


A notification is displayed that the taskflow wass saved successfully.

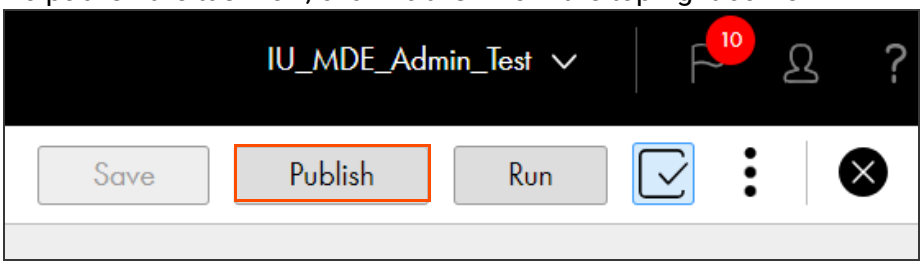


81. Click the **Validation** icon to check the validity of the taskflow. You can see that the taskflow is valid.

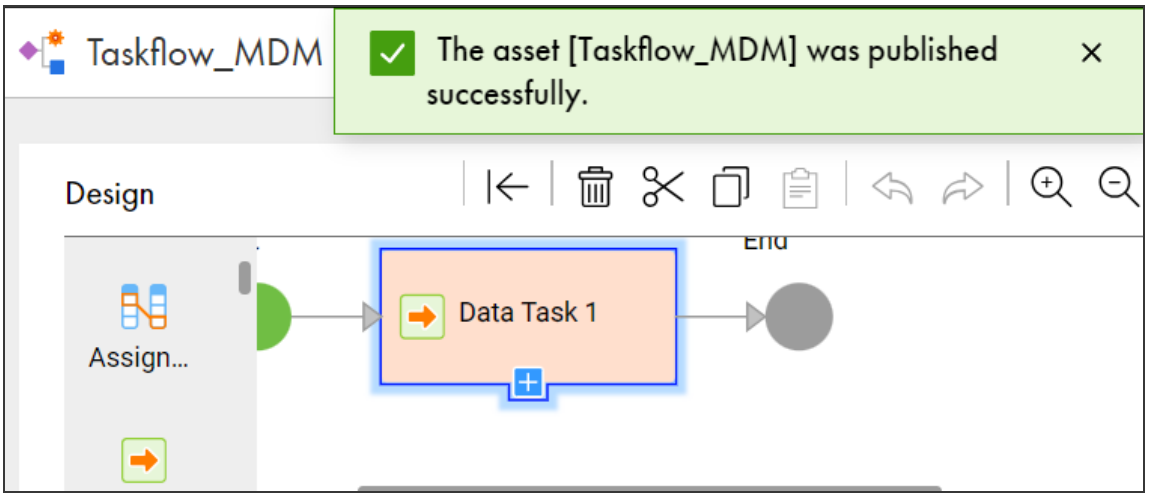




82. To publish the taskflow, click **Publish** from the top right corner.



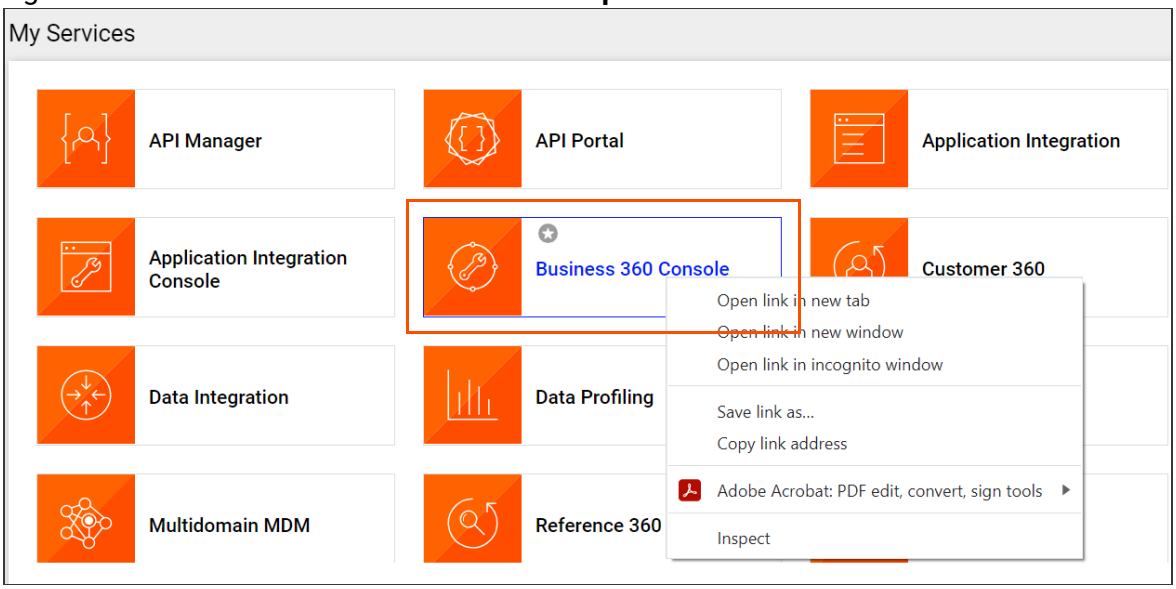
A notification is displayed that the taskflow was published successfully.



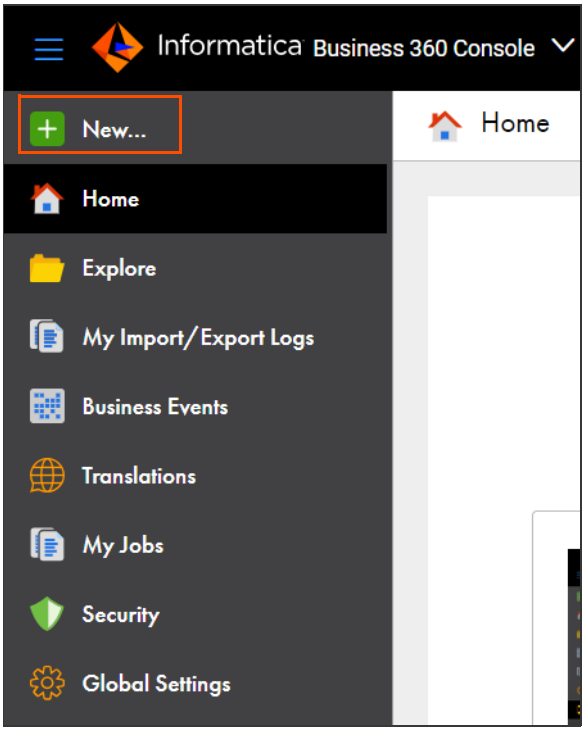
Create Ingress Job in Business 360 Console

The last step is to create an ingess job in Business 360 Console.

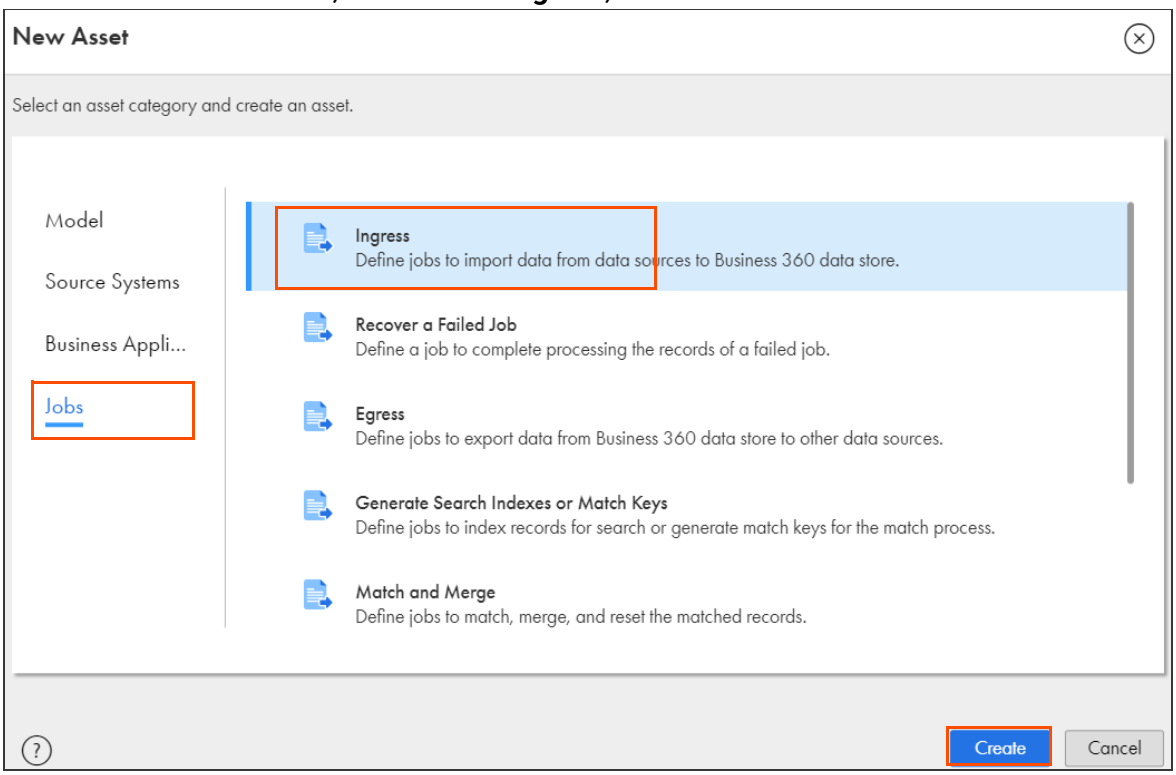
83. Click **Informatica Data Integration** from the top of the page and from the Services window, right-click **Business 360 Console** and select **Open link in new tab**.



84. In the **Business 360 Console** window, click **New**.

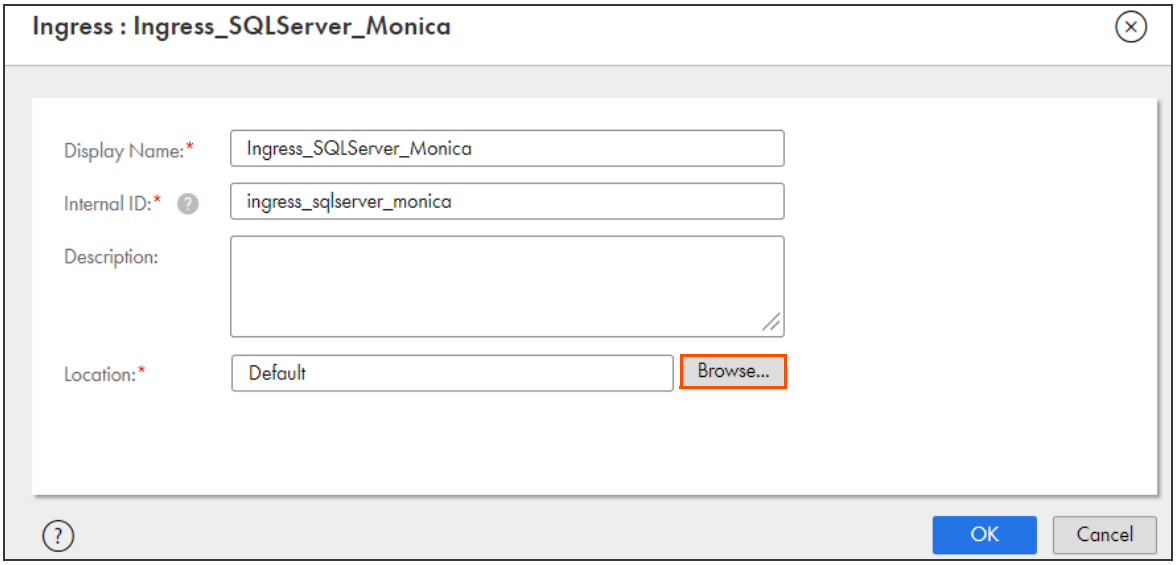


85. In the **New Asset** window, click **Jobs** > **Ingress**, and click **Create**.

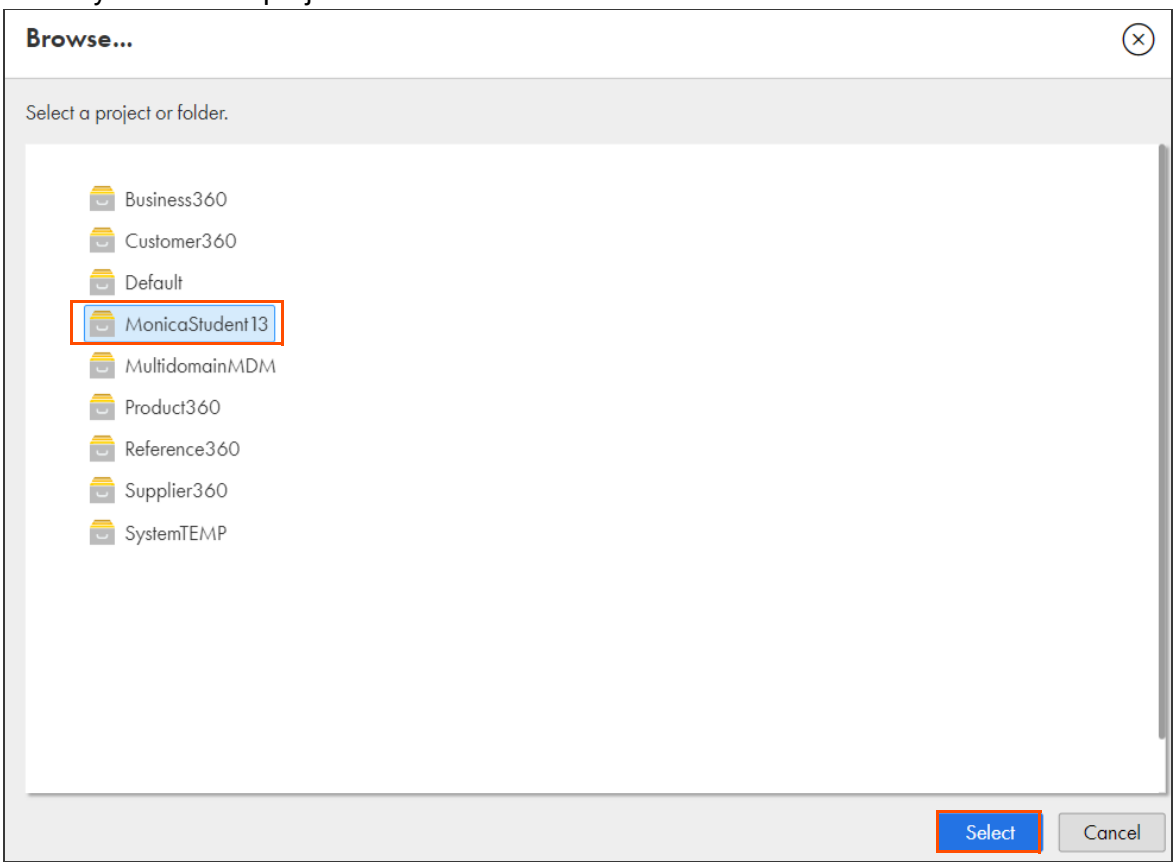


86. Enter **Ingress_SQLServer_<studentName>** as the Display Name, for example Ingress_SQLServer_.

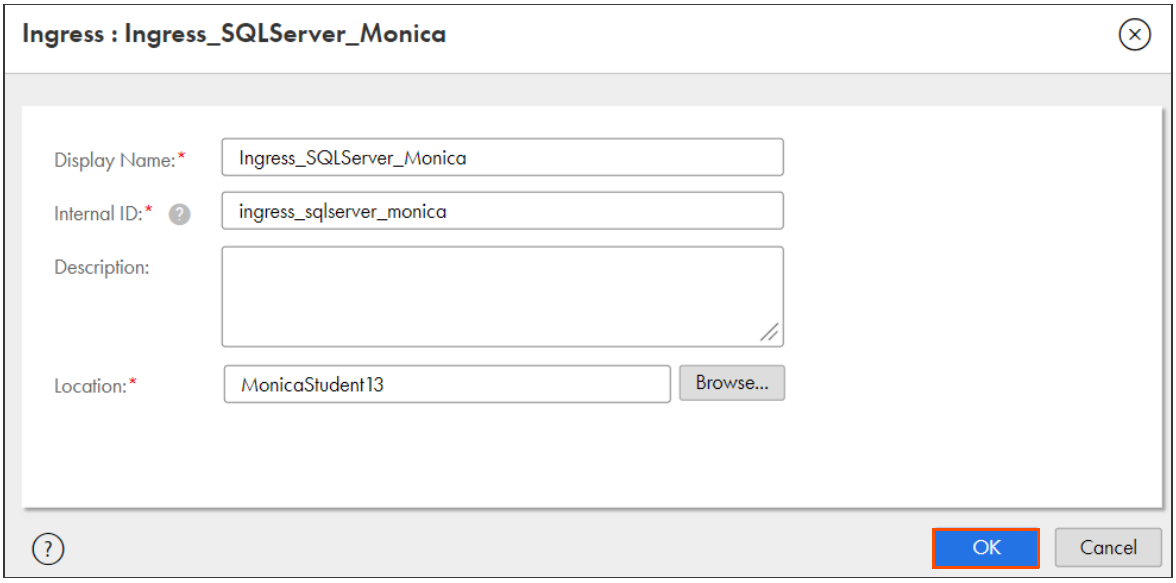
87. For Location, click **Browse**.



88. Select your student project and click **Select**.

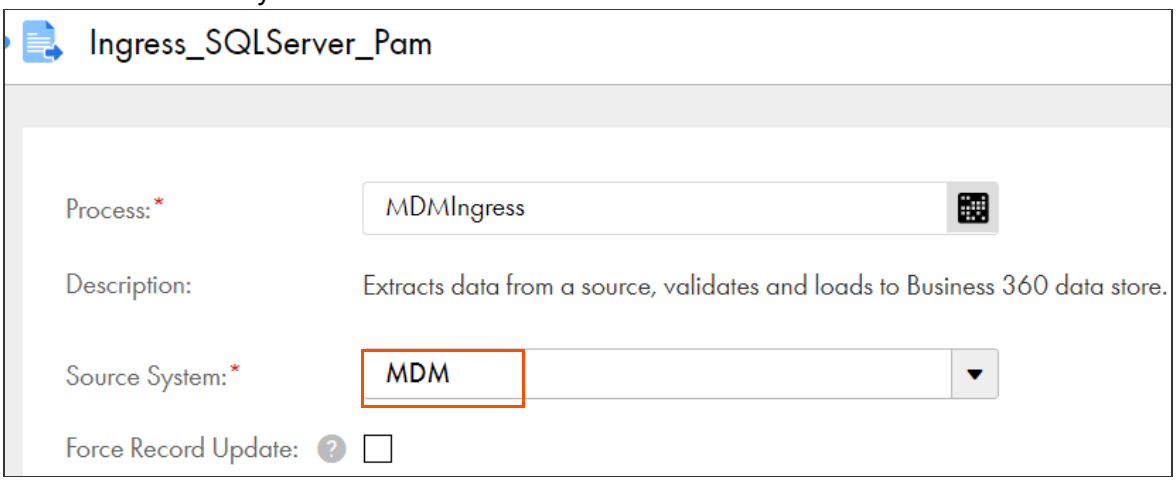


89. Click **OK**.

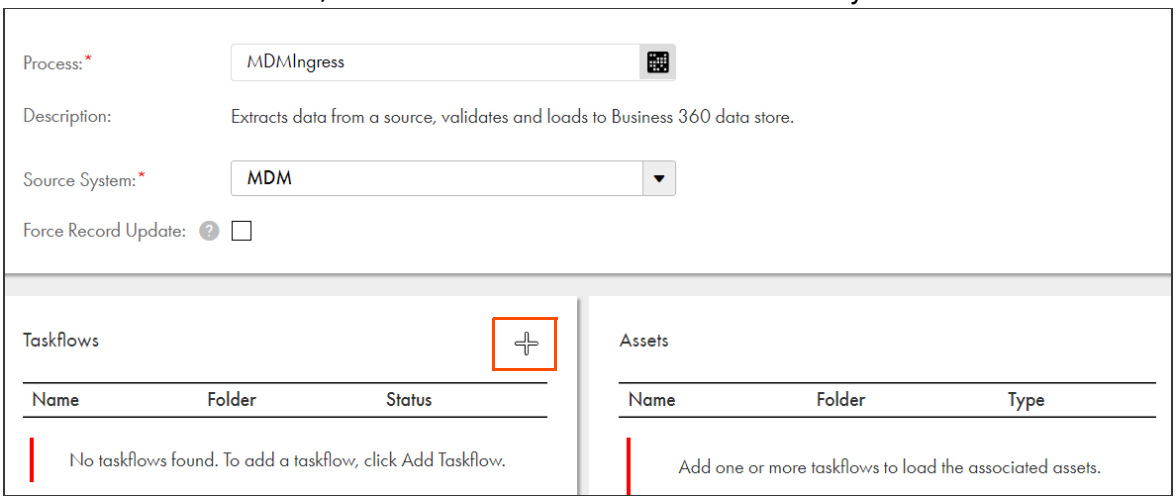


90. The **Process** drop-down would populate the default value a **MDMIngress**.

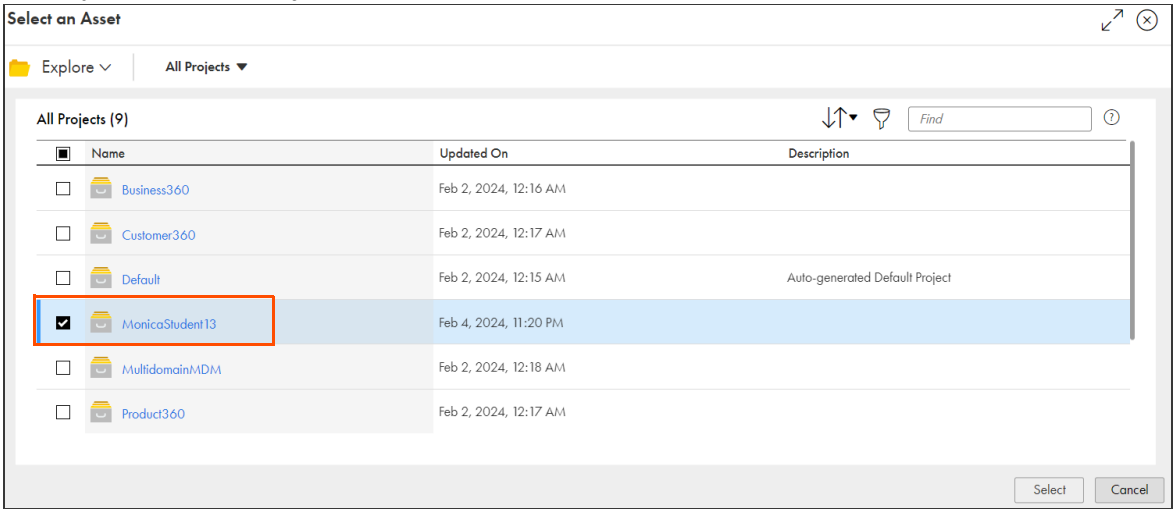
91. Select the Source System as **MDM**.



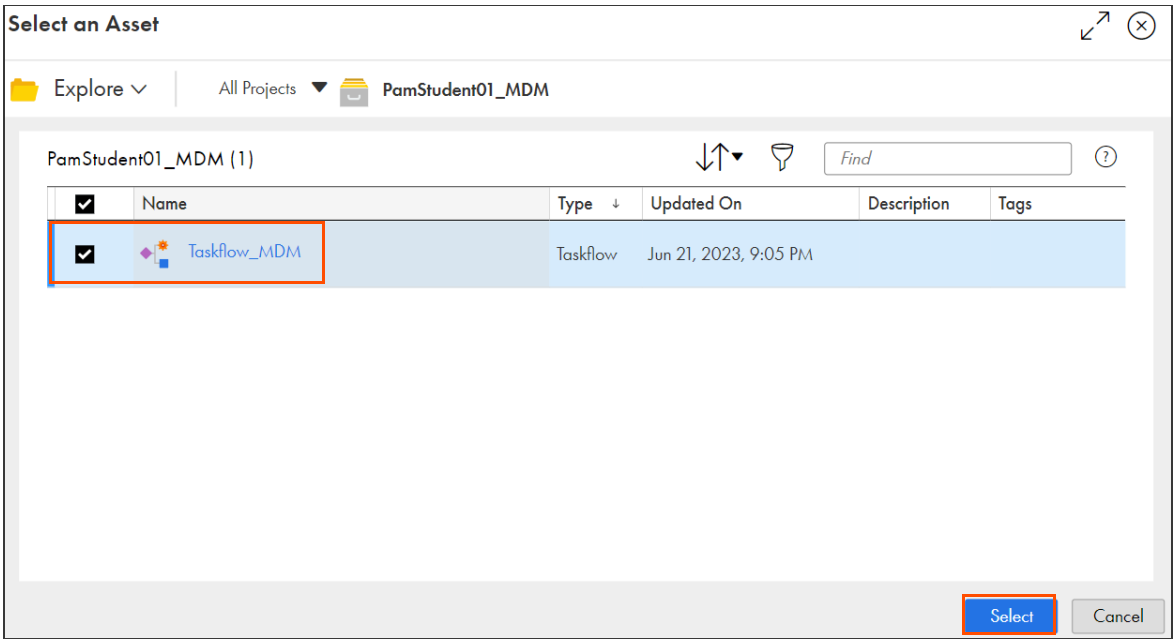
92. In the Taskflows section, click the **+** icon to add the taskflow that you created in CDI.



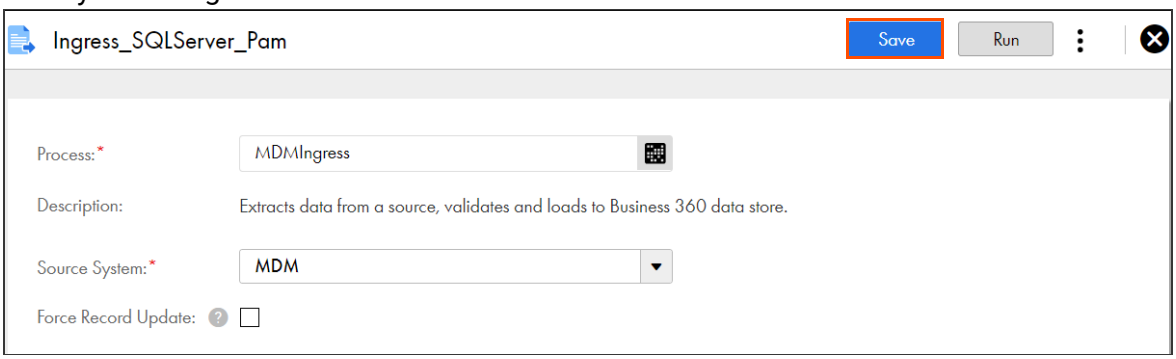
93. Open your student project .



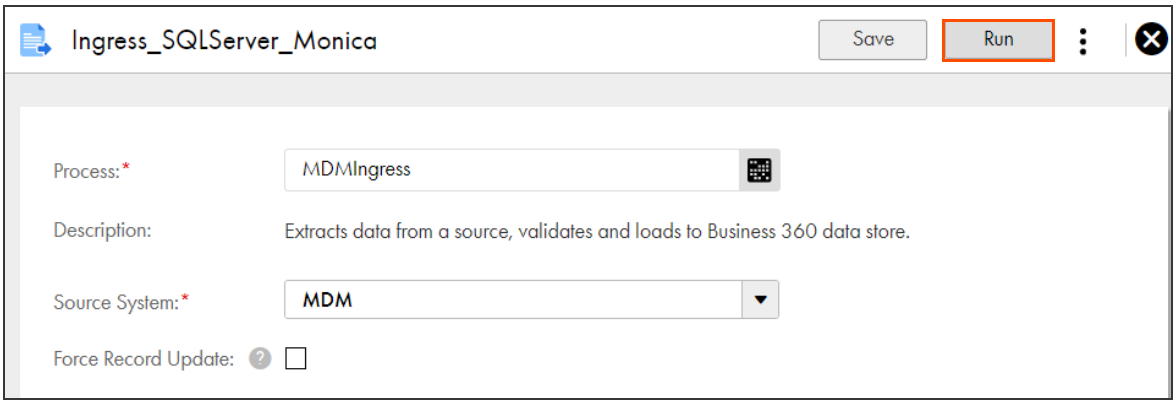
94. Select **Taskflow_MDM** and click **Select**.



95. **Save** your changes.

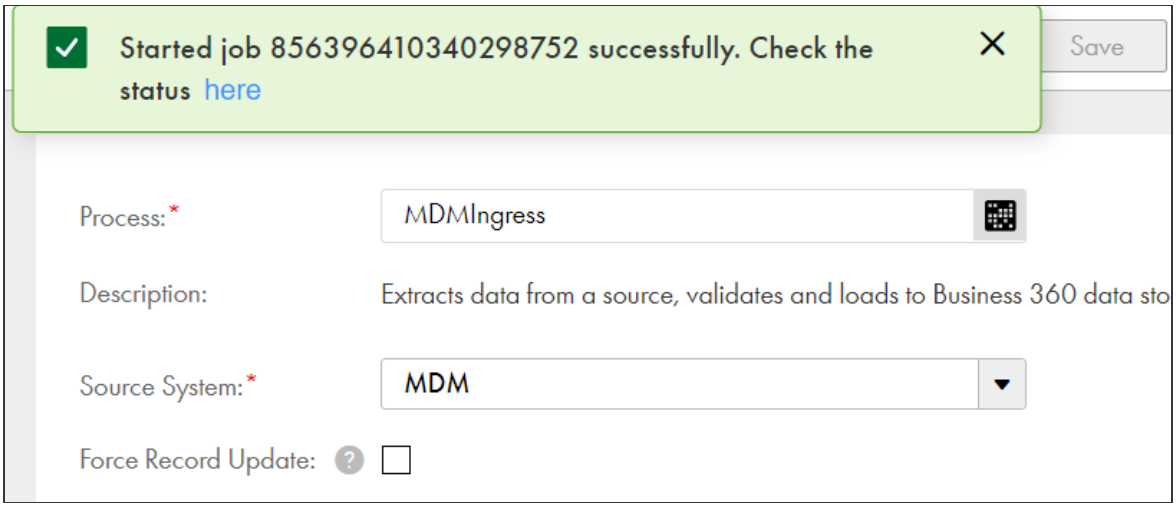


96. Once the save is successful, click **Run**.

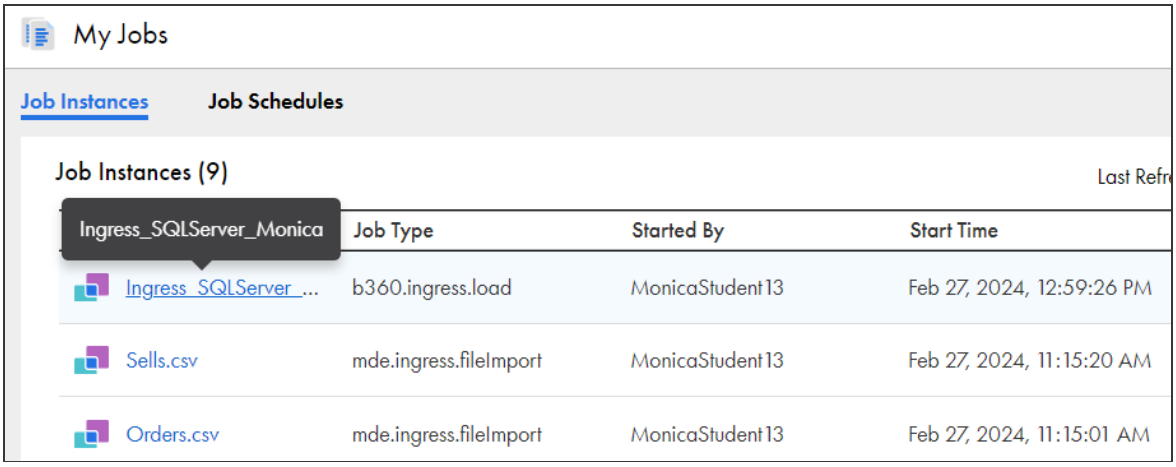


The job run starts. It can be monitored by clicking the link **here**.

If you missed clicking “here”, you can alternatively click **My Jobs** from the left panel.

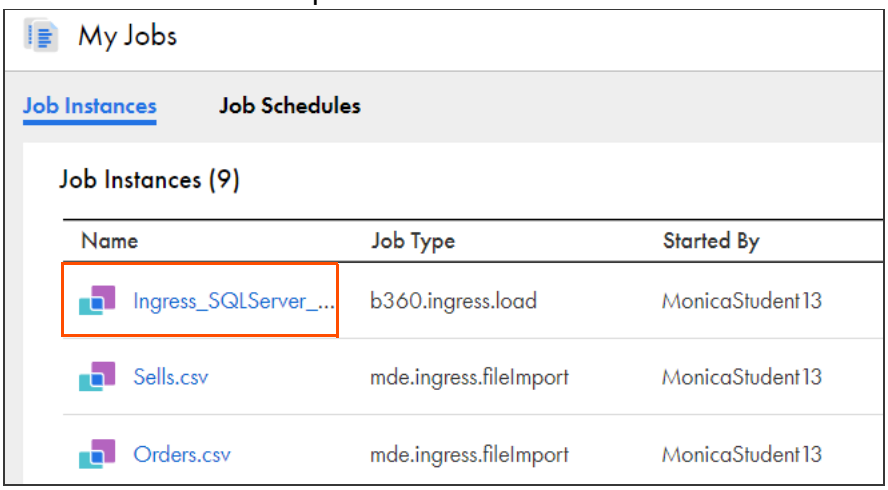


Notice that the job is running.

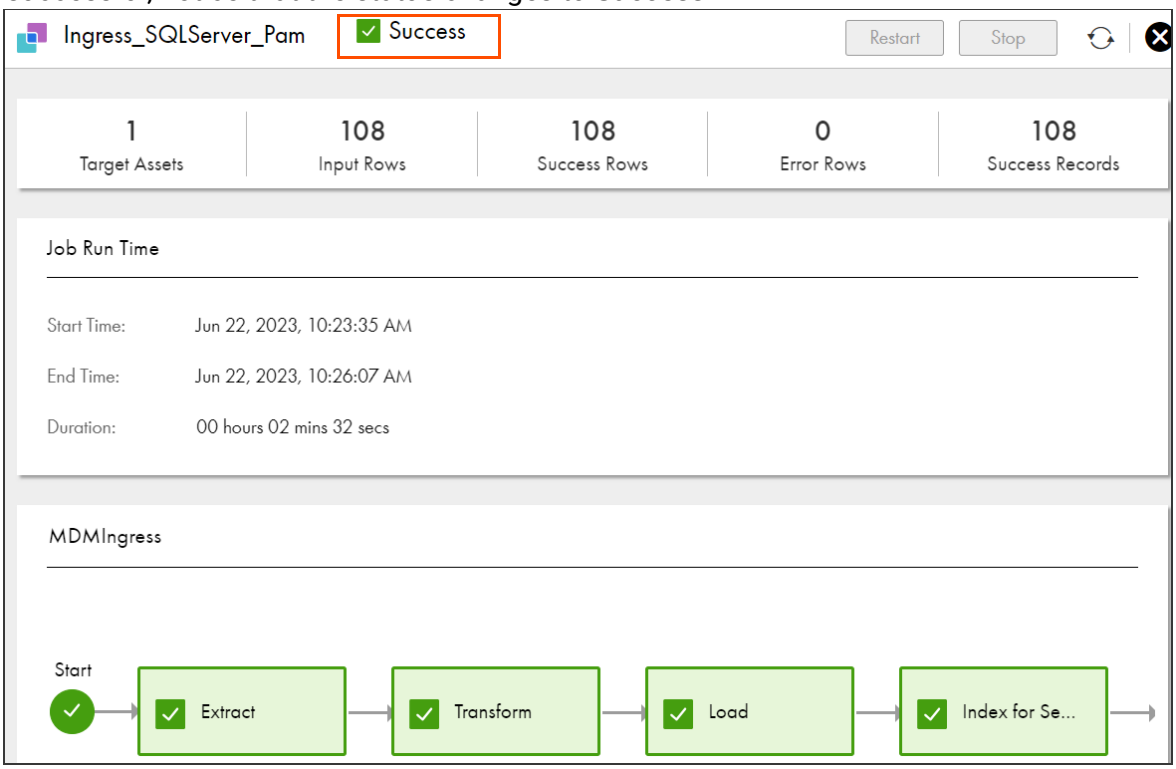


This will take 2-3 minutes to complete.

97. Click the Job name to open and view more details.



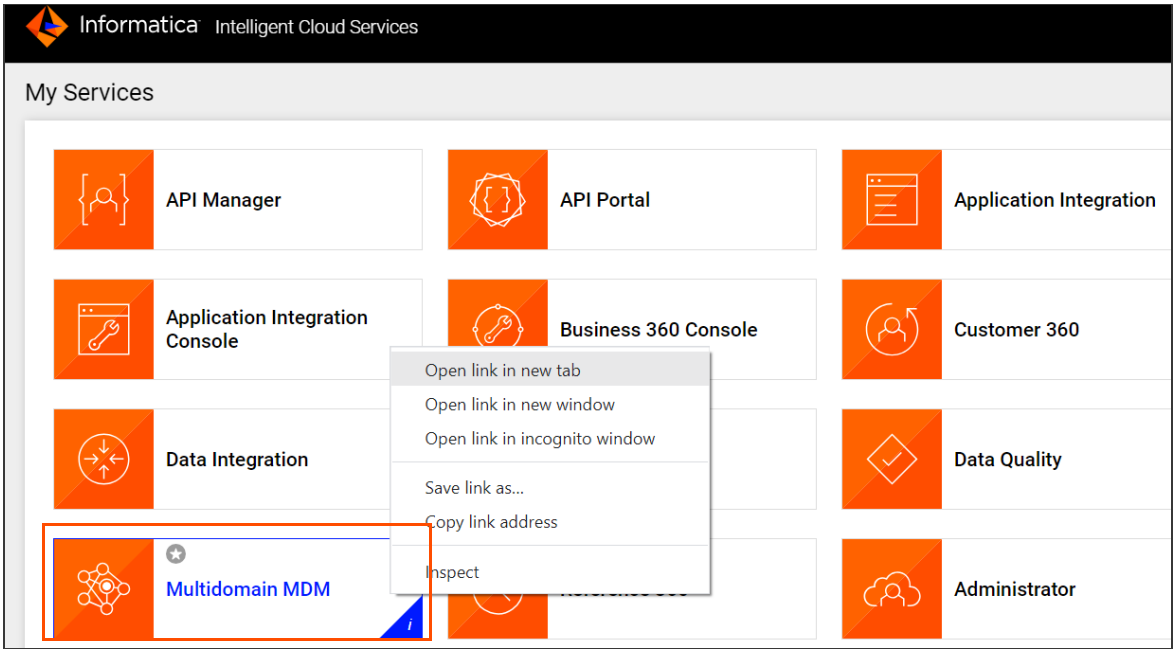
98. You can view the phase in which the job is from this page. Refresh the page.Once the job is successful, notice that the status changes to **Success**.



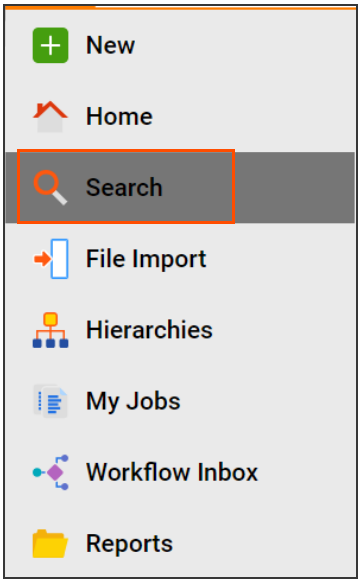
View the Imported Records in Multidomain MDM

Check the imported records in Multidomain MDM to see if the ingress is successful.

99. Click Informatica Business 360 Console and from the Service window, open Multidomain MDM in a new tab.

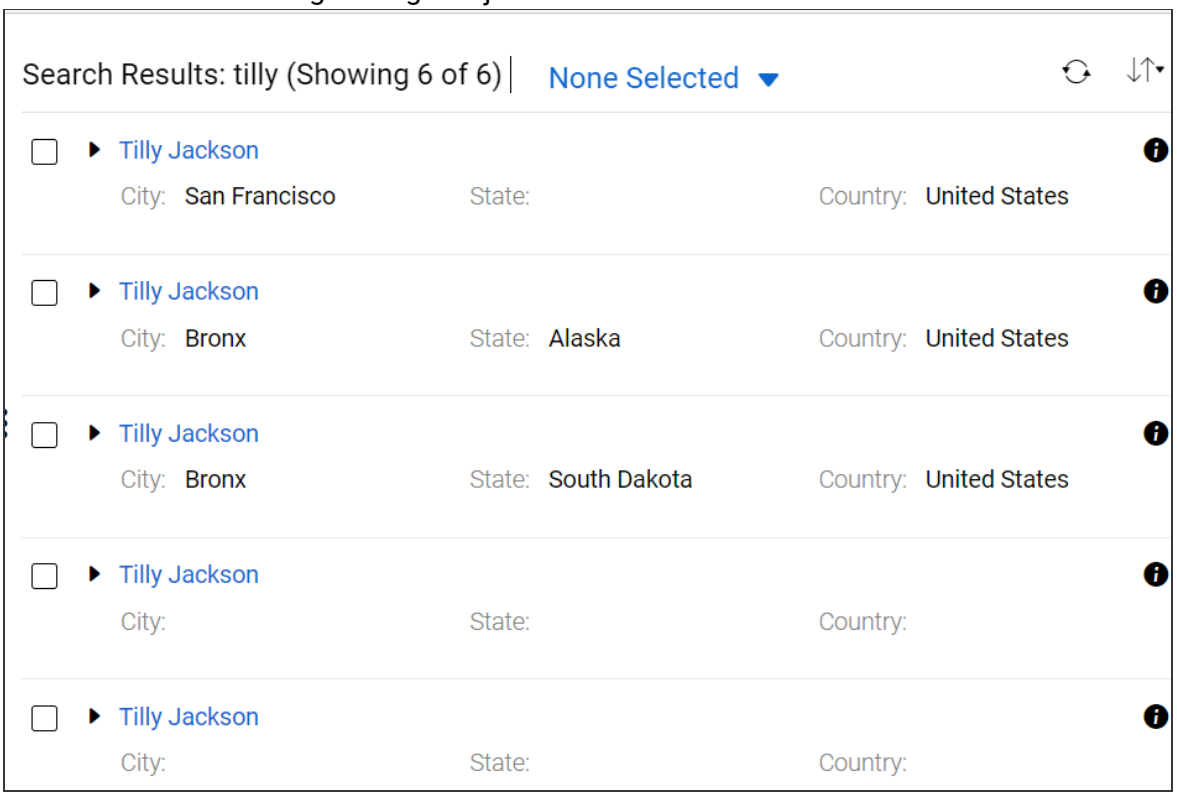


100. In the Multidomain MDM interface, click **Search** from the menu.



101. Select **Customer** from the drop-down and enter **tilly** as the search string. Click the **Search** icon or hit enter on your keyboard.

102. You can view that the records from SQL Server have successfully been imported to Multidomain MDM using the Ingress job.



Number of records may vary in the screenshot.

This concludes the lab.

Lab 9-6: Egressing Data to a Flat File

Overview:

You can export data from the Business 360 data store to an external data source. For example, your organization uses Multidomain MDM to master all domain data. To analyze the purchase behaviour of the customers, you want to load the mastered customer data into Tableau. You can export data from Multidomain MDM to Tableau.

In this lab, you will export customer records from the Business 360 data store to a flat file.

Objectives:

- Create a new project
- Create a DI mapping
- Create a mapping task
- Create a taskflow
- Create and monitor an egress job in Business 360

Scenario:

Mark, from HomeKart, discovers that he has some customer records stored in the Business 360 data store. He intends to export these records into a flat file and creates an egress job for this purpose. In the future, as new customers are added to the Business 360 data store, all he needs to do is simply rerun the egress job to export the updated data.

Duration:

40 minutes

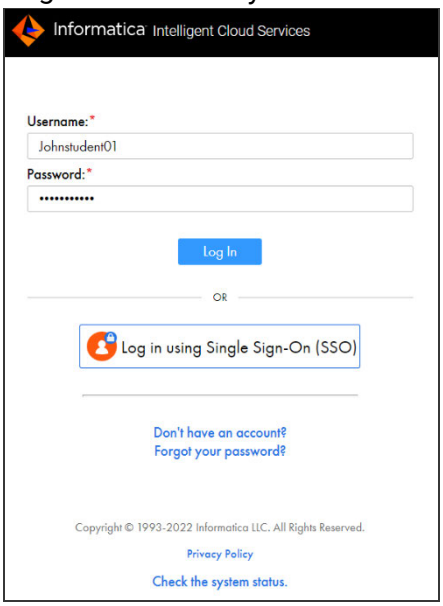
Tasks:

- Create a DI mapping
- Add an expression transformation
- Configure the target connection
- Create In-Out parameter
- Create a mapping task
- Create a taskflow
- Create ingress job in Business 360 Console
- View the imported records in Multidomain MDM

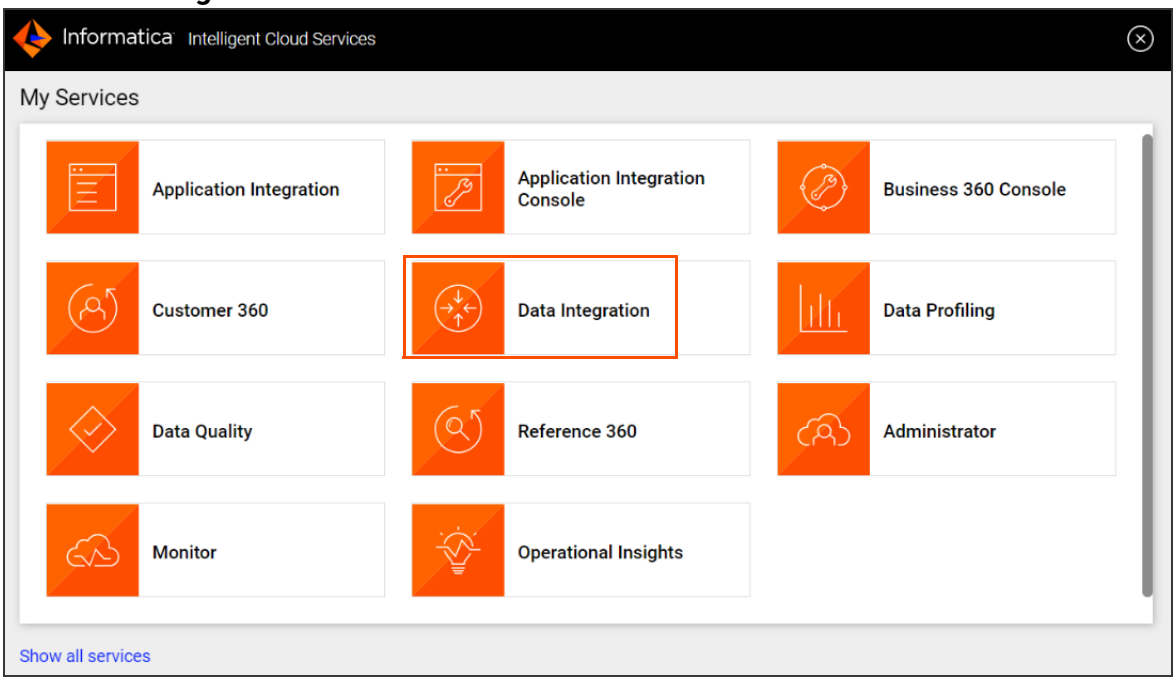
Create a DI Mapping

Create a mapping with the source connection as the Business 360 connection and the flat file connection as the target transformation. Define field mappings for the source and target fields. Ensure that you create an in-out parameter jobInstanceId in the Parameters panel.

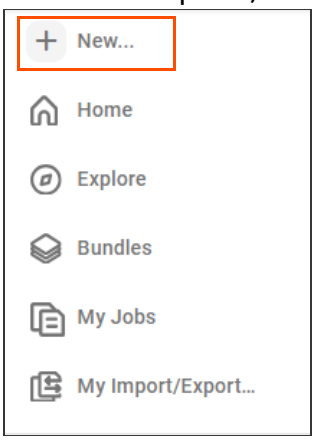
1. Log in to IICS with your credentials as **<studentNamestudent<no.> / Multidomain360**).



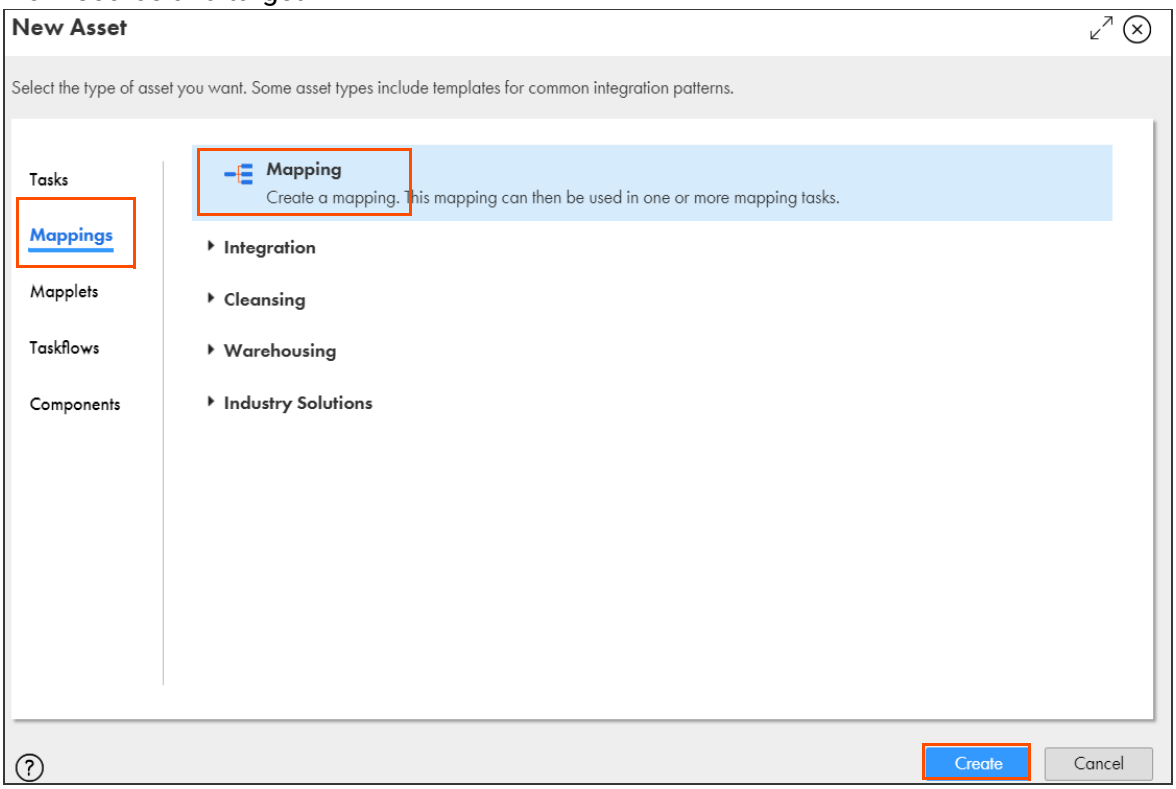
2. Click **Data Integration**.



3. From the left panel, click **New**.

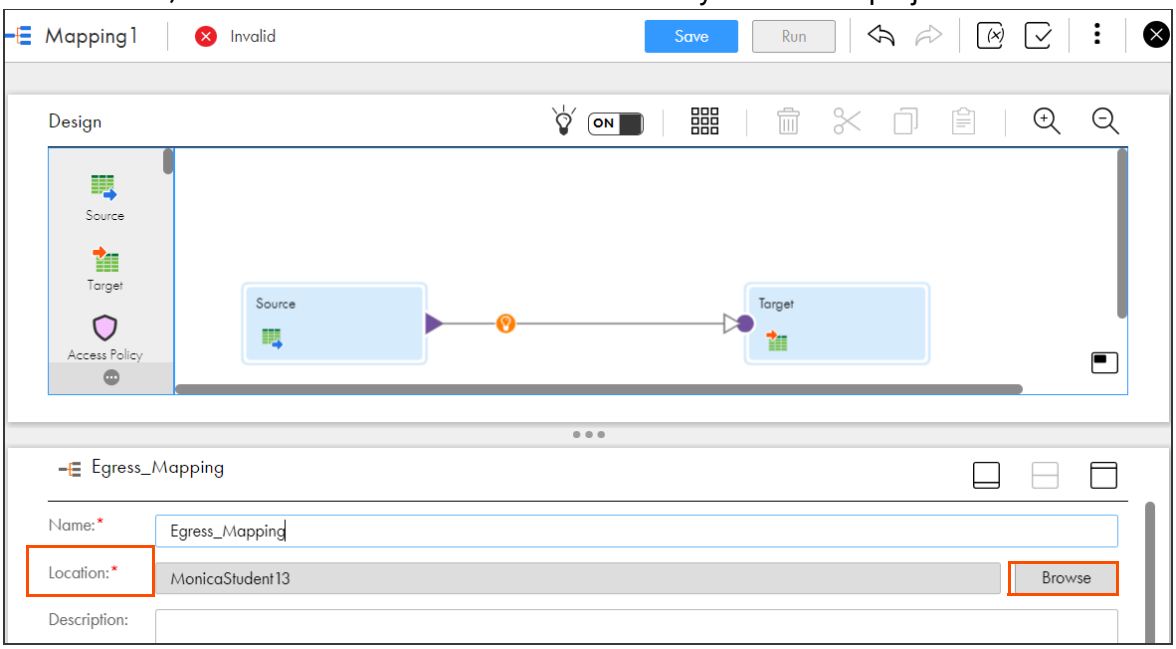


4. In the New Asset window, select **Mappings > Mapping**, and click **Create**. This will open a new mapping designer page. When you configure a mapping, you describe the flow of data from source and target.

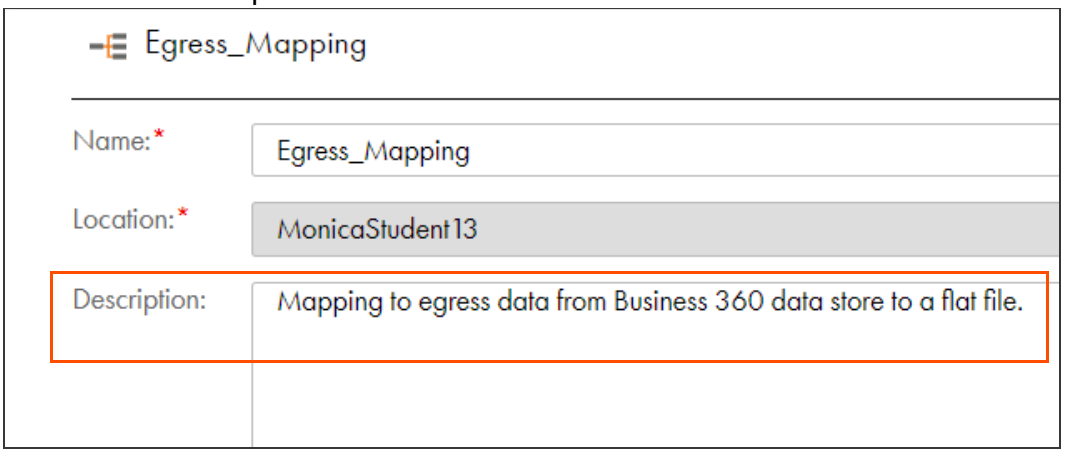


5. The mapping canvas opens with default Source and Target assets in the canvas. In the Properties section, enter the mapping name as **Egress_Mapping**.

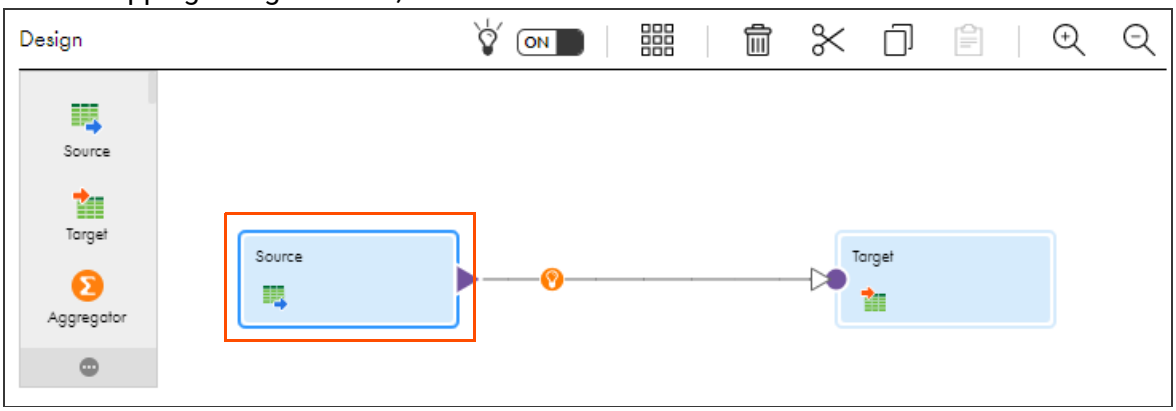
6. For **Location**, click **Browse** and select the Location of your student project.



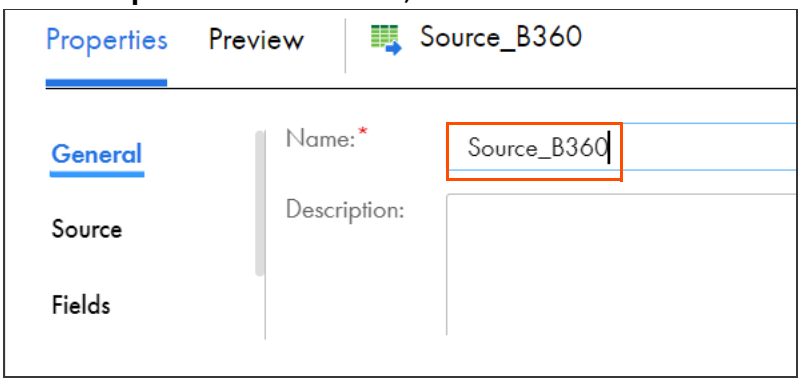
7. Enter a valid description.



8. In the Mapping Design canvas, select the **Source**.

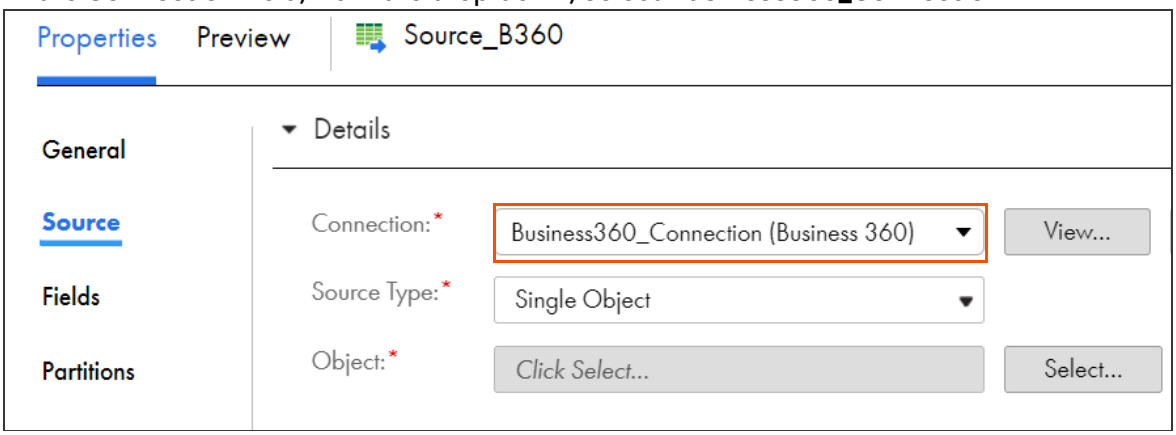


9. In the **Properties** > **General** tab, name the source transformation as **Source_B360**.



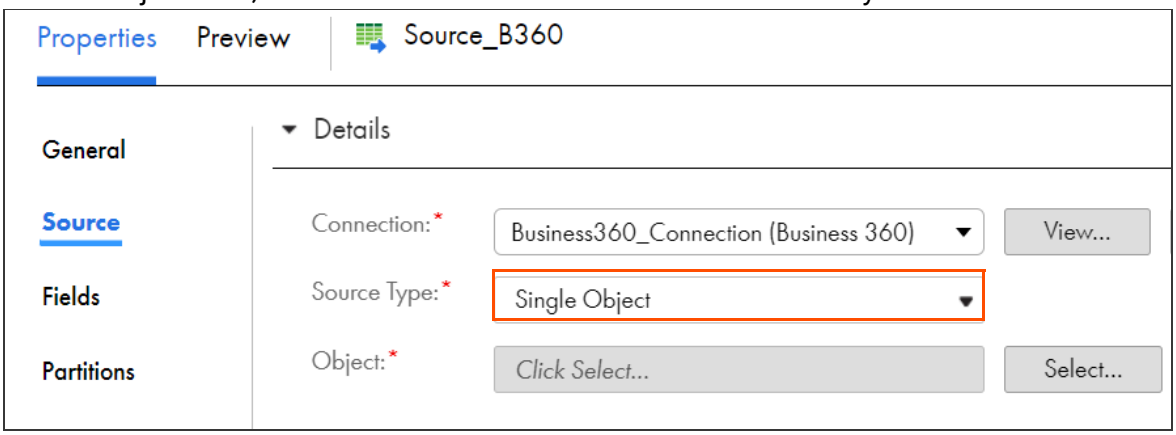
10. Click the **Source** tab.

11. In the **Connection** field, from the drop-down, select **Business360_Connection**.



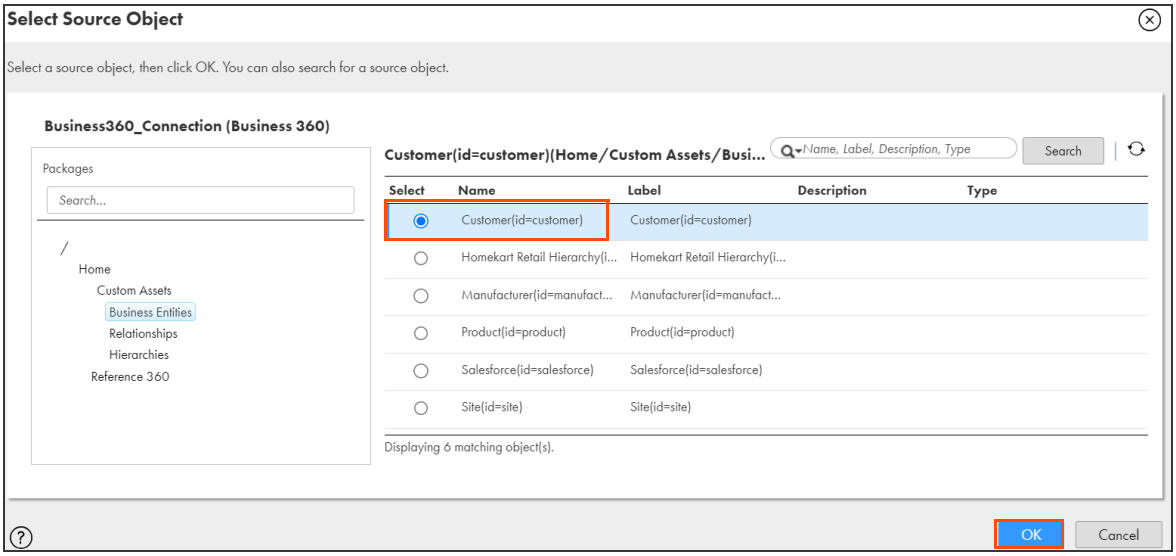
12. You will egress data from a single business entity. Hence, retain the Source Type as **Single Object**.

13. For the Object field, click **Select** button to select the business entity.



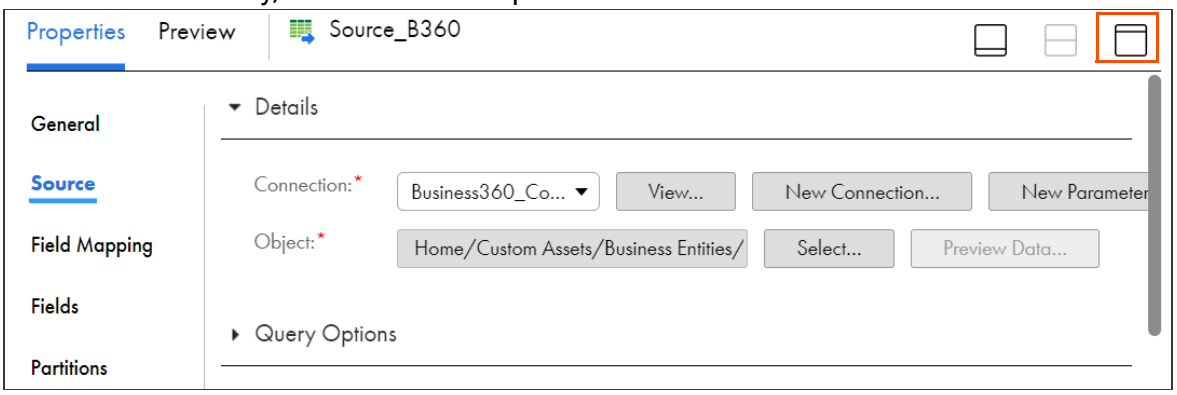
14. Click **Home** > **Custom Assets a**nd select **Business Entities**.

15. Select **Customer** and click **OK**.

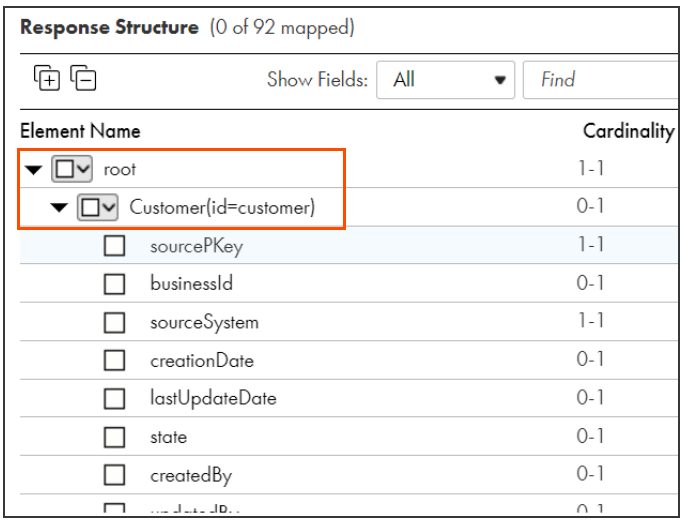


Data will be written from customer object.

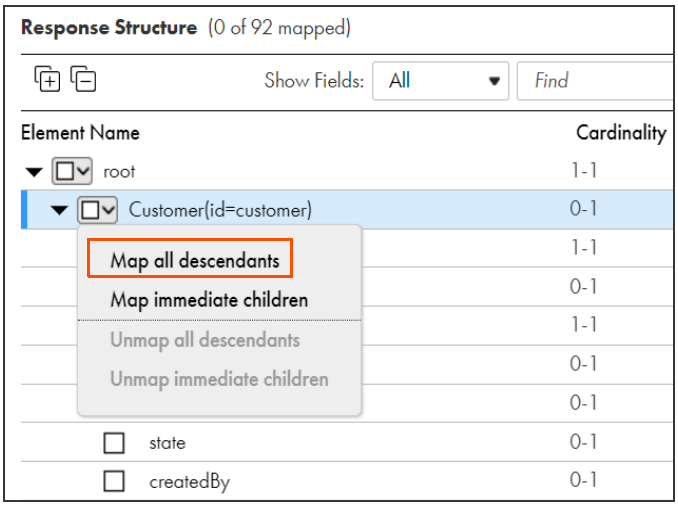
16. For better readability, maximize the Properties window.



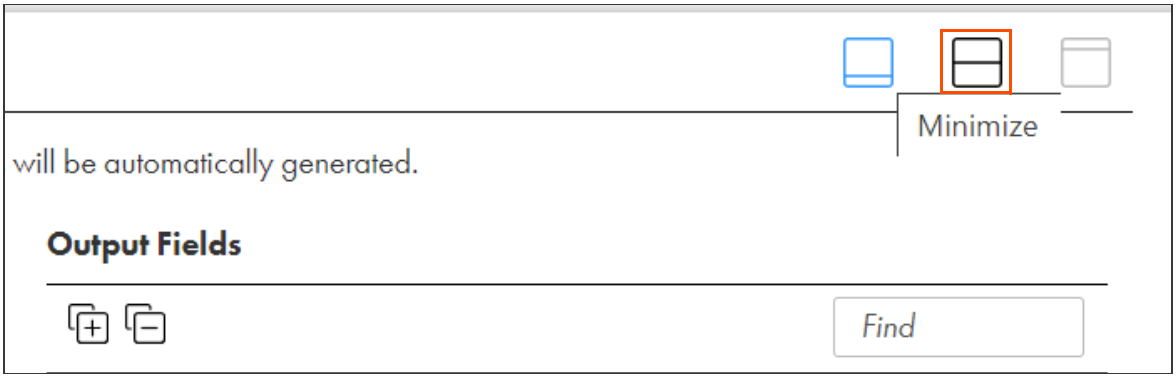
17. Click on **Field Mapping** property. The Response Structure in the Field Mapping tab shows the hierarchical structure of the data that comes from the source. When you select fields in the Response area, the fields appear in the Output Fields area. Expand **root > Customer** to view the hierarchical structure of the customer business entity and the field groups.



18. In the **Response Structure** section, select the check box adjacent to Customer and select **Map all descendants** option. If you expand the customer element, you can see all the fields and field groups of a customer business entity are selected and appear in the Output Fields area, which is on the left side.



19. Click **Restore** icon to retain the properties window to its default size.

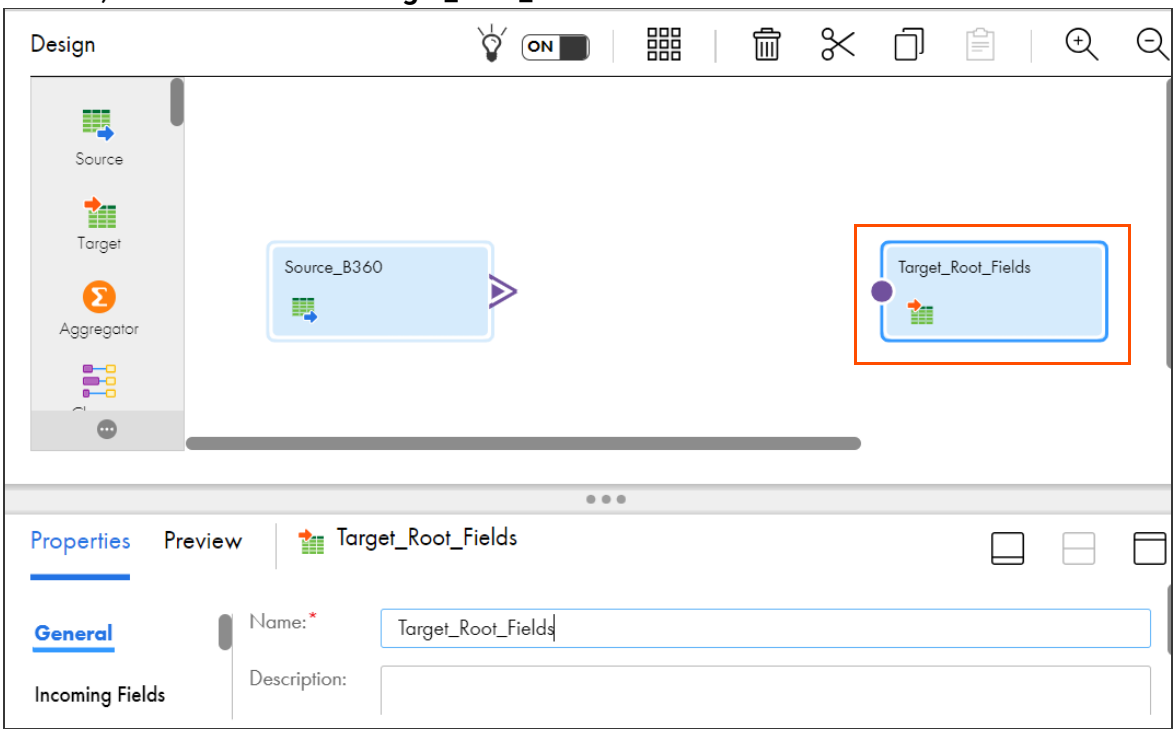


20. Click **Save**.

Configure the Target Connection

Now, you will configure the target connection from which you will be ingressing the data.

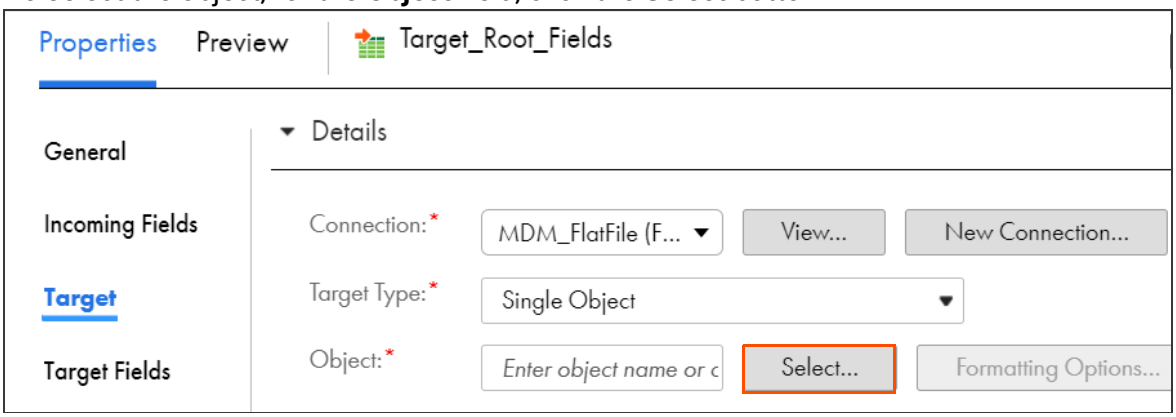
21. In the mapping canvas, select the Target component. From the **Properties** > **General** section, enter the Name as **Target_Root_Fields.**



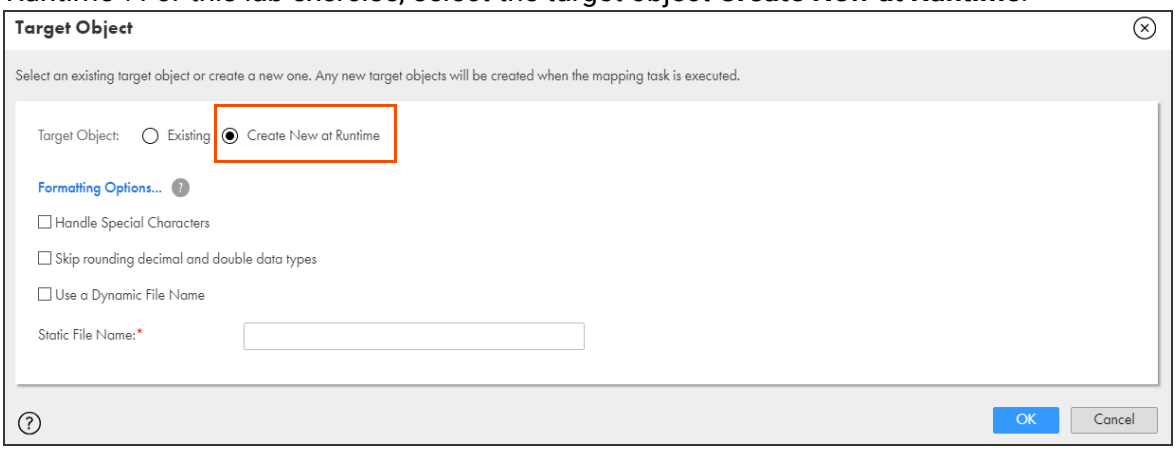
22. Click the **Target** tab.

23. In the Connection field, from the drop-down, select your flat file connection **MDM_Flatfile**.

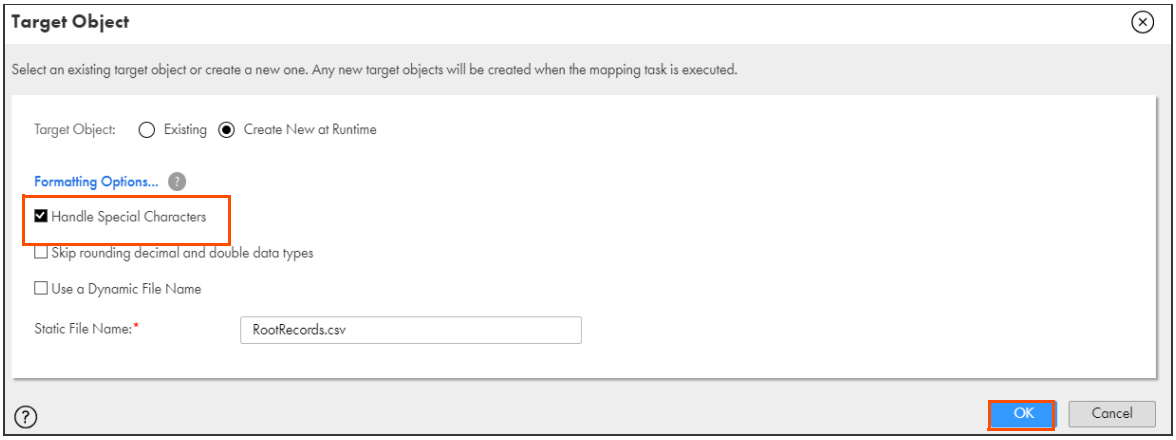
24. To select the object, for the **Object** field, click the **Select** button.



25. In the **Target Object** page, you can select “select an existing target” or “create a new at Runtime”. For this lab exercise, select the target object **Create New at Runtime**.

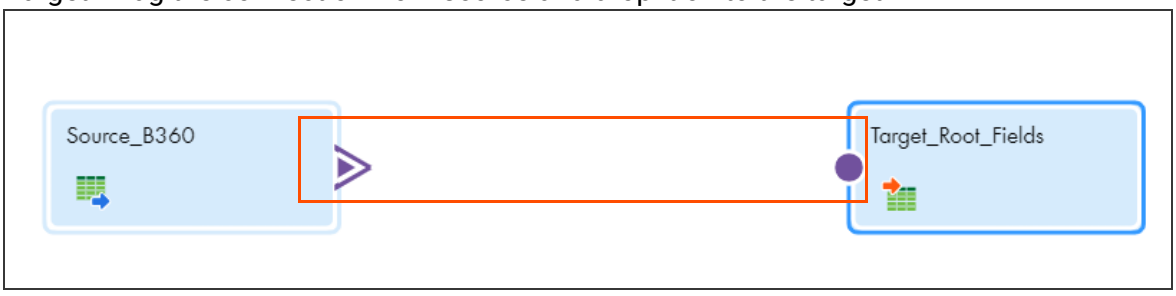


26. Select **Handle Special Characters** check box and enter the Static File Name as **RootRecords.csv**. Click **OK**.

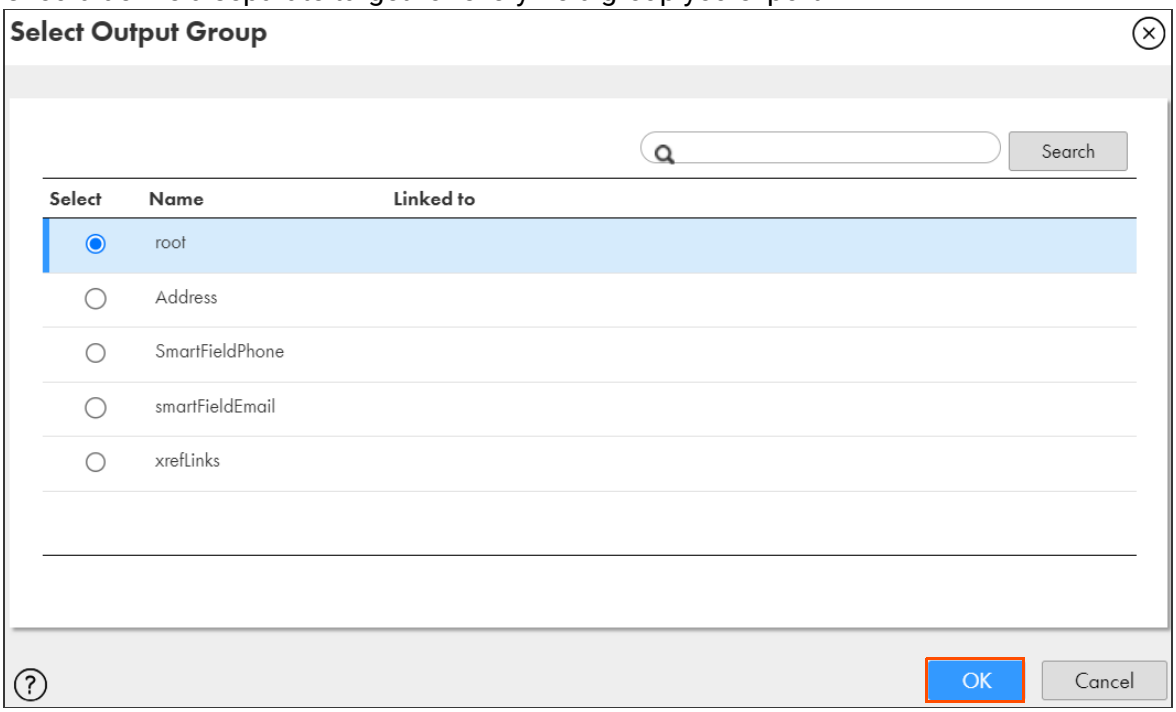


Do not forget to mention the file extension along with the file name.

27. In the mapping canvas, notice that the connections are disconnected between Source to Target. Drag the connection from source and drop it on to the target.

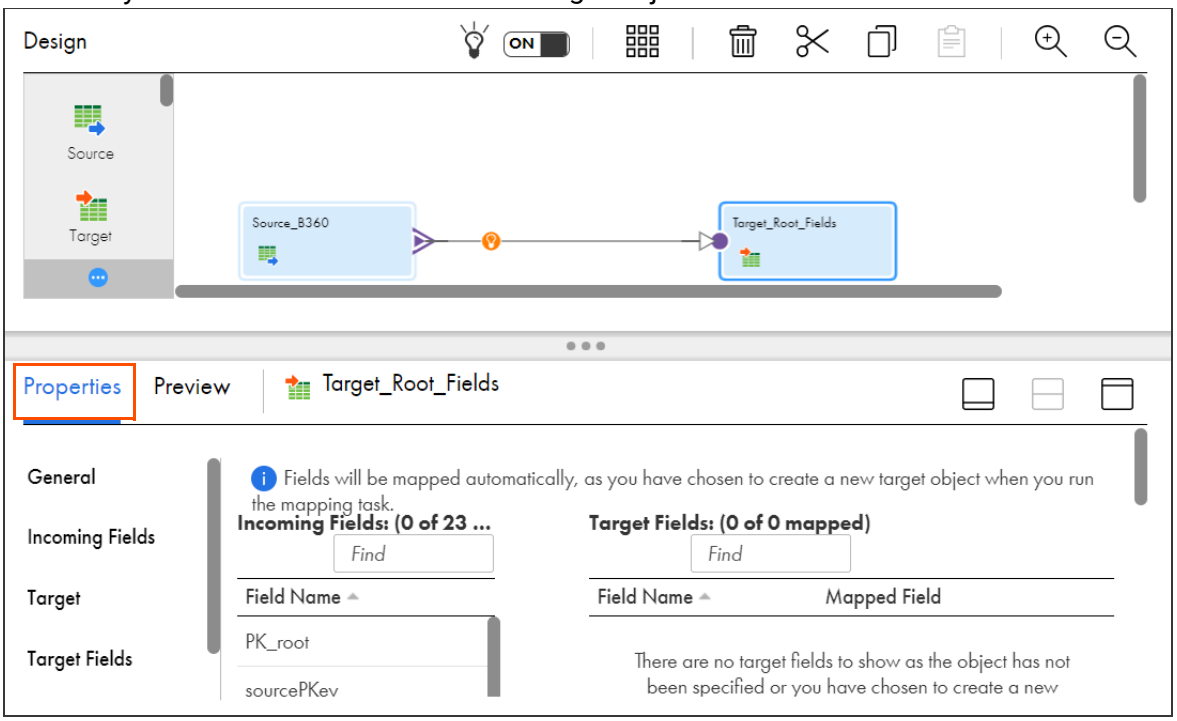


28. Once you drop the connection onto the target, a new page will pop up to select the field group. Select root and click **OK**. Each field group can be exported to only one target. You should define a separate target for every field group you export.

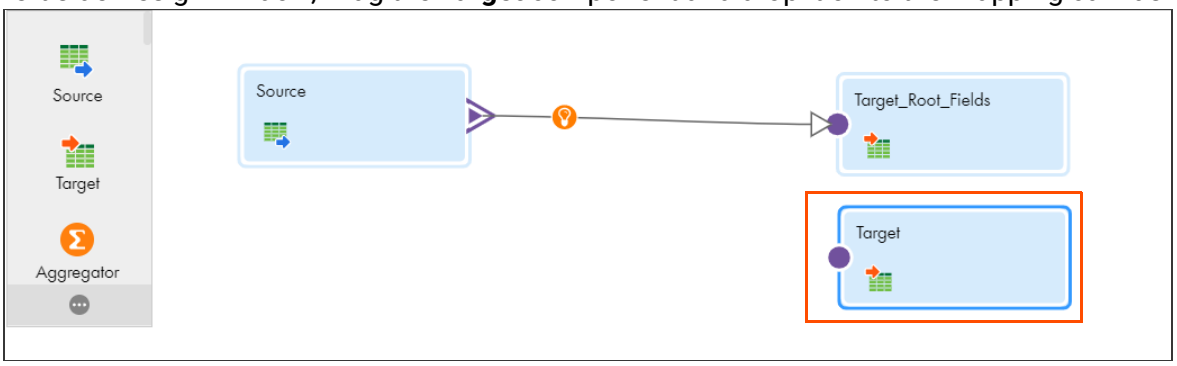


You will export two field groups in this lab: **root** and **PostalAddress**. So the reason you will be defining two targets in the mapping.

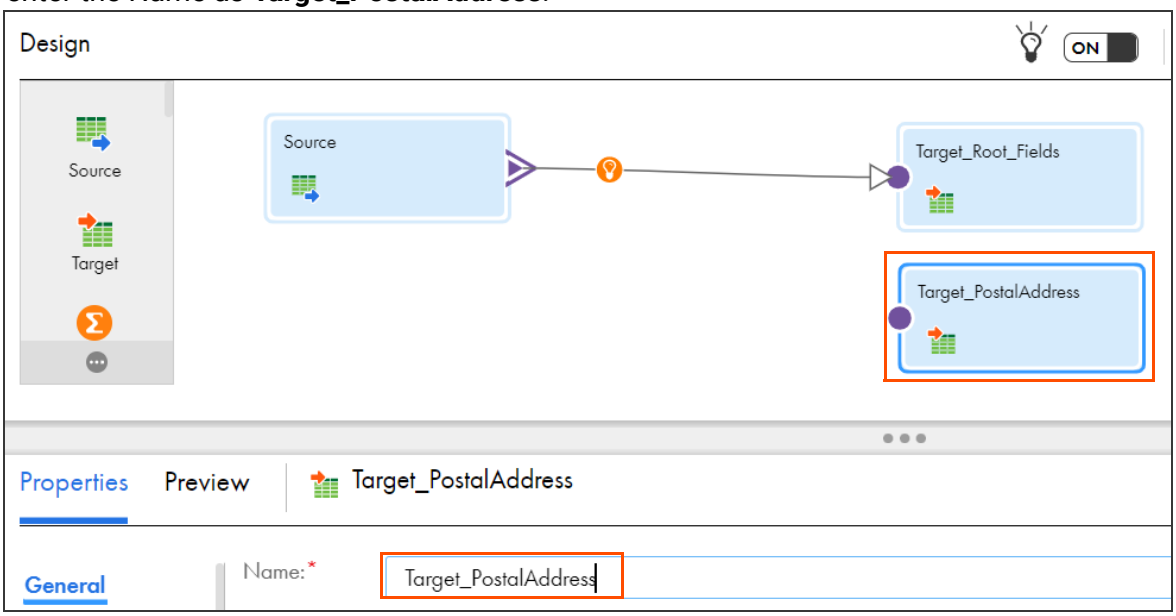
29. Click the target in the mapping canvas, and from the **Properties** section, click the **Field Mapping** tab. In the Incoming Fields section, you can see the root fields as you are exporting root records into this target. On the Target Fields section, you cannot see any fields as you are chosen to create a new target object at runtime.



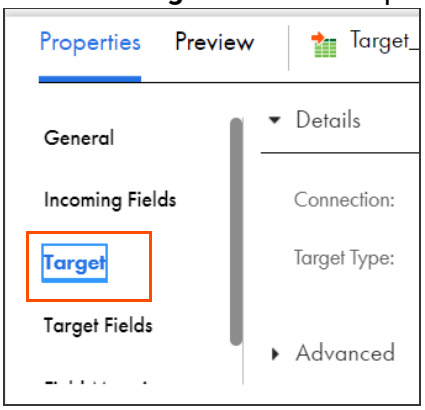
30. To export the **PostalAddress** field group, you need to define the second target. From the left side Design window, Drag the **Target** component and drop it on to the mapping canvas.



31. In the mapping canvas, select the second Target. From the **Properties** > **General** section, enter the Name as **Target_PostalAddress**.

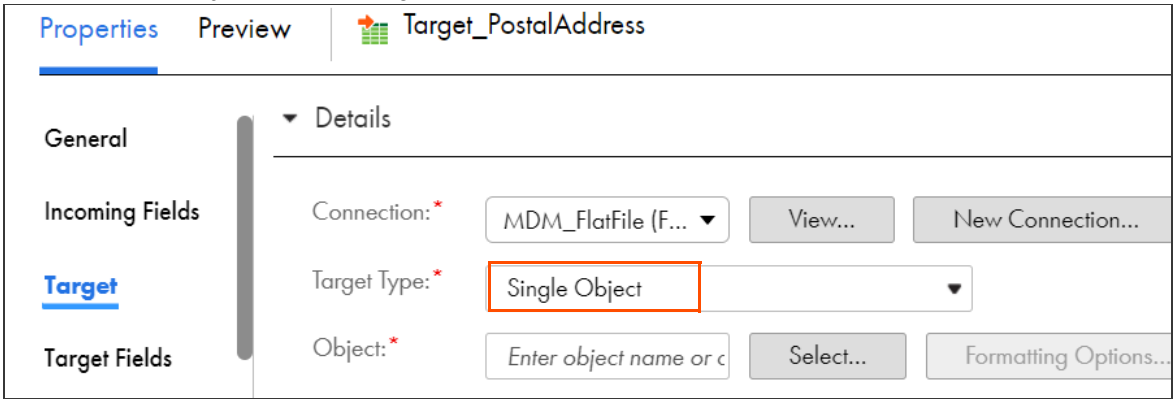


32. Click the **Target** tab from the properties.

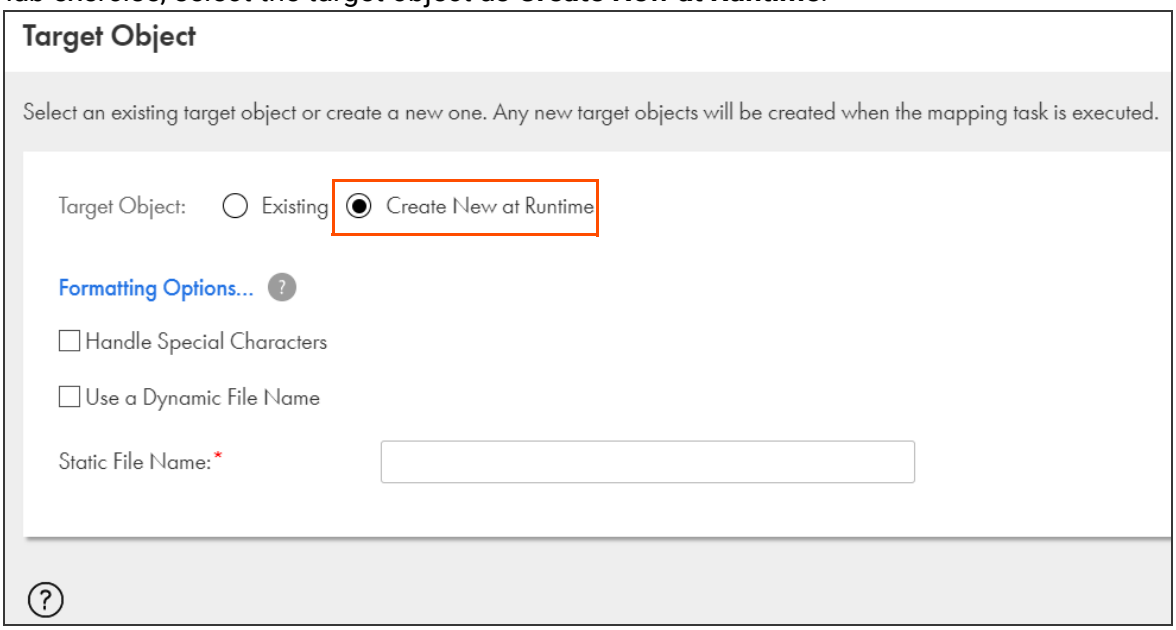


33. In the Connection field, from the drop-down, select your flat file connection **MDM_Flatfile**.

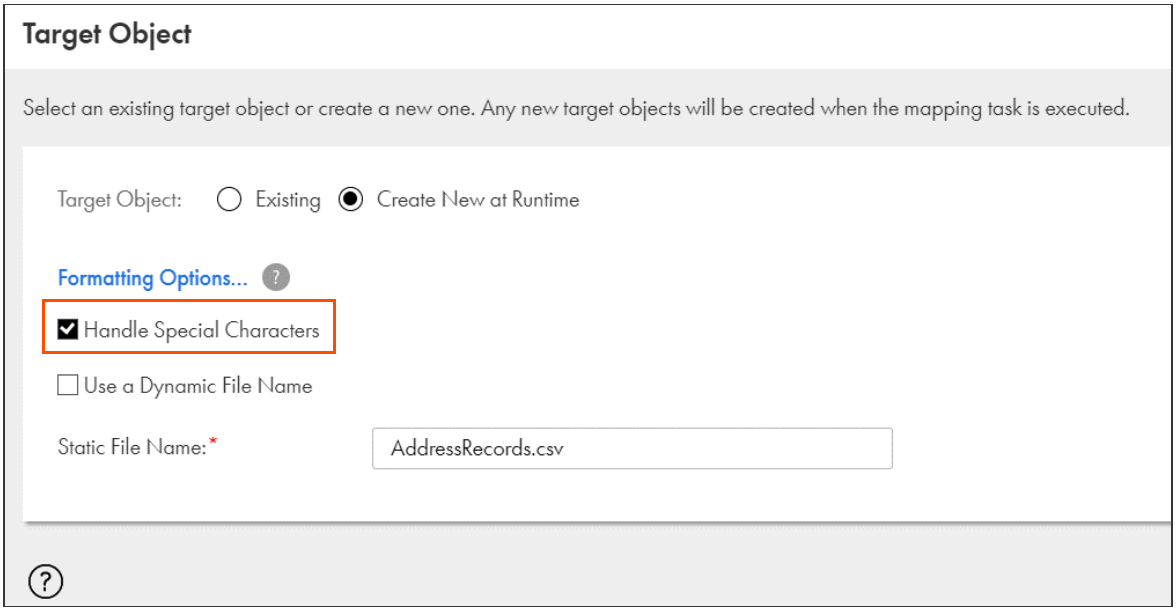
34. To select the object, for the **Object** field, click the **Select** button.



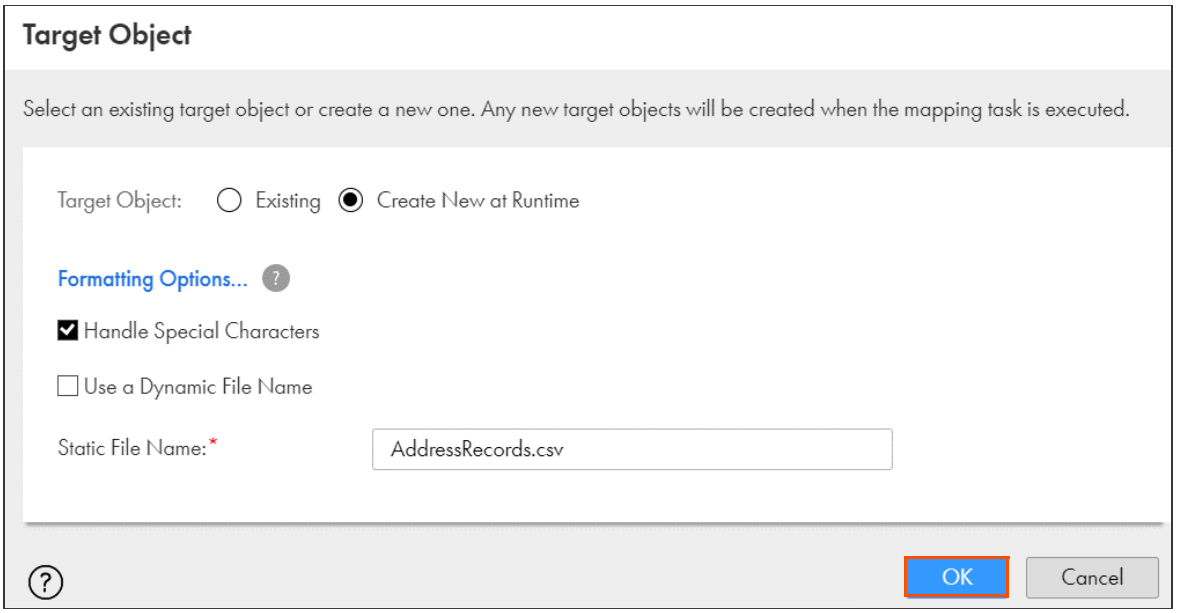
35. In the **Target Object** page, you can select an existing target or create a new one. For this lab exercise, select the target object as **Create New at Runtime**.



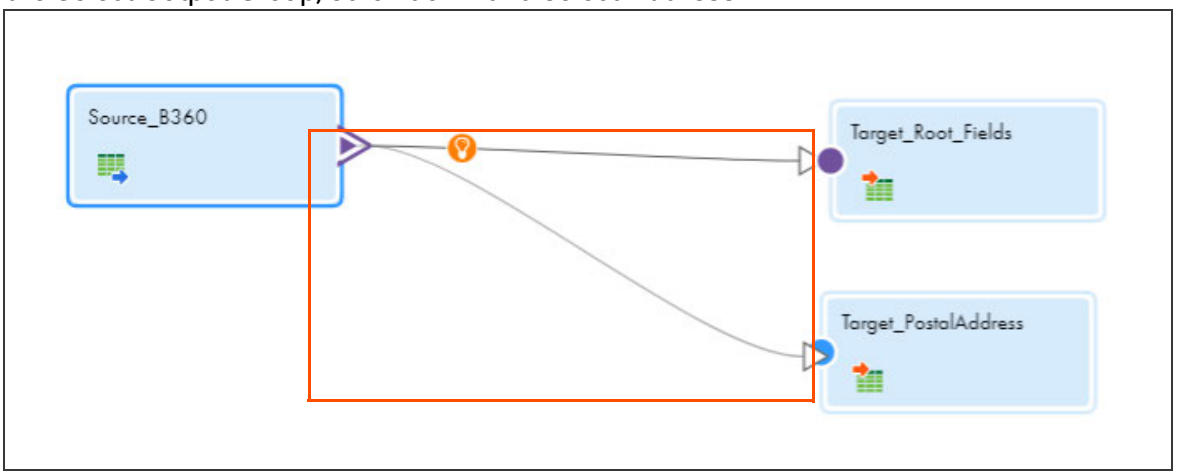
36. Select **Handle Special Characters** check box and enter the Static File Name as **AddressRecords.csv**.

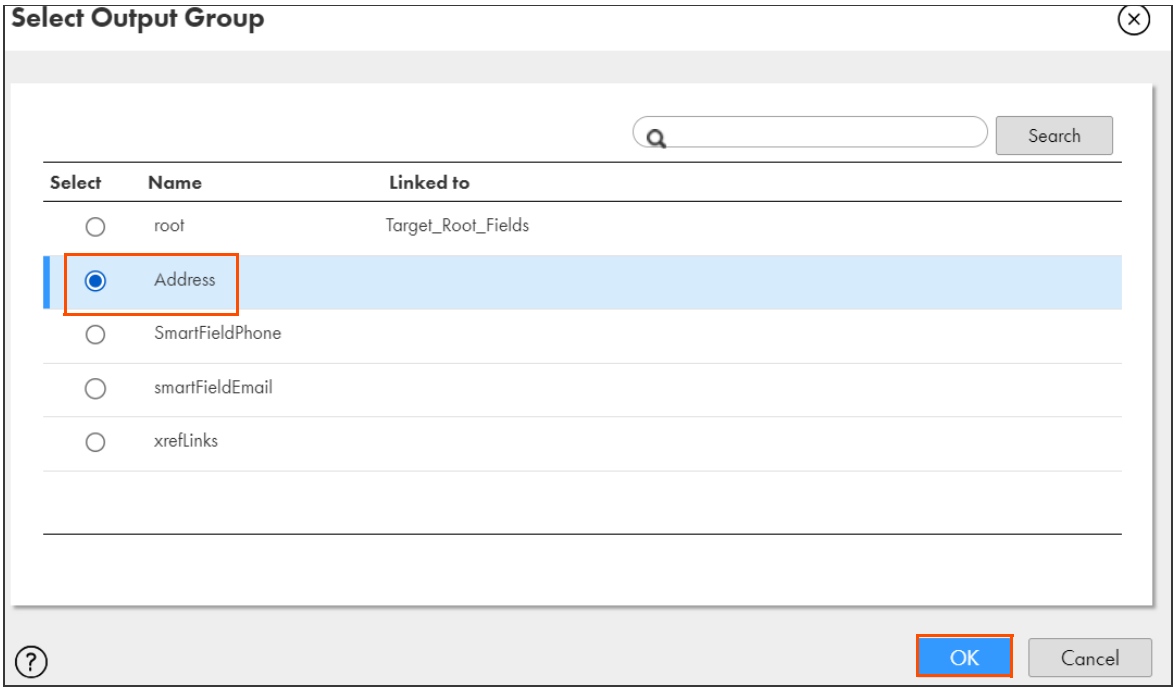


37. Click **OK**.



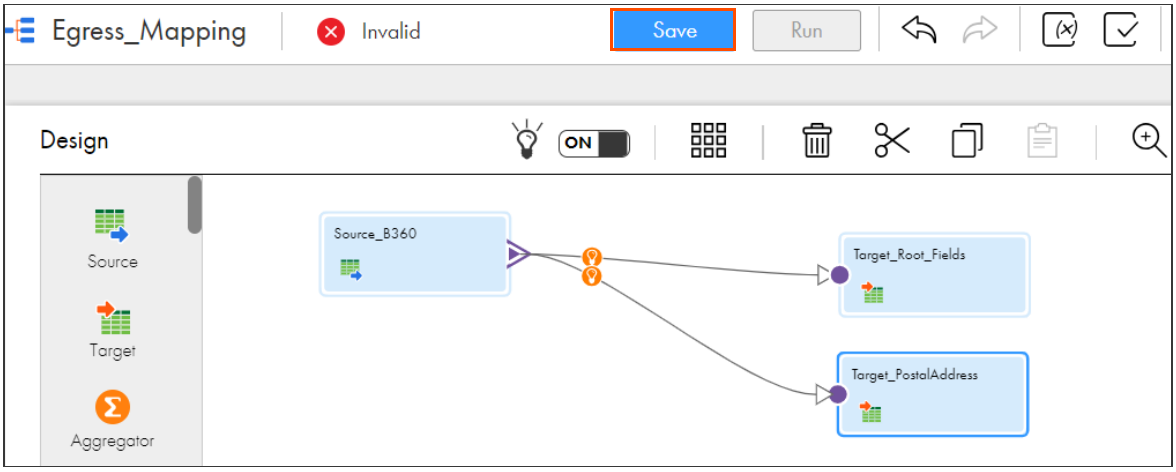
38. Drag the connection from the source and connect it on to the **Target_PostalAddress**. From the Select Output Group, scroll down and select **Address**.





Click the **Arrange All** icon on the top right side of the canvas to arrange the objects in the mapping.

39. **Save** the mapping.

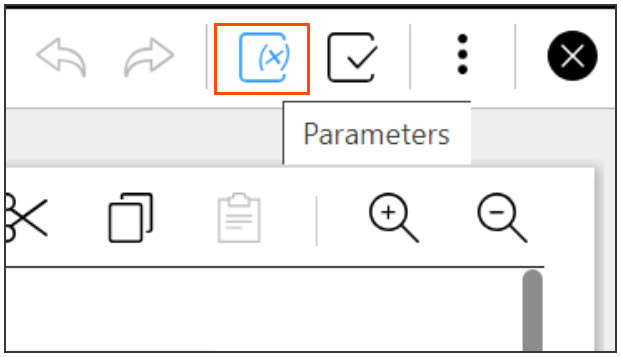


Create In-Out Parameter

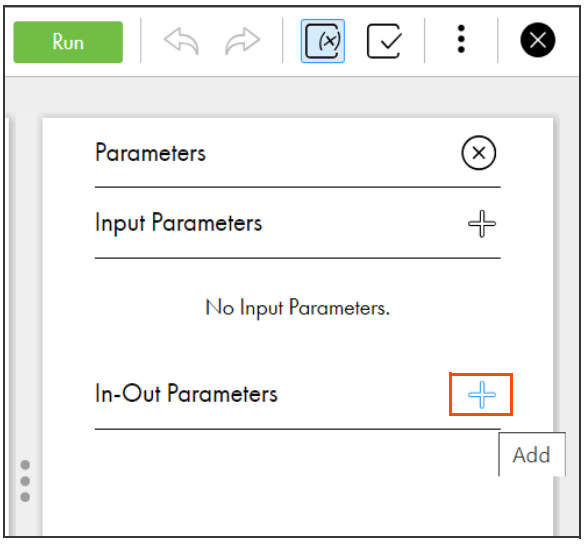
In-out parameters are created when the value of a particular field keeps changing at runtime. You will now create an in-out parameter named jobInstanceID that reads the job instance ID at runtime.

40. In-out parameters are created when the value of a particular field keeps changing at runtime. You will now create an in-out parameter named **jobInstanceID** that reads the job instance ID at runtime.

41. Click the **Parameters** icon from the top right of the page.

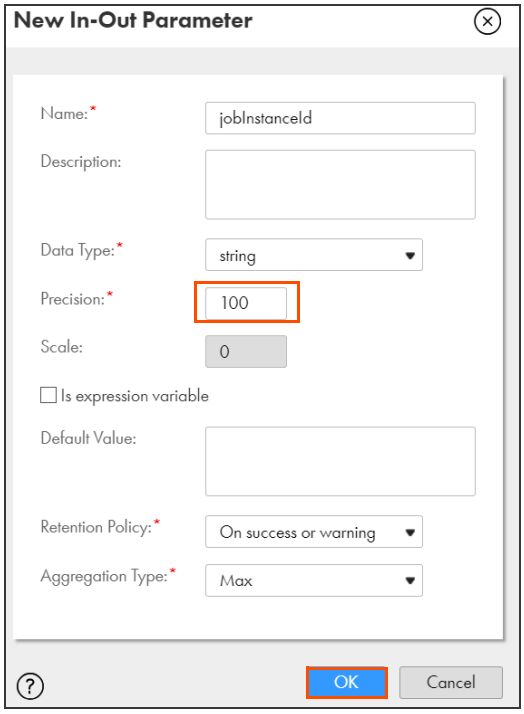


42. In the **Parameters** section on the right pane, click the **+** icon for the **In-Out Parameters**.

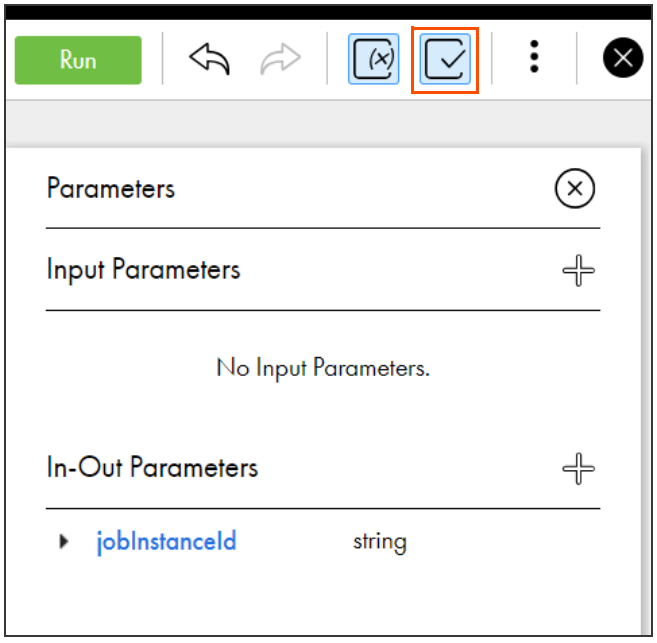


43. Enter **jobInstanceId** as the Name.

44. Set the Precision to **100** and click **OK**.

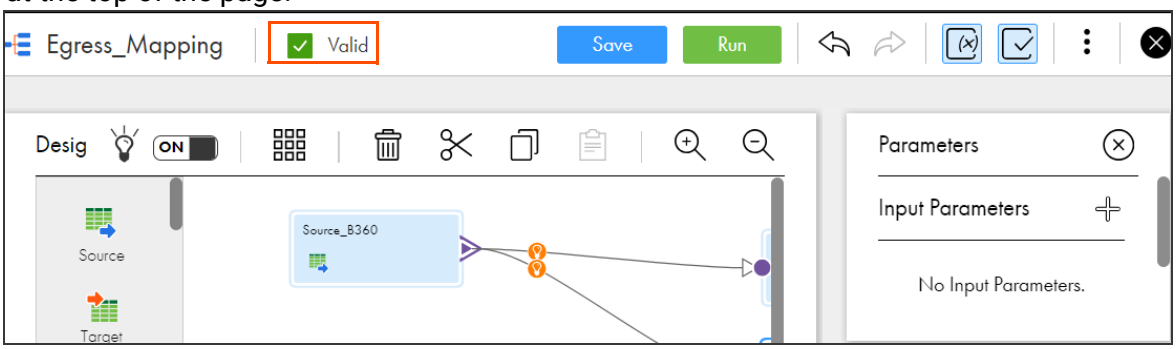


45. The in-out parameter now appears in the Parameters panel. Click the **Validate** icon.



46. **Save** the mapping.

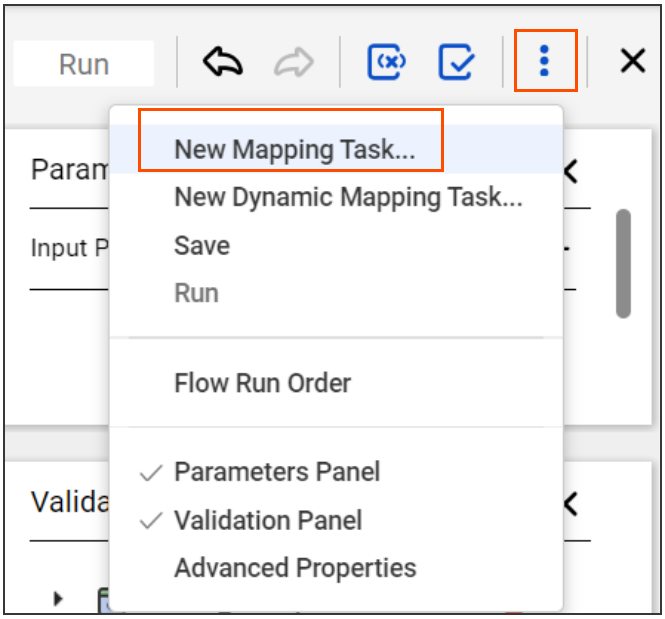
Notice that the mapping is valid in the Validation section. You can also view Valid check mark at the top of the page.



Create a Mapping Task

As the next step, you will create a mapping task and configure the properties.

47. To create a mapping task, click the 3 dots next to the Valid icon and click **New Mapping Task**.

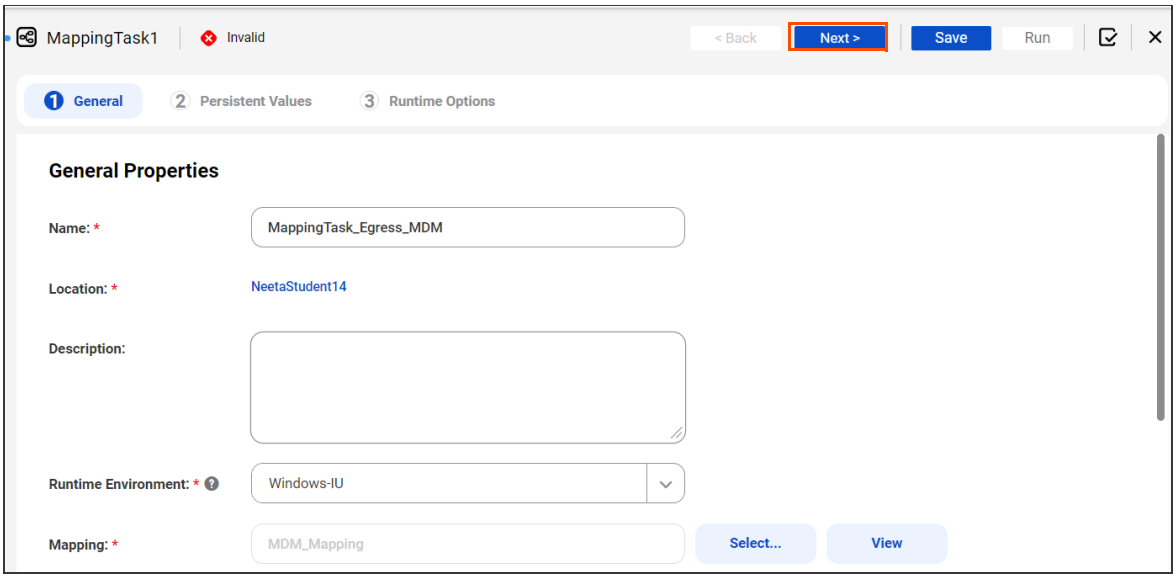


48. In the **New MappingTask** window, enter the Task Name as **MappingTask_Egress_MDM**.

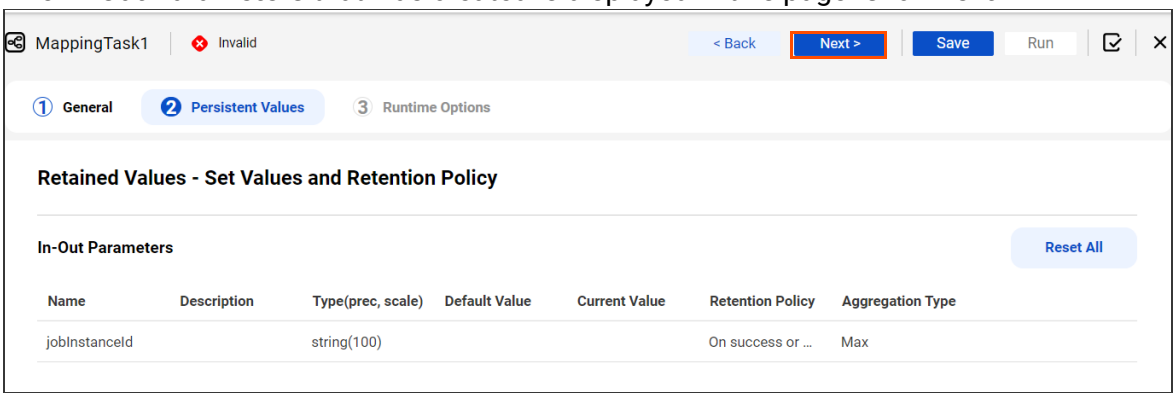
49. The Location and the Mapping are auto populated. Retain the same.

50. Select the **Runtime Environment** (SecureAgent that you have installed).

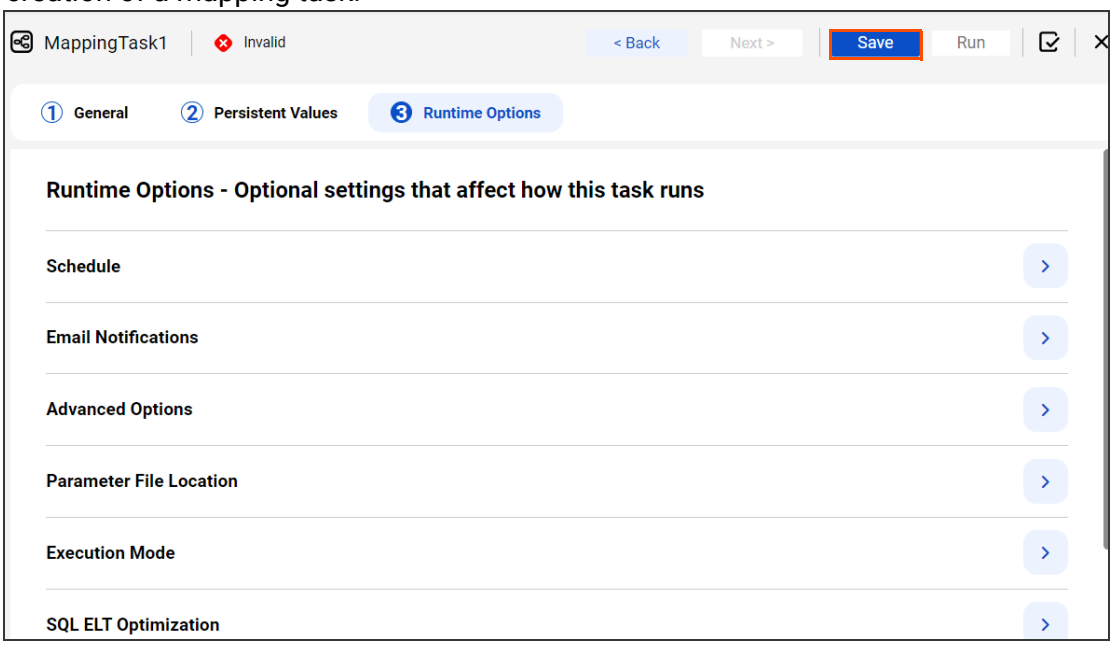
51. Click **Next**.



52. The **In-Out Parameters** that was created is displayed in this page. Click **Next**.



53. Do not make any changes in the **Schedule Details** section. Click **Save** to complete the creation of a mapping task.

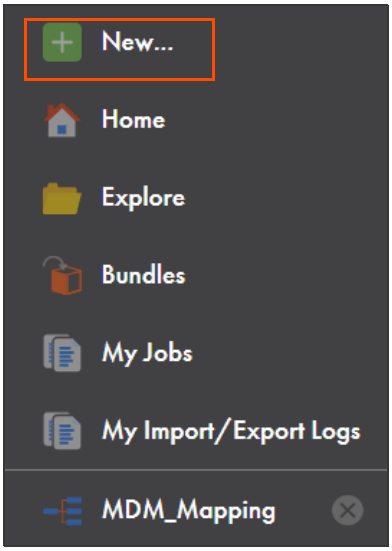


54. The mapping task is saved successfully. Scroll down to see all details.

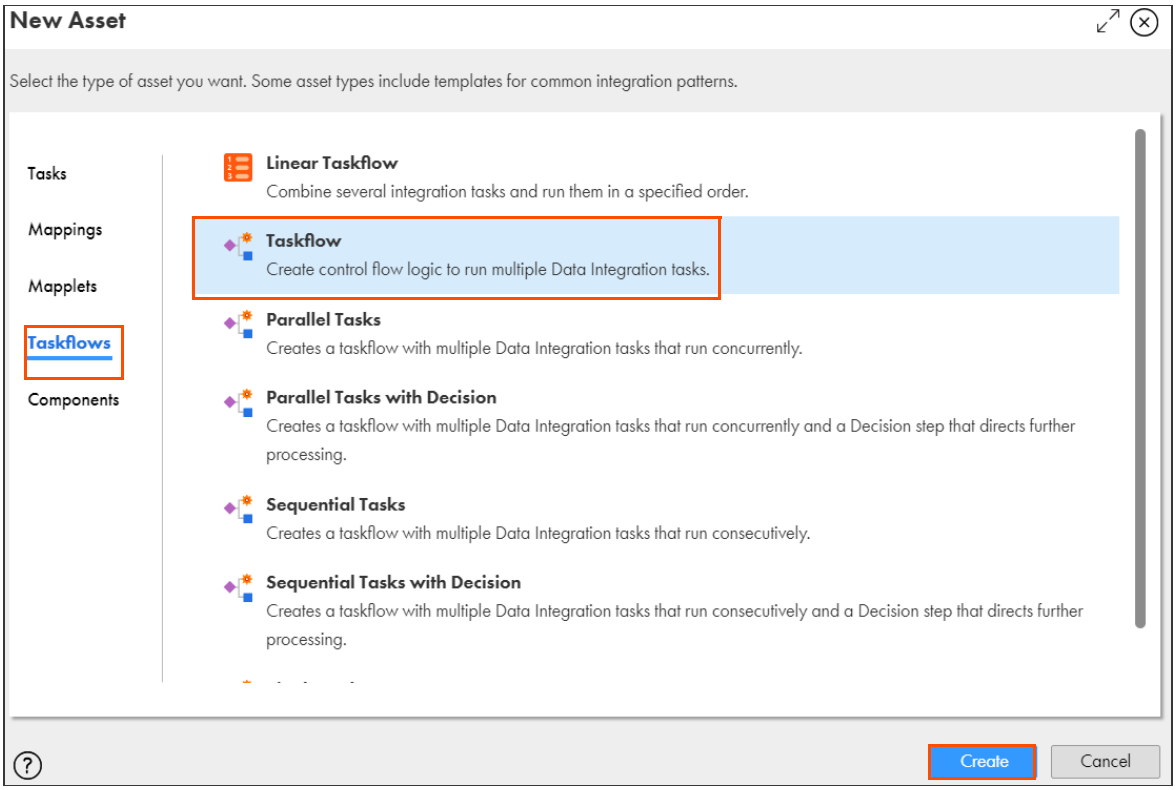
Create a Taskflow

You will now create a taskflow, that you will be publishing later.

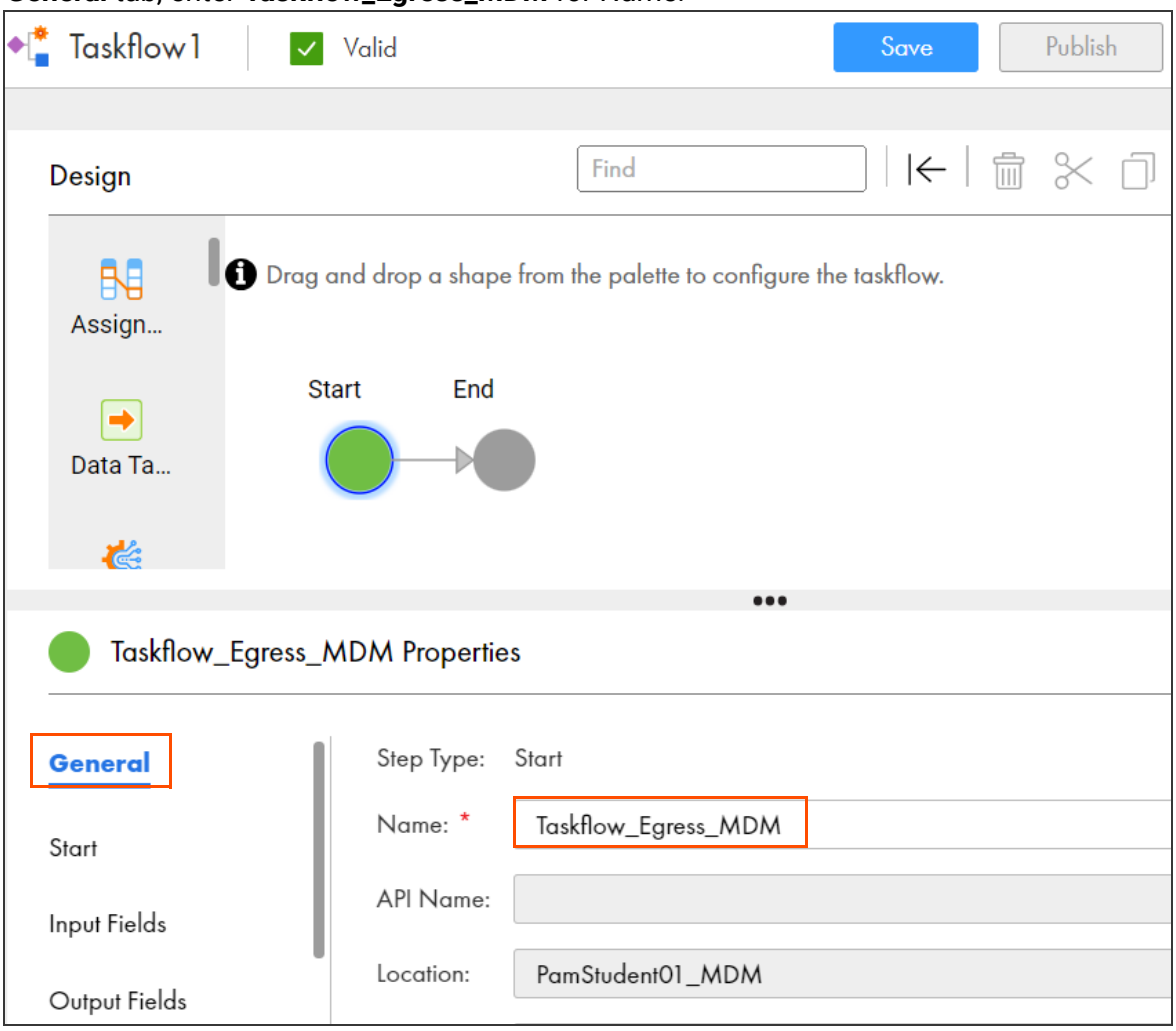
55. From the left panel, click **New**.



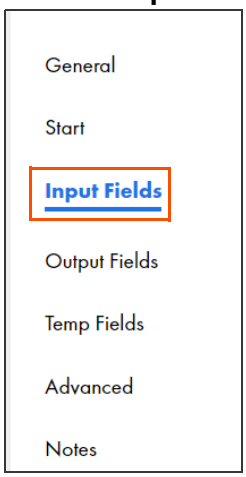
56. Select **Taskflows** > **Taskflow** and click **Create**.



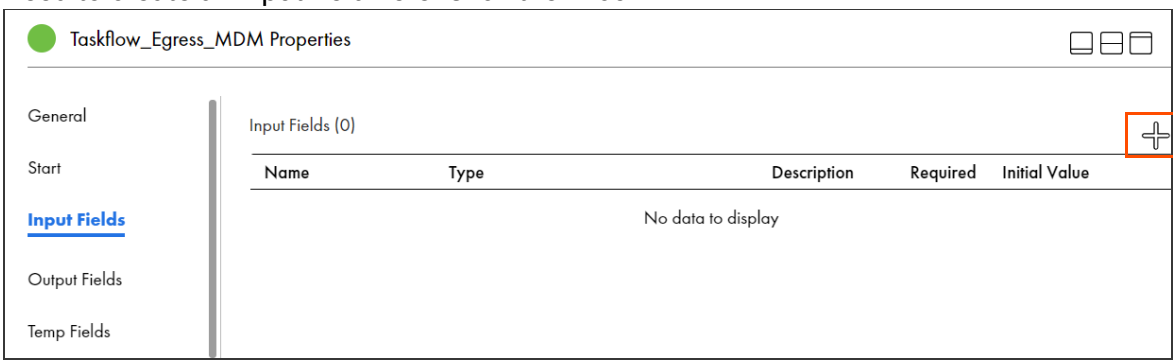
57. The taskflow appears in the canvas. The Start node is selected by default. From the **General** tab, enter **Taskflow_Egress_MDM** for Name.



58. Click the **Input Fields** tab.

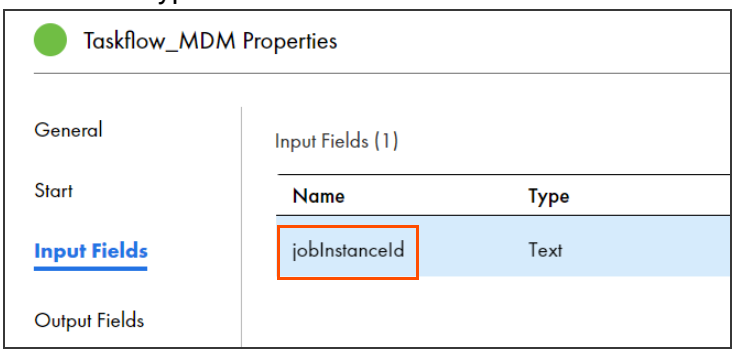


59. The parameter that was created while creating the mapping will be used as an input field here. The mapping will send that instance id as an input to the taskflow. Hence, you will need to create an input field here. Click the **+** icon.

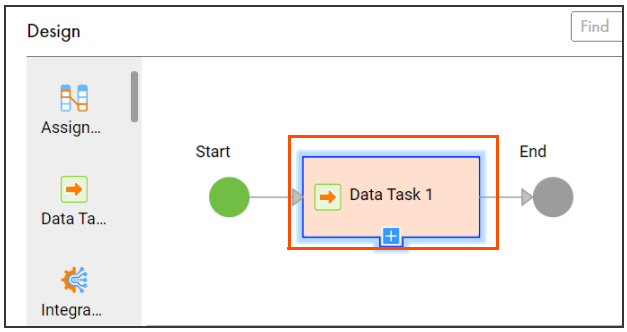


60. Double-click the field and enter the Name as **jobInstanceId**.

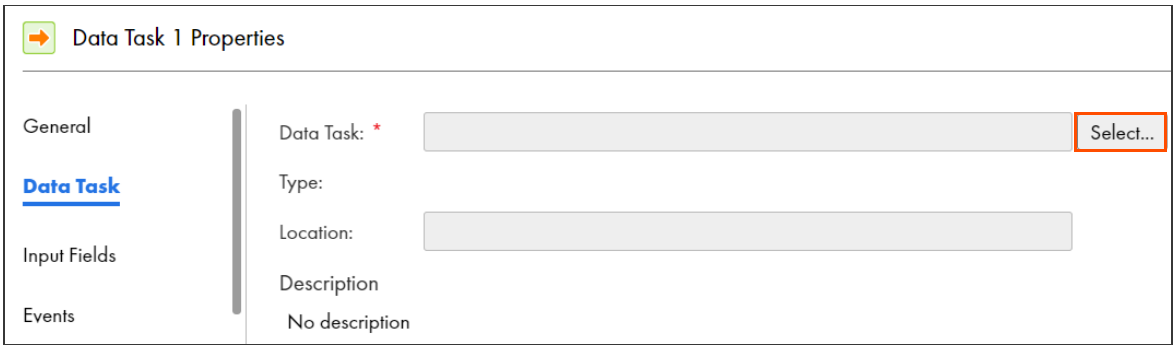
61. Retain the Type as **Text**.



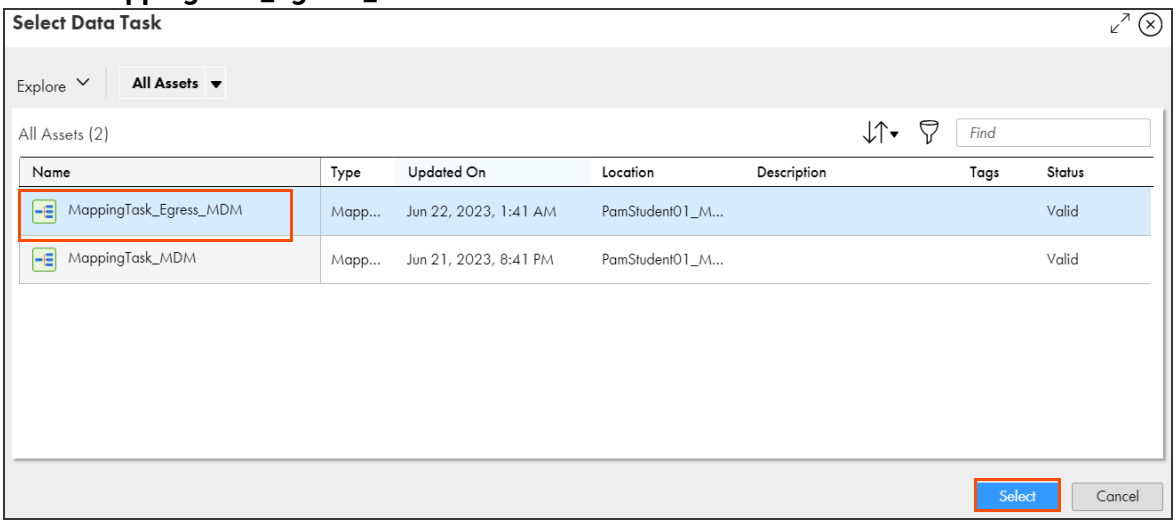
62. To call a mapping task, drag and drop a **Data Task** between the Start and End nodes.



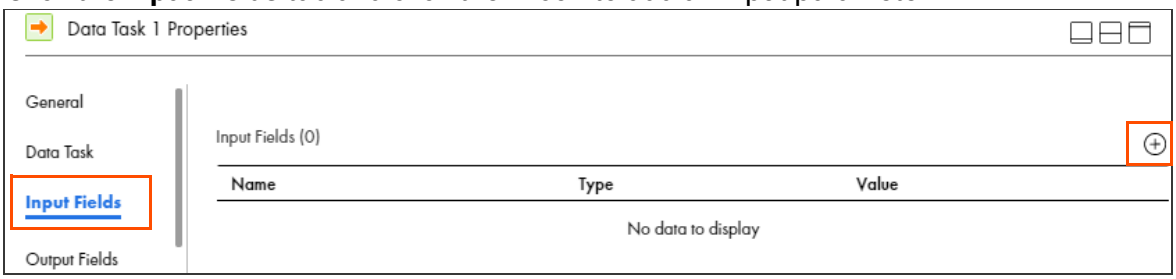
63. For the Data Task field, click **Select**.



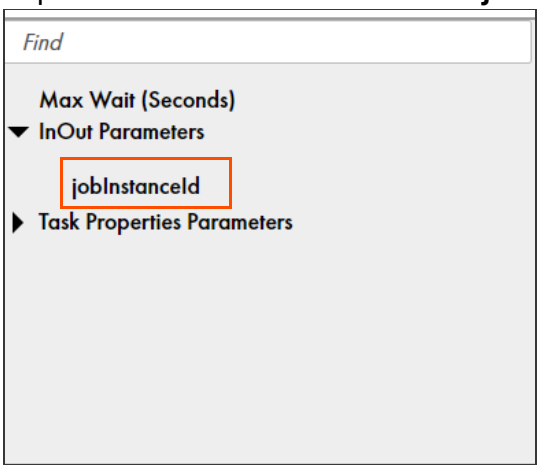
64. Click **MappingTask_Egress_MDM** and click **Select**.



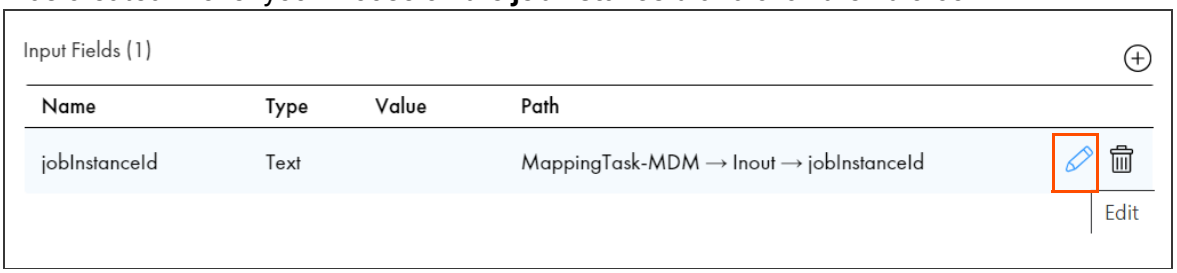
65. Click the **Input Fields** tab and click the **+** icon to add an input parameter.



66. Expand **InOutParameters** and select **jobInstanceId**.

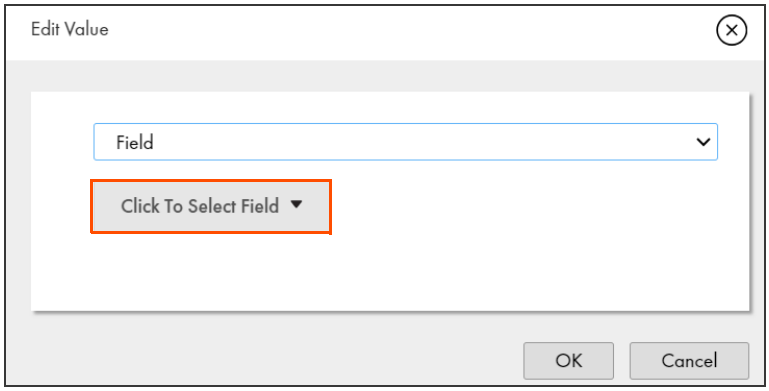


67. By default, the type is **Text**, but "jobInstanceId" is not of type text. It is the input field that was created. Hover your mouse on the **jobInstanceId** and click the **Edit** icon.

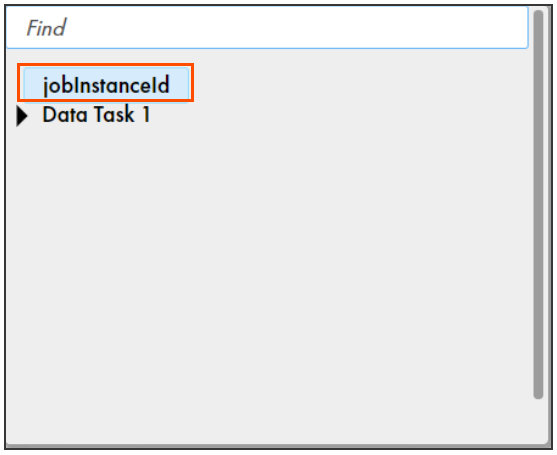


68. In the **Edit Value** window, select the type as **Field** from the drop-down.

69. Click the **Click To Select Field** button.

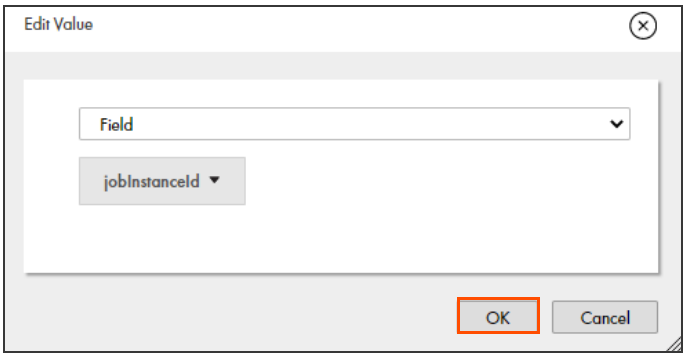


70. Click **jobInstanceId**.

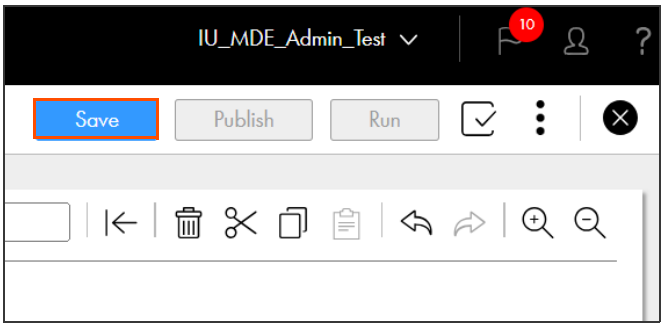


Make sure that you select the jobInstanceId as shown in the above image.

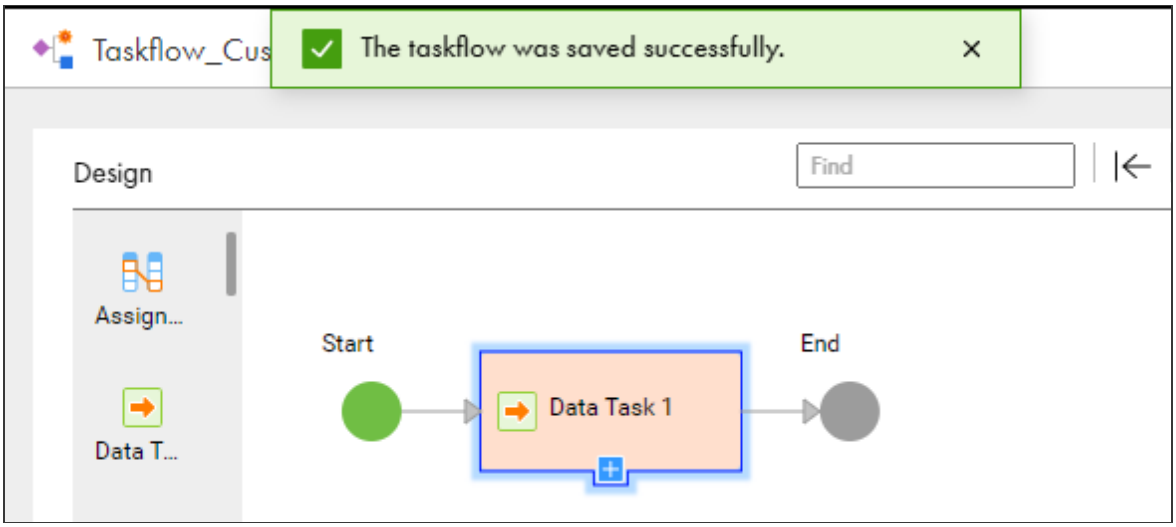
71. Click **OK**.



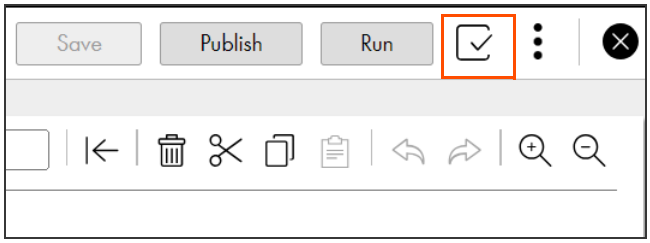
72. **Save** the Taskflow.

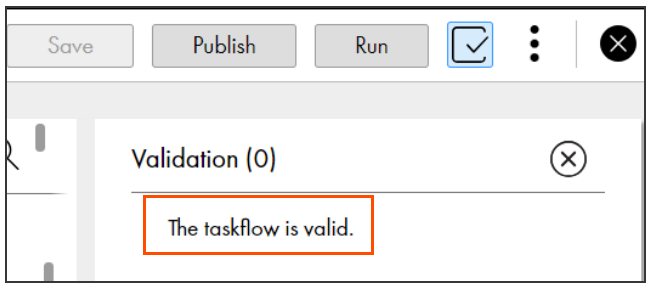


A notification is displayed that the taskflow was saved successfully.

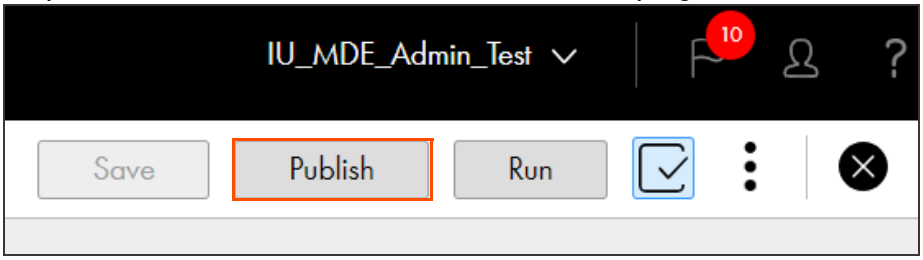


73. Click the **Validation** icon to check the validity of the taskflow. You can see that the taskflow is valid.

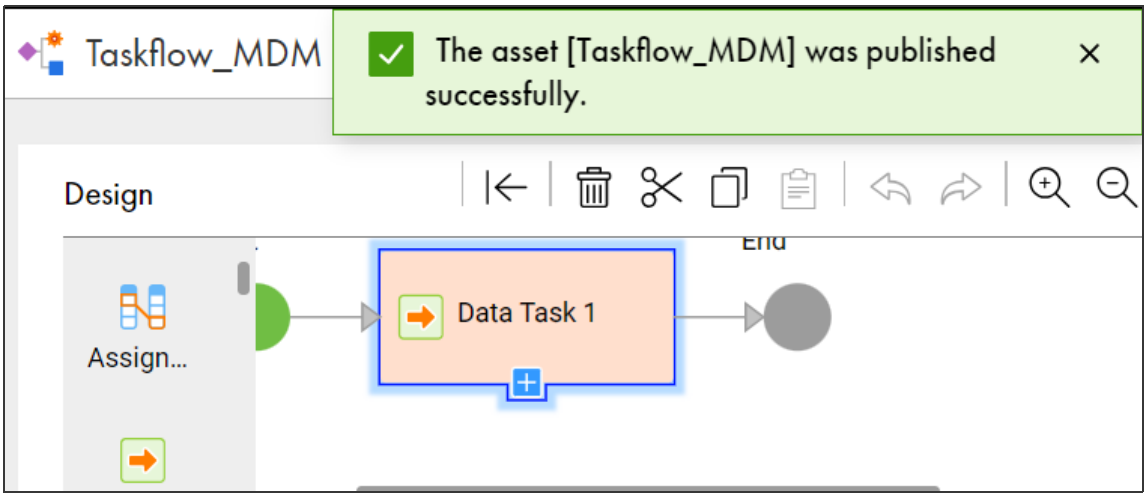




74. To publish the taskflow, click **Publish** from the top right corner.



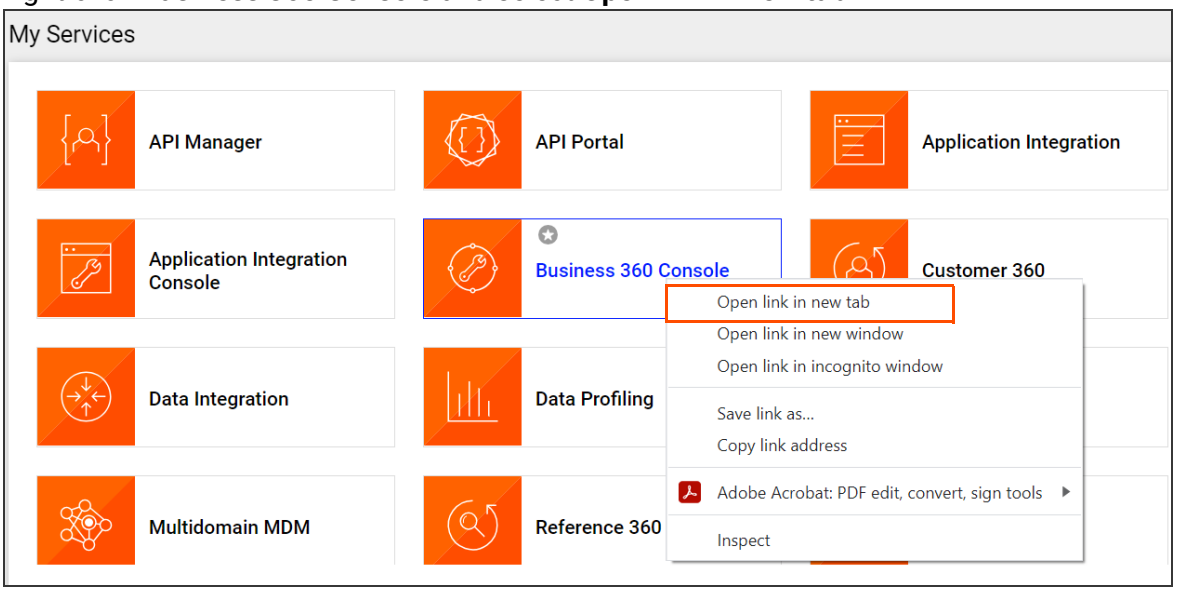
A notification is displayed that the taskflow was published successfully.



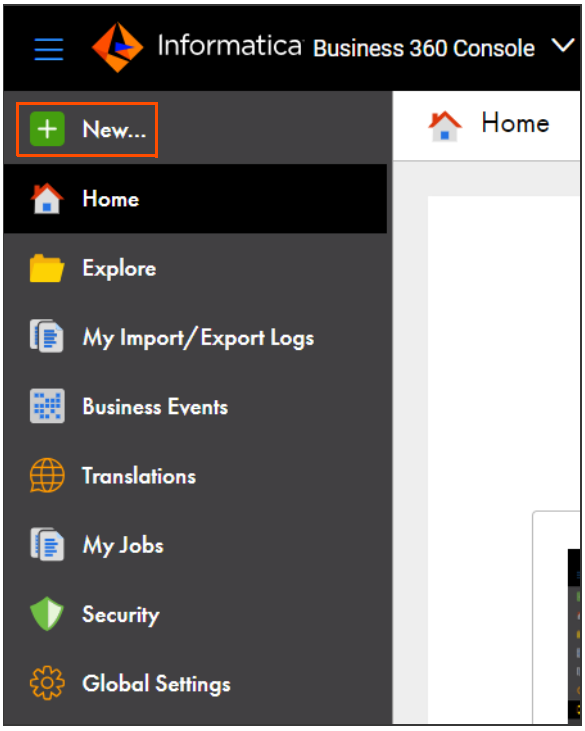
Create Egress Job in Business 360 Console

The last step is to create an egress job in Business 360 Console.

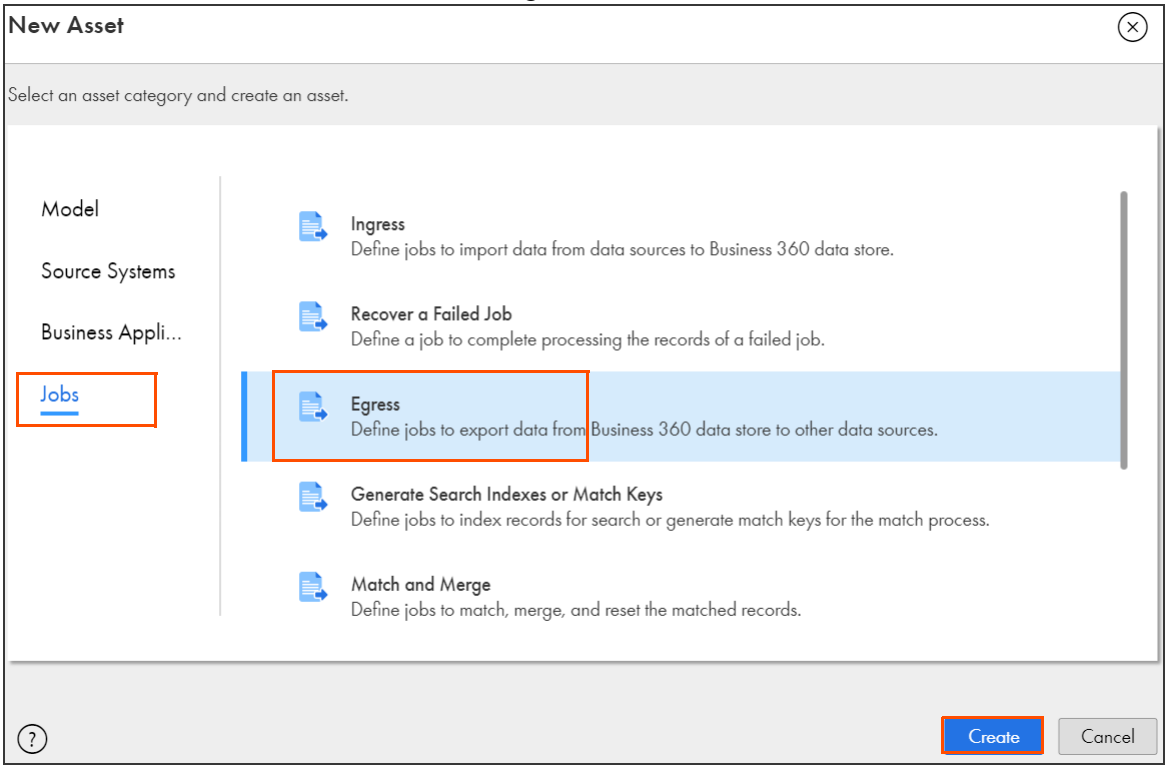
75. Click **Informatica Data Integration** from the top of the page and from the Services window, right-click **Business 360 Console** and select **Open link in new tab**.



76. In the **Business 360 Console** window, click **New**.

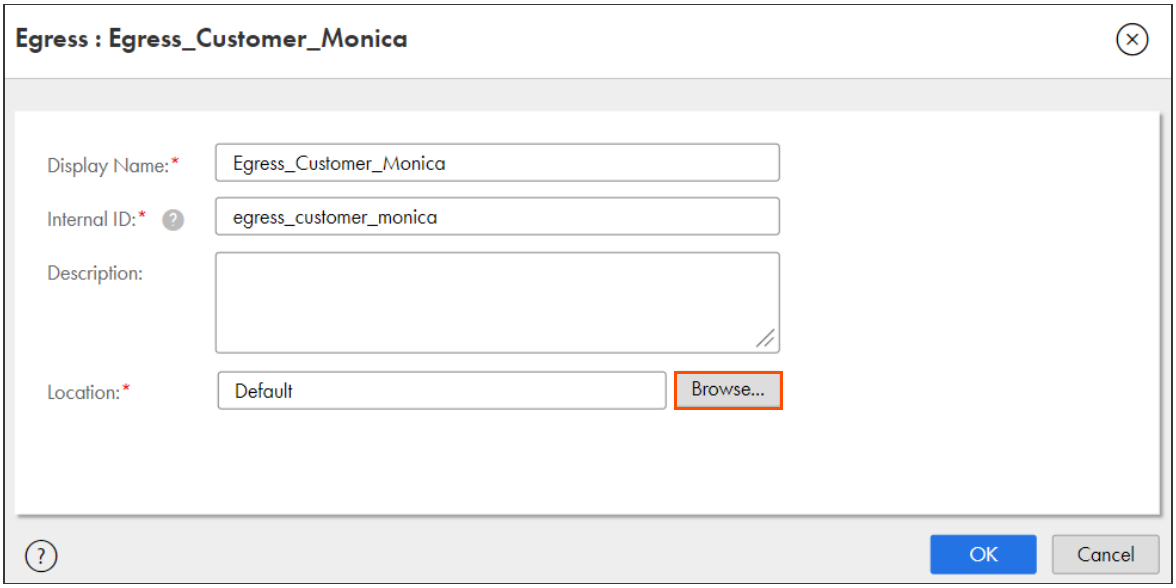


77. In the **New Asset** window, click **Jobs** > **Egress**, and click **Create**.

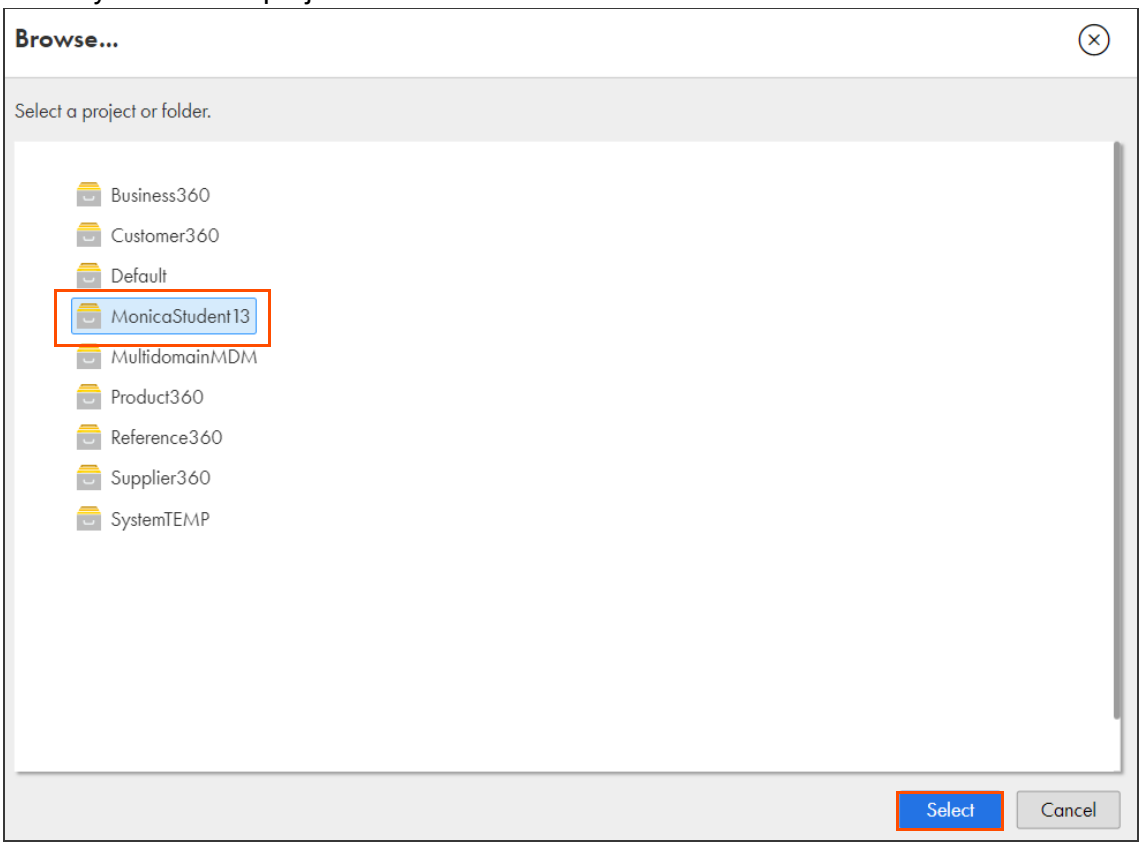


78. Enter **Egress_Customer_<studentName>** as the Display Name, for example Egress_Customer_.

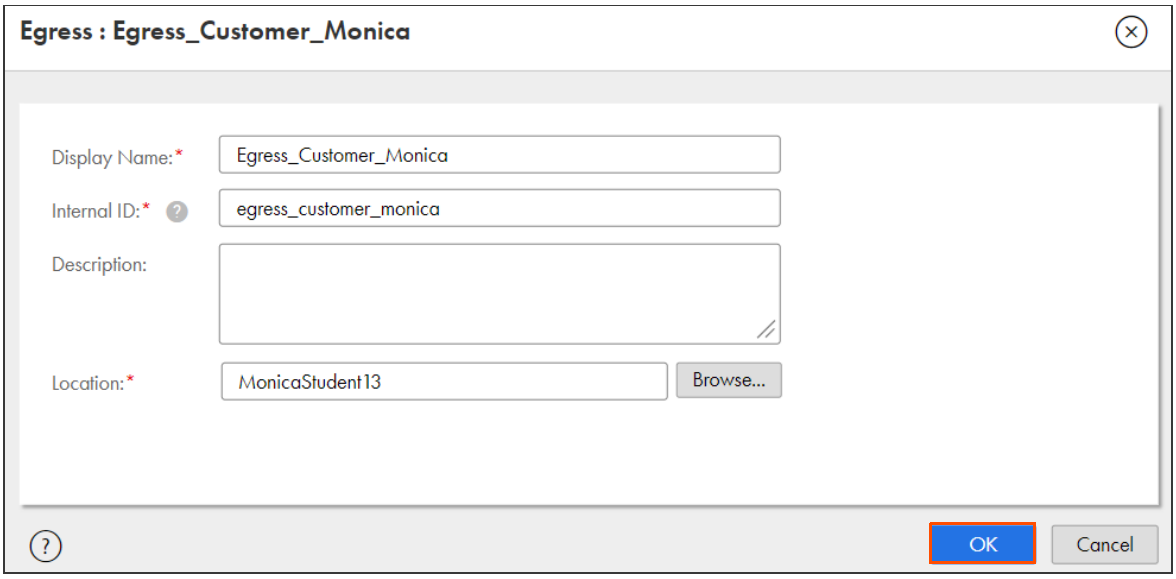
79. For Location, click **Browse**.



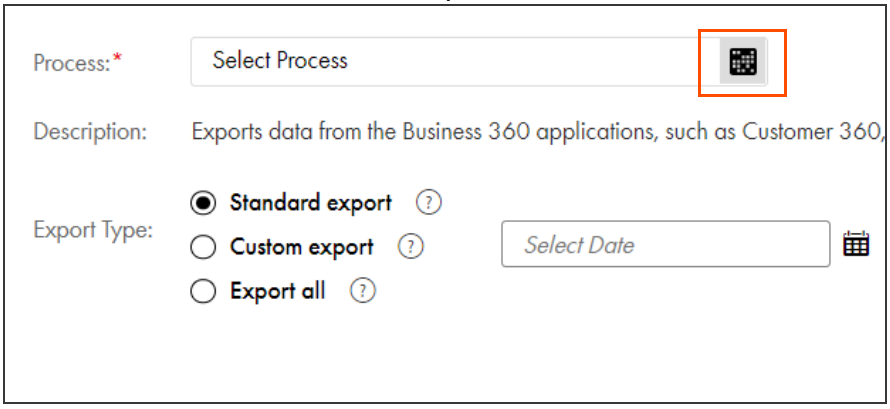
80. Select your student project and click **Select**.



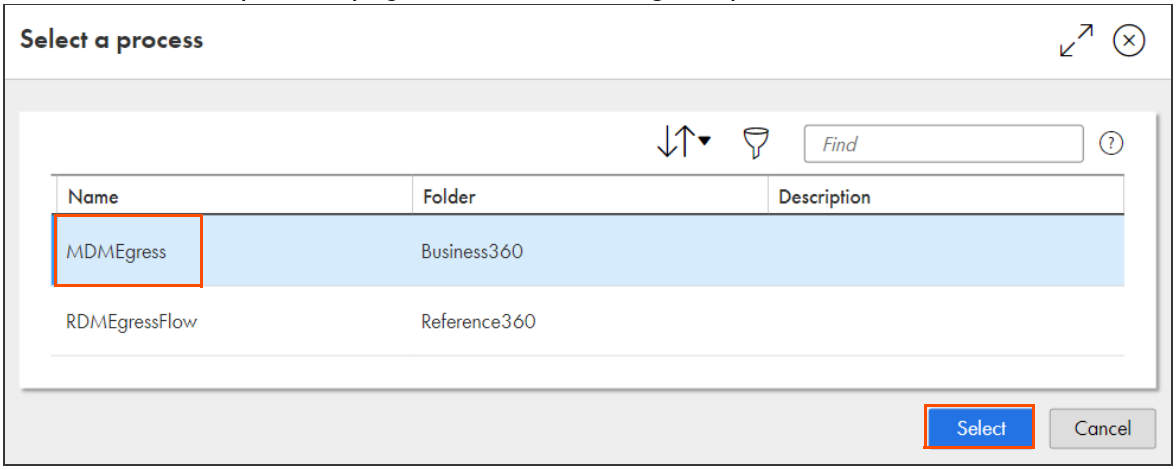
81. Click **OK**.



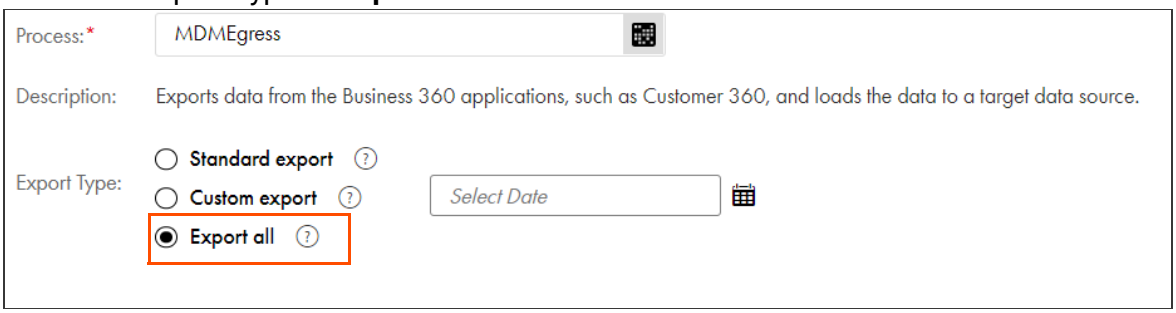
82. To select the **Process**, click asset picker.



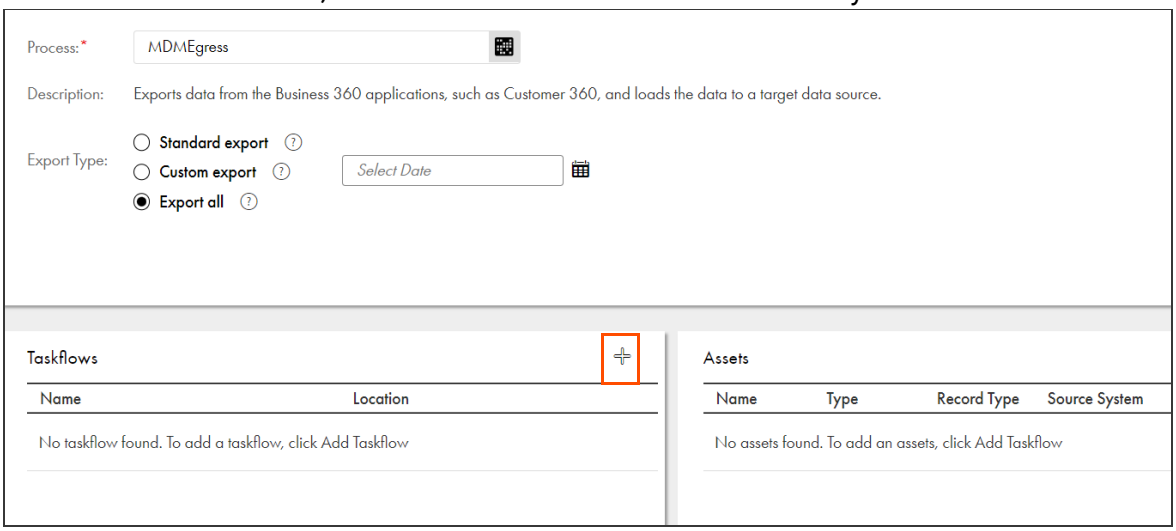
83. From the Select a process page, select the **MDMEgress** process and click Select.



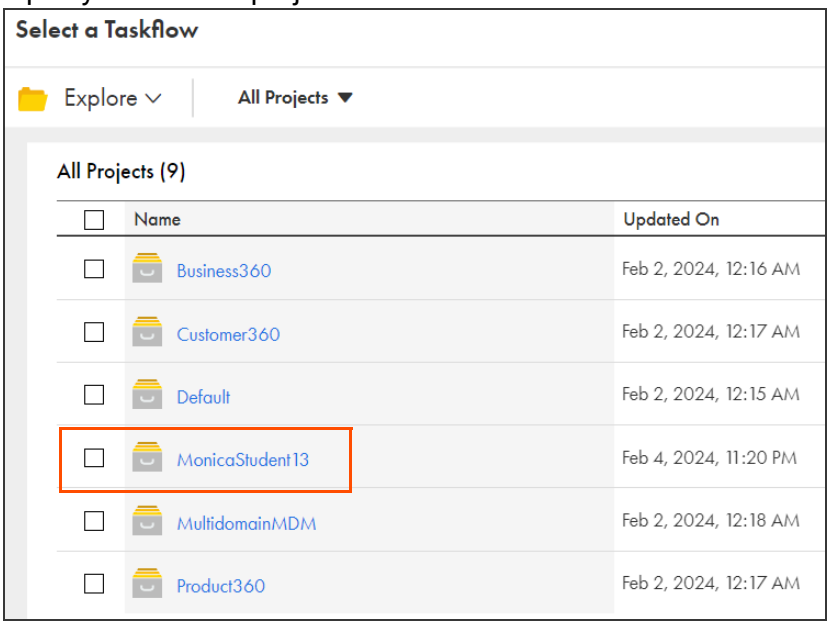
84. Select the Export Type as **Export all**.



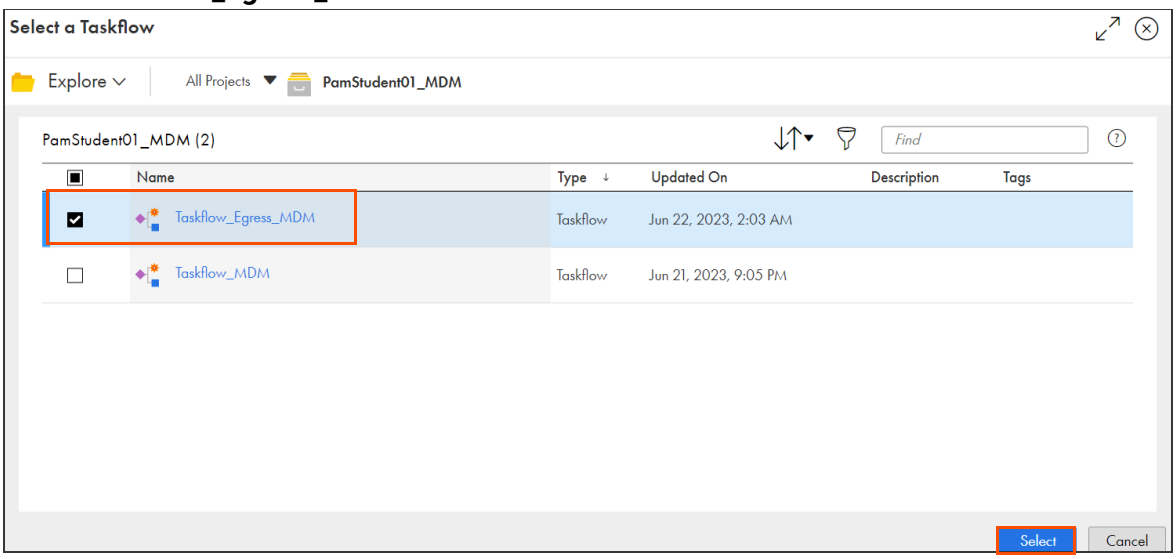
85. In the Taskflows section, click the **+** icon to add the taskflow that you created in CDI.



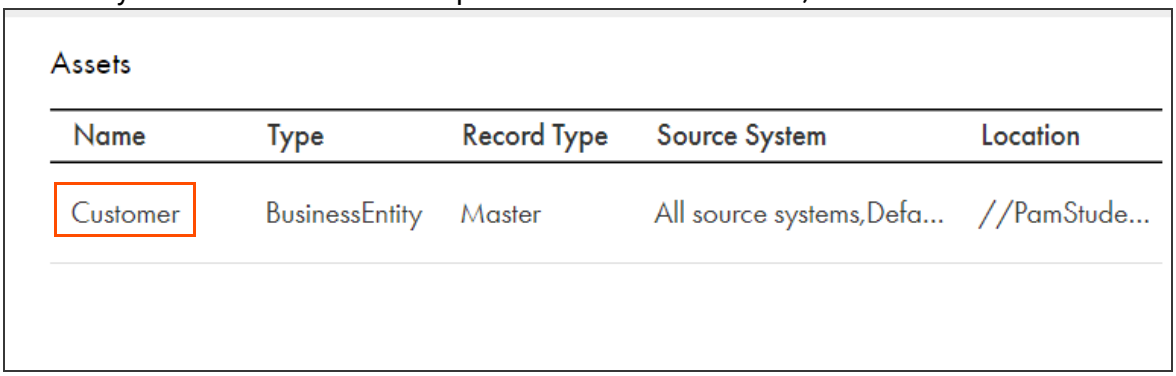
86. Open your student project.



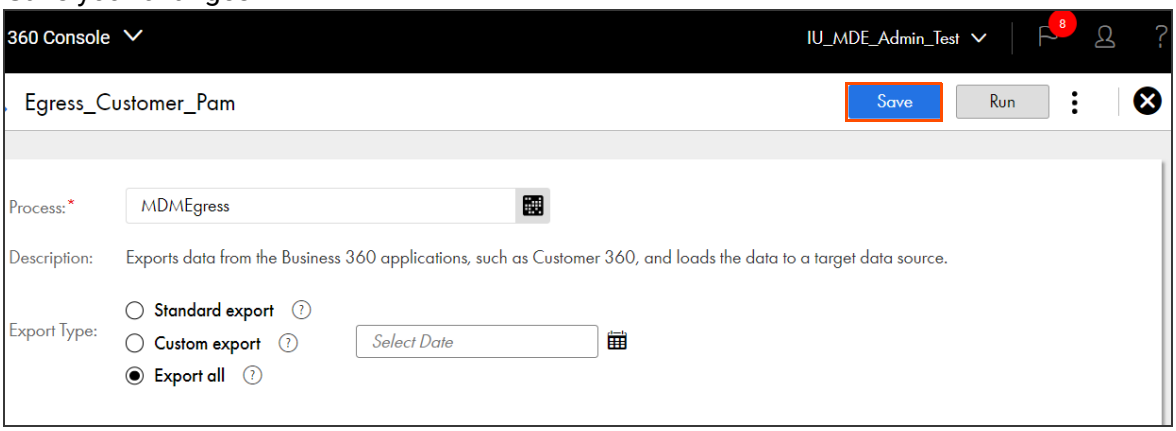
87. Select **Taskflow_Egress_MDM** and click **Select**.



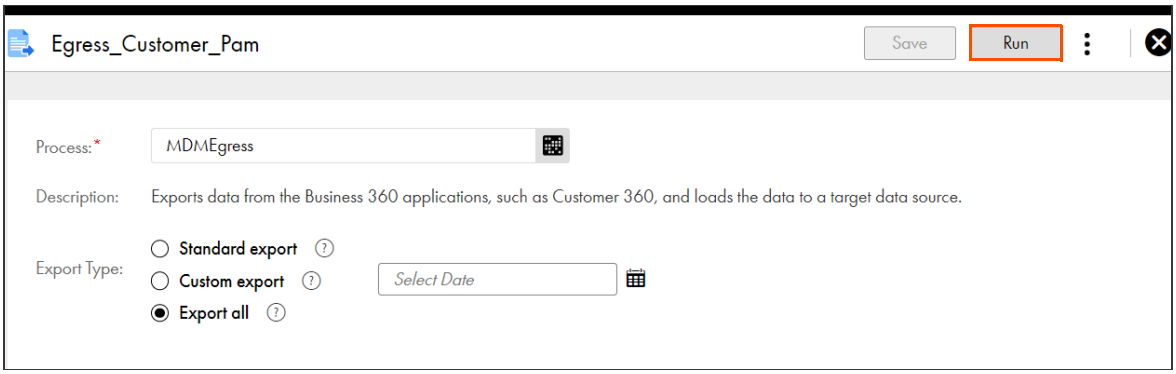
88. In the Assets section, click the data under **Source System** to select the data source from which you want to egress data. The master records from all the source systems are exported by default. To export source records from specific source systems, clear the source systems from the list as required. For this lab exercise, retain the default value.



89. **Save** your changes.



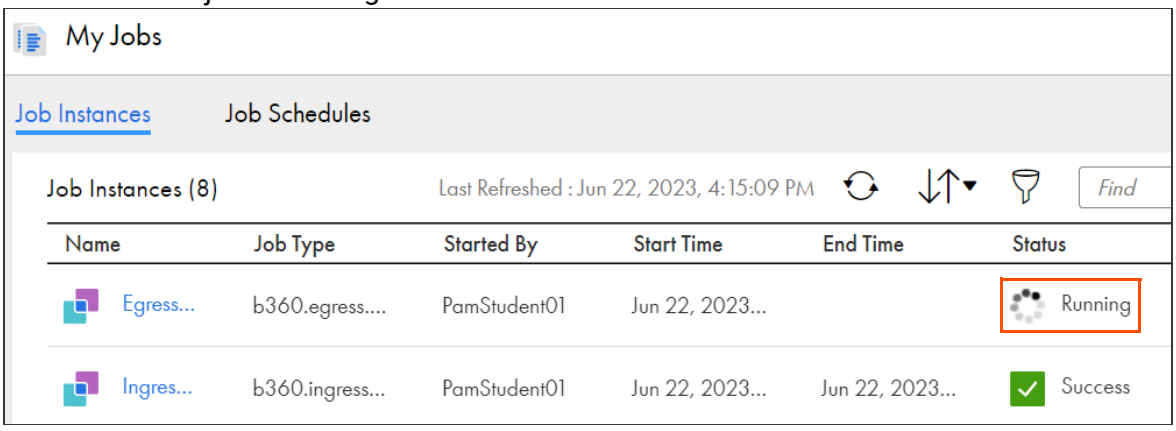
90. Once the save is successful, click **Run**.



91. The job run starts. It can be monitored by clicking the link **here**.

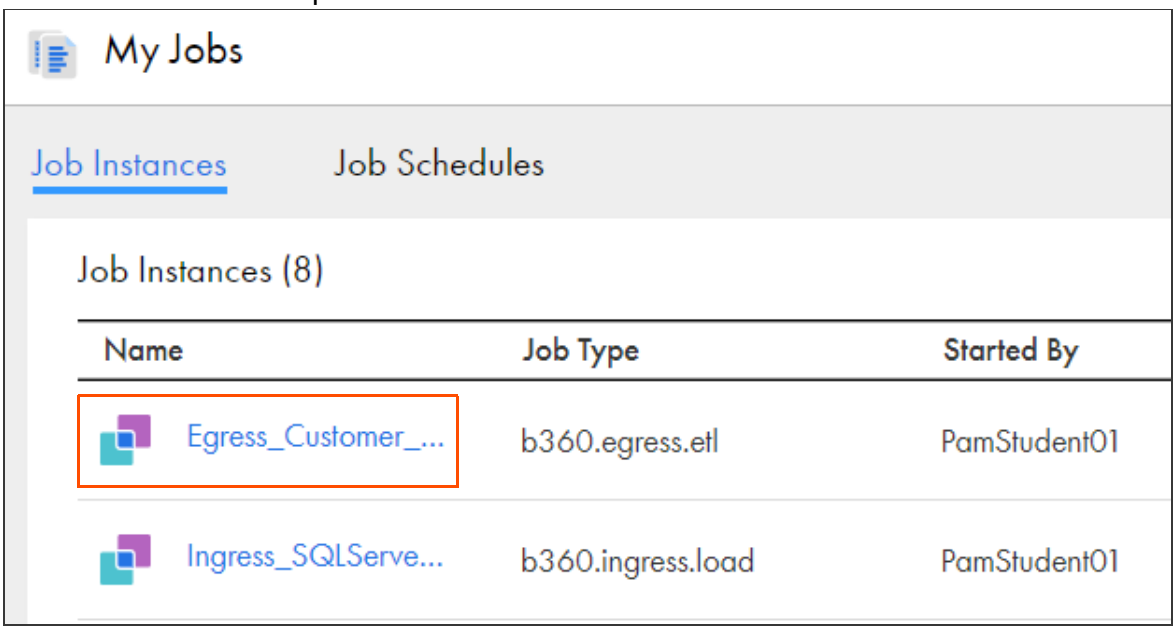
If you missed clicking “here”, you can alternatively click **My Jobs** from the left panel.

Notice that the job is running.



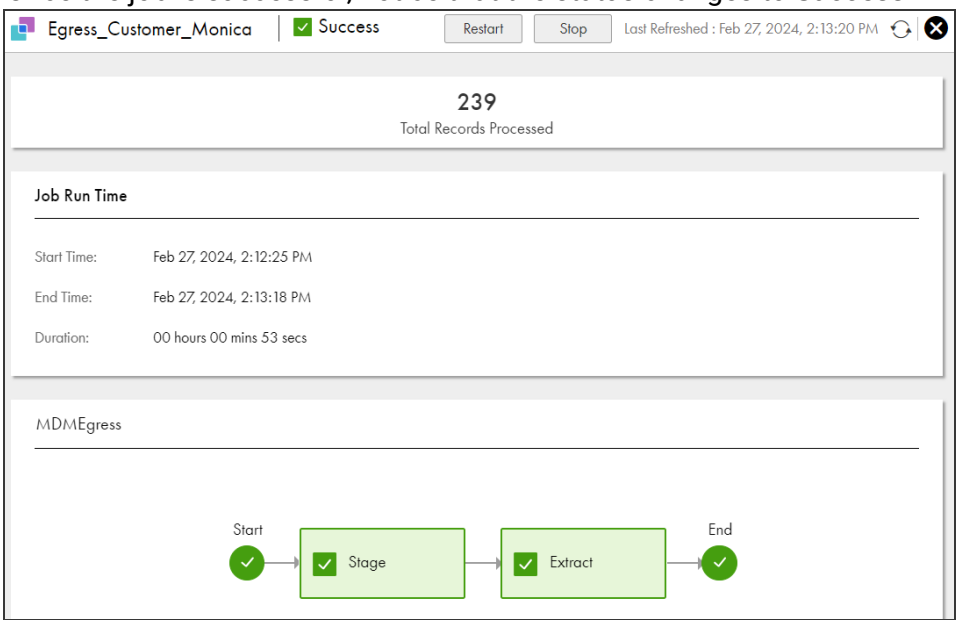
This will take 2-3 minutes to complete.

92. Click the Job name to open and view more details.



93. You can view the phase in which the job is from this page. Refresh the page.

94. Once the job is successful, notice that the status changes to **Success**.

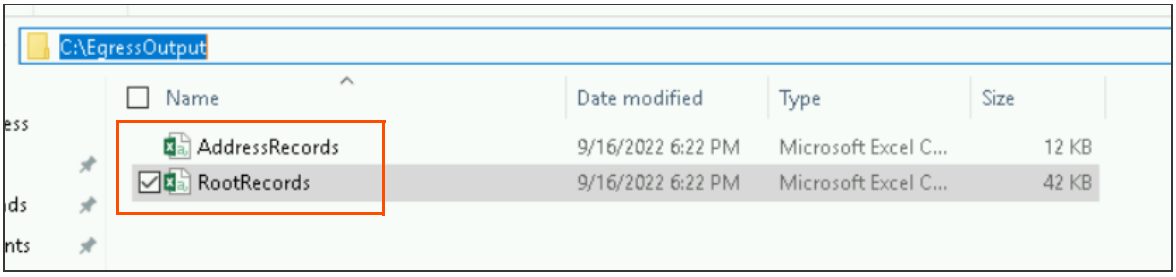


The number of the **Total Records Processed** may vary at your end.

View the Exported Records

View the exported records to see if the egress job is successful.

95. Navigate to the **C:\EgressOutput** folder on your local machine. You can see both Address and Root records files exist in the folder.



96. Double click to open the AddressRecords file. The exported address field records from the Business 360 datastore can be viewed here.

FK_root	sourcePK	parentId	fieldGrou	defaultInd	addressTy	usageTyp	addressSt	addressLi	addressLi	addressLi	country	state	city	county	postalCod
1	a1ec7efaa	MDM0000	ACTIVE	TRUE	Billing	Business	Active	Citrus Heights, 6123 Sunrise Bl			US	CA	San Franci	sco	95610
9	5	MDM0000	ACTIVE		Home			6698 Caldwel			US	AK	Dallas	Taylor	
15	35	MDM0000	ACTIVE		Office			322 Mapleview Drive			US	OK	Kansas Cit	Citrus	
21	56	MDM0000	ACTIVE		Home			94 Horseshoe			US	AK	Fort Wortl	Taylor	
27	64	MDM0000	ACTIVE		Home			762 Murry Street			US	AK	McAllen	Hudson	
33	27	MDM0000	ACTIVE		Office			5214 Forest Avenue			US	OK	St. Louis	Laramie	
39	42	MDM0000	ACTIVE		Home			8199 Clair Street			US	SD	Bronx	Alameda	
45	53	MDM0000	ACTIVE		Home			2246 B Street			US	SD	Fort Wortl	Hancock	
51	51	MDM0000	ACTIVE		Mailing			7777 Aspen Court			US	OK	Louisville	Atlantic	
57	36	MDM0000	ACTIVE		Home			4791 Patterson Fork Road			US	AK	Manhatta	Hancock	
63	61	MDM0000	ACTIVE		Home			7354 Glenwood Avenue			US	SD	McAllen	Laramie	
69	72	MDM0000	ACTIVE		Home			3747 Bassell Avenue			US	AK	Tulsa	Hinds	
75	6	MDM0000	ACTIVE		Mailing			6699 Caldwell St			US	SD	Dallas	Santa Clara	
81	76	MDM0000	ACTIVE		Mailing			8855 Norma Lane			US	OK	Charlesto	Alameda	
87	81	MDM0000	ACTIVE		Home			6309 Radio Park St			US	SD	Charlesto	Hudson	
93	88	MDM0000	ACTIVE		Office			6309 Radio Park Drive			US	AK	Provo	Buchanan	
99	94	MDM0000	ACTIVE		Office			9294 Browning Lane			US	MIN	Wichita	Seminole	
105	85	MDM0000	ACTIVE		Office			4825 Vesta Drive			US	SD	Columbia	Atlantic	
111	82	MDM0000	ACTIVE		Mailing			6309 Radio Avenue			US	MIN	Charlesto	Lackawanna	

97. Similarly, open the other file to check the root records. This is the process of egressing data from the Business 360 data store to external applications.

This concludes the lab.

Lab 9-7: Testing the ABAC Rules in Multidomain MDM

Overview:

In Lab 8-2: Applying Basic Attribute Based Access Control Rule (ABAC) on a Business Entity, you have created two rules, one for Record level and other for Attrbute level. In this lab you will test these rules in Multidomain MDM application and check if they work as desired.

Objectives:

- Test the ABAC rules in Multidomain MDM

Scenario:

Now that Mark has created the ABAC rules, he wants test the rules in Multidomain MDM application and check if the rules are applied to the custom steward role that he has created earlier.

Duration:

10 minutes

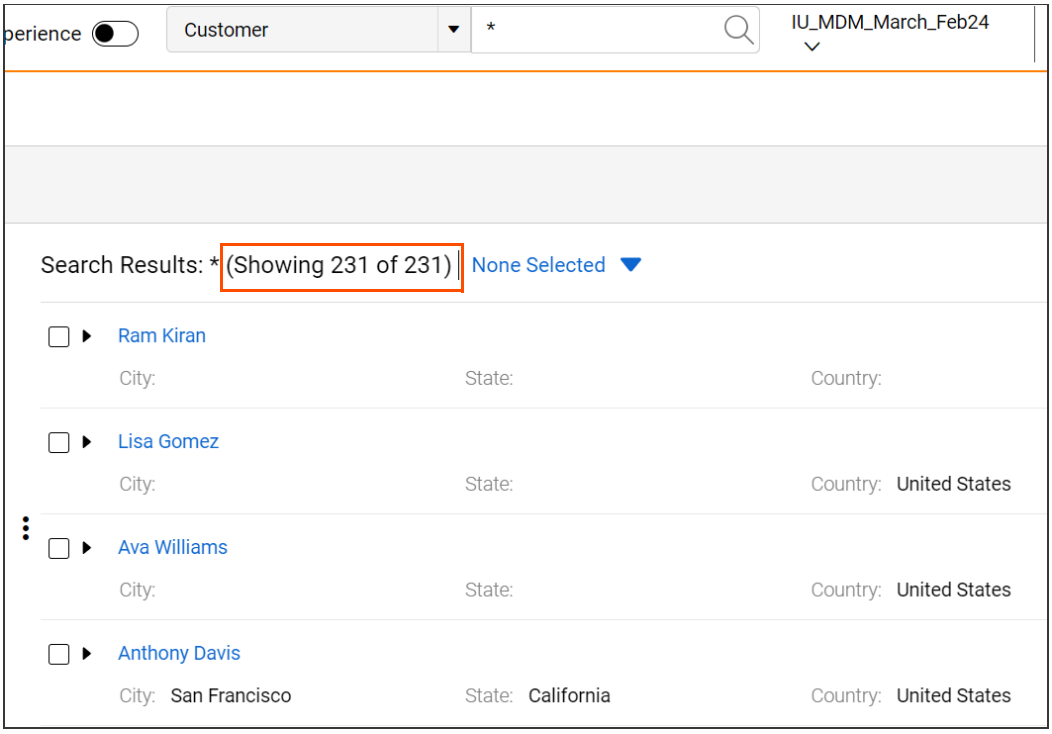
Tasks:

Test the First Rule in Multidomain MDM

Recall Rule 1: Record Level Data Access Rule- Display only US Residents

1. Log in to IICS with your admin credentials (**studentNamestudent<#> / Multidomain360**), if you have logged out.
2. Open **Multidomain MDM** service.
3. Click **Search**, select the **Customer** business entity and enter the search string as ***.

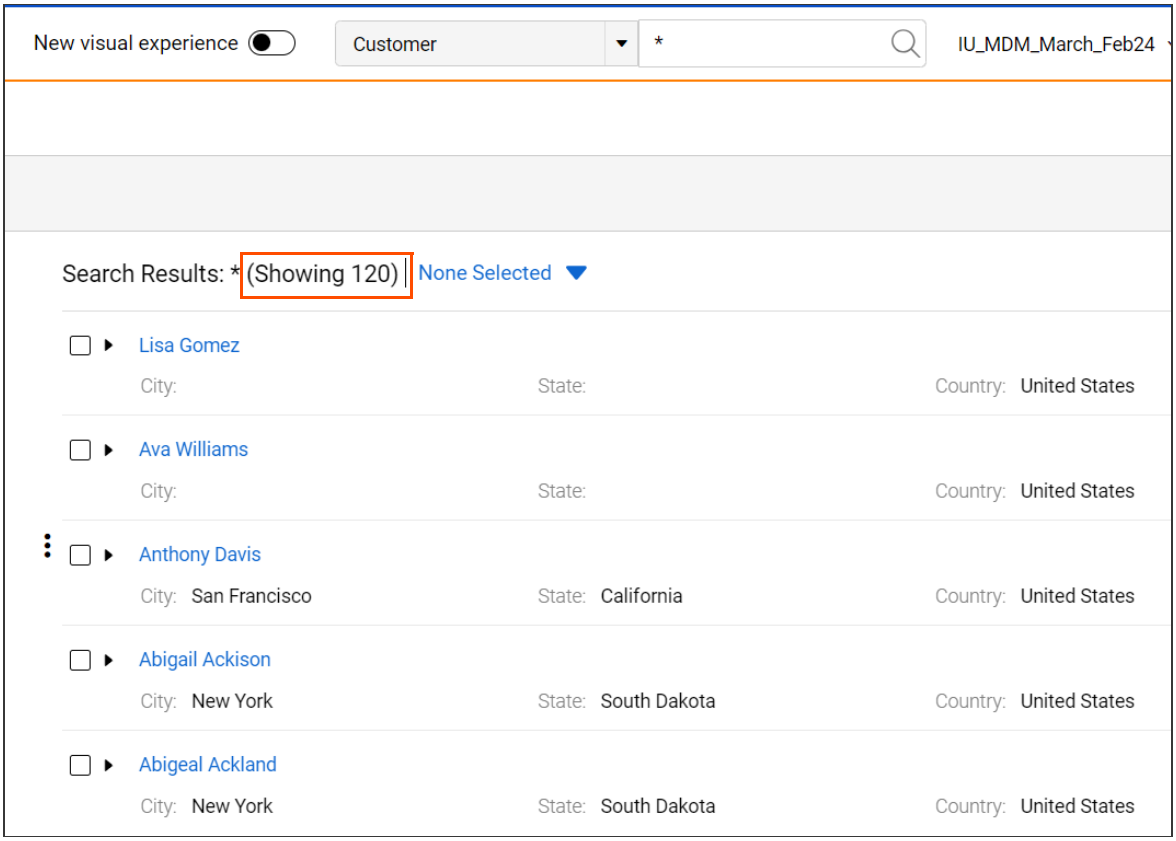
4. You will see all the records present in the Multidomain MDM application, including the one's that do not have United States as their country. Notice the total number of records. Since the Data Access Rule is only applied to Data Steward you are able to see all the records.



5. To confirm whether this rule is being applied to Steward or not, log out as the Admin user and log in as the steward user.

6. Open **Multidomain MDM**.

7. Select the **Customer** business entity in the search tab and enter the search string as ***. This time, you will see the records that only belong to **United States**. Notice the total number of records. They are less than the total number of records that you saw when you logged in as Admin user.



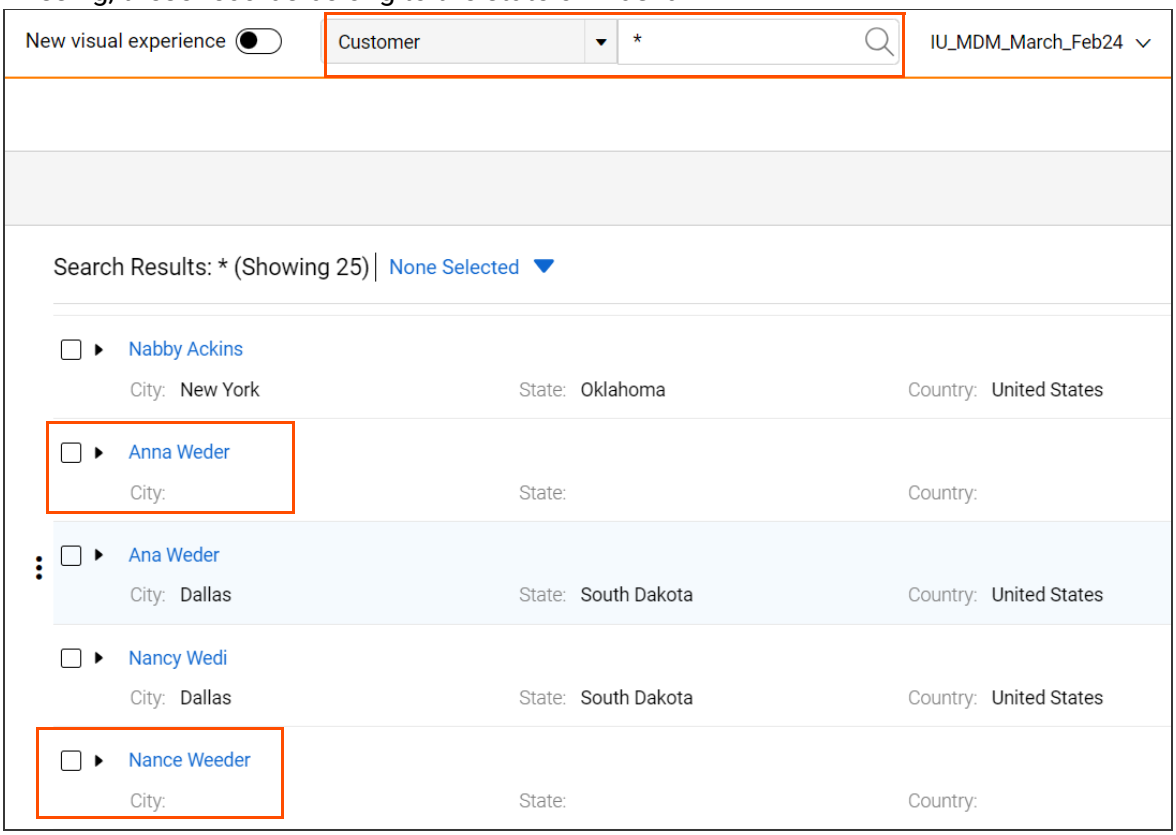
Test the Second Rule in Multidomain MDM

Recall Rule 2: Attribute Level Data Access Rule- HideAlaskaAddress

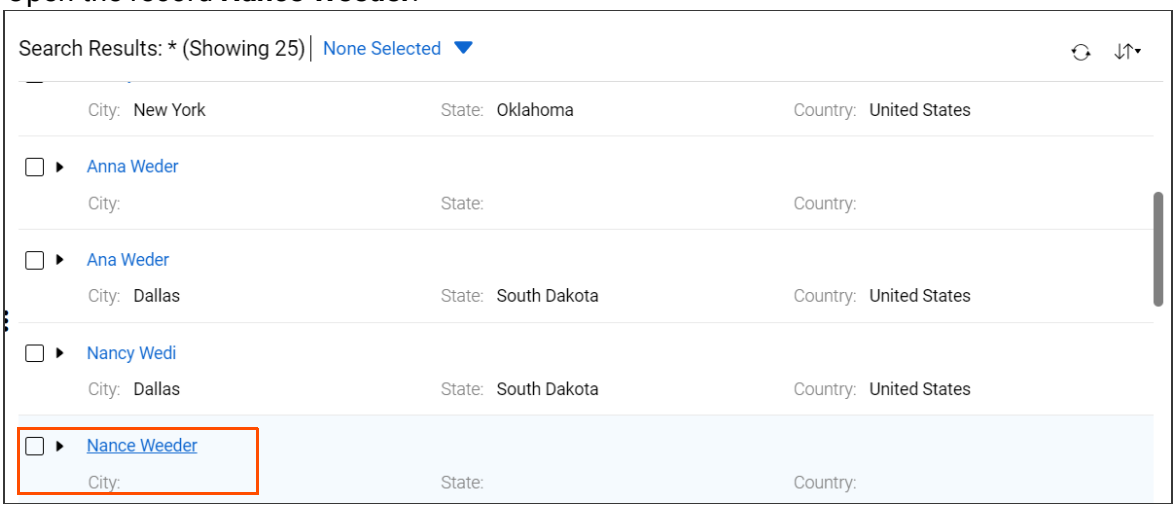
8. Log in using your steward credentials.

9. Open **Multidomain MDM**.

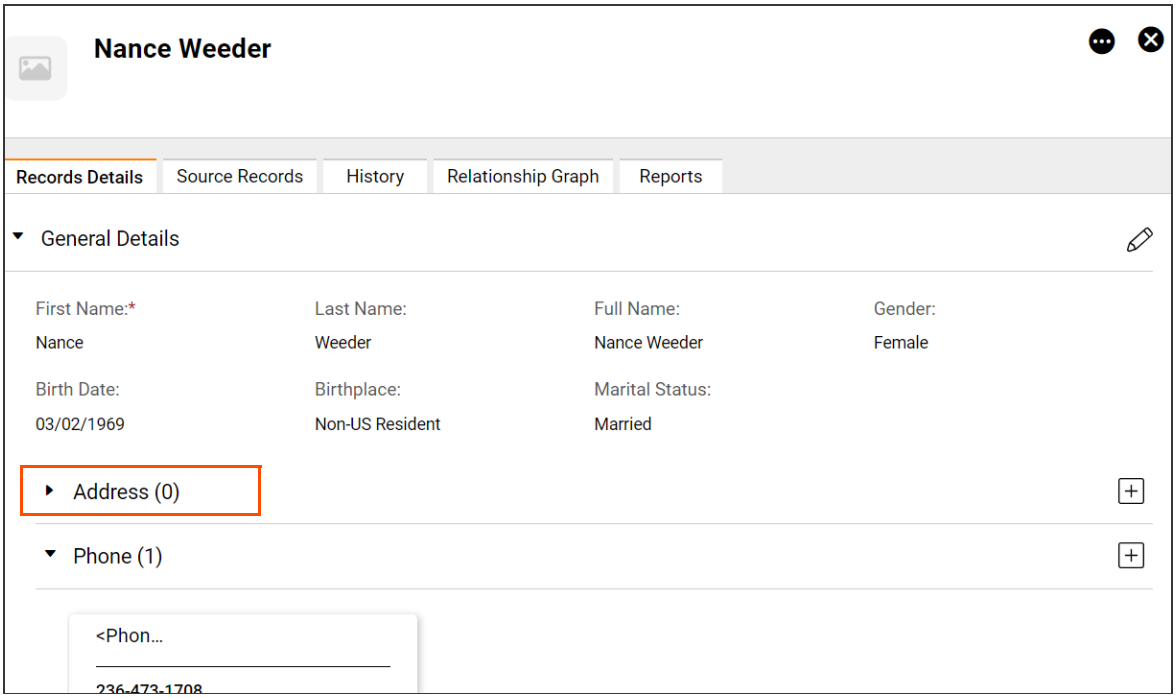
10. Click **Search**, select the **Customer** business entity and enter the search string as ***.** You can notice there are some records like **Anna Weder** and **Nance Weeder**, that have the state field missing, these records belong to the state of **Alaska**.



11. Open the record **Nance Weeder**.

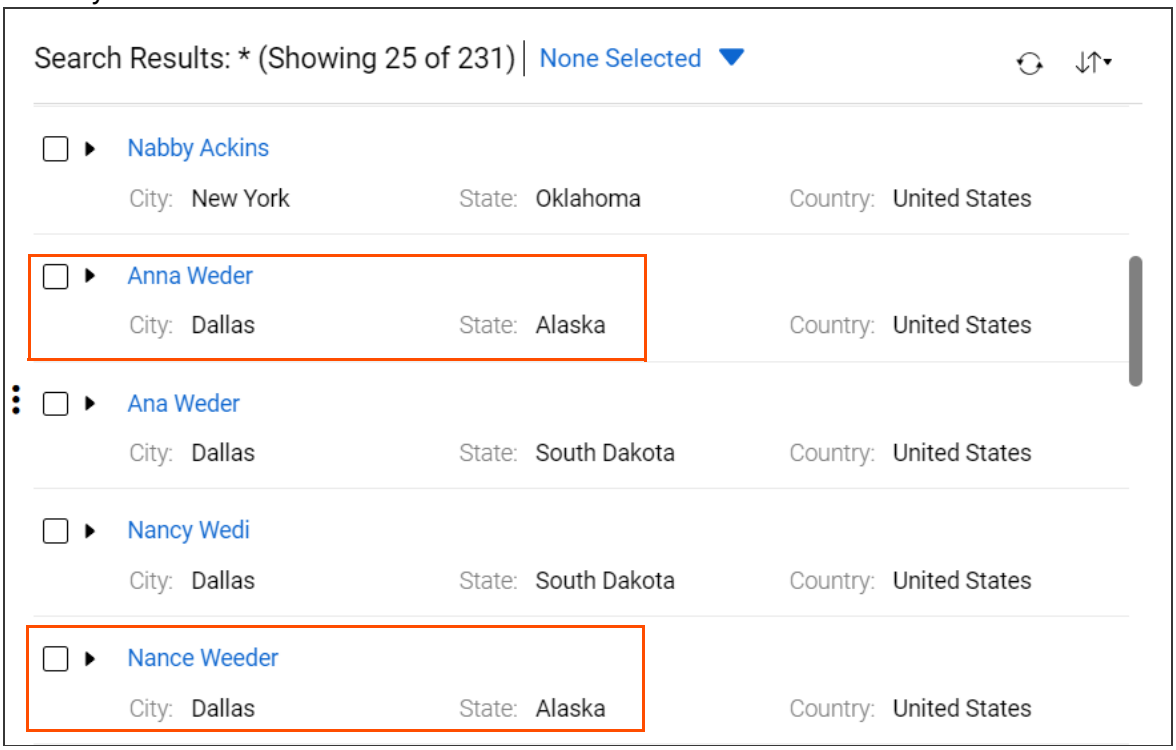


12. You can see that the Address is not shown for this record as this record belongs to **Alaska** state.



You can confirm whether this record belong to Alaska state by logging in as Admin user and searching for the customer records.

13. The following screenshot shows the search results, when you log in as the Admin user, where you can see the state as Aslaska for the Anna Weder and Nance Weeder records.



This concludes the lab.

Key Takeaways

- You can add business entity records to the Multidomain MDM database.
- Related records are the records that contain data associated with a record.
- To find a record, you can either enter a search text in a search box or perform a filtered search.
- Use filtered search to search for records based on one or more field values.
- You can import data, such as business entity records, relationship records, and hierarchies, from a CSV file into Multidomain MDM SaaS.
- Ingress is a process that imports data from external source systems, masters the data, and stores it in MDM SaaS assets.
- You can export data from Business 360 data store to an external data source. For example, your organization uses Multidomain MDM to master customer data.

Module 10: Configuring Match Model

Configuring Match and Merge

You can configure the match and merge process to identify and resolve duplicate records. The match and merge process configuration together constitutes the match model. Based on the outcome of the match process and the merge configuration, the records are either automatically merged, queued for manual merge by a data steward, or skipped as not a candidate for merge.

To resolve duplicates, you can choose to use a machine learning (ML) model, declarative match rules, or both. The declarative match rules are based on an exact or fuzzy match strategy. An exact match strategy identifies identical records, and a fuzzy match strategy identifies similar records. To reduce overmatching and provide a better matching performance, consider adding at least one exact match rule in a match model.

An ML model needs to undergo a training process based on your business needs, where a user labels record pairs to train the ML model. An ML model training reduces the manual effort required for defining declarative match rules.

To configure the match and merge process, use the predefined match model or create a custom model to suit your business needs. You can create a custom match model from scratch or from a copy of an existing match model. It is important to note that you cannot edit the predefined model.

Prerequisites for Configuring Match and Merge

To configure the match and merge process, you need to analyze the characteristics of your data and the specific match and merge requirements.

Before you configure match and merge, ensure that you meet the following prerequisites:

- Understand the business objectives and requirements for matching and merging the duplicate data.
- Analyze the attributes and quality of the data that you want to consider for the match and merge process.
- Determine the size of the data set, which could impact the performance of the match and merge process.
- Determine the match population, which improves match accuracy by accommodating variations and errors that are likely to appear in data for a particular population.

Configuring Match Models

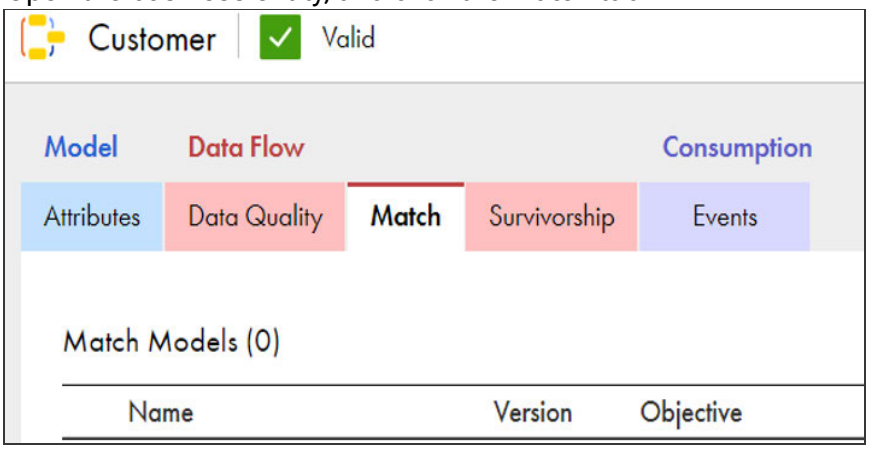
To configure the match and merge process, start by creating a match model. Decide on the method to create a match model based on whether you use a predefined or a custom business entity.

You can use the following methods to create match models:

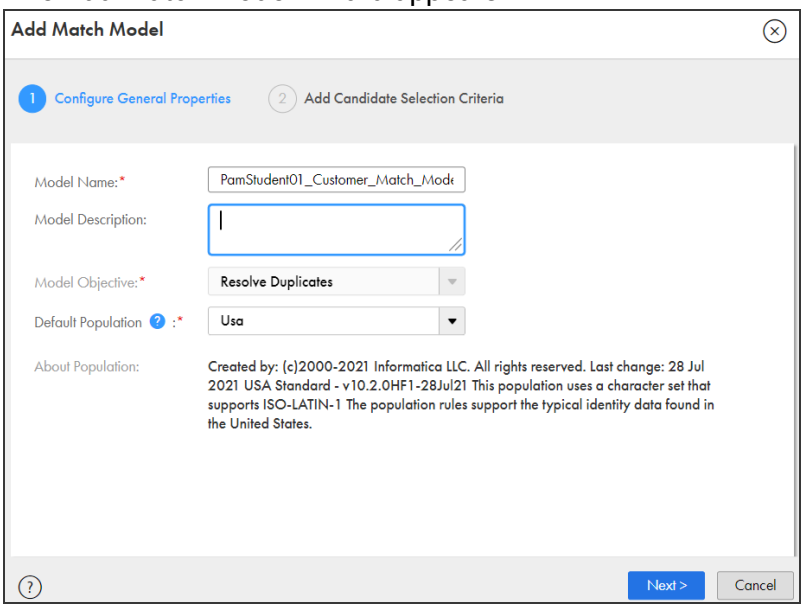
Method for creating match models	Business entity type
From a copy of a match model	Use for predefined business entities.
From scratch	Use for predefined and custom business entities.

Creating a Match Model from Scratch

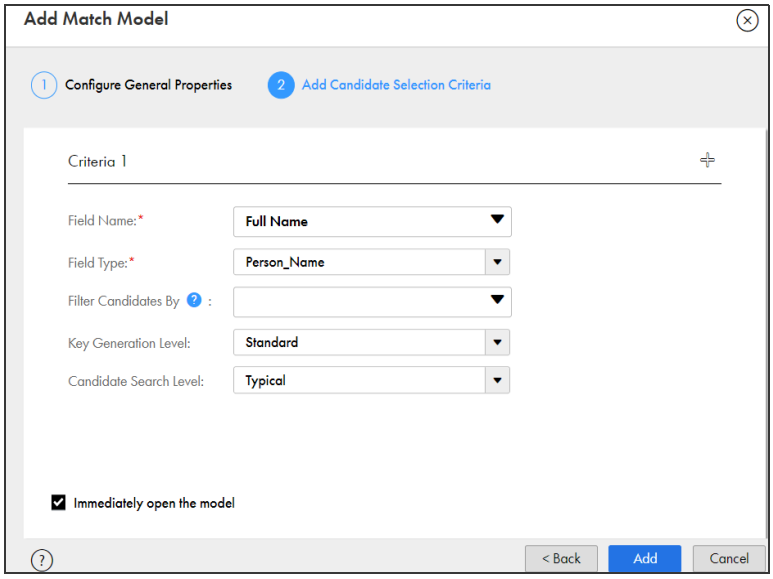
1. Open the business entity, and click the **Match** tab.



2. On the Match Models page, click the **Add Match Model** icon. The Add Match Model wizard appears.



3. Configure the model name and model objective.



4. To configure the maximum number of candidates that the match process can use for each record, select Custom, and enter a value from 1 through 1,000,000.
5. Select a default match population to use if the records don't contain any population information.
   A detailed information about the selected population appears.
6. Click **Next**.
7. On the Add Candidate Selection Criteria page, configure candidate selection criteria.
   - To add a criteria, click the **Add Criteria** icon.
   - Specify the business entity field name, field type, filter candidates by, key generation level, and the candidate search level for candidate selection.
8. To open the match model immediately after the model is created, select **Immediately open the model**.
9. Click **Add**.
   If you selected Immediately open the model, the match model opens. Otherwise, the match model appears on the Match Models page.
10. Next step: Configure the declarative match rules and train the machine learning model.

Declarative Match Rules

You can configure declarative match rules to determine the records that are duplicates and to merge them. You can define a declarative match rule for exact or fuzzy matching. An exact rule matches records that have identical values in match fields. A fuzzy rule matches similar records based on probabilistic match determinations that consider likely variations in data patterns, such as misspellings, transpositions, omissions, and phonetic variations.

Declarative Match Rule Properties

Declarative match rules are a set of conditions and business entity fields that are required to identify duplicate records.

Match strategy

A match strategy indicates the type of match you want to perform. You can decide on an exact or a fuzzy match strategy based on the quality and characteristics of your data.

Exact : A deterministic match that includes records containing identical values. Configure the exact match strategy if the quality of the data is good.
For example, use the exact match strategy if at least some fields contain unique data, such as IDs, postal codes, and industry codes.

Fuzzy : A probabilistic match that includes variations in data patterns, such as spelling variations, misspellings, transpositions, omissions, and phonetic variations. To ensure that the match process does not skip record pairs with data variations, but are a match, configure this match strategy.
For example, use the fuzzy match strategy to match records based on fields that contain addresses, geographic coordinates, and names of people or organizations.

To create a declarative rule, perform the following tasks:

Step 1. Configure match properties and threshold

Step 2. Add required fields

Step 3. Add optional fields

Step 4. Configure exact match field properties

Merge Strategy

- Option 1 - Automated

Automatically merges the records identified as duplicates if their match scores fall within the specified merge threshold range for automated merge.

Usage Recommendation: Use the automated merge strategy for declarative rules that are based on unique identifier fields, such as social security number or Passport number.
Two people cannot have the same Social Security or Passport number, so you do not have to manually review the records.

- Option 2 - Manual

Queues the records identified as duplicates for manual merge if their match scores fall within the specified merge threshold range for manual merge.
Users with the required privileges can review and merge the records in the Business 360 applications.

Usage Recommendation: Use the manual merge strategy for declarative rules that are based on fields, such as the address field. Multiple family members can share the same address and the records cannot be definite matches. You need to manually review the records to confirm whether these are duplicate records.

- Option 3 – Threshold-Based

Available only when you select the fuzzy match strategy. Use the threshold-based merge strategy for any field. Based on the match score of the record pairs, the match and merge process determines whether the result is automated, manual, or skipped merge.

Decide the Merge Threshold

Merge threshold is applicable to declarative rules with the fuzzy match strategy. Merge thresholds are based on match score ranges of records that undergo matching. You can configure merge thresholds for the merge strategy that you select for a declarative rule. When the system identifies records that meet the corresponding merge threshold, the merge strategy that you configure is applied. A higher merge threshold value indicates a higher strength of the merge strategy that you choose.

Match Criteria

Applicable to declarative rules with a fuzzy match strategy. A match criteria is the determining factor behind matching records for your business needs. The types of fields that are required for matching are based on the match criteria that you select. For every fuzzy rule that you define, you must select a criteria from a list of predefined match criteria.

For example, when address is important to determine whether two records are for the same person, use the Resident match criteria. The system compares records based on the Resident match criteria to determine whether the records are a match.

Two declarative rules that have the same properties except for the match criteria result in different sets of matching records.

Match level

Applicable to declarative match rules with a fuzzy match strategy. The match level indicates the level of fuzzy matching that you want to perform on your data. The number of matching records that a declarative match rule generates depends on the match level.

Rank

After you configure all the declarative rules, rank them in the order in which you want to run them for matching record pairs. Based on the ranking, all the declarative match rules complete processing one record pair before processing the next record pair. The declarative match rules are run on one record pair at a time.
First, the declarative rule with rank 1 is used to process a record pair. If the record pair meets the conditions set by the match rule, the merge strategy for the rule is applied to the record pair. The processing for the record pair is complete.
If the record pair does not meet the conditions set by the match rule, then the rule that is next in rank is used to process the record pair.

Match fields

Match fields are business entity fields that you use in declarative rules to identify matching records. You can configure one or more match fields for each declarative rule.

Publishing Match Models

To use a match model for the match and merge process, publish the model after you complete configuring it. After you publish a model, you can define match and merge jobs that use the model. You cannot delete or change a published model back to the draft state. If you publish a match model with changes to the candidate selection criteria, the model is applied to the data that loads after you publish the model.

1. Open the business entity, and click the **Match** tab.
2. Click the name of the match model that you want to publish.
3. On the page that opens, click **Publish**.
4. On the Match Models page, the status of the match model is changed from draft to published.

Lab 10-1: Creating a Match Model and Declarative Match Rules

Overview:

Configure the match and merge process to identify and resolve duplicate records. The match and merge process configuration together constitutes the match model. Based on the outcome of the match process and the merge configuration, the records are either automatically merged, queued for manual merge by a data steward, or skipped as not a candidate for merge.

Objectives:

- Configure a match model
- Configure declarative match rules
- Publish match models

Duration:

30 minutes

Scenario:

After onboarding data, Mark realized that the data was inaccurate, inconsistent, and not up-to-date across various systems. This led to operational inefficiencies, unreliable analytics, and subpar customer experiences. The presence of discrepancies, duplicates, and conflicting information across different data sources can impede decision-making processes and complicate compliance with regulations. Moreover, the absence of a unified data management strategy escalates the costs related to data storage and management, while also hindering the establishment of effective data governance practices. To address these challenges, Mark decided to utilize the Match and Merge feature of MDM SaaS. He set up a match model and initiated Match and Merge jobs on the onboarded data.

Tasks:

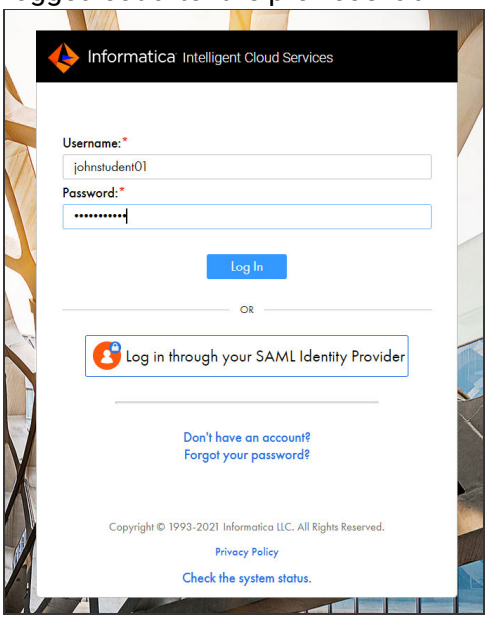
Perform the following tasks to create a match model and add declarative rules to it for Homekart scenario:

- Create declarative match rules
- Publish the match model

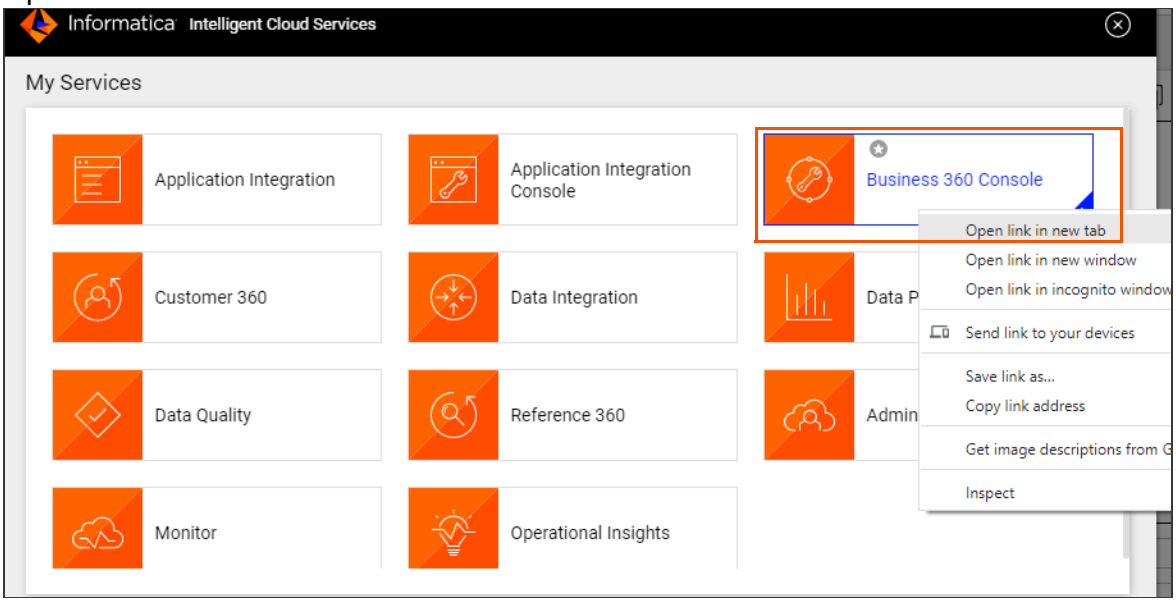
Create Declarative Match Rules

Declarative match rules are a set of conditions and business entity fields that are required to identify duplicate records. As a part of the data model, you will create three declarative rules.

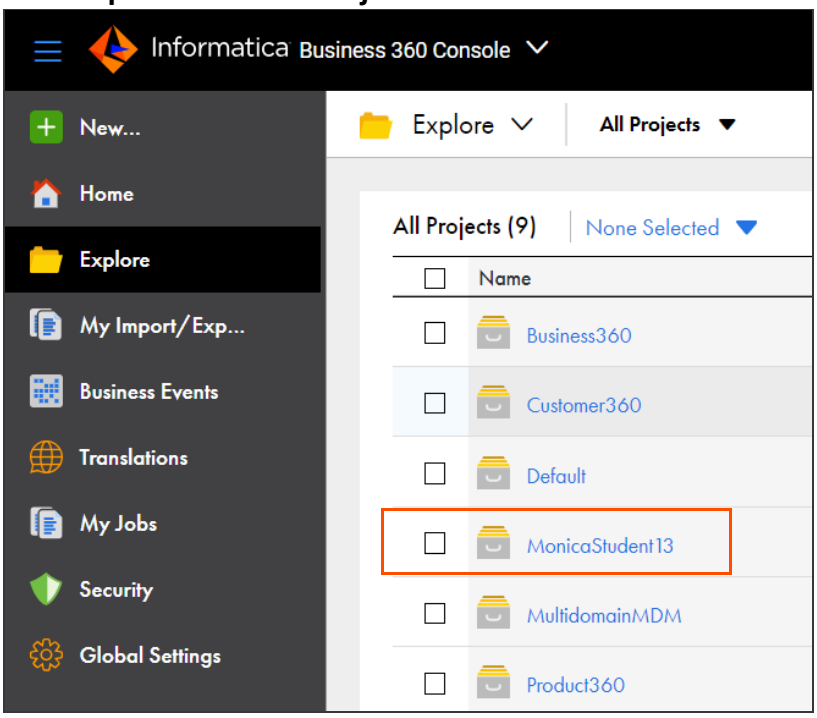
1. Log in to IICS with your credentials (**studentNamestudent<#>/Multidomain360**) if you have logged out after the previous lab.



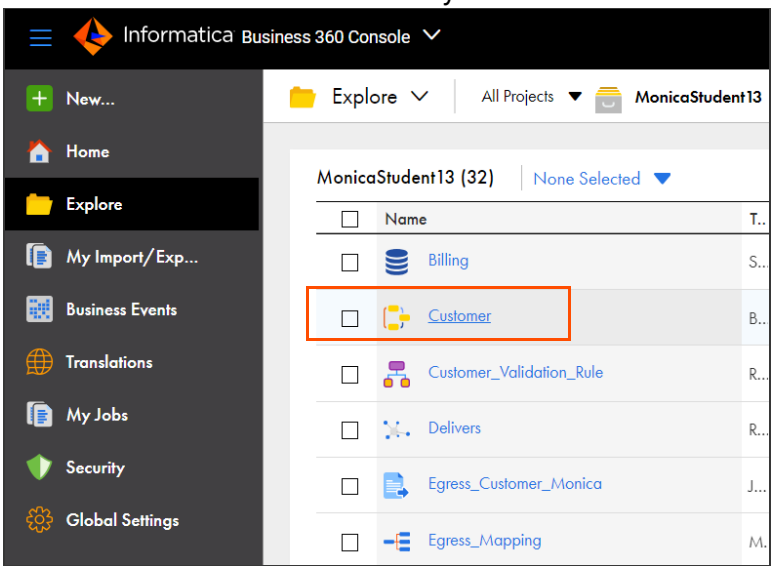
2. Open **Business 360 Console** in a new tab.



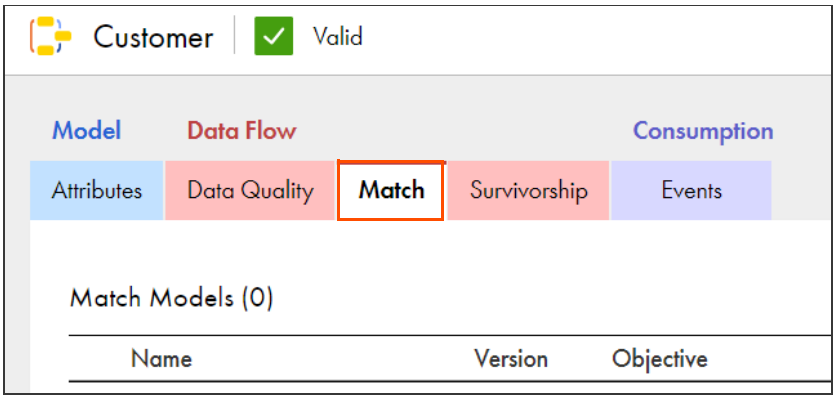
3. Click **Explore** > **Student Project**.



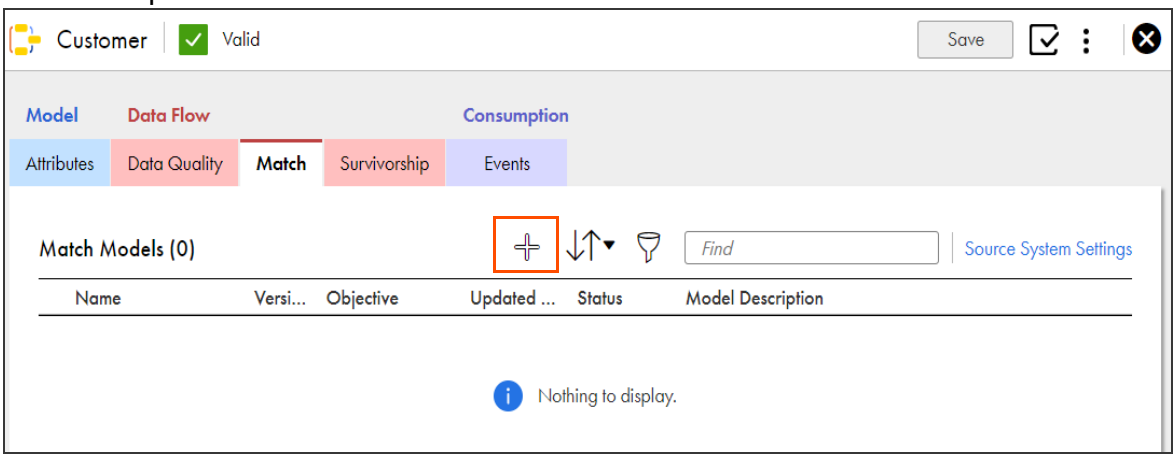
4. Click the **Customer** business entity.



5. Click the **Match** tab.

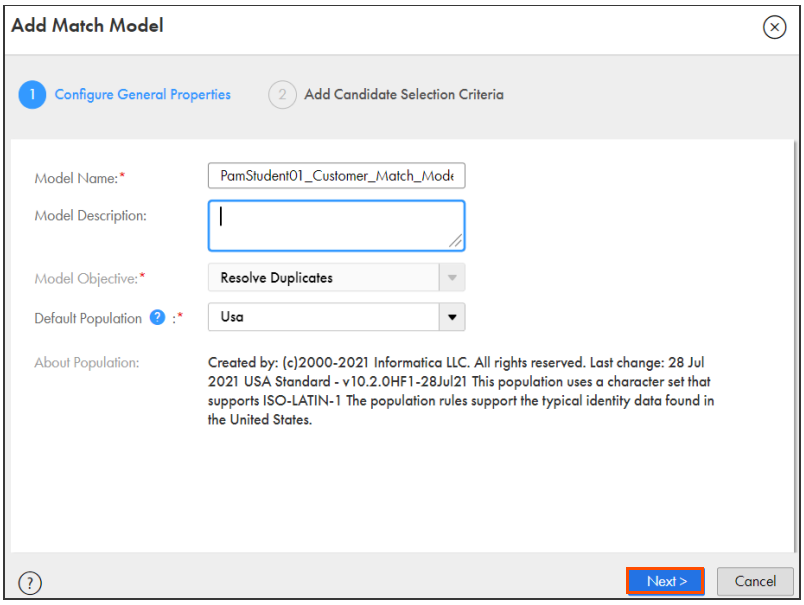


6. Click the **+** plus icon to add match model.



Add Match Model window appears.

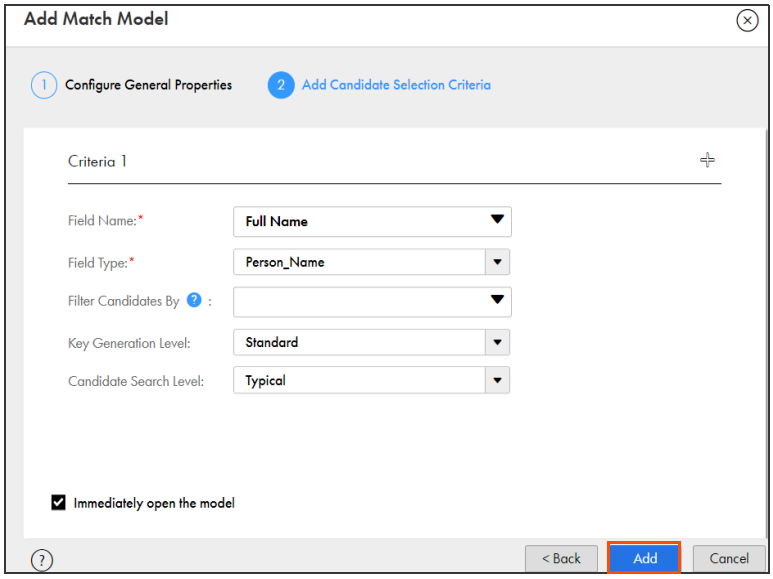
7. Under configure general properties tab, enter the name as **<StudentName>Student<StudentNumber>Customer_Match_Model**, retain the other fields and click **Next**.



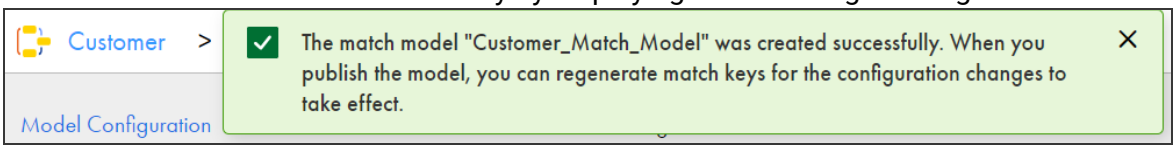
8. In the **Add Candidate Selection Criteria** tab, select the following fields:

Field Name	Value
Field Name	Full Name
Field Type	Person_Name
Key Generation Level	Standard
Candidate Search Level	Typical

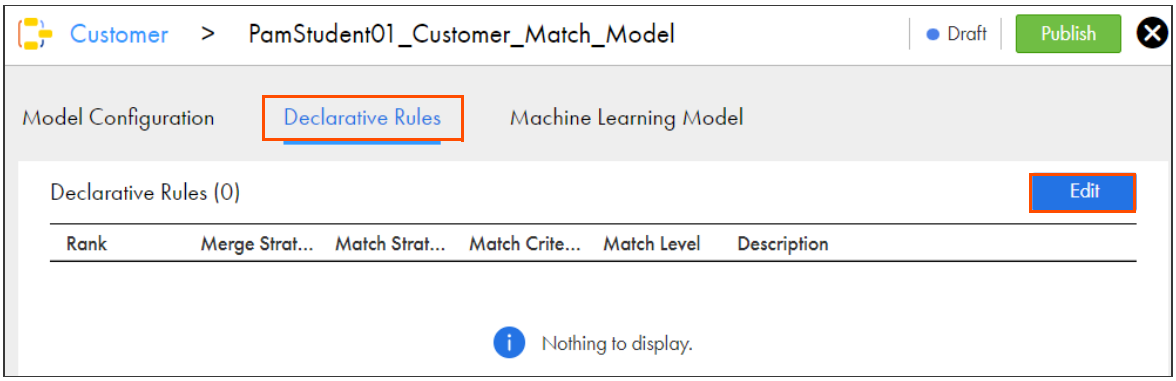
9. Select the check box for **Immediately open the model** and click **Add**.



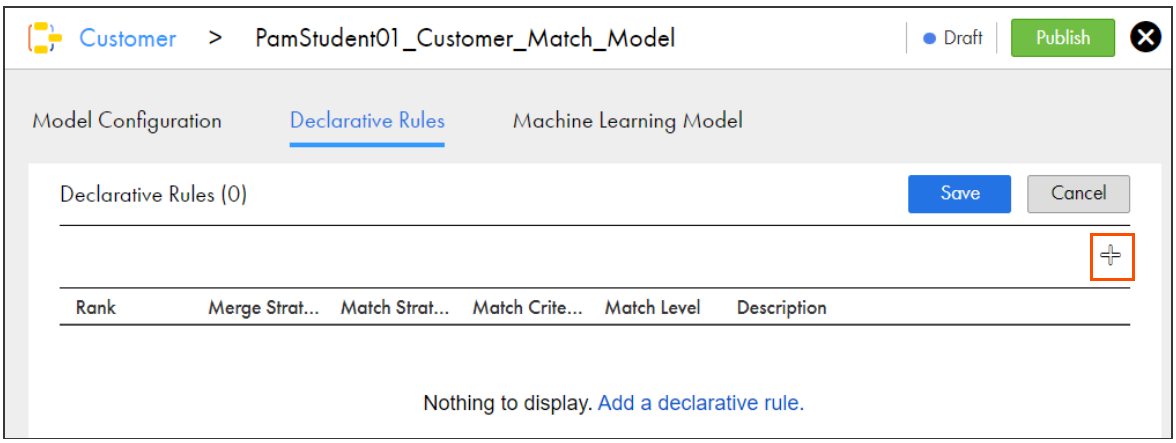
The model will be created successfully by displaying the following message.



10. The draft version of the match model is created. Open the model and click on **Declarative Rules** tab. Click **Edit**.

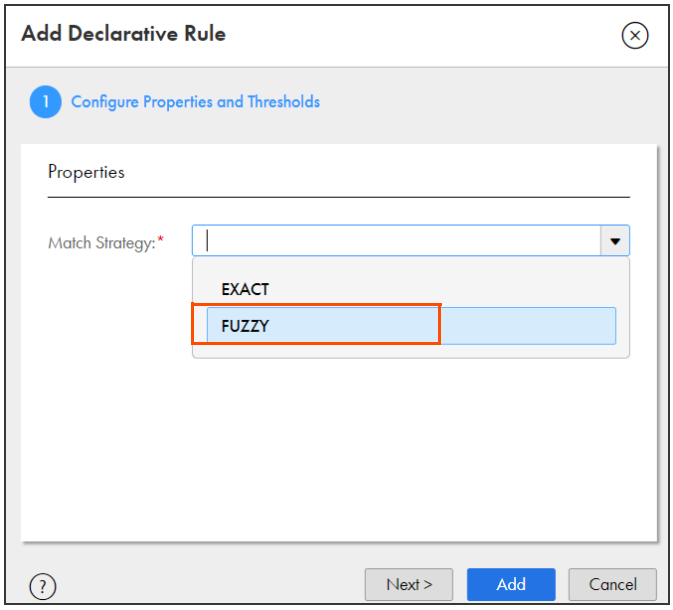


11. Click the **+** icon to add a declarative rule. Alternatively, you can click on **Add a declarative rule**.



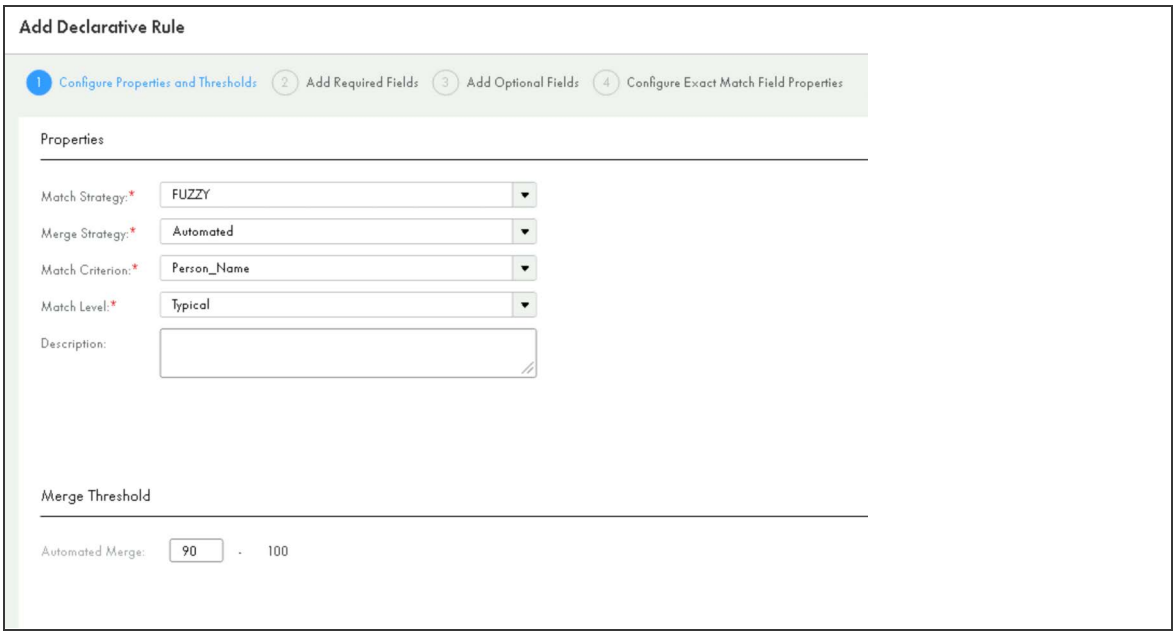
12. In the **Add Declarative Rule** page, for the **Match Strategy** field, click the drop down and select **FUZZY**.

Because you will be dealing with Person_Name, there is a probability of having misspellings, omissions, spelling, and phonetic variations. Hence, the fuzzy match strategy is the best option.

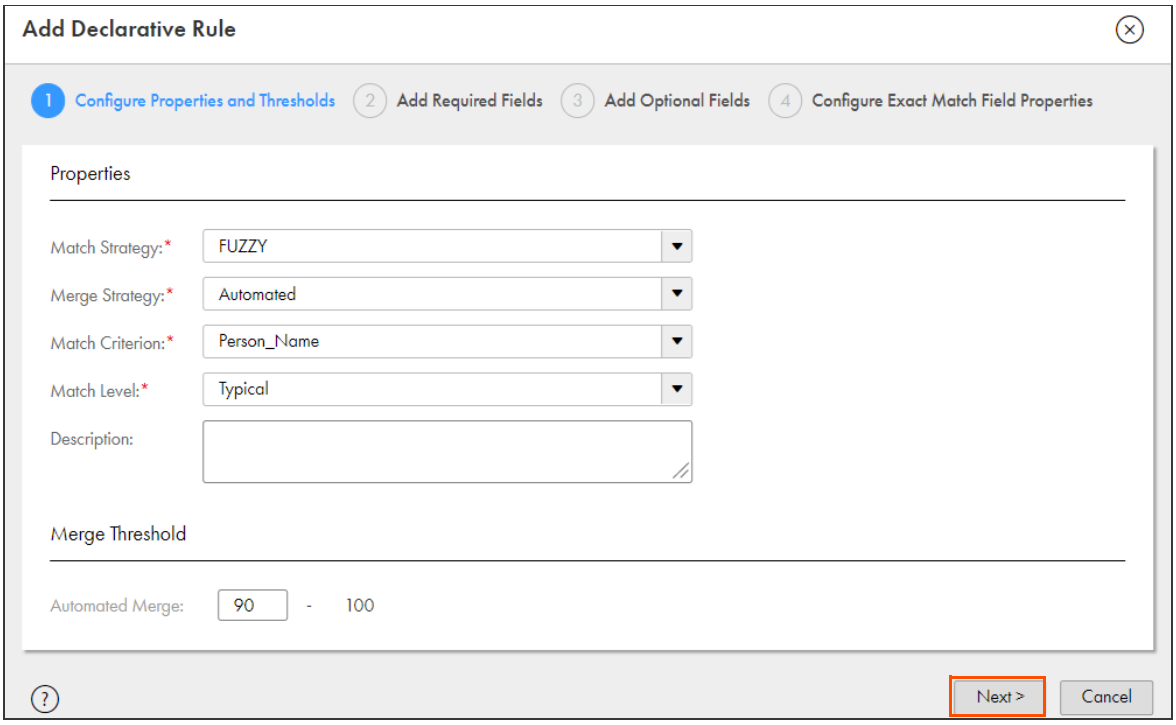


13. Enter the following details:

Field	Value
Merge Strategy	Automated
Match Criterion	Person_Name
Match Level	Typical
Merge Threshold > Automated Merge	90-100



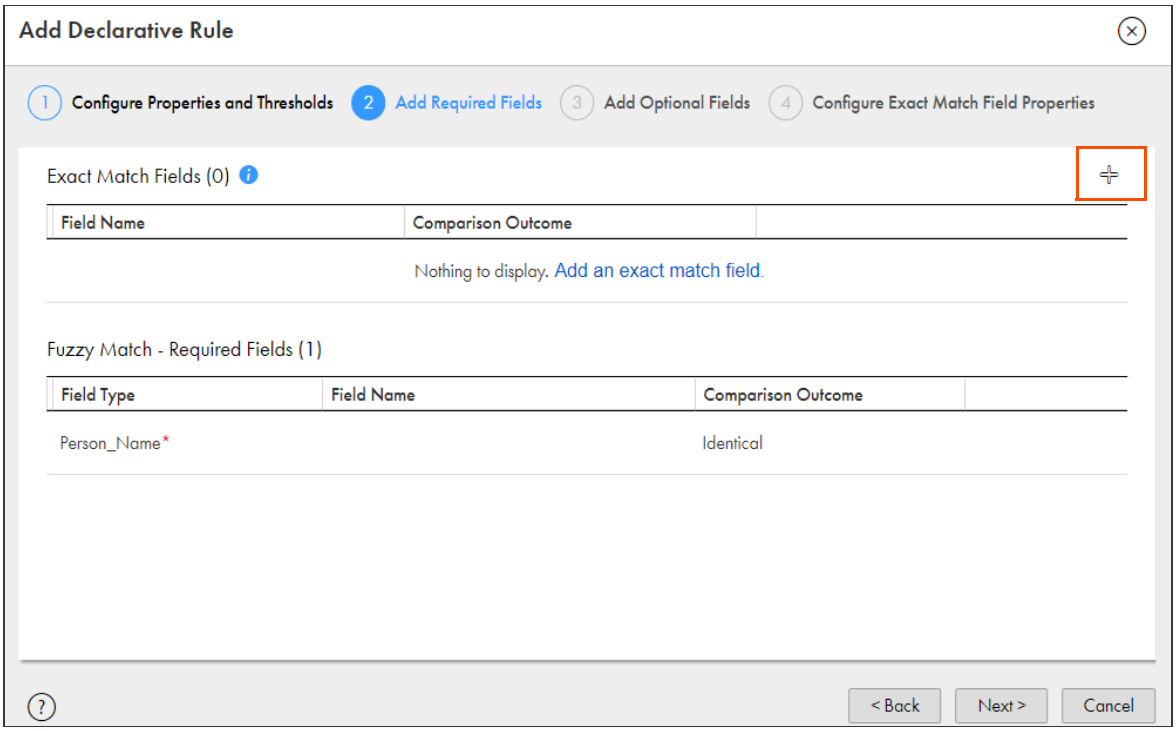
14. Click **Next**.



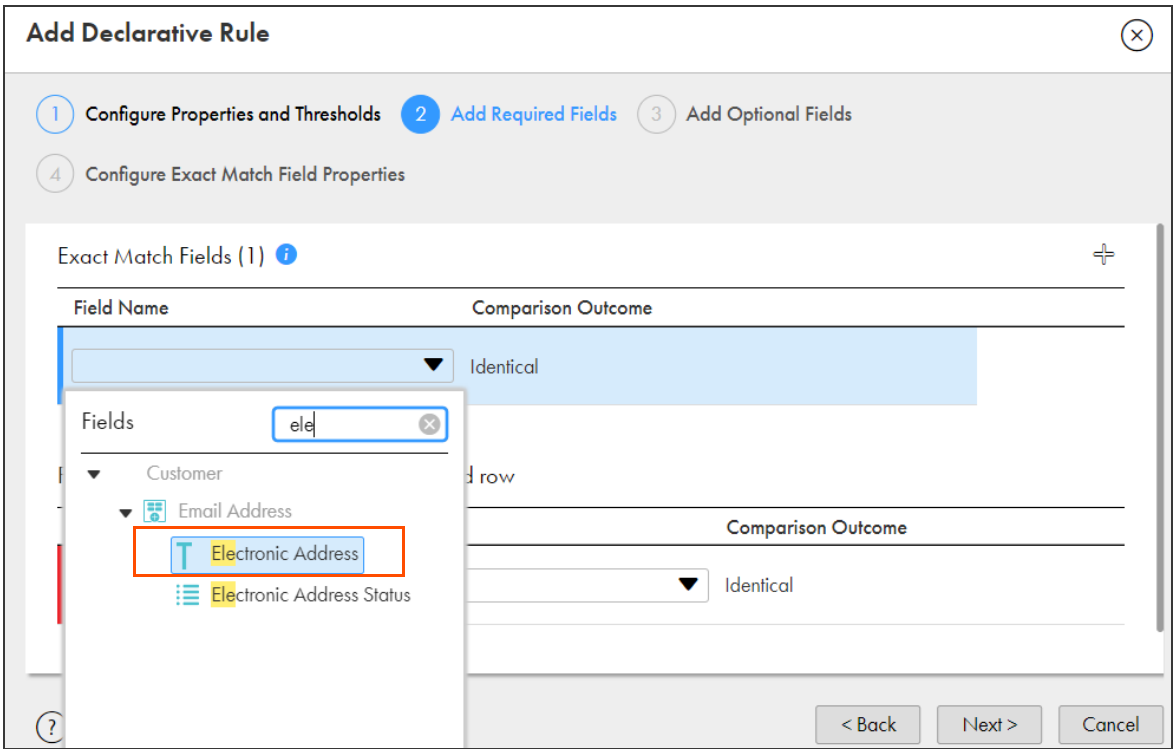
15. You will add the first declarative rule now. For the first rule, the criteria could be:

- Fuzzy match and automated merge strategy
- Full Name, Address Line 1, Address Line 2 – should be similar (Required for Fuzzy match)
- Email, Phone Number, Postal Code, and Date of Birth – should be exact

16. In the **Exact Match Fields** section, click **+** to add an exact match field.

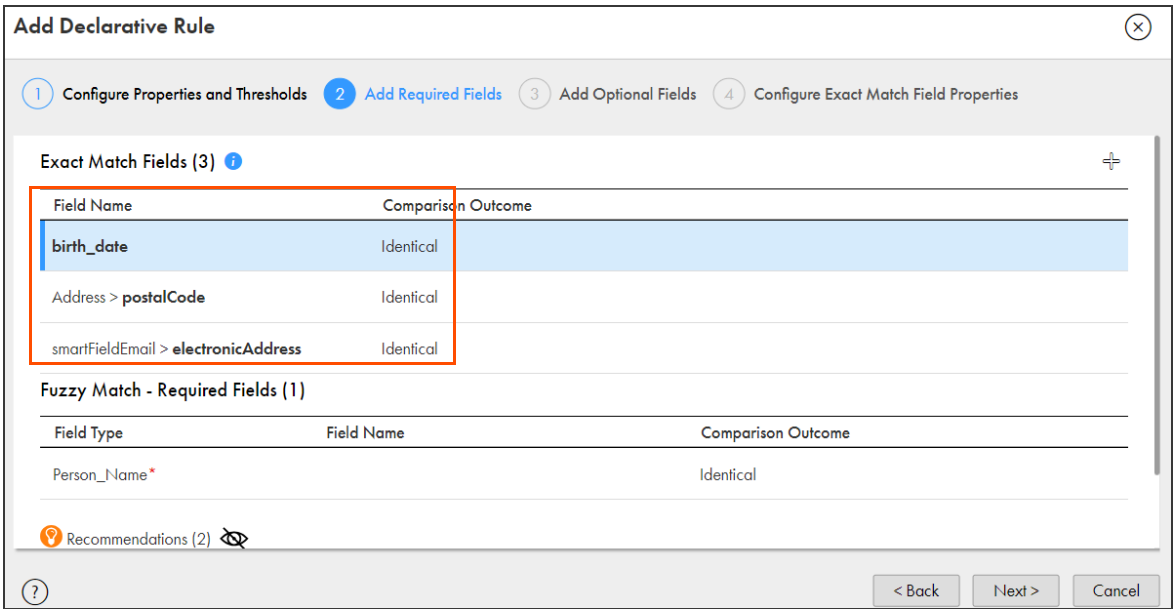


17. From the Field Name drop-down, select **Email: Electronic Address**. You can search for Electronic Address and select from the results.

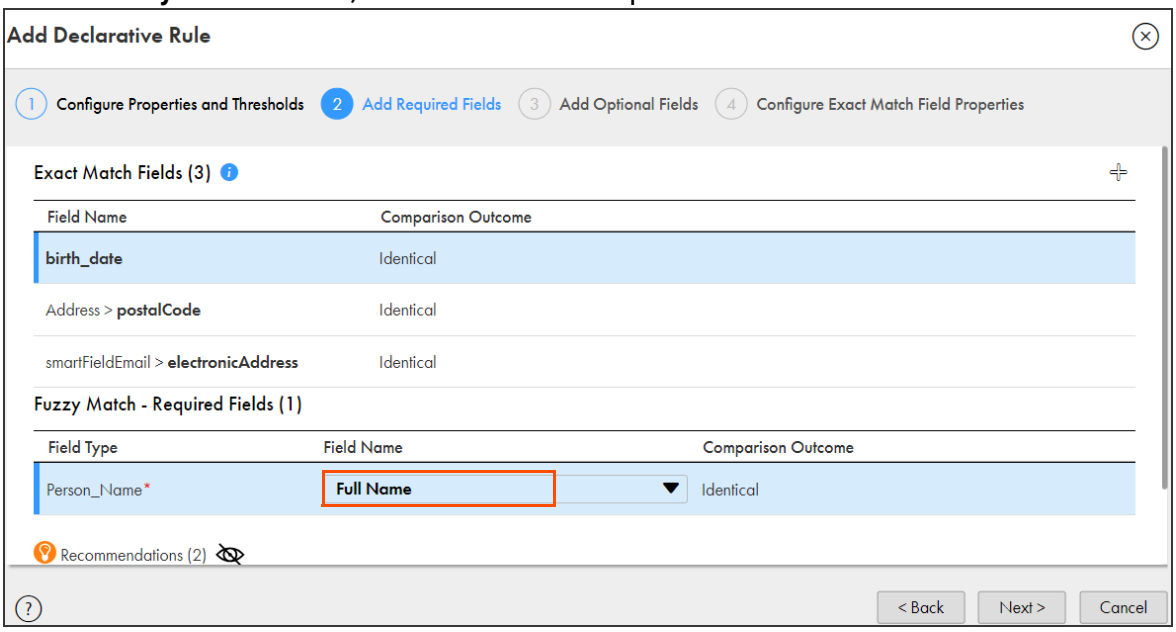


18. Similarly, add the following exact match fields:

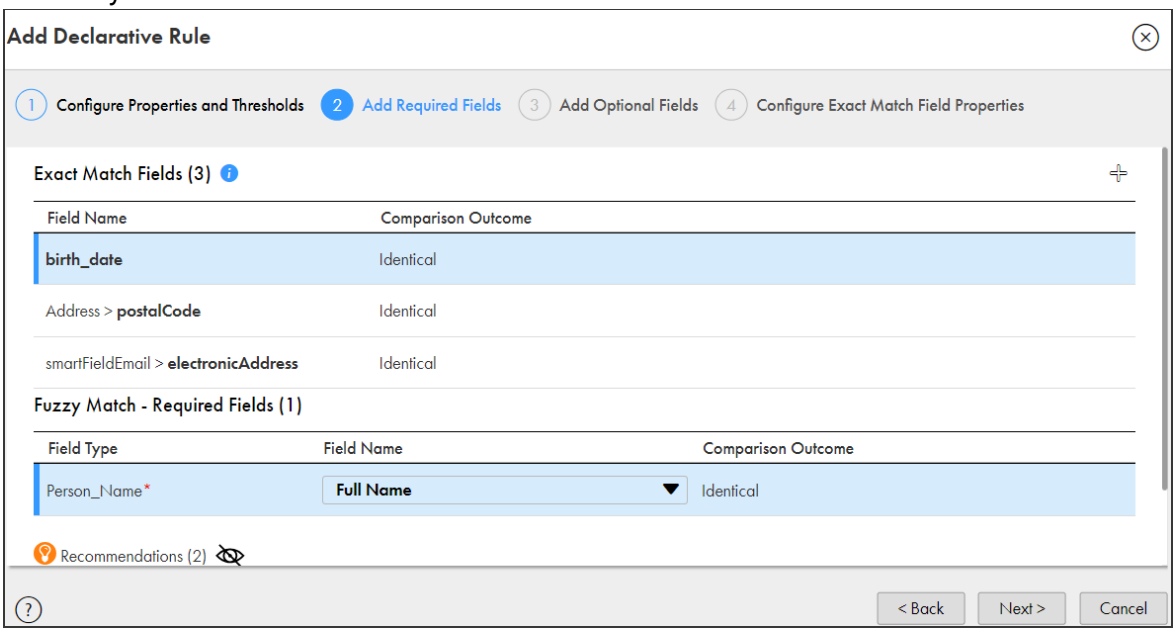
Exact Match Field	Comparison Outcome
Address: Postal Code	Identical
Birth Date	Identical



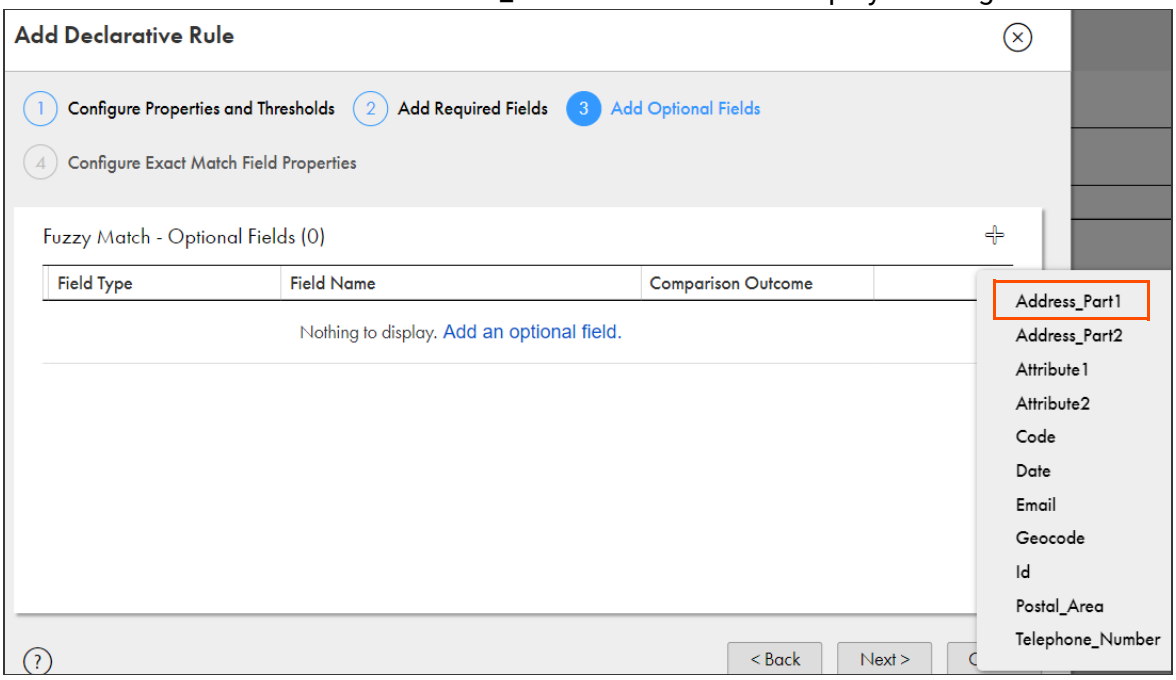
19. Under **Fuzzy Match Fields**, click **Field Name** drop down and select **Full Name**.



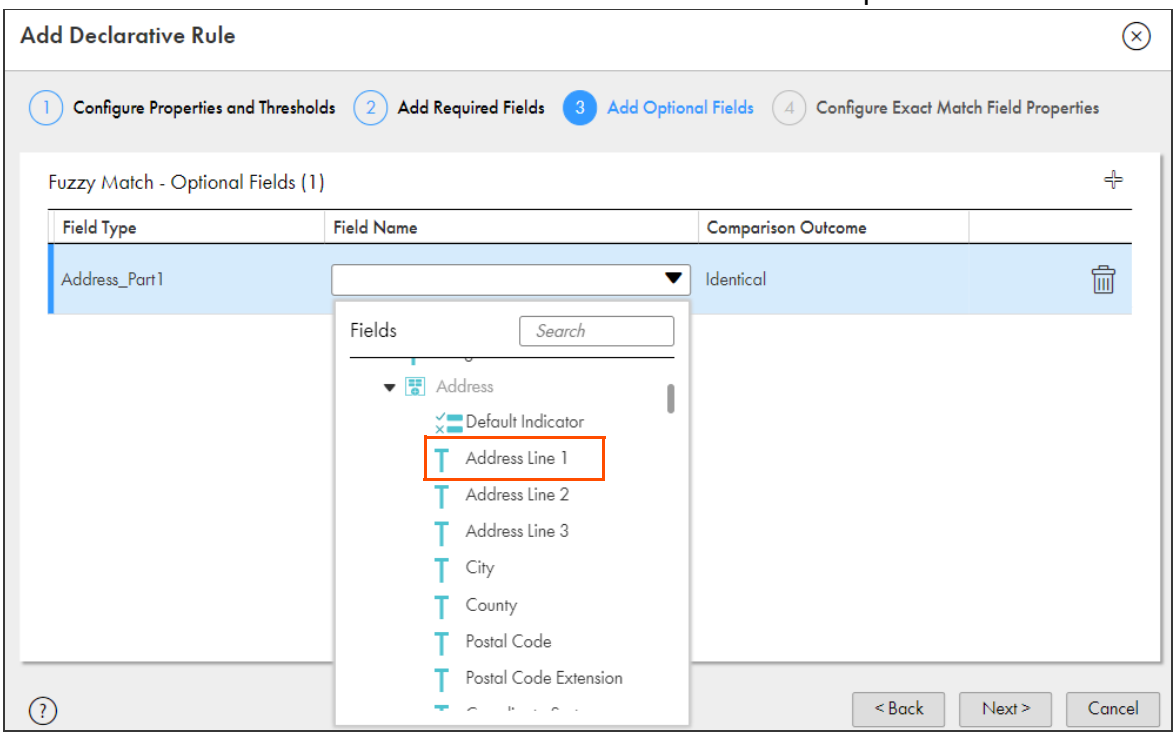
20. Click anywhere in the window to add the record to the table. Click **Next**.



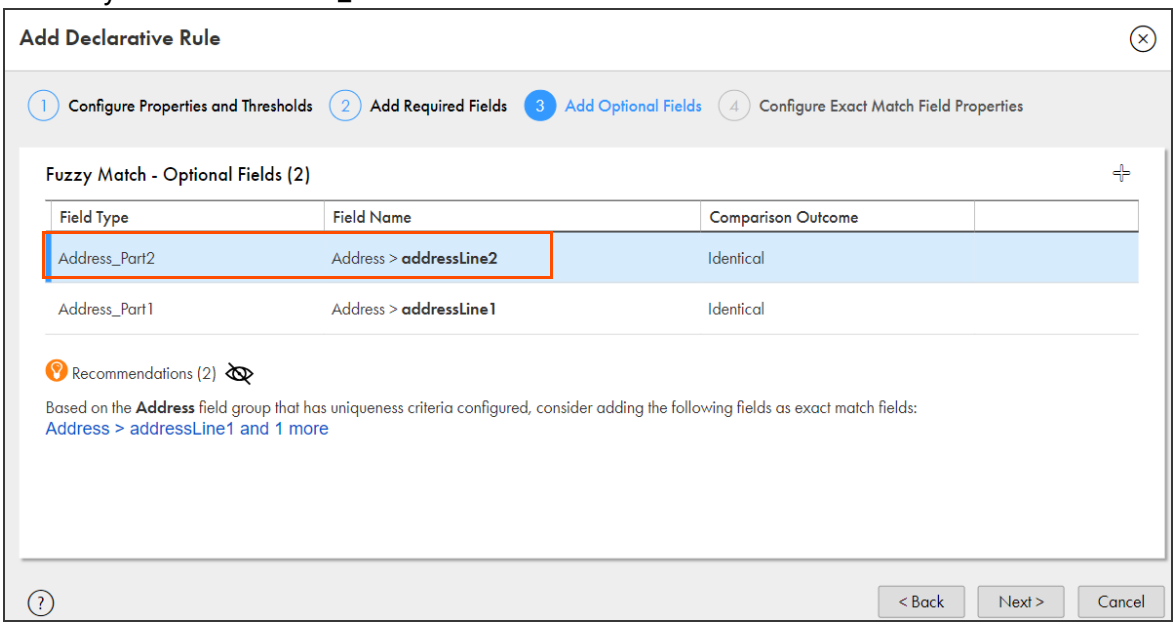
21. To add optional fields, click **Add an optional field** and select a field for the corresponding additional match field. Select **Address_Part1** from the fields displayed on right.



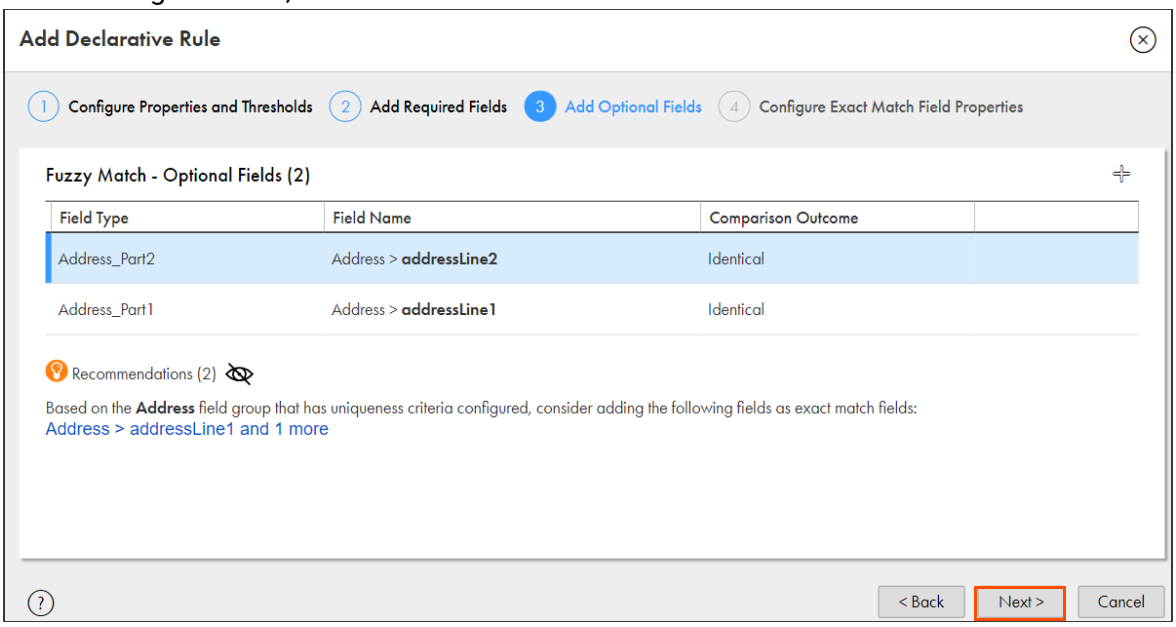
22. Click **Field Name** and select **Address: Address Line 1** from the drop down.



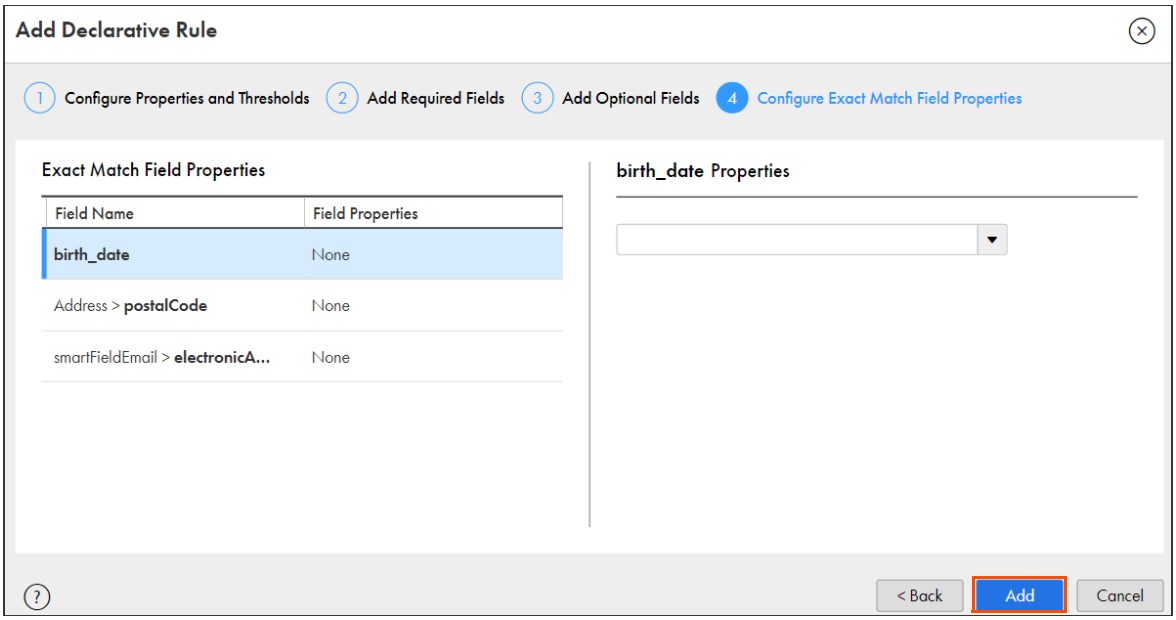
23. Similarly add the **Address_Part2** field and select **Address: Address Line 2** for Field Name.



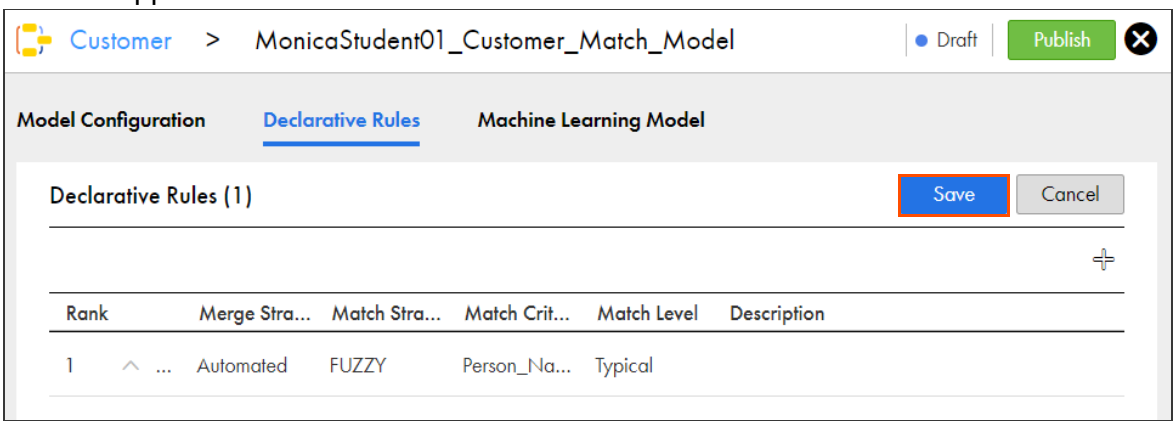
24. After adding the rules, click **Next**.



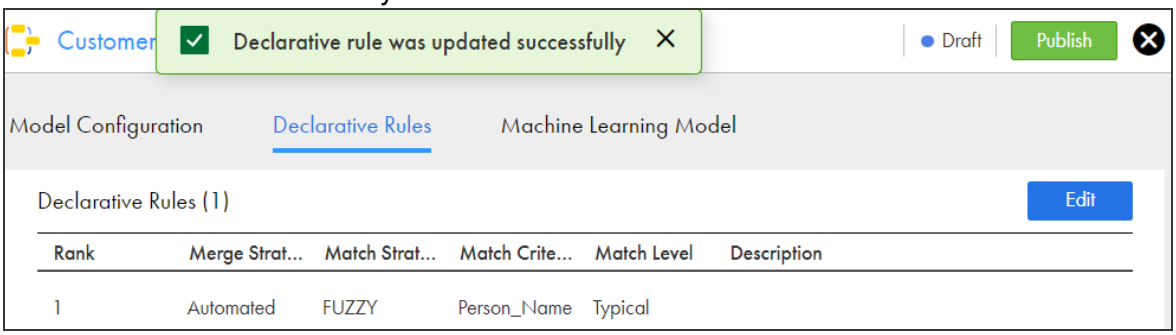
25. In Configure Exact Match Field Properties tab, you can enable segment matching on exact match fields. For this lab exercise, we will not use any segment matching. So, click **Add** to add the declarative match field.



26. The rule appears in the table. Click **Save**.



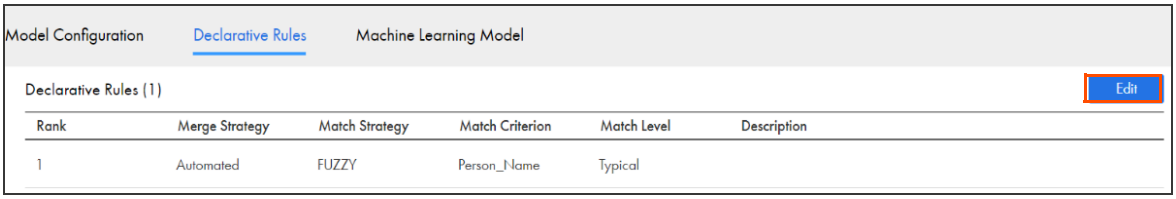
The rule is saved successfully.



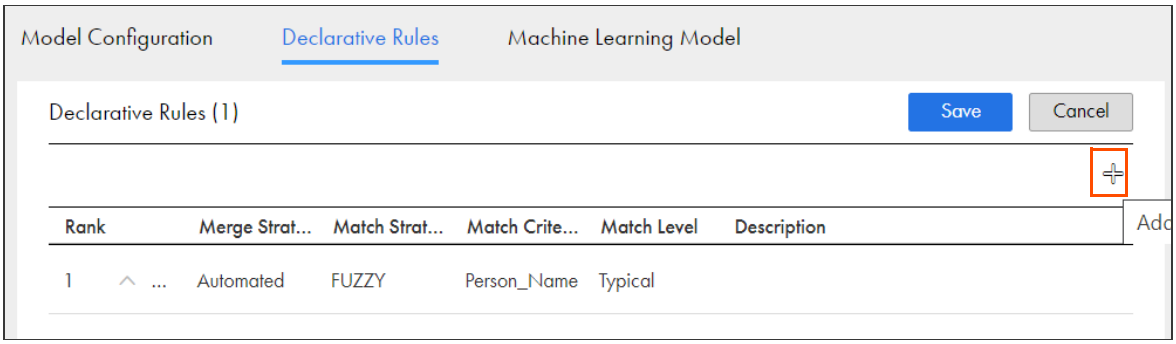
27. You will add the second rule now. The complexity of the second rule decreases when compared to first. The criteria for this rule could be:

- Fuzzy match and a Manual merge.
- Full Name, Address Line 1 – should be similar.
- Postal Code, and Date of Birth – should be exact.

28. Click the **Edit** button to add more declarative rules.



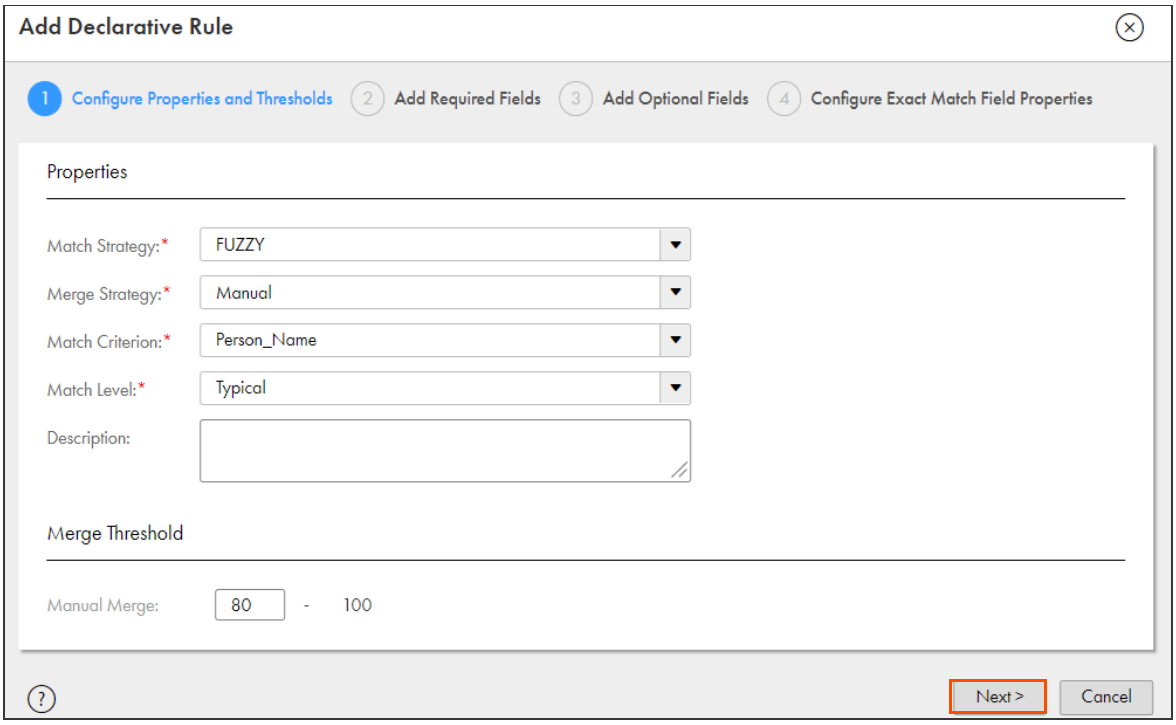
29. Click the **+** icon.



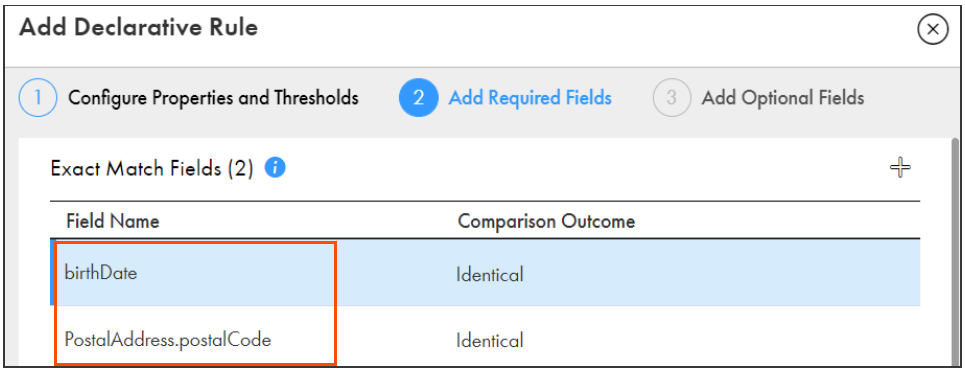
30. In the **Add Declarative Rule** window, enter the following values:

Field	Value
Match Strategy	FUZZY
Merge Strategy	Manual
Match Criteria	Person_Name
Match Level	Typical
Merge Threshold > Manual Merge	80-100

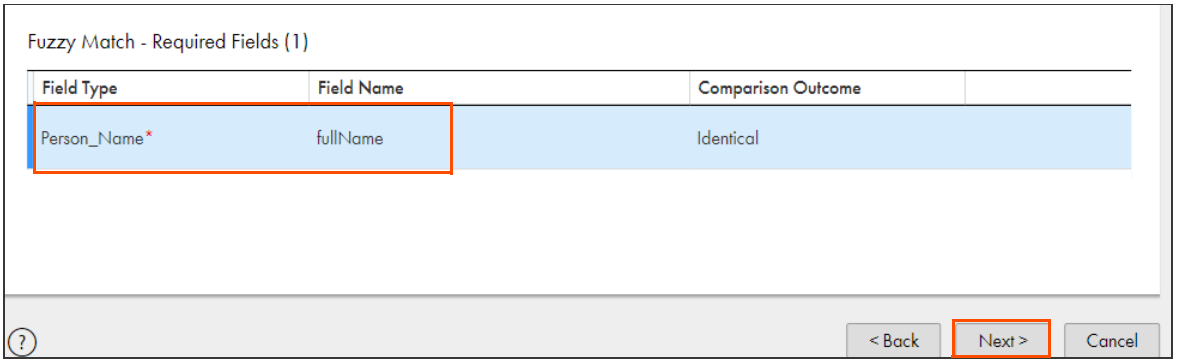
31. Click **Next**.



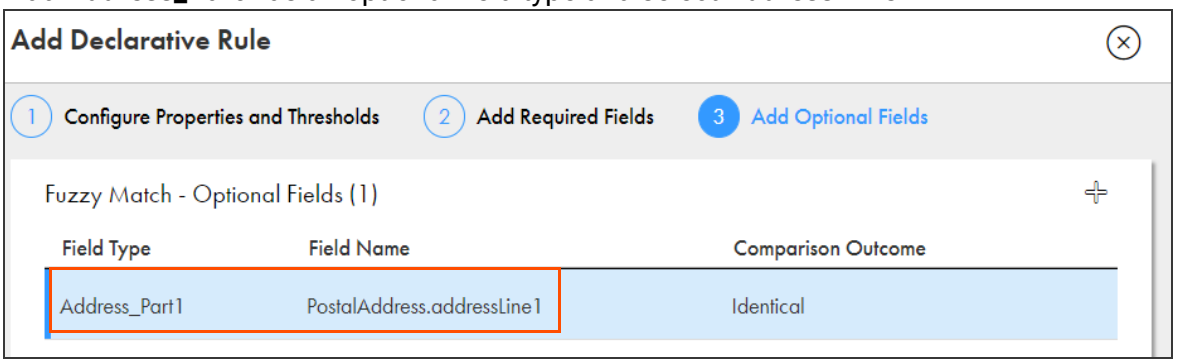
32. In the **Exact Match Fields** section, add the fields **Address**: **Postal Code** and **Birth Date**.



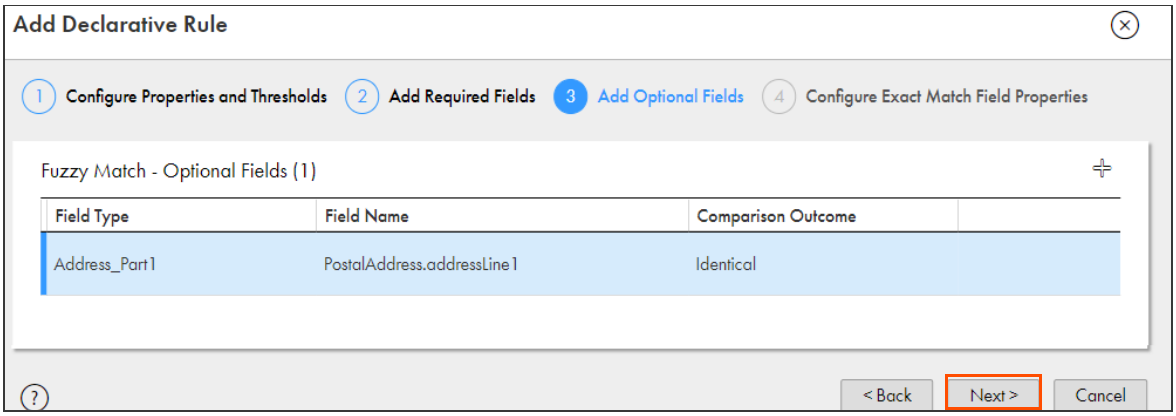
33. In the **Fuzzy Match Fields** section, select **Full Name** as the Field Name for Person_Name. Click **Next**.



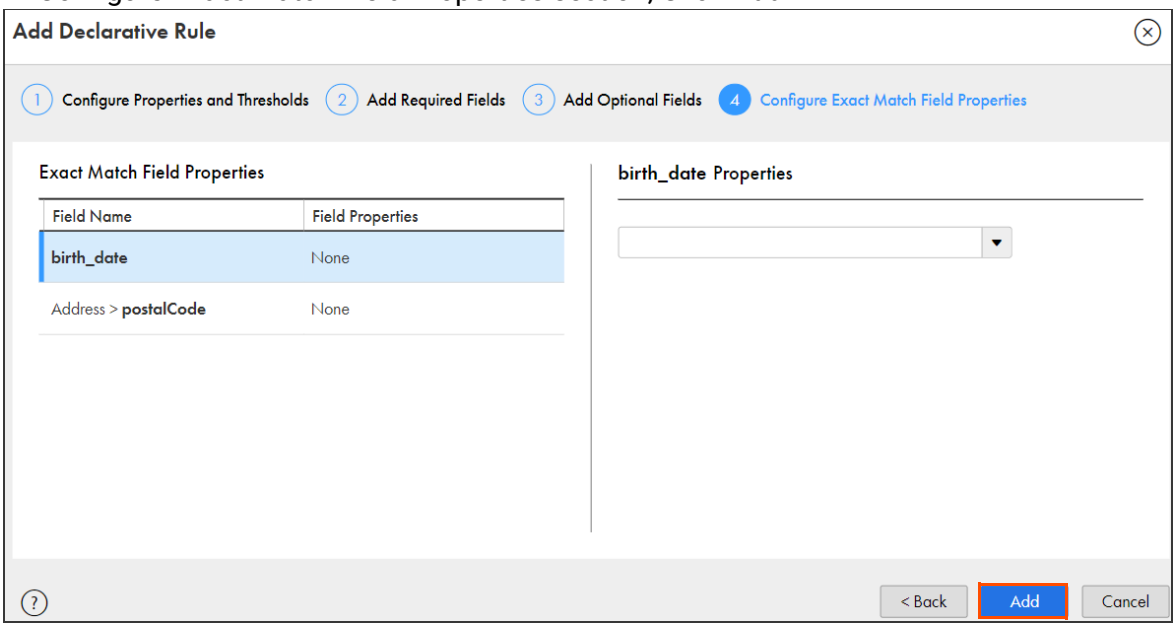
34. Add **Address_Part1** as an optional field type and select **Address Line1**.



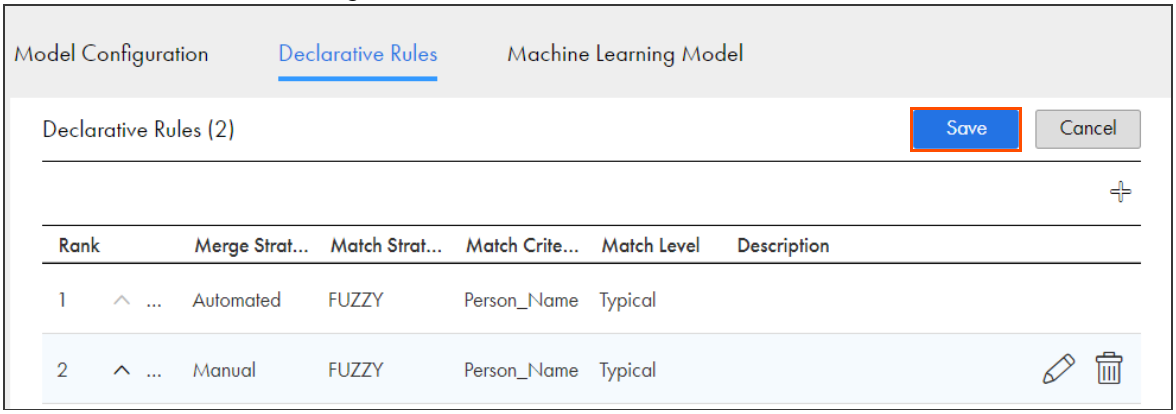
35. Click **Next**.



36. In Configure Exact Match Field Properties section, Click **Add**.



37. Two rules have been configured. Click **Save**.



Self-Try Exercise (Mandatory)

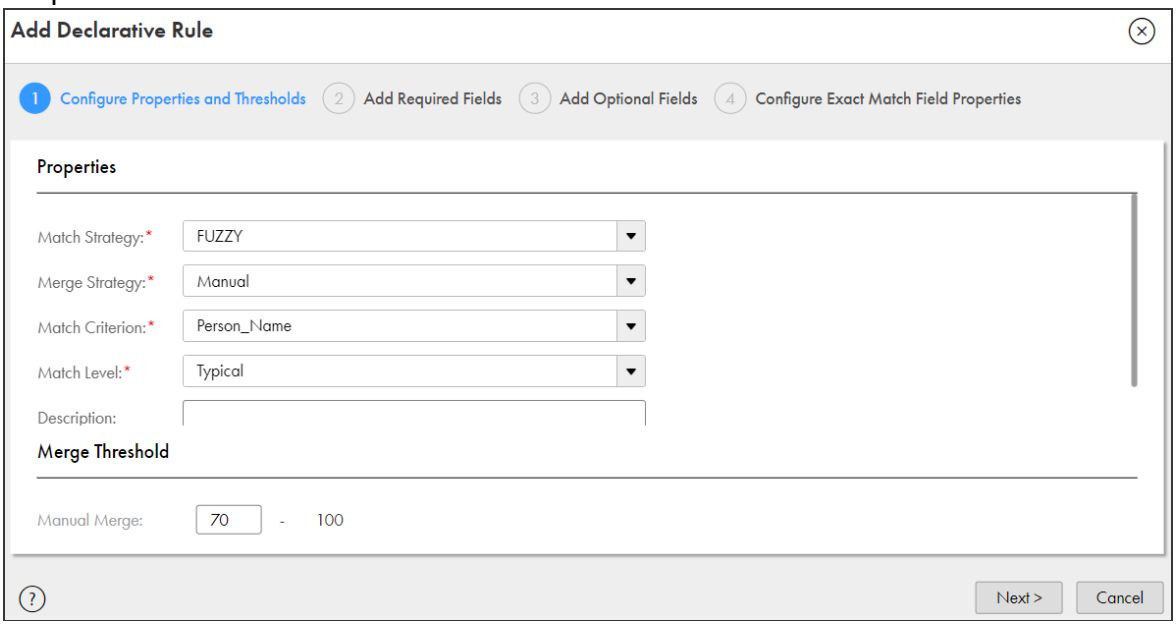
This is a mandatory exercise. Rules 1 and 2 are stringent and you may not get many pairs of matched records. Apply the skills that you learned above and add a third declarative rule. The complexity for the rule further reduces in this case. Apply the following criteria:

- Fuzzy Match and a Manual merge.
- Full Name, Address Line 1 – should be similar.

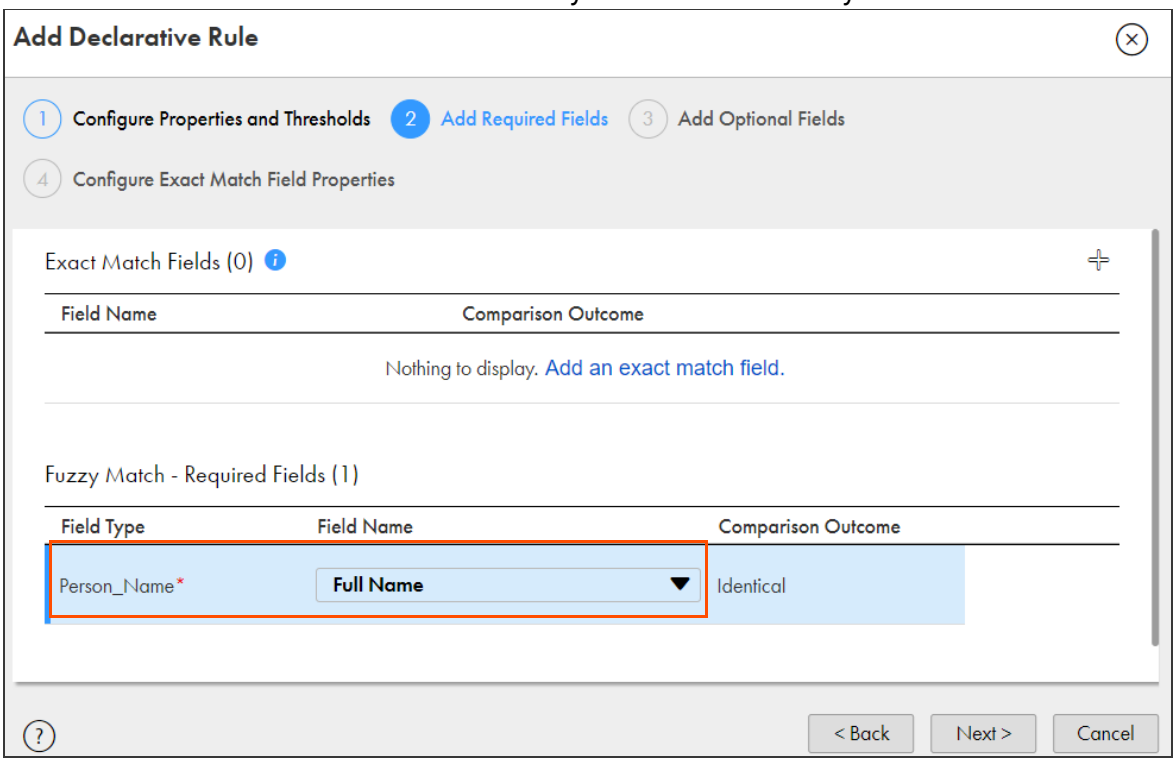
High Level Steps

38. Click **Edit** to add another declarative rule.

39. Properties and Threshold should have the values as shown in the screenshot:

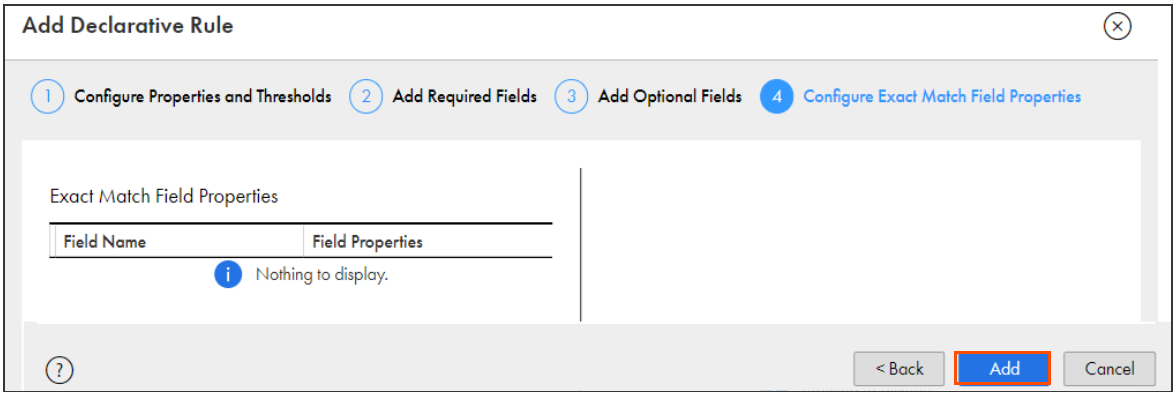


40. There is no need for exact match fields. Only **Full Name** add fuzzy match fields.

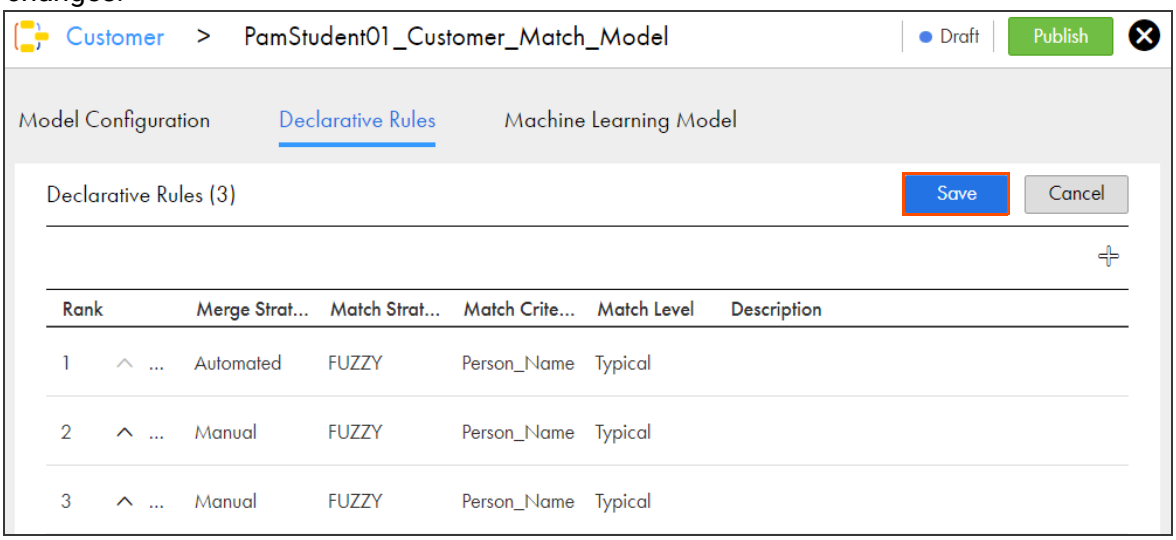


41. In the **Add Optional Fields** options add **Address_Part1** as an optional field type and select **Address Line1** and click **Next**.

42. Click **Add**.



43. All the 3 declarative rules should appear as shown in the screenshot below. **Save** your changes.

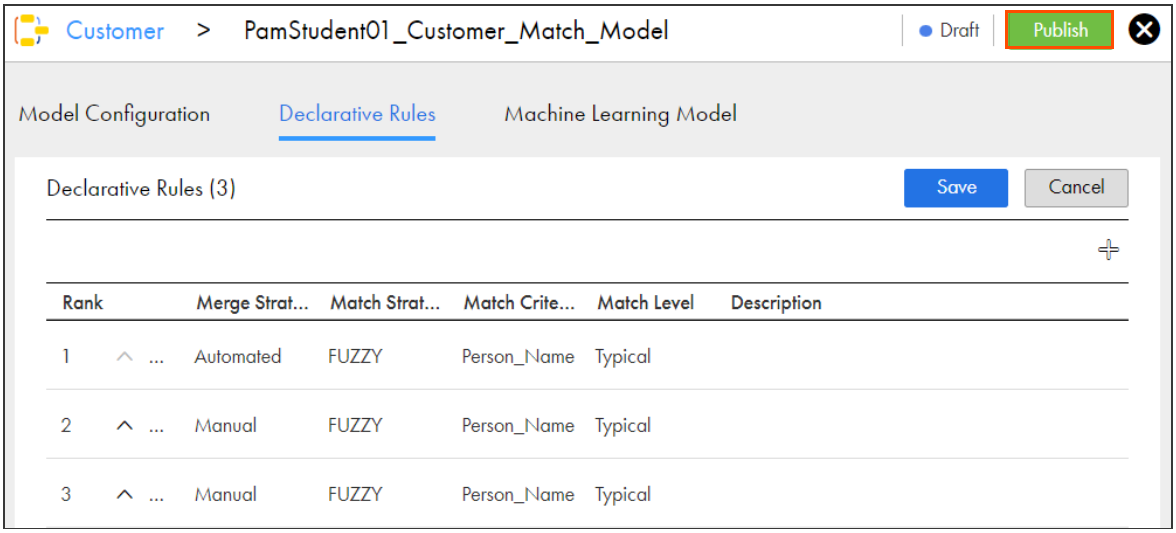


The 2nd and the 3rd rules have lesser match conditions. Hence, the records that do not satisfy the 1st rule condition will move to rule 2 and later to rule 3.

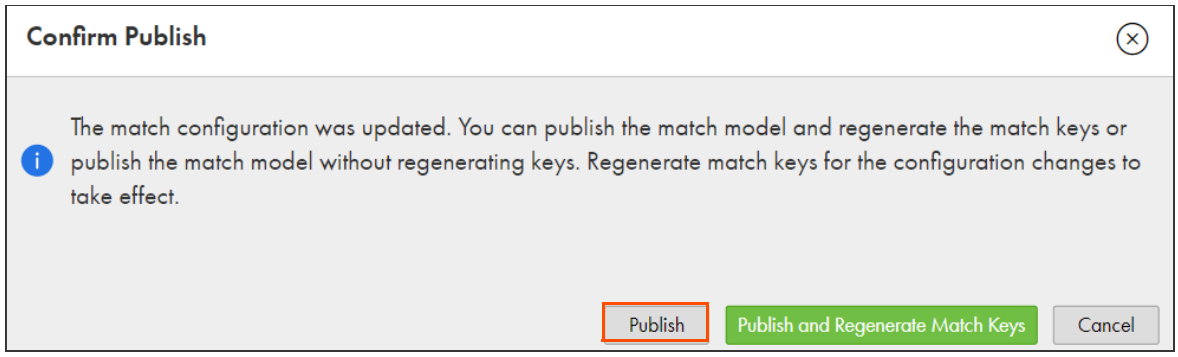
Publish the Match Model

To use a match model for the match and merge process, you need to publish the model after you complete configuring it.

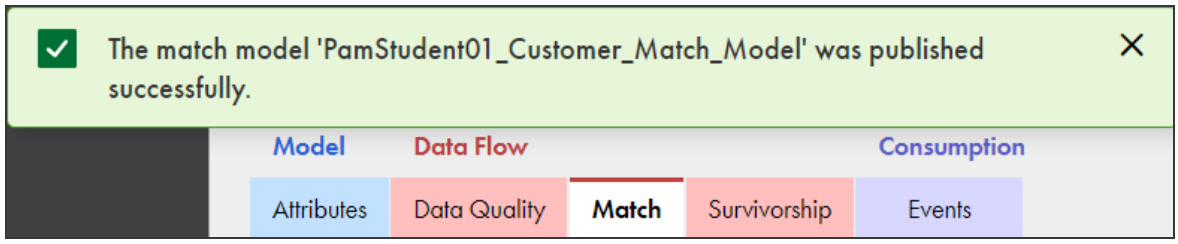
44. Click **Publish**.



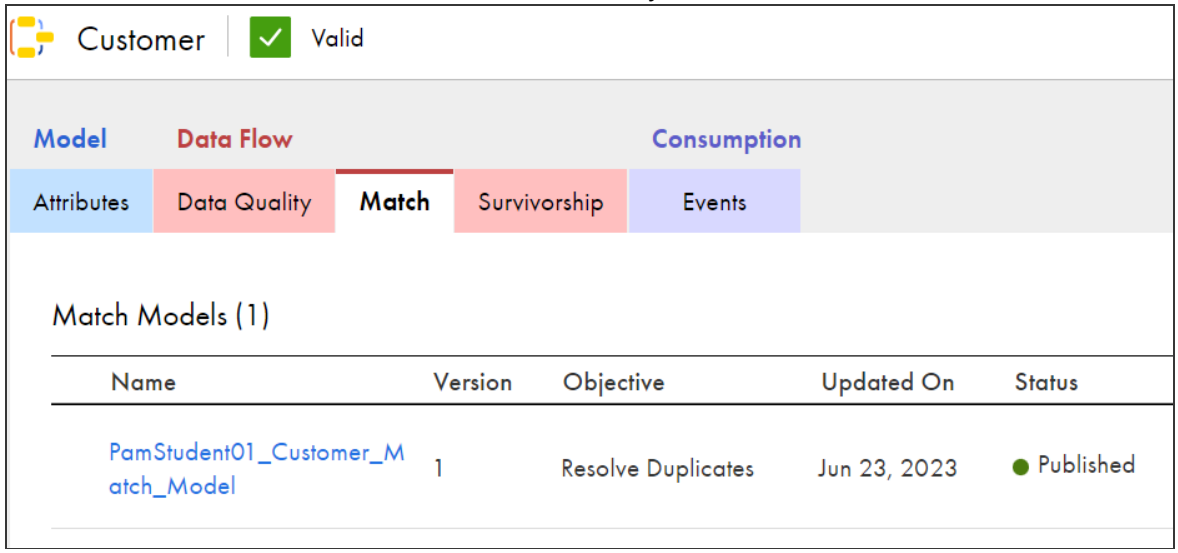
45. In the confirmation window, click **Publish**.



Once published, you cannot delete the model. If you edit, another version of the same model will be saved as a separate model. For example, if you want to change the rule and you edit this model, you will need to publish it again after making the changes, and it will be saved as version 2.



Notice that the Match model is saved successfully, and the status shows as “Published”.



If the status does not show as Published, refresh the browser.

This concludes the lab.

Lab 10-2: Defining Segment Matching

Overview:

When you add declarative rules to a match model, you can now enable segment matching on the exact match fields. Use segment matching to limit matching to specific subsets of data based on the values of exact match fields. If you enable segment matching, you can also choose to match the segment data with the rest of the data.

Objectives:

- Creating a match model from scratch
- Configuring declarative match rules
- Enabling segment match to configure exact match field properties

Duration:

30 minutes

Scenario:

After creating the data model and adding declarative rules to it, Mark wants to enable segment matching for the exact match fields to limit the matching to specific subsets of data.

Tasks:

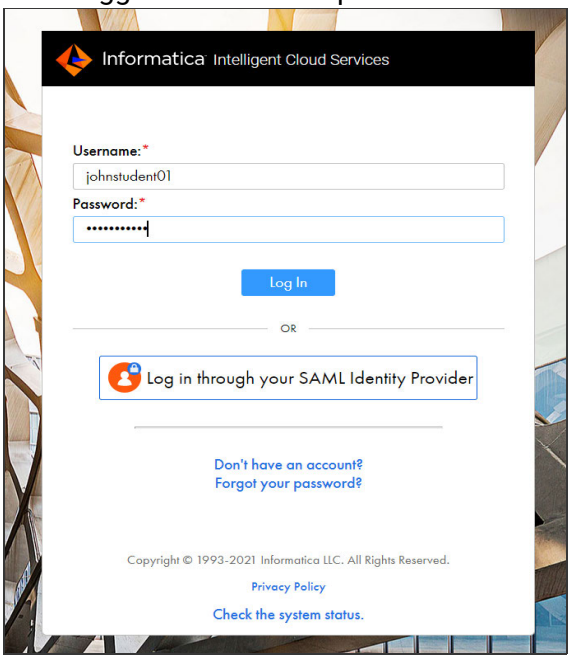
Perform the following tasks to enable segment matching for Homekart scenario:

- Creating a match model
- Create a match and merge job
- Monitor and analyze match

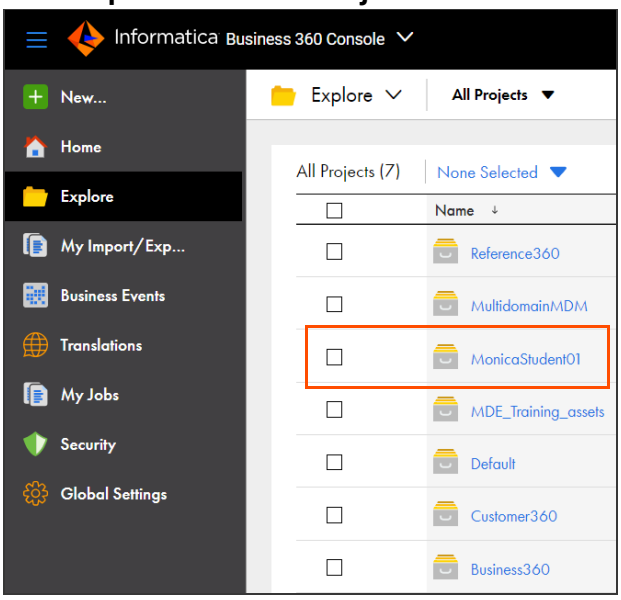
Creating a Match Model

You will create a match model to which you will enable segment matching.

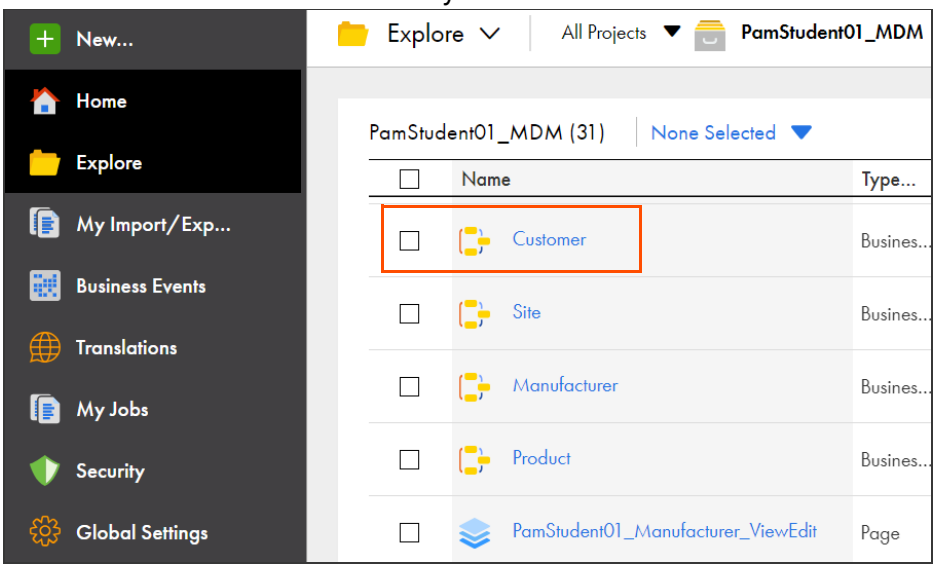
1. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**) if you have logged out after the previous lab.



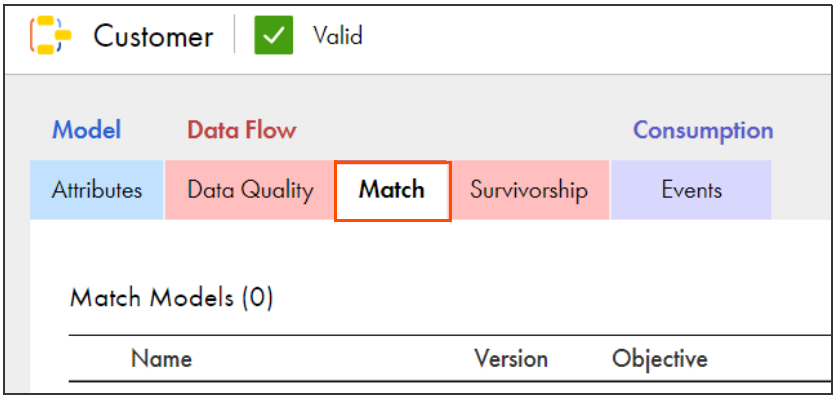
2. Click **Explore** > **Student Project**.



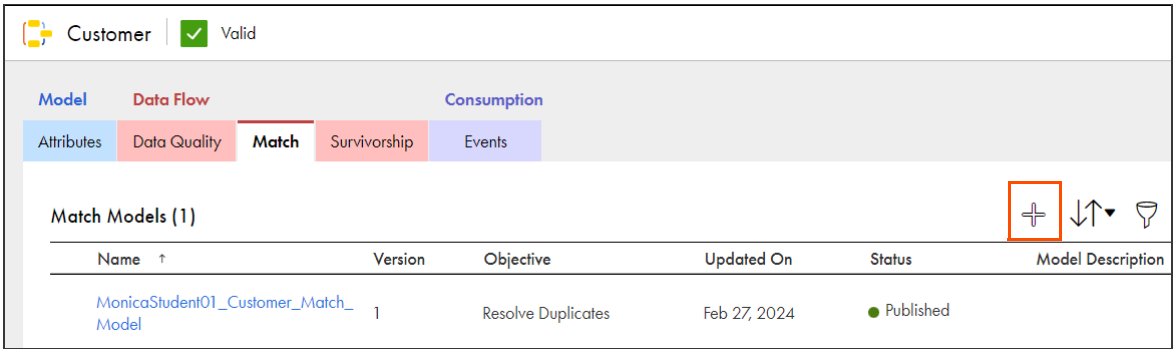
3. Click the **Customer** business entity.



4. Click the **Match** tab.

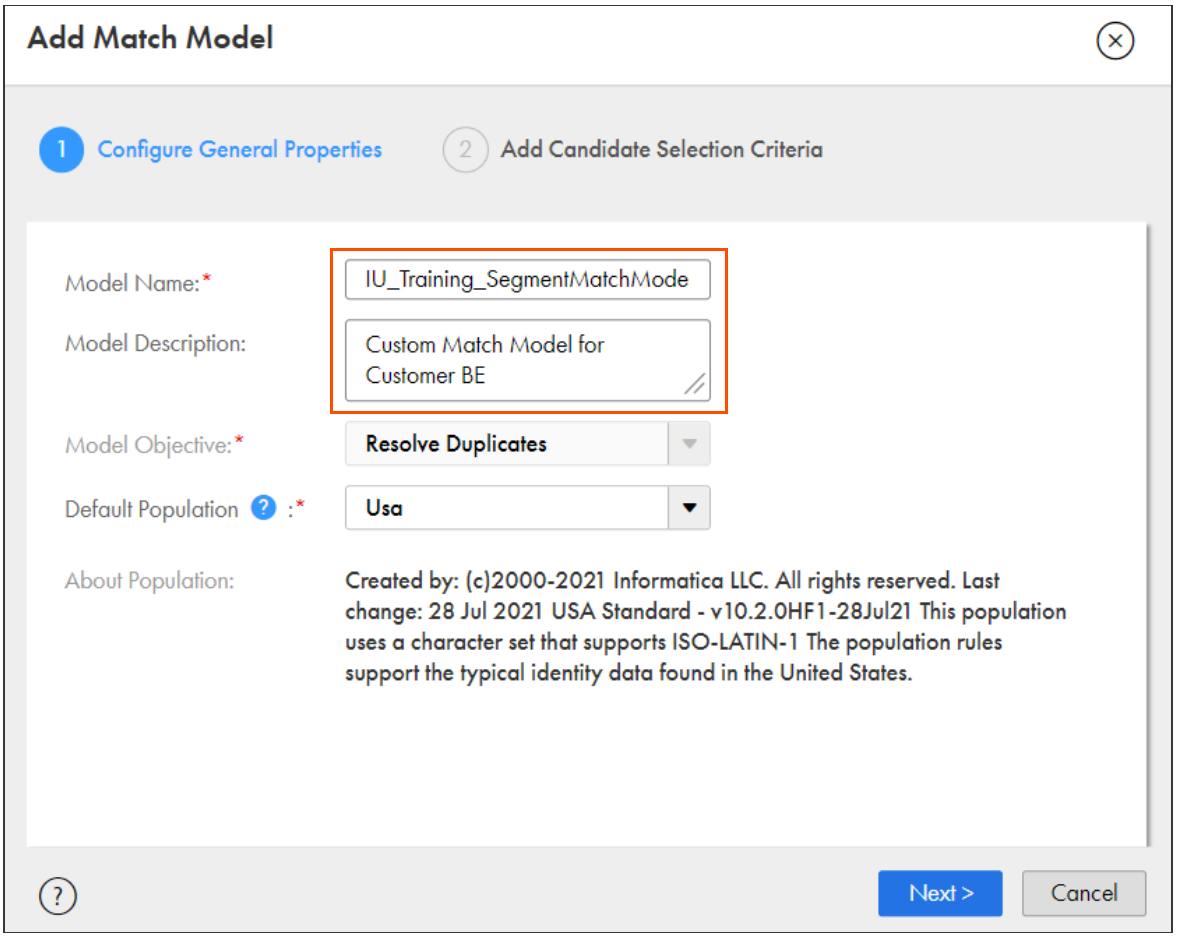


5. Click the **+** icon to add a match model.

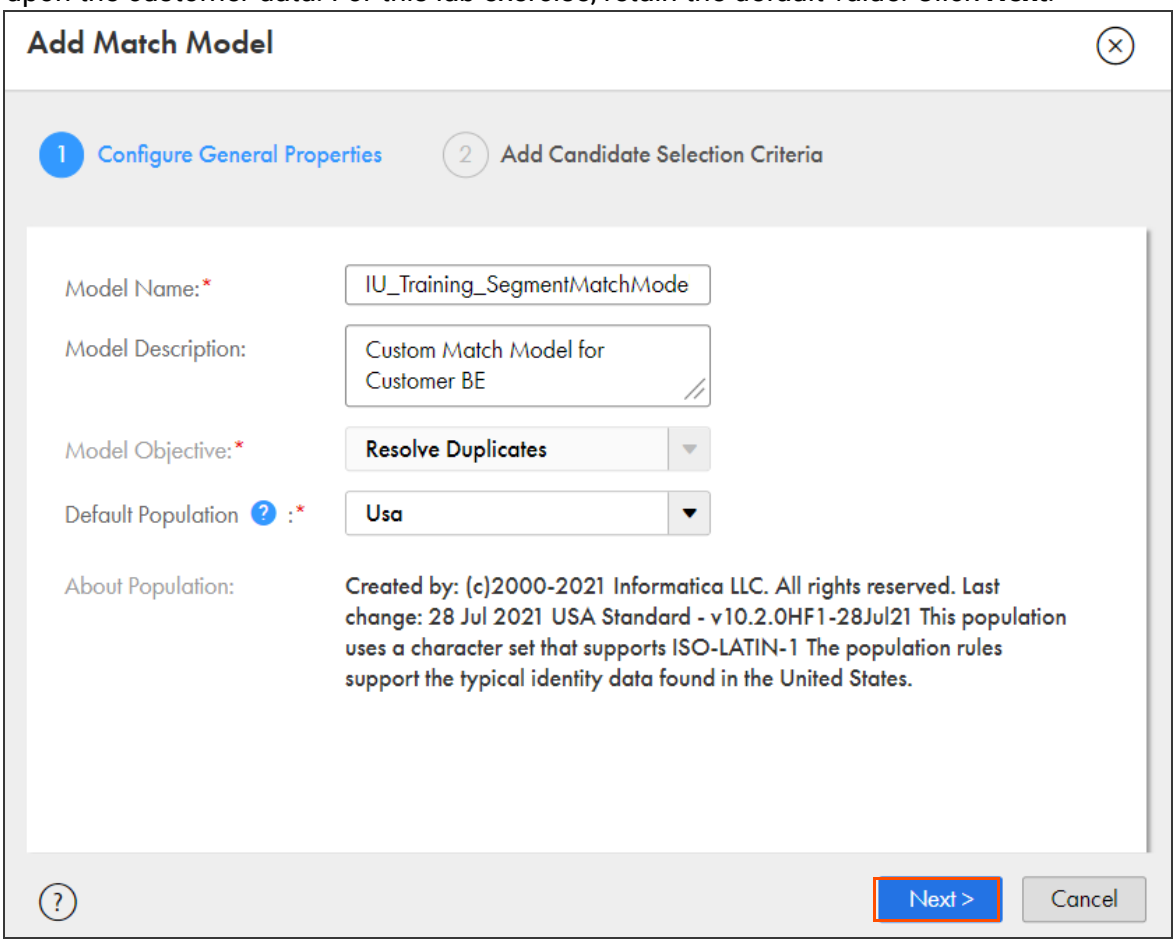


Add Match Model window appears.

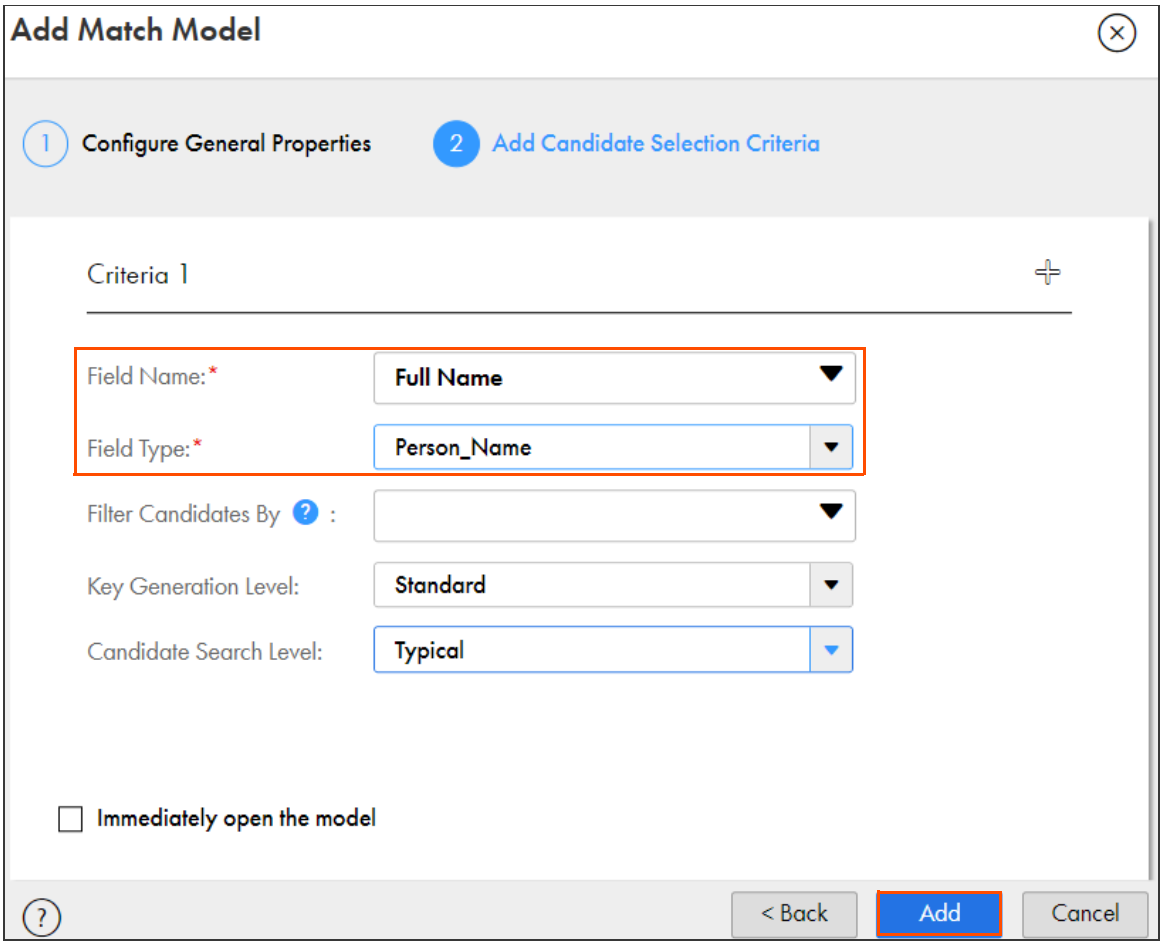
6. The Add Match Model wizard appears. In the Configure General Properties section, enter the Model Name as **IU_Training_SegmentMatchModel**. Provide Description as **Custom Match Model for Customer BE**.



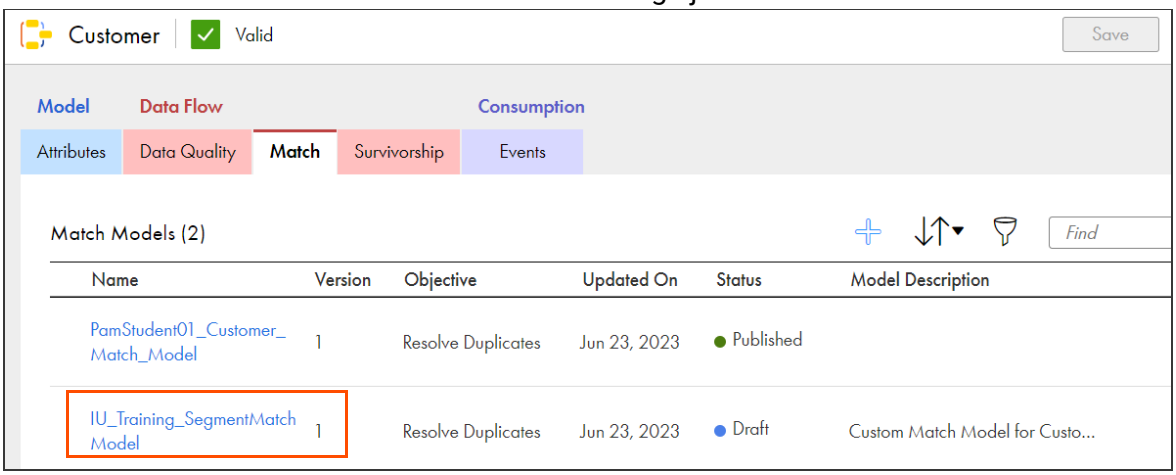
7. Retain the Model Objective. For Default Population, you can select the country depending upon the customer data. For this lab exercise, retain the default value. Click **Next**.



8. Under Criteria 1 section of the Configure General Properties, from the **Field Name** dropdown, select **Full Name**. From the Field Type drop down, select **Person_Name**. Retain the other values and Click **Add**.



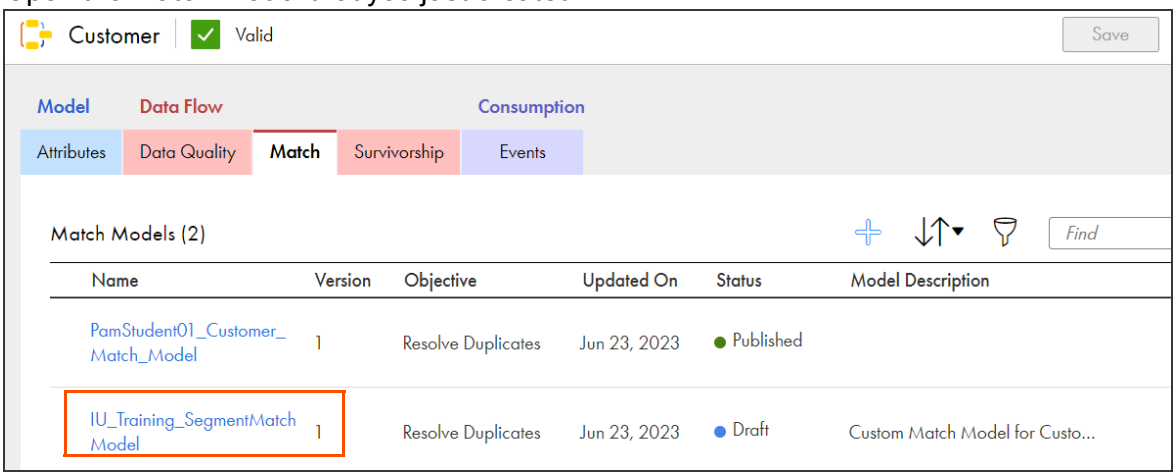
9. You can see that the new match model is created. You can use this model to create declarative match rules and to run match and merge jobs.



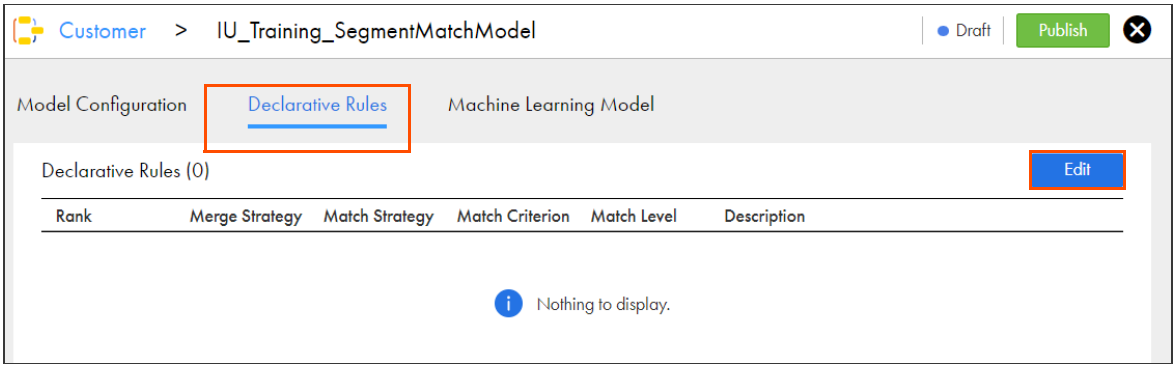
Creating Declarative Rules

The next step is to add declarative rules to the model that you just created.

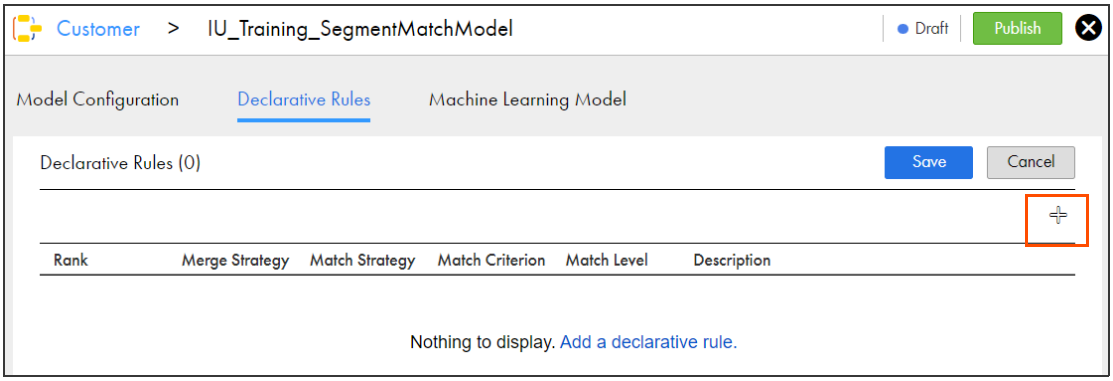
10. Open the match model that you just created.



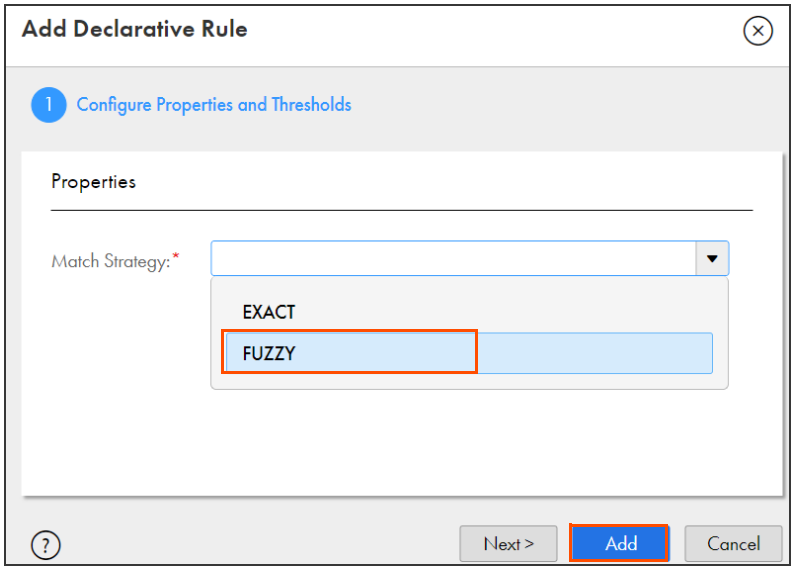
11. Click **Declarative Rules** tab and click **Edit**.



12. Click **+** to add a new declarative rule.

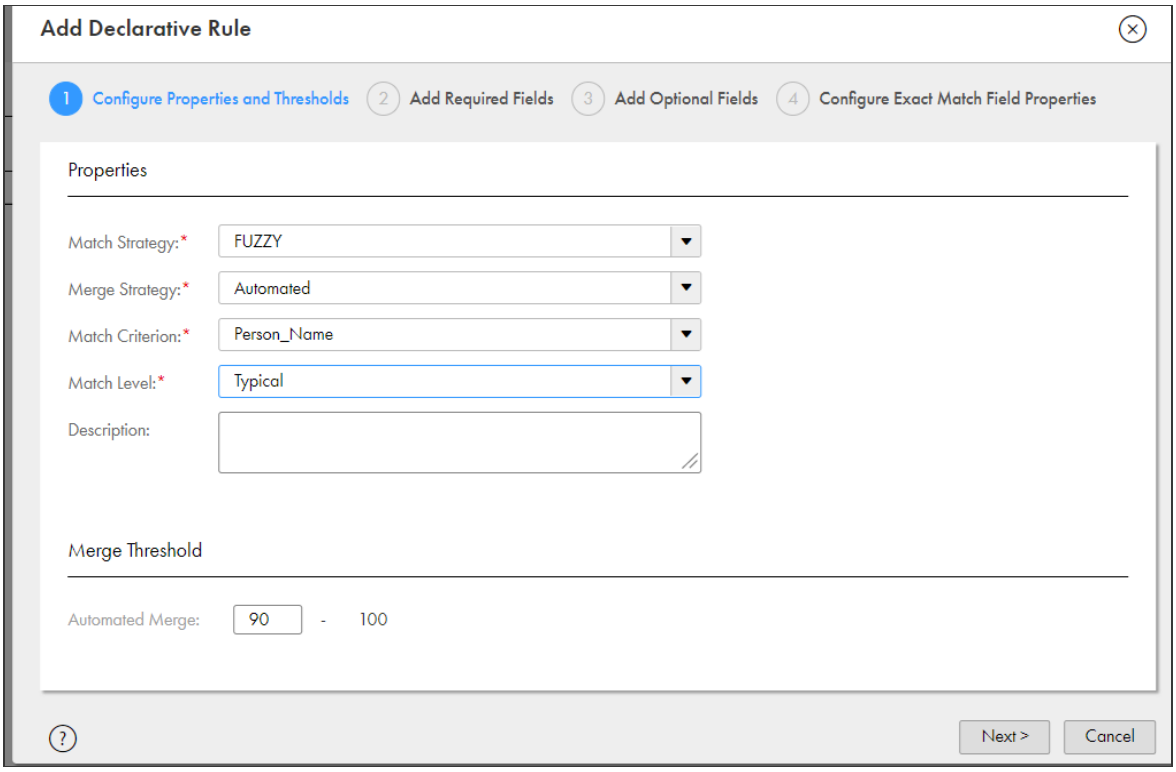


13. In the **Add Declarative Rule** window, from the Match Strategy dropdown select **FUZZY** and click **Add**.

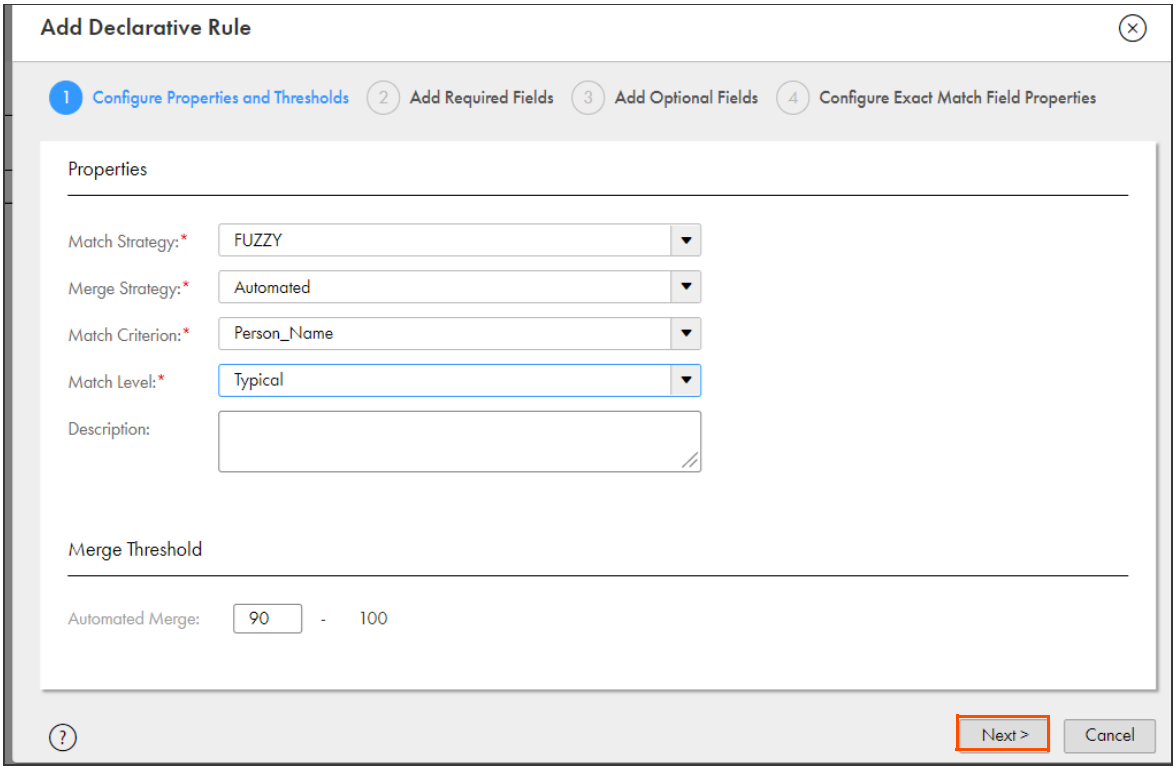


14. A new window will appear. In the **Add Declarative Rule** page, select the following values:

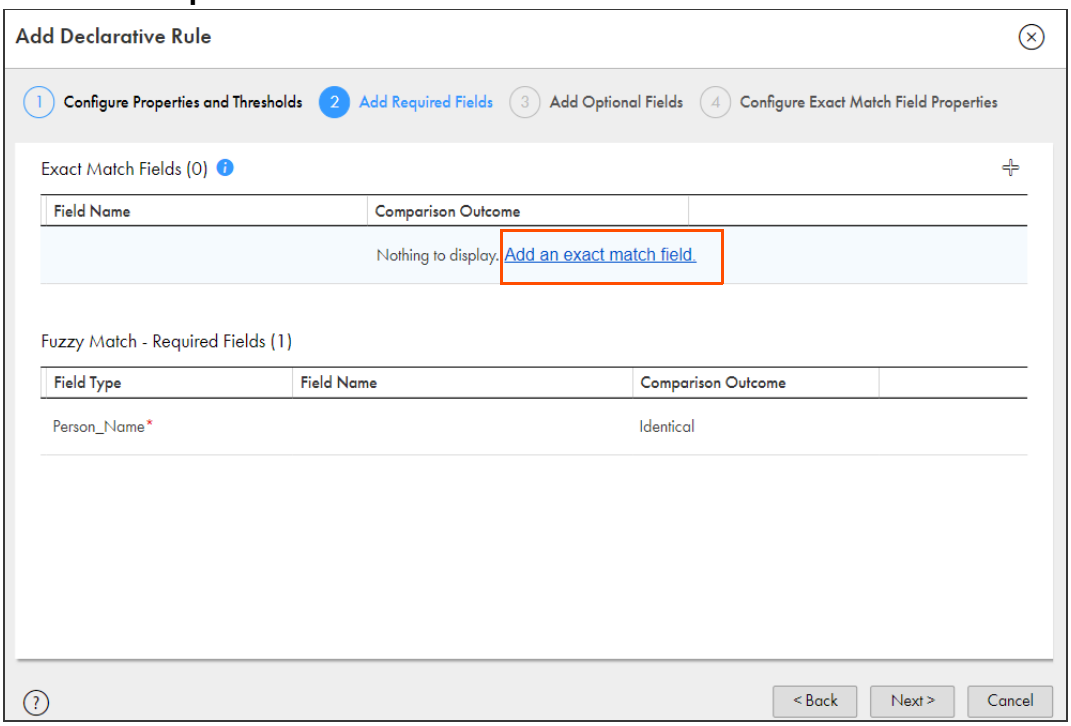
Field	Value
Match Strategy	Fuzzy
Merge Strategy	Automated
Match Criterion	Person_Name
Match Level	Typical
Merge Threshold	90-100



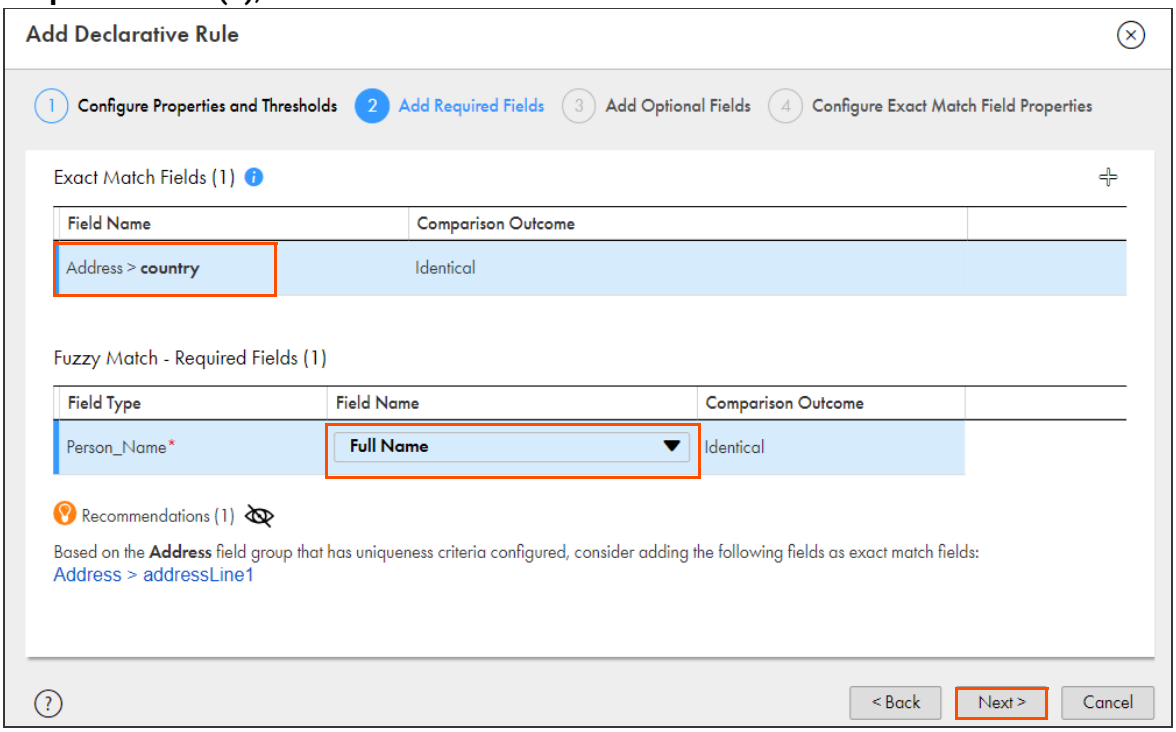
15. Click **Next**.



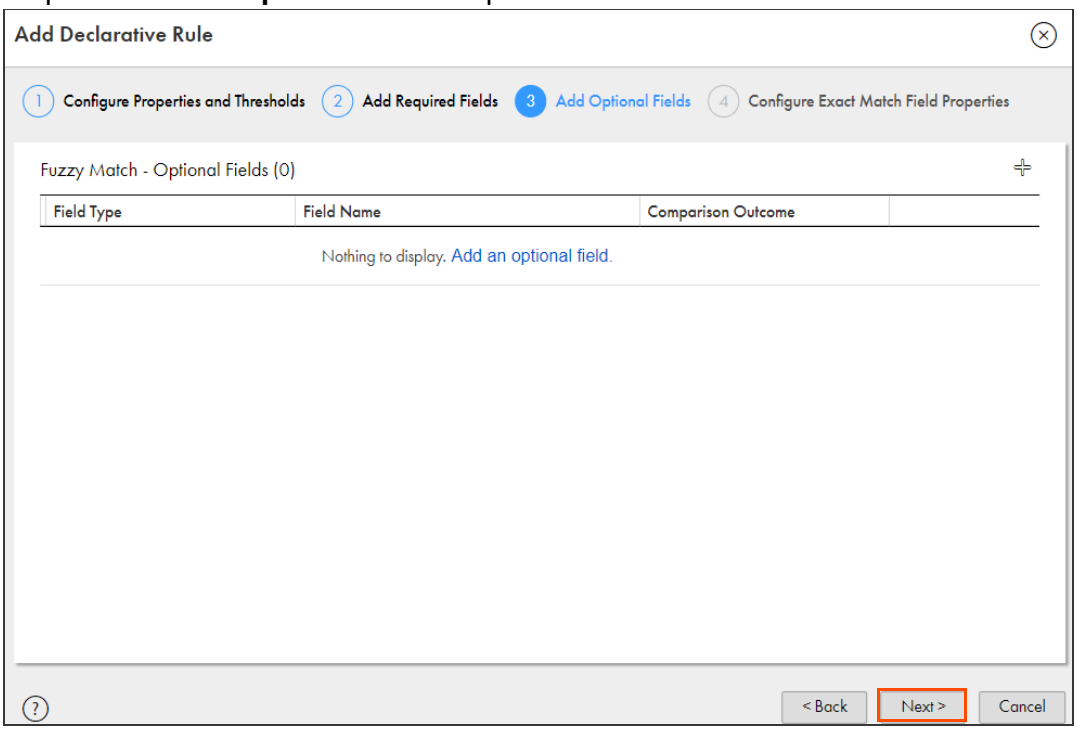
16. In the **Add Required Fields** click **Add an exact match field**.



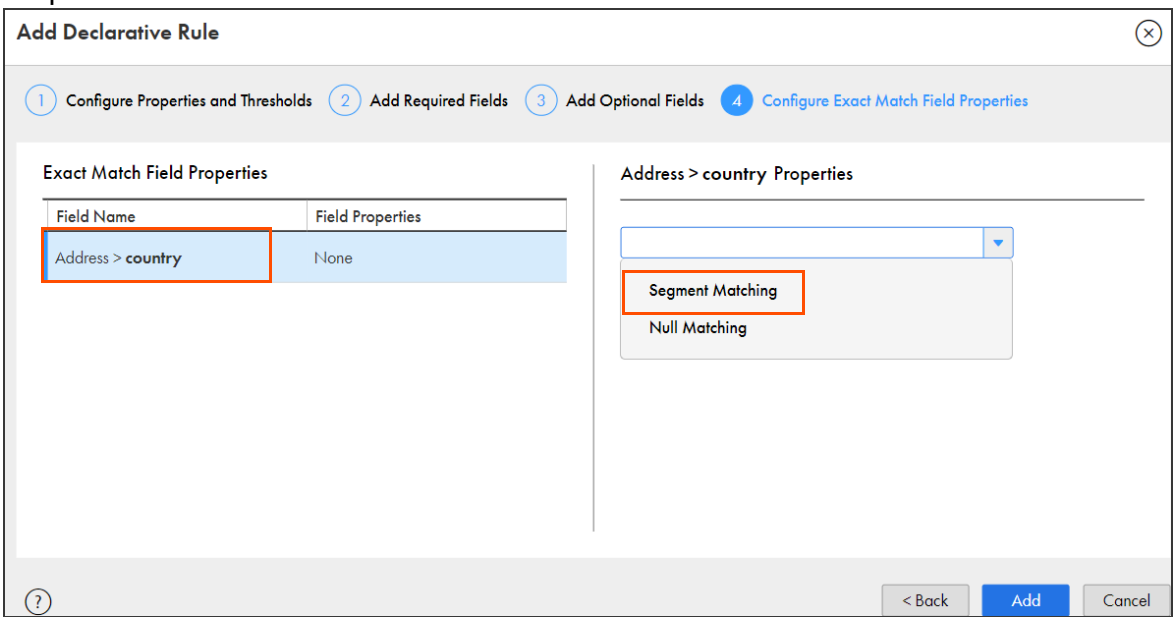
17. From the **Field Name** dropdown, select **Address.Country** and from the **Fuzzy Match – Required Fields (1)**, under the Field Name section select **Full Name**. Click **Next**.



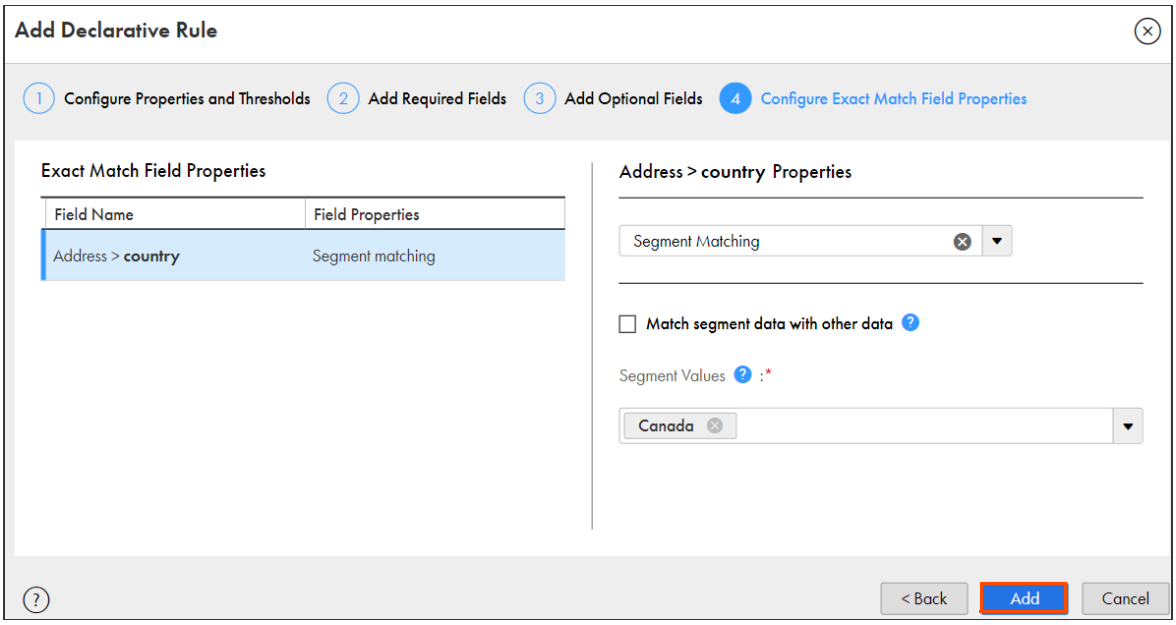
18. Skip the field **Add Optional Fields** step and click **Next**.



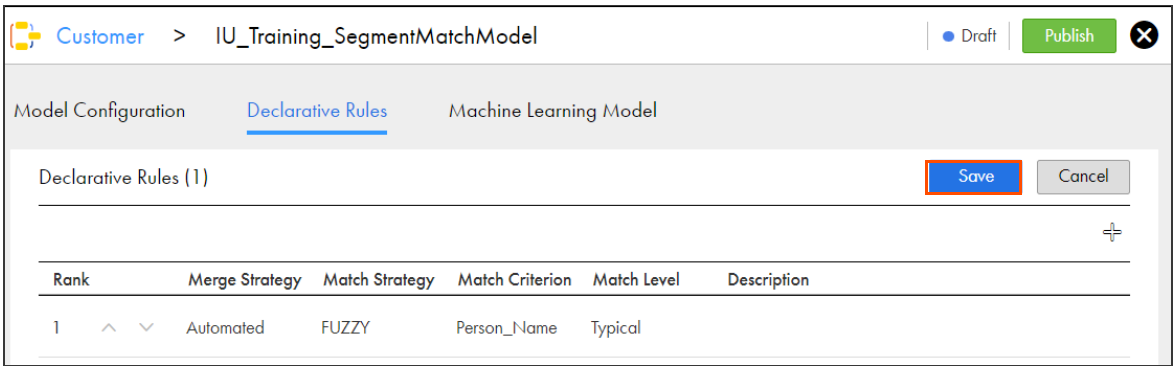
19. In the **Configure Exact Match Field Properties** step, select **Segment Matching** from the drop-down.



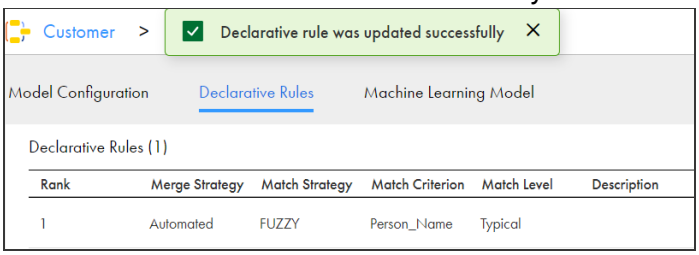
20. Click **Add** to the Add Declarative Rule.



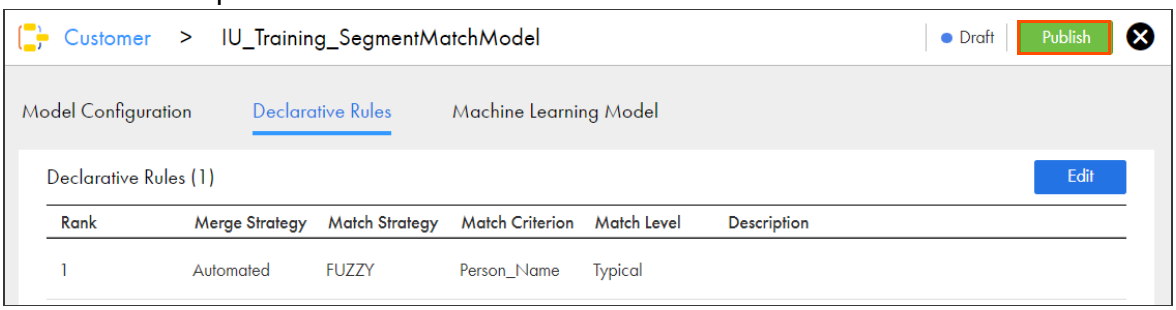
21. The declarative rule is created. Click **Save**.



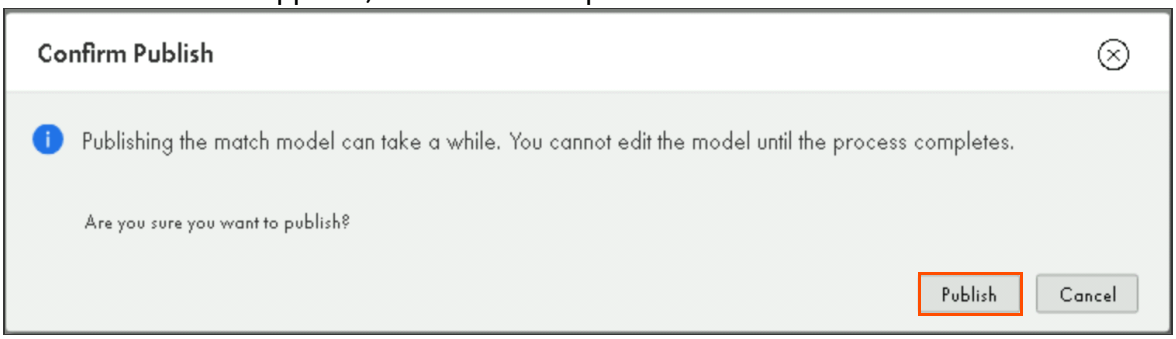
The declarative rule is saved successfully.



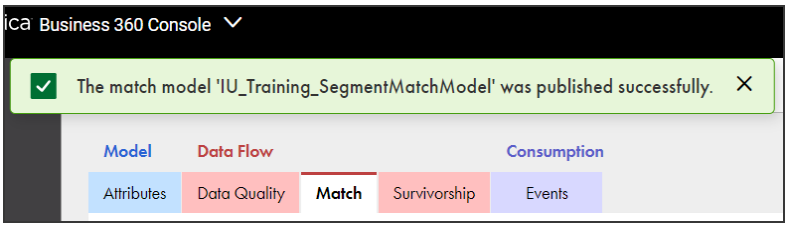
22. Click **Publish** to publish the declarative rule.



23. A confirmation box appears, click **Publish** to publish the model.



The match model will be published.



Self-Try Exercise (Mandatory)

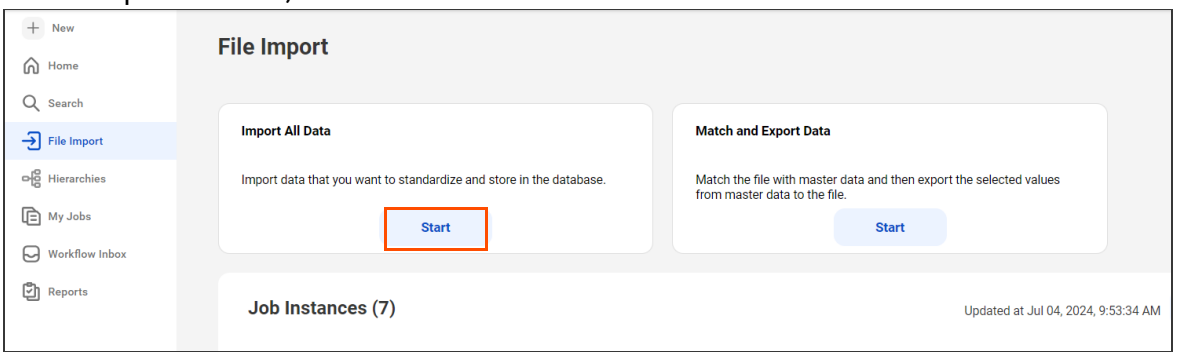
Before running the match and merge job, import the file into **Multidomain MDM** application. The file is present in the location: **C:\Data Files\SegmentMatchData.csv**. The file has records which are candidates for segment match.

High Level Steps:

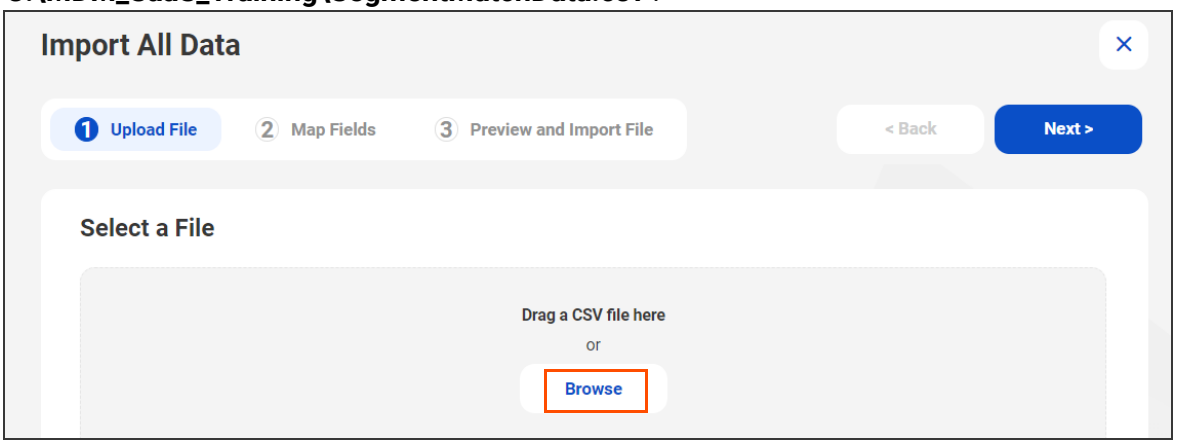
24. Navigate to **Multidomain MDM** application using product switcher.

25. Click **File Import**.

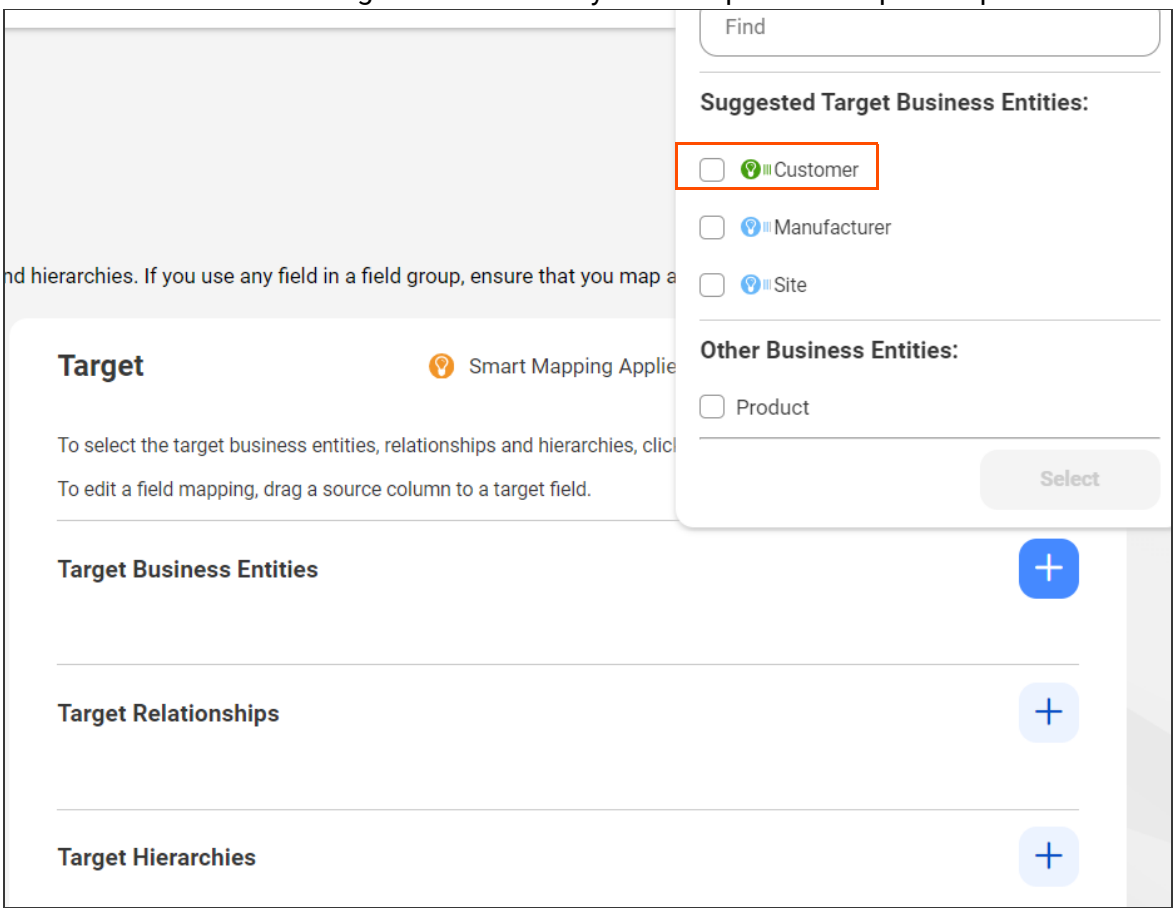
26. Under Import All Data, click **Start**.



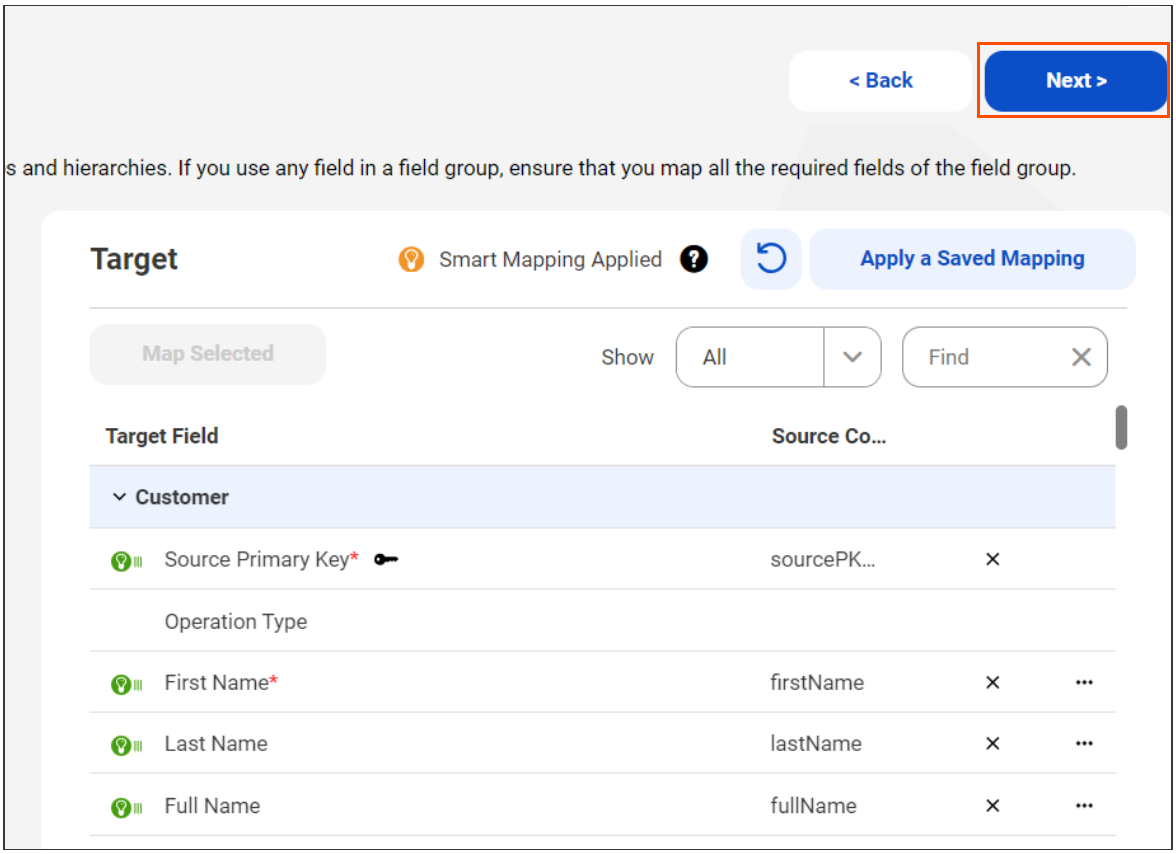
27. Click **Browse** and choose the file from the location **C:\MDM_SaaS_Training\SegmentMatchData.csv** .



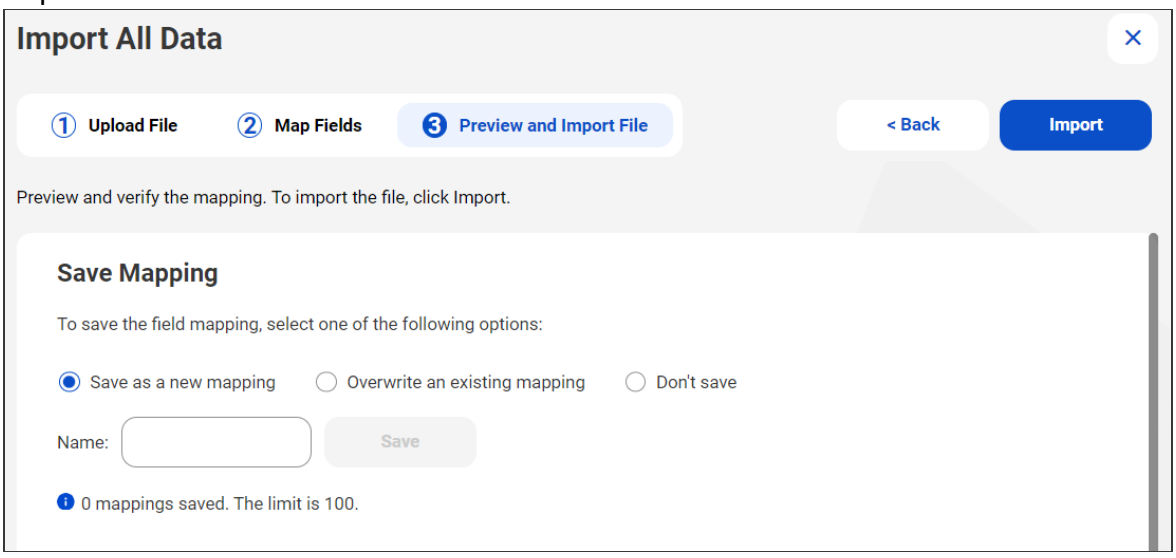
28. Select **Customer** as the Target Business Entity and complete the Import steps.



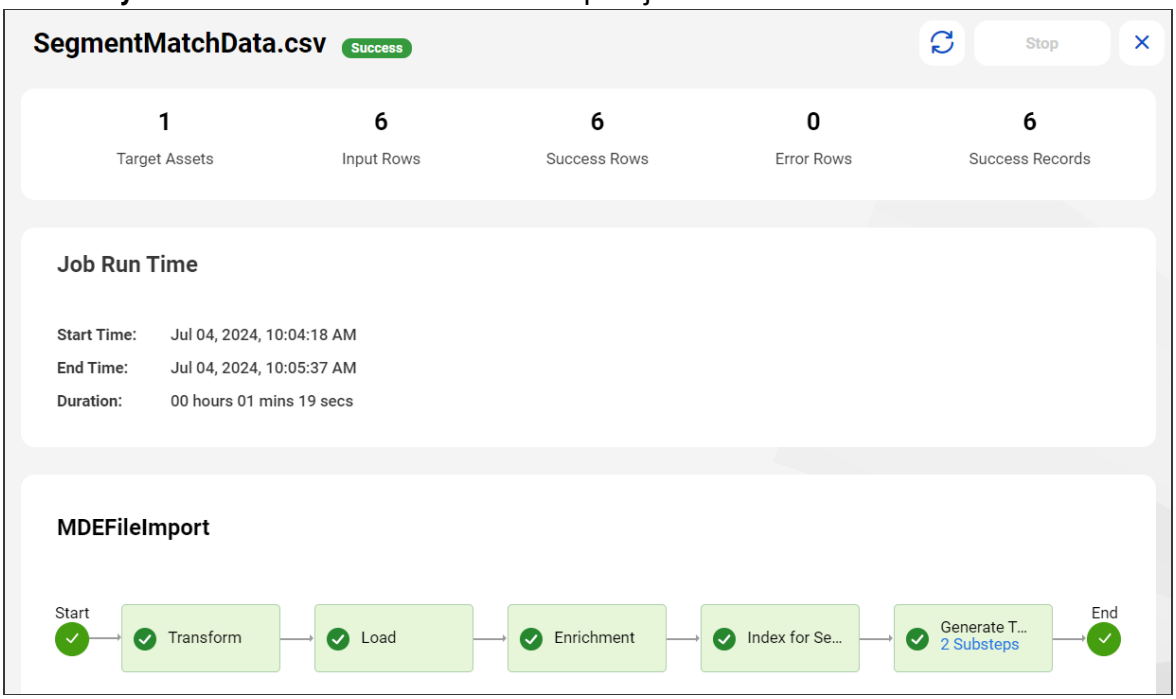
29. Click **Next**.



30. Import the file.



31. Check **My Jobs** to check the status of the import job.



32. Observe the imported CSV file by opening it from the location **C:\Data Files\SegmentMatchData.csv**. There are four records that have the exact same name **David Borean**.

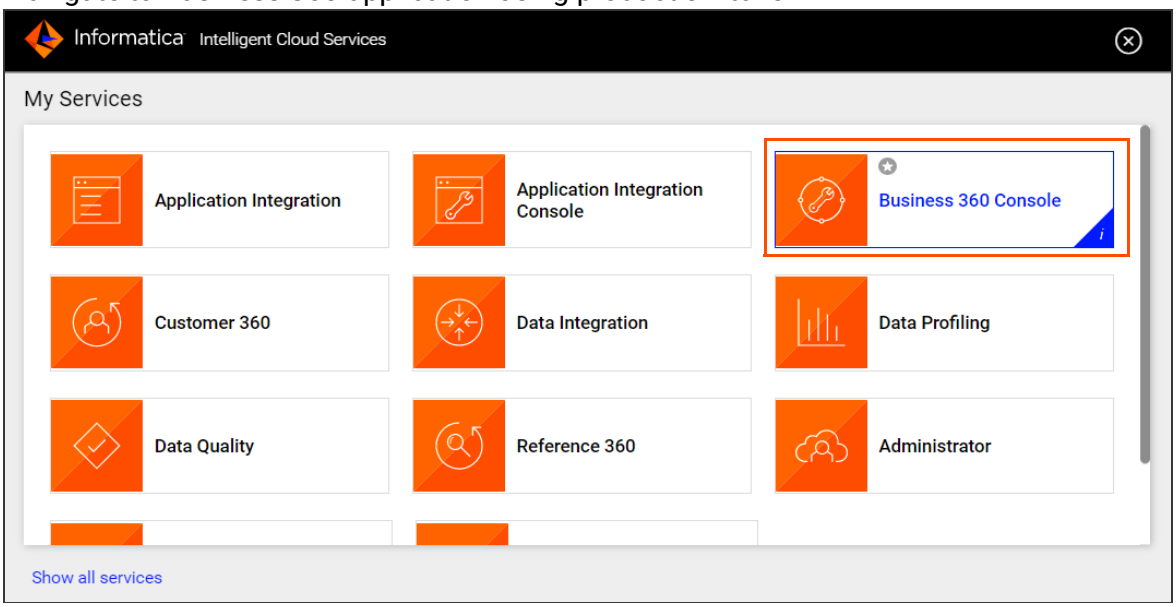
    But if you notice the **PostalAddress_country** of first two and the last two records they belong to two different countries. First two David Borean records belong to the country **Canada** and the last two David Borean records belong to the country **Australia**. After running the match job, David Borean from Canada would be matched since you have configured segment matching to be performed on Canada records.

A	B	C	D	E	F	I	T	W
sourcePKe	firstName	middleNai	lastName	fullName	prefixNam	designatic	PostalAddress_	PostalAddress_
pkey1	David		Borean	David Borean		Analyst	Toronto	CA
pkey2	David		Borean	David Borean		Analyst	Toronto	CA
pkey3	John		Wick	John Wick		Analyst	Toronto	CA
pkey4	Michael		Taylor	Michael Taylor		Manager	Sydney	AU
pkey5	David		Borean	David Borean		Manager	Sydney	AU
pkey6	David		Borean	David Borean		Manager	Sydney	AU

Create a Match and Merge Job

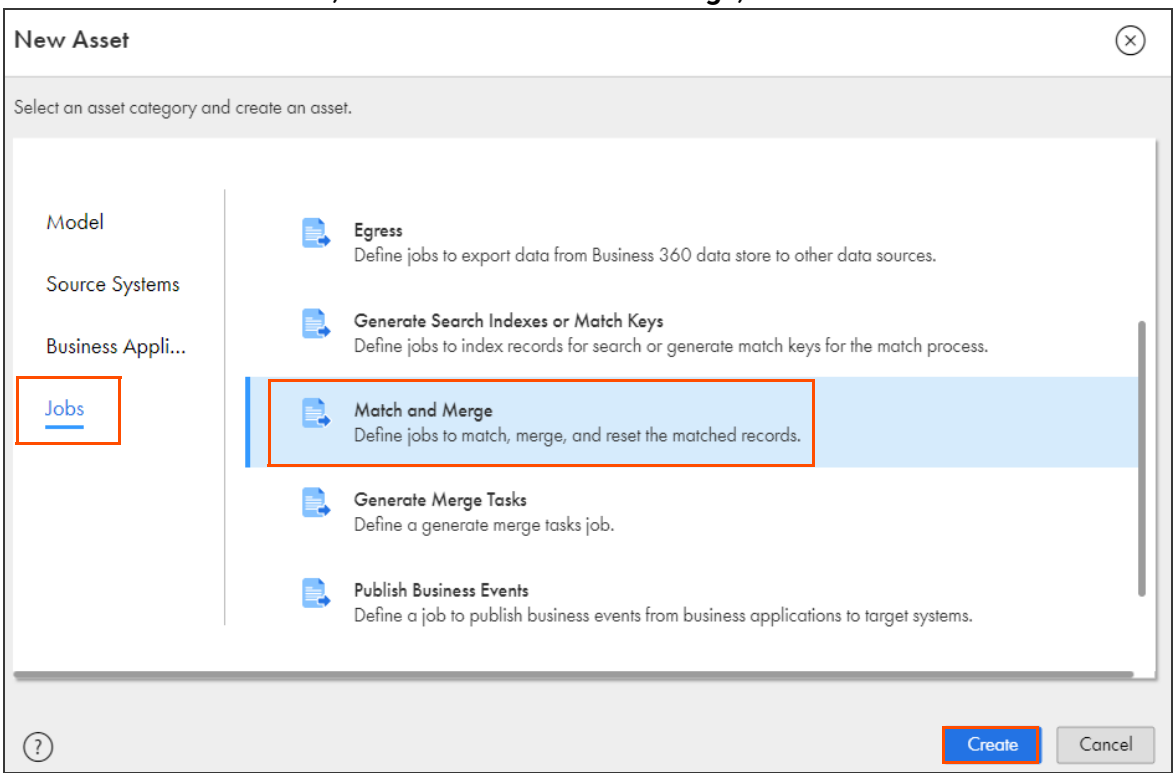
After you configure the match model, you can define a job to match and merge records.

33. Navigate to **Business 360** application using product switcher.

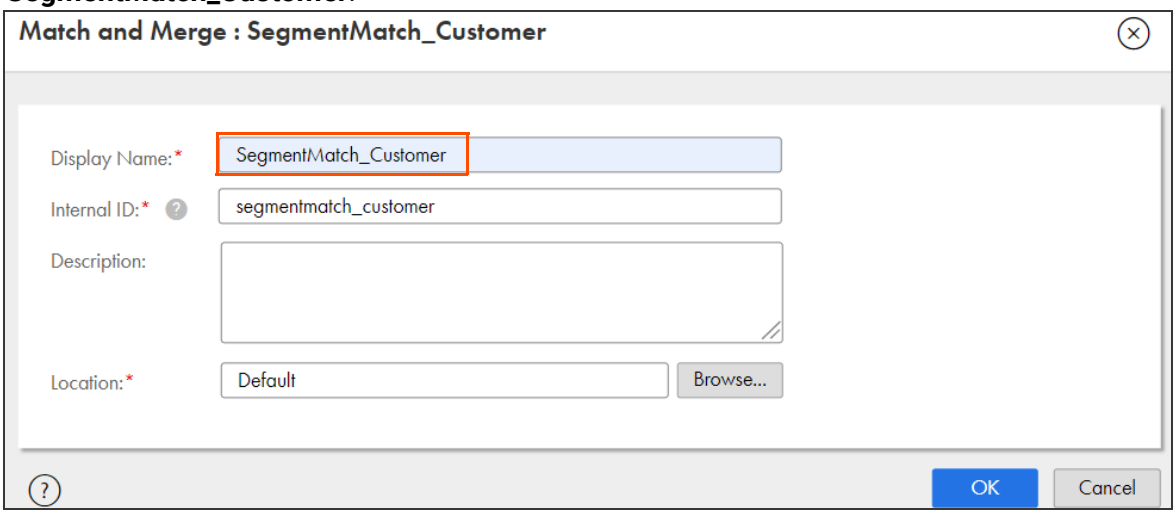


34. Click **New**.

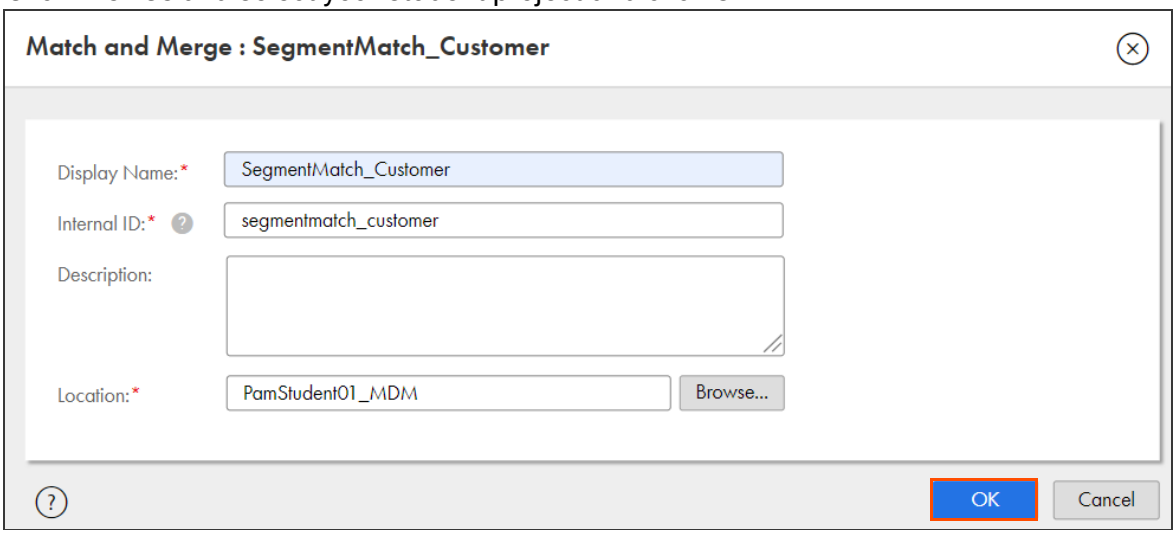
35. In the **New Asset** window, click **Jobs** > **Match and Merge**, and click **Create**.



36. The **Match and Merge** window opens. Enter the Display Name as **SegmentMatch_Customer**.



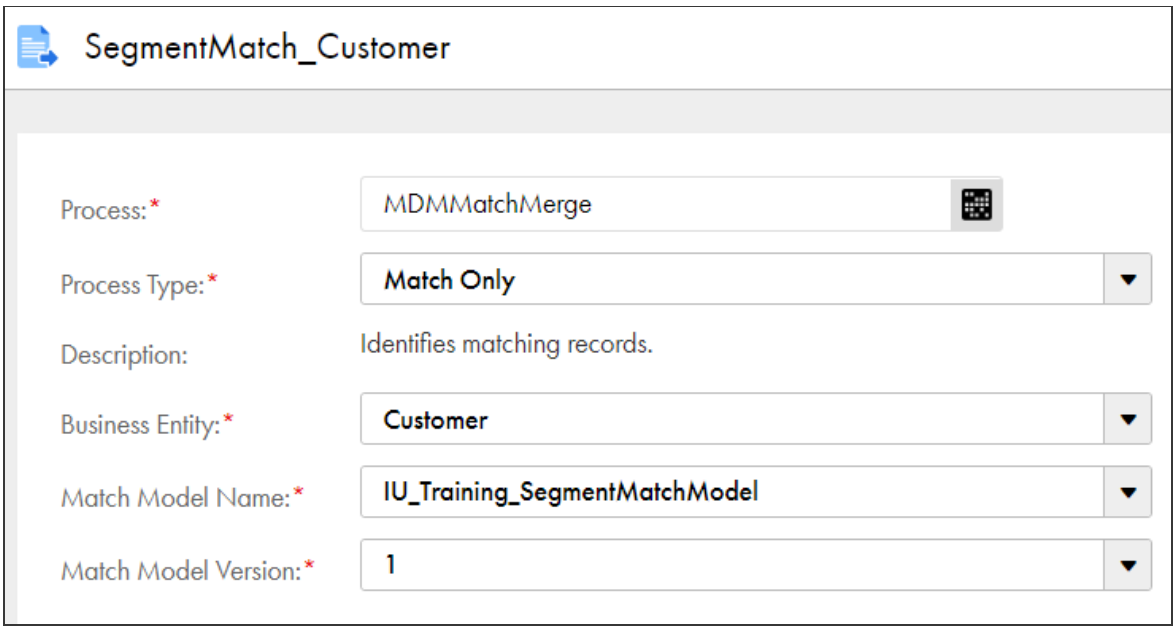
37. Click **Browse** and select your student project and click **OK**.



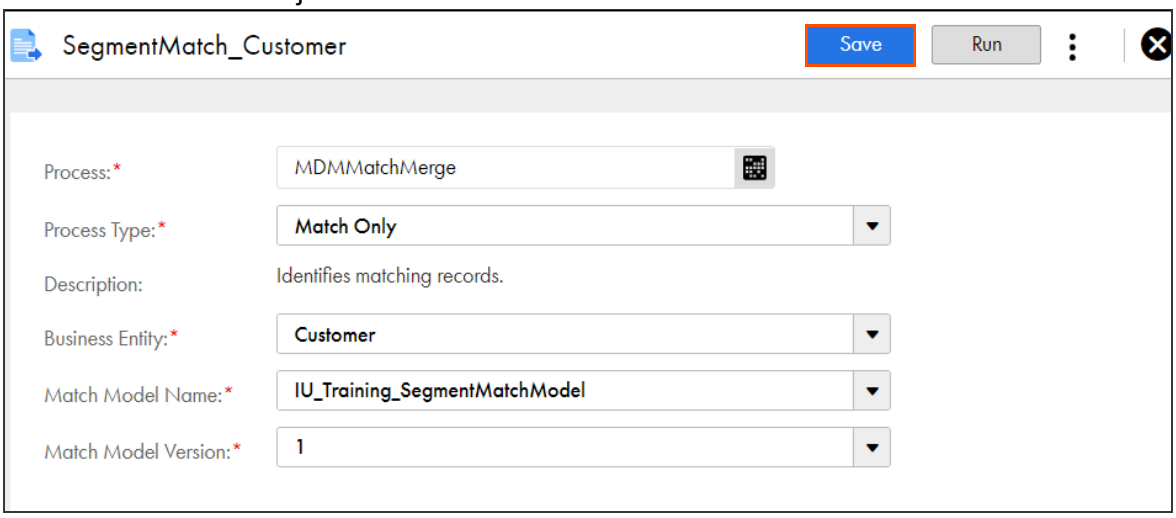
38. In the **SegmentMatch_Person** page, select the following values:

Field	Value
Process	MDMMatchMerge
Process Type	Match Only
Business Entity	Customer
Match Model Name	IU_Training_SegmentMatchModel
Match Model Version	1

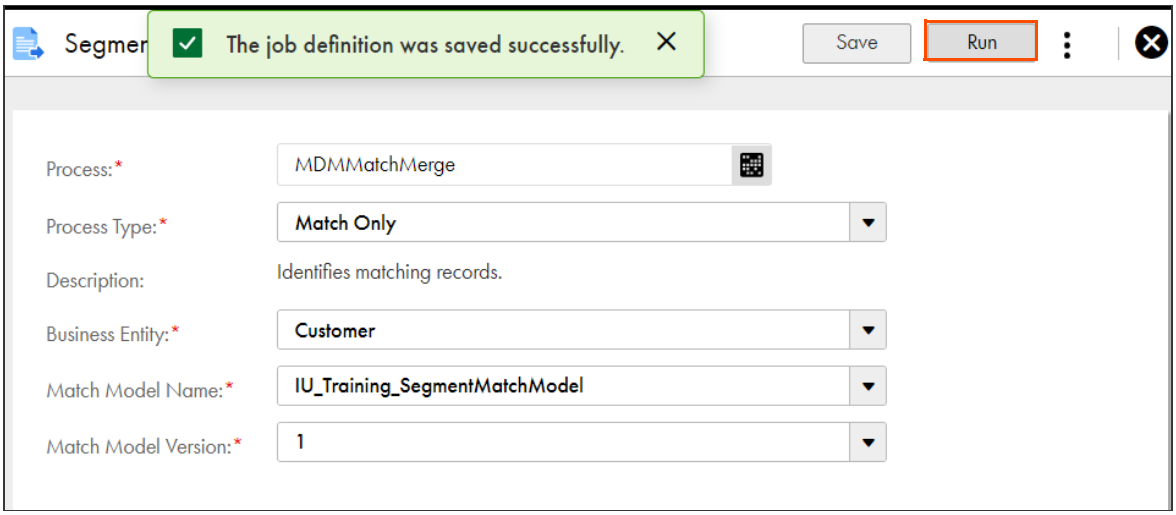
There is no version 2 of this segment match model yet. Hence, it displays only 1.



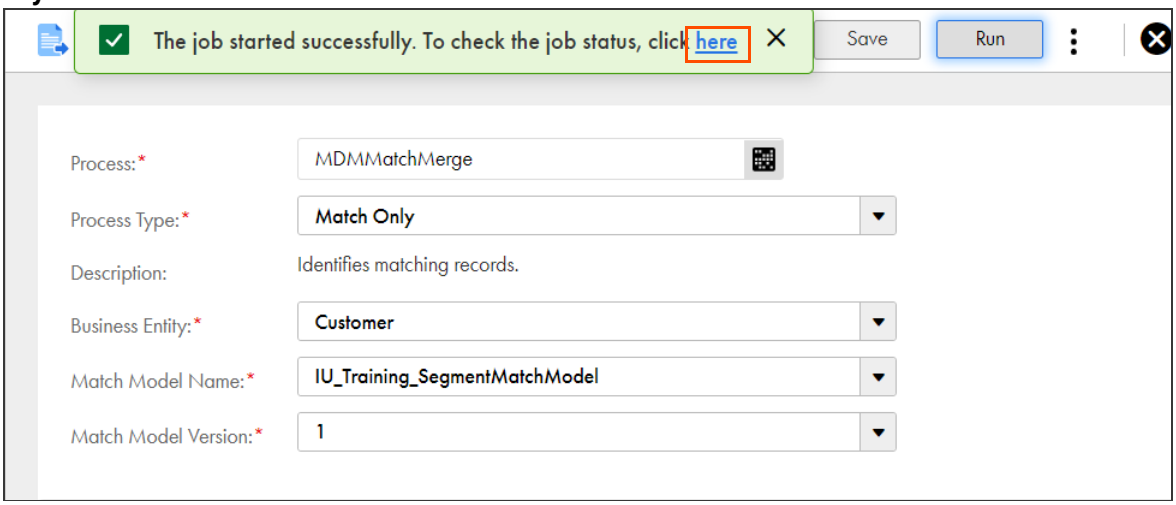
39. Click **Save** to save the job definition.



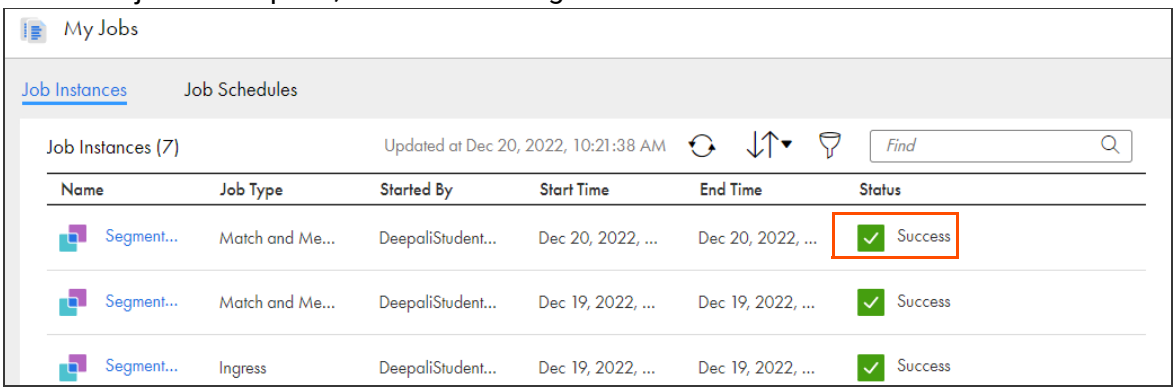
40. Click **Run**.



41. Monitor the status of the job by clicking **here** link that appears in the message box or click **My Jobs** from the menu.



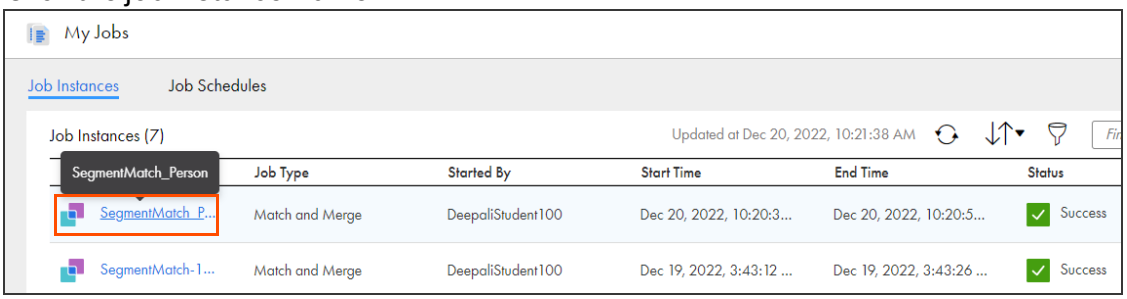
When the job is complete, the status changes to “Success”.



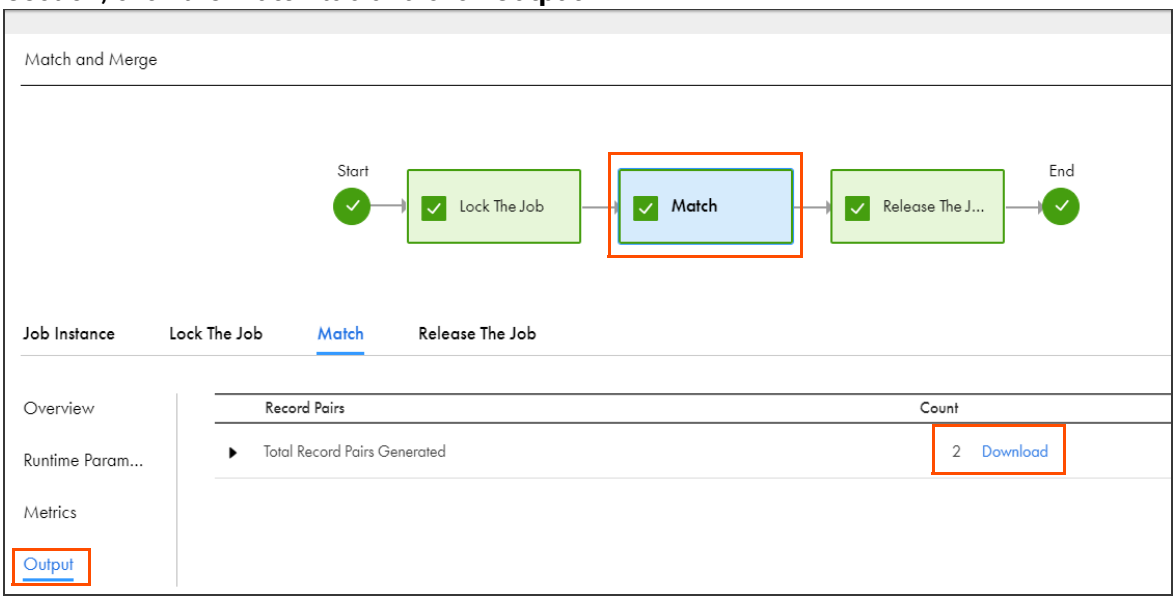
Monitor and Analyze Match

Now that you have run the match and merge job, you will now monitor and analyze the match.

42. Click the job instance Name.

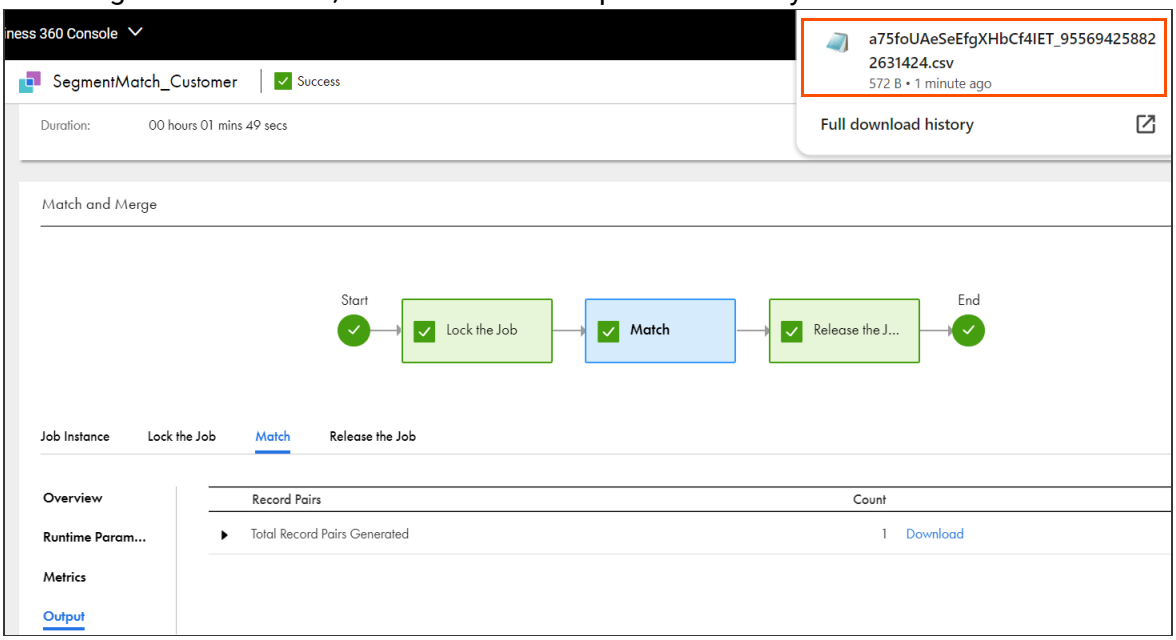


43. Wait for the job to complete to see Metrics under the Match tab. In the Match Merge section, click the **Match** tab and click **Output**.

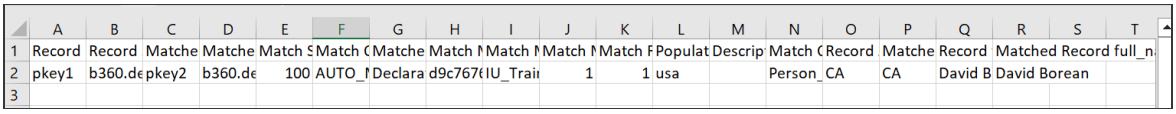


Notice the count under the Count of section and click download to download and analyze the result of the segment merge.

44. The file gets downloaded, click on the file to open it and analyze the results.



Notice the records that are considered for merge.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Record	Record	Matche	Matche	Match S	Match C	Matche	Match M	Match M	Match M	Match F	Populat	Descrip	Match C	Record	Matche	Record	Matched Record		full_n
2	pkey1	b360.de	pkey2	b360.de	100	AUTO_M	Declara	d9c767(	IU_Trai	1	1	usa		Person_	CA	CA	David B	David Borean		
3																				

Recall the rule: While creating the Declarative Rules in the segment match model, in the **Configure Exact Match Field Properties** you have selected Canada as the Segment Values. The segment value identifies the segment data for matching.

With this, a subset of data based on country values is created. Since you have specified the segment value as Canada, the rule uses the records that have Canada as a country value for matching. Hence full name **David Borean** with **Canada** as the country value is considered for matching and not the other two records that have Australia as the country value.

This concludes the lab.

Key Takeaways

- You can configure the match and merge process to identify and resolve duplicate records.
- To configure the match and merge process, you need to analyze the characteristics of your data and the specific match and merge requirements.
- You can configure declarative match rules to determine the records that are duplicates and to merge them.

- A match strategy indicates the type of match you want to perform.
- Merge threshold is applicable to declarative rules with the fuzzy match strategy.
- To use a match model for the match and merge process, publish the model after you complete configuring it.

Module 11: Defining Jobs and Business Events

Define a Match and Merge Job

After you configure the match model, you can define a job to match records and another job to merge records or a single job to match and merge records. The match and merge job depends on the match and merge process. When you run a match and merge job, it completes the match and merge process. A process is a Cloud Application Integration asset that is a set of instructions to automate a business requirement.

Run a Match and Merge Job

When you run a match and merge job, the match and merge process searches for match candidates by applying the match model to the data and identifies duplicate records. Run a match and merge job after you complete running the ingress jobs to load all your data. If you run a match and merge job after every ingress job, data that has already undergone matching goes through the match process again. After you run match and merge jobs, you can monitor the status of the jobs in the My Jobs page.

To match and merge duplicate records in the Business 360 data store, define, and run a match and merge job by following the steps below:

1. Click **New Jobs** > **Match and Merge** > **Create**.
   The Match and Merge dialog box appears.
2. Specify the job properties, and click **OK**.
   The job page appears.
3. Select a process. The supported process is MDMMatchMerge.
4. Select one of the following process types:
   - Match and Merge: Identifies matching records and automatically merges them.
   - Match Only: Identifies matching records.
   - Merge Only: Merges matching records that are identified by the match process.
   - Reset the Matched Records: Resets the matched records that are not yet merged and makes the records available for matching again.
5. Select the business entity.
6. If you selected the Match and Merge or Match Only process, select a match model and the model version to use for the job.
7. Save the job.
8. To run the job, click **Run**.
   You can monitor the status of the job in the My Jobs page.

Define Generate Merge Tasks Job

The match process identifies pairs or records that require manual review. Use the generate merge tasks job to group the pairs or records and generate review tasks for groups that require manual review. Users then review the tasks and manually merge the appropriate records.
After you run a generate merge tasks job, you can monitor the status of the job on the My Jobs page.

Run a Generate Merge Tasks Job

To generate merge tasks for pairs identified by the match process and require a review, define and run a generate merge tasks job. **For example**, you might want to generate merge tasks for pairs identified for the Customer business entity records.

1. Click **Jobs** > **Generate Merge Tasks** > **Create**.
   The Generate Merge Tasks dialog box appears.
2. Specify the job properties, and click **OK**.
   The generate merge tasks job page appears.
3. Select a process. The supported process is MDMGenerateMergeTasks.
4. Optionally, modify the maximum number of tasks created.
5. In the Assets section, click **Add**.
6. In the asset window, select the asset types that you want to run the job on, and click **Select**.
7. Save the job.
8. To run the job, click **Run**.
   You can monitor the status of the job on the My Jobs page.

User triggered vs. System-generated events

When you define a user-triggered event or a system-generated event, you define the details of the event and the workflow that the event triggers. Decide on the user roles, event actions, and assets for which you want to trigger a workflow. Then define the event, workflow tasks, and workflow properties.

You can configure predefined one-step, two-step, or four-step approval workflows. One-step approval workflows require one approver, whereas two-step approval workflows require two approvers. For two-step approval workflows, if the first approver approves the changes, then the second approver can approve or reject the changes. If the first approver rejects the changes, the changes are not sent to the second reviewer and are not published. Four-step approval workflows require four approvers.

If you do not want to use any predefined workflows, then use additional workflows that you create in Cloud Application Integration as a process. Ensure that the process contains at least one Service step with its service action property set to Generic User Task or Originator User Task.

Business Events

A business event can trigger a workflow that consists of multiple tasks linked together. You can define business events related to actions on business entity records, such as create, update, delete, and merge, to trigger a workflow or publish the events to external targets.

To ensure that business managers review and approve the data before it becomes master data, define user-triggered and system-generated events. These events trigger workflows that consist of multiple tasks linked together. A user-triggered event includes actions that users perform, such as creating, updating, and unmerging records, or creating and updating hierarchies. System-generated events are actions that the system performs, such as the system identifying matches that require manual review. To publish events from the business applications to external targets, define publishing events.

User-triggered Events

User-triggered event allows the user to create event scenarios and the user can choose to create, update, or unmerge records. You can create a user-triggered event and define the details of the event that triggers a workflow. **For example**, you might want a workflow triggered when business users create or update a business entity record or hierarchy. You can trigger a workflow for changes related to specific fields and field groups.

Steps to create a User-Triggered Event

1. Click **Business Events**.
2. Click **Add Event** > **User-Triggered Event**.
3. Enter a name for the event.
4. In the **Workflow Trigger** field, select whether to initiate a workflow after the triggering actions are performed.
5. From the **Affected Users Roles** list, select the roles of the users who perform the triggering actions.
6. In the **Event Scenarios** section, click **Add** Event Scenario, and then configure an event scenario.

System-generated Events

System generated triggers the event with only one action – Resolve Potential Matches.
Steps to create a System-Generated Event

1. Click **Business Events**.
2. Click **Add Event** > **System-Generated Event**.
3. Enter a name for the event.
4. In the **Event Scenarios** section, click **Add Event Scenario**, and then configure an event scenario.
5. Click the asset picker icon, and then select the asset to trigger a workflow.
6. Select the only action available – Resolve Potential Matches.
7. Click **Next**.

When you define a user-triggered event or a system-generated event, you define the details of the event and the workflow that the event triggers.

Configure Workflow Tasks

After you create an event, select the workflow that you want the event to trigger and the user roles responsible for each task in the workflow. **For example**, you might want to trigger a two-step-approval workflow where data stewards and then managers must approve the changes.

1. On the Workflow tasks page, in the Name list, select MDM One Step Approval/MDM Two

Step Approval.

2. In the Approvers list, select the user roles responsible for the workflow task.
3. Click Next.
4. On the **Workflow Properties** page, in the Title field, enter the title format for the workflow tasks.
5. From the **Priority** list, select a priority for the task.
6. In the **Due In** field, enter the number of days from the day the task is created from which the task is due.
7. Click **Save**.

Approving Workflow

1. Login as an **approver**.
2. In the Multidomain MDM, click **Workflow Inbox**.
3. Click the **Task Title**.
   You can view the Matching records.
4. Click **Claim**.
5. Click **Approve** / **Stop** / **Release** / **Reject**.
   You will get a message saying that Task action is completed.

Unmerge Records

At any point if the user realizes that the records were incorrectly merged, the user can unmerge the records. To perform the merge, follow the steps below:

1. In Multidomain MDM, click **Search**.
2. Select the Customer entity and enter search string with the name of the record that is merged.
3. In the entity page, click **Source Records**.
4. Click the record and click the **Unmerge** button.

Custom Workflow

Custom workflows are useful when you want to customize the workflows as per the requirement. You can create Sequential as well as Parallel tasks. When you add declarative rules to a match model, you can enable segment matching on the exact match fields. You can see the process running on the CAI console. Task actions and task labels can be customized as per the requirement.

Steps to use custom workflow:

1. Login to Business 360 Console and navigate to **Explore** > **All Projects** > **Business360**
2. Select and clone the MDM Multi-Step Approval Template to your working folder.
3. Open the process template in CAI and customize it as your requirement.
4. Save and Publish the customized template.
5. When defining workflow properties of a business event configuration process, you can import the published CAI process.

Lab 11-1: Running a Match and Merge Job and Executing Business Events

Overview:

The system automatically merges records if the declarative rule satisfies the criteria for auto merge. Else, if the rule is configured for a manual merge, it must go through a workflow, and the MDM Manager or a MDM Steward will have to approve the merge. This requires a business event to be created. The Business event will trigger a workflow and auto merges can be performed.

After you have merged the records, and then realize that the records were either merged incorrectly or should not have been merged, then you can unmerge the records as well.

Objectives:

- Run a match job
- Create a business event
- Execute the workflow
- Observe merge results
- Unmerge records

Duration:

45 minutes

Scenario:

Mark checks the source records randomly and identifies some duplicate records. It would not be possible to manually resolve duplicates, when the number of records are in 100s and 1000s. Hence, Mark uses Multidomain MDM Match rules and Merge strategies to identify and then merge duplicate records. As survivorships are configured, the match and merge algorithms of Multidomain MDM enables resolution of duplicates and merge records of best truth.

MDM Managers and MDM Stewards will approve the merges manually too.

Tasks:

Perform the following tasks to remove duplicate records by creating match and merge job:

- Analyze the onboarded data
- Create a match and merge job
- Monitor and analyze match
- Check source data
- Create a business event for a workflow task
- Perform manual merge
- Approve the workflow
- Unmerge records

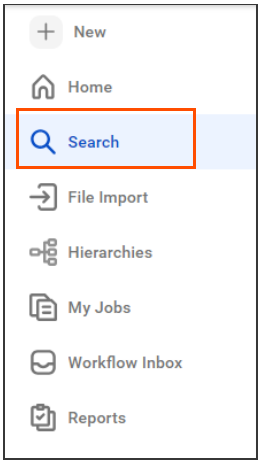
Analyze the Onboarded Data

Notice that the declarative rules created in the previous lab are published. As soon as they are published, and the Match job is created, auto-match and merge take place as rule 1 is automated. Manual merge needs a review process, which you will execute in the next exercise.

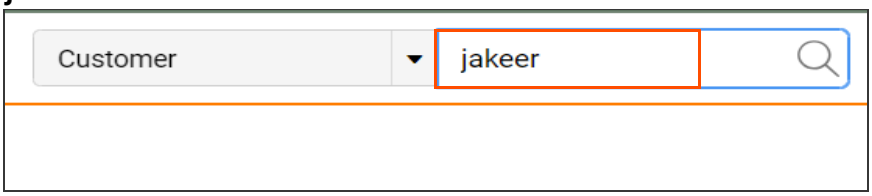
1. Before publishing the rules, it is good to analyze the data that is onboarded to Multidomain MDM so that you can understand the matched and the merged records, once executed.

2. Open **Multidomain MDM** in a new tab.

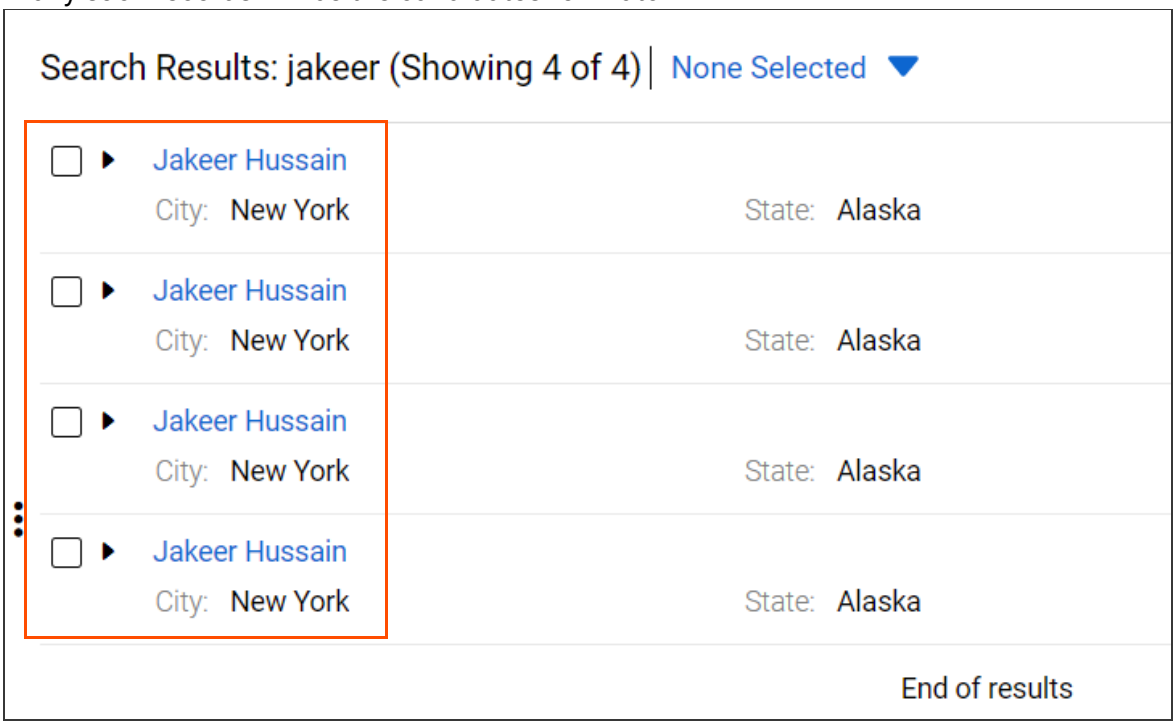
3. Click **Search** from the menu.



4. In the search box, select **Customer** from the drop down and enter the search string as **jakeer**.

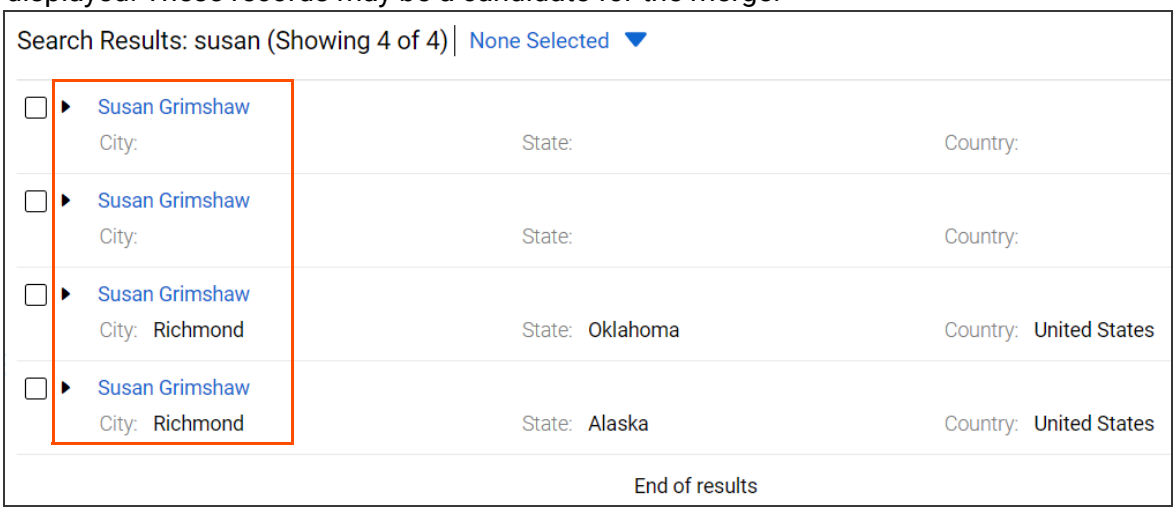


5. The search results are displayed. In the list you will notice matches for the record **Jakeer Hussain**. Observe the Full Names. All the match rules have been configured for Full names. Many such records will be the candidates for Match.



6. Records that satisfy Match rule 1–Automated, will be auto-matched and merged. The ones that do not qualify the rule 1 match criteria will move to manual merges (rules 2 and 3).

7. Perform another search by entering the search string as **Susan** and Notice the records displayed. These records may be a candidate for the merge.



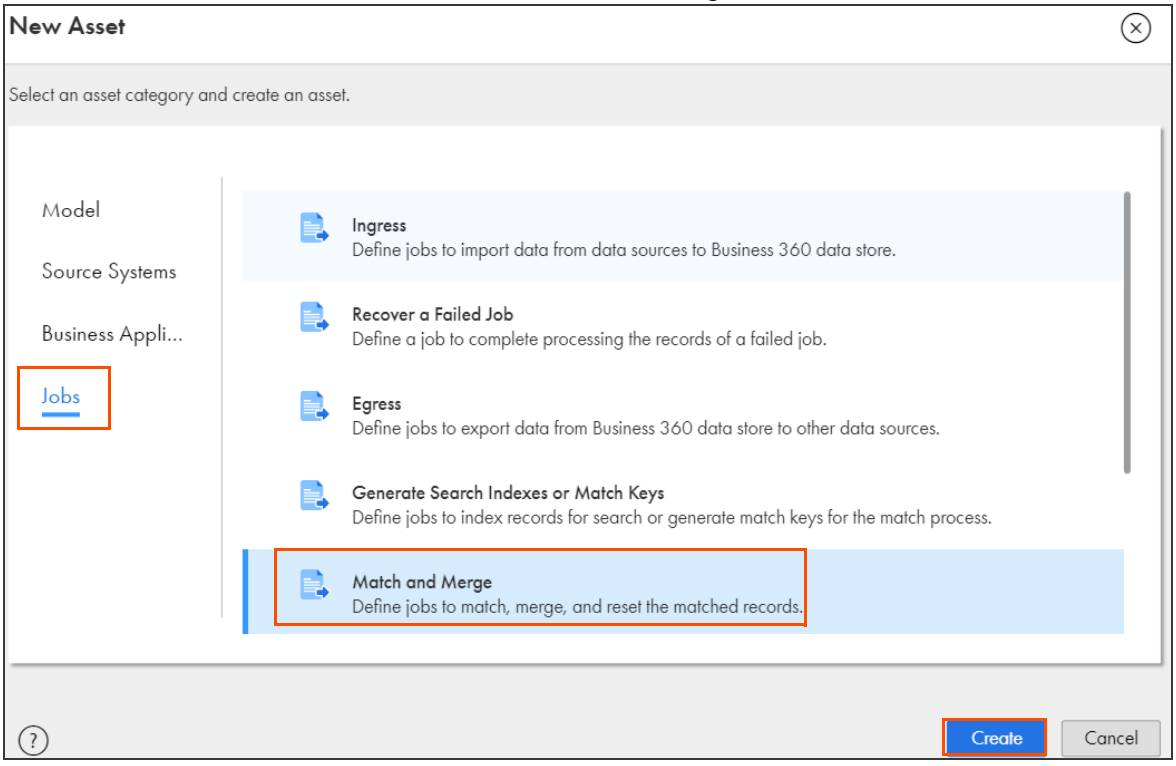
Create a Match and Merge Job

To match and merge duplicate records in the Business 360 data store, define and run a match and merge job.

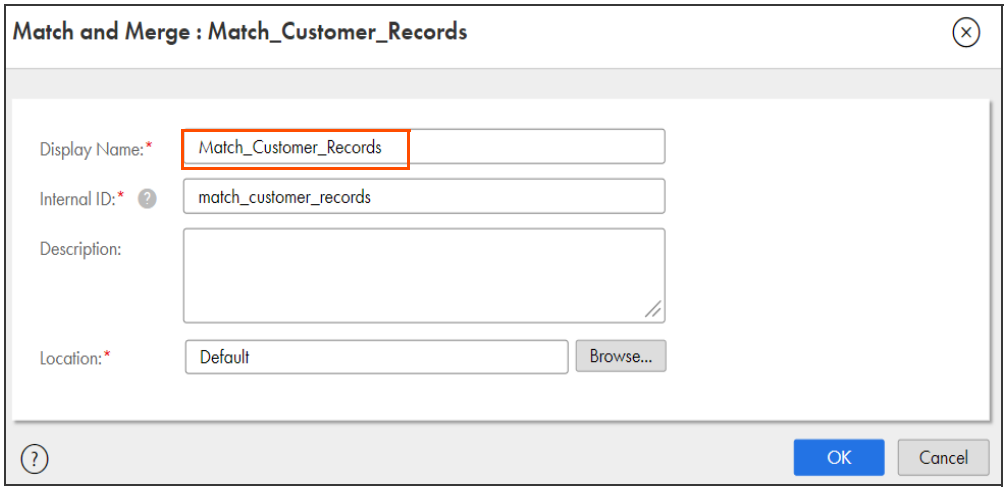
8. Navigate back to **Business 360 Console**.

9. Click **New**.

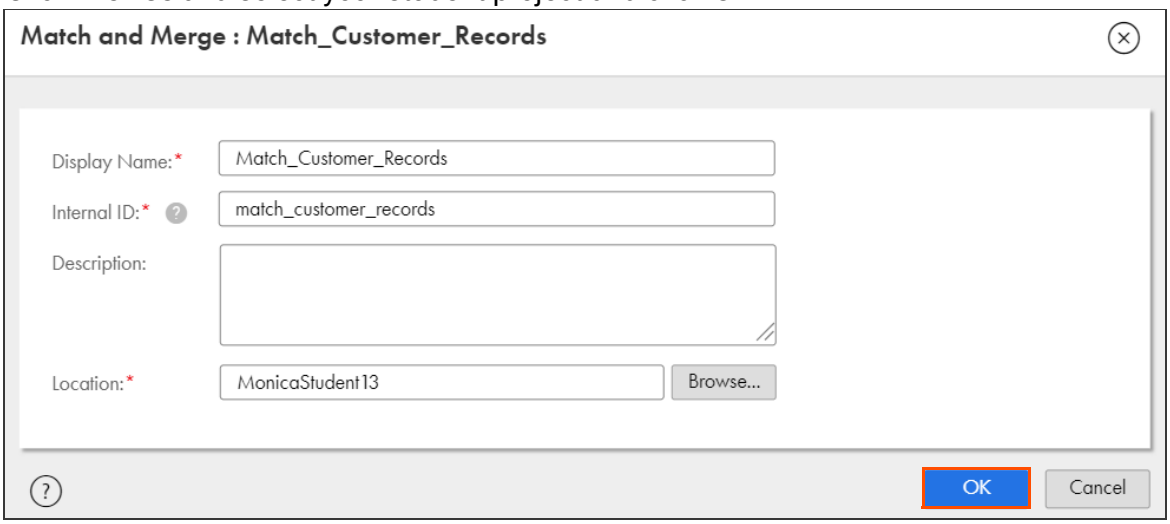
10. In the **New Asset** window, click **Jobs** > **Match and Merge**, and click **Create**.



11. The **Match and Merge** window opens. Enter the Display Name as **Match_Customer_Records**.



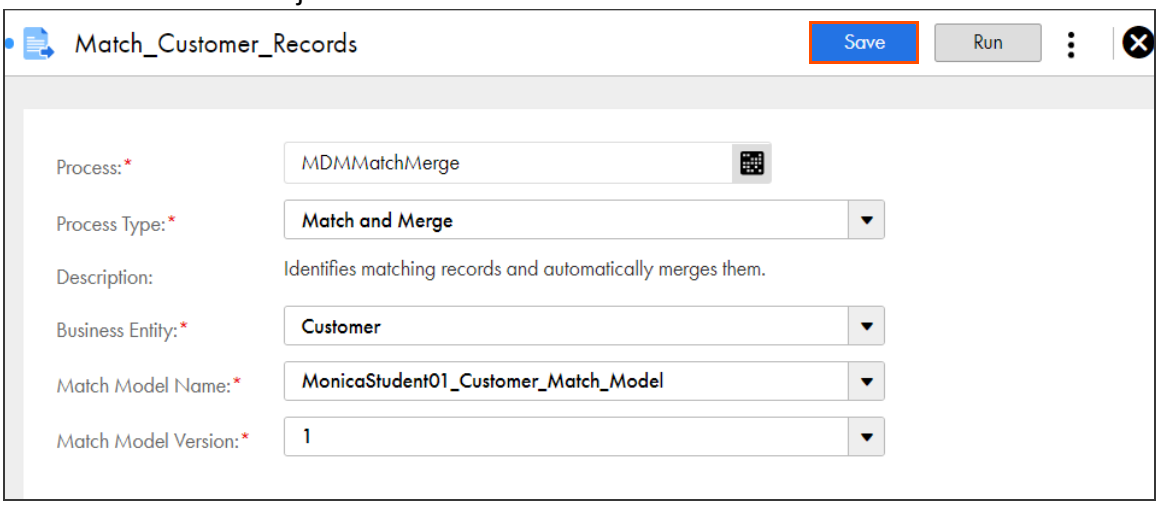
12. Click **Browse** and select your student project and click **OK**.



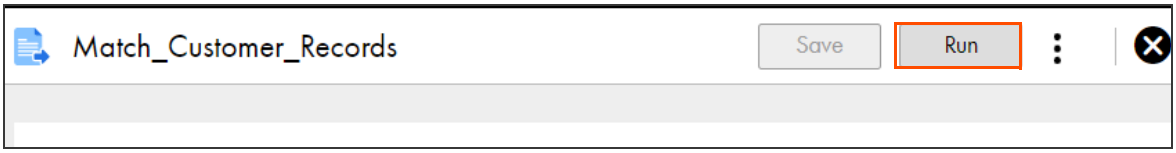
13. In the **Match_Customer_Records** page, select the following values:

Field	Value
Process	MDMMatchMerge
Process Type	Match and Merge
Business Entity	Customer
Match Model Name	StudentNameStudentNumber_Customer_Match_model
Match Model Version	1

14. Click **Save** to save the job definition.

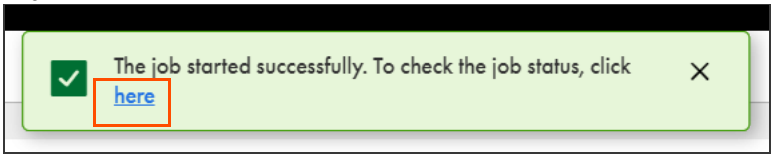


15. Click **Run**.

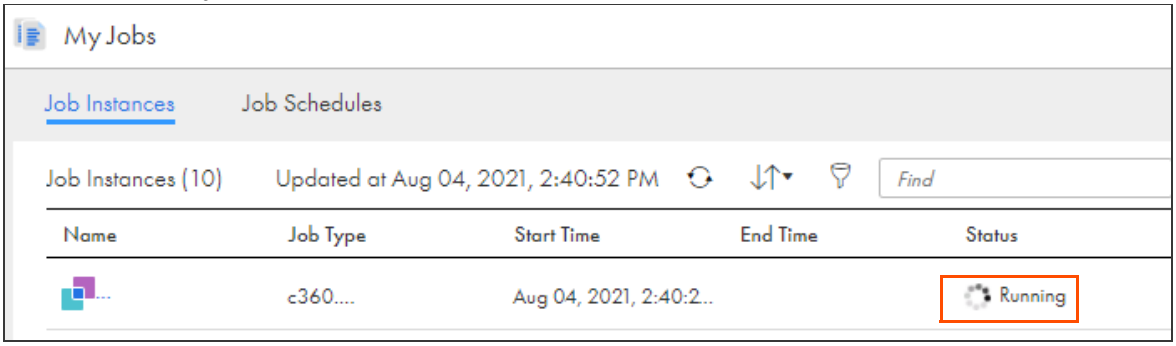


DO NOT click **Run** twice. Accidentally if you click Run twice, the match job will run twice. You can view 2 match jobs running in **My Jobs**. The first job retrieves matched records. The second job will not have any matched records. It displays 0 records, and hence the subsequent manual merges will not generate any tasks. As soon as you click Save, click Run and monitor the job as explained in the steps below.

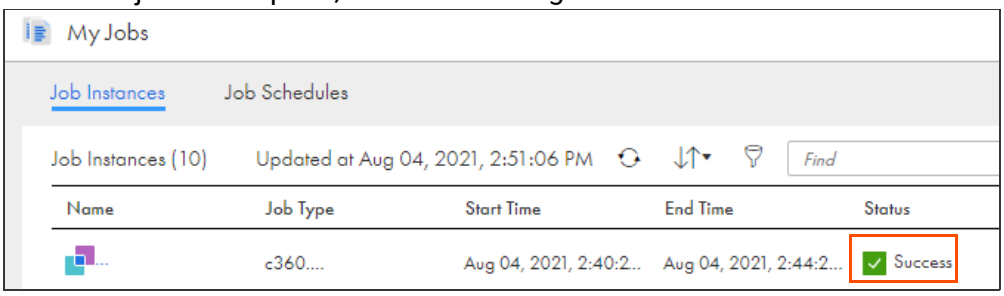
16. Monitor the status of the job by clicking **here** link that appears in the message box or click **My Jobs** from the menu.



17. Notice that the job is in the **Running** state.



18. When the job is complete, the status changes to “Success”.

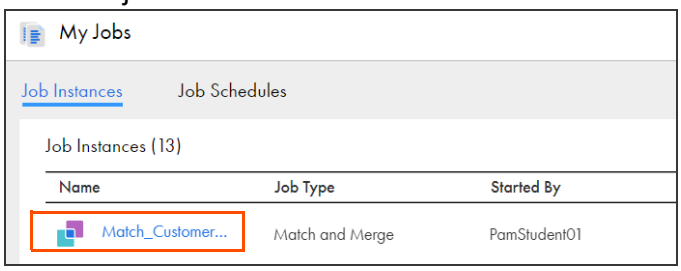


Wait until the job status changes to Success. It will take 5 - 10 min for the job to get completed.

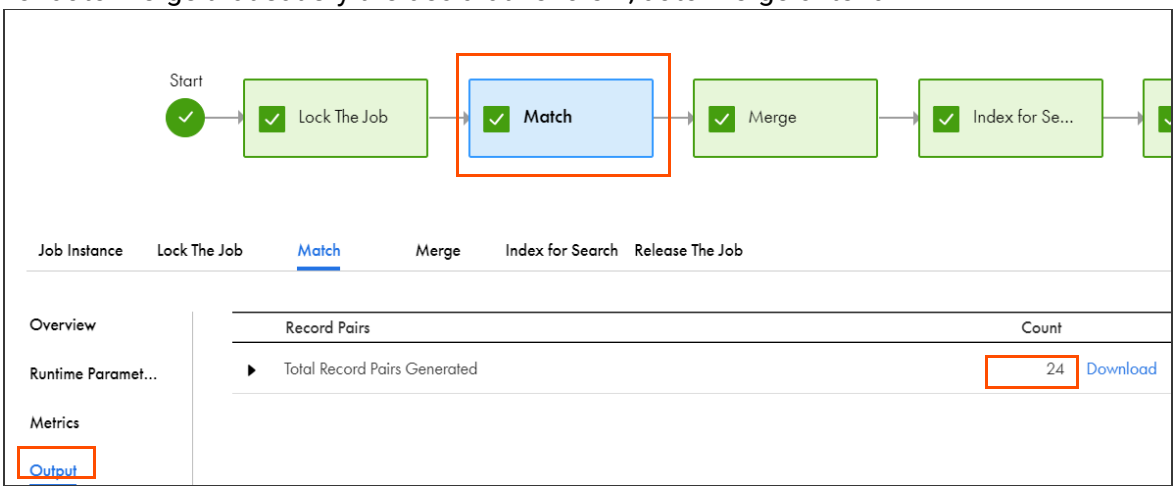
Monitor and Analyze Match

After executing the match and merge job, you should examine the results to determine if any pairs have been created.

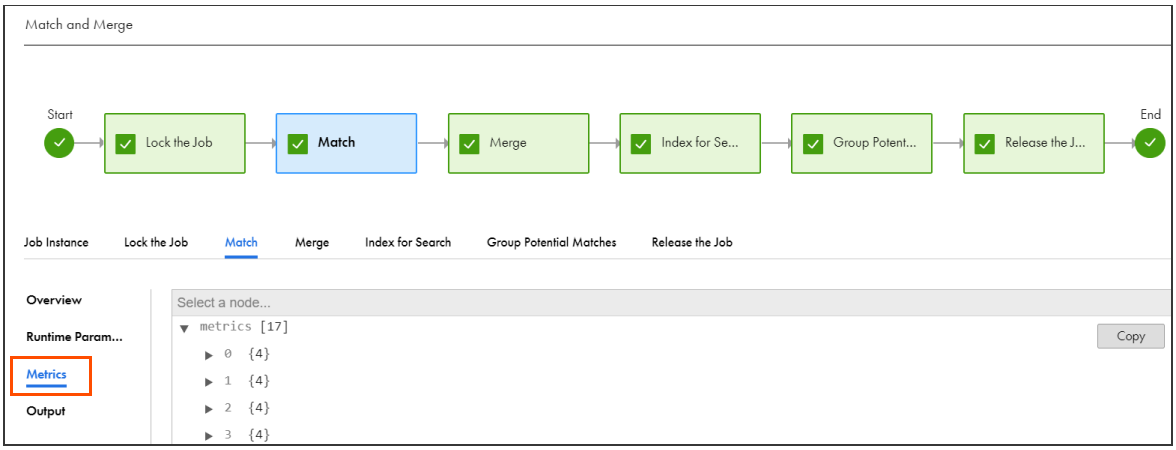
19. Click the job instance Name.



20. Wait for the job to complete to see Metrics under the Match tab. In the Match Merge section, click the **Match** tab. In the **Output** section, you can see that **Total Record Pairs Generated** count is 24. This shows that there are records in the application that are eligible for auto merge that satisfy the declarative rule-1, auto merge criteria.



21. Click **Metrics**.



The metrics may vary depending on the number of records you have in the system. On top of the metrics page, you can see that there are no job metrics to display. The numbers may vary if you have onboarded any additional records to Multidomain MDM. It also depends on the way match rules consider the records for execution, and the way they are configured.

Important: Check the value in **MATCH_DECLARATIVE_STEWARD_REVIEW_PAIRS**. If there are no pairs (0), it means that the match job that you just ran is not successful, and there are no records for manual merge. The data provided to Multidomain MDM in this training has records that will be candidates for merge.

Solution: If you do not see proper values in the match metrics, run the ingress job again. To do this, in Business 360 console, click New > Job > Ingress and select the Taskflow that you created in CDI. You can also import the Customer csv file from Multidomain MDM.

Below Table describes the metrics in detail:

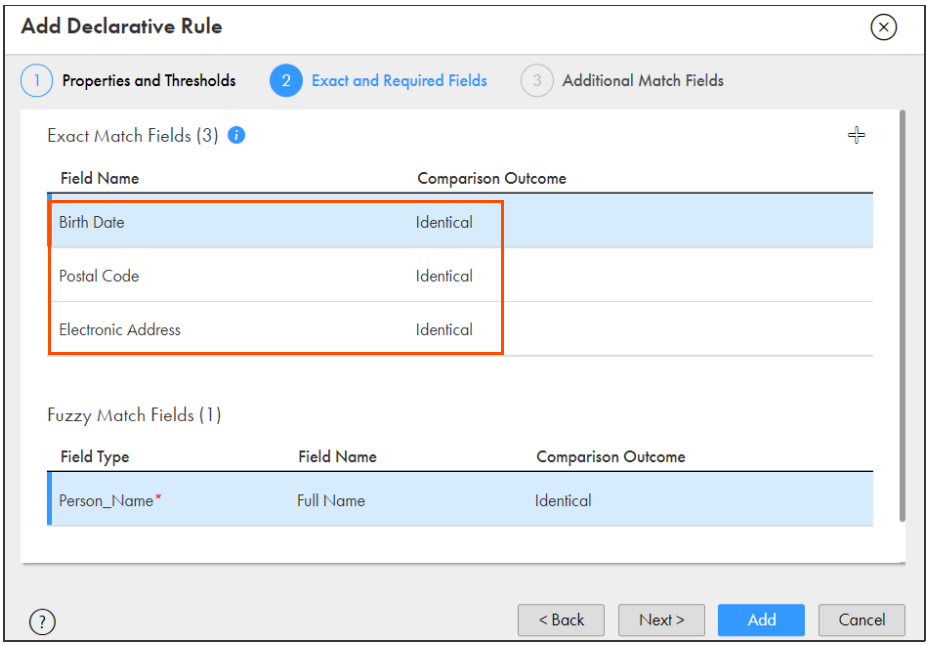
Metric Name	Description
MATCH_TOTAL_RECORDS	Total number of records which were processed
MATCH_FAILED_RECORDS	Total number of records which failed to process
MATCH_DECLARATIVE_NEVER_MERGE_PAIRS	Number of candidate pairs which did not become match pairs from declarative rules evaluation
MATCH_DECLARATIVE_AUTO_MERGE_PAIRS	Number of candidate pairs which were evaluated as match pairs with outcome = auto merge from declarative rule evaluation

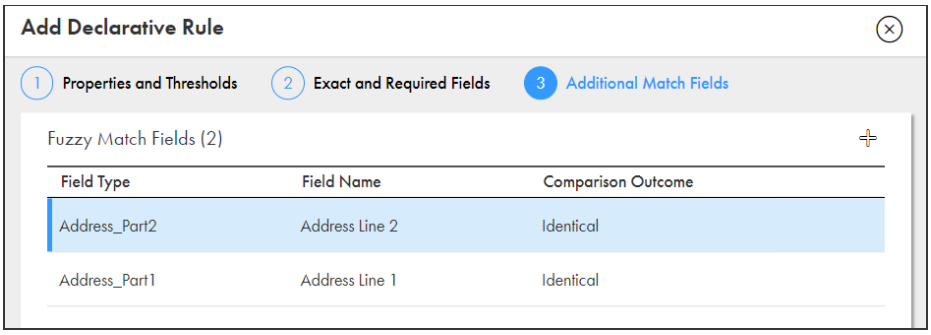
Metric Name	Description
MATCH_DECLARATIVE_STEWARD_REVIEW_PAIRS	Number of candidate pairs which were evaluated as match pairs with outcome = steward review from declarative rule evaluation
MATCH_AI_AUTO_MERGE_PAIRS	Total number of pairs evaluated as Auto Merge by AI trained model
MATCH_AI_NEVER_MERGE_PAIRS	Total number of pairs evaluated as Never Merge by AI trained model
MATCH_AI_MODEL_EVALUATION_PAIRS	Total number of pairs qualified for AI Model evaluation after declarative rule evaluation

MATCH_DECLARATIVE_NEVER_MERGE_PAIRS, MATCH_DECLARATIVE_AUTO_MERGE_PAIRS, MATCH_DECLARATIVE_STEWARD_REVIEW_PAIRS, MATCH_AI_MODEL_EVALUATION_PAIRS contain the total number of candidate pairs evaluated during match.

Rule 1 Recall

- Declarative Rule 1 states that:
  - Full Name, Address Line 1, Address Line 2 should be similar.
  - Birth Date, Postal Code, and Email should be exactly identical.





This seems to be a rigid rule and there few matches and merges.

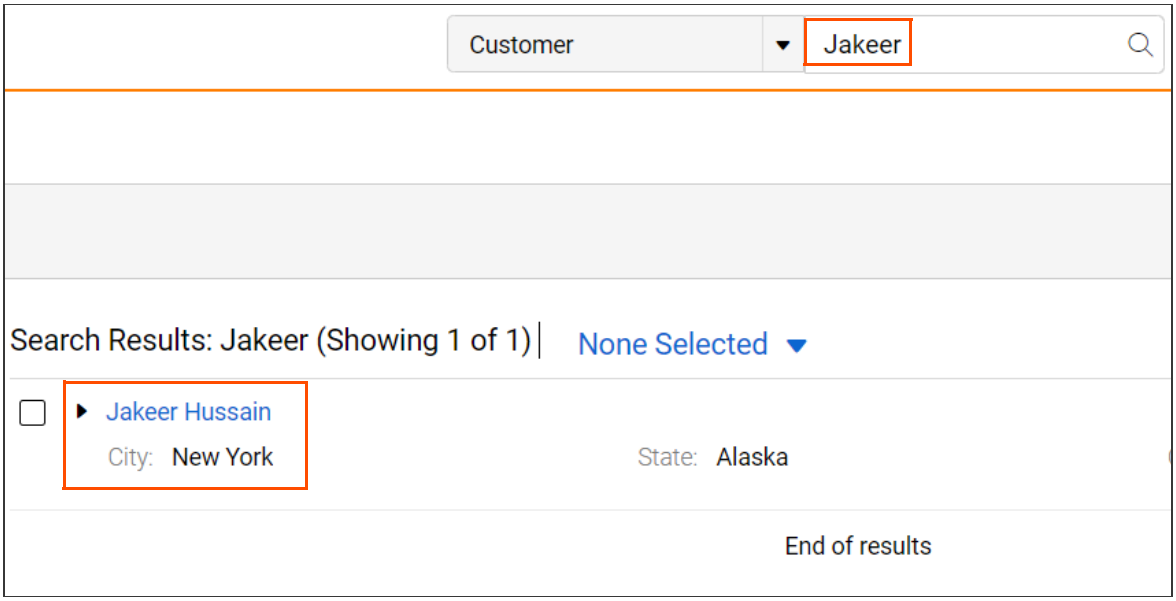
Check Source Data

Now that the match and merge job has been executed, you should verify the source data to confirm whether the records identified as merge candidates were actually merged.

22. Navigate to **Multidomain MDM** and click the **Search** tab.

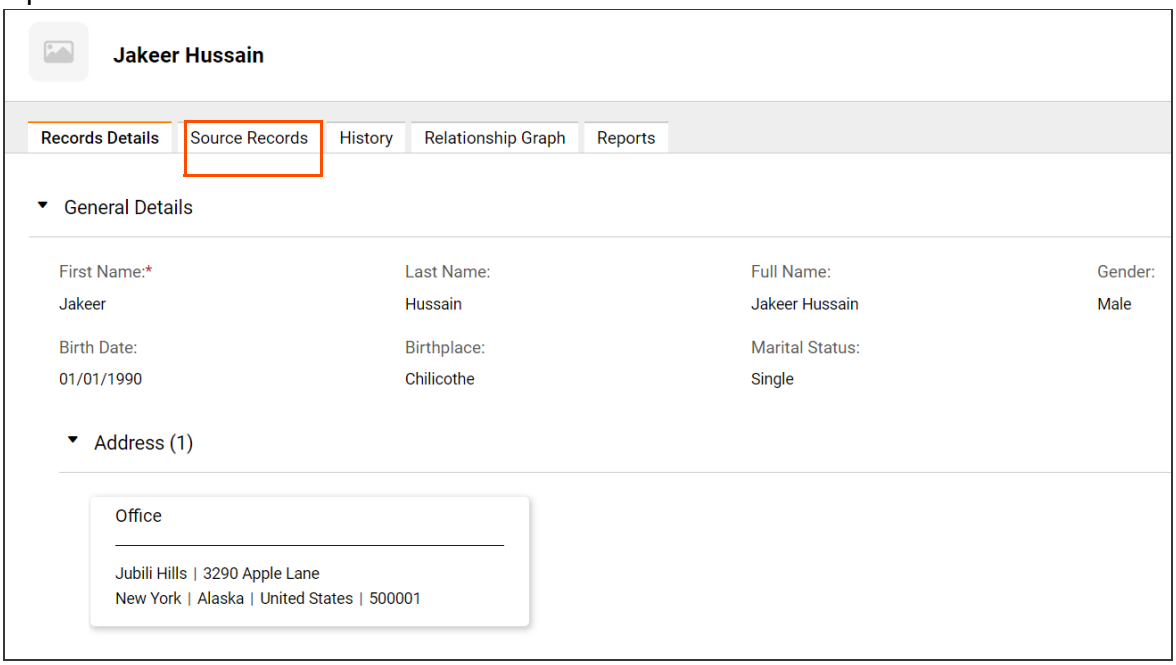
23. Select **Customer** entity and enter the search string as **Jakeer**.

24. Notice that the records have matched, as rule 1 satisfied the requirements.

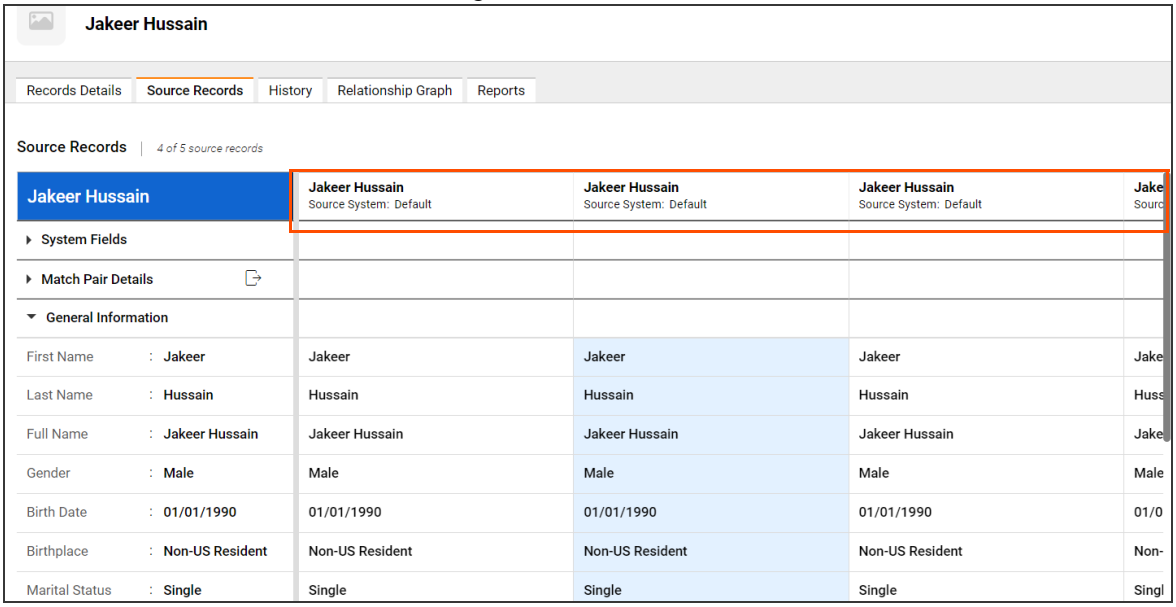


If the records are not merged, and if you still see multiple Jakeer records, then create and run the match and merge job again. Check the data again and confirm the merge.

25. Open the **Jakeer** record. Click the **Source Records** tab.



26. Notice that the four records are merged.



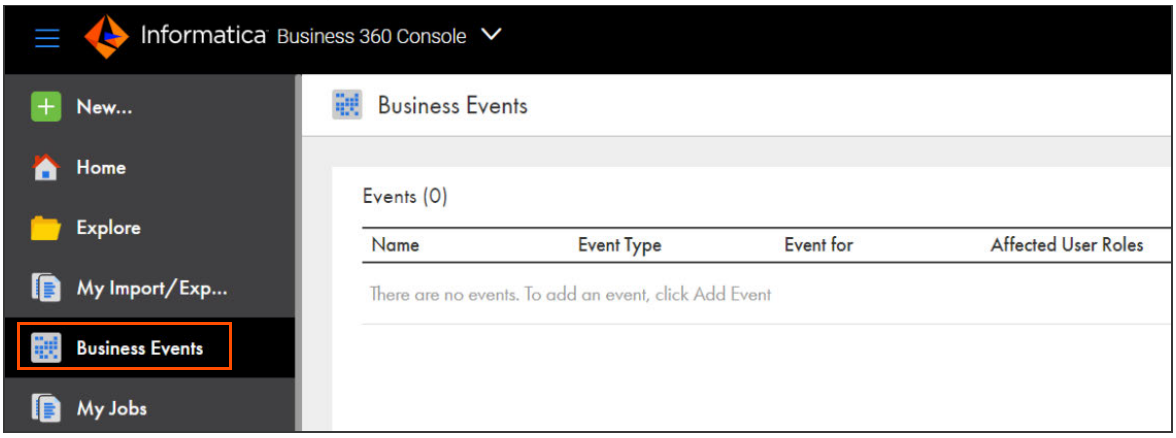
There are some records that might still be the candidate for the merge. As those remaining records does not satisfy the first declarative rule (Auto Match and Merge), we have to create a manual match and merge job to handle those records. This can be handled by creating a business event and sending the task through the workflow task.

Create a Business Event for a Workflow Task

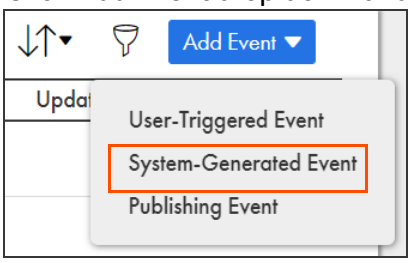
A business event can trigger a workflow that consists of multiple tasks linked together. Use workflows to ensure that business managers review and approve the data before it becomes master data.

In this section, you will create a workflow task for a one-step merge approval. The approvers will be the Multidomain MDM Steward.

27. In Business 360 Console, click **Business Events** from the menu.

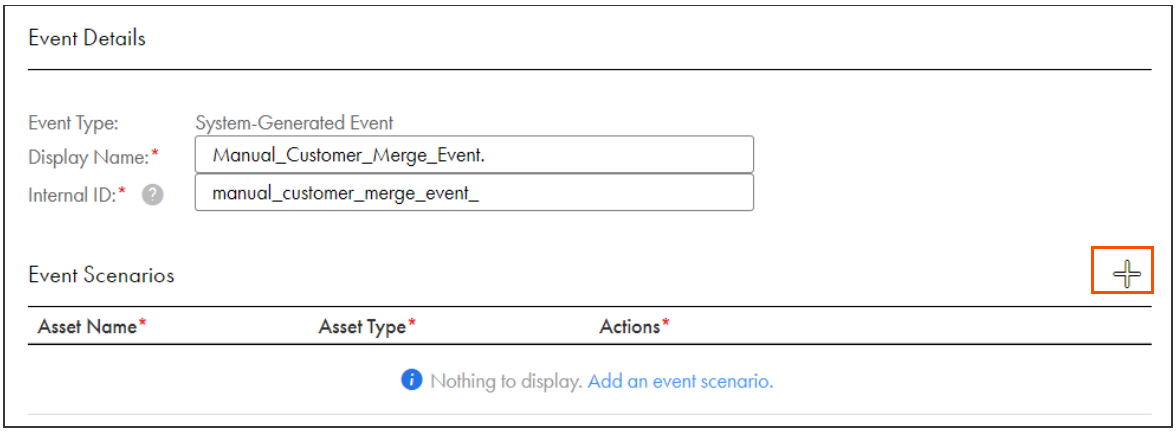


28. Click **Add Event** drop down and select **System-Generated Event**.

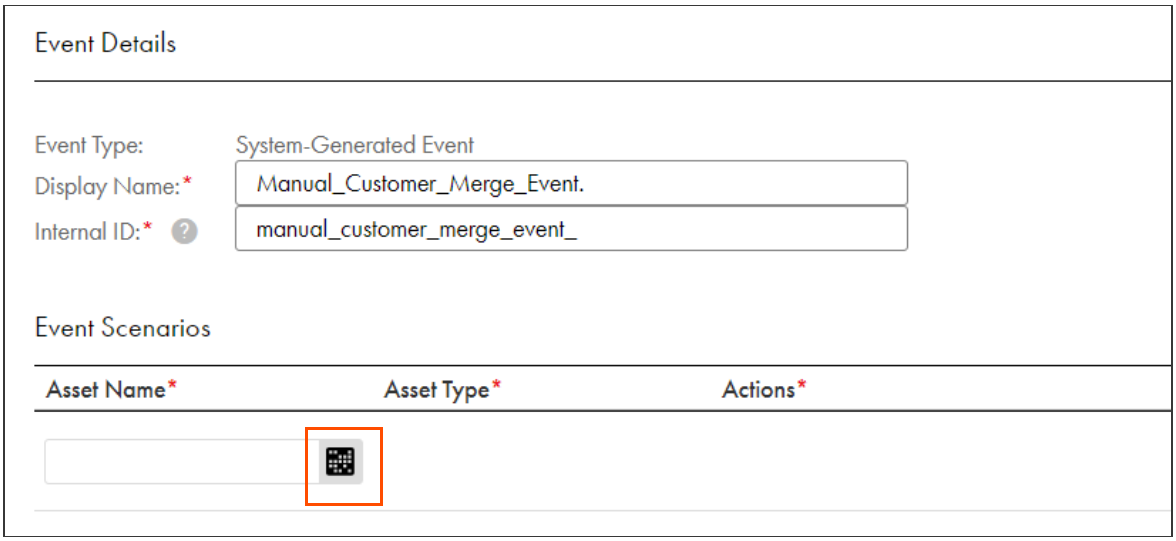


29. Enter the Display Name as **Manual_Customer_Merge_Event**.

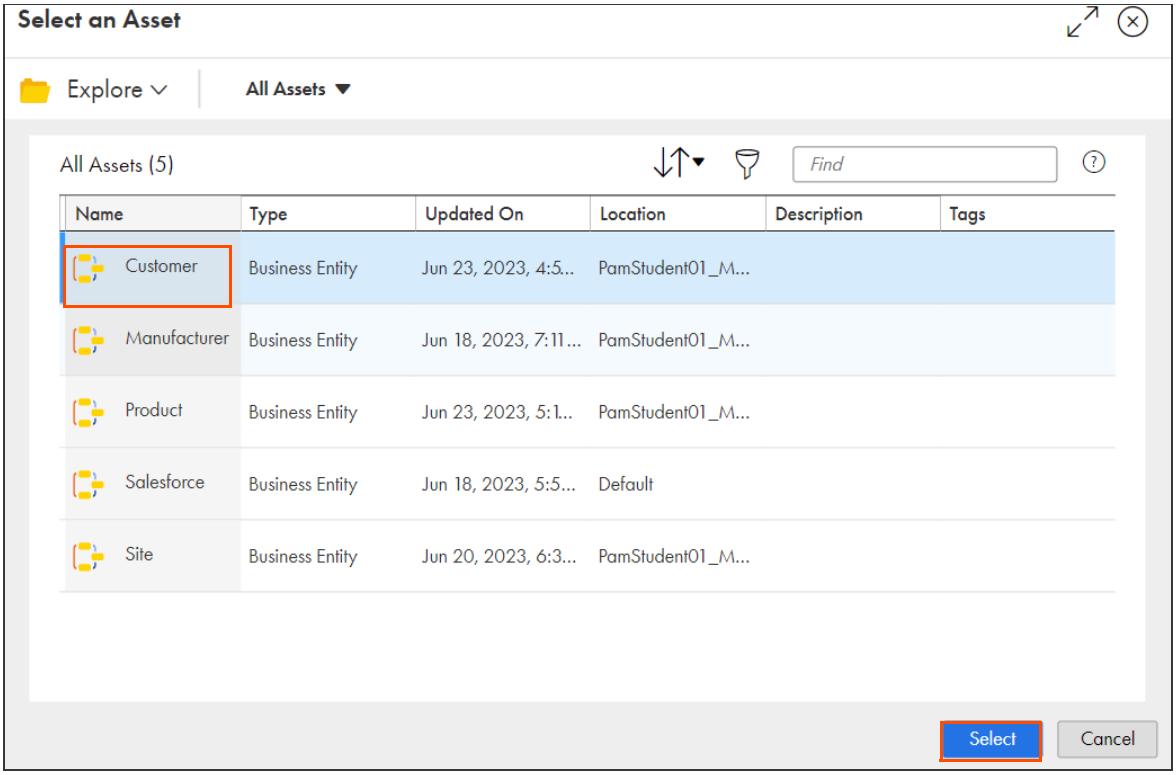
30. In the **Events Scenario** section, click the **+** icon.



31. Click the **Asset Name** selection list.

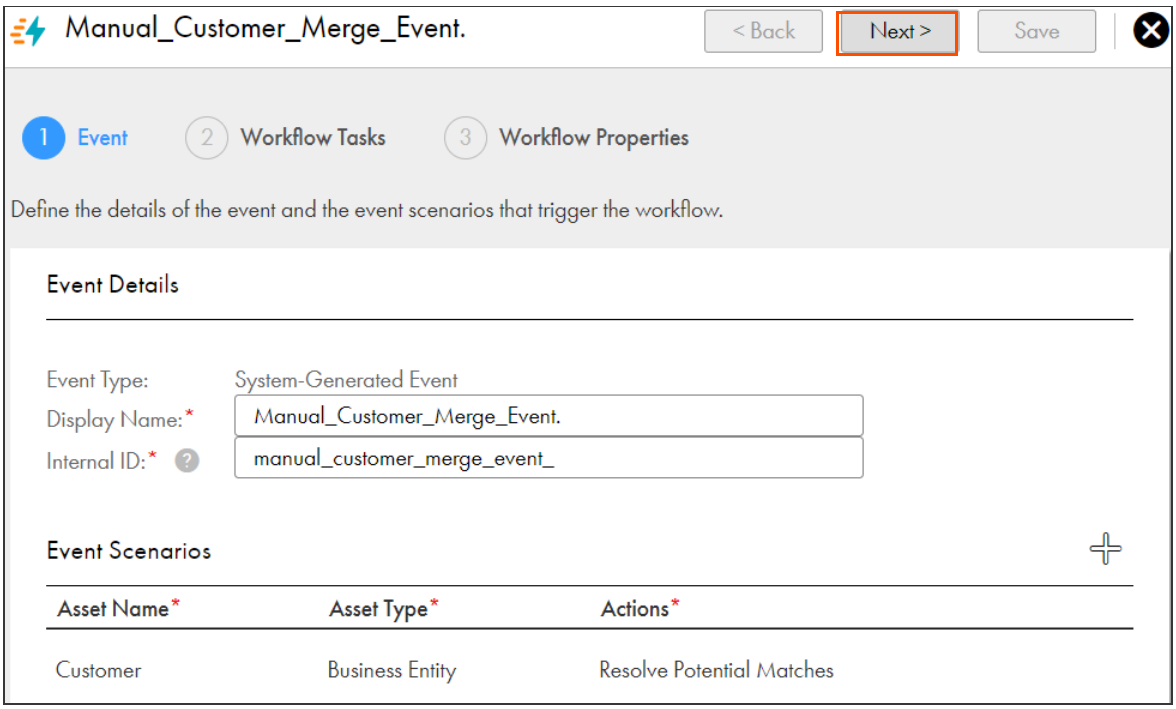


32. From the **Select an Asset** page, select **Customer** and click the **Select** button.

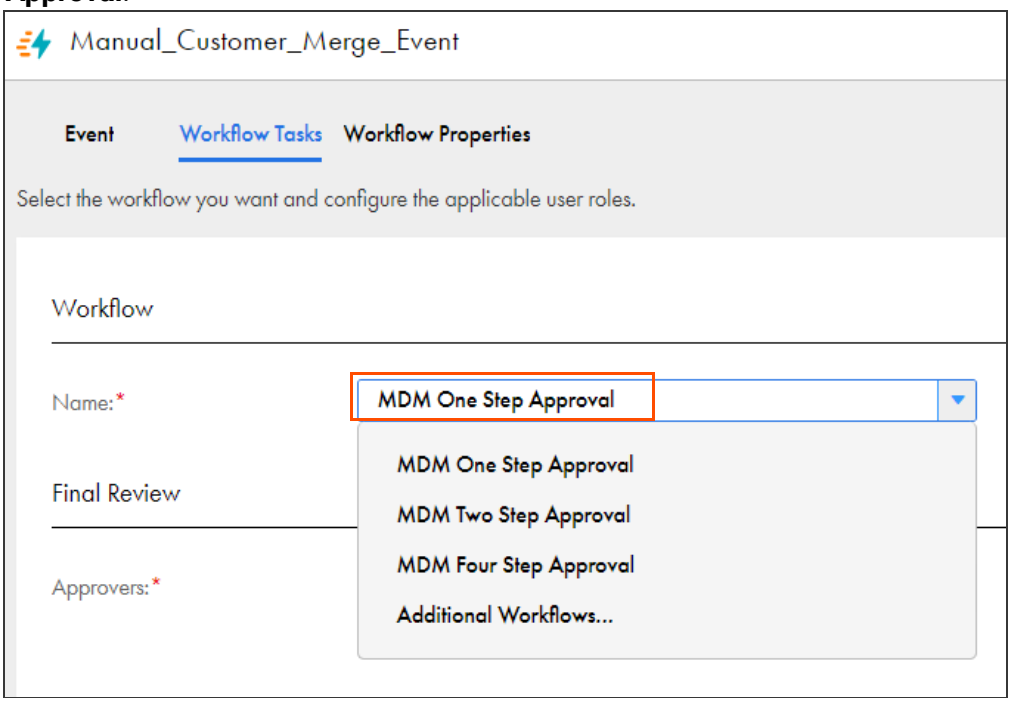


33. Click the **Actions** drop down and select **Resolve Potential Matches**.

34. Click **Next**.

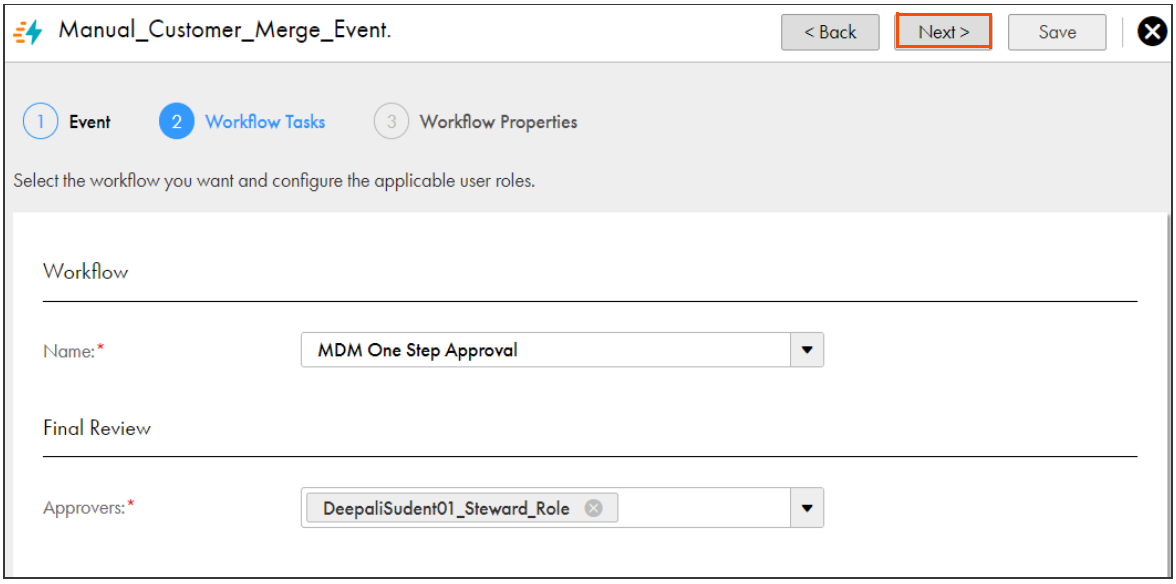


35. In the **Workflow Tasks** window, click the **Name** drop down and select **MDM One Step Approval**.



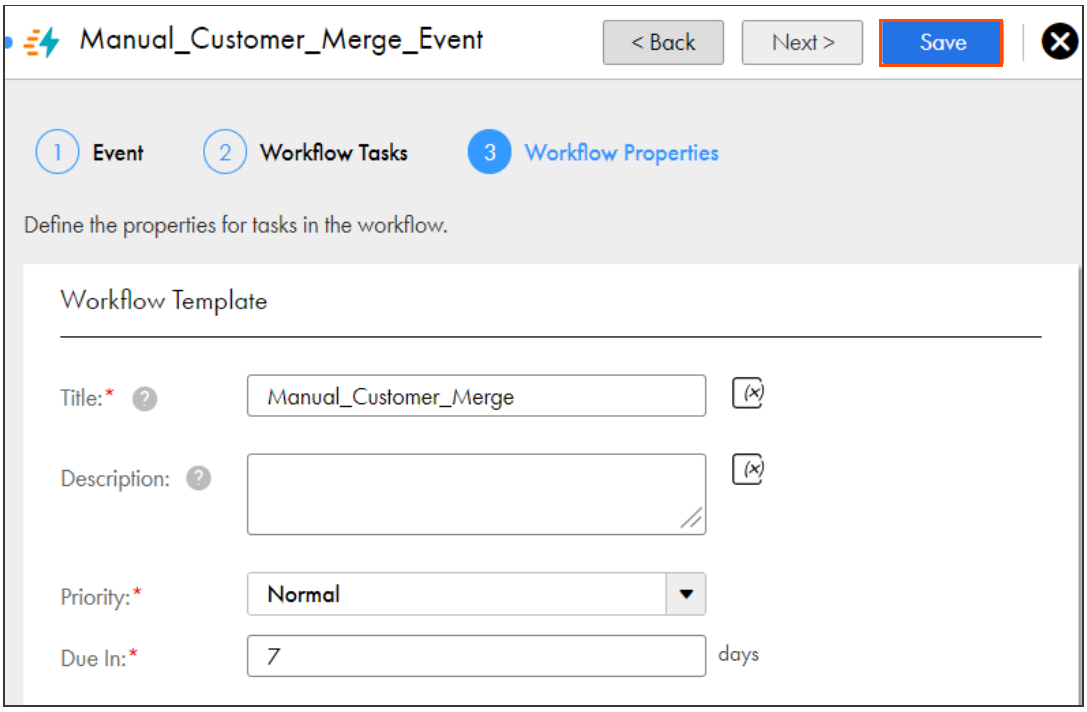
36. Under First Review, select **Multidomain MDM Data Steward** for Approvers.

37. Click **Next**.

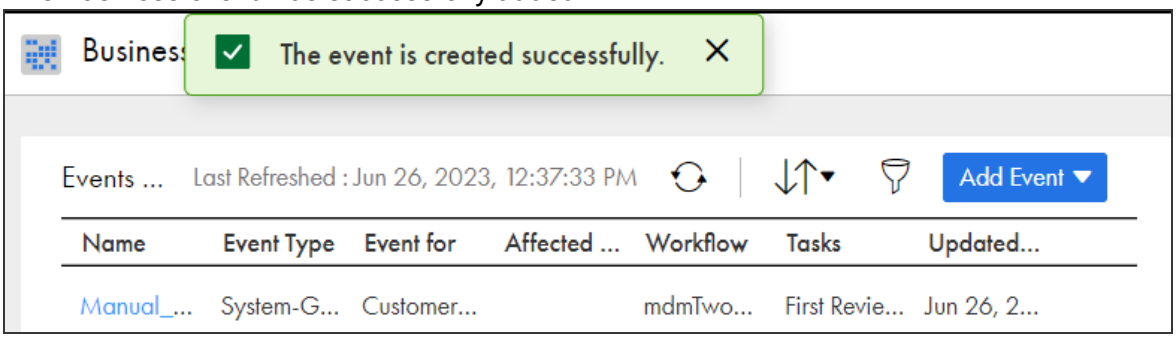


38. In the **Workflow Properties** step, under the **Workflow Template** section, enter the Title as **Manual_Customer_Merge**.

39. Retain the other fields and click **Save**.



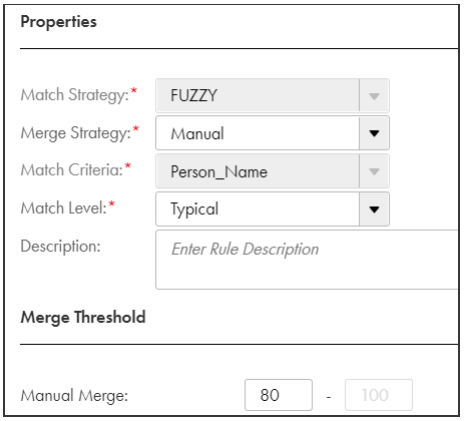
The Business event was successfully added.



Perform Manual Merge

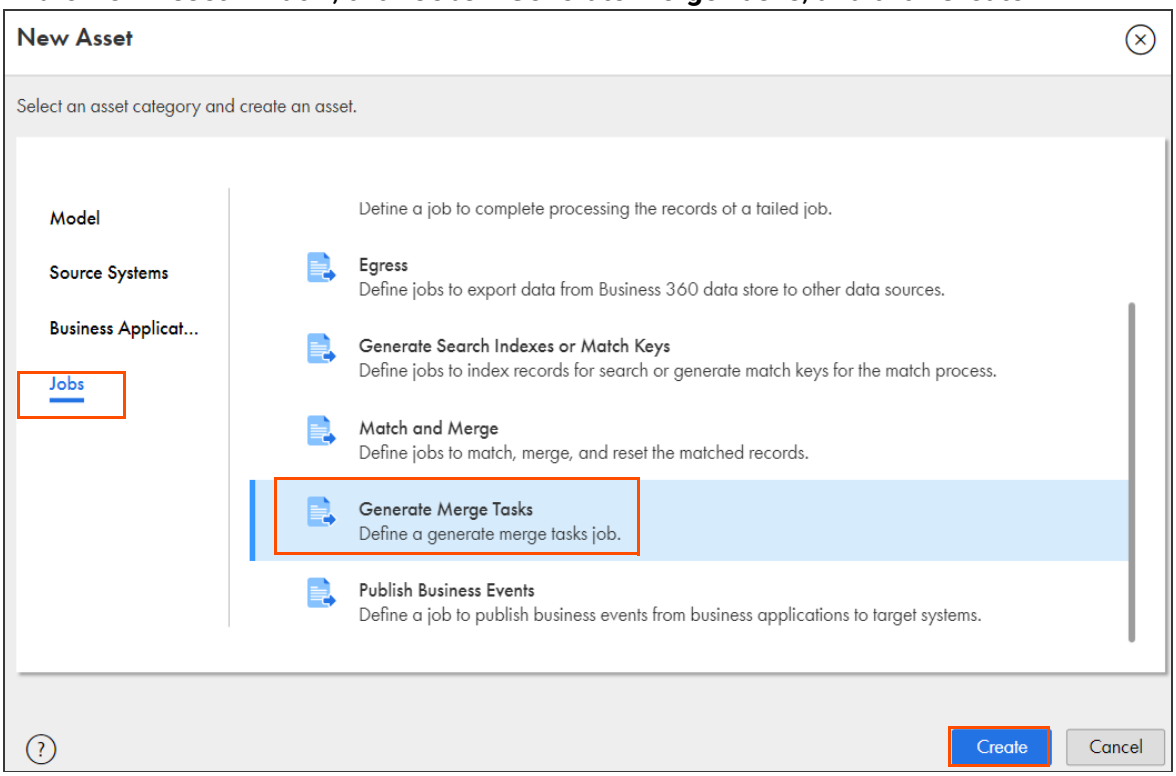
The match process identifies pairs or records that require manual review. Use the generate merge tasks job to grou the pairs or records and generate review tasks for groups that require manual review. Users then review the tasks and manually merge the appropriate records. After you run a generate merge tasks job, you can monitor the status of the job on the My Jobs page. For more information about monitoring jobs, see Monitor Jobs.

You have created two rules where the Merge strategy is set to “Manual”. For these two rules, we need to create a **Generate Merge Tasks** job.

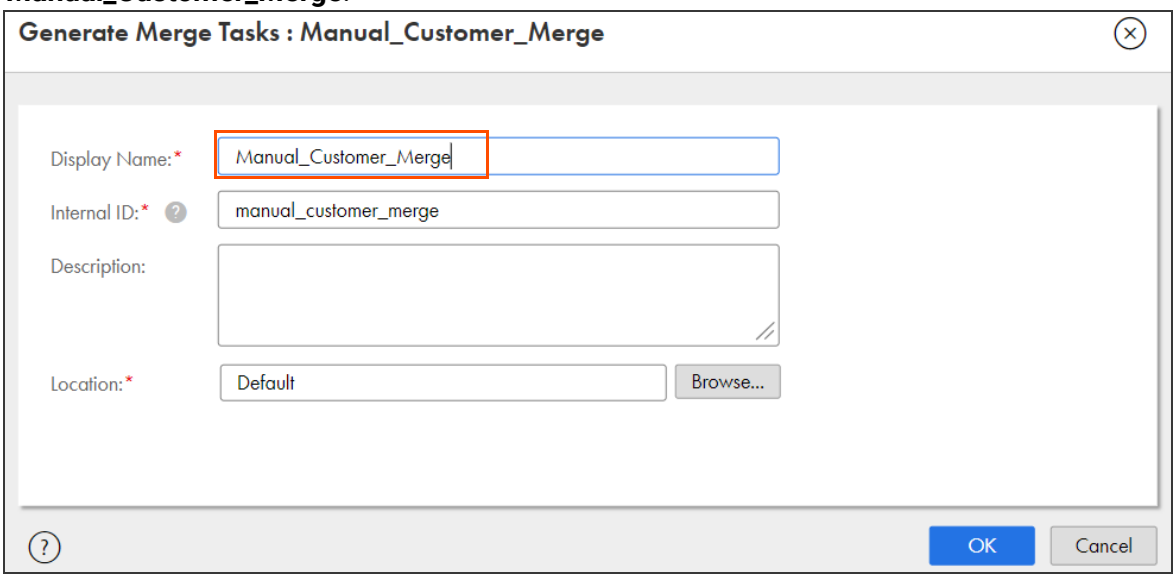


40. In the **Business 360 Console**, click **New**.

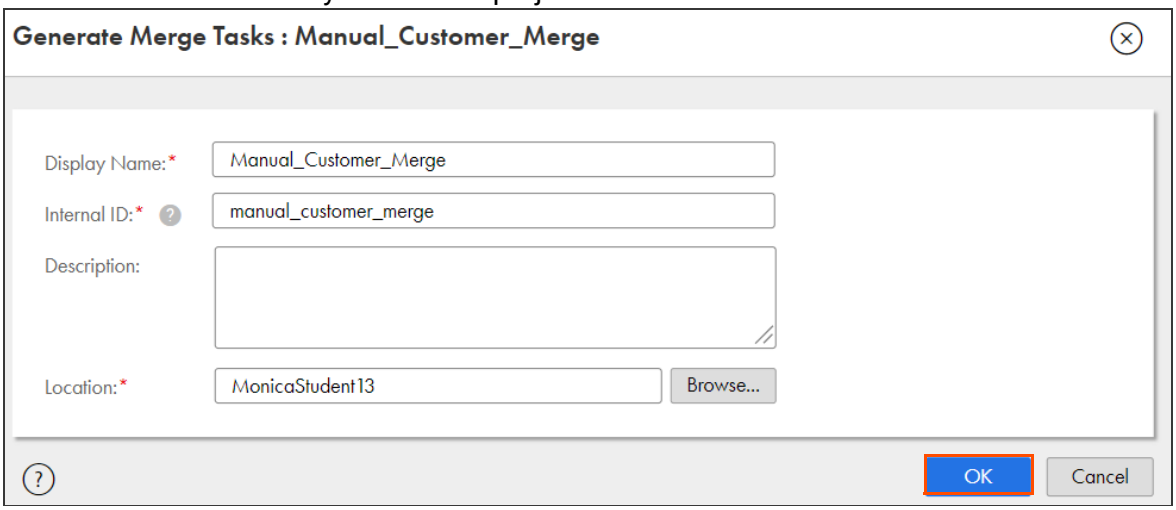
41. In the **New Asset** window, click **Jobs** > **Generate Merge Tasks**, and click **Create**.



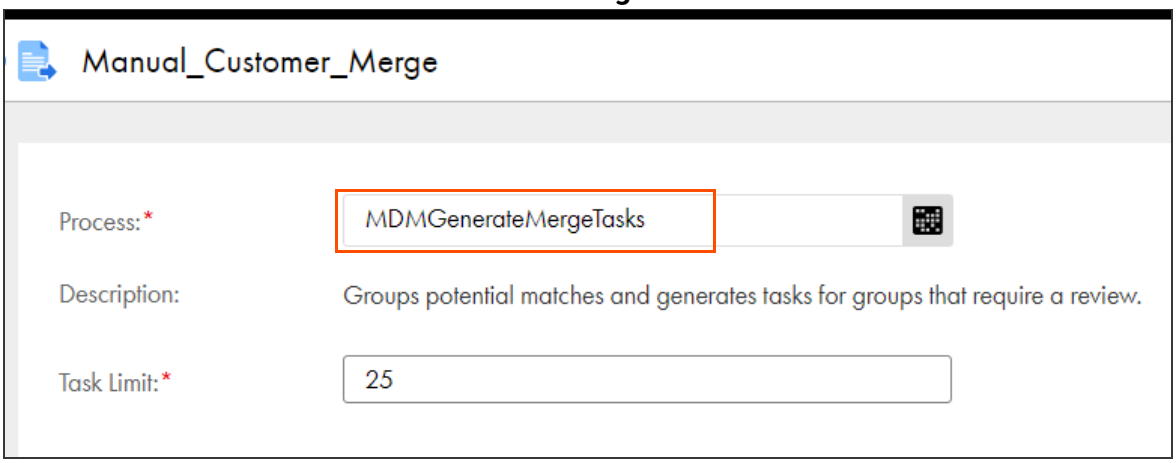
42. In the **Generate Merge Tasks** window, enter the Display Name as **Manual_Customer_Merge**.



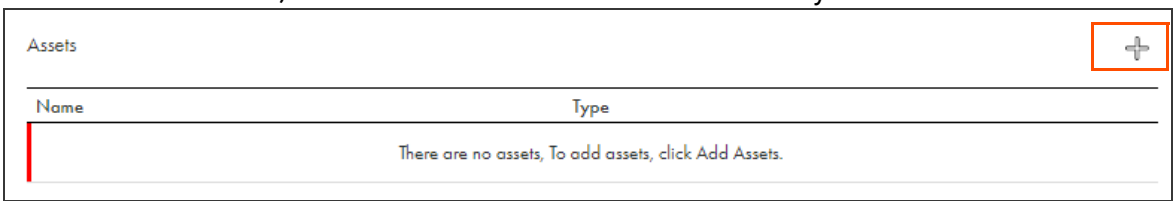
43. Click **Browse** and select your student project and click **OK**.



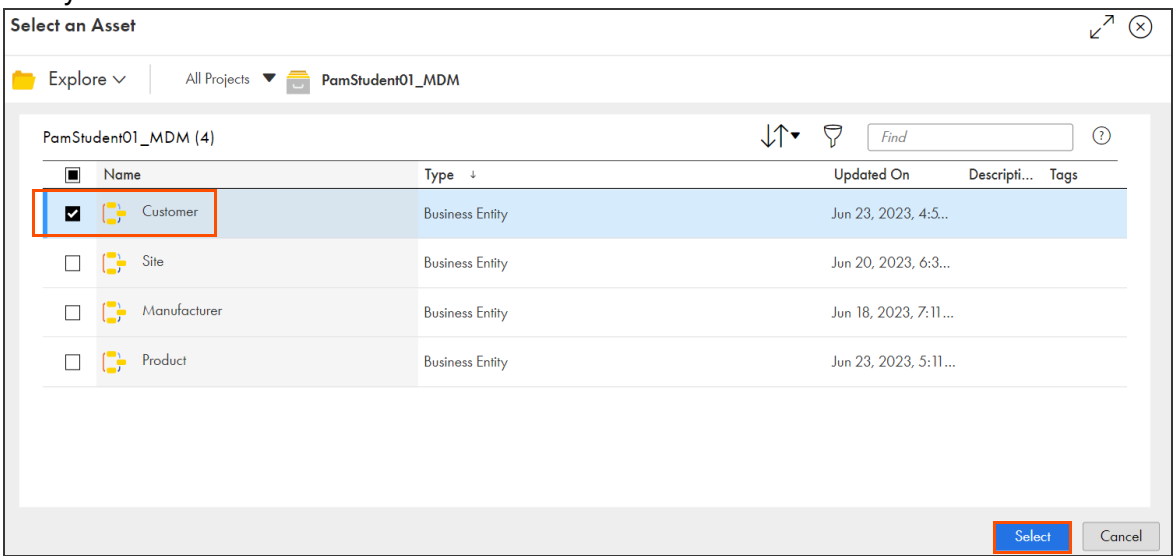
44. Retain the **Process** value as **MDMGenerateMergeTasks**.



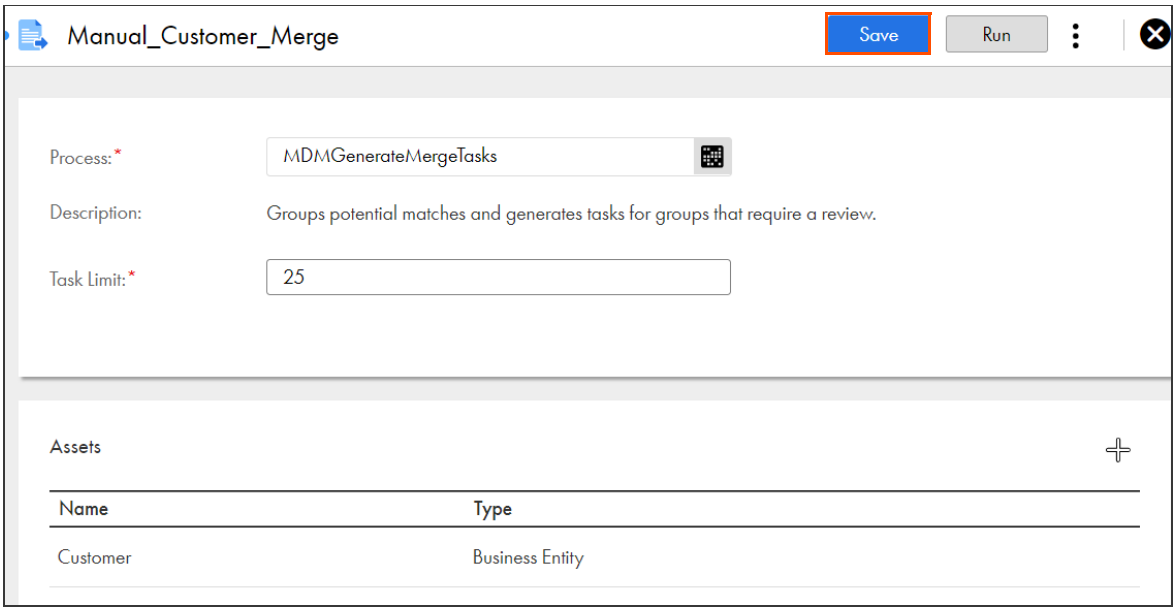
45. In the **Assets** section, click the **+** icon to add the business entity.



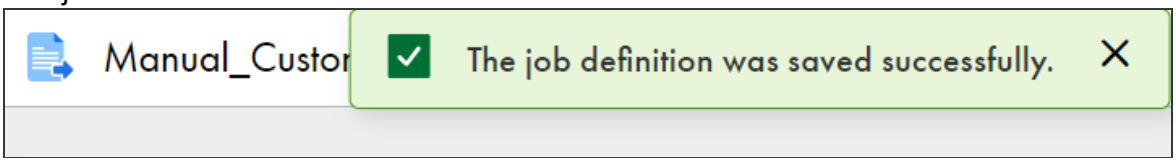
46. In the **Select an Asset** window, under your student project, select the Customer business entity. Click **Select**.



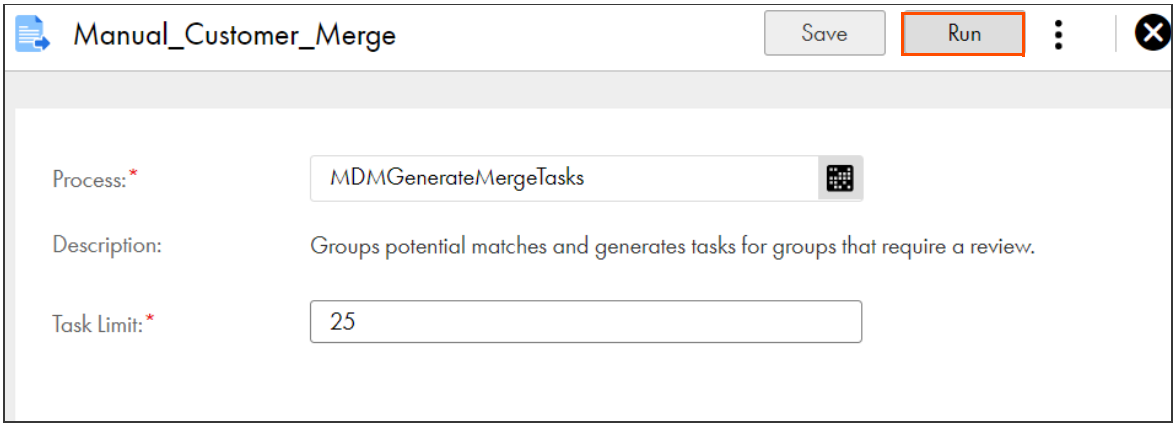
47. Click **Save**.



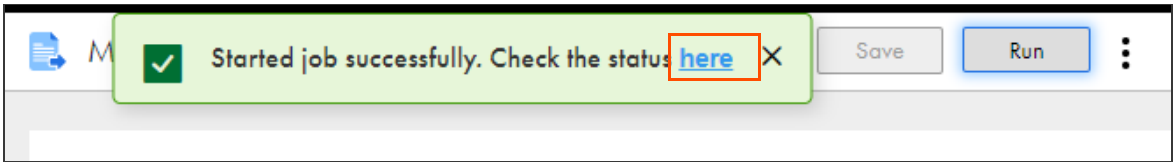
The job definition is saved.



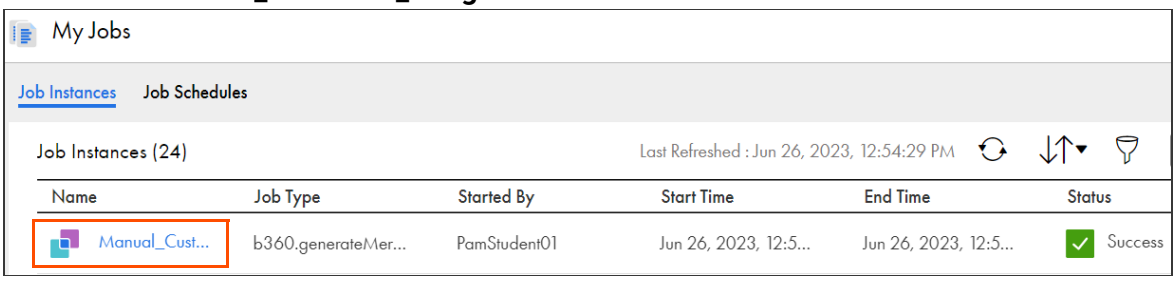
48. Click **Run**.



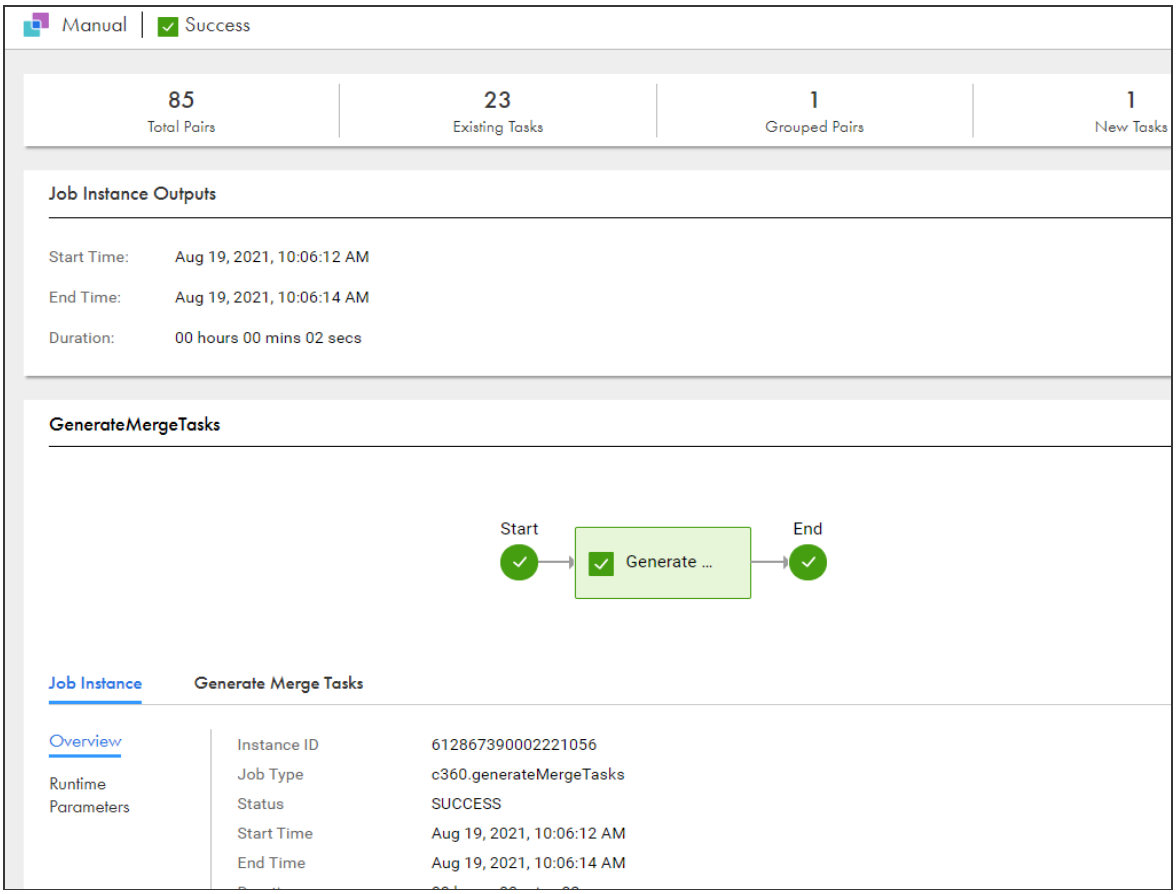
49. Monitor the status of the job by either clicking **here** in the message or navigate to the **My Jobs** page.



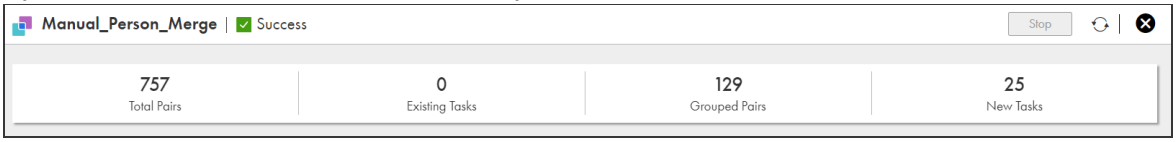
50. Notice that **Manual_Customer_Merge** is successful.



51. Click the Job instance name and view that the task is successful.



52. The match job you ran above has taken the 757 total pairs of records and it grouped them into 129 pairs based on the match criteria we have configured. These 129 grouped pairs generate 25 manual merge tasks as we configured above. These tasks will get approved by the steward and manager before they get merged.



The metrics shown above can change depending on the input data.

Approve the Workflow (Login as an Approver)

You will now login as an approver to approve the workflow that is triggered for the manual merge. For the purpose of this training, you have created a custom user named: **<studentNamestudentno>_Steward / Multidomain360** (johnstudent01_Steward / Multidomain360, for example) in the previous labs.

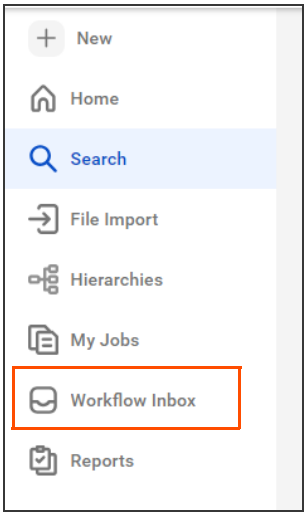
53. You will now need to log in as the MDM Steward and approve the task.

54. In your browser, open an **incognito window**.

55. Log in as **<studentNamestudentno>_Steward /** Multidomain360 (johnstudent01_Steward / Multidomain360).

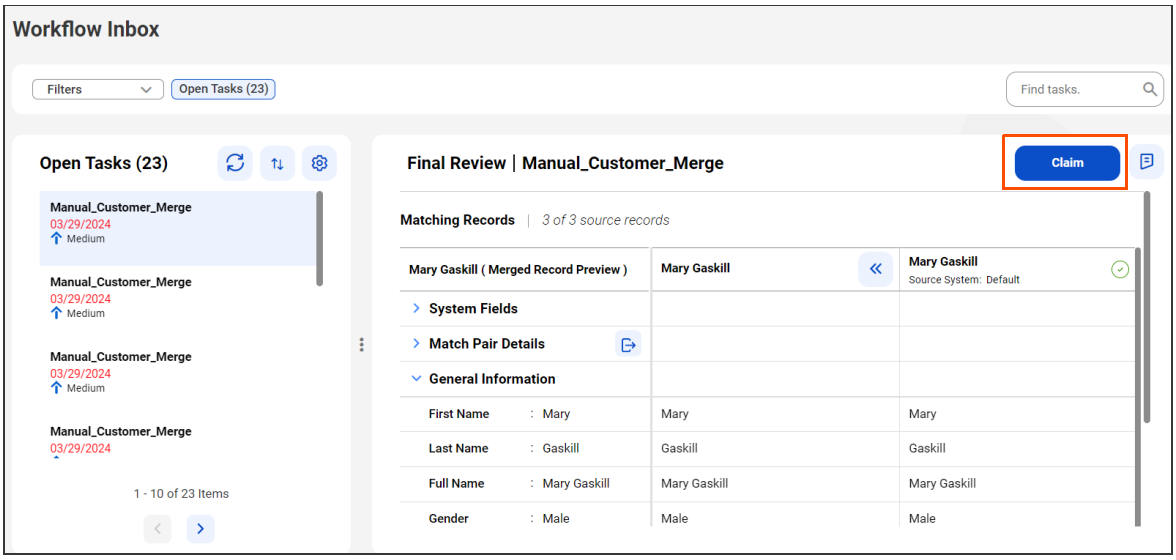
56. From the **Services** window, select **Multidomain MDM**.

57. Click **Workflow Inbox**.



58. Notice that many records have been qualified for manual matches. Click the task title to open the task. Every task displays other sets of matched records. You can either approve the matches, reject, or release them. Once approved, the records will be merged, and the status changes to **Completed**. If it is opened once for viewing, the status changes to **In Progress**. Else, status remains **Unassigned**

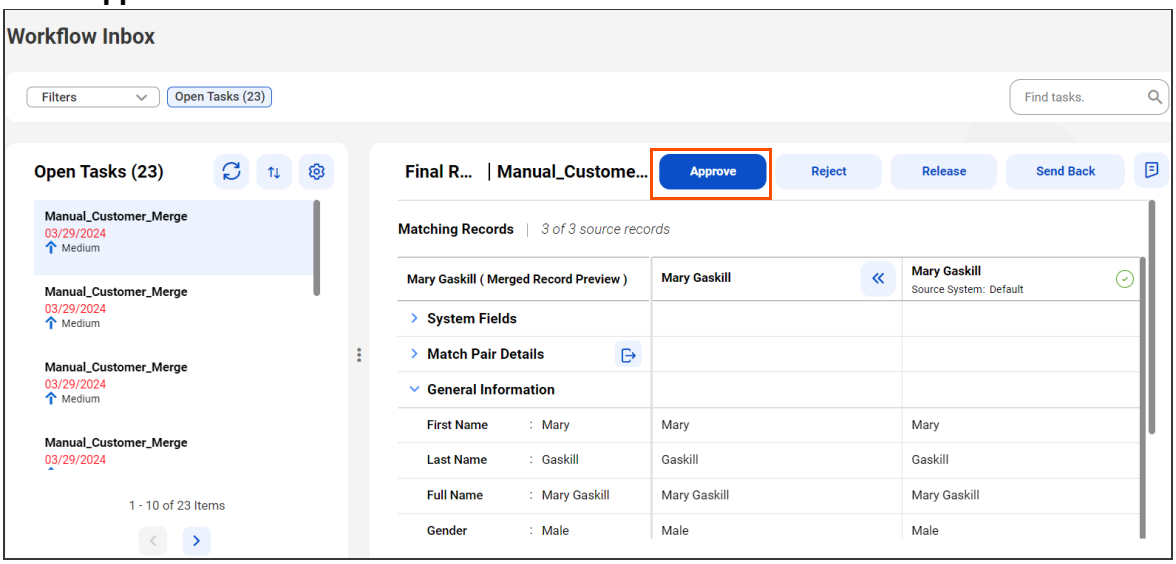
59. You will see the tasks that are under the **Final Review**. Click **Claim**.



The task may contain any of the match pairs that is eligible for merge. You just need to open the first task in the Workflow Inbox, and perform the below steps. In this lab, you will see Mary Gaskill record being approved.

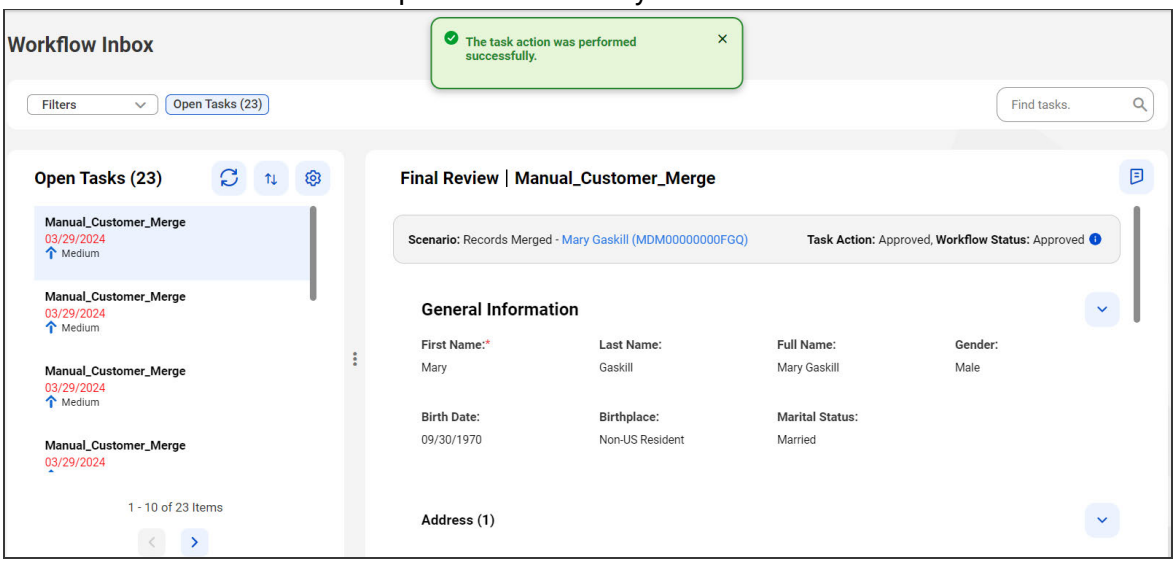
You can also expand the Task Summary to see the details of the task.

60. Click **Approve**.

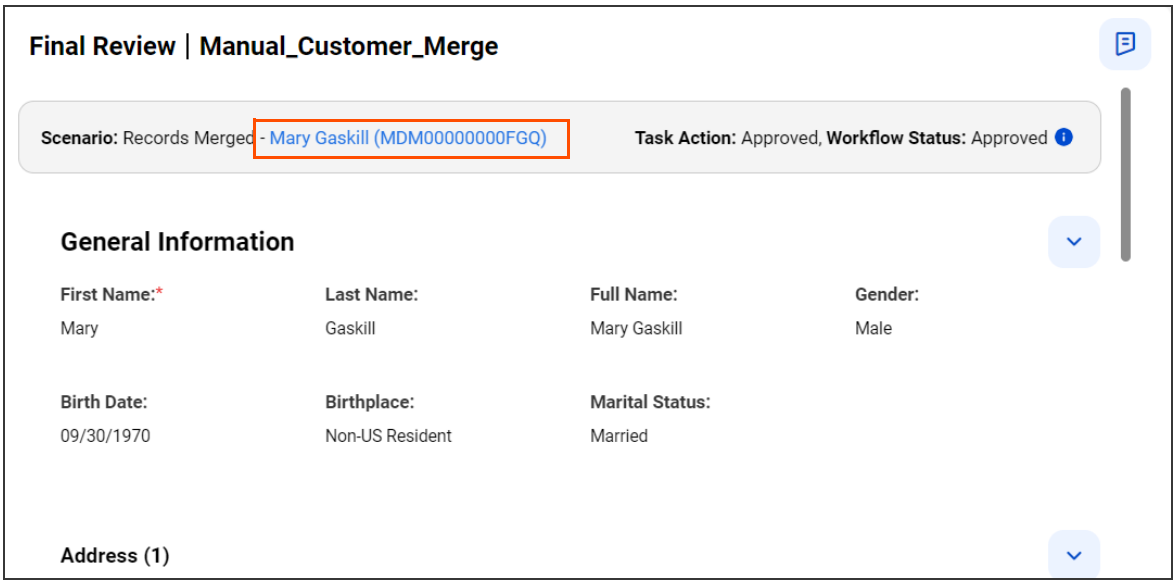


If you **Reject**, the records are not considered for merge. You can also **Release** the task back from the claimed status or **Send Back** to the initiator.

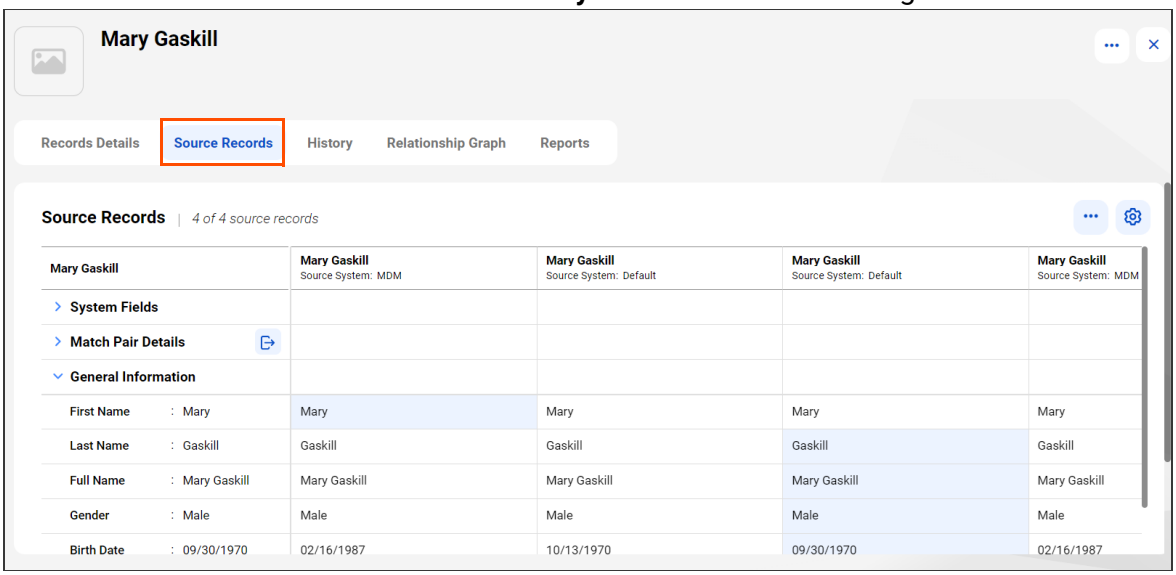
61. You can see that the task completed successfully.



62. Click the **Record Name**.



63. Click **Source Records**. You can see that **Mary Gaskill** records are merged.



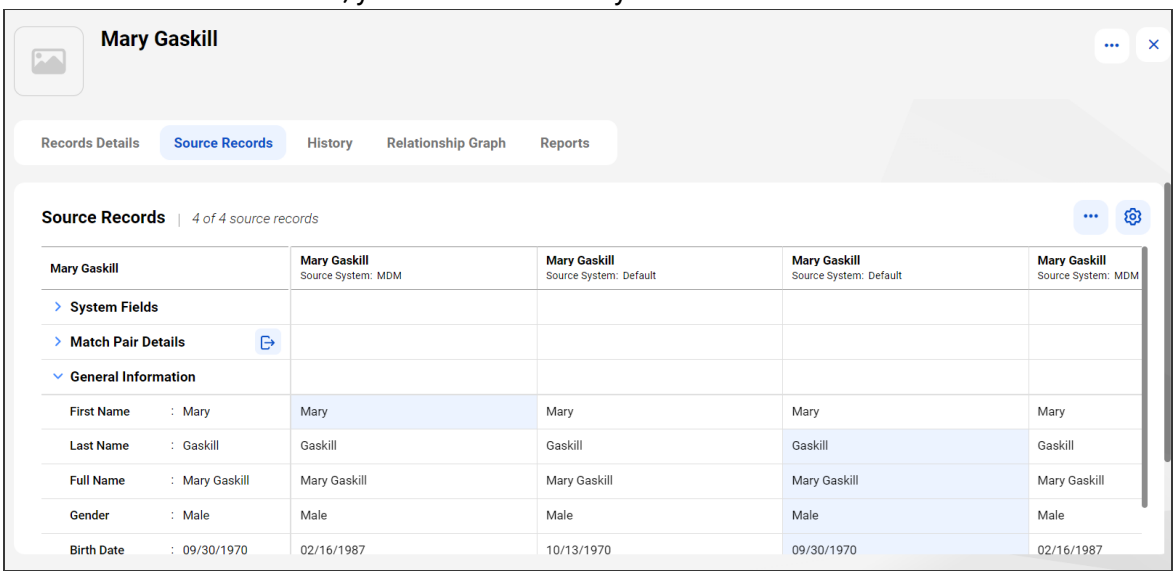
If you are unable to see the Source Records tab, then make sure that you have enabled the The **Source Records** check box for the Steward role as mentioned in **step 56 of lab 8-1**.

You can try opening the other tasks and merge the other records as well.

Unmerge Records

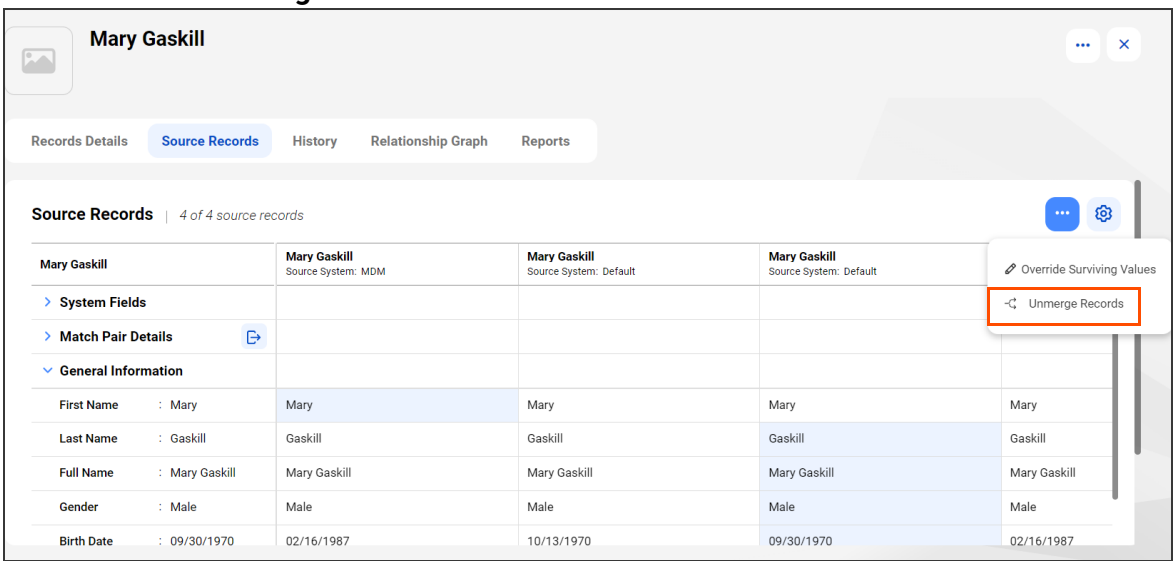
The Business Event that was triggered also allows the Multidomain Steward to unmerge the records. After merging, if you notice that the merges were not correct, you can always unmerge. In this case, if you realize that Susan Grimshaw records that you merged earlier are different and hence they can be unmerged.

64. In the **Source Records** tab, you will see the Mary details.

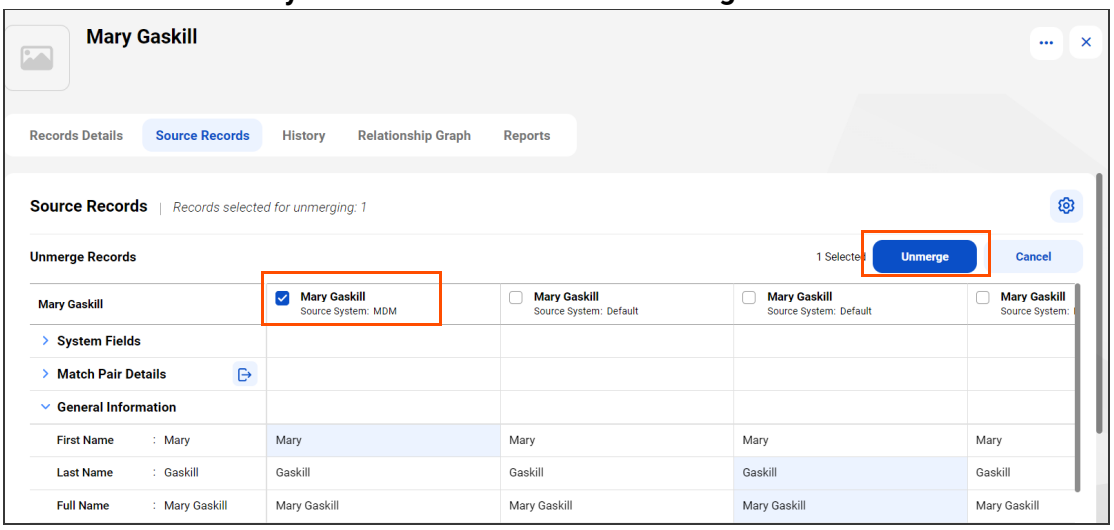


In case you have closed the record, then search for the merged record, in this case **Debra**, and open the record. Then click on **Source Record** tab.

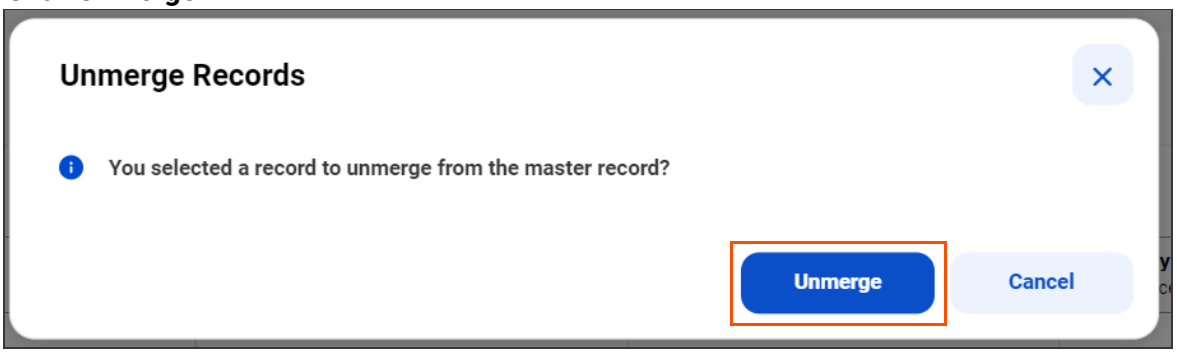
65. Click **Actions** > **Unmerge Records**.



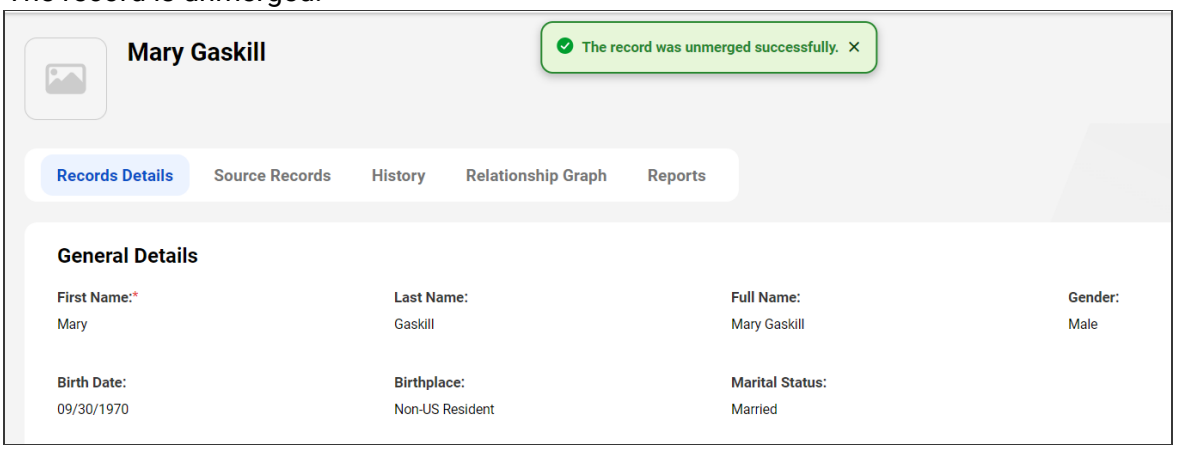
66. Select one of the **Mary Gaskill** records and click **Unmerge**.



67. Click **Unmerge**.



The record is unmerged.



This lab guide includes steps to merge and unmerge for the record “Mary”. You may choose any other set of records and perform these tasks.

This concludes the lab.

Lab 11-2: Configuring Custom Workflows

Overview:

If you do not want to use any predefined approval workflows, define workflows in Cloud Application Integration as processes. Processes contain sets of instructions that automate your business requirements.

To use Cloud Application Integration processes as workflows, add at least one Service step that's a generic user task or an originator user task. These user tasks indicate that the process requires user interaction as part of the workflow. Generic user tasks define actions for users included in the workflow. Originator user tasks define actions for users who trigger a workflow with their activity. You can use the predefined MDM Multi-Step Approval Template to design a process.

Objectives:

- Create a custom workflow
- Analyze the predefined MDM Multi-Step Approval Template
- Configuring business event using the custom workflow

Duration:

30 minutes

Scenario:

To make the customer onboarding process stringent, Mark has opted to implement multiple approvals at each stage. Consequently, he chose to establish a workflow that mandates several approvals prior to storing customer data. The predefined workflows are not suitable for this purpose as they do not support multiple approvers and customized workflow tasks. Thus, Mark decided to utilize custom workflows instead.

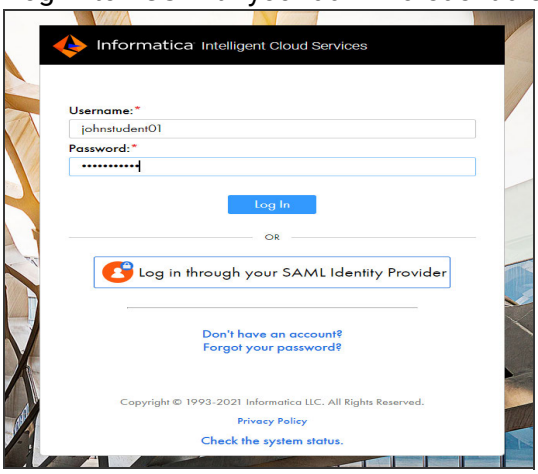
Tasks:

- Analyze MDM Multi-Step approval process template
- Using custom workflow template to create custom workflows
- Approving the workflows

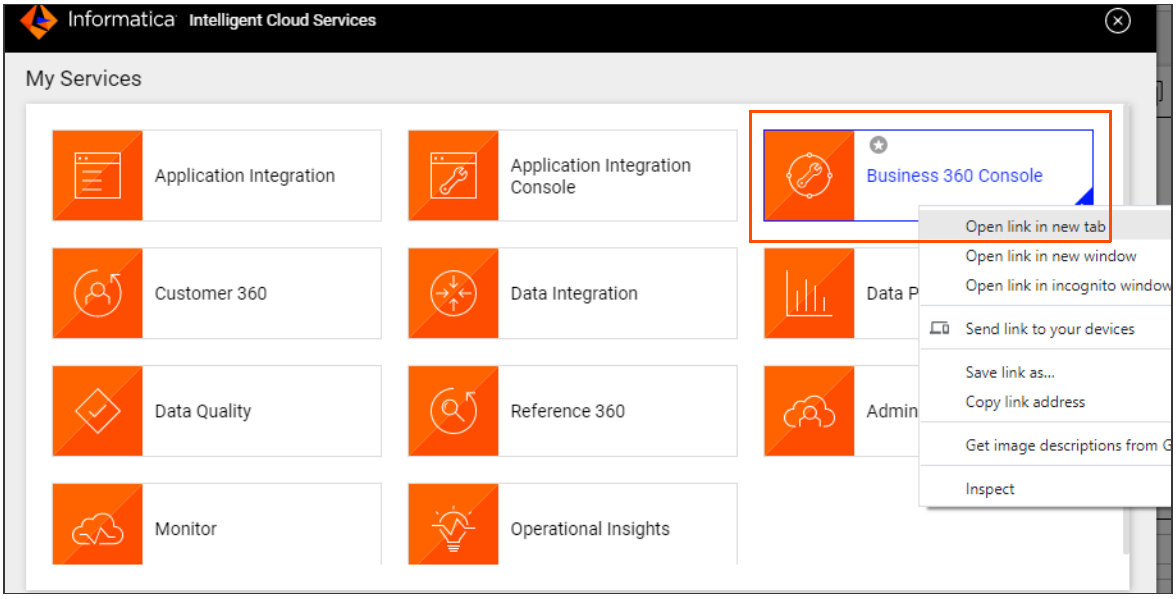
Analyze MDM Multi-Step Approval Process Template

First, you will analyze the MDM multi-step approval process template before making a copy out of it.

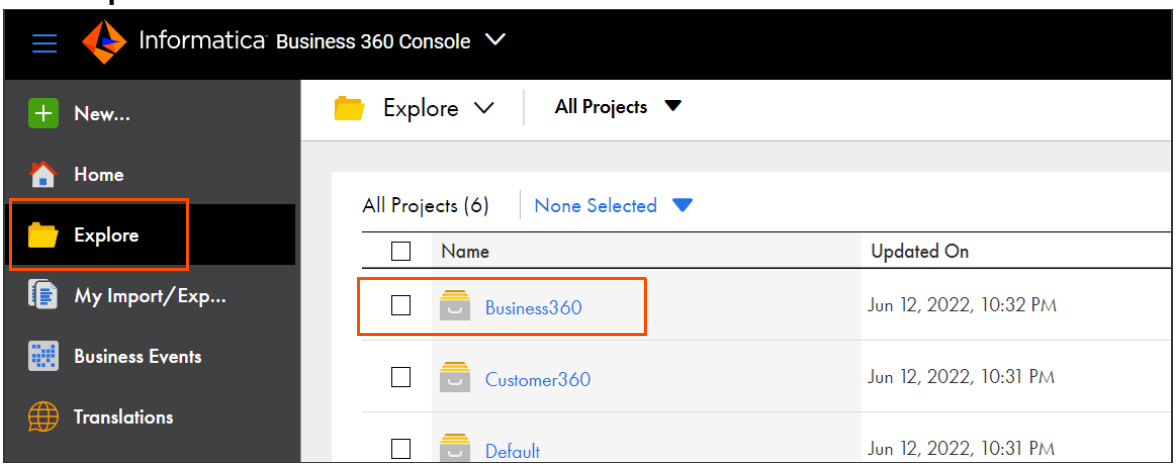
1. Log in to IICS with your admin credentials if you have logged out after the previous lab.



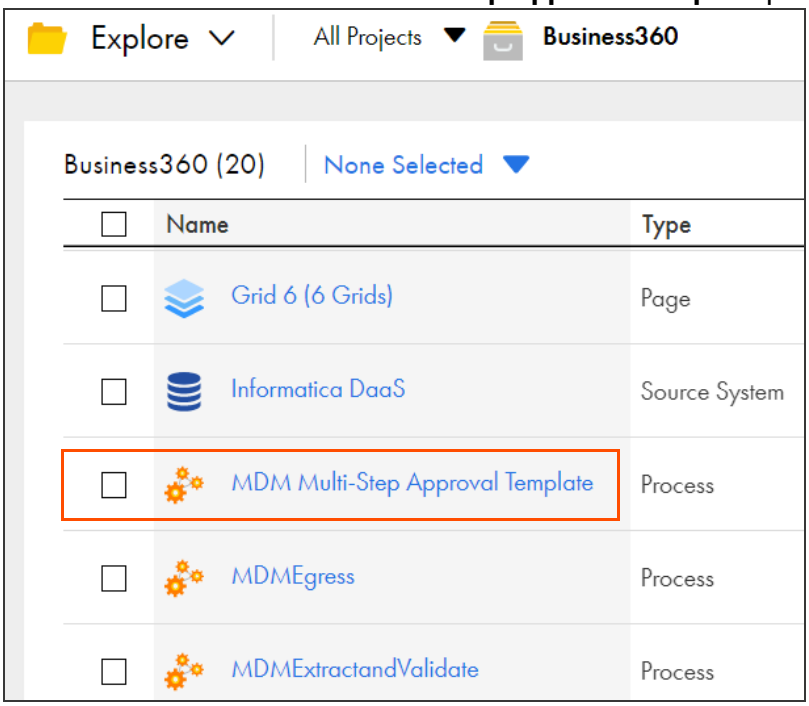
2. Open **Business 360 Console** in a new tab.



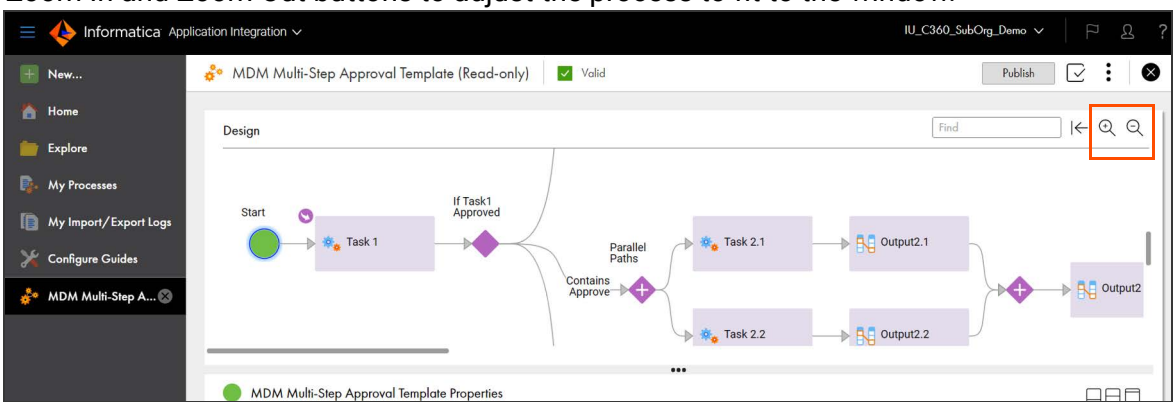
3. Click **Explore** > **Business 360**.



4. Scroll down and click **MDM Multi-Step Approval Template** process to open.

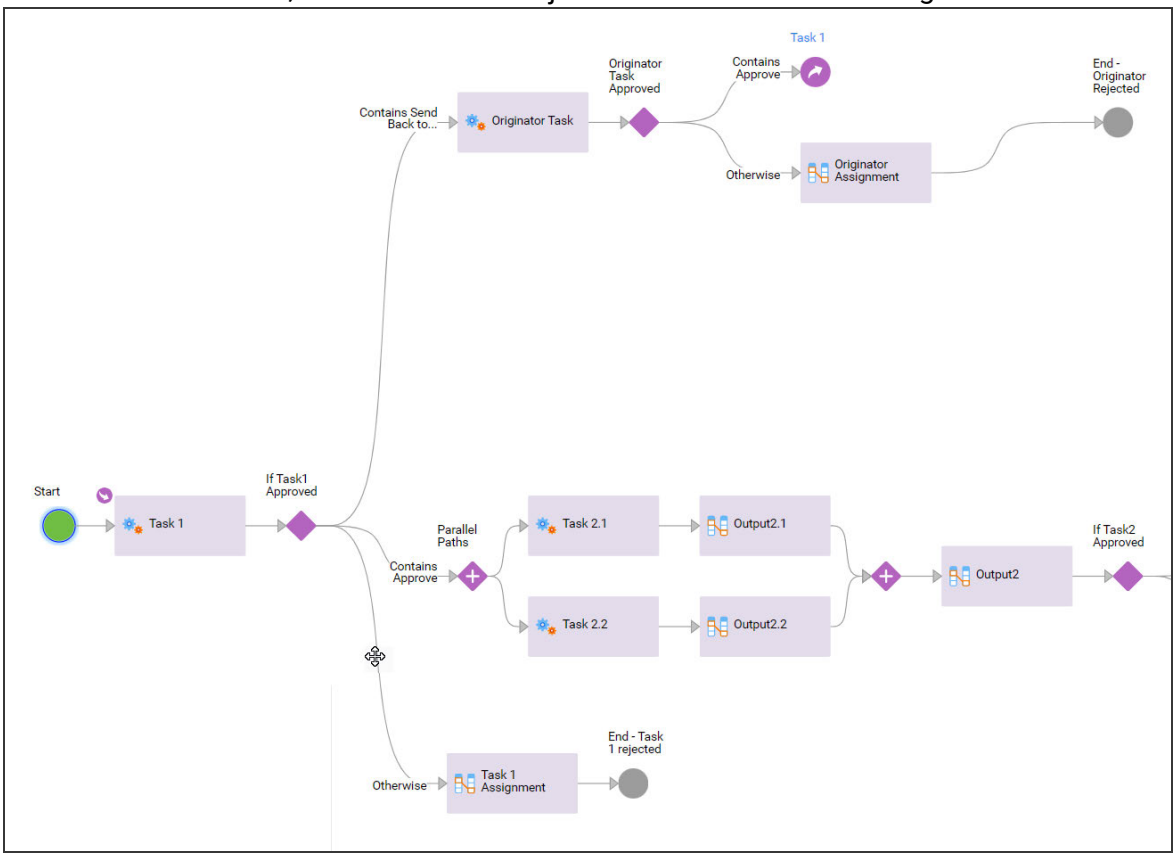


5. You can see a read-only process opened in the Cloud Application Integration console. Click Zoom In and Zoom Out buttons to adjust the process to fit to the window.

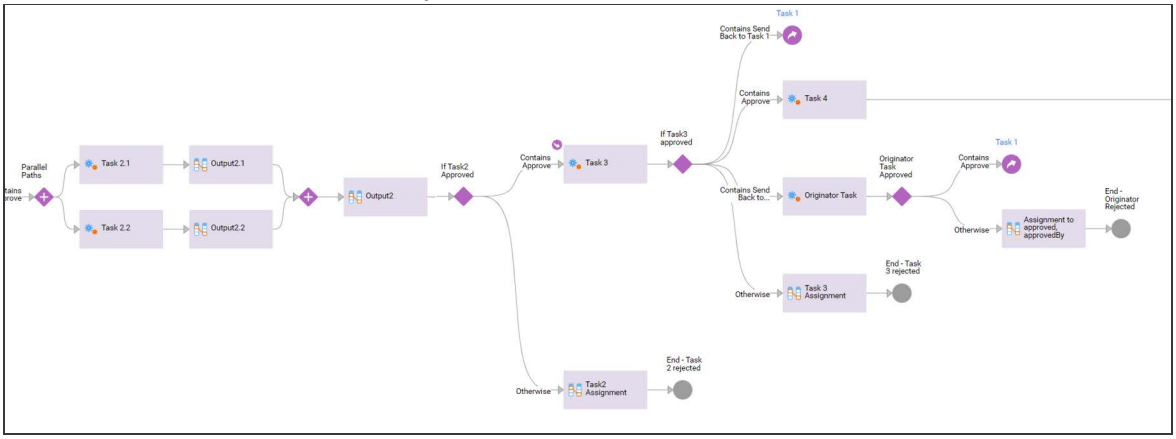


The MDM Multi-Step Approval Template has example steps that demonstrate how to create an approval process with a Start step, Service steps, and additional process steps. You can review the template process and then create a process based on your requirements using the steps mentioned in the documentation.

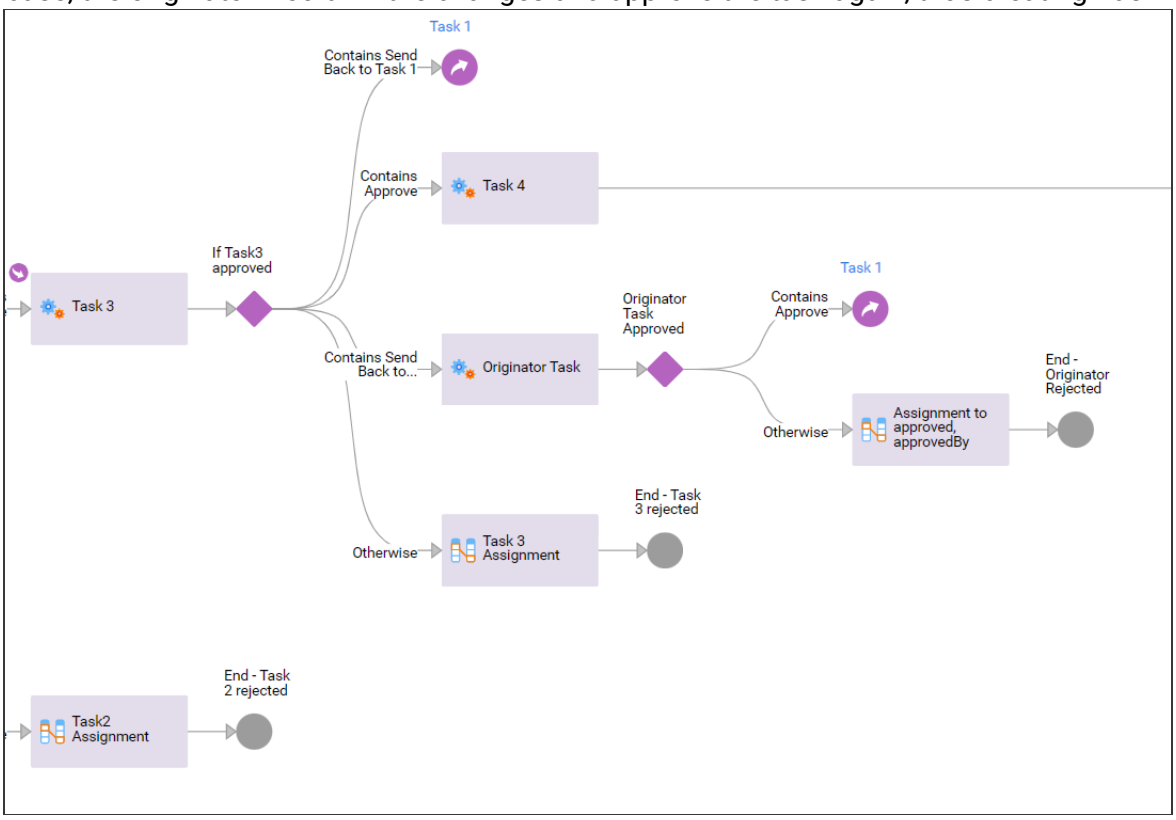
Task 1 would be created when originator starts the workflow. As soon as Task 1 is approved, Task 2 will be created, or Task 1 can be rejected or sent back to the originator.



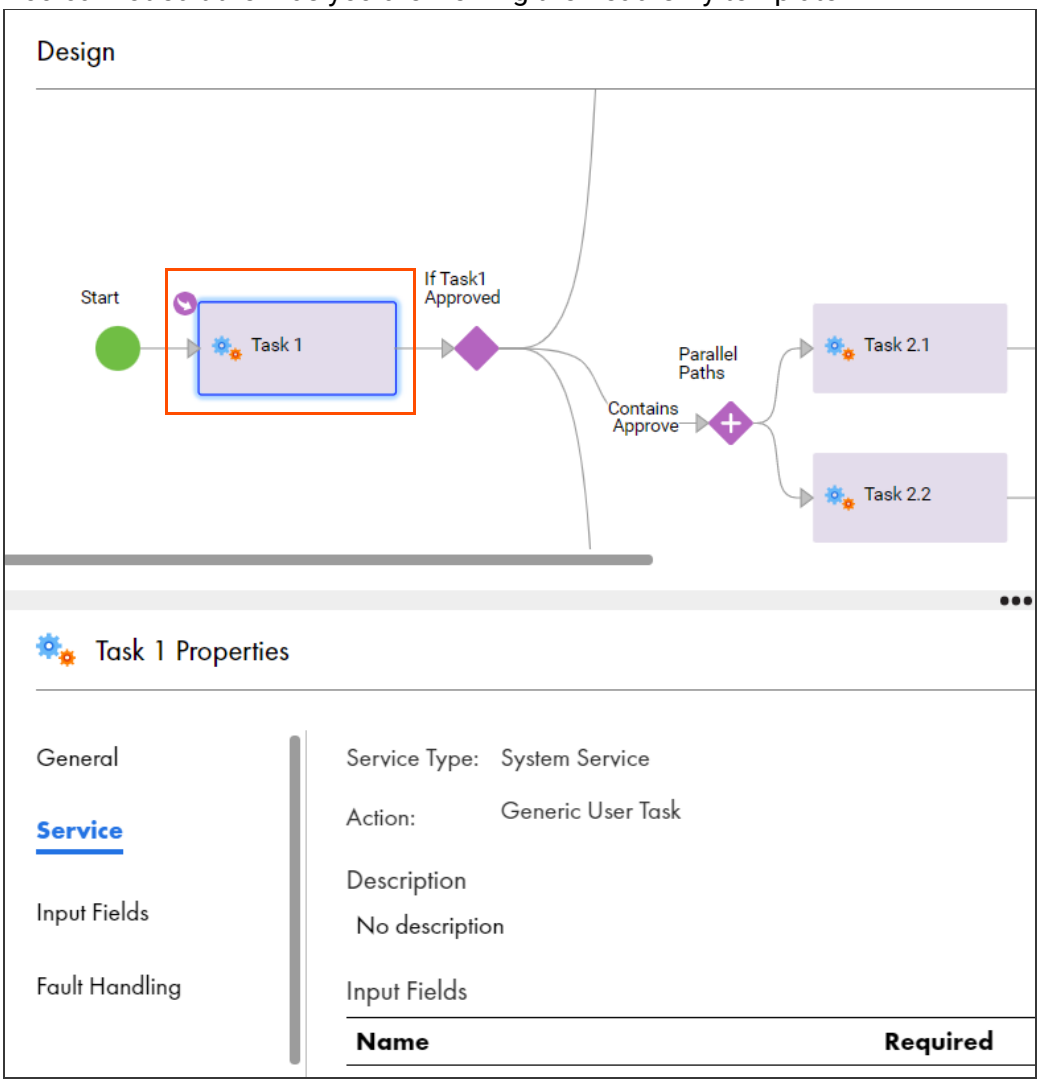
Observe that the Task 2 is a parallel step, and you can see how two parallel steps are combined in the process. The task 2 can either be approved or rejected. Task 3 will be created if it is approved. Otherwise, it will be rejected.



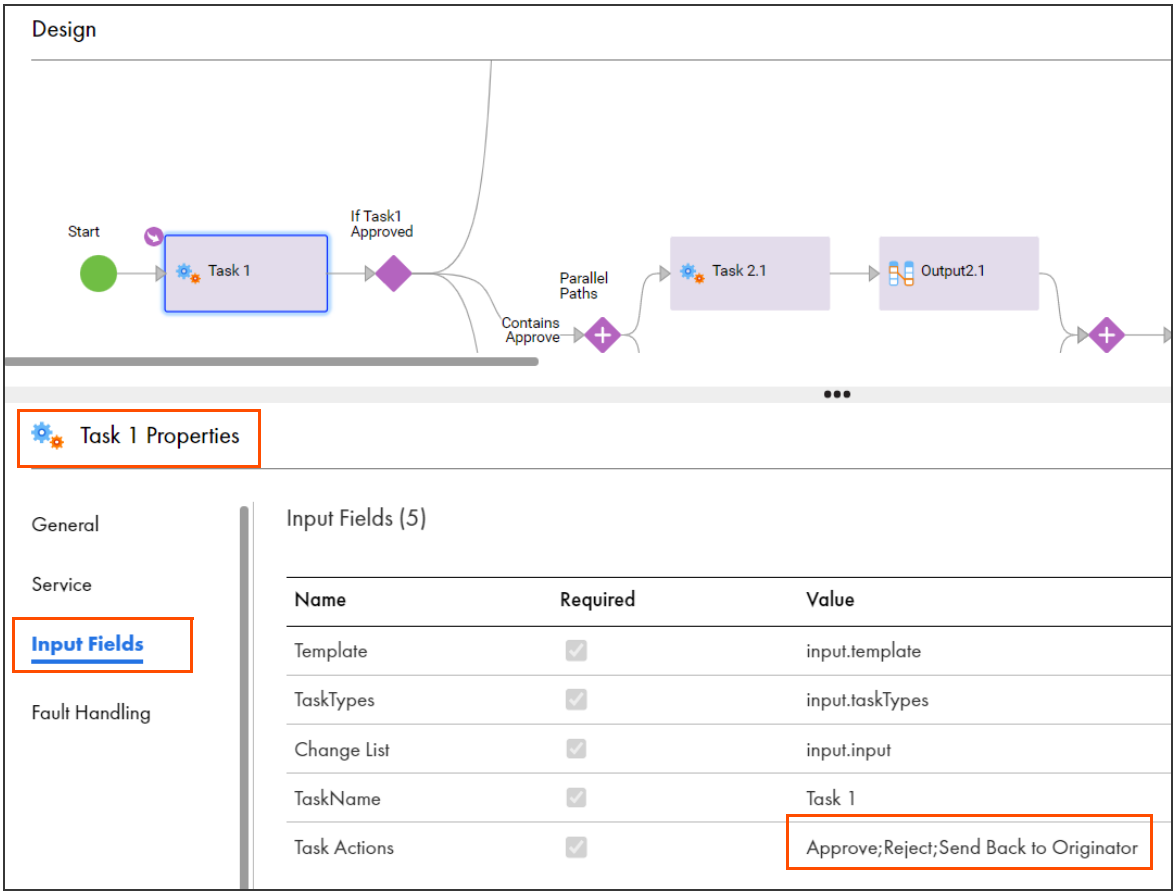
In Task 3, you can either approve the task, send it back to task1, send it back to the originator, or reject it. If it is approved, Task 4 will be created. When it is sent back to originator, in this case, the originator would fix the changes and approve the task again, thus creating Task 1.



6. Select **Task 1** from the Design section and select **Service** Tab under Task 1 Properties section. Observe the properties on the right side configured for the task you have selected. You cannot edit them as you are viewing the Read-Only template.



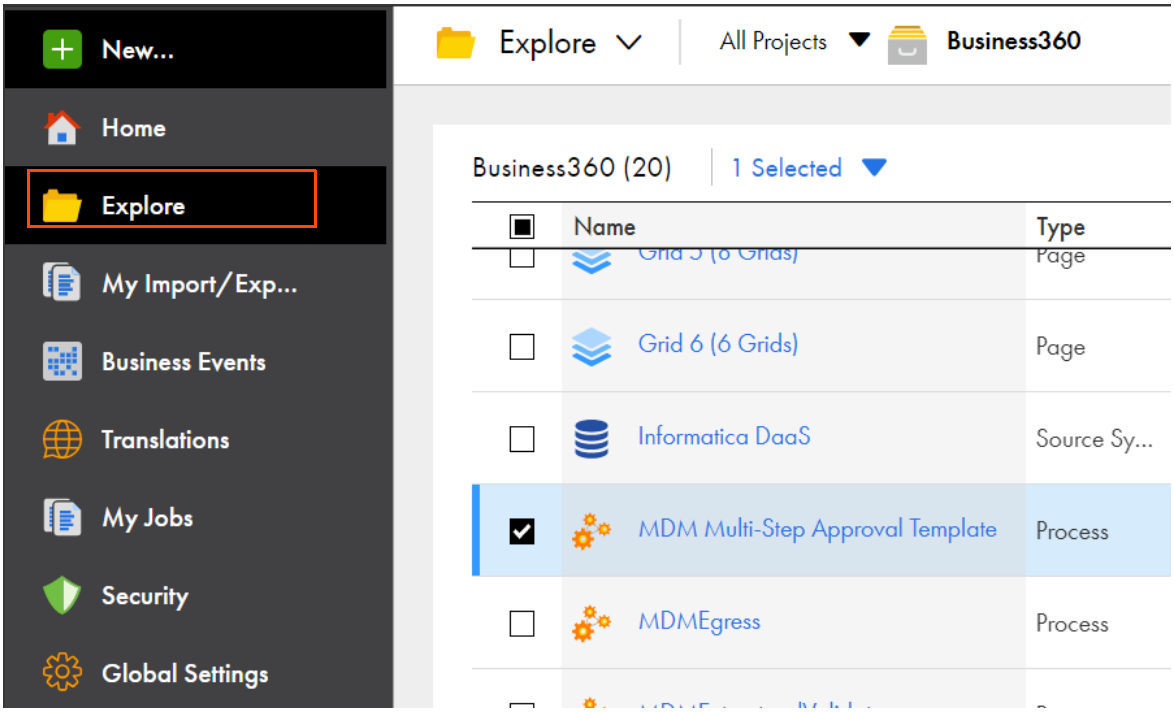
7. Under the **Task 1 Properties** section, select the **Input Fields** tab and observe the properties configured. You can see that the **Task Actions** configured here are options that you can choose when reviewing workflows. You should configure how each task action is handled in the process. For example, what should happen when the reviewer approves or reject or send back the task.



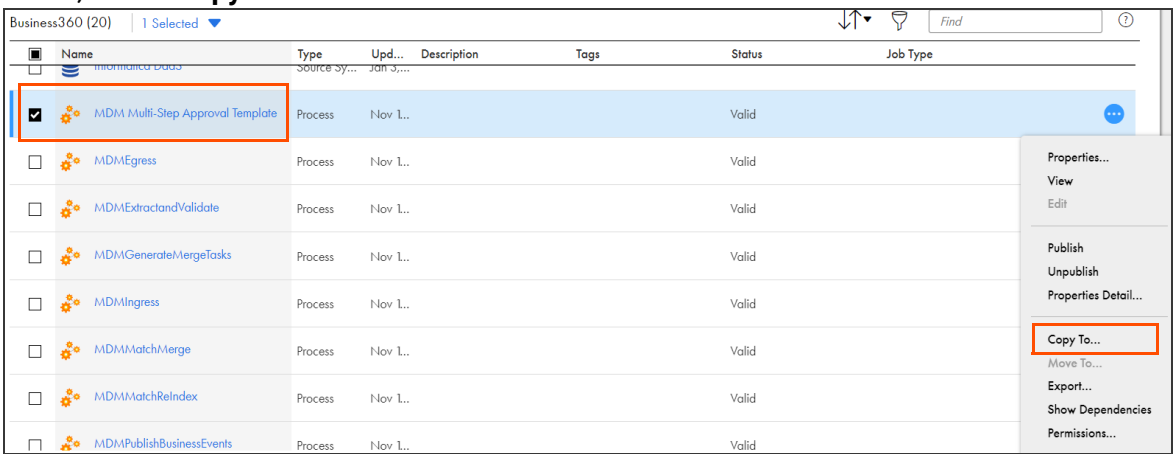
Using Custom Workflow Template to Create Custom Workflows

Create a user-triggered event that creates a task whenever the source records are updated and select the custom workflow you have analyzed in the workflow properties section. The configured custom workflow will be enabled whenever user made edits to the source records. As the approver, log in and approve tasks to ensure the process is running as intended.

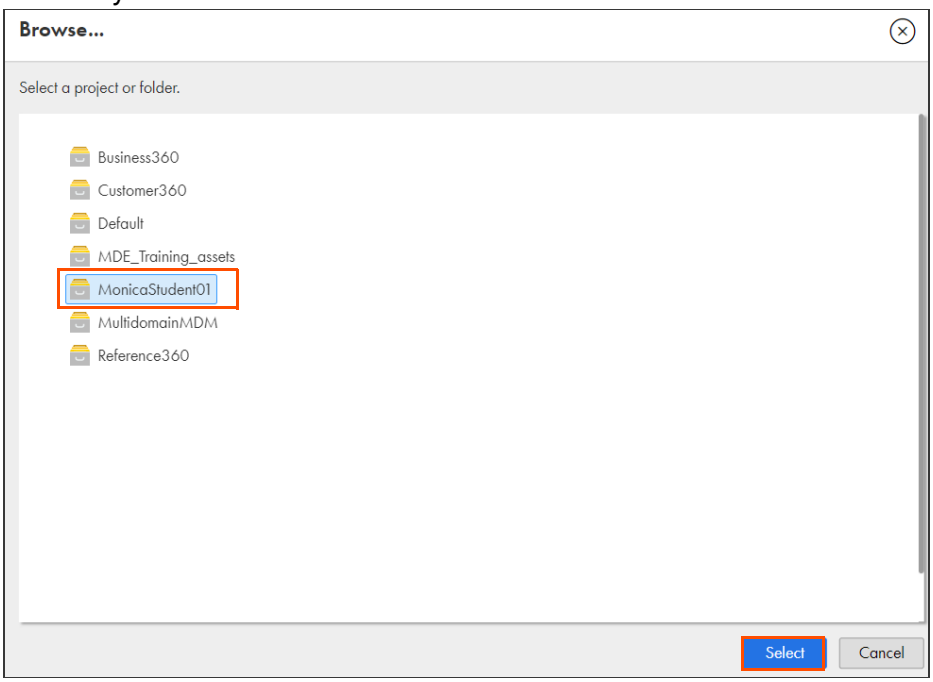
8. Navigate to **Explore** > **Business 360**.



9. Scroll down and select MDM Multi-Step Approval Template and from the actions drop-down, select **Copy To**.

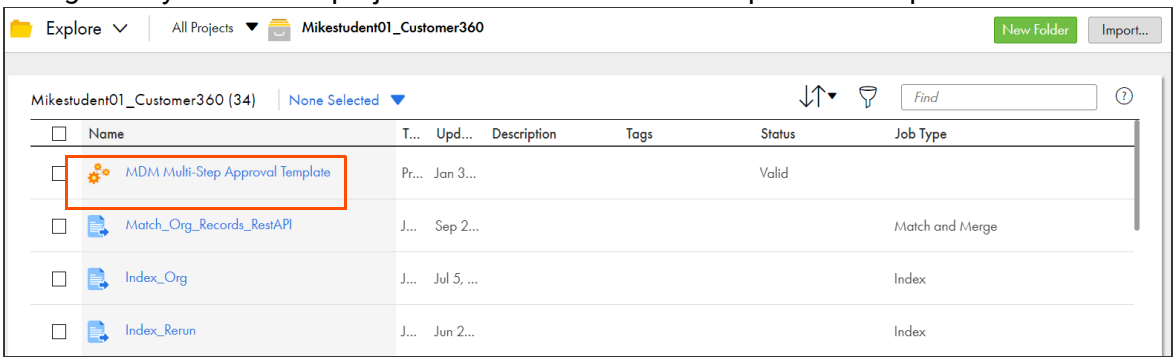


10. Choose your student folder and click **Select**.

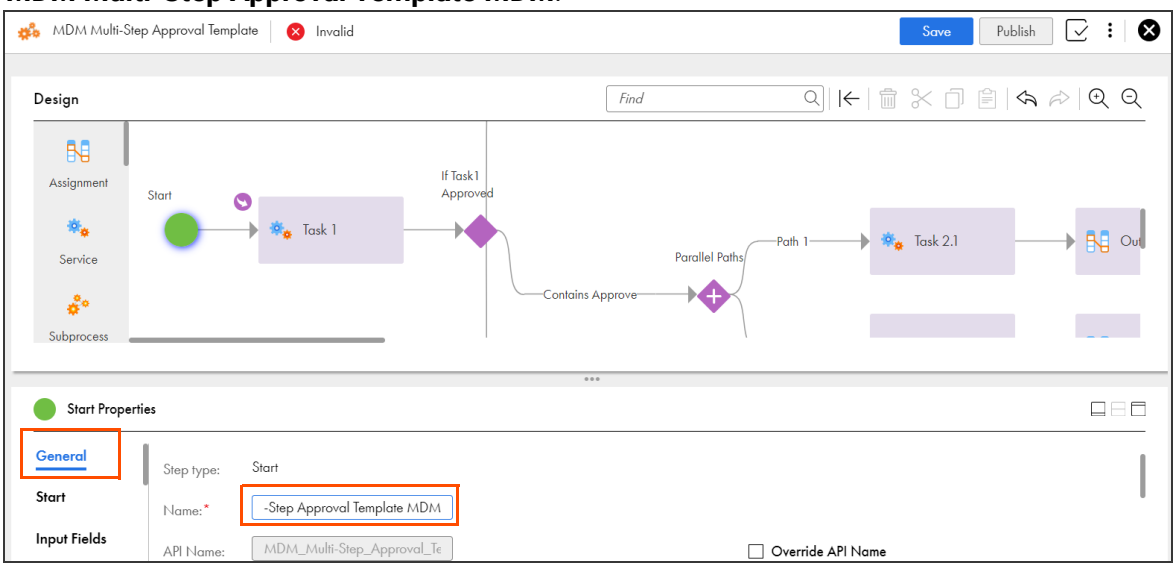


You are cloned the template and it is ready to use in the business event now.

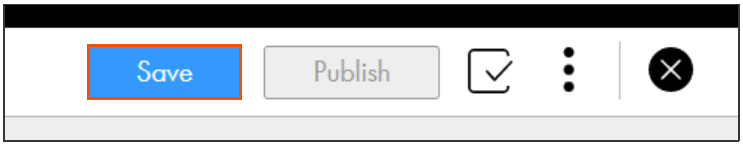
11. Navigate to your student project and click the cloned CAI process to open.



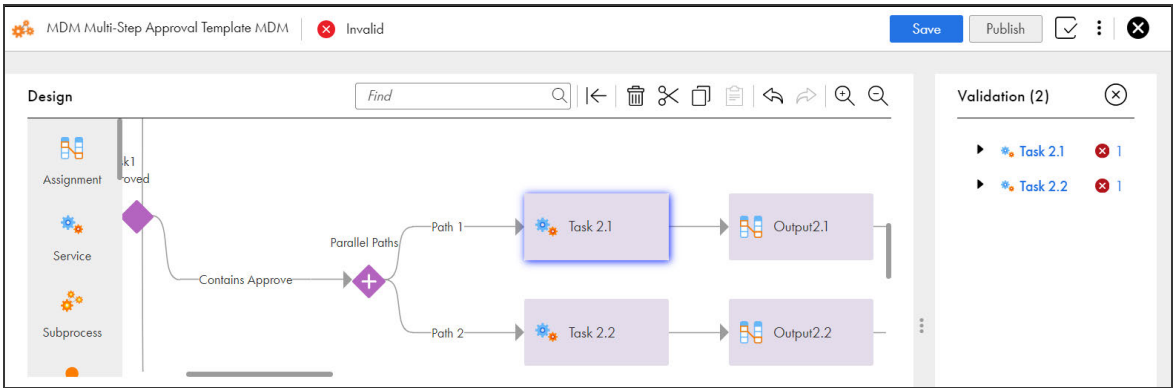
12. The process will open in a new tab. Under **Properties** > **General** section, enter the Name as **MDM Multi-Step Approval Template MDM**.



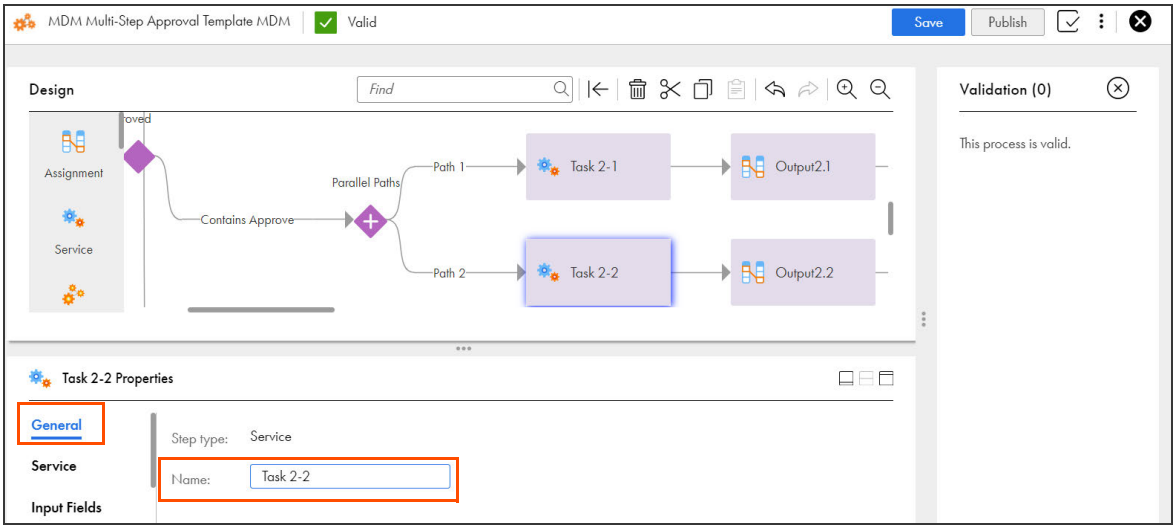
13. Click **Save**.



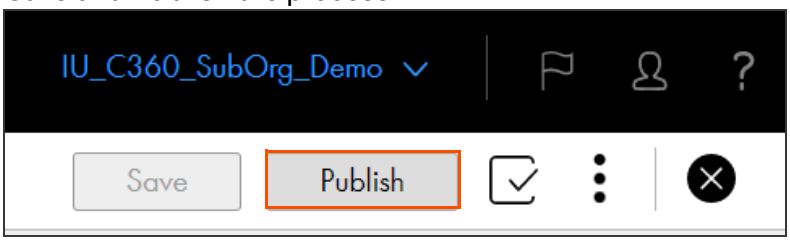
14. The process will show as Invalid. Click on the Validation icon. Here you will see the validation errors that needs to be corrected.



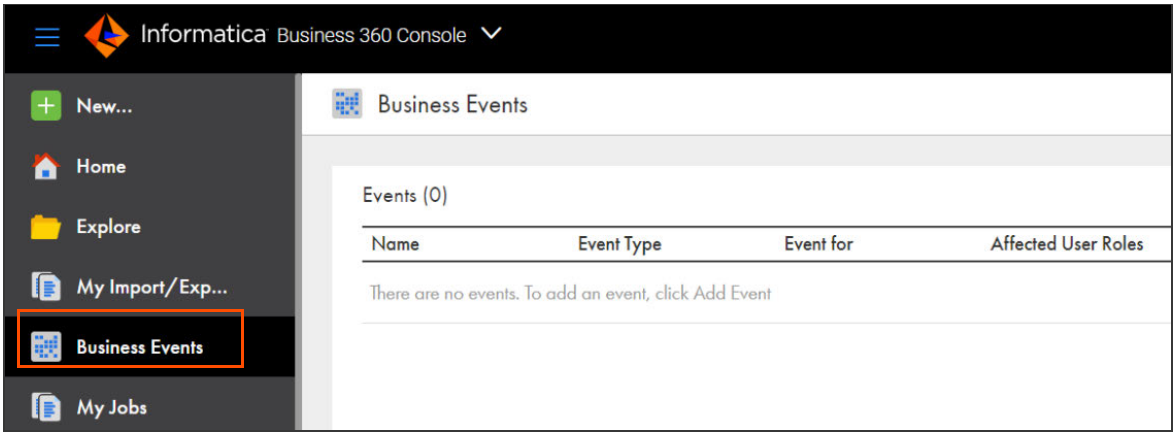
15. Rename the task 2.1 as 2-1 and task 2.2 as 2-2. To do that, go to the General section rename name the Name field.



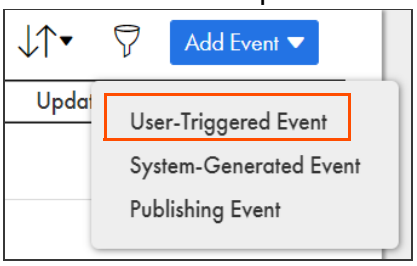
16. Save and **Publish** the process.



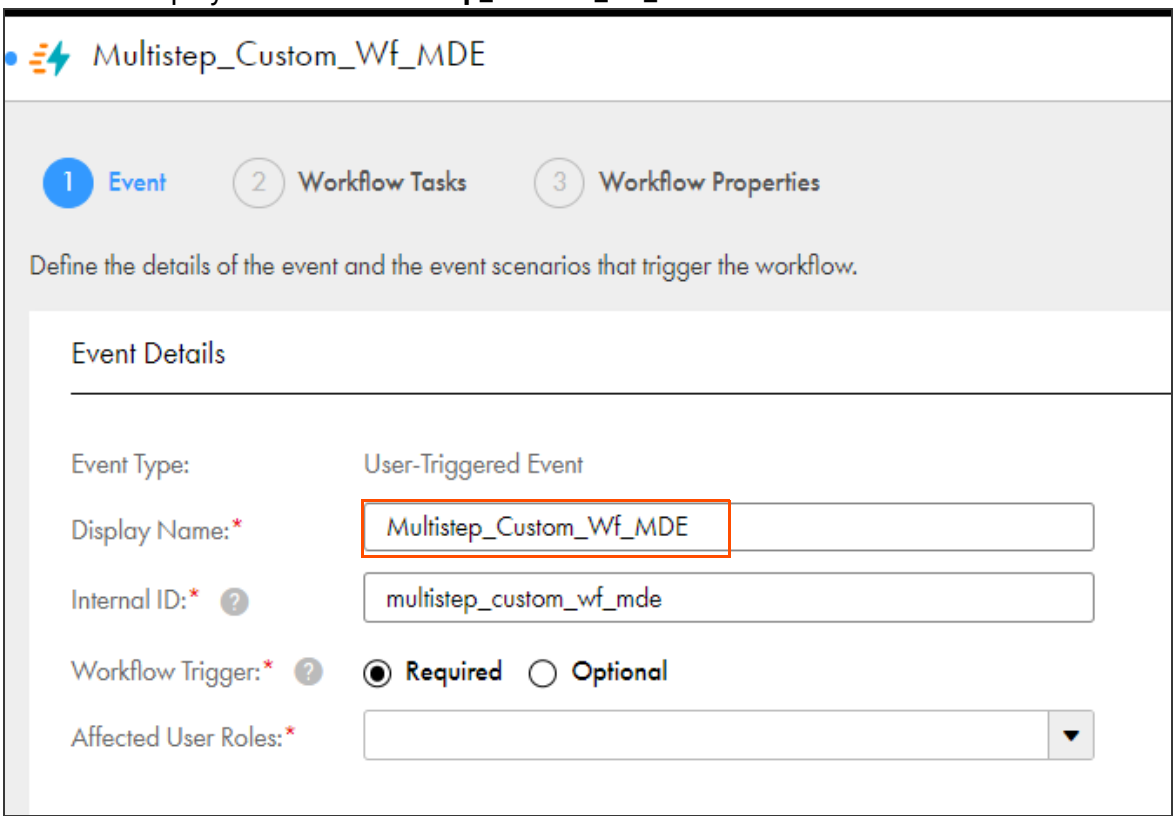
17. Navigate back to Business 360 Console and click **Business Events** from the left side menu.



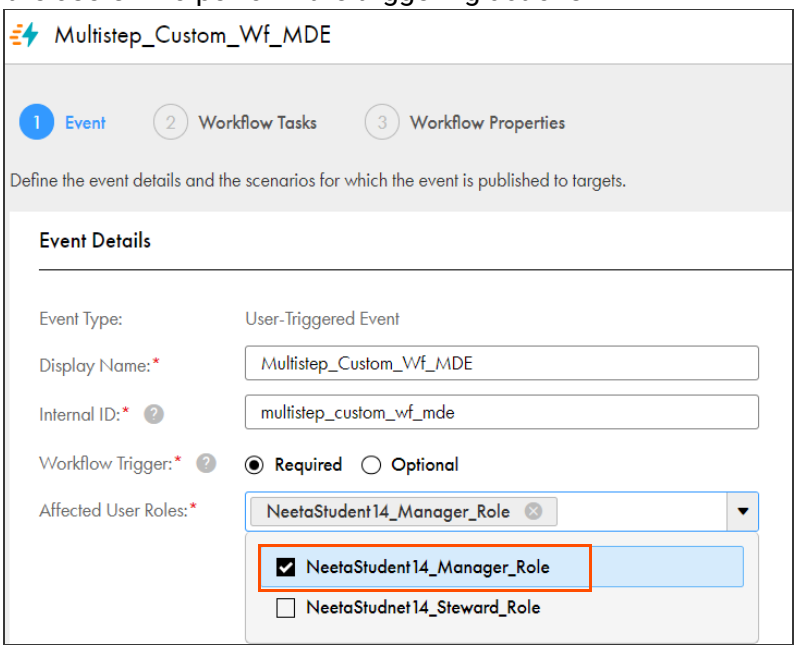
18. Click **Add Event** drop down and select **User-Triggered Event**.



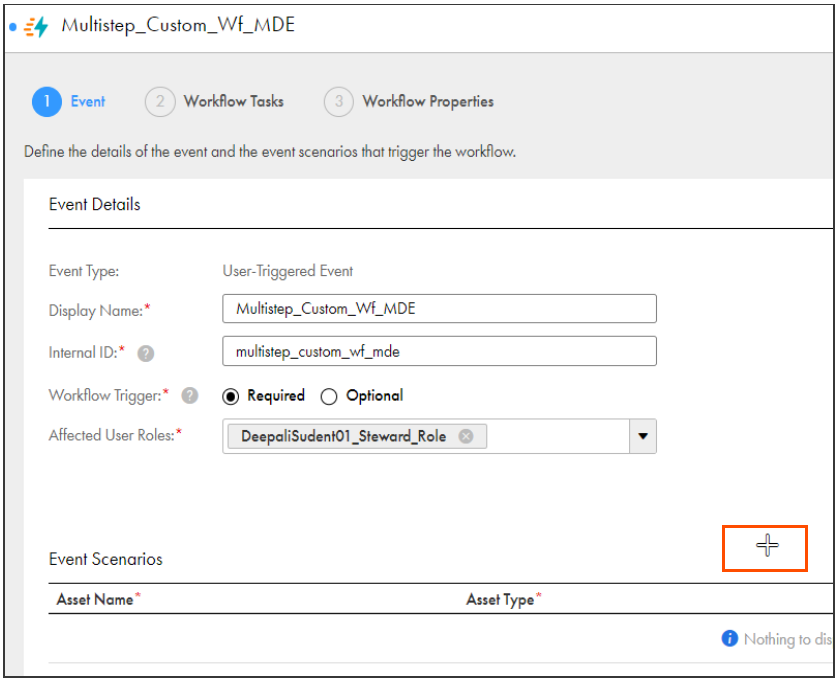
19. Enter the Display Name as **Multistep_Custom_Wf_MDE**.



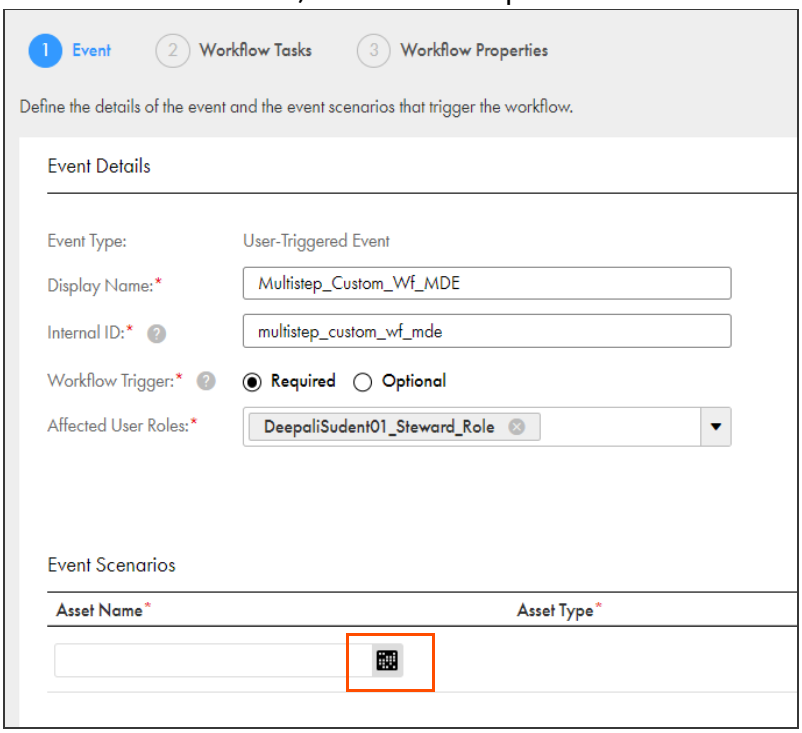
20. From the Affected User Roles drop down, select the custom role you have created for the steward. For example, **Studentnamestudent01_Steward_Role**. You can select any roles of the users who perform the triggering actions.



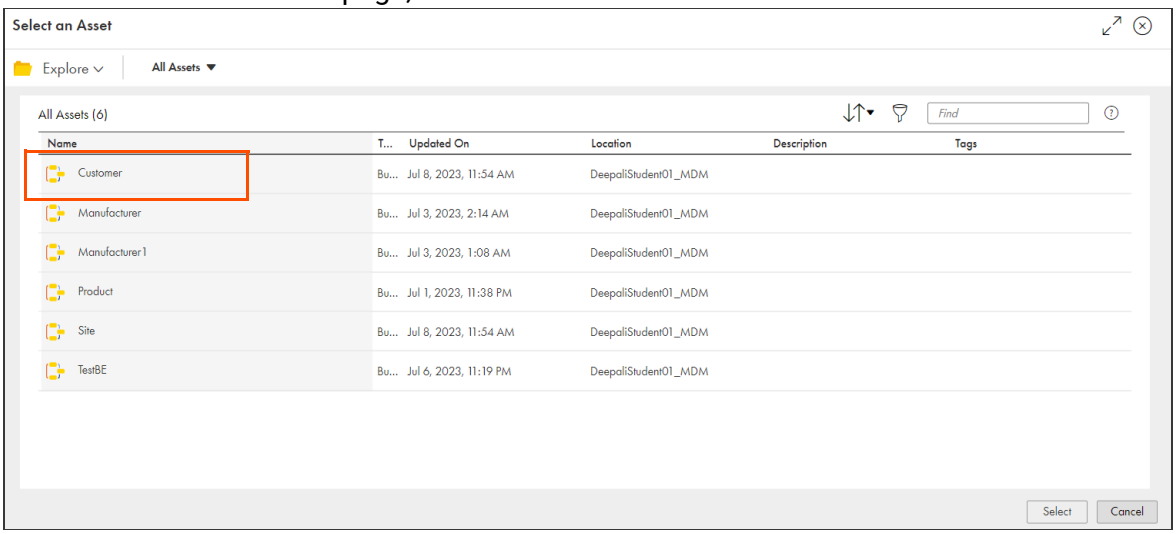
21. From the Event Scenarios section, click the **+** icon to create a new event scenario.



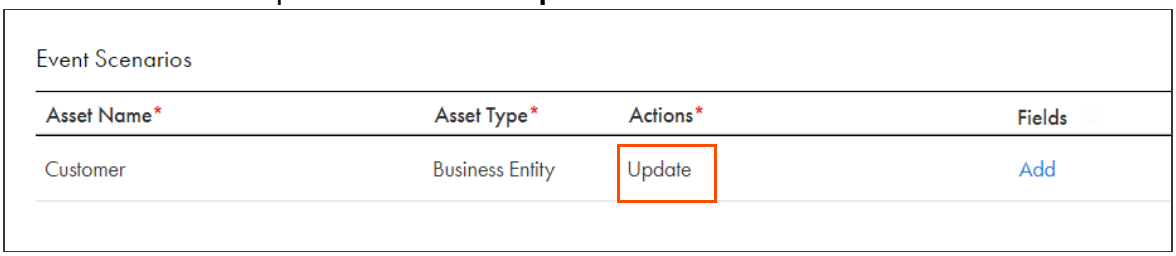
22. Under the Asset Name, click the asset picker icon.



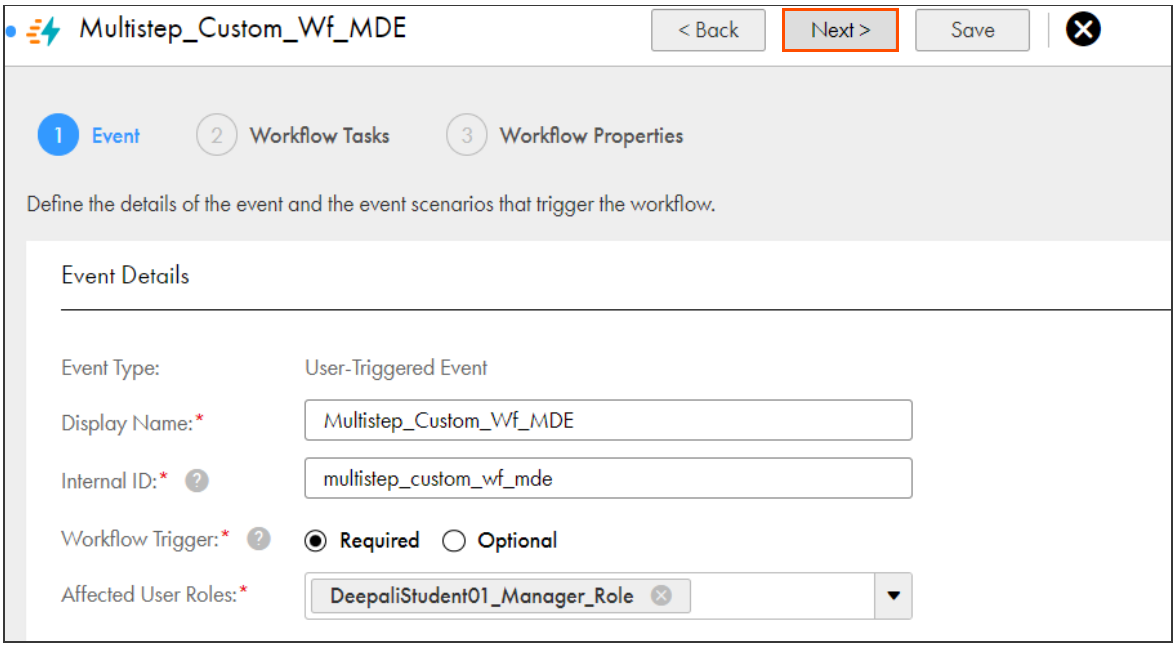
23. From the **Select an Asset** page, select **Customer** and click the **Select** button.



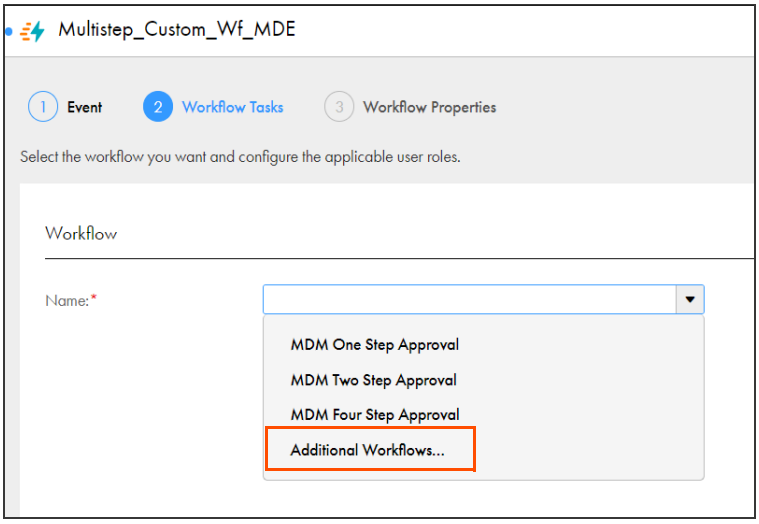
24. Click the **Actions** drop down and select **Update**.



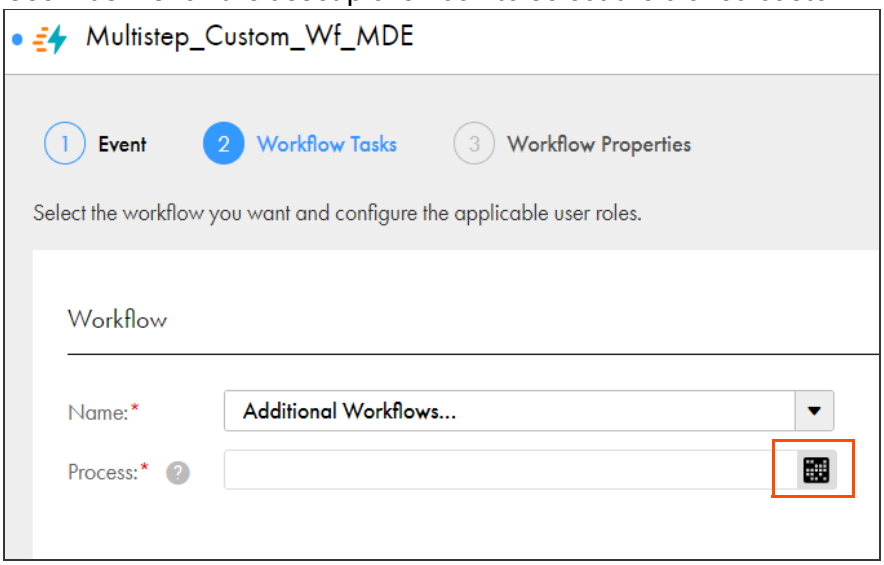
25. Click **Next**.



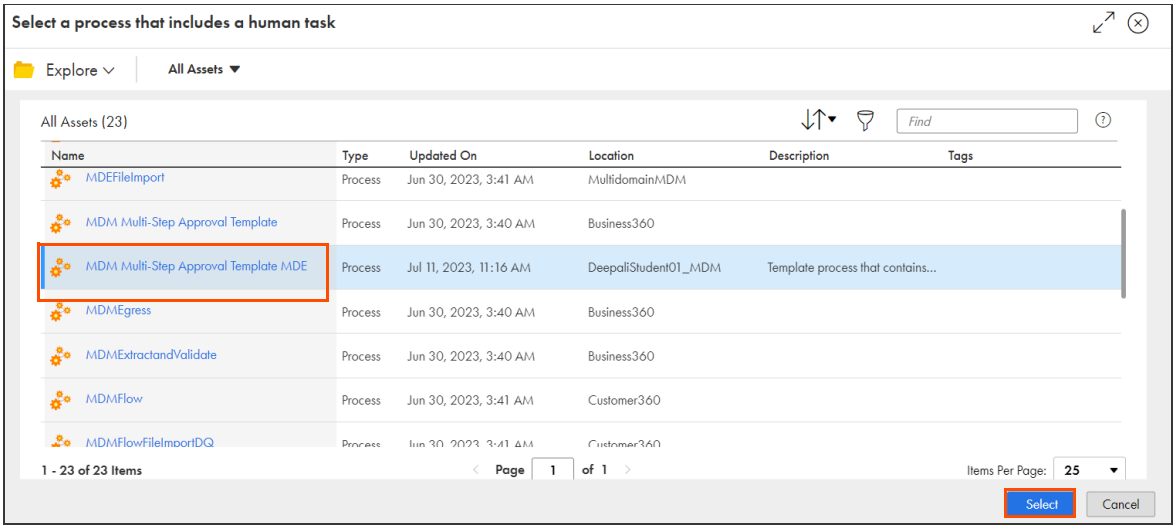
26. In the **Workflow Tasks** window, click the **Name** drop down and select **Additional Workflows**.



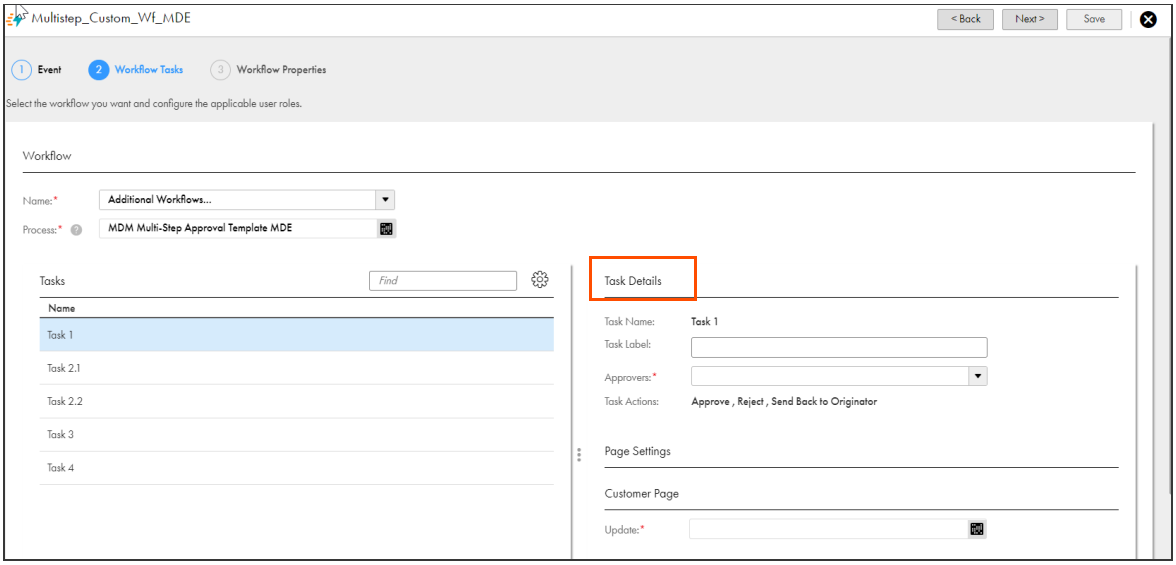
27. Now, you need to select the Cloud Application Integration process that contains at least one Service step with the service action property set to Generic User Task or Originator User Task. Click the asset picker icon to select the cloned custom workflow process.



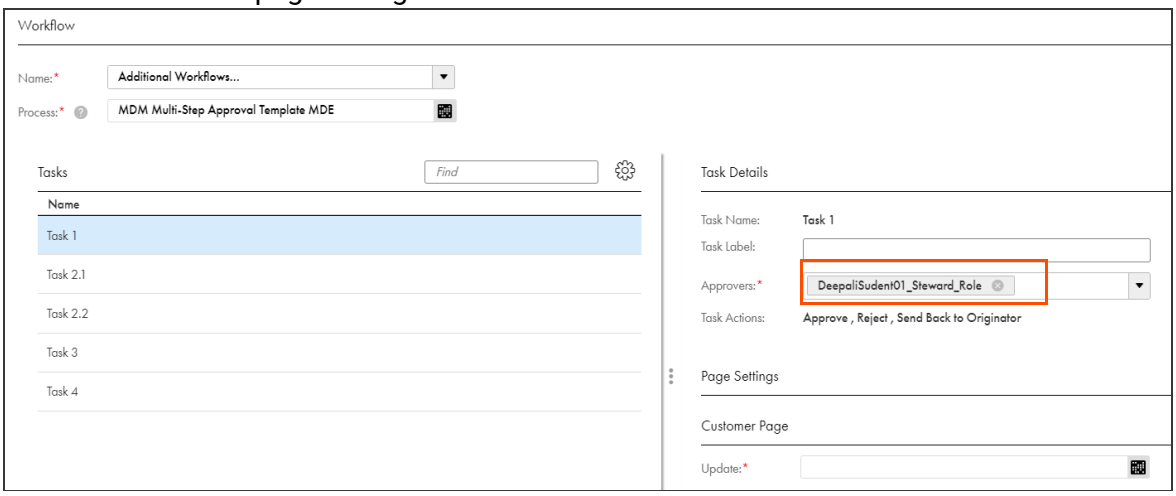
28. Scroll down and select the CAI process that you have copied to your student folder and renamed. Click **Select**.



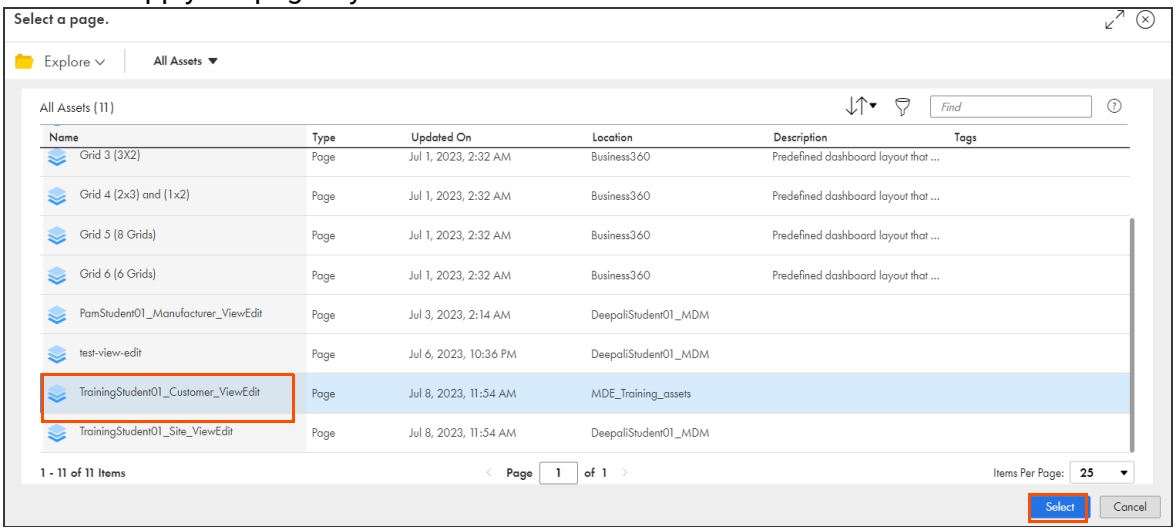
29. You can see the workflow configuration page. Here, you will configure the workflow tasks, task details and page settings. Under the **Tasks** section you can see all the tasks that are defined in the CAI process. In **Task Details** section, you can see the Task Label, Approvers and Task Actions. For example, if the Multidomain MDM Analyst has to review the customer data and update a few fields as part of the first task, we can label it as 'Analyst Approval Task' to make it easier to understand. Select the approvers who will take decisions on tasks in the Approvers section. The Task Actions are the decisions/actions made by the reviewers during the task review. The page settings allow you to specify which page should be visible to the reviewer while reviewing the task. For example, a reviewer may have to review and approve the social media details as part of their task. You can create a page that only includes the social media fields so the reviewer cannot see other details about the customer.



30. Select **Task 1** from the **Tasks** Section. Under **Task Details** section, select **studentnamestudent_Steward_Role** from the Approvers drop down. Under **Page settings**, click Asset Picker icon to select the page. Here you will select the default view page as there is no custom page configured so far.

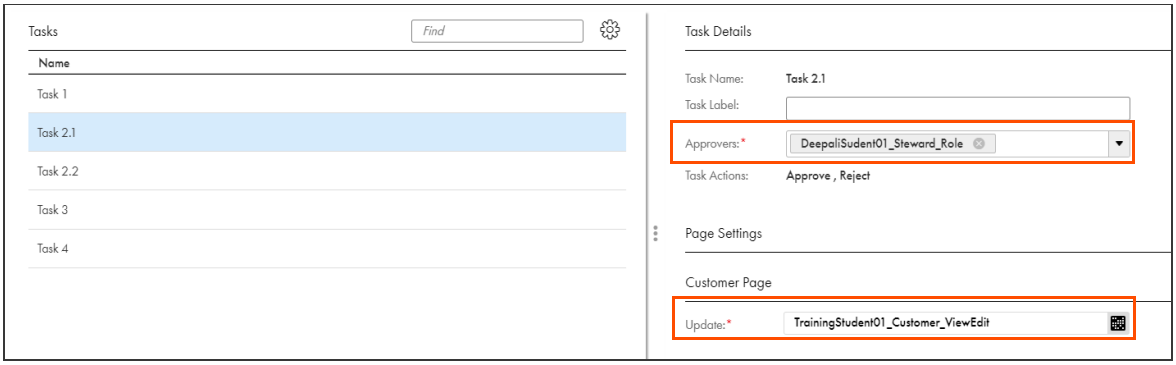


31. Choose **Studentnamestudent01_Customer_ViewEdit** page that you have created and click **Select** to apply the page layout to the task.

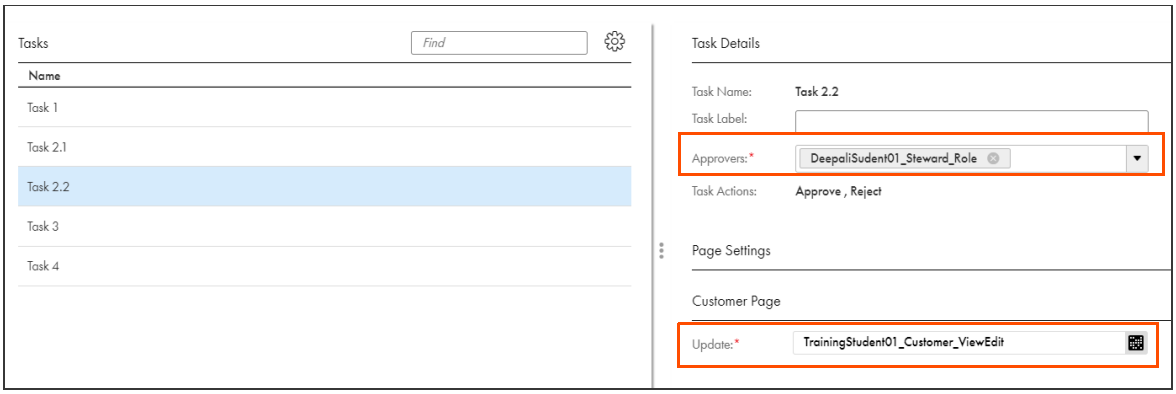


32. Similarly, configure the following properties to the remaining tasks.

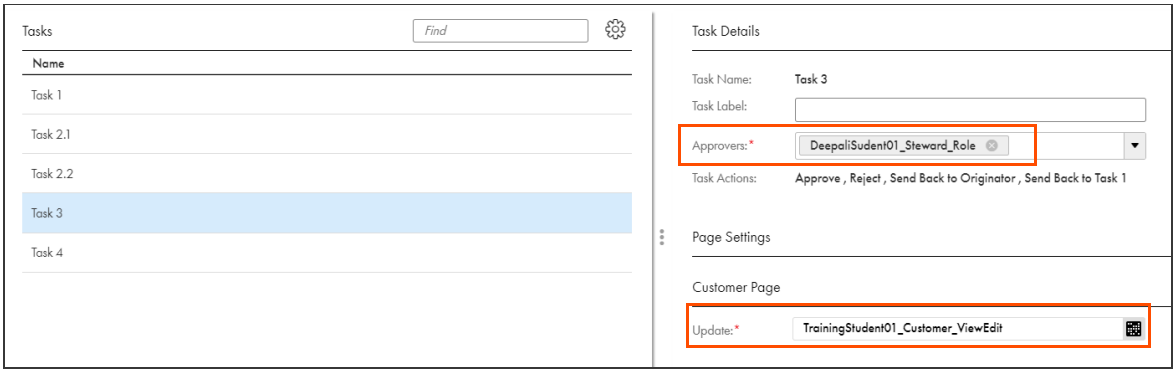
Task 2.1	
Approvers	**StudentnameStudent_Steward_Role**
Page Settings	**StudentnameStudent01_Customer_ViewEdit**



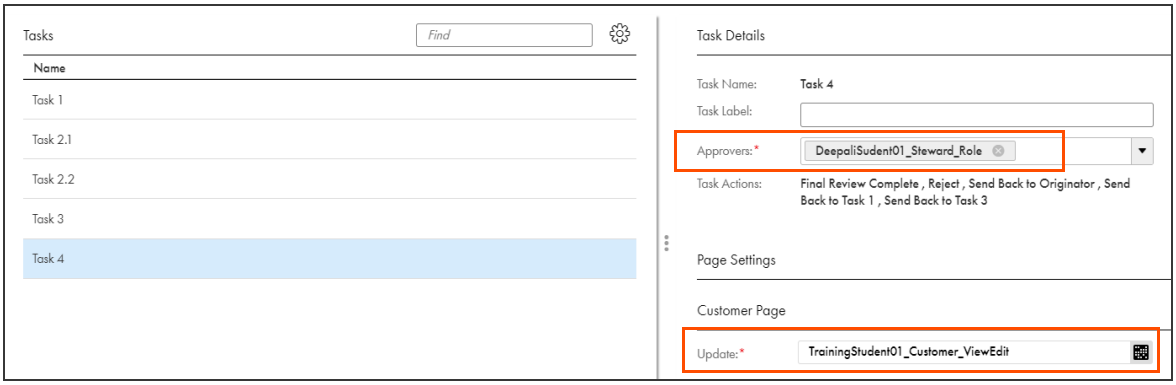
Task 2.2	
Approvers	**StudentnameStudent_Steward_Role**
Page Settings	**Studentnamestudent01_Customer_ViewEdit**



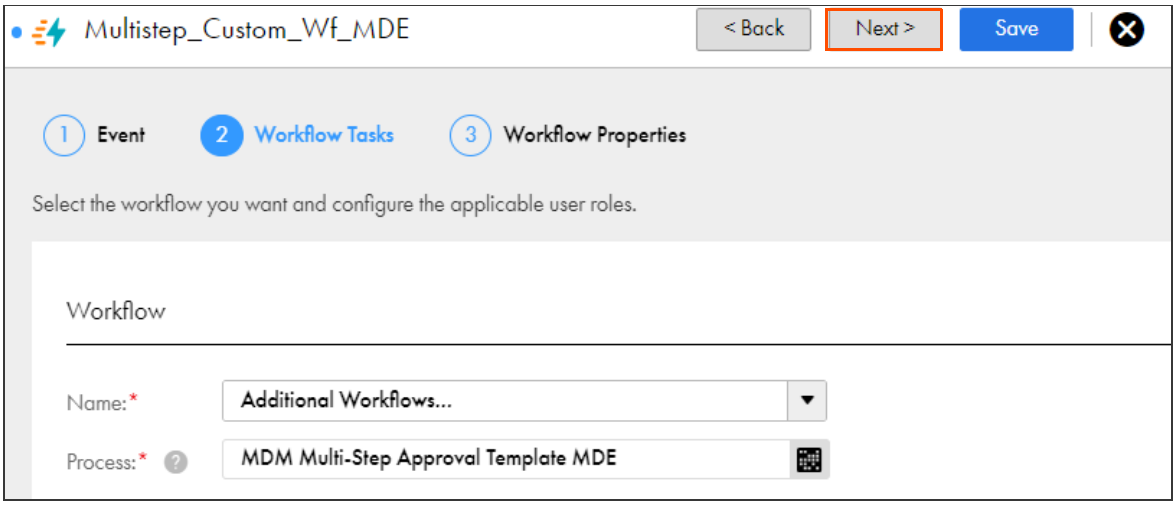
Task 3	
Approvers	**StudentnameStudent_Steward_Role**
Page Settings	**Studentnamestudent01_Customer_ViewEdit**



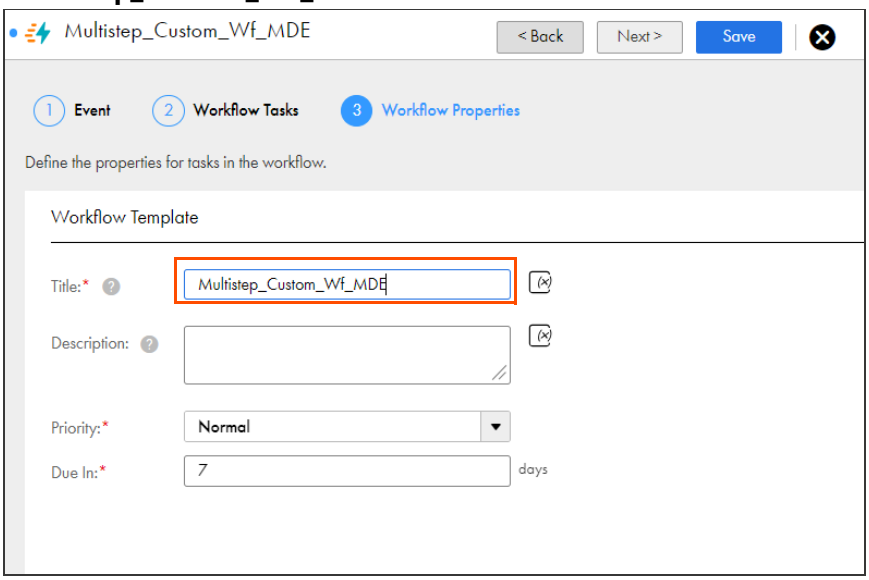
Task 4	
Approvers	**StudentnameStudent_Steward_Role**
Page Settings	**Studentnamestudent01_Customer_ViewEdit**



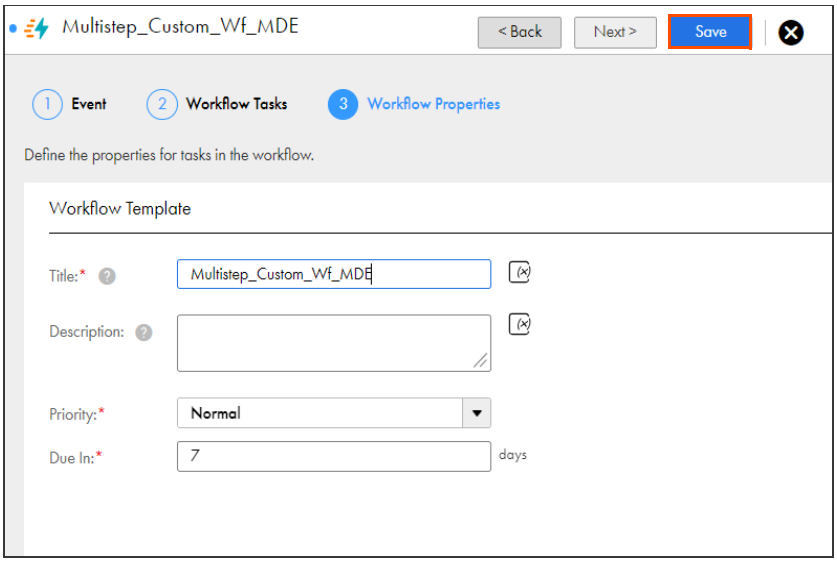
33. Click **Next**.



34. In the **Workflow Properties** step, under the **Workflow Template** section, enter the Title as **Multistep_Custom_WF_MDE**.

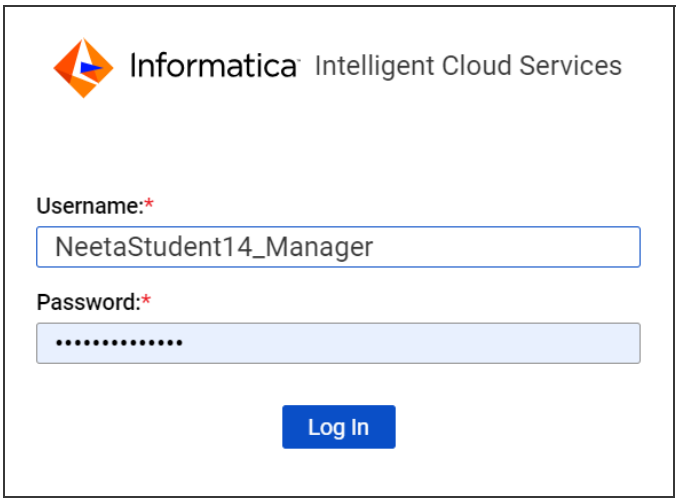


35. Retain the other fields and click **Save**.

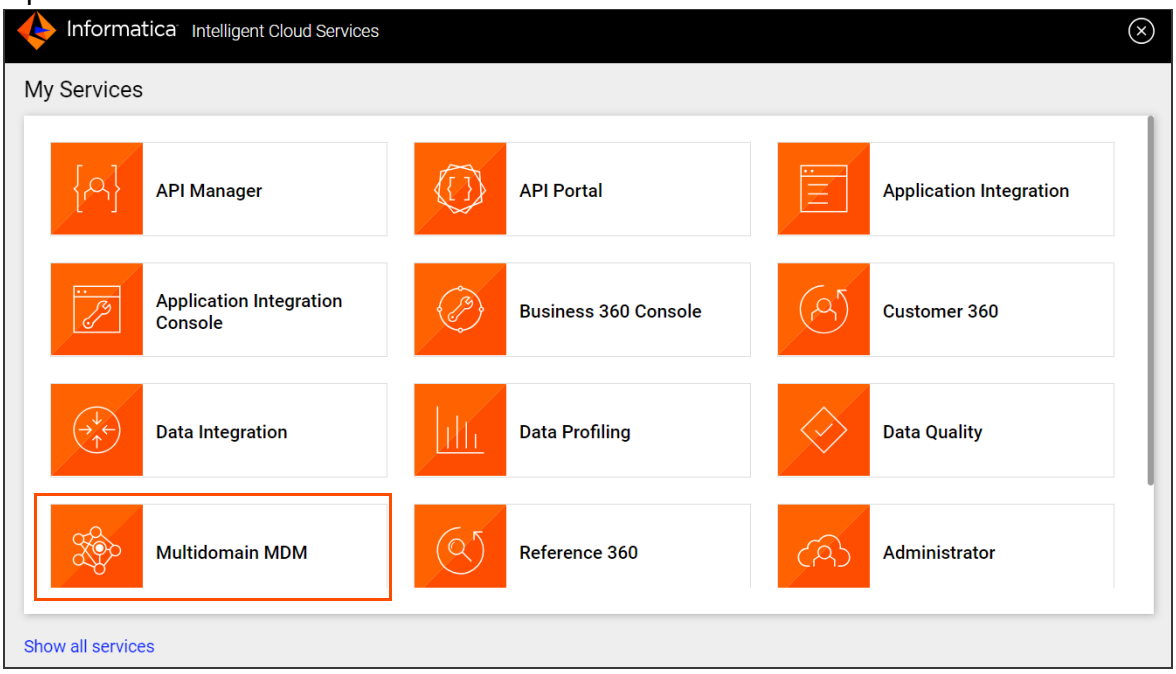


The Business event was successfully added.

36. Log out as the admin user and log in as manager.



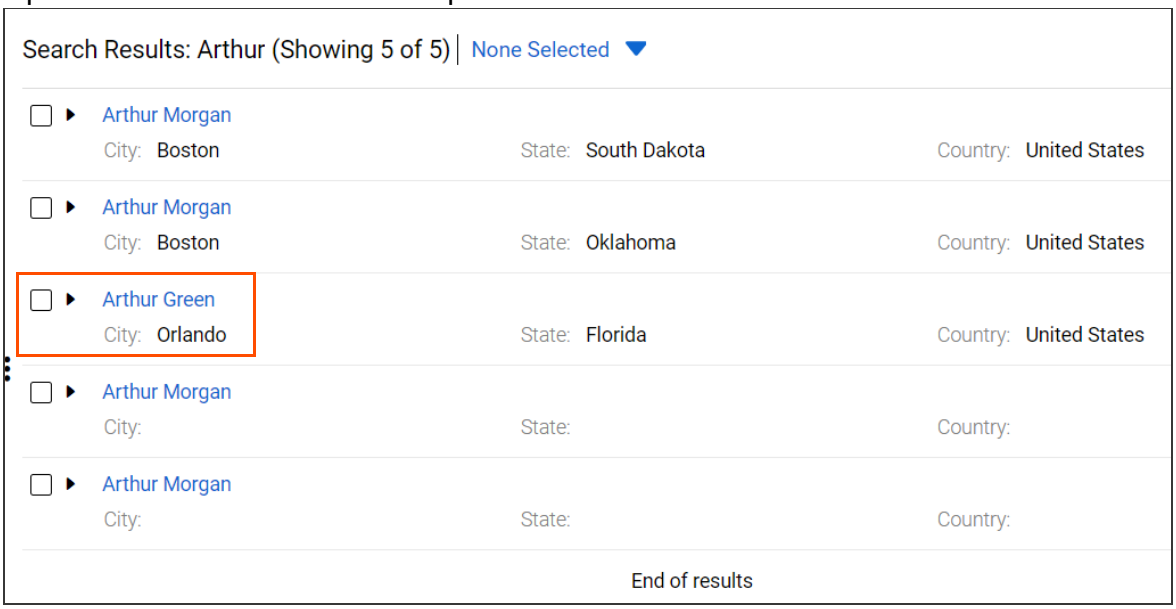
37. Open **Multidomain MDM** service.



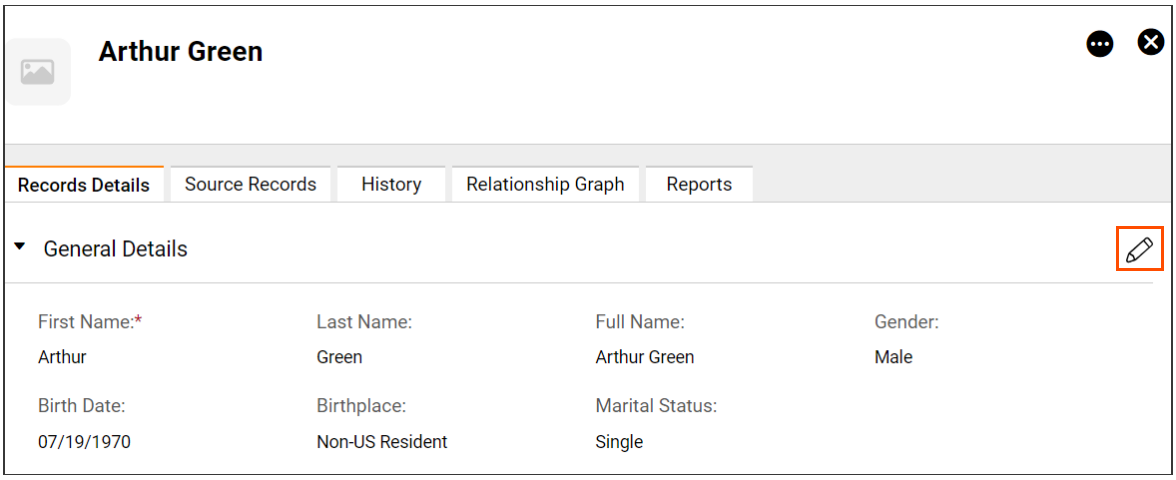
38. Click **Search**.

39. Search for the Customer business entity **Arthur**.

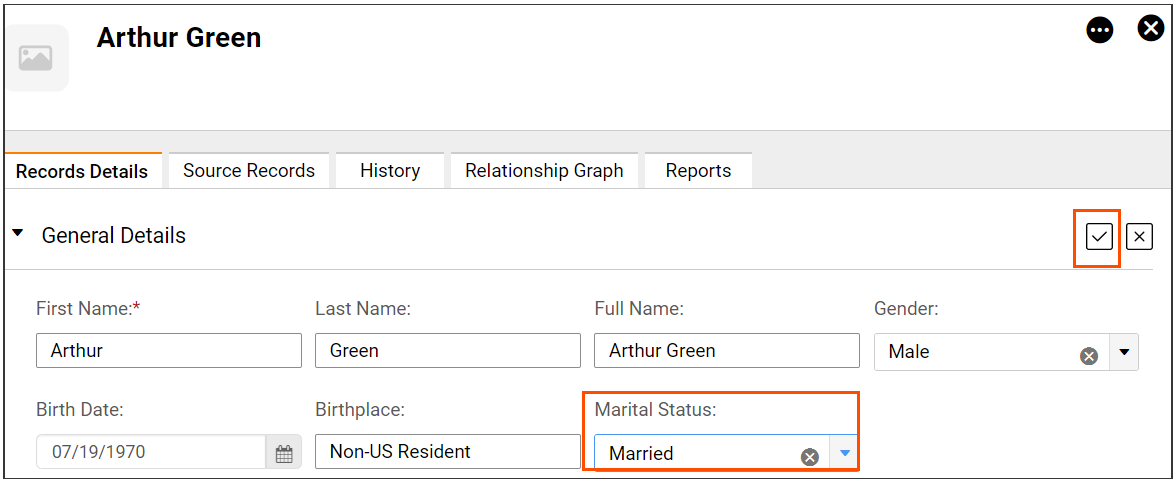
40. Open the **Arthur Green** record to update the details.



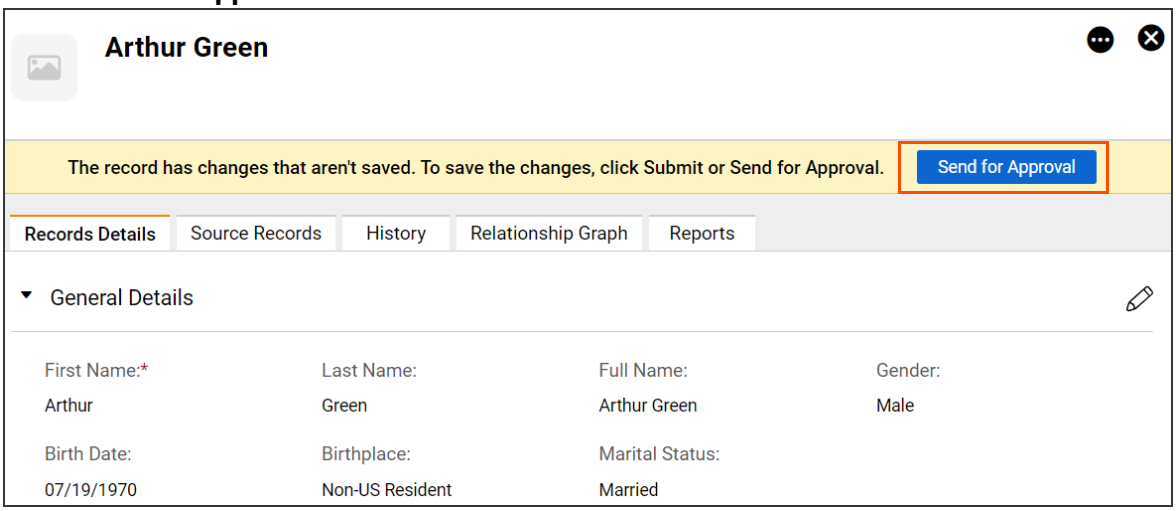
41. Click **Edit** icon to edit the record.



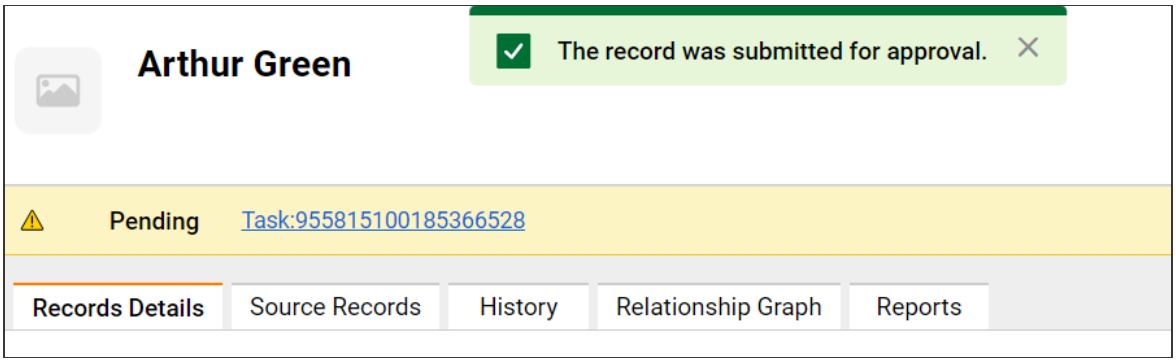
42. From the Marital Status drop-down select **Married**. Click tick icon to save the changes.



43. Click **Send for Approval** to create a task and initiate the workflow.



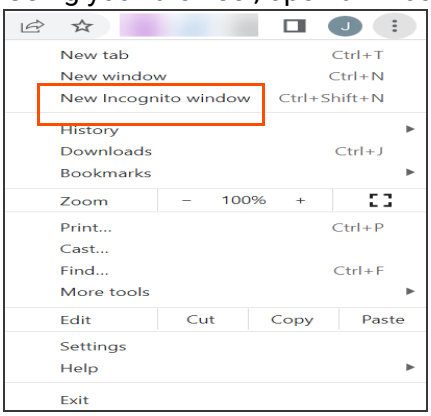
The record will be sent for approval.



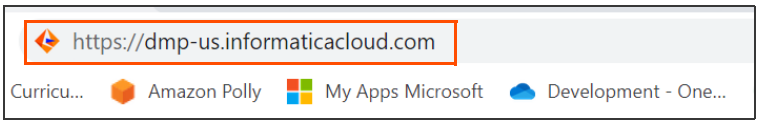
Approving the Workflows

You will now login as the steward user and approve all the tasks.

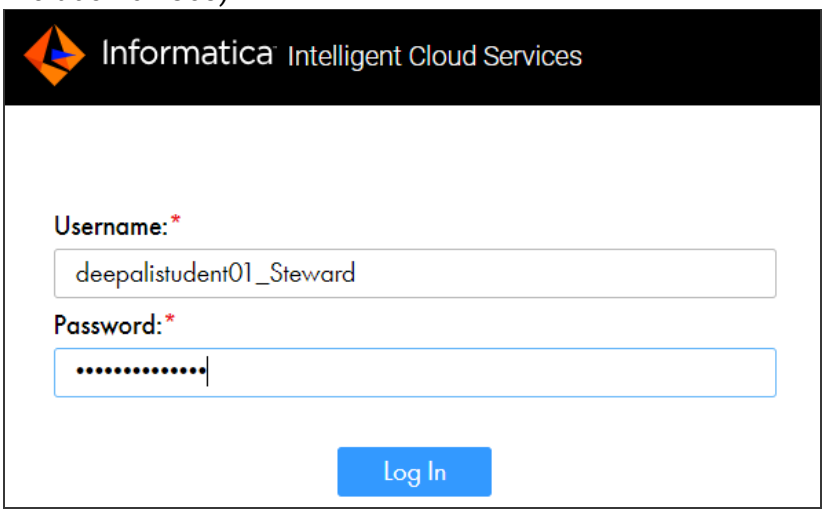
44. Using your browser, open an incognito window.



45. In the browser window, enter the Informatica IICS URL https://dmpus.informaticacloud.com.



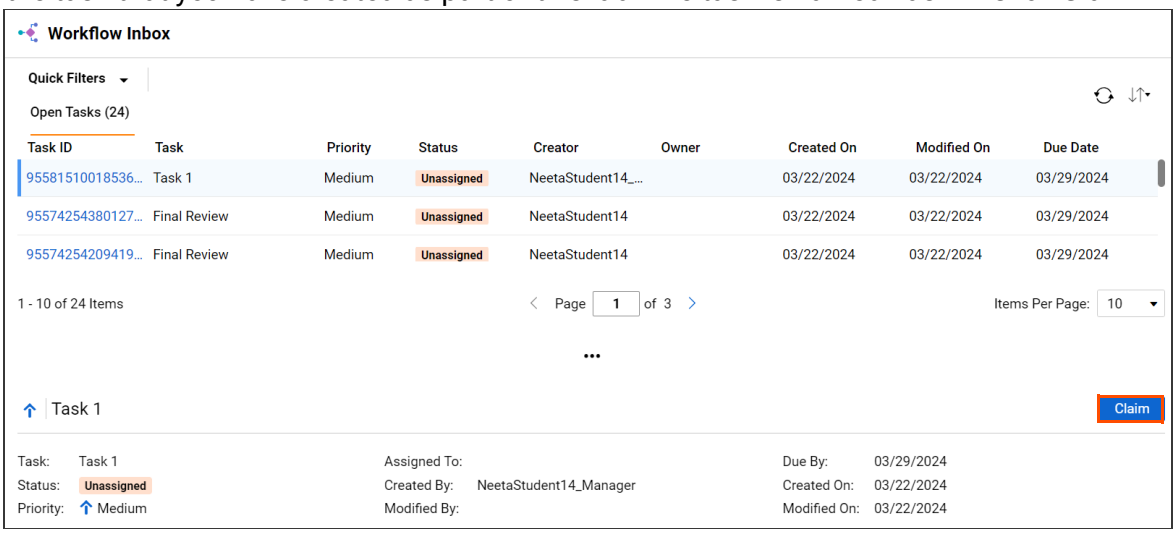
46. The **Informatica Intelligent Cloud Services (IICS)** login window opens. Login as the steward user that you have created. (Studentnamestudent<no>_Steward/ Multidomain360).



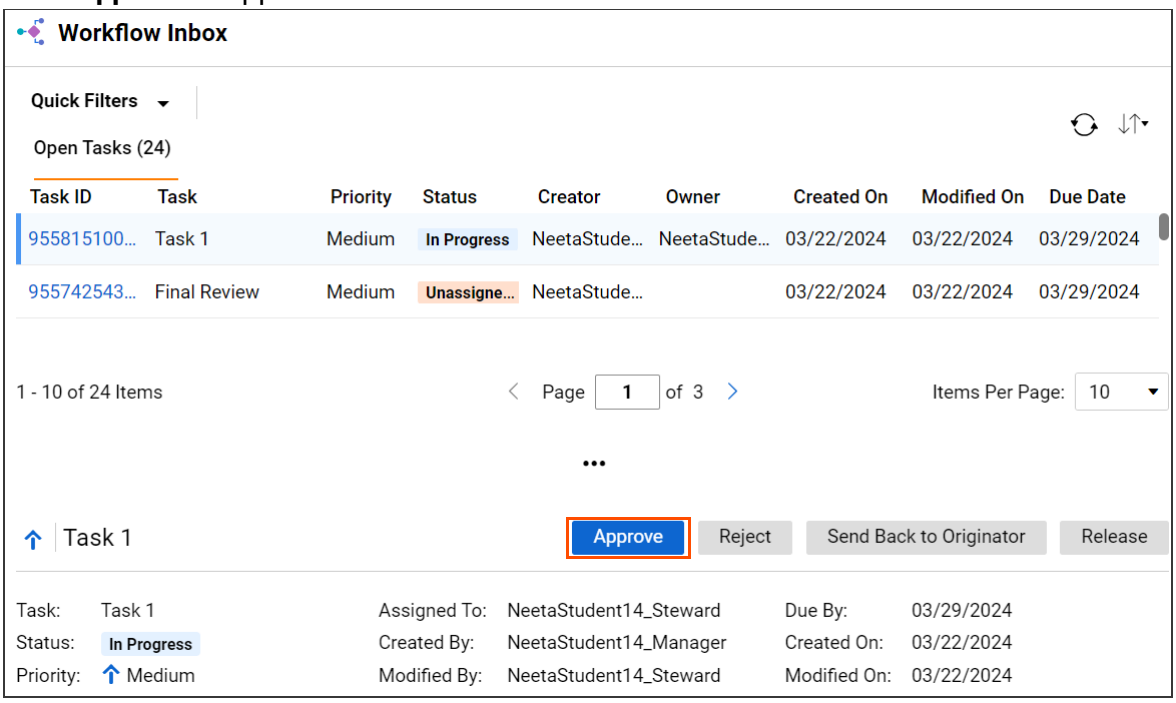
47. From the **Services** window, select **Multidomain MDM**.

48. Click **Workflow Inbox**.

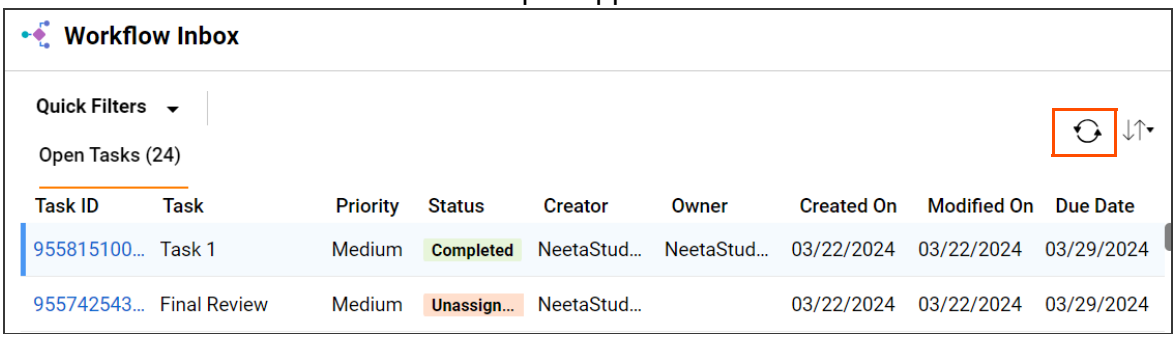
49. There are multiple tasks that are in queue for the steward review. The first one in the row is the task that you have created as part of this lab. The task is named Task 1. Click **Claim**.



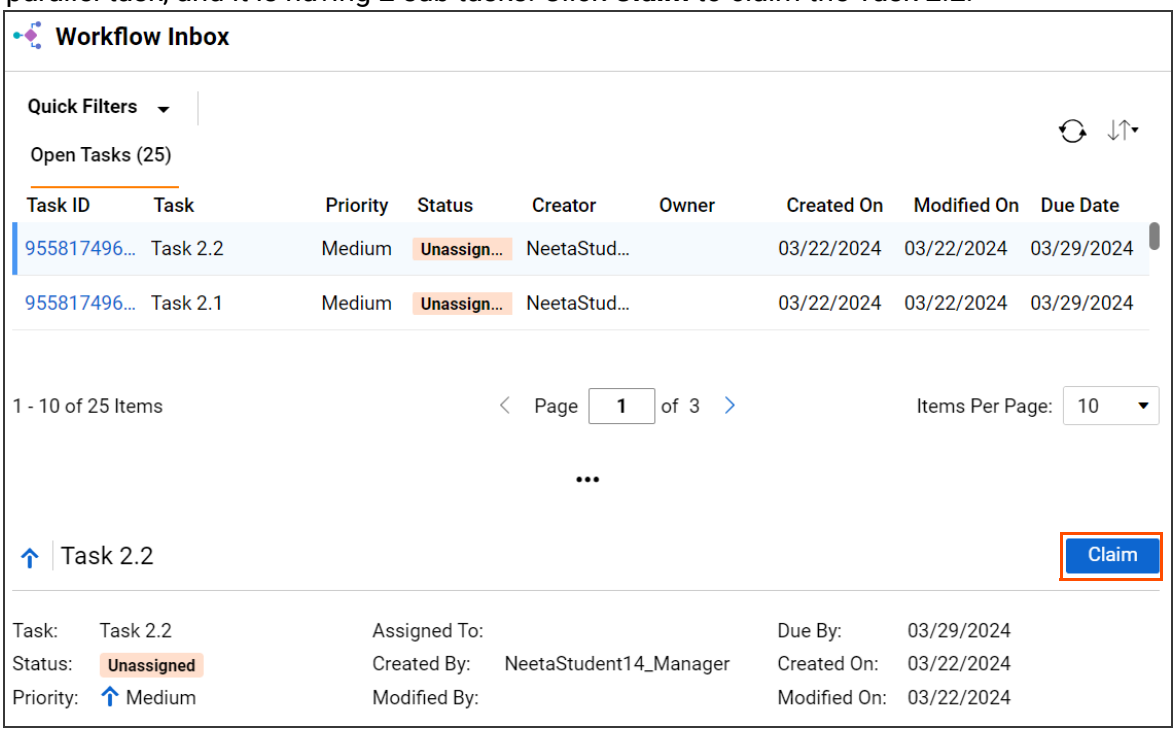
50. Click **Approve** to approve the task.



51. Refresh the task to move to the next step of approval.

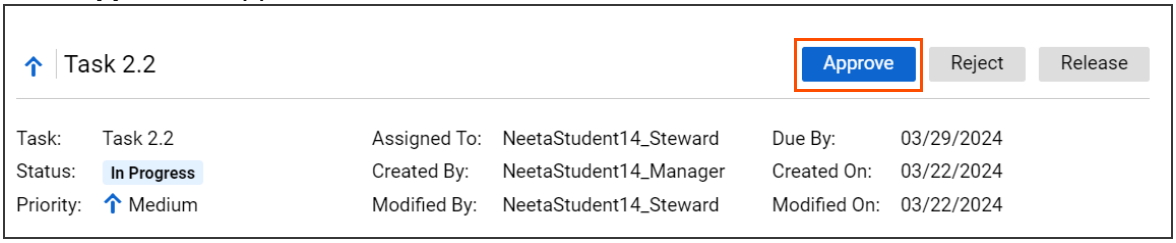


52. You can see that the task is completed, and the second task is created. Second task is a parallel task, and it is having 2 sub tasks. Click **Claim** to claim the Task 2.2.

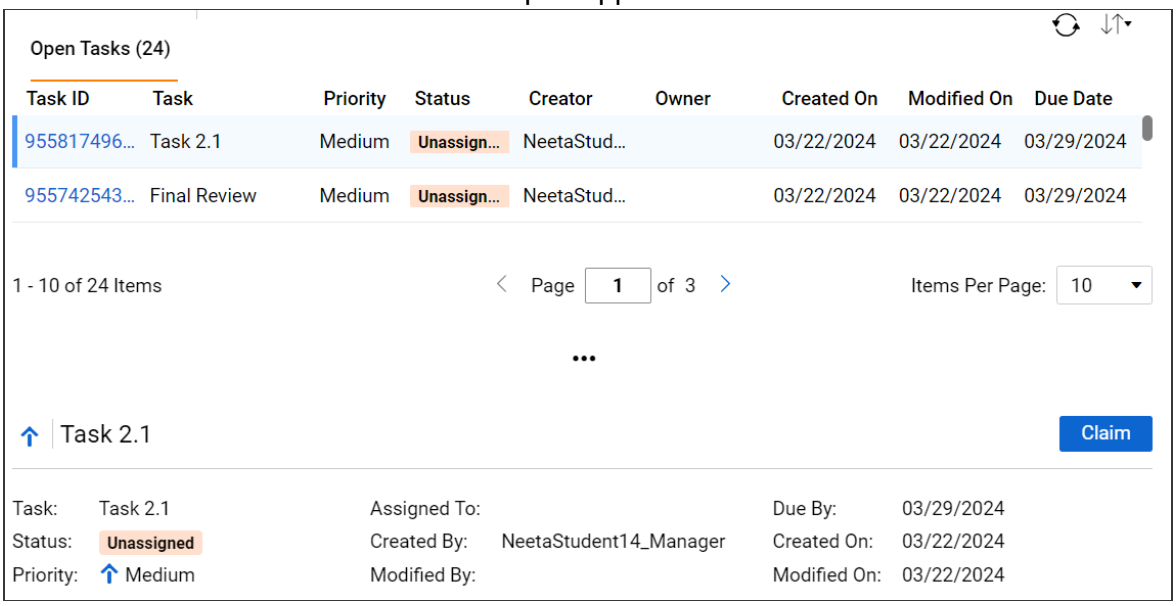


Click refresh icon if you do not see the second task.

53. Click **Approve** to approve the task 2.2.

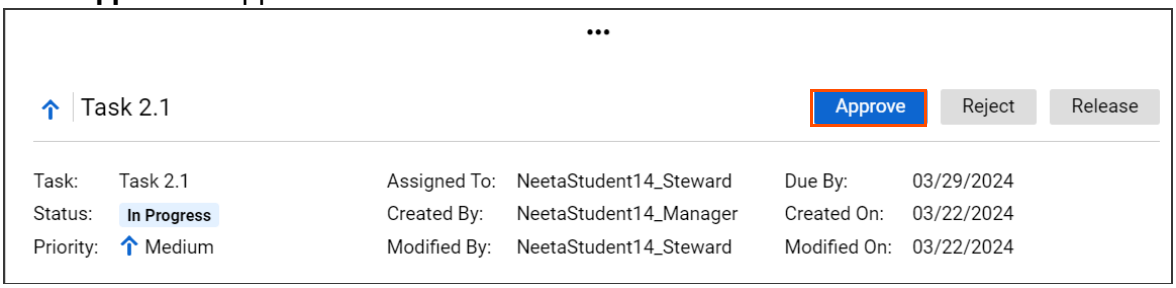


54. **Refresh** the task to move to the next step of approval. Click **Claim** to claim the Task 2.1.

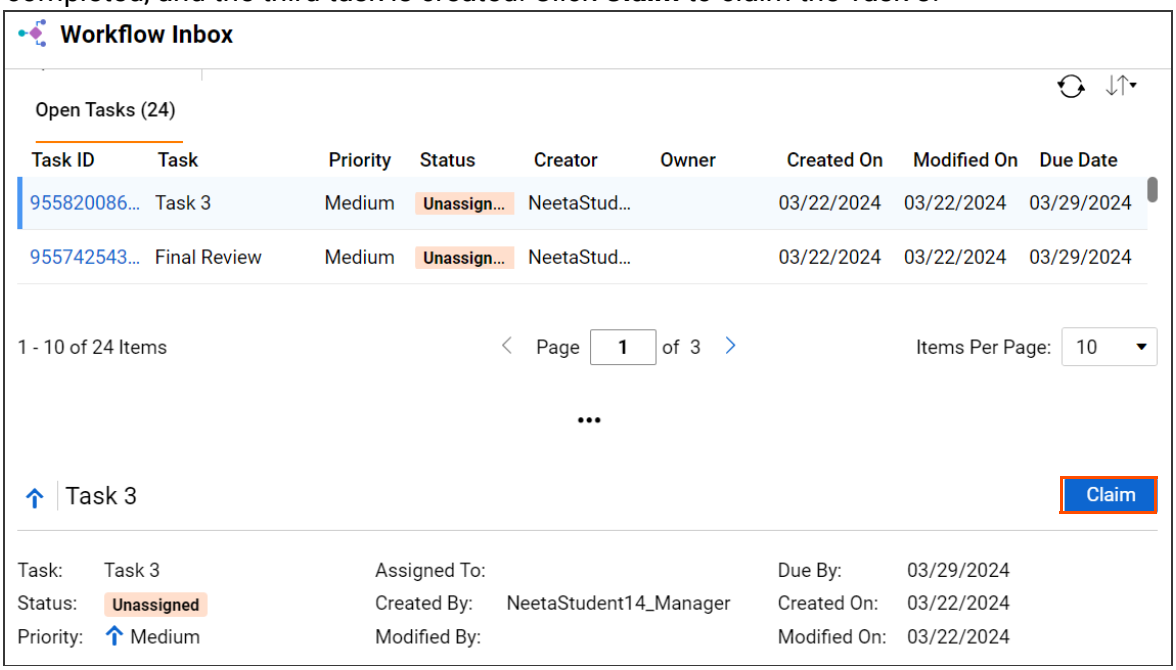


Click refresh icon if you do not see the second task.

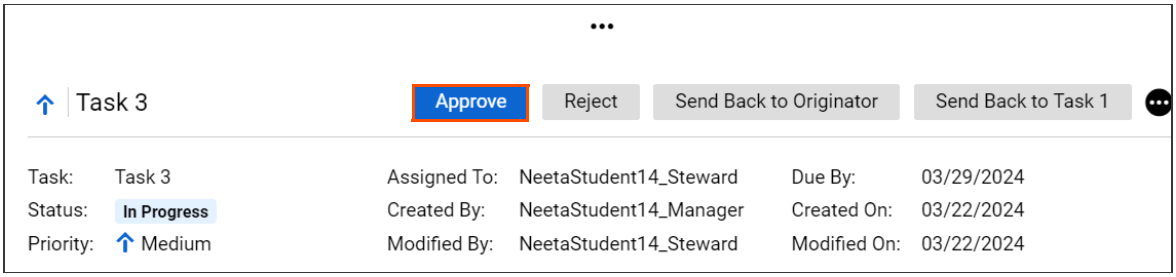
55. Click **Approve** to approve the Task 2.1.



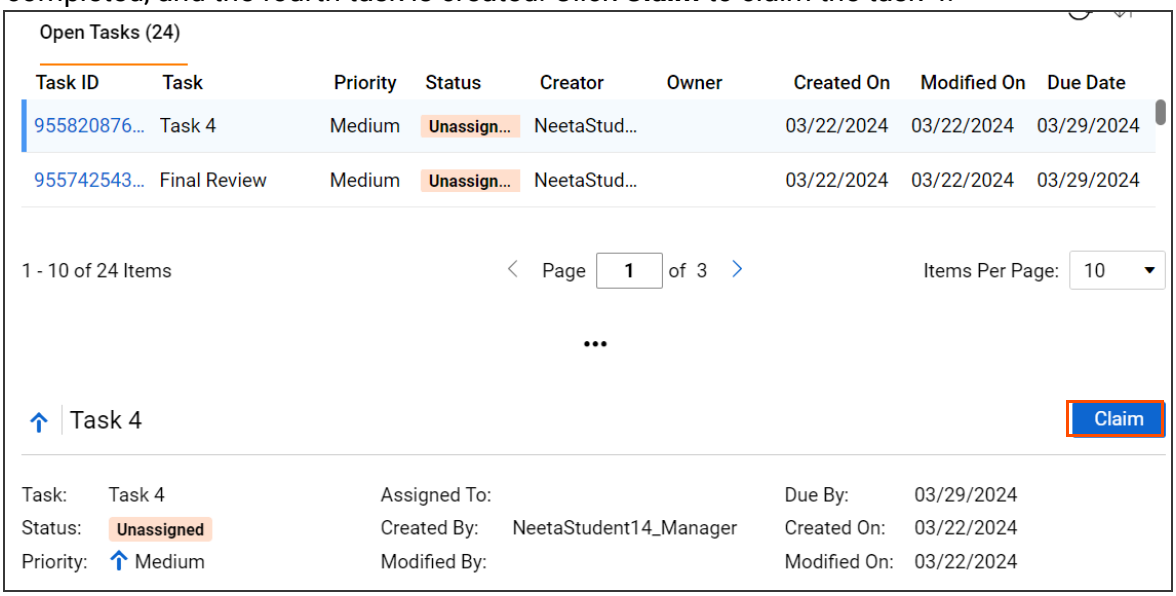
56. Refresh the task to move to the next step of approval. You can see that the Task 2 is completed, and the third task is created. Click **Claim** to claim the Task 3.



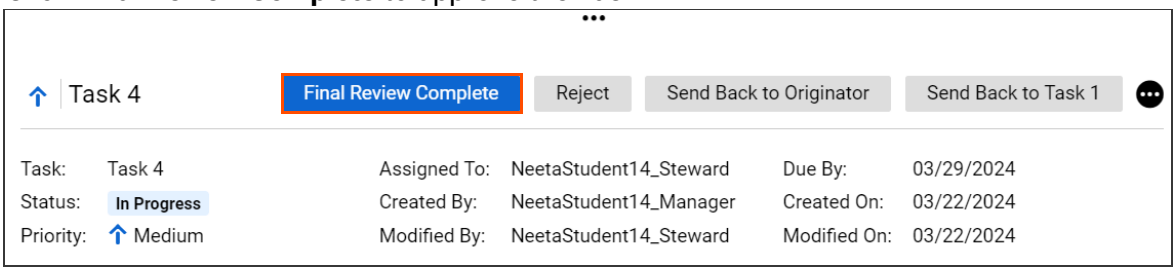
57. Click **Approve** to approve the Task.



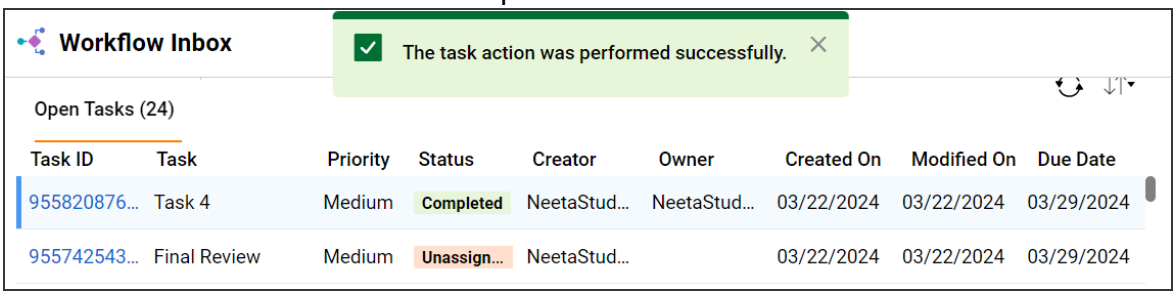
58. Refresh the task to move to the next step of approval. You can see that the task is completed, and the fourth task is created. Click **Claim** to claim the task 4.



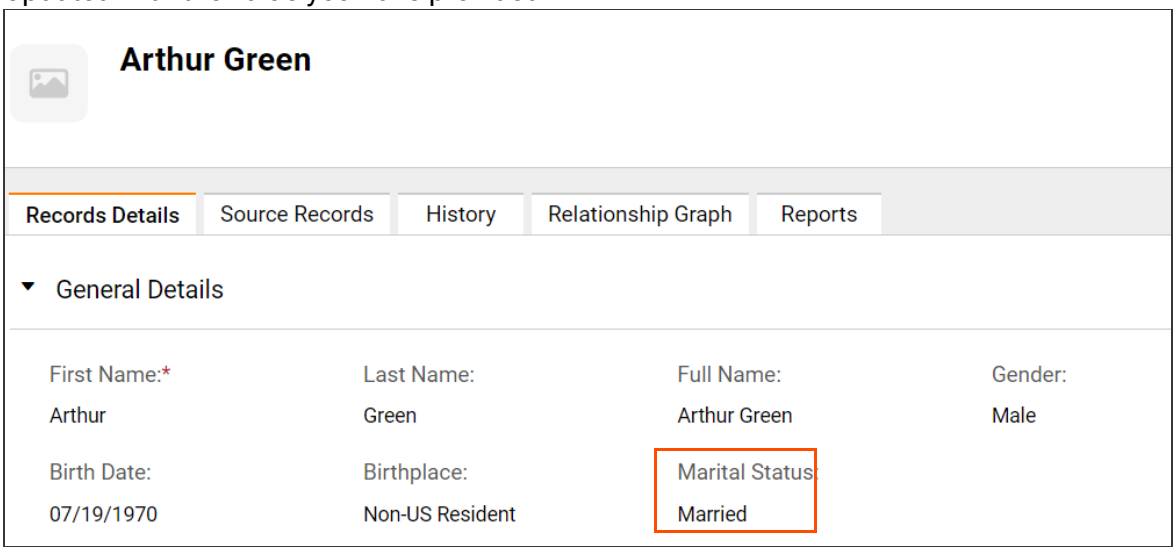
59. Click **Final Review Complete** to approve the Task 4.



You can see that all the tasks are completed.



60. Navigate back to the main browser where you have opened Multidomain MDM application. Search and open the Arthur Green record. You can see that the Marital Status field is updated with the value you have provided.



This concludes the lab.

Key Takeaways

- The match and merge job depends on the match and merge process.
- To generate merge tasks for pairs identified by the match process and require a review, define and run a generate merge tasks job.
- User-triggered event allows the user to create event scenarios and the user can choose to create, update, or unmerge records.
- System-generated triggers the event with only one action – Resolve Potential Matches.
- At any point if the user realizes that the records were incorrectly merged, the user can unmerge the records.
- Custom workflows are useful when you want to customize the workflows as per the requirement

Module 12: Managing Hierarchies and Analyzing Data Model

Managing Hierarchies

Once the hierarchy model is ready, you can create an instance of the hierarchy in Multidomain MDM and add parent and child node to it. Refer to the following steps to create a hierarchy instance.

High-level step to create a hierarchy:

1. Create a hierarchy
2. Add first-level nodes
3. Add and manage child records

Step 1: Create a hierarchy

1. Click **New**.
2. Click **Hierarchy** > **Hierarchy Model** > **Create**.
3. Specify the hierarchy properties.
4. Click **Create**.

Step 2: Add first-level nodes

1. On the hierarchy page, click **Add First Level Nodes**.
2. Find and select the records that you want to define as first-level nodes, and then click Select.

Step 2: Add and manage child records

1. On the hierarchy page, hover over a record, and click **Add**.
2. Select the record type that you want to add and click **Next**.
3. Select the relationship type that you want to add and click **Next**.
4. Find and select the related records that you want to add, and then click **Next**.
5. Click **Add**.
6. To submit your draft hierarchy, click **Submit**.

Analyzing Data

You can visualize and explore data in the Relationship Graphs tab. You can configure a Relationship Graphs component to visualize and explore the relationships of a record.

You can filter relationships that appear on the relationship graph canvas based on business entities or relationship types. You can also filter the records that appear on the relationship list panel by record type, relationship type, and record visibility on the canvas.

To analyze the data perform the following steps:

1. Click **Hierarchies**
   The Hierarchies page appears.
2. Expand the hierarchy model to which the hierarchy is associated.
   The hierarchies associated with the hierarchy model appear.
3. Select the hierarchy that you want to view.
   The hierarchy properties appear.
4. Click **Open**.
   The hierarchy appears on a tab.
5. Optionally, for a hierarchical record, filter the child and parent records that appear on the relationships list panel.

Figure 12-1 shows a sample relationship graph.

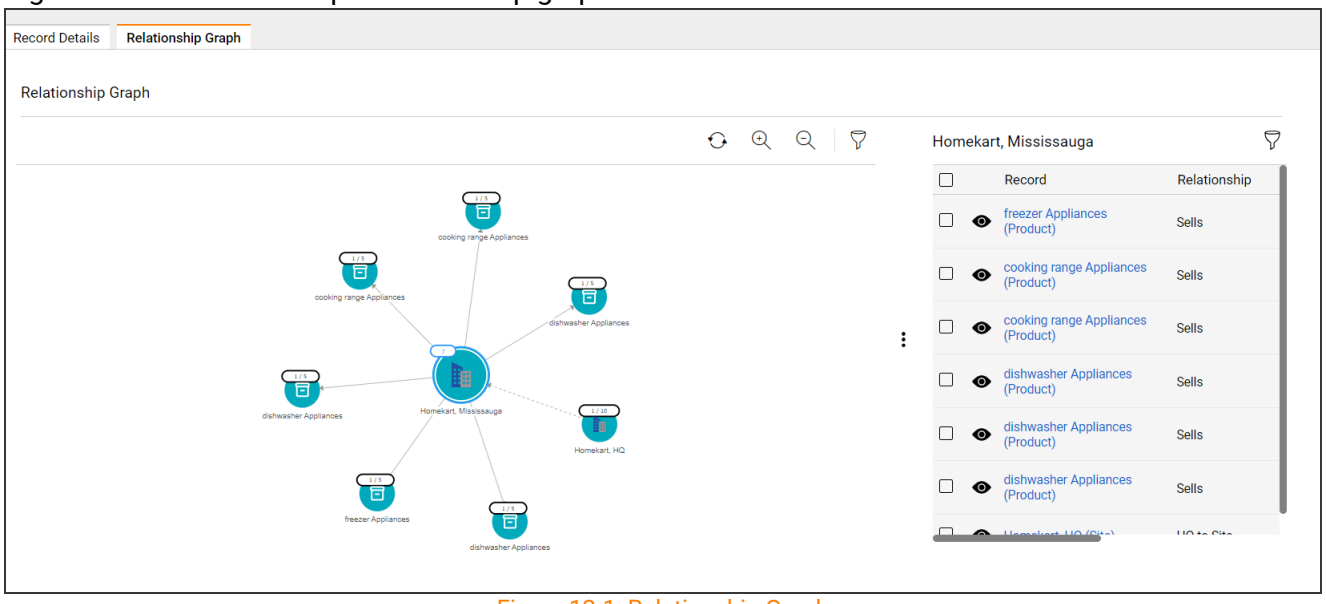


Figure 12-1: Relationship Graph

Recall: Scenario

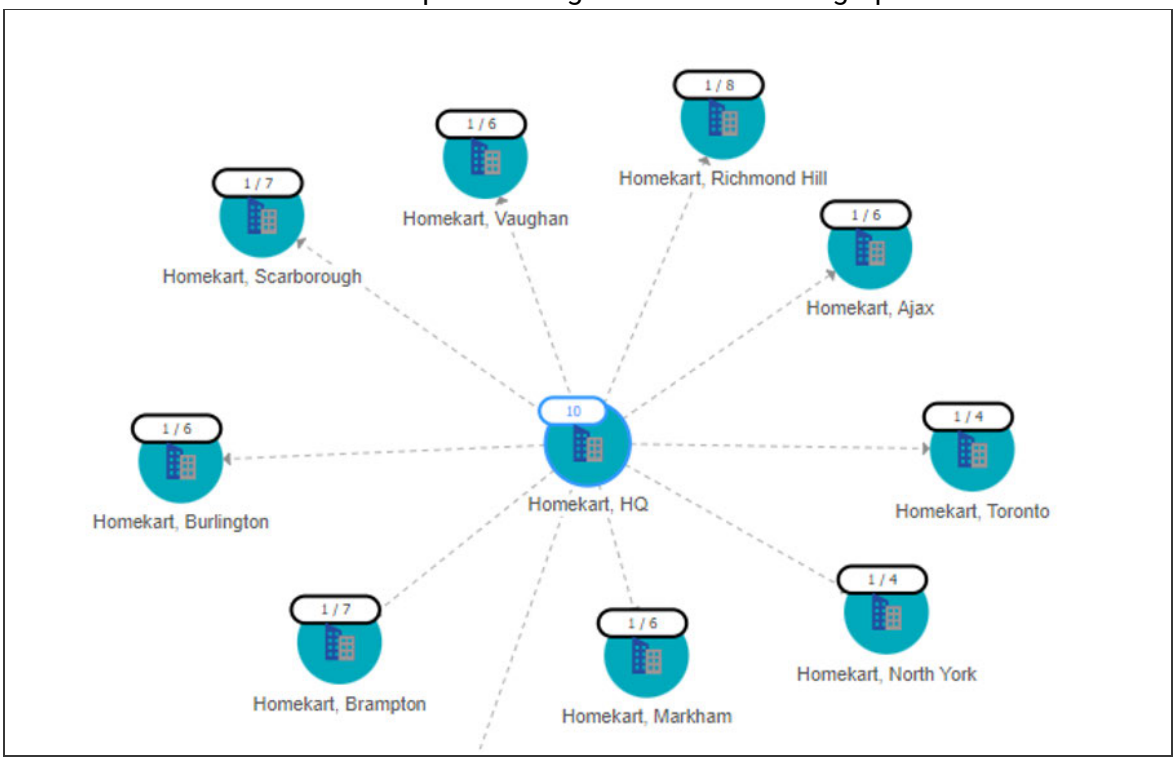
Homekart wants to use Informatica Cloud Multidomain MDM system to get a 360-degree understanding of :

- Storefronts
- Products and inventory per store
- Manufacturer for each product
- Relationship graph to monitor their operational activities, inventory and suppliers

You can visualize and explore the above through the Relationship Graphs.

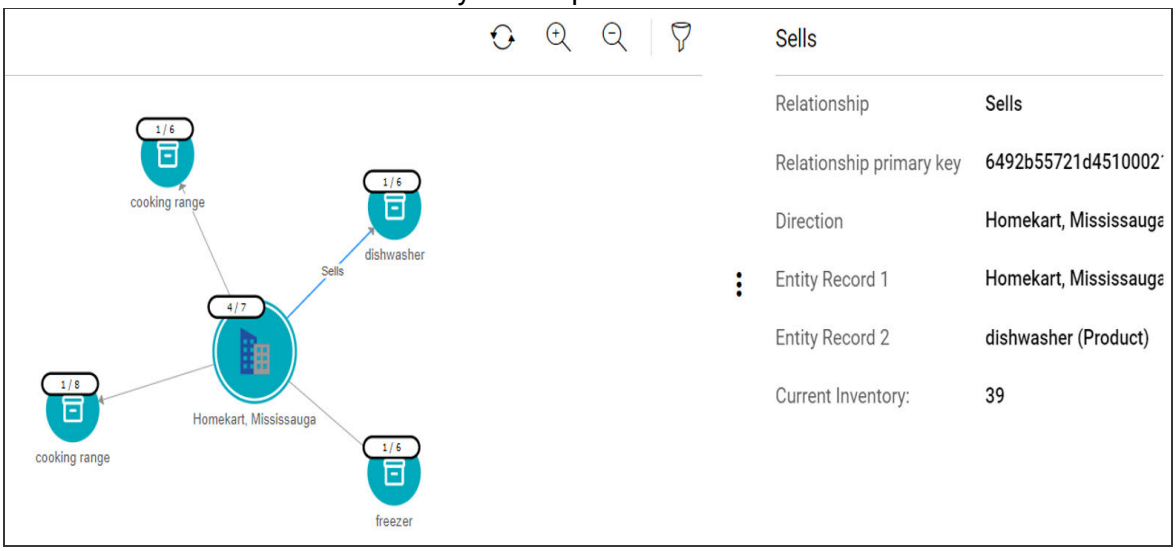
Viewing Storefronts

Here you will see all the storefronts that **Homekart HQ** has all over Canada. You can also see the details of the storefronts and their relationships on the right hand side of the graph.



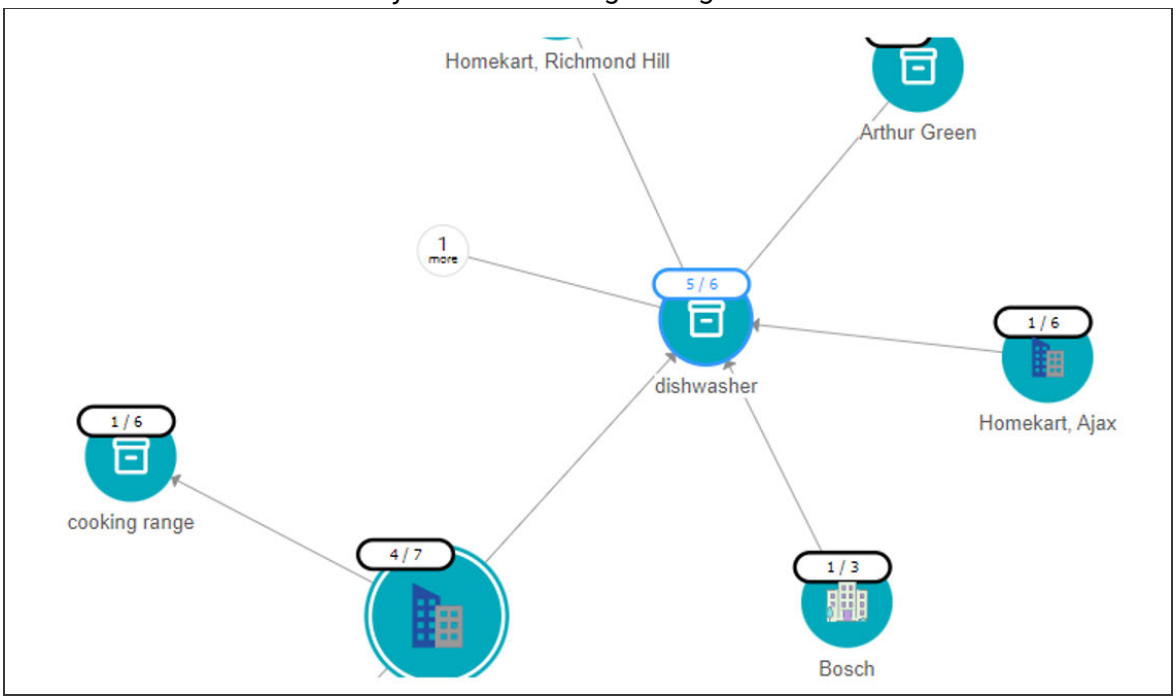
Inventory Details

When you click on the arrow between a site and the product, you will see that the store information, product information and current inventory of that product in that store.



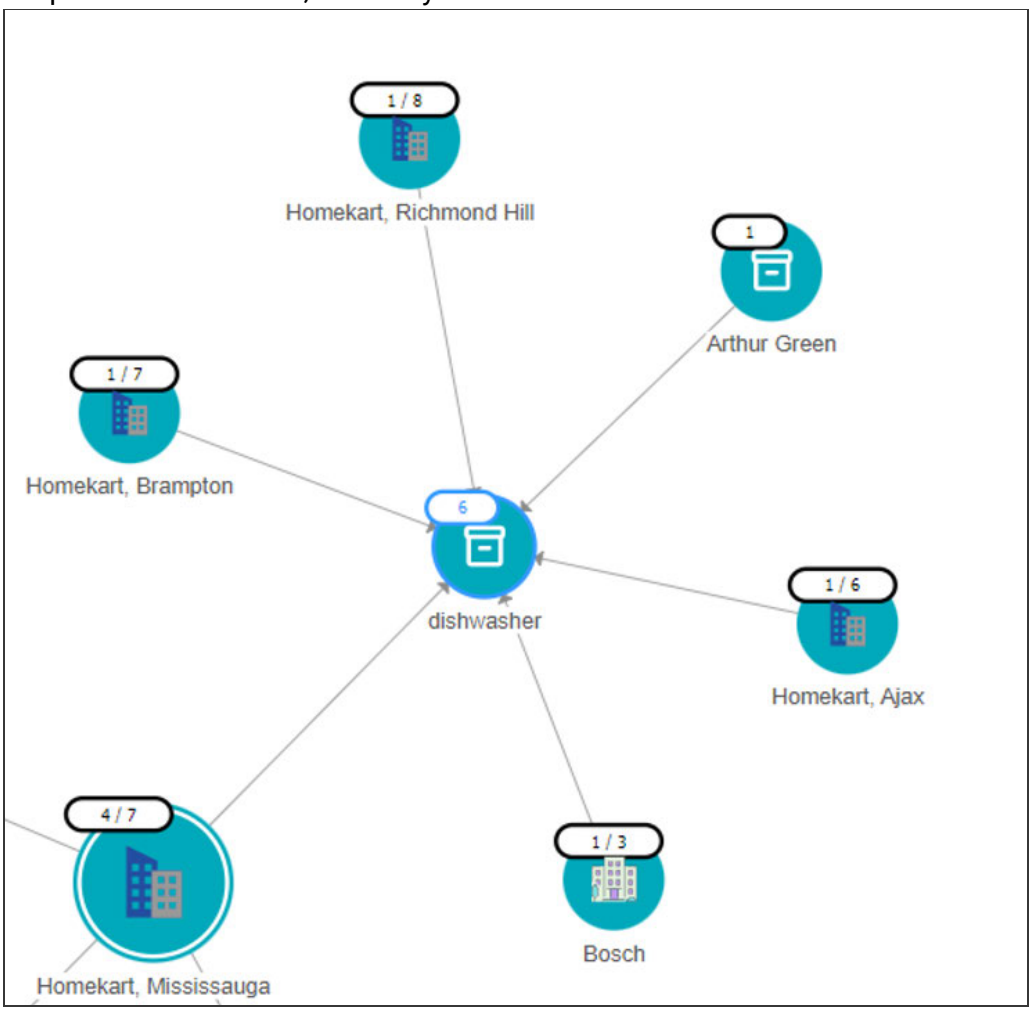
Product Manufacturer

Using the following relationship graph, you can see that Bosch is the manufacture of the dishwasher, where as Homekart,Ajax is the site that is selling the dishwasher. Manufacturer information is important to the homekart, in case there is less inventory of that product in the store or they want to order more inventory in the store. You can also see that Arthur green is the customer who has ordered the product from the site. In this way Homekart can get insight into customer data as well.



360 Degree View of Data

You can see the 360 degree view of the data using the relationship graph and analyze the same to monitor their operational activities, inventory and manufacturer.



Lab 12-1: Managing Hierarchies and Analyzing the Data Model

Overview:

In Business 360 applications, users can create a hierarchy model. Then the users can create a hierarchy instance in Multidomain MDM and add records to it.

In this lab you will create a hierarchy instance and records to it.

Objectives:

- Import data into the site business entity
- Create a hierarchy instance
- Add parent and child records to the hierarchy
- Analyze the data model

Duration:

30 Minutes

Scenario:

Previously, Mark developed a hierarchy model named Homekart, which contained records pertaining to the Canada region. Now, he aims to include records from the India region. To achieve this, he plans to import a CSV file with new site records from India.

Tasks:

Perform the following tasks to analyze the data through relationship graph for Homekart scenario:

- Import data into the site business entity
- Creating a hierarchy Instance
- Deleting the child record
- Analyzing data

Import Data into the Site Business Entity

You have already seen how to import a CSV file into the Multidomain MDM. Apply the same knowledge and import a file to the site business entity.

High-Level Steps:

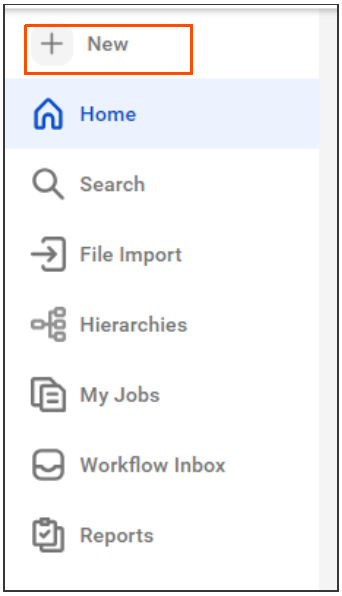
Follow the import steps and import **Site_New_BE.csv** file into the Multidomain MDM.

- Log in to IICS with your admin credentials. (ex: PamStudent01 / Multidomain360).
- File location: **C:\MDM_SaaS_training\File Imports\Site_New_BE.csv.**
- Choose **Site_New_BE.csv** file from the location **C:\MDM_SaaS_training\File Imports**.
- Select the Target Business Entity as **Site** and complete the Import steps.
- Check **My Jobs** to check the status of the import job.

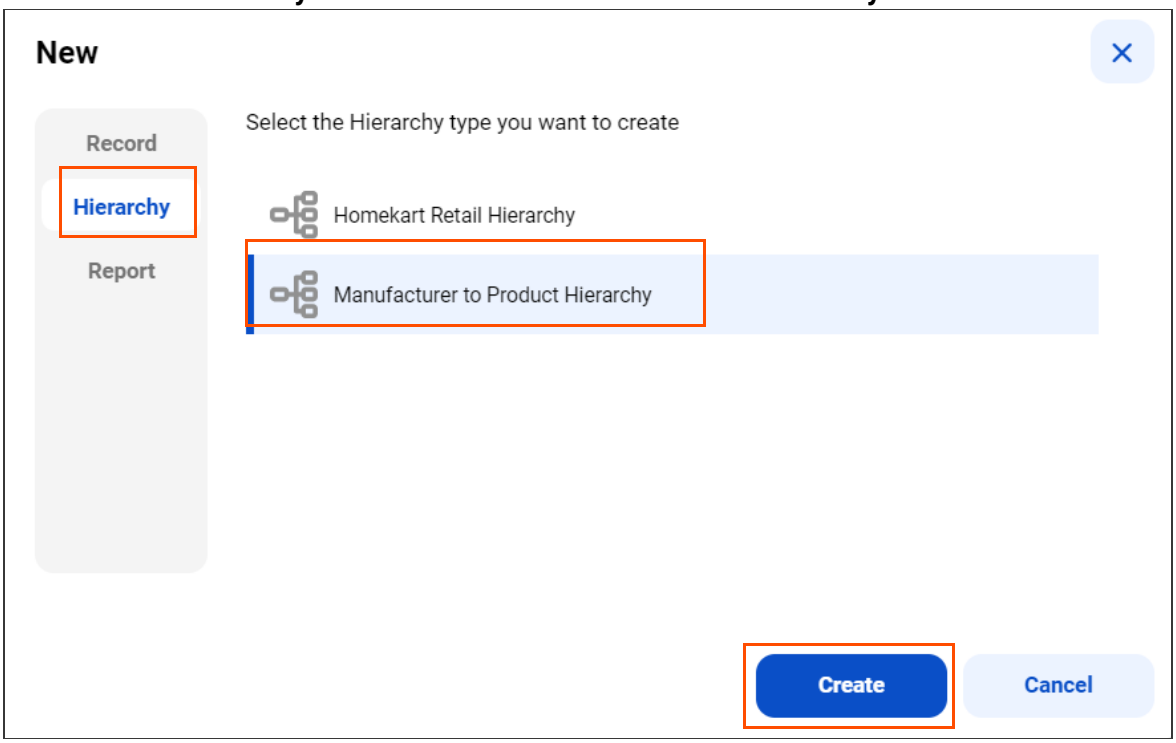
Creating a Hierarchy Instance

Having created a hierarchy model in a previous module, you will now proceed to create an instance of that hierarchy.

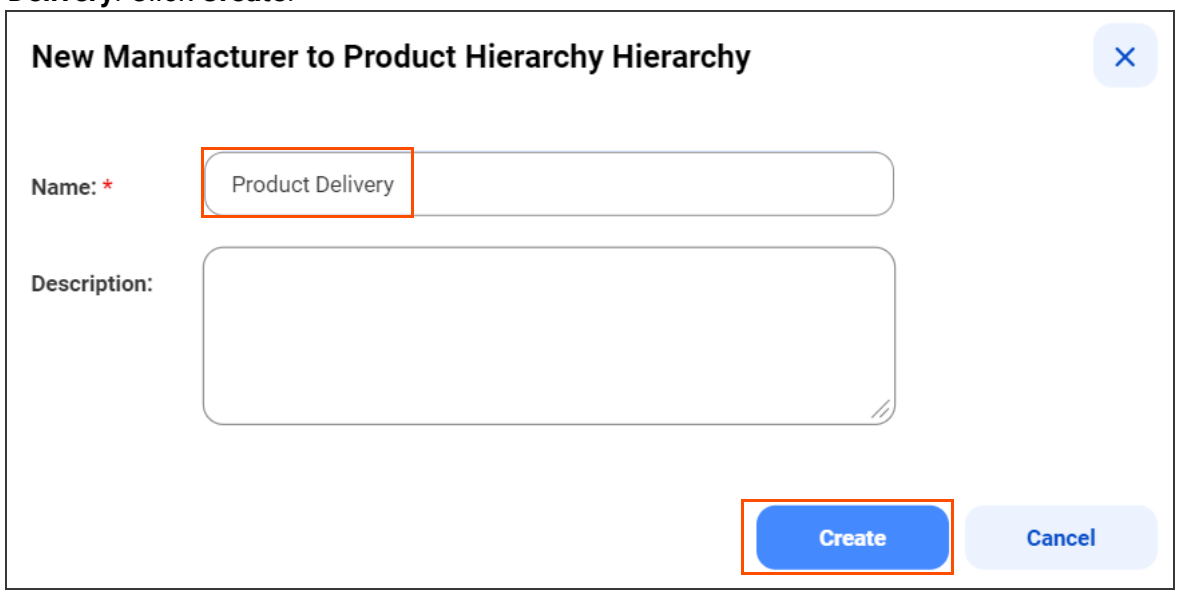
1. In the **Multidomain MDM** service, click **New**.



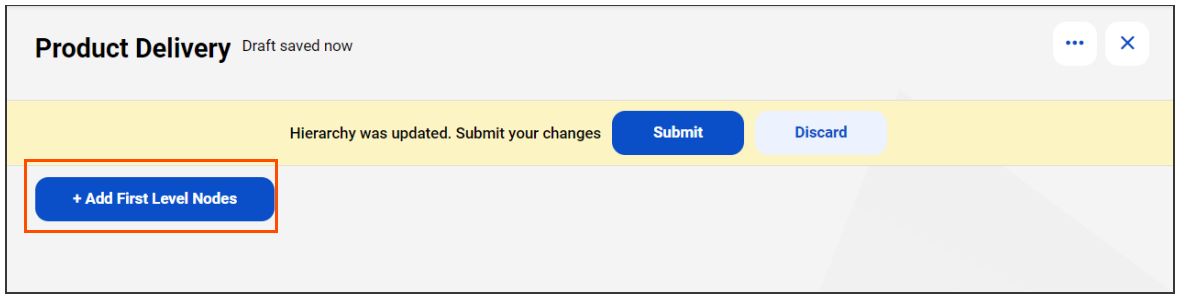
2. Select **New** > **Hierarchy** > **Manufacturer to Product Retail Hierarchy**. Click **Create**.



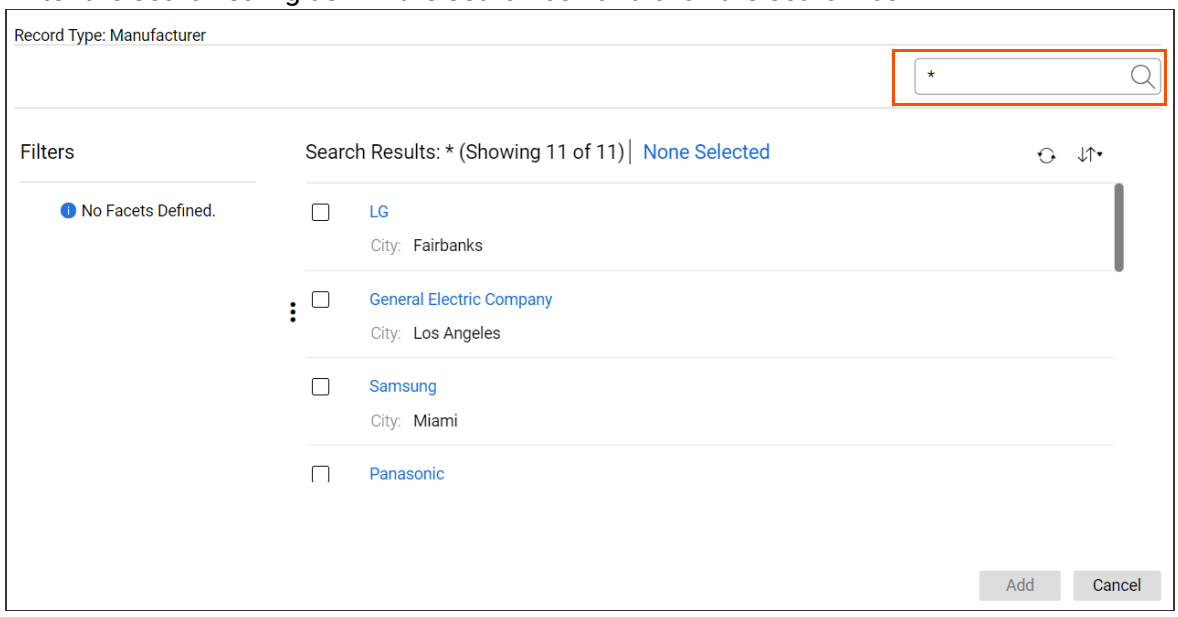
3. In the **Manufacturer to Product Hierarchy** Hierarchy window, enter the name as **Product Delivery**. Click **Create**.



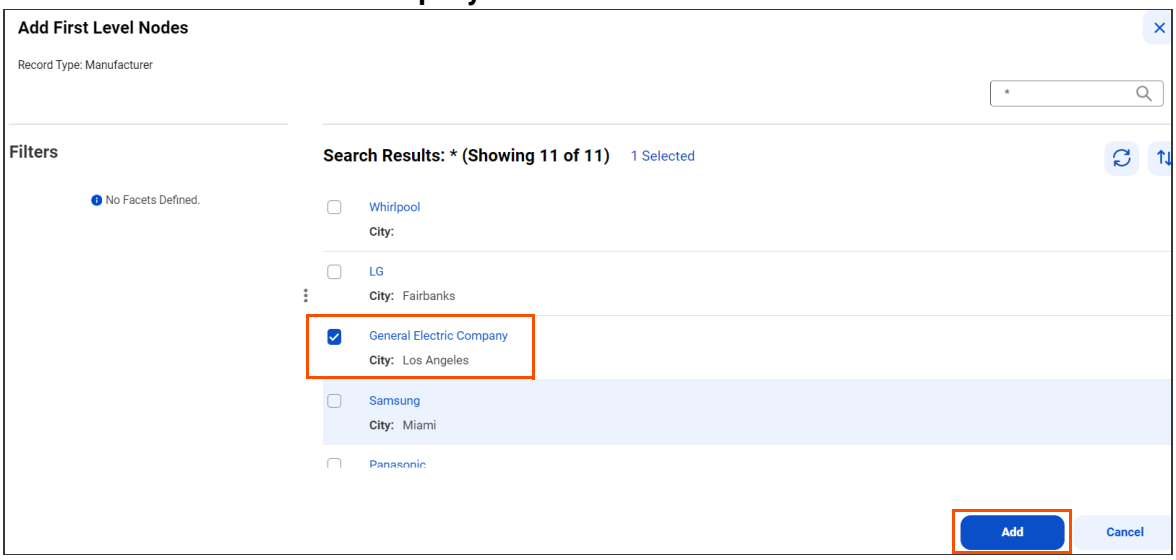
4. Click **+ Add First Level Nodes**.



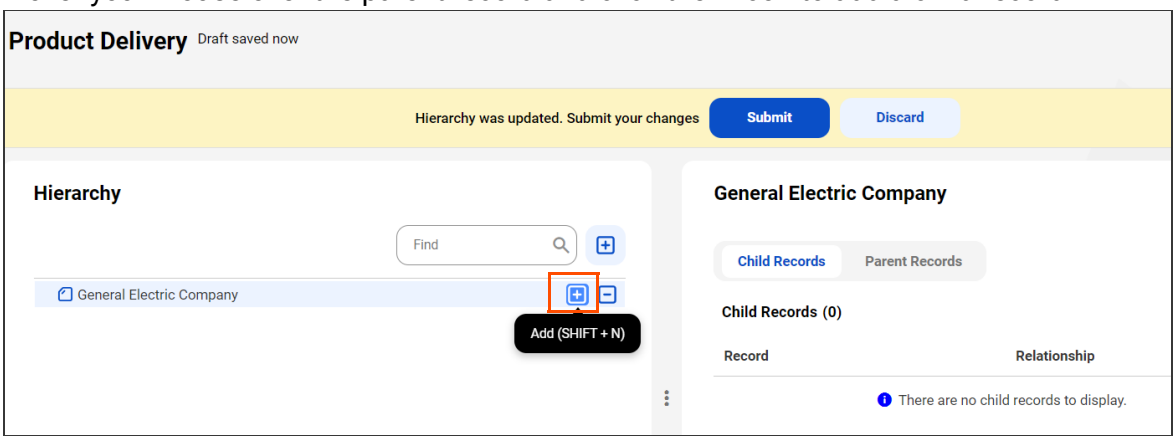
5. Enter the search string as * in the search box and click the search icon.



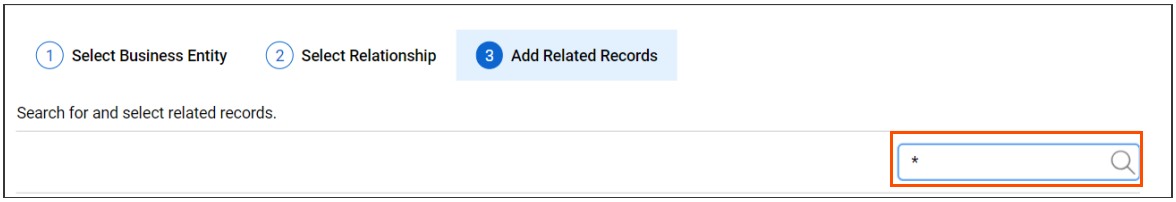
6. Select the **General Electric Company** and click **Add**.



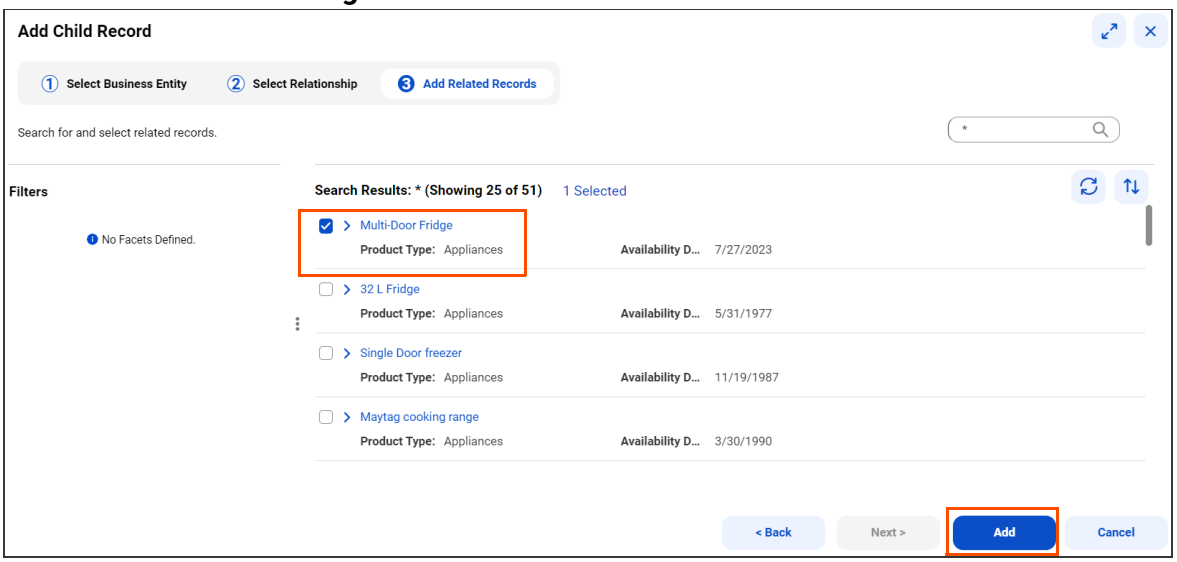
7. Hover your mouse over the parent record and click the **+** icon to add a child record.



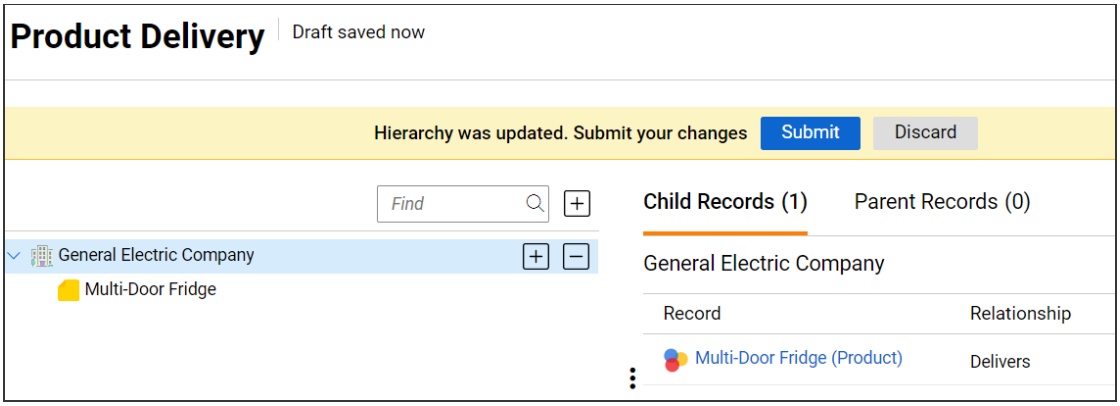
8. In the Add Child Record window, enter the search string as * and click the search icon.



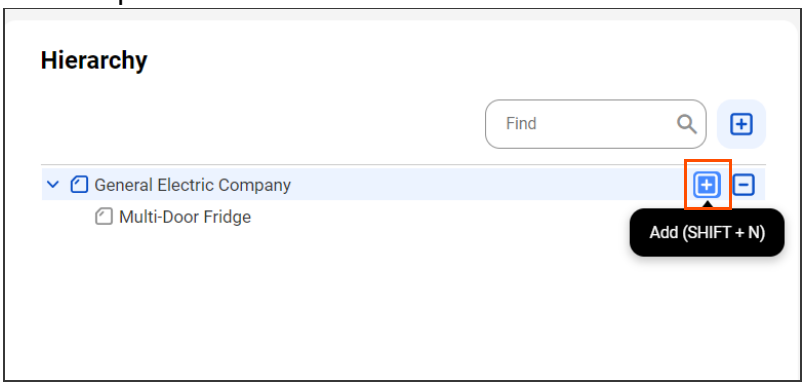
9. Select the **Multi-Door Fridge** record and click **Add**.



You will see that the child record is added.

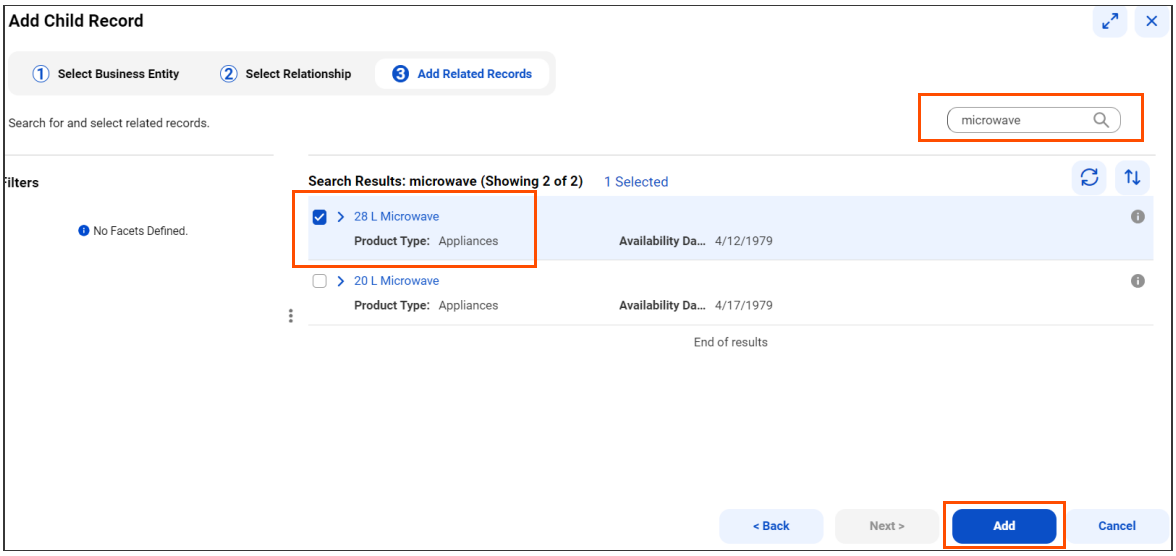


10. Add another child record **Microwave** under the same parent. To do that Hover your mouse over the parent record and click the **+** icon.

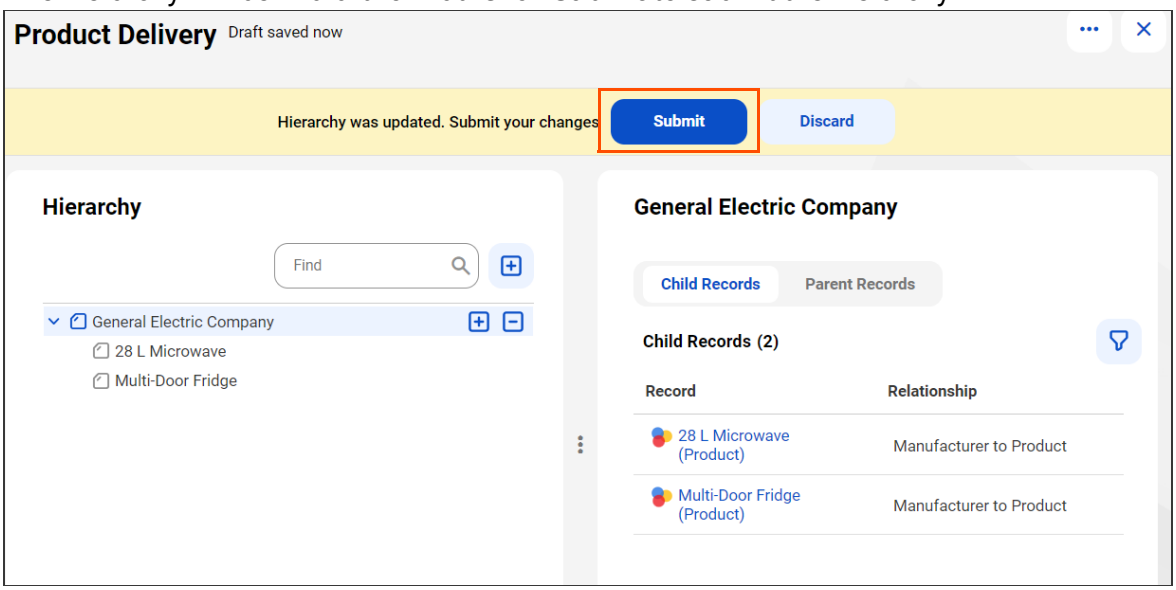


11. In the Add Child Record window, search for the record **Microwave**.

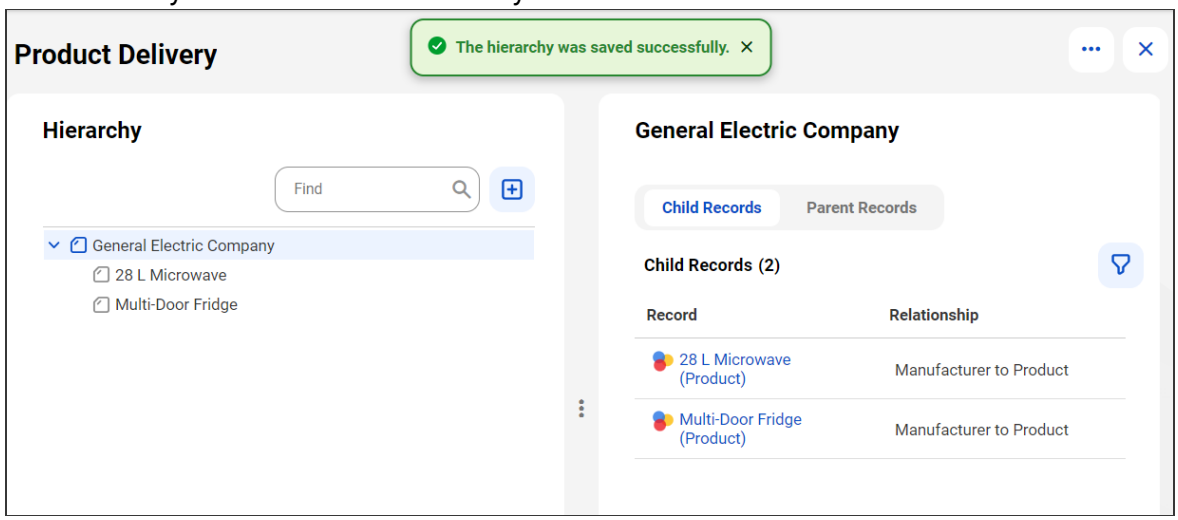
12. Select the **28 L Microwave** record and click **Add**.



13. The hierarchy will be in draft format. Click **Submit** to submit the hierarchy.



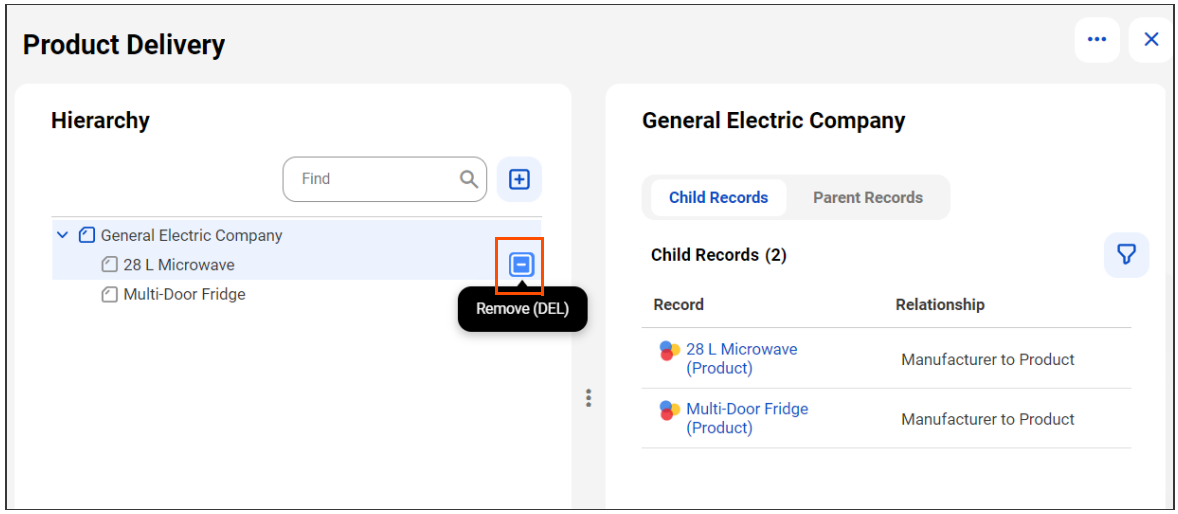
The hierarchy will be saved successfully.



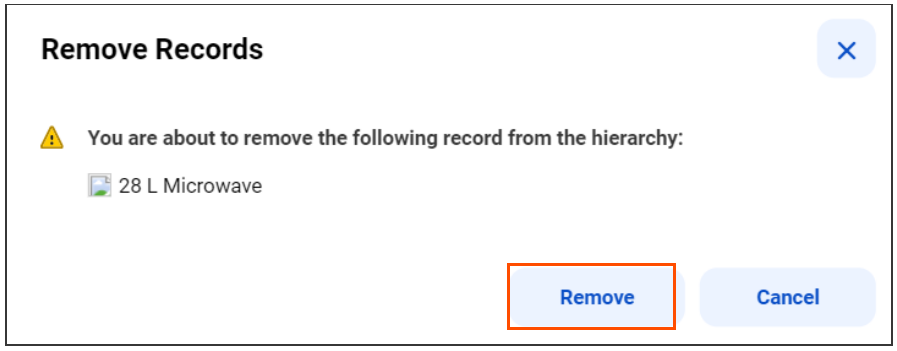
Deleting the Child Record

You will now see how to delete the child node from the hierarchy.

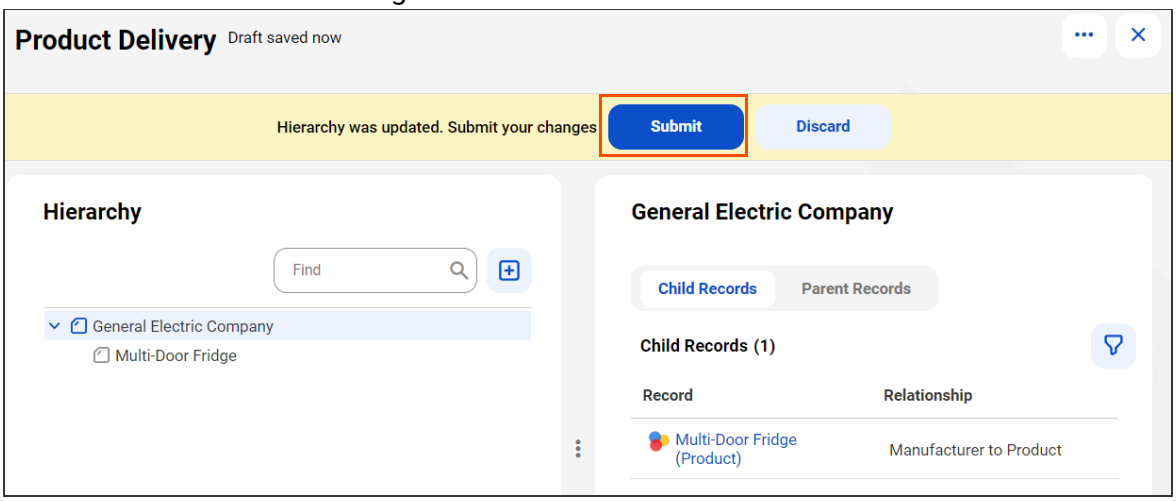
14. Hover over your mouse on the child record that you want to delete. In this case, **28 L Microwave**, and click **Remove**.



15. In the Remove Records dialog box, click **Remove**.



16. Click **Submit** to save the changes.



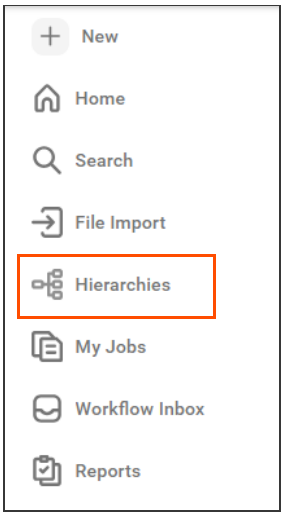
Analyzing Data

Homekart wants to use Informatica Cloud MDM system to get a 360-degree understanding of:

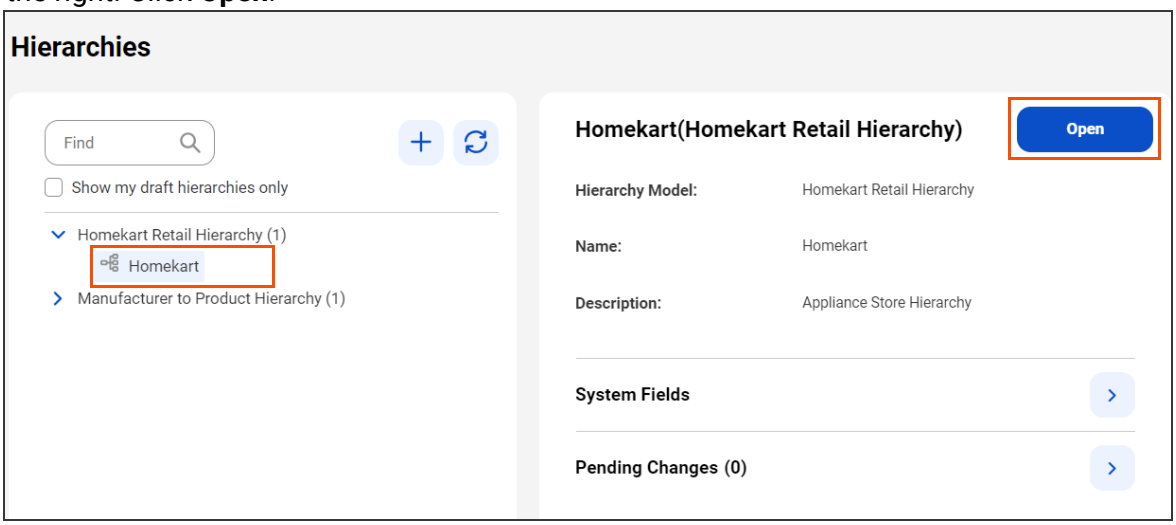
- Storefronts
- Products and inventory per store
- Manufacturer for each product
- Relationship graph to monitor their operational activities, inventory and suppliers

You will now visualize and explore the above through the Relationship Graphs. First you will see the storefronts that Homekart has to offer.

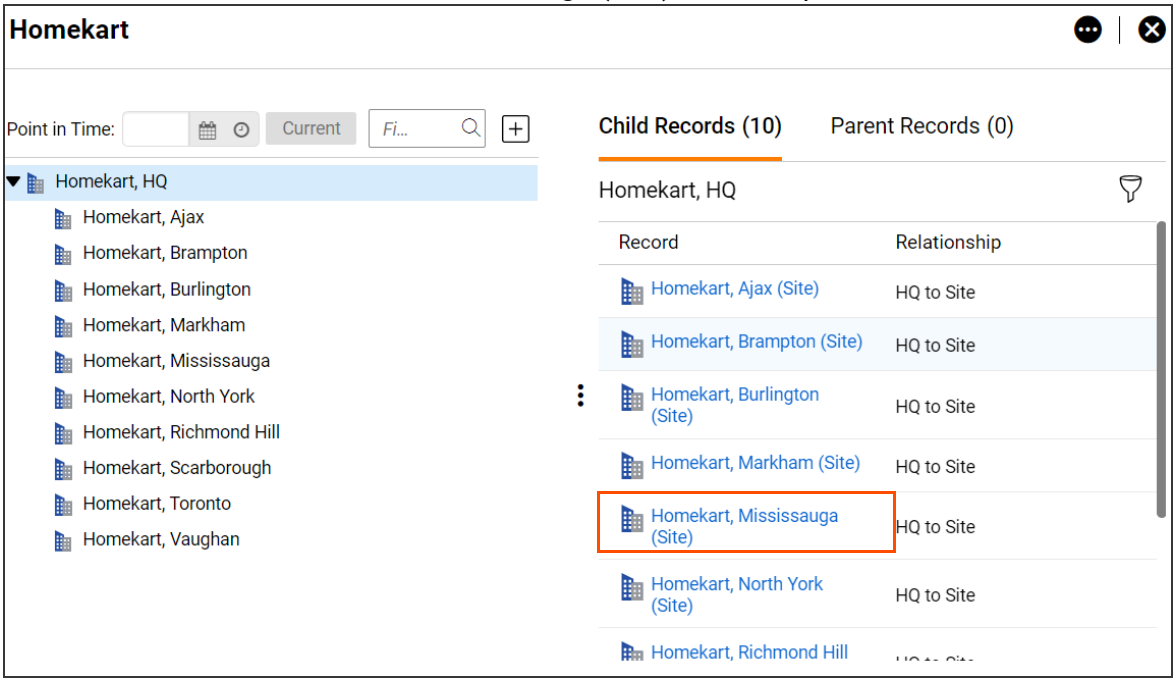
17. From the Menu, Select **Hierarchies** tab.



18. Select **Homekart Retail Hierarchy** > **Homekart.** It will open the hierarchy definition page on the right. Click **Open**.

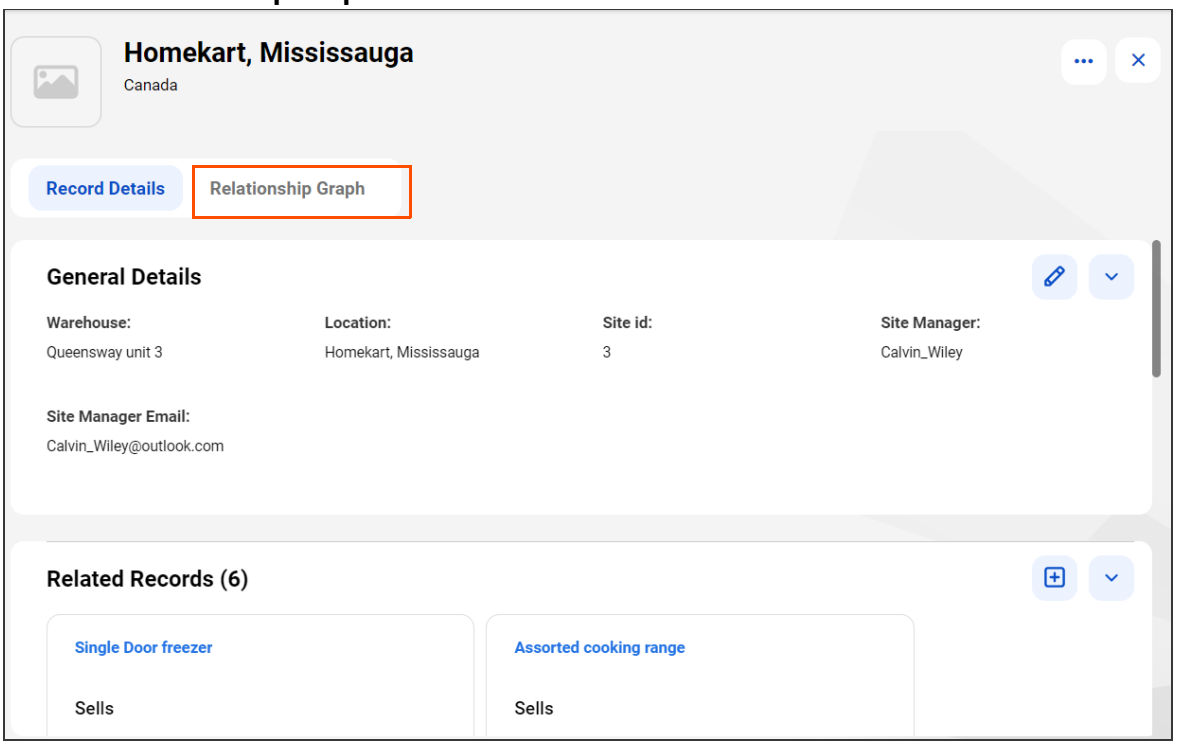


19. You can see the Homekart hierarchy model defined here. It consists of a top-level business entity as **Homekart HQ** and having child records as different Homekart branches. Under Child Records Select **Homekart, Mississauga (Site)** store to open it.

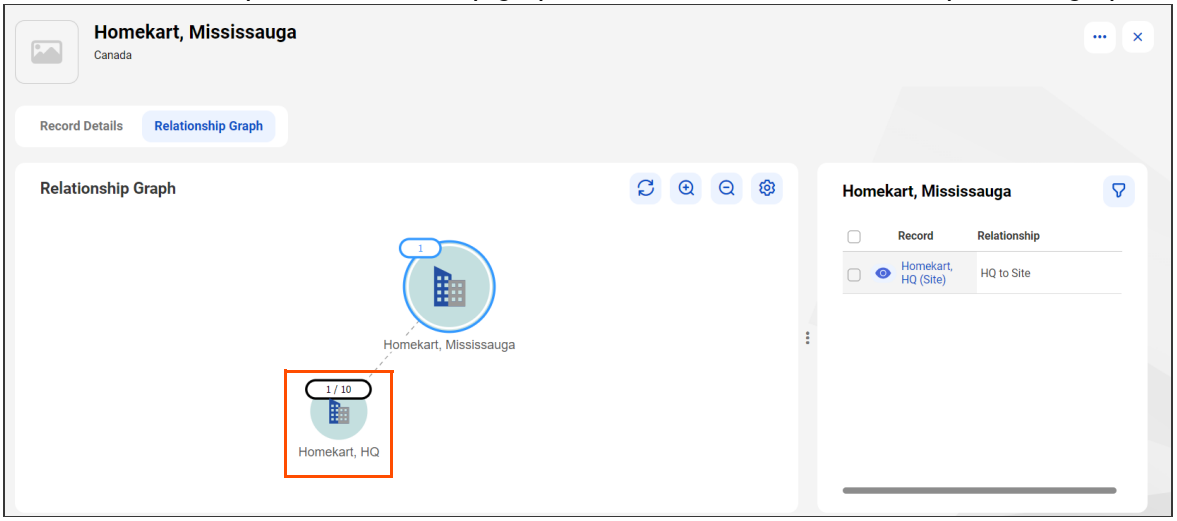


If you are unable to see the name of the sites, just refresh the page.

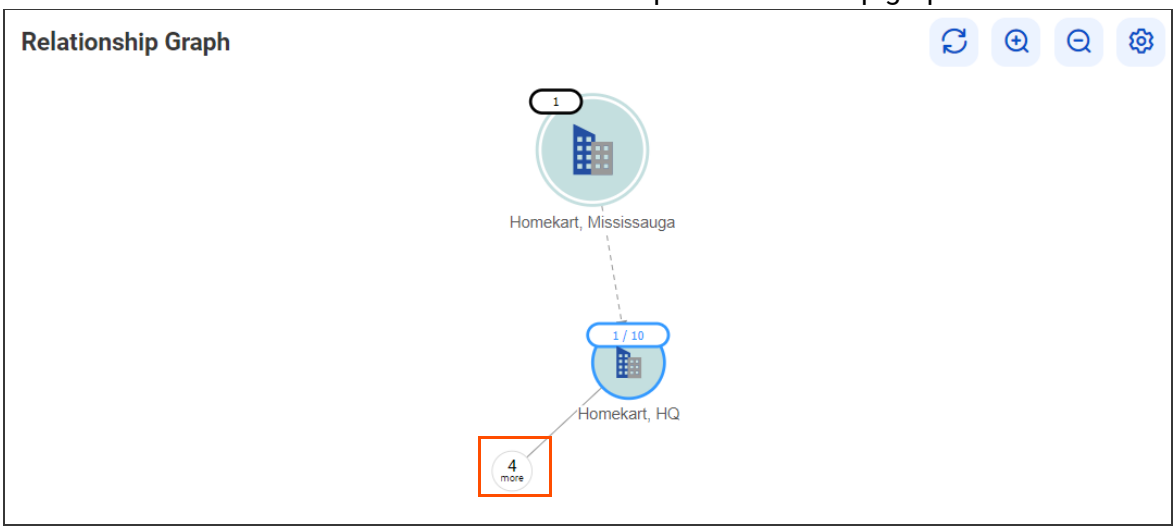
20. Click the **Relationship Graph** tab.



21. You can see the top-level relationship graph. Click on **Homekart HQ** to expand the graph.

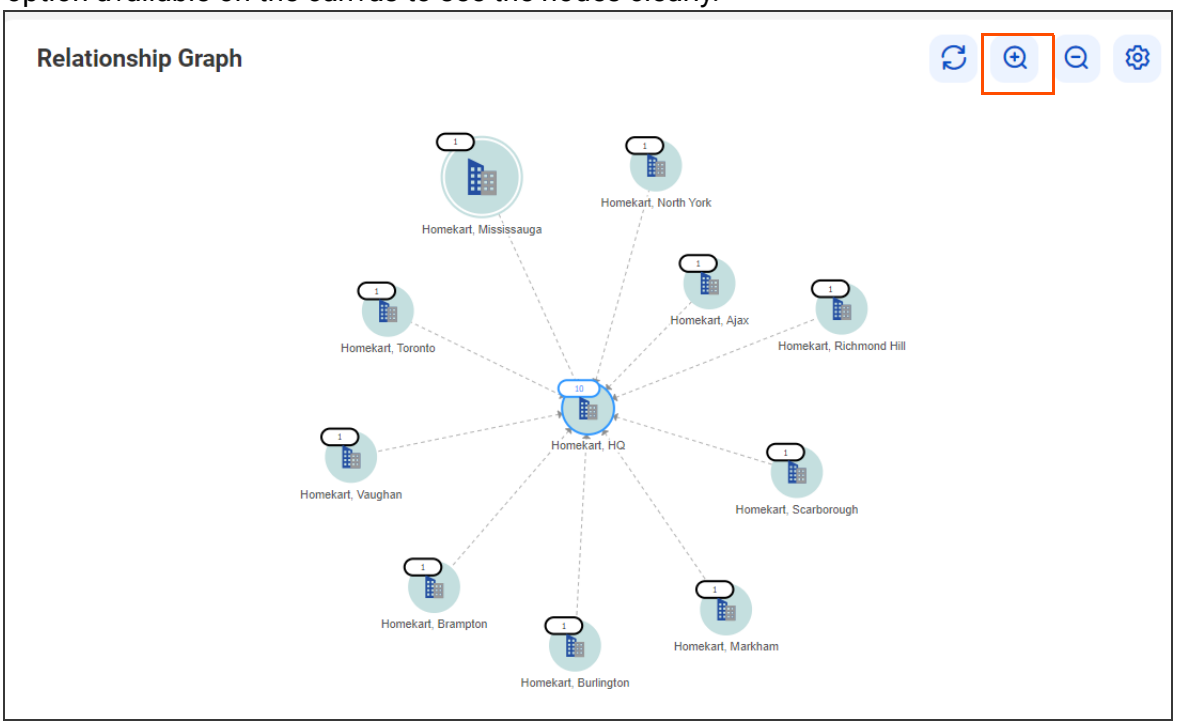


22. Click on **4 more** > **4 more** > **1 more** to see the complete relationship graph.

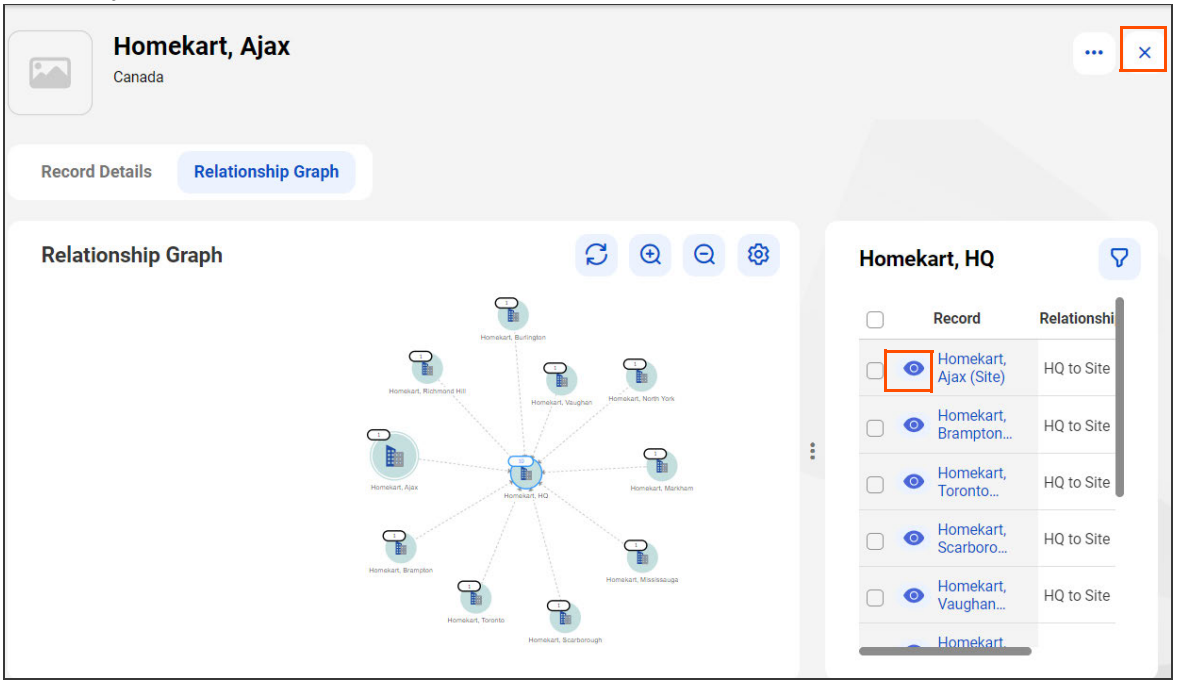


There may be slight variations in the number while viewing the stores. Keep expanding the hierarchy until you see all the storefronts.

23. You will see all the storefronts that **Homekart HQ** has all over Canada. Use the **zoom-in** option available on the canvas to see the nodes clearly.

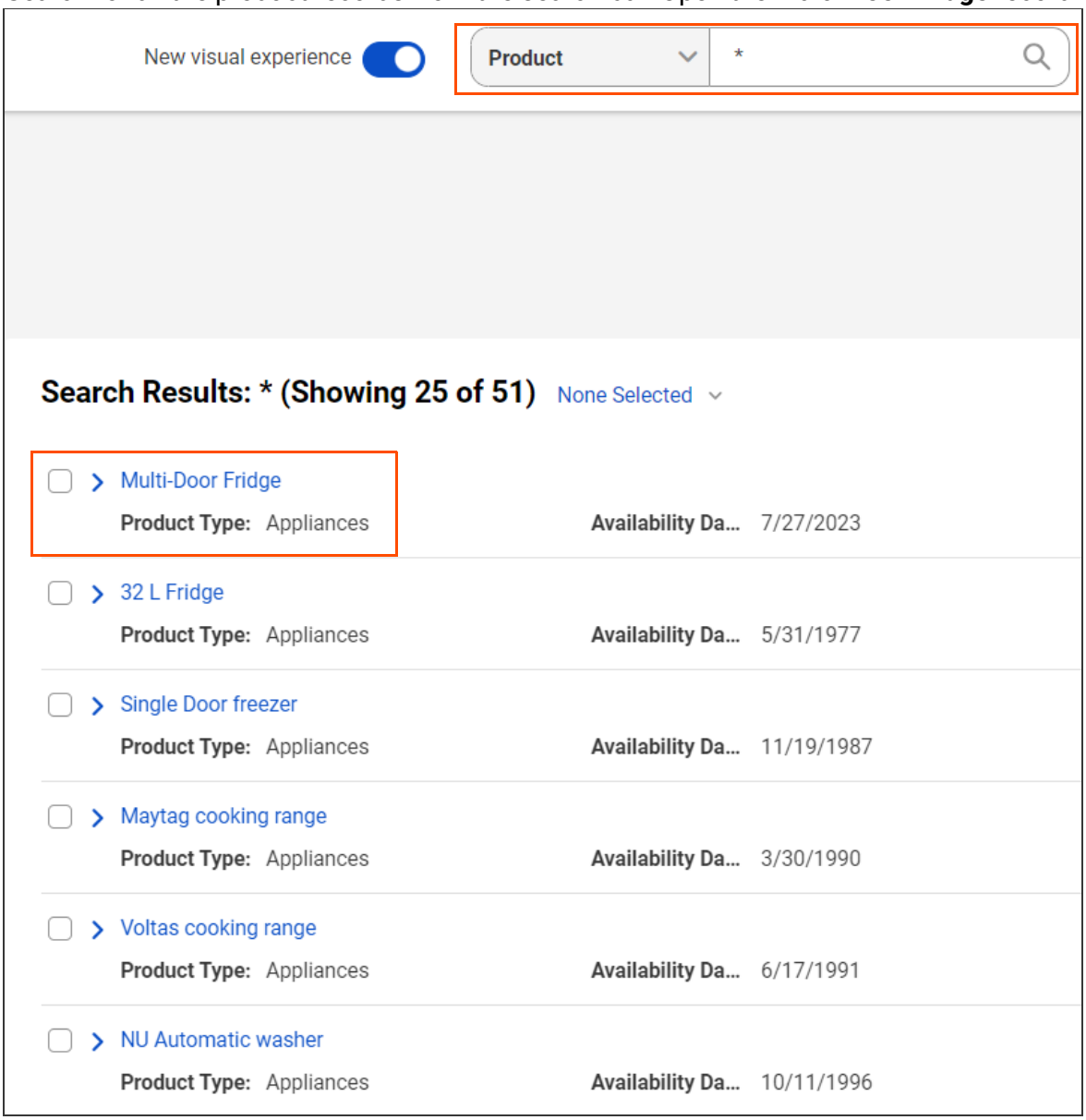


24. On the right-hand side, you can see the information in tabular format. You can show or hide the records on the canvas by clicking the **Show/Hide on canvas** button. **Close** the hierarchy.

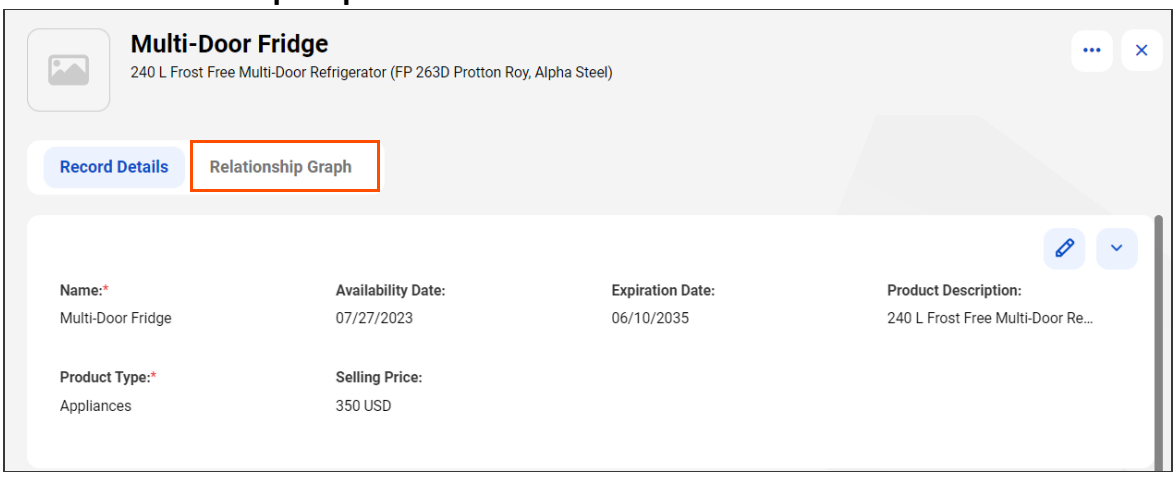


Now you will see the products and inventory per store for Homekart.

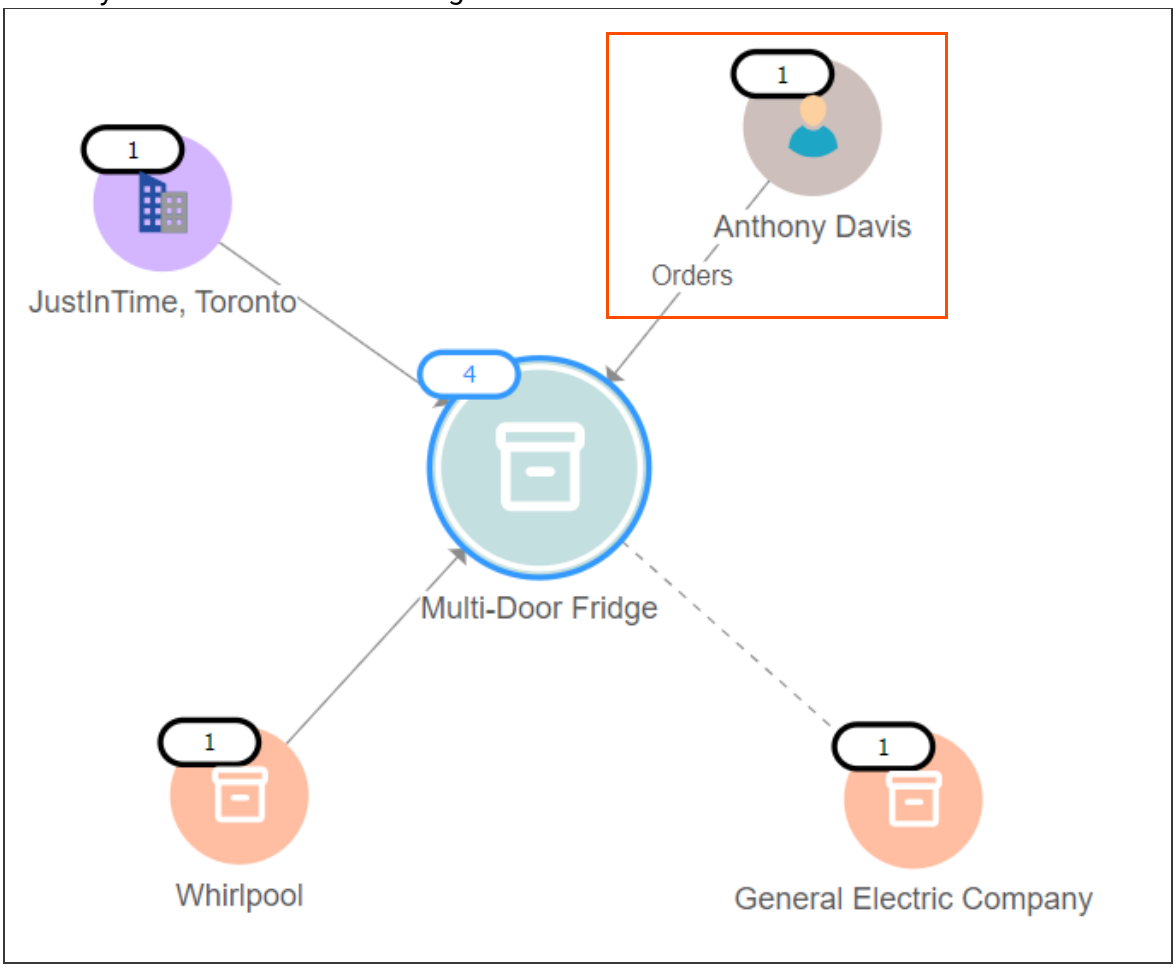
25. Search for all the product records from the search bar. Open the **Multi-Door Fridge** record.



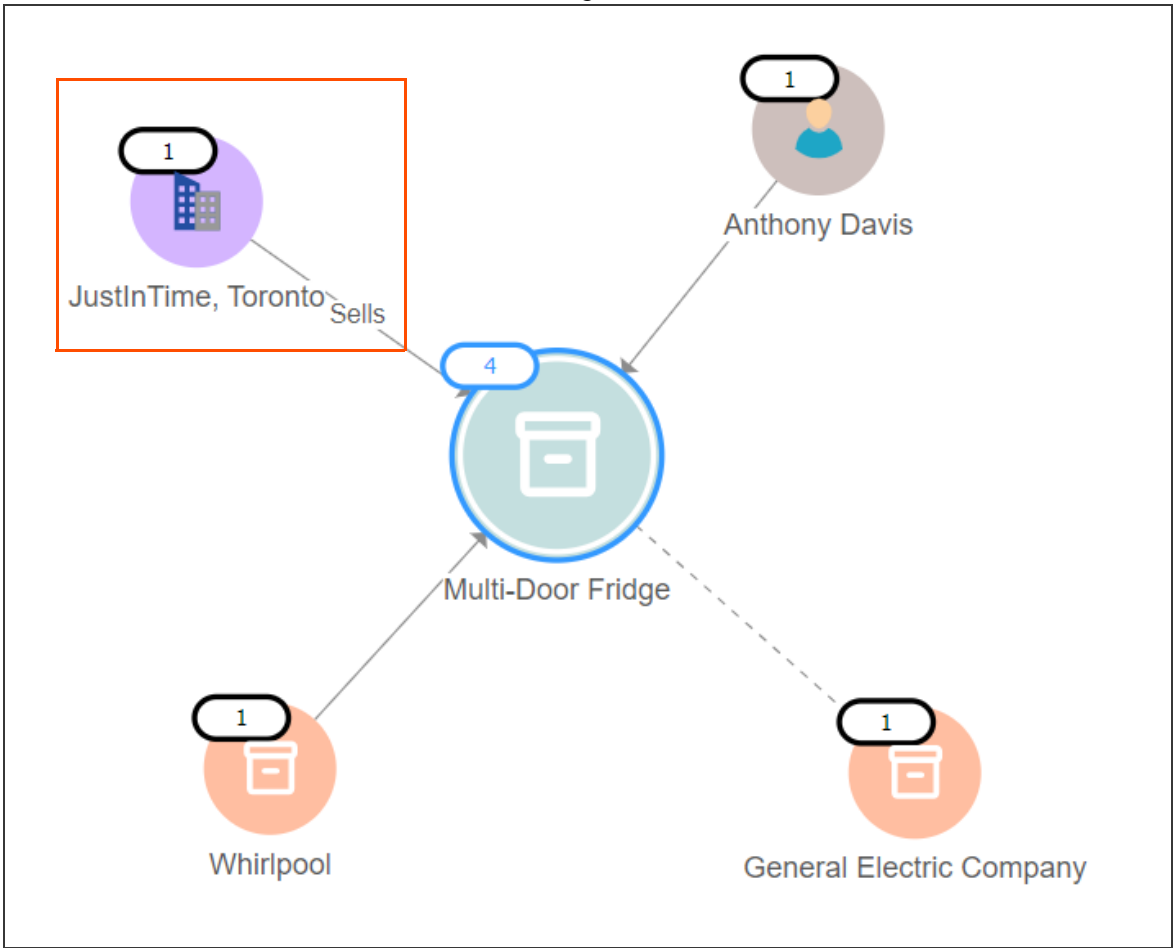
26. Click the **Relationship Graph** tab.



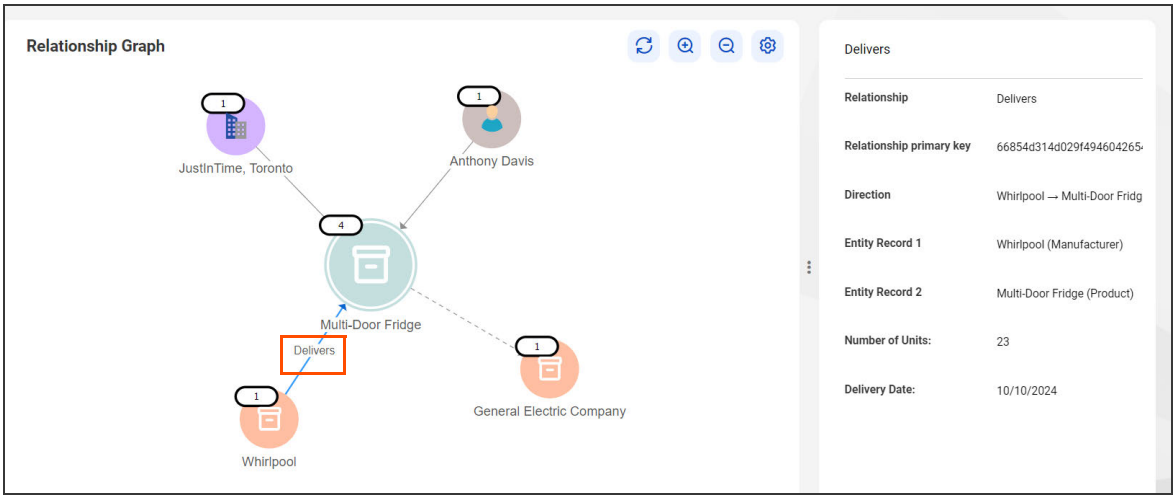
27. You can see that **Multi-Door Fridge** has been **ordered** by **Anthony Davis**. You can see the relationship between the records in the hierarchy by Hovering over the arrow between Anthony Davis and Multi-Door Fridge .



28. Hover over the arrow between **JustInTime, Toronto** and **Multi-Door Fridge**. You can see that the site JustInTime **Sells** Multi-Dorr Fridge.



29. Click on the arrow between **Whirpool** and **Multi-Door Fridge**. The relationship **Delivers** is visible. You will also see the manufacturer and product information and Number of Units that were delivered. The unit count is 23 in this case.



This way you can analyze the data. You can choose to select any site and view the relationship graph associated with that data.

This concludes the lab.

Key Takeaways

- A hierarchy contains hierarchy relationships between records.
- You can create hierarchies to show how records are related to one another.
- You can also import data into the relationship as you did it for the hiearchies.
- You can use relationship graphs to analyze the data and get insights into it.

Module 13: Manage Reports and Dashboard

Managing Reports

A report is a collection of data that is acquired from the master data. You can use reports to gain meaningful insights into your business data and business outcomes. Multidomain MDM allows you to create custom reports to display information about records based on your requirement. After you create a report, you can visually represent the report data as charts. Also, you can customize the look and feel of the charts. You can then add the charts to widgets on the Home page dashboard.

After you create a custom report and chart, you can schedule and publish the report. To create schedules and publish custom reports, you must have an Admin role. All the predefined reports publish every day at 00:00 hours EST without any manual intervention. When you publish a custom report, Multidomain MDM updates the report data at the scheduled intervals. The custom reports you create appear on the Reports page.

Custom Reports

If your user role permits, you can create custom reports. You can then configure a chart type for the report to visualize the data. After you create a custom report and chart, you can schedule and publish the report. When you publish a custom report, Multidomain MDM SaaS updates the report data at the scheduled intervals. The custom reports you create appear on the Reports page.

Custom reports that you create appear for all other users within the organization. If users don't have access to the records or fields that are used in the custom reports, the reports don't appear on their Reports page.

Custom Report Types

You can create different types of custom reports. The reports are categorized based on the data that they use.

You can use the following report types:

- **Reports based on master records**: Use the master records that belong to a business entity as the report data.
- **Reports based on record metadata**: Use the metadata of master records as the report data.

To create meaningful custom reports, configure appropriate fields as reportable when you model a business entity. These fields appear as dimensions and measures when you create reports. Dimensions are the fields that represent the qualitative or categorical values of a data set.

For example, country or lifestyle segment. Measures are the fields that represent the quantitative values of a data set. For example, the average sales or profit count for a month.

Reports Based on Master Records

Custom reports that are based on master records use the fields in a business entity to display the report data. You can use these reports to derive meaningful insights about the master records in a business entity. After you configure the report and apply the changes, the report data appears in the report table and the associated chart. You can view the report table and chart on the report details page.

Additionally, the report details page displays the configured dimensions and measures as search results on the Data Set panel.

The two types of reports are:

- **Standard reports**: Standard reports display report data based on the dimension and measures that you selected for the reports. When you configure a standard report and apply the changes, the report data appears instantly on the report table. These reports can use a single dimension and multiple measures.
- **Drill-down reports**: Drill-down reports use multiple dimensions to provide additional levels of detail. You can add up to three levels of dimensions to the report data. The field values in the report data appear based on the dimension levels.

Reports Based on Record Metadata

Custom reports that are based on record metadata use the metadata of the master records to display the report data. For example, the validation statuses of records or the merge history of source records. You can use these reports to understand the metadata associated with the master records.

Records With Validation Errors

Custom reports that are based on records with validation errors display the validation statuses of master records in a business entity. For example, you can configure a report to view the number of records based on the types of validation errors. After you configure the report and apply the changes, the report data appears in the report table and the associated chart.

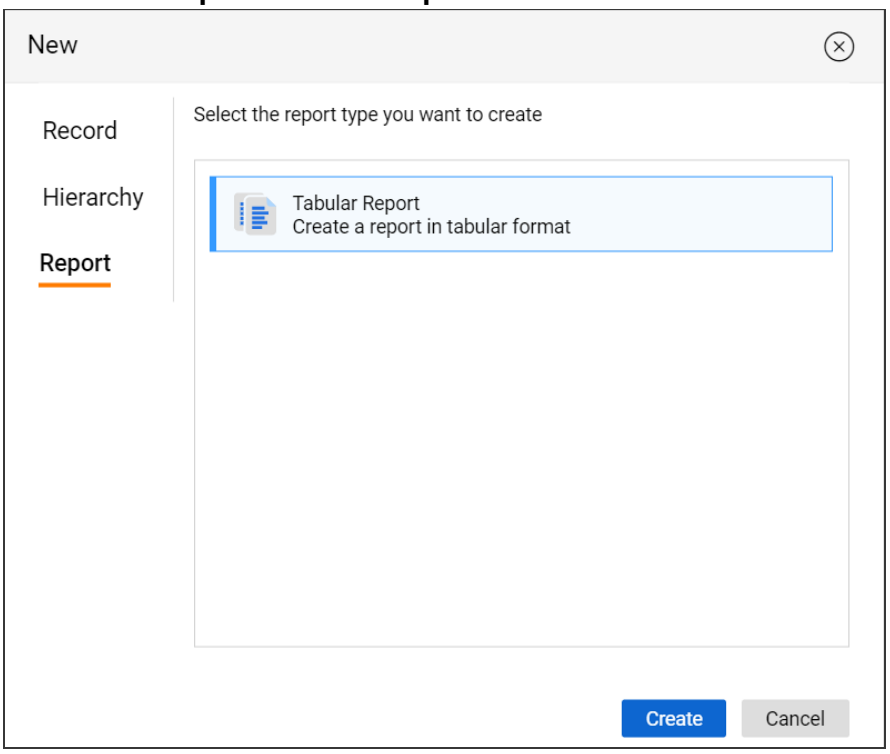
You can view the report table and chart on the report details page. Additionally, the report details page displays the configured dimensions and measures as search results on the Data Set panel.

Creating Custom Reports

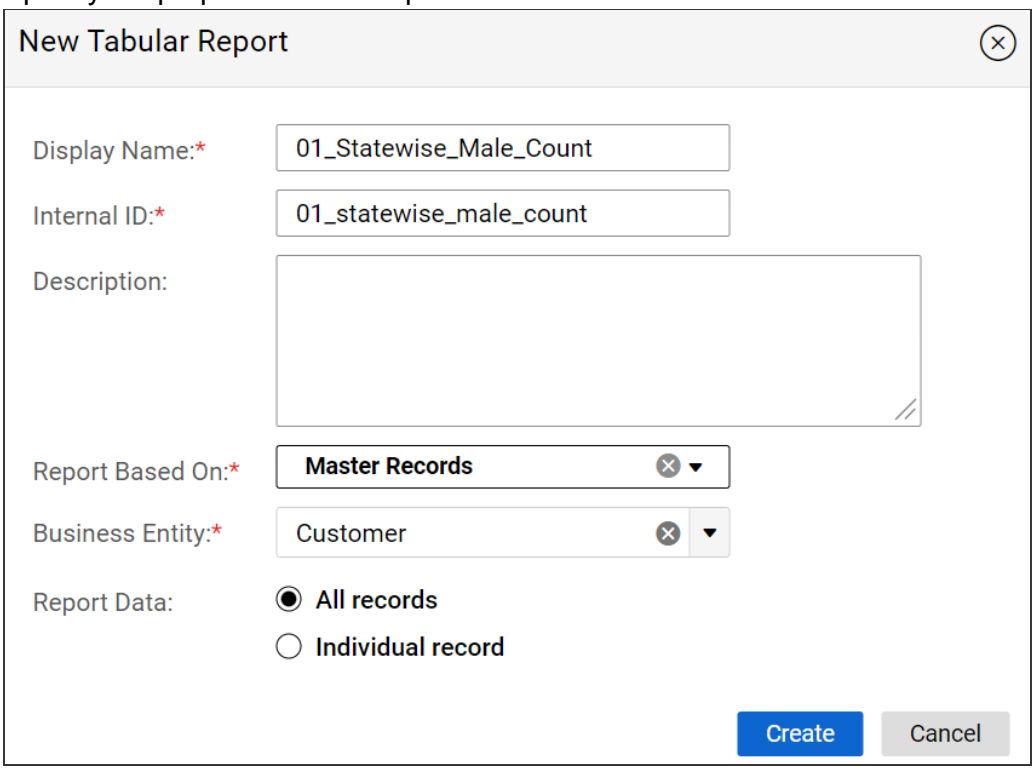
If your user role permits, you can create custom reports. You can then configure a chart type for the report to visualize the data. After you create a custom report and chart, you can schedule and publish the report. The custom reports you create appear on the Reports page.

Refer to the following steps to create a custom report:

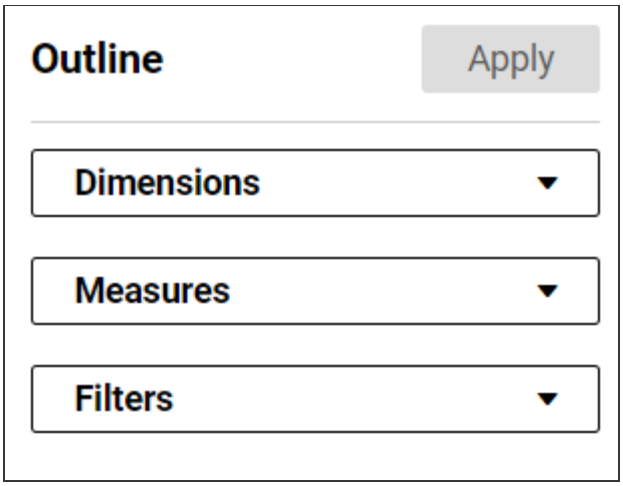
1. Click **New** > **Report** > **Tabular Report** > **Create**.



2. Specify the properties of the report and click **Create**.



3. Select Dimension and Measures and click **Add**.

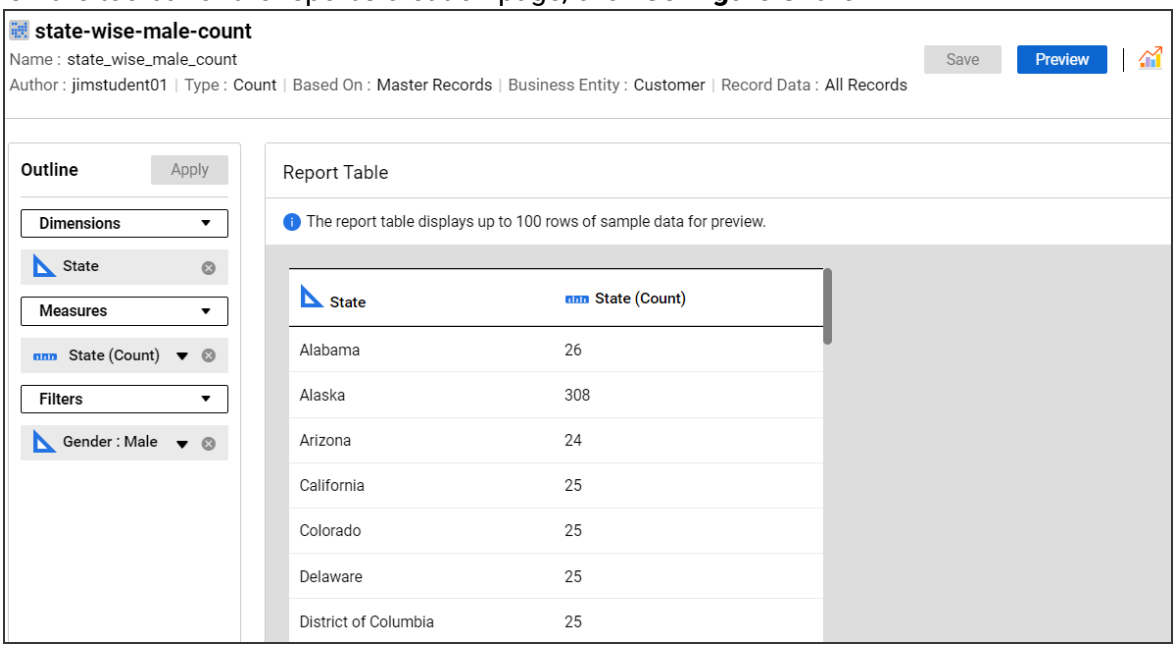


4. To filter the report data based on the field values, select a field from Filters and click **Add**.
5. Click **Apply**.

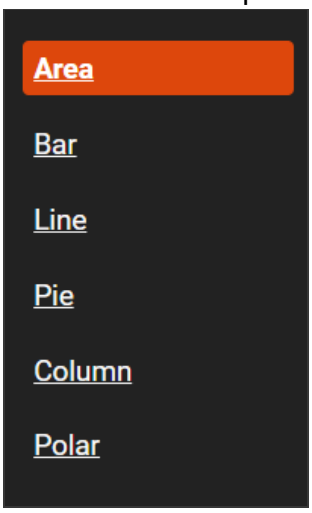
The tabular report displays up to 100 rows of sample data for preview.

Refer to the following steps to add a chart to a custom report:

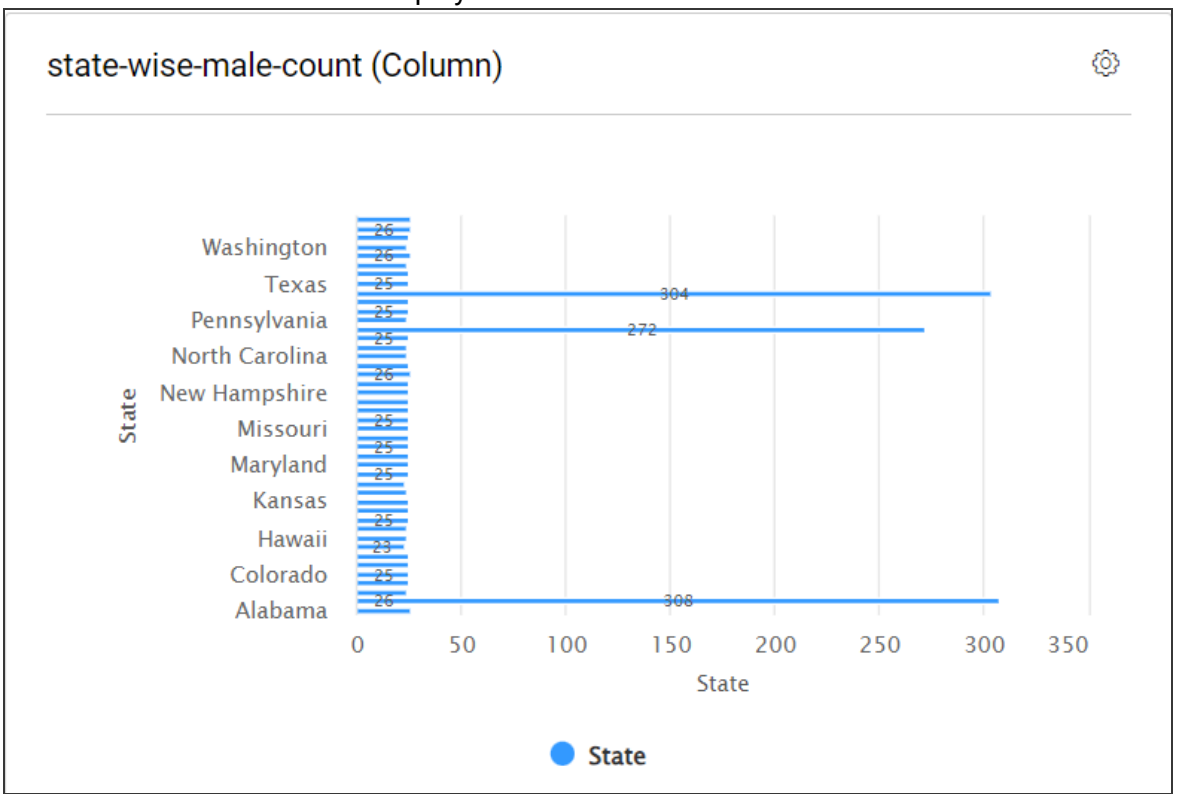
6. On the toolbar of the report's creation page, click **Configure Chart**.



7. Select a chart template that you want to use to display the report data.



8. Click **Save**. The chart will be displayed.



Managing Dashboard

The Home page is a dashboard that can display charts. You can create multiple dashboards on the Home page based on your requirements and business needs. **For example**, you can create a dashboard to visualize record details and another one for sales history. You can create up to six different dashboards using the predefined dashboard layouts

Charts summarize and display data based on reports. You can add charts to the dashboards as widgets. Each widget displays the last updated time and estimated next run time of the associated report. You can add or remove the widgets to customize the dashboard.

You can edit the dashboards that you create to add, edit, or remove the widgets associated with it. If you have multiple dashboards, the first dashboard tab appears by default when you land on the Home page.

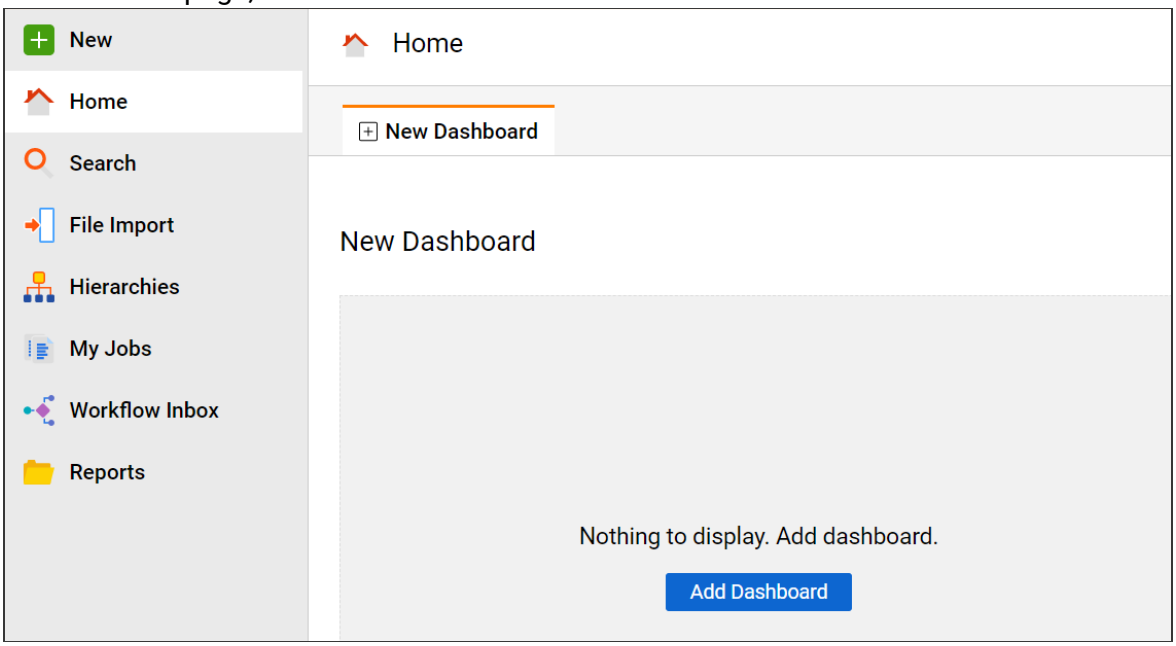
Creating a Dashboard

When you log in for the first time, the Home page displays an empty dashboard. You can create a dashboard and add charts as widgets. You can create dashboards to visualize charts. You can also add predefined dashboards to monitor MDM SaaS.

After you create a custom dashboard, you can add charts as widgets to the dashboard layout. You must add at least one widget to the dashboard layout for the charts to appear on the Home page.

Perform the following steps to create a dashboard:

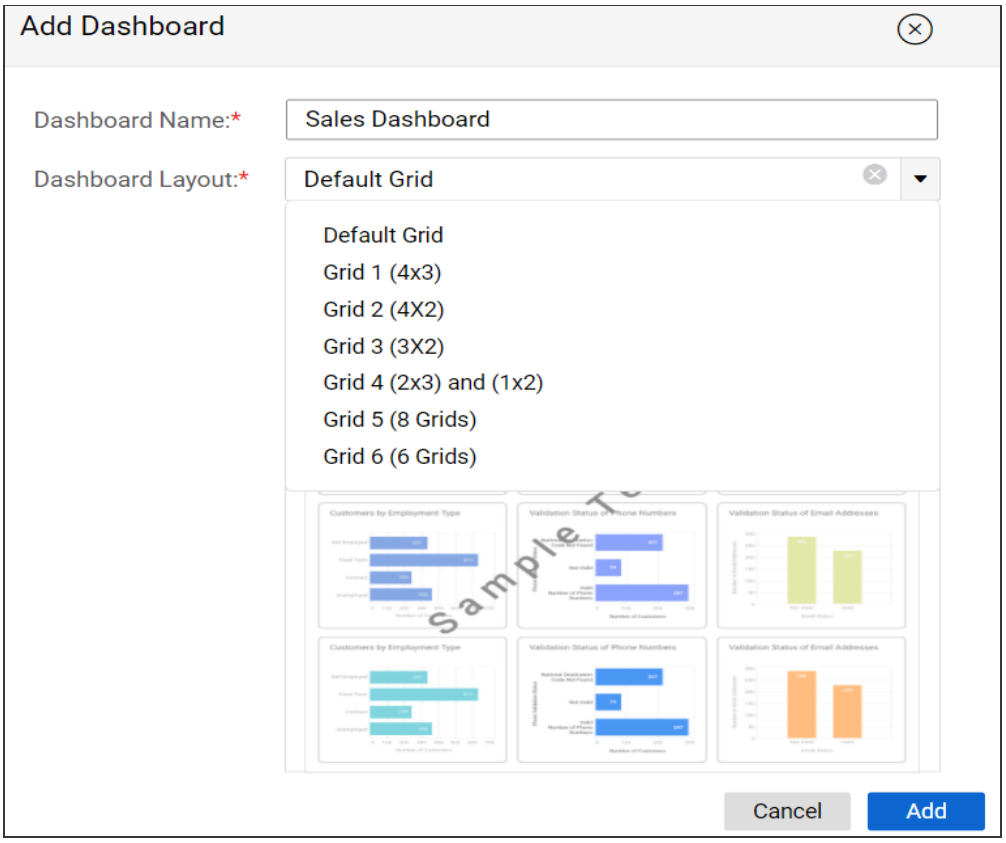
1. On the Home page, click **Add Dashboard**.



   The Add Dashboard dialog box appears.

2. Enter a name for the dashboard and select a layout.

3. Click **Add** to create a dashboard.



You can also edit or delete the dashboard based on the changing business requirements.

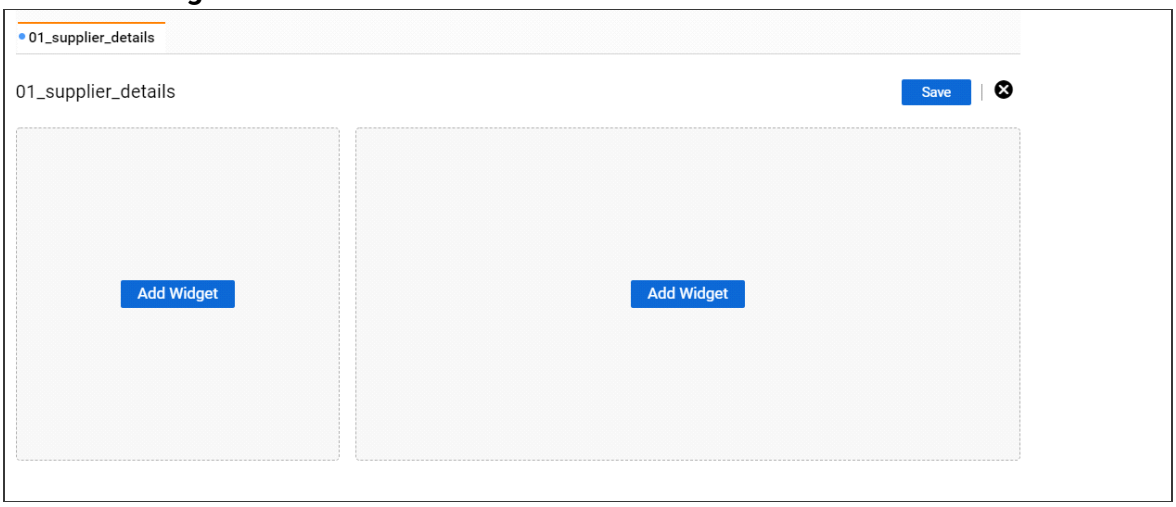
Adding a Widget

After you create a dashboard, add widgets to the dashboard to display charts. When you log in for the first time, the Home page displays an empty dashboard. You can edit the dashboard to add charts as widgets. You can select a report that you want the widget to display as a chart. After you add a widget to the dashboard, you can edit the widget to change the report. You can remove a widget from the dashboard. When you remove a widget, the dashboard layout rearranges to show the remaining widgets on the Home page.

Refer to the following steps to add a widget:

1. On the Home page, click **Edit Dashboard**.

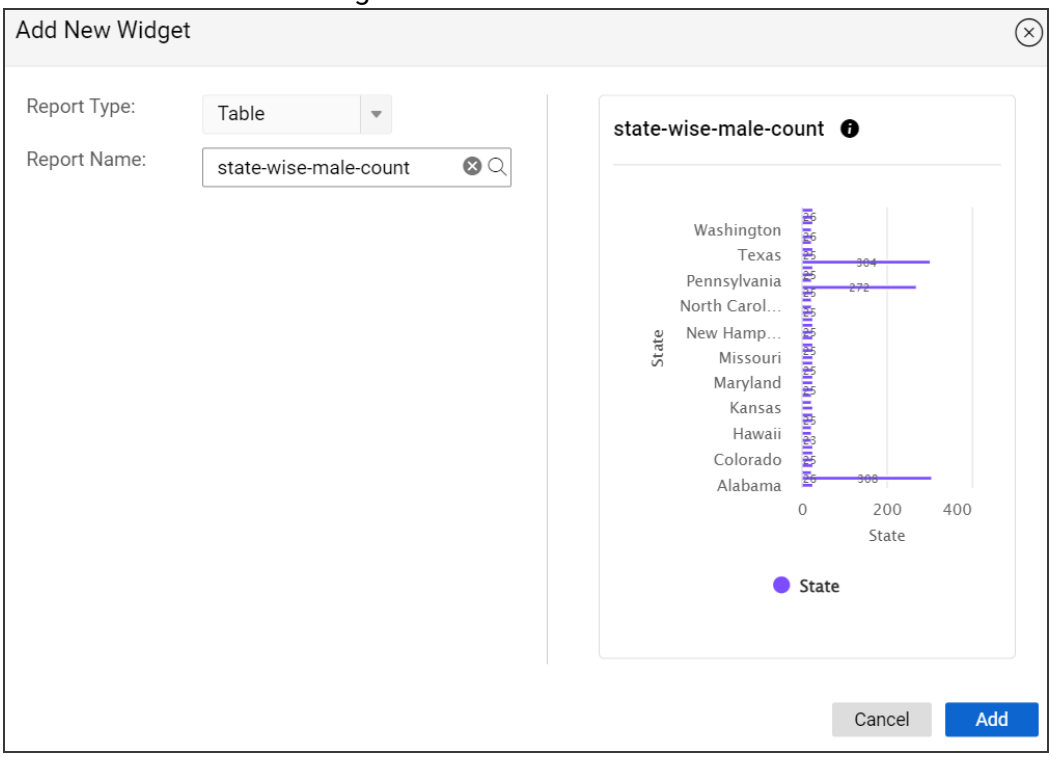
2. Click **Add Widget**.



3. Enter the name of the report to add. The reports with matching names appear in a list.

4. Select the report from the list.

5. Click **Add** and save the widget.

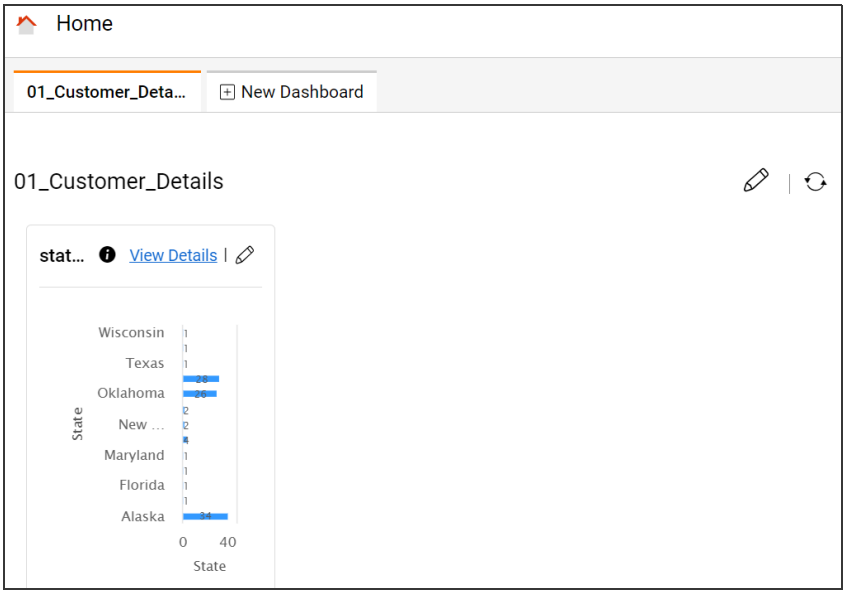


The chart appears on the widget.

Editing Dashboards

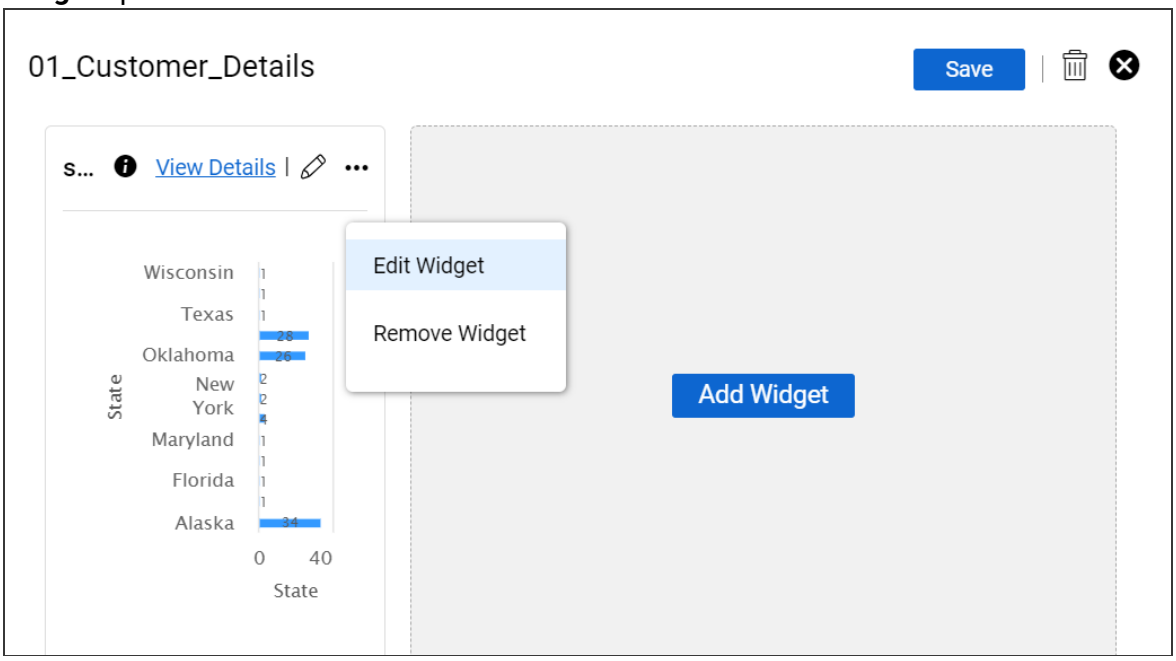
You can edit the dashboards that you create to add, edit, or remove the widgets associated with it.

1. On the home page, select the dashboard that you want to edit and click **Edit**.



The dashboard layout appears in edit mode.

2. You can modify the widget with the **Edit Widget** or remove the widget with the **Remove Widget** option.



Deleting Dashboards

You can delete the dashboards that you create from the Home page using the following steps:

1. On the Home page, select the dashboard that you want to delete.
2. Click **Edit**. The dashboard layout appears in edit mode.
3. Click **Delete**. A confirmation dialog box appears.
4. Click **Delete**. The dashboard is deleted from the Home page.

Lab 13-1: Creating Custom Reports for All Records

Overview:

A report is a collection of data that is acquired from the master data. Use reports to gain meaningful insights into your business data and business outcomes. In Multidomain MDM, you can use the reports to display information about your records. You can also create custom reports to display information about records based on your requirements. After you create a report, you can visually represent the report data as charts. You can then add the charts to widgets on the Home page dashboard.

Objectives:

- Create a new custom report
- Add the report to the Dashboard

Duration:

20 Minutes

Scenario:

Mark plans to generate a report with data from the customer business entity, as specified by a customer requirement. Subsequently, he will enhance the data with charts and then integrate these charts into the dashboard.

Tasks:

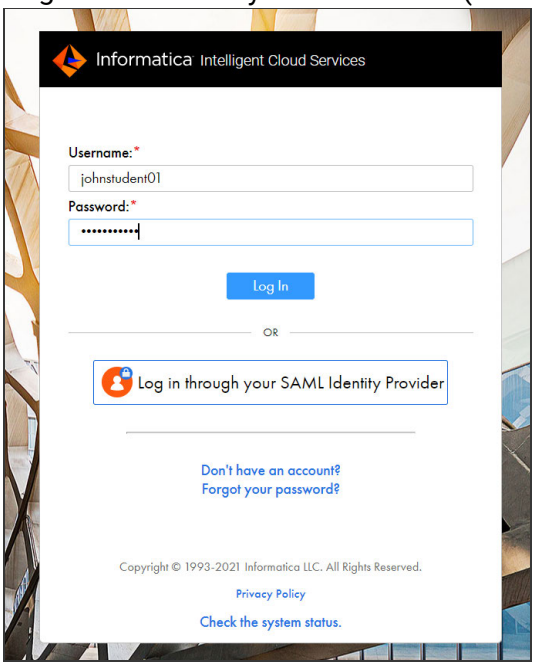
Perform the following tasks to create custom reports for Homekart scenario:

- Making fields reportable
- Create a custom report for State-Wise male population
- Create a dashboard and add reports to It
- Add the new report to the dashboard

Making Fields Reportable

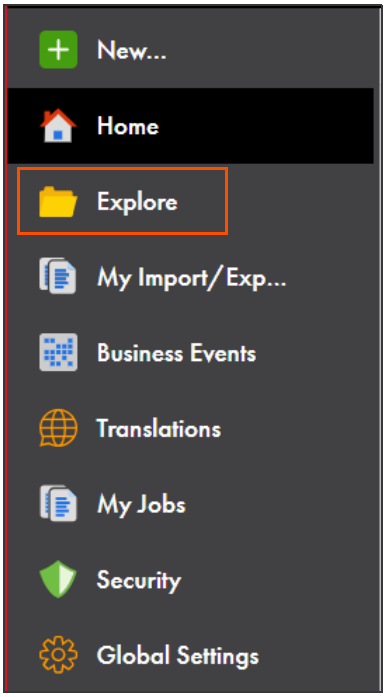
Before you start creating reports in Multidomain MDM, you will have to make the fields reportable if you want to use the fields in the report. This is configured in Business 360 console.

1. Log in to IICS with your credentials (**studentNamestudent<#> / Multidomain360**).

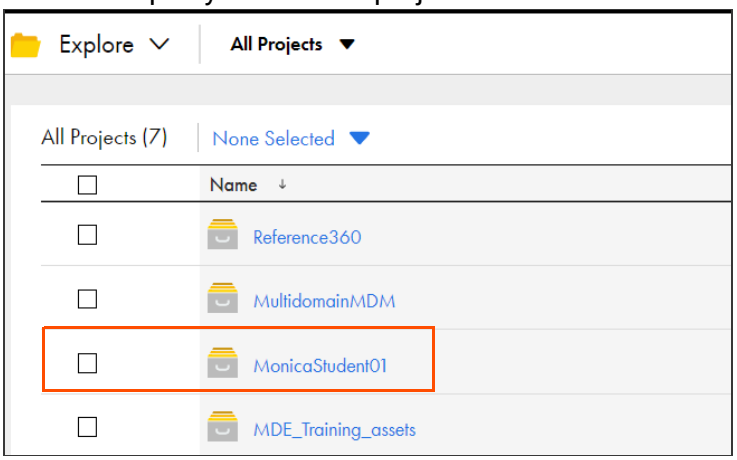


2. Open **Business 360** Console.

3. In the **Business 360 Console** window, from the left panel, click **Explore**.



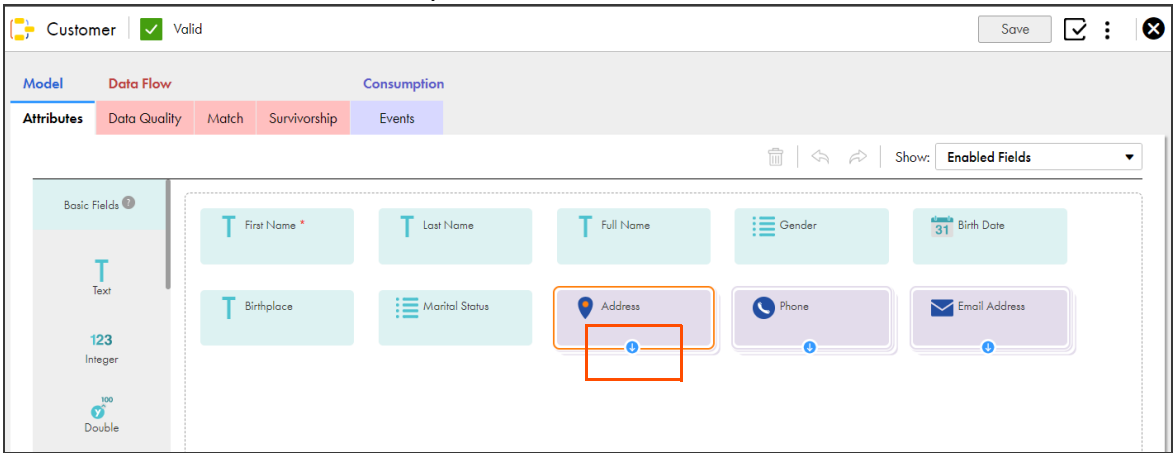
4. Click and open your student project.



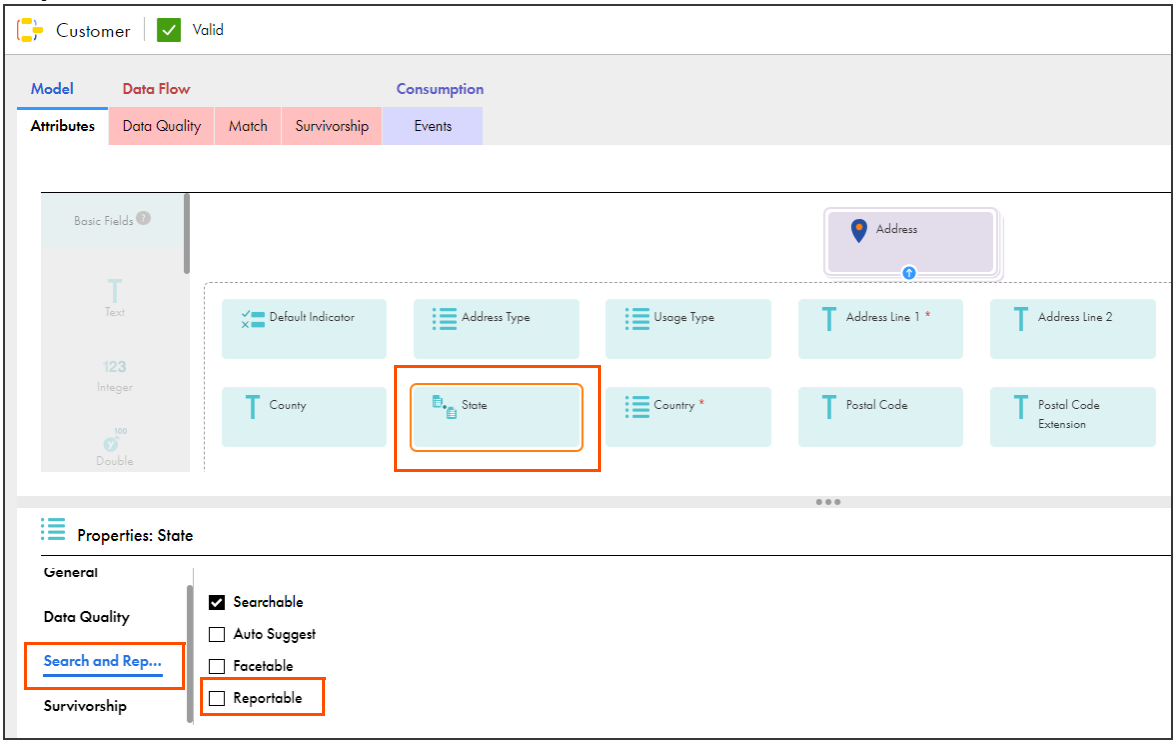
5. Open the **Customer** business entity.

For the first reports you will need fields like state and gender of the Customer business entity to be reportable and for the second report you will need Product Type field of the Product business entity to be reportable.

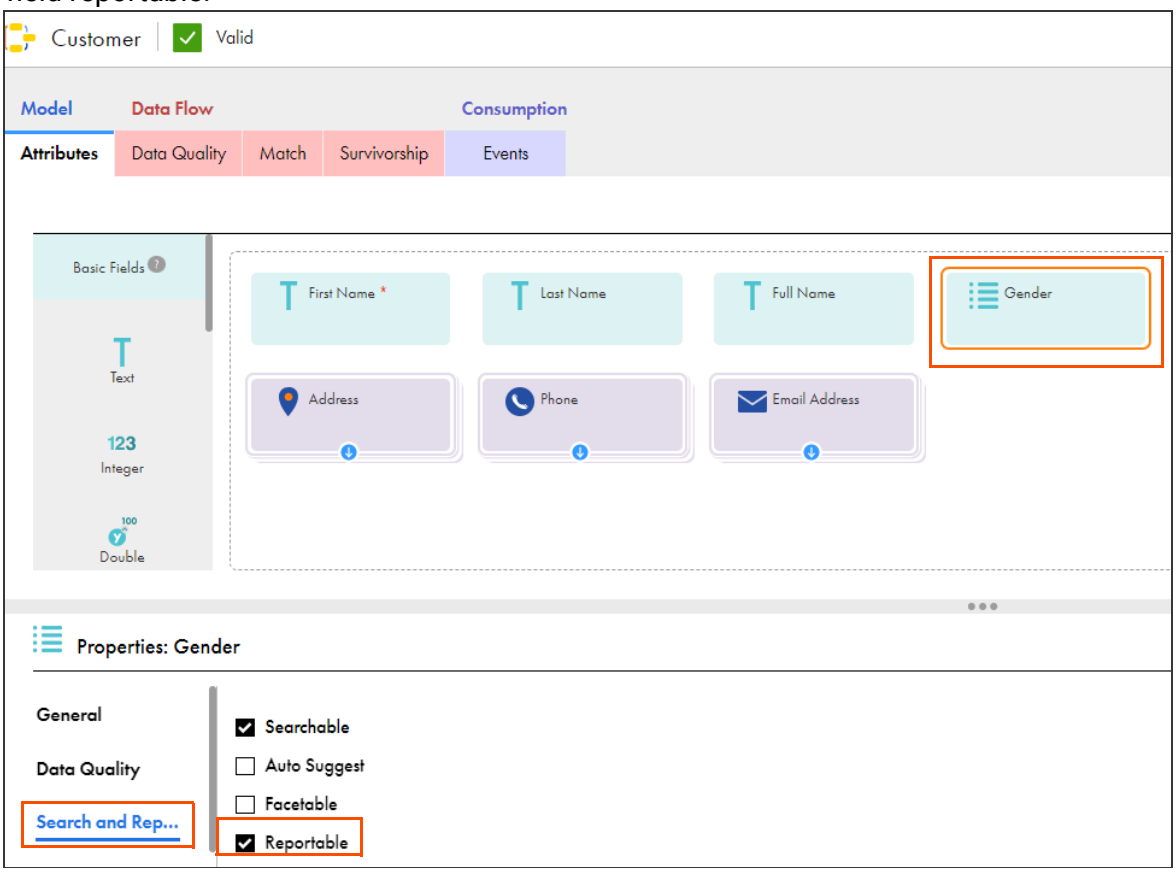
6. The business entity page opens. The canvas displays all the fields that the business entity includes. In the Attributes tab, expand the smart field **Address**.



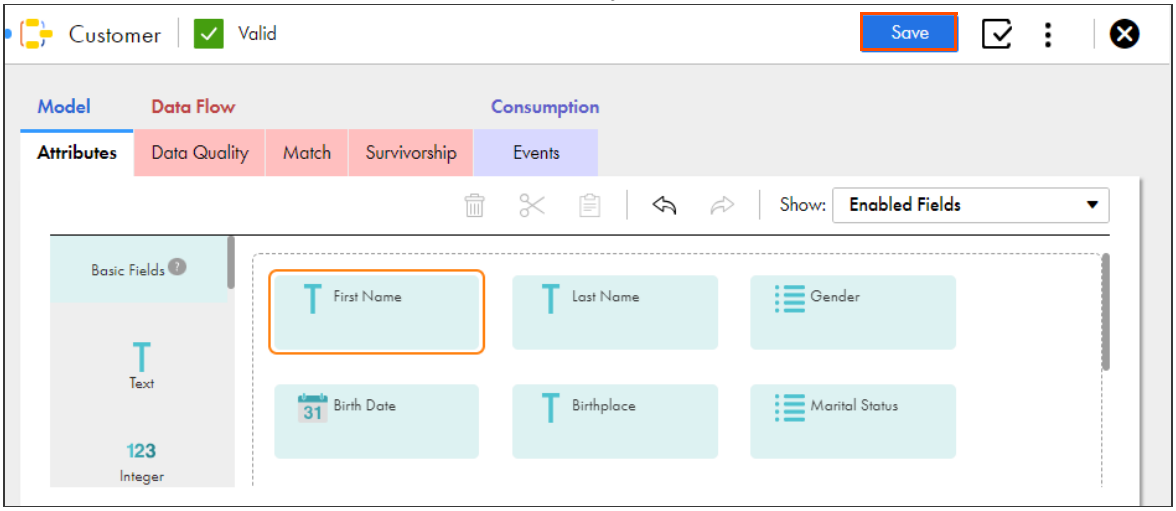
7. Select **State**, under Properties section click **Search and Reports** and enable the option **Reportable**.



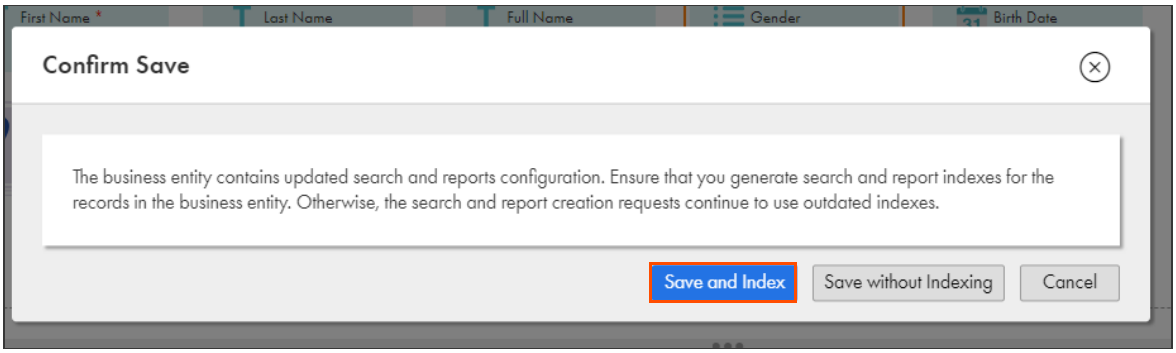
8. Collapse the Address field, select the field **Gender** and under Search and Reports, make the field reportable.



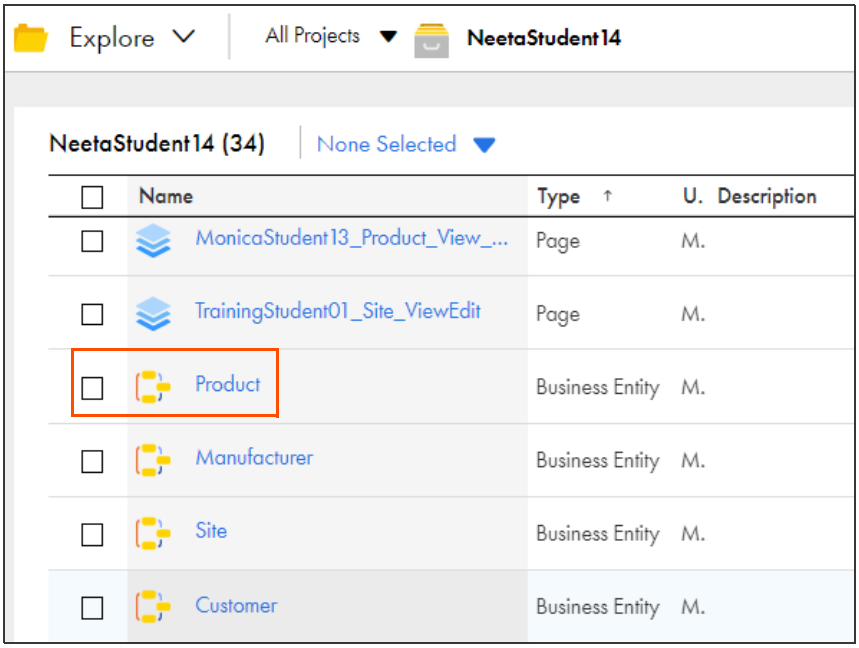
9. Click **Save** to save the customer business entity.



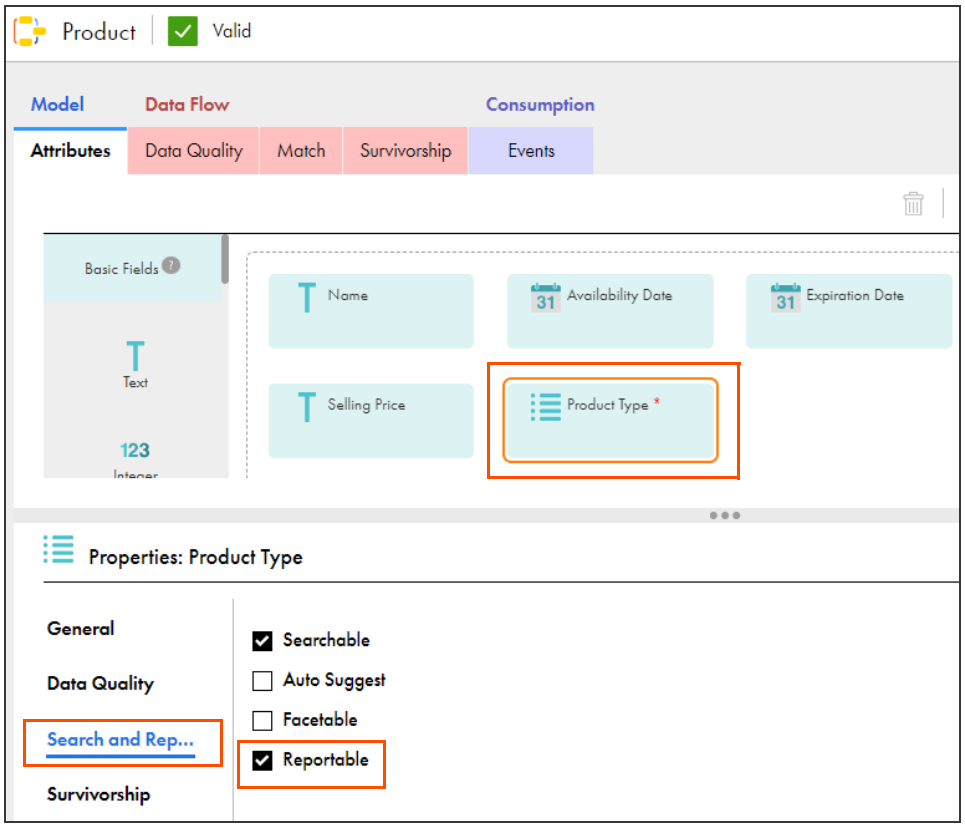
10. Click **Save and Index**.



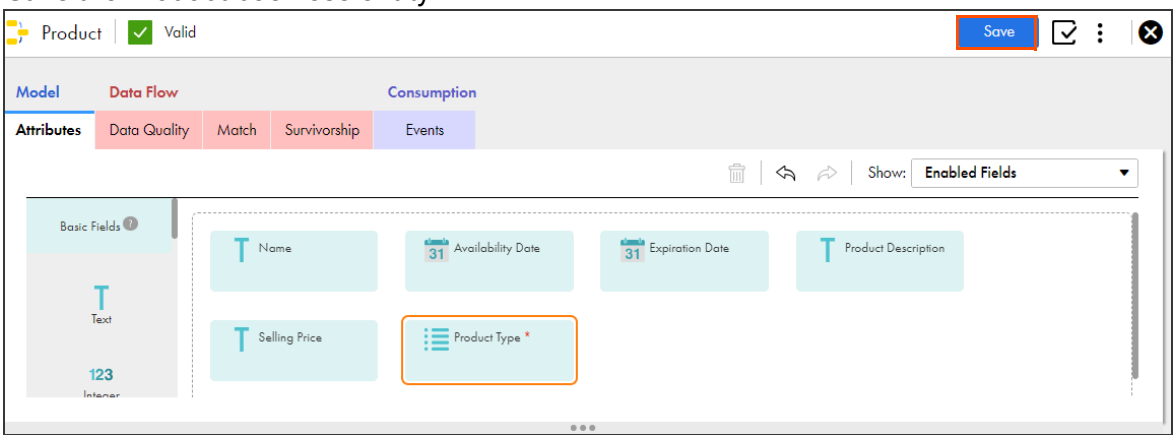
11. Similarly, to make Product Type field of the Product business entity reportable, close the Customer BE and open the **Product** BE from your student project.



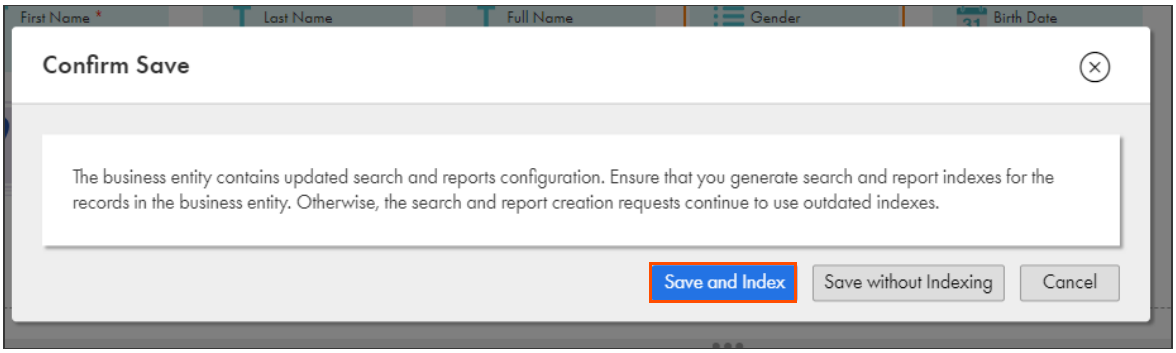
12. Select **Product Type**, under Search and Reports, make the field reportable.



13. **Save** the Product business entity.



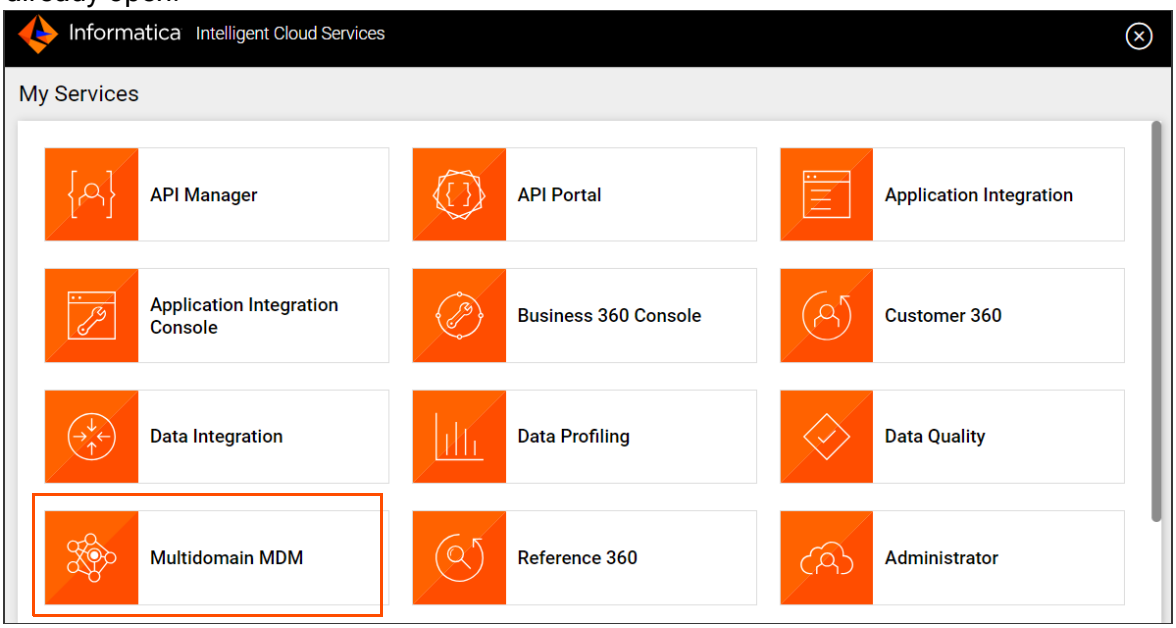
14. Click **Save and Index**.



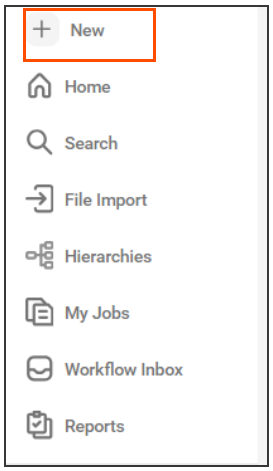
Create a Custom Report for State-Wise Male Population

You will now create a first custom report to see statewise male population.

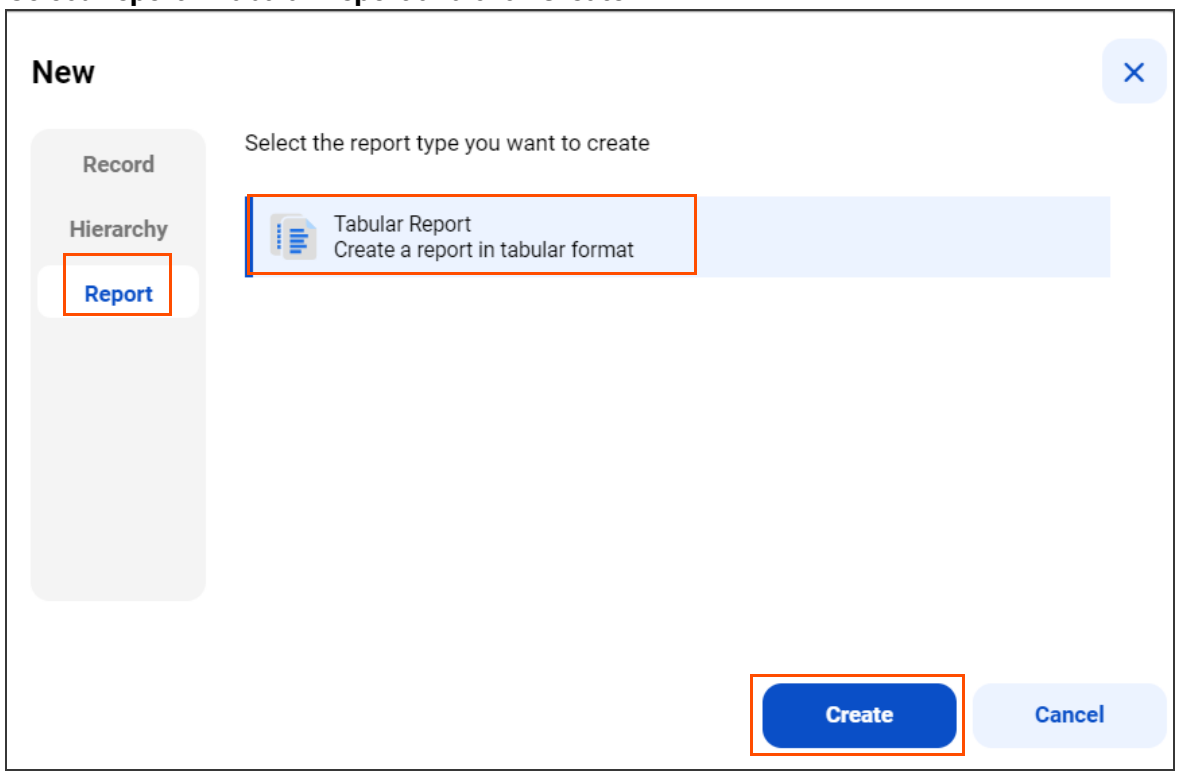
15. Using the product switcher, open **Multidomain MDM**. Refresh the page if the window is already open.



16. From the left menu, click **New**.

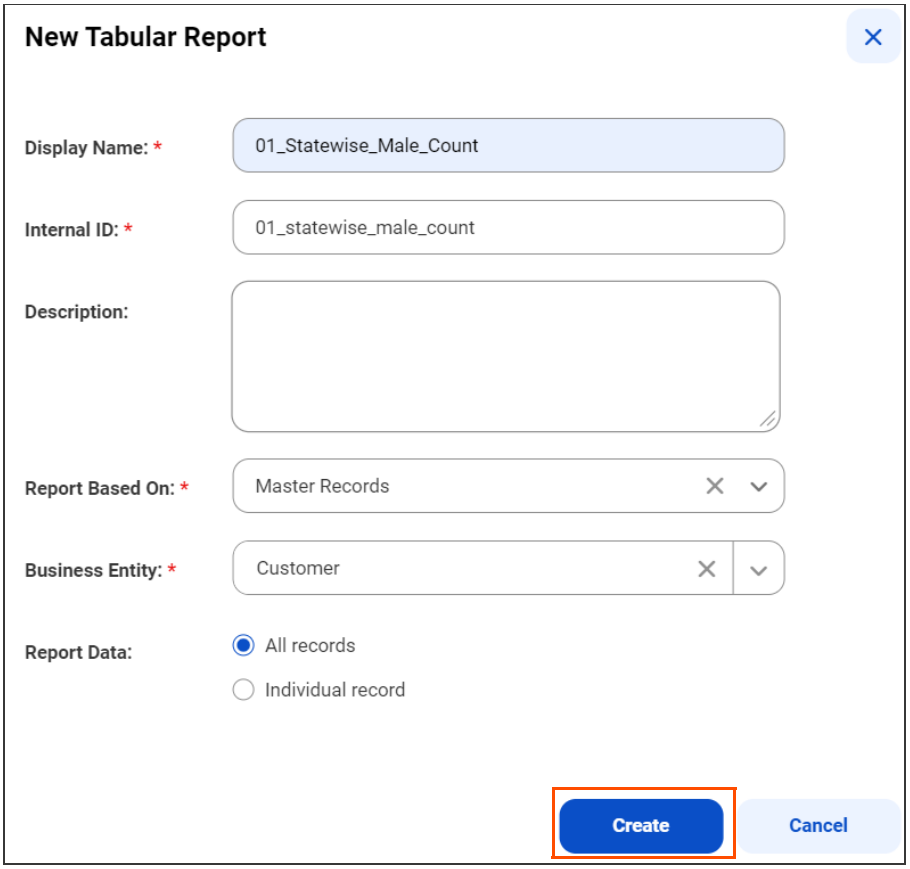


17. Select **Report** > **Tabular Report** and click **Create**.

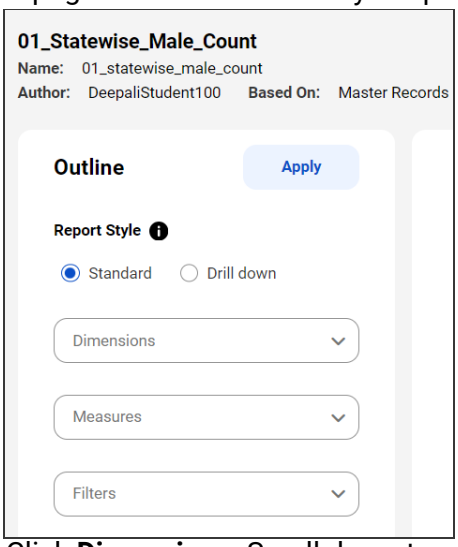


18. In the **New Tabular Report** page, enter the following details and click **Create**.

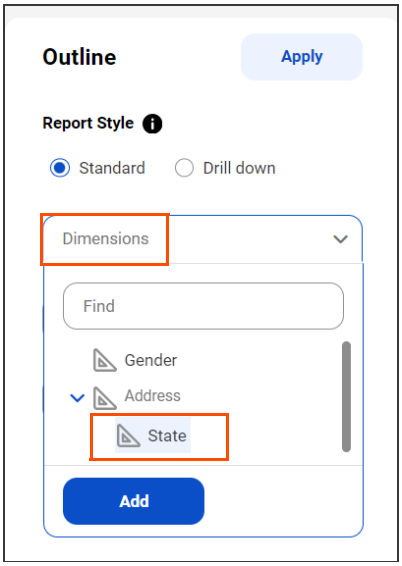
Field	Value
Display Name	<studentNumber>_ Statewise_Male_Count
Report Based On	Master Records
Business Entity	Customer
Report Data	All Records



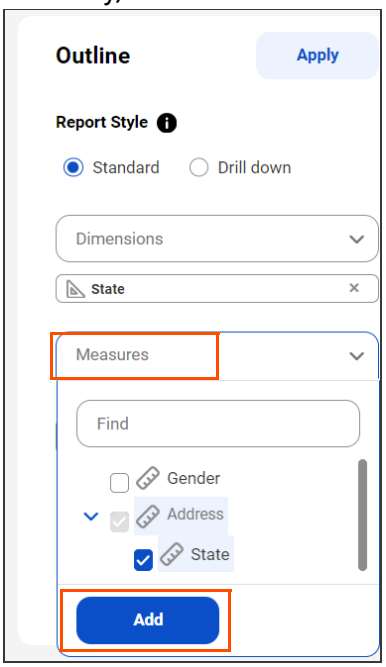
A page with the name that you specify in the Display Name field appears.



19. Click **Dimensions**. Scroll down to select **State** and click **Add**.

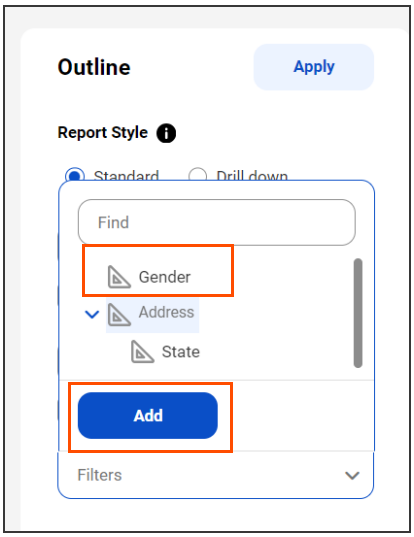


20. Similarly, select **State** as Measures and click **Add**.

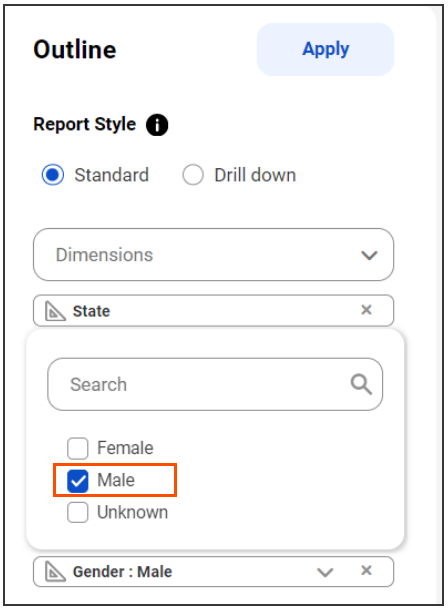


Measures are the fields that represent the quantitative values of a data set. For example, the average sales or profit count for a month.

21. To filter the report data based on the gender field values, click **Filters**. Select **Gender** and click **Add**.

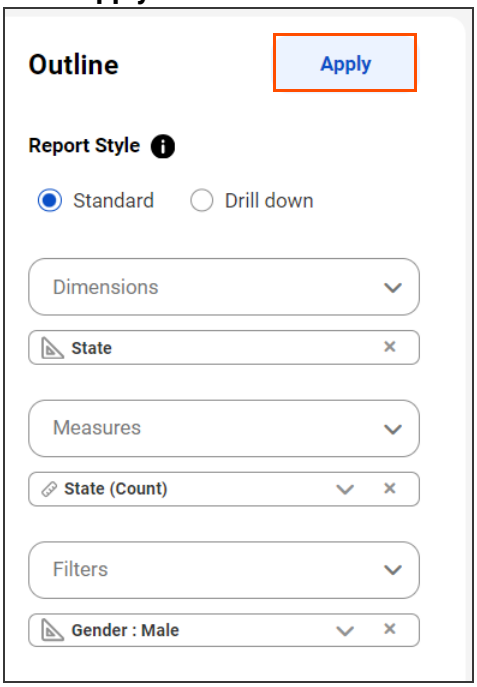


22. From the picklist, select **Male**. Click anywhere on the screen to apply the changes.



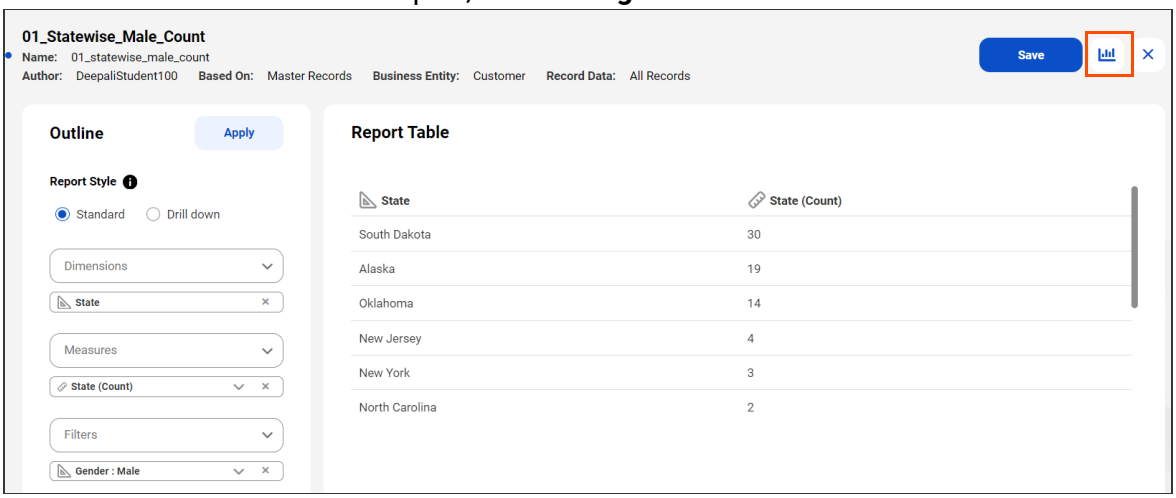
If the picklist values are taking a long time to load, click anywhere on the screen to stop loading and select the filter value again.

23. Click **Apply**.

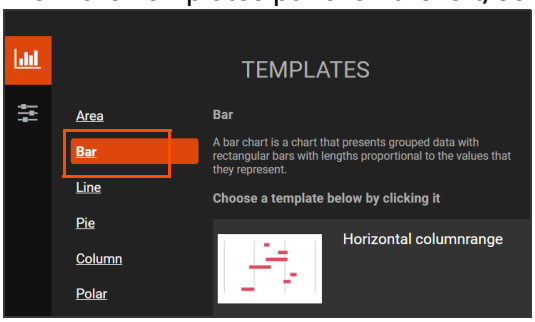


A preview of the report appears in a tabular format. The tabular report displays up to 100 rows of sample data for preview.

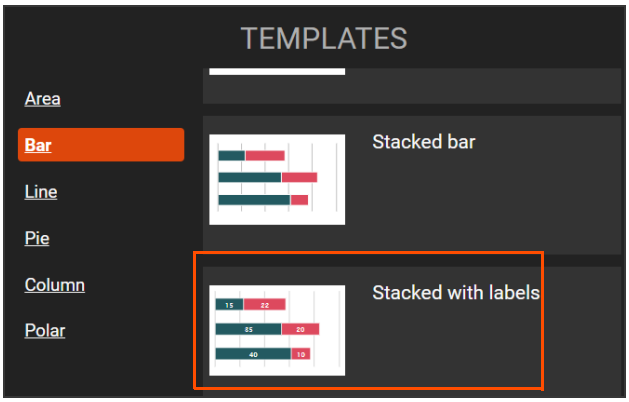
24. To add a chart to the custom report, click **Configure Chart** icon.



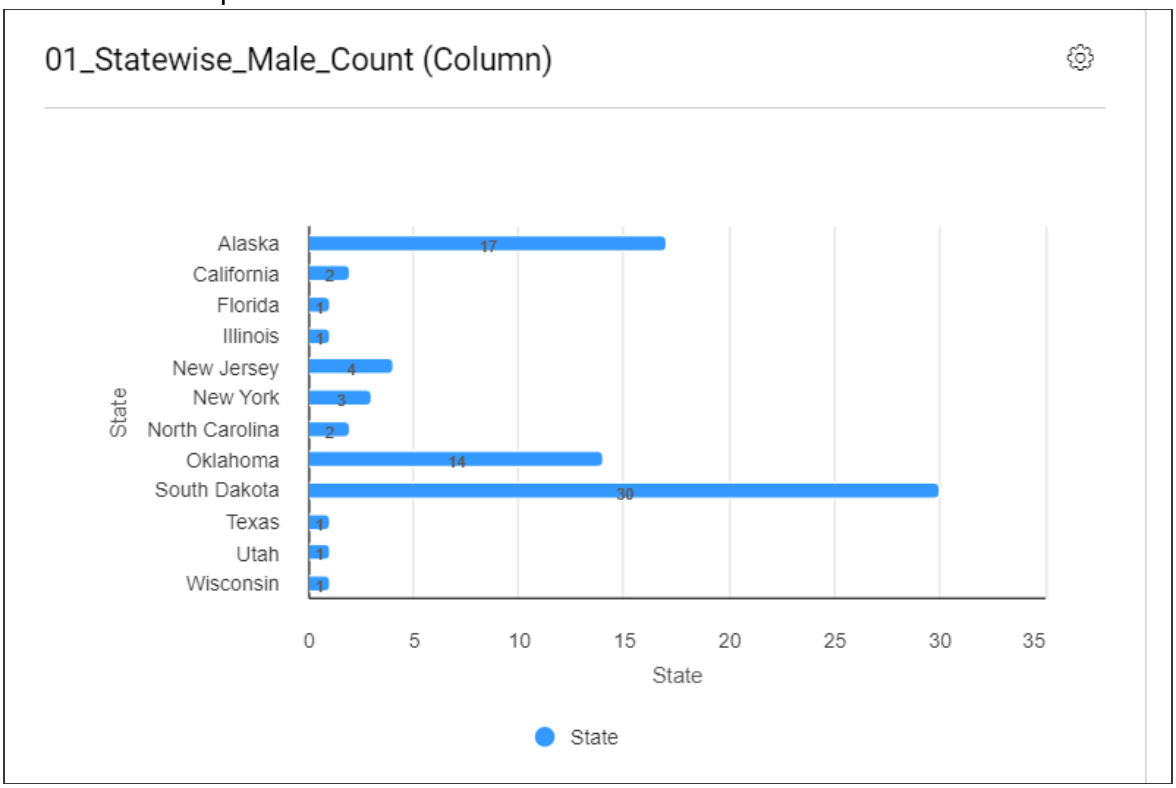
25. From the Templates panel on the left, select **the Bar** chart.



26. Scroll down and select **Stacked with labels**.

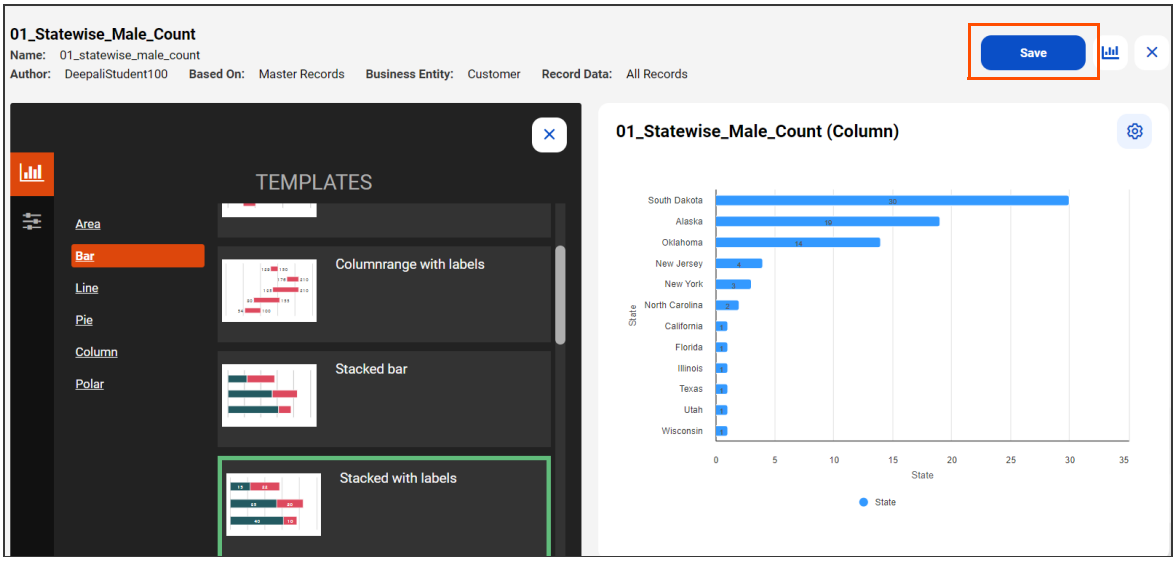


You can see the preview of the chart.



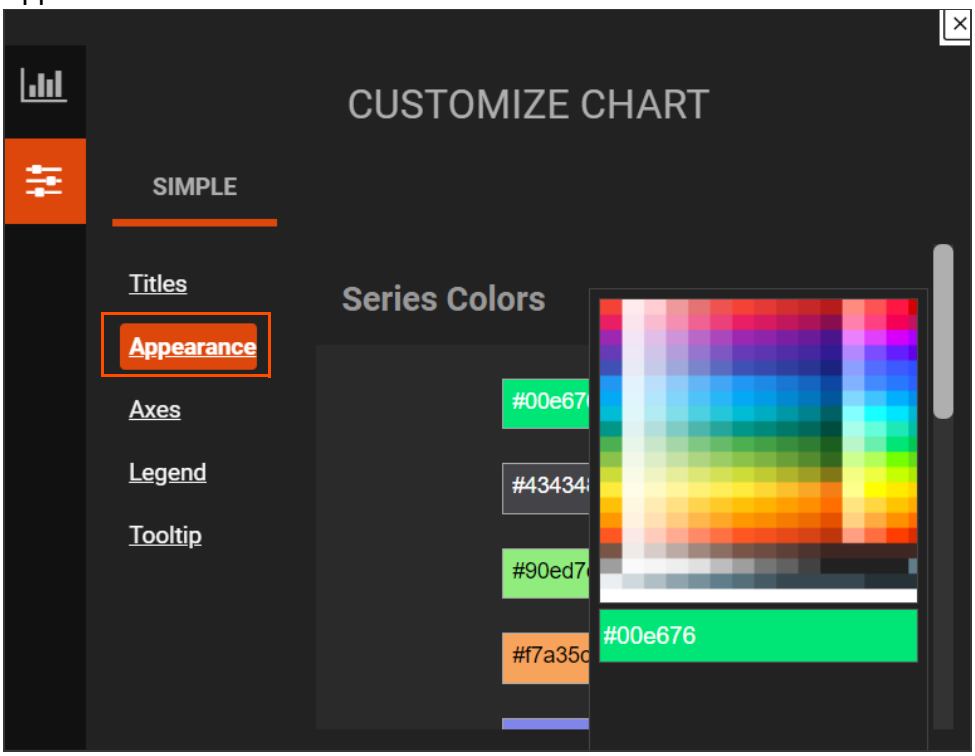
The data on your chart may differ from what is shown in the screenshot.

27. Click **Save**.

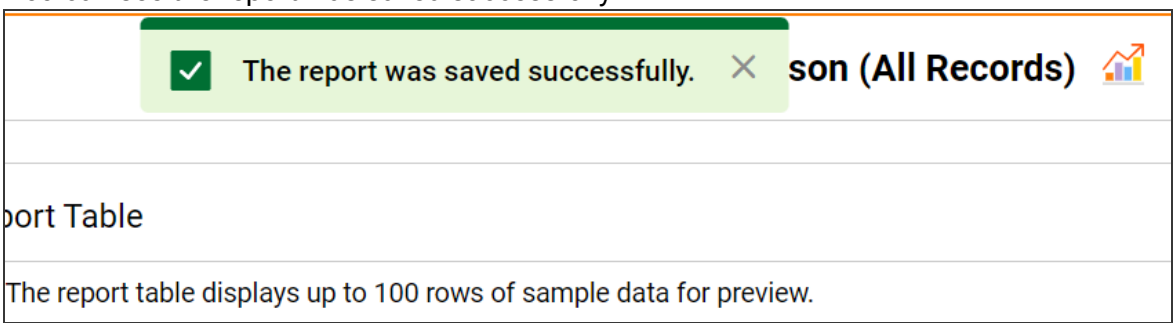


28. You can also choose to customize the chart by clicking on the **Customize Chart** icon and change the appearance like title or color of the chart. For this lab we will retain the existing

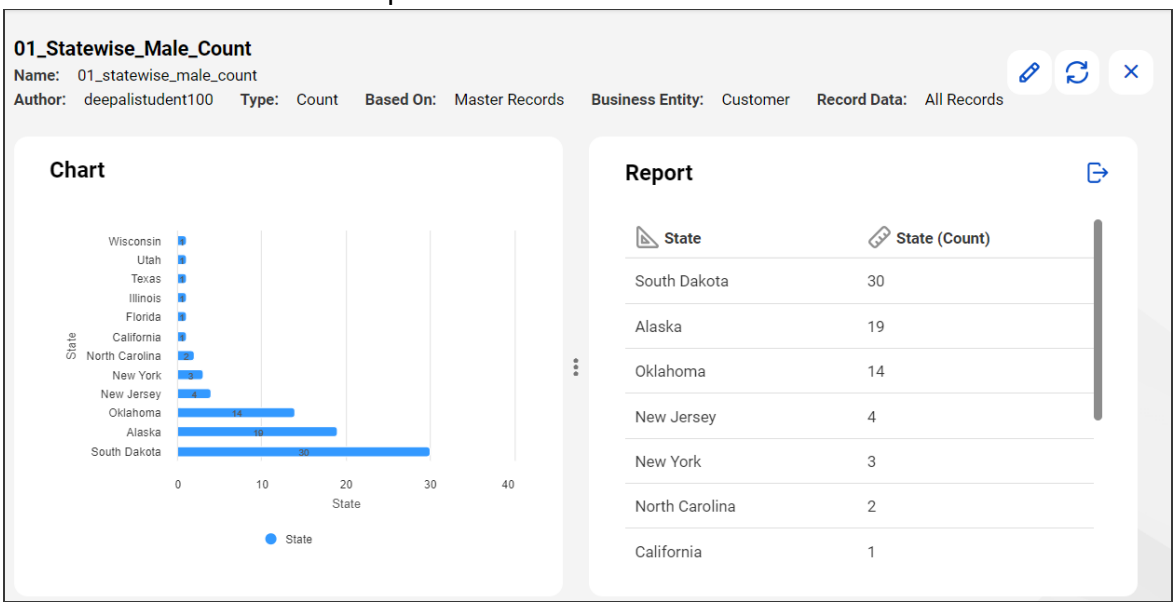
appearance.



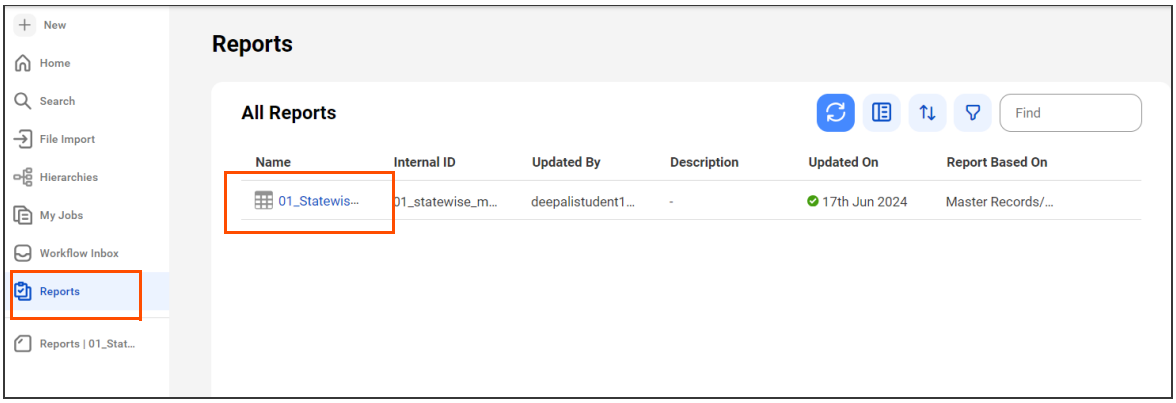
You can see the report was saved successfully.



You can see the Chart and Report.



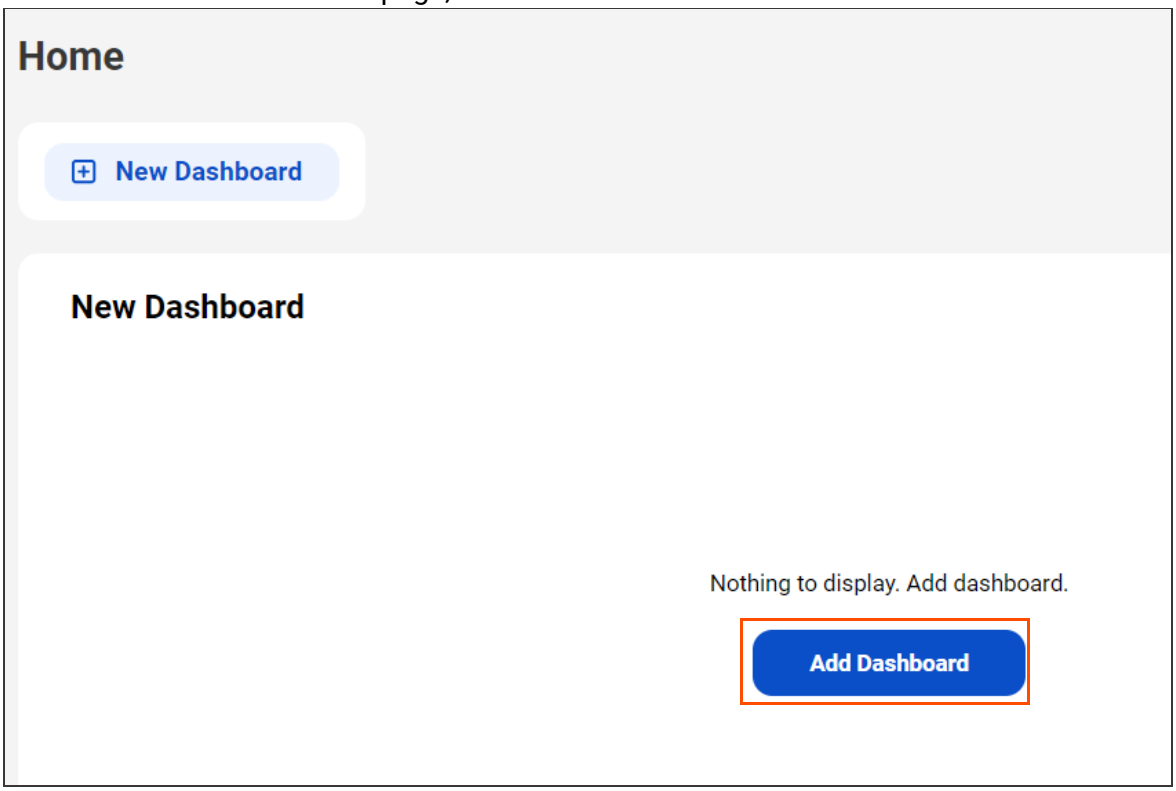
If you are unable to see the report and chart, then click on the **Reports** menu and open the report from the list.



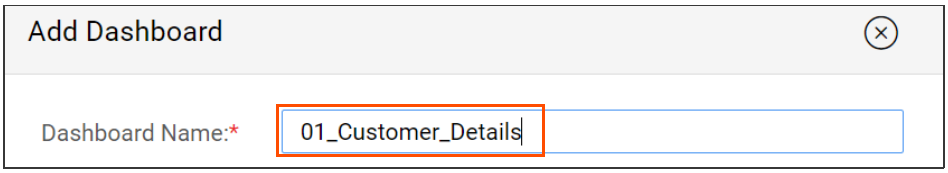
Create a Dashboard and Add Reports to It

The Home page is a dashboard that can display charts. You can add charts to the dashboards as widgets. You can add or remove the widgets to customize the dashboard.

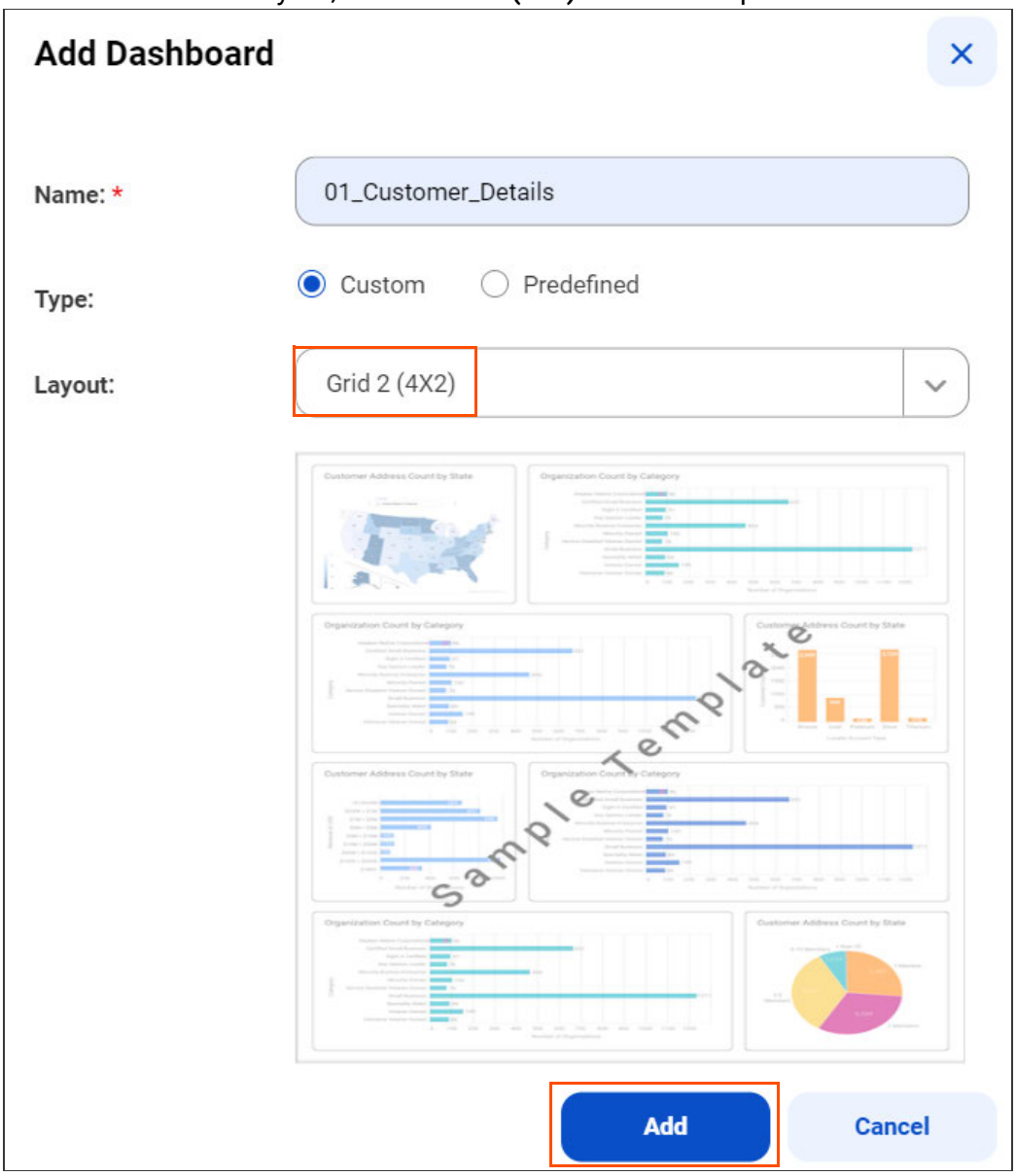
29. Click Home tab. In the home page, click **Add Dashboard**.



30. Enter the Name as **<studentNumber>_ Customer_Details**.

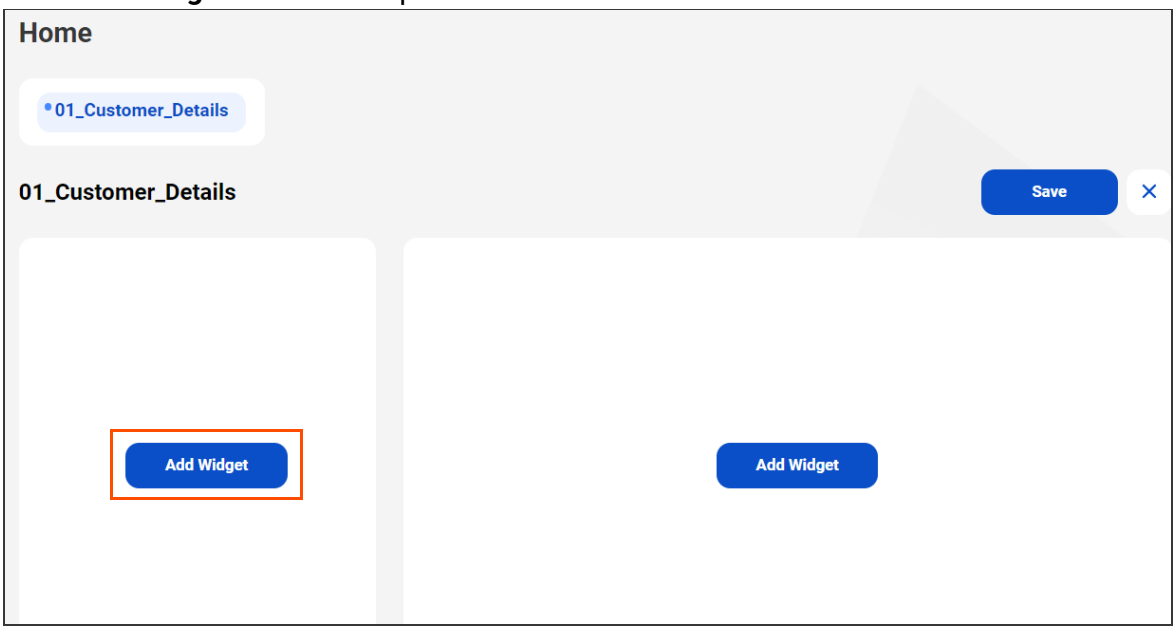


31. For the Dashboard Layout, select **Grid 2 (4*2)** from the drop-down and click **Add**.

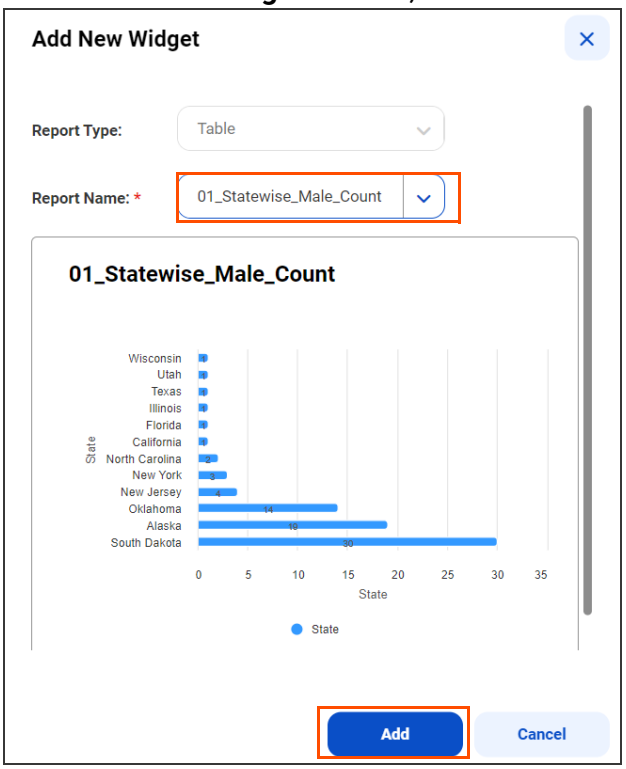


You can see a dashboard created with the widgets. You can edit the dashboard to add charts as widgets. You can select a report that you want the widget to display as a chart. Here, you will add two reports to the dashboard. One will be the predefined report and another one will be the custom report.

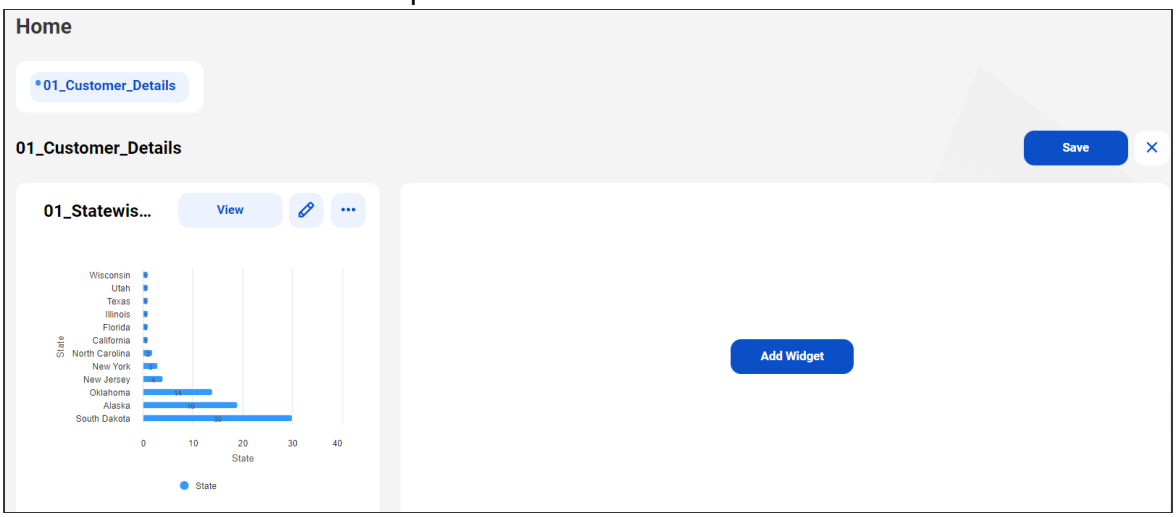
32. Click **Add Widget** to add the report.



33. In the **Add New Widget** window, search and select **State-wise-male-count**. Click **Add**.

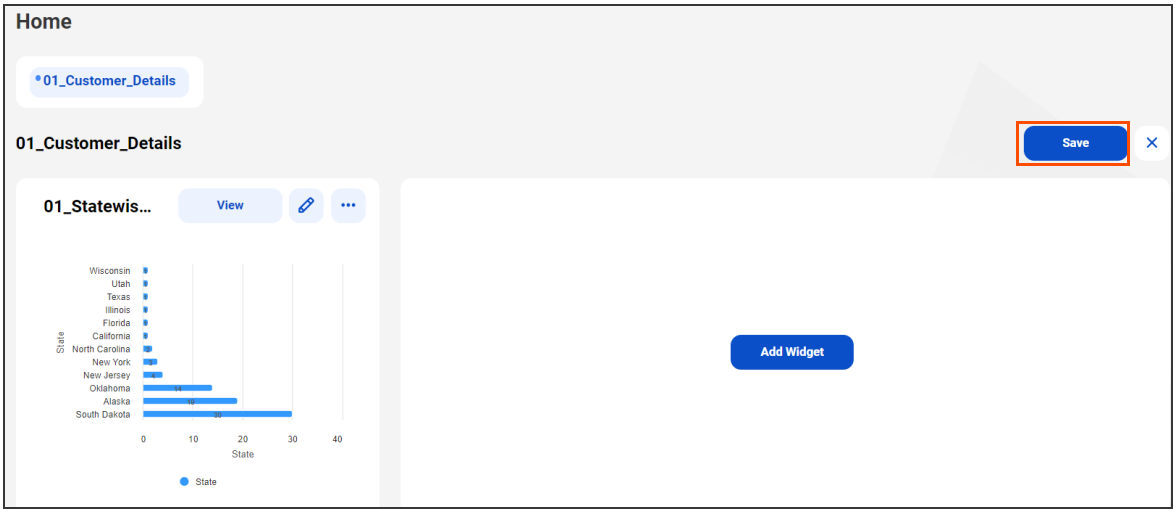


You can see that the custom report is added to the dashboard.



The data on your chart may differ from what is shown in the screenshot.

34. Click **Save**.



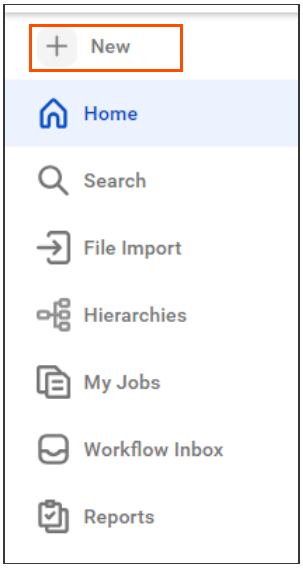
You can see that **The dashboard was created successfully** message.

Self-Try Exercise (Mandatory)

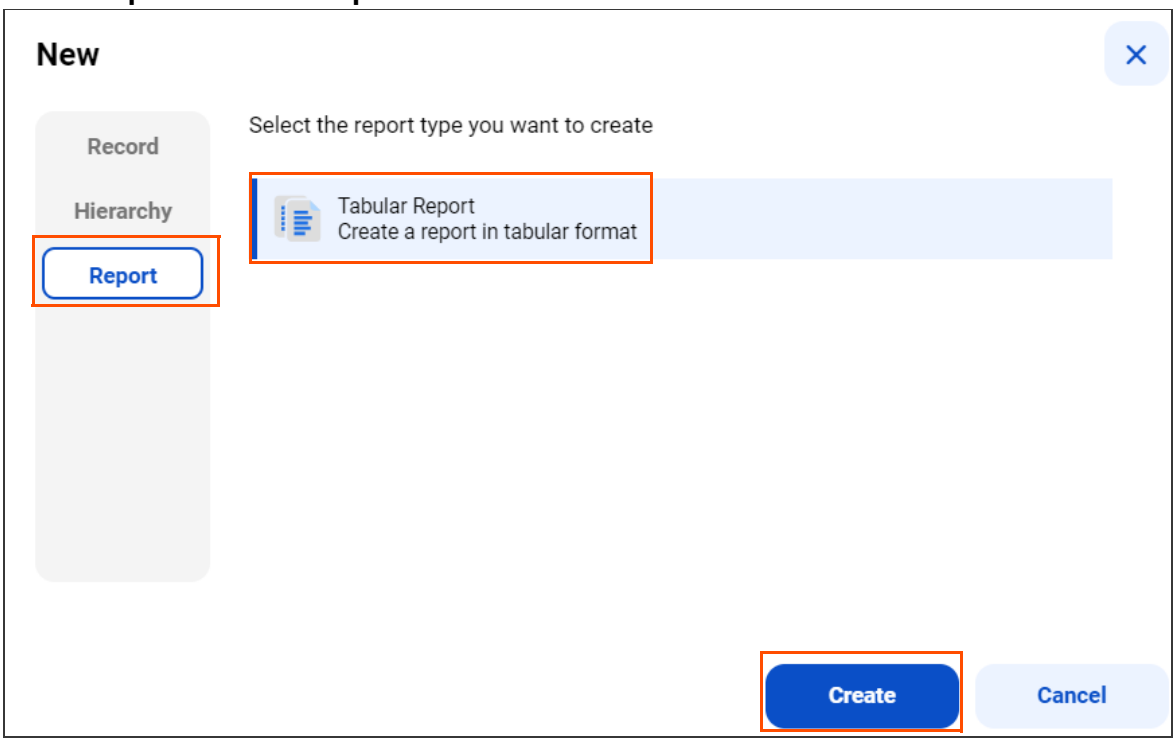
Create a Custom Report to Count Products by Type

You will now create another custom report and add it to the dashboard.

35. From the left menu, click **New**.

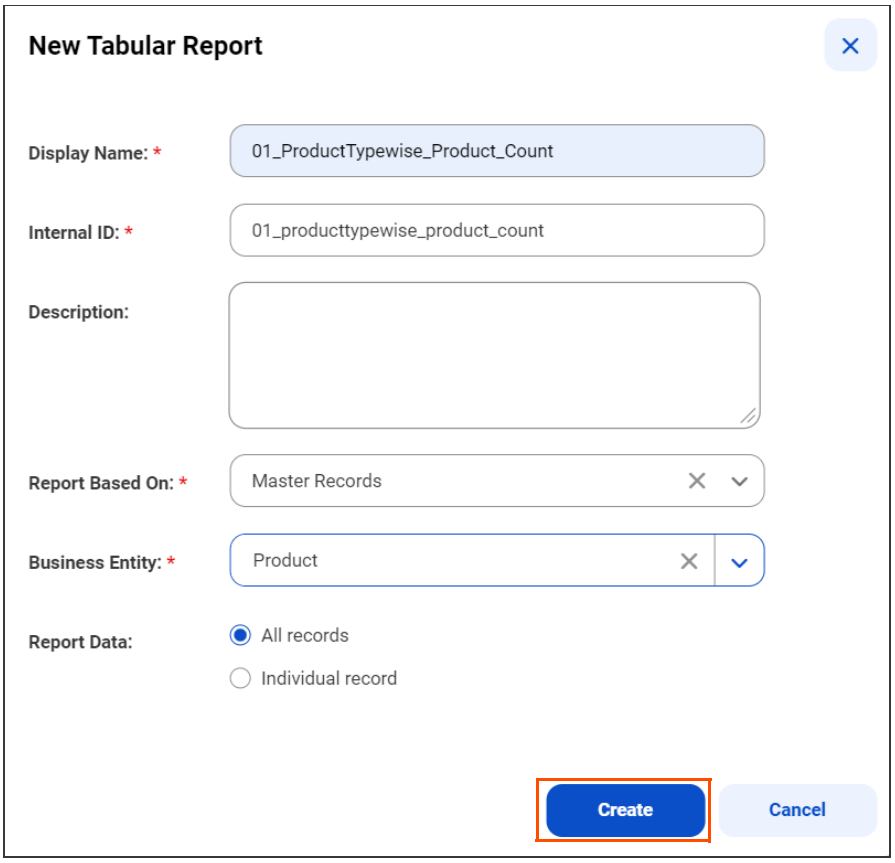


36. Select **Report** > **Tabular Report** and click **Create**.

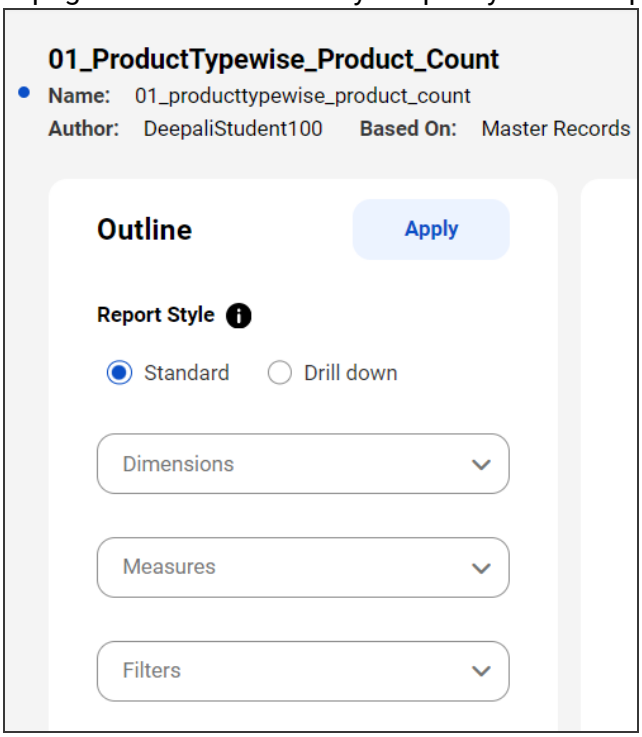


37. In the **New Tabular Report** page, enter the following details and click **Create**.

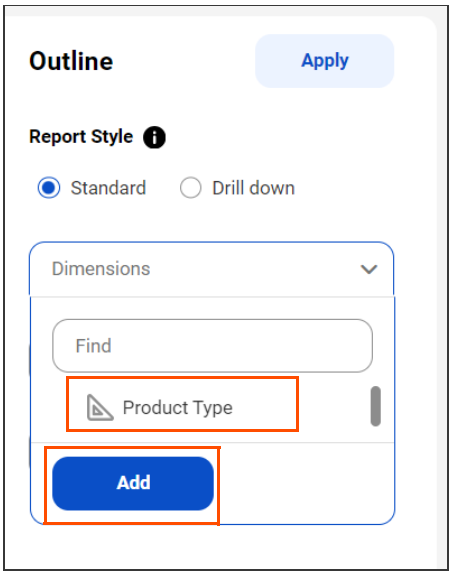
Field	Value
Display Name	<studentNumber>_ ProductTypewise_Product_Count
Report Based On	Master Records
Business Entity	Product
Report Data	All Records



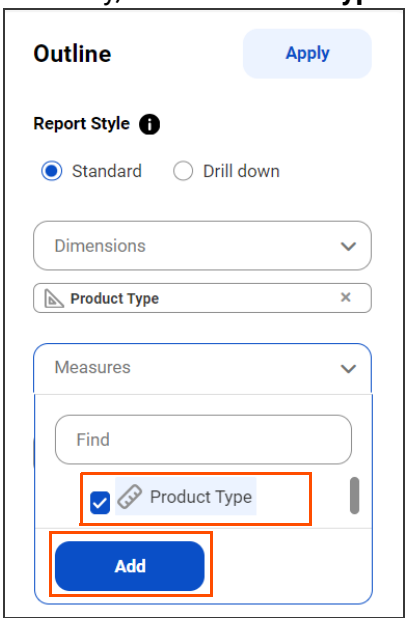
A page with the name that you specify in the Display Name field appears.



38. Click **Dimensions**. Scroll down to select **Product Type** and click **Add**.



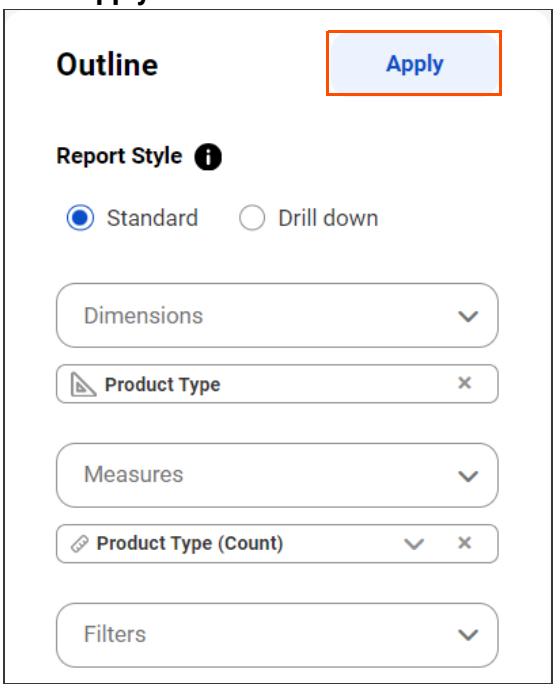
39. Similarly, select **Product Type** as measure and click **Add**.



Measures are the fields that represent the quantitative values of a data set. For example, the average sales or profit count for a month.

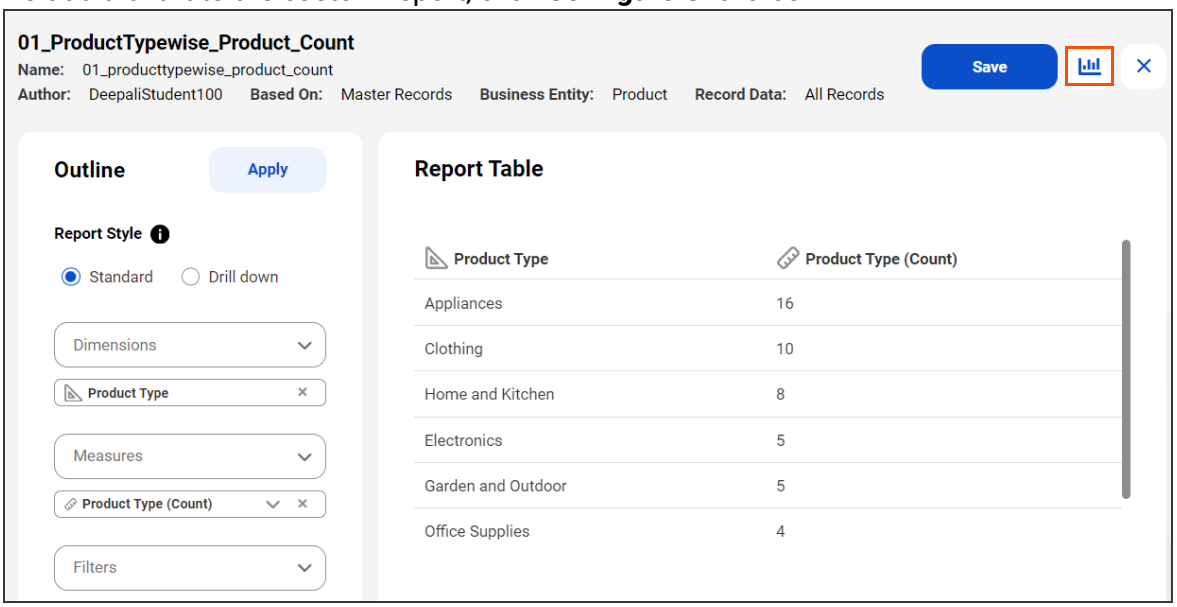
If the picklist values are taking a long time to load, click anywhere on the screen to stop loading and select the filter value again.

40. Click **Apply**.

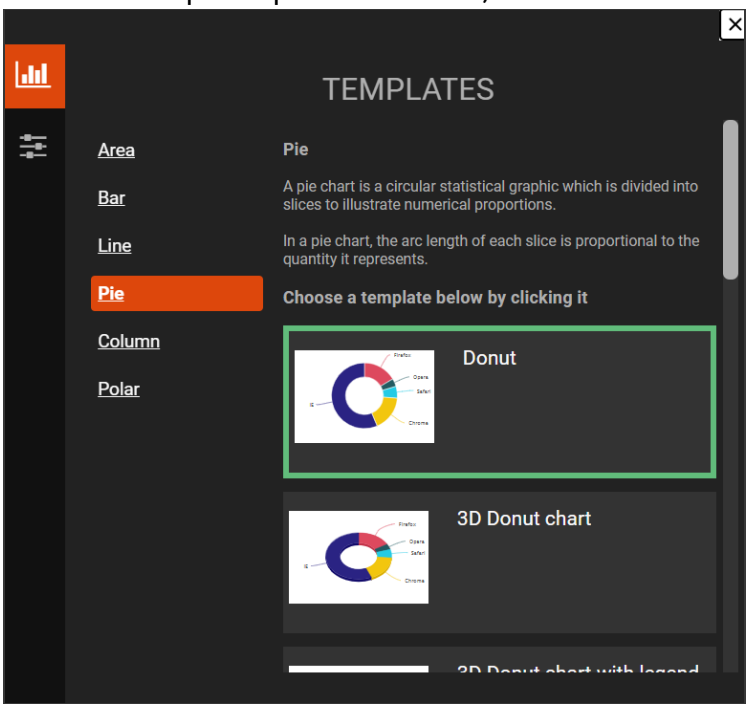


A preview of the report appears in a tabular format. The tabular report displays up to 100 rows of sample data for preview.

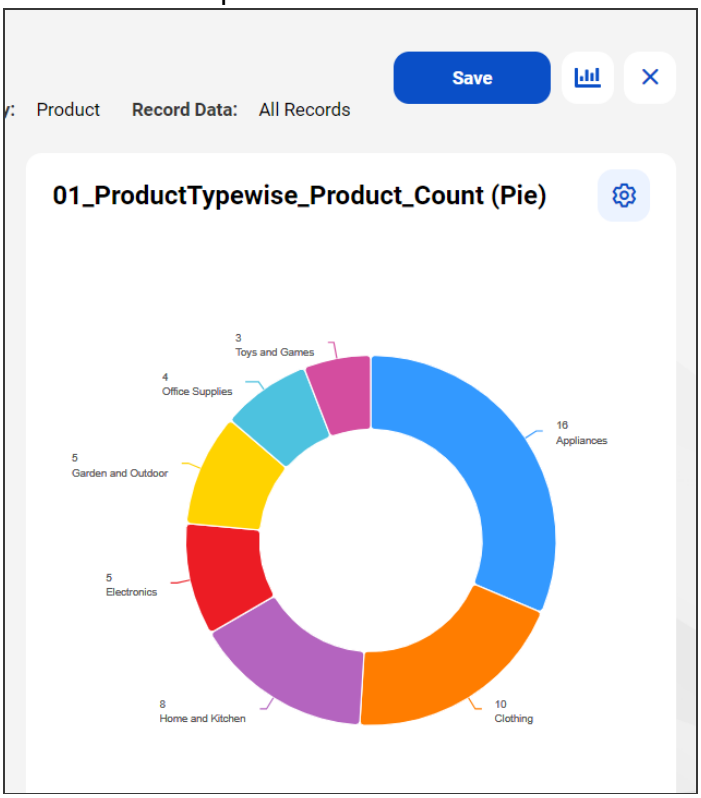
41. To add a chart to the custom report, click **Configure Chart** icon.



42. From the Templates panel on the left, select the **Pie** chart and select **Donut**.

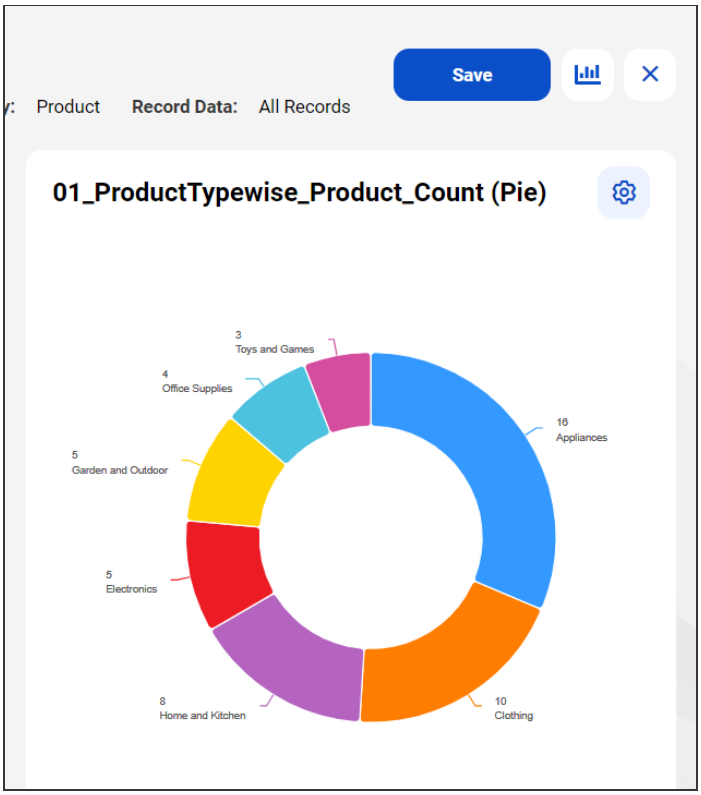


You can see the preview of the chart.



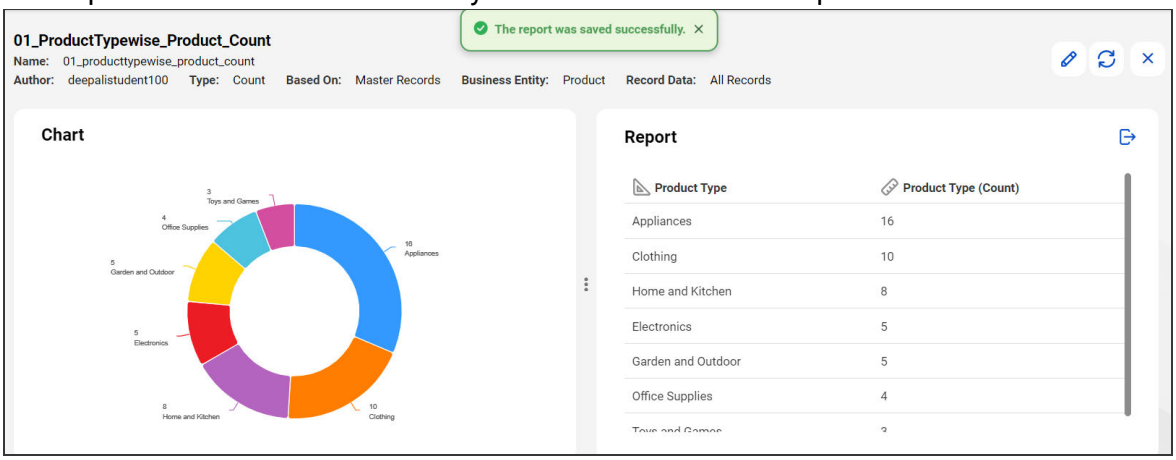
The data on your chart may differ from what is shown in the screenshot.

43. Click **Save**.



Refresh the page if you can't see the report.

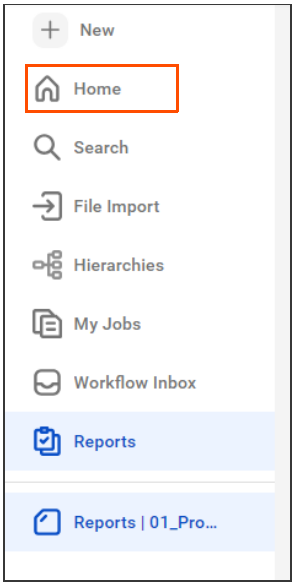
The report will be saved successfully and see the Chart and Report next to each other.



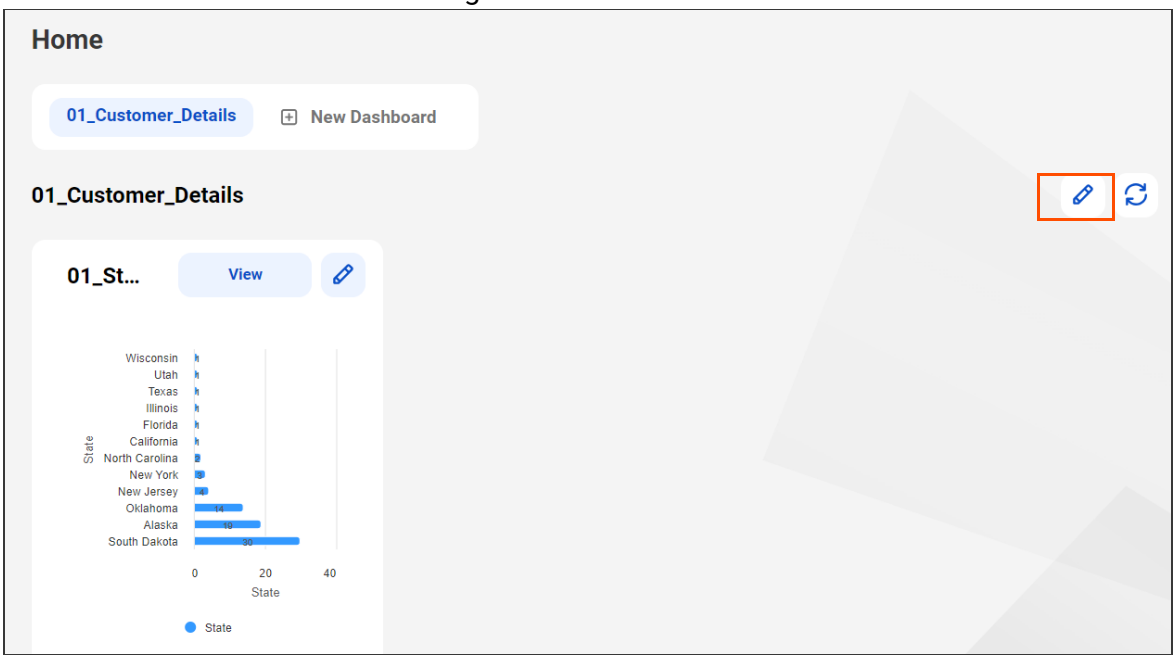
Add the New Report to the Dashboard

Now, you will add the second custom report to the dashboard.

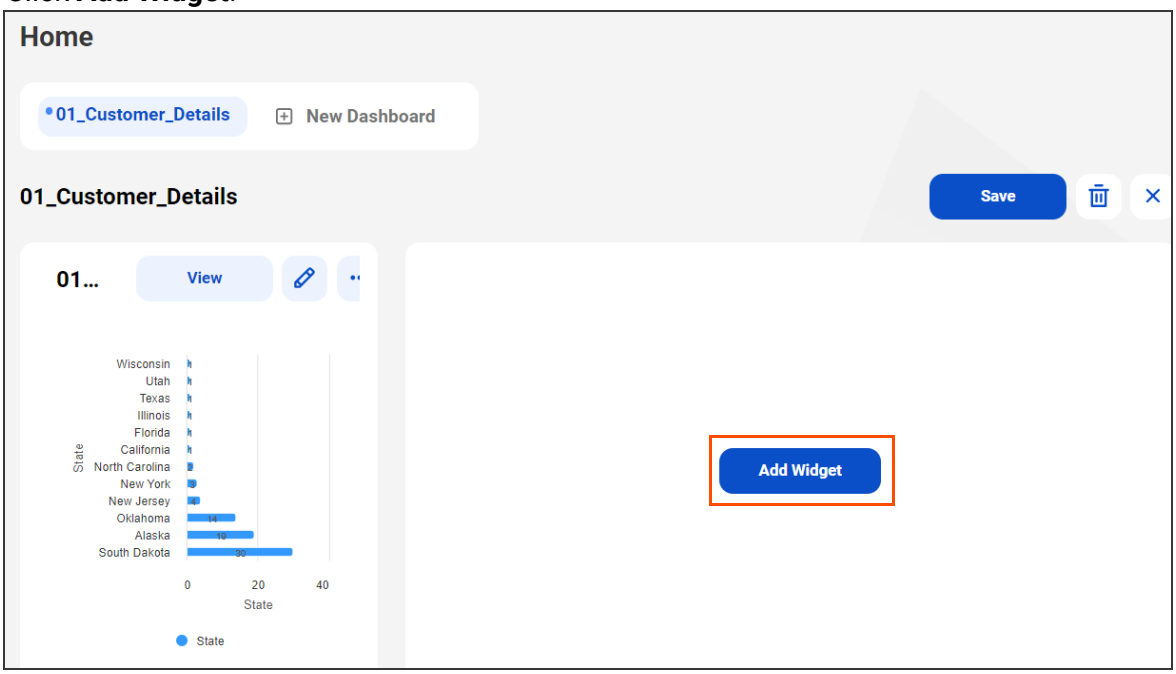
44. Click **Home** from the left menu.



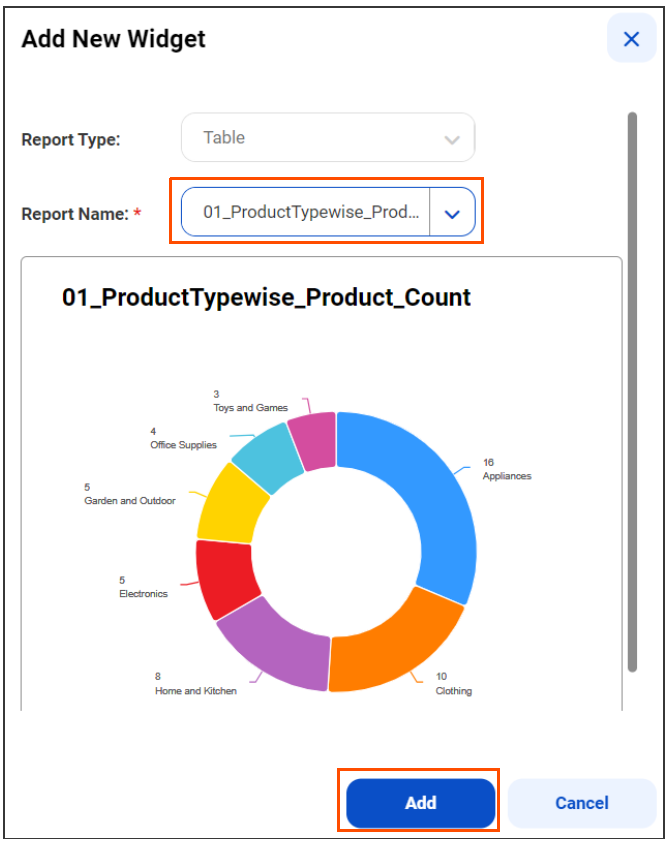
45. Click the **edit** icon to add a new widget to the dashboard.



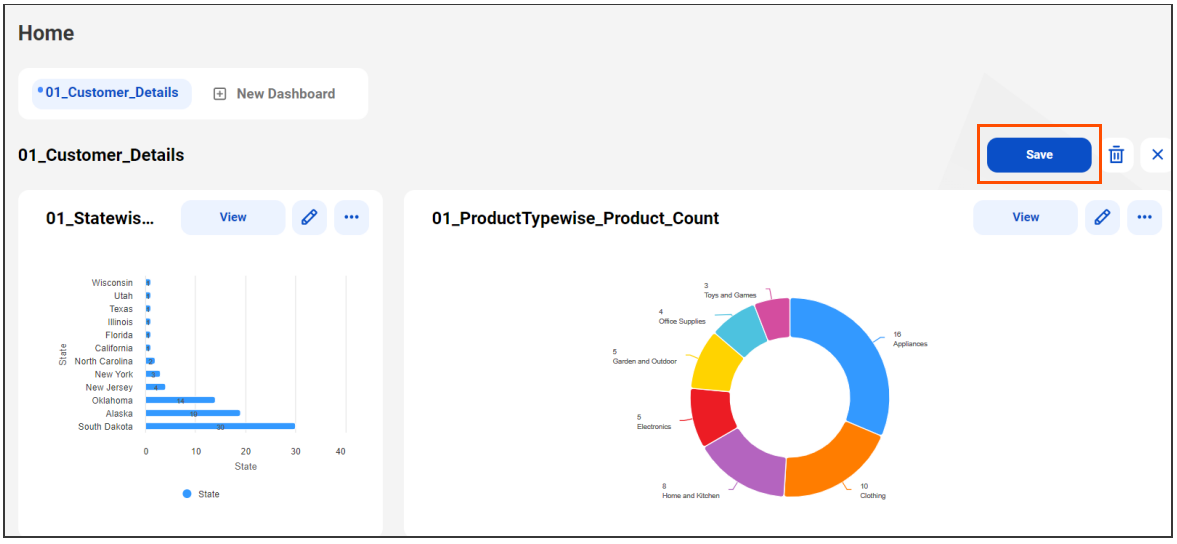
46. Click **Add Widget**.



47. In the **Add New Widget** window, search and select **ProductTypewise_Product_Count** and click **Add**.

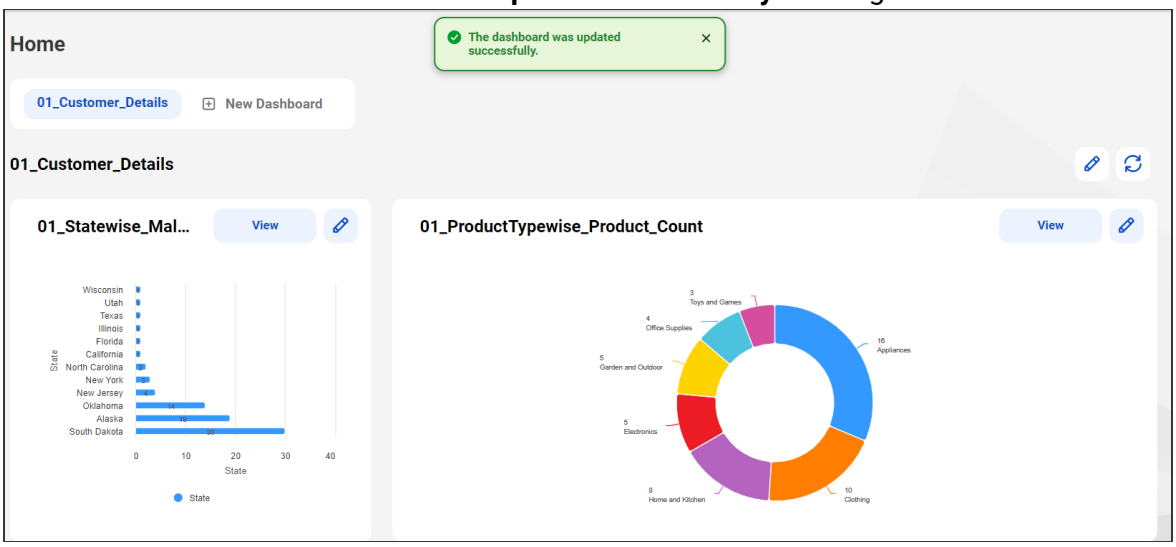


48. You can see that the custom report is added to the dashboard. Click **Save**.



The data on your chart may differ from what is shown in the screenshot.

You can see that **The dashboard was updated successfully** message.



This concludes the lab.

Lab 13-2: Creating Reports for Individual Record

Overview:

A report is a collection of data that is acquired from the master data. Use reports to gain meaningful insights into your business data and business outcomes. In Multidomain MDM 360, you can use the reports to display information about your records.

Objectives:

- Create a new custom report for individual record
- Add report to the records

Duration:

20 Minutes

Scenario:

Mark plans to generate a report for individual records within the customer business entity, as requested by a customer. Following the creation of the report, he will modify the customer Business Entity page to incorporate widgets. Using these widgets, Mark will then integrate the report into an individual record.

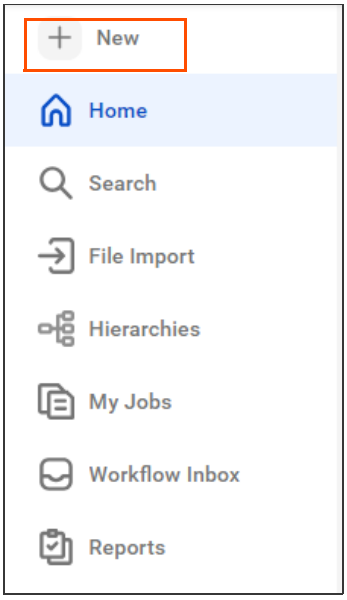
Tasks:

- Creating custom report for individual records
- Creating another custom report
- Add widgets to the record

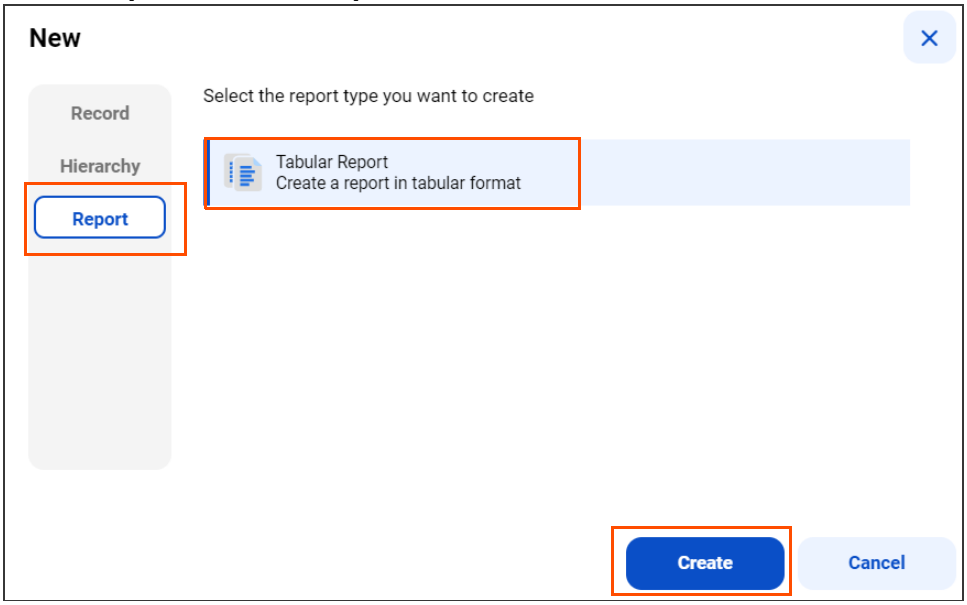
Creating Custom Report for Individual Records

You will create a custom report showing the number of states a customer record has as part of their address.

1. From the left menu, click **New**.

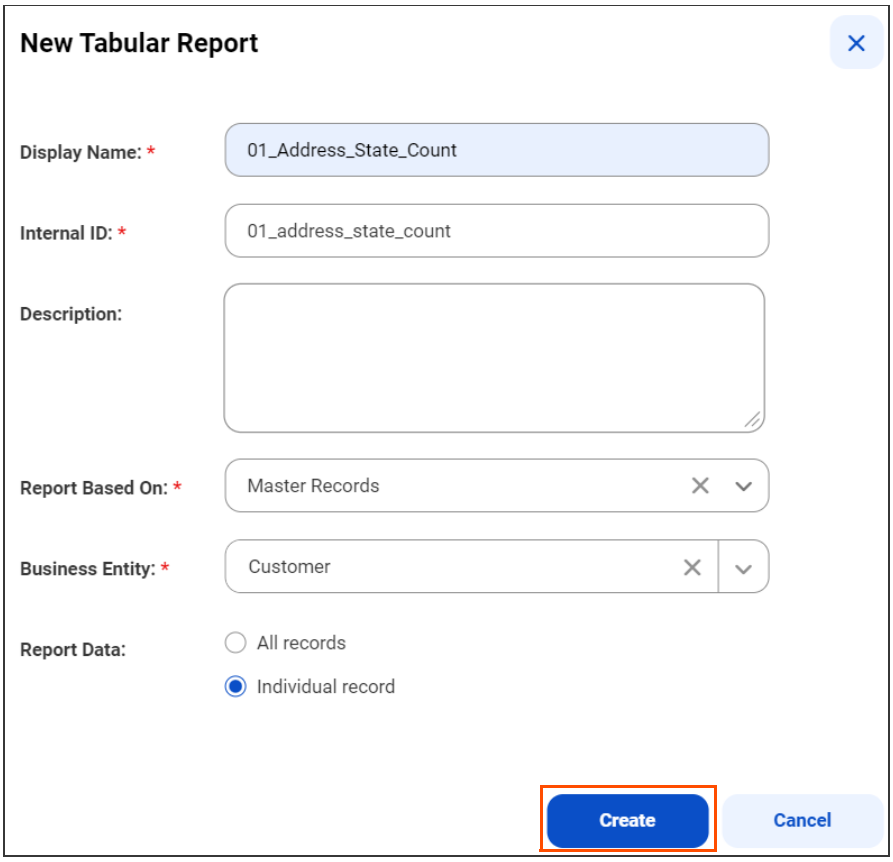


2. Select **Report** > **Tabular Report** and click **Create**.



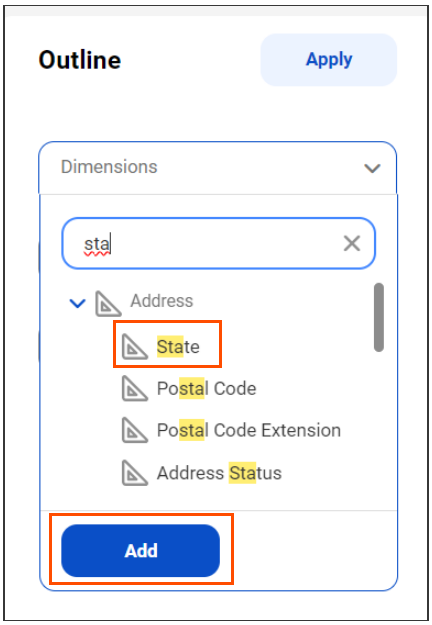
3. In the **New Tabular Report** window, enter the following details and click **Create**.

Field	Value
Display Name	<studentNumber>_ Address_State_Count
Report Based On	Master Records
Business Entity	Customer
Report Data	Individual Records



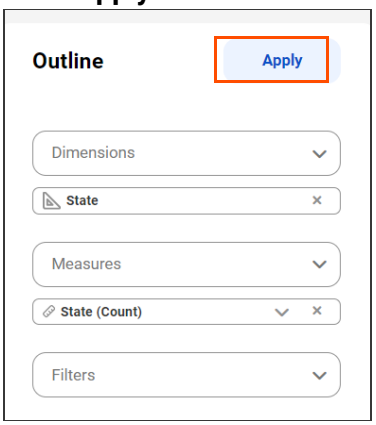
A page with the name that you specify in the Display Name field appears.

4. Click **Dimensions**. Search and under the **Address**, select **State** and click **Add**.



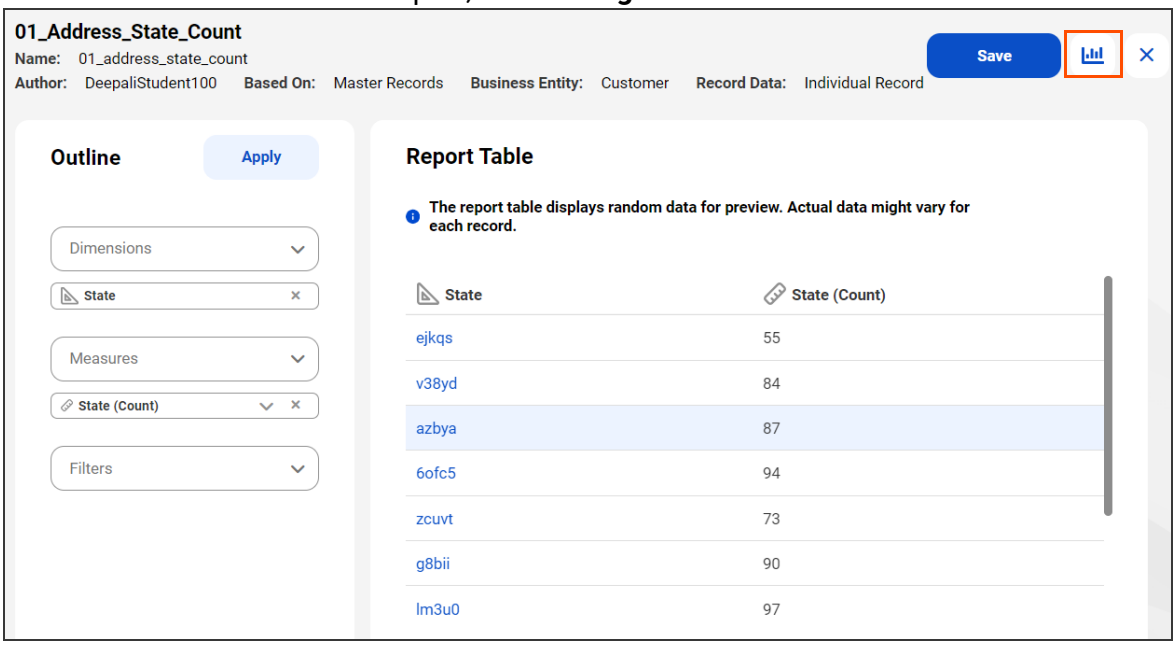
5. Similarly, search and select **State** as Measures and click **Add**.

6. Click **Apply**.



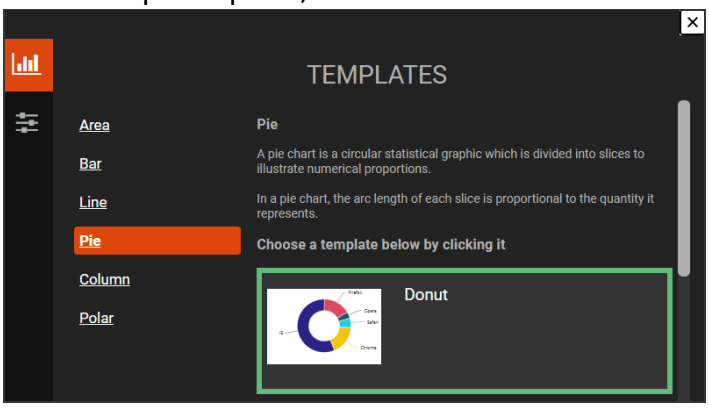
A preview of the report appears in a tabular format. The tabular report displays up to 100 rows of sample data for preview.

7. To add a chart to the custom report, click **Configure Chart** icon.

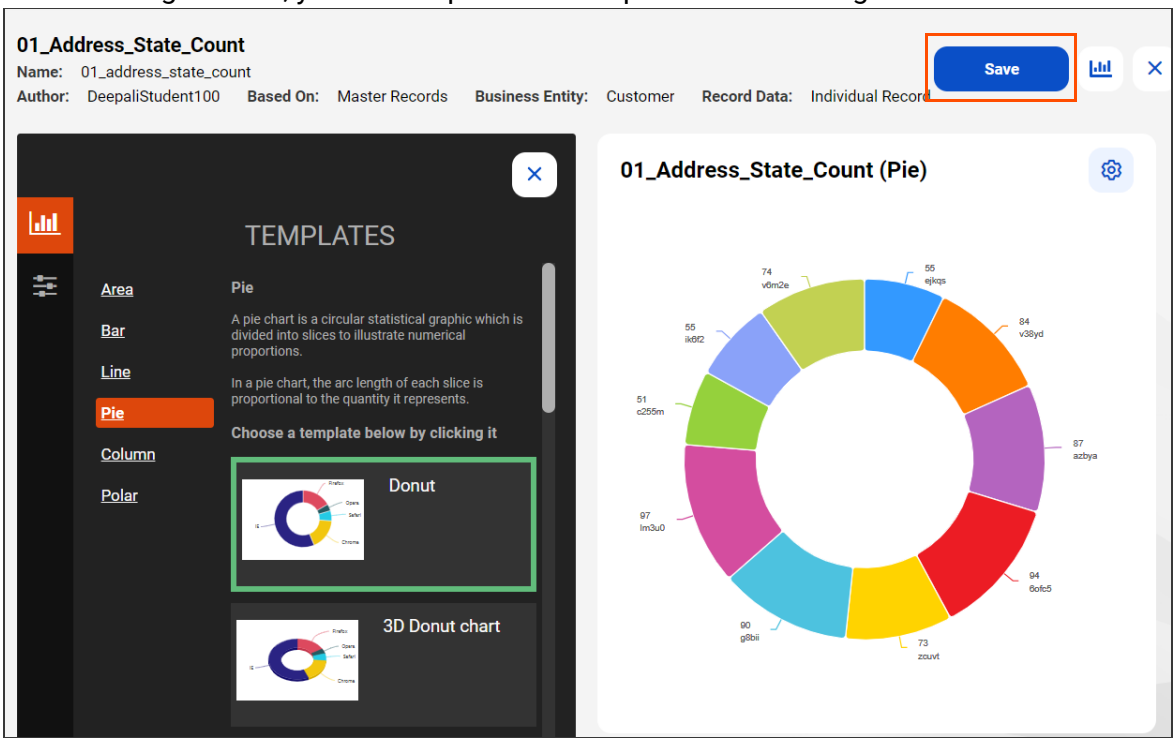


The data in this report comes from a single record. The random values you are seeing in the State field are the metadata of the field values in a business entity.

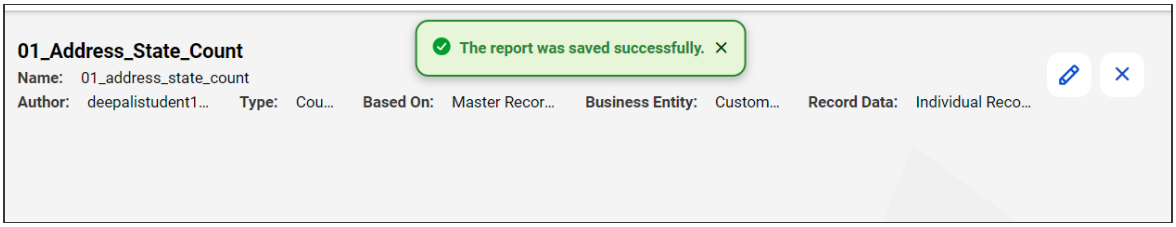
8. In the Templates panel, select **Pie** > **Donut**.



9. Click **Save**.
Chart previews are not available for individual records. Since these charts use real-time data from existing records, you cannot publish the report for scheduling.



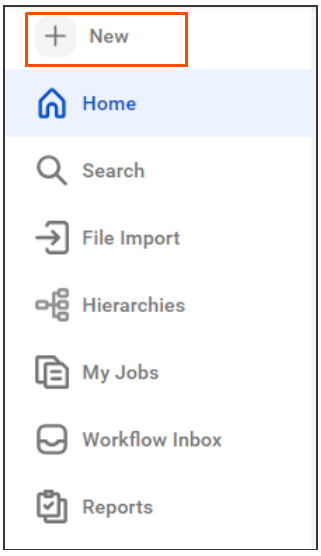
The preview will be unavailable for the individual records as shown below:



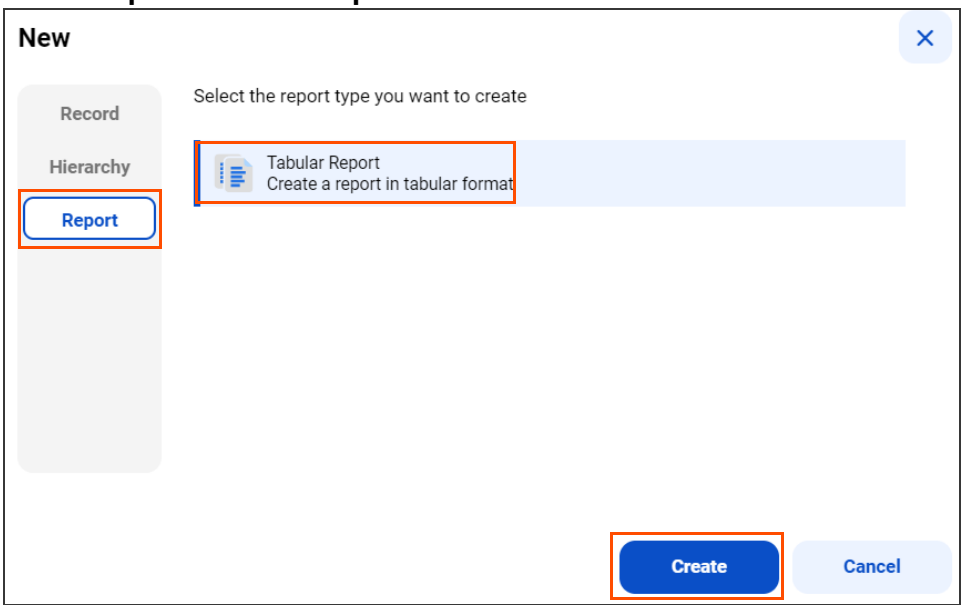
Creating Another Custom Report

Create another custom record that shows the multiple phone records.

10. From the left menu, click **New**.

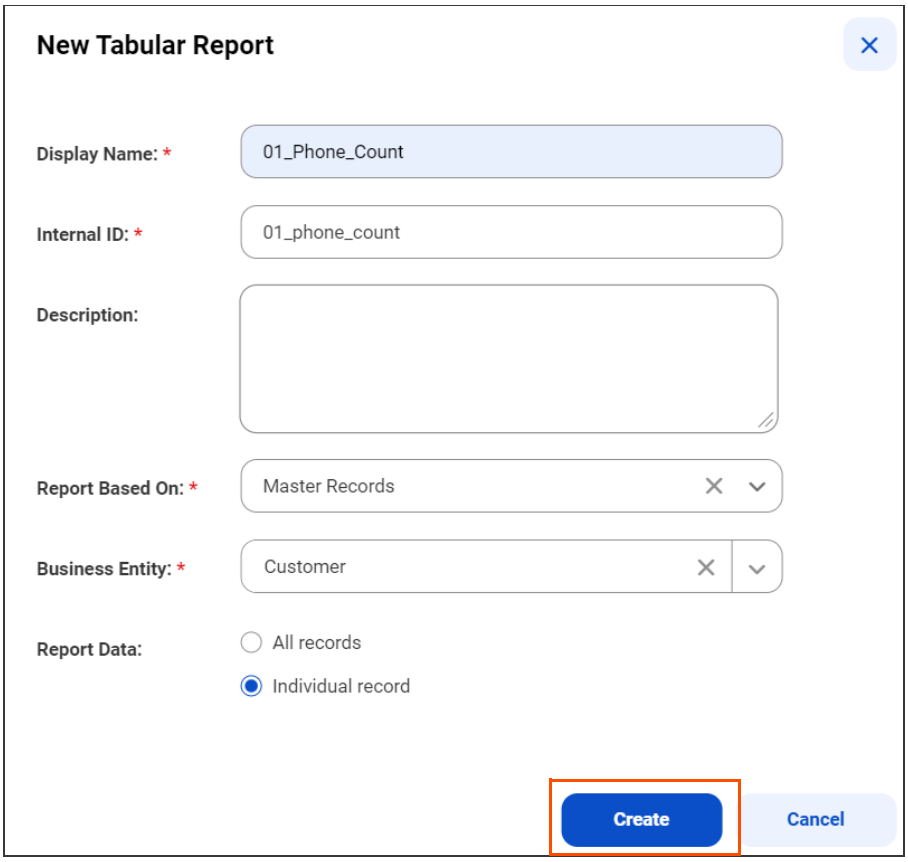


11. Select **Report** > **Tabular Report** and click **Create**.



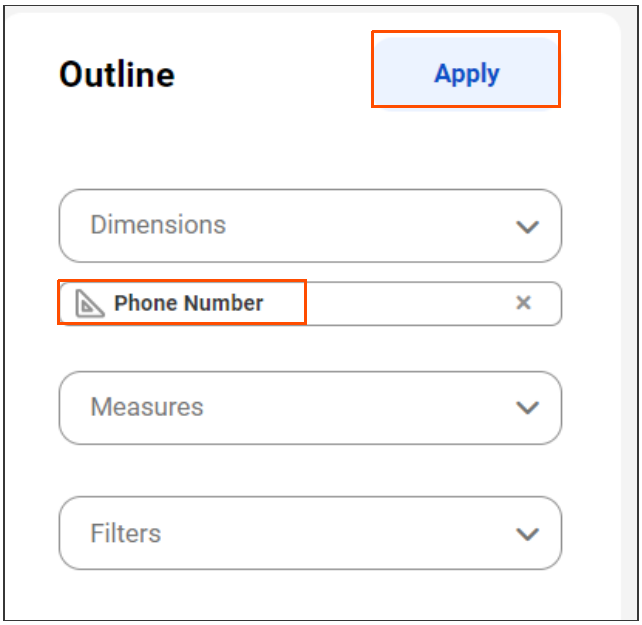
12. In the **New Tabular Report** window, enter the following details and click **Create**.

Field	Value
Display Name	<studentNumber>_ Phone_Count
Report Based On	Master Records
Business Entity	Customer
Report Data	Individual Records

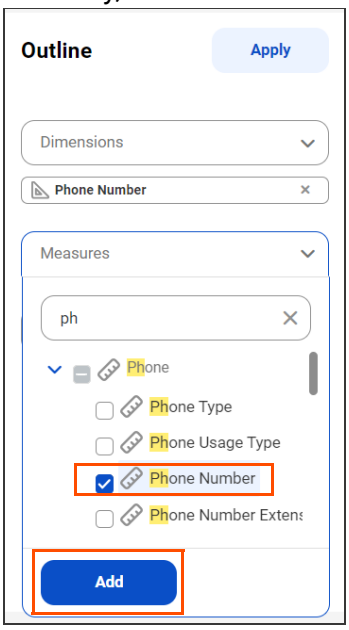


A page with the name that you specify in the Display Name field appears.

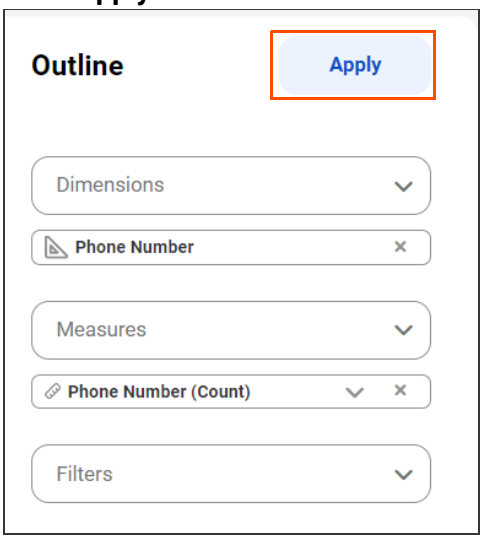
13. Click **Dimensions**. Under the **Phone**, select **Phone Number** and click **Add**.



14. Similarly, search and select **Phone Number** as Measures and click **Add**.

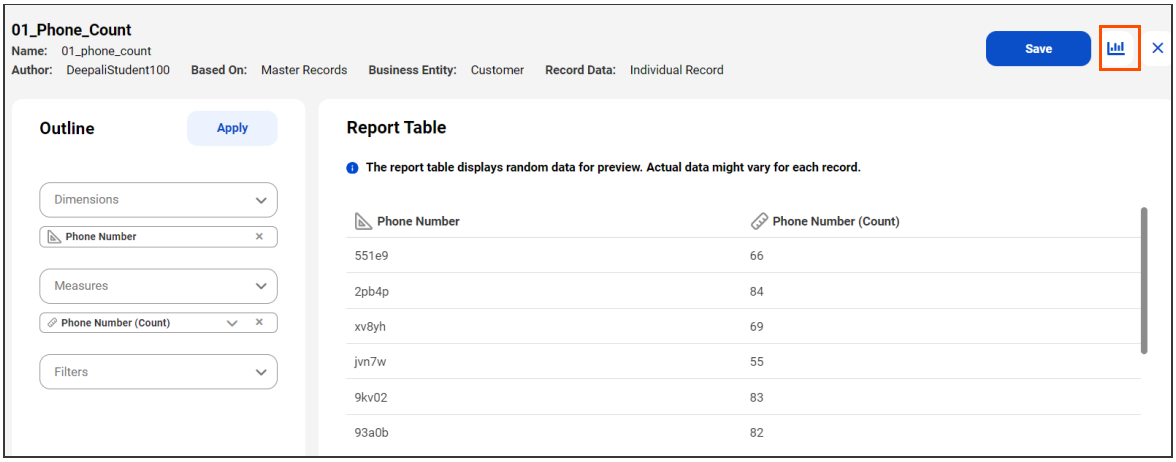


15. Click **Apply**.



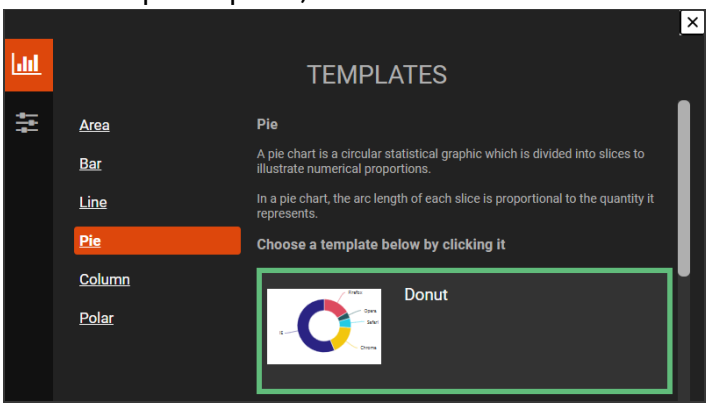
A preview of the report appears in a tabular format. The tabular report displays up to 100 rows of sample data for preview.

16. To add a chart to the custom report, click **Configure Chart** icon.

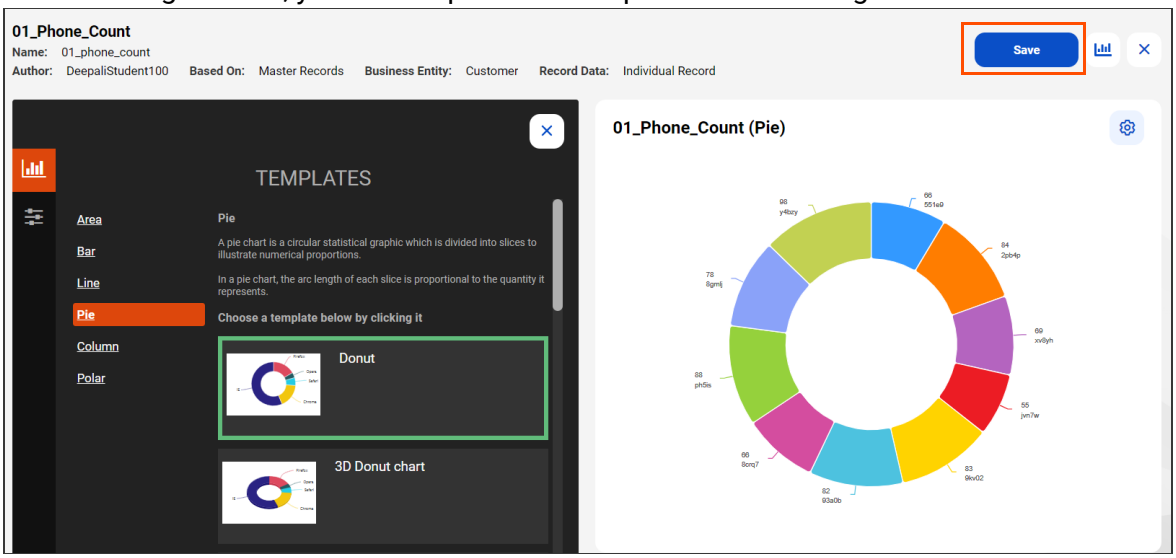


The data in this report comes from a single record. The random values you are seeing in the State field are the metadata of the field values in a business entity.

17. In the Templates panel, select **Pie** > **Donut**.



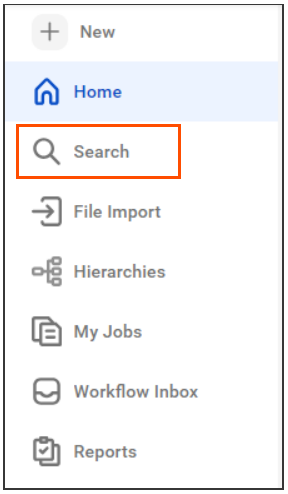
18. Click **Save**.
    Chart previews are not available for individual records. Since these charts use real-time data from existing records, you cannot publish the report for scheduling.



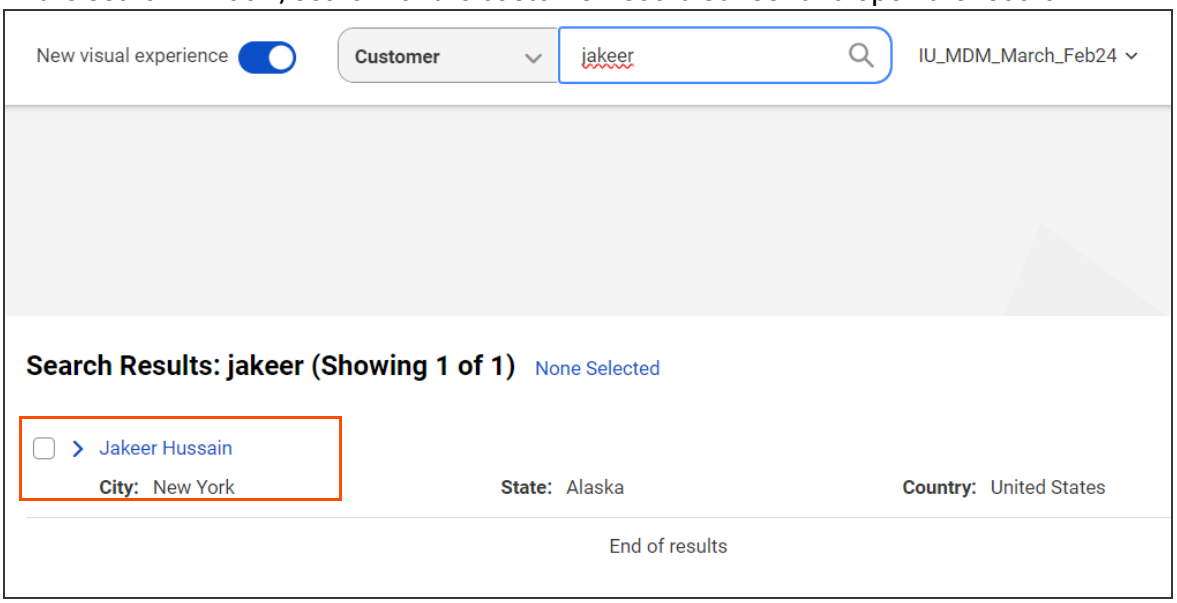
Add Widgets to the Record

A widget is a placeholder user interface element for reports. You will now add a widget to a record to display the reports that you just created.

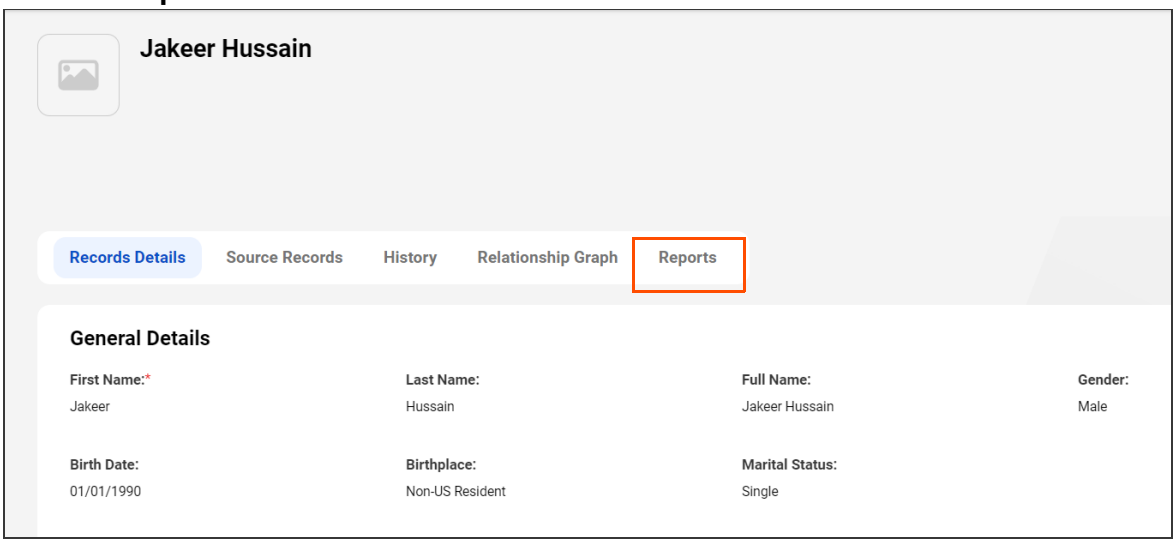
19. In the Multidomain MDM application, click **Search** from the menu.



20. In the search window, search for the customer record **Jakeer** and open the record.

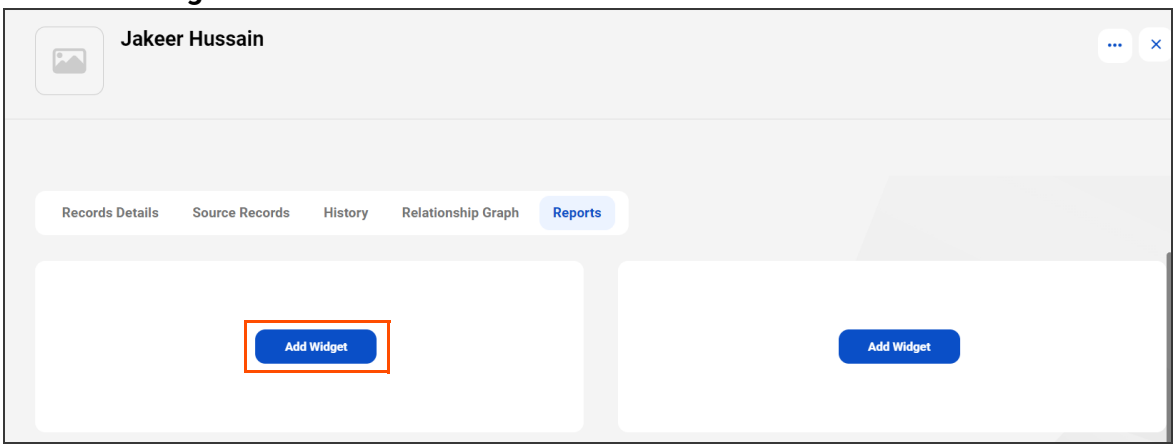


21. Click the **Reports** tab.

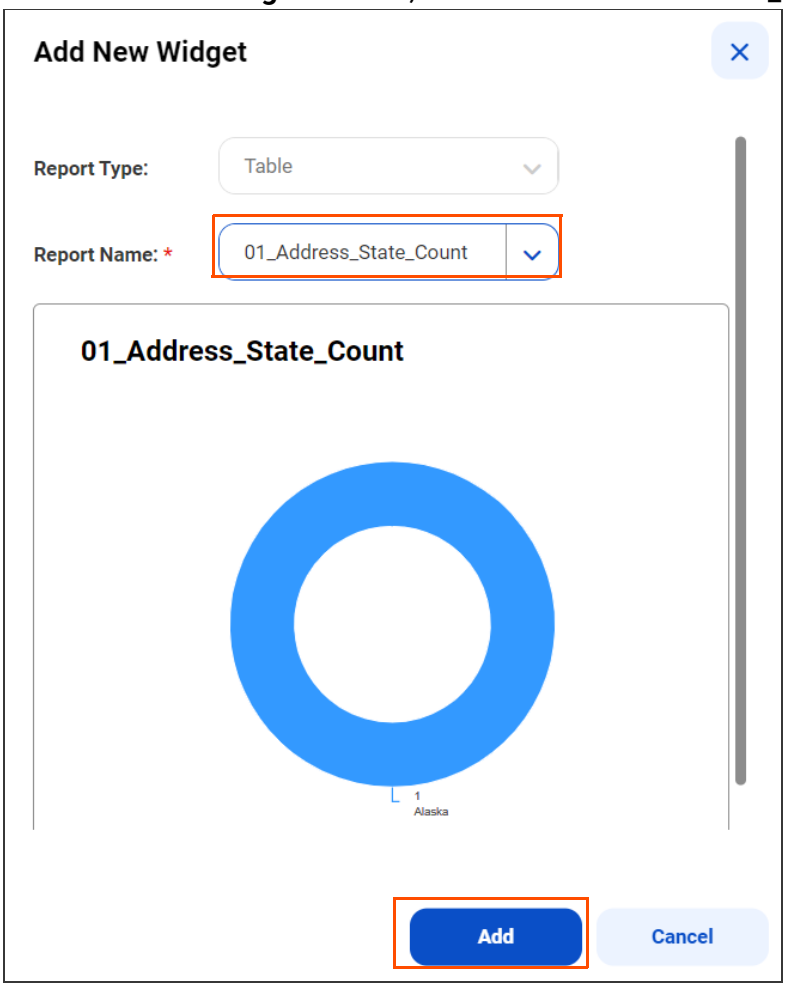


You will see the option to add the widget. You will add the individual reports here.

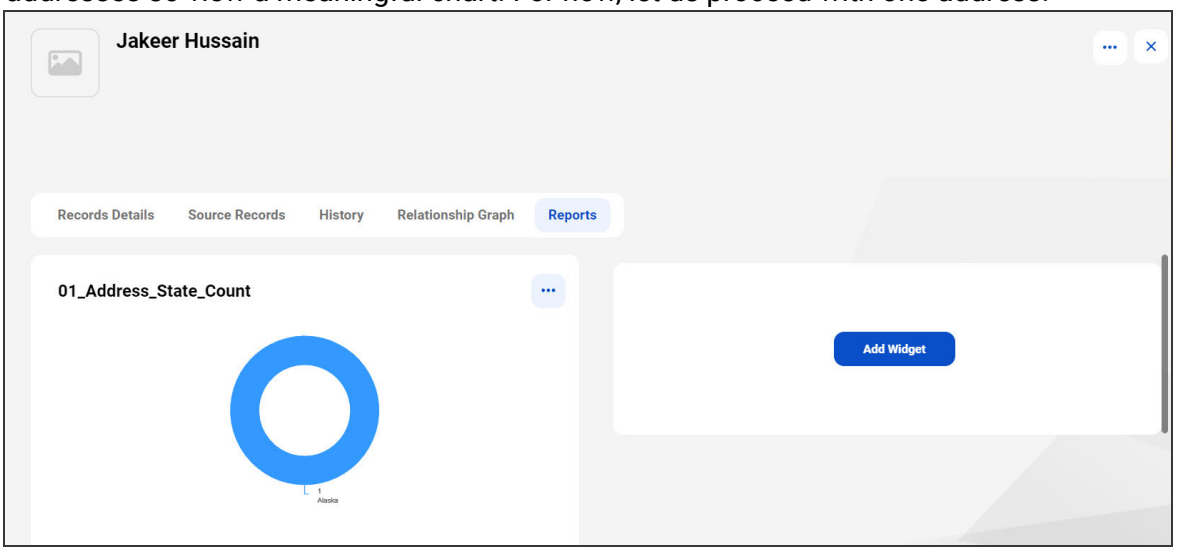
22. Click **Add Widget**.



23. In the **Add New Widget** window, search and select **Address_State_Count** report. Click **Add**.



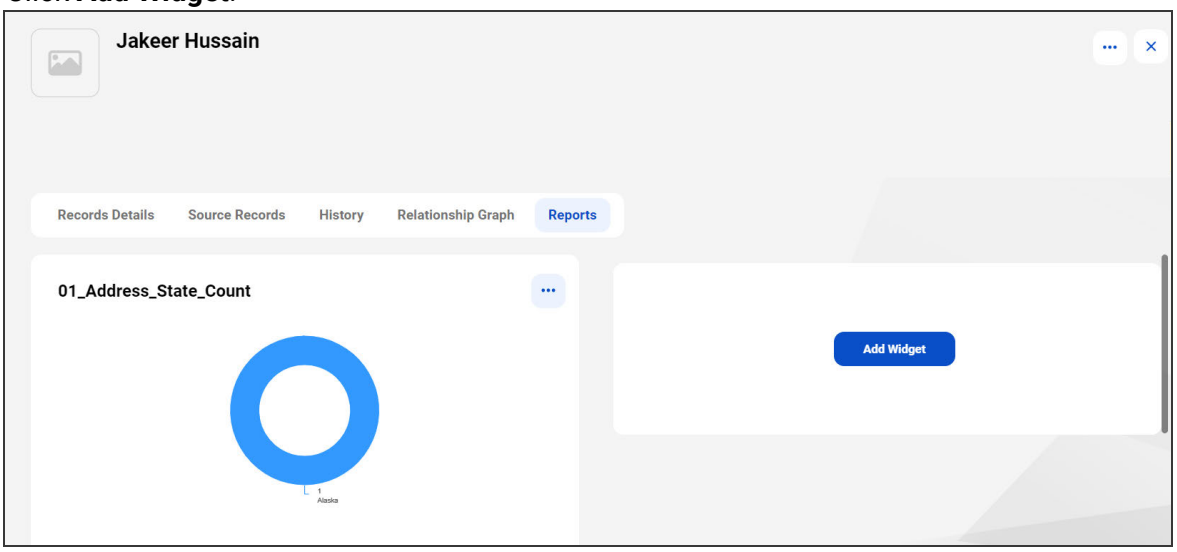
24. You will see that there is only address available for this record. You can add 2 new addresses so view a meaningful chart. For now, let us proceed with one address.



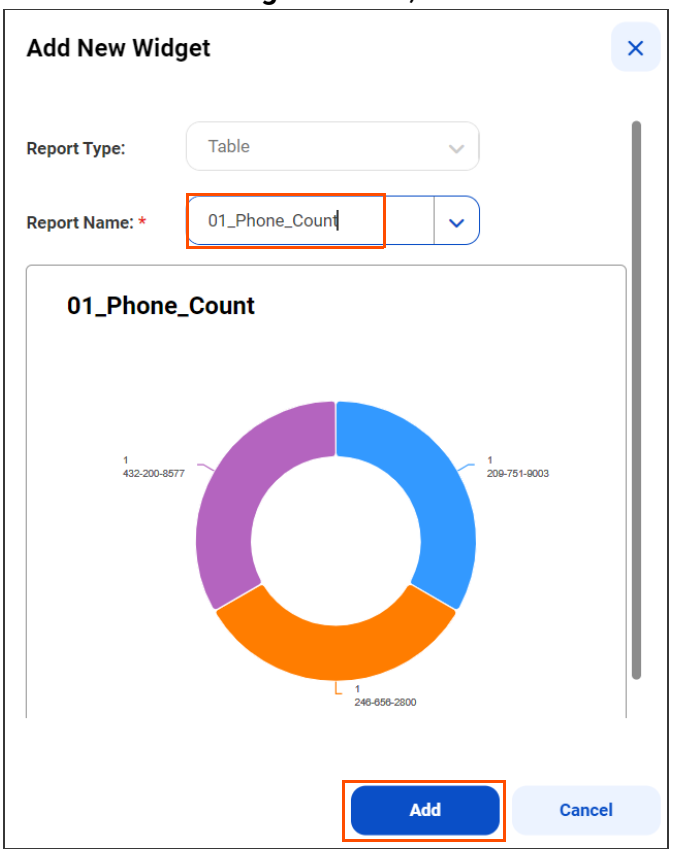
Self-Try Exercise (Mandatory)

Add the second custom report to the record.

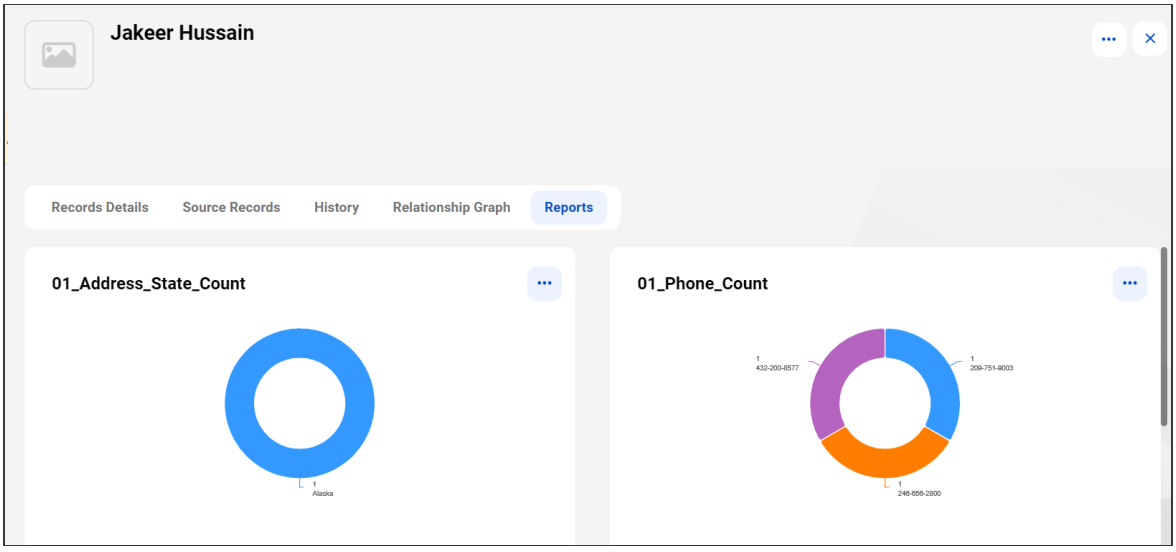
25. Click **Add Widget**.



26. In the **Add New Widget** window, search and select **Address_State_Count** report. Click **Add**.



You can see the second chart updates with the data.



This is how you can create reports for individual record.

This concludes the lab.

Key Takeaways

- Multidomain MDM allows you to gain meaningful insights into your business data and business outcomes with the help of reports.
- You can create a custom report in Multidomain MDM and add a chart to the custom report.
- To create schedules and publish custom reports, you must have an Admin role.
- Multidomain MDM Home page acts as a dashboard that can display charts.
- You can create up to six different dashboards using the predefined dashboard layouts.

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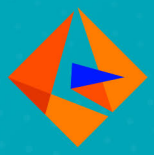

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